Algebraic Topology

— ***EYNTKA*** **Series** —

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Introduction

Algebraic topology is the study of topological spaces by assigning to them an algebraic object (integers, vector spaces, etc) in a "functorial" manner: its primary strategy is to translate topological structures into algebraic information. By assigning algebraic invariants such as groups or rings to topological spaces, we can measure and capture properties of their underlying shapes. Modern algebraic topology has a multitude of ways of assigning algebraic information each creating its own theory, including but not limited to:

1. Homotopy theory: different ways n-spheres can map into a spaces
2. Homology and Cohomology: famously said to formalize the notion of holes in n-dimensions. A different intuition that is arguably more practical is that it measures the failure of a function or section on a space X that can be defined on an (open) subset $U \subseteq X$ to be extended to the entire space X[1].
3. K-theory: measuring how a vector bundle fails to be trivial
4. Cobordism: measuring how much the failure of a n-manifold being the boundary of a $n+1$-dimensional manifold

In this book, our primary focus will be on *homotopy*, *homology*, and *cohomology*, K-theory and cobordism each can be considered *generalized cohomology theories* (see chapter 8). Each of these theories associates an algebraic object $A(X)$ to a given topological space X. Crucially, this assignment is strictly functorial: a continuous map $f : X \to Y$ between spaces induces a corresponding algebraic homomorphism $A(f) : A(X) \to A(Y)$. In fact, all of these theories will be *homotopic* (defined in chapter 1), namely if $f, g : X \to Y$ are continuous and $f \sim g$ are homotopic then $A(f) = A(g)$. Through the power of functoriality, we will see that many difficult geometric questions can be systematically reduced to problems in algebra. Some celebrated examples of theorems we will prove using these methods include:

[1]There are many nuances on what extension means, how independent of a choice from the original set U it is or the choice of simplicial/Simplectic homology, what choice of coefficients we choose, all of which is explored in chapter 3 and chapter 4

1. Invariance of Dimension: $\mathbb{R}^n \ncong \mathbb{R}^m$ for $n \neq m$.
2. Brouwer Fixed Point Theorem: Any continuous map $f : D^n \to D^n$ from the closed unit disk to itself must have a fixed point.
3. Fundamental Theorem of Algebra: Every non-constant complex polynomial has a root.
4. Borsuk-Ulam Theorem: Any continuous map from the sphere S^n to $\mathbb{R}^n$ must map some pair of antipodal points to the same value.
5. The Hairy Ball Theorem: A continuous tangent vector field on an even-dimensional sphere (such as S^2) must vanish at some point. (Colloquially: “you cannot comb a hairy ball flat without creating a cowlick”).
6. Generalized Jordan Curve Theorem: Any continuous embedding of a sphere S^{n-1} into $\mathbb{R}^n$ separates the space into exactly two connected components, an “inside” and an “outside”.
7. The Ham Sandwich Theorem: Given any n measurable “objects” in $\mathbb{R}^n$, there exists a single $(n-1)$-dimensional hyperplane that divides the volume of each object exactly in half.
8. Nielsen-Schreier Theorem: Every subgroup of a free group is itself a free group. (This is a purely algebraic theorem that is rather technical to prove using pure algebra (see [19, appendix] for a purely algebraic proof), but becomes a simple application of covering spaces in topology).
9. Lefschetz Fixed Point Theorem: A vast generalization of the Brouwer fixed point theorem which states that if a certain homological invariant (the Lefschetz number) of a map $f : X \to X$ is non-zero, then f must have a fixed point.
10. H-Space Structures on Spheres: The only spheres that can be equipped with a continuous multiplication operation (making them an H-space) are S^0, S^1, S^3, and S^7, which corresponds to the existence of the real numbers, complex numbers, quaternions, and octonions.

By translating geometric settings into an algebraic framework, we gain access to computational methods that are otherwise unavailable in pure point-set topology or geometry.

A core strategy in computing these invariants is breaking down complicated spaces into simpler, well-understood building blocks. We will extensively study spaces constructed by gluing simple geometric objects such as simplices (generalized triangles) or topological disks-together along their boundaries. Familiar spaces like the torus, spheres, the Klein bottle, and projective spaces all admit such cell structure decompositions. Working with these structures gives our algebraic calculations a geometric content. A central theme of the subject is proving that our algebraic invariants depend only on the intrinsic topology of the space itself, and are independent of any specific choice of decomposition into simpler objects, thus a lot of early hands-on computations chooses a *representation* of the space, while the theory proves these results are *independent* of choice of representation. The independence of choice of representation shows it is a property *intrinsic* to the underlying space and not the choice of representation of the space.

A natural question arises: to what extent are these algebraic invariants preserved? At a minimum we ask that $A(X)$ be a topological invariant: if $X \cong Y$ are homeomorphic, then $A(X) \cong A(Y)$. Homeomorphism is a rigid notion of sameness, namely a homeomorphism is a *bijection* that is continuous in both directions; it is the precise sense in which the common geometric explanation that two spaces are the same if we may bend or stretch them into each others shapes but never tear, glue, or change dimension (like the famous coffee mug to doughnut example)

The algebraic invariants we shall build will be invariant under a greatly weaker conditions: invariant to the coarser relation of *homotopy equivalence* which no longer requires the map to be a bijection, so we may now also *collapse* and *thicken* our shape. For instance, continuously retracting a solid disk to its center, retracting the plane minus a point retracts onto a circle, retracing all of $\mathbb{R}^n$ deforms to a single point shall all be samples of homotopy equivalences. In this setting, a disk and a point are homotopy equivalent even though they are not homeomorphic: homotopy equivalence has forgotten dimension and local detail, remembering only the coarser "hole structure" of a space. Since every homeomorphism is a special case of a homotopy equivalence, this really is a weakening, namely more spaces become equivalent. That flexibility lets us replace a complicated space by a homotopy-equivalent simple one, compute the invariants of the simple model, and read off the answer for the original.

0.0.1 For the Expert There is a structural reason the invariants are homotopic rather than homeomorphic: each is computed from a chain complex, and chain complexes are only well-defined *up to chain homotopy*. A topological homotopy $f \simeq g$ induces a chain homotopy between the maps they induce, so f and g act identically on homology. In the extreme case where a space contracts to a point, this becomes a *contracting homotopy* s with $\partial s + s\partial = \mathrm{id}$, exhibiting the augmented chain complex as split exact.

0.0.2 Images in Book A lot of these images are from either Hatcher's *Algebraic Topology* ([14]) or Munkre's *Topology* ([18]). In the first edition, I shall create my own images. These were put there from when I was learning the material and wanted a quick reference.

0.0.3 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7$\star$; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

1

Combinatorial Topology and Homotopy

Let X be a topological space. Combinatorial topology aims to create a new structure on X that will be thought as a way of interpreting X as build up by simpler spaces (most commonly circles and triangles, along with their higher dimensional counter-parts). How they are glued and how they are chosen to map into our space X will determine the "type" of combinatorial object we are working with. In this chapter, we shall explore the common *CW complex* that will have many important properties in the rest of this book.

We shall also explore a generalization of the notion of a continuous map known as a *homotopy*. As we shall see, a homotopy will be able to distort a shape much more drastically than a continuous map, but as added benifit will be more flexible and give many new maps between spaces that previously where unconnected. These spaces will share many properties that are invariant under the stronger deformation given by homotopies. In the following chapters, we shall see that all the algebraic structure we shall explore shall be invariant under homotopies, showing how they the right notion of morphism in algebraic topology.

1.1 Homotopy and Homotopy Type

For notational simplicity, we'll take the convention that

$$I = [0, 1]$$

Definition 1.1.1: Homotopy

Let X, Y be topological spaces, and suppose we have two maps f, f' which are continuous maps from X to Y. We say that f is *homotopic* to f' if there is a continuous map $F : X \times I \to Y$ such that

$$F(x, 0) = f(x), \qquad F(x, 1) = f'(x)$$

F is called a homotopy between f and f', and we write this as $f \simeq f'$

Thinking of the graphs of functions as shapes, if we can continuously deform one shape into another, the two shapes are homotopic! Note that the topology from which we define continuity ($X \times I$) has *both* the topology of X and the Euclidean topology to define it. The "continuous deformation" is more reliant of the euclidean portion, meaning we can use our usual intuition for the most part of deformations. In terms of I being a closed interval, we can think of this as having "one minute of time to change it, and we must do it in that time" (due to the compactness of the interval). To formalize this intuition, homotopy can be interpreted another way: using the compact-open topology on $[0, 1]$ and the fact that $[0, 1]$ is locally compact, it can be shown that a homotopy can equivalently be defined as a continuous map

$$f : X \to Y^{[0,1]}$$

which sends each point x to a path in Y. Hence, a homotopy between two maps f, g can be thought of as saying saying there is a continuous path from every point $f(x) \in Y$ to a point $g(x) \in Y$. Notice the importance of the domain in this definition: Even if $f(X) = \{y\}$, a homotopy will assign a path for each $x \in X$, hence we shall find that two images of different dimensions are homotopic (as we'll demonstrate).

Example 1.1.2: Homotopies

1. Let $f, g : \mathbb{R} \to \mathbb{R}^2$, $f(x) = (x, x^3)$ and $g(x) = (x, e^x)$. Then the map $F : R \times I \to \mathbb{R}^2$,

   $$F(x, t) = (x, (1 - t)x^3 + te^x)$$

   Is a homotopy. $F(x, 0) = (x, x^3)$, and $F(x, 1) = (x, e^x)$, are easily checked to be continuous. Thus

   $$f \simeq g$$

2. Take $f : I \to I$ $x \mapsto x^2$. Then

   $$F(x, t)((1 - t)x^2 + tx)$$

   is a homotopy. More generally, x^2 can be any polynomial, and this homotopy will work.

3. Take $f : \mathbb{R} \to \mathbb{R}$ be a continuous function, and $g : \mathbb{R} \to \mathbb{R}$, $g(x) = 0$. Then:

   $$F : \{\mathbb{R}\} \times I \to \mathbb{R} \qquad F(x, t) = (1 - t)f(x)$$

   is a continuous function, with $F(x, 0) = f(x)$ and $F(x, 1) = 0$. Thus, every continuous function on $\mathbb{R}$ is homotopic to the constant function.

   If we think of the graphs of the functions, the point $\{0\}$ with the trivial topology any shape of $\mathbb{R}$ with the euclidean topology can be continuously deformed to on another.

   More generally, if F is a homotopy such that it's construction is:

   $$F(x, t) = ((1 - t)f(x) + tg(x))$$

then it is a Straight-line homotopy

4. Let $f, g : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}^2 \setminus \{0\}$ where f is the identity map and $g(x) = \frac{x}{\|x\|}$. Then there is a homotopy between f and g, can you think of it? We shall later show that f is not homotopic to a constant map. However, if the domain and codomain was $\mathbb{R}^2$, then f *would* be homotopic to a consant map (the staright-line homotopy works). If the domain is $\mathbb{R}^2$ and the codomain is $\mathbb{R}^2 \setminus \{0\}$, if f is surjective, can it be homotopic to a constant map?

This last example hints at the importants of holes in our space. Can you repeat the same think in $\mathbb{R}^n$ by making gradually higher dimensional holes (remove a point, a line, a plane, etc)?

It should be verified that the collection of homotopies between two spaces forms an equivalence relation (exercise 1.3.1(1.3.1 (1))). Hence, it is well-defined to take objects up to homotopy, motivating the following definition:

Definition 1.1.3: The Homotopy Category hTop

Let hTop denote the category whose objects are topological spaces and whose morphisms are homotopy equivalence classes of continuous functions.

The pointed version of this category, in which spaces carry a basepoint and both maps and homotopies are required to preserve it, is the setting for the axioms of chapter 8; it is recorded there as definition 8.1.1.

When function's are homotopic to the constant function, we give function a special name:

Definition 1.1.4: Nullhomotopic

If f is homotopic to a constant map, we say that f is *nullhomotopic*. An image of a function that is nullhomotopic is called *contractible*.

The idea of nullhomotopy is saying that if you deform your space in a continuous way such that you can bring it to a constant point, then we can think of the two being homotopically trivial.

Example 1.1.5: Nullhomotopic

Take any conected shape $U \subseteq \mathbb{R}^n$ that does not have a whole (say, a shape that is is homeomorphic to a unit ball). Then the identity map $f : U \to \mathbb{R}^n$ is homotopic to the constant map using the straight-line homotopy. Thus, all such shapes are contractible.

We now strive towards the notion of inverse homotopies and isomorphisms in hTop. The following can be thought of as a first step in that direction as well as an important notion in it of itself:

Definition 1.1.6: Deformation Retraction

Let $f : X \to X$ be the identity map and $A \subseteq X$ a subspace. Then if there exists a homotopy $f_t : X \times I \to X$ such that:

1. $f_0 = \text{id}$
2. $f_1(X) = A$
3. $f_t|_A = \text{id}$ for all $t \in I$

Then F is said to be a *deformation retraction* of the space X to A. If $f_t : X \times I \to Y$ is a family of homotopy where for some subspace $A \subseteq X$, $f_t|_A$ is independent of t, then we call the homotopy *homotopy relative* to A, or *homotopy rel* A.

Think about taking bold-face letters or higher dimensional objects like S^n punctured at the poles and retracting them to their thin counterparts or lower dimensional counterpart. Note that being retractable to a point is slightly stronger than being contractible since the map must restrict to the identity to a point. Every deformation retraction is homotopy relative to a subspace A.

Each deformation retraction that retracts a map f_0 onto a map f_1 called a *retraction*:

Definition 1.1.7: Retraction

Let $f : X \to X$ be continuous map. Then f is said to be a *retraction* on A if $f(X) = A$ and $f|_A = \text{id}$.

Note how the definition of retraction has no notion of a homotopy, in particular there are less conditions that need to hold for a retraction.

Example 1.1.8: Retraction

1. If $X = Y \times Z$, and $A = Y \times \{z_0\}$, then $f(y, z) = (y, z_0)$ is a retraction.
2. if A is an annulus embedded into $\mathbb{R}^3$, and we have a circle in the annulus, then pinching everything to A is a retraction[a].
3. $X = \mathbb{R}$, $A = \mathbb{R}_{\geq 0}$, then $\sigma(t) = |t|$ is a retraction

[a]Note how it was important for A to be embedded in $\mathbb{R}^3$, no such pinching can occur intrinsically in A

Thus, each deformation retraction is homotopic to a retraction map. Note that for each $x \in X$, we have a retraction map to $\{x\}$, however since not all spaces are simply connected, not all spaces have a deformation retraction to a retraction (as we shall shortly see).

Example 1.1.9: Deformation Retraction

1. The following is three different deformation retraction of $\mathbb{R}^2$ with two punctured holes:

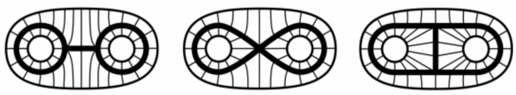


Figure 1.1: $\mathbb{R}^2 \setminus \{a, b\}$ retracted to different curves

2. There is a nice way of visualizing a deformation retraction. For $f_t : X \to Y$, take

$$M_f := (X \times I) \sqcup Y/[((x, 1) \sim y \text{ iff } f_1(x) = y)]$$

which we may visualize as the following:

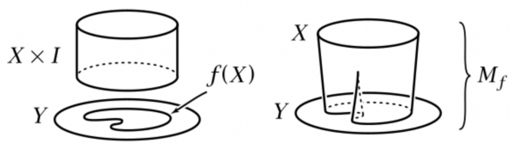


Figure 1.2: Mapping Cylinder of f

Notice that M_f is also a deformation retraction of onto $f(X)$, namely by continuously dragging each (x, t) along $\{x\} \times I \subseteq M_f$ to $f(x)$.

3. Let $f : X \to X$ be a deformation retract onto a point $x \in X$. Show that for each open neighborhoods of x, there exists a $V \subseteq U$ such that $V \hookrightarrow U$ is nullhomotopic. Use this to show the only deformation retract of the space (called the infinite comb or pathological comb)

$$([0, 1] \times \{0\}) \cup (\{r\} \times [0, 1 - r]) \qquad r \in \mathbb{Q} \cap [0, 1]$$

are to point in $[0, 1] \times \{0\}$. Hence, a space like so:

Figure 1.3: Contractible, but not Deformation Retractable

is contractible, but not deformation retractable[a].

[a]Though, there is a *weak deformation retract* from the zig-zagy space above to the darker line

Retraction are useful since they allow us to generalize the notion of invertibility, building us up to isomorphism in hTop:

Definition 1.1.10: Homotopy Equivalence

Let $f : X \to Y$ be a continuous map. Then f is called a *homotopy equivalence* if there exists a g such that $f \circ g \simeq \text{id}$ and $g \circ f \simeq \text{id}$. In this case, we shall write $X \simeq Y$ and call them *homotopy equivalent* or that they have the same *homotopy type*. In hTop, homotopy equivalence are the isomorphisms.

Example 1.1.11: Homotopy Equivalent

1. Let's say the shape X is contractible so that the identity $F : X \to X$ has a homotopy $F_t : X \times I \to X$ to $F_1(X) = \{x\}$. Consider the map $f : X \to \{*\}$ where $f(X) = *$ and homotopy inverse $g : \{*\} \to X$ where $g(*)$ maps to $\{x\}$. Then the map
$$g \circ f : X \to X \qquad g \circ f(X) = \{x\}$$
is homotopy equivalent to $\text{id} : X \to X$ by taking $(g \circ f)_t = F_{1-t}$.

2. A non-trivial example would be that the 3 shapes in figure 1.1 are homotopy equivalent. This also shows that homotopy equivalence need not perserve cardinality, or even dimension, and hence is certainly weaker than a homeomorphism[a].

3. Let $f : X \to X$ be a deformation retract onto A. Then $A \hookrightarrow X$ is a homotopy equivalent (this can be thought as the generalization of the contractible argument). Slightly weaker condition that still works is the following: if $f : X \to X$ has a homotopy such that $f_0 = \text{id}$, $f_1(X) \subseteq A$ and $f_t(A) \subseteq A$, then f is called a *weak deformation retract* onto A. Show that $A \hookrightarrow X$ is still a homotopy equivalence.

[a]The simplest example that only requires a notion of connectivity is showing that $\mathbb{R} \not\cong \mathbb{R}^2$

A large portion of this book will be examining what properties are invariant under homotopy equivalence. As pointed out, cardinality and dimension are not preserved, so in a sense they preserve more "core" topological features. We shall see that number of "holes" will be preserved. Though this might seem like a triviality at first glance, this shall have major implications for finding when certain derivatives exist[1], and more generally for an important rough classification of topological spaces via homotopy and homology/cohomology groups.

1.1.12 Homotopies between Compact Surfaces An interesting point to bring up is that two *closed* surfaces (compact, without boundary, of dimension 2) that are homotopy-equivalent are in fact homeomorphic, so that homotopy adds no extra flexibility when working with closed surfaces. Boundary matters here: the annulus and the Möbius band are homotopy equivalent (both deformation retract to a circle) but not homeomorphic (see [14][a]). This is certainly not the case in general.

[a]This comes down to the classification of compact surfaces; it is up to genus and orientation, and these two characterize compact surfaces and are preserved by homotopies, and hence if they are homotopic, by the characterization they must be homeomorphic

[1]for those who know differential forms, when a closed form is exact

A common theme in geometry and algebraic topology is that a function can be represented by some space. The following demonstrates this principle:

Lemma 1.1.13: Homotopy Equivalence Space Representation

Let $X \simeq Y$. Then there exists Z such that X and Y are deformation retracts of Z

***Proof*:**
Let $Z = M_f$.

We finish off this little section with an example of how being contractible may be eaiser than being a (deformation) retractible:

Example 1.1.14: House With Two Rooms

Stare at this image for a bit:

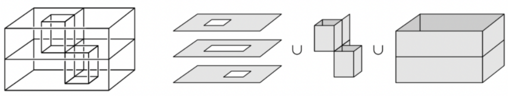


Figure 1.4: House With Two Rooms, Hatcher

If X represents this space, Let X_ϵ be this space with each point thickened with a filled in ϵ-ball sufficinetly small so that X_ϵ can be deformation retraction to X (in fact, is homotopy equivalent to X: if $r : X \to X_\epsilon$ is the retraction and $\iota : X_\epsilon \to X$ is the inclusion, $r\iota \simeq \text{id}$ and $\iota r \simeq \text{id}$). is homeomorphic to the filled in closed ball $D^3 \subseteq \mathbb{R}^3$. Then D^3 is certainly contractible to a point, say $\{*\}$. Hence:

$$X \simeq X_\epsilon \simeq D^3 \simeq \{*\}$$

Hence by transitivity X is contractible. It is certainly possible to show it is also retractible, however this is a bit harder to show.

1.2 Cell-Complex

As is often the case in mathematics, we understand well some simple examples, and would like to create more complicated examples using the simpler family of objects. There are many simple geometric shapes which we may choose that have very simple combinatorial properties; for example the disk, triangles, and so forth. Let us stick to the disk and its higher dimensional counter-parts D^n. The rigid shape of D^n is not important, but its topological properties are. Thus, define an n-cell to be a topological space homeomorphic to D^n. Note that a 1-cell is homeomorphic a closed interval $[a, b]$.

Definition 1.2.1: CW Complex

A *CW complex* or a *cell complex* is a space constructed in the following form:

1. Start with a discrete set X^0 whose points are regarded as 0-cells
2. Inductively, form the n-skeleton X^n from X^{n-1} by attaching n-cells e^n_α via gluing maps
$$\varphi_\alpha : \partial e^n_\alpha \to X^{n-1}$$
where $\partial e^n_\alpha = S^{n-1}$. That is, X^n is the quotient space of the disjoint union $X^{n-1} \sqcup_\alpha D^n_\alpha$ of X^{n-1} with the collection of n-disks D^n_α under the identifications $x \sim \varphi_\alpha(x)$ for $x \in \partial D^n_\alpha$. As a set, $X^n = X^{n-1} \sqcup_\alpha e^n_\alpha / \sim$ where each e^n_α is an open n-disk.
3. The induction stops at a finite step, at which point we set $X = X^n$, or we continue this process indefinitely and set $X = \bigcup_n X^n$ in which case X is given the weak topology via inclusion ($A \subseteq X$ open if and only if $A \cap X^n$ for all n). If $X = X^n$, then X is said to be fintite dimensional with dimension n.

To break down the definition, it is perhaps more intuitively to see that we have

$$X^0 \hookrightarrow X^1 \hookrightarrow \cdots$$

where X_k is obtained by gluing copies of n-cells $(e^n_\alpha)_\alpha$ to X_{k-1} via gluing maps $g^k_\alpha : \partial e^k_\alpha \to X_{k-1}$. Then X is simply the colimit of this diagram with these maps

$$\varinjlim X_i = X$$

The term *CW* stands for "Closure-finite Weak topology".

For future reference, we define the following natural map associated to a cell-complex:

Definition 1.2.2: Characteristic Map

Let X be a cell complex and e^n_α associated to $\varphi_\alpha : \partial e^n_\alpha \to X^{n-1}$ be a cell forming this complex. Then there exists a map $\Phi_\alpha : e^n_\alpha \to X$ extending the map φ_α to be a homeomorphism from the interior of D^n_α to e^n_α, in particular $\Phi_\alpha : e^n_\alpha \hookrightarrow X^{n-1} \sqcup_\alpha D^n_\alpha \to X^n \hookrightarrow X$.

The name characteristic is due to its link to *Eulers characteristic* (see section 3.5.3).

definition 1.2.1 prescribes a construction and a topology, and says nothing about separation. The property one needs first is the weakest one, and it is what lets a compact subset of a cell complex be treated as closed and lets the paracompactness arguments of chapter 7 start.

Theorem 1.2.3: CW Complexes Are Hausdorff

Every CW complex is Hausdorff.

***Proof*:**
Write X^n for the n-skeleton, $\varphi_\alpha\colon S^{n-1} \to X^{n-1}$ for the attaching maps of the n-cells and $\Phi_\alpha\colon D^n_\alpha \to X^n$ for the characteristic maps of definition 1.2.2. Throughout, X^n carries the

quotient topology from $q\colon X^{n-1} \sqcup_\alpha D^n_\alpha \to X^n$, so a subset of X^n is open exactly when its preimage meets X^{n-1} and each D^n_α in an open set.

Step 1: disjoint open sets extend one dimension upward. Let U, V be disjoint open subsets of X^{n-1}. For each n-cell put

$$\hat{U}_\alpha = \left\{ rs \in D^n_\alpha \;\middle|\; s \in \varphi_\alpha^{-1}(U),\ \tfrac{1}{2} < r \le 1 \right\}$$

and define $\hat{V}_\alpha$ from V in the same way, rs being scalar multiplication in $D^n_\alpha \subseteq \mathbb{R}^n$. Each $\hat{U}_\alpha$ is open in D^n_α, being the intersection of D^n_α with the open set $\left\{ x \in \mathbb{R}^n \;\middle|\; \tfrac{1}{2} < |x| < 2,\ x/|x| \in \varphi_\alpha^{-1}(U) \right\}$. The set $U \sqcup_\alpha \hat{U}_\alpha$ is saturated for q: the only identifications are $s \sim \varphi_\alpha(s)$ for $s \in S^{n-1}$, and $s \in \hat{U}_\alpha$ holds exactly when $\varphi_\alpha(s) \in U$. So it descends to a set U' open in X^n with $U' \cap X^{n-1} = U$, and likewise V'. They are disjoint, since $U \cap V = \emptyset$ and $\varphi_\alpha^{-1}(U) \cap \varphi_\alpha^{-1}(V) = \emptyset$ forces $\hat{U}_\alpha \cap \hat{V}_\alpha = \emptyset$.

Step 2: every skeleton is Hausdorff, by induction on n. The set X^0 is discrete. Assume X^{n-1} Hausdorff and take $p \neq q$ in X^n. Each open cell $e^n_\alpha = \Phi_\alpha(D^n_\alpha \setminus S^{n-1})$ is open in X^n, its preimage being $D^n_\alpha \setminus S^{n-1}$ in that disk and empty elsewhere, and Φ_α restricts to a homeomorphism onto it.

1. If p and q lie in distinct open n-cells, those two cells separate them.
2. If they lie in the same open cell, that cell is homeomorphic to $D^n \setminus S^{n-1}$, which is Hausdorff, and the separating sets are open in X^n because the cell is.
3. If $q = \Phi_\alpha(x)$ lies in an open n-cell and $p \in X^{n-1}$, choose ρ with $|x| < \rho < 1$. Then $B = \Phi_\alpha(\{z \mid |z| < \rho\})$ is open in X^n and $C = \Phi_\alpha(\{z \mid |z| \le \rho\})$ is closed in X^n, since the preimage of each meets D^n_α in an open, respectively closed, ball and meets every other disk and X^{n-1} in the empty set. So B and $X^n \setminus C$ separate q and p.
4. If both lie in X^{n-1}, the inductive hypothesis separates them there and Step 1 pushes the separation into X^n.

Step 3: the whole complex. If the construction stops at a finite stage, Step 2 is the claim. Otherwise let $x \neq y$ in $X = \bigcup_n X^n$ and pick N with $x, y \in X^N$. Step 2 gives disjoint open $U_N, V_N \subseteq X^N$ separating them, and Step 1 extends these upward to disjoint open $U_n, V_n \subseteq X^n$ for every $n \ge N$ with $U_{n+1} \cap X^n = U_n$ and $V_{n+1} \cap X^n = V_n$. Put $U = \bigcup_{n \ge N} U_n$ and $V = \bigcup_{n \ge N} V_n$. These are disjoint, and each is open for the weak topology: for $n \ge N$ one has $U \cap X^n = U_n$, and for $n < N$ one has $U \cap X^n = U_N \cap X^n$, open in X^n as a subspace of X^N. Hence U and V separate x and y, completing the proof.

The three cases of Step 2 are the only ways two points of a skeleton can sit relative to the cells, and only the last needs the inductive hypothesis. The same extension device of Step 1 is what makes normality provable by the same induction, with Urysohn functions in place of the radial collars.

The Hausdorff property is the first half of what the later chapters need. The second is paracompactness, and it is proved by the same skeletal induction, now extending a partition of unity rather than a pair of separating open sets. The result matters well beyond this chapter: chapter 7 defines the characteristic classes of a vector bundle over a paracompact Hausdorff base, and without the theorem below those definitions would formally fail to apply to a CW complex, which is the case every computation uses.

Theorem 1.2.4: CW Complexes Are Paracompact

Let X be a CW complex and $\{U_\alpha\}$ an open cover of X. Then there is a partition of unity $\{\varphi_\beta\}$ subordinate to $\{U_\alpha\}$: continuous functions $\varphi_\beta \colon X \to [0,1]$ with locally finite supports, each support contained in some U_α, summing to 1. Consequently X is paracompact in the sense of [12, Paracompact Space], and being Hausdorff by theorem 1.2.3, it is paracompact Hausdorff.

***Proof*:**
The partition of unity is built over the skeleta by induction, following [15, Proposition 1.20].

Step 1: The induction and its extension step. On X^0, a discrete space, choose for each point one U_α containing it and let the corresponding function be the indicator of that point; this is a partition of unity subordinate to the cover, with locally finite supports because X^0 is discrete. Suppose a partition of unity $\{\varphi_\beta\}$ on X^n subordinate to $\{U_\alpha \cap X^n\}$ has been constructed. Let $\Phi_\gamma \colon D^{n+1} \to X$ be the characteristic map of an $(n+1)$-cell (definition 1.2.2). The composites $\{\varphi_\beta\Phi_\gamma\}$ form a partition of unity on $S^n = \partial D^{n+1}$, and since S^n is compact only finitely many of them are not identically zero.

Write points of $D^{n+1} \setminus \{0\}$ in spherical coordinates $(r,x) \in (0,1] \times S^n$, and let $\rho_\varepsilon \colon [0,1] \to [0,1]$ be continuous with $\rho_\varepsilon = 0$ on $[0, 1-\varepsilon]$ and $\rho_\varepsilon = 1$ on $[1-\varepsilon/2, 1]$. Extend each $\varphi_\beta\Phi_\gamma$ over D^{n+1} by $(r,x) \mapsto \rho_\varepsilon(r)\,\varphi_\beta\Phi_\gamma(x)$, and by 0 at the origin; the factor ρ_ε vanishes near 0, so this is continuous. Choosing ε small enough makes these extensions subordinate to $\{\Phi_\gamma^{-1}(U_\alpha)\}$. Indeed, fix one of the finitely many β with $\varphi_\beta\Phi_\gamma \not\equiv 0$ and let $K_\beta = \operatorname{supp}(\varphi_\beta\Phi_\gamma) \subseteq S^n$, a compact set contained in the open set $W_\beta = \Phi_\gamma^{-1}(U_{\alpha(\beta)})$. The support of the extended function lies in the radial collar $\{(r,x) : r \geq 1-\varepsilon,\ x \in K_\beta\}$, whose points are within ε of K_β in the metric of D^{n+1}. Since K_β is compact and $D^{n+1} \setminus W_\beta$ is closed and disjoint from it, their distance is positive, so the collar lies in W_β once ε is smaller than that distance. Let $\varepsilon = \varepsilon_\gamma$ be *half* the minimum of these finitely many distances, which is strictly smaller than each of them.

Step 2: Filling the interior. The disk D^{n+1} is compact Hausdorff, hence paracompact, so the open cover $\{\Phi_\gamma^{-1}(U_\alpha)\}$ admits a subordinate partition of unity $\{\psi_{\gamma j}\}$ ([12, Existence of Subordinate Partitions of Unity]), finite after discarding the identically zero members, the disk being compact. The family

$$\{\, \rho_\varepsilon\, \varphi_\beta\Phi_\gamma \,\} \cup \{\, (1-\rho_\varepsilon)\, \psi_{\gamma j} \,\}$$

is then a partition of unity on D^{n+1} subordinate to $\{\Phi_\gamma^{-1}(U_\alpha)\}$: the first group sums to ρ_ε and the second to $1-\rho_\varepsilon$. On S^n it restricts to $\{\varphi_\beta\Phi_\gamma\}$, because $\rho_\varepsilon = 1$ there kills the second group. Since the topology of X^{n+1} is the quotient topology from $X^n \sqcup_\gamma D^{n+1}_\gamma$, a function on X^{n+1} is continuous exactly when its restriction to X^n and its composite with each Φ_γ are continuous, so these compatible extensions assemble to a partition of unity on X^{n+1} extending $\{\varphi_\beta\}$ and subordinate to $\{U_\alpha \cap X^{n+1}\}$, after adjoining the new functions $(1-\rho_{\varepsilon_\gamma})\psi_{\gamma j}$ pushed forward from each cell.

Step 3: Local finiteness, via thin collars. The neighbourhoods that witness local finiteness must be chosen inside the region where $\rho_{\varepsilon_\gamma} \equiv 1$, because on the outer annulus $1-\varepsilon_\gamma \leq r < 1-\varepsilon_\gamma/2$ the factor $1-\rho_{\varepsilon_\gamma}$ is positive and the $\psi_{\gamma j}$ sum to 1, so some newly adjoined function is nonzero at every point there. Call

$$C_\gamma = \Phi_\gamma\bigl(\{(r,x) : r > 1-\varepsilon_\gamma/2\}\bigr)$$

the *thin* collar of the cell e^{n+1}_γ. On C_γ every $\psi_{\gamma j}$ contributes 0, so the only functions that can be nonzero are the extensions of the φ_β.

Now let $x \in X^n$ and let $N_n \subseteq X^n$ be a neighbourhood of x in X^n meeting the supports of only finitely many φ_β, say those indexed by a finite set S. Put

$$N_{n+1} = N_n \cup \bigcup_\gamma \Big(C_\gamma \cap \Phi_\gamma\big(\{(r,y) : y \in \Phi_\gamma^{-1}(N_n)\}\big)\Big)$$

the union of N_n with the thin collars taken only over N_n. This is a neighbourhood of x in X^{n+1}: its preimage under each Φ_γ is open, and it meets X^n in N_n. By the previous paragraph the only functions nonzero on N_{n+1} are the extensions of the φ_β with $\beta \in S$, the same finite set, because passing to a thin collar adjoins no new function. Iterating gives a nested chain $N_n \subseteq N_{n+1} \subseteq N_{n+2} \subseteq \cdots$ whose union N is open in X, since $N \cap X^m$ is open in X^m for every m and X carries the weak topology determined by its skeleta, and on which only the finitely many functions indexed by S are nonzero. A point in an open cell e_γ^{n+1} is handled by the same construction started from a neighbourhood inside that cell, where the nonzero functions are the finitely many $\psi_{\gamma j}$ together with the finitely many extended $\varphi_\beta \Phi_\gamma$.

This is what the colimit step needs, and the naive argument does not supply it: local finiteness on X^n gives a neighbourhood open in X^n, and an open subset of X^n need not be open in X: a point on the boundary of an attached cell has every X-neighbourhood meeting that cell's interior. The chain above is built precisely to grow one dimension at a time without enlarging S. Continuity of each function and the fact that they sum to 1 do pass to the union pointwise, the first because X carries the weak topology and the second because every point lies in some skeleton.

Step 4: Paracompactness. Given the partition of unity, the open sets $V_\beta = \varphi_\beta^{-1}\big((0,1]\big)$ satisfy $V_\beta \subseteq \mathrm{supp}\varphi_\beta \subseteq U_{\alpha(\beta)}$, so $\{V_\beta\}$ is an open refinement of $\{U_\alpha\}$, and it is locally finite because the supports are. It covers X because the φ_β sum to 1. Hence every open cover of X has a locally finite open refinement, which is [12, Paracompact Space], and X is Hausdorff by theorem 1.2.3, completing the proof.

Two consequences are used repeatedly later and are worth naming here. A CW complex is paracompact Hausdorff, so every statement of chapter 7 whose base is assumed paracompact Hausdorff applies verbatim to a CW complex; and the classification theorem 7.1.9 of vector bundles over a paracompact base therefore subsumes, for vector bundles, the CW classification theorem 2.4.7.

Example 1.2.5: Examples of CW Complexes

1. Let X^0 be some collection of points. Then we may form a graph by attaching a 1-cells (i.e. spaces homeomorphic to closed intervals) to points in X^0. If we so choose, we may form loops in our graphs.

2. $\mathbb{R}$ can be seen as having a CW-structure (i.e. homeomorphic to a CW-complex) by taking the 0-skeleton $\mathbb{Z}$ and attaching the 1-cells $[n, n+1]$. This process can be generalized to give $\mathbb{R}^n$ a CW-structure.

3. The house with two rooms pictures earlier is a CW complex, can you see how? There should be twenty-nine 0-cells, fifty-one 1-cells, and twenty-three 2-cells.

4. One can get the sphere S^{n-1} by attaching to a point the entire boundary $\partial D^n \to e_0$ (each X^k for $1 < k < n$ had no k-cells attaching, meaning $X^1 = X^2 = \cdots = X^{n-1}$). Alternatively,

two lower dimensional disks can be attached to the boundary of D^{n-2} (play with this for S^2)

5. Take $X^0 = \{*\}$ to be a single point. Then we may attach two 1-cells. Then, take one 2-cell, and identify two of it sides with one of the circles, and two of sides with the other circle (to do this, make the map a $2-1$ map onto the two 1-cells). If we straighten out the circle so that it looks like squares, we would get the following visual:

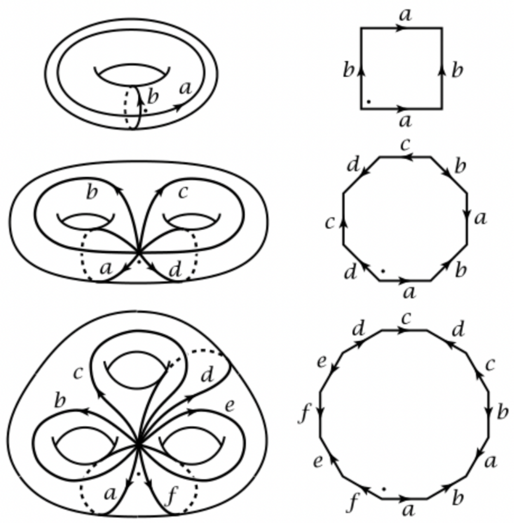


Figure 1.5: n-torus as CW complex, Hatcher

The other examples are when we are attaching a 1-cell to 4 or 6 circles respectively.

6. $\mathbb{R}\mathrm{P}^n$ is a CW-complex. We can see this by interpreting $\mathbb{R}\mathrm{P}^n$ as $S^n/(v \sim -v)$. In this case, $\mathbb{R}\mathrm{P}^n$ is $\mathbb{R}\mathrm{P}^{n-1}$ with the gluing $S^{n-1} \to \mathbb{R}\mathrm{P}^{n-1}$ gluing antipodal points together. This inductive process is well-defined as $\mathbb{R}\mathrm{P}^1 = S^1$. The trick is to glue the boundary at "twice the speed" around S^1. Hence $\mathbb{R}\mathrm{P}^n$ has one cell per dimension.

   From this, we may define $\mathbb{R}\mathrm{P}^\infty = \bigcup_n \mathbb{R}\mathrm{P}^n$ to also be a cell-complex with one cell per dimension. The space $\mathbb{R}\mathrm{P}^\infty$ can be viewed as the space of liens through the origin in $\mathbb{R}^\infty = \bigcup_n \mathbb{R}^n$.

7. $\mathbb{C}\mathrm{P}^n$ is a CW-complex. We shall show how we can create $\mathbb{C}\mathrm{P}^n$ as the quotient of D^{2n} under the relation $v \sim \lambda v$ for $v \in \partial D^{2n}$. Take $S^{2n+1} \subseteq \mathbb{C}^{n+1}$ with last coordinates real and non-negative. Then they are vectors of the form:

$$(w, \sqrt{1-|w^2|}) \in \mathbb{C}^n \times \mathbb{C} \qquad |w| \leq 1$$

   Certainly, these vectors are the graph of the functions $w \mapsto \sqrt{1-|w|^2}$ and form the shape $D_+^{2n} \subseteq S^{2n-1} \subseteq S^{2n+1}$ and consists of vectors of the form $(w, 0) \subseteq \mathbb{C}^n \times \mathbb{C}$ with $|w| = 1$. Then putting the relation $v \sim \lambda v$ on S^{2n+1} can be identified with vectors in D_+^{2n}, and the identification is unique if the last coordinate is nonzero. If the last coordinate is zero, we just take $v \sim \lambda v$ for $v \in S^{2n-1}$.

Thus, w have $\mathbb{C}\mathrm{P}^n$ described as the quotient of D_+^{2n} through $v \sim \lambda v$ for $v \in S^{2n-1}$. Finally, $\mathbb{C}\mathrm{P}^n$ is obtained from $\mathbb{C}\mathrm{P}^{n-1}$ by attaching e^{2n} cells via $S^{2n-1} \mapsto \mathbb{C}\mathrm{P}^{n-1}$. Hence,

$$\mathbb{C}\mathrm{P}^n = e^0 \cup e^2 \cup \cdots e^{2n}$$

We may define $\mathbb{C}\mathrm{P}^\infty$ similarly to $\mathbb{R}\mathrm{P}^\infty$.

In many of our future constructions, we shall need to identify subcomplexes to use in various operations on complexes, we hence single out the following definitions:

Definition 1.2.6: Subcomplex

Let X be a cell complex. Then a *subcomplex* is a closed subspace $A \subseteq X$ that is a union of cells of X. A pair (X, A) consisting of a cell complex X and a subcomplex A is called a *CW pair.*

Since A is closed, the characteristic map of each cell in A has image contained in A, and the image of the attaching maps of A is contained in A, so A is itself a cell-complex!

Note that the closure of each cell of a cell-complex (or the closure of a collection of cells) need not be a subcomplex (though they often will be). For example, take S^1 with the usual one 0-cell and one 1-cell. Attach a 2-cell to S^1 by gluing the boundary of e^2 to a *proper subset* (or sub-arc) of S^1. Then the closure of the 2-cell is not a subcomplex since it contains only a *part* of the 1-cell.

Example 1.2.7: Subcomplexes

1. Each skeleton X^n of a cell-complex X is a subcomplex.
2. Each $\mathbb{R}\mathrm{P}^k \subseteq \mathbb{R}\mathrm{P}^n$ and $\mathbb{C}\mathrm{P}^k \subseteq \mathbb{C}\mathrm{P}^n$ is a subcomplex for $k \leq n$. In fact, these are the only subcomplexes of $\mathbb{R}\mathrm{P}^n$ and $\mathbb{C}\mathrm{P}^n$
3. the natural inclusions $S^0 \subseteq S^1 \subseteq \cdots \subseteq S^n$ are cell complexes given the right CW structure of S^n, namely it would have to be the construction where we glue the two lower dimensional spheres instead of gluing the boundary of D^n.

Cell complexes shall be a fundamental tool we use; many invariance will be defined by showing the CW structure is invariant up to homotopy, and taking advantage of the combinatorial information of the CW structure. There many theorems that characterize when a topological space X admits a CW structure. Milnor proved that every manifold admits a CW complex and for algebraic variety, and the Lundell-Weingram Theorem characterizes some technical requirements. The general rule of thumb should be: if the object is "pretty geometric", it will admit a CW structure. So many "pathological" examples or "'weird infinite dimensional" examples shall not have CW structures, but that is alright as it is already interesting that we can use this structure on essentially all geometric objects.

1.2.1 Operations on Cell Complex

One of the ways we shall take advantage of CW structures is by defining new ways of constructing CW structures given already known ones. Most of these constructions work more generally in the category **Top**; any appropriate adjustments to the definitions can be made.

Definition 1.2.8: Product

Let X, Y be cell-complex. Then $X \times Y$ is a cell-complex with a cells being products $e^m_\alpha \times e^n_\beta$.

Note that if there are infinitely many cells, the product topology on $X \times Y$ may have a finer topology than the CW topology on $X \times Y$, however we shall not concern ourselves with this.

Example 1.2.9: Product Of Cells

Let S^1 have the standard cell structure (attach I to a single point identifying the endpoints). Then $S^1 \times S^1 = T^1$ is a cell-structure. This was demonstrated in figure 1.5.

Definition 1.2.10: Quotient

Let (X, A) be a CW pair. Then X/A has a cell-structure respecting the universal property of quotients given as follows: The cells of X/A are the cells of $X - A$ plus a new 0-cell, the image of A in X/A.

Example 1.2.11: Quotient Of Cells

1. Take S^{n-1} with any cell structure, build D^n from S^{n-1} by attaching an n-cell in the natural way, and consider (D^n, S^{n-1}). Then D^n/S^{n-1} is S^n with the usual cell-structure.

Definition 1.2.12: Suspension

Let X be a cell complex. Then the *suspension* of X, denoted SX is the quotient of $X \times I$ obtained by collapsing $X \times \{0\}$ to a point and $X \times \{1\}$ to a another point.

Visually, a suspension can be seen as so:

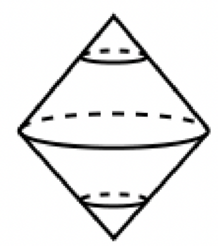


Figure 1.6: Suspension

The prototypical example of a suspension is $S(S^n) = S^{n+1}$. Since suspension are created via quotienting and products, we see that we can create a cell structure on the cone of X:

$$CX = (X \times I)/(X \times \{0\})$$

the cylinder with one end collapsed to a point, so that SX is the union of two copies of CX glued along their common base $X \times \{1\}$.

The presentation of SX as two cones meeting along X underlies every later use of the suspension, and it holds for spaces far more general than cell complexes.

Proposition 1.2.13: Unreduced Suspension as a Union of Two Cones

Let X be a topological space, with cone $CX = (X \times I)/(X \times \{0\})$ and suspension $SX = (X \times I)/(X \times \{0\},\, X \times \{1\})$ (the construction of definition 1.2.12, here applied to an arbitrary space X). Let C_-X and C_+X be the images in SX of the closed halves $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$. Then each of C_-X and C_+X is a cone on X, and

$$SX = C_-X \cup C_+X, \qquad C_-X \cap C_+X = X \times \{\tfrac{1}{2}\} \cong X$$

Each cone is contractible. If X is compact Hausdorff, then CX and SX are compact Hausdorff and the two cones $C_\pm X$ are closed subsets of SX whose union is SX.

***Proof*:**
Write $q\colon X \times I \to SX$ for the quotient map. On $X \times [0, \frac{1}{2}]$ the only identification imposed by q is the collapse of $X \times \{0\}$ to a point, so $C_-X = q(X \times [0, \frac{1}{2}])$ is a copy of the cone CX, and symmetrically $C_+X = q(X \times [\frac{1}{2}, 1])$ is a cone with cone point $q(X \times \{1\})$. Their union is $q(X \times I) = SX$, and q is injective on $X \times \{\frac{1}{2}\}$, where no collapse occurs, so $C_-X \cap C_+X$ is a copy of X. Each cone deformation retracts to its cone point by sliding along the I-coordinate, the retraction of C_-X being $\big(q(x, s), t\big) \mapsto q(x, (1-t)s)$ and that of C_+X its reflection, so both cones are contractible.

Suppose X is compact Hausdorff. Then $X \times I$ is compact Hausdorff, hence normal, and $SX = q(X \times I)$ is compact as a continuous image. Collapsing a closed subspace of a normal space to a point yields a Hausdorff quotient, and SX is obtained from $X \times I$ by collapsing the two disjoint closed subsets $X \times \{0\}$ and $X \times \{1\}$ in turn, their images staying disjoint and closed, so SX is Hausdorff. The same argument applied to the single closed subset $X \times \{0\}$ shows CX is compact Hausdorff. Finally $q^{-1}(C_-X) = X \times [0, \frac{1}{2}]$ and $q^{-1}(C_+X) = X \times [\frac{1}{2}, 1]$ are closed and saturated, so $C_\pm X$ are closed in SX, completing the proof.

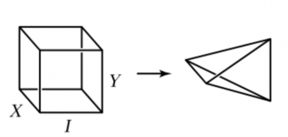
Though this might not seem useful, as we progress this construction will become increasingly more important. Of notable important is that maps can be suspended, if $f : X \to Y$, then we can suspend it to $Sf : SX \to SY$, which is given by the quotient map of $f \times \text{id} : X \times I \to Y \times Y$. Often, the suspension of a "non-simply connected" object shall be simply connected, and the suspension of a map shall make it nullhomotopic.

In a suspension, all the points of X are joined to a single point. We can generalize this by joining every point in X to the points to a space Y (and vice-versa):

Definition 1.2.14: Join

Let X, Y be topological space. Then a *join*, dentoed $X * Y$ is the quotient of the space $X \times Y \times I$ by $(x, y_1, 0) \sim (x, y_2, 0)$ and $(x_1, y, 1) \sim (x_2, y, 1)$. Thus $X \times Y \times \{0\}$ collapses to X while $X \times Y \times \{1\}$ collapses to Y.

If X, Y are CW complexes, then there is a natural CW structure on $X * Y$ with the subspaces X and Y as subcomplexes, and the remaining cells being the product cells of $X \times Y \times (0, 1)$. This process of the join operation can be visualized like so:



In fact, if $X = Y = [0,1]$, then a join would produce the above shape. The join $X * Y$ contains a copy of X and Y, and every point $(x, y, t) \in X * Y$ is on a unique line segment joining the points $x \in X \subseteq X * Y$ and $y \in Y \subseteq X * Y$. Hence, we may think of every point in $X * Y$ as being written in the form

$$t_1 x + t_y y$$

where $0 \leq t_i \leq 1$, $t_1 + t_2 = 1$ and $0x + 1y = y$ and $1x + 0y = x$; that is all possible *convex combinations*. If we iterate and write $X_1 * \cdots * X_n$, we see that this is independent of bracketing and we can write every element in the form:

$$t_1 x_1 + \cdots + t_n x_n \qquad 0 \leq t_i \leq 1, t_1 + \cdots + t_n = 1$$

that is, we get every formal convex linear combinations of the elements. A special but important case is when each X_i is a singleton. In this case, the join of two points is a line segment, the join of 3 is a triangle, the join of 4 is a tetrahedron, and so forth. In general, the join of n points is a convex polyhydron of dimension $n - 1$ known as a *simplex*.

Example 1.2.15: Join

Let X_i be two points, and say they are placed in $\mathbb{R}^n$ on a coordinate axis at unit lengths. Then the join $X_1 * \cdots * X_n$ is the union of 2^n copies of $n - 1$ dimensional simplices. The radial projection from the original gives a homeomorphism from $X_1 * \cdots * X_n$ to S^{n-1}.

Definition 1.2.16: Wedge Sum

Let X, Y be topological spaces. Then given points $x \in X$ and $y \in Y$, the *wedge sum* (with respect to x, y) is

$$X \vee Y = X \sqcup Y / x \sim y$$

If X, Y are pointed spaces, so that by definition there is a disintuished point as part of the definition: a subcomplex to a point. A simple example of a wedge sum is naturally defined (no need to say "with respect to"). If X, Y are cell complexes, $X \vee Y$ is a cell complex by taking the cell complex $X \sqcup Y$ and collapsing a subcomplex to a point. A simple example of a wedge sum would be for any cell complex X, the quotient X^n / X^{n-1} is a wedge sum of the n-spheres $\vee_\alpha S^n_\alpha$ with one sphere for each n-cell of X.

Definition 1.2.17: Smash Product

Let X, Y be topological spaces. Then given $x \in X$ and $y \in Y$, the *smash product* (with respect to x and y) is

$$X \wedge Y := X \times Y / X \vee Y$$

If X, Y are cell complex, then giving $X \times Y$ the cell-complex topology the smash product is a cell-complex. For an example, show that $S^1 \wedge S^1 = S^2$ and more generally $S^n \wedge S^m = S^{n+m}$.

Example 1.2.18: Linking Smash Product And Join
Show that
$$S(X \wedge V) \simeq X * Y$$

1.3 Homotopy Equivalence and Homotopy Extension

In this section, we shall prove some important theorems to find homotopy equivalent spaces. The first is way of collapsing some parts of our space to a point to simplify the problem, and the second varies the way we put subspaces together.

In general, if we take X and X/A, these two spaces may have diferent homotopy types, for example the torus T^2 and $T^2/(S^1 \vee S^1) \cong S^2$. The reason why the homotopy type changes is becasue $S^1 \vee S^1$ is not contractible, and hence change the homotopy; thus to insure an invariance in the homotopy type we must insure that the subspace we are contracting by is contractible. To see the usefulness of this criterion, we start with the following notion:

Definition 1.3.1: Homotopy Extension Property

Let $f_0 : X \to Y$ be a map and $f_t|_A : A \times I \to Y$ be a homotopy on $f_0|_A$. Then if we can extend this to a homotopy on X, $f_t : X \times I \to Y$, we say that (X, A) has the *homotopy extension property.*

Note how the pair (X, A) did not include information about the codomain. Thus, for any Y if $X \times \{0\} \to Y$ is continuous and $A \times I \to Y$ is a homotopy that agrees with $A \times \{0\}$, then it may be extended to $X \times I \to Y$. If this was only valid for certain domains, we would say that it had the homotopy extension proeprty with respect to Y.

Proposition 1.3.2: Condition For Homotopy Extension

A pair (X, A) has the homotopy extension property if and only if $(X \times \{0\}) \cup (A \times I)$ is a retract of $X \times I$

Try visualizing this condition before seeing the formal proof:

***Proof*:**
let (X, A) have the homotopy extension property. Then the identity map $X \times \{0\} \cup A \times I \to X \times \{0\} \cup A \times I$ is homotopic to itself, and so extends to a map $X \times I \to X \times \{0\} \cup A \times I$, so $X \times \{0\} \cup A \times I$ is a retract of $X \times I$.

Conversely, If $X \times \{0\} \cup A \times I$ is a retract of $X \times I$ and X is Haudorff, it suffices to show it for when A must is be closed, since in fact this condition implies that A is closed. For if $r : X \times I \to X \times I$ is a retraction map onto $X \times \{0\} \cup A \times I$, then the image of r is the set of points $z \in X \times I$ with $r(z) = z$, i.e. a closed set if X is Haudorff, so $X \times \{0\} \cup A \times I$ is closed in $X \times I$ and hence A is closed in X.

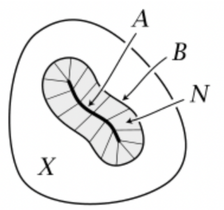
Now, given A is closed, then if we have two continuous maps $X \times \{0\} \to Y$ and $A \times I \to Y$ that agree on $A \times \{0\}$, we can combine them to get a map

$$X \times \{0\} \cup A \times I \to Y$$

which is continuous since it continuous on the closed sets $X \times \{0\}$ and $A \times I$. By assumption, we have a retraction map and so composing to get $X \times I \to X \times \{0\} \cup A \times I \to Y$, we get an extension $X \times I \to Y$, hence (X, A) has the homotopy extension property.

Example 1.3.3: Pairs With the Homotopy Extension Property

1. If A has mapping cylinder neighbourhood in X, then (X, A) has the homotopy extension property. A mapping cylinder neighbourhood is a closed neighbourhood N containing a subspace B (B is the "boundary" of N), with $N \setminus B$ an open neighbourhood of A such that there exists a map $f : B \to A$ and a homeomorphism $h : M_f \to N$ with $h|_{A \cup B} = \text{id}$.



   To see that (X, A) has this property, first not that $I \times I$ retracts onto $I \times \{0\} \cup (\partial I) \times i$, and so $B \times I \times I$ retracts onto $B \times I \times \{0\} \cup B \times (\partial I) \times I$, and hence we induce a retraction of $M_f \times I$ onto $M_f \times \{0\} \cup (A \cup B) \times I$. Thus $(M_f, A \cup B)$ has the homotopy extension property. Thus, the homeomorphic pair $(N, A \cup B)$ does as well.

   Given a continuous map $X \to Y$an a homotopoy restriction to A, we can take the constant homotopy on $X - (N - B)$ and then extend over N by applying the homotopy extension property for $(N, A \cup B)$, to the given homotopy on A and the constant homotopy on B

2. For an example of a pair without the homotopy extension property take (I, A) wehrer $A = \{0, 1, 1/2, 1/3, ...\}$. Then there is no continuous retractions $I \times I \to I \times \{0\} \cup A \times I$.

Proposition 1.3.4: Homotopy Extension For CW Pair

Let (X, A) be a CW pair. Then $X \times \{0\} \cup \times I$ is a deformation retract of $X \times I$, hence (X, A) has the homotopy extension property.

***Proof*:**
We shall first directly work with the cells D^n, and then work with the space X. First, there is a retraction

$$r : D^n \times I \to (D^n \times \{0\}) \cup (\partial D^n \times I)$$

an easy such example would be the radial projection from the point $(0, 2) \in D^n \times \mathbb{R}$. From this retraction, we may construct a deformation retraction from $D^n \times I$ onto $(D^n \times \{0\}) \cup (\partial D^n \times I)$

by setting

$$r_t = tr + (1-t)\mathrm{id}$$

Since $X^n \times I$ is obtained from $(X^n \times \{0\}) \cup (X^{n-1} \cup \mathbb{A}^n) \times I$ by attaching copies of $D^n \times i$ along $(D^n \times \{0\}) \cup (\partial D^n \times I)$, we get a deformation retraction from $(X^n \times \{0\}) \cup (X^{n-1} \cup \mathbb{A}^n) \times I$. If we perform this deformation retraction during the t-interval $[1/2^{n+1}, 1/2^n]$, then the infinite concatenation of these homotopies is a deformation retraction of $X \times I$ onto $(X \times \{0\}) \cup (A \times I)$, since there is no problem with continuity of this deformation retraction at $t = 0$ since it's continuous at $X^n \times I$ (beign stationary in $[0, 1/2^{n+1}]$), and the CW complexes have the weak topology with respect to their skeleton, so the map is continuous if and only if its retraction on each skeleton is continuous. But then we have the retraction.

Theorem 1.3.5: Homotopy Equivalence Of Quotient

Let (X, A) satisfy the homotopy extension property and let A be contractible. Then the quotient map $\pi : X \to X/A$ is a homotopy equivalence.

***Proof*:**
Let $f_t : X \to X$ be a homotopy extending the contraction of A where:

1. $f_0 = \mathrm{id}$
2. $f_t(A) \subseteq A$ for all $t \in I$

Then $\pi \circ f_t : X \to X/A$ sends A to a point, hence there is a well-defined map $\overline{f}_t : X/A \to X/A$, notably the following diagarm commutes for all t:

$$\begin{tikzcd} X \arrow[r, "f_t"] \arrow[d, "\pi"] & X \arrow[d, "\pi"] \\ X/A \arrow[r, "\overline{f}_t"] & X/A \end{tikzcd}$$

In particular $\pi \circ f_t = \overline{f}_t \circ \pi$. When $t = 1$, we have that $f_1(A)$ is a point, hence f_1 induces the map $g : X/A \to X$ in the following diagram:

$$\begin{tikzcd} X \arrow[r, "f_1"] \arrow[d, "\pi"] & X \arrow[d, "\pi"] \\ X/A \arrow[r, "\overline{f}_1"] \arrow[ur, "g"] & X/A \end{tikzcd}$$

this diagram is certainly commutative, in particular $\pi \circ g = \overline{f}_1$, since

$$\pi \circ g(\overline{x}) = \pi \circ g \circ \pi(x) = \pi \circ f_1(x) = \overline{f}_1 \circ \pi(x) = \overline{f}_1(\overline{x})$$

Hence, the maps g and π are homotopy equivalence since

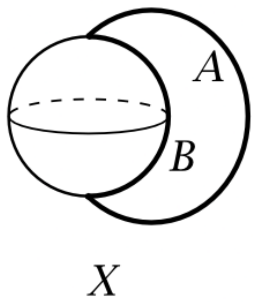
$$g \circ \pi = f_1 \simeq f_0 = \mathrm{id} \qquad \pi \circ g = \overline{f}_1 \simeq \overline{f}_0 = \mathrm{id}$$

via the homotopies f_t and $\overline{f}_t$, giving us the desired result.

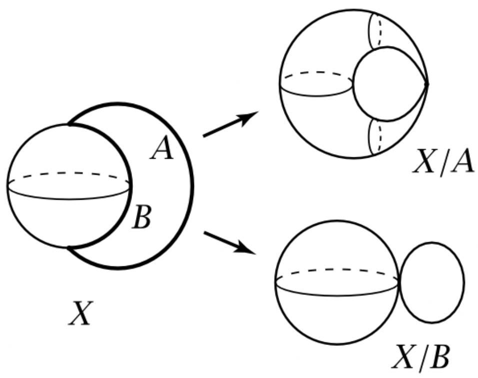
Example 1.3.6: Homotopy Equivalence Using Quotients

1. (Graphs) Let X be any graph with finitely many vertices and edges. If two endpoints of any edge of X are distinct, we can collapse the edge to a point, giving a homotopy equivalent graph with one fewer edge (namely since edges are contractible). This can be repeated until we get a graph X where each component has 1 vertex connected by loops (wedge sum of circles). We shall show in the next chapter that the wedge sum of circles are homotopy equivalent if they have the same number of circles.

2. Let X be the space S^2 along with an interval I attached at two distinct points. Let the interval I be labeled A, and let B be the arc on the sphere connecting the endpoints of A:



Give X a CW complex with the two endpoints of A and B being 0-cells, the interior of A and B beign 1-cells, and the rest of S^2 a 2-cell. Since A, and B are contractible, we get that X/A and X/B are homotopy equivalent to X:

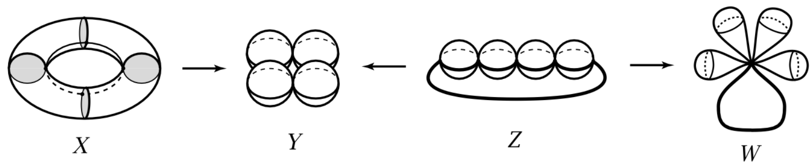


The space X/A is S^2/S^0, that is the sphere with two points identified, while X/B is $S^1 \vee S^2$. Thus,

$$S^2/S^0 \simeq S^1 \vee S^2$$

even though at first glance they may seem quite different

3. Let X be the union of a torus T with n meridional disks. To make a CW complex on X, take a longitudinal circle in the torus, hence intersecting each of the meridional disks at one point. These poitns of intersectio nare the 0-cells, the 1-cells are the rest of hte longitudinal circl an the boundary of the cmeridional disks, and the 2-cells are the remaining regions. We may collapse the meridional disks to a point to yeidl a homotopy equivalent space Y consisten of n 2-spheres, each tangent to it's neighors (we mah think of this as a necklace with n beads). From this intution, we may imagine a space Z which is a line of n 2-spheres each touching each other tangentially (a "strand of n-beads") with a line connecting the ends of the first and last 2-sphere. Then we can collapse the line to get your space Y, showing the two are homotopic. Finally, we may take the union of arcs on the 2-spheres to form a path starting form the endpoint of the first sphere and ending on the endpoint of hte last sphere, which we may then collapse to get a shape that resembles a "bouquet of flowers" with a handle:



4. (reduced suspension) Let X be a CW complex and $x \in X$ a 0-cell. Then in the suspension SX, there is a line segment $\{x\} \times I$ which we can colapse to produce the homotopy-equivalent space ΣX called hte *reduced suspension*. For example, if $X = S^1 \vee S^1$ with x the point of intersection, the SX is the union of two spherses intersection along the arc $\{x\} \times I$, so the reduced suspension is $S^2 \vee S^2$ (which can think of being a simpler space than the suspension). In general, it can be shown that:

$$\Sigma(X \vee Y) = \Sigma X \vee \Sigma Y$$

for arbitrary CW complex X, Y. The reduced suspension also has a simpler CW strucuture: in SX there are two 0-cells (the two suspensions points), and an $(n+1)$-cell $e^n \times (0,1)$ for each n-cell of X other than the 0-cell x. Then the reduced suspension ΣX is the same as the smash product $X \wedge S^1$, wince both spaces are the quotient of $X \times I$ with $(X \times \partial I) \cup \{x\} \times I$ collapsed to a point.

Another application of the extension property is another way of gluing shapes while keeping the homotopy class invariant:

Theorem 1.3.7: Homotopy Equivalence Via Gluing

Let(X_1, A) be a CW pair and elt $f, g : A \to X_0$ be attachings maps that are homotopic, then

$$X_0 \sqcup_f X_1 \simeq X_0 \sqcup_g X_1$$

relative to X_0

When we say that the two spaces are homotopy equivalent relative to X_0, we mean that there are maps between the two spaces that restrict to the identity on X_0 such that they are homotopy equivalent via homotopies that restrict to the identity on Y.

***Proof*:**
Let $F : A \times I \to X_0$ be a homotopy from f to g, and consider the space $X_0 \sqcup_f (X_1 \times I)$. Certainly this space contains $X_0 \sqcup_f X_1$ and $X_0 \sqcup_g X_1$ as subspaces. Then given a deformation retract of $X_1 \times I$ onto $(X_1 \times \{0\}) \cup A \times I$ as given in proposition 1.3.4, , we get an induced deformation retraction of $X_0 \sqcup_f (X_1 \times I)$ onto $X_0 \sqcup_f X_1$. Similarly, $X_0 \sqcup_f (X_1 \times I)$ deformations retracts onto $X_0 \sqcup_g X_1$. Both of these deformation retractions restrict to the identity on X_0, so together they give a homotopy equivalence

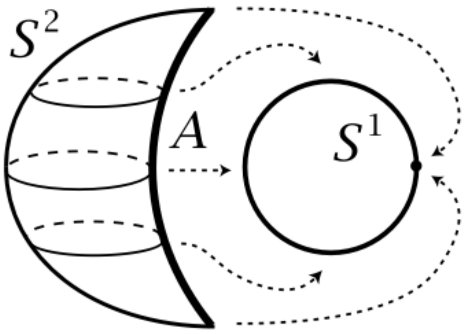
$$X_0 \sqcup_f X_1 \simeq X_0 \sqcup_g X_1$$

relative to X_0, as we sought to show.

see this :https://www.math3ma.com/blog/clever-homotopy-equivalences

Example 1.3.8: Attaching Spaces

1. Let X be a sphere with two points identified. We shall rederive taht this is homotopy equivalent to $S^1 \vee S^2$. Take S^2 and S^1. Take an arc on S^2, and attach it to all of S^1. Since A is contractible, this map is homotopic to a constnant map attaching S^1 to S^1 via a constatn map, giving $S^1 \vee S^2$:



   The result now follows since (S^2, A) is a C Wpair, S^2 being obatine from A by attaching a 2-cell

2. We may do the same with the necklace of beeds as seen in the previous example.a The necklace can be obtained from a circel by attaching n 2-sphers along arcs, so the necklace is homotopy-equivalent to the space obtained by attachign n 2-spheres to a circle at points, then sliding these attaching poitns around the circle until they all coincide, giving us the wedge-sum.

3. Let $CA = (A \times I)/(A \times \{0\})$ be the cone on A. Then if (X, A) is a CW pair,

$$X/A \simeq X \cup CA$$

   For, since CA is contractible subcomplex of $X \cup CA$, we see that:

$$X/A = (X \cup CA)/CA \simeq X \cup CA$$

4. Let (X, A) be a CW pair with A contractible in X (the inclusion map $A \hookrightarrow X$ is homotopic to a constant map). Then we shall show:

$$X/A \simeq X \vee SA$$

By the prevoius example, we have $X/A \simeq \cup CA$. Now, since A is contractible in X, the mapping cone of the inclusion $A \hookrightarrow X$ $(X \cup CA)$ is homotopy equivalent to the mapping cone of a constnat map, which is $X \vee SA$. As an example, $S^n/S^i \cong S^n \vee S^{i+1}$ for $i < n<$ since S^i is contractibel in S^n if $i < n$. As a particular use-case, we get

$$S^2/S^0 \simeq S^2 \vee S^1$$

which we saw earlier, showing us we have two technics to find homotopy-equivalences.

Before moving on, we give a few important technical results when it comes to homotopies relative to a subspace.

Proposition 1.3.9: Homotopy Extension and Homotopy Equivalence

Let (X, A) and (Y, A) satisfy the homotopy extension property and $f : X \to Y$ be a homotopy equivalence with $f|_A = \text{id}$. Then f is a homotopy equivalence relative to A

***Proof*:**
Recall that a homotopy equivalence *relative to A* means a map $f \colon X \to Y$ for which there is $g \colon Y \to X$ with $g|_A = \text{id}$ and homotopies $gf \simeq \text{id}_X$ and $fg \simeq \text{id}_Y$ that are stationary on A. The hypothesis $f|_A = \text{id}$ already fixes f on A; what must be improved is the homotopy inverse g, which a priori need not fix A, and the two round-trip homotopies, which a priori move points of A. The homotopy extension property is exactly the tool that performs both repairs. Fix a homotopy inverse $g \colon Y \to X$, so that $gf \simeq \text{id}_X$ and $fg \simeq \text{id}_Y$. The argument has three stages.

Step 1: Adjusting g so that $g|_A = \text{id}$. Let $h_t \colon X \to X$ be a homotopy from $gf = h_0$ to $\text{id}_X = h_1$. Restricting to A and using $f|_A = \text{id}$, the family $h_t|_A$ is a homotopy of maps $A \to X$ running from $(gf)|_A = g|_A$ (at $t = 0$) to id_A (at $t = 1$). Thus $h_t|_A$ is a homotopy of $g|_A$, viewed as a map $A \to X$. Since (Y, A) satisfies the homotopy extension property (definition 1.3.1), this homotopy of $g|_A$ extends over Y: there is a homotopy $g_t \colon Y \to X$ with $g_0 = g$ and $g_t|_A = h_t|_A$ for all t. Set g_1 to be the end of this homotopy. Then $g_1|_A = h_1|_A = \text{id}_A$, so g_1 restricts to the identity on A, and $g_1 \simeq g$, so g_1 is again a homotopy inverse of f. Replacing the original inverse by g_1, we have arranged $g_1|_A = \text{id}$ while keeping $g_1 f \simeq \text{id}_X$ and $fg_1 \simeq \text{id}_Y$.

Step 2: Upgrading $g_1 f \simeq \text{id}_X$ to a homotopy rel A. A homotopy from $g_1 f$ to id_X is furnished by splicing the reverse of $g_t f$ to the homotopy h_t. Concretely, define $k_t \colon X \to X$ by

$$k_t = \begin{cases} g_{1-2t}\, f, & 0 \le t \le \frac{1}{2}, \\ h_{2t-1}, & \frac{1}{2} \le t \le 1 \end{cases}$$

The two clauses agree at $t = \frac{1}{2}$, where both read $g_0 f = gf = h_0$, so k_t is a well-defined homotopy; it runs from $k_0 = g_1 f$ to $k_1 = h_1 = \text{id}_X$. The point of this particular splicing is its behaviour on A. Since $f|_A = \text{id}$ and $g_t|_A = h_t|_A$, on A the first clause becomes $g_{1-2t}|_A = h_{1-2t}|_A$ and the second is $h_{2t-1}|_A$, so

$$k_t|_A = \begin{cases} h_{1-2t}|_A, & 0 \le t \le \frac{1}{2}, \\ h_{2t-1}|_A, & \frac{1}{2} \le t \le 1 \end{cases}$$

which starts and ends at $h_1|_A = \mathrm{id}_A$ and whose second half retraces its first half in reverse: $k_t|_A = k_{1-t}|_A$ for all t. A homotopy with this palindromic restriction can be deformed, rel its two ends, into one that is stationary on A, as follows.

Build a homotopy of homotopies $k_{t,u}\colon A \to X$ over the parameter square $I \times I$ with coordinates (t, u), the t-axis horizontal and the u-axis vertical. Picture the square cut by the "Λ" whose apex sits at the midpoint $(\frac{1}{2}, 0)$ of the bottom edge and whose two legs run upward to the top corners $(0, 1)$ and $(1, 1)$; at height u the legs sit at $t = \frac{1-u}{2}$ and $t = \frac{1+u}{2}$, so the Λ opens out to span the entire top edge while pinching to a point on the bottom. On the bottom edge set $k_{t,0} = k_t|_A$. Outside the Λ, in the two lower regions flanking the apex, declare $k_{t,u}$ independent of u, equal to its value on the bottom edge directly below; inside the Λ, the upper triangle containing the whole top edge, declare $k_{t,u}$ independent of t, equal to the common value carried up the legs. Concretely this reads

$$k_{t,u}|_A = \begin{cases} k_t|_A, & t \le \frac{1-u}{2} \ \text{ or } \ t \ge \frac{1+u}{2}, \\ k_{\frac{1-u}{2}}|_A, & \frac{1-u}{2} \le t \le \frac{1+u}{2} \end{cases}$$

The two prescriptions agree across the left leg automatically, since the outside value at $t = \frac{1-u}{2}$ is $k_{\frac{1-u}{2}}|_A$, the same as the inside value; across the right leg they agree precisely because $k_t|_A = k_{1-t}|_A$, which forces the outside value $k_{\frac{1+u}{2}}|_A$ at $t = \frac{1+u}{2}$ to equal the inside value $k_{\frac{1-u}{2}}|_A = k_{1-\frac{1+u}{2}}|_A$. So $k_{t,u}$ is well defined and continuous. Now read off the three remaining edges. The left edge $t = 0$ and the right edge $t = 1$ lie outside the Λ for every $u < 1$, hence carry the constant bottom-corner values $k_0|_A = h_1|_A = \mathrm{id}_A$ and $k_1|_A = h_1|_A = \mathrm{id}_A$; and the top edge $u = 1$ lies inside the Λ, carrying the value $k_{\frac{1-u}{2}}|_A$ at $u = 1$, namely $k_0|_A = \mathrm{id}_A$. Thus $k_{t,u}$ is a homotopy of the homotopy $k_t|_A$ (its bottom edge) which is the constant id_A on the other three edges.

Now extend $k_{t,u}$ from A to all of X. Write I_t for the factor carrying the variable t of the homotopy k_t and I_u for the new factor carrying the deformation variable u; the pair to which the extension applies is $(X \times I_t, A \times I_t)$, deformed in the direction u. This pair satisfies the homotopy extension property. Indeed, since (X, A) does, the set $(X \times \{0\}_u) \cup (A \times I_u)$ is a retract of $X \times I_u$ (proposition 1.3.2); let $\rho\colon X \times I_u \to (X \times \{0\}_u) \cup (A \times I_u)$ be such a retraction. Crossing ρ with the identity on the old factor I_t gives a map $\rho \times \mathrm{id}_{I_t}$ on $X \times I_u \times I_t$, which after reordering the factors to $X \times I_t \times I_u$ retracts onto $(X \times I_t \times \{0\}_u) \cup (A \times I_t \times I_u)$. The $\{0\}$ now sits in the u-slot, as it must: this is precisely the retraction criterion for $(X \times I_t, A \times I_t)$ in the direction u. (Equivalently, the product of an (X, A) homotopy extension pair with id_I is again a homotopy extension pair, with the new factor I_u playing the homotopy role.) View $k_{t,u}$ as a homotopy in the variable u of the map $k_t|_A$, the latter regarded as a single map $A \times I_t \to X$ in the variable t. By the homotopy extension property for $(X \times I_t, A \times I_t)$ it extends to a homotopy of the full map $k_t\colon X \to X$, producing $k_{t,u}\colon X \to X$ with bottom edge $k_{t,0} = k_t$ the homotopy of Step 2. Reading off the left, top, and right edges of the (t, u)-square gives a homotopy from $k_0 = g_1 f$ to $k_1 = \mathrm{id}_X$ whose restriction to A is constant equal to id_A throughout (those three edges carry the constant value computed above). Hence

$$g_1 f \simeq \mathrm{id}_X \quad \text{rel } A$$

Step 3: Upgrading $fg_1 \simeq \mathrm{id}_Y$ to a homotopy rel A. It remains to make the other round-trip homotopy stationary on A as well. Since $g_1 \simeq g$, we have $fg_1 \simeq fg \simeq \mathrm{id}_Y$, so f and g_1 are homotopy inverses of each other on both sides. This symmetry lets us run Steps 1 and 2 once more with the two maps interchanged, applying them to the pair g_1, f in place of f, g. The map being shown an equivalence is now $g_1 \colon Y \to X$ with inverse $f \colon X \to Y$, and g_1 already restricts to id on A. The output is an adjusted inverse $f_1 \colon X \to Y$ with $f_1|_A = \mathrm{id}$ together with the relative homotopy produced by Step 2, which here reads

$$f_1 g_1 \simeq \mathrm{id}_Y \quad \mathrm{rel}\ A$$

Two relative homotopies are now in hand: $g_1 f \simeq \mathrm{id}_X$ rel A from Step 2, and $f_1 g_1 \simeq \mathrm{id}_Y$ rel A just obtained. Working throughout with homotopies stationary on A, compute

$$f_1 = f_1\,\mathrm{id}_X \simeq f_1\,(g_1 f) = (f_1 g_1)\,f \simeq \mathrm{id}_Y\,f = f \quad \mathrm{rel}\ A$$

where the first relation postcomposes $g_1 f \simeq \mathrm{id}_X$ rel A by f_1, and the second precomposes $f_1 g_1 \simeq \mathrm{id}_Y$ rel A by f. Hence $f_1 \simeq f$ rel A. Precomposing this relative homotopy with $g_1 \colon Y \to X$, which carries A to A by the identity so that the homotopy stays stationary on A, gives

$$fg_1 \simeq f_1 g_1 \simeq \mathrm{id}_Y \quad \mathrm{rel}\ A$$

Therefore g_1 is a two-sided homotopy inverse of f with $g_1|_A = \mathrm{id}$, and both homotopies $g_1 f \simeq \mathrm{id}_X$ and $fg_1 \simeq \mathrm{id}_Y$ are stationary on A. This is precisely the statement that f is a homotopy equivalence relative to A, completing the proof.

Corollary 1.3.10: homotopy extension and deformation Retraction

Let (X, A) satisfy the homotopy extension proeprty and the inclusion $A \hookrightarrow X$ is a homotopy equivalence, then A is a deformation retraction of X

***Proof*:**
Apply the previous proposition to $A \hookrightarrow X$

Corollary 1.3.11: Homotopy Equivalence via Deformation Retracts

The map $f : X \to Y$ is a homotopy equivalence if and only if X is adeformation retraction of the mapping cylinder M_f. Hence, two spaces X and Y are homotopy equivalent if and only if there is a third space containing both X and Y as deformation retracts.

***Proof*:**
The argument runs through the mapping cylinder $M_f = (X \times I) \sqcup Y/[(x,1) \sim f(x)]$ introduced in example 1.1.9. Identify X with the bottom end $X \times \{0\} \subseteq M_f$ and Y with its image under the gluing, so that both sit inside M_f as subspaces. The map f factors through this cylinder: writing $\iota \colon X \hookrightarrow M_f$ for the inclusion of the bottom end and $r \colon M_f \to Y$ for the map sending $(x,t) \mapsto f(x)$ and fixing Y, the composite $r \circ \iota$ is exactly f. The map r is a deformation retraction of M_f onto Y: the homotopy $r_s(x,t) = (x, (1-s)t + s)$ on the cylinder coordinates, with r_s the identity on Y, slides each point (x,t) along its segment $\{x\} \times I$ up to $(x,1) = f(x) \in Y$, fixing

Y throughout. In particular ι and f differ by the homotopy equivalence r, so ι is a homotopy equivalence if and only if f is.
The pair (M_f, X) satisfies the homotopy extension property. By proposition 1.3.2, it suffices to produce a retraction of $M_f \times I$ onto $(M_f \times \{0\}) \cup (X \times I)$. Write the cylinder coordinates as (x, t, u), where t is the mapping-cylinder coordinate ($X = X \times \{0\}_t$ at the bottom, the glued copy of Y at the top $t = 1$) and u is the homotopy parameter coming from the extra factor of I in $M_f \times I$. On the cylinder portion the target subspace is

$$\big(X \times I_t \times \{0\}_u\big) \cup \big(X \times \{0\}_t \times I_u\big)$$

the first piece being $M_f \times \{0\}$ and the second being $X \times I$, since X sits at $t = 0$. In the square $I_t \times I_u$ this is the L-shape $(I_t \times \{0\}_u) \cup (\{0\}_t \times I_u)$ of the bottom edge together with the $t = 0$ edge.
Retract $I_t \times I_u$ onto this L-shape by radial projection from the apex $(1, 2)$ sitting above the top-right corner: each point is pushed along the ray emanating from $(1, 2)$ until it meets the L-shape. This projection fixes the L-shape pointwise, and it sends the entire right edge $\{1\}_t \times I_u$ to the single corner $(1, 0)$, since the ray from $(1, 2)$ through any $(1, u)$ runs straight down the line $t = 1$ to $(1, 0)$. Performing this retraction on the (t, u)-coordinates of the cylinder gives a retraction of $X \times I_t \times I_u$ onto the L-shape target above, under which the top edge $X \times \{1\}_t \times I_u$ is collapsed by $(x, 1, u) \mapsto (x, 1, 0)$. On the glued copy of Y the target subspace is $Y \times \{0\}_u$, and the retraction collapsing the homotopy parameter, $(y, u) \mapsto (y, 0)$, retracts $Y \times I_u$ onto it.
These two retractions agree on the overlap. Under the gluing $(x, 1) \sim f(x)$, a point $(x, 1, u)$ of the top edge is identified with $(f(x), u) \in Y \times I_u$; the cylinder retraction sends it to $(x, 1, 0)$, identified with $(f(x), 0)$, while the Y-side retraction sends $(f(x), u)$ to $(f(x), 0)$, the same point. Both maps are continuous on their closed domains and agree on the closed overlap, so by the pasting lemma they assemble into a continuous retraction of $M_f \times I$ onto $(M_f \times \{0\}) \cup (X \times I)$. Thus (M_f, X) has the homotopy extension property by proposition 1.3.2.
Suppose first that f is a homotopy equivalence. Then $\iota\colon X \hookrightarrow M_f$ is a homotopy equivalence, by the factorization $f = r \circ \iota$ with r a homotopy equivalence. Since (M_f, X) satisfies the homotopy extension property and the inclusion $X \hookrightarrow M_f$ is a homotopy equivalence, corollary 1.3.10 applies and shows that X is a deformation retract of M_f.
Conversely, suppose X is a deformation retract of M_f. A deformation retraction of M_f onto X exhibits the inclusion $\iota\colon X \hookrightarrow M_f$ as a homotopy equivalence (definition 1.1.10): the retraction $\rho\colon M_f \to X$ satisfies $\rho \circ \iota = \mathrm{id}_X$ and $\iota \circ \rho \simeq \mathrm{id}_{M_f}$. Since $f = r \circ \iota$ with r a homotopy equivalence and ι a homotopy equivalence, f is a composite of homotopy equivalences, hence itself a homotopy equivalence. This establishes the stated equivalence: f is a homotopy equivalence if and only if X is a deformation retract of M_f.
The final assertion follows. If $f\colon X \to Y$ is a homotopy equivalence, then $Z = M_f$ is a space containing both X and Y as deformation retracts, X by the equivalence just proved and Y by the retraction r above. Conversely, if a single space Z contains X and Y as deformation retracts, then the inclusions $X \hookrightarrow Z$ and $Y \hookrightarrow Z$ are homotopy equivalences, and composing one with a homotopy inverse of the other produces a homotopy equivalence $X \simeq Y$, completing the proof.

This last corollary alludes to a more general notion known as *cobordism* which says that tow manifolds are *cobordic* if their union is the boundary of a manifold of one higher dimension. This shall not be further addressed in this book, but those interested can look at EYNTKA differential Geometry.

Exercise 1.3.1

1. Show that homotopies form an equivalence relation.

2

Fundamental Group

As we saw, homotopies allow us to study spaces up to continuous deformation along paths. It is thus natural try and study the paths on a space. In this section, we shall see that paths that are closed (that form a loop) shall naturally form a group structure, and this group structure shall be invariant under homotopy giving our first algebraic invariants on topological spaces. This group shall be called the *fundamental group* or *homotopy group*.

After having defined the fundamental group, we shall give many ways of computing it, how we can piece it together from the homotopy group of smaller spaces, and how homotopy groups of CW complexes can be calculated. Along the way, we shall see that most spaces X shall have a *universal cover*, say $U(X)$ that shall be simply connected (all paths are nullhomotopic) and will be a geometric representation of the fundamental group. We shall see that that there are many intermediary spaces between X and $U(X)$, and the correspondence between the intermediary spaces and the fundamental group of X shall mirror that of the famous *galois correspondence* between the intermediary fields and the subgroups of the galois group.

2.1 Path Homotopy and Fundamental Group

A lot of study is on homotopies, which will eventually bring you to homotopy groups, homology, exact sequences of homotopic groups and Hopf fibration of the sphere. An important class of homotopies worth singling out is homotopies which are "1-dimensional". Working with one-dimensional one's doesn't mean we won't be working with higher-dimensional objects, but the path in the space we use that will be homotopic will be one dimensional.

Definition 2.1.1: Path-Homotopic

Two paths $f, \tilde{f}$ in X are *path-homotopic* if

1. they have the same initial and final points: $f(0) = f'(0)$, $f(1) = f'(1)$
2. There exists a continuous function $F : I \times I \to X$ such that

$$F(s, 0) = f(s), F(s, 1) = f'(s), F(0, t) = x_0, f(1, t) = x_1$$

where $f(0) = f'(0) = x_0$ and $x_1 = f(1) = f'(1)$. We write

$$f \simeq_p f'$$

F is called a *path-homotopy*

Example 2.1.2: Path Homotopy

1. If we take all function $I \to I$ such that $x \mapsto x^n$, then all of these are path-homotopic. In fact, any path such that $f(0) = g(0) = 0$, $f(1) = g(1) = 1$ will be path-homotopic by the straight-line homotopy
2. More generally, if C is a convex set, then any two paths with the same initial and ending points are path-homotopic by the straight-line homotopy
3. "There must be more than the straight-line homotopy" you might be thinking, and you're right; think of previous examples we have worked with in the last chapter and see if you can find a path homotopy that is not a straight line
4. "Are there paths that are not homotopic?". That can be the case too. Take $X = \mathbb{R}^2 - \{0\}$ and

$$f(s) = (\cos(\pi s), \sin(\pi s)) \qquad g(s) = (\cos(\pi s), 2\sin(\pi s))$$

then these two functions are path homotopic using the straight-line homotopy, as you see visually:

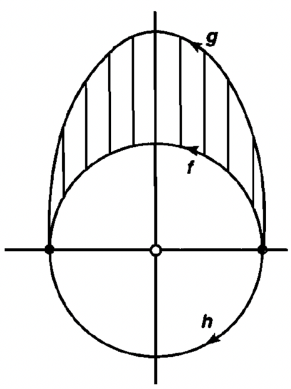


Figure 2.1: Same initial and end points, but one goes under point

In general, if the point 0 was not missing, we can have:

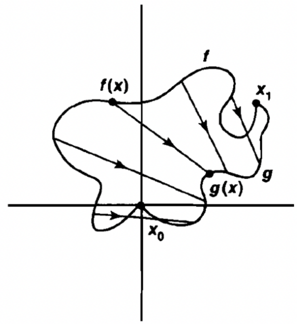


Figure 2.2: Convex space

However, the path $h(s) = (\cos(\pi s), -\sin(\pi s))$ is *not* path-homotopic. In the visual, this is the h path. It should perhaps seem intuitive that we we try to pass a continuous path *through* the point 0, we will introduce a discontinuity. The proof for this is a bit laborious at this point, so being satisfied with this "hole" intuition is useful.

5. Recall all polynomials $x \mapsto x^n$ from earlier? These are in the same equivalence class

$$[f] = [g]$$

And since $F'(x,t) = (x, (t-1)x + t0)$ is also a path-homotopy,

$$[id_0] = [f] = [g]$$

meaning it is also *nullhomotopic*

As may be expected from objects in a category, the morphisms form an object to itself froms a groupoid, and automorphisms form a group. This group is rather complicated and is the study of more recent mathematics[1], and so we shall construct a new much simplier algebraic invariant bassed off of the paths in a space.

Definition 2.1.3: identity path

If $x \in X$, denote e_x to be the constant path $e_x : I \to X$ such that $e_x(t) = x$ for $t \in I$

Therefore, the id in our previous example would be relabeled as e_0.

Next, we need to figure out a way defining an operation between paths. The natural candidate for this would be to extend one path to the next. However, we can extend *arbitrary* paths from one to the next, since that will make discontinuous paths (because the end point of one path might not be the initial point of another). Thus, we we try and define paths by through "combination", we need to impose this limit:

[1]See for example the study of $\mathrm{Diff}(M)$ for a manifold M

Definition 2.1.4: combining paths Operators: $*$

If f, g are paths in X and $f(1) = g(0)$, define $f * g$ by

$$(f * g)(t) = \begin{cases} f(2t) & t \in [0, \frac{1}{2}] \\ g(2t-1) & t \in [\frac{1}{2}, 1] \end{cases}$$

Example 2.1.5: Combining Paths

Try thinking of any two paths in $\mathbb{R}^2$ and combining them into 1.

Lemma 2.1.6: Well-defined: $*$ Preserves Equivalence Relation

if $f \simeq_p f'$, $g \simeq_p g'$, then

$$(f * g) \simeq_p (f' * g')$$

***Proof*:**

Let F, G be homotopies between f and f', g and g' respectively. Basically, we'll just be doing just like in the equivalence relation, doubling the speed appropriately and using the pasting lemma to combine the two continuous functions. Let $H : I \times I \to Y$ such that

$$H(x, t) = \begin{cases} F(2x, t) & t \in [0, \frac{1}{2}] \\ G(2x-1, t) & t \in [\frac{1}{2}, 1] \end{cases}$$

then note that $H(x, 0) = f * g$ and $H(x, 1) = f' * g'$, and H is continuous by the pasting lemma, and so we're done

Because of this, can define $*$ on equivalence classes:

$$[f] * [g] = [f * g]$$

With the operation defined, we can look at the properties of it to see if it would follow all the axioms of a group. Remember that when we defined $*$, it only works on appropriate paths (with end and initial points matching). This will need to be considered in our up-coming examination of its properties:

Theorem 2.1.7: Properties Of $*$

1. (Associativity) Assume f, g, h are paths in X, $f(1) = g(0)$, $g(1) = h(0)$. Then

$$([f] * [g]) * [h] = [f] * ([g] * [h])$$

2. (identity) If f is a path in x, $f(0) = x_0$, $f(1) = x_1$, then

$$[f] * e_{x_1} = [f] = e_{x_0} * [f]$$

3. (invertible) Given a path f in X, define $\tilde{f}(t) = f(1-t)$ Then

$$[f] * [\tilde{f}] = [e_{f(0)}]$$
$$[\tilde{f}] * [f] = [e_{f(1)}]$$

***Proof*:**
Some useful facts that will be used continuously

1. Let $k : X \to Y$ be a continuous map and F be a path homotopy between f and f'. Then $k \circ F$ is a path-homotopy between $k \circ f$ and $k \circ f'$ where

$$(k \circ F)(s, t) = k(F(s, t))$$

2. composition and $*$ are basically distributive: $k \circ (f * g) = (k \circ f) * (k \circ g)$.

As you probably noticed in the construction, the identity wasn't unique! This means that $*$ just falls short of actually being a binary operation. In fact, it is not actually a function, but a *partial function*; you can't take two arbitrary paths, but you need to select paths with the appropriate mid-points (as we noticed when defining $*$ originally!). If we restrict to paths that are always concatinatable, we will indeed have a binary function, but we usually want to think of $*$ being defined on our entire space.

We actually give a name to a "group" which uses a partial function (and hence, has multiple identities depending on how you must restrict the partial function to get a function):

Definition 2.1.8: Groupoid

A *groupoid* is a set (ex. G) with a *partial function* $* : G \times G \to G$ *and* a unary function $^{-1}: G \to G$ which then satisfy the group axioms (associatvitiy, inverse, identity). The slight difference between groups is that we now have:

1. (inverse) $a^{-1} * a$ and $a * a^{-1}$ is always defined
2. if $a * b$, is defined then $a * b * b^{-1} = a$ and $a^{-1} * a * b = b$ is also defined

How exactly the partial function $*$ is "partial" depends on the context - there is no one way of defining this partial function. Furthermore, we choose a unary function for the inverse because the "identity" is now no longer so "unique", as was shown in theorem 2.1.7.

Example 2.1.9: Groupoid Example

f we take $(\mathbb{R}, *, ^{-1})$ to be our groupoid with $*$ as we defined and $(f)^{-1} = \tilde{f}$ where $\tilde{f} = f(1-x)$ (i.e. reverse f), then we defined a groupoid!

With the algebraic structure defined and labeled, we can define the elements of our groupoid:

Definition 2.1.10: Path Space

Let $P(X) = \{\text{equivalence classes of paths } [f]\}$. We will call this the *path space*

there is a natural map $P(X) \to X^2$, $[f] \mapsto (f(0), f(1))$. This map is surjective if and only if X is path connected. You can also think of the groupoid from earlier as

$$(P(X), *, ^{-1})$$

As mentioned, we still don't have a "unique" identity nor a total function, so we are yet to totally define a group. The limiting factor is that we can just combine any function with any function. However, if the initial- an end-points where the same, this would solve our problem! For this reason, we will once again restrict our attention, this time to paths that *loop*:

Definition 2.1.11: Loop

Given $x_0 \in X$ a *loop based at* x_0 is a pth which begins and ends at x_0. that is

$$f : I \to X, \qquad f(0) = f(1) = x_0$$

There is actually an advantage to adding this resctriction and thinking about loops: Imagine you have a plane with a point missing. Then having a loop around that point would mean you can't contract that loop to the constant map (as we've seen before, where we sneakily introduced a loop). This idea of not being able to contract loops will be our key concept! With this definition, we can consider the group structure on a subset of $P(X)$ such that the initial and end points match

Definition 2.1.12: Fundamental Group [at x_0]

Let

$$\pi_1(X, x_0) = \{[f] | f(0) = f(1) = x_0\}$$

be the path space with loops at x_0. Then $\pi_1(X, x_0)$ is a group under $*$, called *fundamental group* of X *based at* x_0, or the *first homotopy group*

2.1.13 Note there *can* be more than one element in $\pi_1(X, x_0)$. We will soon explore spaces where this is the case, and also explore cases where there is only one element.

Sometimes $\pi(X)$ is used to represent the groupoid of the space X. One will naturally as if there is a 1st homotopy group, is there a 2nd? an nth? There is, and it's denoted $\pi_n(X, x_0)$. This higher value of n is where you can think of "dimensions" as I mentioned earlier. These, though, will not be studied here, and are the subject of homotopy theory.

Example 2.1.14: Fundamental Group

Let's consider $\mathbb{R}^n$ with the Euclidean topology and some $x_0 \in \mathbb{R}^n$ to define $\pi_1(\mathbb{R}^n, x_0)$. Then by the straight-line Homotopy, we can collapse every loop to the constant map, so there is only one element:

$$\pi_1(\mathbb{R}^n, x_0) = \{[e_{x_0}]\}$$

More generally, if you have a convex space, then it's fundamental group will be trivial (since you can use the straight-line homotopy).

Note that this group is *not* necessarily abelian! We shall show that $S^1 \vee S^1$ has fundamental group as the free product of two copies of the integers, $\mathbb{Z} * \mathbb{Z}$.

So how many fundamental groups are there on a space? Does every point have a distinct type of group or are some of them isomorphic, or better yet, all of them isomorphic on the same space. The different answers to this will actually give us the topological property we're looking for in groups! That is, when thinking about a space, we usually think about the fundamental groups *up to isomorphism* and then the topologically invariant property are the *types* of fundamental groups. That means that the fundamental group does not rely on the point x_0 on which it was originally defined to get it's structure. To that matter, it will also not rely on the *entire* space X, just the component spaces, so the interesting thing to study are the path-connected components.

To get to that intuition, we first, specify how to define the isomorphism on the group

Proposition 2.1.15: Group Isomorphism Between Fundemental Groups

Let α be a path X $\alpha(0) = x_0$, $\alpha(1) = x_1$, then define the group isomorphism

$$\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$$

where $\hat{\alpha}([f]) = [\alpha] * [f] * [\bar{\alpha}]$ and $\bar{\alpha}$ is the reverse of the funtion.

In words, what this functions does is go back, do the loop, then go back forward.

***Proof*:**
This is basically an Algebra proof. To show it's homomorphic:

$$\begin{aligned}\hat{\alpha}([f] * [g]) = \hat{\alpha}([f * g]) &= [\alpha] * [f * g] * [\alpha] \\ &= [\alpha] * [f * e_{x_1} * g][\bar{\alpha}] \\ &= [\alpha] * [f * \alpha * \tilde{\alpha} * g][\bar{\alpha}] \\ &= ([\alpha] * [f] * [\bar{\alpha}]) * ([\alpha] * [g] * [\bar{\alpha}]) \\ &= \hat{\alpha}([f]) * h(g)\end{aligned}$$

For bijectivity, we can define the two-sided inverse function $\hat{\alpha}^{-1}$ by reverseing $\tilde{\alpha}$, where it can be checked that

$$\hat{\alpha}(\hat{\alpha}^{-1}([f])) = \hat{\alpha}^{-1}(\hat{\alpha}([f])) = [f]$$

Making the homomorphism bijective, making it isomorphic

Corollary 2.1.16: Path-Connectedness And Isomorphism

If X is path-connected, $\pi_1(X, x_0)$ is isomorphic to $\pi_1(X, x_1)$ for any $x_0, x_1 \in X$

This implies that for every path-connected component of X, it will have the *same* fundamental group at each point! Thus, when thinking of Fundamental groups, it is convenient of thinking of X as being a path-connected space (or at least, it will come down to breaking X up to it's path connected components). For the more Categorically inclined, there is also no *natural* isomorphism between points! This will come down to the fact that not every space is commutative[2]. The following corrollary may make this more clear:

Corollary 2.1.17: Fundamental Group Abelian

Let X be a path connectd space. Then $\pi_1(X)$ is abelian if and only if any transition map β_h (where $h(0) = b_0$ and $h(1) = b_1$) only depends on the endpoints of h.

***Proof*:**
Let's say $\pi_1(X)$ is abelian. Consider:

$$\beta_{h_1}([f]) = [h_1 * f * \bar{h}_1]$$

Then:

$$\begin{aligned}
\beta_{h_1}([f]) &= [h_1 * f * \bar{h}_1] \\
&= [h_1 * f * \bar{h}_2] * [h_2 * \bar{h}_1] \\
&= [h_2 * \bar{h}_1] * [h_1 * f * \bar{h}_2] \\
&= [h_2 * f * \bar{h}_2] \\
&= \beta_{h_2}([f])
\end{aligned}$$

Conversely, if $\pi_1(X)$ were not abelian then there exists $[f], [g] \in \pi_1(X)$ such that $[f]*[g] \neq [g]*[f]$, that is $[\bar{g}] * [f] * [g] \neq [f]$. Now, pick some β_h. Since it is an isomorphism and $\beta_{\bar{g}h}$ is also a path-isomrophism:

$$\beta_{\bar{g}h}([f]) = \beta_h([\bar{g}] * [f] * [g]) = [h * \bar{g} * f * g * \bar{h}] \neq [h * g * \bar{h}] = \beta_h([f])$$

showing we have two different paths, completing the proof.

Something we shall not prove but is certainly worth noting is that the fundamental group of a topological group is *always*, hence the topological group of a lie group is always abelian!

Overall, if two path-connected spaces with different fundemental groups, they are topologically different. However, in general, different fundemental groups doesn't mean different spaces (ex. $\mathbb{R}$ and $\mathbb{R}^2$ have the same Fundemental group). The Fundamental group that is simplest to compute is the trivial one we have seen often. If a space has a trivial fundamental group, we give it a name:

[2] see [14] for more details

Definition 2.1.18: Simply-Connected

X is *simply-connected* if X is path-connected, and $\pi_1(X, x_0)$ is the trivial group for some (and hence all) $x_0 \in X$

Note that $\pi_1(X, x_0) = 0$ but a space is *not* contracbie (take for example the warsaw circle, whih is $\sin 1/x$) but turned into a circle, which we shall show is not contractible once we show that $\pi_1(S^1) \cong \mathbb{Z}$). A theorem we shall not show but state right now for reference is Whitehead's theorem: a CW complex whose homotopy groups are all zero is contractible.

Lemma 2.1.19: simply connected simplification

If X is simply connected, $P(X) \cong X^2$ (i.e. a bijection),that is a in X is only determined by its endpoints.

***Proof*:**
First, we'll show all paths are homotopic. Let α, β be paths from x_0 to x_1. Then

$$\begin{aligned}
[\alpha] * [\tilde{\beta}] &\in \pi_1(X, x_0) \\
[\alpha] * [\tilde{\beta}] &= [e_{x_0}] \\
[\alpha] * [\tilde{\beta}] * [\beta] &= [e_{x_0}] * [\beta] \\
[\alpha] &= [\beta]
\end{aligned}$$

With this, we can show injectivity. If $\pi([\alpha]) = \pi([\beta])$, then

$$\alpha(0) = \beta(0), \qquad \alpha(1) = \beta(1)$$

then $[\alpha] = [\beta]$ by proof above

And or surjectivity, given $(x_0, x_1) \in X^2$, there exists a path $\alpha(0) = x_0$, $\alpha(1) = x_1$ such that

$$\pi([\alpha]) = (x_0, x_1)$$

completeing the proof
I will have to look over that surjectivity claim again because it looks like it's breaking well-definedness

2.1.1 Functorial Properties

We've shown that we can define a fundamental group on a space, but what if we have another topological space Y, and a continuous function $f : X \to Y$, what of the group-structure we just build up? Well, we wouldn't of introduced fundamental groups if they weren't topological in nature, which means that continuous maps *preserve* them! This is also how the comment much earlier that in a sense topology is fundamentally the study of objects that don't change under continuation functions comes in.

So if we have a continuous function, can we define an appropriate *homomorphism* (the equivalent of continuous functions). This is what we'll do in this section.

Definition 2.1.20: induced homomorphic function

Suppose $h : X \to Y$ is continuous, with $h(x_0) = y_0$ (i.e. $h(X, x_0) \to (Y, y_0)$) . Then define

$$h_* : \pi_1(x, x_0) \to \pi_1(Y, y_0) \qquad h_*([\alpha]) = [h \circ \alpha]$$

2.1.21 Trivia Notice that it's a topology isomorphism which also specifies a point is a point. This makes our function a little more than continuous, it's continuous *and* preserves that point. Categorically, this means we're working in a slightly different category than Topology (Top): It's technically the Base-topology at x_0 ($\mathbf{Top}_{x_0}$) category.

Proposition 2.1.22: $(-)_*$ is a Functor

1. h_* is a a group-homomorphism
2. If $i : (X, x_0) \to (X, x_0)$ is the identity, then i_* is the identity
3. This map is functorial, that is, if $f : (X, x_0) \to (Y, y_0)$ and $(Y, y_0) \to (Z, z_0)$ then

$$(g \circ f)_* = g_* \circ f_*$$

   Furthermore, if $f : (X, x_0) \to (Y, y_0)$ is a homeomorphism, then f_* is an isomorphism.

Thus, there is a functor between the $\mathbf{Top}_{\{x_0\}}$ Category and the **Grp** Category

***Proof*:**
We'll prove each separately

1.

$$\begin{aligned} h_*([\alpha][\beta]) &= h_*([\alpha * \beta]) \\ &= [h \circ (\alpha * \beta)] \\ &= [(h \circ \alpha) * (h \circ \beta)] \\ &= [(h \circ \alpha)] * [(h \circ \beta)] \\ &= h_*([\alpha]) * h_*([\beta]) \end{aligned}$$

2. $i_*([\alpha]) = [i \circ \alpha] = [\alpha]$, hence it's the identity
3.

$$\begin{aligned} (g \circ f)_*[\alpha] &= [(g \circ f) \circ \alpha] \\ &= [g \circ (f \circ \alpha)] \\ &= g_*([f \circ \alpha]) \\ &= g_*(f_*([\alpha])) \end{aligned}$$

Thus, we've shown that the fundamental group is invariant to homeomorphisms! This is excellent, for now we can properly study them as a topological property of our space. The first indeavour we should seek out is how to reasonably calculate fundamental groups of spaces.

2.1.2 Covering Spaces and Lifting Property

The way we shall seek out fundamental groups is by associating with them a new universal object that shall geometrically represent the loops of a space (hence, geometrically represent the algebraic properties of the fundamental group). The main idea is that each space X can be "covered" by a large space E which will be easier to analyze (usually simply connected). Then we may ask for a "unviersal covering space" that will be a simply connected object that will somehow represent all possible "untangling" of loops, and will exist for almost every space we shall work with, the exception ususally being pathological spaces such as the Hawaian earings (which shall soon be defined)

2.1.23 Where covering-space theory lives The point-set theory of covering spaces , evenly covered neighbourhoods, path and homotopy lifting, the lifting criterion, the universal cover, the classification by subgroups, and deck transformations, is proved in *Topology* [12], which is a prerequisite for this book. The definitions and statements are repeated here because the arguments below read off them constantly, but the proofs are not: each cites the source. What this book proves is the payoff, which the point-set theory cannot reach: Van Kampen's theorem, the homology of covering spaces, orientation coverings, and classifying spaces.

Definition 2.1.24: Evenly Covered and Slices

Let $p : E \to B$ be a continuous surjective map. The open set U of B is said to be *evenly covered* by p if the inverse image $p^{-1}(U)$ can be written as the union of disjoint open sets V_α in E such that for each α, the restriction of p to V_α is a homeomorphism of V_α onto U. The collection $\{V_\alpha\}$ will be called the partition of $p^{-1}(U)$ into *slices*

if p does this to every neighbourhood in B, we call such a map p a covering map:

Definition 2.1.25: Covering Map

Let $p : E \to B$ be continuous and surjective. If every point b of B has a neighborhood U that is evenly covered by p, then p is called a *covering map*, and E is said to be a *covering space* of B.

Example 2.1.26: Covering Maps

1. The identity map $i : X \to X$ is "trivially" a covering map
2. The next most trivial cover is if we have many copies of the object. If $X \times \{1, ..., n\}$ is n disjoint copies of X, then $p(x, i) = x$ for all i is again a covering map, since all the spaces are disjoint and when you limit to any space, it's homeomorphic to X. This function gives a nice visual for how you can think of covering spaces in general:

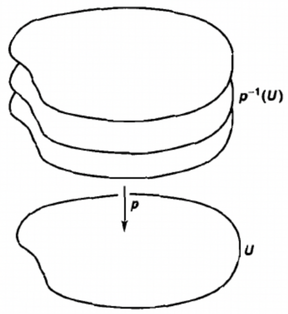


Figure 2.3: Covering space example

3. The map $p : \mathbb{R} \to S^1$ given by the equation

$$p(x) = (\cos(2\pi x), \sin(2\pi, x))$$

is a covering map. This comes down to geometric intuition about these trigonometric functions and circles. As an intuitive check that you get this function, if we take any interval $[n, n + 1] \subseteq \mathbb{R}$, then that interval will wrap itself around the circle completely. Notice that $\mathbb{R}$ is simply connected, while we shall show that S^1 is not simply connected.

4. Another covering space for the circle is $p : S^1 \to S^1$ given by $z \mapsto z^n$ treating $S^1 \subseteq \mathbb{C}$.

5. It may be tempting to say that being a local homeomorphism implies p is a covering map, but this is not the case. Consider:

$$p : \mathbb{R}_{\geq 0} \to S^1 \qquad p(x) = (\cos(2\pi x), \sin(2\pi x))$$

This function is still surjective, and a local homeomorphism (any point $x \in \mathbb{R}_{\geq 0}$ has a neighorhood homeomorphic to it's image), and even is a covering map for *most* points. However, consider the point $b_0 = (1, 0)$ and some open neighborhood U around that point. Then the pre-image of U would be the same as in the third 3rd example, except for V_0 which would be $(0, \epsilon)$ for some ϵ depending on the open neighbourhood U. If we restrict p to $(0, \epsilon)$, then we do not get a homeomorphism, but an *embedding*, that is, a function for which if we restrict the co-domain to it's image, it becomes a homeomorphism. If we restrict to $(0, \epsilon)$ we'd get the open neighborhood starting at 0 and ending at ϵ (relative to where sin and cos sends ϵ.)

Anoher example would be:

$$E = [0, 1] \times \{0\} \cup (0, 1) \times \{1\}, B = [0, 1]$$

which is basically forcing surjectivity on the function $(0, 1) \to [0, 1]$. Then there isn't a neighborhood around $0 \in [0, 1]$, which is evenly covered. Again, this comes down to it being an embedding based on the domain rather than the co-domain.

6. the map $\varphi : \mathbb{C} \to \mathbb{C}^\times$ mapping $z \mapsto e^z$ is a covering map. It is in fact an ∞ to 1 covering map since $z + 2k\pi i \mapsto e^z$. Note that $\mathbb{C}$ is simply connected.

Proposition 2.1.27: Properties of Covering Maps

Let $p : E \to B$ be a covering map.

1. Every $y \in B$, the subspace $p^{-1}(y)$, has the discrete topology
2. The restriction of a covering map by an open $A \subseteq E$ set is a covering map: $p|_A : A \to B$[a]
3. If B_0 is a subspace of B, and if $E_0 = p^{-1}(B_0)$, then the map $p_0 : E_0 \to B_0$ obtained by restricting p is a covering map
4. $p' : E' \to B'$ is also a covering maps, then

$$p \times p' : E \times E' \to B \times B'$$

is a covering map

[a] example 2.1.26 shows why this is not true for general subspaces

***Proof*:**

1. This should be a straight forward test of understanding coverings. Take a neighborhood of y, $y \in U$. Since p is a covering map, then it's pre-image is partitioned into $\{V_\alpha\}$. Restricting our attention to any one slice V_α will lead to a homeomorphism (so for us, bijection) betweeen V_α and U, meaning one point of it maps to U. Since we can do this for every V_α, we get that $p^{-1}(y)$ is a collection of points intersected by open sets, meaning the subspace is discrete.
2. Let $A \subseteq E$. We'll show that $p(A)$ is open. If around every point $x \in p(A)$, there exists an open set V such that $V \subseteq p(A)$, then $p(A)$ is open (basis definition). Let's take a point $x \in p(A)$. Choose an open neighborhood V, $x \in V$. Then since p is an open cover, choose a neighbourhood that is evenly covered by p. Let $\{V_\alpha\}$ be the neighborhood. Since $x \in p(A)$, $p^{-1}(x) \in A$, so choose $y \in p^{-1}(x)$, and the appropraite V_β such that $y \in V_\beta$. Since A is open, $A \cap V_\beta$ is also open. Since p restricted to a V_β is a homeomorphism, $p(A \cap V_\beta)$ is *also* open. Note that $p(A \cap V_\beta) \subseteq p(A)$. Since we found an open neighborhood around x which is contained in $p(A)$, then $p(A)$ is open.
3. exercise
4. Let $b \in B$, $b' \in B$. Then we can take neighbourhoods U, U', $b \in U$, $b' \in U'$ which are evenly covered, so we have $\{V_\alpha\}$ and $\{V'_\alpha\}$. We can take the cross-product of the respective slices. They will still be disjiont, and restricting to slices still a homeomorphism, thus $p \times p'$ is a covering map

Example 2.1.28: Product of Covering Map

1. Take

$$p \times p : \mathbb{R} \times \mathbb{R} \to S^1 \times S^1$$

where p is the same covering map of the circle we introduced earlier. Then this function is a covering map for the torus!

2. Notice that in the Torus we have a hoemomorphic copy of the figure 8 ($S^1 \vee S^1$). We thus have a covering map for it given by proposition 2.1.27(4). There are multiple ways of covering this space. You can keep one circle and unwrap the other, or you can unwarp both. This space will also be a space that will have a universal cover.

As mentioned, the key property about covering spaces that is of interst to us is how it will represent fundamental groups. We now build up to "lifting" paths from our base space up to our covering space in a way that simplifies calculations:

Definition 2.1.29: Lifting

Let $p : E \to B$ be an [arbitrary] continuous function. Let $f : X \to B$ be a continuous map. A *lifting* of f is a continuous map $\tilde{f} : X \to E$ such that $f = p \circ \tilde{f}$. Similarly, we can say that this commutative diagram commutes

$$\begin{tikzcd} & E \arrow[d, "p"] \\ X \arrow[ur, "\tilde{f}"] \arrow[r, "f"] & B \end{tikzcd}$$

If you recall the theory of projective modules, you may recognize homomorphisms always be lifted. Key to us when dealing with homotopies is that lifts through covering maps preserve paths!

Lemma 2.1.30: Paths Lifted Through Covering Map

Let $p : E \to B$ be a covering map such that $p(e_0) = b_0$ and let $f : [0,1] \to B$ be a continuous with inital point $f(0) = b_0$. Then it has a unique lifting to a path $\tilde{f}$ in E with initial point $\tilde{f}(0) = e_0$, that is the following diagram commutes:

$$\begin{tikzcd} & E \arrow[d, "e_0 \mapsto b_0"] \\ {[0,1]} \arrow[ur, "0 \mapsto e_0"] \arrow[r, "0 \mapsto b_0"'] & B \end{tikzcd}$$

***Proof*:**
Proved in *Topology* [12, paths can be lifted]: subdivide $[0,1]$ finely enough that each piece lands in an evenly covered set, using the Lebesgue number of the pulled-back cover, then lift piece by piece into the slice determined by the previous endpoint and glue by the pasting lemma. Uniqueness follows because each piece is connected and the slices are disjoint.

Lemma 2.1.31: Homotopies can be lifted

Let $p : E \to B$ be a covering map, and let $p(e_0) = b_0$. Let $F : I \times I \to B$ be a continuous function such that $F(0,0) = b_0$. Then there exists a unique lift $\tilde{F} : I \times I \to E$ such that $\tilde{F}(0,0) = e_0$. If F is a path homotopy, then $\tilde{F}$ is a path homotopy.

***Proof*:**
Proved in *Topology* [12, Homotopies Can Be Lifted, Path Homotopies can be lifted], by the same subdivision argument as the previous lemma run in two variables over the square, together with the observation that the lift of a path homotopy is again one because the fibres are discrete and the edges are connected.

Note that though paths are lifted, loops *need not be lifted*, in particular it is often the case that a loop is lifted to a path. However, if the path/loops are homotopic, the endpoints of two path-homotopic functions also agree:

Lemma 2.1.32: Path Homotopies can be lifted

Let f, g be paths starting and ending at b_0 (i.e. path-homotopic loops). If $f \simeq_p g$, then $\tilde{f}(1) = \tilde{g}(1)$

***Proof*:**
The path homotopy F between f, g lifts to a path homotopy between $\tilde{f}, \tilde{g}$. In particular, the end points of homotopic paths agree, therefore $\tilde{f}(1) = \tilde{g}(1)$.

Using this, we see that lifiing continuous maps is in fact directly tied with the homotopy group:

Theorem 2.1.33: Conditions for Maps being Lifted

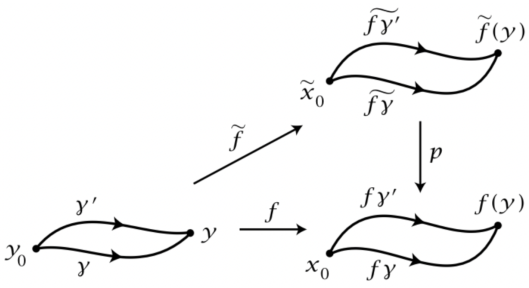
Let $p : (\tilde{X}, e_0) \to (X, x_0)$ be a covering map and $f : (Y, y_0) \to (X, x_0)$ where Y is path conected and locally path connected[a]. Then the lift $\tilde{f}$ exists if and only if $f_*(\pi_1(Y_0)) \subseteq p_*(\pi_1(\tilde{X}, e_0))$

[a] for each $y \in Y$, there exists an open neighbourhood that is path connected, think of a comb-space for a connected, not locally path connected space

***Proof*:**
If $\tilde{f}$ exists, then by functoriality $f_* = p_* \circ \tilde{f}_*$, hence $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, e_0))$.

Conversely, suppoe $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1, \tilde{X}, e_0)$. Choose $y \in Y$, and let γ be a path in Y from y_0 to y. Then $f\gamma$ is a path in X starting at x_0 and and going to $f(y_0)$. Then by lemma 2.1.32, $f\gamma$ lifts to a path in e_0.

Now, define
$$\tilde{f}(y) = \tilde{f\gamma}(1)$$
This map is well-defined: if γ' is another path from y_0 to y, then $h_0 = (f\gamma') * (\overline{f\gamma})$ is a loop at x_0. By assumption $[h_0] \in f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, e_0))$. We shall essentially be mimicking this visual:



Since $[h_0] \in p_*(\pi_1(\tilde{X}, e_0))$, there is a homotopy h_t of h_0 to a loop h_1 in Y that lifts to a loop $\tilde{h}_1$ in $\tilde{X}$ at basepoint e_0. The homotopy h_t can be lifted to $\tilde{h}_t$ by the above lemma. Then since $\tilde{h}_1$ is a loop at e_0, so is $\tilde{h}_0$. By the uniqueness of the lifted paths, the first half of $\tilde{h}_0$ is $\tilde{f\gamma'}$ and the second half is $\tilde{f\gamma}$ traversed backwards, with common midpoint $\tilde{f\gamma}(1) = \tilde{f\gamma'}(1)$. Hence, $\tilde{f}$ is well-defined.

What's left to show is that $\tilde{f}$ is continuous. Let $U \subseteq X$ be open neighbourhood of $f(y)$. Then choose $\tilde{U} \subseteq \tilde{X}$ containing $\tilde{f}(y)$ such that $p : \tilde{U} \to U$ is a homeomorphism. Choose a path-connected open enough V of y such that $f(V)$
$subseteqU$. For paths from y_0 to points $y' \in V$, we can take a fixed path γ from y_0 to y followed by paths η in V from y to the poitns y'. Then the path $(f\gamma) * (f\eta)$ in X have lifts $(\tilde{f\gamma}) * (\tilde{f\eta})$ where $\tilde{f\eta} = p^{-1}f\eta$ and $p^{-1} : U \to \tilde{U}$ is the inverse of $p|_{\tilde{U}}$. Thus, $\tilde{f}(V) \subseteq \tilde{U}$, and $\tilde{f}|_V = p^{-1}f$, hence $\tilde{f}$ is continuous at any y, completing the proof.

Corollary 2.1.34: Unique Lifting Property

Let $p : \tilde{X} \to X$ be a covering map and $f : Y \to X$ be a continuous map. Then if $\tilde{f}_1, \tilde{f}_2 : Y \to \tilde{X}$ are two lifts of f that agree at one point Y and Y is connected, then $\tilde{f}_1$ an $\tilde{f}_2$ agree on all of Y, that is

$$\tilde{f}_1 = \tilde{f}_2$$

***Proof*:**
The strategy is the one that recurs throughout the lifting theory: the locus where the two lifts agree is simultaneously open and closed, so connectedness of Y forces it to be all of Y. Set

$$S = \{y \in Y \mid \tilde{f}_1(y) = \tilde{f}_2(y)\}$$

By hypothesis S is nonempty, since the two lifts agree at some point. It suffices to show that S is both open and closed in Y, for then connectedness of Y leaves $S = Y$ as the only nonempty option, which is exactly the asserted equality $\tilde{f}_1 = \tilde{f}_2$.

Fix $y \in Y$ and write $x = f(y) \in X$. Since $p\colon \tilde{X} \to X$ is a covering map, choose an evenly covered open neighbourhood U of x, so that $p^{-1}(U) = \coprod_\alpha V_\alpha$ is a disjoint union of slices, each mapped homeomorphically onto U by p (definition 2.1.24). The two points $\tilde{f}_1(y)$ and $\tilde{f}_2(y)$ both lie in $p^{-1}(U)$, because $p\tilde{f}_i(y) = f(y) = x \in U$; let V_1 and V_2 be the slices containing them, respectively. By continuity of the lifts there is an open neighbourhood N of y with $\tilde{f}_1(N) \subseteq V_1$ and $\tilde{f}_2(N) \subseteq V_2$; shrinking N if necessary, N may also be taken to satisfy $f(N) \subseteq U$.

Suppose first that $y \in S$, so $\tilde{f}_1(y) = \tilde{f}_2(y)$. This common point lies in both V_1 and V_2, and distinct slices are disjoint, so $V_1 = V_2$; call this common slice V. On N both lifts land in V, and on V the projection p is injective. For any $z \in N$ the relation $p\tilde{f}_1(z) = f(z) = p\tilde{f}_2(z)$ together with this injectivity gives $\tilde{f}_1(z) = \tilde{f}_2(z)$, so $N \subseteq S$. Hence every point of S has a neighbourhood contained in S, and S is open.

Suppose instead that $y \notin S$, so $\tilde{f}_1(y) \neq \tilde{f}_2(y)$. These two points cannot lie in the same slice, for p is injective on each slice while $p\tilde{f}_1(y) = p\tilde{f}_2(y)$; thus $V_1 \neq V_2$, and being distinct slices they are disjoint. For $z \in N$ we then have $\tilde{f}_1(z) \in V_1$ and $\tilde{f}_2(z) \in V_2$ with $V_1 \cap V_2 = \emptyset$, so $\tilde{f}_1(z) \neq \tilde{f}_2(z)$ for every $z \in N$. Thus N is a neighbourhood of y disjoint from S, which shows that the complement $Y \setminus S$ is open, that is, S is closed.

The set S is therefore a nonempty subset of Y that is both open and closed. As Y is connected, $S = Y$, so $\tilde{f}_1(y) = \tilde{f}_2(y)$ for every $y \in Y$ and consequently $\tilde{f}_1 = \tilde{f}_2$, as we sought to show.

As mentioned, the lifts are *not necessarily* loops. When projected using the covering space, they will become loops. The key is that all the paths in B at b_0 lift to paths that start at $e_0 \in p^{-1}(b_0)$ and end at $e_k \in p^{-1}(b_0)$. We may imagine the lifting correspondence as somehow "untangling" the paths. With path-homotopies being conserved with liftings and being homotopy-independent, we can now talk about fundamental groups when their lifted:

Definition 2.1.35: Lifting Correspondence

Let $p : E \to B$ be a covering map. Let $p(e_0) = b_0$. Given $[f] \in \pi_1(B, b_0)$, let $\tilde{f}$ be a lifting of f to E such that $\tilde{f}(0) = e_0$. Then we can define:

$$\varphi : \pi_1(B, b_0) \to p^{-1}(b_0) \qquad \varphi([f]) = \tilde{f}(1)$$

and φ is called the *lifting correspondence* derived from the corvering map p, e_0.

2.1.36 Note it is important to choose a point e_0 since usually the covering space might have many locally homeomorphic neigbhourhoods, so you have to make a choice. The choice, however, doesn't really matter so much since there is a local neigbhourhood around e_0 will be homeomorphic to it's image.

Recall that the inverse of a point of a covering map is a discrete space, and hence we shall get a discrete group for this construction!

Proposition 2.1.37: Lifting Correspondence and Path-Connected Spaces

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$. If E is path-connected, then the lifting correspondence is surjective. If E is also simply-connected, then the lifting correspondence is bijective

***Proof*:**
Assume E is path-connected and Let $e_1 \in p^{-1}(b_0)$. To show surjectivity, choose a path f_E in E such that $f_E(0) = e_0$, $f_E(1) = e_1$. Let $f = p \circ f_E$. Then f_E lifts f, and

$$f(0) = p(f_E(0)) = p(e_0) = b_0 \tag{2.1}$$
$$f(1) = p(f_E(1)) = p(e_1) = b_0 \tag{2.2}$$

Thus, $\varphi([f]) = f_E(1) = e_1$, and so φ is surjective, as we sougth to show.

Now assume E is simply connected and say $\varphi([f]) = \varphi([g])$ so that $\tilde{f}(1) = \tilde{g}(1)$. Since E is simply connected, the paths $\tilde{f} \simeq \tilde{g}$. Then we may take $p \circ \tilde{f}$ and $p \circ \tilde{g}$ which are certainy homotopic. Then certainly $p \circ \tilde{f} = f$ and $p \circ \tilde{g} = g$ are homotopic, but then $[f] = [g]$, showing injectivity.

What we've essentially proven is that we can deduce information of the fundamental group of a space given the covering map, and we can at least deduce all it's elements if the covering map is simply connected. At minimum, if we have a simply connected covering space, then the cardinality of $\pi_1(X, b_0)$ is equal to $p^{-1}(b_0)$!

Example 2.1.38: Lifts

1. Let $B = E = S^1$, and take $p : E \to B\ z \mapsto z^2$. Notice that E is path-connected, hence the fundamental group surjects onto the pre-image of a point of the covering map. Let $b_0 = e_0 = (1, 0)$. Then $\varphi : \pi_1(S^1, b_0) \to p^{-1}(b_0) = \{(1, 0), (-1, 0)\}$.

   We'll exhibit this homomorphism explicitely. Take $e_{b_0} : [0, 1] \to S^1$, $e_{b_0}(t) = b_0 = (1, 0)$. Then the lifting correspondence of e_{b_0} is clearly $\tilde{e_{b_0}} = e_{e_0}$. Then:

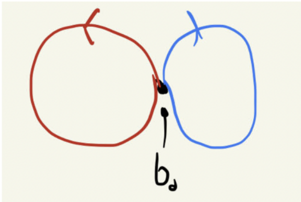
   $$\varphi([e_{b_0}]) = e_{e_0}(1) = e_0 = (1, 0)$$

   now, let:
   $$f : I \to S^1, f(t) = (\cos(2\pi t) \sin(2\pi t))$$

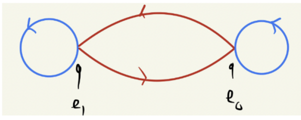
   which is the funciton which will go around the circle once. Then the lifting correspondence is:
   $$\tilde{f} : I \to S^1, \tilde{f}(t) = (\cos(\pi t), \sin(\pi t))$$

   So $\tilde{f}(1) = (-1, 0)$ so $\varphi([f]) = (-1, 0)$ (which also verifies surjectivity). So for the first time, we showed that we have a *non-contractible loop* since the two homotopic classes are different! That is $[f] \neq [e_{b_0}]$

2. Let X be the figure 8 as we described in a pervious example (from the torus). Let the intersection of the circles be b_0. Take one loop going the left circle counter-clockwise, and one for the right circle, clockwise, so that $[f], [g] \in \pi_1(X, b_0)$. Let E represent the following lifting space:



(a) Figure 8



(b) Particular Lifting Space of Figure 8

and take e_0 as in the image. Then

$$\varphi([f]) = e_1$$
$$\varphi([g]) = e_0$$
$$\varphi([f] * [g]) = e_1$$
$$\varphi([g] * [f]) = e_1$$

This told us that $[f]$ and $[g]$ are different paths, but it seems that their product is the same. However, remember that φ is surjective, *not* injective, so φ might've only given us *partial* information on how the elements of this fundamental group interact. We can in fact work wtih a better covering space which will reveal more information.

3. Same X, same b_0, but this time, our lifting space E is different!

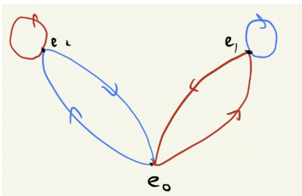


Figure 2.5: Different Lifting Space

Now,

$$\varphi([f] * [g]) = e_1 \tag{2.3}$$
$$\varphi([g] * [f]) = e_2 \tag{2.4}$$

Therefore, $[f] * [g] \neq [g] * [f]$, i.e. $\pi_1(X, b_0)$ is *not* abelian, as we claimed when $\pi_1(X, x_0)$ was introduced!!!

Proposition 2.1.39: Lifting Correspondence And Homomorphisms

Let $p : E \to B$ be a covering map and $p(e_0) = b_0$. Then:

1. The induced homomorphism $p_* : \pi_1(E, e_0) \to \pi_1(B, b_0)$ is injective
2. Let $H = p_*(\pi_1(E, e_0))$. Then the lifting correspondence φ induces an injective map:
$$\Phi : \pi_1(B, b_0)/H \to p^{-1}(b)$$
where $\pi(B, b_0)/H$ is the collection of right cosets of H. The map is bijective if E is path connected
3. If f is a loop in B based at b_0, then $[f] \in H$ if and only if f lifts to a loop in E based at e_0

***Proof*:**

1. Let $\tilde{h}$ be a loop in E at e_0 where $p_*([\tilde{h}])$ is the identity element. Since it's the identity, there exists a path homotopy F between $p \circ \tilde{h}$ and the constant loop. Then let $\tilde{F}$ be the lifting of F to E such that $\tilde{F}(0, 0) = e_0$. But then $\tilde{F}$ is the path homotopy between $\tilde{h}$ and the constant loop at e_0
2. Let f, g be loops in B based at b_0 with corresponding $\tilde{f}$ and $\tilde{g}$ in E that begin at e_0. Then $\varphi([f]) = \tilde{f}(1)$ and $\varphi([g]) = \tilde{g}(1)$. We will show that $\varphi([f]) = \varphi([g])$ if and only if $[f] \in H * [g]$. First, if $[f] \in H * [g]$, then $[f] = [h * g]$ where $h = p \circ \tilde{h}$ for some loop $\tilde{h}$ in E based at e_0. Then the product $\tilde{h} * \tilde{g}$ is defined and is the lifting of $h * g$. Since $[f] = [h * g]$, the liftings $\tilde{f}$ and $\tilde{h} * \tilde{g}$ which begin at e_0 must end at the same point in E. But then $\tilde{f}$ and $\tilde{g}$ end at the same point of E so $\varphi([f]) = \varphi([g])$.

   Conversely, suppose $\varphi([f]) = \varphi([g])$ so that $\tilde{f}$ and $\tilde{g}$ end at the same point in E. Then the product of $\tilde{f}$ and the reverse of $\tilde{g}$ is defiend, and it is a loop in E based at e_0. By directly computing, we get $[\tilde{h} * \tilde{g}] = [\tilde{f}]$. If $\tilde{F}$ is a path homotopy in$E$ between $\tilde{H} * \tilde{g}$ and $\tilde{f}$, $p \circ \tilde{F}$ is a path homotopy in B between $h * g$ and f where $h = p \circ \tilde{h}$. But then $[f] \in H * [g]$, as we sought to show.

   Furthermore, if E is path connected, then φ is surjective, meaning Φ is also surejctive.
3. Injectivity of Φ means $\varphi([f]) = \varphi([g])$ if and only if $[f] \in H * [g]$. Applying this to the constnt loops, we get that $\varphi([f]) = e_0$ if and only if $[f] \in H$. But $\varphi([f]) = e_0$ exactly whenthe lift of f that begins at e_0 ends at e_0.

Finally, we may present the fundamental group of the circle. For the long build-up of it, we'll make it a theorem

Theorem 2.1.40: Fundamental Groups Of The Circle

The fundamental group of S^1 is isomorphic to $(\mathbb{Z}, +)$

***Proof*:**
Let $p : \mathbb{R} \to S^1$, and $p(t) = (\cos(2\pi t), \sin(2\pi t))$. Then p is a covering map such that $p(r) = p(s) \Leftrightarrow r - s \in \mathbb{Z}$. Let $b_0 = (1, 0) \in S^1$. Then $p^{-1}(b_0) = \mathbb{Z}$. Let our starting point in E be $e_0 = 0$. Then the lifting corresponding gives a map $\varphi : \pi_1(S^1, b_0) \to \mathbb{Z}$. Moreover, since $\mathbb{R}$ is simply connected, it follows that φ is a bijetion! So, it remains to show that φ is a homomorphism:

$$\varphi([f] * [g]) = \varphi([f]) + \varphi([g])$$

Let $\tilde{f}, \tilde{g}$ be lifts of f, g starting at 0, so $\varphi([f]) = \tilde{f}(1) \in \mathbb{Z} = p^{-1}(b_0)$. Define $\tilde{g}_2(t) = \tilde{g}(t) + \tilde{f}(1)$. Then

$$p(\tilde{g}_2(t)) = p(\tilde{g}(t) + \tilde{f}(1)) = p(\tilde{g}(t) + k) = p(\tilde{g}(t)) = g(t)$$

since $k \in \mathbb{Z}$, so the covering map projects this path onto g, meaning it is also a lift of g, just shifted over by $f(t)$ to where the path starts. We can see this by $\tilde{g}_2(0) = \tilde{f}(1)$. Thus, we can combine these two paths, $\tilde{f} * \tilde{g}_2$, and by what we've just shown, this is a lifts of $f * g$. Thus

$$\begin{aligned}\varphi([f] * [g]) &= (\tilde{f} * \tilde{g}_2))(1) \\ &= \tilde{g}_2(1) \\ &= \tilde{g}(1) + \tilde{f}(1) = \varphi([f]) + \varphi([g])\end{aligned}$$

completeing the proof.

This also means that the fundamental group of the circle is abelian, which should makes sense if you try and draw loops on a circle. You will get back and forth and will be able to reverse the order no problem. Trying this on the figure 8, you might see how you can get stuck. We just formalized such intuition!

2.1.3 Fundamental Group Invariant Under Homotopy

If f is homeomorphic, then f_* is isomorphic. However, this is a rather rigid condition; it turns out that if $f \simeq g$, then $f_* = g_*$, showing that the fundamental group is invariant under the much looser condition of homotopy. If $f_t : (X, x_0) \to (Y, y_0)$ is a homotopy such that $f_t(x_0) = y_0$, we shall call it a basepoint-preserving homotopy.

Proposition 2.1.41: Invariance Of Induced Homomorphisms Under Homotopy

Let $f_t : X \to Y$ be a homotopy. Then $(f_0)_* = (f_1)_*$

***Proof*:**
If $f_0 \simeq f_1$, then $[f_0] = [f_1]$, and $[f_0 \circ \varphi] = [f_1 \circ \varphi]$ for some loop φ (at any basepoint). Then $(f_0)_*([\varphi]) = [f_0 \circ \varphi] = [f_1 \circ \varphi] = (f_1)_*([\varphi])$

When working with homotopy equivalence, we may in fact drop the basepoint information. We first require the following 2 lemmas:

Lemma 2.1.42: Induced Maps, Retractions, and Deformation Retraction

Let A be a retract of X. Then for $a_0 \in A$, the induced map $\iota_* : \pi_1(A, a_0) \to \pi_1(X, a_0)$ is injective, where $\iota : A \to X$ is the inclusion. If A is a deformation retraction of X, then ι_* is also surective, hence an isomorphism.

This means that X should have *at least* as many elements in it's fundamental group as A

***Proof*:**
Let $r : X \to A$ be a retraction. Then $f \circ \iota = \mathrm{id}_A$. Thus, $(r \circ \iota)_* = (id_A)_* = id|_{\pi_1(A,a_0)}$. Thus

$$[f] = (r_* \circ \iota_*)([f]) = r_*(\iota_*([f]))$$

showing r_* is a left inverse, thus ι_* is injective.

If $r_t : X \to X$ is a deformation retraction of X onto A so that $r_0 = \mathrm{id}$ and $r_t|_A = \mathrm{id}$ and $r_1(X) \subseteq A$, then for any loop $f : I \to X$ based at $x_0 \in A$, the composition $r_t \circ f$ gives a homotopy of f to a loop in A, showing ι_* is surjective, and hence an isomorphism.

The group-theoretic equivalent statement is that a homomorpism $\varphi : G \to H$ restricts to the identity on H. If H is a normal subgroup, Then G is the direct product of H and $\ker(\varphi)$, and if H is not normal then we replace direct product with semi-direct product.

Lemma 2.1.43: Basepoint Path In Homotopy

Let $\varphi_t : X \to Y$ be a homotopy and h the path $\varphi_t(x_0)$ formed by the images of the basepoint $x_0 \in X$. Then there exists a β_h such that the following diagram commutes:

$$\begin{tikzcd}
 & \pi_1(Y, \varphi_1(x_0)) \arrow[d, "\beta_h"] \\
\pi_1(X, x_0) \arrow[ur, "\varphi_{1*}"] \arrow[r, "\varphi_{0*}"'] & \pi_1(Y, \varphi_0(x_0))
\end{tikzcd}$$

***Proof*:**
Let $h_t(s) = h(ts)$ so that h_t is the restriction of h to the interval $[0, t]$ and reparamaterized so that the dmoain is still $[0, 1]$. Let $\overline{h_t}$ be the inverse path. Then if f is a loop pin X at the basepoint x_0, the product $h_t \circ (\varphi_t \circ f) \circ \overline{h_t}$ gives a homotopy of loops at $\varphi_0(x_0)$:

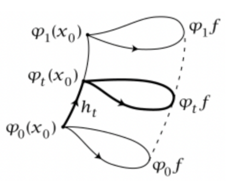


Figure 2.6: Visualization

Restricting this homotopy to $t = 0$ and $t = 1$, we get

$$\varphi_{0*}([f]) = \beta_h(\varphi_{1*}([f]))$$

completing the proof

Theorem 2.1.44: homotopy Equivalence, Then Basepoint Invariant

Let $\varphi_t : X \to Y$ be a homotopy equivalent. Then the induced homomorphism $\varphi_* : \pi_1(X, x_0) \to \pi_1(Y, \varphi(x_0))$ is an isomorphism for all $x_0 \in X$

***Proof*:**
We want to show that φ_* is an isomorphiam. Let $\Psi : Y \to X$ be the homotopy-inverse of φ so that $\varphi \circ \psi \simeq \text{id}$ an $\psi \circ \varphi \simeq \text{id}$. We shall force the necessary condition on φ_* via the following diagram:

$$\pi_1(X, x_0) \xrightarrow{\varphi_*} \pi_1(Y, \varphi(x_0)) \xrightarrow{\psi_*} \pi_1(X, \psi \circ \varphi(x_0)) \xrightarrow{\varphi_*} \pi_1(Y, \varphi\psi\varphi(x_0))$$

The composition of the first two maps is an isomorphism since $\psi \circ \varphi \simeq \text{id}$ implying $\psi_* \circ \varphi_* = \beta_h$ for some h. In particular, since $\psi_* \circ \varphi_*$ is an isomorphism, φ_* is injective. Similarly, the second and third maps shows that ψ_* is injective. Thus, the first two of the three maps are injection and their composition is an isomorphism, hecne φ_* must be surjective as well, completing the proof.

(A word leading for this example)

Example 2.1.45: Fundamental Group Does Not Characterize Spaces
Define the *pseudocircle* to be the set $X = \{a, b, c, d\}$ with the topology:

$$\{\{a, b, c, d\}, \{a, b, c\}, \{a, b, d\}, \{a, b\}, \{a\}, \{b\}, \emptyset\}$$

which can be thought of as the topology induced by the partial oarder $a < c$, $b < c$, $a < d$, $b < d$ where the open sets aer the donward-closed sets. Then we may define a continuous map $f : S^1 \to X$ by

$$f(x, y) = \begin{cases} a & x < 0 \\ b & x > 0 \\ c & (x, y) = (0, 1) \\ d & (x, y) = (0, -1) \end{cases}$$

Then we shall not show this now, but this is a *weak homotopy equivalence*, which we shall show later implies f induces an isomorphism on all homotopy groups! We shall later in this book introduce other concepts such as the singular homology and cohomology, and f will induce an isomorphism between these structures too! See this link for more details:

https://en.wikipedia.org/wiki/Pseudocircle

2.1.4 Interesting Consequences

Theorem 2.1.46: Can't make a whole

Let B^2 be the unit disc in $\mathbb{R}^2$. There is no retraction form B^2 to S^1. You can think of this as you cannot make a whole.

***Proof*:**
The fundamental group of the circle is infinite, but the fundamental group of B^2 has 1 element (because its star-convex). Since the fundamental group of S_1 would need to injectively map to B^2, this implies there is *no* retraction!

Theorem 2.1.47: Brouwer Fixed Point Theorem

if $f : B^2 \to B^2$ is continuous, then there exists $x \in B^2$ such that $f(x) = x$

***Proof*:**
Recall that $B^2 := \left\{(x, y)|x^2 + y^2 \leq 1\right\}$ and thus is $\approx I^2$.

Assume $f : B^2 \to B^2$ is continuous. For the sake of contradiction, let's say that for every $x \in B^2$, $f(x) \neq x$. We will use f to construct a retraction of B^2 to its boundary S^1.

The key idea is that since $f(x) \neq x$, and both are in B^2, we can draw a straight line between x and $f(x)$. We can continue this line to the boundary. If we give a direction (let's say from $f(x)$ to x), we can then make a function $g(x)$ which will take our point x, and given this line, it will bring it to the boundary.

The key idea here is that if we had that $f(x) = x$, then we woouldn't know what ray to draw. for example, if $f(x) = \frac{x}{\|x\|}$ then every point will have a clear ray except $x = 0$. This unclearness of the ray is translated as f being discontinuous (in the case of the concrete f, we divided by 0).

More formally, we'll define $G : B^2 \times \mathbb{R}_{\geq 0} \to \mathbb{R}^2$.

$$G(x, r) = x + r(x - f(x))$$

since Everything is in $\mathbb{R}^2$, mutiplying, adding, subtracting is allowed. Note that

$$\|G(x, r)\|^2 = \|x\|^2 + r^2 \|x - f(x)\|^2 + 2rx \cdot (x - f(x))$$

so for $\|G(x, r)\| = 1$, we use the quadratic formula to solve r:

$$r = \frac{-2x \cdot (x - f(x)) + \sqrt{4(x \cdot (x - f(x)))^2 + 4(1 - \|x\|^2) + \|x - f(x)\|^2}}{2 \|x - f(x)\|^2}$$

This formula is complicated, but it's okay: everything is positive, so everyhing is okay (since f has no fixed points). We can make r into a formula which will make sure that $G(x, r) = 1$ by calling the right hand side $r(x)$.

Thus, we can define

$$g(x) = G(x, r(x)) \in S^1$$

hence, $g(x)$ is continuous. Note that one the circle, $g(x) = x$, showing that it's a retraction. But we showed there can't be a retraction between the ball and the circle - a contradiction!

The following uses the fact that $\pi_1(S^1) = \mathbb{Z}$

Theorem 2.1.48: Fundamental Theorem Of Algebra

let $p(z)$ be a nonconstnat polynomial with coefficients in $\mathbb{C}$. Then $p(z)$ has a root in $\mathbb{C}$

***Proof*:**
Without loss of generailty let $p(z)$ be monic (we may scale the result at the end). Assume that $p(z)$ has no roots in $\mathbb{C}$. Then for each $r \in \mathbb{R}_{\geq 0}$, define

$$f_r(s) : S^1 \to S^1 \qquad f_r(s) = \frac{p(re^{2\pi is})/p(r)}{|p(re^{2\pi is})/p(r)|}$$

which is a loop based at 1. As we let r vary, we get loops f_r that are homotopic based at 1. Since f_0 is the trivial loop, we get that $[f_r] \in \pi_1(S^1)$ is the nullhomotopic element.

Now, fix r larger enough so that it's bigger than $\max |a_1| + \cdots |a_n|, 1$. Then for $|z| = r$, we get

$$|z^n| > (|a_1| + \cdots + |a_n|)|z^{n-1} > |a_1 z^{n-1}| + \cdots + |a_n| \geq |a_1 z^{n-1} + \cdots + a_n|$$

Thus, it follows that

$$p_t(z) = z^n + t(a_1 z^{n-1} + \cdots + a_n)$$

has no roots on $|z| = r$ and $0 \leq t \leq 1$. Now, replacing p by p_t in the formula of f_r and letting $t \to 0$ we get

$$f_{r,0}(s) = \omega_n(s) = e^{2\pi ins}$$

Now, we know that $[w_n] \in \pi_1(S^1)$ maps to n in the isomorphism presented in the proof of the fundametnal group of S^1. Thus, we have:

$$0 = \varphi([f_r]) = \varphi([w_n]) = n$$

and hence $n = 0$, showing that only constant functions don't have roots.

Theorem 2.1.49: Burusk-Ulam Theorem In 2 Dimensions

For every continuous map $f : S^2 \to \mathbb{R}^2$, there exists a pair of antipodal points $x, -x$ such that $f(x) = f(-x)$

An interesting consequence of this is that S^2 is not homeomorphic to a subspace of $\mathbb{R}^2$, an intuitive but not obvious fact to prove directly

***Proof*:**
For the sake of contradiction, let's say there is a continuous map $f : S^2 \to \mathbb{R}^2$ with no equal antipodal images. Then define the map $g : S^2 \to S^1$ by

$$g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$$

Now, imagining the sphere S^2 embeded in $\mathbb{R}^3$, $S^2 \subseteq \mathbb{R}^3$ centered at the origin, take η to be the loop given by the equator

$$\eta(t) = (\cos 2\pi t, \sin 2\pi t, 0)$$

Let $h = g \circ \eta$. Then, $g(-x) = -g(x)$, so

$$h(s + 1/2) = -h(s)$$

for all $s \in [0, 1/2]$. Now, $h : I \to S^1$ can be lifted to a path $\tilde{h} : I \to \mathbb{R}$. The equation $h(s + 1/2) = -h(s)$ then implies

$$\tilde{h}(s + 1/2) = \tilde{h}(s) + q/s \qquad q \in 2\mathbb{Z} + 1$$

This q is independed of s, since by solving the equation $\tilde{h}(s + 1/2) = \tilde{h}(s) + q/2$, we see that q deepends continuosly on $s \in [0, 1/2]$, so q must be a constant since it is constrained to integer values. In particular

$$\tilde{h}(1) = \tilde{h}(1/2) + q/2 = \tilde{h}(0) + q$$

Thus, h represents q times the generator of $\pi_1(S^1)$. Since q is odd, we conclude that h is *not* nullhomotopic. But h is the composition $g \circ \eta : I \to S^2 \to S^1$, and η is certainly nullhomotopic in S^2, so $g \circ \eta$ is nullhomotopic in S^1 by composing nullhomotopy of η with g - a contradiction. Thus, it cannot be that $f(x) \neq f(-x)$ forall x, as we sought to show.

Corollary 2.1.50: Union Of 3 Sets of Sphere contains Antipodal Points

Whenever S^2 is expressed as the union of three closed sets A_1, A_2, A_3, then at least one of these sets must contain a pair of antipodal points $\{x, -x\}$.

***Proof*:**
Let $d_i : S^2 \to \mathbb{R}$, $d_i(x) = \inf_{y \in A_i} |x - y|$ be the distance from x to A_i. This is certainly a continuous function, so we may apply the Burusk-Ulam theorem to

$$S^2 \to \mathbb{R}^2 x \mapsto (d_1(x), d_2(x))$$

namely, we can find a set of points such that $d_1(x) = d_1(-x)$ and $d_2(x) = d_2(-x)$. If either of these two distances is zero, then $x, -x$ both lie in the set A_1 or A_2, since these are closed sets. On the other hand, if the distance from x and $-x$ to A_1 and A_2 are both strictly positive, then x and $-x$ must lie in neither of these sets, but hten it must lie in A_3.

It can be verified that this theorem fails when considering 4 sets (can you come up with a counter example?) Given the higher-dimensional analogue of Burusk-Ulamns theorem, the same theorem works for S^n beign covered by $n + 1$ sets (while being able to cover it with $n + 2$ sets non containing antipodal points). The above corollary even works for S^1: if it is covered by two closed sets, then one must contain antipodal points (closedenss is required, since S^1 can be covered by two half-open sets)

(Some interesting results I've ommited (mainly p. 349 Munkres): extending S^1 to B^2 map, vector field results, A 3×3 positive real matrix, then has positive real eigen value)

(A word on the complexity of coveing spaces of $\pi_n(S^m)$). The homotopy structure also is dependent on information beyond paths, for example $\pi_n(S^2)$ is nonzero for infinitely many values of n.

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}	π_{12}	π_{13}	π_{14}	π_{15}
S^0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S^1	Z	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S^2	0	Z	Z	Z_2	Z_2	Z_{12}	Z_2	Z_2	Z_3	Z_{15}	Z_2	Z_2^2	$Z_{12}\times Z_2$	$Z_{84}\times Z_2^2$	Z_2^2
S^3	0	0	Z	Z_2	Z_2	Z_{12}	Z_2	Z_2	Z_3	Z_{15}	Z_2	Z_2^2	$Z_{12}\times Z_2$	$Z_{84}\times Z_2^2$	Z_2^2
S^4	0	0	0	Z	Z_2	Z_2	$Z\times Z_{12}$	Z_2^2	Z_2^2	$Z_{24}\times Z_3$	Z_{15}	Z_2	Z_2^3	$Z_{120}\times Z_{12}\times Z_2$	$Z_{84}\times Z_2^5$
S^5	0	0	0	0	Z	Z_2	Z_2	Z_{24}	Z_2	Z_2	Z_2	Z_{30}	Z_2	Z_2^3	$Z_{72}\times Z_2$
S^6	0	0	0	0	0	Z	Z_2	Z_2	Z_{24}	0	Z	Z_2	Z_{60}	$Z_{24}\times Z_2$	Z_2^3
S^7	0	0	0	0	0	0	Z	Z_2	Z_2	Z_{24}	0	0	Z_2	Z_{120}	Z_2^3
S^8	0	0	0	0	0	0	0	Z	Z_2	Z_2	Z_{24}	0	0	Z_2	$Z\times Z_{120}$

Figure 2.7: known Higher order Homotopy groups of n-spheres

One last result that shall be stated, but certainly not proven due to it's complexity, is the characterization of all simply connected closed 3-manifolds. The simplest such manifold would be S^3. Up to homeoomorphism, are there any other? This was known as *Poincaré conjecture*, and it was proven that this was *not* the case; all closed simply connected 3-manifolds are homeomorphic to S^3.

2.2 Products, Fibred Products, and Van Kampen's Theorem

We have covered what happens if we have a continuous map, homeomorphism, and homotopy equivalence between two topological spaces. We shall now look at some usual constructions of topological spaces and see how it effects the fundamental group. Most of the products we have seen will require a bit more theory, namely the *van-kanmpen theorem*.

Proposition 2.2.1: Fundamental Group And Product Spaces

Let X, Y be path connected spaces. Then $\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$

***Proof*:**
This follows immediately from the universal property of products: if $f : [0,1] \to X \times Y$ is a loop, then there exists continuous maps g, h such that $f(t) = (g(t), h(t))$. These are certainly loops in X and Y respectively, and similarly a homotopy between two paths f, f' induces a homotopy between g, g' and h, h'. Furthermore, any two paths g, h can be combined to form a path (g, h) in $X \times Y$. Hence, we have a bijection:

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

Showing that this is a homomorphism is immediate.

Example 2.2.2: Fundamental Group Of Torus

Since $T^2 = S^1 \times S^1$, $\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}$. In particular, any loop on T^2 is a loop around the minor axis combined with a loop from the major axis. Interestingly, these loops may knot:

Figure 2.8: Torus Loop Knot

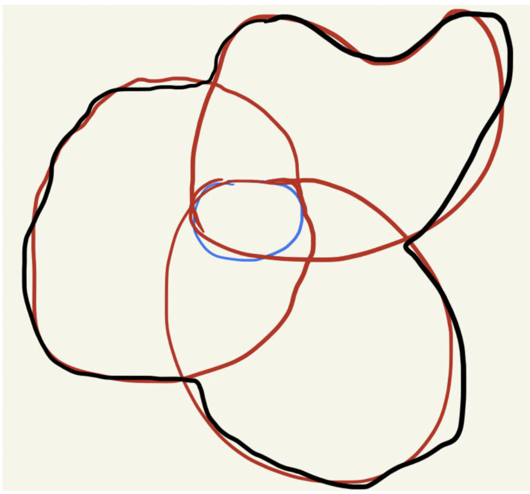
Giving us some interesting non-trivial paths.

A slight generalization of the above proposition is the following:

Proposition 2.2.3: Decomposing Spaces to find Homotopy

Let X be a topological such that it is the union of a collection of path-connected open sets A_α each containing the baspoint $x_0 \in X$ and if each intersection $A_\alpha \cap A_\beta$ is path connected, then every loop in X at x_0 is homotopic to a product of loops each of which is contained in a single A_α.

To visualize:



and a loop f at x_0 is homotopic to $f \simeq (f_1 * f_2 * \cdots * f_n)$ where each $f_i \in A_i$, hence:

$$[f] = [f_1] * [f_2] * \cdots * [f_n]$$

breaking up the problem into one of finding the fundamental group each A_i.

***Proof*:**
Let $f : I \to X$ be a loop with basepoint x_0. First, we may partition I so that each subinterval $[s_{i-1}, s_i]$ is mapped by f to a single A_α, namely since f is continuous each $s \in I$ has an open neighbourhood $V_s \subseteq I$ that gets mapped to A_α by f; we may take these intervals to be sufficinetly small so that the closure is in A_α. Then by compactness finitely many V_s cover I, giving us our desried partition.

Let $f_i = f|_{[i-1,i]}$ and A_i be the A_α containing the iamge of f_i. Then $f = f_1 * f_2 * \cdots f_n$. Since $A_i \cap A_{i+1}$ is path connected for each i, there exists a g_i in $A_i \cap A_{i+1}$ from x_0 to $f(s_i) \in A_i \cap A_{i+1}$. Then

$$(f_1 * \overline{g}_1) * (g_1 * f_1 * \overline{g}_2) * \cdots * (g_{n-1} * f_1)$$

which is certainly homotopic to f, and is a the desired composition, as we sought to show.

Note that we have not given how to find the homotopy group of this space (this shall be done using van-Kampens theorem). In the case where each component is simply connected, we get the following application:

Example 2.2.4: Fundamental Group of n-sphere
Write S^n as the union of A_1 and A_2 each homeomorphic to $\mathbb{R}^n$ and $A_1 \cap A_2$ is homeomorphic to $S^{n-1} \times \mathbb{R}$ (take A_1, A_2 to be the complements of the two antipodal points in S^n). Then choosing a basepoint $x_0 \in A_1 \cap A_2$, if $n \geq 2$ then $A_1 \cap A_2$ is path connected and so by the lemma every loop in S^n based at x_0 is homotopic to a product of loops in A_1 or A_2. Both $\pi_1(A_1)$ and $\pi_1(A_2)$ are zero, and hence S^n is simply connected for $n \geq 2$.

An important consequence that is usually rather difficult to prove classicaly becomes much easie

Corollary 2.2.5: Plane Not Homeomorphic To $\mathbb{R}^n$

$\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \neq 2$

***Proof*:**
The case of $n = 1$ is quickly dispostied of since $\mathbb{R}^2 - \{0\}$ is path connectd, while $\mathbb{R} - \{0\}$ is not. When $n > 2$, then $\mathbb{R}^n - \{x\}$ is homeomorphic to $S^{n-1} \times \mathbb{R}$, hence by proposition 2.2.1

$$\pi_1(\mathbb{R}^n - \{x\}) \cong \pi_1(S^{n-1}) \times \pi_1(\mathbb{R}) \cong \pi_1(S^{n-1})$$

hence, if $n = 2$ we get $\pi_1(\mathbb{R}^n - \{x\}) = 2$ and for $n > 2$ we get id. Then it is a simple application of proposition 2.2.3

For the $\mathbb{R}^m \ncong \mathbb{R}^n$ if $n \neq m$, we can use higher homotopy groups, but there will be an easier proof of this using homology in the next chapter.

Proposition 2.2.3 gave us the paths of a space that can be decomposed into spaces A_α whose intersection is path-connected. However, we did not present what type of group these path can produce, besides some trivial cases. This next theorem expands on the important result to gives us the group structure. It should be remarked that the result is entirely straightfroward, and is simply the application of the existence of pushouts and the functoriality discussed earlier:

Theorem 2.2.6: Van Kampen's Theorem

Let X be a topological space with is the union of two open and path connected subspaces U_1, U_2. Suppose that $U_1 \cap U_2$ is path connected and nonemtpy, and let $x_0 \in U_1 \cap U_2$. Then the push-out of

$$\pi_1(U, x_0) \leftarrow \pi_1(U \cap V, x_0) \rightarrow \pi_1(V, x_0)$$

exists, namely the following diagram commutes:

$$\begin{tikzcd}
& \pi_1(U_1 \cap U_2) \arrow[dl, "\iota_{1*}"'] \arrow[dr, "\iota_{2*}"] & \\
\pi(U_1) \arrow[dr, dashed] \arrow[ddr, bend right, "j_{1*}"'] & & \pi(U_1) \arrow[dl, dashed] \arrow[ddl, bend left, "j_{1*}"] \\
& \pi(U_1) *_{\pi_1(U_1 \cap U_2)} \pi_1(U_2) \arrow[d, "k"] & \\
& \pi_1(X) &
\end{tikzcd}$$

***Proof*:**
The pushout in the category of groups is the amalgamated free product $\pi_1(U_1) *_{\pi_1(U_1 \cap U_2)} \pi_1(U_2)$, the quotient of the free product $\pi_1(U_1) * \pi_1(U_2)$ by the normal subgroup generated by the elements $\iota_{1*}(\omega)\, \iota_{2*}(\omega)^{-1}$ as ω ranges over $\pi_1(U_1 \cap U_2)$. The asserted commutativity of the square is the statement that the induced map k from this quotient to $\pi_1(X)$ is an isomorphism, and the content of the theorem is exactly that k is bijective. It is cleanest to prove the general statement over an arbitrary cover, since the two-set case is the special instance with index set $\{1, 2\}$, and the general form is what the computations of this chapter actually invoke. So let $X = \bigcup_\alpha A_\alpha$ be a union of open path-connected sets, each containing the basepoint x_0, with each pairwise intersection $A_\alpha \cap A_\beta$ and each triple intersection $A_\alpha \cap A_\beta \cap A_\gamma$ path-connected, and consider

$$\varphi : \mathop{\ast}_{\alpha} \pi_1(A_\alpha, x_0) \longrightarrow \pi_1(X, x_0)$$

the homomorphism out of the free product whose restriction to the factor $\pi_1(A_\alpha)$ is the map $j_{\alpha*}$ induced by the inclusion $A_\alpha \hookrightarrow X$. The two-set diagram is recovered by taking the cover $\{U_1, U_2\}$: surjectivity of φ is commutativity of the lower triangles together with surjectivity of k, and the description of $\ker \varphi$ identifies the quotient with the amalgamated product, making k the asserted isomorphism. We prove surjectivity first and then compute the kernel.

Step 1: Surjectivity by subdivision against a Lebesgue number. Let $f : I \to X$ be a loop based at x_0. The sets $f^{-1}(A_\alpha)$ form an open cover of the compact metric space I, so by the Lebesgue number lemma there is a $\delta > 0$ such that every subinterval of I of length less than δ is carried by f into a single A_α. Choose a partition $0 = s_0 < s_1 < \cdots < s_n = 1$ with each $s_i - s_{i-1} < \delta$, and for each i pick an index α_i with $f([s_{i-1}, s_i]) \subseteq A_{\alpha_i}$. Writing $f_i = f|_{[s_{i-1}, s_i]}$ (reparametrized to I), we have a factorization $f = f_1 * f_2 * \cdots * f_n$ in which the consecutive endpoint $f(s_i)$ lies in $A_{\alpha_i} \cap A_{\alpha_{i+1}}$.

This is precisely the situation of proposition 2.2.3, and the construction there converts f into a product of loops at x_0. Concretely, since each intersection $A_{\alpha_i} \cap A_{\alpha_{i+1}}$ is path-connected and contains both x_0 and the point $f(s_i)$, choose a path g_i in $A_{\alpha_i} \cap A_{\alpha_{i+1}}$ from x_0 to $f(s_i)$, taking g_0 and g_n to be the constant path at x_0. Inserting $\bar{g}_i * g_i$ (homotopically trivial by theorem 2.1.7) at each cut point gives

$$f \simeq (f_1 * \bar{g}_1) * (g_1 * f_2 * \bar{g}_2) * \cdots * (g_{n-1} * f_n)$$

rel endpoints. Each parenthesized factor is now a loop based at x_0, and the i-th factor has image inside the single set A_{α_i} (the path g_{i-1}, the subpath f_i, and the path $\bar{g}_i$ all lie in A_{α_i}, using that g_i was chosen inside $A_{\alpha_i} \cap A_{\alpha_{i+1}} \subseteq A_{\alpha_i}$). Hence $[f]$ is the product in $\pi_1(X)$ of the classes $j_{\alpha_i *}[i\text{-th factor}]$, each of which lies in the image of φ. As $[f]$ was an arbitrary loop class, φ is surjective, establishing the first assertion of the theorem.

Before computing the kernel it is worth fixing notation. A word in the free product is a finite sequence $[\gamma_1][\gamma_2]\cdots[\gamma_m]$ in which each γ_k is a loop at x_0 lying in some single set A_{α_k}; the surjectivity argument shows every element of $\pi_1(X)$ arises from such a word, which we call a *factorization* of the loop $\gamma_1 * \gamma_2 * \cdots * \gamma_m$. The element of the free product determined by the word lies in $\ker\varphi$ exactly when this product loop is nullhomotopic in X. We must show that any two factorizations of homotopic loops are connected by a sequence of moves, each of which preserves the image in $\pi_1(X)$ and changes the word only within the normal subgroup N generated by the relators $i_{\alpha\beta}(\omega)\, i_{\beta\alpha}(\omega)^{-1}$.

Here $i_{\alpha\beta} : \pi_1(A_\alpha \cap A_\beta) \to \pi_1(A_\alpha)$ denotes the map induced by the inclusion $A_\alpha \cap A_\beta \hookrightarrow A_\alpha$; for a loop ω in the intersection, $i_{\alpha\beta}(\omega)$ is the class of ω read inside A_α and $i_{\beta\alpha}(\omega)$ the same loop read inside A_β. The relator records that these two readings become equal once mapped to $\pi_1(X)$. Two elementary moves leave the word unchanged modulo N:

1. if a loop γ_k lies in $A_\alpha \cap A_\beta$, the factor $[\gamma_k]$ may be reread from the factor $\pi_1(A_\alpha)$ to the factor $\pi_1(A_\beta)$ of the free product (replacing $i_{\alpha\beta}(\gamma_k)$ by $i_{\beta\alpha}(\gamma_k)$), since the two differ by exactly a generator of N;

2. adjacent factors $[\gamma_k][\gamma_{k+1}]$ lying in the same A_α may be combined into the single factor $[\gamma_k * \gamma_{k+1}]$, and conversely a factor may be split, since this is just multiplication inside the group $\pi_1(A_\alpha)$ and leaves the free-product element unchanged.

Call two words *equivalent* if one is obtained from the other by a finite sequence of these two moves. The moves preserve the element of $*_\alpha\, \pi_1(A_\alpha)$ modulo N, so to prove $\ker\varphi = N$ it suffices to show: *if two factorizations represent loops that are homotopic in X, then the two words are equivalent.* Granting this, a word lying in $\ker\varphi$ represents a nullhomotopic loop, hence is equivalent to the empty word, hence lies in N; and $N \subseteq \ker\varphi$ holds because $j_{\alpha*} \circ i_{\alpha\beta} = j_{\beta*} \circ i_{\beta\alpha}$ (both are the inclusion $A_\alpha \cap A_\beta \hookrightarrow X$ on π_1), so each relator is killed by φ. The remainder of the proof establishes the displayed implication.

Step 2: A staggered Lebesgue grid on the homotopy square. Suppose f and f' are loops at x_0 with chosen factorizations, and let $F : I \times I \to X$ be a homotopy rel endpoints from $f = F(\cdot, 0)$ to $f' = F(\cdot, 1)$, so F carries the left and right edges $\{0\} \times I$ and $\{1\} \times I$ to the basepoint x_0. The sets $F^{-1}(A_\alpha)$ cover the compact square $I \times I$, so the Lebesgue number lemma gives a $\delta > 0$ such that any subset of diameter less than δ maps into a single A_α. Fix first a partition $0 = t_0 < t_1 < \cdots < t_q = 1$ of the vertical coordinate with each $t_j - t_{j-1} < \delta/2$, cutting the square into the horizontal rows $I \times [t_{j-1}, t_j]$, and on each row impose its *own* vertical partition, independently of the other rows: choose $0 = s_0^{(j)} < s_1^{(j)} < \cdots < s_{p_j}^{(j)} = 1$ with each

$s_i^{(j)} - s_{i-1}^{(j)} < \delta/2$, producing the closed rectangles

$$R_{ij} = \left[s_{i-1}^{(j)}, s_i^{(j)}\right] \times [t_{j-1}, t_j] \qquad (1 \le i \le p_j,\ 1 \le j \le q)$$

each of diameter below δ, so $F(R_{ij}) \subseteq A_{ij}$ for some index, written A_{ij}. Here is the one structural choice that the kernel computation turns on, and it is the only place this proof departs from a naive product grid. *Stagger the vertical cuts of adjacent rows*: having chosen the interior cut points $s_i^{(j)}$ $(0 < i < p_j)$ of each row, perturb them by amounts smaller than the minimum gap so that no interior cut point of row j coincides with any interior cut point of row $j+1$, and so that no cut point lands on the boundary in $I \times I$ between two of the assigned sets. The first condition is achievable because the cut points of two rows form two finite subsets of the open interval $(0,1)$, and a generic small shift of one makes them disjoint; the perturbations are small enough that each rectangle still has diameter below δ, so the assignment $R_{ij} \mapsto A_{ij}$ survives. The effect of staggering is the inequality that drives Step 3: with the interior vertical cuts of consecutive rows disjoint, *no point of $I \times I$ lies in more than three rectangles R_{ij}*, equivalently every vertex of the resulting (non-product) grid is a corner of at most three rectangles. This is Hatcher's perturbation ([14, Theorem 1.20]). The top and bottom rows are exempt from any further constraint, so we may additionally require the bottom partition $\{s_i^{(1)}\}$ to refine the partition of the chosen factorization of f and the top partition $\{s_i^{(q)}\}$ to refine that of f'; the splitting move (move (ii)) makes this harmless, replacing each chosen factorization by an equivalent one adapted to the grid.

The deformation proceeds one row at a time, and within a row one rectangle at a time, reading off at each stage a word associated to a path that climbs through the grid and showing that crossing a single rectangle changes the word only by equivalences. We first record the local input each crossing requires.

Step 3: Auxiliary paths land in double or triple intersections. The vertices of the staggered grid are the points $(s_i^{(j)}, t_{j-1})$ and $(s_i^{(j)}, t_j)$, the corners of the rectangles R_{ij}. For each vertex v the point $F(v)$ lies in every assigned set A_{kl} for which v is a corner of R_{kl}: the closed rectangle R_{kl} maps into A_{kl} and contains v, so $F(v) \in A_{kl}$, and the intersection of these assigned sets is therefore nonempty (it contains $F(v)$). The point of staggering is that this intersection involves at most three sets. Classify v by its location:

- if v lies on the left or right edge $\{0\} \times I$ or $\{1\} \times I$, then $F(v) = x_0$ and we take γ_v to be the constant path;
- if v lies on the bottom edge $t = 0$ or the top edge $t = 1$, it is a corner of at most the two rectangles of that single row sharing the vertical cut at v, with assigned sets A, A', and the intersection in play is the double intersection $A \cap A'$;
- if v lies on an interior horizontal line $t = t_j$ $(0 < j < q)$, it is a corner of rectangles drawn from the two rows j and $j+1$ adjacent to that line. Because the interior vertical cuts of rows j and $j+1$ are staggered to be disjoint, v is an interior cut point of at most one of the two rows. If v is an interior cut point of row j (so a shared vertical-edge endpoint of the two row-j rectangles $R_{i,j}, R_{i+1,j}$ with sets A, A'), then v is *not* a cut point of row $j+1$, hence lies in the interior of the bottom edge of a single row-$(j+1)$ rectangle $R_{m,j+1}$ with set A''; the three sets A, A', A'' are all in play and no fourth set occurs. The symmetric case (where v is an interior cut point of row $j+1$ but not of row j) likewise yields three sets. If v is a cut point of neither row (possible only at $v = (0, t_j)$ or $(1, t_j)$, already covered

by the first bullet), or of exactly one row at a position interior to a single rectangle of the other, the count is three or fewer.

Thus at every interior vertex at most three assigned sets are in play. Choose, for each vertex v, a path γ_v in X from x_0 to $F(v)$ lying in the intersection of the assigned sets in play at v: when exactly two sets A, A' occur, take γ_v in $A \cap A'$, which is path-connected by hypothesis and contains both x_0 and $F(v)$; when three sets A, A', A'' occur, take γ_v in $A \cap A' \cap A''$, path-connected by the triple-intersection hypothesis. This is the sole point at which the triple hypothesis enters, and the staggering is exactly what keeps the count at three: without it an interior vertex of a product grid would be a corner of four rectangles carrying four genuinely distinct sets, whose common intersection is a quadruple intersection not assumed path-connected, and no single γ_v serving both adjacent rows could be produced. The failure of the triple hypothesis (as in the suspension of three points) breaks the kernel computation at precisely these vertices while leaving surjectivity (Step 1, which used only double intersections) intact.

Step 4: Reading words off horizontal paths and crossing one rectangle. For $0 \le j \le q$ let h_j denote the horizontal path along the grid line $t = t_j$, that is the loop $s \mapsto F(s, t_j)$ based at x_0 (it is a loop because the endpoints $s = 0, 1$ map to x_0). Then $h_0 = f$ and $h_q = f'$. A single row is now crossed using only that row's own vertical partition, which is a genuine one-dimensional grid, so within a fixed row no four-valent vertex arises and the crossing is a chain of moves across individual rectangles. Fix a row j and write $s_r = s_r^{(j)}$ for its cut points ($0 \le r \le p_j$). For $0 \le r \le p_j$ let h_j^r be the path in X that follows the line $t = t_{j-1}$ from $s = 0$ up to $s = s_r$, then the vertical segment from (s_r, t_{j-1}) to (s_r, t_j), then the line $t = t_j$ from $s = s_r$ to $s = 1$; so $h_j^{p_j}$ traverses the bottom edge $t = t_{j-1}$ of row j and h_j^0 traverses the top edge $t = t_j$, that is $h_j^0 = h_j$ and $h_j^{p_j} = h_{j-1}$. Each such staircase path is cut by the row's grid into subpaths, each lying in a single assigned set A_{kl} (a subpath on $t = t_{j-1}$ lies on the bottom edge of a row-j rectangle, a subpath on $t = t_j$ on a top edge, and the vertical segment on a shared vertical edge, all of row j); inserting the chosen $\bar{\gamma}_v * \gamma_v$ at every grid vertex v it passes through (exactly as in Step 1) turns it into a product of loops at x_0, each loop confined to a single A_{kl}. This product is a word in the free product; call it $w(h_j^r)$.

The passage from h_j^r down across the single rectangle R_{rj} to h_j^{r-1} replaces the lower-left corner path of R_{rj} by its upper-right corner path, the two corner paths agreeing at the two diagonal vertices (s_{r-1}, t_{j-1}) and (s_r, t_j). Since $F(R_{rj}) \subseteq A_{rj}$, the four edges of R_{rj} and the constant-corner loops all map into the single set A_{rj}, and inside A_{rj} the boundary of the square is nullhomotopic (it bounds the disk $F|_{R_{rj}}$); hence the lower-left and upper-right corner loops are homotopic *within* A_{rj}. Reading both through the same factor $\pi_1(A_{rj})$, the corresponding subwords are equal in $\pi_1(A_{rj})$, so they are related by move (ii) (combine/split inside one group) followed by replacing one in-group expression by the equal one. The only subtlety is that the two corner paths attach to the rest of the staircase through the auxiliary paths γ_v at the two diagonal vertices v, and these γ_v were chosen in intersections of several assigned sets; rereading γ_v from the set used by the incoming subpath to the set used by the outgoing subpath is exactly move (i), legitimate because γ_v lies in the relevant double or triple intersection by Step 3 (both the incoming and outgoing sets are among the at-most-three sets in play at v, so the loop recording the change of reading lies in their pairwise intersection inside that triple). Thus

$$w(h_j^{r-1}) \text{ is equivalent to } w(h_j^r)$$

for each rectangle, the equivalence using only moves (i) and (ii).

Step 5: Assembling the rows. Concatenating Step 4 across the rectangles of a single row j from $r = p_j$ down to $r = 0$ gives a chain of equivalences

$$w(h_{j-1}) = w(h_j^{p_j}) \sim w(h_j^{p_j-1}) \sim \cdots \sim w(h_j^0) = w(h_j)$$

so $w(h_{j-1})$ and $w(h_j)$ are equivalent. The auxiliary paths γ_v at the vertices on the line $t = t_j$ are shared between row j (where they close up $h_j = h_j^0$) and row $j+1$ (where they close up $h_j = h_{j+1}^{p_{j+1}}$): a single choice of γ_v per vertex serves both, which is legitimate precisely because Step 3 placed γ_v in the intersection of *all* the at-most-three sets in play at v, those coming from row j below and from row $j+1$ above together. Hence the word $w(h_j)$ read at the top of row j is literally the same word read at the bottom of row $j+1$, and running j from 1 to q chains the row equivalences together:

$$w(f) = w(h_0) \sim w(h_1) \sim \cdots \sim w(h_q) = w(f')$$

It remains to connect the word $w(f) = w(h_0)$ produced by the grid to the originally chosen factorization of f. But the bottom partition $\{s_i^{(1)}\}$ was arranged (Step 2) to refine the partition of the chosen factorization, and the chosen auxiliary paths γ_v at the bottom vertices may be taken to agree with the splitting paths of that factorization (both connect x_0 to the same cut points inside the same double intersections, and any two such choices differ by a loop in a double intersection, absorbed by moves (i) and (ii)); hence $w(h_0)$ is equivalent to the chosen factorization word of f, and likewise $w(h_q)$ to that of f'. Chaining all equivalences shows the chosen factorizations of f and f' are equivalent, which is the displayed implication of Step 1.

Step 6: Conclusion. Apply the implication to a word $W \in \ker\varphi$: it factors a loop f that is nullhomotopic in X, so f is homotopic to the constant loop, whose factorization is the empty word. By Steps 2-5 the word W is equivalent to the empty word, hence $W \in N$ since each equivalence move alters the free-product element only by a generator of N. Combined with $N \subseteq \ker\varphi$ from Step 1, this gives $\ker\varphi = N$, so φ descends to an isomorphism

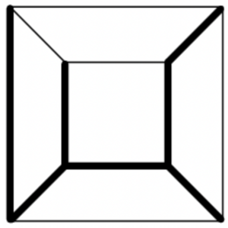
$$\mathop{\ast}_{\alpha} \pi_1(A_\alpha)/N \xrightarrow{\cong} \pi_1(X)$$

Specializing to the cover $\{U_1, U_2\}$, the group N is generated by $\iota_{1*}(\omega)\,\iota_{2*}(\omega)^{-1}$ for $\omega \in \pi_1(U_1 \cap U_2)$, so the quotient is the amalgamated free product $\pi_1(U_1) \ast_{\pi_1(U_1\cap U_2)} \pi_1(U_2)$ and the induced map k to $\pi_1(X)$ is an isomorphism. This is exactly the commutativity and universal property asserted by the pushout diagram, completing the proof.

Example 2.2.7: Use Of Van-Kampen

1. The simplest example would be a wedge-sum. Then if $X = \bigvee_\alpha X_\alpha$ all intersecting at a single point, then $\ast_\alpha \pi_1(X_\alpha) \cong \pi_1(X)$.

2. Let X be the graph:



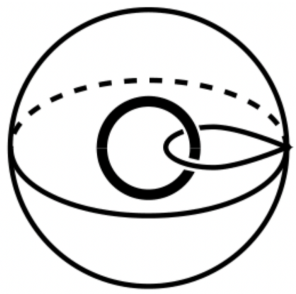
and T be the heavily shaded regions (T is a maximally spanning tree). Notice that T is contractible, and that there are 5 non-shaded edges. I claim that $\pi_1(X) = *_i^5\mathbb{Z}$. We may deduce this using the Van-Kampen Theorem. For each edge $e_\alpha \in X - T$, choose an open neighbourhood of $T \cup e_\alpha$ in X that retracts onto $T \cup e_\alpha$; call it A_α. Then the intersecton $A_\alpha \cap A_\beta$ is T, which is contractible. an path connected. Hence, we get

$$\pi_1(X) \cong *_\alpha \pi_1(A_\alpha)$$

Since each A_α deformation retracts onto a circle, we see that we get that $\pi_1(X)$ is the free product $\mathbb{Z} * \mathbb{Z} * \mathbb{Z} * \mathbb{Z} * \mathbb{Z}$.

3. Let $X = \mathbb{R}^3 - S^1$. First, notice that this deformation retracts to the shape:

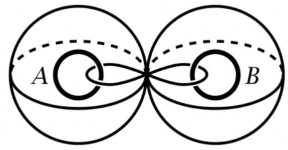
Then one can imagine that we modify this deformation retraction so that we instead have $S^1 \vee S^2$ like so:



Then since we have a deformation retraion, we have that

$$\pi_1(X) \cong \pi_1(S^1 \vee S^2) \cong \mathbb{Z}$$

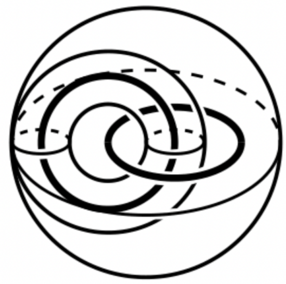
Since $\pi_1(S^2) \cong 0$. Similarly, if $X = \mathbb{R}^3 - (A \cup B)$ where A, B are two unliked circles, then we may deformation retrace to $S^1 \vee S^1 \vee S^2 \vee S^2$ to look like so:



We thus, get that

$$\pi_1(X) \cong \mathbb{Z} * \mathbb{Z}$$

One the other hand, if the two circles where linked, then we get a deformation retraction to $S^2 \vee (S^1 \times S^1)$:



which gives us:

$$\pi_1(X) \cong \mathbb{Z} \times \mathbb{Z}$$

As we see, the compliment of a shape can give us some "knotting" information; this is the first steps into knot theory and knot invariances.

4. (torus knots) For each $m, n \in \mathbb{Z}$ where $\gcd(n, m) = 1$, define $K_{m,n} \subseteq \mathbb{R}^3$ to be the iamge of $f : S^1 \to S^1 \times S^1 \subseteq \mathbb{R}^3$ given by

$$f(z) = (z^m, z^n)$$

The knot $K_{m,n}$ winds around the torus m times in the longitudinal direction and n times in the meridional direction; the relatively prime criterion is to insure f is injective. We shall compute $\pi_1(\mathbb{R}^3 - K_{m,n})$, or equivanently (due to van-kampen) we may take the 1-point compactification and consider $\pi(S^3 - K_{m,n})$.

(quite interesting, Hatcher p. 47, the answer is $(\mathbb{Z}/m\mathbb{Z}) * (\mathbb{Z}/n\mathbb{Z})$)

5. (Hawaian earings) Let $X \subseteq \mathbb{R}^2$ be the union of circles C_n of radius $1/n$ with center $(1/n, 0)$

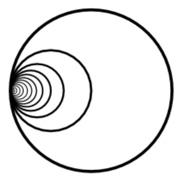


Figure 2.9: Hawaian Earings

> We may think that the fundamental group of this space is an infinite free product of $\mathbb{Z}$, but it is in fact much more complicated. (Hatcher p.48)

An important application of Van-Kampens theorem is when used in tandem with cell-complexes, in particular 2 complexes for they "add relations" by adding homotopies. Let X be a 1-skeleton that we want to attach a collection of 2 cells e_α via the maps $\varphi_\alpha : S^1 \to X$; let the resulting space be labeled Y. These maps define a loop in X. If $s_0 \in S^1$ is a basepoint, then $\varphi_\alpha(s_0)$ is the baspoint of the image of φ_α. For varying α, the basepoints may differ, but we may at $\varphi_\alpha(s_\alpha) \in X$ take γ_α that makes each loop a loop at x_0: $\tilde{\gamma}_\alpha\varphi_\alpha\gamma_\alpha$. These loops may not be nullhomotopic, but after attaching the 2 cells, they will become nullhomotopic. This means that the kernel of the map $\pi_1(X, x_0) \to \pi_1(Y, x_0)$ induced by the inclusion $X \hookrightarrow Y$ contains all $\tilde{\gamma}_\alpha\varphi_\alpha\gamma_\alpha$[3]. Since the kernel represents the added relation, we see that how we attach the cells represent new relations added to Y. The following proposition shows that the kernel consists entirely of paths as we just outlined them, and hence we exactly choose the group we get when working with 2-cells:

Proposition 2.2.8: Van-Kampen and Cells

1. If Y is obtained from X by attaching 2-cells as described above, then the inclusion $X \hookrightarrow Y$ induces a surjection $\pi_1(X, x_0) \to \pi_1(Y, x_0)$ whose kernel is N, hence
$$\pi_1(Y) = \pi_1(X)/N$$
2. If Y is obtained from X by attaching n-cells for fixed $n \geq 3$, then the inclusion $X \hookrightarrow Y$ induces an isomorphism
$$\pi_1(Y, x_0) \cong \pi_1(X, x_0)$$
3. For a path-connected cell complex X, the inclusion of the 2-skeleton $X^2 \hookrightarrow X$ induces an isomorpshim $\pi_1(X^2, x_0) \cong \pi_1(X, x_0)$

This essentially shows that the fundamental group of a CW complex is determine by how we attach 2-cells!

***Proof*:**
All three parts rest on a single device: although Y is built by attaching closed cells, an open cover to which theorem 2.2.6 applies is obtained by fattening each attached cell with a collar so that the pieces are open in Y and their pairwise and triple intersections are path-connected. The numbering below mirrors the three assertions.

1. *Attaching 2-cells.* Write $Y = X \cup \bigcup_\alpha e^2_\alpha$, where e^2_α is attached along $\varphi_\alpha\colon S^1 \to X$. The space Y itself does not split as a union of open sets one of which is each e^2_α, since the cells are closed and their boundaries lie in X; the remedy is to enlarge the picture. Pick a point y_α in the interior of each cell e^2_α, and set
$$B = Y - \{y_\alpha : \alpha\} \qquad \text{and} \qquad A_\alpha = e^2_\alpha \cup (\text{a collar of } e^2_\alpha \text{ inside } X)$$
made precise as follows. Each A_α is the union of the open cell, a small open collar in Y of the attaching circle, and the open cell of e^2_α itself, chosen so that A_α is open in Y and deformation retracts onto the closed cell e^2_α by pushing the collar inward; thus A_α is

[3] Note that it's path-independent: if η_α was a different path, then $\eta_\alpha\varphi_\alpha\tilde{\eta}_\alpha = (\eta_\alpha\tilde{\gamma}_\alpha)\gamma_\alpha\varphi_a\tilde{\gamma}_\alpha(\gamma_\alpha\tilde{\eta}_\alpha)$

contractible (it retracts to a single closed 2-cell, which is a disk), in particular $\pi_1(A_\alpha) = 0$. The set B is open and deformation retracts onto X: each punctured open cell $e^2_\alpha - \{y_\alpha\}$ deformation retracts radially onto its boundary circle in X, and these retractions, being the identity on X, glue to a deformation retraction of B onto X. Hence the inclusion $X \hookrightarrow B$ is a homotopy equivalence and $\pi_1(B, x_0) \cong \pi_1(X, x_0)$.
The intersection $A_\alpha \cap B$ is the collar with y_α removed, an open annular neighbourhood of the attaching circle, which deformation retracts onto that circle and is therefore path-connected with $\pi_1(A_\alpha \cap B) \cong \mathbb{Z}$, generated by the loop φ_α. The triple intersections $A_\alpha \cap A_\beta \cap B = A_\alpha \cap A_\beta$ are empty for $\alpha \neq \beta$ (distinct open cells are disjoint and the collars may be taken disjoint), hence vacuously path-connected. Choosing the path γ_α in X from x_0 to the basepoint $\varphi_\alpha(s_0)$ of the α-th attaching circle realizes the generator of $\pi_1(A_\alpha \cap B, x_0)$, read inside B, as the loop $\gamma_\alpha^{-1}\varphi_\alpha\gamma_\alpha$, that is $\tilde{\gamma}_\alpha\varphi_\alpha\gamma_\alpha$ in the notation preceding the statement.
Apply theorem 2.2.6 to the cover $\{B\} \cup \{A_\alpha\}_\alpha$ of Y, all members containing x_0 after the basepoint is joined into each A_α by a whisker inside its collar (this changes none of the homotopy types just computed). The theorem gives

$$\pi_1(Y, x_0) \;\cong\; \Big(\pi_1(B) * \mathop{\Huge *}_{\alpha} \pi_1(A_\alpha)\Big)/M$$

where M is the normal subgroup generated by the elements $i_{B\alpha}(\omega)\, i_{\alpha B}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap B)$. Since $\pi_1(A_\alpha) = 0$, the free factor $*_\alpha\, \pi_1(A_\alpha)$ is trivial and the free product collapses to $\pi_1(B) \cong \pi_1(X, x_0)$. For each α the group $\pi_1(A_\alpha \cap B)$ is infinite cyclic generated by ω_α, the loop around the collar; reading ω_α into A_α gives $i_{\alpha B}(\omega_\alpha) = 0$ since $\pi_1(A_\alpha) = 0$, while reading it into $B \simeq X$ gives $i_{B\alpha}(\omega_\alpha) = \tilde{\gamma}_\alpha\varphi_\alpha\gamma_\alpha \in \pi_1(X, x_0)$. Thus the defining relator of M is exactly $\tilde{\gamma}_\alpha\varphi_\alpha\gamma_\alpha$, and M, transported along the isomorphism $\pi_1(B) \cong \pi_1(X)$, is the normal subgroup N generated by the loops $\tilde{\gamma}_\alpha\varphi_\alpha\gamma_\alpha$. Consequently

$$\pi_1(Y, x_0) \;\cong\; \pi_1(X, x_0)/N$$

and under this isomorphism the quotient map is induced by the inclusion $X \hookrightarrow B \hookrightarrow Y$, since the composite $\pi_1(X) \cong \pi_1(B) \to \pi_1(Y)$ is the inclusion-induced map and is surjective with kernel N. This is the first assertion.

2. *Attaching cells of dimension* $n \geq 3$. The construction of part (1) goes through verbatim with e^2_α replaced by an n-cell e^n_α attached along $\varphi_\alpha \colon S^{n-1} \to X$: set A_α to be the open n-cell together with an open collar of its boundary sphere, so that A_α deformation retracts onto a closed n-disk and is contractible, and let $B = Y - \{y_\alpha\}$ deformation retract onto X exactly as before. The single change is in the intersection: $A_\alpha \cap B$ now deformation retracts onto the attaching sphere S^{n-1} rather than S^1. For $n \geq 3$ the sphere S^{n-1} is simply connected, so
$$\pi_1(A_\alpha \cap B, x_0) \cong \pi_1(S^{n-1}) = 0$$
by the computation of $\pi_1(S^{n-1})$ for $n - 1 \geq 2$ (example 2.2.4). Feeding this into theorem 2.2.6 with the cover $\{B\} \cup \{A_\alpha\}_\alpha$, every group $\pi_1(A_\alpha \cap B)$ is now trivial, so the normal subgroup M generated by the relators $i_{B\alpha}(\omega)\, i_{\alpha B}(\omega)^{-1}$ is itself trivial; and each $\pi_1(A_\alpha) = 0$, so the free product again collapses to $\pi_1(B) \cong \pi_1(X, x_0)$. Therefore

$$\pi_1(Y, x_0) \cong \pi_1(X, x_0)$$

the isomorphism being induced by the inclusion $X \hookrightarrow Y$. The two excluded ranges are accounted for by the surrounding theory rather than re-proved here: for $n = 2$ the attaching spheres are circles and the intersections have fundamental group $\mathbb{Z}$, which is precisely part (1) and adds the relations N; for $n = 1$ attaching arcs alters the homotopy type of the 1-skeleton and the present argument does not apply. Thus the inclusion induces an isomorphism on π_1 exactly in the stable range $n \geq 3$, which is the case the third assertion invokes.

3. *The* 2*-skeleton suffices.* Let X be a path-connected cell complex with skeleta $X^0 \subseteq X^1 \subseteq X^2 \subseteq \cdots$, and fix the basepoint $x_0 \in X^2$ (a 0-cell, after a path-connectedness adjustment). The inclusion $X^2 \hookrightarrow X$ factors through the skeletal filtration, so it suffices to control the induced map at each stage and then pass to the union. By part (2), for every $n \geq 3$ the inclusion $X^{n-1} \hookrightarrow X^n$, which attaches the n-cells of X, induces an isomorphism $\pi_1(X^{n-1}, x_0) \cong \pi_1(X^n, x_0)$. Composing these isomorphisms from $n = 3$ upward gives, for every finite $N \geq 2$,
$$\pi_1(X^2, x_0) \xrightarrow{\cong} \pi_1(X^N, x_0)$$
each map induced by inclusion.
It remains to pass from the finite skeleta to X itself. The point is that the fundamental group sees only compact data. A loop $f\colon I \to X$ based at x_0 has compact image, and a compact subset of a cell complex meets only finitely many cells, hence lies in some finite skeleton X^N; thus f is a loop in X^N, and by the isomorphism above its class comes from $\pi_1(X^2)$. This shows the inclusion $X^2 \hookrightarrow X$ induces a surjection on π_1. For injectivity, suppose a loop f in X^2 is nullhomotopic in X, witnessed by $H\colon I \times I \to X$. The image $H(I \times I)$ is compact, hence contained in some finite skeleton X^N with $N \geq 2$, so f is already nullhomotopic in X^N; since $\pi_1(X^2) \cong \pi_1(X^N)$ via inclusion, f is nullhomotopic in X^2. Therefore $\pi_1(X^2, x_0) \to \pi_1(X, x_0)$ is injective as well, and so an isomorphism, completing the proof.

The proposition computes what attaching 2-cells does to a fundamental group, which leaves the group it starts from. For a cell complex that group is π_1 of the 1-skeleton, and the 1-skeleton is a graph, so the following is the other half of the presentation recipe. It is used immediately in example 2.2.11.

Definition 2.2.9: Graph and Tree

A *graph* is a CW complex of dimension at most 1; its 0-cells are its *vertices* and its 1-cells its *edges*. A *tree* is a contractible graph. A *spanning tree* of a graph Γ is a subcomplex $T \subseteq \Gamma$ that is a tree and contains every vertex of Γ.

Proposition 2.2.10: The Fundamental Group of a Graph

Let Γ be a connected graph and x_0 a vertex.

1. Γ has a spanning tree.

2. For any spanning tree T, the quotient map $\pi\colon \Gamma \to \Gamma/T$ is a homotopy equivalence, and Γ/T is a wedge of circles with one circle for each edge of Γ not in T.

3. $\pi_1(\Gamma, x_0)$ is a free group with one generator for each edge of Γ not in T. If Γ has finitely many vertices V and edges E, that rank is $E - V + 1$.

***Proof*:**
Step 1: a spanning tree exists. Order by inclusion the subcomplexes of Γ that are trees containing x_0. The union of a chain of such is a subcomplex and is contractible: it is the increasing union of the members of the chain, and a compact subset of a CW complex meets only finitely many cells, so any map of a disk or a sphere into the union lands in a single member, where it is already nullhomotopic. Zorn's lemma gives a maximal such tree T. If some vertex lay outside T, connectedness of Γ would give an edge e with one endpoint in T and the other outside; adjoining e and that endpoint gives a subcomplex deformation retracting onto T by sliding along e, hence contractible, contradicting maximality. So T contains every vertex.

Step 2: collapsing the tree. The pair (Γ, T) is a CW pair, so it has the homotopy extension property by proposition 1.3.4, and T is contractible, so theorem 1.3.5 makes $\pi\colon \Gamma \to \Gamma/T$ a homotopy equivalence. As T contains every vertex, Γ/T has a single 0-cell, and its 1-cells are the edges of Γ outside T, each now attached with both endpoints at that vertex. A 1-cell so attached is a circle, so Γ/T is the wedge sum (definition 1.2.16) of one circle per edge outside T.

Step 3: the fundamental group. A homotopy equivalence induces an isomorphism on fundamental groups, so $\pi_1(\Gamma, x_0) \cong \pi_1(\Gamma/T)$. Applying theorem 2.2.6 to a wedge of circles, covered by open sets each thickening one circle by a short arc of its neighbours, gives the free product of the groups $\pi_1(S^1) \cong \mathbb{Z}$, one factor per circle: every pairwise and triple intersection deformation retracts to the wedge point and is therefore simply connected. A free product of copies of $\mathbb{Z}$ is the free group on the corresponding set. If Γ is finite, a spanning tree on V vertices has $V-1$ edges, so $E-(V-1) = E-V+1$ edges lie outside T, completing the proof.

The count $E-V+1$ is the first Betti number of Γ. Step 2 is why a presentation of the fundamental group of a cell complex takes its generators from the 1-cells: a spanning tree contributes nothing, and each remaining edge contributes one free generator.

Example 2.2.11: Fundamental Group Of M_g
Recall in example 1.2.5(5) that we took the cell complex M_g constructed with one 0-cell, $2g$ 1-cells an one 2-cell. Note that $M_1 = T$ is the torus. We get that the 1-skeleton is the wedge sum of $2g$ circles, giving a fundamental group being the free-product of $2g$ copies of $\mathbb{Z}$. Then the 2-cell attached long the loops gives the commutators, namely $[a_1, b_1], ..., [a_g, b_g]$. Hence:

$$\pi_1(M_g) \cong \langle a_1, b_1, ..., a_g, b_g \mid [a_1, b_1], ..., [a_g, b_g] \rangle$$

Using this, we [finally] see that $M_g \not\simeq M_h$ if $g \neq h$ (hence $M_g \ncong M_h$). Since $\pi_1(M_g) \cong \pi_1(M_h)$ if and only if the number of copies aligned (finitely generated free $\mathbb{Z}$-modules have an invariant basis number).

Corollary 2.2.12: Every Group Appears As A Fundamental Group

Let G be a group. Then there exists a 2-dimensional cell complex X_G such that

$$\pi_1(X_G) \cong G$$

***Proof*:**
Let $G = \langle g_\alpha \mid r_\beta \rangle$ where g_α are the generators and r_β are the relations. Then take skeleton to be $\bigvee_\alpha S^1$ and attach the 2-cells e^2_β specified by the words r_β.

Example 2.2.13: More Fundamental Groups

1. (non-orientable surfaces) Let N_g be the surface where we attach 2-cells to the wedge-sum of g circles by the words $a_1^2, ..., a_g^2$. Note that $N_1 = \mathbb{R}\mathrm{P}^2$ since it is D^2 with the antipodal points of ∂D^2 identified, and N_2 is the Klein bottle (the left image is the usual construciton of the Klein bottle):

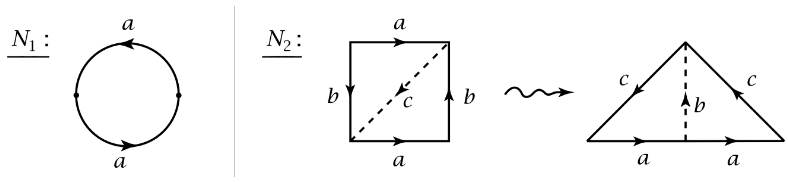


Figure 2.10: $\mathbb{R}\mathrm{P}^2$ and Klein Bottle

Then:
$$\pi_1(N_g) = \langle a_1, ..., a_g \mid a_1^2, ..., a_g^2 \rangle$$
Note that we also can slightly change how we glue the gth 2-cell to get an abelian group $(\mathbb{Z}/2\mathbb{Z})^n$ by choosing generators $a_1, a_2, ..., a_{g-1}, (a_1 + \cdots + a_g)$ with $2(a_1 + \cdots + a_g) = 0$.

Overalll, N_g is not homoopy equivalent to N_h if $g \neq h$, and is not homotopy equivalence to any M_g or M_h.

2. let $G = \langle a \mid a^n \rangle = \mathbb{Z}_n$. Then X_G is S^1 with the 2-cell e^2 attached via the map $z \mapsto z^n$. If $n = 2$, then $X_G = \mathbb{R}\mathrm{P}^2$. Note that when $n \geq 3$, then X_G is not a surface (a smooth 2-manifold). For example, if $n = 3$, then we can have a neighbourhood N of X_G that looks like the product of the tripod (inverse of a triangle, or a triangle in CAT(0)) with the interval I (the tripod can also be thought of as a vertex with three intervals attached to it). Similar results happen for $n \geq 4$. The resulting shape is like a torus with a $1/n$ twist at some point (for $n = 3$, it's a $1/3$ twist):

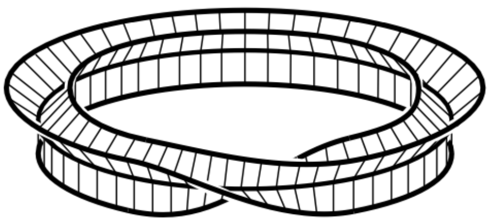


Figure 2.11: visual of X_3

2.3 Classification of Covering Spaces

We finish off this section by classifying covering spaces. let X be a path-connected, locally path-connected space. We shall show there is a similar theory between the varying cover-spaces and galois theory. In particular, every covering map shall be associated with a subgroup $\pi_1(X, x_0)$ (for any given basepoint $x_0 \in X$), where the partial order given by the covering maps is inverse to the ordering of the groups, that is we have a contra-variant functor between the partial orders (giving the galois correspondence).

Recall that if $p : E \to B$ is a covering space where E is simply connectd, then $\pi_1(B, x_0)$ is in bijective correspondence with $p^{-1}(x_0)$. Are there condition conditions on B for which there exist a covering map where E is simply connected? It turns out there is; this type of space a name:

Definition 2.3.1: Semilocally Simply-Connected

Let X be a topological space. Then if for each $x \in X$, there must exist a neighbourhood $U \subseteq X$ such that the map induced by the inclusion map $\iota_* : \pi_1(U, x) \to \pi_1(X, x)$ is trivial, X is called *semilocally simply-connected.*

To see why this is a necesary condition, suppose $p : \tilde{X} \to X$ is a covering space with $\tilde{X}$ simply connecd. Then every point $x \in X$ has a neighbourhood U having a lift $\tilde{U} \subseteq \tilde{X}$ projecting homeomrphically to U by p. Each loop in U lifts to a loop in $\tilde{U}$, and the lifted loop is nullhomotopic in $\tilde{X}$ since $\tilde{X}$ is simply connected. Hence, composing this nullhomotopy with p, the original loop in U is nullhomotopic in X.

All locally simply-conneced spaces are semilocally simply-connected. As an exmaple, CW complexes are locally contractible (appendix of hatcher), and hence shall all have universal covers.

Example 2.3.2: Not Semilocally Connected

Tkae $X \subseteq \mathbb{R}^2$ consisting of circles of radius $1/n$ centered at $(1/n, 0)$ (i.e. the hawaian earings). On the other hand, the cone $CX = (X \times I)/(X \times \{0\})$ on the shrinking wedge of circles is semilocally simply-connected since it is contractible, but is not locally simply-connected

Lemma 2.3.3: Universal Covering Map

Let X be a path connected, locally path connected, semilocally simply connected space. Then there exists a simply connected space $\tilde{X}$ such that $p : \tilde{X} \to X$ is a covering map. Furthermore, this cover is unique up to isomorphism, and any other simply connected covering map commutes with $\tilde{X}$

***Proof*:**

Let X be a path connected, locally path connected, semilocally simply connected space. Define the space

$$\tilde{X} = \{[\gamma] \mid \gamma \text{ is a path in } X \text{ starting at } x_0\}$$

Since X is path-connected, the choice of x_0 is irrelevant. Define $p : \tilde{X} \to X$, $[g] \mapsto \gamma(1)$, which is well-defined since all paths in the homotopy end at the same point. Since X is path-connected, each point $x \in X$ is the endpoint of a path starting at x_0, and hence p is surjective.

We next define a topology on $\tilde{X}$. Let

$$\mathcal{U} = \{U \subseteq X \mid \pi_1(U) \to \pi_1(X) \text{ is trivial}\}$$

Note that is the above map is independent of base point since X is locally path connected (hence U is path-connected). Then since X is semilocally simply-connected, $\mathcal{U}$ forms a basis for X. Given $U \in \mathcal{U}$, and γ a path in X from x_0 to a point in U, elt

$$U_{[}\gamma] = \{[\gamma * \eta] \mid \eta \text{ is a loop in } U\}$$

Clearly, $U_{[\gamma]}$ depends only on the homotopy clas of γ, justifying the notation. The map $p : U_{[\gamma]} \to U$ is surjective since U is path-connectd, and injective since different chocies of η joining $\gamma(1)$ to a fixed $x \in U$ are all homotopic in X since $\pi_1(U) \to \pi_1(X)$ is trivial.

Next, if $[\gamma'] \in U_{[\gamma]}$, then $U_{[\gamma]} = U_{[\gamma']}$. For, if $\gamma' = \gamma * \eta$, then elements of $U_{[\gamma']}$ have the form $[\gamma * \eta * \mu]$, and hecne are in $U_{[\gamma]}$, while conversely elements of $U_{\gamma]}$ ahve hte form:'

$$[\gamma * \mu] = [\gamma * \eta * \overline{\eta} * \mu] = [\gamma' * \overline{\eta} * \mu]$$

and hence aer in $U_{[\gamma']}$. Thus, if $[\gamma''] \in U_{[\gamma]} \cap V_{[\gamma']}$, we get $U_{[\gamma]} = U_{[\gamma'']}$ and $V_{[\gamma']} = V_{[\gamma'']}$. So if $W \subseteq U \cap V$, and contains $\gamma''(1)$, then $[\gamma''] \in W_{[\gamma'']}$ and

$$W_{[\gamma'']} \subseteq U_{[\gamma'']} \cap V_{[\gamma'']}$$

showing that the collection of $U_{[\gamma]}$ forms a basis for a topology on $\tilde{X}$. From this, the bijection $p : U_{[\gamma]} \to U$ is a homeomorphism since it gives a bijection between the subsets $V_{[\gamma']} \subseteq U_{[\gamma]}$, and the sets $V \in \mathcal{U}$ contained in U. This then shows that $p : \tilde{X} \to X$ is continuous. It is also clear that $p : \tilde{X} \to X$ is a covering map. What remains to show si that $\tilde{X}$ is path-connected. For any point $[\gamma] \in \tilde{X}$, let γ_t be the path in X that equals γ on $[0, t]$ and is constant at $\gamma(t)$ on $[t, 1]$. Then the function $t \mapsto [\gamma_t]$ is a path in $\tilde{X}$ lifting γ that starts at $[x_0]$ (the homotopy class of constants paths at x_0) and ends at $[\gamma]$. Since $[\gamma]$ was an arbitrary point in $\tilde{X}$, we see $\tilde{X}$ is path-connected.

Finally, to show $\pi_1(\tilde{X}, [x_0]) = 0$, it suffices to show that hte iamge of htis group under π_* is triial since p_* is injective. Any element in the iamge of p_* are represented by loops γ at $x_8$that lift to loops in $\tilde{X}$ at $[x_0]$. We has seen that hte path $t \mapsto [\gamma_t]$ lifts γ starting at $[x_]$, and for this lifted path to be a loop it means that $[\gamma_1] = [x_0]$. Since $\gamma_1 = \gamma$, w have that $[\gamma] = [x_0]$, and so γ is nullhomotopic, hence the iamge of p_* is trivial, and so $\tilde{X}$ is also simply connected, completing the proof.

(showing uniquness?)

This construction shows that a space with these conditions always has a simple covering space, however most the time there shall be simpler means of constructing covering spaces. For example, we know a simple covering space of S^1, namely $\mathbb{R}$, and any product T^n has covering space $\mathbb{R}^n$ since commute with functors (as they are limits, but we also proved it explicitely).

Example 2.3.4: Simply Connected Covering Spaces
hatcher p. 65

Theorem 2.3.5: Galois Correspondence Of Fundamental Groups

Let X be a path-connectd, locally path-connected, semilocally simply-connected space. Then there is a bijection between the set of basepoint-preserving isomorphsism classes of path-connected covering space $p : (\tilde{X}, e_0) \to (X, b_0)$ and the set of subgroups of $\pi_1(X, x_0)$, obtained by associating the subgroup $p_*(\pi_1(\tilde{X}), e_0)$.

***Proof*:**
This is the classification of coverings, proved in *Topology* [12, Classification of Coverings] for a path-connected, locally path-connected and semilocally simply connected base. Its two halves are the existence of a covering realising a prescribed subgroup, which goes through the universal cover, and the injectivity of $p \mapsto p_*\pi_1(\tilde{X}, e_0)$, which is the lifting criterion applied in both directions.

Theorem 2.3.6: Uniqueness Of Covering Space

Let X be path-connected and locally path-connected. Then two path-connected covering spaces $p_1 : \tilde{X}_1 \to X$ and $p_2 : \tilde{X}_2 \to X$ are basepoint-isomorphic if and only if $\mathrm{im} p_{1_*} = \mathrm{im} p_{2_*}$.

***Proof*:**
This is the injective half of the classification, proved in *Topology* [12, Classification of Coverings]: two coverings with the same image subgroup each lift through the other by the lifting criterion, and the two lifts compose to the identity by uniqueness of lifts, so they are inverse isomorphisms over X.

An isomorphism here would be that the diagram commutes.

***Proof*:**
If there is an isomorphism f betweeen $(\tilde{X}_1, \tilde{x}_1)$ and $(\tilde{X}_2, \tilde{x}_2)$, then we get $p_1 = p_2 f$ and $p_2 = p_1 f^{-1}$, giving us the groups are equal, Conversely, if hte groups are equal, by the lifting criterion, we may lift p_1 to a map $\tilde{p}_1 : (\tilde{X}_1, \tilde{x}_1) \to (\tilde{X}_2, \tilde{x}_2)$ with $p_2\tilde{p}_1 = p_1$. We can do the same the other way to get $p_1\tilde{p}_2 = p_2$. Then by uniqueness of lifting, $\tilde{p}_1\tilde{p}_2 = \mathrm{id}$ and $\tilde{p}_2\tilde{p}_1 = \mathrm{id}$ since the composition fixes the basepoint. These maps are certainly homomorphissm, and they are each other inverses, and hence, each are isomorphissm, as we sought to show.

Proposition 2.3.7: Base-Point Independence

If ignoring the basepoint, the correspondence gives a bijection between isomorphism class of path-connected covering spaces $p : \tilde{X} \to X$ and conjugacy classes of subgroups of $\pi_1(X, x_0)$

***Proof*:**
The basepoint-preserving correspondence of theorem 2.3.5 sends a covering $p\colon (\tilde{X}, e_0) \to (X, b_0)$ to the subgroup $H = p_*(\pi_1(\tilde{X}, e_0)) \subseteq \pi_1(X, b_0)$, and theorem 2.3.6 shows that this assignment is injective on basepoint-preserving isomorphism classes. Forgetting the basepoint amounts to allowing e_0 to range over the fiber $p^{-1}(b_0)$ and identifying two coverings whenever some

such reassignment makes them basepoint-isomorphic. The proof therefore has two halves: first, identify how H changes as e_0 moves within the fiber; second, package the resulting equivalence as the conjugacy relation.

Step 1: Moving the basepoint conjugates the subgroup. Fix the covering $p\colon \tilde{X} \to X$ with X path-connected and locally path-connected, and let $e_0, e_1 \in p^{-1}(b_0)$ be two points of the fiber over b_0. Write $H_0 = p_*(\pi_1(\tilde{X}, e_0))$ and $H_1 = p_*(\pi_1(\tilde{X}, e_1))$. Since $\tilde{X}$ is path-connected, choose a path $\tilde{\eta}$ in $\tilde{X}$ from e_0 to e_1, and let $\eta = p \circ \tilde{\eta}$, a loop in X based at b_0. Set $g = [\eta] \in \pi_1(X, b_0)$. The claim is that

$$H_1 = g^{-1}\, H_0\, g$$

By proposition 2.1.15, the path $\tilde{\eta}$ from e_0 to e_1 induces an isomorphism $\widehat{\tilde{\eta}}\colon \pi_1(\tilde{X}, e_1) \to \pi_1(\tilde{X}, e_0)$ given by $[\tilde{f}] \mapsto [\tilde{\eta}] * [\tilde{f}] * [\overline{\tilde{\eta}}]$. This is correctly typed against definition 2.1.4: for a loop $\tilde{f}$ at e_1, the concatenation $\tilde{\eta} * \tilde{f}$ is defined because $\tilde{\eta}(1) = e_1 = \tilde{f}(0)$, and the further concatenation with $\overline{\tilde{\eta}}$ is defined because $\tilde{f}(1) = e_1 = \overline{\tilde{\eta}}(0)$, so $\tilde{\eta} * \tilde{f} * \overline{\tilde{\eta}}$ traverses $\tilde{\eta}$ from e_0 to e_1, runs $\tilde{f}$, then returns along $\overline{\tilde{\eta}}$ to e_0, giving a loop at e_0. Applying p_* and using functoriality (proposition 2.1.22) together with $p \circ \tilde{\eta} = \eta$, a loop $\tilde{f}$ at e_1 is carried to

$$p_*\big([\tilde{\eta}] * [\tilde{f}] * [\overline{\tilde{\eta}}]\big) = [\eta] * p_*([\tilde{f}]) * [\overline{\eta}] = g\, p_*([\tilde{f}])\, g^{-1}$$

As $[\tilde{f}]$ ranges over $\pi_1(\tilde{X}, e_1)$, the image $p_*([\tilde{f}])$ ranges over H_1. On the other hand $\widehat{\tilde{\eta}}$ is a bijection onto $\pi_1(\tilde{X}, e_0)$, so the classes $[\tilde{\eta}] * [\tilde{f}] * [\overline{\tilde{\eta}}]$ range over all of $\pi_1(\tilde{X}, e_0)$ and their images under p_* range over all of H_0. Comparing the two descriptions of the same set gives $H_0 = g\, H_1\, g^{-1}$, that is $H_1 = g^{-1} H_0\, g$. Thus, as e_1 ranges over $p^{-1}(b_0)$, the subgroup $p_*(\pi_1(\tilde{X}, e_1))$ ranges over conjugates of H_0.

Conversely, every conjugate of H_0 arises this way. Given $g = [\eta] \in \pi_1(X, b_0)$, lift the loop η to the path $\tilde{\eta}$ in $\tilde{X}$ starting at e_0, which exists by lemma 2.1.30, and let $e_1 = \tilde{\eta}(1) \in p^{-1}(b_0)$. Then $\tilde{\eta}$ is a path from e_0 to e_1 projecting to η, so the computation above applies and yields $H_1 = p_*(\pi_1(\tilde{X}, e_1)) = g^{-1} H_0\, g$. As g ranges over $\pi_1(X, b_0)$ the conjugate $g^{-1} H_0\, g$ ranges over the entire conjugacy class of H_0. Therefore the set of subgroups realized by the various choices of basepoint in the fiber over b_0 is precisely the conjugacy class of H_0 in $\pi_1(X, b_0)$.

Step 2: Conjugacy classes classify the unbased coverings. Define a map from basepoint-free isomorphism classes of path-connected coverings to conjugacy classes of subgroups of $\pi_1(X, b_0)$ by sending a covering $p\colon \tilde{X} \to X$ to the conjugacy class of $H_0 = p_*(\pi_1(\tilde{X}, e_0))$ for any chosen $e_0 \in p^{-1}(b_0)$. By Step 1 this conjugacy class does not depend on the choice of e_0, so the map is well-defined on a fixed covering. It is also constant on basepoint-free isomorphism classes: if $f\colon \tilde{X}_1 \to \tilde{X}_2$ is a homeomorphism with $p_2 \circ f = p_1$ (no constraint on basepoints), pick $e_0 \in p_1^{-1}(b_0)$ and set $e_0' = f(e_0) \in p_2^{-1}(b_0)$; then f restricts to a basepoint-preserving isomorphism $(\tilde{X}_1, e_0) \to (\tilde{X}_2, e_0')$, so by functoriality (proposition 2.1.22) $p_{1*}(\pi_1(\tilde{X}_1, e_0)) = p_{2*}(\pi_1(\tilde{X}_2, e_0'))$ as actual subgroups, and a fortiori the two conjugacy classes agree. Hence the assignment descends to basepoint-free isomorphism classes.

The map is surjective: given any subgroup $K \subseteq \pi_1(X, b_0)$, theorem 2.3.5 produces a path-connected covering $p\colon (\tilde{X}, e_0) \to (X, b_0)$ with $p_*(\pi_1(\tilde{X}, e_0)) = K$, and this covering maps to the conjugacy class of K. Since every conjugacy class contains a subgroup, the map hits every conjugacy class.

For injectivity, suppose two path-connected coverings $p_1\colon \tilde{X}_1 \to X$ and $p_2\colon \tilde{X}_2 \to X$ have $H_1 = p_{1*}(\pi_1(\tilde{X}_1, e_1))$ and $H_2 = p_{2*}(\pi_1(\tilde{X}_2, e_2))$ in the same conjugacy class, where $e_1 \in p_1^{-1}(b_0)$ and $e_2 \in p_2^{-1}(b_0)$ are chosen basepoints. By Step 1, moving the basepoint of $\tilde{X}_2$ realizes every conjugate of H_2; in particular there is a point $e_2' \in p_2^{-1}(b_0)$ with $p_{2*}(\pi_1(\tilde{X}_2, e_2')) = H_1$ as subgroups of $\pi_1(X, b_0)$. The two coverings $(\tilde{X}_1, e_1)$ and $(\tilde{X}_2, e_2')$ now have equal images under the induced maps, so theorem 2.3.6 provides a basepoint-preserving isomorphism between them. Forgetting the basepoint, p_1 and p_2 are isomorphic as coverings. Hence coverings with conjugate subgroups lie in the same basepoint-free isomorphism class.

The assignment is therefore a well-defined surjection that is injective on basepoint-free isomorphism classes, hence a bijection between isomorphism classes of path-connected coverings $p\colon \tilde{X} \to X$ and conjugacy classes of subgroups of $\pi_1(X, b_0)$, completing the proof.

Hence, a simply-connected coverign space X is a *universal cover* since it is unique *up to isomorphism.*

(Representing Covering Spaces by Permutations, Hatcher p. 68)

2.3.1 Deck Transformations

You may have noticed in the earlier theorem that the isomorphism between covering spaces that commuted in the diagram

$$\begin{tikzcd} E \arrow[rr, "\sim"] \arrow[dr, "p"'] & & E \arrow[dl, "p"] \\ & B & \end{tikzcd}$$

has the same properties as fiber bundles. There is a relation between fiber bundles, covering maps, and fundamental groups. Since the study of this fiber-bundle morphisms on covering spaces pre-dates categorical nomenclature, we have the following widely-accepted name:

Definition 2.3.8: Deck Transformation

Let $p : E \to B$ be a covering map. Then an isomorphism:

$$\begin{tikzcd} E \arrow[rr, "\sim"] \arrow[dr, "p"'] & & E \arrow[dl, "p"] \\ & B & \end{tikzcd}$$

is called a *Deck transformation*. The collection of deck transformations is denoted $\mathrm{Deck}\,(p)$ or $\mathrm{Aut}(p)$. If we want to specify a particular group for a covering space, we will write $\mathrm{Aut}(p, E)$.

Since the spaces we are working with have unique universal covering maps up to isomorphism (ex. CW complexes), the notation $\mathrm{Deck}\,(p)$ is well-defined given we are always taking universal covers.

Example 2.3.9: Deck Transformation

1. Let $p : \mathbb{R} \to S^1$. Then a deck transformation willl be vertical translations, so

$$\mathrm{Deck}\,(p) = \mathbb{Z}$$

2. Let $p : S^1 \to S^1$ be given by $z \mapsto z^n$. Then $\mathrm{Aut}(p, S^1) = \mathbb{Z}/n\mathbb{Z}$.

By the unique lifting property, a deck transformation is *completely* determined by where it sends a single point (assuming E is path connected). Thus, the only map that fixes points of E is the identity.

Definition 2.3.10: Normal Covering

A covering space $p : E \to B$ is called *normal* if for each $x \in X$, and each pair of lifts $\tilde{x}, \tilde{x}'$ of x, there exists a deck transformation taking $\tilde{x}$ to $\tilde{x}'$.

For example, the covering space $\mathbb{R} \to S^1$ and the n-sheeted covering space $S^1 \to S^1$ are normal. Intuitively, a normal covering space is one with "maximal symmetry".

Proposition 2.3.11: Properties Of Normal Covering Spaces

Let $p : (E, e_0) \to (B, b_0)$ be a path-connected covering space of a path-connected, locally path-connected space B, and let H be the subgroup $H = p_*(\pi_1(E, e_0)) \subseteq \pi_1(B, b_0)$. Then:

1. The covering map is normal if and only if $H \trianglelefteq \pi_1(B, b_0)$
2. $\mathrm{Aut}(p, E)$ is isomorphic to the quotient $N(H)/H$ ($N(H)$ the normalizer of H).

In particular, $\mathrm{Aut}(p, E)$ is isomorphic to $\pi_1(B, b_0)/H$ if E is a normal covering. Hence the universal cover $p : E \to B$ has $\mathrm{Aut}(p) \cong \pi_1(B)$

***Proof*:**
Both clauses are point-set covering-space theory, established in *Topology* [12, The Deck Group as a Quotient of a Normalizer, The Deck Group of the Universal Cover]: the equivalence of normality of the covering with normality of the subgroup H, and the isomorphism $\mathrm{Aut}(p, E) \cong N(H)/H$, are proved there in the chapter on the fundamental group. The specialisation to the universal cover, where H is trivial and the deck group is all of $\pi_1(X, x_0)$, is the corollary recorded there immediately afterwards. They are quoted here in the form the homology of covering spaces uses, as we sought to show.

(more here on some interesting properties)

2.4 Classifying Spaces

The Galois correspondence of theorem 2.3.5 organizes the covering spaces of a fixed base X by the subgroups of $\pi_1(X)$, and the universal cover sits at the top of this lattice as the cover with trivial fundamental group. A covering space is the most rigid kind of bundle with discrete fiber: its deck group acts freely, and a normal cover is recovered from its base as the quotient by that action. This section reverses the perspective. Instead of fixing X and cataloguing the covers above it, one fixes a topological group G and asks for a single space, the *classifying space* BG, that catalogues all the spaces carrying a free G-action of a suitable kind, all at once, over every reasonable base. The objects being catalogued are the *principal G-bundles*, and the catalogue takes the form of a bijection between isomorphism classes of such bundles over a base X and homotopy classes of maps $X \to BG$.

This is the geometric heart of the construction $B(-)$. The same space BG that classifies principal bundles is, when G is discrete, exactly the Eilenberg-MacLane space $K(G,1)$ whose cohomology computes the group cohomology of G; and the same construction, applied to the unitary and orthogonal groups, produces the classifying spaces BU and BO that govern complex and real vector bundles in the companion volume [21]. The unifying mechanism is a universal bundle $EG \to BG$ with EG weakly contractible: every bundle is pulled back from this one, and the pullback is controlled up to isomorphism by the homotopy class of the classifying map. The defining relation $\Omega BG \simeq G$ records that BG is a delooping of G, which is why $B(-)$ shifts homotopy up by one degree and turns the group G into the fundamental group of BG.

Definition 2.4.1: Principal G-Bundle

Let G be a topological group. A *principal G-bundle* is a map $p\colon E \to X$ of spaces together with a continuous right action $E \times G \to E$ of G on E, written $(e,g) \mapsto e \cdot g$, satisfying:

1. the action preserves fibers and is free and transitive on each fiber: $p(e \cdot g) = p(e)$ for all g, and for each $x \in X$ the group G acts simply transitively on $p^{-1}(x)$;
2. the bundle is *locally trivial*: every point of X has an open neighbourhood U admitting a G-equivariant homeomorphism $\varphi_U\colon p^{-1}(U) \xrightarrow{\cong} U \times G$ over U, where G acts on $U \times G$ by right translation in the second factor, $(u,h)\cdot g = (u,hg)$, and "over U" means $\mathrm{pr}_U \circ \varphi_U = p$.

An *isomorphism* of principal G-bundles $p\colon E \to X$ and $p'\colon E' \to X$ over the same base is a G-equivariant homeomorphism $E \to E'$ commuting with the projections to X. The set of isomorphism classes of principal G-bundles over X is written $\mathrm{Prin}_G(X)$.

A principal G-bundle is a fiber bundle whose fiber is the group G itself, but carrying the extra datum of a free fiberwise action that turns each fiber into a *torsor*: a copy of G with no distinguished identity element. Local triviality says the bundle looks like the product $U \times G$ over small opens, exactly as for the fiber bundles of theorem 5.2.4, with the additional requirement that the trivializing homeomorphisms respect the G-action. The transition between two trivializations φ_U and φ_V over $U \cap V$ is left multiplication by a continuous function $g_{UV}\colon U \cap V \to G$, the *transition function*, because an equivariant self-map of $U \cap V \times G$ over $U \cap V$ commuting with right translation is precisely left translation by an element of G varying continuously with the base point. A principal bundle is trivial, that is, isomorphic to the product $X \times G$, if and only if it admits a global section, since a section $s\colon X \to E$ produces the equivariant trivialization $X \times G \to E$, $(x,g) \mapsto s(x)\cdot g$, and conversely the product bundle has the obvious section $x \mapsto (x,e)$. This is the first point of contact with covering-space theory: a covering map with deck group G acting freely and transitively on fibers is a principal G-bundle for the discrete group G.

Example 2.4.2: The Real Line over the Circle

Take $G = \mathbb{Z}$ with the discrete topology and the exponential covering $p\colon \mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$, with $\mathbb{Z}$ acting on $\mathbb{R}$ by $t \cdot n = t + n$. This is a principal $\mathbb{Z}$-bundle. The action is continuous, preserves fibers (since $e^{2\pi i(t+n)} = e^{2\pi i t}$), and is free and transitive on each fiber: the fiber over $e^{2\pi i\theta}$ is the coset $\theta + \mathbb{Z}$, on which $\mathbb{Z}$ acts simply transitively by translation. For local triviality, cover S^1 by the two arcs $U_+ = S^1 \setminus \{-1\}$ and $U_- = S^1 \setminus \{1\}$. Over U_+ the principal branch of $\frac{1}{2\pi i}\log$ gives a continuous section $s_+ : U_+ \to \mathbb{R}$ with values in $(-\frac{1}{2}, \frac{1}{2})$, and the equivariant homeomorphism

$$p^{-1}(U_+) \xrightarrow{\cong} U_+ \times \mathbb{Z}, \qquad t \longmapsto \big(e^{2\pi i t},\, t - s_+(e^{2\pi i t})\big)$$

trivializes the bundle over U_+, the second coordinate landing in $\mathbb{Z}$ because t and $s_+(e^{2\pi i t})$ differ by an integer; the analogous formula works over U_-. The single transition function on the overlap $U_+ \cap U_-$ (two arcs) is the locally constant map to $\mathbb{Z}$ recording the integer jump $s_+ - s_-$, which is 0 on one arc and 1 on the other; this nonconstancy is exactly the obstruction to a global section, so the bundle is nontrivial. The deck group $\mathrm{Deck}\,(p) \cong \mathbb{Z}$ of definition 2.3.8 is the structure group of the bundle, and the principal $\mathbb{Z}$-bundle and the universal cover are the same object viewed two ways.

The next task is to construct, for a fixed G, a single bundle from which every principal G-bundle is pulled back. Pulling back a bundle along a map $f : X \to X'$ replaces the base by X and the total space by the fiber product, and the operation is functorial and homotopy-invariant on a paracompact base. The universal object is characterized by a contractibility condition on its total space.

Definition 2.4.3: Universal Bundle and Classifying Space

Let G be a topological group. A principal G-bundle $p : EG \to BG$ is a *universal bundle*, and its base BG is the *classifying space* of G, if the total space EG is weakly contractible, that is, $\pi_i(EG) = 0$ for all $i \geq 0$. Equivalently (for G acting on a CW complex), EG is a contractible CW complex carrying a free G-action and $BG = EG/G$ is the orbit space.

The defining condition asks the total space to have no homotopy of its own, so that all of the homotopy of the bundle is concentrated in the base and the structure group. When G is discrete, the projection $EG \to BG$ is a covering map with EG simply connected (indeed weakly contractible), hence the universal cover of BG, and the long exact sequence of the covering forces $\pi_1(BG) \cong G$ with all higher homotopy of BG vanishing: BG is then a $K(G, 1)$ in the sense of the Eilenberg-MacLane theory developed in chapter 5 (see definition 5.5.1). For a general topological group G the same long exact sequence of the bundle $G \to EG \to BG$, applied with $\pi_i(EG) = 0$, yields the relation

$$\pi_i(BG) \;\cong\; \pi_{i-1}(G) \qquad \text{for all } i \geq 1$$

which is the homotopy-group form of the delooping $\Omega BG \simeq G$ discussed below. Existence of a universal bundle for every G is given by Milnor's infinite-join construction, recorded in the next proposition; uniqueness up to homotopy will follow from the classification theorem.

Proposition 2.4.4: Existence of the Universal Bundle

For every topological group G there exists a principal G-bundle $p : EG \to BG$ with EG weakly contractible. When G is discrete the spaces EG and BG may be taken to be CW complexes with EG contractible, $BG = EG/G$, and $BG = K(G, 1)$.

***Proof*:**
Step 1: The infinite join. Define EG to be the infinite join

$$EG \;=\; G * G * G * \cdots \;=\; \varinjlim_n \underbrace{G * \cdots * G}_{n}$$

A point of the n-fold join G^{*n} is a formal combination $t_1g_1 \oplus \cdots \oplus t_ng_n$ with $g_i \in G$, weights $t_i \geq 0$ summing to 1, and the identification that a term with weight $t_i = 0$ drops its G-coordinate; the topology is the quotient of $\{(t_i, g_i)\} \subseteq \Delta^{n-1} \times G^n$, and EG carries the colimit (weak) topology over n. The group G acts on the right diagonally, $(t_1g_1 \oplus \cdots \oplus t_ng_n) \cdot g = t_1(g_1g) \oplus \cdots \oplus t_n(g_ng)$, freely because a point with at least one positive weight t_i has a determined coordinate g_i that the action moves unless $g = e$.

Step 2: Weak contractibility of EG. The join G^{*n} is $(n-2)$-connected: a join $A * B$ is always at least $(\mathrm{conn}(A) + \mathrm{conn}(B) + 2)$-connected, and iterating from the (-1)-connected (nonempty) space G gives $\pi_i(G^{*n}) = 0$ for $i \leq n-2$. Any map $S^i \to EG$ has compact image, hence factors through some finite join G^{*n} with $n - 2 \geq i$, where it is null-homotopic; the same compactness argument applied to a null-homotopy shows the class already vanishes in EG. Therefore $\pi_i(EG) = 0$ for every i, so EG is weakly contractible.

Step 3: Local triviality and the bundle. Set $BG = EG/G$ with the quotient topology and p the projection. For each index i let $U_i \subseteq BG$ be the image of the open set $\{t_i > 0\} \subseteq EG$. The coordinate g_i is well defined on $\{t_i > 0\}$ after dividing out the G-action only up to the ambient G, but the map $\{t_i > 0\} \to G$, $x \mapsto g_i$, is G-equivariant and descends to a section of p over U_i, namely $s_i \colon U_i \to EG$ sending an orbit to its representative with $g_i = e$. This section yields the equivariant trivialization $p^{-1}(U_i) \cong U_i \times G$, $e \mapsto (p(e), \text{the unique } g \text{ with } e = s_i(p(e)) \cdot g)$. The opens $\{U_i\}$ cover BG because every point of EG has some positive weight. Hence $p \colon EG \to BG$ is a principal G-bundle with weakly contractible total space.

Step 4: The discrete case. When G is discrete each join G^{*n} is a simplicial complex (a join of discrete sets), so EG is a CW complex on which G acts freely and cellularly; the action being free and cellular, $BG = EG/G$ is again a CW complex and p is a covering map. A weakly contractible CW complex is contractible by Whitehead's theorem (theorem 5.3.5), so EG is contractible. The long exact sequence of the covering (a fibration with discrete fiber G, as in chapter 5) gives $\pi_1(BG) \cong G$ and $\pi_i(BG) \cong \pi_i(EG) = 0$ for $i \geq 2$, so BG is a $K(G, 1)$, completing the proof.

The join model is concrete enough to compute with but is rarely the most efficient representative. Its value is conceptual: it shows a universal bundle exists for every topological group whatsoever, with no hypothesis beyond having a continuous multiplication, and it makes the contractibility of EG transparent through the connectivity of joins. In practice one replaces EG by any convenient weakly contractible space carrying a free G-action, and the classification theorem guarantees the resulting BG is independent of the choice up to homotopy. The next example carries out exactly this replacement for the discrete groups $\mathbb{Z}$ and $\mathbb{Z}/2$ and for the circle group.

Example 2.4.5: Classifying Spaces of $\mathbb{Z}$, $\mathbb{Z}/2$, and $U(1)$

1. For $G = \mathbb{Z}$ take $EG = \mathbb{R}$ with the translation action $t \cdot n = t + n$. The space $\mathbb{R}$ is contractible, the action is free, and $E\mathbb{Z}/\mathbb{Z} = \mathbb{R}/\mathbb{Z} = S^1$. Thus $B\mathbb{Z} = S^1$, recovering the principal bundle of example 2.4.2. Consistently with definition 2.4.3, $\pi_1(S^1) \cong \mathbb{Z}$ and $\pi_i(S^1) = 0$ for $i \geq 2$, so $S^1 = K(\mathbb{Z}, 1)$.

2. For $G = \mathbb{Z}/2$ take $EG = S^\infty = \bigcup_n S^n$ with the antipodal action $v \cdot (-1) = -v$. The infinite sphere is contractible (any compact subset lies in some S^n, which contracts inside S^{n+1} by sliding to a missing point), the antipodal action is free, and $S^\infty/(\mathbb{Z}/2) = \mathbb{R}\mathrm{P}^\infty$. Thus $B(\mathbb{Z}/2) = \mathbb{R}\mathrm{P}^\infty$, so $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$, matching the computation of example 5.5.6.

3. For $G = U(1) = S^1$ take $EG = S^\infty \subseteq \mathbb{C}^\infty$, the unit sphere, with $U(1)$ acting by scalar multiplication $v \cdot \lambda = \lambda v$ for $|\lambda| = 1$. Again S^∞ is contractible, the action is free (no unit vector is fixed by a nontrivial phase), and the orbit space is $S^\infty/U(1) = \mathbb{C}\mathrm{P}^\infty$. Thus $BU(1) = \mathbb{C}\mathrm{P}^\infty$. Here $U(1)$ is not discrete, so $\mathbb{C}\mathrm{P}^\infty$ is not a $K(U(1), 1)$; instead $\pi_i(\mathbb{C}\mathrm{P}^\infty) \cong \pi_{i-1}(U(1)) = \pi_{i-1}(S^1)$, which is $\mathbb{Z}$ for $i = 2$ and 0 otherwise, exhibiting $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z}, 2)$ as predicted by the delooping relation $\pi_i(BG) \cong \pi_{i-1}(G)$.

These three computations display the two faces of the construction in one frame. For the discrete groups $\mathbb{Z}$ and $\mathbb{Z}/2$, the universal bundle is a covering and BG is an aspherical $K(G, 1)$; the higher homotopy of BG vanishes because the cover EG is contractible. For the circle $U(1)$, the same recipe with the same total space S^∞ produces $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z}, 2)$, where the homotopy has been pushed up one degree relative to $U(1) = S^1 = K(\mathbb{Z}, 1)$. The single relation $\pi_i(BG) \cong \pi_{i-1}(G)$ accounts for both: applying B to a $K(A, n)$ yields a $K(A, n + 1)$, so B shifts Eilenberg-MacLane spaces up the dimension ladder, with $B\mathbb{Z} = S^1 = K(\mathbb{Z}, 1)$ and $BU(1) = BK(\mathbb{Z}, 1) = K(\mathbb{Z}, 2) = \mathbb{C}\mathrm{P}^\infty$.

The classification theorem is the structural endpoint of the section. It states that the universal bundle deserves its name: every principal G-bundle over a CW base is pulled back from $EG \to BG$ along a map to BG, and two bundles are isomorphic exactly when their classifying maps are homotopic. The proof rests on two homotopy properties of pullbacks, isolated in the lemma below, together with the weak contractibility of EG.

Lemma 2.4.6: Homotopy Invariance of Pullback Bundles

Let G be a topological group and $p\colon E \to B$ a principal G-bundle. For a map $f\colon X \to B$ the pullback
$$f^*E \;=\; \{(x, e) \in X \times E : f(x) = p(e)\}$$
with the projection to X and the diagonal G-action $(x, e) \cdot g = (x, e \cdot g)$ is a principal G-bundle over X. If X is paracompact (in particular a CW complex) and $f_0 \simeq f_1\colon X \to B$ are homotopic, then $f_0^*E \cong f_1^*E$ as principal G-bundles over X.

***Proof*:**

Step 1: The pullback is a principal bundle. Local triviality transports along f: if $\varphi_U : p^{-1}(U) \cong U \times G$ trivializes E over $U \subseteq B$, then over $f^{-1}(U)$ the map $(x, e) \mapsto (x, \mathrm{pr}_G\varphi_U(e))$ is an equivariant homeomorphism $(f^*E)|_{f^{-1}(U)} \cong f^{-1}(U) \times G$ over $f^{-1}(U)$. The diagonal G-action is free and fiberwise transitive because the E-coordinate already is. Hence $f^*E \to X$ is a principal G-bundle.

Step 2: Reduction to a bundle over $X \times I$. Let $H\colon X \times I \to B$ be a homotopy from f_0 to f_1 and form the principal G-bundle $W = H^*E$ over $X \times I$. Its restrictions to $X \times \{0\}$ and $X \times \{1\}$ are f_0^*E and f_1^*E. It suffices to show that for a principal G-bundle W over $X \times I$ with X paracompact, the restrictions $W|_{X\times\{0\}}$ and $W|_{X\times\{1\}}$ are isomorphic; this is the statement that a bundle over $X \times I$ is determined by its restriction to either end.

Step 3: A bundle over $X \times I$ is a pullback from X. The key fact is that a principal G-bundle $W \to X \times I$ is isomorphic to the pullback $\pi^*(W|_{X\times\{0\}})$ along the projection $\pi\colon X \times I \to X \times \{0\} \cong X$. To see this, use that W admits a trivialization over sets of the form $U \times I$ for a suitable open cover $\{U\}$ of X. Indeed, fixing $x \in X$, compactness of I gives a finite partition $0 = t_0 < \cdots < t_k = 1$ and opens $U_x \ni x$ over each of which W is trivial on $U_x \times [t_{j-1}, t_j]$; the trivializations over consecutive slabs differ by transition functions into G, and these can be straightened, one slab at a time, into a single trivialization over $U_x \times I$ (each adjustment composes the current trivialization with the transition function read at the height t_j, extended constantly in the I-direction). Paracompactness of X then gives a locally finite refinement and a partition of unity subordinate to $\{U_x\}$, which patches the slab-wise trivializations over $U_x \times I$ into an isomorphism $W \cong \pi^*(W|_{X\times\{0\}})$ globally; the partition of unity interpolates the local equivariant isomorphisms exactly as in the proof that a fiber bundle over a paracompact contractible base is trivial. A complete account of the patching is in [14], and in [21, Homotopy Invariant Vector Bundle] for the vector-bundle form.

Step 4: Conclusion. Restricting the isomorphism $W \cong \pi^*(W|_{X\times\{0\}})$ to $X\times\{1\}$ gives $W|_{X\times\{1\}} \cong (\pi^*(W|_{X\times\{0\}}))|_{X\times\{1\}} = W|_{X\times\{0\}}$, since π restricted to $X \times \{1\}$ is again the identification with X. Applied to $W = H^*E$ this reads $f_1^*E \cong f_0^*E$, as we sought to show.

The lemma is what the whole theory uses: it converts the rigid notion of bundle isomorphism into the flexible notion of homotopy of maps. Once a bundle is known to depend on its classifying map only up to homotopy, classifying all bundles reduces to computing a set of homotopy classes. The remaining content of the classification theorem is that the classifying map exists for every bundle (surjectivity onto $[X, BG]$) and is unique up to homotopy (injectivity), and both come from the weak contractibility of EG.

Theorem 2.4.7: Classification of Principal G-Bundles

Let G be a topological group and let X be a CW complex (or any paracompact space of CW homotopy type). Pulling back the universal bundle along a map gives a bijection

$$[X, BG] \xrightarrow{\cong} \mathrm{Prin}_G(X), \qquad [f] \longmapsto f^*EG$$

between the set of homotopy classes of maps $X \to BG$ and the set of isomorphism classes of principal G-bundles over X. The bijection is natural in X: for a map $h\colon X' \to X$ the square relating $[-, BG]$ and $\mathrm{Prin}_G(-)$ commutes, both vertical maps being induced by pulling back along h.

***Proof*:**
Write $p\colon EG \to BG$ for the universal bundle of proposition 2.4.4, with EG weakly contractible, and take EG, BG to be CW complexes (replace them by CW models via CW approximation, theorem 5.3.9, which does not change homotopy classes of maps into BG or the bundle theory, since every principal bundle over a CW base is a fibration and pullbacks are preserved). The assignment $[f] \mapsto f^*EG$ is well defined by the homotopy invariance of lemma 2.4.6, and natural in X because pullback is functorial: $h^*(f^*EG) = (f \circ h)^*EG$. It remains to prove the map is surjective and injective.

Step 1: Surjectivity, constructing a classifying map. Let $q\colon P \to X$ be a principal G-bundle. A map $f\colon X \to BG$ with $f^*EG \cong P$ is the same datum as a G-equivariant map $\tilde{f}\colon P \to EG$ covering it: given $\tilde{f}$, the induced f on orbit spaces satisfies $f^*EG \cong P$ via $e \mapsto (q(e), \tilde{f}(e))$, and conversely an isomorphism $P \cong f^*EG$ composed with the projection $f^*EG \to EG$ yields $\tilde{f}$. So it suffices to build a G-equivariant map $\tilde{f}\colon P \to EG$. This is an extension problem solved skeleton by skeleton, using that EG is weakly contractible.

Build $\tilde{f}$ over the skeleta of X pulled up to P. Trivialize P over the (closed) cells: since each cell e^n_α of X is contractible, $P|_{\overline{e^n_\alpha}}$ is a trivial bundle $\overline{e^n_\alpha} \times G$, by lemma 2.4.6 applied to the contraction of the cell. Define $\tilde{f}$ over the 0-skeleton by sending each point of $P|_{X^0}$ to a chosen point of EG equivariantly (pick the image of the identity section on each trivialized fiber, then extend by equivariance). Suppose $\tilde{f}$ is defined and equivariant over $P|_{X^{n-1}}$. For an n-cell e^n_α with characteristic map $\Phi_\alpha \colon D^n \to X$, the restriction of P to D^n is trivial, $P|_{D^n} \cong D^n \times G$, so an equivariant extension over $P|_{e^n_\alpha}$ is the same as a map $D^n \to EG$ extending the given map on $\partial D^n = S^{n-1}$ (read off from $\tilde{f}$ on X^{n-1} through the trivialization). The obstruction to extending a map $S^{n-1} \to EG$ over D^n is its class in $\pi_{n-1}(EG)$, which vanishes because EG is weakly contractible. Hence the extension exists; performing it on every n-cell and inducting up the skeleta produces an equivariant $\tilde{f} \colon P \to EG$, and the induced $f \colon X \to BG$ satisfies $f^*EG \cong P$. Thus the map $[X, BG] \to \mathrm{Prin}_G(X)$ is surjective.

Step 2: Injectivity. Suppose $f_0, f_1 \colon X \to BG$ satisfy $f_0^*EG \cong f_1^*EG =: P$. Choose equivariant maps $\tilde{f}_0, \tilde{f}_1 \colon P \to EG$ covering f_0, f_1 (their existence is Step 1 read backwards: an isomorphism $P \cong f_i^*EG$ followed by projection gives $\tilde{f}_i$). A homotopy $f_0 \simeq f_1$ is produced by an equivariant homotopy $\tilde{f}_0 \simeq \tilde{f}_1 \colon P \to EG$, since such a homotopy descends to orbit spaces and pulls EG back to the homotopy realizing $f_0 \simeq f_1$. Build the equivariant homotopy $F \colon P \times I \to EG$ by the same skeletal extension as in Step 1, now over the CW complex $X \times I$ with the bundle $P \times I \to X \times I$: F is prescribed on $P \times \{0\}$ and $P \times \{1\}$ by $\tilde{f}_0$ and $\tilde{f}_1$, and the obstruction to extending over an n-cell of $X \times I$ (rel the two ends) again lands in $\pi_{n-1}(EG) = 0$. Hence F exists, $\tilde{f}_0 \simeq \tilde{f}_1$ equivariantly, and $f_0 \simeq f_1$. Therefore the map is injective.

Combining surjectivity and injectivity, $[X, BG] \to \mathrm{Prin}_G(X)$ is a natural bijection, completing the proof.

The theorem realizes the slogan that BG "classifies" principal G-bundles: the entire isomorphism theory of such bundles over a CW complex X is encoded in the homotopy-theoretic invariant $[X, BG]$. The universal bundle $EG \to BG$ is the universal example, the bundle from which all others descend, and the classifying map f records which bundle by recording how X maps into BG. The proof is an obstruction argument of the kind that recurs throughout homotopy theory: the only thing one ever needs of EG is that its homotopy groups vanish, which makes every extension over a cell unobstructed and every two extensions homotopic. Naturality means $\mathrm{Prin}_G(-)$ and $[-, BG]$ are isomorphic as functors on the homotopy category of CW complexes, so BG represents the functor "isomorphism classes of principal G-bundles" in the precise sense of representable functors.

Two structural consequences follow at once. The classification pins down BG up to homotopy, since a representing object of a functor is unique up to homotopy equivalence: any two universal bundles for G have homotopy-equivalent bases. And it identifies the homotopy type of G with the based loop space of BG.

Corollary 2.4.8: Uniqueness of BG and the Delooping

Let G be a topological group.

1. The classifying space BG is well defined up to homotopy equivalence: if $EG \to BG$ and $E'G \to B'G$ are two universal bundles for G, then $BG \simeq B'G$.
2. There is a weak homotopy equivalence $\Omega BG \simeq G$, where Ω denotes the based loop space. In particular $\pi_i(BG) \cong \pi_{i-1}(G)$ for all $i \geq 1$.

***Proof*:**
Part 1. By theorem 2.4.7 both BG and $B'G$ represent the functor $\mathrm{Prin}_G(-)$ on CW complexes. The universal bundle $E'G \to B'G$ is classified by a map $u\colon B'G \to BG$ with $u^*EG \cong E'G$, and the universal bundle $EG \to BG$ is classified by $v\colon BG \to B'G$ with $v^*E'G \cong EG$. Then $(v \circ u)^*E'G \cong E'G$ (the universal bundle over the target $B'G$ is $E'G$), so by the injectivity in theorem 2.4.7 (applied with $X = B'G$ and the identity-classified bundle) $v \circ u \simeq \mathrm{id}_{B'G}$, and symmetrically $u \circ v \simeq \mathrm{id}_{BG}$. Hence u is a homotopy equivalence $B'G \simeq BG$.

Part 2. Apply the long exact sequence of homotopy groups of the bundle $G \to EG \to BG$ (a fibration, since a principal bundle is a fiber bundle and fiber bundles are fibrations as in chapter 5). Weak contractibility $\pi_i(EG) = 0$ collapses the sequence to isomorphisms $\partial\colon \pi_i(BG) \xrightarrow{\cong} \pi_{i-1}(G)$ for every $i \geq 1$. The based path-loop fibration $\Omega BG \to PBG \to BG$ over BG has contractible total space PBG, so its long exact sequence gives $\pi_{i-1}(\Omega BG) \cong \pi_i(BG)$. Composing, $\pi_{i-1}(\Omega BG) \cong \pi_{i-1}(G)$ for all $i \geq 1$, and these isomorphisms are induced by a comparison map $G \to \Omega BG$ (the connecting map of the bundle realized as a loop), which is therefore a weak homotopy equivalence $G \simeq \Omega BG$, as we sought to show.

The delooping relation $\Omega BG \simeq G$ is the reason B is called a delooping functor: it produces a space BG whose loop space recovers G, undoing the loop operation Ω. Read on homotopy groups it says B shifts the homotopy of G up by one degree, which is exactly the behaviour seen in example 2.4.5, where B sent $K(\mathbb{Z}, 1) = S^1$ to $K(\mathbb{Z}, 2) = \mathbb{C}\mathrm{P}^\infty$. For a discrete group G the relation specializes to $\pi_1(BG) \cong \pi_0(G) = G$ with all higher groups zero, recovering $BG = K(G, 1)$; this is the bridge to the Eilenberg-MacLane theory of chapter 5, where the uniqueness clause of theorem 5.5.4 identifies the bundle-theoretic BG with the homotopy-theoretic $K(G, 1)$, since both are aspherical with fundamental group G.

The functoriality of B deserves a word, since it is what makes BG a construction and not just a single space. A continuous homomorphism $\varphi\colon G \to H$ of topological groups induces a map $B\varphi\colon BG \to BH$, well defined up to homotopy: the bundle $EG \times_G H \to BG$, formed by extending the structure group along φ, is a principal H-bundle over BG, hence classified by a map $BG \to BH$, and this assignment respects composition and identities up to homotopy. Thus B is a functor from topological groups to spaces (up to homotopy), and the classification $\mathrm{Prin}_G(X) \cong [X, BG]$ is natural in G as well as in X.

The most consequential application outside the discrete case is to the matrix groups, where BG classifies vector bundles. A rank-n complex vector bundle has an associated principal $U(n)$-bundle of unitary frames, and the two notions of bundle are equivalent: choosing a Hermitian metric reduces the structure group to $U(n)$, and the frame bundle determines the vector bundle by the associated-bundle construction. The classification theorem then identifies complex rank-n bundles over X with $[X, BU(n)]$, and passing to the colimit over n gives the classifying space $BU = \varinjlim_n BU(n)$ for stable complex vector bundles, the representing space for reduced complex K-theory. The orthogonal analogue $BO = \varinjlim_n BO(n)$ classifies real vector bundles. These identifications, and the consequent description of characteristic classes as the cohomology of BU and BO, are the organizing principle of the companion volume [21].

Example 2.4.9: $BU(n)$ as a Grassmannian and Line Bundles
The classifying space $BU(n)$ is the infinite complex Grassmannian $\mathrm{Gr}_n(\mathbb{C}^\infty) = \varinjlim_N \mathrm{Gr}_n(\mathbb{C}^N)$ of n-planes in $\mathbb{C}^\infty$. Its universal bundle is the principal $U(n)$-bundle of unitary n-frames $V_n(\mathbb{C}^\infty) \to \mathrm{Gr}_n(\mathbb{C}^\infty)$, sending a frame to the plane it spans; the total space $V_n(\mathbb{C}^\infty)$, the infinite Stiefel manifold, is weakly contractible by the sliding argument that contracts S^∞, so this is a universal

bundle in the sense of definition 2.4.3. For $n = 1$ the group is $U(1)$ and the Grassmannian is $\mathrm{Gr}_1(\mathbb{C}^\infty) = \mathbb{C}\mathrm{P}^\infty$, recovering $BU(1) = \mathbb{C}\mathrm{P}^\infty$ from example 2.4.5. The classification theorem then reads

$$\mathrm{Vect}^1_{\mathbb{C}}(X) \;\cong\; [X, \mathbb{C}\mathrm{P}^\infty] \;\cong\; \tilde{H}^2(X; \mathbb{Z})$$

where the first bijection is theorem 2.4.7 for $G = U(1)$ (complex line bundles are principal $U(1)$-bundles) and the second is the representability of degree-two cohomology by $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z}, 2)$, the Eilenberg-MacLane representation of chapter 5 (theorem 5.5.9). The composite sends a line bundle to its first Chern class. Thus complex line bundles over X are classified by a single cohomology class, the topological counterpart of the Picard group from algebraic geometry.

The example returns to where the section began. The bundle-classifying space $BU(1) = \mathbb{C}\mathrm{P}^\infty$ is simultaneously the cohomology-representing space $K(\mathbb{Z}, 2)$, so the geometric classification of theorem 2.4.7 and the cohomological representation of chapter 5 agree on the nose for line bundles, and their agreement is the statement that the first Chern class is a complete invariant. The same coincidence at the level of the discrete groups identifies $H^*(BG)$ for a discrete G with the cohomology of $K(G, 1)$, which is by definition the group cohomology of G.

Proposition 2.4.10: Cohomology of BG is Group Cohomology

Let G be a discrete group and M an abelian group with trivial G-action. Then there is a natural isomorphism

$$H^*(BG; M) \;\cong\; H^*(G; M)$$

between the singular cohomology of the classifying space $BG = K(G, 1)$ and the group cohomology of G with coefficients in M. The corresponding statement holds for homology, $H_*(BG; M) \cong H_*(G; M)$.

***Proof*:**
By proposition 2.4.4 the space EG is a contractible CW complex with a free cellular G-action, so its cellular chain complex $C_*(EG)$ is a complex of free $\mathbb{Z}[G]$-modules (free because G permutes the cells freely, so each $C_n(EG)$ is free on the G-orbits of n-cells). Contractibility of EG makes $C_*(EG)$ acyclic: $H_n(C_*(EG)) = H_n(EG) = 0$ for $n > 0$ and $H_0(EG) = \mathbb{Z}$. Hence $C_*(EG) \to \mathbb{Z}$ is a free resolution of the trivial $\mathbb{Z}[G]$-module $\mathbb{Z}$.

The orbit space $BG = EG/G$ has cellular chains $C_*(BG) = C_*(EG) \otimes_{\mathbb{Z}[G]} \mathbb{Z}$, since the cells of BG are the G-orbits of cells of EG. Therefore

$$H_*(BG; M) = H_*\big(C_*(EG) \otimes_{\mathbb{Z}[G]} M\big) = \mathrm{Tor}_*^{\mathbb{Z}[G]}(\mathbb{Z}, M) = H_*(G; M)$$

the middle equality because $C_*(EG)$ is a free resolution of $\mathbb{Z}$ and Tor is computed from any such resolution (the definition of group homology, [8]). Dually,

$$H^*(BG; M) = H^*\big(\mathrm{Hom}_{\mathbb{Z}[G]}(C_*(EG), M)\big) = \mathrm{Ext}^*_{\mathbb{Z}[G]}(\mathbb{Z}, M) = H^*(G; M)$$

using the same resolution to compute Ext, which is the definition of group cohomology ([8]). Naturality in G follows because a homomorphism $G \to G'$ induces a G-equivariant map $EG \to EG'$ (both total spaces being weakly contractible free objects, the map exists and is unique up to equivariant homotopy by the argument of theorem 2.4.7), and this map realizes the induced map on resolutions, completing the proof.

The proposition is the precise sense in which topology computes an algebraic invariant. Group cohomology $H^*(G;M)$ is defined in homological algebra as $\mathrm{Ext}^*_{\mathbb{Z}[G]}(\mathbb{Z},M)$, computed from any free resolution of $\mathbb{Z}$ over the group ring; the classifying space provides a geometric free resolution, the cellular chains of the contractible cover EG, and so the singular cohomology of the space $BG = K(G,1)$ equals the group cohomology of G. This is what makes Eilenberg-MacLane spaces a computational tool: questions about $H^*(G)$ become questions about the topology of a space, and conversely the cohomology of an aspherical space is read off from its fundamental group. The full development of group cohomology, including the bar resolution and the coefficient theory, is the subject of [8].

Example 2.4.11: Cohomology of $B(\mathbb{Z}/2)$ and $B\mathbb{Z}$

1. For $G = \mathbb{Z}/2$ the classifying space is $B(\mathbb{Z}/2) = \mathbb{R}\mathrm{P}^\infty$ by example 2.4.5. Its cohomology with $\mathbb{F}_2 = \mathbb{Z}/2$ coefficients is the polynomial ring on a single degree-one generator,

$$H^*(\mathbb{R}\mathrm{P}^\infty;\mathbb{F}_2) \;\cong\; \mathbb{F}_2[w], \qquad |w| = 1$$

computed from the cellular chain complex of $\mathbb{R}\mathrm{P}^\infty$ (one cell in each dimension, boundary maps alternately 0 and 2, which become 0 after reducing mod 2). By proposition 2.4.10 this equals the group cohomology $H^*(\mathbb{Z}/2;\mathbb{F}_2) \cong \mathbb{F}_2[w]$, the standard computation in [8]. The generator w is the first Stiefel-Whitney class of the tautological line bundle, tying the group-cohomology computation to characteristic classes in [21].

2. For $G = \mathbb{Z}$ the classifying space is $B\mathbb{Z} = S^1$. Its integral cohomology is $H^0(S^1;\mathbb{Z}) = H^1(S^1;\mathbb{Z}) = \mathbb{Z}$ and $H^i(S^1;\mathbb{Z}) = 0$ for $i \geq 2$, matching the group cohomology of $\mathbb{Z}$: a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\mathbb{Z}] = \mathbb{Z}[t,t^{-1}]$ is the two-term complex $0 \to \mathbb{Z}[t,t^{-1}] \xrightarrow{t-1} \mathbb{Z}[t,t^{-1}] \to \mathbb{Z} \to 0$, so $H^*(\mathbb{Z};\mathbb{Z})$ is concentrated in degrees 0 and 1, each $\cong \mathbb{Z}$. The vanishing above degree one reflects that S^1 is one-dimensional, consistent with $\mathbb{Z}$ having cohomological dimension one.

Exercise 2.4.1

1. Let G be a topological group. Prove that a principal G-bundle $p\colon E \to X$ is trivial (isomorphic to $X \times G$ as a principal bundle) if and only if it admits a continuous section $s\colon X \to E$ with $p \circ s = \mathrm{id}_X$. Deduce that the principal $\mathbb{Z}$-bundle $\mathbb{R} \to S^1$ of example 2.4.2 is nontrivial.

2. Using the classification theorem (theorem 2.4.7) and the computation $B\mathbb{Z}/2 = \mathbb{R}\mathrm{P}^\infty$, determine the set $\mathrm{Prin}_{\mathbb{Z}/2}(S^1)$ of isomorphism classes of principal $\mathbb{Z}/2$-bundles over the circle, and identify the two bundles geometrically.

3. Let G be a path-connected topological group with BG its classifying space. Show that the principal G-bundles over the sphere S^n (for $n \geq 1$) are classified by the homotopy group $\pi_{n-1}(G)$, and as a check recover $\mathrm{Prin}_{U(1)}(S^2) \cong \mathbb{Z}$. (The hypothesis that G is path-connected is needed: it forces BG to be simply connected, so that the free homotopy set $[S^n, BG]$ agrees with the based set $\pi_n(BG)$. Without it the count is instead the orbit set of the $\pi_0(G)$-action, as the case $G = \mathbb{Z}/2$, $n = 1$ of exercise 2.4.1 (2) shows.)

4. Prove that for topological groups G and H there is a homotopy equivalence $B(G \times H) \simeq BG \times BH$. Use it to identify the classifying space of the torus $(S^1)^n$ and of $(\mathbb{Z}/2)^n$.

5. Let G be a discrete group. Using proposition 2.4.10, prove that G has cohomological dimension $\leq d$ (every group cohomology $H^k(G; M)$ vanishes for $k > d$ and all M) if and only if there is a $K(G, 1)$ that is a CW complex of dimension $\leq d$ realizable with the property; deduce that a free group has cohomological dimension 1.

3

Homology

When working with the fundamental group $\pi_1(X)$, we must rigidly keep track of a basepoint. The composition of two loops is defined by tracing one path and then the other, a process anchored to their common starting point. This geometric restriction has an algebraic consequence: path concatenation is generally non-commutative, rendering $\pi_1(X)$ a non-abelian group[1]. As we have seen, this non-abelian nature makes computations and group classifications notoriously difficult.

Homology theory offers a shift in perspective by abandoning the basepoint. By stripping away this anchor, we move from studying pointed "loops" to studying unanchored *cycles*. Algebraically, instead of concatenating paths, we will consider formal sums of geometric pieces. To see how this dramatically simplifies the algebra, consider two vertices x and y, with two directed edges a and b traveling between them:

$$\begin{tikzcd} x \arrow[r, bend left, "a"] & y \arrow[l, bend left, "b"] \end{tikzcd}$$

In homotopy, the concatenated loop $a \cdot b$ (based at x) and the loop $b \cdot a$ (based at y) are fundamentally distinct objects living in different groups. In homology, however, we no longer think of traveling along paths in sequence. Instead, we treat a and b as independent geometric building blocks and take their formal commutative sum: $a + b = b + a$. This shift immediately "abelianizes" our theory. All homology groups are abelian, unlocking the highly computable machinery of homological algebra.

Without basepoints and continuous functions as our primary tools, we need a new mechanism to generate our algebraic invariants. Homology achieves this by focusing on the relationship between *cycles* (closed shapes with no boundary, like a circle or a hollow sphere) and *boundaries* (the outer edges of higher-dimensional solid shapes). The entire philosophy of homology can be boiled down to measuring to what extent cycles fail to be boundaries. A cycle represents a "hole" in our space if and only if it is *not* the boundary of some higher-dimensional piece of the space.

To formalize this, our strategy in this chapter will be split into two distinct approaches:

[1] Every finitely generated group can be represented as $\pi_1(BG)$

1. *Simplicial Homology:* We will first introduce a rigid, combinatorial structure known as a Δ-complex (a generalization of simplicial complexes). By building spaces out of discrete n-dimensional triangles (simplices) glued along their faces, we can reduce the topological concepts of "cycles" and "boundaries" to finite matrices and straightforward linear algebra. This will provide us with a powerful tool for explicit computations.

2. *Singular Homology:* While simplicial homology is highly computable, it relies entirely on a specific geometric construction. To guarantee that our invariants depend only on the topological space X itself, we will introduce singular homology. This theory allows continuous maps of simplices directly into X, yielding a theory that is beautifully functorial and theoretically robust, albeit entirely uncomputable by direct definition.

We will conclude the chapter by combining the strengths of both approaches, proving a profound theorem: for spaces that can be triangulated, the singular and simplicial homology groups are isomorphic. This will bridge the gap between pure topological theory and concrete algebraic computation.

(Hatcher gives a fun little historical footnote that the homology used to be numbers, the Betti number and the torsion coefficients, and when Poincaré wrote about homology in the beginning of the 20th century that is what it was. The transition to the modern viewpoint with groups happened gradually for $\tilde{4}0$ years, until essentially solidified in 1952 by Eilenberg and Steenrod, p.131 of Hatcher)

(also see this link here)

3.1 Δ-Complex

In this section, we define the structure we put on our space to define their homology, starting with the building block:

Definition 3.1.1: n-Simplex

A *n-simplex* is the smallest convex set in $\mathbb{R}^m$ containing $n+1$ points $v_0, v_1, ..., v_n$ that do not lie in a hyperplane of dimension less than n along with an ordering of the vertices. Equivalently, $v_0, v_1 ..., v_n$ are $\mathbb{R}$-linearly independent and a tuple form a basis for a vector-space. The points v_i are the vertices of the simplex. If it is important to represent the vertices and their order, we denote it as $(v_0, ..., v_n)$ (called the *barycentric coordinates*).

Note that the ordering is part of the definition, which is why the vertices are in an $(n+1)$-tuple notation; this is to orient the edges. The idea behind the ordering is due to the fact that rotations only produce "half" of the possible rigid-body isomorphisms of a shape, one must reflect to get the other half. In linear algebra, this is represented as the determinant being positive or negative. This shall be important for when we are integrating and must choose a direction to integrate, and for manifolds when we must determine the Stokes orientation of the boundary. We shall also see that there shall be surfaces with no orientation such as the Klein bottle, and these shall be created by gluing edges at opposite orientations. For example, two 3-simplices can be glued among a common edge, and then the other edges could be glued in such a way to produce a torus T^2, a real projective plan $\mathbb{RP}^2$, or a Klein bottle:

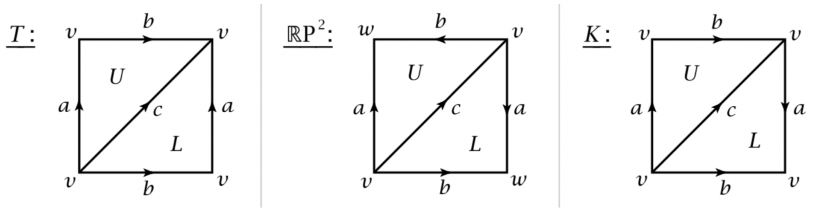


Figure 3.1: Different ways to glue a square

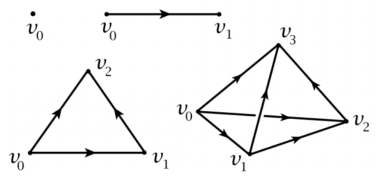
If we delete a vertex from an n-simplex, we get an $(n-1)$-simplex called the *face* of the simplex. if $(v_0, v_1, ..., v_n)$ is an n-complex, we shall represent a face via

$$(v_0, v_1, ..., \hat{v}_i, ..., v_n)$$

where $\hat{v}_i$ indicates the vertex is removed. By convention, the vertices of a face (or any sub-simplex) will always be ordered according to their ordering of the larger simplex.

Example 3.1.2: n-simplex

1. Here is the visual of a $0, 1, 2, 3$ simplex:



2. The standard n-simplex is defined as:

$$\Delta^n = \left\{ (t_0, t_1, ..., t_n) \in \mathbb{R}^{n+1} \;\middle|\; \sum_i t_i = 1,\ t_i \geq 0,\ 0 \leq i \leq n \right\}$$

whose vertices must be, by definition, unit vectors along the coordiante axes. For example, Δ^n looks like:

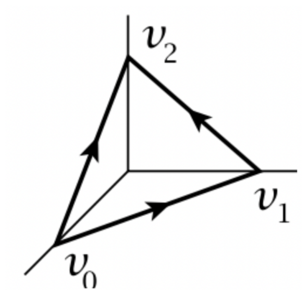


Figure 3.2: Δ^2

Definition 3.1.3: Boundary

Let Δ^n be the standard simplex. Then the union of all the faces of Δ^n is called the *boundary* of Δ^n and is written $\partial\Delta^n$. The *open simplex* $\text{int}\Delta^n$ is the interior of Δ^n

Definition 3.1.4: Δ-Complex

Let X be a topological space. Then a Δ-*complex structure*[a] on X is a collection of continuous maps $\sigma_\alpha : \Delta^{n_\alpha} \to X$ with n_α depending on the index α such that

1. The restriction $\sigma_\alpha|_{\text{int}\Delta^{n_\alpha}}$ is injective, and each point $x \in X$ is in the image of exactly one restriction $\sigma_\alpha|_{\text{int}\Delta^{n_\alpha}}$. We label $\sigma_\alpha|_{\text{int}\Delta^{n_\alpha}}$ as e^n_α.
2. Each restriction σ_α to a face Δ^n is one of the maps $\sigma_\beta : \Delta^{n_\alpha - 1} \to X$.
3. A set $A \subseteq X$ is open if and only if $\sigma_\alpha^{-1}(A)$ is open in Δ^n for each σ_α (the topology of X the initial topology given by the maps σ_α)

[a] Not to be confused with a *Simplicial-complex structure*, see [14, p.107]

Figure 3.1 of (a covering space of) a torus, $\mathbb{R}\text{P}^1$, and a Klein bottle are examples of Δ-complexes; the torus and the Klein bottle each have six maps (one 0-simplex, three 1-simplices, two 2-simplices, each labeled on the diagram), while $\mathbb{R}\text{P}^2$ has two 0-simplices (v, w in the covering space). We see here the importance of the orientation of the edges. We see that X can be represented of disjoint simpices $\Delta^{n_\alpha}_\alpha$ with $\Delta^{n_\alpha - 1}_\beta$ corresponding to the restriction of σ_β of σ_α to the particular face. Hence, any Δ-complexes can be described "combinatorially" as collections of n-simplices together with functions associating each face of an n-simplex to an $(n-1)$-simplex (similar to a cell-complex). In fact, though we will not verify this, the 1st condition insures that Δ-complexes can be thought of as CW complexes (note that a Δ-complex is Hausdorff).

Now, let X a topological space with a Δ-complex. What information can we tell about X given the Δ-complex? The thing we may keep track of is the number of simplices in each dimension:

Definition 3.1.5: Group Of n-Chains

Let X be a Δ-complex. Then $\Delta_n(X)$ is the free abelian group generated by the open n-simplices e^n_α of X. Elements of $\Delta_n(X)$ are called *n-chains*, and can be written as :

$$\sum_\alpha n_\alpha e^n_\alpha$$

or similarly $\Sigma_\alpha n_\alpha \sigma_\alpha$ where $\sigma_\alpha : \Delta^n \to X$ is the characteristic map of e^n_α.

Examples of these groups shall soon be given. These groups generalize the notion of a path (or surface in higher dimension) without a basepoint. How would we get a loop (or closed surface)? This will come down to the boundary "canceling itself out". As the boundary is orientable, we should take a moment to examine how to think about the boundary.

Let $X = \Delta^n$ be the standard Δ-complex so that it can be written as $(v_0, v_1, ..., v_n)$. If $n = 1$, then we shall let:

$$\partial[v_0, v_1] = [v_1] - [v_0]$$

which can be thought similarly to direction of integration. Another way of looking at it is if we take the determinant of

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

placing the coordinates v_0 at $(1, 0)$ and v_1 and $(0, 1)$. If we swap them, then:

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

and hence the orientation is flipped. If $n = 2$, then

$$\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$$

We can visually imagine this as standing inside the triangle and looking at the edges that are left after removing a vertex: two of them go right to left, and one will go left to right. As a determinant, we get:

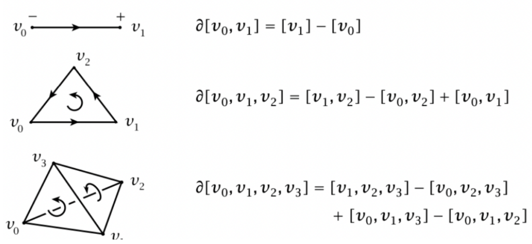
$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \cdot 1$$

We can also think of it as the orientation that is needed to make a *cycle*, in that the result shape is now the path that needs to be taken to form a closed loop; similarly, it is a consistent orientation of the boundary of the inside. Similarly, for $n = 3$, we do the same alternating sum:

$$\partial[v_0, v_1, v_2, v_3] = [v_1, v_2, v_3] - [v_0, v_2, v_3] + [v_0, v_1, v_3] - [v_0, v_1, v_2]$$

visually, we can imagine being inside the box and looking at the triangular faces. Then either the 3 vertices in order will be going clock-wise or anti-clockwise. The same determinant trick will again give us a similar result: if we flip any two positions, we would get a negative. We cannot directly have the same path intuition, but the equivalent in 3-dimension is that each face now has an outwards facing orientation so that all the faces have the same orientation and the interior of Δ^3 now has consistently oriented boundary.

If we consider the following visual may aid in visualizing:



More generally, we see the following emerging pattern:

$$\partial(v_0, v_1, ..., v_n) = \sum_i (-1)^i (v_0, ..., \hat{v}_i, ..., v_n)$$

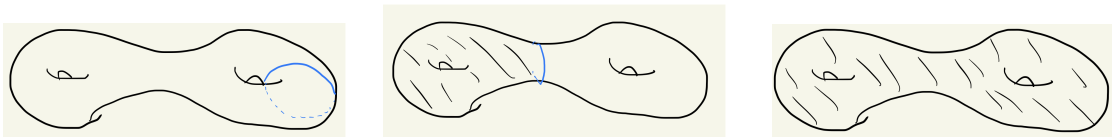
Taking the Δ-complex structure of a space X, we see that if the boundary is zero (for ex. if we have a closed loop), then we have a *cycle*. It can be easily verified that ∂ is a homomorphism, leading to the following important definition:

Definition 3.1.6: Boundary Homomorphism

Let X be a Δ-complex. then the *boundary homomorphism* $\partial_n : \Delta_n(X) \to \Delta_{n-1}(X)$ is defined by:

$$\partial_n(\sigma_\alpha) = \sum_i (-1)^i \ \sigma_\alpha|_{(v_0,\dots,\hat{v}_i,\dots,v_n)}$$

The boundary map may or may not be zero, and not all cycles of dimension lower are the boundary of a higher diensional cycle. Here is a visual that works for CW complexes (we shall see the homology on Δ-complexes is equivalent to CW complexes, it just a bit easier to picture CW complex borders):



(a) Cannot be in image of ∂_2 (b) In image of ∂_2 (c) Image in ∂_2 is Zero

Figure 3.3: Cycles and their Relations

We can then ask what happens if we keep taking boundary maps; will we get a sequence of small n-chains? Due to the orientation, we shall get that the boundary of the boundary must be trivial:

Lemma 3.1.7: Complex Condition

The composition

$$\Delta_n(X) \xrightarrow{\partial_n} \Delta_{n-1}(X) \xrightarrow{\partial_{n-1}} \Delta_{n-2}(X)$$

produces the zero map, $\partial_{n-1} \circ \partial_n = 0$

***Proof*:**
Plugging in, we get:

$$\begin{aligned}\partial_{n-1}(\partial_n(\sigma)) &= \sum_{j<i} (-1)^i(-1)^j \ \sigma|_{(v_0,\dots,\hat{v}_j,\dots,\hat{v}_i,\dots,v_n)} \\ &\quad + \sum_{j>i} (-1)^i(-1)^j \ \sigma|_{(v_0,\dots,\hat{v}_i,\dots,\hat{v}_j,\dots,v_n)}\end{aligned}$$

Then the two summation on the right hand side cancel after switching i, j since the sum becomes the negative of the first.

This algebraic property can certainly be studied independently of the geometric intuitions from which it started and leads to a plethora of interesting results in algebraic geometry and homological algebra. It is thus fruitful to give a purely algebraic definition of the phenomena happening in the above lemma:

Definition 3.1.8: Chain Complex and Exact

Given a chain of homomorphisms

$$\cdots \to C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \to \cdots \to C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

if $\partial_n \partial_{n+1} = 0$ for all n, then the above is called a *chain complex*, in other words:

$$\text{im}\partial_{n+1} \subseteq \ker \partial_n$$

If equality holds, then the sequence is called an *exact sequence*.

As we see, not all cycles are boundary of a chain of one dimension higher. The following measure to what degree cycles "fail" to be boundaries of higher chains:

Definition 3.1.9: nth [Simplicial] Homology Group

Given a chain complex, the *nth homology group* is defined to be

$$H_n = \frac{\ker(\partial_n)}{\text{im}\partial_{n+1}}$$

Elements of $\ker \partial_n$ are called *cycles*, elements of $\text{im}\partial_{n+1}$ are called *boundaries*, elements of H_n are called *homology classes*, and if two elements are in the same cosets, they are called *homologous* (meaning their difference is a boundary)
In the case where $C_n = \Delta_n(X)$, the homology group will be denoted $H_n^\Delta(X)$ and is called the *nth Simplicial homology group* of X

The homology group measures to what degree are cycles *not* the boundary of a higher dimensional chain. If all cycles are boundaries, then the homology is trivial, and if not the homology group is a way of quantifying the obstruction. Since most homology groups in this book will be finitely generated, we shall interpret the torsion-free and torsion part of an abelian group to interpret this obstruction.

Example 3.1.10: Simplicial Homology Group

For reference, here are Δ-complexes for the torus, $\mathbb{R}P^2$, and klein bottle:

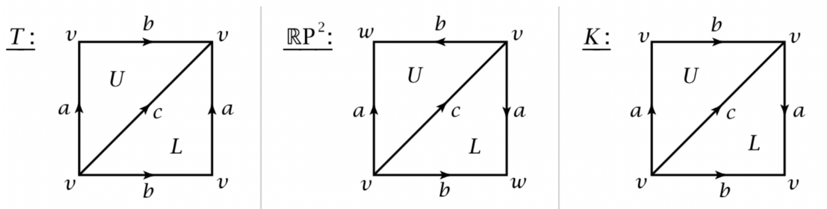


Figure 3.4: Reference Figure for Torus, Projective space, and Klein Bottle

1. Let $X = S^1$, with one vertex v an one edge e. Then $\Delta_0(S^1)$ and $\Delta_1(S^1)$ are both $\mathbb{Z}$. The map $\partial_1 : \Delta_1(S^1) \to \Delta_0(S^1)$ is $\partial_1 \equiv 0$ since
$$\partial e = v - v$$
Hence $\mathrm{im}\partial_1 = 0$. On the other hand, we always have $\ker \partial_0 = \mathbb{Z}$, and so $H_0^\Delta(S^1) = \mathbb{Z}$. Since $\mathrm{im}\partial_1 = 0$, $\ker \partial_1 = \mathbb{Z}$. Certainly for $n \geq 0$, $\Delta_n(S^1) = 0$ since there is no simplices in these dimensions. Hence:
$$H_n^\Delta(S^1) = \begin{cases} \mathbb{Z} & n \in \{0, 1\} \\ 0 & n \geq 2 \end{cases}$$
Notably, if the boundary maps are all zero, the homology groups of the complex are isomorphic to the chain groups $\Delta_n(X)$.

2. Let $X = T$ be the torus given the Δ-complex in figure 3.4, namely it has one vertex, three edges a, b, c, and two 2-simplices U, L, giving us $\Delta_0(T) = \mathbb{Z}$, $\Delta_1(T) = \mathbb{Z}^3$, and $\Delta_2(T) = \mathbb{Z}^2$. Like before, $\mathrm{im}\partial_1 = 0$, so we immediately have $H_0^\Delta(T) = \mathbb{Z}$. Next, $\ker \partial_1 = \mathbb{Z}^3$, and since $\partial_2 U = \partial_2 L = a + b - c$, writing $\Delta_1(T) = \ker \partial_1$ with basis $\{a, b, a + b - c\}$ we have:
$$H_1^\Delta(T) = \mathbb{Z} \oplus \mathbb{Z}$$
with basis $\{[a], [b]\}$. Since there is no 3-simplices, we have that $H_2\Delta(T) = \ker \partial_2$ which is given by $U - L$ since
$$\partial(pU + qL) = (p + q)(a + b - c) = 0 \qquad \iff \qquad p = -q$$
Hence, we overall have:
$$H_n^\Delta(T) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z} \oplus \mathbb{Z} & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

3. Let $X = \mathbb{R}\mathrm{P}^2$ with the Δ-complex strucutre given by figure 3.4, with two vertices v, w, three edges a, b, c and two 2-simpcles U, L. Then $\mathrm{im}\partial_1$ is generated by $w - v$, so
$$H_0^\Delta(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}$$
Since $\partial_2 U = -a + b + c$ and $\partial_2 L = a - b + c$, we get that ∂_2 is injective meaning it's kernel is zero and so:
$$H_2^\Delta(\mathbb{R}\mathrm{P}^2) = 0$$
Next, we see by staring at the image that $\ker \partial_1 = \mathbb{Z} \oplus \mathbb{Z}$ (find all paths that, under the coving map, are closed). By inspecting, we see it's basis is $a - b$ and c. The quotient $\ker \partial_1 / \mathrm{im}\partial_2$ is certainly a finite group since we are quotient two groups with the same rank. To find the quotient, take as bassi for $\ker \partial_1$ $(a - b + c, c)$. Then since $(a - b + c) + (-a + b + c) = 2c$, we see that the quotient must have order 2, thus:
$$H_1^\Delta(\mathbb{R}\mathrm{P}^2) = \mathbb{Z}/2\mathbb{Z}$$
Giving us:
$$H_n(\mathbb{R}\mathrm{P}^2) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z}/2\mathbb{Z} & n = 1 \\ 0 & n \geq 2 \end{cases}$$

The torsion element can be thought of as somehow representing some "non-orientability" or that we have found some "twists" in our complex. We shall calculate $H_k(\mathbb{R}\mathrm{P}^n)$ using Cellular homologies in section 3.5.

4. For a moment assume homeomorphisms preserve homology groups. Let $X = S^n$ with the Δ-complex structure on S^n given by taking two copies of Δ^n and identifying their boundaries via identity maps, giving us a space homeomorphic to S^n. Then to compute $H_n^{\Delta}(S^n)$, we see that $\ker \Delta_n$ is an infinite cyclic group generated by $U - L$ where U, L are the two n-simplices. Since the image is certainly trivial, we have:

$$H_n^{\Delta}(S^n) \cong \mathbb{Z}$$

Calculating explicitly the other dimension is rather difficult wihtout more advanced tools like the Snake lemma, however the fact that $H_n^{\Delta}(S^n) \cong \mathbb{Z}$ will be an important lemma. For now, the case of $n = 2$ can be studied. In particular, we have that any cycles on S^2 is clearly the boundary of of a chain one dimension higher. This should be made rigorous to check that:

$$H_n(S^2) = \begin{cases} \mathbb{Z} & n = 0 \\ 0 & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

5. (Klein Bottle)

As the simplicial homology groups are finitely generated, they are characterized by the free-rank and the torsion. Historically, these were given the following name:

Definition 3.1.11: Betti Number And Torsion Coefficients

The number of $\mathbb{Z}$ summand of $H_n(X)$ is called the nth Betti number of X, and the integers specifying the orders of the finite cyclic summands are called the *torsion coefficients*.

History, in the early development of homology theory in the 1900s, it was not the groups $H_n(X)$ that were studied but the *Betti numbers* and *torsion coefficients*. It wasn't until the 1920s that Emmy Noether brought about the more algebraic point of view.

3.2 Singular Homology

Coming back to the simplicial homology groups, there are natural questions that arise: is the Simplicial homology group independent of homeomorphism? Of homotopy equivalence? These questions are hard to answer with the given Δ-complexes. What we shall do is generalize to a new homology theory known as *singular homology groups*, which is easier to work theoretically, though harder to visualize, but can be defined on more general spaces, and in the end we shall show they produce the same groups as Simplicial homology groups.

Definition 3.2.1: Singular n-Simplex

Let X be a space. Then a *singular n-simplex* in a space is a continuous map $\sigma : \Delta^n \to X$, that is an element of

$$\sigma \in C_n(X) = \mathrm{Hom}_{\mathbf{Top}}(\Delta^n, X)$$

The term singular expresses how the map σ need only be continuous, there may be "singularities" where the image does not look like a simplex. This generality also means that each space X has many more singular n-simplices, often infinite dimensional, and the boundary riles on the domain of σ rather than a geometric intuition in the codomain. Similarly to before, let $C_n(X)$ be the free abelian group with basis being the set of all singular n-simplices in X, that is the elements of $\mathrm{Hom}_{\mathbf{Top}}(\Delta^n, X)$. Elements of $C_n(X)$ are called *[singular] n-chains*, and are finite formal sums $\sum_i n_i\sigma_i$. Just as before, we define a boundary map:

$$\partial_n(\sigma) = \sum_i (-1)^i \; \sigma|_{(v_0,\dots,\hat{v}_i,\dots,v_n)}$$

Naturally, there is implicitly an identification of $(v_0, ..., \hat{v}_i, ..., v_0)$ with ∂_{n-1} preserving ordering of vertices so that we can think of the map $\sigma|_{(v_0,\dots,\hat{v}_i,\dots,v_n)}$ as being from $\Delta^{n-1} \to X$ (i.e. of a singular $(n-1)$-simplex). When not ambiguous, we shall write the border map $\partial_n : C_n(X) \to C_{n-1}(X)$ simply as ∂. Then we may concisely write the condition $\partial_n\partial_{n+1} = 0$ as $\partial^2 = 0$, and again the image of ∂ is called the boundary, while everything within the kernel is called the cycle. Thus, we again have the following definition:

Definition 3.2.2: Singular Homology Group

Let X be a topological spaces with singular n-simplices. Then the nth *singular homology group* is:

$$H_n(X) = \frac{\ker \partial_n}{\mathrm{im}\partial_{n+1}}$$

Unlike Simplicial Homology groups, singular homology groups are preserved under homeomorphisms, for there is a natural bijection $\varphi : C_n(X) \to C_n(Y)$ $f \mapsto \varphi \circ f$ that works well with the boundary condition (namely it shall map the shapes of the right dimensions to themselves[2]). Note that $C_n(X)$ is usually uncountable and may seem hard to work with, but as we shall see this flexibility gives us many more easily provable results. In fact, the singular homology group can be identified with the simplicial homology group by a special construction on a space X (labeled $S(X)$), by taking for each singular n-simplex σ a n-simplex Δ^n_σ with Δ^n_σ attached in the obvious way to the $(n-1)$-simplices of $S(X)$ that are restrictions of σ to the appropriate faces. Then $H^\Delta_n(S(X)) = H_n(X)$ for all x, though $S(X)$ is usually huge.

In the case where X is a smooth manifold, a rough idea of an element of $H_n(X)$ is that it can be interpreted as the collection of all possible oriented sub-manifolds. The orientation information comes from the borer map which "preserves" orientation information. Then $H_1(X)$ are represented by oriented loops, elements in $H_2(X)$ are represented oriented surface into X, and so forth. The term represented here can be seen as saying that the domain when mapping into X can be "thought of" as a loop, surface e,tc. With some work, it can be shown that an oriented 1-cycle $\coprod_\alpha S^1\alpha \to X$ is zero in $H_1(X)$ if and only if it extends to a map of a compact oriented surface with boundary $\coprod_\alpha S^1_\alpha$

[2]This result shall be proven later, but it can also be seen through DeRham Theory if you have done some Differential Geometry

into X, and similarly for 2-cycles. It was hopped in the early days of homology theory that the close connection with boundary of manifolds would have continued to higher dimensions, but that was in fact not the case (ex. $\mathbb{RP}^3$ is not null-cobordant)! These notions would further be explored in a theory known as *cobordism*.

With these intuition in mind, we may look at some useful properties of singular homologies:

Proposition 3.2.3: Homology group and path-connected components

Corresponding to the decomposition of a space X into its path-components X_α,

$$H_n(X) \cong \bigoplus_\alpha H_n(X_\alpha)$$

***Proof*:**
This comes down to properties of continuous maps. Since singular simplex is always path-connected, the image is too, so $C_n(X)$ splits as the direct sum of subgroups $C_n(X_\alpha)$. Then the boundary maps ∂_n preserves direct sums of decompositions taking $C_n(X_\alpha)$ to $C_{n-1}(X_\alpha)$, hence $\ker \partial_n$ and $\mathrm{im}\partial_{n+1}$ split similarly as direct sums, hence the homology group splits, that is:

$$H_n(X) \cong \bigoplus_\alpha H_n(X_\alpha)$$

as we sought to show.

We shall later see that $H_n(\bigvee_\alpha X_\alpha) = \bigoplus_\alpha H_n(X_\alpha)$. We may wish that $H_n(X \times Y)$ also has an easy structure, however this becomes a good bit more complicated, and shall be addressed in section 4.3.

Proposition 3.2.4: 0th Singular Homology Group for Path-Connected Space

If X is nonempty and path-connected, then $H_0(X) \cong \mathbb{Z}$. Hence, for any space X, $H_0(X)$ is a direct sum of $\mathbb{Z}$ for each path component.

***Proof*:**
By definition, $H_0(X) = C_0(X)/\mathrm{im}\partial_1$ (since $\partial_0 = 0$). What we shall do is define the surjective homomorphism:

$$\epsilon : C_0(X) \to \mathbb{Z} \qquad \sum_i n_i\sigma_i \mapsto \sum_i n_i$$

and show that $\ker \epsilon = \mathrm{im}\partial_1$, giving us the structure of $H_0(X)$.

($\mathrm{im}\partial_1 \subseteq \ker \epsilon$) Note that each singular 1-simplex $\sigma : \Delta^1 \to X$ has

$$\epsilon(\partial_1(\sigma)) = \epsilon(\sigma|_{(v_1)} - \sigma|_{(v_0)}) = 1 - 1 = 0$$

hence, they are conained in the kernel.

($\mathrm{im}\partial_1 \supseteq \ker \epsilon$) Suppose $\epsilon\left(\sum_i n_i\sigma_i\right) = \sum_i n_i = 0$. If we can find an element in $C_1(X)$ for which $\sum_i n_i\sigma_i$ is the boundary, then we're done. Since the σ_i' are singular 0-simplices, their image is simply a point in X. For each i, choose a path $\tau_i : I \to X$ where $\tau_i(0) = x_0$ and $\tau_i(1) = \sigma_i(v_0)$, and let σ_0 be the singular 0-simplex with iamge x_0. Then τ_i can be seen as a singular 1-simplex, namely $\tau_1 : [v_0, v_1] \to X$, and

$$\partial \tau_i = \sigma_i - \sigma_0$$

Thus:

$$\partial \left(\sum_i n_i \tau_i \right) = \sum_i n_i \sigma_i - \sum_i n_i \sigma_0 = \sum_i n_i \sigma_i$$

But then we see that $\sum_i n_i \sigma_i$ is a boundary, giving $\ker \epsilon \subseteq \mathrm{im} \partial_1$, as we sought to show.

Proposition 3.2.5: Homology Group of a Point

If X is a point, then

$$H_0(X) \cong \mathbb{Z} \qquad H_n(X) = 0, \ \forall n > 0$$

***Proof*:**
There is a unique singular n-simplex σ_n for each n, and $\partial(\sigma_n) = \sum_i (-1)^i \sigma_{n-1}$ is a sum of $n+1$ terms, and thus is 0 when n is odd and σ_{n-1} when n is even ($n \neq 0$). Thus, we get

$$\cdots \to \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$$

with the boundary maps alternating between being an isomorphism and the trivial map except at the last $\mathbb{Z}$. Then it can be read off that the homology groups of this complex are all trivial except for $H_0 \cong \mathbb{Z}$.

For future reference, it is sometimes useful to define a modified version of homology for which the homology is trivial in all dimensions (including 0).

Definition 3.2.6: Reduced Homology Group

Define the *reduced homology groups* $\tilde{H}_n(X)$ to be the homology groups of the augmented chain complex:

$$\cdots \to C_2(X) \xrightarrow{\partial_2} C_1(X) \xrightarrow{\partial_1} C_0(X) \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where

$$\epsilon \left(\sum_i n_i \sigma_i \right) = \sum_i n_i$$

Certainly, we should require X to be nonempty, or else we would have a nontrivial homology group in dimension -1. Since $\epsilon \circ \partial_1 = 0$, ϵ vanishes on $\mathrm{im}(\partial_1)$, an hence induces the map $H_0(X) \to \mathbb{Z}$ with kernel $\tilde{H}_0(X)$. Thus:

$$H_0(X) \cong \tilde{H}_0(X) \oplus \mathbb{Z} \qquad H_n(X) \cong \tilde{H}_n(X), \ n > 0$$

We can think of the extra $\mathbb{Z}$ in the augmented chain as representing the unique map $(\emptyset) \to X$ where

$\emptyset$ is the empty simplex with no vertices. Then the augmentation map ϵ is the boundary map since:

$$\partial(v_0) = (\hat{v}_0) = (\emptyset)$$

It is worth noting that $H_1(X)$ is the abelianization of $\pi_1(X)$ whenever X is path connected. This shall not be proven, but see [14, 2.A] for reference.

3.3 Homotopy Invariance

In this section, we shall show that homotopy equivalent spaces have isomorphic homology groups, in particular we shall show there is a functor so that the continuous map $f : X \to Y$ induces a homomorphism $f_* : H_n(X) \to H_n(Y)$ for each n and that f_* is an isomorphism if f is a homotopy equivalence.

Definition 3.3.1: Pushforward

Let $f : X \to Y$ be a continuous map. Then f induces a homomorphsim $f_\sharp : C_n(X) \to C_n(Y)$ by mapping

$$\sigma \mapsto f \circ \sigma$$

which then is extended linearly:

$$f_\sharp\left(\sum_i n_i\sigma_i\right) = \sum_i n_i f_\sharp(\sigma_i) = \sum_i n_i f \circ \sigma_i$$

Since $f_\sharp\partial = \partial f_\sharp$, namely:

$$\begin{aligned} f_\sharp(\partial(\sigma)) &= f_\sharp\left(\sum_i (-1)^i \ \sigma_i|_{(v_0,\dots,\hat{v}_i,\dots,v_n)}\right) \\ &= \sum_i (-1)^i f(\sigma_i|_{(v_0,\dots,\hat{v}_i,\dots,v_n)}) \\ &= \partial(f_\sharp(\sigma)) \end{aligned}$$

the following diagram commute:

$$\begin{tikzcd} \cdots \arrow[r] & C_{n+1}(X) \arrow[r, "\partial"] \arrow[d, "f_\sharp"] & C_{n+1}(X) \arrow[r, "\partial"] \arrow[d, "f_\sharp"] & C_{n+1}(X) \arrow[r] \arrow[d, "f_\sharp"] & \cdots \\ \cdots \arrow[r] & C_{n+1}(X) \arrow[r, "\partial"] & C_{n+1}(X) \arrow[r, "\partial"] & C_{n+1}(X) \arrow[r] & \cdots \end{tikzcd}$$

In terms of categorie, f induces a homomorphism between complexes. In particular, it takes cycles to cycles since if $\partial(\alpha) = 0$ then $\partial(f_\sharp(\alpha)) = f_\sharp(\partial(\alpha)) = 0$, and $f_\sharp$ takes boundary to boundaries since $f_\sharp(\partial\beta) = \partial(f_\sharp\beta)$. Hence $f_\sharp$ induces a homomorphism:

$$f_* : H_n(X) \to X_n(Y)$$

Proposition 3.3.2: Functor between Top and Homology

let f, g be continuous maps. Then:

1. $(fg)_* = f_* g_*$
2. If f is a homeomorphism, f_* is an isomorhpism.
3. id_* is the identity map

Hence $(-)_* : \mathbf{Top} \to C(\mathbf{Ab})$ is a functor where $C(\mathbf{Ab})$ is the category of complexes with objects with complexes with objects $\mathbf{Ab}$ and morphisms being commutative diagrams as seen above.

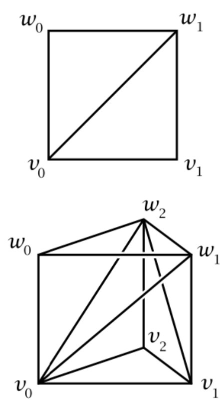
***Proof*:**
exercise, note that f being a homeomorphism inducing f_* to be an isomorphism was already mentioned under the definition of singular homology.

Theorem 3.3.3: Homotopy, Then Isomorphism

Let $f, g : X \to Y$ be homotopic maps. Then they induce the same homomorpism:

$$f_* = g_*$$

***Proof*:**
The key is to subdivide $\Delta^n \times I$ into simplices. In $\Delta^n \times I$, let $\Delta^n \times \{0\} = [v_0, ..., v_n]$, and $\Delta^n \times \{1\} = [w_0, ..., w_n]$, where v_i, w_i have the same image under the projection from $\Delta^n \times I \twoheadrightarrow \Delta^n$. To construct the $(n+1)$-simplex, we move a vertex v_i up to w_i, where we start at v_n and work backwards to v_0. In the $n \in \{1, 2\}$ case, this looks like so:



Concretely, we start by moving

$$[v_0, ..., v_n] \text{ to } [v_0, ..., v_{n-1}, w_n]$$

then move it to $[v_0, ..., v_{n-2}, w_{n-2}, w_n]$ and so on where we move $[v_0, ..., v_i, w_{i+1}, ..., w_n]$ up to $[v_0, ..., v_{i-1}, w_i, ..., w_n]$. Now, the region between the $(n+1)$-simplex $[v_0, ..., v_i, w_i, ..., w_n]$ has as lower face $[v_0, ..., v_i, w_{i+1}, ..., w_n]$ and as upper face $[v_0, ..., v_{i-1}, w_i, ..., w_n]$. Overall, $\Delta^n \times I$ is the union of these $(n+1)$-simplices, each intersecting the next in an n-simplex face.

Now, let $F : X \times I \to Y$ be a homotopy from f to g, and let $\sigma : \Delta^n \to X$ be a singular simplex. Define:

$$F \circ (\Sigma \times \mathrm{id})\Delta^n \times I \to X \times I \to Y$$

From this, we can define the following operator, called the *prism operator* as follows:

$$P : C_n(X) \to C_{n+1}(Y)$$
$$P(\sigma) = \sum_i (-1)^i F \circ (\sigma \times \mathrm{id})|_{[v_0,...,v_i,w_i,...,w_n]}$$

We shall show taht the prism operator satisfies

$$\partial P = g_\# - f_\# - P\partial$$

where the left hand side represnts the boundary of the prism, and the right hand side represents the top $\Delta^n \times \{1\}$ face, the bottom $\Delta^n \times \{0\}$ face, and the $\partial\Delta^n \times I$ sides of the prism. For this, we simply calculate:

$$\begin{aligned}\partial P(\sigma) &= \sum_{j\le i} (-1)^i(-1)^j F \circ (\sigma \times \mathrm{id})|_{[v_0,...,\hat{v_j},...,v_i,w_i,...,w_n]} \\ &+ \sum_{j\le i} (-1)^i(-1)^{j+1} F \circ (\sigma \times \mathrm{id})|_{[v_0,...,v_i,w_i,...,\hat{w}_j,...,w_n]}\end{aligned}$$

The terms where $i = j$ in the sum cancel out, with the exception of $F \circ (\sigma \times \mathrm{id})|_{\hat{v}_0,w_0,...,w_n]}$, which gives $g \circ \sigma = g_\#(\sigma)$, and the term $-F \circ (\sigma \circ \mathrm{id})|_{[v_0,...,v_n,\hat{w_n}]}$ which gives $-f \circ \sigma = -f_\$(\sigma)$. When $i \neq j$, then we get $-P\partial(\sigma)$ since

$$\begin{aligned}\partial P(\sigma) &= \sum_{j>i} (-1)^i(-1)^j F \circ (\sigma \times \mathrm{id})|_{[v_0,...,\hat{v_j},...,v_i,w_i,...,w_n]} \\ &+ \sum_{j>i} (-1)^i(-1)^{j+1} F \circ (\sigma \times \mathrm{id})|_{[v_0,...,v_i,w_i,...,\hat{w}_j,...,w_n]}\end{aligned}$$

But now, if $\alpha \in C_n(X)$ is a cycle, then

$$g_\#(\alpha) - f_\#(\alpha) = \partial P(\alpha) + P\partial(\alpha) = \partial P(\alpha)$$

since $\partial\alpha = 0$. Hence, $g_\#(\alpha) - f_\#(\alpha)$ is a boundary, hence $g_\#(\alpha), f_\#(\alpha)$ determines the homology class, but then g_* and f_* are equal on the homology class of α, completing the proof.

Corollary 3.3.4: Homotopy Equivalence, Then Isomorphism

Let $f : X \to Y$ be a homotopy equivalence. Then $f_* : H_n(X) \to H_n(Y)$ are isomorphisms for all $n \in \mathbb{N}$

Corollary 3.3.5: Homology Group Of Simply Connected Spaces

Let X be simply-connected path-connected space. Then:

$$H_n(X) = \begin{cases} \mathbb{Z} & n = 0 \\ 0 & n \geq 1 \end{cases}$$

***Proof*:**
Since $X \sim \{x_0\}$, $H_n(X) = H_n(\{x_0\})$, this follows from proposition 3.2.5

The key property that allowed us to derive the conclusion is the following

Definition 3.3.6: Chain Homotopy

The relationship

$$\partial P + P\partial = g_\# - f_\#$$

says that P is a *chain homotopy* between the chains $f_\#$ and $g_\#$

This property is more general than that of homotopy, and is the key fact in the proof to induce the homomorphism:

Proposition 3.3.7: chain homotopic chain maps

Chain homotopic chain maps induce the same homomorphism on homology

This shall be important when we generalize homology to more algebraic structures.

***Proof*:**
The relation $\partial P + P\partial = g_\sharp - f_\sharp$ from the above theorem shows exactly this.

Since $f_\# \epsilon = \epsilon f_\#$ (where $f_\#$ is the identity map on the added groups $\mathbb{Z}$) gives us the induced homomorphisms $f_* : \tilde{H}_n(X) \to \tilde{H}_n(Y)$ for the reduced homology groups, and the properties above hold equally well for reduced homologies.

3.4 Relative Homology: Quotient Complexes

In geometry, we very often construct more complicated shapes given simpler ones. Often, we may decompose a information about a shape X into:

1. information on $A \subseteq X$ and X/A
2. information on $A, B \subseteq X$ where $A \cup B = X$

We shall show how we can in the right cases find $A \subseteq X$ and X/A that let's us use the homology group of $H_n(A)$ and $H_n(X/A)$ to produce $H_n(X)$, and similarly for $A, B \subseteq X$. This builds up to two important theorems, the 2nd a direct application of the first:

1. Snakes Lemma Sequences
2. Mayer-Vietoris Sequences

An important simplification in homology comes when taking a subspace $A \subseteq X$, and then we want to consider $C_n(X)/C_n(A)$, which essentially shows to consider all homology information on the space A as trivial. We give this group a name:

Definition 3.4.1: Relative Chain Groups

Let X be a topological space and $A \subseteq X$ a subspace. Then

$$C_n(X, A) := C_n(X)/C_n(A)$$

is called the *relative homology groups* of X with respect to A.

Notice that the boundary map $\partial : C_n(X) \to C_{n-1}(X)$ takes $C_n(A)$ to $C_{n-1}(A)$, hence we have an induced map $\partial_n(X, A) \to C_{n-1}(X, A)$, and so $\partial^2 = 0$. Thus, we have the following:

Definition 3.4.2: Relative Homology Groups

Define

$$H_n(X, A) = \frac{\ker \partial_n}{\mathrm{im}\partial_{n+1}}$$

to be the *relative homology group* where ∂_n and ∂_{n+1} are defined as above. Elements of $H_n(X, A)$ are represented by *relative cycles* (n-chains $\alpha \in C_n(X)$ such that $\partial\alpha \in C_{n-1}(A)$), and relative cycles α is trivial in $H_n(X, A)$ if and only it is a *relative boundary* $\alpha = \partial\beta + \gamma$, where $\beta \in C_{n+1}(X)$ and $\gamma \in C_n(A)$

Note that the quoteint $C_n(X)/C_n(A)$ *can* be viewed as a subgroup of $C_n(X)$ by considering it as having basis the singular n-simplcies $\sigma : \Delta^n \to X$ whose image is *not* contained in A, however the boundary map does not take this subgroup of $C_n(X)$ to the coresponding subgroup of $C_{n-1}(X)$, hence this is not something we shall be considering for it does not work well with the algebra we have built up.

We would like to link $H_n(X, A)$ with $H_n(X)$. We shall do so in the following long exact sequence:

$$\cdots \to H_n(A) \to H_n(X) \to H_n(X,A) \to H_{n-1}(A) \to H_{n-1}(X) \to H_{n-1}(X,A) \to \cdots \to H_0(X,A) \to 0$$

This should naturally bring into mind the notion of a short exact sequence of complexes:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
\cdots \arrow[r] & A_{n+2} \arrow[r] \arrow[d, "\iota"] & A_{n+1} \arrow[r] \arrow[d, "\iota"] & A_n \arrow[r] \arrow[d, "\iota"] & A_{n-1} \arrow[r] \arrow[d, "\iota"] & \cdots \\
\cdots \arrow[r] & B_{n+2} \arrow[r] \arrow[d, "\pi"] & B_{n+1} \arrow[r] \arrow[d, "\pi"] & B_n \arrow[r] \arrow[d, "\pi"] & B_{n-1} \arrow[r] \arrow[d, "\pi"] & \cdots \\
\cdots \arrow[r] & C_{n+2} \arrow[r] \arrow[d] & C_{n+1} \arrow[r] \arrow[d] & C_n \arrow[r] \arrow[d] & C_{n-1} \arrow[r] \arrow[d] & \cdots \\
 & 0 & 0 & 0 & 0 &
\end{tikzcd} \tag{3.1}$$

where when working with chain complexes we get each long exact sequence has maps $\partial_k : X_{k+1} \to X$. If $A_k \subseteq B_k$ where B_k is a topological space and $C_k = B_k/A_k$, then the above diagram commutes.

We shall now show that this short exact sequence of chain complexes stretches into a long exact sequence of homology groups:

Theorem 3.4.3: Algebraic Snake Lemma

Given a short exact sequence of chain complex as in equation (3.1):

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{\pi_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \xrightarrow{\pi_*} H_{n-1}(C) \to \cdots$$

Visually, we can see the theorem shows that we make chain together equation 3.1 into:

$$\begin{tikzcd}
 & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & 0 \arrow[d, dotted] & \\
\cdots \arrow[r, dotted] & H_{n+2}(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_{n+1}(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_n(A) \arrow[r, dotted] \arrow[d, "\iota"] & H_{n-1}(A) \arrow[r, dotted] \arrow[d, "\iota"] & \cdots \\
\cdots \arrow[r, dotted] & H_{n+2}(B) \arrow[r, dotted] \arrow[d, "\pi"] & H_{n+1}(A) \arrow[r, dotted] \arrow[d, "\pi"] & H_n(B) \arrow[r, dotted] \arrow[d, "\pi"] & H_{n-1}(B) \arrow[r, dotted] \arrow[d, "\pi"] & \cdots \\
\cdots \arrow[r, dotted] & H_{n+2}(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_{n+1}(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_n(C) \arrow[r, dotted] \arrow[d, dotted] \arrow[uur] & H_{n-1}(C) \arrow[r, dotted] \arrow[d, dotted] & \cdots \\
 & 0 & 0 & 0 & 0 &
\end{tikzcd}$$

Note that this theorems work in the general context of homological algebra, for the notion of exact sequence of complexes is independent of any geometric considerations.

***Proof*:**
We first define the map $\partial : H_n(C) \to H_{n-1}(A)$. We should keep in mind the following diagram:

$$\begin{tikzcd} & A_{n-1} \arrow[d, "\iota"] \\ B_n \arrow[r, "\partial"] \arrow[d, "\pi"] & B_{n-1} \\ C_n & \end{tikzcd}$$

Let $c \in C_n$ be a cycle so that $c \in \ker \partial_n$. Since π is surjective, $c = \pi(b)$ for some $b \in B_n$. Then by definition $\partial : B_n \to B_{n-1}$ so $\partial(b) \in B_{n-1}$. Then by commutativity $j(\partial(b)) = \partial(j(b)) = \partial(c) = 0$ (since c is a cycle), thus $\partial b \in \ker(\pi)$. Since $\ker j = \mathrm{im} i$, $\partial b = i(a)$ for some $a \in A_{n-1}$. Since $i(\partial a) = \partial i(a) = \partial\partial b = 0$ since i is injective, $\partial a = 0$. Thus, define $\partial : H_n(C) \to H_{n-1}(A)$ by $[c] \mapsto [a]$. This is well defined since:

1. The element a is uniquely determined by ∂b since i is injective
2. for differnet choice b' for b, we get $j(b') = j(b)$, os $b' - b \in \ker(j)$ which is equal to $\mathrm{im}(i)$. Thus, $b' - b = i(a')$ for some a', ehcen $b' = b + i(a')$. But then this means we just chanage a to another representative (a homologous element) $a + \partial a'$, namely since
$$i(a + \partial a') = i(a) + i(\partial a') = \partial b + \partial i(a') = \partial(b = i(a'))$$
3. A difference choice of c within its homology class would have hte form $c = \partial c'$. To see this, since $c' = j(b')$ for some b',
$$c + \partial c' = c\partial j(b') = c + j(\partial b') = j(b + \partial b')$$
thus b is replaced by $b = \partial b'$, which gives ∂b and thus a is unchanged.

Finally, $\partial : H_n(C) \to H_{n-1}(A)$ is a homomorphism, for if $\partial[c_1] = [a_1]$ and $\partial[c_2] = [a_2]$ via elements b_1, b_2, then
$$j(b_1 + b_2) = j(b_1) + j(b_2) = c_1 = c_2$$
and
$$i(a_1 + a_2) = i(a_1) + i(a_2) = \partial b_1 + \partial b_2 = \partial(b_1 + b_2)$$
hence
$$\partial([c_1] + [c_2]) = [a_1] + [a_2]$$
giving us the map ∂ is indeed a homomorphism. What's left to show is the following inclusions:

1. $\mathrm{im} i_* \subseteq \ker j_*$. Since $ji = 0$, $j_* i_* = 0$
2. $\mathrm{im} j_* \subseteq \ker \partial$. By definition of ∂, $\partial b = 0$ so $\partial j_* = 0$.
3. $\ker j_* \subseteq \mathrm{im} i_*$. Let $b \in B_n$ with boundary represent a homology class in $\ker j_*$, so $j(b) = \partial c'$ for some $c' \in C_{n+1}$. Since j is surjective, $c' = j(b')$ for some $b' \in B_{n+1}$. Then since $\partial j(b') = \partial c' = j(b)$,
$$j(b - \partial b') = j(b) - j(\partial b') = j(b) - \partial j(b') = 0$$

Thus, $b - \partial b' = i(a)$ for some $a \in A - n$. Since $i(\partial a) = \partial i(a) = \partial(b - \partial b') = \partial b = 0$ and i is injective, a is a cycle. Thus:

$$\iota_*[a] = [b - \partial b'] = [b]$$

showing that i_* maps onto $\ker j_*$

4. $\ker \partial \subseteq \mathrm{im} j_*$. By definition of ∂, if c represents a homology class in $\ker \partial$, then $a = \partial a'$ for some $a' \in A_n$. The element $b - i(a')$ is a cycle since $\partial(b - i(a')) = \partial b - \partial i(a') = \partial b - i(\partial a') = \partial b - i(a) = 0$. Since

$$j(b - i(a')) = j(b) - ji(a') = j(b) = c$$

we get that $j_*[b - i(a')] = [c]$

5. $\ker i_* \subseteq \mathrm{im} \partial$. Given a cycle $a \in A_{n-1}$ such that $i(a) = \partial b$ for seem $b \in B_n$, then $j(b)$ is a cycle since $\partial j(b) = j(\partial b) = ji(a) = 0$, hence $\partial[j(b)] = [a]$

Thus, as an immediate consequence in the current topological settings we have:

$$\cdots \to H_n(A) \to H_n(X) \to H_n(X,A) \to H_{n-1}(A) \to H_{n-1}(X) \to H_{n-1}(X,A) \to \cdots \to H_0(X,A) \to 0$$

Concretely, the map $\partial : H_n(X,A) \to H_{n-1}(A)$ take a class $[\alpha]$, represented by a relative cycle α, and maps it to the class represented by the cycle $\partial\alpha$ in $H_{n-1}(A)$. Essentially, the long exact seuqences measures the difference between $H_n(X)$ and $H_n(A)$. In particular, exactness at all n would imply $H_n(X,A) = 0$, then the inclusion $A \hookrightarrow X$ indcues an isomorphism $H_n(A) \cong H_n(X)$ for all n (the converse is also true, left as an exercise).

Note that we may augment these results to reduced homology groups when $A \neq \emptyset$ by applying the above to the short exact sequence $0 \to C_n(A) \to C_n(X) \to C_n(X,A) - 0$ to non-negative diemnsions, and add $0 \to \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \to 0 \to 0$ for the -1 dimension. Thus, $\tilde{H}_n(X,A)$ is the same as $H_n(X,A)$ for all n (given $A \neq \emptyset$)

The construction of the connecting homomorphism involved choices, a lift b of the cycle c among them, and the proof of theorem 3.4.3 verified that the resulting class depends on none of them. That independence gives more than well-definedness: it makes the connecting homomorphism compatible with maps of short exact sequences, so that entire long exact sequences can be compared term by term.

Proposition 3.4.4: Naturality of the Connecting Homomorphism

Let

$$\begin{tikzcd} 0 \arrow[r] & A_\bullet \arrow[r, "i"] \arrow[d, "\alpha"] & B_\bullet \arrow[r, "j"] \arrow[d, "\beta"] & C_\bullet \arrow[r] \arrow[d, "\gamma"] & 0 \\ 0 \arrow[r] & A'_\bullet \arrow[r, "i'"] & B'_\bullet \arrow[r, "j'"] & C'_\bullet \arrow[r] & 0 \end{tikzcd}$$

be a commutative diagram of chain maps whose rows are short exact sequences of chain complexes. Then the long exact sequences of theorem 3.4.3 associated to the two rows form a commuting ladder

$$\begin{tikzcd} \cdots \arrow[r] & H_n(A) \arrow[r, "i_*"] \arrow[d, "\alpha_*"] & H_n(B) \arrow[r, "j_*"] \arrow[d, "\beta_*"] & H_n(C) \arrow[r, "\partial"] \arrow[d, "\gamma_*"] & H_{n-1}(A) \arrow[r, "i_*"] \arrow[d, "\alpha_*"] & \cdots \\ \cdots \arrow[r] & H_n(A') \arrow[r, "i'_*"] & H_n(B') \arrow[r, "j'_*"] & H_n(C') \arrow[r, "\partial'"] & H_{n-1}(A') \arrow[r, "i'_*"] & \cdots \end{tikzcd}$$

***Proof*:**
There are three squares to check in each period of the ladder, and only the one containing the connecting homomorphisms has content.

For the square with i_* and i'_*, let $a \in A_n$ be a cycle. Then

$$i'_*\alpha_*[a] = [i'\alpha(a)] = [\beta i(a)] = \beta_* i_*[a]$$

using the chain-level commutativity $i'\alpha = \beta i$ in the middle equality. The square with j_* and j'_* commutes by the same computation, with $j'\beta = \gamma j$ in place of $i'\alpha = \beta i$.

For the connecting square, the claim is $\alpha_*\partial = \partial'\gamma_*$. Let $[c] \in H_n(C)$ be represented by the cycle $c \in C_n$, and unwind the definition of ∂ from the proof of theorem 3.4.3: choose $b \in B_n$ with $j(b) = c$, observe that $\partial b \in \ker j = \operatorname{im} i$, write $\partial b = i(a)$ for a unique cycle $a \in A_{n-1}$, and set $\partial[c] = [a]$. Now push the entire chase through the vertical maps. The element $\beta(b) \in B'_n$ satisfies

$$j'(\beta(b)) = \gamma(j(b)) = \gamma(c)$$

so $\beta(b)$ is a legitimate lift of $\gamma(c)$, which is a cycle since γ is a chain map. Its boundary is

$$\partial(\beta(b)) = \beta(\partial b) = \beta(i(a)) = i'(\alpha(a))$$

using first that β is a chain map and then the commutativity $\beta i = i'\alpha$. Thus the element of A'_{n-1} produced by the chase for $\gamma(c)$, run with the lift $\beta(b)$, is $\alpha(a)$. Since the proof of theorem 3.4.3 showed the resulting homology class to be independent of the choice of lift,

$$\partial'\gamma_*[c] = \partial'[\gamma(c)] = [\alpha(a)] = \alpha_*[a] = \alpha_*\partial[c]$$

which is the claimed commutativity, completing the proof.

Example 3.4.5: Application of Snake Lemma
Look at lecture notes for CW complexes and their faces.

1. Consider $(D^n, \partial D^n)$ and a long exact sequence of reduced homology groups. Then the ∂ map
$$\cdots \to H_i(D^n) \to H_i(D^n, \partial D^n) \xrightarrow{\partial} \tilde{H}_{i-1}(S^{n-1}) \to H_{i-1}(D^n) \to \cdots$$
are isomorphisms for all $i > 0$ since the remaining terms $\tilde{H}_i(D^n)$ are all zero for all i. Thus assuming the homology of the sphere is $H_n(S^n) = \mathbb{Z}$ and $H_k(S^n) = 0$ for $k \neq n$:
$$H_i(D^n, \partial D^n) \cong \begin{cases} \mathbb{Z} & i = n \\ 0 & i \neq n \end{cases}$$

2. Consider (X, x_0) for $x_0 \in X$ and a long exact sequence of reduced homology groups. Then
$$H_n(X, x_0) \cong \tilde{H}_n(X)$$
for all n since $\tilde{H}_n(x_0) = 0$ for all $n \in \mathbb{N}$.

A map $f : X \to Y$ with $f(A) \subseteq B$ (we may write $f : (X, A) \to X(B)$) induces homomorphisms $f_\# : C_n(X, A) \to C_n(Y, B)$. The relation $f_\# \partial = \partial f_\#$ holds for relative chains (since it holds for chains), hence we have the induced homomorphisms $f_* : H_n(X, A) \to H_n(Y, B)$

Proposition 3.4.6: Homotopy Between Relative Maps

Let $f, g : (X, A) \to (Y, B)$ be two maps where $f(A) \subseteq B$. If these maps are homotopic through maps of pairs $f_t : (X, A) \to (Y, B)$, then $f_* = g_* : H_n(X, A) \to H_n(Y, B)$

***Proof*:**
This is just about checking that the previous work is still valid. Since the prism operator P form the above prove take $C_n(A)$ to $C_{n+1}(B)$, we have an induced prism operator $P : C_n(X, A) \to C_{n+1}(Y, B)$. When passing through the quotient, we still have
$$\partial P - P\partial = g_\# - f_\#$$
Hence, $f_\#$, $g_\#$ on relative chain groups are chain homotopic, hence they induce the smae homomorphisms on relative homology groups.

For future reference, we may generalize the long exact sequence given by the pair (X, A) to one by the triple (X, A, B), $B \subseteq A \subseteq X$ via
$$\cdots H_n(A, B) \to H_n(X, B) \to H_n(X, A) \to H_{n-1}(A, B) \to \cdots$$
associated to the chain complexes from by short exact sequences:
$$0 \to C_n(A, B) \to C_n(X, B) \to C_n(X, A) \to 0$$
An easy example would be if B is a point, then the long exact sequence formed by the triple (X, A, B) is just a long exact sequence of reduced homology for the pair (X, A).

We now build up to relating X with X/A. The key result we built up to is when we can delete, or excise, a part of the space and keep the same homology group:

Theorem 3.4.7: Excision Theorem

Let $Z \subseteq A \subseteq X$ such that the closure of Z is contained in the interior of A ($\overline{Z} \subseteq \text{int}A$). Then the inclusion $(X - Z, A - Z) \hookrightarrow (X, A)$ induces isomorphisms

$$H_n(X - Z, A - Z) \cong H_n(X, A) \qquad \forall n \in \mathbb{N}$$

Equivalently, for subspaces $A, B \subseteq X$ whose interiors cover X $\text{int}A \cup \text{int}B = X$, then inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces isomorphisms

$$H_n(B, A \cap B) \cong H_n(X, A) = H_n(A \cup B, A) \qquad \forall n \in \mathbb{N}$$

The equivalence is obtained by setting $B = X - Z$ and $Z = X - B$ so that $A \cap B = A - Z$ and $\bar{Z} \subseteq \text{int}A$ is equivalent to $X = \text{int}A \cup \text{int}B$ since $X - \text{int}B = \bar{Z}$.

To prove this theorem, we require a few lemmas. First Let X be a topological space and U a collection of subsest of X whose interior covers X. Let $C_n^U(X) \leqslant C_n(X)$ be chains $\sum_i n_i\sigma_i$ such that $\text{im}\sigma_i \subseteq U_j \in U$ for some element in U. Then the boundary map takes $C_n^U(X)$ to $C_{n-1}^U(X)$, hence $C_n^U(X)$ forms a chain complex; let the respective homology group be denoted $H_n^U(X)$

Lemma 3.4.8: Homology Of Covered Space

The inclusion map $\iota : C_n^U(X) \hookrightarrow C_n(X)$ is a chain homotopy equivalence, in particular there exists a ma p$\rho : C_n(X) \to C_n^U(X)$ such that $\iota \circ \rho$ and $\rho \circ \iota$ are chain homotopic to the identity. Thus, ι induces an isomorphism:

$$H_n^U(X) \cong H_n(X) \qquad \forall n \in \mathbb{N}$$

The proof is the technical heart of this chapter. The strategy is the method of *small simplices*: every singular chain can be cut into pieces small enough that each piece lands inside a single member of the cover, and the cutting is performed by an operator (barycentric subdivision) that is chain homotopic to the identity, so it changes nothing on homology. The argument has four stages. The first defines subdivision on the affine chains of a single convex set by coning from the barycentre, and computes its effect on boundaries. The second transports subdivision to arbitrary singular chains and produces an explicit chain homotopy between the subdivision operator and the identity. The third shows that iterating subdivision drives the geometric size of the pieces to zero, so that after finitely many iterations every piece is U-small. The fourth assembles these into the operator ρ and the required chain homotopies.

> ***Proof*:**
> Throughout, an *affine simplex* in a convex set $Y \subseteq \mathbb{R}^N$ is written $[w_0, \ldots, w_n]$, the singular simplex $\Delta^n \to Y$ sending the ith vertex of Δ^n to w_i and extending affinely on barycentric coordinates. The free abelian group on the affine n-simplices of Y is denoted $LC_n(Y)$, the *linear chains*, a subcomplex of $C_n(Y)$ on which the boundary map acts by the usual face formula $\partial[w_0, \ldots, w_n] = \sum_i (-1)^i [w_0, \ldots, \hat{w}_i, \ldots, w_n]$. We augment this complex by a copy of $\mathbb{Z} = LC_{-1}(Y)$ generated by the empty simplex $[\varnothing]$, with $\partial[w_0] = [\varnothing]$ for every 0-simplex, so

that the augmented linear chain complex $LC_\bullet(Y)$ runs down to degree -1.

Step 1: The cone operator and subdivision of linear chains. Fix a point $b \in Y$. The *cone* operator $b \cdot (-)\colon LC_n(Y) \to LC_{n+1}(Y)$ prepends b as a new initial vertex,

$$b \cdot [w_0, \dots, w_n] = [b, w_0, \dots, w_n]$$

and is extended linearly; on the empty simplex it gives $b \cdot [\varnothing] = [b]$. A direct expansion of the face formula, separating the term that drops the apex b from the terms that drop one of the w_i, yields the cone identity

$$\partial(b \cdot \lambda) = \lambda - b \cdot (\partial\lambda) \qquad (\lambda \in LC_n(Y)) \tag{3.2}$$

valid for all $n \geq 0$, where the augmentation makes the identity hold in degree 0 as well (there $\partial(b \cdot [w_0]) = \partial[b, w_0] = [w_0] - [b]$ and $b \cdot \partial[w_0] = b \cdot [\varnothing] = [b]$). The identity says geometrically that the boundary of a cone is the base minus the cone on the boundary; it is the algebraic form of the fact that a cone is contractible, and every computation below uses it.

Define the *barycentric subdivision* operator $S\colon LC_n(Y) \to LC_n(Y)$ recursively. On the augmentation set $S[\varnothing] = [\varnothing]$. For an affine simplex $\lambda = [w_0, \dots, w_n]$ with barycentre $b_\lambda = \frac{1}{n+1}\sum_i w_i$, set

$$S\lambda = b_\lambda \cdot (S\partial\lambda) \tag{3.3}$$

and extend linearly. Unwinding the recursion, $S\lambda$ is the signed sum of the $(n+1)!$ affine simplices whose vertices are the barycentres of an increasing chain of faces of λ; this is the classical barycentric subdivision of the geometric simplex λ, oriented coherently. We claim S is a chain map, $\partial S = S\partial$. This is an induction on n. In degree 0 subdivision is the identity and the claim is trivial. For $n \geq 1$, using (3.3), then (3.2), then the inductive hypothesis $\partial S\partial\lambda = S\partial\partial\lambda = 0$,

$$\partial S\lambda = \partial(b_\lambda \cdot S\partial\lambda) = S\partial\lambda - b_\lambda \cdot (\partial S\partial\lambda) = S\partial\lambda - b_\lambda \cdot (S\partial\partial\lambda) = S\partial\lambda$$

so S commutes with ∂.

Step 2: A chain homotopy from S to the identity on linear chains, then on singular chains. We construct an operator $T\colon LC_n(Y) \to LC_{n+1}(Y)$ with

$$\partial T + T\partial = \mathrm{id} - S \tag{3.4}$$

again by induction and again by coning. Set $T = 0$ on $LC_{-1}(Y)$ and on $LC_0(Y)$ (where $S = \mathrm{id}$ already, so the right side vanishes). For an affine n-simplex λ with barycentre b_λ, define

$$T\lambda = b_\lambda \cdot (\lambda - T\partial\lambda) \tag{3.5}$$

The verification of (3.4) is the bookkeeping computation; we record it. Applying (3.2) to (3.5),

$$\partial T\lambda = (\lambda - T\partial\lambda) - b_\lambda \cdot \partial(\lambda - T\partial\lambda) = \lambda - T\partial\lambda - b_\lambda \cdot (\partial\lambda - \partial T\partial\lambda)$$

By the inductive form of (3.4) applied to the $(n-1)$-chain $\partial\lambda$, namely $\partial T\partial\lambda = \partial\lambda - S\partial\lambda - T\partial\partial\lambda = \partial\lambda - S\partial\lambda$, the parenthesised term becomes $\partial\lambda - (\partial\lambda - S\partial\lambda) = S\partial\lambda$. Substituting and recalling $S\lambda = b_\lambda \cdot S\partial\lambda$ from (3.3),

$$\partial T\lambda = \lambda - T\partial\lambda - b_\lambda \cdot S\partial\lambda = \lambda - T\partial\lambda - S\lambda$$

which rearranges to $\partial T\lambda + T\partial\lambda = \lambda - S\lambda$, that is (3.4). (The explicit closed form of T as a signed sum over flags, together with the same sign accounting, is in [14, pp. 119-124]; the recursion above determines it uniquely.)

The operators S and T now transfer from linear chains to singular chains of an arbitrary space X by naturality, the standard device of evaluating on the universal model. The identity map $\mathrm{id}_{\Delta^n} \in LC_n(\Delta^n)$ is itself an affine n-simplex (the canonical one). For any singular simplex $\sigma\colon \Delta^n \to X$, define

$$S\sigma = \sigma_\#(S\,\mathrm{id}_{\Delta^n}), \qquad T\sigma = \sigma_\#(T\,\mathrm{id}_{\Delta^n})$$

where $\sigma_\#\colon C_\bullet(\Delta^n) \to C_\bullet(X)$ is the chain map induced by σ, and extend linearly to $C_n(X)$. Because $\sigma_\#$ commutes with ∂ and the construction on Δ^n is the linear one above, the relations established for linear chains pass through: $S\colon C_\bullet(X) \to C_\bullet(X)$ is a chain map, and

$$\partial T + T\partial = \mathrm{id} - S \qquad \text{on } C_\bullet(X) \tag{3.6}$$

Naturality also gives $f_\# S = S f_\#$ and $f_\# T = T f_\#$ for any map $f\colon X \to Y$, since both sides are determined by their value on id_{Δ^n}. This naturality is what will let subdivision respect the subspaces appearing in the cover.

Step 3: Iterated subdivision makes chains U-small. The mesh of a linear chain is controlled by a single geometric estimate on the standard simplex. If λ is an affine n-simplex with diameter $\operatorname{diam}\lambda$ (the maximal distance between two of its points, attained at a pair of vertices since λ is the convex hull of its vertices), then every simplex appearing in $S\lambda$ has diameter at most $\frac{n}{n+1}\operatorname{diam}\lambda$. To see this, a vertex of a simplex of $S\lambda$ is the barycentre b_τ of some face $\tau = [w_{i_0}, \ldots, w_{i_k}]$ of λ. The distance from b_τ to any vertex w_j of λ satisfies

$$|b_\tau - w_j| = \Big|\tfrac{1}{k+1}\sum_l (w_{i_l} - w_j)\Big| \le \tfrac{1}{k+1}\sum_l |w_{i_l} - w_j| \le \tfrac{k}{k+1}\operatorname{diam}\lambda \le \tfrac{n}{n+1}\operatorname{diam}\lambda$$

where the term $w_{i_l} = w_j$ contributes zero, leaving at most k nonzero summands each bounded by $\operatorname{diam}\lambda$. Since any two vertices of a simplex of $S\lambda$ are each barycentres of faces, and a barycentre is a convex combination of vertices, the distance between two such barycentres is bounded by the maximum of $|b_\tau - w_j|$ over vertices w_j, hence also by $\frac{n}{n+1}\operatorname{diam}\lambda$. The diameter of a simplex equals the maximal vertex-to-vertex distance, giving the claim.

Iterating, the m-fold subdivision $S^m\lambda$ consists of simplices of diameter at most $\left(\frac{n}{n+1}\right)^m \operatorname{diam}\lambda$, which tends to 0 as $m \to \infty$ because $\frac{n}{n+1} < 1$. Now fix a singular simplex $\sigma\colon \Delta^n \to X$. The open sets $\{\sigma^{-1}(\mathrm{int}U_j) : U_j \in U\}$ cover the compact metric space Δ^n, so the Lebesgue number lemma furnishes $\delta > 0$ such that every subset of Δ^n of diameter less than δ lies in a single $\sigma^{-1}(\mathrm{int}U_j)$. Choose m with $\left(\frac{n}{n+1}\right)^m \operatorname{diam}\Delta^n < \delta$. Then every simplex of $S^m\mathrm{id}_{\Delta^n}$ has image (in Δ^n) of diameter less than δ, hence contained in some $\sigma^{-1}(\mathrm{int}U_j)$. Applying $\sigma_\#$, every simplex of $S^m\sigma = \sigma_\#(S^m\mathrm{id}_{\Delta^n})$ has image inside some U_j. Thus $S^m\sigma \in C_n^U(X)$: a high enough iterate of subdivision carries σ into the U-small subcomplex. The integer $m = m(\sigma)$ depends on σ, which is the point requiring care in the next step.

Step 4: Assembling ρ and the chain homotopies. The iterates of the homotopy T telescope into a homotopy from id to S^m. Define, for each $m \ge 1$,

$$D_m = \sum_{0 \le i < m} TS^i\colon C_n(X) \to C_{n+1}(X)$$

Since S is a chain map and T satisfies (3.6), summing the relations $\partial TS^i + TS^i\partial = S^i - S^{i+1}$ over $0 \le i < m$ telescopes to

$$\partial D_m + D_m\partial = \mathrm{id} - S^m \tag{3.7}$$

so D_m is a chain homotopy from id to S^m for each fixed m. The difficulty is that no single m works for all simplices at once; the remedy is to let m vary simplex by simplex and absorb the variation into a well-defined operator.

For each singular n-simplex σ let $m(\sigma)$ be the least integer for which $S^{m(\sigma)}\sigma \in C_n^U(X)$, finite by Step 3, and set

$$D\sigma = D_{m(\sigma)}\sigma = \sum_{0\le i<m(\sigma)} TS^i\sigma$$

extended linearly to $C_n(X)$. This D is a well-defined natural operator $C_n(X) \to C_{n+1}(X)$ since $m(\sigma)$ is determined by σ. Define the candidate retraction by

$$\rho = \mathrm{id} - \partial D - D\partial \colon C_n(X) \to C_n(X) \tag{3.8}$$

Being id plus a boundary-type correction, ρ is automatically a chain map ($\partial\rho = \partial - \partial D\partial = \rho\partial$). Two facts remain: that ρ lands in $C_n^U(X)$, and that ι and ρ are mutually inverse up to chain homotopy.

ρ lands in $C_n^U(X)$. For a single simplex σ we compute $\rho\sigma$ directly. Because D agrees with $D_{m(\sigma)}$ on σ but $D\partial\sigma$ uses the (smaller) values $m(\sigma')$ on the faces σ' of σ, expand

$$\rho\sigma = \sigma - \partial D\sigma - D\partial\sigma = \big(\sigma - \partial D_{m(\sigma)}\sigma - D_{m(\sigma)}\partial\sigma\big) + \big(D_{m(\sigma)}\partial\sigma - D\partial\sigma\big)$$

By (3.7) the first bracket is $S^{m(\sigma)}\sigma$, which lies in $C_n^U(X)$ by the choice of $m(\sigma)$. The second bracket is $\sum_{\sigma'} \pm(D_{m(\sigma)} - D_{m(\sigma')})\sigma'$ summed over the faces σ' of σ; each difference $D_{m(\sigma)} - D_{m(\sigma')} = \sum_{m(\sigma')\le i<m(\sigma)} TS^i$ is a sum of terms $TS^i\sigma'$ with $i \ge m(\sigma')$, and $S^i\sigma'$ is U-small for $i \ge m(\sigma')$, hence so is $TS^i\sigma'$ since T is natural and sends a U-small simplex (one with image in a single U_j) to chains with image in the same U_j. Therefore $\rho\sigma \in C_n^U(X)$, and ρ corestricts to a chain map $\rho\colon C_n(X) \to C_n^U(X)$.

ι and ρ are chain homotopy inverse. By construction (3.8), D is a chain homotopy between $\iota\circ\rho$ and $\mathrm{id}_{C_\bullet(X)}$: indeed $\iota\rho = \rho = \mathrm{id} - (\partial D + D\partial)$ as operators into $C_\bullet(X)$, which is exactly the statement $\partial D + D\partial = \mathrm{id} - \iota\rho$. For the other composite, note first that ρ restricts to the identity-up-to-homotopy already at the level of $C_n^U(X)$: if σ is itself U-small then $m(\sigma) = 0$, so $D\sigma = 0$, while $D\partial\sigma$ involves only faces of σ, which are also U-small with $m = 0$, hence $D\partial\sigma = 0$; thus $\rho\sigma = \sigma$ for U-small σ. Consequently $\rho\circ\iota = \mathrm{id}_{C_n^U(X)}$ on the nose. Restricting the homotopy D to $C_n^U(X)$ (where it vanishes) shows $\iota\circ\rho$ is chain homotopic to the identity on $C_\bullet(X)$ via D, and $\rho\circ\iota$ equals the identity on $C_\bullet^U(X)$. Hence ι is a chain homotopy equivalence with homotopy inverse ρ.

A chain homotopy equivalence induces isomorphisms on homology (proposition 3.3.7 applied to both composites), so

$$\iota_*\colon H_n^U(X) \xrightarrow{\cong} H_n(X) \qquad \forall n \in \mathbb{N}$$

completing the proof.

With the small-simplices lemma in hand, the excision theorem follows by applying it to a two-element cover.

***Proof Of the Excision Theorem*:**
By the equivalence recorded after the statement, it suffices to prove the second form: for $A, B \subseteq X$ with $\text{int}A \cup \text{int}B = X$, the inclusion induces isomorphisms $H_n(B, A \cap B) \cong H_n(X, A)$ for all n. Take the cover $U = \{A, B\}$, whose interiors cover X by hypothesis.

Step 1: Comparison of relative complexes. Form the U-small chains $C_n^U(X) = C_n(A) + C_n(B)$, the subgroup of chains expressible as a sum of a chain in A and a chain in B, and pass to the relative complex $C_n^U(X)/C_n(A)$. The composite

$$C_n(B) \hookrightarrow C_n^U(X) \twoheadrightarrow C_n^U(X)/C_n(A)$$

has kernel exactly $C_n(B) \cap C_n(A) = C_n(A \cap B)$, since a singular simplex with image in B that also lies in $C_n(A)$ is a simplex with image in $A \cap B$ (the basis of $C_n(X)$ being the singular simplices, an intersection of free subgroups generated by simplices is generated by the common simplices). The map is surjective because every generator of $C_n^U(X)/C_n(A)$ is represented by a simplex in B. Therefore it descends to an isomorphism of chain complexes

$$C_n(B)/C_n(A \cap B) \xrightarrow{\cong} C_n^U(X)/C_n(A) \tag{3.9}$$

natural in n and commuting with the boundary maps, hence an isomorphism $H_n(B, A \cap B) \cong H_n\big(C_\bullet^U(X)/C_\bullet(A)\big)$ on homology.

Step 2: The small-simplices lemma kills the difference between C^U and C. Consider the short exact sequences of chain complexes

$$0 \to C_\bullet(A) \to C_\bullet^U(X) \to C_\bullet^U(X)/C_\bullet(A) \to 0, \qquad 0 \to C_\bullet(A) \to C_\bullet(X) \to C_\bullet(X)/C_\bullet(A) \to 0$$

The inclusion $C_\bullet^U(X) \hookrightarrow C_\bullet(X)$ is the identity on the common subcomplex $C_\bullet(A)$ and is the map ι of lemma 3.4.8, which is a chain homotopy equivalence inducing isomorphisms $H_n^U(X) \cong H_n(X)$. The induced maps on the quotients fit into a morphism of the two short exact sequences, hence into a map of their long exact sequences (definition 3.4.2), commuting with all connecting homomorphisms. In that ladder the maps on the $H_n(A)$ terms are identities and the maps on the $H_n^U(X) \to H_n(X)$ terms are the lemma's isomorphisms. By the five lemma the induced map on the relative homology of the quotient complexes is an isomorphism,

$$H_n\big(C_\bullet^U(X)/C_\bullet(A)\big) \xrightarrow{\cong} H_n\big(C_\bullet(X)/C_\bullet(A)\big) = H_n(X, A) \tag{3.10}$$

The five lemma is the algebraic statement that a map of long exact sequences which is an isomorphism on four out of every five consecutive terms is an isomorphism on the fifth (see [10, Five Lemma]).

Step 3: Conclusion. Composing the isomorphism (3.9) on homology with (3.10),

$$H_n(B, A \cap B) \cong H_n\big(C_\bullet^U(X)/C_\bullet(A)\big) \cong H_n(X, A) \qquad \forall n \in \mathbb{N}$$

and tracing the maps shows the composite is induced by the inclusion $(B, A \cap B) \hookrightarrow (X, A)$, since at the chain level it is exactly the inclusion $C_\bullet(B) \hookrightarrow C_\bullet(X)$ read modulo $C_\bullet(A)$. Translating back through $B = X - Z$, $A \cap B = A - Z$ gives the first form $H_n(X - Z, A - Z) \cong H_n(X, A)$, as we sought to show.

Finally to finish the snakes lemma, we need to replace the relative homology groups with absolute homology groups:

Corollary 3.4.9: Snakes Lemma With Absolute Homology Groups

Let (X, A) be a pair where A is a nonempty closed subspace that is a deformation retract of some neighbourhood of X, $\pi : (X, A) \to (X/A, A/A)$ the natural quotient map, and $\pi_* : H_n(X, A) \to H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$ the induced map. Then π_* is an isomorphism:

$$H_n(X, A) \cong H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$$

***Proof*:**
Let V be a neighbourhood of A in X that deformation retracts onto A, which exists by the hypothesis that (X, A) is a good pair. The proof assembles three isomorphisms: one provided by the deformation retraction, one by excision (theorem 3.4.7), and one by passing to the quotient, where the same two devices apply because $(X/A, A/A)$ inherits a good neighbourhood from V. Throughout, all maps between relative homology groups are those induced by inclusions or by the quotient map $\pi\colon (X, A) \to (X/A, A/A)$, and the diagrams below commute because they commute already at the level of singular chains.

Step 1: The deformation retraction collapses V to A. Consider the triple $A \subseteq V \subseteq X$ and the long exact sequence of the triple (theorem 3.4.3 applied to the short exact sequence of chain complexes $0 \to C_\bullet(V, A) \to C_\bullet(X, A) \to C_\bullet(X, V) \to 0$),

$$\cdots \to H_n(V, A) \to H_n(X, A) \to H_n(X, V) \xrightarrow{\partial} H_{n-1}(V, A) \to \cdots$$

Let $r_t\colon V \to V$ be the deformation retraction, so $r_0 = \mathrm{id}_V$, $r_1(V) = A$, and $r_t|_A = \mathrm{id}_A$ for all t. Each r_t is a map of pairs $(V, A) \to (V, A)$, and $r_0 = \mathrm{id}$ is homotopic through such maps to the composite $(V, A) \xrightarrow{r_1} (A, A) \hookrightarrow (V, A)$. By the homotopy invariance of relative homology (proposition 3.4.6) the identity on $H_n(V, A)$ therefore factors through $H_n(A, A) = 0$, so $H_n(V, A) = 0$ for every n. Exactness of the displayed sequence then forces the inclusion-induced map

$$H_n(X, A) \xrightarrow{\cong} H_n(X, V) \qquad \forall n \in \mathbb{N}$$

to be an isomorphism.

Step 2: Excise the subspace A. The set $Z = A$ satisfies $\overline{Z} = A \subseteq \mathrm{int} V$, since V is a neighbourhood of A and A is closed. Excision (theorem 3.4.7) applied to $Z \subseteq V \subseteq X$ gives that the inclusion $(X - A, V - A) \hookrightarrow (X, V)$ induces an isomorphism

$$H_n(X - A, V - A) \xrightarrow{\cong} H_n(X, V) \qquad \forall n \in \mathbb{N}$$

Step 3: Transport the chain to the quotient. Write $q\colon X \to X/A$ for the quotient map and set $\bar{A} = A/A$, the single point onto which A collapses, and $\bar{V} = V/A = q(V)$. Because A is closed and V deformation retracts onto A, the pair $(X/A, \bar{A})$ is again a good pair: composing r_t with q and using that $r_t|_A = \mathrm{id}_A$ shows r_t descends to a deformation retraction $\bar{r}_t\colon \bar{V} \to \bar{V}$ of the neighbourhood $\bar{V}$ of $\bar{A}$ onto $\bar{A}$. Steps 1 and 2, applied verbatim to $(X/A, \bar{A})$ with the neighbourhood $\bar{V}$, yield isomorphisms

$$H_n(X/A, \bar{A}) \xleftarrow{\cong} H_n\big((X/A) - \bar{A},\ \bar{V} - \bar{A}\big) \cong H_n(X/A, \bar{V})$$

the left arrow induced by inclusion and excision, the right by the deformation retraction.

Step 4: Identify the two complements and conclude. The quotient map q restricts to a homeomorphism of pairs

$$q\colon (X - A,\ V - A) \xrightarrow{\cong} \big((X/A) - \bar{A},\ \bar{V} - \bar{A}\big)$$

since q is a bijection away from A, is continuous, and is open there: a set $W \subseteq X - A$ is open in $X - A$ if and only if $q(W)$ is open in $(X/A) - \bar{A}$, as $q^{-1}(q(W)) = W$ for W disjoint from A. A homeomorphism of pairs induces an isomorphism on relative homology, so the induced map $q_*\colon H_n(X - A, V - A) \to H_n\big((X/A) - \bar{A},\ \bar{V} - \bar{A}\big)$ is an isomorphism.

Name the inclusion-induced isomorphisms of Steps 1 and 2 and their quotient counterparts: in the source, $a\colon H_n(X, A) \to H_n(X, V)$ from the deformation retraction of Step 1 and $e\colon H_n(X-A, V-A) \to H_n(X, V)$ from the excision of Step 2; in the quotient, $\bar{a}\colon H_n(X/A, \bar{A}) \to H_n(X/A, \bar{V})$ and $\bar{e}\colon H_n\big((X/A) - \bar{A},\ \bar{V} - \bar{A}\big) \to H_n(X/A, \bar{V})$ from the corresponding maps of Step 3. Each of $a, e, \bar{a}, \bar{e}$ is an isomorphism. Write $\pi'_*\colon H_n(X, V) \to H_n(X/A, \bar{V})$ for the map induced by the quotient map of pairs $(X, V) \to (X/A, \bar{V})$. These maps assemble into the diagram

$$\begin{tikzcd}
H_n(X, A) \arrow[r, "a", "\cong"'] \arrow[d, "\pi_*"'] & H_n(X, V) \arrow[d, "\pi'_*"] & H_n(X - A, V - A) \arrow[l, "e"', "\cong"] \arrow[d, "\cong"', "q_*"] \\
H_n(X/A, \bar{A}) \arrow[r, "\cong", "\bar{a}"'] & H_n(X/A, \bar{V}) & H_n\big((X/A) - \bar{A},\ \bar{V} - \bar{A}\big) \arrow[l, "\cong"', "\bar{e}"]
\end{tikzcd}$$

in which every horizontal edge is inclusion-induced: the left-hand pair $a, \bar{a}$ are the deformation-retraction isomorphisms of Step 1, the right-hand pair $e, \bar{e}$ the excision isomorphisms of Step 2, while the three vertical maps π_*, π'_*, q_* are induced by the quotient. Both squares commute by naturality: each is the image under q of a commuting square of inclusions of pairs, so q-compatibility of the inclusions gives $\bar{a} \circ \pi_* = \pi'_* \circ a$ on the left and $\bar{e} \circ q_* = \pi'_* \circ e$ on the right. From the right square, $\pi'_* = \bar{e} \circ q_* \circ e^{-1}$ is a composite of three isomorphisms, hence an isomorphism. From the left square, $\bar{a} \circ \pi_* = \pi'_* \circ a$, and since $a, \bar{a}, \pi'_*$ are isomorphisms,

$$\pi_* = \bar{a}^{-1} \circ \pi'_* \circ a$$

is an isomorphism as well. This is exactly the map on relative homology induced by π, since at the chain level all of the inclusions and the quotient compose to the chain map of $\pi\colon (X, A) \to (X/A, \bar{A})$. Thus

$$\pi_*\colon H_n(X, A) \xrightarrow{\cong} H_n(X/A, \bar{A}) = H_n(X/A, A/A) \qquad \forall n \in \mathbb{N}$$

is an isomorphism.

Finally, $\bar{A}$ is a single point, so the long exact sequence of the pair $(X/A, \bar{A})$ in reduced homology (theorem 3.4.3, applied to reduced chains since $\bar{A} \neq \emptyset$) has the terms $\tilde{H}_n(\bar{A}) = 0$, whence its connecting maps give $H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$ for all n by the identification of relative homology with a point against reduced homology (definition 3.2.6). Chaining the isomorphisms,

$$H_n(X, A) \cong H_n(X/A, A/A) \cong \tilde{H}_n(X/A) \qquad \forall n \in \mathbb{N}$$

with the first induced by π_*, as we sought to show.

This motivates the following definition

Definition 3.4.10: Good Pair

Let $A \subseteq X$ be a closed subspace that is a deformation retract of some neighbourhood of X. Then (X, A) is called a good pair

With this, we may upgrade the snake lemma to the following:

Theorem 3.4.11: Topological Snake Lemma

Let (X, A) be a good pair. Then there is an exact sequence:

$$\cdots \to \tilde{H}_n(A) \xrightarrow{i_*} \tilde{H}_n(X) \xrightarrow{\pi_*} H_n(X/A) \xrightarrow{\partial} \tilde{H}_{n-1}(A) \xrightarrow{i_*} \tilde{H}_{n-1}(X) \xrightarrow{\pi_*} \tilde{H}_{n-1}(X/A) \to \cdots$$

where i is the inclusion and j the quotient.

***Proof*:**
Immedaite consequence of above results.

Corollary 3.4.12: Homology Groups of Spheres

$$\tilde{H}_n(S^n) \cong \mathbb{Z} \qquad \tilde{H}_i(S^n) = 0 \quad (i \neq n)$$

***Proof*:**
For $n > 0$, take $(X, A) = (D^n, S^{n-1})$ so that $X/A = S^n$. Then the terms $\tilde{H}_i(D^n)$ in the long exact sequence for this pair are zero since D^n is contractible. Exactness of the sequence implies the maps

$$\tilde{H}_i(S^n) \xrightarrow{\partial} \tilde{H}_{i-1}(S^{n-1})$$

are isomorphisms for $i > 0$. Since $\tilde{H}_0(S^{n-i}) = 0$ for $n \neq i$ and $\tilde{H}_0(S^0) \cong \mathbb{Z}$ since S^0 has two connected components, we get the homology groups of S_n to be:

$$\tilde{H}_k(S^n) = \begin{cases} \mathbb{Z} & k = n \\ 0 & k \neq n \end{cases}$$

as we sought to show.

Corollary 3.4.13: No Retract For Border Of Spheres

The border of D^n does not retract, that is ∂D^n is not a retract of D^n. Hence, every map $f : D^n \to D^n$ has a fixed point

***Proof*:**
If $r : D^n \to \partial D^n$ is a retraction, then $r \circ \iota = \text{id}$, ι the inclusion map. Then the composition

$$\tilde{H}_{n-1}(\partial D^n) \xrightarrow{\iota_*} \tilde{H}_{n-1}(D^n) \xrightarrow{r_*} \tilde{H}_{n-1}(\partial D^n)$$

is the identity map on $\tilde{H}_{n-1}(\partial D^n) \cong \mathbb{Z}$. But ι_* and r_* are zero since $\tilde{H}_{n-1}(D^n) = 0$, a contradiction. Then the fixed point follow from Brouwers fixed point theorem (theorem 2.1.47).

(Hatcher elaboartes on how to generalize from (X, A) to arbitrary pairs on p.124-125)

Example 3.4.14: Finding Explicit Generators of $H_n(S^n)$
Hatcher p.125; it shall be the inclusion map Consider the homotopic equivalent $(\Delta^n, \partial\Delta^n)$.

Corollary 3.4.15: Excisions For CW Complex

Let X be a CW complex that is the union of subcomplexes A and B. Then the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ indcues an isomorphism

$$H_n(B, A \cap B) \cong H_n(X, A) \qquad \forall n \in \mathbb{N}$$

***Proof*:**
Let $A \cap B \neq \emptyset$. Since CW pairs are good, by corollary 3.4.9 we may pass to the quotient space $B/(A \cap B)$ and X/A, which are homeomrophic.

Corollary 3.4.16: Wedge Sum Homology Group

Let X_α be topological spaces and $\bigvee \alpha X_\alpha$ their wedge sum. Then the inclusion $\iota_\alpha : X_\alpha \hookrightarrow \bigvee X_\alpha$ indcues an isomorphism:

$$\bigoplus_\alpha : (\iota_\alpha)_* : \bigoplus_\alpha \tilde{H}_n(X_\alpha) \to \tilde{H}_n\left(\bigvee_\alpha X_\alpha\right)$$

given the wedge sum is formed at the same basepoint $x_\alpha \in X_\alpha$ such that the pairs (X_α, x_α) are good.

***Proof*:**
This follows from the proposition, since reducd homology is the same as the homology relative to a basepoint, giving us:

$$(X, A) = \left(\cup_\alpha X_\alpha, \coprod_\alpha \{x_\alpha\}\right)$$

The following conseuqence is of great importance:

Theorem 3.4.17: Invariance Of Dimension

Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be nonempty open nsubsets. Then if $U \cong V$, then $m = n$.

***Proof*:**
Take $x \in U$, and let $A = \mathbb{R}^n - \{x\}$, $B = U$, so that $A \cup B = \mathbb{R}^n$ and $A \cap B = U - \{x\}$. Since $(B, A \cap B)$ is a good pair of a CW complex, by excision:

$$H_k(U, U - \{x\}) \cong H_k(\mathbb{R}^n, \mathbb{R}^n - \{x\})$$

By the long exact seuqence for the pair $(\mathbb{R}^n, \mathbb{R}^n - \{x\})$, we get

$$H_k(\mathbb{R}^n, \mathbb{R}^n - \{x\}) \cong \tilde{H}_{k-1}(\mathbb{R}^n - \{x\})$$

Since $\mathbb{R}^n - \{x\}$ deformation retracts onto S^{n-1}, we get that

$$H_k(U, U - \{x\}) \cong \begin{cases} \mathbb{Z} & k = n \\ 0 & k \neq n \end{cases}$$

Thus, by the same reasoning:

$$H_k(V, V - \{y\}) \cong \begin{cases} \mathbb{Z} & k = m \\ 0 & k \neq m \end{cases}$$

Since a homeomorphism $\varphi : U \to V$ indcuces an isomorphism $\varphi_* : H_k(U, U - \{x\}) \to H_k(V, V - \{h(x)\})$, we get that it must be that $n = m$, as we sought to show.

For a statement that may at fisrt seem rather complicated to prove explicitely using classical tools of topology ,the proof turns out to be rather simple. To me, this is a good indicator, since it shows that the notion of dimension is not in this context a very "subtle" idea, but instead quite a natural one.

Note that the converse is not true for general open sets (take a punctured intD^n and intD^n, they have the same dimension but different homology groups). Limiting to simply connected sets, the case of $n = 1$ is easy to check as an exercise, the case of $n = 2$ is true by the Riemann mapping theorem (which states the stronger condition that simply connected bounded sets are in fact biholomorphic, to D^2). For $n \geq 3$, it starts getting more complicated; If U is contractible and still simply connected when adding the point at infinity. The extra condition is necessary, being contractible does not imply being homeomorphic to D^3 (whiteheads manifold is a counter-example). These explorations continue in a class on Geometry.

The idea of the above theorem can be generalized to find when two spaces are not locally homeomorphic:

Definition 3.4.18: Local Homology Groups

Let X be a topological space and $x \in X$ be a [closed] point. Then

$$H_n(X, X - \{x\})$$

is called the *local homology group* of the space X at x.

If U is an open neighbourhood of X, then by excision we get

$$H_n(X, X - \{x\}) \cong H_n(U, U - \{x\})$$

meaning the local homology groups only depends on the local topology of X near x. Then a homeomorphism induces:

$$H_n(X, X - \{x\}) \cong H_n(Y, Y - \{f(x)\})$$

for all x and n. Hence, we see that the local homology group can be used to study spaces locally, and determine whether a space is locally homeomorphic at certain points.

3.4.1 Naturality

The long exact sequence of a pair is natural in the pair. A map of pairs $f : (X, A) \to (Y, B)$ induces chain maps on subspace, ambient, and relative chains, and these assemble into a commutative diagram of short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & C_\bullet(A) \arrow[r] \arrow[d, "f_\#"] & C_\bullet(X) \arrow[r] \arrow[d, "f_\#"] & C_\bullet(X, A) \arrow[r] \arrow[d, "f_\#"] & 0 \\
0 \arrow[r] & C_\bullet(B) \arrow[r] & C_\bullet(Y) \arrow[r] & C_\bullet(Y, B) \arrow[r] & 0
\end{tikzcd}$$

the left square commuting since $f(A) \subseteq B$, and the right square since the relative $f_\#$ is defined by passing to quotients. Proposition 3.4.4 then yields the commuting ladder

$$\begin{tikzcd}
\cdots \arrow[r] & H_n(A) \arrow[r] \arrow[d, "f_*"] & H_n(X) \arrow[r] \arrow[d, "f_*"] & H_n(X, A) \arrow[r, "\partial"] \arrow[d, "f_*"] & H_{n-1}(A) \arrow[r] \arrow[d, "f_*"] & \cdots \\
\cdots \arrow[r] & H_n(B) \arrow[r] & H_n(Y) \arrow[r] & H_n(Y, B) \arrow[r, "\partial"] & H_{n-1}(B) \arrow[r] & \cdots
\end{tikzcd}$$

relating the long exact sequences of the pairs (X, A) and (Y, B). The same argument applies to the reduced sequences and to triples. This ladder is the standard device for transporting information between pairs, and it returns below in the comparison of simplicial with singular homology and in the cellular boundary formula (proposition 3.5.14).

3.4.2 Mayer-Vietoris Sequences

As we saw, the snake lemma provided a useful way of linking information between homology groups and relative homology groups. There is another long exact sequence that gives us more information, and is in many cases easier to computer.

Theorem 3.4.19: Mayer-Vietoris Sequence

Let $A, B \subseteq X$ such that X is the union of the interior of A and B. Then we have the following exact sequence

$$\cdots \to H_n(A\cap B) \xrightarrow{\varphi} H_n(A) \oplus H_n(B) \xrightarrow{\psi} H_n(X) \xrightarrow{\partial} H_{n-1}(A\cap B) \xrightarrow{\varphi} H_{n-1}(A) \oplus H_{n-1}(B) \to \cdots \to H_0(X) \to 0$$

called the *Mayer-Vietoris sequence.*

The idea behind this sequence is that we often use it for induction proofs, where we know information about A, B and $A \cap B$, and build up to show information about the homology of $A \cup B$.

(look at commented link for another proof)

***Proof*:**
The whole sequence is the long exact sequence of homology attached, via the algebraic snake lemma (theorem 3.4.3), to a short exact sequence of chain complexes. The work is to produce that short exact sequence and to identify its outer terms with $H_n(A \cap B)$ and $H_n(A) \oplus H_n(B)$ and its middle term, after one further isomorphism, with $H_n(X)$. The argument has three stages: first the chain group $C_n(A+B)$ and the verification that the inclusion $C_n(A+B) \hookrightarrow C_n(X)$ computes the homology of X; then the short exact sequence and its exactness; and finally the assembly.

Stage 1: small chains compute $H_n(X)$. Let $C_n(A+B) \subseteq C_n(X)$ be the subgroup of chains $\sum_i n_i \sigma_i$ each of whose singular simplices σ_i has image contained in A or in B. Equivalently, $C_n(A+B) = C_n(A) + C_n(B)$ as subgroups of $C_n(X)$, the sum of the images of the inclusions of A and B. The boundary $\partial \colon C_n(X) \to C_{n-1}(X)$ carries each σ_i to an alternating sum of its faces, and a face of a simplex with image in A (respectively B) again has image in A (respectively B); hence ∂ maps $C_n(A+B)$ into $C_{n-1}(A+B)$, so the groups $C_\bullet(A+B)$ form a subcomplex of $C_\bullet(X)$. Write $H_n(A+B)$ for its homology.

Apply lemma 3.4.8 to the collection $U = \{A, B\}$. Its interiors $\mathrm{int}A$ and $\mathrm{int}B$ cover X by hypothesis, and the small-chain complex $C_\bullet^U(X)$ of that lemma is by construction $C_\bullet(A+B)$. The lemma provides a chain map $\rho \colon C_\bullet(X) \to C_\bullet(A+B)$ with $\iota\rho$ and $\rho\iota$ chain homotopic to the identity, where $\iota \colon C_\bullet(A+B) \hookrightarrow C_\bullet(X)$ is the inclusion. Consequently ι induces an isomorphism

$$\iota_* \colon H_n(A+B) \xrightarrow{\sim} H_n(X) \qquad \forall n \in \mathbb{N}$$

This is the single point at which the covering hypothesis $X = \mathrm{int}A \cup \mathrm{int}B$ enters; everything else is formal.

Stage 2: the short exact sequence of complexes. For each n define homomorphisms

$$0 \longrightarrow C_n(A \cap B) \xrightarrow{\varphi} C_n(A) \oplus C_n(B) \xrightarrow{\psi} C_n(A+B) \longrightarrow 0$$

by $\varphi(x) = (x, -x)$ and $\psi(a,b) = a + b$, where each chain is regarded inside $C_n(X)$ through the relevant inclusion. Both are restrictions of maps built from the inclusion-induced chain maps $C_\bullet(A \cap B) \to C_\bullet(A)$, $C_\bullet(A \cap B) \to C_\bullet(B)$, $C_\bullet(A) \to C_\bullet(X)$ and $C_\bullet(B) \to C_\bullet(X)$, each of which commutes with ∂; hence φ and ψ are chain maps. Exactness is checked degreewise.

1. *φ is injective.* If $\varphi(x) = (x, -x) = 0$ then $x = 0$ already in the first coordinate, so $\ker \varphi = 0$.
2. *ψ is surjective.* An element of $C_n(A+B)$ is by definition a sum $a + b$ with $a \in C_n(A)$ and $b \in C_n(B)$, and $\psi(a,b) = a + b$; so ψ hits every generator.
3. $\ker \psi = \mathrm{im}\varphi$. The composite $\psi\varphi(x) = x + (-x) = 0$, so $\mathrm{im}\varphi \subseteq \ker \psi$. Conversely, suppose $\psi(a,b) = a + b = 0$ as a chain in $C_n(X)$, with $a \in C_n(A)$ and $b \in C_n(B)$. Then $a = -b$ as elements of $C_n(X)$, an equality of formal $\mathbb{Z}$-linear combinations of singular simplices. Two such combinations are equal precisely when they assign the same coefficient to each singular simplex of X, so every simplex occurring in a with nonzero coefficient occurs in $-b$ with the same coefficient. Such a simplex has image in A (because it occurs in a) and in B (because it occurs in b), hence in $A \cap B$. Therefore $a \in C_n(A \cap B)$, and $(a,b) = (a,-a) = \varphi(a) \in \mathrm{im}\varphi$. This proves $\ker \psi \subseteq \mathrm{im}\varphi$.

Since φ and ψ are chain maps and the displayed sequence is exact in every degree, the diagram is a short exact sequence of chain complexes.

Stage 3: assembly. Feed the short exact sequence of Stage 2 into the algebraic snake lemma (theorem 3.4.3). It produces the long exact sequence

$$\cdots \to H_n(A\cap B) \xrightarrow{\varphi_*} H_n(A)\oplus H_n(B) \xrightarrow{\psi_*} H_n(A+B) \xrightarrow{\partial} H_{n-1}(A\cap B) \to \cdots$$

in which the homology of the middle complex $C_\bullet(A)\oplus C_\bullet(B)$ is $H_n(A)\oplus H_n(B)$, because homology commutes with the direct sum of complexes. Now replace $H_n(A+B)$ throughout by $H_n(X)$ using the isomorphism ι_* of Stage 1. Because ι is an inclusion of subcomplexes of $C_\bullet(X)$, the map ψ followed by ι is exactly the chain map $(a,b)\mapsto a+b\in C_n(X)$, so after this substitution the map $H_n(A)\oplus H_n(B)\to H_n(X)$ is the one induced by $(a,b)\mapsto a+b$, that is, the difference of the two inclusion-induced maps $H_n(A)\to H_n(X)$ and $H_n(B)\to H_n(X)$ up to the chosen sign on φ. Likewise φ_* becomes $x\mapsto(x,-x)$ on homology, the pair of inclusion-induced maps $H_n(A\cap B)\to H_n(A)$ and $H_n(A\cap B)\to H_n(B)$. Substituting gives the exact sequence

$$\cdots\to H_n(A\cap B)\xrightarrow{\varphi} H_n(A)\oplus H_n(B)\xrightarrow{\psi} H_n(X)\xrightarrow{\partial} H_{n-1}(A\cap B)\xrightarrow{\varphi} H_{n-1}(A)\oplus H_{n-1}(B)\to\cdots$$

The sequence terminates on the right in $\cdots\to H_0(X)\to 0$, since $H_{-1}(A\cap B)=0$ forces the snake lemma's tail to end at H_0. This is the Mayer-Vietoris sequence, completing the proof.

Naturally, we also get a formally identical Mayer-Vietoris sequence for redcued homology groups, namely by augmenting the previosu short exact sequence of complexes in the natural way:

$$\begin{tikzcd}
0 \arrow[r] & C_0(A\cap B) \arrow[r] \arrow[d, "\epsilon"] & C_0(A)\oplus C_0(B)) \arrow[r] \arrow[d, "\epsilon\oplus\epsilon"] & C_0(A+B) \arrow[r] \arrow[d, "\epsilon"] & 0 \\
0 \arrow[r] & \mathbb{Z} \arrow[r] & \mathbb{Z}\oplus\mathbb{Z} \arrow[r] & \mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

If $A\cap B$ is path-connected, then we may think of Mayer-Vietoris sequences as an analog of van-Kapmen's theorem. In particular, the reduced Mayer-Vietoris sequence gives an isomorphism:

$$H_1(X)\cong\frac{H_1(A)\oplus H_1(B)}{\mathrm{im}\varphi}$$

which is exactly the abelianization of the van-Kapempen theorem, where H_1 is the abelianization of π_1 for path connected spaces (shown in 2.A of Hatcher)

(a bit on if A, B, and $A\cap B$ deformation retract onto U,V, and $U\cap V$ respectively, and how that works with CW complex, Hatcher p.150)

Example 3.4.20: Mayer-Vietoris Sequence

1. Let $X=S^n$, A,B the northern and southern hemispheres so that $A\cap B=S^{n-1}$. Then the reduced Mayer-Vietoris sequence of

$$\tilde{H}_i(A)\oplus\tilde{H}_i(B)=0$$

hence:

$$\tilde{H}_i(S^n)\cong\tilde{H}_{i-1}(S^{n-1})$$

This gives another way of calculating the homology of S^n using induction.

2. Let K be the Klein bottle, and see it as the union of two Möbius bans A and B glued at thier boundary by a homeomorphism. Then A, B, and $A \cap B$ are homotopy equivalence to circles. Then the important part of the Mayer-Vietoris sequence for the decomposition $K = A \cup B$ is

$$0 \to K_2(K) \to H_1(A \cap B) \xrightarrow{\varphi} H_1(A) \oplus H_1(B) \to H_1(K) \to 0$$

The map $\varphi : \mathbb{Z} \to \mathbb{Z} \oplus \mathbb{Z}$ is given by $1 \mapsto (2, -2)$, since the boundary circle of a Möbius band wraps twice around the core circle. Since φ is injective, we get

$$H_2(K) = 0$$

Furthermore, $H_1(K) = \mathbb{Z} \oplus \mathbb{Z}_2$, since we can choose $(1, 0)$ and $(1, -1)$ as a basis for $\mathbb{Z} \oplus \mathbb{Z}$. The higher homologys are all trivial as we saw earlier. Hence:

$$\tilde{H}_n(K) = \begin{cases} 0 & n = 0 \\ \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

3. (A generalization of Mayer-Vietoris in some cases, p. 151 Hatcher)

Finally, the relative version of Mayer-Vietoris is sometimes useful. Let

$$(X, Y) = (A \cup B, C \cup D) \qquad C \subseteq A, \quad D \subseteq B$$

Then we would get the realtive Mayer-Vietoris sequence to be:

$$\cdots \to H_n(A \cap B, C \cap D) \xrightarrow{\varphi} H_n(A, C) \oplus H_n(B, D) \xrightarrow{\psi} H_n(X, Y) \xrightarrow{\partial}\to \cdots$$

which is a good exercise to check.

Example 3.4.21: Product With Sphere
Let X be a path-connected topological space. We shall show that

$$H_k(X \times S^n) \cong H_k(X) \oplus H_{k-n}(X)$$

3.4.3 Equivalence of Simplicial and Singular Homology

We have now developed the theory of homology through singular homologies and have developed computational techniques via simplicial homologies. We now aim to show the two homology groups are isomorphic. We shall also want to draw parallel between relative homology groups, so to that end if X Is a Δ-complex with $A \subseteq X$ a subcomplex, we may define the relative groups $H_n^{\Delta}(X, A)$ give $\Delta_n(X, A) = \Delta_n(X)/\Delta_n(A)$ (Since A is the Δ complex formed by any union of simplices of X). Next, there is a canonical homomorphism $H_n^{\Delta}(X, A) \to H_n(X, A)$ induced by the chain map $\Delta_n(X, A) \to C_n(X, A)$ by sending each n-simplex of$X$ to its characteristic map $\sigma : \Delta^n \to X$. If $A = \emptyset$, the relative homology is the absolute homology.

For it, we shall require an important lemma

Lemma 3.4.22: The Five Lemma

Given the following commutative diagram:

$$\begin{tikzcd} A \arrow[r, "a"] \arrow[d, "\alpha"] & B \arrow[r, "b"] \arrow[d, "\beta"] & C \arrow[r, "c"] \arrow[d, "\gamma"] & D \arrow[r, "d"] \arrow[d, "\delta"] & E \arrow[d, "\epsilon"] \\ A' \arrow[r, "a'"] & B' \arrow[r, "b'"] & C' \arrow[r, "c'"] & D' \arrow[r, "d'"] & E' \end{tikzcd}$$

if the two rows are exact, and α, β, δ, and ϵ are isomorphisms, then so is γ.

***Proof*:**
The argument is a diagram chase, and it is the prototype for all the comparison arguments that follow. Throughout, "the rows are exact" means ker = im at each interior node (B, C, D on top and B', C', D' on the bottom), and "the diagram commutes" means each of the four squares commutes, so that primed maps applied after vertical maps agree with vertical maps applied after unprimed maps. The hypotheses are deliberately economical: only $\alpha, \beta, \delta, \epsilon$ are assumed to be isomorphisms, and in fact each half of the proof uses strictly less than that. Splitting the claim into surjectivity and injectivity of γ isolates exactly which hypotheses each direction uses.

Step 1: Surjectivity of γ (the four-lemma using β, δ surjective and ϵ injective). Let $c' \in C'$ be arbitrary; the goal is to produce $c \in C$ with $\gamma(c) = c'$. Push c' forward along the bottom row to $d' := c'(c') \in D'$. Since δ is surjective, choose $d \in D$ with $\delta(d) = c'(c')$. Apply the bottom map d' and use commutativity of the rightmost square:

$$\epsilon(d(d)) = d'(\delta(d)) = d'(c'(c')) = 0$$

the last equality because $\mathrm{im} c' = \ker d'$ by exactness of the bottom row at D'. As ϵ is injective, $d(d) = 0$, so $d \in \ker d = \mathrm{im} c$ by exactness of the top row at D; pick $\tilde{c} \in C$ with $c(\tilde{c}) = d$. The element $\tilde{c}$ need not map to c', but it is off by something killed by c'. To see this, compute, using commutativity of the third square,

$$c'(\gamma(\tilde{c})) = \delta(c(\tilde{c})) = \delta(d) = c'(c')$$

so $c'(\gamma(\tilde{c}) - c') = 0$, that is, $\gamma(\tilde{c}) - c' \in \ker c' = \mathrm{im} b'$ by exactness of the bottom row at C'. Choose $b' \in B'$ with $b'(b') = \gamma(\tilde{c}) - c'$. Since β is surjective, choose $b \in B$ with $\beta(b) = b'$, and set

$$c := \tilde{c} - b(b) \in C$$

Then, using commutativity of the second square ($\gamma \circ b = b' \circ \beta$),

$$\gamma(c) = \gamma(\tilde{c}) - \gamma(b(b)) = \gamma(\tilde{c}) - b'(\beta(b)) = \gamma(\tilde{c}) - b'(b') = \gamma(\tilde{c}) - (\gamma(\tilde{c}) - c') = c'$$

Hence c' lies in the image of γ. As c' was arbitrary, γ is surjective.

Step 2: Injectivity of γ (the four-lemma using β, δ injective and α surjective). Let $c \in C$ satisfy $\gamma(c) = 0$; the goal is $c = 0$. Push c along the top row to $d := c(c) \in D$ and use commutativity of the third square:

$$\delta(c(c)) = c'(\gamma(c)) = c'(0) = 0$$

Since δ is injective, $c(c) = 0$, so $c \in \ker c = \mathrm{im} b$ by exactness of the top row at C; choose $b \in B$ with $b(b) = c$. Its image $\beta(b)$ is not yet zero, but it is killed by b': by commutativity of the

second square,
$$b'(\beta(b)) = \gamma(b(b)) = \gamma(c) = 0$$
so $\beta(b) \in \ker b' = \mathrm{im} a'$ by exactness of the bottom row at B'. Choose $a' \in A'$ with $a'(a') = \beta(b)$. Since α is surjective, choose $a \in A$ with $\alpha(a) = a'$. Then, by commutativity of the leftmost square ($\beta \circ a = a' \circ \alpha$),
$$\beta(a(a)) = a'(\alpha(a)) = a'(a') = \beta(b)$$
and since β is injective this forces $a(a) = b$. Therefore
$$c = b(b) = b(a(a)) = 0$$
because $b \circ a = 0$ by exactness of the top row at B (the composite of two consecutive maps in an exact sequence vanishes, $\mathrm{im} a = \ker b$). Hence $\ker \gamma = 0$ and γ is injective.

Combining the two steps, γ is a bijective homomorphism, hence an isomorphism, as we sought to show. The accounting deserves a closing remark: surjectivity used β, δ surjective and ϵ injective, while injectivity used β, δ injective and α surjective. The full strength of the hypotheses (all four maps isomorphisms) is more than is strictly needed, and the two "four-lemma" halves recorded here are exactly the forms invoked when comparing two long exact sequences term by term, completing the proof.

Theorem 3.4.23: Singular And Simplicial Homology

The homomorphism $H_n^\Delta(X, A) \to H_n(X, A)$ is an isomorphism for all $n \in \mathbb{N}$ and all Δ-complex pairs (X, A)

***Proof*:**
The strategy is to prove the absolute case ($A = \emptyset$) for finite-dimensional X by induction on the dimension, comparing the skeletal long exact sequences of simplicial and singular homology through the canonical chain map and invoking the five lemma (lemma 3.4.22) on the resulting ladder. The base of the induction is the computation of the relative groups $H_n(X^k, X^{k-1})$, which turn out to be free abelian on the k-simplices in *both* theories; once this is in hand the five lemma propagates the isomorphism up the skeleta. Two further reductions, a direct-limit argument for infinite-dimensional X and a second five-lemma comparison for the relative case, complete the theorem.

Throughout, write X^k for the k-skeleton of X, the union of all simplices of dimension $\leq k$. The canonical map of the statement is induced by the chain map $\Delta_n(X) \to C_n(X)$ sending each open n-simplex e_α^n to its characteristic map $\sigma_\alpha \colon \Delta^n \to X$, a singular n-simplex; this commutes with the boundary operators because the simplicial boundary $\partial e_\alpha^n = \sum_i (-1)^i e_\alpha^n|_{[v_0,\ldots,\hat{v}_i,\ldots,v_n]}$ is precisely the singular boundary of σ_α under the identification of the i-th face of σ_α with the characteristic map of the corresponding $(n-1)$-simplex. It therefore descends to homology and, by passing to quotients, to the relative groups.

Step 1: The relative groups of a skeletal pair, simplicially. Fix $k \geq 0$. By construction $\Delta_n(X^k, X^{k-1}) = \Delta_n(X^k)/\Delta_n(X^{k-1})$ is the free abelian group on the open k-simplices when $n = k$ and is zero for $n \neq k$, since the only simplices of X^k not already in X^{k-1} are the k-simplices, and a Δ-complex has simplices in dimension n generating Δ_n. The relative simplicial boundary maps therefore vanish (their source or target is 0), so the relative simplicial homology is the relative chain group itself:

$$H_n^{\Delta}(X^k, X^{k-1}) = \begin{cases} \Delta_k(X^k) & n = k \\ 0 & n \neq k \end{cases}$$

free abelian on the k-simplices of X.

Step 2: The relative groups of a skeletal pair, singularly. The same groups are now computed in singular homology. Each k-simplex e_α^k has characteristic map $\sigma_\alpha \colon \Delta^k \to X^k$ restricting to a map $\partial\Delta^k \to X^{k-1}$, so the characteristic maps assemble into a map of pairs

$$\Phi \colon \Big(\coprod_\alpha \Delta_\alpha^k, \ \coprod_\alpha \partial\Delta_\alpha^k \Big) \longrightarrow (X^k, X^{k-1})$$

one copy $(\Delta_\alpha^k, \partial\Delta_\alpha^k)$ of $(D^k, \partial D^k)$ for each k-simplex. This Φ induces a homeomorphism of quotients $\coprod_\alpha \Delta_\alpha^k/\partial\Delta_\alpha^k \xrightarrow{\cong} X^k/X^{k-1}$, the latter being the wedge of k-spheres recorded in the cellular discussion. The pair (X^k, X^{k-1}) is a good pair (definition 3.4.10): X^{k-1} is closed in X^k and is a deformation retract of a neighbourhood, namely the union of X^{k-1} with the complements of the barycentres in the open k-simplices, which retracts radially onto X^{k-1}. The topological snake lemma for good pairs (theorem 3.4.11) then identifies the relative groups with the reduced homology of the quotient,

$$H_n(X^k, X^{k-1}) \cong \tilde{H}_n\Big(\bigvee_\alpha S_\alpha^k \Big) \cong \bigoplus_\alpha \tilde{H}_n(S_\alpha^k)$$

the second isomorphism because reduced homology takes wedges to direct sums (each S_α^k contributing a summand, via excision and the additivity of homology on the disjoint union of the disks). The disk computation $H_n(D^k, \partial D^k) \cong \mathbb{Z}$ for $n = k$ and 0 otherwise (example 3.4.5), equivalently $\tilde{H}_n(S^k) \cong \mathbb{Z}$ for $n = k$ and 0 otherwise (corollary 3.4.12), gives

$$H_n(X^k, X^{k-1}) = \begin{cases} \bigoplus_\alpha \mathbb{Z} & n = k \\ 0 & n \neq k \end{cases}$$

with one $\mathbb{Z}$ summand for each k-simplex. Comparing with Step 1, the canonical map sends the simplicial generator e_α^k to the relative singular class of σ_α, which by the disk computation is exactly the generator $\tilde{H}_k(S_\alpha^k) \cong \mathbb{Z}$ of the corresponding summand. The canonical map

$$H_n^{\Delta}(X^k, X^{k-1}) \longrightarrow H_n(X^k, X^{k-1})$$

carries a basis bijectively to a basis, hence *is an isomorphism for every n and every k*. This is the geometric heart of the theorem: in both theories the relative homology of a skeletal pair is free on the simplices, and the canonical map matches the two bases.

Step 3: The induction on skeleta for finite-dimensional X. Assume first that X is finite-dimensional and that $X = X^k$ for some k; argue by induction on k that $H_n^{\Delta}(X^k) \to H_n(X^k)$ is an isomorphism for all n. The base case $k = 0$ is immediate: X^0 is a discrete set of points, and both $H_0^{\Delta}(X^0)$ and $H_0(X^0)$ are free abelian on those points with the canonical map the identity on generators, while all higher groups vanish. For the inductive step, the inclusion of pairs gives a commuting ladder relating the long exact sequence of the simplicial pair (X^k, X^{k-1})

(the algebraic long exact sequence of theorem 3.4.3 applied to $0 \to \Delta_\bullet(X^{k-1}) \to \Delta_\bullet(X^k) \to \Delta_\bullet(X^k, X^{k-1}) \to 0$) to that of the singular pair (definition 3.4.2):

$$\begin{array}{ccccccccc} H^\Delta_{n+1}(X^k, X^{k-1}) & \longrightarrow & H^\Delta_n(X^{k-1}) & \longrightarrow & H^\Delta_n(X^k) & \longrightarrow & H^\Delta_n(X^k, X^{k-1}) & \longrightarrow & H^\Delta_{n-1}(X^{k-1}) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ H_{n+1}(X^k, X^{k-1}) & \longrightarrow & H_n(X^{k-1}) & \longrightarrow & H_n(X^k) & \longrightarrow & H_n(X^k, X^{k-1}) & \longrightarrow & H_{n-1}(X^{k-1}) \end{array}$$

The squares commute by proposition 3.4.4: the canonical chain map defines a morphism between the two short exact sequences of chain complexes (it is natural for inclusions of pairs and intertwines the simplicial and singular boundaries, as noted at the outset), and the proposition gives the ladder, including the connecting squares. The two outer vertical maps adjacent to the centre, those on the relative groups, are isomorphisms by Step 2, and the two vertical maps on $H^\Delta_*(X^{k-1}) \to H_*(X^{k-1})$ are isomorphisms by the inductive hypothesis. The five lemma (lemma 3.4.22) applied to the five-term window centred at $H^\Delta_n(X^k) \to H_n(X^k)$, with the four flanking maps isomorphisms, forces the central map $H^\Delta_n(X^k) \to H_n(X^k)$ to be an isomorphism. Since n was arbitrary, the inductive step is complete, and the theorem holds for all finite-dimensional X (in the absolute case).

Step 4: The passage to infinite-dimensional X. Suppose now X is arbitrary. The point is that homology, simplicial and singular alike, is detected on finite-dimensional skeleta, because both simplicial and singular chains are *compactly supported*: a chain is a finite sum of (characteristic maps of) simplices, and the image of each simplex is a compact subset meeting only finitely many open simplices of X, hence contained in some skeleton X^k. Concretely, a class in $H^\Delta_n(X)$ or $H_n(X)$ is represented by a cycle whose finitely many simplices all lie in X^k for k large enough, and a boundary witnessing triviality likewise lies in some X^m. Therefore the natural maps

$$\varinjlim_k H^\Delta_n(X^k) \xrightarrow{\cong} H^\Delta_n(X), \qquad \varinjlim_k H_n(X^k) \xrightarrow{\cong} H_n(X)$$

over the skeletal filtration are isomorphisms in both theories: surjectivity because every cycle is supported in some X^k, and injectivity because every bounding chain is supported in some X^m. (In the singular case this is the statement that singular homology commutes with the directed union $X = \bigcup_k X^k$, the directed system having the weak topology with respect to its skeleta.) The canonical map is compatible with the filtration, so it induces a map of directed systems; by Step 3 each map $H^\Delta_n(X^k) \to H_n(X^k)$ is an isomorphism, and the colimit of isomorphisms is an isomorphism. Passing to the colimit gives that $H^\Delta_n(X) \to H_n(X)$ is an isomorphism for every n, removing the finite-dimensionality hypothesis.

Step 5: The relative case. Finally, let (X, A) be an arbitrary Δ-complex pair, $A \subseteq X$ a subcomplex. The absolute case just proved applies to both X and A. The simplicial relative chain complex $\Delta_\bullet(X, A) = \Delta_\bullet(X)/\Delta_\bullet(A)$ sits in the short exact sequence $0 \to \Delta_\bullet(A) \to \Delta_\bullet(X) \to \Delta_\bullet(X, A) \to 0$, and the singular one in $0 \to C_\bullet(A) \to C_\bullet(X) \to C_\bullet(X, A) \to 0$; the canonical chain map is a map between these short exact sequences. Their associated long exact sequences (theorem 3.4.3) form a commuting ladder (proposition 3.4.4)

$$\begin{array}{ccccccccc} H^\Delta_n(A) & \longrightarrow & H^\Delta_n(X) & \longrightarrow & H^\Delta_n(X, A) & \longrightarrow & H^\Delta_{n-1}(A) & \longrightarrow & H^\Delta_{n-1}(X) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ H_n(A) & \longrightarrow & H_n(X) & \longrightarrow & H_n(X, A) & \longrightarrow & H_{n-1}(A) & \longrightarrow & H_{n-1}(X) \end{array}$$

in which the four vertical maps flanking the central one, namely those on $H_*^\Delta(A) \to H_*(A)$ and $H_*^\Delta(X) \to H_*(X)$, are isomorphisms by Steps 3 and 4. The five lemma (lemma 3.4.22) then forces the central map $H_n^\Delta(X, A) \to H_n(X, A)$ to be an isomorphism for every n. As (X, A) was an arbitrary Δ-complex pair, this establishes the theorem in full, completing the proof.

Some interesting consequences can immediately be deduced. A finitely generated abelian group must have finitely generated subgroups[3], hence if $\Delta_n(X)$ is finitely generated, so must $H_n^\Delta(X)$ be, hence $H_n(X)$ would be too.

3.4.4 Retractions

Next, we look at how the induced map form retraction behaves. It turns to induce really simple group structures, as perahps can be expected given its a section map. We first require a lemma.

Lemma 3.4.24: Splitting Lemma

Let $0 \to A \to B \to C \to 0$ be a short exact sequence of abliean groups. Then the following are equivalent:

1. There is a splitting homomorphism $p : B \to A$ (i.e. a section map)
2. there is a splitting homomorphism $s : C \to B$ (i.e. a section map)
3. There is an isomorphism $B \cong A \oplus C$

If any of these are satisfied, the exact sequence is said to split

***Proof*:**
See any book on projective modules, for ex. Dummit and Foote, or EYTNKA Algebra.

Proposition 3.4.25: Retraction And Split Sequence

Let $r : X \to A$ be a retraction. Then:

$$H_n(X) \cong H_n(A) \oplus H_n(X, A)$$

***Proof*:**

3.5 Application and Computation of Homology

We shall now take some time to give some applications and computations of important spaces, in particular we shall take the time to develop new methods of calculating homologies of CW complexes. For these computations, we shall explore the notion of *degree* which generalizes the notion of winding number and brings the idea of Euler Characteristic. Next define Cellular homology for CW complex

[3]This need not be the case for non-abelian groups

and show it's isomorphic to singular (and hence simplicial) homology groups. We then briefly mention the induced homology of a retraction, and prove the important theorem of *invariance of domain*.

3.5.1 Degree

In complex analysis, the winding number of a function gives a lot of information about the number of roots as well as the values of path integrals. The following is a generalization of this concept:

Definition 3.5.1: Degree

A continuous map $f : S^n \to S^n$ induces a map $f_* : H_n(S^n) \to H_n(S^n)$ which must be of the form $f_*(\alpha) = d\alpha$ for some $d \in \mathbb{Z}$ depending on f. Then d is called the *degree* of f, and is denoted $\deg f$.

Proposition 3.5.2: Properties Of Degree

1. $\deg \mathrm{id} = 1$
2. If s is not surjective, then $\deg f = 0$. Note the converse is not true
3. $f \simeq g$, then $\deg f = \deg g$ (the converse is more difficult to prove, see [14, Corollary 4.25][a])
4. $\deg fg = \deg f \deg g$. Hence $\deg f = \pm 1$, f is a homotopy equivalence since $fg \simeq \mathrm{id}$ implies $\deg f \deg g = \deg \mathrm{id} = 1$
5. $\deg f = -1$ if f is a reflect onto S^n, fixing the points in a subsphere S^{n-1}, and inerchanging the two complementary hemispheres
6. The antipodal points $-\mathrm{id} : S^n \to S^n$ given by $x \mapsto -x$ has degree $(-1)^{n+1}$
7. If $f : S^n \to S^n$ has no fixed points, then $\deg f = (-1)^{n+1}$, in particualr it is antipodal.

[a] This was first proved by Hopf in 1925

***Proof*:**

1. Immediate since $\mathrm{id}_* = \mathrm{id}$
2. If $x_0 \in S^n - f(S^n)$, then f can be factored as
$$S^n \to (S^n - \{x_0\}) \hookrightarrow S^n$$
and $H_n(S^n - \{x_0\}) = 0$ since $S^n - \{x_0\}$ is contractible, hence $f_* = 0$
3. $f \simeq g$ then $f_* = g_*$ so $\deg f = \deg g$
4. Consequence of $(fg)_* = f_* g_*$
5. We can give S^n a Δ-compelx structure with these two hemisphers as its two n-simplcies Δ_1^n and D_2^n and n-chains $\Delta_1^n - \Delta_2^n$ represetns generators of $H_n(S^n)$, as seen in a preivou example. THus, the relfection interchangign Δ_1^n and Δ_2^n send this generator to its negative

6. this maps is a composition of $n+1$ reflections, each chagnign the sign of one coordiante in $\mathbb{R}^{n+1}$

7. If $f(x) \neq x$, then the line segment from $f(x)$ to $-x$ defined by $t \mapsto (1-t)(f(x)-tx)$ does not pass through the origin. Hence if f has no fixed poitns, the formula

$$f_t(x) = \frac{(1-t)f(x)-tx}{|(1-t)f(x)-tx|}$$

defines a homotopy from f to the antipodal points. The antipodal maps have no fixed points, so the fact that maps without fixed oints are homotopic to the antipodal points are a converse of that

For (2), can you think of a counter-example for the converse? Degree's immediately give a few interesting and famous results:

Theorem 3.5.3: Hairy ball theorem

Let $X = S^n$. Then X has a continuous field of nonzero tangent vectors if and only if n is odd

***Proof*:**
Let $V : S^n \to TS^n$ where $x \mapsto v(x)$ be the vector-field map. Thinking of $v(x)$ as a vector at the origin and as x as a vector, we see that x and $v(x)$ are orthogonal in $\mathbb{R}^{n+1}$.

Now, if $v(x) \neq 0$ for all x, then

$$\frac{v(x)}{|v(x)|}$$

is well-defined. Replacing $v(x)$ by this normalization, the vectors

$$\cos(t)x + \sin(t)v(x)$$

are in teh unit circle in the plane spanned by x and $v(x)$. Taking t from 0 to π, we get a homotopy:

$$f_t(x) = \cos(t)x + \sin(t)v(x)$$

from S^n to the antipodal map $-\mathrm{id}$. But then:

$$(-1)^{n+1} = \deg(-\mathrm{id}) = \deg(\mathrm{id}) = 1$$

implying n is odd. Conversely if n is odd so that $n = 2k-1$, we can take

$$v(\vec{x}) = (-x_2, x_1, ..., -x_{2k}, x_{2k-1})$$

Then $v(x)$ is orthogonal to x, so v is a tangent vector field on S^n and $|v(x)| = 1$ for all $x \in S^n$.

Proposition 3.5.4: Free Action on Sphere

$\mathbb{Z}/2\mathbb{Z}$ is the only nontrivial group that can act freely on S^n if n is even

***Proof*:**
Let $G \curvearrowright S^n$ so that there is a homomorphism $\rho : G \to \mathrm{Aut}_{\mathbf{Top}}(S^n)$. Since homeomorphisms have degree ± 1, an action by a group G on S^n induces a "degree" function $d : G \to \{\pm 1\}$, which is a homomorphism since $\deg fg = \deg f \deg g$. If the action is free, d sends nontrivial elements of G to $(-1)^{n+1}$ by property 7 above. But then, when n is even, d has trivial kernel, hence $G \subseteq \mathbb{Z}_2$.

We now proceed how to compute the degree of a map in the most common case, that is when for some $y \in S^n$, $f^{-1}(y)$ is finite. Say $x_1, x_2, ..., x_n$ are the points with disjoint neighbourhoods $U_1, U_2, ..., U_n$. Then $f(U_i - x_i) \subseteq V - y$ for each i, and we have:

$$\begin{tikzcd}
& H_n(U_i, U_i - x_i) \arrow[r, "f_*"] \arrow[dl, "\cong"'] \arrow[d, "k_i"] & H_n(V, V - y) \arrow[d, "\cong"] \\
H_n(S^n, S^n - x_i) & H_n(S^n, S^n - f^{-1}(y)) \arrow[l, "\pi_i"'] \arrow[r, "f_*"] & H_n(S^n, S^n - x_i) \\
& H_n(S^n) \arrow[ul, "\cong"] \arrow[u, "j"] \arrow[r, "f_*"] & H_n(S^n) \arrow[u, "\cong"']
\end{tikzcd} \tag{3.11}$$

where k_i and π_i are induced by inclusion, and all triangles and square commute. The two isomorphisms in the upper half are given by excision, the lower two isomorphism from exact sequences of pairs. Given these four isomorphisms, the top two groups in the diagram being identified as $H_n(S^n) \cong \mathbb{Z}$ and the top homomorphism f_* becomes multiplication by an integer, and is given a name:

Definition 3.5.5: Local Degree

Given the above setup, the *local degree* at x_i is the degree of the map f_* and is written as:

$$\deg f|_{x_i}$$

For example, if f is a homeomorphism, then $f^{-1}(y) = x$ is a single point, and so $\deg f|_{x_i} = \deg f = \pm 1$. More generally, if f maps to each U_i homeomorphically onto V (if f is a local homeomorphism), then we have

$$\deg f|_{x_i} = \pm 1 \qquad 1 \leq i \leq n$$

where the sign is usually rather easy to determine. From this, we can get the general degree:

Proposition 3.5.6: Computing Degree

Given the setup as above, we have:

$$\deg f = \sum_i (\deg f|_{x_i})$$

***Proof*:**
In the diagram at equation (3.11), the term $H_n(S^n, S^n - f^{-1}(y))$ is the direct sum of $H_n(U_i, U_i - x_i) \cong \mathbb{Z}$ and $k - i$, the inclusion of the ith summand. The π_i is the projection onto the ith

summand since the upper triangle commutes and

$$\pi_i k_i = 0 \qquad j \neq i$$

since $\pi_i k_i$ factors through $H_n(U_j, U_j) = 0$. Identifying the outer group in the diagram with $\mathbb{Z}$, commutativity of the lower triangles gives us that

$$\pi_{ij}(1) = 1 \implies j(1) = (1, ..., 1) = \sum_i k_i(1)$$

Commutativity of the upper square gives that the middle f_* takes $k_i(1)$ to $\deg f|_{x_i}$, thus

$$\sum_i k_i(1) = j(1) \mapsto \sum_i \deg f|_{x_i}$$

Commutativity of the lower square finally gives us:

$$\deg f = \sum_i \deg f|_{x_i}$$

as we sought to show.

Proposition 3.5.7: Suspension And Degree

Let $f : S^n \to S^n$ be a continuous map, $Sf : S^{n+1} \to S^{n+1}$ its suspension. Then:

$$\deg Sf = \deg f$$

***Proof*:**
The statement follows once the suspension is shown to induce, naturally in X, an isomorphism between the reduced homology of X and that of SX shifted up one degree; applying this to S^n, where $SS^n = S^{n+1}$, converts the equality of degrees into the equality of two integers read off the same isomorphism.

Step 1: The cone decomposition of the suspension. Recall from definition 1.2.12 that SX is the quotient of $X \times I$ collapsing $X \times \{0\}$ to a point p_- and $X \times \{1\}$ to a point p_+. Write $q\colon X \times I \to SX$ for the quotient map and set

$$C_+ = q\big(X \times (0, 1]\big), \qquad C_- = q\big(X \times [0, 1)\big)$$

so that $SX = C_+ \cup C_-$. Each piece is a cone on X: the formula $h_+\big(q(x, s), t\big) = q\big(x, t + (1-t)s\big)$ is well-defined on C_+ (it fixes the cone point $p_+ = q(X \times \{1\})$ and is continuous because it descends from a continuous map on $X \times (0, 1] \times I$) and gives a deformation retraction of C_+ onto p_+, so C_+ is contractible; symmetrically C_- is contractible. The interiors $q(X \times (0, 1])$ and $q(X \times [0, 1))$ already cover SX, since their union is $q(X \times [0, 1]) = SX$ and each is open (its preimage under q is the saturated open set $X \times (0, 1]$, respectively $X \times [0, 1)$). Thus (C_+, C_-) satisfies the hypothesis of theorem 3.4.19: SX is the union of the interiors of C_+ and C_-.

Step 2: The suspension isomorphism. The intersection $C_+ \cap C_- = q(X \times (0, 1))$ deformation retracts onto the equatorial copy $X_{1/2} = q(X \times \{1/2\})$ via $\big(q(x, s), t\big) \mapsto q\big(x, (1-t)s + t/2\big)$, and $X_{1/2}$ is homeomorphic to X through $x \mapsto q(x, 1/2)$; hence $C_+ \cap C_-$ is homotopy equivalent to X. The reduced Mayer-Vietoris sequence for the decomposition $SX = C_+ \cup C_-$ reads, in each degree,

$$\tilde{H}_{n+1}(C_+) \oplus \tilde{H}_{n+1}(C_-) \to \tilde{H}_{n+1}(SX) \xrightarrow{\partial} \tilde{H}_n(C_+ \cap C_-) \to \tilde{H}_n(C_+) \oplus \tilde{H}_n(C_-)$$

Since C_+ and C_- are contractible, $\tilde{H}_k(C_\pm) = 0$ for all k, so the two outer terms vanish and the connecting homomorphism ∂ is an isomorphism. Composing with the isomorphism $\tilde{H}_n(C_+ \cap C_-) \cong \tilde{H}_n(X)$ induced by the homotopy equivalence of definition 1.2.12's intersection with X, define

$$s_X \colon \tilde{H}_{n+1}(SX) \xrightarrow{\cong} \tilde{H}_n(X)$$

the *suspension isomorphism*, an isomorphism in every degree n.

Step 3: Naturality. A continuous map $f \colon X \to Y$ suspends to $Sf \colon SX \to SY$, the map induced by $f \times \mathrm{id}_I$ on the quotients. It carries C_+^X into C_+^Y and C_-^X into C_-^Y, hence $C_+^X \cap C_-^X$ into $C_+^Y \cap C_-^Y$, and on the equatorial copies it restricts to f under the identifications $X_{1/2} \cong X$, $Y_{1/2} \cong Y$. The connecting homomorphism of the Mayer-Vietoris sequence is natural for maps of such triads, and the homotopy equivalences of Step 2 are compatible with f, so the square

$$\begin{tikzcd}
\tilde{H}_{n+1}(SX) \arrow[r, "s_X"] \arrow[d, "(Sf)_*"'] & \tilde{H}_n(X) \arrow[d, "f_*"] \\
\tilde{H}_{n+1}(SY) \arrow[r, "s_Y"'] & \tilde{H}_n(Y)
\end{tikzcd}$$

commutes; that is, $f_* \circ s_X = s_Y \circ (Sf)_*$.

Step 4: Reading off the degree. Now take $X = Y = S^n$ and the given map $f \colon S^n \to S^n$, so that $Sf \colon S^{n+1} \to S^{n+1}$ since $SS^n = S^{n+1}$. By corollary 3.4.12 we have $\tilde{H}_n(S^n) \cong \mathbb{Z}$ and $\tilde{H}_{n+1}(S^{n+1}) \cong \mathbb{Z}$, and for $n \geq 1$ the reduced and unreduced groups agree, so the degree of definition 3.5.1 is multiplication by the same integer whether computed on H_n or on $\tilde{H}_n$: $f_* = \deg f$ on $\tilde{H}_n(S^n) \cong \mathbb{Z}$ and $(Sf)_* = \deg Sf$ on $\tilde{H}_{n+1}(S^{n+1}) \cong \mathbb{Z}$. Specializing the commuting square of Step 3 to $X = Y = S^n$ gives

$$\deg f \cdot s_{S^n} = f_* \circ s_{S^n} = s_{S^n} \circ (Sf)_* = \deg Sf \cdot s_{S^n}$$

as endomorphisms of the group $\tilde{H}_{n+1}(S^{n+1}) \cong \mathbb{Z}$, where $s_{S^n} \colon \tilde{H}_{n+1}(S^{n+1}) \to \tilde{H}_n(S^n)$ is the suspension isomorphism. Since s_{S^n} is an isomorphism between infinite cyclic groups, it is injective and nonzero, so the two integers acting through it must agree, namely $\deg Sf = \deg f$.

For $n = 0$, the map $f \colon S^0 \to S^0$ acts on $\tilde{H}_0(S^0) \cong \mathbb{Z}$ and its degree is again the integer by which f_* multiplies; the suspension isomorphism $s_{S^0} \colon \tilde{H}_1(S^1) \to \tilde{H}_0(S^0)$ and the same square give $\deg Sf = \deg f$ verbatim, so the identity holds in every dimension, completing the proof.

Example 3.5.8: Computing Degree

1. For a map $f : S^n \to S^n$, let $q : S^n \to \bigvee_k S^n$ be the quotient map given by collapsing teh complement of k disjoint open balls B_i in S^n to a point, and let $\pi : \bigvee_k S^n \to S^n$ identify all the summands to a single sphere. Then take $f = \pi \circ q$. Then teh local degree of f at x_i is ± 1 since f i a homeomorphism near x_i. By precomposing p iwth reflection of the summands of $\bigvee_k S^n$ if necessary, we can make each loca ldegree either 1 or-1 giving us a map with

degree $\pm k$

2. Let $n = 1$ so that we have S^1. Show that the map $f(z) = z^k$ has degree k for $k \in \mathbb{Z}$. Since suspension preserve degrees, another way of getting a degree k map from S^n to S^n is by taking repeated suspensions of the map $z \mapsto z^k$.

Note that for a map $f : S^n \to S^n$, the suspension Sf maps only 1 point to each of the poles of S^{n+1}, hence the local degree of Sf at each point must equal the global degree of Sf. hence, the local degree of a map can be any integer if $n \geq 2$, just like for the global degree

3.5.2 Cellular Homology: Equivalent to Singular

We now move onto computing the homology groups of cellular complex. These shall take advantage of the above degree calculations. Before computing, we establish some important preliminary results:

Lemma 3.5.9: Cellular Homology Lemma

Let X be a CW complex. Then:

1. $H_k(X^n, X^{n-1}) = 0$ for $k \neq n$ and a free abelian group for $k = n$, with basis in injective correspondence with the n-cells of X

$$H_n(X^n, X^{n-1}) = \mathbb{Z}^m$$

2. $H_k(X^n) = 0$ for all $k > n$. In particular, If X is finite-dimensional, then $H_k(X) = 0$ for $k > \dim X$

3. The map $H_k(X^n) \to H_k(X)$ induced by the inclusion $X^n \hookrightarrow X$ is an isomorphism for $k < n$, and surjects for $k = n$

$$\begin{aligned} H_k(X^n) &\cong H_k(X) & \qquad k < n \\ H_n(X^n) &\twoheadrightarrow H_n(X) & \qquad k = n \end{aligned}$$

Note that X can be finite or infinite dimensional.

***Proof*:**

1. The pair (X^n, X^{n-1}) is a good pair (definition 3.4.10): the $(n-1)$-skeleton is closed in X^n and is a deformation retract of the neighbourhood obtained by deleting the centre point of each open n-cell and retracting radially outward to the cell's boundary. The quotient X^n/X^{n-1} is the wedge sum $\bigvee_\alpha S^n_\alpha$ of n-spheres, one for each n-cell e^n_α of X, since collapsing X^{n-1} collapses the attaching image ∂e^n_α of every n-cell to the common basepoint. By the topological snake lemma for good pairs (theorem 3.4.11) and its absolute form (corollary 3.4.9),

$$H_k(X^n, X^{n-1}) \cong \tilde{H}_k(X^n/X^{n-1}) = \tilde{H}_k\left(\bigvee_\alpha S^n_\alpha\right)$$

The pairs $(S^n_\alpha, \text{basepoint})$ are good, so corollary 3.4.16 splits the reduced homology of the wedge as a direct sum,

$$\tilde{H}_k\left(\bigvee_\alpha S^n_\alpha\right) \cong \bigoplus_\alpha \tilde{H}_k(S^n_\alpha)$$

and $\tilde{H}_k(S^n) \cong \mathbb{Z}$ for $k = n$ and 0 otherwise by corollary 3.4.12. Hence $H_k(X^n, X^{n-1}) = 0$ for $k \neq n$, while for $k = n$ it is the free abelian group $\bigoplus_\alpha \mathbb{Z}$ with one $\mathbb{Z}$-summand for each n-cell. The summand attached to e^n_α is generated by the image of a chosen generator of $\tilde{H}_n(S^n_\alpha)$, which gives the asserted injective correspondence between a basis and the n-cells, so $H_n(X^n, X^{n-1}) \cong \mathbb{Z}^m$ when there are m such cells (and a free abelian group of the appropriate rank in general).

2. Apply the long exact sequence of the pair (X^n, X^{n-1}) from the algebraic snake lemma (theorem 3.4.3),

$$\cdots \to H_{k+1}(X^n, X^{n-1}) \to H_k(X^{n-1}) \to H_k(X^n) \to H_k(X^n, X^{n-1}) \to \cdots$$

By part (1) the relative term $H_k(X^n, X^{n-1})$ vanishes unless $k = n$. Fix k and suppose $n \neq k$ and $n \neq k+1$: then both $H_k(X^n, X^{n-1})$ and $H_{k+1}(X^n, X^{n-1})$ vanish, so exactness forces the inclusion-induced map to be an isomorphism,

$$H_k(X^{n-1}) \xrightarrow{\cong} H_k(X^n) \qquad n \notin \{k, k+1\}$$

Reading these isomorphisms along the skeletal filtration, the only places where the map $H_k(X^{n-1}) \to H_k(X^n)$ can fail to be an isomorphism are $n = k$ and $n = k+1$. Thus for k fixed the sequence of inclusion-induced maps

$$H_k(X^0) \to H_k(X^1) \to \cdots \to H_k(X^{k-1}) \to H_k(X^k) \to H_k(X^{k+1}) \to \cdots$$

consists of isomorphisms except possibly at $H_k(X^{k-1}) \to H_k(X^k)$ (where the relative term $H_k(X^k, X^{k-1})$ sits to the right and may obstruct surjectivity) and at $H_k(X^k) \to H_k(X^{k+1})$ (where $H_{k+1}(X^{k+1}, X^k)$ sits to the left and may obstruct injectivity). In particular,

$$H_k(X^{k-1}) \cong H_k(X^{k-2}) \cong \cdots \cong H_k(X^0)$$

Now assume $k > 0$. The space X^0 is a discrete set of 0-cells, so $H_k(X^0) = 0$ for $k > 0$, whence $H_k(X^{k-1}) = 0$. This already delivers the vanishing claimed in part (2). Fix n with $k > n$, equivalently $n \leq k-1$, and run the inclusion-induced maps from X^n up to X^{k-1},

$$H_k(X^n) \to H_k(X^{n+1}) \to \cdots \to H_k(X^{k-1})$$

Each map here is $H_k(X^{m-1}) \to H_k(X^m)$ for some m with $n+1 \leq m \leq k-1$, so $m < k$ and in particular $m \notin \{k, k+1\}$; by the display $H_k(X^{m-1}) \cong H_k(X^m)$ valid for $m \notin \{k, k+1\}$, every map in the chain is an isomorphism. The terminal group $H_k(X^{k-1})$ has just been shown to vanish, so every group in the chain is zero, and in particular $H_k(X^n) = 0$. Hence $H_k(X^n) = 0$ for all $k > n$. If X is finite-dimensional with $\dim X = d$, then $X = X^d$, and taking $n = d$ gives $H_k(X) = H_k(X^d) = 0$ for all $k > d$.

3. The inclusion $X^n \hookrightarrow X$ factors through every higher skeleton, and we treat the finite- and infinite-dimensional cases in turn.

Finite-dimensional X. Suppose $\dim X = d < \infty$, so $X = X^d$. For $k < n$ the isomorphisms of part (2) read off the right end of the filtration give

$$H_k(X^n) \cong H_k(X^{n+1}) \cong \cdots \cong H_k(X^d) = H_k(X)$$

since each map $H_k(X^m) \to H_k(X^{m+1})$ with $m \geq n > k$ avoids both exceptional positions (the obstructing relative terms $H_k(X^{m+1}, X^m)$ and $H_{k+1}(X^{m+1}, X^m)$ vanish for $m \geq n > k$ by part (1)). For $k = n$ the map $H_n(X^n) \to H_n(X^{n+1})$ is the one possibly failing injectivity but is surjective, because the term to its right, $H_n(X^{n+1}, X^n)$, vanishes by part (1) ($n \neq n+1$); composing the surjection $H_n(X^n) \twoheadrightarrow H_n(X^{n+1})$ with the isomorphisms $H_n(X^{n+1}) \cong \cdots \cong H_n(X^d) = H_n(X)$ for the higher skeleta (each map there has $k = n < m$, hence is an isomorphism) shows $H_n(X^n) \twoheadrightarrow H_n(X)$.

Infinite-dimensional X. When X is infinite-dimensional the chain $H_k(X^n) \to H_k(X^{n+1}) \to \cdots$ no longer terminates, and the comparison with $H_k(X)$ rests on the fact that singular homology is detected on finite skeleta. A singular simplex $\sigma\colon \Delta^j \to X$ has compact image, and a compact subset of a CW complex meets only finitely many open cells, hence lies in some skeleton X^m; consequently any singular chain, being a finite sum of simplices, is supported in a single skeleton. Therefore every cycle representing a class in $H_k(X)$ lies in some $H_k(X^m)$, and every chain witnessing that such a cycle bounds lies in some $H_k(X^{m'})$. This is exactly the statement that the natural map from the colimit over the skeletal filtration is an isomorphism,

$$\varinjlim_{m} H_k(X^m) \xrightarrow{\cong} H_k(X)$$

surjective because each class is supported in some skeleton and injective because each bounding relation is supported in some skeleton (this is the same compact-support argument used for the comparison of cellular and singular homology in section 3.5, where the colimit-over-skeleta isomorphism for singular homology under the weak topology is established; the purely algebraic input, that a filtered colimit of abelian groups is computed termwise and preserves exactness, is [10]). The colimit is computed along the maps $H_k(X^m) \to H_k(X^{m+1})$. By part (2), for $m \geq n > k$ in the case $k < n$ every such map is an isomorphism, so the colimit stabilizes at $H_k(X^n)$ and the composite $H_k(X^n) \to \varinjlim_m H_k(X^m) \cong H_k(X)$ is an isomorphism. For $k = n$ the first map $H_n(X^n) \to H_n(X^{n+1})$ is surjective and the subsequent maps $H_n(X^m) \to H_n(X^{m+1})$ for $m \geq n+1$ are isomorphisms (each has $k = n < m$), so the colimit is a quotient of $H_n(X^n)$ and the composite $H_n(X^n) \to H_n(X)$ is surjective. Thus in both the finite- and infinite-dimensional cases the inclusion induces an isomorphism for $k < n$ and a surjection for $k = n$, completing the proof.

By using the above lemma, we get the following commutative diagram:

$$\begin{tikzcd}
 & & & & 0 & & \\
 & 0 \arrow[dr] & & H_n(X^{n+1}) \cong H_n(X) \arrow[ur] & & & \\
 & & H_n(X^n) \arrow[ur] \arrow[dr, "j_n"] & & & & \\
\cdots \arrow[r] & H_{n+1}(X^{n+1}, X^n) \arrow[ur, "\partial_{n+1}"] \arrow[rr, "d_{n-1}"] & & H_n(X^n, X^{n-1}) \arrow[rr, "d_n"] \arrow[dr, "\partial_n"] & & H_{n-1}(X^{n-1}, X^{n-2}) \arrow[r] & \cdots \\
 & & & & H_{n-1}(X^{n-1}) \arrow[ur, "j_{n-1}"] & & \\
 & & & 9 \arrow[ur] & & &
\end{tikzcd}$$

where $d_{n+1} = j_n \partial_{n+1}$ and $d_n = j_{n-1}\partial_n$. These are the "relative boundary maps" of ∂_{n+1} and ∂_n;

notice that $d_n d_{n+1} = 0$. We thus give a name to the horizontal row:

Definition 3.5.10: Cellular Chain Complex

In the above diagram, the chain complex is called the *cellular chain complex*.

Hence, it has a homology group, let's label it as $H_n^{CW}(X)$. The key fact is the following:

Theorem 3.5.11: Cellular Homology

$$H_n^{CW}(X) \cong H_n(X)$$

***Proof*:**
The cellular chain complex (definition 3.5.10) has n-th group $H_n(X^n, X^{n-1})$ and boundary maps $d_n = j_{n-1}\partial_n$, where $\partial_n \colon H_n(X^n, X^{n-1}) \to H_{n-1}(X^{n-1})$ is the connecting map of the pair (X^n, X^{n-1}) and $j_{n-1} \colon H_{n-1}(X^{n-1}) \to H_{n-1}(X^{n-1}, X^{n-2})$ is the map in the long exact sequence of the pair (X^{n-1}, X^{n-2}). By definition

$$H_n^{CW}(X) = \frac{\ker d_n}{\mathrm{im} d_{n+1}}$$

so the claim amounts to identifying this subquotient of $H_n(X^n, X^{n-1})$ with $H_n(X)$. The argument reads both $\ker d_n$ and $\mathrm{im} d_{n+1}$ off the diagram preceding definition 3.5.10, whose commutativity and exactness along its diagonals are exactly the content of lemma 3.5.9 and the long exact sequences of the pairs (X^n, X^{n-1}).

Step 1: The maps j_n are injective and $\mathrm{im} d_{n+1} = \mathrm{im}(j_n \partial_{n+1})$. Fix n and consider the long exact sequence of the pair (X^n, X^{n-1}) (theorem 3.4.3 applied to the short exact sequence of relative chains, definition 3.4.2):

$$H_n(X^{n-1}) \to H_n(X^n) \xrightarrow{j_n} H_n(X^n, X^{n-1}) \xrightarrow{\partial_n} H_{n-1}(X^{n-1})$$

By lemma 3.5.9(2) the group $H_n(X^{n-1})$ vanishes, since $n > n-1$. Exactness at $H_n(X^n)$ then forces $\ker j_n = \mathrm{im}\big(H_n(X^{n-1}) \to H_n(X^n)\big) = 0$, so

$$j_n \colon H_n(X^n) \hookrightarrow H_n(X^n, X^{n-1})$$

is injective. The same applies one degree up: j_{n+1} is injective. Now $d_{n+1} = j_n \partial_{n+1}$, so $\mathrm{im} d_{n+1} = j_n\big(\mathrm{im}\partial_{n+1}\big)$ inside $H_n(X^n, X^{n-1})$, where $\partial_{n+1} \colon H_{n+1}(X^{n+1}, X^n) \to H_n(X^n)$ is the connecting map of (X^{n+1}, X^n).

Step 2: Computing $\mathrm{im} d_{n+1}$. Exactness of the long exact sequence of (X^{n+1}, X^n) at $H_n(X^n)$,

$$H_{n+1}(X^{n+1}, X^n) \xrightarrow{\partial_{n+1}} H_n(X^n) \to H_n(X^{n+1})$$

identifies $\mathrm{im}\partial_{n+1}$ with the kernel of the inclusion-induced map $H_n(X^n) \to H_n(X^{n+1})$. Applying the injective j_n of Step 1 to this equality of subgroups of $H_n(X^n)$ gives

$$\mathrm{im}d_{n+1} = j_n\big(\mathrm{im}\partial_{n+1}\big) = j_n\Big(\ker\big(H_n(X^n) \to H_n(X^{n+1})\big)\Big)$$

By lemma 3.5.9(3) the inclusion induces an isomorphism $H_n(X^{n+1}) \cong H_n(X)$, and the surjection $H_n(X^n) \twoheadrightarrow H_n(X)$ of part (3) factors through $H_n(X^{n+1})$, so the kernel of $H_n(X^n) \to H_n(X^{n+1})$ coincides with the kernel of the surjection $H_n(X^n) \to H_n(X)$. Writing $q\colon H_n(X^n) \twoheadrightarrow H_n(X)$ for that surjection, we have recorded

$$\mathrm{im}d_{n+1} = j_n(\ker q)$$

Step 3: Computing $\ker d_n$. Here $d_n = j_{n-1}\partial_n$ with $\partial_n\colon H_n(X^n, X^{n-1}) \to H_{n-1}(X^{n-1})$ and $j_{n-1}\colon H_{n-1}(X^{n-1}) \to H_{n-1}(X^{n-1}, X^{n-2})$. By Step 1 (one degree lower) the map j_{n-1} is injective, so $d_n(x) = j_{n-1}(\partial_n x) = 0$ if and only if $\partial_n x = 0$. Therefore

$$\ker d_n = \ker \partial_n$$

Exactness of the long exact sequence of (X^n, X^{n-1}) at $H_n(X^n, X^{n-1})$,

$$H_n(X^n) \xrightarrow{j_n} H_n(X^n, X^{n-1}) \xrightarrow{\partial_n} H_{n-1}(X^{n-1})$$

gives $\ker \partial_n = \mathrm{im}j_n$. Combining,

$$\ker d_n = \mathrm{im}j_n = j_n\big(H_n(X^n)\big)$$

where the last equality uses that j_n is injective, so its image is a faithful copy of $H_n(X^n)$.

Step 4: Identifying the subquotient. By Steps 2 and 3, both $\ker d_n$ and $\mathrm{im}d_{n+1}$ lie inside $H_n(X^n, X^{n-1})$ as images under the injection j_n of subgroups of $H_n(X^n)$, namely of $H_n(X^n)$ itself and of $\ker q$ respectively. An injective homomorphism induces an isomorphism onto its image and carries the quotient by a subgroup to the quotient of the image by the image of that subgroup, so j_n descends to an isomorphism

$$\frac{\ker d_n}{\mathrm{im}d_{n+1}} = \frac{j_n\big(H_n(X^n)\big)}{j_n(\ker q)} \cong \frac{H_n(X^n)}{\ker q}$$

The map $q\colon H_n(X^n) \to H_n(X)$ is surjective by lemma 3.5.9(3), so by the first isomorphism theorem ([7, First Isomorphism Theorem]) it induces an isomorphism $H_n(X^n)/\ker q \cong H_n(X)$. Composing,

$$H_n^{CW}(X) = \frac{\ker d_n}{\mathrm{im}d_{n+1}} \cong \frac{H_n(X^n)}{\ker q} \cong H_n(X)$$

The composite isomorphism is $j_n(z) + \mathrm{im}d_{n+1} \mapsto q(z)$, natural in X for cellular maps, since every map in its construction (the maps j_n, ∂_n, and the inclusion-induced q) is induced by inclusions of skeleta and hence commutes with maps of CW pairs. This gives the asserted isomorphism $H_n^{CW}(X) \cong H_n(X)$, as we sought to show.

Corollary 3.5.12: Consequence Of CW Homology

1. $H_n(X) = 0$ if X is a CW complex with no n-cells
2. If X is a CW complex with k n-cells, then $H_n(X)$ is generated by at most k elements.
3. If X is a CW complex having no two of its in adjacent dimensions, then $H_n(X)$ is a free abelian group with basis in one-to-one correspondence with the n-cells of X.

***Proof*:**

1. immediate
2. Since $H_n(X^n, X^{n-1})$ is free abelian on k-generators, the subgroup $\ker d_n$ must be generated by at most k elements, hence similarly for the quotient $\ker d_n / \mathrm{im} d_{n+1}$
3. The cellular boundary maps d_n are automatically zero, giving the result.

Example 3.5.13: CW Homology

Take $X = \mathbb{C}\mathrm{P}^n$, which has a CW structure with one cell each even dimension as we saw way back. Thus:

$$H_k(\mathbb{C}\mathrm{P}^n) = \begin{cases} \mathbb{Z} & k \in \{0, 2, 4, ..., 2n\} \\ 0 & \text{otherwise} \end{cases}$$

In general, our CW complex can be more complicated, and hence we need a way to compute ∂_n:

Proposition 3.5.14: Cellular Boundary Formula

Let X be a CW complex, and consider the cellular homology with boundary map $\partial_n : C_n(X^n, X^{n-1}) \to C_{n-1}(X^{n-1}, X^{n-1})$. For e^n_α is a cell with attaching map:

$$\varphi_\alpha : \partial e^n_\alpha \to X^{n-1}$$

and

$$\partial_{\alpha\beta} : S^{n-1} \xrightarrow{\cong} \partial e^n_\alpha \xrightarrow{\varphi_\alpha} X^{n-1} \xrightarrow{\pi} X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta) \xrightarrow{\cong} S^{n-1}$$

representing the attaching map φ_α with respect to an $(n-1)$-cell e^{n-1}_β, then ∂_n is determined by how it acts on each e_α and

$$\partial_n(e^n_\alpha) = \sum_\beta \deg(\partial_{\alpha\beta}) e^{n-1}_\beta$$

***Proof*:**

Since ∂_n is a homomorphism out of the free abelian group $H_n(X^n, X^{n-1})$, it is determined by its values on the basis elements e^n_α given by lemma 3.5.9. Fix one n-cell e^n_α with characteristic map $\Phi_\alpha \colon (D^n, \partial D^n) \to (X^n, X^{n-1})$ (definition 1.2.2), whose restriction to $\partial D^n = S^{n-1}$ is the attaching map φ_α. The cellular boundary is $d_n = j_{n-1}\partial_n$, where $\partial_n \colon H_n(X^n, X^{n-1}) \to$

$H_{n-1}(X^{n-1})$ is the connecting homomorphism of the pair (X^n, X^{n-1}) and $j_{n-1}\colon H_{n-1}(X^{n-1}) \to H_{n-1}(X^{n-1}, X^{n-2})$ is induced by the inclusion of pairs. Computing $d_n(e^n_\alpha)$ means computing the coefficient of each basis element e^{n-1}_β of $H_{n-1}(X^{n-1}, X^{n-2})$, and the claim is that this coefficient is $\deg(\partial_{\alpha\beta})$. The argument assembles one large commutative diagram from the naturality of the connecting homomorphism and an excision that isolates the cell e^{n-1}_β; the identification of the relevant map of spheres with $\partial_{\alpha\beta}$ then converts the coefficient into a degree, which the local-degree calculus evaluates.

Step 1: The master diagram. For each $(n-1)$-cell e^{n-1}_β let $q_\beta\colon X^{n-1} \to X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta)$ be the quotient collapsing everything outside the open cell to a point; the target is a sphere S^{n-1}_β, since $(X^{n-1}, X^{n-1} \setminus e^{n-1}_\beta)$ is, up to the characteristic map of e^{n-1}_β, the pair $(D^{n-1}/\partial D^{n-1})$. Consider the diagram

$$\begin{tikzcd}
H_n(D^n, \partial D^n) \arrow[r, "\partial"] \arrow[d, "\Phi_\alpha"'] & \tilde{H}_{n-1}(\partial D^n) \arrow[d, "(\varphi_\alpha)_*"] \arrow[dr, "(q_\beta\varphi_\alpha)_*"] & \\
H_n(X^n, X^{n-1}) \arrow[r, "\partial_n"] & H_{n-1}(X^{n-1}) \arrow[r, "(q_\beta)_*"] \arrow[d, "j_{n-1}"'] & \tilde{H}_{n-1}(S^{n-1}_\beta) \\
& H_{n-1}(X^{n-1}, X^{n-2}) \arrow[ur, "p_\beta", "\cong"'] &
\end{tikzcd}$$

The left square commutes by naturality of the connecting homomorphism (proposition 3.4.4) applied to the map of pairs $\Phi_\alpha\colon (D^n, \partial D^n) \to (X^n, X^{n-1})$: the connecting map is natural for maps of pairs, and Φ_α restricts on ∂D^n to φ_α. The top triangle commutes by functoriality of the induced map, since $q_\beta\varphi_\alpha = q_\beta \circ \varphi_\alpha$. The bottom triangle is the definition of p_β: it is the composite

$$H_{n-1}(X^{n-1}, X^{n-2}) \cong \bigoplus_\gamma \tilde{H}_{n-1}(S^{n-1}_\gamma) \xrightarrow{\mathrm{pr}_\beta} \tilde{H}_{n-1}(S^{n-1}_\beta)$$

of the basis identification of lemma 3.5.9 with the projection onto the summand indexed by β, and $(q_\beta)_* = p_\beta \circ j_{n-1}$ because both realize the passage from a class on X^{n-1} to its component supported on the single cell e^{n-1}_β. That last equality is the content of the cellular excision: collapsing X^{n-2} does not change the homology supported on the open cell e^{n-1}_β (corollary 3.4.15), so reading off the β-coordinate of $j_{n-1}(z)$ agrees with applying q_β directly.

Step 2: Reading off a single coefficient. Write $[D^n]$ for the generator of $H_n(D^n, \partial D^n) \cong \mathbb{Z}$ (corollary 3.4.12) corresponding under Φ_α to the basis element e^n_α, and $[\partial D^n] = \partial[D^n]$ for the resulting generator of $\tilde{H}_{n-1}(\partial D^n) \cong \mathbb{Z}$. The coefficient of e^{n-1}_β in $d_n(e^n_\alpha) = j_{n-1}\partial_n(e^n_\alpha)$ is, by definition of the basis, the integer $m_{\alpha\beta}$ with

$$p_\beta\big(j_{n-1}\partial_n(e^n_\alpha)\big) = m_{\alpha\beta}\,[S^{n-1}_\beta]$$

where $[S^{n-1}_\beta]$ is the chosen generator of $\tilde{H}_{n-1}(S^{n-1}_\beta) \cong \mathbb{Z}$. Chasing $[D^n]$ through the master diagram of Step 1 in two ways gives this integer. Down-then-right-then-project sends

$$[D^n] \longmapsto e^n_\alpha \longmapsto \partial_n(e^n_\alpha) \longmapsto p_\beta\big(j_{n-1}\partial_n(e^n_\alpha)\big) = m_{\alpha\beta}\,[S^{n-1}_\beta]$$

using $p_\beta j_{n-1} = (q_\beta)_*$ from the bottom triangle and the commuting left square. Right-then-diagonal sends

$$[D^n] \longmapsto [\partial D^n] \longmapsto (q_\beta \varphi_\alpha)_*[\partial D^n]$$

using the top triangle. Commutativity of the whole diagram forces these two outputs to agree:

$$m_{\alpha\beta}\,[S^{n-1}_\beta] = (q_\beta \varphi_\alpha)_*[\partial D^n]$$

Thus $m_{\alpha\beta}$ is exactly the integer by which the map $q_\beta\varphi_\alpha \colon \partial D^n \to S^{n-1}_\beta$ multiplies on $\tilde{H}_{n-1} \cong \mathbb{Z}$.

Step 3: Identifying the map with $\partial_{\alpha\beta}$ and its degree. The composite $q_\beta\varphi_\alpha$ is, after the canonical identifications $\partial D^n \cong S^{n-1}$ and the target $X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta) \cong S^{n-1}$, precisely the map

$$\partial_{\alpha\beta} \colon S^{n-1} \xrightarrow{\cong} \partial e^n_\alpha \xrightarrow{\varphi_\alpha} X^{n-1} \xrightarrow{\pi} X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta) \xrightarrow{\cong} S^{n-1}$$

of the statement, since q_β is the quotient map π followed by the homeomorphism to S^{n-1}. A self-map of S^{n-1} acts on $\tilde{H}_{n-1}(S^{n-1}) \cong \mathbb{Z}$ as multiplication by its degree (definition 3.5.1), so the integer extracted in Step 2 is

$$m_{\alpha\beta} = \deg(\partial_{\alpha\beta})$$

once the generators $[\partial D^n]$ and $[S^{n-1}_\beta]$ are taken to correspond under the chosen identifications. This is the coefficient we sought.

Step 4: Finiteness of the sum and assembly. For fixed α the integers $\deg(\partial_{\alpha\beta})$ vanish for all but finitely many β. The image $\varphi_\alpha(S^{n-1})$ is compact, hence meets only finitely many open $(n-1)$-cells e^{n-1}_β (the open cells form a locally finite partition by compactness of the image against the weak topology of X^{n-1}); for any β whose open cell is disjoint from this image, the map $q_\beta\varphi_\alpha$ is constant, so $\partial_{\alpha\beta}$ is null-homotopic and $\deg(\partial_{\alpha\beta}) = 0$ by proposition 3.5.2. Therefore the sum below is finite. Reassembling the coordinates computed in Step 3,

$$d_n(e^n_\alpha) = \sum_\beta m_{\alpha\beta}\, e^{n-1}_\beta = \sum_\beta \deg(\partial_{\alpha\beta})\, e^{n-1}_\beta$$

which is the cellular boundary formula. (When several distinct points of S^{n-1} map into the same cell e^{n-1}_β, the degree $\deg(\partial_{\alpha\beta})$ is computed as the sum of the local degrees of $\partial_{\alpha\beta}$ over the preimages of a regular value, by proposition 3.5.6 and definition 3.5.5; the formula already records the total degree, so no separate bookkeeping is needed.) Writing ∂_n for the cellular boundary d_n as in the statement, this establishes

$$\partial_n(e^n_\alpha) = \sum_\beta \deg(\partial_{\alpha\beta})\, e^{n-1}_\beta$$

for every n-cell e^n_α, completing the proof.

The above proposition now gives us all we need to calculate the homology of CW structures, it is fully based on the gluing maps:

Example 3.5.15: Computing Cellular Boundary Map

1. Let $X = M_g$ be a closed orientable surface of genus g with the usual CW structure of one 0-cell, $2g$ 1-cells, and one 2-cell attached by the product of the commutative $[a_1, b_1], ..., [a_g, b_g]$. Then the associated celular chain is:

$$0 \to \mathbb{Z} \xrightarrow{d_2} \mathbb{Z}^{2g} \xrightarrow{d_1} \mathbb{Z} \to 0$$

Then d_1 must be zero since there is only one 0-cell. d_2 is also zero since each a_i or b_i appears with its inverse in $[a_1, b_1], ..., [a_g, b_g]$, hence the map $\Delta_{\alpha\beta}$ are homotopic to constants maps. Since d_1, d_2 ar eboth zero, teh homology groups of M_g are the same as the cellular chain groups, hence:

$$H_n(M_g) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z}^{2g} & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

2. Let $X = N_g$ be closed nonorientable surface N_g of genus g which has one 0-cell, g 1-cells ,and one 2-cell attached by the word $a_1^2 \cdots a_g^2$. We again ahve $d_1 = 0$ and $d_2 : \mathbb{Z} \to \mathbb{Z}^g$ is given by $d_2(1) = (2, ..., 2)$ since ach a_i appeas in teh attaching words of the 2-cell with total exponent 2, which means $\Delta_{\alpha\beta}$ is homotopic to the map $z \mapsto z^2$ of degree 2. Since $d_2(1) = (2, ..., 2)$, we get that d_2 is injective, hence $H_2(N_g) = 0$. If we change the basis for $\mathbb{Z}^g$ by replacing the last standard basis element $(0, ..., 0, 1)$ by $(1, ..., 1)$ we see that

$$H_1(N_g) \cong \mathbb{Z}^{g-1} \oplus \mathbb{Z}_2$$

Hence:

$$H_n(N_g) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z}^{g-1} \oplus \mathbb{Z}_2 & n = 1 \\ 0 & n \geq 2 \end{cases}$$

Notice that we have detected orientability of a closed connected n-manifold by looking at $H_n(M)$, which was $\mathbb{Z}$ when M is orientable and 0 otherwise, this pattern generalizes, see [14, Theorem 3.26]

3. (3-dimensional Torus, Hatcher p.142)

4. (Moore space) Let G be an abelian group and take $n \in \mathbb{N}$. Then we can construct a CW complex such that $H_n(X) \cong G$ and $\tilde{H}_i(X) = 0$ for all $i \neq 0$. The construction we will be doing is called the *Moore space*, and is often denoted $M(G, n)$

By some group theory, we know that $G = F \oplus T$ for some torsion part T and free part F. The simplest case is $G = \mathbb{Z}$. Then for any n we have $X = S^n$. If $G + \mathbb{Z}/m\mathbb{Z}$, then for any n take S^n and attach e^{n+1} via $\varphi : S^n \to S^n$ of degree m (you can inductively take the suspension on $z \mapsto z^m$). For the general case (not necessarily finitely generated), take $F \to G$ to be the free map from the free abelian group F to G. The kernel of this map is free since F is free. Then if $\{x_\alpha\}$ is a basis for F, $\{y_\beta\}$ is a basis for the kernel K, and $y_\beta = \sum_\alpha d_{\beta\alpha} x_\alpha$, we define

$$X^n = \bigvee_\alpha S^n_\alpha \quad \implies \quad H_n(X^n) \cong F$$

Now, to construct X from $^n<$ we shall attach cells e_β^{n+1} via $f_\beta : S^N \to X^n$ such that f_b projects onto the summand S^n_α with degree $d_{\beta\alpha}$. The cellular boundary map d_{n+1} will be the inclusion $K \hookrightarrow F$, hence X has the dissed homology.

Now, by taking the wedge sum of Moore spaces for varying n, we can get a connected CW complex with arbitrary abelian groups G_n at every dimension.

5. Let $X = \mathbb{R}\mathrm{P}^n$. Then we saw that $\mathbb{R}\mathrm{P}^n$ has a CW structure of one cell, e^k for each $k \leq n$, and the attaching map e^k is the 2-sheeted covering $\varphi : S^{k-1} \to \mathbb{R}\mathrm{P}^{k-1}$. Th compute d_k, we compute the degree so

$$S^{k-1} \xrightarrow{\varphi} \mathbb{R}\mathrm{P}^{k-1} \to q\mathbb{R}\mathrm{P}^{k-1}/\mathbb{R}\mathrm{P}^{k-2} \cong S^{k-1}$$

The map $q \circ \varphi$ restricts to a homeomorphism from each $S^{k-1} - S^{k-1}$ onto $\mathbb{R}\mathrm{P}^{k-1} - \mathbb{R}\mathrm{P}^{k-2}$, and these two homeomorphism are given by pre-composing with antipodal maps of S^{k-1}, which as degree $(-1)^k$. Thus:

$$\deg q \circ \varphi = \deg \mathrm{id} + \deg(-\mathrm{id}) + 1 + (-1)^k$$

Hence, d_k is either 0 or multiplication by 2, depending on whether k is even or odd. Thus, we get:

$$0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \cdots \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0 \qquad n \text{ is even}$$
$$0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \cdots \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0 \qquad n \text{ is odd}$$

Thus, if n is *odd*:

$$H_k(\mathbb{R}\mathrm{P}^n) = \begin{cases} \mathbb{Z} & k = 0,\ k = n \\ \mathbb{Z}/2\mathbb{Z} & k \text{ odd},\ 0 < k < n \\ 0 & \text{otherwise} \end{cases}$$

and if n is *even*:

$$H_k(\mathbb{R}\mathrm{P}^n) = \begin{cases} \mathbb{Z} & k = 0 \\ \mathbb{Z}/2\mathbb{Z} & k \text{ odd},\ 0 < k < n \\ 0 & \text{otherwise} \end{cases}$$

6. (Lens space p.144)

As we noted under definition 2.1.18, having a trivial fundamental group does not imply the space is contractible. The same is true for when a space has only trivial homology groups, this does not imply it's contractible. We first give a name to such spaces:

Definition 3.5.16: Acyclic Space

Let X be a space. Then if $\tilde{H}_i(X) = 0$ for all i, we call such a space *acyclic*.

Theorem 3.5.17: Excised Sphere Is Acyclic

$S^n - B$ is acyclic where B is a k-cell

***Proof*:**
We do induction on k. Suppose it is true for $k-1$ cells. We ahve shown taht under at least one homomorhpism, i_* orj_* induced by $\iota : (S^n - B) \to (S^n - B_1)$ or $j : (S^N - B) \to (S^n - B_2)$, the iamge of a non-trivial 2 is not trivial.

Suppoe there exists a non-trivial $d \in \tilde{H}_i(S^n - B)$. Without loss of generailty suppose $\iota_*(\alpha) \neq 0$ in $H_i(S^n - B_i)$. Let $B_1 = B_{1,1} \cup B_{1,2}$. Think:

$$[0,1] \supseteq [a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$$

Then $\cap_i [a_i, b_i] = \{e\}$, Let $D_i I^{k-1} \times [a_i, b_i]$. Then:

$$E = h(I^{k-1} \times \{e\}) \qquad E = \cap D_i$$

The iamge of α under the indcued hom is trivial, so

$$\tilde{H}_i(S^n - E) = \{0\}$$

Now $S^n - E = \cup_i (S^n - D_i)$. Now there exitss a compset subset A such that the iamge of α vanishes, but $A \subseteq S^n - D_i$ for some i, a contradiction

Another example given by Hatcher is the space X obtained from $S^1 \vee S^1$ by attaching two 2-cells through $a^5 b^{-3}$ and $b^4 (ab)^{-2}$, see Hatcher p.142

Theorem 3.5.18: More Homology calculations

Let $0 \leq k < n$ and let $h : S^k \to S^n$ an embedding. Then

$$\tilde{H}_i(S^n - h(S^k)) \cong \begin{cases} \mathbb{Z} & i = n-k-1 \\ 0 & \text{otherwise} \end{cases}$$

***Proof*:**
We do induction on K, if $k = 0$, then $S^k \{p, q\}$. If we remove two points from S^n, then

$$S^n - h(S^k) \cong \mathbb{R}^n - \{0\}$$

hence, it deformation retracts to S^{n-1} Hence

$$\tilde{H}_i(S^n - S^-) \cong \begin{cases} \mathbb{Z} & i = n-1 \\ 0 & \text{otherwise} \end{cases}$$

Now, by induction suppose the theorem holds true in dim $= k-1$. Consider $h : S^k \to S^n$. Take:

$$X = S^n - h(S^{k-1}) = X_1 \cup X_2$$

Let $X = S^n - h(E_+^k)$ and $X_2 = S^n - h(E_-^k)$ so that

$$X = X_1 \cup X_2 = S^n - h(S^{k-1}) \qquad X_1 \cap X_2 = A = S^n - h(S^k)$$

Then by Mayer-Viatoris:

$$\tilde{H}_{i+1}(X_1) \oplus \tilde{H}_{i+1}(X_2) \to \tilde{H}_{i+1}(X) \to \tilde{H}_i(A) \to \tilde{H}_i(X_1) \oplus H_i(X_2)$$

First, we know the term on the left and right are zero, an have:

$$0 \to \tilde{H}_{i+1}(X) \to \tilde{H}_i(A) \to 0$$

This gives

$$\tilde{H}_{i+1}(X) = \tilde{H}_i(A)$$

then $i+1 = n-(k-1)-1$, gives the non-trivial result, and 0 otherwise.

skipped for now, since it required information from appendix 1.B, not covered curently in class.

Theorem 3.5.19: Generalized Jordan Curve Theorem

Let $n > 0$, Let C be a subset of S^n homeomorphic to the $n-1$ sphere. Then $S^n - C$ has precisely two components, of which C is a common topological boundary.

***Proof*:**
As we know, we want to calculate

$$\tilde{H}_0(S^n - C) = \mathbb{Z}$$

(uhh not done?)

Theorem 3.5.20: Invariance Of Domain

Let $U \subseteq \mathbb{R}^n$ be open, and $f : U \to \mathbb{R}^n$ an injective continuous map. Then $f(U)$ is open in $\mathbb{R}^n$, and f is an embedding onto its image.

***Proof*:**
let $f : U \to S^n$. be continuous. want to show $f(U)$ is open in the usual topological way, for every point there is an open neighbourhood. Choose an open ball in U for which the closure is contained in U (which we may do since $U \subseteq \mathbb{R}^n$[a]). Then

$$S_\epsilon = \overline{B_\epsilon(x)} - B_\epsilon(x)$$

is a sphere of dim $n-1$> Then

$$f(S_\epsilon) \cong S^{n-1}$$

Thus, it separates S^n into two connected component: W_1, W_2. each W_i is open in S^n. Sicne $f(B_\epsilon)$ is connected, then $f(B_\epsilon) \cap f(S_\epsilon) = \emptyset$. So $f(B_\epsilon)$ is either in W_1 or W_2. If it sits in W_1, it must be equal to W_1, so $f(B_\epsilon)$ must be open, and is the neighbourhood we were looking for, completing the proof.

[a] since $\mathbb{R}^n$ is normal, or by direct metric manipulation

3.5.3 Euler-Characteristic

Recall the famous equation for polygons in $\mathbb{R}^2$:

$$V - E + F = 2$$

We can define a similar constant for any CW complex X, namely its *Euler Characteristic.* defiend to e

$$\chi(X) = \sum_n (-1)^n c_n$$

where c_n is the number of n-cells of X. We shall see that this quantity can in fact be defined in a purely homological way, showing it will be independent of CW structure:

Theorem 3.5.21: Euler's Characteristic

$$\chi(X) = \sum_n (-1)^n \operatorname{rank} H_n(X)$$

***Proof*:**
Hathcer p.146 (A purely algebraic proof!)

For a finite CW complex the n-th cellular chain group has rank equal to the number of n-cells, and the alternating sum of ranks is unchanged by passing to homology, so the formula is also the alternating sum of the cell counts. Using this, prove that $\chi(X \times Y) = \chi(X)\chi(Y)$

3.5.4 Lefschetz Fixed Point Theorem

Next, let G be a free-abelian group. Take the group homomorphism $\varphi : G \to G$. Then we can rerpesent φ in a matrix form, say A. Then if we have a finite simplicial complex K[4] and we have a chain map $\varphi : C_p(K) \to C_p(K)$ (same G). Then we can write $\operatorname{tr}(\varphi, C_p(K))$. Then $H_p(K)$ is not necessarily free, but we can factor the torsion part out and make it free: $H_p(K)/T_p(K)$. These ranks are called *Betti numbers*. If φ is as above. it induces a map

$$\varphi_* : H_p(X)/T_p(K) \to H_p(K)/T_p(K)$$

Theorem 3.5.22: Hopf Trace Theorem

Let K be a finite (simplicial or CW) complex. Let $\varphi : C_n(K) \to C_n(K)$ be a chain map. Then

$$\sum_{n=0} (-1)^n \operatorname{tr}(\varphi, C_n(K)) = \sum_n (-1)^n \operatorname{tr}\left(\varphi_*, \frac{H_n(K)}{T_p(K)}\right)$$

***Proof*:**
see Munkres algebraic topology or Hatcher

[4]the theorem does generalize to CW complexes

From this, we can define the Euler characteristic as:

$$\chi(K) = \sum (-1)^p \operatorname{rank}(C_p(K))$$

For a triangle with a line down the middle we get

$$\chi(X) = 4 - 5 + 2 = 1$$

Theorem 3.5.23: Euler Characteristic from Class

Let K be a simplicial complex. Let β_p be the rank of $H_p(K)$. Then:

$$\chi(K) = \sum_p (-1)^p \beta_p$$

***Proof*:**
We use hopf trace formula. Let id : $K \to K$. Then we get $i_\# : C_p(K) \to C_p(K)$ and a homomorphsim $i_* : H_P(K) \to H_p(K)$. Then the matrix representation give a trace of rank of C_p, and the trace of ι_* is the rank of H_p, but now we are done

Definition 3.5.24: Lefschetz Number

Let K be a finite simplicial complex, $h : K \to K$ a continuous map. Then

$$\Lambda(h) = \sum_{i=0}^{n} (-1)^i \operatorname{tr}(h_*(H_i(K)/T_i(K))$$

is called the *lefschetz* number of h

Theorem 3.5.25: Lefschetz Fixed Point Theorem

Let K be a finite complex, $h : K \to K$ continuous. If $\Lambda(h) \neq 0$, then h has a fixed point

***Proof*:**
Say $h : K \to K$ where K is connected. Then $h_* : H_0(K) \to H_0(K)$ is an isomorphism. (conclude?)

here is a consequence: Let H be a finite acyclic complex. If $h : K \to K$ is continuous, then h has a fixed point

Proof. $\Lambda(h) = 1$, so it has a fixed point □

The antipodal map $a : S^n \to S^n$ has degree $(-1)^{n+1}$. and

$$\Lambda(a) = 1 + (-1)^n \operatorname{tr}(a, \mathbb{Z}) = 0$$

so $\operatorname{tr}(a, \mathbb{Z}) = (-1)^{n+1}$, but tha that is the degree of the map. Now consider $h : \mathbb{R}\mathrm{P}^2 \to \mathbb{R}\mathrm{P}^2$, so

$$H_0(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) = \mathbb{Z}, \qquad H_1(\mathbb{R}\mathrm{P}^2, \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z} \qquad H_2(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) = 0$$

so the Lefschetz number is not zero, so there must be a fixed point.

3.5.5 Homology of Groups

Definition 3.5.26: Chain Over Groups

Consider chains $\sigma_i n_i \sigma_i$ where each σ_i is a singular n-simplex, andthe coeffiients n_i are taken in some fixed group G (rather than just $\mathbb{Z}$.)Thne the collectio nis called

$$C_n(X;G)$$

and the relative version is:

$$C_n(X,A;G) = C_n(X;G)/C_n(A;G)$$

The smae boundary map works as before, that is

$$\sum_i n_i\sigma_i \overset{\partial}{\mapsto} \sum_{i,j}(-1)^j\sigma|_i$$

The reducd homology is now ending with:

$$\cdots \to C_0(X;G) \xrightarrow{\epsilon} G \to 0$$

and as before, is X is a point, then:

$$H_n(X;G) \cong \begin{cases} G & n=0 \\ 0 & \text{otherwise} \end{cases}$$

Furthermore:

$$H_n^{CW}(X;G) \cong H_n(X;G) \qquad d_n\left(\sum_\alpha n_\alpha e_\alpha^n\right) = \sum_\alpha(\cdots)$$

Example 3.5.27: Homology With Coefficients

Take $\mathbb{R}\mathrm{P}^n$ and coefficients in $\mathbb{Z}/2\mathbb{Z}$. Recall it's cellular strucutre is given by $\mathbb{R}\mathrm{P}^n = e_1 \cup \cdots \cup e_n$. Now recall:

$$H_k(\mathbb{R}\mathrm{P}^k, \mathbb{R}\mathrm{P}^{k-1}) \xrightarrow{d_k} H_{k-1}(\mathbb{R}\mathrm{P}^{k-1}, \mathbb{R}\mathrm{P}^{k-1})$$

now we have $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$. Then $d_k = 1 + (-1)^k$, so if k is even we get 2, and if k is odd we get 0, but in $\mathbb{Z}/2\mathbb{Z}$, that means we always bet 0, hence d_k is a zero map. Hence, we get

$$H_k(\mathbb{R}\mathrm{P}^n) = \mathbb{Z}/2\mathbb{Z} \qquad 0 \leq k \leq n$$

Lemma 3.5.28: Degree For Abelian Groups

let $f : S^k \to S^k$ be of degree m. Then $f_* : H_k(S^k;G) \to H_k(S^k;G)$ is multiplication by m.

***Proof*:**
Hatcher p.154. Keep this commutative diagram in mind:

$$\begin{tikzcd} \mathbb{Z} \approx \widetilde{H}_k(S^k;\mathbb{Z}) \arrow[r, "J_*"] \arrow[d, "\varphi_*"] & \widetilde{H}_k(S^k;\mathbb{Z}) \approx \mathbb{Z} \arrow[d, "\varphi_*"] \\ G \approx \widetilde{H}_k(S^k;G) \arrow[r, "f_*"] & \widetilde{H}_k(S^k;G) \approx G \end{tikzcd}$$

3.6 Summary

(better summary here, and I want this foreshadowing:)

The singular, simplicial, and cellular homologies of this chapter are not three theories that happen to coincide but three models of one theory, pinned down by the *Eilenberg-Steenrod axioms*: a homology theory is a sequence of functors $H_n(-)$ on pairs of spaces with natural boundary maps, satisfying homotopy invariance, exactness, excision, additivity, and the *dimension axiom* fixing the homology of a point. Any two theories satisfying these axioms agree on all CW complexes, which is what turns the coincidence of the models into a theorem. Relaxing the dimension axiom enlarges the notion to a *generalized* (or extraordinary) homology theory, with K-theory and cobordism as the prototypes. These axioms, the uniqueness theorem characterizing ordinary theories, and the generalized theories obtained by dropping the dimension axiom are developed in full in chapter 8 (see definition 8.2.4, theorem 8.2.14, and definition 8.2.16).

4

Cohomology

We shall now develop a dual theory to homological algebra focusing on the functionals on the simplices for the favourable algebraic advantages. A homology was defined given a complex $C_\bullet$. We always have access to the contravariant $\mathrm{Hom}(C_\bullet, G)$ functor that we may wish to apply to study some properties of $C_\bullet$. Often the dual category doesn't bring us much direct new information.

The main thing that cohomology gives us is a new product called the *cup product*. This product is technically present in homology, but is not natural and is computationally much harder to find: the natural function $X \to X \times X$ given by the diagonal map $x \mapsto (x, x)$ exists, but for a general topological space X, there is no natural map $X \times X \to X$; it is not clear how we can make a natural choice here. This shall introduce a new algebraic structure, a graded-commutative R-algebra structure on $H^\bullet(X, R)$ (given R is a commutative ring) which shall give us more insights on our spaces.

4.1 Cohomology of Spaces

To intuitively understand how to think of $\mathrm{Hom}(C^\bullet, G)$, let X be a 1-dimensional Δ-complex, i.e. a graph. Given a fixed abelian group G (or really any abelian object), the set $\mathrm{Hom}_{\mathbf{Set}}(X, G)$ forms an abelian group, which we denote $\Delta^0(X; G)$ (if $G = \mathbb{Z}$, in the finite dimensional case, this group is [not necessarily naturally] isomorphic to $\Delta_0(X)$). We may do the same for edges defining $\Delta^1(X; G)$ and more generally $\Delta^n(X; G)$. Given these function groups, we shall define a function δ between them.

Starting by understanding $\Delta^i(X; G)$ where X is a simplicial complex[1]. Then $\varphi \in \Delta^0(X; G)$ is a function from the vertices of X to G, and $\psi \in \Delta^1(X; G)$ is a function from the edges of X to G, and similarly for higher dimensions. Consider the map:

[1] just like for homology, the singular and simplicial will be shown to have the same homology, and hence we use simplicial complexes for simpler visualizations

$$\delta : \Delta^0(X;G) \to \Delta^1(X;G) \tag{4.1}$$

where φ maps to $\delta(\varphi)$ where if $e = [v_0, v_1]$ is and:

$$\delta(\varphi)(e) = \varphi(v_1) - \varphi(v_0)$$

Note how the orientability was also important to have a natural choice of which vertex to take the difference of. If we take equation (4.1) and embedded into a chain complex with zero's before and after it:

$$\cdots \to 0 \to \Delta^0(X;G) \xrightarrow{\delta} \Delta^1(X;G) \to 0 \to \cdots$$

Then

$$H^0(X;G) = \ker\delta \qquad \text{and} \qquad H^1(X;G) = \Delta^1(X;G)/\mathrm{im}\delta$$

Now, $H^0(X;G)$ describes maps that have the same value on the two vertices of an edge, that is

$$\delta(\varphi) = 0 \qquad \text{iff} \qquad \varphi(v_i) = \varphi(v_j),\ e = [v_i, v_j]$$

that is φ is *constant* on each component of G (so φ is “locally constant” with respect to edges). Hence, $H^0(X;G)$ is all functions from the sets of component of X to G (similarly to how $H_0(X)$ counts the number of path components). Notice how this is similar to how $\mathrm{d}\,f = 0$ if f is constant.

Next, take $\overline{\psi} \in H^1(X;G)$ where $\psi : X \to G$ is a function where $\psi(e) \in G$ for each edge. Consider the case where $\overline{\psi} = 0$ so that $\psi \in \mathrm{im}\delta$. If $\psi = \delta(\varphi)$ for some $\varphi \in \Delta^0(X;G)$, then ψ is purely determined by it's values at it's “extrema”, in this case the value of ψ is determine by the change of value between each each corresponding to a function φ. For future reference, notice how this is similar to finding a function F given a change-in-value function f (usually the derivative). This connection is made explicit with *deRham cohomology* which we do in EYTNKA Differential Topology. This parallels goes even further; if there exists a φ such that $\delta\varphi = \psi$, then it is unique up to adding an element of the kernel of δ.

When X is a tree, this equation is always solvable; in this case we have $H^1(X;G) = 0$. If X is not a tree (hence it has cycles), then the equation may fail (ex. consider the each edge in a closed loop having difference 1). To measure to what degree this fails, we first pick a maximal tree. Then a ψ can be determined by a φ on this maximal tree, however the value of ψ on each edge not in the maximal tree must equal the different of already-determined values of φ at two ends of edges, which may not work since ψ can have, in general, arbitrary values on edges. Thus, $H^1(X;G)$ can be thought of as the direct product of G's, one for each edge of X not in the chosen maximal tree. This is similar to $H_1(X;G)$ which consists of a direct sum of copies of G for each edge X not in the maximal tree as seen in example 2.2.7 (abelianized).

Now, moving up a dimension, let $\Delta^2(X;G)$ be all functions from faces to G, and define

$$\delta : \Delta^1(X;G) \to \Delta^2(X;G)$$

$$\delta\psi([v_0, v_1, v_2]) = \psi([v_0, v_1]) + \psi([v_1, v_2]) - \psi([v_0, v_2])$$

i.e. a signed sum of the values of ψ on the edges. Now, combining to form a chain:

$$\cdots \to 0 \to \Delta^0(X;G) \xrightarrow{\delta} \Delta^1(X;G) \xrightarrow{\delta} \Delta^2(X;G) \to 0 \to \cdots \tag{4.2}$$

Notice that

$$\delta\delta(\varphi) = (\varphi(v_1) - \varphi(v_0)) + (\varphi(v_2) - \varphi(v_1)) - (\varphi(v_2) - \varphi(v_0)) = 0$$

Showing that $\mathrm{im}\delta_1 \subseteq \ker \delta_2$. Again making a chain complex by adding 0 before and after in equation (4.2) we may define $H^2(X;G)$. There are many ways of thinking of elements of $H^2(X;G)$, each giving a useful insight.

First, notice that is $\delta\psi = 0$, then:

$$\psi([v_0, v_2]) = \psi([v_0, v_1]) + \psi([v_1, v_2])$$

showing that the cohomology groups measures the failure of ψ being additive. Another more "geometric" interpretation is that it measure the degree of failure of finding a $\varphi \in \Delta^0(X;G)$ such that $\psi = \delta\varphi$, since if $\psi = \delta\varphi$, then $\delta\psi = 0$ (since $\delta\delta\varphi = 0$). In particular, since the value of the equation $\psi = \delta\varphi$ only depends on values defined on elements, we may think of it as measuring the local obstruction. But since $H^1(X;G)$ is zero if and only if $\psi = \delta\varphi$, we can think of $H^1(X;G)$ as measuring global obstruction. Think of it as trying to find when a vector fields is the gradient of a differentiable function.

Now, instead of working with $\Delta^n(X;G) = \mathrm{Hom}_{\mathbf{Set}}(X^n, G)$, we shall work with

$$C^1(X;G) = \mathrm{Hom}_{\mathbf{Grp}}(\Delta_n(X), G)$$

This is equivalent, since $\Delta_i(X)$ is a free abelian group with basis the n-simplices of X, and a homomorphism from a free object is uniquely determined by where it maps its basis (which, naturally, can be mapped arbitrarily). The group $C^n(X;G)$ is given a name:

Definition 4.1.1: Dual Group

Let X be a topological space and $C_\bullet(X)$ the singular chain groups. Then:

$$C^\bullet(X;G) := \mathrm{Hom}(C_\bullet(X), G)$$

is the *singular cochain groups* with coefficients in G.

The boundary homomorphism ∂_n generalizes to:

$$\delta\varphi([v_0, ..., v_{n+1}]) = \sum_i (-1)^i \varphi([v_0, ..., \hat{v}_i, ..., v_n + 1]) = \varphi(\partial[v_0, ..., v_{n+1}])$$

Hence:

$$\delta(\varphi) = \varphi \circ \partial$$

meaning δ is a dual map, since the $\mathrm{Hom}(-, G)$ functor maps $C_n \mapsto \mathrm{Hom}(C_n, G)$ and $\partial \mapsto (\varphi \mapsto \varphi \circ \partial)$. Now, to form a complex, let X be a topological. We may then form a chain:

$$\cdots \to C_n \xrightarrow{\partial_n} C_{n-1} \to \cdots$$

whether singular, simplicial, or cellular. If we take the homology, we get $H_n(X)$. On the other hand, if we apply the [contravariant] hom functor:

$$\cdots \leftarrow \mathrm{Hom}(C_n, G) \xleftarrow{\delta_n} \mathrm{Hom}(C_{n-1}, X) \leftarrow \cdots$$

In particular, each $\varphi \in C^n(X;G)$ assigned to $\sigma : \Delta^n \to X$ a value $\varphi(\sigma) \in G$. Then $\delta = \partial^* : C^n(X;G) \to C^{n+1}(X;G)$ is given by composition:

$$\delta(\varphi) = \partial^*(\varphi) = \varphi \circ \partial$$

In particular:

$$\delta(\varphi)(\sigma) = \sum_i (-1)^i \varphi(\sigma|_{v_0,\dots,\hat{v_i},\dots,v_{n+1}})$$

hence $\delta^2 = 0$, hence the cohomology groups are well-defined.

Definition 4.1.2: Cocycles And Coboundaries

Let δ be defined as above, then element of $\ker\delta$ are called *cocycles* and element of $\mathrm{im}\delta$ are called *coboundaries*. A cochain φ is a cocycle if

$$\delta(\varphi) = \varphi \circ \partial = 0$$

i.e. $\varphi \in \mathrm{Hom}(X, G)$ vanishes on the boundary.

We give the image of $\mathrm{Hom}(-, G)$ a name:

Definition 4.1.3: Cochain Complex

Let X be a topological and $C_\bullet$ a complex. Then $C_\bullet^* = \mathrm{Hom}(C_\bullet, G)$ is a *cochain complex* with respect to G.

From this we get the comology group:

Definition 4.1.4: Cohomology Group

Let $C_\bullet$ be a chain complex. Then for some abelian group G, we may form a chain and dualize with $\mathrm{Hom}(-, G)$ to form the cochain $C^\bullet$. Then:

$$H^n(C; G) = \frac{\ker(\delta)}{\mathrm{im}\delta}$$

is the nth *cohomology group* of X.

We will show that the cohomology group is algebraically determined by the homology group and G. A first hope may be that $\mathrm{Hom}(-, G)$ commutes with taking the quotient:

$$H^n(C; G) \cong \mathrm{Hom}(H_n(C), G)$$

However, this need not be the case:

Example 4.1.5: Hom Functor Not Commuting With Quotienting

Take the chain:

$$0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$$

where the 2 map sends $x \mapsto 2x$. Then dualizing with $\mathrm{Hom}_{\mathbb{Z}}(-, \mathbb{Z})$ we get the (isomorphic) cochain:

$$0 \leftarrow \mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{2} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \leftarrow 0$$

Then $H_2(C) = 0$, but $H^2(C) = \mathbb{Z}/2\mathbb{Z}$, while $\mathrm{Hom}_G(H_2(C), \mathbb{Z}/2\mathbb{Z}) = 0$.

Though this is not the csae, this is not a bad first guess, and there is at minimum a homomorphism $h : H^n(C;G) \to \mathrm{Hom}_G(H_n(C), G)$. In fact, we can be more precise. First, we require a lemma:

Lemma 4.1.6: Free Resolution Homomorphism

Let F, F' be free resolutions of abelian groups H, H'. Then every homomorphism $\alpha : H \to H'$ can be extended a homomorphism of chain maps F to F':

$$\begin{tikzcd}
\cdots \arrow[r] & F_2 \arrow[r] \arrow[d, "\alpha_2"] & F_1 \arrow[r] \arrow[d, "\alpha_1"] & F_0 \arrow[r] \arrow[d, "\alpha_0"] & H \arrow[r] \arrow[d, "\alpha"] & 0 \\
\cdots \arrow[r] & F_2' \arrow[r] & F_1' \arrow[r] & F_0' \arrow[r] & H' \arrow[r] & 0
\end{tikzcd}$$

Proof:
Hatcher p.194, or Aluffi

Recall every abelian group has a free resolution of *length one*, that is one with two terms $0 \to F_1 \to F_0 \to A \to 0$: take F_0 free on a generating set and $F_1 = \ker(F_0 \to A)$, which is free because $\mathbb{Z}$ is a PID. (The ranks are unconstrained; $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$ has both of rank one.)

Theorem 4.1.7: Universal Coefficient Theorem For Homology

Let $C_\bullet$ be a chain complex of free abelian groups and G an abelian group. For each n there is a short exact sequence

$$0 \to H_n(C) \otimes G \to H_n(C;G) \to \mathrm{Tor}(H_{n-1}(C), G) \to 0$$

where $H_n(C;G)$ is the homology of $C_\bullet \otimes G$. The sequence is natural in $C_\bullet$ and in G. It splits, but the splitting is not natural.

Proof:
Write $Z_n = \ker(\partial\colon C_n \to C_{n-1})$ and $B_n = \mathrm{im}(\partial\colon C_{n+1} \to C_n)$, both subgroups of the free abelian group C_n, hence themselves free, a submodule of a free module over the PID $\mathbb{Z}$ being free. No finite-generation hypothesis is needed, which matters because the chain complexes this is applied to are usually of infinite rank. The strategy is the one used for cohomology below: replace $C_\bullet$ by the cycles and boundaries, tensor there, and only then take homology.

Step 1: a free resolution of the homology. For each n the sequence

$$0 \to B_n \to Z_n \to H_n(C) \to 0$$

is exact by definition of $H_n(C) = Z_n/B_n$, and B_n, Z_n are free. So it is a free resolution of $H_n(C)$ of length one, which is what computes Tor: tensoring with G gives

$$0 \to \mathrm{Tor}(H_n(C), G) \to B_n \otimes G \to Z_n \otimes G \to H_n(C) \otimes G \to 0$$

the left-hand term being Tor precisely because the resolution has length one. That the answer does not depend on the chosen resolution is [7, The Derived Functors Are Well Defined](1); lemma 4.1.6 supplies only the *existence* of a comparison chain map, which is one half of what independence needs, the other being that any two such lifts are chain homotopic.

Step 2: the complex of cycles and boundaries. Regard $Z_\bullet$ and $B_\bullet$ as chain complexes with zero differential. Since C_n is free and $B_{n-1} \subseteq C_{n-1}$ is free, the sequence

$$0 \to Z_n \to C_n \xrightarrow{\partial} B_{n-1} \to 0$$

splits, so tensoring with G keeps it exact:

$$0 \to Z_n \otimes G \to C_n \otimes G \to B_{n-1} \otimes G \to 0$$

This is a short exact sequence of chain complexes, the outer two having zero differential.

Step 3: the long exact sequence. Its homology long exact sequence reads

$$B_n \otimes G \xrightarrow{i_n \otimes 1} Z_n \otimes G \to H_n(C;G) \to B_{n-1} \otimes G \xrightarrow{i_{n-1} \otimes 1} Z_{n-1} \otimes G$$

writing $i_k \colon B_k \rightarrowtail Z_k$ for the inclusions. The connecting map is $i_{n-1} \otimes 1$: an element of $B_{n-1} \otimes G$ lifts to $c \otimes g$ with $\partial c = b$, and applying the differential of $C_\bullet \otimes G$ returns $b \otimes g$ viewed inside $Z_{n-1} \otimes G$. Breaking the sequence up, $H_n(C;G)$ is an extension of $\ker(i_{n-1} \otimes 1)$ by $\operatorname{coker}(i_n \otimes 1)$. By Step 1 those are exactly $\operatorname{Tor}(H_{n-1}(C), G)$ and $H_n(C) \otimes G$, which gives the displayed sequence. Naturality in $C_\bullet$ and G holds for the middle and left terms by naturality of the long exact sequence (proposition 3.4.4) and of $\otimes$. For the right-hand term it needs one thing more: the identification $\ker(i_{n-1} \otimes 1) \cong \operatorname{Tor}(H_{n-1}(C), G)$ was made *through* the resolution $0 \to B_{n-1} \to Z_{n-1} \to H_{n-1}(C) \to 0$. A chain map $C_\bullet \to C'_\bullet$ carries boundaries to boundaries and cycles to cycles, so it restricts to a comparison chain map between these resolutions lifting the induced map on H_{n-1}. The canonical map on Tor is computed from any such lift: any two lifts are chain homotopic ([10, Projective Resolution As Initial Objects]), the homotopy persists after tensoring with G, and chain homotopic maps induce the same map on homology (proposition 3.3.7). So the map the chain map induces on $\ker(i_{n-1} \otimes 1)$ is the canonical one, and the sequence is natural in the right-hand term as well.

Step 4: splitting. The sequence $0 \to Z_n \to C_n \to B_{n-1} \to 0$ of Step 2 splits, so choose a retraction $p_n \colon C_n \to Z_n$ with $p_n|_{Z_n} = \mathrm{id}$. It is tempting to say this retraction splits the displayed sequence directly, but it does not: p is *not* a chain map $C_\bullet \to Z_\bullet$ when $Z_\bullet$ carries the zero differential, since for $c \in C_n$ the element ∂c lies in $B_{n-1} \subseteq Z_{n-1}$, where p_{n-1} is the identity, so $p_{n-1}(\partial c) = \partial c \neq 0$.

Composing with the projection fixes it. Put $\varphi_n := \pi_n \circ p_n \colon C_n \to H_n(C)$, where $\pi_n \colon Z_n \twoheadrightarrow Z_n/B_n = H_n(C)$. Now φ *is* a chain map into $H_\bullet(C)$ with zero differential, because $\varphi_{n-1}(\partial c) = \pi_{n-1}(\partial c) = 0$, the class of a boundary being zero. Tensoring with G and passing to homology gives $(\varphi \otimes 1)_* \colon H_n(C;G) \to H_n(C) \otimes G$, and for a cycle $z \in Z_n$ one has $\varphi_n(z) = [z]$, so this map restricts to the identity on the image of $H_n(C) \otimes G \hookrightarrow H_n(C;G)$, that inclusion being $[z] \otimes g \mapsto [z \otimes g]$ under the Step 1 identification $\operatorname{coker}(i_n \otimes 1) \cong H_n(C) \otimes G$. It is therefore a retraction, and the sequence splits.

The splitting is not natural. Non-canonicity of the choice does not by itself show this, so here is a counterexample. Take $G = \mathbb{Z}/2$, let $C_\bullet$ be $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ in degrees $1, 0$ and $C'_\bullet$ be $\mathbb{Z} \to 0$ in degrees $1, 0$, and let φ be the identity in degree 1 and zero in degree 0. Then $H_1(C) \otimes G = 0$, so any splitting s_C of $H_1(C;G) \to \operatorname{Tor}(H_0(C), G)$ is an isomorphism $\mathbb{Z}/2 \to \mathbb{Z}/2$; while $\operatorname{Tor}(H_0(C'), G) = 0$ forces $s_{C'} = 0$. But $\varphi_* \colon H_1(C;G) \to H_1(C';G)$ is an isomorphism $\mathbb{Z}/2 \to \mathbb{Z}/2$, so naturality would demand $\varphi_* s_C = s_{C'} \operatorname{Tor}(\varphi_*) = 0$, which is false. As we sought to show.

The two coefficient theorems answer the same question in opposite variances: Tor measures the failure of $\otimes G$ to commute with homology, and Ext the failure of $\mathrm{Hom}(-, G)$ to do so. The cohomology statement is the one used most often below, and is next.

Theorem 4.1.8: Universal Coefficient Theorem For Cohomology

Let C be a chain complex of free abelian groups with homology $H_\bullet(C; G)$. Then the cohomology groups $H^n(C; G)$ of the cochain complex $\mathrm{Hom}_G(C_\bullet, G)$ is given by:

$$0 \to \mathrm{Ext}(H_{n-1}(C), G) \to H^n(C; G) \xrightarrow{h} \mathrm{Hom}_G(H_n(C), G) \to 0$$

***Proof*:**
Throughout write $Z_n = \ker(\partial\colon C_n \to C_{n-1})$ for the cycles and $B_n = \mathrm{im}(\partial\colon C_{n+1} \to C_n)$ for the boundaries, both subgroups of the free abelian group C_n. The strategy is to first replace the chain complex $C_\bullet$ by the much simpler complexes of cycles and boundaries, dualize there, and only then reassemble. The single algebraic fact that makes this possible is that a subgroup of a free abelian group is free (a submodule of a free $\mathbb{Z}$-module is free, since $\mathbb{Z}$ is a PID), so that Z_n and B_n are themselves free.

Step 1: Splitting the cycles and boundaries. For each n the boundary map $\partial\colon C_n \to C_{n-1}$ has image B_{n-1} and kernel Z_n, giving the short exact sequence

$$0 \to Z_n \to C_n \xrightarrow{\partial} B_{n-1} \to 0 \tag{4.3}$$

Its right-hand term B_{n-1} is a subgroup of the free abelian group C_{n-1}, hence free, so (4.3) splits: there is a retraction $p\colon C_n \to Z_n$ with $p|_{Z_n} = \mathrm{id}_{Z_n}$, equivalently a direct sum decomposition $C_n \cong Z_n \oplus B_{n-1}$. This splitting is the only non-canonical choice in the proof, and it is the source of the (non-natural) splitting of the final sequence. Regard $Z_\bullet$ and $B_\bullet$ as chain complexes with *zero* differential. The inclusions $Z_n \hookrightarrow C_n$ and the boundary maps $C_n \to B_{n-1}$ assemble into a short exact sequence of chain complexes

$$0 \to Z_\bullet \to C_\bullet \xrightarrow{\partial} B_{\bullet-1} \to 0 \tag{4.4}$$

(in which the degree of the boundary complex is shifted down by one, matching $\partial\colon C_n \to B_{n-1}$). The point of (4.4) is that $Z_\bullet$ and $B_\bullet$ carry the zero differential, so their cochain complexes are transparent.

Step 2: Dualizing. Apply the contravariant functor $\mathrm{Hom}_G(-, G) = \mathrm{Hom}(-, G)$ to (4.4). Because each sequence (4.3) splits, $\mathrm{Hom}(-, G)$ carries it to a (split) short exact sequence of abelian groups in every degree, so dualizing the complexes is exact:

$$0 \to \mathrm{Hom}(B_{\bullet-1}, G) \xrightarrow{\partial^*} \mathrm{Hom}(C_\bullet, G) \to \mathrm{Hom}(Z_\bullet, G) \to 0 \tag{4.5}$$

is a short exact sequence of cochain complexes. Here $\mathrm{Hom}(Z_\bullet, G)$ and $\mathrm{Hom}(B_\bullet, G)$ carry the zero differential, since $Z_\bullet$ and $B_\bullet$ do. The cochain complex $\mathrm{Hom}(C_\bullet, G)$ is the one whose cohomology is $H^n(C; G)$, and (4.5) expresses it as an extension of two complexes with vanishing differential, whose cohomology is computed without any further work: $H^n(\mathrm{Hom}(Z_\bullet, G)) = \mathrm{Hom}(Z_n, G)$ and $H^n(\mathrm{Hom}(B_\bullet, G)) = \mathrm{Hom}(B_n, G)$.

Step 3: The long exact sequence and its connecting map. The short exact sequence of cochain complexes (4.5) produces a long exact sequence in cohomology. Using the identification of the outer cohomology groups from Step 2, and accounting for the degree shift by one in the boundary complex, the long exact sequence reads

$$\cdots \to \mathrm{Hom}(Z_{n-1}, G) \xrightarrow{i^*_{n-1}} \mathrm{Hom}(B_{n-1}, G) \to H^n(C; G) \to \mathrm{Hom}(Z_n, G) \xrightarrow{i^*_n} \mathrm{Hom}(B_n, G) \to \cdots \tag{4.6}$$

The connecting homomorphism $i^*_n \colon \mathrm{Hom}(Z_n, G) \to \mathrm{Hom}(B_n, G)$ is, by the standard identification of the connecting map of (4.5), the map induced by the inclusion $i \colon B_n \hookrightarrow Z_n$ of boundaries into cycles, that is $i^*_n = \mathrm{Hom}(i, G)$ (restriction of a homomorphism on Z_n to the subgroup B_n). This is the one computational point of the proof and is verified by chasing a representative cochain through (4.5): a cocycle of $\mathrm{Hom}(C_\bullet, G)$ whose image in $\mathrm{Hom}(Z_n, G)$ is a prescribed homomorphism on cycles has connecting image equal to the restriction of that homomorphism along $B_n \hookrightarrow Z_n$.

Step 4: Extracting the short exact sequence. The long exact sequence (4.6) breaks into short exact sequences in the usual way, by splitting off the cokernel of the incoming map and the kernel of the outgoing map:

$$0 \to \mathrm{coker}(i^*_{n-1}) \to H^n(C; G) \xrightarrow{h} \ker(i^*_n) \to 0 \tag{4.7}$$

where $i^*_{n-1} \colon \mathrm{Hom}(Z_{n-1}, G) \to \mathrm{Hom}(B_{n-1}, G)$ and $i^*_n \colon \mathrm{Hom}(Z_n, G) \to \mathrm{Hom}(B_n, G)$ are the dual restriction maps of Step 3. It remains to identify the two outer terms with $\mathrm{Ext}(H_{n-1}(C), G)$ and $\mathrm{Hom}(H_n(C), G)$ respectively, and to check that the right-hand map of (4.7) is the map h named in the statement.

Step 5: The free resolution of H_n and the identification of the outer terms. Fix n and consider the exact sequence

$$0 \to B_n \xrightarrow{i} Z_n \to H_n(C) \to 0 \tag{4.8}$$

It is exact by the very definition $H_n(C) = Z_n/B_n$, and both B_n and Z_n are free by the freeness fact of Step 1. Hence (4.8) is a free resolution of $H_n(C)$ of length one (every abelian group has such a resolution, since $\mathbb{Z}$ is a PID, as recalled before lemma 4.1.6). The functor $\mathrm{Ext}(H_n(C), G)$ is computed by dropping the term $H_n(C)$, applying $\mathrm{Hom}(-, G)$, and taking cohomology of the resulting two-term complex $0 \to \mathrm{Hom}(Z_n, G) \xrightarrow{i^*_n} \mathrm{Hom}(B_n, G) \to 0$ (this is the definition of Ext as a derived functor of Hom, computed from any free resolution, and is independent of the resolution by lemma 4.1.6 and [10, Unique Projection Resolution]). By left exactness of Hom this two-term complex has

$$\ker(i^*_n) = \mathrm{Hom}(H_n(C), G), \qquad \mathrm{coker}(i^*_n) = \mathrm{Ext}(H_n(C), G) \tag{4.9}$$

The first equality holds because $\mathrm{Hom}(-, G)$ is left exact: applying it to the right-exact sequence $B_n \xrightarrow{i} Z_n \to H_n(C) \to 0$ gives the left-exact sequence $0 \to \mathrm{Hom}(H_n(C), G) \to \mathrm{Hom}(Z_n, G) \xrightarrow{i^*_n} \mathrm{Hom}(B_n, G)$, so a homomorphism $Z_n \to G$ lies in $\ker(i^*_n)$ (vanishes on B_n) precisely when it factors through the quotient $Z_n/B_n = H_n(C)$. The second equality is the definition of $\mathrm{Ext}(H_n(C), G)$ as the first cohomology of the dualized resolution.

Step 6: Assembling the sequence. Substituting (4.9) into (4.7) (using the index $n-1$ for the cokernel term and n for the kernel term) yields

$$0 \to \mathrm{Ext}(H_{n-1}(C), G) \to H^n(C; G) \xrightarrow{h} \mathrm{Hom}(H_n(C), G) \to 0$$

which is the asserted sequence. The cokernel of i^*_{n-1} is $\mathrm{Ext}(H_{n-1}(C), G)$ by (4.9) with n replaced by $n-1$, and the kernel of i^*_n is $\mathrm{Hom}(H_n(C), G)$, sitting in degree n. The map h is the composite of the map $H^n(C;G) \to \ker(i_n^*)$ from (4.7) (induced by the surjection $\mathrm{Hom}(C_\bullet, G) \to \mathrm{Hom}(Z_\bullet, G)$ of (4.5), that is, restriction of a cochain to cycles) with the identification $\ker(i_n^*) = \mathrm{Hom}(H_n(C), G)$, and it is exactly the evaluation map of the statement: a cocycle $\varphi\colon C_n \to G$ (with $\varphi \circ \partial = 0$, so φ vanishes on B_n) restricts to $\varphi|_{Z_n}$, which factors through $Z_n/B_n = H_n(C)$ to give $\overline{\varphi}\colon H_n(C) \to G$, and $h([\varphi]) = \overline{\varphi}$. This is well-defined on cohomology classes: a coboundary $\varphi = \psi \circ \partial$ vanishes on Z_n, so its associated $\overline{\varphi}$ is zero. Thus h is the homomorphism $[\varphi] \mapsto \overline{\varphi|_{Z_n}}$, as named in the statement.

Step 7: The splitting. The sequence (4.7) splits because the retraction $p\colon C_n \to Z_n$ of (4.3) chosen in Step 1 furnishes a section of h directly. Given a homomorphism $\bar{\psi} \in \mathrm{Hom}(H_n(C), G)$, let $\psi_0\colon Z_n \to G$ be its composite with the quotient $Z_n \twoheadrightarrow Z_n/B_n = H_n(C)$, so that ψ_0 vanishes on B_n and descends to $\bar{\psi}$. Set $\varphi = \psi_0 \circ p\colon C_n \to G$. This φ is a cocycle: for $c \in C_{n+1}$ the element ∂c lies in Z_n, so the retraction property $p|_{Z_n} = \mathrm{id}_{Z_n}$ gives $\varphi(\partial c) = \psi_0(p(\partial c)) = \psi_0(\partial c)$, and $\partial c \in B_n = \ker(Z_n \to H_n(C))$ forces $\psi_0(\partial c) = 0$; hence $\varphi \circ \partial = 0$. Moreover $\varphi|_{Z_n} = \psi_0 \circ p|_{Z_n} = \psi_0$, which descends to $\bar{\psi}$, so by the description of h in Step 6, $h([\varphi]) = \overline{\varphi|_{Z_n}} = \bar{\psi}$. Thus

$$k\colon \mathrm{Hom}(H_n(C), G) \to H^n(C;G), \qquad \bar{\psi} \mapsto [\psi_0 \circ p]$$

is a homomorphism with $h \circ k = \mathrm{id}$, a section of h. A section of the surjection in (4.7) splits the sequence, so

$$H^n(C;G) \cong \mathrm{Ext}(H_{n-1}(C), G) \oplus \mathrm{Hom}(H_n(C), G)$$

as abelian groups. The section k depends on the chosen retraction p, equivalently on the chosen splitting of (4.3); a different choice gives a different k, and no choice is preserved by an arbitrary chain map. Hence the splitting is not natural, even though the short exact sequence itself is natural in C (naturality being inherited from the naturality of the long exact sequence (4.6), that is proposition 3.4.4 read for cochain complexes after reindexing, and of the resolution (4.8)), completing the proof.

To comput Ext, we have a couple of nice shorhands:

Proposition 4.1.9: Computing Exterior Group

Let H be finitely generated Then

1. $\mathrm{Ext}(H \oplus H', G) \cong \mathrm{Ext}(H, G) \oplus \mathrm{Ext}(H', G)$
2. $\mathrm{Ext}(H, G) = 0$ if H is free
3. $\mathrm{Ext}(\mathbb{Z}_n, G) \cong G/nG$

In particular, if we have homology groups H_n, H_{n-1} with torsion subgroups $T_n \subseteq H_n$ and $T_{n-1} \subseteq H_{n-1}$, then:

$$H^n(C;\mathbb{Z}) \cong (H_n/T_n) \oplus T_{n-1}$$

***Proof*:**
[14, p.196]

Naturally, the short exact sequences in the universal coefficient theorem are natural, that is a chain map $\alpha : C \to C'$ of free abelian groups induces a commutative diagram:

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Ext}(H_{n-1}(C), G) \arrow[r] & H^n(C; G) \arrow[r] & \mathrm{Hom}(H_n(C), G) \arrow[r] & 0 \\
0 \arrow[r] & \mathrm{Ext}(H_{n-1}(C'), G) \arrow[r] \arrow[u, "(\alpha_*)^*"] & H^n(C'; G) \arrow[r] \arrow[u, "\alpha^*"] & \mathrm{Hom}(H_n(C'), G) \arrow[r] \arrow[u, "(\alpha_*)^*"] & 0
\end{tikzcd}$$

On the other hand, the splitting is not natural, since it depends on the projection $\pi : C_n \to Z_n$. Together with the short-five lemma we get:

Corollary 4.1.10: Homology Isomorphism, then Cohomology Isomorphism

Let α be a map between chains. Then if α induces an isomorphism on homology groups, it induces an isomorphism on cohomology groups with any coefficient group G.

***Proof*:**
immediate

Note that instead of using abelian groups (i.e. $\mathbb{Z}$-modules), we could use general R-modules. If R is a PID, then then the universal coefficient theorem is exactly the same, while if R is not then we may have longer free resolutions. If $R = F$ is a field, then any F-module is always free, hence we always have length 0 free resolutions.

Some remarks are in order: when $n = 0$, then $\mathrm{Ext} = 0$, and so we simply get

$$H^0(X; G) \cong \mathrm{Hom}(H_0(X), G)$$

Explicitly, we can see this since singular 0-simplices are just points, and a cochain in $C^0(X; G)$ is an arbitrary function $\varphi : X \to G$, not necessarily continuous. For this to be a cocycle, we have that each 1-simplex $\sigma[v_0, v_1] \to X$ w have

$$\delta(\varphi)(\sigma) = \varphi(\partial\sigma) = \varphi(\sigma(v_1)) - \varphi(\sigma(v_1)) = 0$$

Hence, it is equivalent to saying that φ is constant on path components of X. Thus, $H^0(X; G)$ is all the function from path-components of X to G, which is the same as $\mathrm{Hom}(H_0(X), G)$.

Similarly, since $H_0(X)$ is free, $\mathrm{Ext}(H_0(X), G)0$, so

$$H^1(X; G) \cong \mathrm{Hom}(H_1(X), G)$$

Finally, when $G = F$ is a field, then Ext_F vanishes, and so

$$H^n(X; F) \cong \mathrm{Hom}_F(H_n(X; F), F)$$

Over a field the theorem becomes a duality between homology and cohomology.

Corollary 4.1.11: Cohomology with Field Coefficients is the Dual of Homology

Let F be a field and X a topological space. For every n the Kronecker evaluation pairing induces a natural isomorphism of F-vector spaces

$$H^n(X;F) \xrightarrow{\cong} \operatorname{Hom}_F\big(H_n(X;F),F\big)$$

so cohomology with coefficients in F is the F-linear dual of homology with coefficients in F. When $H_n(X;F)$ is finite-dimensional the two spaces have the same dimension,

$$\dim_F H^n(X;F) = \dim_F H_n(X;F)$$

so the n-th Betti number is computed equally by homology or cohomology over F.

***Proof*:**
Over the field F the singular chain groups $C_\bullet(X;F)$ are F-vector spaces and the cochain complex is their F-dual, $C^\bullet(X;F) = \operatorname{Hom}_F(C_\bullet(X;F),F)$. Dualizing a linear map is exact on vector spaces, since every short exact sequence of F-vector spaces splits, so $\operatorname{Hom}_F(-,F)$ commutes with passage to homology and $H^n\big(\operatorname{Hom}_F(C_\bullet(X;F),F)\big) \cong \operatorname{Hom}_F\big(H_n(X;F),F\big)$. Equivalently, in the universal coefficient theorem (theorem 4.1.8) the term $\operatorname{Ext}_F(H_{n-1}(X;F),F)$ vanishes because every F-module is free, leaving the evaluation map $H^n(X;F) \to \operatorname{Hom}_F(H_n(X;F),F)$ an isomorphism whose naturality is that of the coefficient sequence displayed above. A linear map and its dual have the same rank, so the two dimensions agree whenever they are finite, completing the proof.

Example 4.1.12: Cohomology Groups
For reference, we recall the following Δ-complexes from example 3.1.10

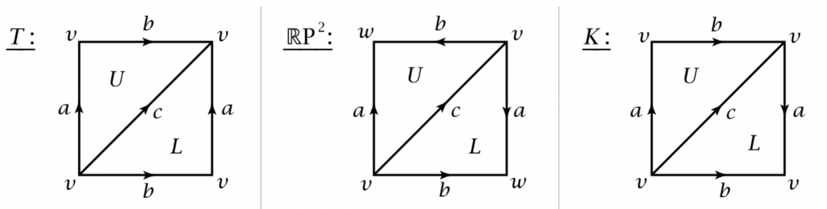


Figure 4.1: Reference Figure for Torus, Projective space, and Klein Bottle

In the proceeding calculations, we may either use the Δ-complex or the cellular complex (we shall later go over how they are all equivalent, which is a rather routine algebraic argument)

1. Let $X = T = S^1 \times S^1$ and

$$\Delta_0(T) = \mathbb{Z} \qquad \Delta_1(T) = \mathbb{Z}^3 \qquad \Delta_2(T) = \mathbb{Z}^2$$

with the chain:

$$0 \to \Delta_2 \xrightarrow{\partial_2} \Delta_1 \xrightarrow{\partial 1} \Delta_1 \to 0$$

taking the $\mathrm{Hom}(-, \mathbb{Z})$ functor, we get:

$$0 \to \mathrm{Hom}(\Delta_0, \mathbb{Z}) \xrightarrow{\partial^1} \mathrm{Hom}(\Delta_1, \mathbb{Z}) \xrightarrow{\partial^2} \mathrm{Hom}(\Delta_2, \mathbb{Z}) \to 0$$

By some basic algebra, if the generators of Δ_0 is v, the generators ofΔ_1 are a, b, c and the generators of Δ_2 are U, L, then the generators of the hom-dual are v^*, a^*, b^*, c^*, and U^*, L^* which are all the usual δ function (for example $v^*(x) = 1$ if $x = v$ and 0 otherwise). Hence, we have:

$$0 \to \mathbb{Z} \xrightarrow{\partial^1} \mathbb{Z}^3 \xrightarrow{\partial^2} \mathbb{Z}^2 \to 0$$

Now, to find these mapping first notice that:

$$\partial^1(v^*)(e) = v^*(v) - v^*(v) = 0$$

hence $\partial^1 = 0$ is trivial. Next

$$\begin{aligned} \partial^2(a^*)(U) &= a^*(a) + a^*(b) - a^*(c) = 1 \\ \partial^2(a^*)(L) &= a^*(c) - a^*(a) - a^*(b) = -1 \end{aligned}$$

Thus

$$\partial^2(a^*) = U^* - L^*$$

Similarly, we get:

$$\begin{aligned} \partial^2(b^*) &= U^* - L^* \\ \partial^2(c^*) &= L^* - U^* \end{aligned}$$

We now have all the values needed to compute the cohomology groups. This is the same type of linear algebra we have been already doing, the result of which is:

$$H^n(X; \mathbb{Z}) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z}^2 & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

Now, if $G = \mathbb{Z}/2\mathbb{Z}$, then the dual is isomorphic to:

$$0 \to \mathbb{Z}/2\mathbb{Z} \to (\mathbb{Z}/2\mathbb{Z})^3 \to \mathbb{Z}/2\mathbb{Z}^2 \to 0$$

then we still have $\partial^1 = 0$ (the coefficients do not effect the maps), however we now have that $= -1$, hence $\partial^2(a^*) = \partial^2(a^*) = \partial^2(a^*) = U^* + L^*$. (Put down your computations)

$$H^n(X) = \begin{cases} \mathbb{Z}/2\mathbb{Z} & n = 0 \\ (\mathbb{Z}/2\mathbb{Z})^2 & n = 1 \\ (\mathbb{Z}/2\mathbb{Z}) & n = 2 \\ 0 & n \geq 3 \end{cases}$$

as we sought to show.

2. Let $X = K$ be the Klein bottle. Then the cellular complex:

$$0 \to \mathbb{Z} \to \mathbb{Z}^2 \to \mathbb{Z} \to 0$$

With homology group:

$$H_n(K) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z} \oplus (\mathbb{Z}/2\mathbb{Z}) & n = 1 \\ 0 & n \geq 2 \end{cases}$$

When dualizing the chain respect to $\mathbb{Z}$ we get:

$$0 \leftarrow \mathbb{Z} \overset{\delta}{\leftarrow} \mathbb{Z}^3 \overset{\delta}{\leftarrow} \leftarrow 0$$

since $(\mathbb{Z}^n)^* \cong \mathbb{Z}^n$. Let $H^n = H^n(K;\mathbb{Z})$. Now:

$$\delta^0(f)(a,b) = f(\delta_1(a,b)) = 0$$

Hence $\delta^0 = 0$. Next:

$$\delta^1(g)(n) = g(\partial_2(n)) = g(2n, 0)$$

Choosing the right basis for C^1, this map becomes

$$(a,b) \mapsto 2a$$

Hence, the cohomology groups are:

$$H^n(X;\mathbb{Z}) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z} & n = 1 \\ \mathbb{Z}/2\mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

3. Assuming cellular complexes are isomorphic to singular complexes, we have that:

$$\tilde{H}^k(S^n; G) = \begin{cases} G & n = k \\ 0 & \text{otherwise} \end{cases}$$

Also thought this link was cool: many ways of computing the homology of a torus here

Finally, we quickly go through how many of the concepts from homology, such as relative homology or reduced homology and others, naturally translate to cohomology

(hatcher p.199; reduced is naturally defined, it is homotopy invariant excision still works, and the axioms of cohomology translate. Simplicial cohomology is naturally isomorphic, cellular cohomology follows too)

For cellular homology, in particular their skeletons

Proposition 4.1.13: Cellular Cohomology

$H^n(X;G) \cong \ker d^n/\mathrm{im} d^{n-1}$. Furthermore

$$(H^\bullet(X^n, X^{n-1}; G), \delta_\bullet) \cong (\mathrm{Hom}(H^\bullet(X^n, X^{n-1}; G), G), d_\bullet \circ h)$$

i.e. the cellular complex is isomorphic to the dual of the cellular chain complex after applying the $\mathrm{Hom}(-, G)$ functor.

***Proof*:**
[14, p.203]

One translation deserves a full treatment, since later chapters lean on it: the Mayer-Vietoris sequence of theorem 3.4.19 has a cohomology counterpart, with the arrows reversed and the connecting homomorphism raising degree. The proof dualizes the chain-level machinery already assembled for the homology version, and the dualization rests on two small algebraic facts, recorded first.

Lemma 4.1.14: Dualizing Split Exactness and Chain Homotopies

Let G be an abelian group.

1. If $0 \to A \xrightarrow{i} B \xrightarrow{j} C \to 0$ is a split short exact sequence of abelian groups, then
$$0 \to \mathrm{Hom}(C, G) \xrightarrow{j^*} \mathrm{Hom}(B, G) \xrightarrow{i^*} \mathrm{Hom}(A, G) \to 0$$
is a short exact sequence, and it splits.
2. If $f, g : C_\bullet \to D_\bullet$ are chain homotopic chain maps, then the dual maps $f^*, g^* : \mathrm{Hom}(D_\bullet, G) \to \mathrm{Hom}(C_\bullet, G)$ are cochain homotopic cochain maps. In particular, $\mathrm{Hom}(-, G)$ carries a chain homotopy equivalence to a cochain homotopy equivalence, and hence induces isomorphisms on cohomology.

***Proof*:**
For (1), exactness needs the splitting only at the last step. The map j^* is injective: if $\theta \circ j = 0$ then $\theta = 0$, since j is surjective. For exactness at $\mathrm{Hom}(B, G)$, the composite $i^* j^* = (ji)^* = 0$ gives $\mathrm{im} j^* \subseteq \ker i^*$. Conversely, if $\eta : B \to G$ vanishes on $\mathrm{im} i = \ker j$, then $\eta(b)$ depends only on $j(b)$, so η factors as $\theta \circ j$ for a homomorphism $\theta : C \to G$, that is $\eta \in \mathrm{im} j^*$. For surjectivity of i^*, choose a retraction $r : B \to A$ with $ri = \mathrm{id}_A$, which the splitting provides (lemma 3.4.24 converts a section of the quotient map into such a retraction). Then for any $h : A \to G$ the homomorphism $h \circ r$ restricts along i to h, so $i^*(h \circ r) = h$. The same r shows that the dual sequence splits: $r^* : \mathrm{Hom}(A, G) \to \mathrm{Hom}(B, G)$ satisfies $i^* r^* = (ri)^* = \mathrm{id}$, a section of i^*.

For (2), a dual map is a cochain map: for $\theta \in \mathrm{Hom}(D_n, G)$,

$$\delta(f^*\theta) = \theta \circ f_n \circ \partial_{n+1} = \theta \circ \partial_{n+1} \circ f_{n+1} = f^*(\delta\theta)$$

using that f is a chain map in the middle equality. Now let P be a chain homotopy from f to g

(definition 3.3.6), a collection of homomorphisms $P_n : C_n \to D_{n+1}$ with

$$g_n - f_n = \partial_{n+1} P_n + P_{n-1} \partial_n$$

Define $Q : \mathrm{Hom}(D_n, G) \to \mathrm{Hom}(C_{n-1}, G)$ by $Q\theta = \theta \circ P_{n-1}$. Then for $\theta \in \mathrm{Hom}(D_n, G)$,

$$g^*\theta - f^*\theta = \theta \circ (g_n - f_n) = \theta \circ \partial_{n+1} \circ P_n + \theta \circ P_{n-1} \circ \partial_n = Q(\delta\theta) + \delta(Q\theta)$$

since $\theta \circ \partial_{n+1} = \delta\theta$ and $(Q\theta) \circ \partial_n = \delta(Q\theta)$. So Q is a cochain homotopy from f^* to g^*. Finally, if ρ is a homotopy inverse of a chain homotopy equivalence ι, so that $\rho\iota$ and $\iota\rho$ are chain homotopic to identities, then dualizing gives that $\iota^*\rho^* = (\rho\iota)^*$ and $\rho^*\iota^* = (\iota\rho)^*$ are cochain homotopic to identities, so ι^* is a cochain homotopy equivalence. A cochain complex is a chain complex after reindexing, so proposition 3.3.7 applies: ι^* and ρ^* induce mutually inverse maps on cohomology, so the maps ι^* induces are isomorphisms, completing the proof.

Theorem 4.1.15: Mayer-Vietoris Sequence in Cohomology

Let $A, B \subseteq X$ with $X = \mathrm{int}A \cup \mathrm{int}B$, and let G be an abelian group. Then there is a long exact sequence

$$0 \to H^0(X;G) \to \cdots \to H^n(X;G) \xrightarrow{\psi^*} H^n(A;G) \oplus H^n(B;G) \xrightarrow{\varphi^*} H^n(A \cap B;G) \xrightarrow{\delta} H^{n+1}(X;G) \to \cdots$$

in which ψ^* is the pair of restrictions induced by the inclusions $A \hookrightarrow X$ and $B \hookrightarrow X$, and φ^* is the difference of the restrictions induced by the inclusions $A \cap B \hookrightarrow A$ and $A \cap B \hookrightarrow B$.

***Proof*:**
The proof dualizes the two chain-level ingredients of theorem 3.4.19, the short exact sequence of chain complexes built there and the small-chain comparison of lemma 3.4.8, with lemma 4.1.14 giving exactly what remains after dualization.

Stage 1: the chain-level sequence and its degreewise splitting. Write $C_n(A+B) = C_n(A) + C_n(B) \subseteq C_n(X)$ for the subcomplex of chains each of whose simplices has image in A or in B, as in Stage 1 of the proof of theorem 3.4.19. Stage 2 of that proof established the short exact sequence of chain complexes

$$0 \to C_\bullet(A \cap B) \xrightarrow{\varphi} C_\bullet(A) \oplus C_\bullet(B) \xrightarrow{\psi} C_\bullet(A+B) \to 0 \tag{4.10}$$

with $\varphi(x) = (x, -x)$ and $\psi(a,b) = a+b$. In each degree this is a sequence of free abelian groups: $C_n(A \cap B)$ is free on the simplices with image in $A \cap B$, the direct sum $C_n(A) \oplus C_n(B)$ is free on the two embedded copies of the simplices of A and of B, and $C_n(A+B)$ is free on the small simplices themselves, these being a subset of the basis of $C_n(X)$. A short exact sequence of abelian groups whose quotient is free splits: choose a preimage under ψ of each basis element of $C_n(A+B)$ and extend freely to a section of ψ. So (4.10) splits degreewise.

Stage 2: dualizing to a short exact sequence of cochain complexes. Apply $\mathrm{Hom}(-, G)$ to (4.10). By lemma 4.1.14(1) each degree yields a short exact sequence of abelian groups, and the dual maps are cochain maps by lemma 4.1.14(2), so the result is a short exact sequence of cochain complexes

$$0 \to C^\bullet_{A+B}(X;G) \xrightarrow{\psi^*} C^\bullet(A;G) \oplus C^\bullet(B;G) \xrightarrow{\varphi^*} C^\bullet(A \cap B;G) \to 0 \tag{4.11}$$

where $C^n_{A+B}(X;G) := \mathrm{Hom}(C_n(A+B), G)$ and the middle term uses the canonical identification $\mathrm{Hom}(M \oplus N, G) \cong \mathrm{Hom}(M, G) \oplus \mathrm{Hom}(N, G)$, a homomorphism on a direct sum being the pair of its restrictions to the summands, an identification compatible with the coboundaries since the boundary of the middle term of (4.10) is $\partial \oplus \partial$. Under this identification the dual maps are explicit: $\psi^*\theta = (\theta|_{C_\bullet(A)}, \theta|_{C_\bullet(B)})$ restricts a small cochain to the chains lying in A and to those lying in B, and $\varphi^*(\alpha, \beta) = \alpha|_{C_\bullet(A\cap B)} - \beta|_{C_\bullet(A\cap B)}$, the sign coming from $\varphi(x) = (x, -x)$.

Stage 3: small cochains compute $H^n(X;G)$. By lemma 3.4.8 applied to the collection $U = \{A, B\}$, whose interiors cover X by hypothesis, the inclusion $\iota : C_\bullet(A+B) \hookrightarrow C_\bullet(X)$ is a chain homotopy equivalence. By lemma 4.1.14(2) its dual

$$\iota^* : C^\bullet(X;G) \to C^\bullet_{A+B}(X;G)$$

is a cochain homotopy equivalence and induces isomorphisms $\iota^* : H^n(X;G) \xrightarrow{\sim} H^n_{A+B}(X;G)$ for every n, writing $H^n_{A+B}(X;G)$ for the cohomology of $C^\bullet_{A+B}(X;G)$. As in the homology proof, this is the single point where the covering hypothesis enters.

Stage 4: assembly. A cochain complex is a chain complex after reindexing, so the algebraic snake lemma (theorem 3.4.3) applies to (4.11), with the connecting homomorphism now raising degree by one. The resulting long exact sequence reads

$$\cdots \to H^n_{A+B}(X;G) \xrightarrow{\psi^*} H^n(A;G) \oplus H^n(B;G) \xrightarrow{\varphi^*} H^n(A\cap B;G) \xrightarrow{\delta} H^{n+1}_{A+B}(X;G) \to \cdots$$

the middle term being identified by the fact that cohomology commutes with direct sums of cochain complexes, kernels and images being computed coordinatewise. The sequence begins with $0 \to H^0_{A+B}(X;G)$, since there are no cochains in negative degrees. Now substitute $H^n(X;G)$ for $H^n_{A+B}(X;G)$ throughout, using the isomorphisms ι^* of Stage 3. Replacing groups along isomorphisms, with the adjacent maps composed with the isomorphisms and their inverses, preserves exactness, so it remains only to identify the substituted maps. The composite $\psi^*\iota^*$ sends a cochain $\theta \in C^n(X;G)$ to the pair of its restrictions to chains in A and to chains in B, which is exactly the pair of maps induced by the inclusions $A \hookrightarrow X$ and $B \hookrightarrow X$, and the new connecting map is the composite $(\iota^*)^{-1}\delta$. The result is the long exact sequence of the statement, completing the proof.

Corollary 4.1.16: Naturality of the Mayer-Vietoris Sequence in Cohomology

The sequence of theorem 4.1.15 is natural in the triple (X, A, B): if $f : X \to Y$ is continuous with $f(A) \subseteq A'$ and $f(B) \subseteq B'$, where $X = \mathrm{int}A \cup \mathrm{int}B$ and $Y = \mathrm{int}A' \cup \mathrm{int}B'$, then the diagram

$$\begin{tikzcd}
\cdots \arrow[r] & H^n(Y;G) \arrow[r] \arrow[d, "f^*"] & H^n(A';G) \oplus H^n(B';G) \arrow[r] \arrow[d] & H^n(A'\cap B';G) \arrow[r, "\delta"] \arrow[d] & H^{n+1}(Y;G) \arrow[r] \arrow[d, "f^*"] & \cdots \\
\cdots \arrow[r] & H^n(X;G) \arrow[r] & H^n(A;G) \oplus H^n(B;G) \arrow[r] & H^n(A\cap B;G) \arrow[r, "\delta"] & H^{n+1}(X;G) \arrow[r] & \cdots
\end{tikzcd}$$

commutes, the unlabelled vertical maps being induced by the restrictions of f.

***Proof*:**
The chain map $f_\# : C_\bullet(X) \to C_\bullet(Y)$ carries $C_\bullet(A)$ into $C_\bullet(A')$, $C_\bullet(B)$ into $C_\bullet(B')$, $C_\bullet(A\cap B)$ into $C_\bullet(A'\cap B')$, and hence $C_\bullet(A+B)$ into $C_\bullet(A'+B')$. The resulting vertical maps form a commutative diagram from the sequence (4.10) of (X,A,B) to that of (Y,A',B'), both squares commuting at the level of chains: for instance $\psi(f_\# a, f_\# b) = f_\# a + f_\# b = f_\#\psi(a,b)$. Applying $\mathrm{Hom}(-,G)$ reverses all arrows and preserves commutativity, by functoriality, giving a commutative diagram from the dualized sequence (4.11) of (Y,A',B') to that of (X,A,B). Proposition 3.4.4, applied after the reindexing of Stage 4 above, yields a commuting ladder between the two long exact sequences in the $H^\bullet_{A+B}$ form. It remains to transport the ladder along the isomorphisms ι^* of Stage 3, and for that it suffices that the square

$$\begin{tikzcd}
H^n(Y;G) \arrow[r, "\iota_Y^*"] \arrow[d, "f^*"] & H^n_{A'+B'}(Y;G) \arrow[d] \\
H^n(X;G) \arrow[r, "\iota_X^*"] & H^n_{A+B}(X;G)
\end{tikzcd}$$

commutes. At the chain level $f_\#\iota_X = \iota_Y f_\#$, both composites being $f_\#$ applied to small chains, so the dual square commutes already at the level of cochains. Every square of the transported ladder is then a composite of commuting squares, including those containing the connecting maps $(\iota^*)^{-1}\delta$, completing the proof.

This is the sequence that the de Rham comparison in [20, The de Rham Map is a Morphism of Mayer-Vietoris Sequences] runs against the Mayer-Vietoris sequence of differential forms.

4.2 Cup Product

Let $G = R$ now be a ring. We shall show that we can put a graded ring structure on $C^\bullet(X;R)$. If you have done DeRham cohomology, the product we shall define can be thought of as the generalization of the wedge product and shall be behave very similarly:

Definition 4.2.1: Cup Product [on chains]

Let $f \in C^k(X;R)$ and $g \in C^\ell(X;R)$. Then the *cup product*, denoted $f \smile g \in C^{k+\ell}(X;R)$ is the cochain that evaluates on a singular simplex $\sigma : \Delta^{k+\ell} \to X$ is given by:

$$(f\smile g)(\sigma) = f(\sigma|_{[v_0,\dots,v_k]})g(\sigma|_{[v_k,\dots,v_{k+\ell}]}) \in R$$

To see that the cup product works on cohomology classes, we require the following

Lemma 4.2.2: Derivation Property Of Cup Product

$$\delta(f\smile g) = (\delta f)\smile g + (-1)^k f \smile (\delta g)$$

***Proof*:**
Let $\sigma : \Delta^{k+\ell+1} \to X$. Then:

$$(\delta f) \smile g(\sigma) = \sum_{i}^{k+1} (-1)^i f(\sigma|_{v_0,\dots,\hat{v}_i,\dots,v_{k+1}}) g(\sigma|_{v_{k+1},\dots,v_{k+\ell+1}})$$
$$(-1)^k (f \smile \delta g)(\sigma) = \sum_{i=k}^{k+\ell+1} (-1)^i f(\sigma|_{v_0,\dots,\hat{v}_i,\dots,v_k}) g(\sigma|_{v_k,\dots,v_{k+\ell+1}})$$

Adding these together, the last term of the first some cancels with the first term of the second sum, the remaining terms are then exactly

$$\delta(f \smile g)(\sigma) = (f \smile g)(\partial\sigma)$$

since

$$\partial\sigma = \sum_{i}^{k+\ell+1} (-1)^i \sigma|_{[v_0,\dots,\hat{v}_i,\dots,v_{k+\ell+1}]}$$

completing the proof.

From this, we see that the cup product of two cocycles is a cocycles, and the product of a cocycles and a coboundary (or a coboundary and a cocycle) is a coboundary since

$$f \smile (\delta g) = \pm\delta(f \smile g)$$

if $\delta f = 0$ and

$$(\delta f) \smile g = \delta(f \smile g)$$

if $\delta g = 0$. Thus, the cup-product is well-defined on cohomology groups:

$$H^k(X;R) \times H^\ell(X;R) \xrightarrow{\smile} H^{k+\ell}(X;R)$$

Since on the level of chains the cup product is associative and distributive, so too it is on cohomologies. Furthermore, since R has an identity, so too does the cup product with the class $1 \in H^0(X;R)$ which is the 0-cocycle taking the value 1 and each singular 0-simplex.

Since the cup product for simplicial cohomology can be defined by the exact same formula for singular cohomologies, the canonical isomorphism between simplicial and singular cohomologies respect cup products.

Proposition 4.2.3: Cup Product Commutes With Continuous Map

Let $f : X \to Y$ be a continuous map, and let $f^* : H^n(Y;R) \to H^n(X;R)$ the induced cohomology maps. Then

$$f^*(\alpha \smile \beta) = f^*(\alpha) \smile f^*(\beta)$$

and similarly for the relative case.

***Proof*:**
This is a simple computation with $f^\#$ (see [14, p.210] if you need help)

Theorem 4.2.4: Graded-Commutative Structure

When R is commutative,
$$\alpha \smile \beta = (-1)^{k\ell} \beta \smile \alpha$$
for all $\alpha \in H^k(X, A; R)$ and $\beta \in H^\ell(X, A; R)$.

In particular, this means that $2(\alpha \smile \alpha) = 0$, which can more concisely be written as $2\alpha^2 = 0$.

***Proof*:**
Hatcher p.210

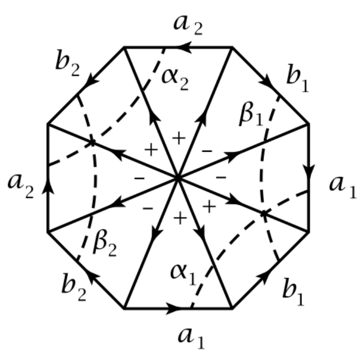
Definition 4.2.5: Cohomology Ring and Algebra

Let X be a topological space. Then $H^\bullet(X; R) = \bigoplus_k H^k(X; R)$ is a ring with the $+$ and $\smile$ operations, and is called the *cohomology ring* with respect to R and X. Multiplying by the scalars R, we get the *cohomology algebra* with respect to R and X.

Naturally, these rings have a graded structure. If A is a graded ring and $a \in A$, then we usually write $|a| = k$ to say that $a \in A_k$, and call it the *dimension* or *degree* of a. Note that by proposition 4.2.3,
$$f^* : H^\bullet(X; R) \to H^\bullet(f(X); R)$$
is a ring homomorphism.

Example 4.2.6: Cup Product and Cohomology Ring

1. Let M^g be a closed orientable surface of genus $g \geq 1$ with the usual Δ-complex structure. To enforce orientability and for reference, here is the Δ-complex when $g = 2$:



Naturally, grading of interest is:
$$H^1(M^g) \times H^1(M^g) \to H^2(M)$$
Let $\mathbb{Z}$ be the coefficients. By the universal coefficient theorem, $H^1(M) \cong \mathrm{Hom}(H_1(M), \mathbb{Z})$; given any basis for $H_1(M)$, let $\alpha_1, \alpha_2, ..., \alpha_g, \beta_1, \beta_2, ..., \beta_g$ be the basis for $H^1(M)$. For each α_i or β_j, we can choose a representative cocycle, namely a $\varphi_i \in H^1(M)$ such that $\delta\varphi_i = 0$.

Let α_i be the loop in M meeting a_i in one point and disjoint form all other basis elements a_j and b_j. Then define φ_i to have 1 on edges meeting the arc α_i and the value 0 on all other edges. Then φ_i counts the number of intersections for each edges with the arc α_i. We may similarly find representatives ψ_i for each arc β_i. We thus have a list of representatives $\varphi_1, \varphi_2, ..., \varphi_g$ and $\psi_1, \psi_2, ..., \psi_g$.

Now, applying we see that $\varphi_1 \smile \psi_1(\sigma)$ is zero when σ does not take have the outer edge b_1, as seen in the lower-right of the above figure. Now, if c is the 2-chain formed by the sum of all the 2-simplices with the signs indicated in the center of the figure, we see that

$$\varphi_1 \smile \psi_1(c) = 1$$

Since M is 2-dimensional, $\partial c = 0$ (since c is not a boundary), $0 \neq c \in H_2(M)$. Since $(\varphi_1 \smile \psi_1)(c)$ is a generators of $\mathbb{Z}$, c represents a generator of $H_2(M) \cong \mathbb{Z}$, hence $\varphi_1 \smile \psi_1$ sends (c) to the dual $\gamma \in H^2(M) \cong \mathrm{Hom}(H_2(M), \mathbb{Z}) \cong \mathbb{Z}$ which generates it. Hence:

$$\alpha_1 \smile \alpha_2 = \gamma$$

Doing a similar computation, we get that :

$$-(\beta_i \smile \alpha_j) = \alpha_i \smile \beta_j = \begin{cases} \gamma & i = j \\ 0 & i \neq j \end{cases}, \qquad \alpha_i\alpha_j = 0, \qquad \beta_i\beta_j = 0$$

Which gives us all the relations we need to determine the cup product since it it distributive. This cup product is *not* commutative since $\alpha_i \smile \beta_i = -(\beta_i \smile \alpha_i)$.

The cohomology ring we have described is called the *exterior algebra*, if $\Lambda_{\mathbb{Z}}[\alpha_1, \alpha_2, ..., \alpha_n]$ is the exterior algebra on n generator respecting the relation

$$\alpha_i\alpha_j = -\alpha_j\alpha_i, \ i \neq j \qquad \alpha_i^2 = 0$$

Then

$$H^\bullet(T; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[\alpha, \beta] \qquad |\alpha| = |\beta| = 1$$

More generally, we can see that

$$H^\bullet(T^n; R) \cong \Lambda_R[\alpha_1, \alpha_2, ..., \alpha_n]$$

The same is true for any product of odd-dimensional spheres where $\alpha_i|$ is the dimension of the ith sphere.

2. Computation for N^g, in particular $N^1 = \mathbb{R}\mathrm{P}^2$ (Hatcher p.208)

   From this, we see that

$$H^\bullet(\mathbb{R}\mathrm{P}^2; \mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha]/(\alpha^3)$$

3. Let X be a 2-dimensional CW complex by attaching the boundary of the 2-cell S^1 by the map $z \mapsto z^m$. We see using cellular cohomology (or cellular homology and the universal coefficient theorem) that

$$H^n(X; \mathbb{Z}) = \begin{cases} \mathbb{Z} & n = 0 \\ 0 & n = 1 \\ \mathbb{Z}_m & n = 2 \\ 0 & n \geq 3 \end{cases}$$

which might suggest the cup product must be trivial. However, if we take $\mathbb{Z}_m$ coefficients, we get:

$$H^n(X;\mathbb{Z}_m) = \begin{cases} \mathbb{Z}_m & n \in \{0,1,2\} \\ 0 & n \geq 3 \end{cases}$$

suggesting in this case that the cup product may not be trivial. (Hatcher p.209)

Naturally, the cup product works for relative cohomologies (Hatcher elaborates on all the different combinations, which I shall for now omit)

Naturally, it may be asked what graded commutative R-algebras occurs as cup product algebras on $H^\bullet(X;R)$ for some space X. This is called the *Realization problem*, and is in general quite difficult. If $R = \mathbb{Q}$, the problem is almost solved, if $R = \mathbb{Z}/p\mathbb{Z}$, there is some progress, and for $R = \mathbb{Z}$, very little is known.

An interesting application of cohomology rings is showing a space cannot be split into a wedge-sum up to homotopy equivalent. For example:

Example 4.2.7: Cohomology Ring And Products
There is a natural isomorphism

$$H^\bullet(\coprod_\alpha X_\alpha; R) \cong \prod_\alpha H^\bullet(X_\alpha; R)$$

and:

$$\tilde{H}^\bullet\left(\bigvee_\alpha X_\alpha; R\right) \cong \prod_\alpha \tilde{H}^\bullet(X_\alpha; R)$$

where we take the reduced cohomology in the 2nd isomorphism to be relative to a basepoint and take the relative cup product[a]

Let's now show that $\mathbb{C}\mathrm{P}^2$ is not homotopy equivalent to $S^2 \vee S^4$. In particular, with $\mathbb{Z}$ coefficients, we have:

$$H^\bullet(S^2;\mathbb{Z}) \cong H^\bullet(S^4;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^2)$$

while

$$H^\bullet(\mathbb{C}\mathrm{P}^2) \cong \mathbb{Z}[\alpha]/(\alpha^3)$$

But certainly:

$$H^\bullet(S^2 \vee S^4;\mathbb{Z}) \cong \left(\mathbb{Z}[\alpha]/(\alpha)^2 \oplus \mathbb{Z}[\alpha]/(\alpha)^2\right) \not\cong \mathbb{Z}[\alpha]/(\alpha)^3 \cong H^\bullet(\mathbb{C}\mathrm{P}^2;\mathbb{Z})$$

Giving us the result. Can you use this fact to show that no map $f : S^4 \to S^2$ is nullhomotopic? (hint, reduce to showing that if f is null-homotopic, then $C_f \simeq SX \vee Y$ where $C_f = X \cup_f Y$) (see comment for details)

Note that taking the cohomology ring over different coefficients may yield different results; sometimes two spaces will have the same cohomology ring over one ring but not over another. For example, consider if $\mathbb{R}\mathrm{P}^5$ is homotopic to $\mathbb{R}\mathrm{P}^4 \vee S^5$. Notice that these have the same cohomology groups (exercise), and hence can't be distinguished that way. They even have the same cohomology ring over $\mathbb{Z}$. However, they do not have isomorphic cohomology rings over $\mathbb{Z}/2\mathbb{Z}$.

[a] Naturally, we also assume that the base-point are deformation retracts of some neighborhoods in order for the isomorphism to hold

4.3 The Künneth Formula

The homology of a disjoint union splits as a direct sum, $H_n(\coprod_\alpha X_\alpha) \cong \bigoplus_\alpha H_n(X_\alpha)$, and this was settled in chapter 3 by a direct argument on chains. The question deferred there was the homology of a *product* $X \times Y$, and the answer is genuinely more delicate. A singular simplex in $X \times Y$ is a map $\Delta^n \to X \times Y$, equivalently a pair of maps into X and into Y sharing a domain; it carries no canonical decomposition into a simplex of X times a simplex of Y, because a simplex is not a product of simplices. The resolution proceeds in two stages, one topological and one algebraic. The topological input is the Eilenberg-Zilber theorem, which compares the singular chains of $X \times Y$ with the algebraic tensor product of the singular chains of X and of Y; the two chain complexes are chain homotopy equivalent even though they are not equal. The algebraic input is the Künneth theorem for chain complexes over a principal ideal domain, which computes the homology of a tensor product of complexes from the homologies of the factors, with a correction term measured by Tor. Combining the two produces the Künneth formula for spaces, the statement that $H_*(X \times Y)$ is built from $H_*(X)$ and $H_*(Y)$ by a tensor product and a single Tor correction.

Throughout this section R is a principal ideal domain (a PID), and homology is taken with coefficients in R; the abelian-group case is $R = \mathbb{Z}$. All tensor products and Tor groups are over R unless decorated otherwise. The graded modules $H_*(X;R) = \bigoplus_n H_n(X;R)$ and $H_*(Y;R)$ are the homologies of the singular chain complexes $C_*(X;R)$ and $C_*(Y;R)$, which are complexes of free R-modules: $C_n(X;R)$ is free on the singular n-simplices of X.

The first object to construct is the map that turns a homology class on X and one on Y into a class on the product. On chains it is induced by the algebraic tensor product of complexes, transported to the product space by Eilenberg-Zilber.

Theorem 4.3.1: Eilenberg-Zilber Theorem

For topological spaces X and Y there are natural chain maps

$$\mathrm{EZ}\colon C_*(X;R)\otimes_R C_*(Y;R) \longrightarrow C_*(X\times Y;R), \qquad \mathrm{AW}\colon C_*(X\times Y;R) \longrightarrow C_*(X;R)\otimes_R C_*(Y;R)$$

that are mutually inverse up to natural chain homotopy. Consequently $C_*(X \times Y;R)$ and $C_*(X;R) \otimes_R C_*(Y;R)$ are naturally chain homotopy equivalent, and EZ, AW induce mutually inverse isomorphisms on homology. In degree zero EZ is the identification of a pair of points with the corresponding point of $X \times Y$, and EZ is associative and commutative up to the evident chain homotopies.

Here $C_*(X;R) \otimes_R C_*(Y;R)$ denotes the tensor product of chain complexes: its degree-n term is $\bigoplus_{i+j=n} C_i(X;R) \otimes_R C_j(Y;R)$, with differential $\partial(a \otimes b) = \partial a \otimes b + (-1)^{|a|} a \otimes \partial b$. The map EZ admits an explicit formula, the *shuffle product*, sending a tensor of a singular i-simplex with a singular j-simplex to a signed sum over the $\binom{i+j}{i}$ ways of interleaving their vertices into a singular $(i+j)$-simplex of the product; the map AW is the *Alexander-Whitney map*, sending a singular n-simplex $\sigma = (\sigma_X, \sigma_Y)$ of $X \times Y$ to $\sum_{i+j=n} (\sigma_X|_{[v_0,\dots,v_i]}) \otimes (\sigma_Y|_{[v_i,\dots,v_n]})$, the front-face/back-face splitting already used to define the cup product (definition 4.2.1). That these are inverse up to homotopy is the substance of the theorem.

***Proof Key ideas*:**
The proof is by the method of acyclic models and is purely formal once the two explicit maps are written down; the verification that the shuffle and Alexander-Whitney maps are chain maps inverse up to a natural chain homotopy is a bookkeeping computation with no conceptual obstruction, but it is long. The acyclic-models mechanism runs as follows. Both functors $X, Y \mapsto C_*(X \times Y; R)$ and $X, Y \mapsto C_*(X; R) \otimes_R C_*(Y; R)$ are free on the models (Δ^i, Δ^j) (the representing pairs of simplices) and acyclic on those models (the chains of $\Delta^i \times \Delta^j$ and the tensor product of the chains of two simplices are both resolutions of R in degree zero, since simplices are contractible). A natural transformation agreeing with a prescribed map in degree zero exists and is unique up to natural chain homotopy whenever the source is free on models and the target is acyclic on those models; applying this in both directions and to the two composites produces EZ, AW and the homotopies $\mathrm{EZ} \circ \mathrm{AW} \simeq \mathrm{id}$, $\mathrm{AW} \circ \mathrm{EZ} \simeq \mathrm{id}$. The full acyclic-models theorem and the explicit homotopies are carried out in [14, §3.B] and in [10]; the argument uses only contractibility of simplices and the freeness of singular chains, both established in chapter 3. The remaining assertions (degree-zero behaviour, associativity, commutativity up to homotopy) follow from the same uniqueness clause applied to the relevant pairs of natural maps.

The content of Eilenberg-Zilber is that, on the level of homology, the product space behaves exactly as the algebraic tensor product of complexes does, despite the absence of a simplexwise product decomposition. This converts the topological problem into an algebraic one: to compute $H_*(X \times Y; R)$, compute the homology of the tensor product complex $C_*(X; R) \otimes_R C_*(Y; R)$. The map EZ is the geometric source of the cross product, and the map AW is the source of the cup product; the two products are dual aspects of the single comparison.

With the comparison in hand, the cross product is the composite of the obvious map from a tensor of homology classes into the homology of the tensor complex with the Eilenberg-Zilber isomorphism.

Definition 4.3.2: Homology Cross Product

Let X and Y be spaces and R a PID. For $\alpha \in H_i(X; R)$ and $\beta \in H_j(Y; R)$, choose cycles $a \in Z_i(X; R)$, $b \in Z_j(Y; R)$ representing them. Then $a \otimes b$ is a cycle in $\big(C_*(X; R) \otimes_R C_*(Y; R)\big)_{i+j}$, and its image under the homology isomorphism induced by EZ of theorem 4.3.1 is the *homology cross product*

$$\alpha \times \beta := \mathrm{EZ}_*[a \otimes b] \in H_{i+j}(X \times Y; R)$$

The cross product is R-bilinear, natural in X and Y, and assembles into a homomorphism of graded R-modules

$$\times\colon \bigoplus_{i+j=n} H_i(X; R) \otimes_R H_j(Y; R) \longrightarrow H_n(X \times Y; R)$$

The assignment $[a] \otimes [b] \mapsto [a \otimes b]$ is well defined on homology because the differential of the tensor complex satisfies the graded Leibniz rule: if a, b are cycles then so is $a \otimes b$, and altering a by a boundary ∂c changes $a \otimes b$ by $\partial(c \otimes b) \mp c \otimes \partial b = \partial(c \otimes b)$, a boundary (using $\partial b = 0$). The cross product is the universal recipient of bilinear, natural pairings $H_i(X) \times H_j(Y) \to H_{i+j}(X \times Y)$, and the Künneth theorem is the determination of how far it is from being an isomorphism. Naturality means that for maps $f\colon X \to X'$ and $g\colon Y \to Y'$ one has $(f \times g)_*(\alpha \times \beta) = f_*\alpha \times g_*\beta$, which follows from naturality of EZ.

Example 4.3.3: Cross Product of Sphere Classes

Take $X = S^1$ and $Y = S^1$, each with its standard generator. Write $\mu \in H_1(S^1;\mathbb{Z}) \cong \mathbb{Z}$ for the fundamental class of the circle, represented by a 1-cycle wrapping once around. The cross product $\mu \times \mu \in H_2(S^1 \times S^1;\mathbb{Z})$ is represented by $\mathrm{EZ}(a \otimes b)$ where a, b are wrapping 1-cycles. Concretely, give S^1 the Δ-complex with one vertex v and one edge, and the torus $T^2 = S^1 \times S^1$ the standard square model of example 3.1.10, with two triangular 2-simplices U, L; the shuffle product of the two edge cycles is, up to sign, the cycle $U - L$, which is exactly the generator of $H_2(T^2;\mathbb{Z}) \cong \mathbb{Z}$ computed there (the only 2-cycle, since $\partial(pU + qL) = (p+q)(a+b-c)$ vanishes iff $p = -q$). Thus $\mu \times \mu$ generates $H_2(T^2;\mathbb{Z})$: the product of the two circle classes is the top class of the torus. In degree zero, with $p \in H_0(S^1;\mathbb{Z})$ the class of a point, $p \times \mu$ and $\mu \times p$ are the two coordinate circle classes generating $H_1(T^2;\mathbb{Z}) \cong \mathbb{Z}^2$, and $p \times p$ is the point class generating H_0.

The example exhibits the cross product as an isomorphism for the torus: every homology class of T^2 is a cross product of classes from the two circle factors, and there is no correction term. The general theorem identifies exactly when this happens and what the correction is. The mechanism is entirely algebraic, so the statement and proof are first given for chain complexes.

Theorem 4.3.4: Algebraic Künneth Theorem

Let R be a PID, let C_* be a chain complex of free R-modules, and let D_* be any chain complex of R-modules. Then for each n there is a short exact sequence of R-modules

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D) \xrightarrow{\times} H_n(C \otimes_R D) \longrightarrow \bigoplus_{i+j=n-1} \mathrm{Tor}_R\big(H_i(C), H_j(D)\big) \longrightarrow 0$$

natural in C and D. The sequence splits, but the splitting is not natural. If in addition R is a field, the Tor term vanishes and the cross product is an isomorphism $H_n(C \otimes_R D) \cong \bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D)$.

The map labelled $\times$ is the same construction as in definition 4.3.2: a cycle of C tensored with a cycle of D is a cycle of $C \otimes_R D$, and $[a] \otimes [b] \mapsto [a \otimes b]$. The theorem says this map is injective with cokernel measured by a single Tor group, shifted down by one in degree. Over a field every module is flat and Tor vanishes, recovering the clean statement that homology commutes with tensor product; the Tor term is the precise obstruction to that over a general PID, and it is supported on torsion, since $\mathrm{Tor}_R(A, B) = 0$ whenever either argument is free (equivalently, torsion-free over a PID).

***Proof*:**

Step 1: Splitting the cycles and boundaries. For each n write $Z_n = \ker(\partial\colon C_n \to C_{n-1})$ and $B_n = \mathrm{im}(\partial\colon C_{n+1} \to C_n)$, both submodules of the free module C_n. Over a PID a submodule of a free module is free, so Z_n and B_n are free, hence projective. The short exact sequence

$$0 \longrightarrow Z_n \longrightarrow C_n \xrightarrow{\partial} B_{n-1} \longrightarrow 0 \tag{4.12}$$

has free right-hand term B_{n-1}, so it splits: $C_n \cong Z_n \oplus B_{n-1}$ as R-modules. Regard Z_* and B_* as chain complexes with *zero* differential. The inclusion $Z_* \hookrightarrow C_*$ and the boundary map $C_* \to B_{*-1}$ (lowering degree by one onto the boundaries) then assemble into a short exact sequence of chain complexes

$$0 \longrightarrow Z_* \longrightarrow C_* \xrightarrow{\partial} B_{*-1} \longrightarrow 0 \tag{4.13}$$

in which Z_* and B_* carry the zero differential. This is the key step of the proof: it presents C_* as an extension of complexes whose homology is transparent.

Step 2: Tensoring with D. Each $C_n \cong Z_n \oplus B_{n-1}$ is free, hence flat, so tensoring (4.13) over R with the complex D_* preserves exactness in every degree: because B_{*-1} is a complex of free (hence flat) modules, the sequence

$$0 \longrightarrow Z_* \otimes_R D_* \longrightarrow C_* \otimes_R D_* \xrightarrow{\partial\otimes 1} (B_{*-1}) \otimes_R D_* \longrightarrow 0 \tag{4.14}$$

is a short exact sequence of chain complexes. Since Z_* and B_* have zero differential, the differential on $Z_* \otimes_R D_*$ is $\pm 1 \otimes \partial_D$, so this complex is just D_* with its degree shifted and grouped: in each total degree n, $(Z_* \otimes_R D_*)_n = \bigoplus_i Z_i \otimes_R D_{n-i}$, and because Z_i is free,

$$H_n(Z_* \otimes_R D_*) = \bigoplus_i Z_i \otimes_R H_{n-i}(D), \qquad H_n(B_* \otimes_R D_*) = \bigoplus_i B_i \otimes_R H_{n-i}(D) \tag{4.15}$$

the freeness of Z_i and B_i being what lets the functor $Z_i \otimes_R (-)$ commute with taking homology of D_*.

Step 3: The long exact sequence and its connecting map. The short exact sequence of complexes (4.14) yields a long exact homology sequence. Using (4.15) to identify the outer terms (and accounting for the degree shift by one in the B-complex), it reads

$$\cdots \longrightarrow \bigoplus_i B_i \otimes_R H_{n-i}(D) \xrightarrow{\iota} \bigoplus_i Z_i \otimes_R H_{n-i}(D) \longrightarrow H_n(C \otimes_R D) \longrightarrow \bigoplus_i B_i \otimes_R H_{n-1-i}(D) \xrightarrow{\iota} \cdots$$

The connecting homomorphism ι is, degreewise, the map $B_i \otimes_R H_{n-i}(D) \to Z_i \otimes_R H_{n-i}(D)$ induced by the inclusion $j\colon B_i \hookrightarrow Z_i$ of boundaries into cycles tensored with the identity, that is $\iota = (j \otimes 1)$. This identification of the connecting map is the one computational point and is checked by chasing a representative through (4.14).

Step 4: Extracting the Künneth sequence. The long exact sequence breaks into short exact sequences

$$0 \longrightarrow \operatorname{coker}(\iota_n) \longrightarrow H_n(C \otimes_R D) \longrightarrow \ker(\iota_{n-1}) \longrightarrow 0 \tag{4.16}$$

where $\iota_n\colon \bigoplus_i B_i \otimes_R H_{n-i}(D) \to \bigoplus_i Z_i \otimes_R H_{n-i}(D)$. It remains to identify $\operatorname{coker}(\iota_n)$ and $\ker(\iota_{n-1})$. Fix i and consider the free resolution

$$0 \longrightarrow B_i \xrightarrow{j} Z_i \longrightarrow H_i(C) \longrightarrow 0 \tag{4.17}$$

of $H_i(C)$: it is exact by definition of $H_i(C) = Z_i/B_i$, and both B_i, Z_i are free, so this is a length-one free resolution of $H_i(C)$ over the PID R. Tensoring (4.17) with the module $H_{n-i}(D)$ and taking homology is the definition of Tor_R: the complex $0 \to B_i \otimes_R H_{n-i}(D) \xrightarrow{j\otimes 1} Z_i \otimes_R H_{n-i}(D) \to 0$ has

$$\operatorname{coker}(j\otimes 1) = Z_i \otimes_R H_{n-i}(D)/\operatorname{im}(j\otimes 1) = H_i(C) \otimes_R H_{n-i}(D), \qquad \ker(j\otimes 1) = \operatorname{Tor}_R\big(H_i(C), H_{n-i}(D)\big)$$

the first because $\otimes$ is right exact, the second because the kernel of the tensored inclusion is exactly the first derived functor computed from this resolution. Summing over i gives

$$\operatorname{coker}(\iota_n) = \bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D), \qquad \ker(\iota_n) = \bigoplus_{i+j=n} \operatorname{Tor}_R\big(H_i(C), H_j(D)\big)$$

Substituting into (4.16) produces the asserted short exact sequence, with the cokernel term sitting in degree n and the Tor term, coming from $\ker(\iota_{n-1})$, sitting in degree $n-1$. That the left map is the cross product $\times$ follows from tracing the construction: the image of $[a] \otimes [b]$ under $\operatorname{coker}(\iota_n) \hookrightarrow H_n(C \otimes_R D)$ is $[a \otimes b]$.

Step 5: The splitting. For the (non-natural) splitting, return to the split short exact sequence $0 \to Z_n \to C_n \xrightarrow{\partial} B_{n-1} \to 0$ of (4.12). Its right-hand term B_{n-1} is free, so the sequence splits, and a choice of splitting is the same datum as a retraction $r_n\colon C_n \twoheadrightarrow Z_n$ of the inclusion $Z_n \hookrightarrow C_n$ (that is, $r_n|_{Z_n} = \mathrm{id}_{Z_n}$). Such a retraction exists for the same reason the sequence split in Step 1, and the choice is non-canonical. Compose r_n with the projection $\pi_n\colon Z_n \twoheadrightarrow Z_n/B_n = H_n(C)$ to obtain $\varphi_n := \pi_n \circ r_n\colon C_n \to H_n(C)$. The maps φ_* assemble into a chain map $\varphi\colon C_* \to H_*(C)$, where $H_*(C)$ carries the zero differential: for $c \in C_n$ the element ∂c already lies in Z_{n-1}, so $r_{n-1}(\partial c) = \partial c$ (the retraction restricts to the identity on Z_{n-1}), whence $\varphi_{n-1}(\partial c) = \pi_{n-1}(\partial c) = 0$ since $\partial c \in B_{n-1} = \ker \pi_{n-1}$. On homology φ induces the identity $H_n(C) \to H_n(C)$, because for a cycle $z \in Z_n$ one has $r_n(z) = z$ and hence $\varphi_n(z) = \pi_n(z) = [z]$. Choosing such a φ for C and the analogous chain map $\psi\colon D_* \to H_*(D)$ for D, the composite

$$C_* \otimes_R D_* \xrightarrow{\varphi \otimes \psi} H_*(C) \otimes_R H_*(D)$$

is a chain map to a complex with zero differential, and on degree-n homology its restriction to the image of the cross product is the identity of $\bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D)$. Thus $(\varphi \otimes \psi)_*$ is a retraction of the injection $\bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D) \hookrightarrow H_n(C \otimes_R D)$, which splits the short exact sequence (4.16). The splitting depends on the choice of retraction r_*, equivalently on the choice of splitting of (4.12). That the choice is non-canonical does not by itself show no natural choice exists, so the non-naturality needs a counterexample rather than an appeal to the construction. One is already available: taking $D_\bullet$ to be G concentrated in degree 0 turns this theorem into theorem 4.1.7, whose proof exhibits a chain map for which no natural splitting can exist.

Step 6: The field case. If R is a field, every R-module is free, hence flat, so $\operatorname{Tor}_R(-,-) = 0$ identically. The Tor term in the short exact sequence vanishes, the sequence degenerates to the isomorphism $H_n(C \otimes_R D) \cong \bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D)$, and the cross product is that isomorphism, as we sought to show.

The proof isolates exactly why the answer has two pieces. The tensor product of the homologies accounts for cycles that are products of cycles; the Tor term accounts for the failure of the inclusion of boundaries into cycles to remain injective after tensoring, which is a torsion phenomenon and disappears over a field. The degree shift in the Tor term, with Tor of degree-i and degree-j classes contributing in total degree $i+j+1$, is forced by the connecting homomorphism of the long exact sequence and is the source of the most common errors in computation: a class of $H_i(X)$ with $H_j(Y)$ produces a tensor contribution in degree $i+j$ and a Tor contribution in degree $i+j+1$.

Feeding the Eilenberg-Zilber equivalence into the algebraic theorem gives the topological Künneth formula at once. This is the central result of the section.

Theorem 4.3.5: Künneth Formula for Homology

Let X and Y be topological spaces and R a PID. Then for each n the homology cross product fits into a natural short exact sequence

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(X;R)\otimes_R H_j(Y;R) \xrightarrow{\times} H_n(X\times Y;R) \longrightarrow \bigoplus_{i+j=n-1} \mathrm{Tor}_R\big(H_i(X;R), H_j(Y;R)\big) \longrightarrow 0$$

The sequence splits, though not naturally. If R is a field, the Tor term vanishes and the cross product is an isomorphism of graded R-modules

$$H_*(X \times Y;R) \cong H_*(X;R) \otimes_R H_*(Y;R)$$

***Proof*:**
Apply theorem 4.3.4 with $C_* = C_*(X;R)$ and $D_* = C_*(Y;R)$. The complex $C_*(X;R)$ is free in each degree, since $C_n(X;R)$ is free on the singular n-simplices of X, so the hypothesis of theorem 4.3.4 is met; $D_* = C_*(Y;R)$ is an arbitrary complex of R-modules, which is all the theorem requires of the second factor. This produces, for each n, the natural short exact sequence

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(X;R)\otimes_R H_j(Y;R) \longrightarrow H_n\big(C_*(X;R)\otimes_R C_*(Y;R)\big) \longrightarrow \bigoplus_{i+j=n-1} \mathrm{Tor}_R\big(H_i(X;R), H_j(Y;R)\big) \longrightarrow 0$$

using $H_i(C_*(X;R)) = H_i(X;R)$ and likewise for Y. By the Eilenberg-Zilber theorem (theorem 4.3.1), EZ induces a natural isomorphism $H_n\big(C_*(X;R) \otimes_R C_*(Y;R)\big) \cong H_n(X \times Y;R)$. Transporting the middle term along this isomorphism replaces $H_n\big(C_*(X;R) \otimes_R C_*(Y;R)\big)$ by $H_n(X \times Y;R)$, and the composite of the left map with EZ_* is by construction the homology cross product of definition 4.3.2. The splitting and its non-naturality, and the field case, are inherited verbatim from theorem 4.3.4, completing the proof.

The formula reduces the homology of any product to the homologies of its factors, mechanically. The only subtlety is the Tor term, which is nonzero precisely when both factors carry torsion in complementary degrees; for spaces with free homology, such as spheres, tori, and orientable surfaces, the cross product is an isomorphism on the nose and $H_*(X \times Y) \cong H_*(X) \otimes H_*(Y)$. The non-naturality of the splitting is a genuine feature: the extension can be nonsplit as a sequence *varying* with X and Y, even though each individual extension splits because the relevant groups are over a PID. The two worked computations below display both behaviours.

Example 4.3.6: Künneth Computations

Throughout, recall from chapter 3 that $\tilde{H}_*(S^n;\mathbb{Z})$ is $\mathbb{Z}$ in degree n and zero otherwise, so the unreduced homology of S^n $(n \geq 1)$ is $\mathbb{Z}$ in degrees 0 and n and zero elsewhere.

1. *The torus* $T^2 = S^1 \times S^1$. The homology of S^1 is $H_0 = H_1 = \mathbb{Z}$, free in every degree, so the Tor term in theorem 4.3.5 vanishes and the cross product is an isomorphism $H_*(T^2;\mathbb{Z}) \cong H_*(S^1;\mathbb{Z}) \otimes_{\mathbb{Z}} H_*(S^1;\mathbb{Z})$. Writing $1 \in H_0(S^1)$ and $\mu \in H_1(S^1)$ for the generators, the tensor

product is free on $1 \otimes 1$, $\mu \otimes 1$, $1 \otimes \mu$, $\mu \otimes \mu$ in degrees $0, 1, 1, 2$, so

$$H_n(T^2; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z}^2 & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

The two degree-one generators are the coordinate circles $\mu \times 1$ and $1 \times \mu$, and the degree-two generator is $\mu \times \mu$, the top class computed in example 4.3.3. This matches the direct Δ-complex computation of chapter 3.

2. *Products of spheres* $S^m \times S^n$. Both factors have free homology ($\mathbb{Z}$ in degrees 0 and m, respectively 0 and n), so again Tor vanishes and $H_*(S^m \times S^n; \mathbb{Z}) \cong H_*(S^m; \mathbb{Z}) \otimes_{\mathbb{Z}} H_*(S^n; \mathbb{Z})$. The tensor product is free on $1 \otimes 1$, $a \otimes 1$, $1 \otimes b$, $a \otimes b$ in degrees $0, m, n, m+n$, where a, b generate $H_m(S^m)$, $H_n(S^n)$. When $m \neq n$,

$$H_k(S^m \times S^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & k \in \{0, m, n, m+n\} \\ 0 & \text{otherwise} \end{cases}$$

and when $m = n$ the two middle generators coincide in degree, giving $H_m(S^m \times S^m; \mathbb{Z}) \cong \mathbb{Z}^2$ with the other nonzero groups $\mathbb{Z}$ in degrees 0 and $2m$. For instance $H_*(S^2 \times S^2; \mathbb{Z})$ is $\mathbb{Z}, 0, \mathbb{Z}^2, 0, \mathbb{Z}$ in degrees 0 through 4.

3. *A nonzero* Tor *term:* $\mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^2$. The integral homology of $\mathbb{R}\mathrm{P}^2$ is $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}/2$, $H_2 = 0$. The cross product accounts for the tensor contributions, and the Tor term contributes in one higher degree. In degree 2,

$$H_2(\mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^2; \mathbb{Z}) \;\cong\; \underbrace{(H_1 \otimes H_1)}_{\mathbb{Z}/2 \otimes \mathbb{Z}/2 = \mathbb{Z}/2} \;\oplus\; \underbrace{\bigoplus_{i+j=1} \mathrm{Tor}_{\mathbb{Z}}(H_i, H_j)}_{0}$$

is $\mathbb{Z}/2$, the Tor summands in degree 2 vanishing because $H_0(\mathbb{R}\mathrm{P}^2) = \mathbb{Z}$ is free. The Tor correction from the two degree-one classes instead appears in degree 3:

$$H_3(\mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^2; \mathbb{Z}) \;\cong\; \mathrm{Tor}_{\mathbb{Z}}\big(H_1(\mathbb{R}\mathrm{P}^2), H_1(\mathbb{R}\mathrm{P}^2)\big) = \mathrm{Tor}_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$$

so $\mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^2$ has homology in degree 3 even though each factor is two-dimensional. This is the Tor term made visible: a product of two-dimensional complexes acquires three-dimensional homology purely from the interaction of torsion classes.

The third computation shows that the Tor term is not a formal nuisance but produces homology classes that are present in no factor and in no tensor product, arising solely from torsion crossing torsion. It also displays the degree bookkeeping concretely: two torsion classes in degree 1 produce a Tor class in degree $1 + 1 + 1 = 3$, the extra unit being the degree shift built into the connecting map.

Cohomology carries a dual cross product, and over a field or under a finiteness hypothesis it satisfies the analogous formula. The cohomology cross product is built from the cup product (definition 4.2.1) and the two projections of the product space; it is what makes the cohomology Künneth theorem a statement about *rings*, not just modules.

Definition 4.3.7: Cohomology Cross Product

Let X, Y be spaces with projections $p\colon X \times Y \to X$ and $q\colon X \times Y \to Y$, and let R be a commutative ring. For $\alpha \in H^k(X;R)$ and $\beta \in H^\ell(Y;R)$, the *cohomology cross product* is

$$\alpha \times \beta := p^*\alpha \smile q^*\beta \in H^{k+\ell}(X \times Y; R)$$

where $\smile$ is the cup product of definition 4.2.1. This is R-bilinear and natural, and assembles into a homomorphism of graded R-modules

$$\times\colon \bigoplus_{k+\ell=n} H^k(X;R) \otimes_R H^\ell(Y;R) \longrightarrow H^n(X \times Y; R)$$

which is moreover a homomorphism of graded rings $H^\bullet(X;R) \otimes_R H^\bullet(Y;R) \to H^\bullet(X \times Y;R)$, where the source carries the graded tensor-product ring structure $(\alpha \otimes \beta)(\alpha' \otimes \beta') = (-1)^{|\beta||\alpha'|}(\alpha \smile \alpha') \otimes (\beta \smile \beta')$.

That the cross product is a ring map records the interaction of cup and cross: cupping in the product corresponds to cupping in each factor, with a Koszul sign from moving a degree-$|\beta|$ class past a degree-$|\alpha'|$ class. The sign is exactly the one forced by graded commutativity of the cup product (theorem 4.2.4); it is what allows the source to be a graded-commutative ring isomorphic, when the cross product is an isomorphism, to the cohomology ring of the product. The verification that $\times$ is a ring map is the identity $p^*(\alpha \smile \alpha') \smile q^*(\beta \smile \beta') = \pm(p^*\alpha \smile q^*\beta) \smile (p^*\alpha' \smile q^*\beta')$, which follows from p^*, q^* being ring homomorphisms (proposition 4.2.3) and the graded commutativity used to interleave the four factors.

Example 4.3.8: The Cohomology Cross Product on the Torus

Take $X = Y = S^1$ with $R = \mathbb{Z}$. Then $H^\bullet(S^1;\mathbb{Z}) = \mathbb{Z}[\alpha]/(\alpha^2)$ with $|\alpha| = 1$ (an exterior algebra on one degree-one generator). The graded tensor product ring is $\Lambda_{\mathbb{Z}}[\alpha] \otimes_{\mathbb{Z}} \Lambda_{\mathbb{Z}}[\alpha'] \cong \Lambda_{\mathbb{Z}}[\alpha, \alpha']$, the exterior algebra on two degree-one generators, with $\alpha\alpha' = -\alpha'\alpha$ from the Koszul sign. Setting $a = \alpha \times 1 = p^*\alpha$ and $b = 1 \times \alpha = q^*\alpha$ in $H^\bullet(T^2;\mathbb{Z})$, the cross product gives $a \smile b = \alpha \times \alpha$ generating $H^2(T^2;\mathbb{Z}) \cong \mathbb{Z}$, while $a^2 = p^*(\alpha^2) = 0$ and $b^2 = q^*(\alpha^2) = 0$. This reproduces, from the Künneth viewpoint, the exterior-algebra structure $H^\bullet(T^2;\mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[a, b]$ found by the direct cup-product computation of example 4.2.6: the generator of H^2 is the cross product of the two circle generators, and the relations $a^2 = b^2 = 0$, $ab = -ba$ are the tensor-product relations.

The cohomology Künneth theorem requires a finiteness hypothesis absent from the homology statement, because the cohomology of a product is built from the homology of the product, and dualizing a direct sum produces a direct product unless the sum is finite.

Theorem 4.3.9: Künneth Formula for Cohomology

Let R be a PID and let X, Y be spaces such that $H_k(X;R)$ is a finitely generated R-module for every k (one says X is of *finite type* over R). Then for each n the cohomology cross product fits into a natural short exact sequence

$$0 \longrightarrow \bigoplus_{k+\ell=n} H^k(X;R)\otimes_R H^\ell(Y;R) \xrightarrow{\times} H^n(X\times Y;R) \longrightarrow \bigoplus_{k+\ell=n+1} \mathrm{Tor}_R\big(H^k(X;R), H^\ell(Y;R)\big) \longrightarrow 0$$

which splits, not naturally. If R is a field, the Tor term vanishes and the cross product is an isomorphism of graded R-algebras

$$H^\bullet(X\times Y;R) \cong H^\bullet(X;R)\otimes_R H^\bullet(Y;R)$$

the tensor product carrying its graded-commutative ring structure as in definition 4.3.7.

***Proof*:**
Step 1: Reduction to a tensor product of cochain complexes. The cohomology of $X \times Y$ is the cohomology of $\mathrm{Hom}_R\big(C_*(X\times Y;R), R\big)$. By the Eilenberg-Zilber theorem (theorem 4.3.1), $C_*(X\times Y;R) \simeq C_*(X;R)\otimes_R C_*(Y;R)$ by a natural chain homotopy equivalence, and applying the homotopy-invariant functor $\mathrm{Hom}_R(-,R)$ gives a cochain homotopy equivalence $C^*(X\times Y;R) \simeq \mathrm{Hom}_R\big(C_*(X;R)\otimes_R C_*(Y;R), R\big)$. The finite-type hypothesis on X enters here: when $C_*(X;R)$ has finitely generated homology, the cochain complex $C^*(X;R) = \mathrm{Hom}_R(C_*(X;R), R)$ is, up to chain homotopy equivalence, a complex of finitely generated free R-modules, and there is a natural cochain map

$$C^*(X;R)\otimes_R C^*(Y;R) \longrightarrow \mathrm{Hom}_R\big(C_*(X;R)\otimes_R C_*(Y;R), R\big), \qquad (f\otimes g)(a\otimes b) = (-1)^{|g||a|} f(a)\, g(b)$$

which is a chain homotopy equivalence precisely because $H_*(X;R)$ is finitely generated in each degree (so that the natural map $\mathrm{Hom}_R(M,R)\otimes_R \mathrm{Hom}_R(N,R) \to \mathrm{Hom}_R(M\otimes_R N, R)$ is an isomorphism for the relevant finitely generated free M). Thus $H^n(X\times Y;R) \cong H^n\big(C^*(X;R)\otimes_R C^*(Y;R)\big)$.

Step 2: The algebraic Künneth theorem applied to cochains. The cochain complex $C^*(X;R)$ may be replaced, up to chain homotopy equivalence, by a complex of free R-modules with cohomology $H^*(X;R)$ (using the finite-type hypothesis to make these free of finite rank), and $C^*(Y;R)$ is an arbitrary complex of R-modules. Re-indexing cohomological degree as homological degree, theorem 4.3.4 applied to these two complexes yields the natural short exact sequence

$$0 \longrightarrow \bigoplus_{k+\ell=n} H^k(X;R)\otimes_R H^\ell(Y;R) \xrightarrow{\times} H^n\big(C^*(X;R)\otimes_R C^*(Y;R)\big) \longrightarrow \bigoplus_{k+\ell=n+1} \mathrm{Tor}_R\big(H^k(X;R), H^\ell(Y;R)\big) \longrightarrow$$

the Tor term landing in cohomological degree $n+1$ (one *higher*, since the connecting map of the algebraic theorem raises cohomological degree by one). Composing the middle term with the isomorphism of Step 1 and checking that the resulting left-hand map is $p^*\alpha \smile q^*\beta$ identifies it with the cohomology cross product of definition 4.3.7.

Step 3: Ring structure and the field case. When R is a field, Tor vanishes and the cross product is an isomorphism of graded R-modules; that it is a ring isomorphism is definition 4.3.7, since the cross product is always a ring map and here it is bijective. The splitting and its non-naturality are inherited from theorem 4.3.4, completing the proof.

The cohomology Künneth theorem is the statement that the cohomology *ring* of a product is the graded tensor product of the rings, when one factor has finitely generated cohomology and (for the clean isomorphism) coefficients lie in a field. Over $\mathbb{Z}$ with finite type and free cohomology, the same isomorphism holds and is the standard route to computing cup-product structures on products: the relation $H^\bullet(T^n;\mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[\alpha_1,\ldots,\alpha_n]$ recorded in example 4.2.6 is the n-fold iterate of example 4.3.8, and the structure of $H^\bullet$ of any finite product of spheres follows the same way. The finiteness hypothesis cannot be dropped: for an infinite wedge or a space with infinitely generated homology in some degree, the natural map $H^\bullet(X) \otimes H^\bullet(Y) \to H^\bullet(X \times Y)$ can fail to be surjective, as the cohomology of the product involves a product where the tensor product involves a sum.

The finite-type hypothesis appears only on cohomology because cochains dualize chains: a direct sum of homology groups dualizes to a direct *product* of cohomology groups, and only finite generation forces sum and product to agree. Homology has no such constraint, which is why theorem 4.3.5 holds for all spaces while theorem 4.3.9 requires finite type. With these two theorems the homology and cohomology of any product, and the cup-product ring of a finite-type product, are computable from the factors.

Exercise 4.3.1

1. Compute $H_*(S^2 \times S^4;\mathbb{Z})$ and $H_*(S^3 \times S^3;\mathbb{Z})$ in every degree using theorem 4.3.5, and identify a generator of each nonzero group as a cross product of sphere classes.
2. Let $X = S^1 \times \mathbb{R}\mathrm{P}^2$. Compute $H_n(X;\mathbb{Z})$ for all n, being careful with the Tor term. In which degree, if any, does a Tor contribution appear, and why does it (or does it not) vanish?
3. Using the cohomology Künneth theorem over the field $\mathbb{Z}/2$, compute $H^\bullet(\mathbb{R}\mathrm{P}^\infty \times \mathbb{R}\mathrm{P}^\infty;\mathbb{Z}/2)$ as a graded ring, given that $H^\bullet(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}/2) \cong \mathbb{Z}/2[x]$ with $|x| = 1$. (Recall $\mathbb{R}\mathrm{P}^\infty$ has finite type over $\mathbb{Z}/2$.)
4. Show directly from theorem 4.3.5 that if X and Y both have free R-homology in every degree, then the cross product $H_*(X;R) \otimes_R H_*(Y;R) \to H_*(X \times Y;R)$ is an isomorphism. Where is freeness used?
5. The Künneth splitting is not natural. Exhibit the issue concretely: explain why a single choice of splitting of the short exact sequence for $H_*(X \times Y)$ cannot be made compatible with all maps $X \to X'$ at once, by reference to the dependence of the splitting on the chosen retraction $r_n\colon C_n \twoheadrightarrow Z_n$ (equivalently the splitting of (4.12)) in the proof of theorem 4.3.4.
6. Compute the cohomology ring $H^\bullet(S^2 \times S^2;\mathbb{Z})$ using theorem 4.3.9, writing the generators and all relations. Contrast its ring with that of $\mathbb{C}\mathrm{P}^2$ from example 4.2.7, and conclude that $S^2 \times S^2$ and $\mathbb{C}\mathrm{P}^2$ are not homotopy equivalent.

4.4 Orientations and the Fundamental Class

The symmetry between homology and cohomology that Poincaré duality records is not a property of an arbitrary space. It is a property of *manifolds*, and it depends on an extra piece of geometric data that a bare topological space does not carry: an *orientation*. Before duality can be stated, this data must be defined, its existence analyzed, and the single homology class it produces (the *fundamental class*) constructed. That is the work of this section. The strategy is local-to-global throughout. The local homology groups of definition 3.4.18 already see the manifold one point at a time, and the first result computes them: at every point of an n-manifold the local homology with coefficients in a ring R is a free R-module of rank one. A choice of generator at a point is a local orientation; the problem is to choose generators at all points at once, compatibly, and then to package the compatible choice into a global homology class. The compatibility is governed by a covering space, the obstruction to a global choice is a $\mathbb{Z}/2$ phenomenon, and the packaging is carried out by a compactness argument that glues the local data over the whole manifold.

Throughout, an *n-manifold* is a Hausdorff space M in which every point has an open neighbourhood homeomorphic to $\mathbb{R}^n$, and *closed* means compact without boundary. The coefficient object is a commutative ring R with unit; the two cases the reader should keep in mind are $R = \mathbb{Z}$ and $R = \mathbb{Z}/2$. All homology is singular homology with coefficients in R, as in chapter 3.

Theorem 4.4.1: Local Homology of a Manifold

Let M be an n-manifold and $x \in M$. Then the local homology group (definition 3.4.18) with coefficients in a ring R is

$$H_i\big(M, M \setminus \{x\}; R\big) \cong \begin{cases} R & i = n \\ 0 & i \neq n \end{cases}$$

More generally, if $U \subseteq M$ is open and homeomorphic to $\mathbb{R}^n$ and $x \in U$, then the inclusion induces an isomorphism $H_i(U, U \setminus \{x\}; R) \cong H_i(M, M \setminus \{x\}; R)$ for all i.

***Proof*:**
Step 1: Reduction to a Euclidean neighbourhood. Choose an open set $U \ni x$ with $U \cong \mathbb{R}^n$. Set $Z = M \setminus U$ and $A = M \setminus \{x\}$. Then $Z \subseteq A$, and $\overline{Z} \subseteq Z \subseteq A = \mathrm{int} A$ since A is open, so the closure of Z lies in the interior of A. The excision theorem (theorem 3.4.7) applied to $Z \subseteq A \subseteq M$ gives

$$H_i\big(M \setminus Z, A \setminus Z; R\big) \cong H_i\big(M, A; R\big)$$

Now $M \setminus Z = U$ and $A \setminus Z = (M \setminus \{x\}) \cap U = U \setminus \{x\}$, so

$$H_i\big(U, U \setminus \{x\}; R\big) \cong H_i\big(M, M \setminus \{x\}; R\big)$$

This is the second assertion, and it reduces the computation to the model pair $(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ after transporting x to 0 by the homeomorphism $U \cong \mathbb{R}^n$.

Step 2: The local homology of $\mathbb{R}^n$ at the origin. Consider the long exact sequence of the pair $(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ with coefficients in R (definition 3.4.2):

$$\cdots \to H_i(\mathbb{R}^n; R) \to H_i\big(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R\big) \xrightarrow{\partial} H_{i-1}\big(\mathbb{R}^n \setminus \{0\}; R\big) \to H_{i-1}(\mathbb{R}^n; R) \to \cdots$$

The space $\mathbb{R}^n$ is contractible, so $H_i(\mathbb{R}^n; R) = 0$ for $i > 0$ and $H_0(\mathbb{R}^n; R) = R$. For $i \geq 2$ the two outer terms vanish, so the connecting map ∂ is an isomorphism

$$H_i\big(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R\big) \cong H_{i-1}\big(\mathbb{R}^n \setminus \{0\}; R\big) \cong \widetilde{H}_{i-1}\big(\mathbb{R}^n \setminus \{0\}; R\big)$$

the reduced and unreduced groups agreeing in positive degree. The radial deformation retraction $\mathbb{R}^n \setminus \{0\} \simeq S^{n-1}$ identifies $\widetilde{H}_{i-1}(\mathbb{R}^n \setminus \{0\}; R) \cong \widetilde{H}_{i-1}(S^{n-1}; R)$, which is R for $i - 1 = n - 1$ and 0 otherwise. Hence $H_i(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R) \cong R$ for $i = n$ and 0 for $i \geq 2$, $i \neq n$.

Step 3: The low-degree end. It remains to treat $i = 0$ and $i = 1$, and the case $n = 1$ where $S^{n-1} = S^0$ is disconnected. For $i = 0$ the relative group $H_0(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R)$ is the cokernel of $H_0(\mathbb{R}^n \setminus \{0\}; R) \to H_0(\mathbb{R}^n; R)$; for $n \geq 1$ both $\mathbb{R}^n \setminus \{0\}$ and $\mathbb{R}^n$ are nonempty and the induced map on H_0 is surjective (an inclusion of nonempty spaces into a path-connected one is onto H_0), so the relative H_0 vanishes. For $i = 1$ the relevant segment is

$$H_1(\mathbb{R}^n; R) \to H_1\big(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R\big) \xrightarrow{\partial} \widetilde{H}_0\big(\mathbb{R}^n \setminus \{0\}; R\big) \to \widetilde{H}_0(\mathbb{R}^n; R) = 0$$

with $H_1(\mathbb{R}^n; R) = 0$, so $H_1(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R) \cong \widetilde{H}_0(\mathbb{R}^n \setminus \{0\}; R) \cong \widetilde{H}_0(S^{n-1}; R)$. Now $\widetilde{H}_0(S^{n-1}; R)$ is R when $n = 1$ (where S^0 is two points) and 0 when $n \geq 2$ (where S^{n-1} is path-connected), so the degree-one group $H_1(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R)$ is R for $n = 1$ and 0 for $n \geq 2$, agreeing in both cases with "R in degree n, zero otherwise." Combining the three steps, $H_i(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R)$ is R in degree $i = n$ and 0 in every other degree, for all $n \geq 1$, as we sought to show.

The computation says that a manifold looks, homologically and one point at a time, exactly like Euclidean space: the only nonzero local homology sits in the top degree n, where it is a free R-module on a single generator. The degree n is the dimension, recovered intrinsically; this is the homological content of invariance of dimension (theorem 3.4.17), now with arbitrary coefficients. The single rank-one group in degree n is what makes orientation a binary choice: a free R-module of rank one has exactly the generators that R permits, and choosing one at each point is the data to be organized.

Example 4.4.2: Local Orientations of the Plane
Take $M = \mathbb{R}^2$, $x = 0$, and $R = \mathbb{Z}$. By theorem 4.4.1, $H_2(\mathbb{R}^2, \mathbb{R}^2 \setminus \{0\}; \mathbb{Z}) \cong \mathbb{Z}$. A generator is produced concretely as follows. Let $\Delta \subseteq \mathbb{R}^2$ be a small triangle with 0 in its interior, regarded as a singular 2-simplex $\sigma\colon \Delta^2 \to \mathbb{R}^2$ via an affine homeomorphism onto Δ. Its boundary $\partial\sigma$ is a 1-cycle encircling 0, a loop in $\mathbb{R}^2 \setminus \{0\}$ representing a generator of $H_1(\mathbb{R}^2 \setminus \{0\}; \mathbb{Z}) \cong \mathbb{Z}$. Under the connecting isomorphism $\partial\colon H_2(\mathbb{R}^2, \mathbb{R}^2 \setminus \{0\}; \mathbb{Z}) \xrightarrow{\cong} \widetilde{H}_1(\mathbb{R}^2 \setminus \{0\}; \mathbb{Z})$ of Step 2 above, the relative class $[\sigma]$ maps to $[\partial\sigma]$, so $[\sigma]$ is a generator. Reversing the vertex order of σ (a transposition, of sign -1) replaces $[\sigma]$ by $-[\sigma]$, the other generator. The two generators $\pm[\sigma]$ are the two local orientations at 0: the triangle traversed counterclockwise and clockwise. With $R = \mathbb{Z}/2$ instead, $H_2(\mathbb{R}^2, \mathbb{R}^2 \setminus \{0\}; \mathbb{Z}/2) \cong \mathbb{Z}/2$ has a unique nonzero element, $[\sigma] = -[\sigma]$, so there is exactly one local $\mathbb{Z}/2$-orientation: over $\mathbb{Z}/2$ clockwise and counterclockwise are the same.

The example isolates the binary choice at a single point. An orientation of the whole manifold must make this choice everywhere, and the choices at nearby points must agree in the only sense that makes sense: through a chart that identifies the local homology groups of neighbouring points. The device that records this agreement is a covering space whose fibre over x is the local homology group at x.

Definition 4.4.3: Orientation Covering

Let M be an n-manifold and R a ring. The *R-orientation covering* of M is the set

$$\widetilde{M}_R = \coprod_{x \in M} H_n(M, M \setminus \{x\}; R)$$

with projection $p\colon \widetilde{M}_R \to M$ sending an element of $H_n(M, M \setminus \{x\}; R)$ to x. Its topology is generated by the following basic open sets. For each open $U \subseteq M$ homeomorphic to $\mathbb{R}^n$, each ball $B \subseteq U$ with $\overline{B}$ compact and contained in U, and each $\alpha_B \in H_n(M, M \setminus B; R)$, let

$$\langle \alpha_B \rangle = \left\{ \rho_x(\alpha_B) \;:\; x \in B \right\}$$

where $\rho_x\colon H_n(M, M \setminus B; R) \to H_n(M, M \setminus \{x\}; R)$ is the map induced by the inclusion $M \setminus B \hookrightarrow M \setminus \{x\}$.

The map $p\colon \widetilde{M}_R \to M$ is a covering space whose fibre over x is the R-module $H_n(M, M \setminus \{x\}; R) \cong R$. The reason the sets $\langle \alpha_B \rangle$ form a basis is that over a Euclidean ball B the restriction maps ρ_x are all isomorphisms with a common source $H_n(M, M \setminus B; R) \cong R$ (this is the $A = \overline{B}$ case of the gluing lemma below), so each $\langle \alpha_B \rangle$ is a sheet over B and these sheets partition $p^{-1}(B)$ into $|R|$ copies of B. A global orientation is exactly a continuous choice of one element from each fibre, made into a generator: a section of p landing in generators. This reformulates the everywhere-compatible local choice as a single continuous map.

Definition 4.4.4: Orientation

Let M be an n-manifold and R a ring. A *local R-orientation of M at x* is a choice of generator μ_x of the free rank-one R-module $H_n(M, M \setminus \{x\}; R)$ (theorem 4.4.1). An *R-orientation of M* is a function $x \mapsto \mu_x$ assigning to each $x \in M$ a local R-orientation, subject to the local consistency condition: each $x \in M$ has an open neighbourhood $U \cong \mathbb{R}^n$ and a ball $B \ni x$ with $\overline{B} \subseteq U$ compact, together with a class $\mu_B \in H_n(M, M \setminus B; R)$, such that $\rho_y(\mu_B) = \mu_y$ for every $y \in B$. The manifold M is *R-orientable* if an R-orientation exists. An R-orientation is the same datum as a section $x \mapsto \mu_x$ of the orientation covering $\widetilde{M}_R$ (definition 4.4.3) taking values in generators of the fibres; for $R = \mathbb{Z}$ one speaks simply of an *orientation* and of M being *orientable*.

The consistency condition is the precise meaning of "the choice varies continuously." Over a single chart $U \cong \mathbb{R}^n$ all the local homology groups are canonically identified through a class μ_B defined on a ball, and consistency requires that the chosen generators be the restrictions of one such μ_B; equivalently, the function $x \mapsto \mu_x$ is a continuous section of $\widetilde{M}_R$, since the basic open set $\langle \mu_B \rangle$ is exactly a neighbourhood through which the section is constant. For connected M a section taking generator values is determined by its value at a single point, because $\widetilde{M}_R$ restricted to generators is a covering of M with fibre the unit group worth of generators and a section of a covering over a connected base is rigid once a sheet is chosen. The case $R = \mathbb{Z}$ is the classical one: each fibre $H_n(M, M \setminus \{x\}; \mathbb{Z}) \cong \mathbb{Z}$ has exactly two generators $\pm\mu_x$, so an orientation is a consistent choice of sign, and the obstruction to making it is whether the two-sheeted covering of generators is connected (no orientation) or split (orientation exists). This corrects the earlier provisional definition: orientability is not a statement about the torsion of H_{n-1}, but about the existence of a consistent generator of the top local homology at every point.

Example 4.4.5: The Möbius Band is Non-Orientable
Let M be the open Möbius band, the quotient of $\mathbb{R} \times (-1, 1)$ by $(s, t) \sim (s + 1, -t)$. It is a 2-manifold. Fix the point $x_0 = [(0, 0)]$ on the core circle and a local orientation μ_{x_0}, realized as in example 4.4.2 by a small counterclockwise triangle in a chart around x_0. Transport μ_{x_0} along the core circle $t = 0$, parametrized by $s \in [0, 1]$: at each stage extend the chosen generator continuously into the next overlapping chart using a class μ_B as in definition 4.4.4. After traversing the full circle, the parameter returns from $s = 0^+$ to $s = 1^-$, but the gluing $(s, t) \sim (s + 1, -t)$ reflects the second coordinate, sending a counterclockwise triangle to a clockwise one. The transported generator that returns to x_0 is therefore $-\mu_{x_0}$, not μ_{x_0}. A $\mathbb{Z}$-orientation would require a globally consistent choice, contradicting the sign reversal along the loop, so the Möbius band admits no $\mathbb{Z}$-orientation: it is non-orientable. The same loop traversed with $R = \mathbb{Z}/2$ returns μ_{x_0} to itself, since $-\mu_{x_0} = \mu_{x_0}$ in $\mathbb{Z}/2$; the Möbius band is $\mathbb{Z}/2$-orientable, as every manifold is.

The example exhibits the orientation covering directly: the two-sheeted covering of generators of $\widetilde{M}_{\mathbb{Z}}$ over the Möbius band is the connected double cover (the annulus mapping onto the band), and the failure of orientability is the failure of that double cover to split into two sheets. For an orientable manifold the double cover is two disjoint copies of M, and choosing one is choosing an orientation. With $R = \mathbb{Z}/2$ there is only one generator per fibre, so the covering of generators is a copy of M itself, always split, which is why $\mathbb{Z}/2$-orientations always exist. This last point is what makes mod-2 Poincaré duality unconditional.

The local data is now organized. The remaining task is global: to convert an orientation, a compatible family of local generators, into a single homology class on M living in $H_n(M; R)$. The mechanism is a gluing lemma that builds a class over any compact set out of its pointwise restrictions, controlling at the same time the higher-degree groups. This is the technical heart of the subject.

Lemma 4.4.6: Compact Gluing of Local Homology

Let M be an n-manifold, R a ring, and $A \subseteq M$ compact. Then:

1. $H_i(M, M \setminus A; R) = 0$ for all $i > n$.
2. A class $\alpha \in H_n(M, M \setminus A; R)$ is zero if and only if its restriction $\rho_x(\alpha) \in H_n(M, M \setminus \{x\}; R)$ is zero for every $x \in A$.
3. Given any section $x \mapsto \mu_x$ of the orientation covering over A (a function with $\rho_y(\mu_B) = \mu_y$ locally, as in definition 4.4.4), there is a unique class $\mu_A \in H_n(M, M \setminus A; R)$ with $\rho_x(\mu_A) = \mu_x$ for all $x \in A$.

***Proof*:**
The argument is an induction on the number of pieces in a finite cover of A by Euclidean balls, with the Mayer-Vietoris sequence (theorem 3.4.19) stitching the pieces together. Write $\alpha_x := \rho_x(\alpha)$ for the restriction of a relative class to the local homology at x.

Step 1: Two reductions. First, the three statements are stable under the Mayer-Vietoris induction in the following sense. Suppose $A = A_1 \cup A_2$ with $A_1, A_2, A_1 \cap A_2$ all compact and the lemma known for each of A_1, A_2, $A_1 \cap A_2$. The relative Mayer-Vietoris sequence for the pair of open sets $M \setminus A_1$, $M \setminus A_2$ (whose union is $M \setminus (A_1 \cap A_2)$ and whose intersection is $M \setminus A$) reads

$$\cdots \to H_i(M, M \setminus A; R) \to H_i(M, M \setminus A_1; R) \oplus H_i(M, M \setminus A_2; R) \to H_i(M, M \setminus (A_1 \cap A_2); R) \to \cdots$$

For $i > n$ the two middle and the right terms vanish by inductive hypothesis (1), so the left term vanishes, proving (1) for A. For (2) and (3), the segment

$$0 \to H_n(M, M \setminus A; R) \xrightarrow{\Phi} H_n(M, M \setminus A_1; R) \oplus H_n(M, M \setminus A_2; R) \xrightarrow{\Psi} H_n(M, M \setminus (A_1 \cap A_2); R)$$

is exact on the left because the preceding term $H_{n+1}(M, M \setminus (A_1 \cap A_2); R)$ vanishes by (1). Here $\Phi(\alpha) = (\rho^{A_1}\alpha, \rho^{A_2}\alpha)$ and $\Psi(\beta_1, \beta_2) = \rho^{A_1 \cap A_2}\beta_1 - \rho^{A_1 \cap A_2}\beta_2$, the maps induced by the inclusions of complements. Injectivity of Φ gives statement (2): a class restricting to zero at every point of A restricts to zero on each A_j (by inductive (2) for A_j), hence is zero in each summand, hence zero by injectivity. For statement (3), the given section over A restricts to sections over A_1 and A_2, producing μ_{A_1} and μ_{A_2} by inductive (3); these agree over $A_1 \cap A_2$ by uniqueness of $\mu_{A_1 \cap A_2}$, so $\Psi(\mu_{A_1}, \mu_{A_2}) = 0$ and exactness yields a class μ_A with $\Phi(\mu_A) = (\mu_{A_1}, \mu_{A_2})$, restricting correctly at every point, and unique by (2). Thus the lemma propagates across unions.

Second, by compactness A is covered by finitely many Euclidean balls, and the union reduction lets one induct on their number once the base case (a single compact convex set in a chart) is settled.

Step 2: The base case, A a compact convex subset of a chart. Let $A \subseteq U \cong \mathbb{R}^n$ be compact and convex. By excision (the Euclidean reduction of theorem 4.4.1 applied to the pair, replacing the point by A), $H_i(M, M \setminus A; R) \cong H_i(\mathbb{R}^n, \mathbb{R}^n \setminus A; R)$. Since A is convex, $\mathbb{R}^n \setminus A$ deformation retracts onto a sphere S^{n-1} at large radius, so the pair $(\mathbb{R}^n, \mathbb{R}^n \setminus A)$ has the same relative homology as $(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$: namely R in degree n and 0 elsewhere, by the computation in the proof of theorem 4.4.1. This gives (1), with vanishing above degree n, and shows $H_n(M, M \setminus A; R) \cong R$. For (2) and (3), the restriction map $\rho_x \colon H_n(M, M \setminus A; R) \to H_n(M, M \setminus \{x\}; R)$ is an isomorphism for each $x \in A$, because both groups are computed by the same retraction of $\mathbb{R}^n \setminus A \simeq \mathbb{R}^n \setminus \{x\} \simeq S^{n-1}$ and ρ_x is the induced identity. Hence a class is zero iff its (single, up to the isomorphism) restriction is zero, giving (2); and a section over A, being the consistent family of images of one generator $\mu_A \in H_n(M, M \setminus A; R) \cong R$, is realized by that unique μ_A, giving (3).

Step 3: The general compact set. Cover A by finitely many balls $B_1, \ldots, B_k$ contained in charts, with each $\overline{B_j}$ compact. Replacing each B_j by a compact convex set $A_j = A \cap \overline{B_j}$ inside its chart (intersections of A with closed balls, which are compact, and which one further subdivides into convex pieces inside a single chart if necessary), write $A = A_1 \cup \cdots \cup A_k$ with each A_j a finite union of compact convex chart-pieces. The base case settles each A_j and each of their pairwise (and higher) intersections, which are again compact convex chart-pieces. Inducting on k via the union reduction of Step 1 establishes all three statements for A. The hardest bookkeeping is the verification that the iterated Mayer-Vietoris assembly produces a genuinely well-defined section-to-class correspondence independent of the chosen cover; this is carried out in full in [14, Lemma 3.27], whose argument is exactly the induction outlined here. This completes the proof.

The lemma is what the whole theory rests on. Part (1) says a manifold has no homology above its dimension, even relative to the complement of a compact set; part (2) says top-degree relative classes are detected pointwise, so the local homology groups separate them; part (3) says a compatible family of local generators over a compact set always assembles into a single class. Applying part (3) to $A = M$ when M itself is compact converts an orientation into a homology class, which is exactly the fundamental class. The next theorem carries this out and, in the process, reads off the entire top homology of a closed connected manifold.

Theorem 4.4.7: Top Homology of a Closed Manifold

Let M be a closed connected n-manifold and R a ring. For each $x \in M$ the restriction map

$$\rho_x \colon H_n(M;R) \longrightarrow H_n\big(M, M \setminus \{x\}; R\big) \cong R$$

is injective, and:

1. if M is R-orientable, ρ_x is an isomorphism, so $H_n(M;R) \cong R$;
2. if M is not R-orientable, the image of ρ_x is $\{\, r \in R : 2r = 0 \,\}$, the 2-torsion of R.

Moreover $H_i(M;R) = 0$ for all $i > n$. In particular $H_n(M;\mathbb{Z}) \cong \mathbb{Z}$ when M is orientable and $H_n(M;\mathbb{Z}) = 0$ when M is not, while $H_n(M;\mathbb{Z}/2) \cong \mathbb{Z}/2$ always.

***Proof*:**
Apply lemma 4.4.6 with $A = M$, which is compact since M is closed; here $M \setminus A = \varnothing$, so $H_i(M, M \setminus A; R) = H_i(M, \varnothing; R) = H_i(M;R)$, and the restriction ρ_x of the lemma is the map in the statement.

Step 1: Vanishing above the dimension. Part (1) of lemma 4.4.6 gives $H_i(M;R) = H_i(M, M \setminus M; R) = 0$ for $i > n$, the last assertion.

Step 2: Injectivity. Let $\alpha \in H_n(M;R)$ with $\rho_x(\alpha) = 0$ for the given x. Because M is connected, the function $y \mapsto \rho_y(\alpha)$ is a section of the orientation covering $\widetilde{M}_R$ over the connected base M; sections are locally constant (the consistency neighbourhoods $\langle \mu_B \rangle$ of definition 4.4.3 are exactly the sheets through which $y \mapsto \rho_y(\alpha)$ is constant), hence constant on the connected M. If $\rho_x(\alpha) = 0$ then $\rho_y(\alpha) = 0$ for all $y \in M$, and part (2) of lemma 4.4.6 forces $\alpha = 0$. Thus ρ_x is injective for every x.

Step 3: The image. Fix x and identify $H_n(M, M \setminus \{x\}; R) \cong R$. By Step 2 and the constancy of $y \mapsto \rho_y(\alpha)$, the image $\rho_x(H_n(M;R)) \subseteq R$ consists exactly of those $r \in R$ that extend to a constant section $y \mapsto r$ (under the local identifications $H_n(M, M \setminus \{y\}; R) \cong R$) which lifts to a global class. By part (3) of lemma 4.4.6 applied to $A = M$, a section over M lifts to a class precisely when it *is* a section, that is, when $y \mapsto r$ is consistent across overlaps. Two charts overlap with their local identifications differing by the sign of the Jacobian: an orientation-preserving overlap identifies r with r, an orientation-reversing overlap identifies r with $-r$. Consistency of the constant section $y \mapsto r$ therefore requires $r = -r$, that is $2r = 0$, exactly along every reversing overlap.

1. If M is R-orientable, all overlaps can be taken orientation-preserving (the orientation μ trivializes the covering), so the constraint $2r = 0$ is vacuous and every $r \in R$ extends to a global class: ρ_x is onto R, hence an isomorphism by Step 2, and $H_n(M;R) \cong R$.
2. If M is not R-orientable, the generator-covering of $\widetilde{M}_R$ is nontrivial, so there is a loop along which the local identification reverses sign; a constant section $y \mapsto r$ is consistent around this loop only if $r = -r$. Hence the image is contained in $\{r : 2r = 0\}$. Conversely

every such r does extend, since the sign ambiguity $r = -r$ is then immaterial and the section is globally consistent. So the image is exactly the 2-torsion of R.

Step 4: Specializations. For $R = \mathbb{Z}$ the 2-torsion is $\{0\}$, so ρ_x has image $\mathbb{Z}$ (orientable, giving $H_n(M;\mathbb{Z}) \cong \mathbb{Z}$) or image 0 (non-orientable, giving $H_n(M;\mathbb{Z}) = 0$, since ρ_x is injective with zero image). For $R = \mathbb{Z}/2$ every element satisfies $2r = 0$, so M is automatically $\mathbb{Z}/2$-orientable and $H_n(M;\mathbb{Z}/2) \cong \mathbb{Z}/2$ in all cases, as we sought to show.

The theorem reads the orientability of M directly off its top homology: a closed connected n-manifold is orientable exactly when $H_n(M;\mathbb{Z}) \cong \mathbb{Z}$, and non-orientable exactly when $H_n(M;\mathbb{Z}) = 0$. The injectivity of ρ_x is the structural fact underneath: top homology of a closed connected manifold injects into a single copy of R, so it is at most rank one, and orientability decides whether the rank is one or zero. The mod-2 statement explains why $\mathbb{Z}/2$-coefficients are the universal setting for duality on possibly-non-orientable manifolds: $H_n(M;\mathbb{Z}/2) \cong \mathbb{Z}/2$ gives a fundamental class no matter what, though it forgets signs. With the top homology pinned down, the distinguished generator can be named.

Definition 4.4.8: [Manifold] Fundamental Class

Let M be a closed connected R-orientable n-manifold equipped with an R-orientation $x \mapsto \mu_x$ (definition 4.4.4). The *fundamental class* of M with respect to this orientation is the unique class

$$[M] \in H_n(M;R)$$

whose restriction $\rho_x([M]) = \mu_x$ for every $x \in M$. When $R = \mathbb{Z}$ and an orientation is fixed, $[M]$ is *the* fundamental class; for $R = \mathbb{Z}/2$ the (unique nonzero) class $[M] \in H_n(M;\mathbb{Z}/2) \cong \mathbb{Z}/2$ is the mod-2 fundamental class, requiring no choice of orientation.

Theorem 4.4.9: Existence and Uniqueness of the Fundamental Class

With M as in definition 4.4.8, the fundamental class $[M]$ exists and is unique. It is a generator of $H_n(M;R) \cong R$, and reversing the orientation $\mu_x \mapsto -\mu_x$ replaces $[M]$ by $-[M]$.

***Proof*:**
The orientation $x \mapsto \mu_x$ is, by definition 4.4.4, a section of the orientation covering over M taking generator values. Apply part (3) of lemma 4.4.6 with $A = M$ (compact, since M is closed): there is a unique class $\mu_M = [M] \in H_n(M, M \setminus M; R) = H_n(M;R)$ with $\rho_x([M]) = \mu_x$ for all x. Uniqueness is part of the lemma and also follows from the injectivity of ρ_x in theorem 4.4.7: two classes with the same restriction at one point are equal. By that theorem $\rho_x\colon H_n(M;R) \to R$ is an isomorphism (the R-orientable case), and $\rho_x([M]) = \mu_x$ is a generator of R, so $[M]$ is a generator of $H_n(M;R)$. Replacing the orientation by its negative replaces each μ_x by $-\mu_x$; by uniqueness the new fundamental class restricts to $-\mu_x$ everywhere, and $-[M]$ does exactly this, so the reversed orientation has fundamental class $-[M]$, completing the proof.

The fundamental class is the algebraic embodiment of the manifold as a whole: a single homology cycle that simultaneously orients every point and generates the top homology. Its existence is what allows Poincaré duality to be phrased as cap product with one class, $D(\alpha) = [M] \frown \alpha$ in the notation of the next section. The sign ambiguity $[M] \leftrightarrow -[M]$ is precisely the two orientations; over $\mathbb{Z}/2$, where $[M] = -[M]$, the ambiguity disappears and the fundamental class is canonical. The next example computes $[M]$ in the cases that recur throughout duality theory.

Example 4.4.10: Fundamental Classes of Spheres and Projective Planes

1. *The sphere* S^n. The n-sphere is a closed connected orientable n-manifold. Its homology is $H_n(S^n;\mathbb{Z}) \cong \mathbb{Z}$ and $H_i(S^n;\mathbb{Z}) = 0$ for $0 < i < n$ (chapter 3), consistent with theorem 4.4.7. A generator of $H_n(S^n;\mathbb{Z}) \cong \mathbb{Z}$ is a fundamental class $[S^n]$; the two choices $\pm[S^n]$ are the two orientations, and the standard one is given by the degree-one generator under which the identity map $\mathrm{id}\colon S^n \to S^n$ has degree $+1$. Concretely, writing $S^n = \partial\Delta^{n+1}$ as the boundary of a simplex, the alternating sum of the n-faces with the boundary orientation is a cycle representing $[S^n]$. Restriction $\rho_x([S^n])$ at any x is the local orientation μ_x, recovering the local generator of example 4.4.2 in the $n = 2$ case.

2. *The projective plane* $\mathbb{R}\mathrm{P}^2$. The real projective plane is a closed connected non-orientable 2-manifold. Its integral homology is $H_0(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) = \mathbb{Z}$, $H_1(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) = \mathbb{Z}/2$, and $H_2(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) = 0$ (chapter 3). The vanishing $H_2(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) = 0$ is exactly the non-orientability predicted by theorem 4.4.7(2): there is no integral fundamental class. Non-orientability can be seen geometrically as in example 4.4.5, since $\mathbb{R}\mathrm{P}^2$ contains a Möbius band (the complement of an open disk), and transporting a local orientation around its core circle reverses the sign. With $\mathbb{Z}/2$ coefficients, however, $H_2(\mathbb{R}\mathrm{P}^2;\mathbb{Z}/2) \cong \mathbb{Z}/2$, and its nonzero element is the mod-2 fundamental class $[\mathbb{R}\mathrm{P}^2]_2$, which exists with no choice required. This is the class on which mod-2 Poincaré duality for $\mathbb{R}\mathrm{P}^2$ rests.

3. *The torus* $T^2 = S^1 \times S^1$. The torus is closed, connected, and orientable, with $H_2(T^2;\mathbb{Z}) \cong \mathbb{Z}$ (chapter 3). Its fundamental class $[T^2]$ is the generator corresponding to the product orientation; under the cellular decomposition with one 2-cell, $[T^2]$ is the class of that oriented 2-cell, with $\rho_x([T^2]) = \mu_x$ the product of the two circle orientations at each point. Reversing either circle's orientation reverses $[T^2]$.

These three computations are the templates for everything that follows. Spheres and tori carry honest integral fundamental classes and obey integral Poincaré duality; $\mathbb{R}\mathrm{P}^2$ has only a mod-2 fundamental class and obeys only the mod-2 duality. The dichotomy is exactly the orientability dichotomy of theorem 4.4.7, and the fundamental class is the bridge from this section's local analysis to the global duality isomorphism developed next.

Exercise 4.4.1

1. Let M be an n-manifold and $x \in M$. Using theorem 4.4.1, compute $H_i(M, M \setminus \{x\};\mathbb{Z}/2)$ in every degree i, and explain why the result is independent of whether M is orientable. Then state the analogous computation for $H_i(M, M \setminus \{x\};\mathbb{Q})$ and verify that the rank-one conclusion is unchanged.

2. Prove that a connected n-manifold M is orientable if and only if its orientation covering $\widetilde{M}_{\mathbb{Z}}$, restricted to the two generators $\pm\mu_x$ in each fibre, is a *disconnected* (equivalently, trivial two-sheeted) covering of M. Deduce that every simply connected manifold is orientable.

3. Let M be a closed connected n-manifold that is *not* $\mathbb{Z}$-orientable. Using theorem 4.4.7, determine $H_n(M;\mathbb{Z})$, $H_n(M;\mathbb{Z}/2)$, and $H_n(M;\mathbb{Z}/3)$. Explain in each case whether a fundamental class exists, and reconcile your answer with the Möbius-band sign-reversal mechanism of

example 4.4.5.

4. Let M and N be closed connected oriented n- and m-manifolds with fundamental classes $[M]$ and $[N]$. Assuming the homology cross product $H_n(M) \otimes H_m(N) \to H_{n+m}(M \times N)$ of chapter 3, argue that $M \times N$ is a closed connected orientable $(n+m)$-manifold and identify its fundamental class $[M \times N]$ in terms of $[M]$ and $[N]$. Use this to write down $[T^n]$ for the n-torus $T^n = (S^1)^n$.

5. The orientability of a closed connected n-manifold can fail only through a sign. Prove that for any closed connected n-manifold M, the group $H_n(M; \mathbb{Z})$ is either $\mathbb{Z}$ or 0, and that $H_n(M; \mathbb{Z}) \cong \mathbb{Z}$ if and only if M is orientable. (Use the injectivity in theorem 4.4.7 and the structure of the 2-torsion of $\mathbb{Z}$.)

6. Let M be a closed connected orientable n-manifold with fundamental class $[M]$ and let $f\colon M \to M$ be a continuous map. The *degree* of f is the integer $\deg f$ with $f_*[M] = (\deg f)[M]$ in $H_n(M; \mathbb{Z}) \cong \mathbb{Z}$. Show this is well defined, that $\deg(\mathrm{id}) = 1$, that $\deg(g \circ f) = \deg g \cdot \deg f$, and that a map of nonzero degree is surjective. (For surjectivity, suppose f misses a point y and factor f_* through $H_n(M \setminus \{y\}; \mathbb{Z})$.)

4.5 The Cap Product and Poincaré Duality

The cup product of definition 4.2.1 multiplies two cochains to produce a cochain, turning $H^\bullet(X; R)$ into the graded ring of definition 4.2.5. There is a companion operation that lets cohomology act on homology: a cochain and a chain combine to produce a chain of lower degree. This is the *cap product*, and it is the algebraic engine behind the symmetry between the homology and cohomology of a manifold. On a closed oriented n-manifold M there is a distinguished homology class, the fundamental class $[M] \in H_n(M; R)$ of definition 4.4.8, and capping with $[M]$ converts a degree-k cohomology class into a degree-$(n-k)$ homology class. The content of Poincaré duality is that this single operation $\alpha \mapsto [M] \frown \alpha$ is an isomorphism in every degree. The dimensions k and $n-k$ pair off, the Betti numbers of M are symmetric about the middle dimension, and the cup product becomes a perfect pairing. This section constructs the cap product at the chain level, proves it descends to (co)homology, establishes its formal properties, and proves the duality theorem in full for the compact case, deferring only the direct-limit bookkeeping of the non-compact case to a named source.

The starting point is a chain-level formula. A singular n-simplex has $n+1$ ordered vertices, and a cochain of degree k can be fed the front k-face $[v_0, \ldots, v_k]$ while the back $(n-k)$-face $[v_k, \ldots, v_n]$ is retained as a chain. The first records a scalar in R, the second a simplex of dimension $n-k$, and their product is the cap.

Definition 4.5.1: Cap Product [on chains]

Let X be a space and R a commutative ring. For a singular n-simplex $\sigma\colon \Delta^n \to X$ and a cochain $\varphi \in C^k(X;R)$ with $k \le n$, the *cap product* $\sigma \frown \varphi \in C_{n-k}(X;R)$ is the chain

$$\sigma \frown \varphi = \varphi\big(\sigma|_{[v_0,\dots,v_k]}\big)\ \sigma|_{[v_k,\dots,v_n]}$$

where $\sigma|_{[v_0,\dots,v_k]}$ is the restriction of σ to the front k-face and $\sigma|_{[v_k,\dots,v_n]}$ the restriction to the back $(n-k)$-face. Extending R-bilinearly over chains and cochains gives an R-bilinear map

$$\frown\colon C_n(X;R) \otimes_R C^k(X;R) \longrightarrow C_{n-k}(X;R)$$

When $k > n$ the cap product $\sigma \frown \varphi$ is declared to be 0.

The cap product runs against the grain of degree: it lowers homological dimension by the cohomological dimension of the cochain. The cochain φ "eats" the front face of the simplex and leaves the back face behind, scaled by the value it read off. The asymmetry of front and back is the source of the sign that will appear in the boundary formula, and it mirrors the splitting $[v_0,\dots,v_k] \cup [v_k,\dots,v_n]$ that already governed the cup product of definition 4.2.1: the two operations are built from the same combinatorial subdivision of a simplex into a front and a back face sharing the vertex v_k. The convention places the chain on the left and the cochain on the right, so that the preserved expression $[M] \frown \alpha$ of the duality theorem is the cap of the fundamental cycle with a cohomology class.

Example 4.5.2: The Cap Product on a Single Simplex

Take $X = \Delta^2$, the standard 2-simplex with vertices v_0, v_1, v_2, and let $\sigma = \mathrm{id}\colon \Delta^2 \to \Delta^2$ be the identity, a singular 2-simplex. Work with coefficients in $R = \mathbb{Z}$.

Degree 0. For $\varphi \in C^0(\Delta^2;\mathbb{Z})$ the front 0-face is the vertex $[v_0]$, and

$$\sigma \frown \varphi = \varphi([v_0])\ \sigma|_{[v_0,v_1,v_2]} = \varphi(v_0)\,\sigma$$

so capping with a 0-cochain rescales the simplex by the value of φ at the initial vertex. The unit cochain $1 \in C^0$, taking the value 1 on every vertex, gives $\sigma \frown 1 = \sigma$.

Degree 1. For $\varphi \in C^1(\Delta^2;\mathbb{Z})$ the front 1-face is the edge $[v_0, v_1]$ and the back face is the edge $[v_1, v_2]$, so

$$\sigma \frown \varphi = \varphi([v_0, v_1])\ [v_1, v_2]$$

a 1-chain. If φ is the cochain dual to the edge $[v_0, v_1]$ (value 1 there, 0 on the other edges), then $\sigma \frown \varphi = [v_1, v_2]$.

Degree 2. For $\varphi \in C^2(\Delta^2;\mathbb{Z})$ the front 2-face is all of $[v_0, v_1, v_2]$ and the back face is the vertex $[v_2]$, so

$$\sigma \frown \varphi = \varphi([v_0, v_1, v_2])\ [v_2] \in C_0(\Delta^2;\mathbb{Z})$$

If φ is dual to the 2-simplex itself, this is the 0-chain $[v_2]$. In each case the cap lowered the dimension from 2 to $2-k$, as the definition requires.

For the cap product to descend to homology and cohomology, it must interact with the boundary ∂ and coboundary δ in a controlled way. The cup product obeyed the Leibniz rule of lemma 4.2.2; the cap product obeys a dual identity that is again a Leibniz rule, now mixing ∂ on the chain side and δ on the cochain side.

Lemma 4.5.3: Boundary Formula for the Cap Product

For σ a singular n-simplex and $\varphi \in C^k(X;R)$ with $k \leq n$,

$$\partial(\sigma \frown \varphi) = (-1)^k(\partial\sigma \frown \varphi - \sigma \frown \delta\varphi)$$

***Proof*:**
Write $\sigma_{[a,\dots,b]}$ for $\sigma|_{[v_a,\dots,v_b]}$. By definition $\sigma \frown \varphi = \varphi(\sigma_{[0,\dots,k]})\,\sigma_{[k,\dots,n]}$, so its boundary is

$$\partial(\sigma \frown \varphi) = \varphi(\sigma_{[0,\dots,k]})\sum_{i=k}^{n}(-1)^{i-k}\,\sigma_{[k,\dots,\hat{v}_i,\dots,n]}$$

the sign $(-1)^{i-k}$ arising because v_i is the $(i-k)$-th vertex of the back face $[v_k,\dots,v_n]$. Pulling the scalar $(-1)^{-k} = (-1)^k$ out front,

$$\partial(\sigma \frown \varphi) = (-1)^k\varphi(\sigma_{[0,\dots,k]})\sum_{i=k}^{n}(-1)^{i}\,\sigma_{[k,\dots,\hat{v}_i,\dots,n]} \tag{4.18}$$

Now expand the right-hand side. First,

$$\partial\sigma \frown \varphi = \sum_{i=0}^{n}(-1)^i\,(\sigma_{[0,\dots,\hat{v}_i,\dots,n]} \frown \varphi)$$

Capping $\sigma_{[0,\dots,\hat{v}_i,\dots,n]}$, an $(n-1)$-simplex, with the k-cochain φ reads off its front k-face. For $i \leq k$ the deleted vertex lies among the front $k+1$ vertices, so the front k-face of $\sigma_{[0,\dots,\hat{v}_i,\dots,n]}$ is $\sigma_{[0,\dots,\hat{v}_i,\dots,k+1]}$ and the back face is $\sigma_{[k+1,\dots,n]}$; for $i \geq k+1$ the front k-face is the unchanged $\sigma_{[0,\dots,k]}$ and the deletion happens in the back face, giving back face $\sigma_{[k,\dots,\hat{v}_i,\dots,n]}$. Splitting the sum at $i = k$,

$$\partial\sigma \frown \varphi = \sum_{i=0}^{k}(-1)^i\varphi(\sigma_{[0,\dots,\hat{v}_i,\dots,k+1]})\,\sigma_{[k+1,\dots,n]} + \sum_{i=k+1}^{n}(-1)^i\varphi(\sigma_{[0,\dots,k]})\,\sigma_{[k,\dots,\hat{v}_i,\dots,n]} \tag{4.19}$$

the index $i = k$ being placed with the first block by convention. Second, the coboundary $\delta\varphi \in C^{k+1}$ evaluated on the front $(k+1)$-face gives

$$\sigma \frown \delta\varphi = (\delta\varphi)(\sigma_{[0,\dots,k+1]})\,\sigma_{[k+1,\dots,n]} = \left(\sum_{i=0}^{k+1}(-1)^i\varphi(\sigma_{[0,\dots,\hat{v}_i,\dots,k+1]})\right)\sigma_{[k+1,\dots,n]}$$

using $(\delta\varphi)(\tau) = \varphi(\partial\tau)$ on the simplex $\tau = \sigma_{[0,\dots,k+1]}$. Therefore

$$\sigma \frown \delta\varphi = \sum_{i=0}^{k+1}(-1)^i\varphi(\sigma_{[0,\dots,\hat{v}_i,\dots,k+1]})\,\sigma_{[k+1,\dots,n]} \tag{4.20}$$

Now form $\partial\sigma \frown \varphi - \sigma \frown \delta\varphi$ using (4.19) and (4.20). The first block of (4.19) runs i from 0 to k; the sum (4.20) runs i from 0 to $k+1$. Their difference cancels the overlapping terms $i = 0,\dots,k$ exactly, leaving from (4.20) only the single term $i = k+1$,

$$-(-1)^{k+1}\varphi(\sigma_{[0,\dots,k]})\,\sigma_{[k+1,\dots,n]} = (-1)^k\varphi(\sigma_{[0,\dots,k]})\,\sigma_{[k+1,\dots,n]}$$

(noting $\sigma_{[0,\dots,\widehat{v_{k+1}},\dots,k+1]} = \sigma_{[0,\dots,k]}$). Combined with the surviving second block of (4.19), the difference equals

$$\varphi(\sigma_{[0,\dots,k]}) \left(\sum_{i=k+1}^{n} (-1)^i \, \sigma_{[k,\dots,\hat{v}_i,\dots,n]} + (-1)^k \, \sigma_{[k+1,\dots,n]} \right)$$

The trailing term is precisely the $i = k$ term of the sum $\sum_{i=k}^{n} (-1)^i \sigma_{[k,\dots,\hat{v}_i,\dots,n]}$ (deleting v_k from the back face leaves $\sigma_{[k+1,\dots,n]}$ with sign $(-1)^k$), so the parenthesized expression is exactly $\sum_{i=k}^{n} (-1)^i \sigma_{[k,\dots,\hat{v}_i,\dots,n]}$. Multiplying by $(-1)^k$ recovers the right-hand side of (4.18), that is

$$(-1)^k (\partial\sigma \frown \varphi - \sigma \frown \delta\varphi) = \partial(\sigma \frown \varphi)$$

as we sought to show.

The boundary formula is the single computation on which everything rests. Read it as a derivation property: the boundary of a cap distributes over the two factors, with ∂ acting on the chain and δ on the cochain, up to the global sign $(-1)^k$ and the minus sign between the two terms. From it the descent to (co)homology is immediate.

Proposition 4.5.4: The Cap Product Descends to (Co)homology

The chain-level cap product of definition 4.5.1 induces an R-bilinear map

$$\frown : H_n(X; R) \otimes_R H^k(X; R) \longrightarrow H_{n-k}(X; R)$$

***Proof*:**
Let $z \in C_n(X; R)$ be a cycle ($\partial z = 0$) and $\varphi \in C^k(X; R)$ a cocycle ($\delta\varphi = 0$). The boundary formula of lemma 4.5.3, extended R-bilinearly from simplices to the cycle z, gives

$$\partial(z \frown \varphi) = (-1)^k (\partial z \frown \varphi - z \frown \delta\varphi) = (-1)^k (0 - 0) = 0$$

so $z \frown \varphi$ is a cycle and represents a class in $H_{n-k}(X; R)$. It remains to check that this class depends only on the homology class of z and the cohomology class of φ. Replacing z by $z + \partial w$ for $w \in C_{n+1}(X; R)$, and using $\delta\varphi = 0$,

$$(z + \partial w) \frown \varphi = z \frown \varphi + \partial w \frown \varphi = z \frown \varphi + (-1)^k \partial(w \frown \varphi)$$

where the last equality is again lemma 4.5.3 applied to w (the term $w \frown \delta\varphi$ vanishes). Thus $(z + \partial w) \frown \varphi$ and $z \frown \varphi$ differ by a boundary, hence represent the same class. Replacing φ by $\varphi + \delta\psi$ for $\psi \in C^{k-1}(X; R)$, and using $\partial z = 0$,

$$z \frown (\varphi + \delta\psi) = z \frown \varphi + z \frown \delta\psi = z \frown \varphi + (-1)^{k-1} (\partial z \frown \psi - \partial(z \frown \psi)) = z \frown \varphi - (-1)^{k-1} \partial(z \frown \psi)$$

again differing by a boundary. Hence the induced map on classes is well defined, and it is R-bilinear because the chain-level operation is, completing the proof.

With the cap product established on homology and cohomology, its formal properties follow. Three are needed for duality: a module structure making $H_\bullet(X; R)$ a module over the cohomology ring

$H^\bullet(X;R)$, an adjunction tying the cap to the cup product and the Kronecker evaluation pairing, and a naturality (projection) formula governing the behaviour under continuous maps. The Kronecker pairing referenced below is the evaluation $\langle \varphi, z\rangle = \varphi(z) \in R$ of a cochain on a chain, descending to $\langle\cdot,\cdot\rangle\colon H^k(X;R)\otimes H_k(X;R)\to R$ exactly as in the universal coefficient theorem theorem 4.1.8.

Proposition 4.5.5: Properties of the Cap Product

Let X, Y be spaces and R a commutative ring. The cap product of proposition 4.5.4 satisfies the following.

1. *Module structure.* For $\varphi \in H^k(X;R)$, $\psi \in H^\ell(X;R)$ and $z \in H_n(X;R)$,
$$z \frown (\varphi \smile \psi) = (z \frown \varphi) \frown \psi$$
and $z \frown 1 = z$ for the unit $1 \in H^0(X;R)$. Thus $H_\bullet(X;R)$ is a graded right module over the cohomology ring $H^\bullet(X;R)$ of definition 4.2.5.

2. *Adjunction with the Kronecker pairing.* For $\varphi \in H^k(X;R)$, $\psi \in H^\ell(X;R)$ and $z \in H_{k+\ell}(X;R)$,
$$\langle \psi, z \frown \varphi\rangle = \langle \varphi \smile \psi, z\rangle$$
where $\langle\cdot,\cdot\rangle$ is the Kronecker evaluation pairing.

3. *Naturality (projection formula).* For a continuous $f\colon X \to Y$, $z \in H_n(X;R)$ and $\psi \in H^k(Y;R)$,
$$f_*(z \frown f^*(\psi)) = f_*(z) \frown \psi$$

***Proof*:**
Each identity is established at the chain level on a single singular simplex $\sigma\colon \Delta^n \to X$ and then extended R-bilinearly; proposition 4.5.4 guarantees passage to classes.

1. *Module structure.* Let $\varphi \in C^k$, $\psi \in C^\ell$ and σ an n-simplex with $n = k+\ell+m$. Computing the right side,
$$\sigma \frown \varphi = \varphi(\sigma_{[0,\dots,k]})\,\sigma_{[k,\dots,n]}$$
and capping this $(\ell{+}m)$-chain with ψ reads off the front ℓ-face of $\sigma_{[k,\dots,n]}$, namely $\sigma_{[k,\dots,k+\ell]}$, leaving $\sigma_{[k+\ell,\dots,n]}$,
$$(\sigma \frown \varphi) \frown \psi = \varphi(\sigma_{[0,\dots,k]})\,\psi(\sigma_{[k,\dots,k+\ell]})\,\sigma_{[k+\ell,\dots,n]}$$
Computing the left side, the cup product of definition 4.2.1 is $(\varphi\smile\psi)(\tau) = \varphi(\tau_{[0,\dots,k]})\psi(\tau_{[k,\dots,k+\ell]})$, so capping σ with $\varphi\smile\psi$ reads the front $(k+\ell)$-face,
$$\sigma \frown (\varphi\smile\psi) = (\varphi\smile\psi)(\sigma_{[0,\dots,k+\ell]})\,\sigma_{[k+\ell,\dots,n]} = \varphi(\sigma_{[0,\dots,k]})\,\psi(\sigma_{[k,\dots,k+\ell]})\,\sigma_{[k+\ell,\dots,n]}$$
The two expressions agree. For the unit, $1 \in C^0$ takes the value 1 on every vertex, so $\sigma \frown 1 = 1(\sigma_{[0]})\,\sigma_{[0,\dots,n]} = \sigma$. Associativity and the unit law make $H_\bullet(X;R)$ a graded right module over $H^\bullet(X;R)$, the degree of $z \frown \varphi$ being $n-k$ when $|z| = n$, $|\varphi| = k$.

2. *Adjunction.* Let $\varphi \in C^k$, $\psi \in C^\ell$ and σ a $(k+\ell)$-simplex. The cap $\sigma \frown \varphi = \varphi(\sigma_{[0,\dots,k]})\,\sigma_{[k,\dots,k+\ell]}$ is an ℓ-chain, so evaluating ψ on it,
$$\langle \psi, \sigma\frown\varphi\rangle = \psi(\sigma\frown\varphi) = \varphi(\sigma_{[0,\dots,k]})\,\psi(\sigma_{[k,\dots,k+\ell]})$$

On the other side, the cup product evaluated on σ is

$$\langle \varphi \smile \psi, \sigma\rangle = (\varphi \smile \psi)(\sigma) = \varphi(\sigma_{[0,\dots,k]})\,\psi(\sigma_{[k,\dots,k+\ell]})$$

which is the same scalar in R. Extending bilinearly over chains gives $\langle \psi, z \frown \varphi\rangle = \langle \varphi \smile \psi, z\rangle$ for all $z \in C_{k+\ell}$, and the identity passes to (co)homology because both sides are defined there by proposition 4.5.4 and the descent of the Kronecker pairing.

3. *Naturality.* Let $f\colon X \to Y$ be continuous with chain map $f_\#\colon C_\bullet(X) \to C_\bullet(Y)$ and cochain map $f^\# = \mathrm{Hom}(f_\#, R)$, so $f^\#(\psi) = \psi \circ f_\#$ on cochains. For an n-simplex σ in X and $\psi \in C^k(Y;R)$, write $f \circ \sigma$ for the composite simplex in Y. Front and back faces are preserved by composition, $(f\circ\sigma)_{[a,\dots,b]} = f \circ (\sigma_{[a,\dots,b]})$, so

$$f_\#(\sigma \frown f^\#\psi) = f_\#\big((f^\#\psi)(\sigma_{[0,\dots,k]})\,\sigma_{[k,\dots,n]}\big) = \psi\big(f\circ\sigma_{[0,\dots,k]}\big)\ (f\circ\sigma_{[k,\dots,n]})$$

using $(f^\#\psi)(\tau) = \psi(f\circ\tau)$ and that $f_\#$ is R-linear, sending the simplex $\sigma_{[k,\dots,n]}$ to $f\circ\sigma_{[k,\dots,n]}$. On the other hand,

$$f_\#(\sigma) \frown \psi = (f\circ\sigma) \frown \psi = \psi\big((f\circ\sigma)_{[0,\dots,k]}\big)\ (f\circ\sigma)_{[k,\dots,n]} = \psi\big(f\circ\sigma_{[0,\dots,k]}\big)\ (f\circ\sigma_{[k,\dots,n]})$$

The two right-hand sides are identical, so $f_\#(\sigma \frown f^\#\psi) = f_\#(\sigma) \frown \psi$ on chains. Passing to classes gives $f_*(z \frown f^*\psi) = f_*(z) \frown \psi$, completing the proof.

The three properties say, in turn, that capping is an action, that it is adjoint to cupping, and that it commutes with maps in the only way the variances allow. The module structure of part (1) makes the entire homology of X into a module over its cohomology ring, so that Poincaré duality can be read as an isomorphism of $H^\bullet(M;R)$-modules. The adjunction of part (2) is the bridge between the two products: it says the cap product is the operation $H^\bullet \otimes H_\bullet \to H_\bullet$ uniquely determined by the requirement that pairing the result against any further cohomology class reproduces a cup product. The projection formula of part (3) has the cohomology class pulled back along f on the left and pushed nowhere on the right, which is forced because cohomology is contravariant and homology covariant; it is the formula that lets duality be checked on the pieces of an open cover.

The adjunction has an immediate consequence worth isolating, because it is what converts the duality isomorphism into a statement about pairings. Capping with a fixed class $z \in H_n(X;R)$ and then pairing is the same as cupping and pairing with z.

Corollary 4.5.6: Duality Map and the Cup Pairing

Let $z \in H_n(X;R)$ and let $D_z\colon H^k(X;R) \to H_{n-k}(X;R)$ be $D_z(\varphi) = z \frown \varphi$. For all $\varphi \in H^k(X;R)$ and $\psi \in H^{n-k}(X;R)$,

$$\langle \psi, D_z(\varphi)\rangle = \langle \varphi \smile \psi, z\rangle$$

so the composite $H^k \otimes H^{n-k} \xrightarrow{\smile} H^n \xrightarrow{\langle\,\cdot\,,z\rangle} R$ equals the pairing of ψ against $D_z(\varphi)$ via the Kronecker pairing.

***Proof*:**
This is proposition 4.5.5(2) with $\ell = n - k$ and the substitution $D_z(\varphi) = z \frown \varphi$, read directly: $\langle \psi, z \frown \varphi \rangle = \langle \varphi \smile \psi, z \rangle$, as required.

The reading of corollary 4.5.6 is that the duality map D_z and the cup-product pairing $(\varphi, \psi) \mapsto \langle \varphi \smile \psi, z \rangle$ carry the same information: D_z is an isomorphism if and only if, over a field, the cup pairing is nondegenerate. This is the form of duality that controls the cohomology rings of manifolds, and it is recorded among the consequences below.

Before proving the theorem, the geometric input is recalled. A closed manifold is a compact manifold without boundary. For such an M of dimension n, the local homology groups of definition 3.4.18 satisfy $H_i(M, M - \{x\}; R) \cong R$ for $i = n$ and 0 otherwise, since M is locally Euclidean and $H_i(\mathbb{R}^n, \mathbb{R}^n - \{0\}; R) \cong H_{i-1}(S^{n-1}; R)$ by the long exact sequence of the pair. An R-orientation is a choice, continuous in x, of generator of each $H_n(M, M - \{x\}; R) \cong R$; this is the orientation of definition 4.4.4. For M closed and R-orientable there is a unique class $[M] \in H_n(M; R)$, the fundamental class of definition 4.4.8, restricting to the chosen generator in every $H_n(M, M - \{x\}; R)$. The duality theorem asserts that capping with $[M]$ is an isomorphism.

Theorem 4.5.7: Poincaré Duality

Let M be a closed R-orientable n-manifold with fundamental class $[M] \in H_n(M; R)$ of definition 4.4.8. Then the map $D\colon H^k(M; R) \to H_{n-k}(M; R)$ given by $D(\alpha) = [M] \frown \alpha$ is an isomorphism for all k, that is

$$H^k(M; R) \cong H_{n-k}(M; R)$$

The strategy is local-to-global. Duality holds trivially for the open ball $\mathbb{R}^n$ (in the appropriate compactly-supported formulation), the Mayer-Vietoris sequences of theorem 3.4.19 for homology and for cohomology with compact supports glue two pieces on which duality holds into their union, and an induction over a finite cover of M by Euclidean opens assembles the global statement. The one ingredient that must be widened beyond ordinary cohomology is the cohomology theory: gluing open sets forces the use of cohomology with compact supports, for which the duality map is contravariant in inclusions of opens and admits its own Mayer-Vietoris sequence.

Definition 4.5.8: Cohomology with Compact Supports

Let X be a space and R a ring. A cochain $\varphi \in C^k(X; R)$ has *compact support* if there is a compact $K \subseteq X$ with φ vanishing on every simplex in $X - K$, equivalently $\varphi \in C^k(X, X - K; R) \subseteq C^k(X; R)$. The compactly supported cochains form a subcomplex, and its cohomology is the *cohomology with compact supports*

$$H_c^k(X; R) = \varinjlim_{K} H^k(X, X - K; R)$$

the colimit over compact $K \subseteq X$ ordered by inclusion, with the maps induced by $H^k(X, X - K) \to H^k(X, X - K')$ for $K \subseteq K'$.

For compact X one may take $K = X$, so $H_c^k(X; R) = H^k(X; R)$ and the compactly supported theory agrees with ordinary cohomology. The point of the definition is its behaviour for open subsets and inclusions of opens. If $U \subseteq V$ is an inclusion of open subsets of M, a compactly supported cochain on U extends by zero to one on V, giving a covariant map $H^k_c(U;R) \to H^k_c(V;R)$, the opposite variance to ordinary cohomology. This is exactly the variance of homology, which is why H^k_c pairs with homology in the duality map and why the inductive gluing works.

Example 4.5.9: Compactly Supported Cohomology of $\mathbb{R}^n$

Take $X = \mathbb{R}^n$ with coefficients R. Every compact $K \subseteq \mathbb{R}^n$ is contained in a closed ball B, and the restriction map $H^k(\mathbb{R}^n, \mathbb{R}^n - K; R) \to H^k(\mathbb{R}^n, \mathbb{R}^n - B; R)$ is an isomorphism for all such $K \subseteq B$ once B is large (both compute the local cohomology at the ball), so the colimit defining H^k_c stabilizes at the value for a single large ball B. By excision theorem 3.4.7 applied to the good pair $(\mathbb{R}^n, \mathbb{R}^n - B)$ and the deformation retraction of $\mathbb{R}^n - B$ onto a sphere,

$$H^k_c(\mathbb{R}^n;R) \cong H^k(\mathbb{R}^n, \mathbb{R}^n - B; R) \cong \widetilde{H}^k(S^n;R) = \begin{cases} R & k = n \\ 0 & k \neq n \end{cases}$$

Contrast this with ordinary cohomology $H^k(\mathbb{R}^n;R)$, which is R for $k=0$ and 0 otherwise: the two theories differ sharply on non-compact spaces, the compactly supported version concentrating the class in the top degree n rather than the bottom degree 0. The single nonzero group $H^n_c(\mathbb{R}^n;R) = R$ is generated by a compactly supported n-cocycle, the orientation class of the ball, and it is this class that the Euclidean case of duality pairs against the local fundamental class.

For an open subset U of an oriented M the fundamental class restricts to a compatible family of local classes, defining a duality map

$$D_U \colon H^k_c(U;R) \longrightarrow H_{n-k}(U;R)$$

by capping with these local fundamental classes; for $U = M$ compact this is the map D of the theorem. The proof of the theorem is the assertion that D_U is an isomorphism for every open U that is a finite union of Euclidean opens, in particular for M itself.

***Proof of theorem 4.5.7*:**

The argument establishes that $D_U \colon H^k_c(U;R) \to H_{n-k}(U;R)$ is an isomorphism for all open $U \subseteq M$ that admit a finite cover by open sets homeomorphic to $\mathbb{R}^n$; taking $U = M$, compact, gives the theorem with $H^k_c(M) = H^k(M)$ and $D_M = D$.

Step 1: The Euclidean case. Let $U \cong \mathbb{R}^n$, an open ball. Then $H_{n-k}(U;R) = H_{n-k}(\mathbb{R}^n;R)$ is R for $k=n$ and 0 otherwise, since $\mathbb{R}^n$ is contractible. For compactly supported cohomology, $H^k_c(\mathbb{R}^n;R) \cong H^k(\mathbb{R}^n, \mathbb{R}^n - B; R)$ for a large closed ball B (the colimit stabilizes once K contains a ball, as any compact $K \subseteq \mathbb{R}^n$ sits inside such a B and the restriction maps are isomorphisms). By excision theorem 3.4.7 and the long exact sequence of the pair $(\mathbb{R}^n, \mathbb{R}^n - B)$,

$$H^k(\mathbb{R}^n, \mathbb{R}^n - B; R) \cong \widetilde{H}^k(\mathbb{R}^n/(\mathbb{R}^n - B); R) \cong \widetilde{H}^k(S^n;R)$$

which is R for $k=n$ and 0 otherwise. Thus both $H^k_c(\mathbb{R}^n;R)$ and $H_{n-k}(\mathbb{R}^n;R)$ vanish except at $k=n$, where both equal R. At $k=n$ the map D_U caps the generator of $H^n_c(\mathbb{R}^n;R)$ against the local fundamental class; under the identifications above this is the evaluation of a generator of $H^n(\mathbb{R}^n, \mathbb{R}^n - B; R) \cong R$ against the generator of $H_n(\mathbb{R}^n, \mathbb{R}^n - B; R) \cong R$ defining the orientation, which is the Kronecker pairing $R \times R \to R$ sending $(1,1) \mapsto 1$ by the choice of orientation. Hence D_U is an isomorphism (an isomorphism $R \to R$ at $k=n$, and $0 \to 0$ elsewhere).

Step 2: The Mayer-Vietoris gluing. Suppose $U = A \cup B$ with A, B open and the theorem known for A, B, and $A \cap B$. There is a Mayer-Vietoris sequence for homology, theorem 3.4.19,

$$\cdots \to H_{n-k}(A \cap B) \to H_{n-k}(A) \oplus H_{n-k}(B) \to H_{n-k}(U) \to H_{n-k-1}(A \cap B) \to \cdots$$

and a Mayer-Vietoris sequence for cohomology with compact supports, with the variance reversed so that the maps run in the same direction as in homology,

$$\cdots \to H^k_c(A \cap B) \to H^k_c(A) \oplus H^k_c(B) \to H^k_c(U) \to H^{k+1}_c(A \cap B) \to \cdots$$

This compactly supported sequence arises from the short exact sequence of compactly supported cochain complexes

$$0 \to C^\bullet_c(A \cap B) \to C^\bullet_c(A) \oplus C^\bullet_c(B) \to C^\bullet_c(U) \to 0$$

the first map the difference of extensions by zero, the second the sum, exactness following because a compactly supported cochain on U supported in the compact $K \subseteq A \cup B$ splits, by a partition of the support into the parts inside A and inside B, into a sum of compactly supported cochains on A and on B agreeing on $A \cap B$. The duality maps $D_A, D_B, D_{A\cap B}, D_U$ commute with these two sequences: the squares relating them commute by the naturality of the cap product, proposition 4.5.5(3), applied to the inclusions $A \cap B \hookrightarrow A, B \hookrightarrow U$, together with the compatibility of the local fundamental classes under restriction. (The squares involving the connecting homomorphisms commute up to sign; replacing D by $\pm D$ on alternate groups, a change of sign that does not affect being an isomorphism, makes the ladder strictly commute, as detailed in [14, Lemma 3.36].) The result is a commuting ladder between the two long exact sequences with $D_A \oplus D_B$ and $D_{A\cap B}$ isomorphisms by hypothesis. By the five lemma (lemma 3.4.22), D_U is an isomorphism.

Step 3: Induction on a convex cover. The induction must be set up so that the intersections that arise stay inside the class being inducted on. The class of arbitrary Euclidean opens is not closed under intersection: an open subset of $\mathbb{R}^n$ need not be Euclidean, nor even a finite union of Euclidean opens, since in general it is a countably infinite union of convex opens (for instance a disjoint union of infinitely many balls). Convex opens behave better, because a finite intersection of convex sets is convex. The induction is therefore run over finite unions of convex opens within a single chart, and the passage to arbitrary opens and to M is made afterward by a colimit.

Fix a chart $\mathbb{R}^n \cong V \subseteq M$ and consider opens of the form $W = C_1 \cup \cdots \cup C_m$ with each $C_i \subseteq V$ a convex open subset of the chart. The claim is that D_W is an isomorphism for every such W, by induction on m. For $m = 1$, $W = C_1$ is convex, hence homeomorphic to $\mathbb{R}^n$, and Step 1 applies. For the inductive step set $A = C_1 \cup \cdots \cup C_{m-1}$ and $B = C_m$. Then A is a union of $m - 1$ convex opens, so D_A is an isomorphism by induction; D_B is an isomorphism by Step 1; and

$$A \cap B = (C_1 \cap C_m) \cup \cdots \cup (C_{m-1} \cap C_m)$$

is a union of the $m - 1$ opens $C_i \cap C_m$, each of which is convex (a finite intersection of convex sets), hence a union of $m - 1$ convex opens. By the inductive hypothesis $D_{A\cap B}$ is an isomorphism. Step 2 then gives that D_W is an isomorphism, completing the induction over finite unions of convex opens.

The extension to an arbitrary open $U \subseteq M$ that admits a finite cover by Euclidean opens is by colimit. Each Euclidean chart, and hence U, is an increasing union $U = \bigcup_j U_j$ where each U_j is a finite union of convex opens drawn from finitely many charts (take U_j to be the convex pieces of a fixed locally finite convex refinement of the cover that meet a compact exhaustion

$K_1 \subseteq K_2 \subseteq \cdots$ of U); the case where a U_j spans several charts is itself handled by one more pass of the Mayer-Vietoris induction of Step 2 across the charts. Duality for U follows from duality for the U_j by passing to the colimit: both $H_c^k(-;R)$ and $H_{n-k}(-;R)$ commute with such increasing unions, H_c^k by its defining colimit over compact supports and H_{n-k} because every singular chain has compact image, and the duality maps D_{U_j} assemble into D_U. This colimit-commutes-with-(co)homology bookkeeping is the direct-limit argument carried out in [14, Theorem 3.35 and Lemma 3.36], the same argument deferred for the non-compact assembly of Step 4 below. Taking $U = M$, which is compact and so covered by finitely many Euclidean charts, gives that

$$D\colon H^k(M;R) = H_c^k(M;R) \xrightarrow{\cong} H_{n-k}(M;R)$$

is an isomorphism for all k.

Step 4: The non-compact case. When M is non-compact (and R-orientable, not assumed compact), the same local argument applies, but the global statement reads $D\colon H_c^k(M;R) \to H_{n-k}(M;R)$, and assembling it requires the same passage to the colimit over compact subsets that appeared in Step 3, now over an exhaustion of the whole manifold rather than of a single open. The conceptual content is entirely the local case of Step 1 and the Mayer-Vietoris gluing of Step 2, both proved here in full; the only deferral, for the compact M of the theorem statement as well as for the non-compact case, is the direct-limit bookkeeping (commuting the colimit past homology and checking that the duality maps assemble) of [14, Theorem 3.35 and Lemma 3.36], which is routine but lengthy. This completes the proof.

The proof is the local-to-global pattern in its purest form. Duality is a statement about a single ball, where it is the self-pairing of R recording the orientation; Mayer-Vietoris glues balls; induction exhausts the manifold. The only place where ordinary cohomology had to be replaced by the compactly supported version is the gluing, where the extension-by-zero of a compactly supported cochain provides the covariant maps that match the variance of homology. For compact M the two cohomologies coincide and the theorem is the clean isomorphism $H^k(M;R) \cong H_{n-k}(M;R)$.

The consequences of duality are statements about the shape of the (co)homology of a closed manifold: a symmetry of Betti numbers, a nondegenerate pairing on cohomology, and a vanishing of the Euler characteristic in odd dimensions.

Corollary 4.5.10: Consequences of Poincaré Duality

Let M be a closed R-orientable n-manifold.

1. *Symmetry of Betti numbers.* If $R = \mathbb{F}$ is a field, then $\dim_{\mathbb{F}} H^k(M;\mathbb{F}) = \dim_{\mathbb{F}} H^{n-k}(M;\mathbb{F})$ for all k. In particular, over $\mathbb{Q}$ the Betti numbers satisfy $b_k(M) = b_{n-k}(M)$.

2. *Nondegeneracy of the cup pairing.* If $R = \mathbb{F}$ is a field, the bilinear pairing

$$H^k(M;\mathbb{F}) \otimes_{\mathbb{F}} H^{n-k}(M;\mathbb{F}) \xrightarrow{\smile} H^n(M;\mathbb{F}) \xrightarrow{\langle\,\cdot\,,[M]\rangle} \mathbb{F}$$

is nondegenerate.

3. *Vanishing Euler characteristic in odd dimensions.* If n is odd, the Euler characteristic $\chi(M) = 0$.

***Proof*:**

1. *Symmetry of Betti numbers.* Over a field $\mathbb{F}$ the universal coefficient theorem theorem 4.1.8 has vanishing Ext term, so $H^j(M;\mathbb{F}) \cong \mathrm{Hom}_{\mathbb{F}}(H_j(M;\mathbb{F}),\mathbb{F})$ and $\dim_{\mathbb{F}} H^j(M;\mathbb{F}) = \dim_{\mathbb{F}} H_j(M;\mathbb{F})$ for every j. Poincaré duality theorem 4.5.7 gives $H^k(M;\mathbb{F}) \cong H_{n-k}(M;\mathbb{F})$, so
$$\dim_{\mathbb{F}} H^k(M;\mathbb{F}) = \dim_{\mathbb{F}} H_{n-k}(M;\mathbb{F}) = \dim_{\mathbb{F}} H^{n-k}(M;\mathbb{F})$$
For $\mathbb{F} = \mathbb{Q}$ these dimensions are the Betti numbers $b_k(M)$, giving $b_k(M) = b_{n-k}(M)$.

2. *Nondegeneracy.* By corollary 4.5.6 with $z = [M]$, the pairing in question equals $(\varphi,\psi) \mapsto \langle \psi, D(\varphi)\rangle$, where $D(\varphi) = [M] \frown \varphi$. Fix $\varphi \neq 0$ in $H^k(M;\mathbb{F})$. By theorem 4.5.7, $D(\varphi) \in H_{n-k}(M;\mathbb{F})$ is nonzero. Over a field the Kronecker pairing $\langle\,\cdot\,,\cdot\,\rangle\colon H^{n-k}(M;\mathbb{F}) \otimes H_{n-k}(M;\mathbb{F}) \to \mathbb{F}$ is the evaluation of $\mathrm{Hom}_{\mathbb{F}}(H_{n-k},\mathbb{F})$ on H_{n-k}, which is nondegenerate, so there exists $\psi \in H^{n-k}(M;\mathbb{F})$ with $\langle \psi, D(\varphi)\rangle \neq 0$. Hence no nonzero φ pairs trivially with all ψ. Symmetrically, since the pairing $\langle\,\cdot\,,[M]\rangle \circ \smile$ has the cup product graded-commutative up to sign (theorem 4.2.4), no nonzero ψ pairs trivially with all φ; thus the pairing is nondegenerate on both sides.

3. *Euler characteristic.* Compute $\chi(M)$ over $\mathbb{Q}$, where $\chi(M) = \sum_{k=0}^{n}(-1)^k b_k(M)$ with $b_k = \dim_{\mathbb{Q}} H_k(M;\mathbb{Q})$ (the Euler characteristic is independent of the field, and over $\mathbb{Q}$ it is the alternating sum of Betti numbers). If M is $\mathbb{Q}$-orientable, part (1) gives $b_k = b_{n-k}$. With n odd, the substitution $k \mapsto n-k$ sends $(-1)^k$ to $(-1)^{n-k} = -(-1)^k$, so pairing the terms k and $n-k$,
$$\chi(M) = \sum_{k=0}^{n}(-1)^k b_k = \sum_{k=0}^{n}(-1)^{n-k} b_{n-k} = -\sum_{k=0}^{n}(-1)^k b_k = -\chi(M)$$
whence $2\chi(M) = 0$ and $\chi(M) = 0$. For a non-orientable closed M of odd dimension the same conclusion holds by applying the orientable case to the orientable double cover $\widetilde{M}$, for which $\chi(\widetilde{M}) = 2\chi(M)$ and $\chi(\widetilde{M}) = 0$ force $\chi(M) = 0$; this completes the proof.

The symmetry of part (1) is the original form in which Poincaré stated his duality, before the cohomological machinery existed: the k-th and $(n-k)$-th Betti numbers of a closed orientable manifold coincide. Part (2) is the form that constrains cohomology rings. Over a field the cup product makes $H^\bullet(M;\mathbb{F})$ a graded-commutative ring carrying a nondegenerate pairing between complementary degrees, which forces, for instance, that a nonzero class in H^k has a partner in H^{n-k} multiplying with it to the generator of H^n. Part (3) is the cleanest numerical consequence: an odd-dimensional closed manifold has Euler characteristic zero, recovered here from the Betti-number symmetry alone, and extended to the non-orientable case through the orientable double cover.

The theorem and its corollary are best absorbed through computation. The following example verifies duality directly for the spheres and reads off the cohomology pairings of the torus and of complex projective space from the duality isomorphism and the rings of definition 4.2.5.

Example 4.5.11: Poincaré Duality for Spheres, Tori, and Projective Spaces

1. *The sphere S^n.* The sphere is a closed orientable n-manifold with $H_k(S^n;\mathbb{Z}) = \mathbb{Z}$ for $k \in \{0,n\}$ and 0 otherwise; by the universal coefficient theorem the cohomology is the same.

Duality predicts $H^k(S^n) \cong H_{n-k}(S^n)$. Checking degrees: $H^0(S^n) = \mathbb{Z} \cong H_n(S^n) = \mathbb{Z}$ and $H^n(S^n) = \mathbb{Z} \cong H_0(S^n) = \mathbb{Z}$, while all intermediate groups vanish on both sides. The duality map sends the generator $1 \in H^0(S^n)$ to $[S^n] \frown 1 = [S^n] \in H_n(S^n)$ by the unit law of proposition 4.5.5(1), the fundamental class itself, and sends the generator $\alpha \in H^n(S^n)$ to $[S^n] \frown \alpha \in H_0(S^n)$, the class whose Kronecker pairing computes $\langle \alpha, [S^n] \rangle$, a generator of $H_0(S^n) = \mathbb{Z}$. Both maps are isomorphisms, confirming the theorem.

2. *The torus T^2.* The torus is a closed orientable surface with $H^\bullet(T^2;\mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[\alpha,\beta]$, the exterior algebra on generators $\alpha, \beta \in H^1(T^2;\mathbb{Z})$ of example 4.2.6, with $\alpha \smile \beta = \gamma$ the generator of $H^2(T^2;\mathbb{Z}) \cong \mathbb{Z}$ and $\alpha^2 = \beta^2 = 0$. Here $n = 2$ and the significant pairing is in the middle degree $k = n - k = 1$,

$$H^1(T^2;\mathbb{Z}) \otimes H^1(T^2;\mathbb{Z}) \xrightarrow{\smile} H^2(T^2;\mathbb{Z}) \xrightarrow{\langle \cdot, [T^2] \rangle} \mathbb{Z}$$

On the basis $\{\alpha, \beta\}$ this pairing has matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, since $\alpha \smile \alpha = 0$, $\beta \smile \beta = 0$, $\alpha \smile \beta = \gamma$ and $\beta \smile \alpha = -\gamma$ by graded-commutativity theorem 4.2.4, and $\langle \gamma, [T^2] \rangle = 1$. This matrix is invertible over $\mathbb{Z}$ (determinant 1), so the pairing is nondegenerate, exactly as corollary 4.5.10(2) requires. The skew-symmetry is forced by the odd middle degree.

3. *Complex projective space $\mathbb{C}\mathrm{P}^n$.* The space $\mathbb{C}\mathrm{P}^n$ is a closed orientable $2n$-manifold with cohomology ring $H^\bullet(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}[u]/(u^{n+1})$, where $|u| = 2$, so $H^{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = \mathbb{Z}\langle u^k \rangle$ for $0 \le k \le n$ and the odd groups vanish. (The additive groups are the CW computation of example 3.5.13, and the ring structure is derived in theorem 4.5.12 below.) Duality pairs degree $2k$ with degree $2n - 2k$,

$$H^{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \otimes H^{2n-2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \xrightarrow{\smile} H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \xrightarrow{\langle \cdot, [\mathbb{C}\mathrm{P}^n] \rangle} \mathbb{Z}$$

Here $u^k \smile u^{n-k} = u^n$, the generator of $H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$, and $\langle u^n, [\mathbb{C}\mathrm{P}^n] \rangle = 1$, so each complementary pair (u^k, u^{n-k}) multiplies to the top generator and pairs to 1. The pairing matrix in the bases $\{u^k\}$ and $\{u^{n-k}\}$ is therefore the 1×1 identity in each complementary pair of degrees, nondegenerate, and the duality map $D(u^k) = [\mathbb{C}\mathrm{P}^n] \frown u^k$ generates $H_{2n-2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$. The ring structure $\mathbb{Z}[u]/(u^{n+1})$ is precisely the structure that the nondegenerate duality pairing forces: u^k must have a multiplicative partner reaching the top class, and u^{n-k} is that partner.

The three computations show duality in its two registers. For the sphere it is the bare statement that $H^k \cong H_{n-k}$, with the fundamental class as the image of the unit. For the torus and for $\mathbb{C}\mathrm{P}^n$ it is the nondegeneracy of the cup pairing, which reads the multiplicative structure of the cohomology ring off the requirement that complementary degrees pair perfectly: in the torus the middle pairing is the symplectic form $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, in $\mathbb{C}\mathrm{P}^n$ it is the statement that u^k and u^{n-k} multiply to the generator. The cohomology ring and the duality pairing determine one another.

The last computation took the ring $\mathbb{Z}[u]/(u^{n+1})$ as given. Poincaré duality in fact forces it, and the same cell structure identifies the additive generators with the projective subspaces they bound.

Theorem 4.5.12: The Cohomology Ring of Complex Projective Space

For $n \geq 0$ the integral cohomology ring of complex projective space is the truncated polynomial ring

$$H^\bullet(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \cong \mathbb{Z}[u]/(u^{n+1}), \qquad |u| = 2$$

on a generator $u \in H^2(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$, so that u^k generates $H^{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \cong \mathbb{Z}$ for $0 \leq k \leq n$ and every other group vanishes. Dually, $H_{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \cong \mathbb{Z}$ is generated by the class $[\mathbb{C}\mathrm{P}^k]$ carried by a linearly embedded projective subspace $\mathbb{C}\mathrm{P}^k \subseteq \mathbb{C}\mathrm{P}^n$, and u^{n-k} is Poincaré dual to $[\mathbb{C}\mathrm{P}^k]$, in the sense that $[\mathbb{C}\mathrm{P}^n] \frown u^{n-k}$ generates $H_{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$.

***Proof*:**
Additive structure and the geometric generators. The standard CW structure of example 3.5.13 gives $\mathbb{C}\mathrm{P}^n$ one cell in each even dimension $0, 2, \ldots, 2n$ and none in odd dimensions, so $H_{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \cong \mathbb{Z}$, the odd groups vanish, and by the universal coefficient theorem (theorem 4.1.8) the cohomology groups match. In that structure the $2k$-skeleton is the linearly embedded $\mathbb{C}\mathrm{P}^k = \{[z_0 : \cdots : z_k : 0 : \cdots : 0]\}$, whose single top cell is the unique $2k$-cell of $\mathbb{C}\mathrm{P}^n$. Hence the inclusion $\iota\colon \mathbb{C}\mathrm{P}^k \hookrightarrow \mathbb{C}\mathrm{P}^n$ carries the fundamental class of the closed orientable $2k$-manifold $\mathbb{C}\mathrm{P}^k$ to a generator $[\mathbb{C}\mathrm{P}^k]$ of $H_{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$.

Ring structure, by induction on n. For $n = 0$ the ring is $\mathbb{Z}$, and for $n = 1$ the space $\mathbb{C}\mathrm{P}^1 = S^2$ has ring $\mathbb{Z}[u]/(u^2)$ with $|u| = 2$. Assume the claim for $n-1$. The pair $(\mathbb{C}\mathrm{P}^n, \mathbb{C}\mathrm{P}^{n-1})$ has a single relative cell, in dimension $2n$, so $H^i(\mathbb{C}\mathrm{P}^n, \mathbb{C}\mathrm{P}^{n-1}; \mathbb{Z}) \cong \tilde{H}^i(S^{2n}; \mathbb{Z})$ vanishes for $i \neq 2n$, and the cohomology long exact sequence of the pair makes restriction $H^i(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \to H^i(\mathbb{C}\mathrm{P}^{n-1}; \mathbb{Z})$ an isomorphism for $i \leq 2n-2$. Let u generate $H^2(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$. It restricts to a generator $\bar{u}$ of $H^2(\mathbb{C}\mathrm{P}^{n-1}; \mathbb{Z})$. By the inductive hypothesis $\bar{u}^k$ generates $H^{2k}(\mathbb{C}\mathrm{P}^{n-1}; \mathbb{Z})$ for $0 \leq k \leq n-1$, and since restriction is a ring homomorphism and an isomorphism in these degrees, u^k generates $H^{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$ for $0 \leq k \leq n-1$. Only $H^{2n}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \cong \mathbb{Z}$ remains. Since $\mathbb{C}\mathrm{P}^n$ is a closed orientable $2n$-manifold and $H_*(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$ is torsion-free (so the Kronecker pairing $H^{2n}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \otimes H_{2n}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \to \mathbb{Z}$ is perfect), Poincaré duality (theorem 4.5.7), through the cup-cap adjunction $\langle \alpha \smile \beta, [\mathbb{C}\mathrm{P}^n] \rangle = \langle \alpha, \beta \frown [\mathbb{C}\mathrm{P}^n] \rangle$, makes the cup-product pairing

$$H^2(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \otimes H^{2n-2}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \xrightarrow{\smile} H^{2n}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \cong \mathbb{Z}$$

nonsingular. As u generates the first factor and u^{n-1} the second, the product $u \smile u^{n-1} = u^n$ generates $H^{2n}(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$. The odd groups vanish and $H^i(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) = 0$ for $i > 2n$, so the ring is $\mathbb{Z}[u]/(u^{n+1})$.

The duality identification. Capping with the fundamental class is the duality isomorphism $D\colon H^{2n-2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \xrightarrow{\cong} H_{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$ of theorem 4.5.7, so $D(u^{n-k}) = [\mathbb{C}\mathrm{P}^n] \frown u^{n-k}$ generates $H_{2k}(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$, exhibiting u^{n-k} as the Poincaré dual of $[\mathbb{C}\mathrm{P}^k]$, completing the proof.

4.5.1 Manifolds with Boundary

The proof just given establishes more than the theorem states. What the induction verifies is that $D_U\colon H^k_c(U; R) \to H_{n-k}(U; R)$ is an isomorphism for every open $U \subseteq M$, and compactness of M entered only at the last step, to identify $H^k_c(M; R)$ with $H^k(M; R)$. Recording the general statement needs no extra work and is what the boundary case will run on.

Corollary 4.5.13: Duality for Non-Compact Manifolds

Let M be an R-orientable n-manifold, not necessarily compact. Then

$$D_M\colon H_c^k(M;R) \longrightarrow H_{n-k}(M;R)$$

is an isomorphism for every k.

***Proof*:**
This is the conclusion of the induction in the proof of theorem 4.5.7 applied to $U = M$, which nowhere used compactness of M: the base case is example 4.5.9, the inductive step is the Mayer-Vietoris comparison, and the direct-limit step covers an arbitrary open by Euclidean pieces.

A manifold with boundary is not a manifold, so duality cannot apply to it directly; but its interior is one, and the boundary is glued on in a way that changes neither the homotopy type nor the compactly supported cohomology in a manner one cannot control. The control is given by a collar.

Definition 4.5.14: Collared Boundary

Let M be a topological n-manifold with boundary ∂M. A *collar* is an embedding $c\colon \partial M \times [0,1) \to M$ onto a neighbourhood of ∂M with $c(x,0) = x$ for every $x \in \partial M$. The boundary is *collared* when such a c exists.

Every smooth compact manifold with boundary has a collar, obtained by flowing along an inward-pointing vector field; that construction is differential-topological rather than algebraic and is not carried out here. In this section the collar is a hypothesis.

Duality needs a class to cap with, and definition 4.4.8 produces one only for closed manifolds. The boundary case has its own, living in the relative group.

Proposition 4.5.15: Relative Fundamental Class

Let M be a compact R-orientable n-manifold with collared boundary, with an R-orientation $x \mapsto \mu_x$ on its interior. There is a unique class

$$[M, \partial M] \in H_n(M, \partial M; R)$$

whose image under $H_n(M, \partial M; R) \to H_n(M, M - x; R) \cong R$ is μ_x for every interior point x.

***Proof*:**
Existence. Let c be a collar and let $DM = M \cup_{\partial M} M$ be the double, two copies of M glued along the boundary. The two collars fit together into an embedding $\partial M \times (-1,1) \to DM$, so DM is a closed n-manifold, and it is R-orientable: orient one copy by μ and the other by its mirror image, the two agreeing on the overlap collar. By theorem 4.4.9 it carries a fundamental class $[DM]$. Excising the interior of the second copy from the pair $(DM, \text{second copy})$ gives

$$H_n(DM, \text{second copy}; R) \cong H_n(M, \partial M; R),$$

and the image of $[DM]$ under $H_n(DM;R) \to H_n(DM, \text{second copy}; R) \cong H_n(M, \partial M; R)$ restricts at each interior $x \in M$ to μ_x, because restriction to $H_n(DM, DM - x; R)$ factors through the same map.

Uniqueness. A difference of two such classes restricts to 0 at every interior point. The local-to-global argument of theorem 4.4.9 shows that a class in $H_n(M, \partial M; R)$ is determined by its restrictions to $H_n(M, M - x; R)$ over interior x: that argument is a statement about the compactly supported sections of the orientation covering over the interior, and the collar makes ∂M contribute nothing to it. Hence the difference is 0.

Theorem 4.5.16: Poincaré-Lefschetz Duality

Let M be a compact R-orientable n-manifold with collared boundary ∂M, with relative fundamental class $[M, \partial M] \in H_n(M, \partial M; R)$. Then capping with $[M, \partial M]$ gives isomorphisms

$$H^k(M;R) \xrightarrow{\cong} H_{n-k}(M, \partial M; R), \qquad H^k(M, \partial M; R) \xrightarrow{\cong} H_{n-k}(M;R)$$

for every k.

***Proof*:**
Write $\mathring{M} = M - \partial M$ for the interior, an R-orientable n-manifold without boundary, and let $c\colon \partial M \times [0,1) \to M$ be a collar.

Step 1: the collar retracts M onto its interior. Pushing in along the collar, $r_t(c(x,s)) = c(x, s + t(1-s)/2)$ on the image of c and $r_t = \mathrm{id}$ outside it, defines a homotopy from id_M to a map into $\mathring{M}$; it is continuous because the two formulas agree near $s = 1$. The inclusion $\mathring{M} \hookrightarrow M$ is therefore a homotopy equivalence, so $H_{n-k}(\mathring{M};R) \cong H_{n-k}(M;R)$.

Step 2: the collar computes compactly supported cohomology of the interior. The compact subsets $K_\epsilon = M - c(\partial M \times [0,\epsilon))$ are cofinal among compact subsets of $\mathring{M}$, so by definition 4.5.8

$$H_c^k(\mathring{M};R) = \varinjlim_{\epsilon} H^k(\mathring{M}, \mathring{M} - K_\epsilon; R).$$

Each pair $(\mathring{M}, \mathring{M} - K_\epsilon)$ includes into $(M, \partial M)$, and both inclusions induce isomorphisms: on the first factor by Step 1, and on the second because $\mathring{M} - K_\epsilon = c(\partial M \times (0,\epsilon))$ deformation retracts to a slice of the collar, as does the open collar neighbourhood of ∂M in M. Excising the closed complement of the collar and passing to the colimit gives $H_c^k(\mathring{M};R) \cong H^k(M, \partial M; R)$.

Step 3: assemble. Apply corollary 4.5.13 to $\mathring{M}$ and substitute Steps 1 and 2:

$$H^k(M, \partial M; R) \cong H_c^k(\mathring{M};R) \xrightarrow[\cong]{D_{\mathring{M}}} H_{n-k}(\mathring{M};R) \cong H_{n-k}(M;R).$$

Under these identifications $D_{\mathring{M}}$ is cap product with $[M, \partial M]$, since the local fundamental classes assembling $[M, \partial M]$ restrict to those of $\mathring{M}$ at every interior point. The other isomorphism is the same argument with the roles of the absolute and relative theories exchanged, using that $H^k(M;R) \cong H^k(\mathring{M};R)$ by Step 1 and that the long exact sequences of the pair on the two sides match under cap product by proposition 4.5.5.

The corollary below is the form in which the theorem is usually used, and the one that matters for intersection theory on an even-dimensional manifold with boundary: the middle-dimensional pairing is not merely nondegenerate over a field but *unimodular* over $\mathbb{Z}$ once torsion is discarded.

Corollary 4.5.17: Integral Unimodularity of the Intersection Pairing

Let M be a compact oriented $2m$-manifold with collared boundary. Then the intersection pairing

$$H_m(M;\mathbb{Z})/\mathrm{tors} \;\times\; H_m(M,\partial M;\mathbb{Z})/\mathrm{tors} \longrightarrow \mathbb{Z}$$

is unimodular: it identifies each factor with the full $\mathbb{Z}$-dual of the other. Equivalently, in bases of the two free quotients its matrix has determinant ± 1.

***Proof*:**
The Kronecker pairing $H^m(M,\partial M;\mathbb{Z})\times H_m(M,\partial M;\mathbb{Z}) \to \mathbb{Z}$ induces, by the universal coefficient theorem theorem 4.1.8, an isomorphism

$$H^m(M,\partial M;\mathbb{Z})/\mathrm{tors} \xrightarrow{\cong} \mathrm{Hom}_{\mathbb{Z}}\big(H_m(M,\partial M;\mathbb{Z}),\mathbb{Z}\big),$$

the Ext term being torsion and so dying in the free quotient. Composing its inverse with the duality isomorphism $H^m(M,\partial M;\mathbb{Z}) \cong H_m(M;\mathbb{Z})$ of theorem 4.5.16 identifies $H_m(M;\mathbb{Z})/\mathrm{tors}$ with $\mathrm{Hom}_{\mathbb{Z}}(H_m(M,\partial M;\mathbb{Z}),\mathbb{Z})$, and the pairing implementing that identification is cap-then-evaluate, which is the intersection pairing. A perfect pairing of finitely generated free $\mathbb{Z}$-modules has unimodular matrix, since the induced map to the dual is an isomorphism of free modules of the same rank.

Exercise 4.5.1

1. Using the boundary formula of lemma 4.5.3, verify directly that if $z \in C_n(X;R)$ is a cycle and $\varphi \in C^k(X;R)$ is a cocycle, then for any $\psi \in C^{k-1}(X;R)$ the chains $z \frown \varphi$ and $z \frown (\varphi + \delta\psi)$ are homologous. (This is the cochain-independence half of proposition 4.5.4; reproduce it in detail.)

2. Let $M = S^1 \times S^1$ be the torus with its standard cohomology ring $\Lambda_{\mathbb{Z}}[\alpha,\beta]$ from example 4.2.6. Compute the duality images $D(\alpha) = [M] \frown \alpha$ and $D(\beta) = [M] \frown \beta$ as elements of $H_1(M;\mathbb{Z}) \cong \mathbb{Z}^2$, and verify that $D\colon H^1(M;\mathbb{Z}) \to H_1(M;\mathbb{Z})$ is an isomorphism by exhibiting the matrix of D in suitable bases.

3. Let M be a closed orientable n-manifold with n even, say $n = 2m$. Show that the middle cup pairing $H^m(M;\mathbb{Q}) \otimes H^m(M;\mathbb{Q}) \to \mathbb{Q}$ is symmetric when m is even and skew-symmetric when m is odd, and deduce that when m is odd, $\dim_{\mathbb{Q}} H^m(M;\mathbb{Q})$ is even.

4. Prove that no closed orientable n-manifold M with n odd can have $\chi(M) \neq 0$, and compute $\chi(S^2 \times S^1)$ from corollary 4.5.10(3), confirming it against a cell-count. (Contrast: $\mathbb{RP}^2$ is a closed surface with $\chi(\mathbb{RP}^2) = 1 \neq 0$, consistent with its even dimension; explain why this nonzero Euler characteristic is not in conflict with the result just proved.)

5. Using the projection formula proposition 4.5.5(3), show that if $f\colon M \to N$ is a map of closed oriented n-manifolds with $f_*[M] = d\,[N]$ for an integer d (the degree of f), then for every $\psi \in H^k(N;\mathbb{Z})$ one has $f_*([M] \frown f^*\psi) = d\,\big([N] \frown \psi\big)$ in $H_{n-k}(N;\mathbb{Z})$.

6. Let $\mathbb{F}$ be a field and M a closed $\mathbb{F}$-orientable n-manifold. Show that if $H^k(M;\mathbb{F}) = 0$ for some $0 < k < n$, then $H^{n-k}(M;\mathbb{F}) = 0$ as well, and give an example with $M = S^2$ where this forces a vanishing group from a nonvanishing one in the complementary degree.

4.6 Homology and Cohomology with Local Coefficients

The (co)homology theories built so far attach to a space X a single fixed abelian group of coefficients, the same copy of G over every point. The orientation theory of this chapter already pressed against that rigidity. The local orientations of an n-manifold are the groups $H_n(M, M \setminus \{x\}) \cong \mathbb{Z}$ of theorem 4.4.1, one over each point, and dragging a generator around an orientation-reversing loop returns it negated (definition 4.4.4); the Möbius band (example 4.4.5) is the first space where no global choice is consistent. For a closed non-orientable manifold the cost is exact: the top homology $H_n(M;\mathbb{Z})$ vanishes (theorem 4.4.7), so there is no fundamental class and the Poincaré duality of section 4.5 has nothing to pair against. The defect is not in the manifold but in the coefficients: they were required to be constant, while the geometry asks for a group that the fundamental group is allowed to permute. This section builds the resulting twisted (co)homology in full, and shows that it restores both a fundamental class and Poincaré duality in the non-orientable case.

The organizing idea is that the coefficients should be a representation of $\pi_1(X)$ rather than a bare abelian group. Such a representation is a *local coefficient system*, and the construction that turns it into homology and cohomology runs entirely through one object already in hand: the universal cover $\widetilde{X}$, on which $\pi_1(X)$ acts by deck transformations (definition 2.3.8). The cellular chains of $\widetilde{X}$ form a complex of free modules over the group ring $\mathbb{Z}[\pi_1(X)]$, and tensoring or homming this complex against the chosen representation produces homology and cohomology with local coefficients; when the representation is trivial, both reduce to the ordinary theories of chapter 3 and chapter 4. The orientation system is the motivating case, but the same construction is exactly what two later developments will call for: the Serre spectral sequence of a fibration whose fundamental group acts nontrivially on the fiber, where the E^2 page is the homology of the base with coefficients in the local system $H_q(F)$ (chapter 6), and obstruction theory for a non-simple fiber, where the obstruction to extending a map lives in a twisted group $H^{n+1}(X;\pi_n(F))$ (chapter 5). Those are recorded here as forward applications, not as prerequisites for the construction.

Definition 4.6.1: Local Coefficient System

Let X be a path-connected space with basepoint x_0 and fundamental group $\pi = \pi_1(X, x_0)$. A *local coefficient system* (or *local system*) of abelian groups on X is a functor

$$\mathcal{M} \colon \Pi_1(X) \to \mathbf{Ab}$$

from the fundamental groupoid $\Pi_1(X)$ (the category whose objects are the points of X and whose morphisms $x \to y$ are homotopy classes rel endpoints of paths from x to y) to the category of abelian groups. Equivalently, since X is path-connected the groupoid $\Pi_1(X)$ is equivalent to the one-object category π, and a local system is the same datum as a *left* $\mathbb{Z}[\pi]$*-module* $M = \mathcal{M}(x_0)$: an abelian group M together with an action of π by automorphisms, the *monodromy action*, extended $\mathbb{Z}$-linearly to the group ring $\mathbb{Z}[\pi]$. The system is *trivial* (or *constant*) if the monodromy action is trivial, so that every $[\gamma] \in \pi$ acts as id_M.

The two descriptions are equivalent. The functorial description assigns an abelian group $\mathcal{M}(x)$ to each point and an isomorphism $\mathcal{M}(x) \to \mathcal{M}(y)$ to each homotopy class of paths, compatibly with concatenation; this is the form in which local systems arise geometrically, as a family of groups varying continuously over X. The module description collapses the family to its value $M = \mathcal{M}(x_0)$ at the basepoint, where the loops at x_0, which are the morphisms $x_0 \to x_0$ in $\Pi_1(X)$, act as automorphisms and make M a representation of π. Path-connectedness is what makes the two equivalent: any two points are joined by a path, so every $\mathcal{M}(x)$ is isomorphic to M, and the only invariant left is the action of the loops at the basepoint. The equivalence of categories $\Pi_1(X) \simeq \pi$ underlying this is the groupoid form of the statement that π_1 is independent of basepoint up to the conjugation ambiguity already recorded in chapter 2; passing to the value at x_0 rigidifies the choice.

There is a third face, and it is the one under which local systems arise in geometry. A family of groups varying over X is, before anything else, a sheaf; the functor on $\Pi_1(X)$ appears only after paths are chosen, and the π-module only after a basepoint is. Consumers of this theory — a variation of Hodge structure, a family of étale sheaves, a D-module — hand us the sheaf and expect the representation, so the passage between them is worth recording once here rather than in each of them.

Definition 4.6.2: Locally Constant Sheaf

Let X be a space and R a ring. A sheaf $\mathcal{L}$ of R-modules on X is *locally constant*, or a *local system of R-modules*, if every point has an open neighbourhood V with $\mathcal{L}|_V \cong \underline{M}_V$ for some R-module M, where $\underline{M}_V$ is the constant sheaf on V with value M. When each such M may be taken to be a finite-dimensional vector space over a field, $\mathcal{L}$ has finite rank.

Theorem 4.6.3: Locally Constant Sheaves Are Local Systems

Let X be path connected and locally simply connected with basepoint x_0. Sending a locally constant sheaf $\mathcal{L}$ to the assignment $x \mapsto \mathcal{L}_x$, together with transport of stalks along paths, is an equivalence between the category of locally constant sheaves of R-modules on X and the category of functors $\Pi_1(X) \to R\text{-}\mathbf{Mod}$; composing with definition 4.6.1 it is an equivalence with the category of $R[\pi]$-modules.

***Proof*:**
Transport. Let $\gamma\colon [0,1] \to X$ be a path. Cover $\gamma([0,1])$ by opens on which $\mathcal{L}$ is constant and choose $0 = t_0 < \cdots < t_N = 1$ with $\gamma([t_{i-1}, t_i]) \subseteq V_i$ for such an open V_i; this is possible by compactness of $[0,1]$ and the Lebesgue number lemma. On V_i the isomorphism $\mathcal{L}|_{V_i} \cong \underline{M_i}$ identifies every stalk over V_i canonically with M_i, so composing along the chain gives an isomorphism $\tau_\gamma\colon \mathcal{L}_{\gamma(0)} \to \mathcal{L}_{\gamma(1)}$. Two chains admit a common refinement, on which the two composites visibly agree, so τ_γ is independent of the choices.

Homotopy invariance. Let $H\colon [0,1]^2 \to X$ be a homotopy rel endpoints. Subdivide the square into finitely many closed subsquares each mapping into a single open on which $\mathcal{L}$ is constant, again by compactness. Transport around the boundary of one such subsquare is the identity, since all four stalk identifications factor through the single module attached to that open. Composing subsquares across the subdivision shows transport along the two horizontal edges agrees, so τ_γ depends only on the homotopy class. Functoriality under concatenation holds by construction, so $x \mapsto \mathcal{L}_x$ is a functor on $\Pi_1(X)$.

Fully faithful. A morphism $\mathcal{L} \to \mathcal{L}'$ induces stalk maps commuting with transport. Conversely a compatible family of stalk maps is, over each open on which both sheaves are constant, a single module map, and these agree on overlaps by compatibility; a morphism of sheaves may be glued from local data, so the family comes from a unique morphism.

Essentially surjective. Given a functor $\mathcal{M}$ on $\Pi_1(X)$, cover X by simply connected opens V_i — available because X is locally simply connected. On V_i any two paths between the same points are homotopic, so the transport isomorphisms of $\mathcal{M}$ give a canonical identification of all its values, and $\underline{\mathcal{M}(x_i)}_{V_i}$ is a well-defined constant sheaf for a chosen $x_i \in V_i$. On $V_i \cap V_j$ transport gives isomorphisms between these constant sheaves, locally constant hence constant on each component, and the cocycle condition holds by functoriality of $\mathcal{M}$. Gluing produces a locally constant sheaf whose associated functor is $\mathcal{M}$.

The equivalence is the reason the words "local system" are used for all three objects without comment. It also explains a phenomenon that looks surprising from the sheaf side: a locally constant sheaf on a simply connected space is constant, since the corresponding representation of the trivial group is trivial.

Example 4.6.4: The Orientation System of a Manifold

Let M be a connected n-manifold and R a commutative ring. theorem 4.4.1 computes the local homology $H_n(M, M \setminus \{x\}; R) \cong R$ at each point, a free R-module of rank one. As x varies these groups assemble into a local system on M: the orientation system

$$\mathcal{O}_R \colon \Pi_1(M) \to \mathbf{Ab}, \qquad x \mapsto H_n(M, M \setminus \{x\}; R)$$

A homotopy class of paths from x to y transports a local orientation μ_x to μ_y by the continuation that already governs definition 4.4.4, and this transport is the structure map $\mathcal{O}_R(x) \to \mathcal{O}_R(y)$. As a $\mathbb{Z}[\pi_1(M)]$-module the system is R with $\pi_1(M)$ acting through the sign of the determinant of the transition along a loop: a loop preserving the local orientation acts as $+1$, a loop reversing it acts as -1. Concretely, the action factors through a homomorphism

$$w_1 \colon \pi_1(M) \to \{\pm 1\}$$

acting on R through $\pm\mathrm{id}_R \in \mathrm{Aut}(R)$, the first Stiefel–Whitney class read as a representation. The system $\mathcal{O}_R$ is trivial precisely when M is R-orientable: a global orientation in the sense of definition 4.4.4 is exactly a $\pi_1(M)$-invariant generator, which exists if and only if every loop acts as $+1$. For the open Möbius band of example 4.4.5 the core circle reverses orientation, so the generator of $\pi_1 \cong \mathbb{Z}$ acts as -1 on $\mathbb{Z}$ and the system is nontrivial; this is the nonorientability of the band expressed as a nontrivial monodromy representation. With $R = \mathbb{Z}/2$ the only automorphism of $\mathbb{Z}/2$ is the identity, so $\mathcal{O}_{\mathbb{Z}/2}$ is always trivial, recovering the fact that every manifold is $\mathbb{Z}/2$-orientable.

Example 4.6.5: The Fiber-Homology System of a Fibration

Let $p \colon E \to B$ be a Serre fibration (definition 5.2.2) with path-connected base B and fiber $F = p^{-1}(b_0)$. Lifting a loop γ in B to a fiber-preserving homotopy of E produces a self-homotopy-equivalence of F, the *monodromy* of the fibration around γ, well defined up to homotopy and depending only on $[\gamma] \in \pi_1(B, b_0)$. Applying $H_q(-)$ gives, for each q, an action of $\pi_1(B)$ on $H_q(F)$ by automorphisms, hence a local system

$$\mathcal{H}_q(F) \colon \Pi_1(B) \to \mathbf{Ab}, \qquad b \mapsto H_q(p^{-1}(b))$$

on the base. This is exactly the coefficient system that will drive the twisted Serre spectral sequence of box 6.2.8, where the E^2 page $H_p(B; \mathcal{H}_q(F))$ is homology of B with coefficients in this system. The system is trivial when $\pi_1(B)$ acts trivially on $H_*(F)$, the hypothesis under which the untwisted spectral sequence of theorem 6.2.4 holds. The smallest example where it is not is the Klein bottle K, realized as the S^1-bundle over S^1 obtained from $S^1 \times [0,1]$ by gluing the ends through the reflection $z \mapsto \bar{z}$ of the circle fiber. The reflection reverses orientation of $F = S^1$, so transport around the base circle acts on $H_1(F) = \mathbb{Z}$ by -1 and on $H_0(F) = \mathbb{Z}$ trivially; the degree-one system $\mathcal{H}_1(F)$ is the sign representation of $\pi_1(S^1) \cong \mathbb{Z}$ on $\mathbb{Z}$, nontrivial, and it is precisely the E^2 input that produces the torsion in $H_1(K; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ in exercise 4.6.1 (5).

With the coefficients defined, the chain-level construction can be assembled. The single geometric input is the deck action on the universal cover, which makes the cellular chains into free $\mathbb{Z}[\pi]$-modules.

Proposition 4.6.6: The Cellular Chains of the Universal Cover are Free over $\mathbb{Z}[\pi]$

Let X be a connected CW complex with universal cover $q\colon \widetilde{X} \to X$ and $\pi = \pi_1(X, x_0)$, identified with the deck group $\mathrm{Deck}\,(q)$ via proposition 2.3.11. Give $\widetilde{X}$ the CW structure lifting that of X, in which the cells of $\widetilde{X}$ are the connected components of the preimages $q^{-1}(e)$ of the open cells e of X. Then the deck action of π permutes the cells of $\widetilde{X}$ freely, and for each n the cellular chain group $C_n^{CW}(\widetilde{X})$ is a free left $\mathbb{Z}[\pi]$-module with one basis element for each n-cell of X. The cellular boundary maps are $\mathbb{Z}[\pi]$-linear, so $C_*^{CW}(\widetilde{X})$ is a complex of free $\mathbb{Z}[\pi]$-modules.

***Proof*:**

Step 1: The cells of $\widetilde{X}$ over a cell of X form a free π-set. A cell e of X has characteristic map $\Phi\colon D^n \to X$ whose restriction to the open disk is a homeomorphism onto e. Because D^n is simply connected, the lifting criterion of chapter 2 lifts Φ to $\widetilde{X}$, and the lifts are exactly the maps $\sigma \cdot \widetilde{\Phi}$ for $\sigma \in \pi$ and one fixed lift $\widetilde{\Phi}$, since two lifts of the same map differ by a deck transformation. Distinct σ give distinct lifts, as a nontrivial deck transformation of the universal cover fixes no point (definition 2.3.8); hence the open cells of $\widetilde{X}$ lying over e are in bijection with π, and π acts on them by left translation, freely and transitively.

Step 2: Freeness of the chain group. Choosing one lifted cell $\widetilde{e}$ over each cell e of X selects a basis. By Step 1 every n-cell of $\widetilde{X}$ is $\sigma \cdot \widetilde{e}$ for a unique $\sigma \in \pi$ and a unique cell e of X, so the cellular chain group $C_n^{CW}(\widetilde{X})$, free abelian on the n-cells of $\widetilde{X}$, is free abelian on the symbols $\{\sigma \cdot \widetilde{e}\}$. Collecting the translates of each fixed $\widetilde{e}$, this is precisely the free $\mathbb{Z}[\pi]$-module on the chosen lifts $\{\widetilde{e}\}$, one generator per n-cell of X. The deck action on chains is the left regular action permuting the σ, which is the $\mathbb{Z}[\pi]$-module structure.

Step 3: Equivariance of the boundary. The cellular boundary of definition 3.5.10 is induced by the attaching maps and the quotient maps to spheres, all of which are deck-equivariant, since the CW structure on $\widetilde{X}$ was lifted from X and the deck group acts cellularly. Concretely, $\partial(\sigma \cdot \widetilde{e}) = \sigma \cdot \partial(\widetilde{e})$ because applying a deck transformation commutes with restricting to a subcomplex and collapsing, so ∂ is $\mathbb{Z}[\pi]$-linear. Thus $C_*^{CW}(\widetilde{X})$ is a chain complex of free $\mathbb{Z}[\pi]$-modules, completing the proof.

The proposition is the structural fact that powers the whole section. The universal cover is the maximal symmetric cover, and its symmetry is exactly the action of π that the cellular chains record. Choosing a lift of each cell of X trivializes the chains into a free $\mathbb{Z}[\pi]$-module, with rank in each degree equal to the number of cells of X in that degree, no more and no less than the ordinary cellular chains, but now remembering the deck symmetry as a module structure. A local system is a $\mathbb{Z}[\pi]$-module; pairing the two through $\otimes$ and Hom over $\mathbb{Z}[\pi]$ is what produces the twisted theories.

Definition 4.6.7: Homology and Cohomology with Local Coefficients

Let X be a connected CW complex with universal cover $\widetilde{X}$ and $\pi = \pi_1(X, x_0)$, and let $\mathcal{M}$ be a local coefficient system, regarded as a $\mathbb{Z}[\pi]$-module M as in definition 4.6.1. The *homology of X with local coefficients in $\mathcal{M}$* is the homology of the chain complex obtained by tensoring the free $\mathbb{Z}[\pi]$-complex $C_*^{CW}(\widetilde{X})$ of proposition 4.6.6 with M over $\mathbb{Z}[\pi]$,

$$H_*(X;\mathcal{M}) \;=\; H_*\big(C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi]} M\big)$$

where π acts on $C_*^{CW}(\widetilde{X})$ on the right through the deck action (using $\sigma \cdot c = c \cdot \sigma^{-1}$ to convert the left deck action to a right action) and on M on the left through the monodromy. The *cohomology of X with local coefficients in $\mathcal{M}$* is the cohomology of the cochain complex of $\mathbb{Z}[\pi]$-module homomorphisms,

$$H^*(X;\mathcal{M}) \;=\; H^*\big(\mathrm{Hom}_{\mathbb{Z}[\pi]}(C_*^{CW}(\widetilde{X}), M)\big)$$

with coboundary dual to the cellular boundary. When $\mathcal{M}$ is the trivial system on an abelian group M, these are written $H_*(X;M)$ and $H^*(X;M)$ in the ordinary notation.

The construction is the ordinary cellular recipe with the deck symmetry inserted as a tensor and a Hom. Ordinary homology with coefficients in M tensors the cellular chains $C_*^{CW}(X)$ with M over $\mathbb{Z}$; the local version tensors the *lifted* chains $C_*^{CW}(\widetilde{X})$ with M over $\mathbb{Z}[\pi]$, so that the deck action and the monodromy action are identified in the tensor product. The effect is to twist the boundary maps by the monodromy: where the ordinary boundary multiplies a coefficient by an integer incidence number, the local boundary multiplies it by an element of $\mathbb{Z}[\pi]$, whose action on M depends on how π_1 moves the coefficients. Cohomology is the dual story, with $\mathrm{Hom}_{\mathbb{Z}[\pi]}$ in place of $\otimes_{\mathbb{Z}[\pi]}$. The definition uses a CW structure and the universal cover, both of which are choices; that the resulting groups are independent of these choices, and agree with the singular-chain definition of local (co)homology, is the cover-independence statement whose proof requires the comparison of free resolutions deferred to [14]. The construction and all the properties used in this book are established here in full.

The first property to check is that the twisted theory specializes to the ordinary one when the twist is switched off. This is what makes the local theory a genuine extension rather than a parallel construction.

Proposition 4.6.8: Reduction to Ordinary (Co)homology for a Trivial System

Let X be a connected CW complex and M an abelian group, regarded as a trivial local system on which $\pi = \pi_1(X)$ acts by the identity. Then there are natural isomorphisms

$$H_*(X;\mathcal{M}) \cong H_*(X;M), \qquad H^*(X;\mathcal{M}) \cong H^*(X;M)$$

with the ordinary cellular homology and cohomology of section 3.5 and chapter 4.

***Proof*:**
Write $\varepsilon\colon \mathbb{Z}[\pi] \to \mathbb{Z}$ for the augmentation, the ring map sending every group element to 1. The quotient of $\widetilde{X}$ by the deck action is X, and on cellular chains this quotient is exactly the base change along ε: the chosen $\mathbb{Z}[\pi]$-basis $\{\widetilde{e}\}$ of $C_*^{CW}(\widetilde{X})$ from proposition 4.6.6 maps to the cellular basis $\{e\}$ of $C_*^{CW}(X)$, and the deck-equivariant boundary descends to the cellular boundary of X under ε. Hence there is a natural isomorphism of chain complexes

$$C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi]} \mathbb{Z} \;\cong\; C_*^{CW}(X)$$

When π acts trivially on M, the $\mathbb{Z}[\pi]$-module M is the base change $\mathbb{Z} \otimes_{\mathbb{Z}} M$ with π acting through ε, so

$$C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi]} M \;\cong\; \big(C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi]} \mathbb{Z}\big) \otimes_{\mathbb{Z}} M \;\cong\; C_*^{CW}(X) \otimes_{\mathbb{Z}} M$$

the right-hand complex being the one whose homology is ordinary $H_*(X;M)$. Taking homology gives the first isomorphism. For cohomology, a $\mathbb{Z}[\pi]$-linear map $C_*^{CW}(\widetilde{X}) \to M$ into a module with trivial action is the same as a $\mathbb{Z}$-linear map $C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi]} \mathbb{Z} \to M$, that is a $\mathbb{Z}$-linear map $C_*^{CW}(X) \to M$, by the adjunction between base change and restriction along ε; this identifies the cochain complex $\mathrm{Hom}_{\mathbb{Z}[\pi]}(C_*^{CW}(\widetilde{X}), M)$ with the ordinary cochain complex $\mathrm{Hom}_{\mathbb{Z}}(C_*^{CW}(X), M)$, and taking cohomology gives the second isomorphism. Both identifications are natural in M and in X, as the augmentation and the basis correspondence are, completing the proof.

The reduction confirms that the local theory deserves the same notation: a trivial system reproduces the ordinary groups, so writing $H_*(X;M)$ for the trivial system creates no ambiguity. The mechanism is the augmentation $\mathbb{Z}[\pi] \to \mathbb{Z}$, which collapses the deck symmetry and returns the chains of X from the chains of $\widetilde{X}$. Every twisting phenomenon to come is a deviation from this collapse, controlled by how far the monodromy action is from trivial.

Functoriality is the second property. A local system pulls back along a map, and the twisted groups are functorial for the pullback.

Proposition 4.6.9: Functoriality of Local Coefficients

Let $f\colon X \to Y$ be a cellular map of connected CW complexes inducing $f_*\colon \pi_1(X) \to \pi_1(Y)$, and let $\mathcal{N}$ be a local system on Y given by a $\mathbb{Z}[\pi_1(Y)]$-module N. The *pullback system* $f^*\mathcal{N}$ is the $\mathbb{Z}[\pi_1(X)]$-module N with $\pi_1(X)$ acting through f_*, that is $\sigma \cdot n = f_*(\sigma) \cdot n$. Then f induces natural homomorphisms

$$f_*\colon H_*(X; f^*\mathcal{N}) \to H_*(Y;\mathcal{N}), \qquad f^*\colon H^*(Y;\mathcal{N}) \to H^*(X; f^*\mathcal{N})$$

compatible with composition: $(g \circ f)_* = g_* \circ f_*$ and $(g \circ f)^* = f^* \circ g^*$, and reducing to the ordinary induced maps when $\mathcal{N}$ is trivial.

***Proof*:**
The map f lifts to a f_*-equivariant map of universal covers $\widetilde{f}\colon \widetilde{X} \to \widetilde{Y}$, unique once basepoint lifts are fixed, by the lifting criterion of chapter 2; equivariance means $\widetilde{f}(\sigma \cdot \widetilde{x}) = f_*(\sigma) \cdot \widetilde{f}(\widetilde{x})$ for

$\sigma \in \pi_1(X)$. Since f is cellular, $\widetilde{f}$ is cellular, and it induces a chain map

$$\widetilde{f}_\#\colon C_*^{CW}(\widetilde{X}) \to C_*^{CW}(\widetilde{Y})$$

which is f_*-equivariant: $\widetilde{f}_\#(\sigma \cdot c) = f_*(\sigma) \cdot \widetilde{f}_\#(c)$. Regard $C_*^{CW}(\widetilde{Y})$ as a $\mathbb{Z}[\pi_1(X)]$-complex by restriction along f_*; then $\widetilde{f}_\#$ is $\mathbb{Z}[\pi_1(X)]$-linear. Tensoring with $N = f^*\mathcal{N}$ over $\mathbb{Z}[\pi_1(X)]$ and using that restriction-along-f_* followed by $\otimes_{\mathbb{Z}[\pi_1(X)]}N$ agrees with $\otimes_{\mathbb{Z}[\pi_1(Y)]}N$ on $C_*^{CW}(\widetilde{Y})$, the chain map descends to

$$C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi_1(X)]} N \to C_*^{CW}(\widetilde{Y}) \otimes_{\mathbb{Z}[\pi_1(Y)]} N$$

whose induced map on homology is f_*. Dually, precomposition with $\widetilde{f}_\#$ sends a $\mathbb{Z}[\pi_1(Y)]$-linear map $C_*^{CW}(\widetilde{Y}) \to N$ to the $\mathbb{Z}[\pi_1(X)]$-linear map $C_*^{CW}(\widetilde{X}) \to N$ obtained by restriction along f_*, giving the cochain map and the induced f^*. Composition compatibility follows because lifts compose, $\widetilde{g \circ f} = \widetilde{g} \circ \widetilde{f}$ for compatible basepoint choices, so the chain maps compose; and when $\mathcal{N}$ is trivial the equivariance is vacuous and the construction is the ordinary induced map under proposition 4.6.8, completing the proof.

Functoriality makes the twisted theory a genuine invariant: a map of spaces with a local system on the target induces maps on the twisted groups in the expected variance, homology covariantly and cohomology contravariantly, and the pullback of coefficients tracks how the fundamental groups are related. The one subtlety absent from the untwisted case is that the coefficient system on the source is forced to be the pullback $f^*\mathcal{N}$; one cannot compare twisted groups for unrelated systems, just as one cannot compare ordinary groups for unrelated coefficient groups without a map between them. With the formal properties in place, the smallest nontrivial example can be computed in full.

Example 4.6.10: Twisted Homology of the Circle

Take $X = S^1$, with $\pi = \pi_1(S^1) \cong \mathbb{Z}$ generated by the loop t, and universal cover $q\colon \mathbb{R} \to S^1$ with deck group $\mathbb{Z}$ acting by integer translations (example 2.3.9). Give S^1 the CW structure with one 0-cell and one 1-cell; the lifted structure on $\mathbb{R}$ has 0-cells at the integers and 1-cells between consecutive integers. By proposition 4.6.6 the cellular chain complex of $\mathbb{R}$ over $\mathbb{Z}[\pi] = \mathbb{Z}[t, t^{-1}]$ is free of rank one in each of degrees 0 and 1,

$$C_*^{CW}(\mathbb{R})\colon \quad 0 \to \mathbb{Z}[t, t^{-1}] \xrightarrow{\cdot(t-1)} \mathbb{Z}[t, t^{-1}] \to 0$$

with the boundary of the chosen 1-cell $\widetilde{e}_1$ equal to $t \cdot \widetilde{e}_0 - \widetilde{e}_0 = (t-1)\widetilde{e}_0$, the difference of its two endpoints, which are the integer 0-cell and its translate by the generator. This is the standard free resolution of $\mathbb{Z}$ over the group ring of $\mathbb{Z}$.

The twisted system. Let $M = \mathbb{Z}$ with $\pi = \mathbb{Z}$ acting through the generator t by multiplication by -1, so that $t \cdot m = -m$. This is the sign representation of $\mathbb{Z}$, the orientation system of example 4.6.4 for the Möbius band realized over its core circle. Tensoring the resolution with M over $\mathbb{Z}[\pi]$ replaces each free rank-one module by $M = \mathbb{Z}$ and the boundary $\cdot(t-1)$ by the action of $t-1$ on M, which is $t \cdot m - m = -m - m = -2m$. Thus

$$C_*^{CW}(\mathbb{R}) \otimes_{\mathbb{Z}[\pi]} \mathbb{Z}\colon \quad 0 \to \mathbb{Z} \xrightarrow{\cdot(-2)} \mathbb{Z} \to 0$$

with the $\mathbb{Z}$ in degrees 1 and 0. The homology with local coefficients is

$$H_0(S^1; \mathcal{M}) = \mathbb{Z}/2\mathbb{Z}, \qquad H_1(S^1; \mathcal{M}) = \ker(\cdot(-2)) = 0$$

since multiplication by -2 on $\mathbb{Z}$ has cokernel $\mathbb{Z}/2\mathbb{Z}$ and trivial kernel. For cohomology, $\mathrm{Hom}_{\mathbb{Z}[\pi]}(C_*^{CW}(\mathbb{R}), \mathbb{Z})$ is again $\mathbb{Z}$ in degrees 0 and 1, with coboundary the dual of $\cdot(t-1)$, which is once more multiplication by -2, so

$$H^0(S^1; \mathcal{M}) = \ker(\cdot(-2)) = 0, \qquad H^1(S^1; \mathcal{M}) = \mathbb{Z}/2\mathbb{Z}$$

the degrees of the nonzero group swapped relative to homology, as the universal coefficient pattern of theorem 4.1.8 predicts.

Contrast with the untwisted circle. For the trivial system $M = \mathbb{Z}$ with t acting as $+1$, the boundary $\cdot(t-1)$ acts on M as $1 - 1 = 0$, so the complex is $0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$ and

$$H_0(S^1; \mathbb{Z}) = \mathbb{Z}, \qquad H_1(S^1; \mathbb{Z}) = \mathbb{Z}$$

recovering the ordinary homology of the circle from chapter 3, in agreement with proposition 4.6.8. The twist has collapsed the free $H_0 = \mathbb{Z}$ to the torsion group $\mathbb{Z}/2\mathbb{Z}$ and killed H_1 entirely: dragging a coefficient once around the circle multiplies it by -1, so an invariant 0-cycle must satisfy $m = -m$, forcing $2m = 0$ on classes and $m = 0$ on cycles. The single sign flip is the whole difference between the cylinder ($H_* = H_*(S^1; \mathbb{Z})$ of the trivial bundle) and the Möbius band, whose twisted H_0 records that the band has no consistent transverse orientation.

The example exhibits the mechanism in its purest form. A free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\pi]$, here the length-one resolution coming from the contractible cover $\mathbb{R}$, is the engine; tensoring with the coefficient module evaluates the boundary on the monodromy, and a single nontrivial sign converts the homology of the circle into the group cohomology of $\mathbb{Z}$ with sign coefficients. The computation is identical in form to a group-cohomology computation, and indeed for an aspherical space $X = K(\pi, 1)$ the twisted (co)homology of definition 4.6.7 is exactly the group (co)homology of π with coefficients in M, since the universal cover is contractible and $C_*^{CW}(\widetilde{X})$ is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\pi]$; the circle is the case $\pi = \mathbb{Z}$, treated in the companion development of group cohomology in [8]. For a general X the cover need not be contractible, so $C_*^{CW}(\widetilde{X})$ is no longer a resolution of $\mathbb{Z}$; what carries over is the same algebraic pattern with the resolution replaced by the free $\mathbb{Z}[\pi]$-complex of the cover tensored against the representation, the hyper-derived reading recorded in box 4.6.12.

Two structural remarks place the construction in the wider theory and close the loop with the deferrals that opened the section.

4.6.11 Remark: Local Coefficients as Twisted Coefficients in the Two Deferred Theorems The twisted Serre spectral sequence of box 6.2.8 has E^2 page $H_p(B; \mathcal{H}_q(F))$, the homology of the base B with coefficients in the fiber-homology system $\mathcal{H}_q(F)$ of example 4.6.5. definition 4.6.7 is exactly the definition of this group: $H_p(B; \mathcal{H}_q(F)) = H_p\big(C_*^{CW}(\widetilde{B}) \otimes_{\mathbb{Z}[\pi_1(B)]} H_q(F)\big)$, the cellular chains of the universal cover of B tensored against the monodromy representation on $H_q(F)$. When $\pi_1(B)$ acts trivially, proposition 4.6.8 returns the ordinary E^2 page $H_p(B; H_q(F))$ of theorem 6.2.4. Likewise the obstruction theory of theorem 5.6.10 and box 5.6.13, applied to a fibration with non-simple fiber F, has its obstruction cochain valued in the local system $\pi_n(F)$ on X, and its obstruction class lives in $H^{n+1}(X; \pi_n(F))$ in the sense of definition 4.6.7. The simplicity hypothesis of definition 5.1.13 is precisely what trivializes this system and returns the untwisted obstruction group used throughout chapter 5. Both deferrals are now coefficient settings of one construction.

4.6.12 Remark: The Derived Interpretation and Cover-Independence The complex $C_*^{CW}(\widetilde{X})$ is a complex of free, hence projective, $\mathbb{Z}[\pi]$-modules. Its $\mathbb{Z}[\pi]$-module structure makes precise the sense in which the twisted theory is a derived functor, but the form of that statement depends on whether X is aspherical.
When X is aspherical, that is $X = K(\pi, 1)$, the universal cover $\widetilde{X}$ is contractible, so $C_*^{CW}(\widetilde{X}) \to \mathbb{Z}$ is a free resolution of the trivial $\mathbb{Z}[\pi]$-module $\mathbb{Z}$. Tensoring or homming this resolution against M computes the classical derived functors of a *module*, and the displayed identity has $\mathbb{Z}$ (not the complex) in the resolved slot, exactly as in the computation of $H_*(BG; M)$ at proposition 2.4.10,

$$H_*(X; \mathcal{M}) = \mathrm{Tor}_*^{\mathbb{Z}[\pi]}(\mathbb{Z}, M) = H_*(\pi; M), \qquad H^*(X; \mathcal{M}) = \mathrm{Ext}^*_{\mathbb{Z}[\pi]}(\mathbb{Z}, M) = H^*(\pi; M)$$

the group homology and cohomology of π with coefficients in M from [8]. The circle of example 4.6.10 is the case $\pi = \mathbb{Z}$, where the length-one resolution is the one displayed there.
For a general X the complex $C_*^{CW}(\widetilde{X})$ is *not* a resolution of $\mathbb{Z}$: its homology is $H_*(\widetilde{X})$, which is nonzero in positive degrees as soon as $\widetilde{X}$ is not contractible, so $\mathbb{Z}$ does not sit in the resolved slot and the classical $\mathrm{Tor}_*^{\mathbb{Z}[\pi]}(\mathbb{Z}, M)$ no longer computes the twisted theory. What survives is the hyper-derived functor of the *complex*: writing $\otimes^{\mathbf{L}}_{\mathbb{Z}[\pi]}$ for the derived tensor product and $\mathbf{R}\mathrm{Hom}_{\mathbb{Z}[\pi]}$ for the derived Hom of [10], the definitions of definition 4.6.7 read

$$H_*(X; \mathcal{M}) = H_*\big(C_*^{CW}(\widetilde{X}) \otimes^{\mathbf{L}}_{\mathbb{Z}[\pi]} M\big), \qquad H^*(X; \mathcal{M}) = H^*\big(\mathbf{R}\mathrm{Hom}_{\mathbb{Z}[\pi]}(C_*^{CW}(\widetilde{X}), M)\big)$$

the underived $\otimes_{\mathbb{Z}[\pi]}$ and $\mathrm{Hom}_{\mathbb{Z}[\pi]}$ of the definition agreeing with their derived counterparts here precisely because $C_*^{CW}(\widetilde{X})$ is already a complex of *free* modules in every degree, so no further flatness or projectivity hypothesis on M is needed. These hyper-Tor and hyper-Ext groups reduce to the classical $\mathrm{Tor}_*^{\mathbb{Z}[\pi]}(\mathbb{Z}, M)$ and $\mathrm{Ext}^*_{\mathbb{Z}[\pi]}(\mathbb{Z}, M)$ exactly when $C_*^{CW}(\widetilde{X})$ is quasi-isomorphic to $\mathbb{Z}$, that is in the aspherical case above.
That $H_*(X; \mathcal{M})$ and $H^*(X; \mathcal{M})$ depend only on X and $\mathcal{M}$, and not on the CW structure or the chosen lifts of cells, is the statement that any two free $\mathbb{Z}[\pi]$-complexes arising from cell structures on X are chain homotopy equivalent over $\mathbb{Z}[\pi]$ (both compute the same hyper-derived functor); the comparison is the quasi-isomorphism invariance of the derived tensor product and derived Hom from [10], carried out for the singular model in [14] §3.H. The construction and every property used in this book are established above; the cover-independence bookkeeping is the one external citation.

Exercise 4.6.1

1. Let $X = S^1$ and $\mathcal{M}$ the local system on which the generator $t \in \pi_1(S^1) \cong \mathbb{Z}$ acts on $M = \mathbb{Z}$ by multiplication by a fixed integer k. Compute $H_0(S^1; \mathcal{M})$ and $H_1(S^1; \mathcal{M})$ as a function of k, and identify the values $k = 1$ and $k = -1$ with chapter 3 and example 4.6.10.

2. Show that for any connected CW complex X and any local system $\mathcal{M}$ with module M, the degree-zero homology is the group of coinvariants $H_0(X; \mathcal{M}) \cong M_\pi = M/\langle \sigma m - m : \sigma \in \pi, m \in M\rangle$, and the degree-zero cohomology is the group of invariants $H^0(X; \mathcal{M}) \cong M^\pi = \{m \in M : \sigma m = m \text{ for all } \sigma \in \pi\}$.

3. Let $\mathcal{M}$ be a local system on X and let $q\colon \widetilde{X} \to X$ be the universal cover, with the trivial system $q^*\mathcal{M}$ on $\widetilde{X}$ obtained by pulling back along q (so $\pi_1(\widetilde{X}) = 0$ acts trivially). Identify

$H_*(\widetilde{X}; q^*\mathcal{M})$ with the ordinary homology $H_*(\widetilde{X}; M)$, and explain why the local coefficients on X become ordinary coefficients upstairs.

4. Prove that if $\pi = \pi_1(X)$ is finite of order n and M is a $\mathbb{Z}[\pi]$-module that is a $\mathbb{Q}$-vector space (so n is invertible), then $H_k(X; \mathcal{M}) \cong H_k(\widetilde{X}; M)_\pi$ and the twisted homology agrees with the π-coinvariants of the ordinary homology of the cover. (Hint: averaging over π splits the relevant sequences; use that $\mathbb{Q}[\pi]$ is semisimple.)

5. The Klein bottle K is the total space of an S^1-bundle over S^1 with monodromy acting on $H_1(S^1) = \mathbb{Z}$ by -1. Using example 4.6.10 as the E^2 input, identify the row $H_p(S^1; \mathcal{H}_1(F))$ of the twisted Serre spectral sequence box 6.2.8 for this bundle, and compare $H_1(K; \mathbb{Z})$ computed this way with the known value $H_1(K; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

6. Let $\mathcal{M}$ be the orientation system $\mathcal{O}_\mathbb{Z}$ of example 4.6.4 on a closed connected non-orientable n-manifold M. Show that $H_n(M; \mathcal{M}) \cong \mathbb{Z}$ even though the ordinary top homology $H_n(M; \mathbb{Z}) = 0$, so the twisted top homology recovers a fundamental class with coefficients in the orientation system. (This is the twisted Poincaré duality that repairs duality for non-orientable manifolds.)

5

Homotopy Theory

The fundamental group of chapter 2 measures failure of contractions of loops. Generalising, mapping the n-sphere into a space and remembering the base produces the higher homotopy groups $\pi_n(X, x_0)$, the central objects of this chapter. These groups are at once more powerful and more complicated than π_1:

1. More powerful as they detect phenomena invisible to homology and to the fundamental group; the failure of a high-dimensional sphere to retract onto a lower one, the linking of spheres inside spheres, the obstruction to extending a map across a cell
2. More complicated as they are notoriously hard to compute, even for the spheres themselves, where as of writing this book the answer still is unknown in general and contains the entire theory of stable homotopy.[1]

This chapter builds the machinery of higher homotopy theory. It start by defining π_n and the induced long exact sequence of a pair, the tool that, together with fibrations in the next section, converts geometric decompositions into algebra. It then develops the comparison between homotopy and homology through the *Hurewicz theorem* which says that the first nonvanishing homotopy group agrees with the first nonvanishing homology group. The chapter culminates in the Eilenberg-MacLane spaces $K(\pi, n)$, spaces constructed to have a *single* nonzero homotopy group, and using these spaces to show [ordinary] cohomology is a *representable homotopy functor*: cohomology classes on X are but homotopy classes of maps from X into a $K(\pi, n)$ (section 5.5). That identification bridges the geometry of spaces to the algebra of cohomology, and it sets the stage for the theory of spectra in chapter 6.

[1] see [14] for the known homotopy groups of the sphere

5.1 Higher Homotopy Groups

Throughout, $I = [0,1]$ denotes the unit interval, I^n the n-cube, and ∂I^n its boundary, the set of points with at least one coordinate equal to 0 or 1. A based space is a pair (X, x_0) with $x_0 \in X$ a chosen basepoint. A based map $f\colon (X, x_0) \to (Y, y_0)$ is a continuous map with $f(x_0) = y_0$, and a based homotopy is a homotopy through such maps.

Definition 5.1.1: Higher Homotopy Groups

Let (X, x_0) be a based space and $n \geq 1$. The *n-th homotopy group* $\pi_n(X, x_0)$ is the set of homotopy classes of maps

$$f\colon (I^n, \partial I^n) \to (X, x_0)$$

that is, continuous maps $f\colon I^n \to X$ sending the entire boundary ∂I^n to x_0, where two such maps are identified when they are homotopic through maps carrying ∂I^n to x_0 at every stage. The group operation is

$$(f+g)(t_1, \dots, t_n) = \begin{cases} f(2t_1, t_2, \dots, t_n) & t_1 \in [0, \frac{1}{2}] \\ g(2t_1 - 1, t_2, \dots, t_n) & t_1 \in [\frac{1}{2}, 1] \end{cases}$$

concatenating in the first coordinate.

Collapsing ∂I^n to a point turns the cube I^n into the quotient $I^n/\partial I^n$, which is homeomorphic to the sphere S^n with the image of ∂I^n as basepoint. A map $(I^n, \partial I^n) \to (X, x_0)$ is therefore the same as a based map $(S^n, *) \to (X, x_0)$, and $\pi_n(X, x_0)$ is equally the set of based-homotopy classes of maps $S^n \to X$. The cube description makes the group law visible (concatenation glues two cubes along a face), while the sphere description makes the geometry visible (an element of π_n is a way of wrapping a sphere through the space). For $n = 1$ this is the fundamental group of definition 2.1.12: a map $(I, \partial I) \to (X, x_0)$ is a loop at x_0, and the operation is concatenation of loops.

The sum $f + g$ is continuous by the pasting lemma ([12, Pasting Lemma]), and $f + g$ that carries ∂I^n to x_0 is immediate, since each branch does.

Example 5.1.2: The Homotopy Groups of Euclidean Space

Take $X = \mathbb{R}^m$ with basepoint the origin 0. Any map $f\colon (I^n, \partial I^n) \to (\mathbb{R}^m, 0)$ is null-homotopic through the straight-line homotopy

$$F(t_1, \dots, t_n, s) = (1 - s)\, f(t_1, \dots, t_n)$$

At $s = 0$ this is f, at $s = 1$ it is the constant map 0, and at every stage the boundary ∂I^n is sent to $(1 - s) \cdot 0 = 0$, so the homotopy is through based maps. Every element of $\pi_n(\mathbb{R}^m, 0)$ therefore equals the class of the constant map, and

$$\pi_n(\mathbb{R}^m, 0) = 0 \qquad \text{for all } n \geq 1$$

The same argument works verbatim for any convex subset of $\mathbb{R}^m$, and more generally for any contractible space: if H contracts X to x_0, composing f with H produces a based null-homotopy. Said geometrically, Euclidean space has no holes in any dimension.

Proposition 5.1.3: π_n is a Group

For every based space (X, x_0) and every $n \geq 1$, the operation $+$ of definition 5.1.1 is well defined on homotopy classes and makes $\pi_n(X, x_0)$ a group. The identity is the class 0 of the constant map c_{x_0}, and the inverse of $[f]$ is the class of $\bar{f}(t_1, t_2, \ldots, t_n) = f(1 - t_1, t_2, \ldots, t_n)$.

***Proof*:**

1. *Well-definedness.* Suppose $f \simeq f'$ and $g \simeq g'$ through based homotopies $F, G\colon I^n \times I \to X$, each carrying $\partial I^n \times I$ to x_0. Define $H\colon I^n \times I \to X$ by concatenating in the first coordinate at each homotopy time s,

$$H(t_1, \ldots, t_n, s) = \begin{cases} F(2t_1, t_2, \ldots, t_n, s) & t_1 \in [0, \frac{1}{2}] \\ G(2t_1 - 1, t_2, \ldots, t_n, s) & t_1 \in [\frac{1}{2}, 1] \end{cases}$$

The two branches agree on $t_1 = \frac{1}{2}$ (both equal x_0 there), so H is continuous by the pasting lemma, it carries $\partial I^n \times I$ to x_0, and $H(\,\cdot\,, 0) = f + g$ while $H(\,\cdot\,, 1) = f' + g'$. Hence $f + g \simeq f' + g'$ and $[f] + [g] := [f + g]$ is well defined.

2. *Associativity.* For $[f], [g], [h]$ the two bracketings $(f + g) + h$ and $f + (g + h)$ differ only by a reparametrization of the first coordinate. Explicitly, $(f + g) + h$ runs f on $[0, \frac{1}{4}]$, g on $[\frac{1}{4}, \frac{1}{2}]$, h on $[\frac{1}{2}, 1]$, while $f + (g + h)$ runs f on $[0, \frac{1}{2}]$, g on $[\frac{1}{2}, \frac{3}{4}]$, h on $[\frac{3}{4}, 1]$. Let $\varphi\colon I \to I$ be the piecewise-linear homeomorphism with $\varphi(\frac{1}{2}) = \frac{1}{4}$ and $\varphi(\frac{3}{4}) = \frac{1}{2}$, fixing 0 and 1. Then $\big((f + g) + h\big) \circ (\varphi \times \mathrm{id}_{I^{n-1}})$ equals $f + (g + h)$, since φ carries the breakpoints $\frac{1}{2}, \frac{3}{4}$ of the right bracketing to the breakpoints $\frac{1}{4}, \frac{1}{2}$ of the left and is affine between them. Writing $\varphi_s = (1 - s)\mathrm{id} + s\,\varphi$ (a homeomorphism of I fixing 0 and 1 for each s), the assignment $s \mapsto \big((f + g) + h\big) \circ (\varphi_s \times \mathrm{id})$ is a based homotopy from $(f + g) + h$ (at $s = 0$) to $f + (g + h)$ (at $s = 1$), the boundary mapping to x_0 throughout because $\varphi_s \times \mathrm{id}$ preserves ∂I^n.

3. *Identity* Let c_{x_0} be the constant map. Then $f + c_{x_0}$ runs f at double speed on $[0, \frac{1}{2}]$ and is constant at x_0 on $[\frac{1}{2}, 1]$. The homotopy

$$J(t_1, \ldots, t_n, s) = \begin{cases} f\left(\frac{2t_1}{1+s}, t_2, \ldots, t_n\right) & t_1 \in [0, \frac{1+s}{2}] \\ x_0 & t_1 \in [\frac{1+s}{2}, 1] \end{cases}$$

deforms $f + c_{x_0}$ (at $s = 0$) to f (at $s = 1$) through based maps, the reparametrization $\frac{2t_1}{1+s}$ stretching the active interval $[0, \frac{1+s}{2}]$ back onto $[0, 1]$. The boundary stays at x_0 throughout. The mirror homotopy gives $c_{x_0} + f \simeq f$, so $0 = [c_{x_0}]$ is a two-sided identity.

4. *Inverses* With $\bar{f}(t_1, \ldots, t_n) = f(1 - t_1, t_2, \ldots, t_n)$, consider $f + \bar{f}$, which runs f forward in the first coordinate and then f backward. The contraction pulls the turning point in from $t_1 = 1$ to $t_1 = 0$ as s increases, holding the cube constant on a growing middle band. Define

$$K(t_1, \ldots, t_n, s) = \begin{cases} f(2t_1, t_2, \ldots, t_n) & t_1 \in [0, \frac{1-s}{2}], \\ f(1 - s, t_2, \ldots, t_n) & t_1 \in [\frac{1-s}{2}, \frac{1+s}{2}], \\ f(2 - 2t_1, t_2, \ldots, t_n) & t_1 \in [\frac{1+s}{2}, 1] \end{cases}$$

At $s = 0$ the middle interval degenerates and $K(\cdot, 0) = f + \bar{f}$; at $s = 1$ the outer intervals degenerate and $K(\cdot, 1)$ is the constant map at $f(0, t_2, \ldots, t_n) = x_0$ (since $(0, t_2, \ldots, t_n) \in \partial I^n$). On the overlaps the branches agree (at $t_1 = \frac{1-s}{2}$ both give $f(1-s, \ldots)$, similarly on the right), so K is continuous, and it carries $\partial I^n \times I$ to x_0 because at $t_1 \in \{0, 1\}$ the first and third branches give f of a boundary point. Hence $f + \bar{f} \simeq c_{x_0}$. Applying this to $\bar{f}$ in place of f (and noting $\bar{\bar{f}} = f$) gives $\bar{f} + f \simeq c_{x_0}$ as well, so $[\bar{f}] = -[f]$, completing the proof.

The proof is the one-variable theory of chapter 2 carried out in the first coordinate, with the remaining coordinates carried along unchanged. The reparametrizations of associativity and the identity are exactly the loop reparametrizations from π_1; the inverse is the path run backwards in the first coordinate. Nothing in this argument used $n \geq 2$, which is why π_n is a group for every $n \geq 1$ and reduces to the fundamental group when $n = 1$.

The key way $\pi_n(X)$ differs for for $n \geq 2$ that these groups are *abelian*, requiring the second coordinate to prove. For $n \geq 2$ one can concatenate in any of them. Concatenation in the second coordinate:

$$(f +_2 g)(t_1, t_2, t_3, \ldots, t_n) = \begin{cases} f(t_1, 2t_2, t_3, \ldots, t_n) & t_2 \in [0, \frac{1}{2}], \\ g(t_1, 2t_2 - 1, t_3, \ldots, t_n) & t_2 \in [\frac{1}{2}, 1] \end{cases}$$

By the same proof as above applied to the second slot, another group operation on the same set $\pi_n(X, x_0)$, with the same identity 0. The two operations $+$ and $+_2$ are not a priori related, but they satisfy a single compatibility, and that compatibility forces both to coincide and to be commutative:

Theorem 5.1.4: Higher Homotopy Groups are Abelian

For every based space (X, x_0) and every $n \geq 2$, the group $\pi_n(X, x_0)$ is abelian.

This proof is sometimes called the *Eckmann-Hilton argument.*

***Proof*:**
Write $+_1 := +$ for concatenation in the first coordinate and $+_2$ for concatenation in the second; by proposition 5.1.3 applied to the respective slot both are group operations on $\pi_n(X, x_0)$ with the same identity element $0 = [c_{x_0}]$.

1. *The interchange law.* Fix four classes represented by maps $a, b, c, d \colon (I^n, \partial I^n) \to (X, x_0)$. Consider the map $w \colon I^n \to X$ that subdivides the unit square in the (t_1, t_2)-coordinates into four quadrants and places a in the lower-left, b in the lower-right, c in the upper-left, d in the upper-right, each precomposed with the linear rescaling of its quadrant back to I^n in the first two coordinates and the identity in the remaining coordinates. Concretely, on the lower-left quadrant $[0, \frac{1}{2}] \times [0, \frac{1}{2}]$ the map is $(t_1, t_2, \ldots) \mapsto a(2t_1, 2t_2, t_3, \ldots, t_n)$, and similarly for the others. The boundary between quadrants is sent to x_0 (each map sends its quadrant boundary to x_0), so w is continuous and represents an element of π_n.

   Reading w as "first combine left-right, then top-bottom" computes it as $(a +_1 b) +_2 (c +_1 d)$: the lower half is $a +_1 b$, the upper half is $c +_1 d$, and the two halves are stacked in the t_2-direction. Reading w as "first combine top-bottom, then left-right" computes it as $(a +_2 c) +_1 (b +_2 d)$: the left half is $a +_2 c$, the right half is $b +_2 d$, stacked in the t_1-direction. The two readings are the same map w, so

$$(a +_1 b) +_2 (c +_1 d) = (a +_2 c) +_1 (b +_2 d)$$

holds in $\pi_n(X, x_0)$.

2. *The two operations agree and are commutative.* Apply the interchange law with $b = c = c_{x_0}$, so that their classes are the common identity 0. Using that 0 is the identity for both $+_1$ and $+_2$, the left side becomes $(a +_1 0) +_2 (0 +_1 d) = a +_2 d$, and the right side becomes $(a +_2 0) +_1 (0 +_2 d) = a +_1 d$. Hence

$$a +_2 d = a +_1 d \qquad \text{for all } [a], [d]$$

so the two operations coincide; write $+$ for the common operation. Now apply the interchange law again, this time with $a = d = c_{x_0}$. The left side becomes $(0 +_1 b) +_2 (c +_1 0) = b +_2 c = b + c$, and the right side becomes $(0 +_2 c) +_1 (b +_2 0) = c +_1 b = c + b$. Hence

$$b + c = c + b$$

for all classes $[b], [c]$, so $\pi_n(X, x_0)$ is abelian, as we sought to show.

The above proof is purely algebraic: any set carrying two unital binary operations that satisfy the interchange law has both operations equal, associative, and commutative. The geometry enters only in producing the second operation and the interchange law, and for π_n with $n \geq 2$ both come for free from the two independent cube directions. This is the structural reason higher homotopy groups are abelian while π_1 need not be: a one-dimensional cube has a single direction, so there is no second operation to interchange with. The same argument explains why the fundamental group of a topological group is abelian (multiplication of loops provides a second operation), and it recurs throughout topology wherever two compatible compositions meet.

Example 5.1.5: π_2 of the 2-Sphere is Nontrivial

The identity map $\mathrm{id}: S^2 \to S^2$, viewed as a based map $(S^2, *) \to (S^2, *)$, represents a class in $\pi_2(S^2, *)$. This class is nonzero: were id based null-homotopic, S^2 would be contractible, contradicting $H_2(S^2) \cong \mathbb{Z} \neq 0$ from chapter 3, since a contractible space has vanishing reduced homology. In fact $\pi_2(S^2, *) \cong \mathbb{Z}$, generated by [id], with an element's integer being its degree as a self-map of S^2 in the sense of section 3.5. The group is abelian, infinite cyclic, and the addition is concatenation of cube maps, the integer recording how many times (with sign) the sphere is wrapped. The full computation that $\pi_2(S^2) \cong \mathbb{Z}$ follows from the Hurewicz theorem, theorem 5.4.9.

Homotopy groups would be of little use as invariants if continuous maps did not act on them: a based map $\varphi: (X, x_0) \to (Y, y_0)$ sends an n-cube $f: (I^n, \partial I^n) \to (X, x_0)$ to the composite $\varphi \circ f: (I^n, \partial I^n) \to (Y, y_0)$, and this respects homotopy and the group law. The resulting assignment is functorial, turning a homotopy equivalence into an isomorphism of homotopy groups, making π_n a topological invariance:

Proposition 5.1.6: Functoriality of π_n

Let $n \geq 1$.

1. A based map $\varphi\colon (X, x_0) \to (Y, y_0)$ induces a group homomorphism
$$\varphi_*\colon \pi_n(X, x_0) \to \pi_n(Y, y_0), \qquad \varphi_*[f] = [\varphi \circ f]$$
2. Functoriality holds: $(\mathrm{id}_X)_* = \mathrm{id}$ and $(\psi \circ \varphi)_* = \psi_* \circ \varphi_*$ for composable based maps.
3. If $\varphi \simeq \varphi'$ through based maps, then $\varphi_* = \varphi'_*$. Consequently a based homotopy equivalence induces an isomorphism on every π_n, and π_n is an invariant of based homotopy type.

***Proof*:**

1. If $f \simeq f'$ through a based homotopy F, then $\varphi \circ F$ is a based homotopy $\varphi \circ f \simeq \varphi \circ f'$, so φ_* is well defined on classes. For the group law, $\varphi \circ (f + g) = (\varphi \circ f) + (\varphi \circ g)$ as maps, because φ acts pointwise and concatenation in the first coordinate commutes with postcomposition: on $t_1 \in [0, \frac{1}{2}]$ both sides equal $\varphi(f(2t_1, \dots))$, and on $t_1 \in [\frac{1}{2}, 1]$ both equal $\varphi(g(2t_1 - 1, \dots))$. Hence $\varphi_*([f] + [g]) = \varphi_*[f] + \varphi_*[g]$.
2. For $\varphi = \mathrm{id}_X$, $\varphi \circ f = f$, so $(\mathrm{id}_X)_* = \mathrm{id}$. For composites, $(\psi \circ \varphi) \circ f = \psi \circ (\varphi \circ f)$, so $(\psi \circ \varphi)_*[f] = [\psi \circ \varphi \circ f] = \psi_*[\varphi \circ f] = (\psi_* \circ \varphi_*)[f]$.
3. Suppose $G\colon X \times I \to Y$ is a based homotopy from φ to φ', so $G(x_0, s) = y_0$ for all s. For a representative $f\colon (I^n, \partial I^n) \to (X, x_0)$, the map
$$(t, s) \mapsto G(f(t), s)\colon I^n \times I \to Y$$
is a homotopy from $\varphi \circ f$ to $\varphi' \circ f$, and it is based: when $t \in \partial I^n$ one has $f(t) = x_0$, hence $G(f(t), s) = G(x_0, s) = y_0$ for all s. Therefore $[\varphi \circ f] = [\varphi' \circ f]$, that is $\varphi_* = \varphi'_*$. If φ is a based homotopy equivalence with based homotopy inverse ψ, then $\psi_*\varphi_* = (\psi\varphi)_* = (\mathrm{id}_X)_* = \mathrm{id}$ and likewise $\varphi_*\psi_* = \mathrm{id}$ by parts (2) and (3), so φ_* is an isomorphism, completing the proof.

Thus, to compute $\pi_n(X)$ one may replace X by any homotopy-equivalent space. It also delivers the standard nonexistence results in the same form as for π_1 (ex. if S^n retracted onto a contractible subspace the retraction would split a nonzero homotopy group through zero, an impossibility).

Example 5.1.7: Homotopy Groups of a Product

A projection-based computation shows that π_n converts products into products. Let (X, x_0) and (Y, y_0) be based spaces, give $X \times Y$ the basepoint (x_0, y_0), and let $p\colon X \times Y \to X$ and $q\colon X \times Y \to Y$ be the projections. The claim is
$$\pi_n(X \times Y, (x_0, y_0)) \;\cong\; \pi_n(X, x_0) \times \pi_n(Y, y_0)$$
for every $n \geq 1$. Define $\Phi = (p_*, q_*)$, sending $[h]$ to $([p \circ h], [q \circ h])$; this is a homomorphism by proposition 5.1.6. A continuous map into a product is exactly a pair of continuous maps into the factors, so $h \mapsto (p \circ h, q \circ h)$ is a bijection from maps $I^n \to X \times Y$ to pairs of maps

$(I^n \to X, I^n \to Y)$, and it carries $\partial I^n \to (x_0, y_0)$ to the pair of conditions $\partial I^n \to x_0$ and $\partial I^n \to y_0$. The same bijection applied to homotopies (a homotopy into $X \times Y$ is a pair of homotopies) shows Φ is injective: if $p \circ h \simeq p \circ h'$ and $q \circ h \simeq q \circ h'$ through based homotopies F, G, then $(F, G)\colon I^n \times I \to X \times Y$ is a based homotopy $h \simeq h'$. For surjectivity, given $[f] \in \pi_n(X)$ and $[g] \in \pi_n(Y)$, the map $h = (f, g)\colon I^n \to X \times Y$, $h(t) = (f(t), g(t))$, sends ∂I^n to (x_0, y_0) and satisfies $p \circ h = f$, $q \circ h = g$, so $\Phi[h] = ([f], [g])$. Thus Φ is an isomorphism. For instance, the n-torus $T^n = (S^1)^n$ has $\pi_1(T^n) \cong \mathbb{Z}^n$ and $\pi_k(T^n) = 0$ for $k \geq 2$, the latter because $\pi_k(S^1) = 0$ for $k \geq 2$ (the universal cover $\mathbb{R} \to S^1$ identifies $\pi_k(S^1)$ with $\pi_k(\mathbb{R}) = 0$ once covering-space theory from chapter 2 is in hand).

The product formula is the higher-dimensional version that π_1 of a product is the product of the fundamental groups, and its proof is the universal property of the product respecting homotopy (i.e. working in the homotopy category). It is the first useful computation tool: combined with the homotopy groups of spheres it determines π_n for tori, products of spheres, and any finite product whose factors are understood.

So far every group has been computed from contractibility or from products. Already the homotopy groups of spheres, even the lowest degrees, are not elementary. Two facts will be proved in section 5.4 as consequences of the Hurewicz theorem and are recorded here so that they can be used for examples. First, $\pi_k(S^n) = 0$ for $k < n$: a sphere of dimension n has no nontrivial maps from lower-dimensional spheres up to homotopy, reflecting that S^n is $(n-1)$-connected. Second, $\pi_n(S^n) \cong \mathbb{Z}$, generated by the identity, with the isomorphism given by the degree. The off-diagonal groups $\pi_k(S^n)$ with $k > n$ are mostly nonzero, often finite, and their determination is a major open problem (as of writing this book); the first such, $\pi_3(S^2) \cong \mathbb{Z}$ generated by the Hopf map, appears in section 5.2.

The absolute groups $\pi_n(X, x_0)$ measure a single space. To compare a space with a subspace, and to set up the inductive computations that follow, one needs a relative version similar to what was worked out from theorem 3.4.7. The relative homotopy group $\pi_n(X, A, x_0)$ measures maps of a cube that send most of the boundary to the basepoint but are allowed to land anywhere in A along one distinguished face. The precise device is to single out, inside ∂I^n, the face I^{n-1} where the last coordinate vanishes, and to require that the rest of the boundary go to x_0 while I^{n-1} maps into A.

Write $I^n = I^{n-1} \times I$ with last coordinate t_n. Let

$$I^{n-1} = \{t \in \partial I^n : t_n = 0\}$$

be the bottom face, and let $J^{n-1} = \overline{\partial I^n \setminus I^{n-1}}$ be the closure of the rest of the boundary, the union of the top face $t_n = 1$ with the $2(n-1)$ side faces where some coordinate t_i $(i < n)$ is 0 or 1. Then $\partial I^n = I^{n-1} \cup J^{n-1}$ and $I^{n-1} \cap J^{n-1} = \partial I^{n-1}$.

Definition 5.1.8: Relative Homotopy Groups

Let $A \subseteq X$ be a subspace containing the basepoint x_0, and let $n \geq 1$. The *n-th relative homotopy group* $\pi_n(X, A, x_0)$ is the set of homotopy classes of maps

$$f\colon (I^n,\, \partial I^n,\, J^{n-1}) \to (X,\, A,\, x_0)$$

that is, maps $f\colon I^n \to X$ with $f(I^{n-1}) \subseteq A$ and $f(J^{n-1}) = x_0$, where the homotopies are through maps of the same form. For $n \geq 2$ the operation is concatenation in the first coordinate, exactly as in definition 5.1.1, and $\pi_n(X, A, x_0)$ is a group, abelian for $n \geq 3$. For $n = 1$ the set $\pi_1(X, A, x_0)$ is pointed but not in general a group.

Unwinding the conditions: f maps the bottom face I^{n-1} into the subspace A, sends the entire rest of the boundary to the basepoint, and is otherwise unconstrained inside X. When $A = \{x_0\}$ the condition $f(I^{n-1}) \subseteq A$ forces all of ∂I^n to x_0, and the relative group collapses to the absolute group $\pi_n(X, x_0)$. The relative group thus interpolates between X and the pair (X, A), recording maps of a disk into X whose boundary sphere lands in A. The operation concatenates in the first coordinate t_1, which is one of the directions *not* singled out as t_n; this is why concatenation is available for $n \geq 2$ (two free directions: t_1 exists and is distinct from t_n) but $\pi_1(X, A, x_0)$ carries no natural group structure, since for $n = 1$ the first and last coordinates coincide. The shift in the commutativity threshold, abelian for $n \geq 3$ rather than $n \geq 2$, comes from the same count: one coordinate is reserved for the A-face, leaving two free directions only when $n \geq 3$, so the Eckmann-Hilton argument applies one dimension later.

The geometric content is cleaner in the sphere-and-disk picture. Collapsing J^{n-1} to a point turns (I^n, I^{n-1}) into (D^n, S^{n-1}) with a basepoint $* \in S^{n-1}$, so an element of $\pi_n(X, A, x_0)$ is a based-homotopy class of maps $(D^n, S^{n-1}, *) \to (X, A, x_0)$: a disk in X whose boundary circle (more precisely boundary sphere S^{n-1}) is constrained to lie in A, the basepoint of that sphere pinned at x_0.

Example 5.1.9: Relative Homotopy of a Pair with $A = X$

When $A = X$ the relative group is trivial in every degree: $\pi_n(X, X, x_0) = 0$ for all $n \geq 1$. Given a relative cube $f\colon (I^n, \partial I^n, J^{n-1}) \to (X, X, x_0)$, set

$$F(t, s) = f\big(r(t, s)\big)$$

Then $F(\,\cdot\,, 0) = f$, and at every stage F is a relative cube: on J^{n-1}, r fixes the point, so $F(J^{n-1} \times \{s\}) = f(J^{n-1}) = x_0$; on the bottom face I^{n-1} there is no constraint to violate, since $A = X$. At $s = 1$, $r(I^n, 1) \subseteq J^{n-1}$ forces $F(\,\cdot\,, 1) = f|_{J^{n-1}} \circ r(\,\cdot\,, 1) = x_0$ everywhere, the constant map. Hence $[f] = 0$, and since f was arbitrary, $\pi_n(X, X, x_0) = 0$.

The relative group comes equipped with a boundary homomorphism that restricts a relative cube to its A-face. Given $f\colon (I^n, \partial I^n, J^{n-1}) \to (X, A, x_0)$, its restriction to the bottom face I^{n-1} is a map $I^{n-1} \to A$ sending $\partial I^{n-1} = I^{n-1} \cap J^{n-1}$ to x_0, hence a representative of an element of $\pi_{n-1}(A, x_0)$. This restriction is the connecting map of the pair.

Definition 5.1.10: Boundary Map of a Pair

The *boundary homomorphism*

$$\partial\colon \pi_n(X, A, x_0) \longrightarrow \pi_{n-1}(A, x_0)$$

is defined by $\partial[f] = [f|_{I^{n-1}}]$, the homotopy class of the restriction of f to the bottom face I^{n-1}, viewed as a map $(I^{n-1}, \partial I^{n-1}) \to (A, x_0)$. For $n \geq 2$ it is a group homomorphism.

That ∂ is well defined and a homomorphism is a matter of computation: a homotopy of relative cubes restricts to a homotopy of their bottom faces, and concatenation in the first coordinate of I^n restricts to concatenation in the first coordinate of I^{n-1} (the first coordinate is never the distinguished last one), so $\partial(f + g) = \partial f + \partial g$. The boundary map, the inclusion $A \hookrightarrow X$, and the map $\pi_n(X) \to \pi_n(X, A)$ that views an absolute cube as a relative one assemble into the long exact sequence, the key computational instrument of homotopy theory:

Proposition 5.1.11: Long Exact Sequence of a Pair

Let $A \subseteq X$ with basepoint $x_0 \in A$, let $i\colon A \hookrightarrow X$ be the inclusion and $j\colon (X, x_0, x_0) \hookrightarrow (X, A, x_0)$ the map sending an absolute cube to the same cube regarded as relative. Then the sequence

$$\cdots \to \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial} \pi_{n-1}(A, x_0) \xrightarrow{i_*} \cdots$$

is exact, terminating in $\cdots \to \pi_1(X, A, x_0) \xrightarrow{\partial} \pi_0(A, x_0) \xrightarrow{i_*} \pi_0(X, x_0)$, where the last three terms are pointed sets and exactness means equality of image and preimage of the basepoint.

***Proof*:**
Exactness must be verified at each of the three types of term. Throughout, the identity element (or basepoint) of each group is the class 0 of the constant map, and a class is 0 exactly when its representative is null-homotopic through maps of the prescribed type. The composite of any two consecutive maps is checked to be zero first; then the harder containment, that a class killed by the next map lies in the image of the previous one, is established by an explicit homotopy.

Step 1: Exactness at $\pi_n(X, x_0)$. First $j_*i_* = 0$. If $[g] \in \pi_n(A, x_0)$, then $i_*[g]$ is g regarded as a map into X, and $j_*i_*[g]$ is that same map viewed as a relative cube $(I^n, \partial I^n, J^{n-1}) \to (X, A, x_0)$, which is legitimate because g has image in A (so the bottom face lands in A) and $g(J^{n-1}) = x_0$. This relative cube is null: with $r\colon I^n \times I \to I^n$ the deformation retraction of I^n onto J^{n-1} from example 5.1.9, the homotopy $(t, s) \mapsto g(r(t, s))$ is through relative cubes, since g maps the whole cube into A (so the bottom-face constraint is met at every stage) and r fixes J^{n-1} pointwise (so $g \circ r$ equals x_0 on J^{n-1} throughout); at $s = 1$ it is constant at x_0. Hence $j_*i_*[g] = 0$, and $\operatorname{im} i_* \subseteq \ker j_*$.

Conversely suppose $[f] \in \pi_n(X, x_0)$ has $j_*[f] = 0$. The kernel of j_* is characterized as follows: $j_*[f] = 0$ if and only if f is homotopic, rel ∂I^n, to a map whose image lies in A. Granting this characterization, such a homotopy keeps ∂I^n at x_0 (since $f(\partial I^n) = x_0$), so the end map f' has $f'(\partial I^n) = x_0$ and image in A, hence represents a class $[g] \in \pi_n(A, x_0)$ with $i_*[g] = [f' \text{ in } X] = [f]$, giving $[f] \in \operatorname{im} i_*$ and $\ker j_* \subseteq \operatorname{im} i_*$.

The characterization itself is proved from the relative null-homotopy. In one direction, if f is homotopic rel ∂I^n to a map f' into A, then $j_*[f] = j_*[f']$, and f' is a relative cube with image in A, hence null in $\pi_n(X, A, x_0)$ by the deformation-retraction argument of the forward direction (apply $g \mapsto g \circ r$ with $g = f'$); so $j_*[f] = 0$. In the other direction, let $F\colon I^n \times I \to X$ be a relative null-homotopy from f to the constant map, with $F(I^{n-1} \times \{s\}) \subseteq A$ and $F(J^{n-1} \times \{s\}) = x_0$ for all s. On the boundary of the solid $I^n \times I$, the map F takes values in A on the face $I^{n-1} \times I$, equals x_0 on the top $I^n \times \{1\}$ and on the sides $J^{n-1} \times I$, and equals f on the bottom $I^n \times \{0\}$. The union C of all boundary faces of $I^n \times I$ other than the open bottom is a closed n-disk and a deformation retract of the solid $I^n \times I$, by a retraction ρ_u that pushes radially away from the center of the bottom face and fixes C pointwise. Then $(t, u) \mapsto F(\rho_u(t, 0))$ is a homotopy from f (at $u = 0$, where ρ_0 is the identity on the bottom) to $F|_C$ reparametrized over I^n (at $u = 1$), and it is rel ∂I^n, because $\partial I^n \times \{0\} \subseteq J^{n-1} \times \{0\} \subseteq C$ is fixed by every ρ_u and there $F = x_0$. The end map has image in $F(C) \subseteq A \cup \{x_0\} = A$, since C consists of the A-valued face together with x_0-valued faces. This deforms f rel ∂I^n into A, completing the characterization and the proof that $\ker j_* \subseteq \operatorname{im} i_*$.

Step 2: Exactness at $\pi_n(X, A, x_0)$*.* First $\partial j_* = 0$. For $[f] \in \pi_n(X, x_0)$, the relative cube $j_* f$ has the same underlying map f, whose restriction to the bottom face I^{n-1} is constant at x_0 (because $f(\partial I^n) = x_0$ and $I^{n-1} \subseteq \partial I^n$). Thus $\partial j_*[f] = [c_{x_0}] = 0$, so $\mathrm{im} j_* \subseteq \ker \partial$.

Conversely suppose $[f] \in \pi_n(X, A, x_0)$ has $\partial[f] = 0$, so the restriction $g = f|_{I^{n-1}} \colon (I^{n-1}, \partial I^{n-1}) \to (A, x_0)$ is null-homotopic in A: there is $G \colon I^{n-1} \times I \to A$ with $G(\,\cdot\,, 0) = g$, $G(\,\cdot\,, 1) = x_0$, and $G(\partial I^{n-1} \times I) = x_0$. Use G to homotope f rel J^{n-1} so that its bottom face becomes constant: extend G to a homotopy of f supported in a collar $I^{n-1} \times [0, \epsilon]$ of the bottom face, replacing the values of f on the collar by a copy of G that slides the A-face down to x_0. The result f' has $f'(I^{n-1}) = x_0$, hence $f'(\partial I^n) = x_0$, so f' is an absolute cube and represents a class in $\pi_n(X, x_0)$ with $j_*[f'] = [f' \text{ relative}] = [f]$ (the homotopy was through relative cubes). Thus $[f] \in \mathrm{im} j_*$, giving $\ker \partial \subseteq \mathrm{im} j_*$.

Step 3: Exactness at $\pi_{n-1}(A, x_0)$*.* First $i_* \partial = 0$. For $[f] \in \pi_n(X, A, x_0)$, the class $\partial[f] = [f|_{I^{n-1}}]$ becomes, under i_*, the class of $f|_{I^{n-1}}$ regarded as a map into X. But f itself is a homotopy, rel basepoint conditions, contracting $f|_{I^{n-1}}$ inside X: slide the bottom face up through the cube I^n. Precisely, $H(t_1, \ldots, t_{n-1}, s) = f(t_1, \ldots, t_{n-1}, s)$ defines a homotopy in X from $f|_{I^{n-1}}$ (at $s = 0$) to $f|_{t_n=1}$, and the latter lies in J^{n-1}, hence is constant at x_0. This H keeps ∂I^{n-1} at x_0 for every s (those points lie in J^{n-1} at every height $t_n = s$). Thus $i_* \partial[f] = [f|_{I^{n-1}} \text{ in } X] = 0$, giving $\mathrm{im}\partial \subseteq \ker i_*$.

Conversely suppose $[g] \in \pi_{n-1}(A, x_0)$ has $i_*[g] = 0$ in $\pi_{n-1}(X, x_0)$, witnessed by a null-homotopy $H \colon I^{n-1} \times I \to X$ with $H(\,\cdot\,, 0) = g$, $H(\,\cdot\,, 1) = x_0$, and $H(\partial I^{n-1} \times I) = x_0$. View H as a map on I^n by setting $t_n =$ the homotopy parameter: define $f \colon I^n \to X$ by $f(t_1, \ldots, t_{n-1}, t_n) = H(t_1, \ldots, t_{n-1}, t_n)$. Then $f(I^{n-1}) = H(\,\cdot\,, 0) = g(I^{n-1}) \subseteq A$, f on the top face $t_n = 1$ is x_0, and on the side faces ($t_i \in \{0, 1\}$, $i < n$) is $H(\partial I^{n-1} \times I) = x_0$; together these say $f(J^{n-1}) = x_0$. Hence f is a legitimate relative cube, $[f] \in \pi_n(X, A, x_0)$, and $\partial[f] = [f|_{I^{n-1}}] = [g]$. Thus $[g] \in \mathrm{im}\partial$, giving $\ker i_* \subseteq \mathrm{im}\partial$.

The three steps establish exactness at all three term types and at every degree, and the low-degree tail involving π_0 is the same statement read as an equality of based sets, completing the proof.

Thus, whenever a space X contains a subspace A whose homotopy is understood, the sequence relates $\pi_n(X)$, $\pi_n(A)$, and the relative groups $\pi_n(X, A)$ in a single exact chain, and exactness lets one solve for an unknown group from its neighbors. Three uses recur:

1. When the relative groups vanish in a range, i_* is an isomorphism there, so A and X have the same homotopy in that range.
2. When A is contractible, $\pi_n(A) = 0$ and the sequence forces $\pi_n(X) \cong \pi_n(X, A)$.
3. when the pair (X, A) is highly connected, the sequence measures exactly how far the inclusion $A \hookrightarrow X$ is from inducing isomorphisms. This same sequence reappears in section 5.2 as the long exact sequence of a fibration, where A is replaced by a fiber, and it is the core of the Hurewicz and Whitehead theorems later in the chapter.

One dependence on the basepoint remains to be addressed. In chapter 2 a path γ from x_0 to x_1 produced an isomorphism $\pi_1(X, x_0) \to \pi_1(X, x_1)$ by conjugating loops with γ. The same construction acts on higher homotopy groups, and when γ is a loop (so $x_1 = x_0$) it yields an action of $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$. Given a loop γ at x_0 and a cube $f \colon (I^n, \partial I^n) \to (X, x_0)$, shrink the cube toward its center and use γ to carry the now-free boundary along the loop: there is a map $\gamma \cdot f \colon (I^n, \partial I^n) \to (X, x_0)$ that on a small central subcube agrees with f and on the collar between that subcube and ∂I^n

traverses γ radially outward. The construction is the higher-dimensional analogue of basepoint change: classes involved.

Proposition 5.1.12: The Action of π_1 on π_n

Let (X, x_0) be a based space and $n \geq 1$. The assignment $([\gamma], [f]) \mapsto [\gamma \cdot f]$ above is well defined on homotopy classes and defines a left action of the group $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$ by group automorphisms, that is, $[\gamma] \cdot ([\gamma'] \cdot [f]) = ([\gamma][\gamma']) \cdot [f]$, the constant loop acts trivially, and each $[\gamma]$ acts by a group automorphism of $\pi_n(X, x_0)$. For $n \geq 2$ the group $\pi_n(X, x_0)$ is abelian, so this action makes it a left module over the group ring $\mathbb{Z}[\pi_1(X, x_0)]$. For $n = 1$ the action is conjugation: $[\gamma] \cdot [f] = [\gamma][f][\gamma]^{-1}$.

We shall often abbreviate and just say $\pi_n(X, x_0)$ is a module over $\pi_1(X, x_0)$ instead of $\mathbb{Z}[\pi_1(X, x_0)]$.

***Proof*:**
For well-definedness, a homotopy $\gamma \simeq \gamma'$ relative to endpoints together with a homotopy $f \simeq f'$ relative to ∂I^n combine, applied collarwise, into a homotopy $\gamma \cdot f \simeq \gamma' \cdot f'$ relive to ∂I^n; the construction used only homotopy classes of the radial path and of the central cube. The module identities are basepoint-change identities transported from chapter 2: traversing γ then γ' along the collar is, up to reparametrizing the collar, the same as traversing the concatenation $\gamma * \gamma'$, giving $[\gamma] \cdot ([\gamma'] \cdot [f]) = [\gamma * \gamma'] \cdot [f]$; the constant loop leaves the collar at x_0 and returns f; and $[\gamma] \cdot ([f] + [g]) = [\gamma] \cdot [f] + [\gamma] \cdot [g]$ because the collar construction is performed independently on the two halves of $f + g$ in the first coordinate, each dragged along the same γ. Hence each $[\gamma]$ acts as an automorphism. For $n = 1$, dragging the two endpoints of a loop f outward along γ produces $\gamma * f * \bar{\gamma}$, whose class is $[\gamma][f][\gamma]^{-1}$, the conjugation formula, completing the proof.

The action records how the higher homotopy of a space is twisted by its fundamental group. When the action is nontrivial, $\pi_n(X, x_0)$ is a module over the (possibly nonabelian) group ring $\mathbb{Z}[\pi_1(X, x_0)]$, not just an abelian group, and many statements (the Hurewicz theorem among them) require the action to be trivial to take their cleanest form. Spaces where this complication is absent deserve a name:

Definition 5.1.13: n-Simple and Simple Spaces

A path-connected space X is *n-simple* if the action of $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$ from proposition 5.1.12 is trivial, that is, $[\gamma] \cdot [f] = [f]$ for every loop $[\gamma] \in \pi_1(X, x_0)$ and every $[f] \in \pi_n(X, x_0)$. The space is *simple* (or *abelian*) if it is n-simple for all $n \geq 1$. A space is 1-simple exactly when $\pi_1(X, x_0)$ is abelian, since for $n = 1$ the action is conjugation.

Simplicity is automatic in the most common situations. Any simply connected space is simple, since π_1 is trivial and acts trivially on everything. Any topological group, and more generally any H-space, is simple: the multiplication makes basepoint change inessential, as the abelian conclusion for π_1 already signaled. For a simple space the homotopy groups $\pi_n(X, x_0)$ are independent of the basepoint not just up to isomorphism but canonically, and the comparison maps between homotopy and homology in section 5.4 will hold without correction terms. When the action is nontrivial, those corrections become the source of the local-coefficient systems studied alongside the spectral sequences of chapter 6.

Example 5.1.14: The Action for the Wedge of Circles

Let $X = S^1 \vee S^1$. By section 2.2, $\pi_1(X, x_0) \cong \mathbb{Z} * \mathbb{Z} = \langle a, b \rangle$, the free group on the two circle classes. The action of π_1 on π_1 is conjugation, so $a \cdot b = aba^{-1} \neq b$ in the free group (the two are distinct reduced words). Hence X is not 1-simple. By contrast, the circle S^1 itself is simple: $\pi_1(S^1) \cong \mathbb{Z}$ is abelian, so the conjugation action is trivial, and $\pi_n(S^1) = 0$ for $n \geq 2$ leaves nothing to act on. The torus $S^1 \times S^1$ is likewise simple, being a topological group; its fundamental group $\mathbb{Z}^2$ is abelian and acts trivially on every π_n.

The action also measures the difference between the based and the free homotopy classes of a map out of a sphere. A based map $S^n \to X$ represents a class in $\pi_n(X, x_0)$, while forgetting the basepoint remembers only its *free* homotopy class in $[S^n, X]$, the set of homotopy classes of unbased maps. The two are related exactly by the action of proposition 5.1.12.

Proposition 5.1.15: Free and Based Homotopy Classes of Maps from a Sphere

Let X be path-connected and $n \geq 1$. The map that forgets the basepoint,

$$\pi_n(X, x_0) = [S^n, X]_* \longrightarrow [S^n, X]$$

is surjective, and two based classes have the same image if and only if they lie in one orbit of the action of $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$. It therefore descends to a bijection

$$[S^n, X] \;\cong\; \pi_n(X, x_0)/\pi_1(X, x_0)$$

from the free homotopy classes onto the set of orbits of the action. In particular, if X is n-simple (definition 5.1.13), and so if X is simply connected, the forgetful map is itself a bijection $[S^n, X] \cong \pi_n(X, x_0)$.

***Proof*:**

Surjectivity. Let $f \colon S^n \to X$ be any map and $s_0 \in S^n$ the basepoint. Since X is path-connected, choose a path from $f(s_0)$ to x_0. The inclusion $\{s_0\} \hookrightarrow S^n$ is a cofibration, so this path extends to a homotopy of f ending at a map that sends s_0 to x_0. That end map is based and freely homotopic to f, so every free class has a based representative.

Fibers are orbits. If $[g] = [\gamma] \cdot [f]$ for a loop γ at x_0, the map $\gamma \cdot f$ built in proposition 5.1.12 carries the basepoint of f around γ, and that construction is a homotopy of unbased maps from f to a representative of $[g]$, so f and g are freely homotopic. Conversely, let $H \colon S^n \times I \to X$ be a free homotopy between based maps f and g. Its trace $\gamma(t) = H(s_0, t)$ is a loop at x_0, and restricting H to a collar of the basepoint exhibits g as the map obtained from f by carrying the basepoint along γ, that is $[g] = [\gamma] \cdot [f]$. Two based classes therefore agree freely exactly when one carries to the other under the action.

The bijection onto the orbit set follows. When X is n-simple the action is trivial, so each orbit is a single class and the forgetful map is a bijection. A simply connected space is n-simple because its fundamental group is trivial, completing the proof.

The case $n = 1$ names a classical set. The action of π_1 on itself is conjugation, so its orbits are the conjugacy classes.

Corollary 5.1.16: Free Homotopy Classes of Loops are Conjugacy Classes

Let X be path-connected. The forgetful map induces a bijection

$$[S^1, X] \cong \{\text{conjugacy classes of } \pi_1(X, x_0)\}$$

between the free homotopy classes of loops in X and the conjugacy classes of the fundamental group. Two based loops are freely homotopic if and only if their classes are conjugate in $\pi_1(X, x_0)$.

***Proof*:**
Apply proposition 5.1.15 with $n = 1$: the free homotopy classes are the orbits of the action of $\pi_1(X, x_0)$ on $\pi_1(X, x_0)$. By proposition 5.1.12 that action is conjugation, $[\gamma] \cdot [f] = [\gamma][f][\gamma]^{-1}$, whose orbits are the conjugacy classes, completing the proof.

Exercise 5.1.1

1. Let X be a path-connected topological group with identity e, with multiplication providing a second composition on loops. Using the Eckmann-Hilton argument, prove that $\pi_1(X, e)$ is abelian. Identify precisely which two operations play the roles of $+_1$ and $+_2$ and verify the interchange law.

2. Compute $\pi_n(S^1 \times S^1, *)$ for all $n \geq 1$, citing the product formula and the homotopy groups of S^1. Then compute $\pi_n(S^1 \times S^2, *)$ for $n = 1, 2$, assuming the stated values $\pi_k(S^1) = 0$ for $k \geq 2$, $\pi_1(S^2) = 0$, and $\pi_2(S^2) \cong \mathbb{Z}$.

3. Let $A \subseteq X$ with A a deformation retract of X (so the inclusion is a homotopy equivalence). Use the long exact sequence of the pair (X, A) to prove that $\pi_n(X, A, x_0) = 0$ for all $n \geq 1$. Conversely, if $\pi_n(X, A, x_0) = 0$ for all n and all basepoints, what does the sequence tell you about $i_* \colon \pi_n(A) \to \pi_n(X)$?

4. Prove that a map $f \colon (I^n, \partial I^n) \to (X, x_0)$ represents 0 in $\pi_n(X, x_0)$ if and only if it extends to a continuous map $D^{n+1} \to X$ that is constant at x_0 on a hemisphere, equivalently if and only if the corresponding map $S^n \to X$ extends over the disk D^{n+1}. (Use the sphere description $I^n / \partial I^n \cong S^n$.)

5. Show that for $n \geq 2$ the operation $+_2$ (concatenation in the second coordinate) of theorem 5.1.4 defines the same group structure on $\pi_n(X, x_0)$ as $+_1$, without invoking commutativity. That is, prove $[f] +_1 [g] = [f] +_2 [g]$ directly using the Eckmann-Hilton interchange law with two of the four entries set to the identity.

6. Recall the notion of an n-simple space from definition 5.1.13, where $\pi_1(X, x_0)$ acts trivially on $\pi_n(X, x_0)$. Give an example of a space that is *not* 1-simple, that is, where π_1 acts nontrivially on itself by conjugation, and explain why the condition "π_1 acts trivially on π_1" is the same as "π_1 is abelian."

5.2 Fibrations and the Long Exact Sequence

The higher homotopy groups of section 5.1 are easy to define and notoriously difficult to compute by hand: even $\pi_3(S^2)$ resists every elementary attack, namely since a map $S^3 \to S^2$ cannot be analyzed simplex-by-simplex the way a 1-dimensional loop can. To breaks this impasse, we shall break down computations into a *long exact sequence of a fibration*. A *fibration* $F \to E \to B$ is a map $p\colon E \to B$ flexible enough that homotopies in the base B can always be lifted to homotopies in the total space E; the fiber $F = p^{-1}(b_0)$ then sits inside E in a controlled way, and the relative homotopy groups $\pi_n(E, F)$ of section 5.1 collapse onto $\pi_n(B)$. Splicing this collapse into the relative long exact sequence produces a single exact sequence relating $\pi_*(F)$, $\pi_*(E)$, and $\pi_*(B)$. When two of the three spaces are understood, the third is forced. This is the engine behind the first non-trivial computation of a nonzero higher homotopy group, $\pi_3(S^2) \cong \mathbb{Z}$.

The whole apparatus rests on the ability to lift homotopies. The path-lifting and homotopy-lifting lemmas for covering spaces (lemma 2.1.30, lemma 2.1.31) are the prototype, and the definition below abstracts exactly the feature of covering maps that those lemmas used:

Definition 5.2.1: Homotopy Lifting Property

Let $p\colon E \to B$ be a continuous map and let X be a topological space. The map p has the *homotopy lifting property* with respect to X if, for every homotopy $G\colon X \times I \to B$ and every map $\tilde{g}_0\colon X \to E$ lifting the initial stage (that is, $p \circ \tilde{g}_0 = G(\,\cdot\,, 0)$), there exists a homotopy $\tilde{G}\colon X \times I \to E$ with

$$p \circ \tilde{G} = G \qquad \text{and} \qquad \tilde{G}(\,\cdot\,, 0) = \tilde{g}_0$$

Equivalently, every commutative square of solid arrows admits a diagonal filler $\tilde{G}$ making both triangles commute:

$$\begin{tikzcd} X \times \{0\} \arrow[r, "\tilde{g}_0"] \arrow[d, hook] & E \arrow[d, "p"] \\ X \times I \arrow[r, "G"'] \arrow[ur, dashed, "\tilde{G}"] & B \end{tikzcd}$$

The square encodes the only thing being asked: a homotopy downstairs in B, together with a chosen lift of its starting position, can be promoted to a homotopy upstairs in E that projects back down to the original. The lifted homotopy is not required to be unique (covering spaces are the special case where it is). What matters is existence, because existence is what lets one transport homotopical information from B to E and back. The class of test spaces X for which one requires this property is what distinguishes the two notions of fibration introduced next:

Definition 5.2.2: Hurewicz and Serre Fibrations

A continuous map $p\colon E \to B$ is a

1. *Hurewicz fibration* if it has the homotopy lifting property (definition 5.2.1) with respect to *every* topological space X.
2. *Serre fibration* (or *weak fibration*) if it has the homotopy lifting property with respect to every disk D^n, $n \geq 0$.

For a fibration $p\colon E \to B$ with B path-connected, the *fiber* over a basepoint $b_0 \in B$ is the subspace $F = p^{-1}(b_0) \subseteq E$. The data (E, p, B, F) is represented bythe notation

$$F \hookrightarrow E \xrightarrow{p} B$$

with F included into E and E projected onto B.

The fiber F depends on the choice of b_0 only up to homotopy equivalence when B is path-connected (a path from b_0 to b_1 lifts, via the homotopy lifting property, to a homotopy equivalence $p^{-1}(b_0) \simeq p^{-1}(b_1)$), so one speaks of *the* fiber.

Every Hurewicz fibration is a Serre fibration, since disks are among all spaces, but the converse fails: an example (a generalized covering map of the Hawaiian earring, which lifts homotopies parametrized by disks but not those parametrized by an arbitrary space) is constructed in [1].[2]

The gap between Serre and Hurewicz is harmless for the homotopy theory of this book, because the homotopy groups are built from maps out of spheres and disks and the Serre condition lifts exactly those; both notions nonetheless survive in the literature because Serre fibrations are far easier to recognize (one tests only against disks, which are compact and contractible) while Hurewicz fibrations have cleaner formal properties.

Example 5.2.3: The Real Line over the Circle, and Covering maps

1. The exponential covering map $p\colon \mathbb{R} \to S^1$, $p(t) = (\cos 2\pi t, \sin 2\pi t)$, is a Hurewicz fibration with fiber $F = p^{-1}(1,0) = \mathbb{Z} \subseteq \mathbb{R}$, a discrete set. To see the homotopy lifting property against an arbitrary space X, take a homotopy $G\colon X \times I \to S^1$ and an initial lift $\tilde{g}_0\colon X \to \mathbb{R}$. For each fixed $x \in X$ the path $t \mapsto G(x,t)$ is a path in S^1 starting at $p(\tilde{g}_0(x))$, so by unique path lifting (lemma 2.1.30) it has a unique lift $t \mapsto \tilde{G}(x,t)$ in $\mathbb{R}$ with $\tilde{G}(x,0) = \tilde{g}_0(x)$. Setting $\tilde{G}(x,t)$ to be this lift defines a function $X \times I \to \mathbb{R}$ with $p \circ \tilde{G} = G$ and the correct initial stage. Continuity in (x,t) jointly follows from the same evenly-covered-neighborhood argument used in lemma 2.1.31: near any point the lift is $\tilde{G} = (p|_V)^{-1} \circ G$ for the sheet V containing the current value, and $(p|_V)^{-1}$ is a homeomorphism. Thus the square fills, and p is a Hurewicz fibration. The fiber $\mathbb{Z}$ is discrete because the covering is evenly covered with countably many sheets.
2. Every covering map is a Hurewicz fibration by the identical argument: unique path lifting

[2]The gap closes under mild hypotheses: by Dold's local-global theorem [13], a map that is a Hurewicz fibration locally over a *numerable* open cover of the base (one admitting a subordinate partition of unity, e.g. any open cover of a paracompact Hausdorff base, by [12, Existence of Subordinate Partitions of Unity]) is a Hurewicz fibration globally; in particular a fiber bundle over a paracompact Hausdorff base is a Hurewicz fibration, sharpening the remark after theorem 5.2.4. Moreover, in the convenient category of compactly generated spaces, a Serre fibration whose base *and* total space are both CW complexes is automatically a Hurewicz fibration [22, 2]

promotes any base homotopy to a total-space homotopy. Covering spaces are precisely the fibrations whose fibers are discrete. In general, the fibers may be large and connected.

The first substantial theorem of the section certifies a large, geometrically natural class of Serre fibrations: the fiber bundles. Recall that a *fiber bundle* with fiber F is a map $p: E \to B$ that is *locally trivial*, meaning B is covered by open sets U over each of which p looks like a product, i.e. there is a homeomorphism $\varphi_U : p^{-1}(U) \to U \times F$ with $\mathrm{pr}_U \circ \varphi_U = p$, where $\mathrm{pr}_U : U \times F \to U$ is the projection. The smooth refinement, with a structure group acting on the fiber, is [20, Fiber Bundle]. Projections $U \times F \to U$ trivially lift homotopies (lift in the U-coordinate using G, hold the F-coordinate fixed at its initial value), so a bundle lifts homotopies locally over the base; the following theorem shows these local lifts can be patched into a single global lift when the base of the homotopy is a cube:

Theorem 5.2.4: Fiber Bundles are Serre Fibrations

Every fiber bundle $p\colon E \to B$ is a Serre fibration. That is, p has the homotopy lifting property with respect to every cube I^n (equivalently every disk D^n).

***Proof*:**
Fix $n \geq 0$ and write $X = I^n$. Let $G\colon X \times I \to B$ be a homotopy and $\tilde{g}_0\colon X \to E$ a lift of $G(\,\cdot\,, 0)$. The cube $X \times I = I^{n+1}$ is compact, and $\{p^{-1}(U)\}$ for trivializing opens $U \subseteq B$ pulls back under G to an open cover $\{G^{-1}(U)\}$ of $X \times I$.

Step 1: A Lebesgue grid adapted to the cover. By the Lebesgue number lemma applied to the compact metric space $X \times I$, there is $m \in \mathbb{N}$ so large that every subcube of the subdivision of $X \times I = I^{n+1}$ into m^{n+1} equal closed subcubes of side $1/m$ is mapped by G into a single trivializing open set U. Subdivide the last (homotopy) coordinate at the points $0 = t_0 < t_1 < \cdots < t_m = 1$ with $t_j = j/m$, and subdivide $X = I^n$ into subcubes $C_1, \ldots, C_{m^n}$ of side $1/m$. For each pair $(C_i, [t_{j-1}, t_j])$ the box $C_i \times [t_{j-1}, t_j]$ lands in a single trivializing $U_{i,j}$ under G.

Step 2: Lifting one box at a time, in lexicographic order. We construct the lift $\tilde{G}$ on $X \times I$ by lifting it over the boxes one homotopy-layer at a time, $j = 1, 2, \ldots, m$, and within each layer cube by cube. Suppose inductively that $\tilde{G}$ has been defined and is continuous on

$$A \;=\; (X \times [0, t_{j-1}]) \;\cup\; \left(\textstyle\bigcup_{k<i} C_k \times [t_{j-1}, t_j]\right)$$

lifting $G|_A$ and extending $\tilde{g}_0$, and that we wish to extend it across the next box $B_0 = C_i \times [t_{j-1}, t_j]$. The lift is already defined on the part of B_0 that meets A, namely

$$L \;=\; \left(C_i \times \{t_{j-1}\}\right) \;\cup\; \left((\partial C_i \cap \textstyle\bigcup_{k<i} C_k) \times [t_{j-1}, t_j]\right)$$

a closed subset of the cube B_0. Concretely L is the bottom face of B_0 together with some of its vertical side faces, which is a retract of B_0: there is a retraction $r\colon B_0 \to L$ (the cube retracts onto its bottom face union any union of vertical faces).

Step 3: Trivialize and lift over the box. Over B_0 the bundle is trivial: choose a trivialization $\varphi\colon p^{-1}(U_{i,j}) \xrightarrow{\sim} U_{i,j} \times F$ with $\mathrm{pr}_U \circ \varphi = p$. Write the already-defined lift on L in trivialized coordinates,

$$\varphi \circ \tilde{G}|_L \;=\; \left(G|_L,\, \sigma\right)\colon L \to U_{i,j} \times F$$

where the first component is forced to be $G|_L$ (because $\tilde{G}$ lifts G) and $\sigma\colon L \to F$ is the fiber component. Extend σ to all of B_0 by precomposing with the retraction, $\bar{\sigma} = \sigma \circ r\colon B_0 \to F$, and define

$$\tilde{G}|_{B_0} \;=\; \varphi^{-1} \circ \big(G|_{B_0},\, \bar{\sigma}\big)$$

This is continuous (a composite of continuous maps), satisfies $p \circ \tilde{G}|_{B_0} = \mathrm{pr}_U \circ \varphi \circ \tilde{G}|_{B_0} = G|_{B_0}$, and agrees with the old lift on L because $r|_L = \mathrm{id}_L$ forces $\bar{\sigma}|_L = \sigma$ and the first coordinate already matched. By the pasting lemma the extended map is continuous on $A \cup B_0$, since A and B_0 are closed and the two definitions agree on the closed overlap L.

Step 4: Conclusion. Iterating over the m^n cubes of layer j and then over the m layers exhausts all of $X \times I$, producing a continuous $\tilde{G}\colon X \times I \to E$ with $p \circ \tilde{G} = G$ and $\tilde{G}(\,\cdot\,, 0) = \tilde{g}_0$ (the latter because the very first layer started from $\tilde{g}_0$ on $X \times \{0\}$ and every extension preserved already-defined values). Thus the lifting square fills for $X = I^n$, and p is a Serre fibration, as we sought to show.

The proof is the prototype for lifting argument in homotopy theory: compactness chops the problem into pieces small enough to see local triviality, and a retraction extends a partial lift across each piece. The only place local triviality is used is Step 3, and the only place compactness is used is the Lebesgue grid of Step 1. With more care (a partition of unity in place of the Lebesgue grid) the same conclusion holds for Hurewicz fibrations:

Theorem 5.2.5: Fiber Bundles over Paracompact are Hurewicz Fibrations

Every fiber bundle $p\colon E \to B$ over a paracompact Hausdorff base is a Hurewicz fibration.

***Proof*:**
see [13] for Dold's proof via partitions of unity

Serre fibrations are also stable under pullback, and the fiber is unchanged. This is the form in which fibrations are transported along a comparison map, and it is used that way in the proof of theorem 6.5.1.

Proposition 5.2.6: Pullbacks of Serre Fibrations

Let $p\colon E \to B$ be a Serre fibration and $f\colon B' \to B$ a continuous map. Form the pullback

$$f^*E = \{(b', e) \in B' \times E \mid f(b') = p(e)\}$$

with projection $q\colon f^*E \to B'$, $(b', e) \mapsto b'$. Then q is a Serre fibration, and for every $b'_0 \in B'$ the map $(b'_0, e) \mapsto e$ is a homeomorphism from $q^{-1}(b'_0)$ onto the fiber $p^{-1}(f(b'_0))$ of p.

***Proof*:**
Let $G\colon D^n \times I \to B'$ be a homotopy and $\tilde{g}_0\colon D^n \to f^*E$ a lift of $G(\,\cdot\,, 0)$ under q. Write $\tilde{g}_0(x) = (G(x, 0), \eta(x))$ for a map $\eta\colon D^n \to E$ with $p \circ \eta = f \circ G(\,\cdot\,, 0)$, which is forced by membership in f^*E together with $q \circ \tilde{g}_0 = G(\,\cdot\,, 0)$. The composite $f \circ G\colon D^n \times I \to B$ is a homotopy in B whose initial stage is lifted by η, so, p being a Serre fibration, it lifts to $H\colon D^n \times I \to E$ with $H(\,\cdot\,, 0) = \eta$ and $p \circ H = f \circ G$. Define $\tilde{G}(x, t) = (G(x, t), H(x, t))$. It lands in f^*E because $p(H(x, t)) = f(G(x, t))$, it is continuous, it satisfies $q \circ \tilde{G} = G$, and its initial stage is $(G(\,\cdot\,, 0), \eta) = \tilde{g}_0$. So q is a Serre fibration.

For the fiber, $q^{-1}(b_0') = \{(b_0', e) \mid p(e) = f(b_0')\}$, and $(b_0', e) \mapsto e$ is a continuous bijection onto $p^{-1}(f(b_0'))$ with continuous inverse $e \mapsto (b_0', e)$, hence a homeomorphism, completing the proof.

For a.e. homotopy-group computations of this book, the Serre version above is sufficient since those groups are detected by spheres and disks. Before extracting the long exact sequence, one example of a fiber bundle that is not a covering space deserves mention, as its long exact sequence will give $\pi_3(S^2)$.

Example 5.2.7: The Hopf Fibration

Regard $S^3 = \{(z_0, z_1) \in \mathbb{C}^2 : |z_0|^2 + |z_1|^2 = 1\}$ and $S^2 = \mathbb{C}\mathrm{P}^1$, the complex projective line, which is the set of complex lines through the origin in $\mathbb{C}^2$. The *Hopf map* is

$$p\colon S^3 \longrightarrow \mathbb{C}\mathrm{P}^1 = S^2, \qquad p(z_0, z_1) = [z_0 : z_1]$$

sending a unit vector to the complex line it spans. The preimage of a point $[z_0 : z_1]$ is the set of unit vectors on that line, namely $\{(\lambda z_0, \lambda z_1) : \lambda \in \mathbb{C}, |\lambda| = 1\}$, a circle. Thus the fiber is $F = S^1$. The map is a fiber bundle: over the standard chart $U_0 = \{[z_0 : z_1] : z_0 \neq 0\}$ of $\mathbb{C}\mathrm{P}^1$, the trivialization sends $(z_0, z_1) \in p^{-1}(U_0)$ to the pair (the affine coordinate $z_1/z_0 \in \mathbb{C} \cong U_0$, the phase $z_0/|z_0| \in S^1$), with inverse rebuilding the unit vector; similarly over $U_1 = \{z_1 \neq 0\}$. These two charts cover $\mathbb{C}\mathrm{P}^1$, so p is locally trivial with fiber S^1. By theorem 5.2.4 it is a Serre fibration, written

$$S^1 \hookrightarrow S^3 \xrightarrow{p} S^2$$

Identifying $\mathbb{C}\mathrm{P}^1$ with S^2 uses stereographic coordinates: U_0 and U_1 are the two hemispheres' coordinate charts, glued along the equator by $w \mapsto 1/w$, which is exactly the gluing of S^2 from two disks.

The Hopf fibration is the first map $S^3 \to S^2$ that fails to be nullhomotopic, and its existence is the geometric content of $\pi_3(S^2) \neq 0$. The long exact sequence will show $\pi_3(S^2) \cong \mathbb{Z}$, generated by the Hopf class.

The remaining section build up to the long exact sequence. Recall from section 5.1 the relative homotopy group $\pi_n(E, F, e_0)$, defined for the pair (E, F) with basepoint $e_0 \in F$ as homotopy classes of maps $(D^n, S^{n-1}, s_0) \to (E, F, e_0)$, where the boundary sphere lands in F and the basepoint $s_0 \in S^{n-1}$ lands at e_0. The projection p collapses F to the point b_0, so p carries such a map to a map $(D^n, S^{n-1}, s_0) \to (B, b_0, b_0)$, that is, a map of D^n sending all of ∂D^n to b_0, which is the same as a map $S^n = D^n/\partial D^n \to B$ based at b_0. This defines a homomorphism $p_*\colon \pi_n(E, F, e_0) \to \pi_n(B, b_0)$, and the next proposition says it is an isomorphism:

Proposition 5.2.8: The Lifting Isomorphism

Let $p\colon E \to B$ be a Serre fibration with fiber $F = p^{-1}(b_0)$ and basepoint $e_0 \in F$. For every $n \geq 1$ the map induced by p,

$$p_*\colon \pi_n(E, F, e_0) \xrightarrow{\cong} \pi_n(B, b_0)$$

is an isomorphism (a bijection of pointed sets when $n = 1$, of groups when $n \geq 2$).

***Proof*:**
Throughout, model $\pi_n(B, b_0)$ by maps $(I^n, \partial I^n) \to (B, b_0)$ and $\pi_n(E, F, e_0)$ by maps $(I^n, I^{n-1} \times \{0\}, J^{n-1}) \to (E, F, e_0)$, where I^n is the n-cube, $I^{n-1} \times \{0\}$ is its bottom face, and $J^{n-1} \subseteq \partial I^n$ is the union of all the *other* faces (the top face together with all side faces). In this model a relative class is a map of the cube whose bottom face lands in F and whose remaining boundary faces J^{n-1} go to the basepoint e_0; applying p sends the bottom face into F to the point b_0 as well, hence sends all of ∂I^n to b_0, recovering an element of $\pi_n(B, b_0)$ and defining the homomorphism p_* to be analyzed. The pair (I^n, J^{n-1}) is homeomorphic to $(I^n, I^{n-1} \times \{0\})$ as a pair of a cube with a chosen face, because a self-homeomorphism of the cube carries the union of all-but-the-bottom faces onto a single face; in particular J^{n-1} is a retract of I^n, with retraction $r\colon I^n \to J^{n-1}$ used below.

Step 1: Surjectivity. Let $f\colon (I^n, \partial I^n) \to (B, b_0)$ represent a class in $\pi_n(B, b_0)$. View f as a homotopy parametrized by the bottom-face direction: write $I^n = I^{n-1} \times I$ with bottom face $I^{n-1} \times \{0\}$ and set $G = f\colon I^{n-1} \times I \to B$. The initial stage $G(\,\cdot\,, 0)\colon I^{n-1} \to B$ is the constant map at b_0 (the bottom face lies in ∂I^n), which lifts to the constant map $\tilde{g}_0 \equiv e_0\colon I^{n-1} \to E$. Since p is a Serre fibration and I^{n-1} is a cube, the homotopy lifting property (definition 5.2.1) yields a lift $\tilde{f}\colon I^{n-1} \times I \to E$ with $p \circ \tilde{f} = f$ and $\tilde{f}(\,\cdot\,, 0) = e_0$. The bottom face goes to $e_0 \in F$ by construction, and the remaining faces J^{n-1} satisfy $p \circ \tilde{f} = f \equiv b_0$ there, so $\tilde{f}(J^{n-1}) \subseteq p^{-1}(b_0) = F$. Thus $\tilde{f}$ is a map $(I^n, I^{n-1} \times \{0\}, J^{n-1}) \to (E, F, F)$ sending the bottom face to e_0 and J^{n-1} into F. Such a map is homotopic, through maps of the same triple, to one with $J^{n-1} \to e_0$: since J^{n-1} is contractible and contains the corner $J^{n-1} \cap (I^{n-1} \times \{0\})$ that already maps to e_0, the restriction $\tilde{f}|_{J^{n-1}}\colon J^{n-1} \to F$ is nullhomotopic in F to the constant map e_0 (its image lies in the path-component of e_0). Extending this nullhomotopy across the cube via the retraction r deforms $\tilde{f}$, rel the bottom face, to a map sending J^{n-1} to e_0, the whole deformation taking place over ∂I^n and hence inside $F = p^{-1}(b_0)$. The homotoped map represents a class in $\pi_n(E, F, e_0)$, and $p_*[\tilde{f}] = [p \circ \tilde{f}] = [f]$ because the deformation occurred in $F = p^{-1}(b_0)$ and so does not change the projection b_0. Hence p_* is surjective.

Step 2: Injectivity. Suppose $g_0, g_1\colon (I^n, \partial I^n, J^{n-1}) \to (E, F, e_0)$ are relative classes with $p_*[g_0] = p_*[g_1]$ in $\pi_n(B, b_0)$. Then $p \circ g_0$ and $p \circ g_1$ are homotopic rel ∂I^n through maps into B; let $H\colon I^n \times I \to B$ be such a homotopy, with $H(\,\cdot\,, 0) = p \circ g_0$, $H(\,\cdot\,, 1) = p \circ g_1$, and H constant at b_0 on $\partial I^n \times I$. We build a lift $\tilde{H}\colon I^n \times I \to E$ with prescribed behavior on the part of the boundary where the answer is already forced, then read off a relative homotopy from g_0 to g_1.

Consider the subspace

$$A \;=\; \big(I^n \times \{0\}\big) \cup \big(I^n \times \{1\}\big) \cup \big(J^{n-1} \times I\big) \;\subseteq\; I^n \times I$$

and define $h\colon A \to E$ by $h = g_0$ on $I^n \times \{0\}$, $h = g_1$ on $I^n \times \{1\}$, and $h \equiv e_0$ on $J^{n-1} \times I$. These agree on overlaps (on $J^{n-1} \times \{0\}$ and $J^{n-1} \times \{1\}$ both prescriptions give e_0, since g_0, g_1 send J^{n-1} to e_0), so h is a well-defined continuous map lifting $H|_A$. The pair $(I^n \times I, A)$ is homeomorphic to $(I^n \times I, I^n \times \{0\})$, because A is the union of the bottom, top, and the J-faces of the prism $I^n \times I$, and that union is a face-of-a-cube configuration carried to a single face by a self-homeomorphism of I^{n+1}. Under such a homeomorphism the lifting problem "extend h on A to a lift of H" becomes a standard homotopy lifting problem against the cube I^n, solved by the Serre property. The resulting lift $\tilde{H}\colon I^n \times I \to E$ satisfies $p \circ \tilde{H} = H$, restricts to g_0 and g_1 on the two ends, and is e_0 on $J^{n-1} \times I$. Finally, on the bottom face $I^{n-1} \times \{0\} \subseteq \partial I^n$ across all time, $p \circ \tilde{H} = H$ is constant at b_0, so $\tilde{H}$ maps $(\partial I^n) \times I$ into F. Therefore $\tilde{H}$ is a homotopy of relative classes from g_0 to g_1, giving $[g_0] = [g_1]$ in $\pi_n(E, F, e_0)$.

Both directions established, p_* is a bijection, and it is a homomorphism for $n \geq 2$ because p respects the concatenation operation on the first coordinate that defines the group structure (section 5.1), completing the proof.

The proposition says that, as far as homotopy groups can see, projecting to the base is the same as taking the pair (E, F) and forgetting the fiber. The relative group $\pi_n(E, F)$ measures maps of disks into E whose boundary lies in F; lifting says every map of a sphere into B is the projection of exactly one such disk, up to homotopy. This is the precise sense in which a fibration "unwinds" the base into the total space relative to the fiber. With this identification in hand, the relative long exact sequence of (E, F) from section 5.1 becomes a sequence in $\pi_*(F)$, $\pi_*(E)$, $\pi_*(B)$:

Theorem 5.2.9: Long Exact Sequence of a Fibration

Let $p\colon E \to B$ be a Serre fibration with B path-connected, fiber $F = p^{-1}(b_0)$, and basepoint $e_0 \in F$. Let $i\colon F \hookrightarrow E$ be the inclusion. Then there is a long exact sequence of homotopy groups

$$\cdots \to \pi_n(F, e_0) \xrightarrow{i_*} \pi_n(E, e_0) \xrightarrow{p_*} \pi_n(B, b_0) \xrightarrow{\partial} \pi_{n-1}(F, e_0) \xrightarrow{i_*} \cdots$$

terminating in $\cdots \to \pi_1(B, b_0) \xrightarrow{\partial} \pi_0(F) \xrightarrow{i_*} \pi_0(E)$, exact as a sequence of groups for $n \geq 2$, of groups and pointed sets at $n = 1$. The map ∂ is the composite of the boundary map $\pi_n(B, b_0) \xrightarrow{(p_*)^{-1}} \pi_n(E, F, e_0) \to \pi_{n-1}(F, e_0)$ from the relative long exact sequence.

***Proof*:**
The relative long exact sequence of the pair (E, F) with basepoint e_0 from section 5.1 reads

$$\cdots \to \pi_n(F, e_0) \xrightarrow{i_*} \pi_n(E, e_0) \xrightarrow{j_*} \pi_n(E, F, e_0) \xrightarrow{\delta} \pi_{n-1}(F, e_0) \xrightarrow{i_*} \cdots$$

and is exact (proposition 5.1.11), where $j_*\colon \pi_n(E) \to \pi_n(E, F)$ is induced by the inclusion of pairs $(E, e_0) \hookrightarrow (E, F)$ and δ is the connecting map sending a relative class $(D^n, S^{n-1}, s_0) \to (E, F, e_0)$ to the restriction of its boundary $(S^{n-1}, s_0) \to (F, e_0)$.

Step 1: Substitute the lifting isomorphism. By proposition 5.2.8, $p_*\colon \pi_n(E, F, e_0) \to \pi_n(B, b_0)$ is an isomorphism for every $n \geq 1$. Replace each occurrence of $\pi_n(E, F, e_0)$ in the relative sequence by $\pi_n(B, b_0)$ via p_*. The two adjacent maps transform as follows. The composite $\pi_n(E, e_0) \xrightarrow{j_*} \pi_n(E, F, e_0) \xrightarrow{p_*} \pi_n(B, b_0)$ equals p_* itself, because for a map $a\colon (I^n, \partial I^n) \to (E, e_0)$ the class $j_*[a]$ is a viewed as a relative map (its boundary is already at $e_0 \in F$), and applying p gives $p \circ a$, which is precisely the image of $[a]$ under the homomorphism $p_*\colon \pi_n(E) \to \pi_n(B)$ induced by p. So the arrow out of $\pi_n(E)$ becomes the genuine $p_*\colon \pi_n(E, e_0) \to \pi_n(B, b_0)$ induced by the map p.

Step 2: The connecting map. Define $\partial\colon \pi_n(B, b_0) \to \pi_{n-1}(F, e_0)$ as the composite $\partial = \delta \circ (p_*)^{-1}$. Because p_* is an isomorphism, replacing the segment $\pi_n(E, F) \xrightarrow{\delta} \pi_{n-1}(F)$ by $\pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F)$ preserves exactness at every node: exactness is a statement about kernels and images, and pre/post-composing a map with an isomorphism alters neither ker nor im in a way that breaks the equalities im = ker at each spot. Concretely, exactness of the original sequence at $\pi_n(E, F, e_0)$ ($\mathrm{im} j_* = \ker \delta$) transports under the isomorphism p_* to exactness of the new sequence at $\pi_n(B, b_0)$ ($\mathrm{im} p_* = \ker \partial$), since $p_*(\mathrm{im} j_*) = \mathrm{im}(p_* j_*) = \mathrm{im} p_*$ and $p_*(\ker \delta) = \ker(\delta(p_*)^{-1}) = \ker \partial$. Likewise exactness at $\pi_n(E, e_0)$ and at $\pi_{n-1}(F, e_0)$ is unchanged, the maps there being i_*, p_*, and the transported ∂.

Step 3: Assemble and read off the low end. Splicing the transported segments for all n yields the asserted sequence

$$\cdots \to \pi_n(F, e_0) \xrightarrow{i_*} \pi_n(E, e_0) \xrightarrow{p_*} \pi_n(B, b_0) \xrightarrow{\partial} \pi_{n-1}(F, e_0) \to \cdots$$

exact at each term by Step 2, of groups for $n \geq 2$ and of pointed sets through π_0 at the bottom, exactly as the relative sequence terminates. The terminal segment $\pi_1(B) \xrightarrow{\partial} \pi_0(F) \xrightarrow{i_*} \pi_0(E)$ records that two points of the fiber lie in the same path component of E if and only if they differ by the boundary action of a loop in B. This is the long exact sequence claimed, completing the proof.

Read in either direction, this sequences converts information of two of the three spaces into knowledge of the third. If the fiber F has vanishing homotopy in a range, then p_* is an isomorphism there, so E and B have the same homotopy groups in that range; if instead the total space E is highly connected, the boundary map ∂ becomes an isomorphism and reads off $\pi_n(B)$ as $\pi_{n-1}(F)$. The three running examples below exercise both readings. The concrete meaning of each map is worth keeping in view: i_* includes a sphere in the fiber into the total space; p_* projects a sphere in the total space down to the base; and ∂ takes a sphere $S^n \to B$, lifts it to a disk $D^n \to E$ as in proposition 5.2.8, and returns the boundary sphere $S^{n-1} \to F$, which is the obstruction to the lift closing up into a sphere.

The simplest application recovers, and extends to all dimensions, a fact the covering-space theory of chapter 2 could only see in dimension one:

Example 5.2.10: Higher Homotopy of the Circle

For the covering $p\colon \mathbb{R} \to S^1$ of example 5.2.3, the fiber is $F = \mathbb{Z}$, a discrete space, so $\pi_n(F) = 0$ for all $n \geq 1$ (a discrete space has no nonconstant maps from a connected S^n, $n \geq 1$, up to homotopy) and the total space $\mathbb{R}$ is contractible, so $\pi_n(\mathbb{R}) = 0$ for all $n \geq 1$. The long exact sequence (theorem 5.2.9) at level $n \geq 2$ reads

$$\underbrace{\pi_n(\mathbb{Z})}_{0} \to \underbrace{\pi_n(\mathbb{R})}_{0} \to \pi_n(S^1) \xrightarrow{\partial} \underbrace{\pi_{n-1}(\mathbb{Z})}_{0}$$

so $\pi_n(S^1)$ is squeezed between two zero groups and is therefore 0. Hence

$$\pi_n(S^1) = 0 \qquad \text{for all } n \geq 2$$

At $n = 1$ the sequence gives $0 = \pi_1(\mathbb{R}) \to \pi_1(S^1) \xrightarrow{\partial} \pi_0(\mathbb{Z}) \to \pi_0(\mathbb{R}) = 0$, and exactness with $\pi_0(\mathbb{Z}) = \mathbb{Z}$ as a set recovers the bijection $\pi_1(S^1) \cong \mathbb{Z}$ established by covering-space theory in chapter 2.

The circle is an Eilenberg-MacLane space $K(\mathbb{Z}, 1)$: it has a single nonzero homotopy group, $\pi_1 = \mathbb{Z}$, and nothing above. The contractibility of its universal cover is exactly what kills every higher group. The same mechanism applies to any space with contractible universal cover, the *aspherical* spaces, whose higher homotopy vanishes identically. Surfaces of genus at least one, and more generally any space whose universal cover is contractible, are aspherical for this reason.

The second example is not a single fibration but a construction that turns *any* pointed space into a fibration, and in doing so reveals that the higher homotopy groups are iterated π_1's of loop spaces.

Definition 5.2.11: Path Space and Loop Space

Fix a pointed space (X, x_0). The *path space* is

$$PX = \{\gamma\colon I \to X : \gamma(0) = x_0\}$$

the space of paths in X starting at x_0, topologized by the compact-open topology. The *loop space* is

$$\Omega X = \{\gamma\colon I \to X : \gamma(0) = \gamma(1) = x_0\}$$

the subspace of loops based at x_0. The endpoint map $p\colon PX \to X$, $p(\gamma) = \gamma(1)$, evaluates a path at its far end, and its fiber over x_0 is exactly ΩX.

Example 5.2.12: The Path-Loop Fibration

For a pointed space (X, x_0) the endpoint map $p\colon PX \to X$, $\gamma \mapsto \gamma(1)$, is a Hurewicz fibration with fiber ΩX, fitting into

$$\Omega X \hookrightarrow PX \xrightarrow{p} X$$

To verify the homotopy lifting property against a space Y, take a homotopy $G\colon Y \times I \to X$ and a lift $\tilde{g}_0\colon Y \to PX$ of $G(\,\cdot\,, 0)$; write $w_y = \tilde{g}_0(y) \in PX$, a path from x_0 to $G(y, 0)$. The idea is to traverse w_y and then follow the trace $t \mapsto G(y, t)$ of the homotopy out to time s, but the two pieces must be reparametrized so that at $s = 0$ the trailing piece has length zero and the lift returns to w_y exactly. Define, for $(y, s) \in Y \times I$, the path $\tilde{G}(y, s) \in PX$ by its value at $u \in [0, 1]$,

$$\tilde{G}(y,s)(u) \;=\; \begin{cases} w_y\!\left(\dfrac{2u}{2-s}\right), & 0 \le u \le 1 - \frac{s}{2} \\ G(y,\, 2u - 2 + s)\,, & 1 - \frac{s}{2} \le u \le 1 \end{cases}$$

The two branches agree at the join $u = 1 - \frac{s}{2}$, where both give $w_y(1) = G(y, 0)$, so $\tilde{G}(y, s)$ is a continuous path; it starts at $w_y(0) = x_0$ and ends at $G(y, s)$, hence lies in PX with $p(\tilde{G}(y, s)) = G(y, s)$. At $s = 0$ the first branch covers all of $[0, 1]$ and reduces to $w_y(u)$, while the second branch has degenerate domain, so $\tilde{G}(y, 0) = w_y = \tilde{g}_0(y)$ exactly. Joint continuity of $(y, s, u) \mapsto \tilde{G}(y, s)(u)$ gives continuity of $\tilde{G}\colon Y \times I \to PX$ in the compact-open topology. So the square fills for every Y, and p is a Hurewicz fibration. The total space PX is contractible: the homotopy $\gamma_r(t) = \gamma(rt)$ for $r \in I$ slides every path back along itself to the constant path at x_0, giving a deformation retraction of PX onto $\{x_0\}$.

Feeding this into theorem 5.2.9 with $E = PX$ contractible kills the middle terms: for every $n \ge 1$,

$$\underbrace{\pi_n(PX)}_{0} \to \pi_n(X) \xrightarrow{\partial} \pi_{n-1}(\Omega X) \to \underbrace{\pi_{n-1}(PX)}_{0}$$

is exact, so ∂ is an isomorphism, yielding the dimension shift

$$\pi_n(X, x_0) \;\cong\; \pi_{n-1}(\Omega X, e_{x_0}) \qquad (n \ge 1)$$

Iterating, $\pi_n(X) \cong \pi_{n-1}(\Omega X) \cong \pi_{n-2}(\Omega^2 X) \cong \cdots \cong \pi_0(\Omega^n X)$.

The path-loop fibration is the source of the slogan that "homotopy groups are abelian above degree one because they are double loop spaces". The isomorphism $\pi_n(X) \cong \pi_{n-1}(\Omega X)$ trades a sphere of dimension n in X for a sphere of dimension $n-1$ in the loop space, and after two iterations one reaches $\pi_{n-2}(\Omega^2 X)$; the two independent loop coordinates of $\Omega^2 X$ commute, which is the conceptual reason behind the commutativity proved combinatorially in theorem 5.1.4. Concretely, $\pi_2(X) \cong \pi_1(\Omega X)$ identifies the second homotopy group of X with the ordinary fundamental group of its loop space.

The third example the first nonzero higher homotopy group of a sphere, where we shall assume $\pi_k(S^m) = 0$ for $k < m$ (proven in section 5.4):

Example 5.2.13: The Homotopy of S^2 via the Hopf Fibration

Apply theorem 5.2.9 to the Hopf fibration $S^1 \hookrightarrow S^3 \xrightarrow{p} S^2$ of example 5.2.7. The relevant inputs are $\pi_n(S^1) = 0$ for $n \geq 2$ (example 5.2.10) and the connectivity of spheres, $\pi_k(S^m) = 0$ for $k < m$ (anticipating corollary 5.4.11 in section 5.4), in particular $\pi_1(S^3) = \pi_2(S^3) = 0$ since S^3 is 2-connected. The sequence around degrees 2 and 3 reads, with $F = S^1$ and $E = S^3$,

$$\pi_3(S^1) \to \pi_3(S^3) \xrightarrow{p_*} \pi_3(S^2) \xrightarrow{\partial} \pi_2(S^1) \to \pi_2(S^3) \xrightarrow{p_*} \pi_2(S^2) \xrightarrow{\partial} \pi_1(S^1) \to \pi_1(S^3)$$

Substituting the known zeros $\pi_3(S^1) = 0$, $\pi_2(S^1) = 0$, $\pi_2(S^3) = 0$, and $\pi_1(S^3) = 0$, this becomes

$$0 \to \pi_3(S^3) \xrightarrow{p_*} \pi_3(S^2) \to 0 \qquad \text{and} \qquad 0 \to \pi_2(S^2) \xrightarrow{\partial} \pi_1(S^1) \to 0$$

Exactness of the first short sequence (a map flanked by zeros is an isomorphism) gives $p_* \colon \pi_3(S^3) \xrightarrow{\cong} \pi_3(S^2)$, and $\pi_3(S^3) \cong \mathbb{Z}$ by the Hurewicz theorem (S^3 is 2-connected, so $\pi_3(S^3) \cong H_3(S^3) \cong \mathbb{Z}$, anticipating section 5.4; the generator is $[\mathrm{id}_{S^3}]$). Therefore

$$\pi_3(S^2) \;\cong\; \mathbb{Z}$$

generated by the class of the Hopf map $[p]$, since $p_*[\mathrm{id}_{S^3}] = [p \circ \mathrm{id}] = [p]$. Exactness of the second short sequence gives $\partial \colon \pi_2(S^2) \xrightarrow{\cong} \pi_1(S^1) \cong \mathbb{Z}$, so

$$\pi_2(S^2) \;\cong\; \mathbb{Z}$$

generated by $[\mathrm{id}_{S^2}]$, recovering by a fibration argument the degree isomorphism $\pi_2(S^2) \cong \mathbb{Z}$.

The computation $\pi_3(S^2) \cong \mathbb{Z}$ is the first example that homotopy groups *don't necessarily* vanish once the dimension exceeds that of the space. A 3-sphere can wrap *nontrivially* around a 2-sphere, and the Hopf map is the generating wrapping; its $\mathbb{Z}$ worth of homotopy classes are detected by the linking number of the preimages of two distinct points (each a circle in S^3), the *Hopf invariant* (if it still exists, see this animation on youtube of Hopf Fibrations). That two great circles in S^3 corresponding to distinct points of S^2 are linked exactly once is the geometric content of the generator $[p]$. This single example already shows that the homotopy groups of spheres, the $\pi_k(S^m)$, are more intricate than their homology, and computing them is a central thread of the chapters that follow.

Exercise 5.2.1

1. Let $p \colon E \to B$ and $p' \colon E' \to B'$ be Serre fibrations. Show that the product map $p \times p' \colon E \times E' \to B \times B'$, $(e, e') \mapsto (p(e), p'(e'))$, is a Serre fibration, and identify its fiber over (b_0, b_0') in terms of the fibers of p and p'.

2. Let $p\colon E \to B$ be a Serre fibration, $f\colon B'' \to B'$ and $g\colon B' \to B$ continuous. Using proposition 5.2.6, show that the two ways of pulling p back to B'' agree: $f^*(g^*E) \cong (g \circ f)^*E$ as spaces over B'', compatibly with the projections. Deduce that a composite of pullbacks of a Serre fibration is again one.

3. Suppose the Serre fibration $p\colon E \to B$ admits a *section*, a map $s\colon B \to E$ with $p \circ s = \mathrm{id}_B$ and $s(b_0) = e_0$. Using the long exact sequence (theorem 5.2.9), prove that for every $n \geq 1$ the sequence splits, giving $\pi_n(E, e_0) \cong \pi_n(F, e_0) \oplus \pi_n(B, b_0)$ for $n \geq 2$.

4. Use the Hopf fibration and example 5.2.13 to show that S^3 is *not* homotopy equivalent to $S^2 \times S^1$, even though both are closed 3-manifolds, by comparing π_2 of each.

5. Let $X = S^m$ with $m \geq 2$. Using the path-loop fibration (example 5.2.12), express $\pi_k(\Omega S^m)$ in terms of homotopy groups of spheres, and deduce the value of $\pi_{m-1}(\Omega S^m)$.

6. Let $p\colon \tilde{X} \to X$ be a covering map with $\tilde{X}$ path-connected. Using that a covering map is a Serre fibration with discrete fiber, derive from theorem 5.2.9 that $p_*\colon \pi_n(\tilde{X}) \to \pi_n(X)$ is an isomorphism for all $n \geq 2$.

5.3 Whitehead's Theorem and CW Approximation

A natural question to ask is how much of a space do homotopy groups detect. A continuous map $f\colon X \to Y$ induces homomorphisms $f_*\colon \pi_n(X, x_0) \to \pi_n(Y, f(x_0))$ for every n and every basepoint; if all of these are isomorphisms, the two spaces are (homotopy-groups)–indistinguishable. The central result of this section, due to Whitehead, is that for the class of CW complexes this indistinguishability is a homotopy equivalence. The CW hypothesis cannot be dropped due to the Warsaw circle. The section closes with the converse construction, CW approximation, which replaces an arbitrary space by a CW complex with the same homotopy groups, so that the homotopy-theoretic study of *any* space reduces to the study of CW complexes.

Definition 5.3.1: Weak Homotopy Equivalence

A map $f\colon X \to Y$ is a *weak homotopy equivalence* if the induced map

$$f_*\colon \pi_n(X, x_0) \longrightarrow \pi_n\big(Y, f(x_0)\big)$$

is a bijection for every $n \geq 0$ and every choice of basepoint $x_0 \in X$. (For $n = 0$ the condition is that f induces a bijection on path components; for $n \geq 1$ it is that f_* is a group isomorphism.)

The quantifier over all basepoints is not redundant. Homotopy groups based at points in different path components carry independent information, and a map can be an isomorphism on the homotopy groups of one component while failing on another. The condition on π_0 forces f to match up the components of X and Y, and within each matched pair the higher conditions then compare the groups based at a point of that component. A homotopy equivalence is always a weak homotopy equivalence, since a homotopy inverse induces a two-sided inverse on each π_n by the functoriality of proposition 5.1.6. The content of Whitehead's theorem is the converse for CW complexes; the content of the Warsaw circle example is that the converse fails in general.

Example 5.3.2: The Inclusion of a Deformation Retract

Let $A \subseteq X$ be a deformation retract, so there is a retraction $r\colon X \to A$ and a homotopy $H\colon X \times I \to X$ from id_X to $\iota \circ r$ fixing A pointwise, where $\iota\colon A \hookrightarrow X$ is the inclusion. Then ι is a weak homotopy equivalence; in fact it is a homotopy equivalence. Concretely, take $A = S^1 \subseteq \mathbb{R}^2 \setminus \{0\} = X$, the punctured plane, with $r(v) = v/|v|$ and $H(v,t) = (1-t)v + t\,v/|v|$, a straight-line homotopy that never crosses the origin. On homotopy groups, $r_* \circ \iota_* = (\mathrm{id}_A)_* = \mathrm{id}$ and $\iota_* \circ r_* = (\iota r)_* = (\mathrm{id}_X)_*$ because $\iota r \simeq \mathrm{id}_X$, so ι_* is an isomorphism on every π_n at every basepoint of S^1. For instance $\pi_1(S^1) \cong \mathbb{Z}$ maps isomorphically onto $\pi_1(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{Z}$, the winding number being preserved because the deformation retraction does not change how a loop wraps around the puncture. Every higher group vanishes on both sides, since S^1 is a $K(\mathbb{Z}, 1)$.

The key result behind both proceeding theorems is the compression lemma, which says that maps into a target can be pushed off the cells of a CW pair as soon as the relevant relative homotopy groups vanish. Everything in this section is an application of that lemma together with the homotopy extension property of CW pairs established in proposition 1.3.4.

Lemma 5.3.3: Compression Lemma

Let (X, A) be a CW pair and let (Y, B) be any pair with $B \neq \varnothing$. Suppose that for every n such that $X \setminus A$ has cells of dimension n, the relative homotopy group $\pi_n(Y, B, y_0)$ is trivial for all basepoints $y_0 \in B$. Then every map $f\colon (X, A) \to (Y, B)$ is homotopic rel A to a map $X \to Y$ whose image lies entirely in B.

In short, one wants to homotope a map so that it avoids a chosen subspace B of the target, and the obstruction to pushing the map off each n-cell of the domain lives in the relative homotopy group $\pi_n(Y, B)$. When those groups vanish, every obstruction disappears and the compression goes through cell by cell.

***Proof*:**

The strategy is to push f into B one skeleton at a time, using the vanishing relative homotopy groups to clear the cells of each dimension, and assembling the partial homotopies with the homotopy extension property. Throughout, f already maps A into B, and the homotopy will fix A.

Step 1: Setting up the induction on skeleta. Write X^k for the k-skeleton of X. The plan is to construct, by induction on k, a homotopy of f rel A that moves f to a map carrying $A \cup X^k$ into B. Since the homotopy will be performed rel A and the cells of A are already sent into B, only the cells of $X \setminus A$ require attention. Assume inductively that f has already been homotoped rel A so that $A \cup X^{k-1}$ maps into B; the base case $k = 0$ requires clearing the 0-cells of $X \setminus A$, which is handled by the same argument below with $n = 0$ (a relative homotopy group $\pi_0(Y, B)$ being trivial means every point of Y in the relevant component can be joined to B by a path).

Step 2: Clearing a single n-cell. Let e^n_α be an n-cell of $X \setminus A$ with characteristic map $\Phi_\alpha\colon D^n \to X$ and attaching map $\varphi_\alpha = \Phi_\alpha|_{\partial D^n}\colon S^{n-1} \to X^{n-1}$. By the inductive hypothesis f already sends X^{n-1}, hence the image of φ_α, into B. Therefore the composite

$$f \circ \Phi_\alpha\colon (D^n, \partial D^n) \longrightarrow (Y, B)$$

is a map of pairs, and as such it represents an element of the relative homotopy group $\pi_n(Y, B, y_0)$, where $y_0 = f(\Phi_\alpha(s_0))$ for a chosen basepoint $s_0 \in \partial D^n$. By hypothesis this group is trivial, so

$f \circ \Phi_\alpha$ is homotopic, as a map of pairs $(D^n, \partial D^n) \to (Y, B)$, to a map with image contained in B. The triviality of a relative homotopy group means precisely that every map of the pair $(D^n, \partial D^n)$ into (Y, B) can be deformed, through maps of pairs, into one landing in B; this is the standard characterisation of the zero element established for relative homotopy groups in definition 5.1.8.

Step 3: Assembling the homotopy over all n-cells. Performing the homotopy of Step 2 simultaneously over all n-cells of $X \setminus A$ defines a homotopy

$$G\colon \big(A \cup X^{n-1}\big) \times I \;\cup\; \big(n\text{-cells of } X \setminus A\big) \times I \longrightarrow Y$$

The homotopy is constant on $A \cup X^{n-1}$ (those cells already lie in B and need not be moved) and on each new n-cell it is the cell-by-cell homotopy of Step 2, which agrees with the constant homotopy on the boundary ∂D^n because that boundary maps into X^{n-1}, already cleared. Thus the pieces match on overlaps and define a homotopy on $A \cup X^n$, starting at $f|_{A\cup X^n}$ and ending at a map sending $A \cup X^n$ into B.

Step 4: Extending over X via the homotopy extension property. The pair $(X, A \cup X^n)$ is a CW pair, hence has the homotopy extension property by proposition 1.3.4. The map $f\colon X \to Y$ together with the homotopy G just constructed on $A \cup X^n$ therefore extends to a homotopy $\widetilde{G}\colon X \times I \to Y$ of f. Replacing f by $\widetilde{G}(\,\cdot\,, 1)$ advances the induction: the new f sends $A \cup X^n$ into B, and the homotopy has been performed rel A.

Step 5: Passing to the colimit. Carrying out Steps 2-4 for $k = 0, 1, 2, \ldots$ produces a sequence of homotopies, the k-th of which is performed on the time interval $[1 - 2^{-k}, 1 - 2^{-(k+1)}]$ and is stationary on $A \cup X^{k-1}$. Concatenating them gives a single homotopy $X \times I \to Y$, rel A. Continuity at the final time $t = 1$ holds because X carries the weak topology with respect to its skeleta (definition 1.2.1): a map out of $X \times I$ is continuous if and only if its restriction to each $X^k \times I$ is, and on $X^k \times I$ the homotopy is eventually stationary, so it is continuous there. The end map of this concatenated homotopy sends all of $X = A \cup \bigcup_k X^k$ into B, completing the proof.

Example 5.3.4: Compressing a Map into a Highly Connected Subspace

Take $X = D^2$ with the standard CW structure $X = e^0 \cup e^1 \cup e^2$ (one 0-cell and one 1-cell forming $\partial D^2 = S^1$, and one 2-cell), and let $A = \varnothing$. Suppose (Y, B) is a pair with $\pi_0(Y, B) = \pi_1(Y, B) = \pi_2(Y, B) = 0$ for all basepoints in B, meaning $B \hookrightarrow Y$ is 2-connected: every component of Y meets B, and the relative groups up to degree 2 vanish. The compression lemma then says any map $f\colon D^2 \to Y$ is homotopic to a map landing in B. Concretely, let $Y = \mathbb{R}^3$ and $B = \mathbb{R}^2 \times \{0\}$. Since $\mathbb{R}^3$ deformation retracts onto $\mathbb{R}^2 \times \{0\}$, every relative group $\pi_n(\mathbb{R}^3, \mathbb{R}^2)$ vanishes, so the hypotheses hold in all dimensions. Any map $f\colon D^2 \to \mathbb{R}^3$ is compressed into the plane by the straight-line homotopy $(v, t) \mapsto (f(v)_1, f(v)_2, (1-t)f(v)_3)$, which collapses the third coordinate to zero.

With the compression lemma in hand, Whitehead's theorem is almost immediate. The idea is to replace the map f by an inclusion (its mapping cylinder), observe that the vanishing of all relative homotopy groups of the inclusion is exactly the statement that f is a weak homotopy equivalence, and then compress the identity of the mapping cylinder onto the source. This produces a deformation retraction, which is the homotopy equivalence sought.

Theorem 5.3.5: Whitehead's Theorem

Let X and Y be CW complexes and let $f\colon X \to Y$ be a weak homotopy equivalence. Then f is a homotopy equivalence. If f is moreover the inclusion of a subcomplex, then X is a deformation retract of Y.

***Proof*:**
The proof reduces the general case to the case of an inclusion by means of the mapping cylinder, then applies the compression lemma to that inclusion.

Step 1: Reduction to an inclusion via the mapping cylinder. Recall the mapping cylinder $M_f = \big((X \times I) \sqcup Y\big)/\big((x,1) \sim f(x)\big)$. There is a deformation retraction of M_f onto Y, so the inclusion $Y \hookrightarrow M_f$ is a homotopy equivalence, and f factors as

$$X \xhookrightarrow{\iota} M_f \xrightarrow{r} Y$$

where $\iota(x) = (x,0)$ is the inclusion of X as the top of the cylinder and r is the retraction onto Y, with $r \circ \iota = f$. The retraction r is a homotopy equivalence, so f is a homotopy equivalence if and only if ι is. Moreover, when X and Y are CW complexes the mapping cylinder M_f is a CW complex containing X as a subcomplex (after a homotopy of f to a cellular map, which by the cellular approximation theorem ([14, Theorem 4.8]) changes nothing up to homotopy), so (M_f, X) is a CW pair. Thus it suffices to prove the second assertion: *if $\iota\colon X \hookrightarrow Z$ is an inclusion of CW complexes that is a weak homotopy equivalence, then X is a deformation retract of Z*, with $Z = M_f$. The first assertion follows because a deformation retract inclusion is in particular a homotopy equivalence.

Step 2: The relative homotopy groups of the pair vanish. Let $\iota\colon X \hookrightarrow Z$ be an inclusion of CW complexes that is a weak homotopy equivalence. The long exact sequence of the pair (Z, X) from proposition 5.1.11 reads

$$\cdots \longrightarrow \pi_n(X,x_0) \xrightarrow{\iota_*} \pi_n(Z,x_0) \longrightarrow \pi_n(Z,X,x_0) \longrightarrow \pi_{n-1}(X,x_0) \xrightarrow{\iota_*} \pi_{n-1}(Z,x_0) \longrightarrow \cdots$$

for each basepoint $x_0 \in X$. Because ι is a weak homotopy equivalence, every map $\iota_*\colon \pi_n(X,x_0) \to \pi_n(Z,x_0)$ is an isomorphism. A short exactness argument then forces the relative groups to vanish: the map $\pi_n(Z,X) \to \pi_{n-1}(X)$ has image contained in the kernel of the injective ι_*, hence is zero, so $\pi_n(Z,X) \to \pi_{n-1}(X)$ is the zero map; and the map $\pi_n(Z) \to \pi_n(Z,X)$ is surjective onto the kernel of that zero map, which is all of $\pi_n(Z,X)$, while its source $\pi_n(Z)$ is hit isomorphically from $\pi_n(X)$ by ι_*, so the composite $\pi_n(Z) \to \pi_n(Z,X)$ is zero. A surjective zero map has trivial target, so $\pi_n(Z,X,x_0) = 0$ for all n and all basepoints $x_0 \in X$.

Step 3: Compressing the identity of Z onto X. Apply the compression lemma (lemma 5.3.3) to the identity map

$$\mathrm{id}_Z\colon (Z,X) \longrightarrow (Z,X)$$

with $(X,A) = (Z,X)$ in the notation of the lemma and $(Y,B) = (Z,X)$. The pair (Z,X) is a CW pair, the subspace X is nonempty, and the relative homotopy groups $\pi_n(Z,X)$ vanish in every dimension by Step 2, so the hypotheses are met in all dimensions. The lemma produces a homotopy, rel X, from id_Z to a map $\rho\colon Z \to Z$ whose image lies in X.

Step 4: Identifying the deformation retraction. The map $\rho\colon Z \to X$ satisfies $\rho|_X = \mathrm{id}_X$, since the homotopy was performed rel X and id_Z restricts to the identity on X. Thus ρ is a retraction of Z onto X, and the homotopy from id_Z to ρ is a homotopy rel X from the identity of Z to a map into X. This is exactly a deformation retraction of Z onto X. Therefore X is a deformation retract of Z, the inclusion $X \hookrightarrow Z$ is a homotopy equivalence, and tracing through Step 1 the original map f is a homotopy equivalence, completing the proof.

Whitehead's theorem is the statement that for CW complexes the homotopy groups are a *complete* invariant of homotopy type, in the sense that a map realising isomorphisms on all of them is invertible up to homotopy. The qualification "a map realising" is essential: two CW complexes can have abstractly isomorphic homotopy groups in every degree without being homotopy equivalent, because no single map need induce those isomorphisms simultaneously. The spaces $\mathbb{R}\mathrm{P}^2 \times S^3$ and $\mathbb{R}\mathrm{P}^3 \times S^2$ are the standard example of this phenomenon, having identical homotopy groups but distinct homotopy types.

Example 5.3.6: $\mathbb{R}^n$ Is Contractible by Whitehead

Let $X = \{*\}$, a single point with its evident CW structure, and let $Y = \mathbb{R}^n$ with the CW structure of example 1.2.5 (a product of the line's cell structure). The constant inclusion $f\colon \{*\} \hookrightarrow \mathbb{R}^n$ is a weak homotopy equivalence: $\mathbb{R}^n$ is path connected so π_0 agrees, and $\mathbb{R}^n$ is convex so every map of a sphere into it is null-homotopic by a straight-line homotopy to a constant, giving $\pi_k(\mathbb{R}^n) = 0 = \pi_k(\{*\})$ for all $k \geq 1$. Both source and target are CW complexes, so Whitehead's theorem applies and f is a homotopy equivalence: $\mathbb{R}^n$ is contractible. This particular conclusion is visible directly from the contraction $(v,t) \mapsto (1-t)v$, but the point is methodological. Whitehead's theorem reduces the assertion "$\mathbb{R}^n$ is contractible" to the purely homotopy-group computation $\pi_k(\mathbb{R}^n) = 0$, with no need to exhibit a contraction; in less transparent examples no explicit contraction is available and the homotopy-group route is the only one.

Example 5.3.7: The House With Two Rooms Is Contractible

The previous example settled the contractibility of $\mathbb{R}^n$ by computing homotopy groups, but there the conclusion already followed from the straight-line contraction. The house with two rooms of example 1.1.14, Bing's house, is the promised case where no contraction presents itself, yet the same two-line homotopy-group argument applies verbatim. Regard it as a finite, 2-dimensional CW complex B: two stacked chambers, each entered only through a vertical tube that passes through the floor of the other, with the supporting walls, partition, and tube walls glued along their shared edges into a complex of 0-cells, 1-cells, and 2-cells. The aim is to prove B contractible without ever writing down a deformation; the route is to show B is simply connected with vanishing reduced homology and to let theorem 5.4.9 and theorem 5.3.5 do the rest.

B is simply connected. The fundamental group of a connected 2-complex is read off from its 2-skeleton by the presentation-complex recipe of section 2.2: choosing a spanning tree T of the 1-skeleton $B^{(1)}$, the group $\pi_1(B^{(1)})$ is free on the 1-cells of $B^{(1)}$ not in T (collapsing T identifies $B^{(1)}$ with a wedge of circles, whose group is free by Van Kampen's theorem, theorem 2.2.6), and each 2-cell e^2_β, attached along a loop w_β in $B^{(1)}$, imposes the single relation $w_\beta = 1$ on the group. This is the same mechanism used to realize an arbitrary group as π_1 of a 2-complex in theorem 5.5.4: attaching a 2-cell along a loop w kills exactly the class $[w]$ and nothing more. The outcome is the finite presentation

$$\pi_1(B) = \big\langle\, x_e \ (e \notin T) \;\big|\; w_\beta \ (\beta = 1, \ldots, c_2) \,\big\rangle$$

in which the generators x_e are indexed by the 1-cells outside T and the relators w_β are the attaching words of the c_2 two-cells.[a] For Bing's house this presentation collapses to the trivial group: each generator threading a tunnel appears, in some relator w_β, as the attaching word of a supporting-wall 2-cell whose boundary runs once along that tunnel edge and otherwise along edges already expressible through other relators, so successive substitution eliminates the generators one by one until none remain. Verifying this collapse for the full structure is a finite but lengthy word computation over the 23 generators and 23 relators, with no conceptual content beyond the substitution just described; the cell structure itself, with $c_0 = 29$ zero-cells, $c_1 = 51$ one-cells, and $c_2 = 23$ two-cells, is enumerated in [14, Ex. 0.2]. The generator count is $c_1 - (c_0 - 1) = 51 - 28 = 23$, the number of 1-cells left once a spanning tree of the connected 1-skeleton accounts for $c_0 - 1 = 28$ of them, equivalently the first Betti number $c_1 - c_0 + 1$ of $B^{(1)}$ (the rank of the free group $\pi_1(B^{(1)})$); the relators therefore exactly match the generators in number, the balanced presentation forced on an acyclic 2-complex, where the 23 relators must kill the rank-23 free group $\pi_1(B^{(1)})$. The presentation therefore yields $\pi_1(B) = 0$.

B has trivial reduced homology. Because B is 2-dimensional, $\tilde{H}_i(B) = 0$ for $i \geq 3$ automatically. In degree one, B is path connected and now known to be simply connected, so the dimension-one Hurewicz theorem (theorem 5.4.5) gives $H_1(B) \cong \pi_1(B)^{\mathrm{ab}} = 0$. The remaining group $H_2(B)$ is settled by an Euler-characteristic count rather than by inspecting which 2-cells fit together. Since B is 2-dimensional, its cellular chain complex is $0 \to C_2(B) \to C_1(B) \to C_0(B) \to 0$, so $H_2(B) = \ker \partial_2$ is a subgroup of the free abelian group $C_2(B)$ and is therefore itself free abelian; its rank is the only unknown. Euler's characteristic theorem (theorem 3.5.21) computes that rank homologically,

$$\chi(B) = \sum_n (-1)^n \mathrm{rank} H_n(B) = \mathrm{rank} H_0(B) - \mathrm{rank} H_1(B) + \mathrm{rank} H_2(B) = 1 - 0 + \mathrm{rank} H_2(B)$$

using the established $H_0(B) \cong \mathbb{Z}$ from path connectivity and $H_1(B) = 0$ from the previous paragraph. On the other hand $\chi(B)$ is computed directly from the cell counts of Bing's house by its definition $\chi(B) = \sum_n (-1)^n c_n$ as the alternating sum of the numbers c_n of n-cells, and the enumeration $c_0 = 29$, $c_1 = 51$, $c_2 = 23$ recorded above ([14, Ex. 0.2]) gives $\chi(B) = 29 - 51 + 23 = 1$. Comparing the two expressions for $\chi(B)$ forces $\mathrm{rank} H_2(B) = 0$, and since $H_2(B)$ is free abelian this gives $H_2(B) = 0$. Together with $\tilde{H}_0(B) = 0$ from path connectivity, every reduced homology group vanishes, so B is acyclic in the sense of definition 3.5.16.

Hurewicz and Whitehead finish the argument. A simply connected space with vanishing reduced homology is weakly contractible, by induction on the connectivity using theorem 5.4.9. Indeed B is 1-connected; assuming B is $(n-1)$-connected for some $n \geq 2$, the Hurewicz isomorphism $\pi_n(B) \xrightarrow{\cong} H_n(B) = 0$ gives $\pi_n(B) = 0$, so B is n-connected, and the induction yields $\pi_n(B) = 0$ for every n. The unique map $B \to \{*\}$ is therefore a weak homotopy equivalence: a π_0-bijection by path connectivity and an isomorphism on each π_n, both sides being trivial. Both B and $\{*\}$ are CW complexes, so Whitehead's theorem (theorem 5.3.5) upgrades this weak equivalence to a genuine homotopy equivalence, and $B \simeq \{*\}$ is contractible, with no contraction ever exhibited.

The contrast with $\mathbb{R}^n$ of example 5.3.6 is the whole point. For $\mathbb{R}^n$ the contraction $(v,t) \mapsto (1-t)v$ is written on sight and the homotopy-group computation only confirms what the eye already sees; for B no contraction is available to inspection, yet the identical argument, vanishing π_1 and vanishing reduced homology feeding Hurewicz and then Whitehead, settles the contractibility of both spaces by the same two lines.

[a]This route to $\pi_1(B) = 0$ is deliberately independent of contractibility. Hatcher establishes contractibility of B directly, by thickening B to a neighborhood homeomorphic to the ball D^3 and retracting ([14, Ch. 0]); quoting that for $\pi_1(B) = 0$ would make the present example circular, since contractibility is exactly the conclusion being

re-derived here through Hurewicz and Whitehead. The presentation above derives $\pi_1(B) = 0$ from the cell structure alone, with no contraction used.

Without the CW assumption, a weak homotopy equivalence need not be a homotopy equivalence, and the standard witness to this is the Warsaw circle. The point of the example is that a space can be *weakly contractible* (path connected with all homotopy groups trivial, so the map to a point is a weak homotopy equivalence) while being far from contractible, because homotopy groups simply fail to register certain pathologies of non-CW spaces:

Example 5.3.8: The Warsaw Circle

The *Warsaw circle* W is the subspace of $\mathbb{R}^2$ formed from the closed topologist's sine curve together with an arc joining its ends. Explicitly, let

$$S = \{(x, \sin(1/x)) \mid 0 < x \le 1/\pi\}, \qquad V = \{0\} \times [-1, 1]$$

so that the closure $\overline{S} = S \cup V$ is the topologist's sine curve, and let C be a simple arc in $\mathbb{R}^2$ from the point $(1/\pi, 0) \in S$ to the point $(0, -1) \in V$ that meets $\overline{S}$ only at its two endpoints and otherwise lies outside the region swept by S. Set $W = \overline{S} \cup C$.

W is path connected. The bare topologist's sine curve $\overline{S}$ is not path connected: no path in $\overline{S}$ joins a point of the oscillating part S to a point of the limit segment V, because such a path would have to traverse the infinitely many oscillations of $\sin(1/x)$ as $x \to 0$, which no continuous parametrisation can do. The arc C repairs this. A point of S is joined to a point of V by going the "other way" along C, which lies outside $\overline{S}$ and reaches V at $(0, -1)$ without ever entering the oscillating region. Thus W is path connected, and $\pi_0(W)$ is a single point.

All homotopy groups of W are trivial. The argument rests on a confinement lemma: any continuous map from a compact space into W stays clear of a tail of the oscillations. Precisely, for $a > 0$ write

$$S_a = \{(x, \sin(1/x)) \mid 0 < x < a\}$$

for the part of the oscillating graph with x-coordinate below a, an open arc of S accumulating onto the limit segment V. The claim is that for any compact, locally connected, metrizable space K and any continuous map $\beta\colon K \to W$ there exists $a > 0$ with $\beta(K) \cap S_a = \varnothing$, so β avoids S_a entirely. (Both paths $\beta\colon [0, 1] \to W$ and sphere maps $\beta\colon S^k \to W$ are instances, $[0, 1]$ and S^k being compact, locally connected, and metrizable.)

Proof of the confinement lemma. It suffices to treat K locally connected and metrizable, which covers the only cases used below, $K = [0, 1]$ and $K = S^k$. Suppose the conclusion fails: then for each i there is a point $p_i \in K$ with $\beta(p_i) \in S_{1/i}$, so the x-coordinate of $\beta(p_i)$ tends to 0. By compactness of K pass to a subsequence with $p_i \to p_*$; by continuity $\beta(p_*) = \lim_i \beta(p_i)$, and since the x-coordinates of $\beta(p_i)$ tend to 0 while the y-coordinates lie in $[-1, 1]$, the limit has x-coordinate 0, so $\beta(p_*) \in V$, say $\beta(p_*) = (0, c)$. Continuity of β at p_* together with local connectedness of K gives a connected open neighbourhood U of p_* in K with $\beta(U)$ contained in the ball B of radius $\frac{1}{2}$ about $(0, c)$ in W. Within B the set W consists of the segment of V through $(0, c)$ together with countably many disjoint vertical strands of the graph: for $c \in (-1, 1)$ the curve $y = \sin(1/x)$ crosses the height $y = c$ at a discrete set of x-values $x_1 > x_2 > \cdots \to 0$, and near V the strands run almost vertically and are separated from one another and from V by gaps bounded below, so each strand is a separate connected component of $W \cap B$ and the component containing $(0, c)$ is exactly the V-segment (the cases $c = \pm 1$ are identical, the extremal strands being even more sharply separated). The image $\beta(U)$ is connected and contains $(0, c)$, so it lies in that single V-segment and meets no strand of the graph. But $\beta(p_i) \in S_{1/i}$ lies on the graph for $p_i \in U$ with i

large, a contradiction. This proves the lemma.
The lemma forces every loop and every higher sphere into a contractible part of W. Given $k \geq 1$ and a map $g\colon S^k \to W$, the lemma applied to $\beta = g$ on the compact space $K = S^k$ produces a single $a > 0$ with $g(S^k) \cap S_a = \varnothing$. Thus g lands in $W_a := W \setminus S_a$. Removing the tail S_a deletes the accumulation, leaving three pieces: the segment V, the finite-length graph arc $G_a := \{(x, \sin(1/x)) \mid a \leq x \leq 1/\pi\}$ (each a homeomorphic copy of a closed interval, now disjoint from one another since V has $x = 0$ while G_a has $x \geq a > 0$), and the joining arc C, which runs from the endpoint $(1/\pi, 0)$ of G_a to the endpoint $(0, -1)$ of V. Concatenated end to end, V, then C, then G_a form a single embedded arc, a homeomorphic copy of $[0, 1]$ with no oscillation barrier remaining, and an arc is contractible. A map of S^k into a contractible space is null-homotopic, so $[g] = 0$ in $\pi_k(W, w_0)$. Hence $\pi_n(W, w_0) = 0$ for all $n \geq 1$. For the complete topologist's-sine-curve and Warsaw-circle computations underlying the confinement lemma and the contractibility of W_a, see [18, §24, §27] and [14, Ch. 1]. Together with path connectivity, the unique map $W \to \{*\}$ is therefore a weak homotopy equivalence: a π_0-bijection and an isomorphism on every π_n, $n \geq 1$, since both sides are trivial.

But W is not contractible, hence not homotopy equivalent to a point. The non-contractibility is detected by Čech cohomology, which (unlike singular homotopy) is built from open covers and registers the global "hole" that W shares with the ordinary circle: $\check{H}^1(W; \mathbb{Z}) \cong \mathbb{Z} \cong \check{H}^1(S^1; \mathbb{Z})$. A contractible space has trivial reduced Čech cohomology in all positive degrees, so $\check{H}^1(W) \neq 0$ rules out contractibility. The map $W \to \{*\}$ is thus a weak homotopy equivalence that is *not* a homotopy equivalence, and W is a space with the singular homotopy groups of a point but the Čech cohomology of a circle. By Whitehead's theorem (theorem 5.3.5) W cannot have the homotopy type of a CW complex, for if it did, weak contractibility would force genuine contractibility. This is exactly the obstruction the CW hypothesis removes: for CW complexes, triviality of all homotopy groups forces contractibility, whereas the non-CW space W shows it need not in general.

The Warsaw circle shows that homotopy groups are blind to certain non-CW pathologies, and so weak homotopy equivalence is strictly weaker than homotopy equivalence in general. The remedy to confine to spaces where they suffice. The mechanism for doing so is CW approximation: every space, however pathological, is weakly equivalent to a CW complex, and once the comparison is made the CW model can be studied with all the tools that the CW hypothesis unlocks, Whitehead's theorem foremost among them.

Theorem 5.3.9: CW Approximation

For every topological space X there exists a CW complex Z and a weak homotopy equivalence $f\colon Z \to X$. Given moreover a point $x_0 \in X$ with X path-connected, the pair may be chosen with a 0-cell $z_0 \in Z$ satisfying $f(z_0) = x_0$, since the construction below starts from a wedge of circles at a chosen basepoint and only ever extends that map. The pair (Z, f) is called a *CW approximation* to X, and it is unique up to homotopy equivalence: if $f'\colon Z' \to X$ is another CW approximation, there is a homotopy equivalence $h\colon Z \to Z'$ with $f' \circ h \simeq f$.

***Proof*:**
The construction builds Z by attaching cells in increasing dimension, alternately creating the homotopy classes needed for f_* to be surjective and then killing the classes needed for f_* to be injective. For clarity the construction is given for path-connected X with a chosen basepoint; the general case is the disjoint union of the constructions over the path components of X.

Step 1: The base of the construction (surjectivity in degree 1*).* Fix a basepoint $x_0 \in X$. Begin with $Z_1 = \bigvee_\alpha S^1_\alpha$, a wedge of circles, one for each generator in a chosen set of generators of $\pi_1(X, x_0)$ (if π_1 is trivial, take $Z_1 = \{z_0\}$ a point). Define $f_1 \colon Z_1 \to X$ on each circle S^1_α to be a loop representing the corresponding generator of $\pi_1(X, x_0)$. By construction $(f_1)_* \colon \pi_1(Z_1) \to \pi_1(X)$ is surjective, since its image contains a generating set. Both Z_1 and X are path connected, so the inductive invariant holds at $n = 1$:

$$(f_1)_* \colon \pi_k(Z_1) \longrightarrow \pi_k(X) \quad \text{is an isomorphism for } k < 1 \text{ and surjective for } k = 1$$

(the $k < 1$ condition being the π_0-bijection, which holds as both spaces are connected).

Step 2: The inductive step, killing the kernel in degree n (injectivity). Suppose inductively that a CW complex Z_n and a map $f_n \colon Z_n \to X$ have been constructed so that $(f_n)_*$ is an isomorphism on π_k for $k < n$ and surjective on π_n. The map $(f_n)_*$ on π_n may have a kernel. For each element β in a generating set of $\ker\big((f_n)_* \colon \pi_n(Z_n) \to \pi_n(X)\big)$, represented by a map $\gamma_\beta \colon S^n \to Z_n$, attach an $(n+1)$-cell to Z_n along γ_β, and call the result Z'_n. Attaching an $(n+1)$-cell along γ_β sets the class $[\gamma_\beta]$ to zero in $\pi_n(Z'_n)$, by the relation that a sphere bounding a disk is null-homotopic, while leaving π_k unchanged for $k < n$ (cells of dimension $n+1$ do not affect homotopy below dimension n). The map f_n extends over each attached $(n+1)$-cell because $f_n \circ \gamma_\beta$ is null-homotopic in X (its class lies in $\ker(f_n)_*$, hence is trivial in $\pi_n(X)$), and a null-homotopy is precisely an extension of $f_n \circ \gamma_\beta$ over the disk D^{n+1}. The extended map $f'_n \colon Z'_n \to X$ now induces an *isomorphism* on π_n: surjectivity is inherited from f_n, and injectivity holds because the kernel has been killed. The lower-degree isomorphisms are preserved, so $(f'_n)_*$ is an isomorphism on π_k for $k \le n$.

Step 3: The inductive step, surjectivity in degree $n+1$*.* To make $(f'_n)_*$ surjective in degree $n+1$ while preserving the isomorphisms in degrees $\le n$, attach $(n+1)$-spheres to create the missing classes. Choose a set of generators of $\pi_{n+1}(X)$; for each such generator δ, represented by a map $S^{n+1} \to X$, form the wedge with an $(n+1)$-sphere S^{n+1}_δ and extend f'_n to send S^{n+1}_δ by a representative of δ. Wedging on spheres of dimension $n+1$ does not change π_k for $k \le n$, so the isomorphisms in degrees $\le n$ are preserved, and the new spheres surject onto $\pi_{n+1}(X)$. Call the result Z_{n+1} with map f_{n+1}. By construction

$$(f_{n+1})_* \colon \pi_k(Z_{n+1}) \longrightarrow \pi_k(X) \quad \text{is an isomorphism for } k \le n \text{ and surjective for } k = n+1$$

which is the inductive invariant with n replaced by $n+1$, advancing the induction.

Step 4: Passing to the colimit. The complexes $Z_1 \subseteq Z_2 \subseteq Z_3 \subseteq \cdots$ form an increasing sequence of CW complexes, each obtained from the previous by attaching cells, and the maps f_n are compatible with the inclusions ($f_{n+1}|_{Z_n} = f_n$). Let $Z = \bigcup_n Z_n$ with the weak topology, a CW complex, and let $f \colon Z \to X$ be the map determined by the f_n (well-defined and continuous because Z has the weak topology of definition 1.2.1 and f is continuous on each Z_n). For each fixed k, the homotopy group $\pi_k(Z)$ is computed from a compact image, so any map $S^k \to Z$ and any homotopy $S^k \times I \to Z$ land in some finite stage Z_n with $n > k$; consequently $\pi_k(Z) = \pi_k(Z_n)$ for all $n > k$. At such a stage $(f_n)_*$ is an isomorphism on π_k, so $f_* \colon \pi_k(Z) \to \pi_k(X)$ is an isomorphism for every k. Therefore f is a weak homotopy equivalence and Z is a CW complex, establishing existence.

Step 5: Uniqueness up to homotopy equivalence. Let $f \colon Z \to X$ and $f' \colon Z' \to X$ be two CW approximations. Form the mapping cylinder $M_{f'}$ of f' and regard X as a deformation retract of it; then $f' \colon Z' \hookrightarrow M_{f'}$ is, up to the homotopy equivalence $M_{f'} \simeq X$, the inclusion of a subcomplex. The composite $Z \to X \hookrightarrow M_{f'}$ is a map from the CW complex Z into $M_{f'}$ that is a weak homotopy equivalence (both f and the inclusion $X \hookrightarrow M_{f'}$ are). Since Z' is a CW complex and $Z' \hookrightarrow M_{f'}$ is a weak homotopy equivalence with $M_{f'} \simeq X$, both Z and Z' are CW complexes weakly equivalent over X; the map $Z \to M_{f'}$ may be homotoped into the subcomplex Z' and, being a weak homotopy equivalence between CW complexes, is a homotopy equivalence $h\colon Z \to Z'$ by Whitehead's theorem (theorem 5.3.5), with $f' \circ h \simeq f$ by construction. This gives uniqueness up to homotopy equivalence, completing the proof.

CW approximation is the structural reason that algebraic topology can restrict to CW complexes without loss of generality at the level of homotopy theory. Any space whatsoever is replaced by a CW complex with the same homotopy groups, the comparison being a single map inducing all the isomorphisms at once. Combined with Whitehead's theorem, this means the homotopy category of all spaces, localised at the weak homotopy equivalences, is equivalent to the homotopy category of CW complexes. The cellular structure of the approximation also makes the construction compatible with cellular homology of section 3.5: the homology of the CW approximation Z computes the singular homology of X, because a weak homotopy equivalence induces isomorphisms on singular homology (a fact that follows from the Hurewicz theorem and induction, or directly from the homotopy invariance of singular homology applied to the cells).

The construction admits a relative refinement that is constantly used in obstruction theory and the comparison of pairs. One often needs not just a CW approximation to a space but a CW approximation to a map, or an approximation that already agrees with a given CW structure on a subspace, controlled by a connectivity parameter.

Proposition 5.3.10: Relative and n-Connected CW Approximation

Let (X, A) be a pair of spaces and let A already be a CW complex. Then there exists a CW pair (Z, A) extending the CW structure of A and a weak homotopy equivalence $f\colon Z \to X$ restricting to the inclusion $A \hookrightarrow X$ on A. Moreover, given any integer $n \geq 0$, the approximation may be taken to be *n-connected*: Z can be built so that f induces isomorphisms on π_k for $k > n$ and a surjection on π_n, with Z obtained from A by attaching cells of dimension greater than n only. Such a pair (Z, A) is called an *n-connected CW model* of (X, A).

***Proof*:**
The argument is the relative form of the construction in theorem 5.3.9, started not from a wedge of spheres but from the given subcomplex A, with all cell attachments performed away from A.

Step 1: The relative construction. Run the cell-attachment procedure of theorem 5.3.9 with A as the initial stage in place of Z_1, and with the map $f|_A$ taken to be the inclusion $A \hookrightarrow X$. At each stage the new cells are attached to the part of the complex built so far without touching A, exactly as in Steps 3 and 4 of that proof: surjectivity in each degree is arranged by wedging on spheres (attached at the basepoint, away from A), and injectivity by attaching cells of one higher dimension along maps that miss the interior of A. The inductive hypothesis is now that f restricts to the inclusion on A and induces an isomorphism on π_k for k below the current dimension and a surjection at it. Passing to the colimit as in Step 5 gives a CW pair (Z, A) with A a subcomplex and a weak homotopy equivalence $f\colon Z \to X$ extending $A \hookrightarrow X$.

Step 2: The connectivity refinement. For the n-connected version, begin the attachment process only in dimensions exceeding n: take Z to agree with A through dimension n and attach cells of dimension $n+1$ and higher to fix surjectivity and injectivity of f_* in degrees $\geq n+1$, never adding cells of dimension $\leq n$. Then (Z, A) has no relative cells of dimension $\leq n$, so the pair is n-connected: the relative homotopy groups $\pi_k(Z, A)$ vanish for $k \leq n$ because a CW pair with cells only in dimensions $> n$ has $\pi_k(Z, A) = 0$ for $k \leq n$ (any map $(D^k, \partial D^k) \to (Z, A)$ with $k \leq n$ compresses into A by the compression lemma lemma 5.3.3, since Z has no relative cells in those dimensions). The induced map f_* is then an isomorphism on π_k for $k > n$ and a surjection on π_n, as the lower cells were left untouched and only the higher homotopy was adjusted, as we sought to show.

The relative version turns CW approximation from a statement about single spaces into a tool for comparing spaces along a map, while the connectivity parameter lets one approximate only the part of the homotopy above a chosen dimension. Both features are exactly what the next sections require: the Hurewicz theorem (theorem 5.4.9) is phrased for n-connected spaces, and the construction of Eilenberg-MacLane spaces and Postnikov towers proceeds by attaching cells in prescribed dimensions to control homotopy degree by degree, which is the n-connected model applied repeatedly.

Example 5.3.11: An n-Connected Model Kills Low Homotopy

Take $X = S^2$ and $A = \{*\}$ a basepoint, and ask for a 1-connected CW model. The construction starts from the point $A = \{*\}$ and attaches cells only in dimensions ≥ 2. The first stage attaches a single 2-cell to realise the generator of $\pi_2(S^2) \cong \mathbb{Z}$ (giving S^2 itself), then higher cells in dimensions ≥ 3 to correct $\pi_3, \pi_4, \ldots$, which are nontrivial for S^2 (for instance $\pi_3(S^2) \cong \mathbb{Z}$, generated by the Hopf map). The resulting model Z is a CW complex with no 1-cells, so $(Z, \{*\})$ is 1-connected, and $f\colon Z \to S^2$ is an isomorphism on π_k for $k \geq 2$ and the trivial (hence surjective) map on $\pi_1 = 0$. The point of the example is that the connectivity parameter controls the dimensions in which cells appear: demanding a 1-connected model forbids 1-cells, which is consistent with S^2 being simply connected, and the entire homotopical content lives in dimension 2 and above, exactly where the cells of Z sit.

Exercise 5.3.1

1. Let $f\colon X \to Y$ be a map of path-connected spaces inducing an isomorphism on π_n for all $n \geq 1$ but where X and Y are not assumed to be CW complexes. Show by example that f need not be a homotopy equivalence, and explain which hypothesis of Whitehead's theorem fails.

2. Let (X, A) be a CW pair such that the inclusion $A \hookrightarrow X$ is a weak homotopy equivalence. Using the compression lemma, prove directly that A is a deformation retract of X, without invoking the mapping-cylinder reduction.

3. Suppose X is a CW complex with $\pi_n(X, x_0) = 0$ for all $n \geq 0$ (a single path component, all higher groups trivial). Prove that X is contractible. Where does the argument use that X is a CW complex?

4. In the CW approximation construction, the first stage attaches 1-cells to surject onto $\pi_1(X)$ and the second attaches 2-cells to kill the kernel. Carry out these first two stages explicitly for X a space with $\pi_1(X) \cong \mathbb{Z}/3\mathbb{Z}$ and identify the resulting 2-complex Z_2 up to homotopy.

5. Let $f\colon Z \to X$ be a weak homotopy equivalence with Z a CW complex. Prove that f induces an isomorphism on singular homology $H_n(Z) \to H_n(X)$ for all n, assuming the Hurewicz theorem (theorem 5.4.9) for simply connected spaces and reducing to that case. (Hint: relate H_n to π_n through the approximation and use induction on n.)

6. Construct an n-connected CW model of the pair (D^{n+1}, S^n) for $n \geq 1$, where $S^n = \partial D^{n+1}$ carries its standard CW structure. Identify the relative cells and verify that the model is n-connected.

5.4 The Hurewicz Theorem

The higher homotopy groups $\pi_n(X)$ of definition 5.1.1 encode the homotopy classes of maps from spheres into X, and the homology groups $H_n(X)$ of chapter 3 encode cycles modulo boundaries. These two families of invariants are manufactured from completely different starting points. The Hurewicz theorem connects them: in the lowest dimension where a space carries any homotopy, the homotopy and homology groups agree, and the agreement is realized by an explicit map. In dimension 1 this map exhibits $H_1(X)$ as the abelianization of $\pi_1(X)$; in higher dimensions, where the homotopy groups are already abelian, it is an isomorphism on the first nonvanishing level. The theorem is the principal computational tool linking the two theories.

The construction begins with a canonical generator of the homology of a sphere. By section 3.5 the reduced homology $\tilde{H}_n(S^n)$ is infinite cyclic, so it has exactly two generators; fixing one of them once and for all is what turns a homotopy class into a homology class.

Definition 5.4.1: Fundamental Class of the Sphere

For each $n \geq 1$ fix a generator

$$\iota_n \in H_n(S^n) \cong \mathbb{Z}$$

called the *fundamental class* of S^n. Concretely, give S^n the Δ-complex structure with two n-simplices Δ_1^n and Δ_2^n glued along their common boundary; then ι_n is the homology class of the cycle $\Delta_1^n - \Delta_2^n$, oriented so that ι_n is the image of ι_{n-1} under the connecting isomorphism $H_n(D^n, S^{n-1}) \xrightarrow{\partial} \tilde{H}_{n-1}(S^{n-1})$ of the long exact sequence of the pair (D^n, S^{n-1}).

The choice of ι_n is the choice of an orientation of S^n. Compatibility under the boundary map ∂ ties the fundamental classes in consecutive dimensions together, which is precisely what makes the Hurewicz map behave well under suspension. Nothing in the theory depends on the particular orientation, only on having fixed one; reversing the choice in every dimension replaces the Hurewicz map by its negative.

With a fixed generator in hand, every map $f\colon S^n \to X$ produces a homology class in X: push ι_n forward along f. Homotopic maps induce the same map on homology, so this depends only on the homotopy class of f. This is the Hurewicz homomorphism.

Definition 5.4.2: Hurewicz Homomorphism

Let X be a space with basepoint x_0 and let $n \geq 1$. The *Hurewicz homomorphism* is the map

$$h\colon \pi_n(X, x_0) \longrightarrow H_n(X), \qquad h([f]) = f_*(\iota_n)$$

sending the homotopy class of a basepoint-preserving map $f\colon (S^n, s_0) \to (X, x_0)$ to the image of the fundamental class ι_n of definition 5.4.1 under $f_*\colon H_n(S^n) \to H_n(X)$.

The Hurewicz map is the device that converts the spherical, homotopy-theoretic information in π_n

into the linear-algebraic information in H_n; everything in this section is an analysis of how much is lost in the conversion.

The map h is well defined because a homotopy $f \simeq g$ of based maps is in particular a free homotopy, hence $f_* = g_*$ on homology by homotopy invariance (chapter 3), so $f_*(\iota_n) = g_*(\iota_n)$. That h is a homomorphism requires an argument: the group operation on π_n is induced by the pinch map $c\colon S^n \to S^n \vee S^n$ that collapses an equator to a point, and on homology this pinch sends ι_n to (ι_n, ι_n) because the two halves of S^n map to the two wedge summands by maps of degree 1:

Proposition 5.4.3: The Hurewicz Map is a Homomorphism

For every $n \geq 1$ the map $h\colon \pi_n(X, x_0) \to H_n(X)$ of definition 5.4.2 is a group homomorphism. It is natural in X: a based map $\varphi\colon (X, x_0) \to (Y, y_0)$ gives a commuting square $\varphi_* \circ h = h \circ \varphi_*$.

***Proof*:**
Step 1: The pinch map on homology. Recall from definition 5.1.1 that the sum $[f] + [g]$ in $\pi_n(X, x_0)$ is represented by the composite

$$S^n \xrightarrow{c} S^n \vee S^n \xrightarrow{f \vee g} X$$

where c collapses the equatorial $S^{n-1} \subset S^n$ to the wedge point, sending the upper hemisphere onto the first summand and the lower hemisphere onto the second. Write $q_1, q_2\colon S^n \vee S^n \to S^n$ for the two maps that collapse one summand to the basepoint and are the identity on the other. By the wedge axiom for homology (a consequence of the good-pair long exact sequence of theorem 3.4.11, since $S^n \vee S^n$ is the quotient of $S^n \sqcup S^n$ by a basepoint pair) the map

$$(q_{1*}, q_{2*})\colon H_n(S^n \vee S^n) \xrightarrow{\cong} H_n(S^n) \oplus H_n(S^n)$$

is an isomorphism. Each composite $q_i \circ c\colon S^n \to S^n$ collapses one hemisphere to a point and is a homeomorphism on the interior of the other, hence has degree ± 1; orienting consistently, $q_i \circ c$ has degree 1, so $(q_i \circ c)_*(\iota_n) = \iota_n$. Therefore

$$c_*(\iota_n) = (\iota_n, \iota_n) \in H_n(S^n) \oplus H_n(S^n)$$

under the identification above.

Step 2: Adding the two maps. Under the same identification the inclusions of the two summands realize $H_n(S^n) \oplus H_n(S^n)$ inside $H_n(S^n \vee S^n)$, and $(f \vee g)_*$ acts on the two copies of ι_n by f_* and g_* respectively. Hence

$$h([f] + [g]) = (f \vee g)_* \, c_*(\iota_n) = (f \vee g)_*(\iota_n, \iota_n) = f_*(\iota_n) + g_*(\iota_n) = h([f]) + h([g])$$

so h is a homomorphism.

Step 3: Naturality. For a based map $\varphi\colon (X, x_0) \to (Y, y_0)$ and $[f] \in \pi_n(X, x_0)$ one has $\varphi_*[f] = [\varphi \circ f]$ by definition of the induced map on π_n, and functoriality of homology gives

$$h(\varphi_*[f]) = (\varphi \circ f)_*(\iota_n) = \varphi_* \, f_*(\iota_n) = \varphi_* \, h([f])$$

so the square commutes, completing the proof.

Naturality is what makes h a transformation of functors $\pi_n \Rightarrow H_n$ on the category of based spaces, and it is the key result for every computation in this section: once h is known to be an isomorphism for one space, naturality transports the conclusion across any map. The remaining work is to identify the kernel and cokernel of h in the cases that matter.

Example 5.4.4: The Hurewicz Map for the Circle

Take $X = S^1$ and $n = 1$. By the computation of the fundamental group in chapter 2, $\pi_1(S^1, s_0) \cong \mathbb{Z}$ generated by the class $[\omega]$ of the loop $\omega(t) = (\cos 2\pi t, \sin 2\pi t)$, which is the identity map of S^1. Homology gives $H_1(S^1) \cong \mathbb{Z}$ generated by the fundamental class ι_1. The Hurewicz map sends

$$h([\omega]) = \omega_*(\iota_1) = \mathrm{id}_*(\iota_1) = \iota_1$$

a generator, and more generally the class $[\omega]^k$ is represented by the degree-k map $z \mapsto z^k$, whose induced map on H_1 is multiplication by k, so $h([\omega]^k) = k\,\iota_1$. Thus $h\colon \mathbb{Z} \to \mathbb{Z}$ is the identity, an isomorphism. Since $\pi_1(S^1)$ is already abelian, no abelianization is needed here, and the circle is the simplest instance of the dimension-1 Hurewicz theorem proved next.

The circle suggests the general dimension-1 statement, but a subtlety appears as soon as the fundamental group is nonabelian: $H_1(X)$ is always abelian, while $\pi_1(X)$ need not be, so the best one can hope for in dimension 1 is that h collapses exactly the commutator subgroup. This is the abelianization theorem.

Theorem 5.4.5: Hurewicz Theorem in Dimension One

Let X be a path-connected space with basepoint x_0. Then the Hurewicz homomorphism

$$h\colon \pi_1(X, x_0) \longrightarrow H_1(X)$$

is surjective, and its kernel is the commutator subgroup $[\pi_1, \pi_1]$. Consequently h induces an isomorphism

$$\pi_1(X, x_0)^{\mathrm{ab}} = \pi_1(X, x_0)/[\pi_1, \pi_1] \xrightarrow{\cong} H_1(X)$$

from the abelianization of the fundamental group onto the first homology group.

***Proof*:**

Throughout, regard a loop $\gamma\colon I \to X$ based at x_0 simultaneously as a singular 1-simplex. Since $\gamma(0) = \gamma(1) = x_0$, this 1-simplex is a cycle: $\partial\gamma = \gamma(1) - \gamma(0) = 0$ in the singular chain group $C_0(X)$. For $n = 1$ the fundamental class $\iota_1 \in H_1(S^1)$ is represented by the identity loop, so $h([\gamma])$ is exactly the homology class $[\gamma] \in H_1(X)$ of this 1-cycle. The whole proof is the translation between loop concatenation and chain addition.

Step 1: h is well defined as a homomorphism into H_1, restated for cycles. By proposition 5.4.3 the map h is a homomorphism. Made explicit, this asserts that for loops α, β based at x_0 the 1-chains are homologous,

$$\alpha \cdot \beta \;\sim\; \alpha + \beta$$

where $\alpha \cdot \beta$ is the concatenated loop. To see this directly, consider the singular 2-simplex $\sigma\colon \Delta^2 \to X$ obtained as follows. Label the vertices of Δ^2 as v_0, v_1, v_2. Let $\rho\colon \Delta^2 \to [v_0, v_2]$ be the affine map collapsing v_1 to the midpoint of $[v_0, v_2]$ and fixing v_0, v_2, and set $\sigma = (\alpha \cdot \beta) \circ \rho$ after identifying $[v_0, v_2] \cong I$ affinely. Then ρ sends $[v_0, v_1]$ onto the first half $[v_0, \mathrm{mid}]$ and $[v_1, v_2]$

onto the second half $[\mathrm{mid}, v_2]$, so the edge $[v_0, v_1]$ of σ is α, the edge $[v_1, v_2]$ is β, and the edge $[v_0, v_2]$ is $\alpha \cdot \beta$. Hence

$$\partial\sigma = \beta - (\alpha \cdot \beta) + \alpha$$

so $(\alpha \cdot \beta) - \alpha - \beta = -\partial\sigma$ is a boundary, giving $\alpha \cdot \beta \sim \alpha + \beta$ in $H_1(X)$. In particular h is a homomorphism, and since $H_1(X)$ is abelian the commutator subgroup $[\pi_1, \pi_1]$ lies in $\ker h$.

Step 2: Two reductions. Record three elementary homology relations among loops, all proved by exhibiting an explicit 2-chain.

1. If γ is the constant loop e_{x_0}, then $\gamma \sim 0$: the constant singular 2-simplex at x_0 has boundary $e_{x_0} - e_{x_0} + e_{x_0} = e_{x_0}$, so e_{x_0} is a boundary.
2. If $\gamma \simeq \gamma'$ rel endpoints, then $\gamma \sim \gamma'$ as 1-cycles: a path homotopy is a map $\Delta^1 \times I \to X$, and subdividing the prism $\Delta^1 \times I$ into two 2-simplices expresses $\gamma - \gamma'$ as a boundary plus chains supported at the (constant) endpoints, which vanish by (a).
3. For the reverse loop $\bar{\gamma}(t) = \gamma(1-t)$ one has $\bar{\gamma} \sim -\gamma$: by (a) and Step 1, $\gamma \cdot \bar{\gamma} \sim \gamma + \bar{\gamma}$ and $\gamma \cdot \bar{\gamma} \simeq e_{x_0}$ rel endpoints, so $\gamma + \bar{\gamma} \sim 0$.

Relations (a) and (b) say h is the well-defined map of definition 5.4.2; relation (c) records inverses.

Step 3: Surjectivity. Let $z = \sum_i n_i \sigma_i$ be a singular 1-cycle representing a class in $H_1(X)$, where each $\sigma_i \colon I \to X$ is a singular 1-simplex (a path) and $n_i \in \mathbb{Z}$. Splitting σ_i with $n_i < 0$ into $-n_i$ copies of its reverse, using relation (c), one may assume every $n_i = 1$, so $z = \sum_i \sigma_i$ is a sum of paths. The cycle condition $\partial z = 0$ reads

$$\sum_i \big(\sigma_i(1) - \sigma_i(0)\big) = 0 \in C_0(X)$$

meaning that, counted with multiplicity, each point of X appears as an endpoint exactly as often as it appears as a starting point among the σ_i. Since X is path-connected, choose for every point p occurring among the $\sigma_i(0), \sigma_i(1)$ a path λ_p from x_0 to p, with $\lambda_{x_0} = e_{x_0}$ the constant path. Replace each σ_i by the loop

$$\tilde{\sigma}_i = \lambda_{\sigma_i(0)} \cdot \sigma_i \cdot \overline{\lambda_{\sigma_i(1)}}$$

based at x_0. By Step 1, $\tilde{\sigma}_i \sim \lambda_{\sigma_i(0)} + \sigma_i - \lambda_{\sigma_i(1)}$ as 1-chains (using relation (c) of Step 2 for the reversed path). Summing over i,

$$\sum_i \tilde{\sigma}_i \ \sim\ \sum_i \sigma_i + \sum_i \big(\lambda_{\sigma_i(0)} - \lambda_{\sigma_i(1)}\big)$$

and the second sum is $-\sum_p \big(\#\{p \text{ as terminal}\} - \#\{p \text{ as initial}\}\big)\lambda_p$, which is zero by the cycle condition $\partial z = 0$. Hence $z = \sum_i \sigma_i \sim \sum_i \tilde{\sigma}_i$. The latter is a sum of loops based at x_0, so by Step 1 it equals $h\big(\big[\prod_i \tilde{\sigma}_i\big]\big)$. Thus every homology class is in the image of h, and h is surjective.

Step 4: The kernel. Let $\Phi \colon \pi_1(X, x_0)^{\mathrm{ab}} \to H_1(X)$ be the homomorphism induced by h (well defined by Step 1, since $[\pi_1, \pi_1] \subseteq \ker h$). Step 3 shows Φ is surjective; it remains to construct an inverse, which forces Φ to be injective. Define

$$\Psi \colon C_1(X) \longrightarrow \pi_1(X, x_0)^{\mathrm{ab}}$$

on a singular 1-simplex σ (a path) by $\Psi(\sigma) = \left[\lambda_{\sigma(0)} \cdot \sigma \cdot \overline{\lambda_{\sigma(1)}}\right]$, the class of the loop obtained by closing σ up through the chosen paths λ_p as in Step 3, and extend additively to $C_1(X)$, landing in the abelian group π_1^{ab}. Two facts pin down Ψ.

Step 4a: Ψ kills boundaries. For a singular 2-simplex $\tau\colon \Delta^2 \to X$ with edges $a = \tau|_{[v_0,v_1]}$, $b = \tau|_{[v_1,v_2]}$, $c = \tau|_{[v_0,v_2]}$, the boundary is $\partial\tau = b - c + a$. Closing each edge up through the λ_p and using $\Psi(a)\Psi(b) = \Psi(c)$ in π_1^{ab} (because the closed-up loops satisfy $\hat{a} \cdot \hat{b} \simeq \hat{c}$ rel endpoints, the homotopy being τ itself restricted across the 2-simplex) gives

$$\Psi(\partial\tau) = \Psi(a) + \Psi(b) - \Psi(c) = 0$$

in the additive notation of the abelian group. Hence Ψ descends to a homomorphism $\Psi\colon H_1(X) \to \pi_1(X, x_0)^{\mathrm{ab}}$.

Step 4b: Ψ inverts Φ. For a loop γ based at x_0, viewed as a 1-cycle, $\Psi([\gamma]) = [\lambda_{x_0} \cdot \gamma \cdot \overline{\lambda_{x_0}}] = [\gamma]$ since $\lambda_{x_0} = e_{x_0}$. Therefore $\Psi \circ \Phi = \mathrm{id}$ on π_1^{ab}, because $\Phi([\gamma]) = [\gamma] \in H_1(X)$ and Ψ sends it back to $[\gamma]$. A surjective homomorphism with a left inverse is an isomorphism, so Φ is an isomorphism and $\ker h = [\pi_1, \pi_1]$, completing the proof.

Example 5.4.6: First Homology of the Figure Eight and the Genus-Two Surface
Two applications of theorem 5.4.5 compute H_1 from a known fundamental group.

1. For the wedge $X = S^1 \vee S^1$, van Kampen (section 2.2) gives $\pi_1(X) \cong F_2$, the free group on two generators a, b. Its abelianization is $\mathbb{Z}^2$, so

$$H_1(S^1 \vee S^1) \cong F_2^{\mathrm{ab}} \cong \mathbb{Z}^2$$

   recovering the homology of the figure eight without any chain computation.

2. For the closed orientable surface Σ_2 of genus 2, the fundamental group has presentation

$$\pi_1(\Sigma_2) = \langle a_1, b_1, a_2, b_2 \,\big|\, [a_1, b_1][a_2, b_2] = 1\rangle$$

   Abelianizing kills all commutators, so the single relation $[a_1, b_1][a_2, b_2] = 1$ becomes trivial, and the abelianization is the free abelian group on the four generators:

$$H_1(\Sigma_2) \cong \mathbb{Z}^4$$

   in agreement with the homology computed via the Δ-complex structure in chapter 3.

Dimension 1 is special precisely because π_1 may fail to be abelian. In every higher dimension π_n is already abelian (theorem 5.1.4), and one expects no loss of information at the first level where homotopy appears. The general Hurewicz theorem confirms this. The notion of connectivity isolates “the first level where homotopy appears.”

Definition 5.4.7: n-Connected Space

A space X is *n-connected* (for $n \geq 0$) if it is nonempty and $\pi_i(X, x_0) = 0$ for all $0 \leq i \leq n$ and all basepoints x_0. Equivalently, every map $S^i \to X$ with $i \leq n$ extends to a map $D^{i+1} \to X$. A 0-connected space is a path-connected space; a 1-connected space is a simply connected space.

To say X is $(n-1)$-connected is to say the first non-trivial homotopy is in dimension n. The Hurewicz theorem asserts that the first nontrivial homology group appears in the same dimension, and that the two agree there. Before the theorem, one preparatory fact: connectivity in homotopy forces vanishing of low-dimensional homology, which is half of the statement and is proved by a direct cellular argument.

Lemma 5.4.8: Connectivity Kills Low Homology

Let $n \geq 1$ and let X be an $(n-1)$-connected space. Then the reduced homology groups satisfy $\tilde{H}_i(X) = 0$ for all $i < n$.

***Proof*:**
Step 1: Reduce to a CW complex. By the CW approximation theorem (theorem 5.3.9) there is a CW complex Z and a weak homotopy equivalence $\varphi\colon Z \to X$, that is, a map inducing isomorphisms on all homotopy groups. A weak homotopy equivalence induces isomorphisms on all singular homology groups; this is a standalone consequence of CW approximation, proved by mapping-cylinder reduction and excision without any appeal to the Hurewicz theorem ([14, Proposition 4.21]), so no circularity arises in using it here. Since $\pi_i(X) = 0$ for $i \leq n-1$ and φ is a π_*-isomorphism, Z is also $(n-1)$-connected and has the same homology as X. It therefore suffices to prove the lemma when $X = Z$ is a CW complex.

Step 2: Replace X by a CW complex with no low-dimensional cells. A refinement of the CW approximation construction (theorem 5.3.9) produces, for any $(n-1)$-connected CW complex, a homotopy-equivalent CW complex Y with a single 0-cell and no cells in dimensions $1, 2, \ldots, n-1$. One realizes Y by the same cell-attaching procedure used to build a CW approximation, choosing at each stage generators in the lowest dimension and cancelling redundant cells; the precise statement is [14, Corollary 4.16, Proposition 4.15]. Homotopy equivalent spaces have isomorphic homology, so it suffices to treat Y.

Step 3: Cellular homology has nothing in low degrees. For the CW complex Y the cellular chain complex computes homology (section 3.5), and its degree-i group is free abelian on the i-cells of Y. Since Y has a single 0-cell and no cells in dimensions 1 through $n-1$, the cellular chain groups satisfy $C_0^{\text{cell}}(Y) = \mathbb{Z}$ and $C_i^{\text{cell}}(Y) = 0$ for $1 \leq i \leq n-1$. Hence $H_i(Y) = 0$ for $1 \leq i \leq n-1$, and $\tilde{H}_0(Y) = 0$ because Y is path-connected. Transporting back through the homotopy equivalences of Steps 1 and 2, $\tilde{H}_i(X) = 0$ for all $i < n$, as we sought to show.

The lemma gives the vanishing half of the Hurewicz theorem; the substantive half is the isomorphism at the first nonvanishing level. The proof of the full theorem proceeds by induction on connectivity, reducing the general case to the model situation of a wedge of spheres, where both sides can be computed by hand.

Theorem 5.4.9: Hurewicz Theorem

Let $n \geq 2$ and let X be an $(n-1)$-connected space with basepoint x_0. Then:

1. the reduced homology groups vanish, $\tilde{H}_i(X) = 0$ for all $i < n$; and
2. the Hurewicz homomorphism

$$h\colon \pi_n(X, x_0) \xrightarrow{\cong} H_n(X)$$

is an isomorphism.

Together with theorem 5.4.5 (the case $n = 1$, where h induces an isomorphism on abelianization), this identifies the first nonvanishing homotopy group of a simply connected space with its first nonvanishing homology group.

***Proof*:**
Claim (1) is lemma 5.4.8. The work is claim (2). By CW approximation (theorem 5.3.9) and naturality of h (proposition 5.4.3), it suffices to prove the isomorphism for a CW complex, and by Step 2 of lemma 5.4.8 one may take X to be a CW complex with a single 0-cell and no cells in dimensions $1, \ldots, n-1$. The proof is induction on the connectivity, organized around the structure of such a complex.

Step 1: The case of a wedge of n-spheres. Suppose first $X = \bigvee_\alpha S^n_\alpha$ is a wedge of n-spheres, $n \geq 2$. Each summand is simply connected (as $n \geq 2$), and a wedge of simply connected spaces is simply connected, so $\pi_1(X) = 0$ and the action of π_1 on π_n is trivial. By the wedge axiom for homology (theorem 3.4.11),

$$H_n(X) = \bigoplus_\alpha H_n(S^n_\alpha) = \bigoplus_\alpha \mathbb{Z}\,\iota^\alpha_n$$

is free abelian on the fundamental classes of the summands (theorem 3.4.11 applied to the wedge). On the homotopy side, the inclusions $S^n_\alpha \hookrightarrow X$ define classes $[j_\alpha] \in \pi_n(X)$, and $\pi_n\big(\bigvee_\alpha S^n_\alpha\big)$ is free abelian on the $[j_\alpha]$ in this first nontrivial dimension. To see this for a finite wedge, the inclusion $\bigvee_\alpha S^n_\alpha \hookrightarrow \prod_\alpha S^n_\alpha$ into the product differs from a homeomorphism only by cells of dimension $\geq 2n$ (the wedge is the n-skeleton of the product CW structure), so it induces an isomorphism on π_n for $n \geq 2$; since $\pi_n\big(\prod_\alpha S^n_\alpha\big) = \bigoplus_\alpha \pi_n(S^n_\alpha)$ is free abelian on the factor inclusions (homotopy groups commute with products), the same holds for the wedge. An arbitrary wedge reduces to the finite case because any map $S^n \to \bigvee_\alpha S^n_\alpha$ has compact image meeting only finitely many summands. The Hurewicz map sends

$$h([j_\alpha]) = (j_\alpha)_*(\iota_n) = \iota^\alpha_n$$

a generator of the α-summand of $H_n(X)$, so h carries a basis to a basis and is an isomorphism. The single sphere $X = S^n$ is the case of a one-element wedge.

Step 2: Pairs and the relative reduction. For the inductive step, present X by its cells: write $X = X^{(n)} \cup (\text{cells of dimension} > n)$ where, by the normalization above, the n-skeleton $X^{(n)} = \bigvee_\alpha S^n_\alpha$ is a wedge of n-spheres (a CW complex with one 0-cell and only n-cells), and X is obtained from it by attaching cells of dimension $\geq n+1$. Cells of dimension $\geq n+2$ affect neither π_n nor H_n (attaching a k-cell with $k \geq n+2$ changes only π_i and H_i for $i \geq k-1 \geq n+1$), so for computing

π_n and H_n one may assume X is obtained from $W := \bigvee_\alpha S^n_\alpha$ by attaching $(n+1)$-cells along maps $g_\beta\colon S^n \to W$.

Step 3: Both groups as the same cokernel. Attaching the $(n+1)$-cells e_β along $g_\beta\colon S^n \to W$ gives, on homotopy and on homology, parallel exact sequences. On homology, the cellular chain complex of X in the relevant range is

$$\mathbb{Z}^{\{\beta\}} = C^{\text{cell}}_{n+1}(X) \xrightarrow{d_{n+1}} C^{\text{cell}}_n(X) = \bigoplus_\alpha \mathbb{Z}\,\iota^\alpha_n \to 0$$

where $d_{n+1}(e_\beta) = \sum_\alpha (\deg_{\beta\alpha})\,\iota^\alpha_n$ and $\deg_{\beta\alpha}$ is the degree of the composite $S^n \xrightarrow{g_\beta} W \to S^n_\alpha$ collapsing the other summands. Hence

$$H_n(X) = \operatorname{coker}\Big(\mathbb{Z}^{\{\beta\}} \xrightarrow{d_{n+1}} \bigoplus_\alpha \mathbb{Z}\,\iota^\alpha_n\Big)$$

On homotopy, the relative homotopy long exact sequence of the pair (X, W) (proposition 5.1.11) reads

$$\pi_{n+1}(X, W) \xrightarrow{\partial} \pi_n(W) \xrightarrow{i_*} \pi_n(X) \longrightarrow \pi_n(X, W)$$

Since X is built from W by attaching $(n+1)$-cells, the relative group $\pi_n(X, W) = 0$: every map $(D^n, S^{n-1}) \to (X, W)$ compresses into W because, after cellular approximation, its image meets no cell of $X \setminus W$ in dimension $\le n$ (by the compression lemma lemma 5.3.3). Hence $i_*\colon \pi_n(W) \to \pi_n(X)$ is surjective. Its kernel is computed directly by cellular approximation, with no appeal to the Hurewicz theorem: a based map $S^n \to X$ that is nullhomotopic in X can be pushed into W and its nullhomotopy made cellular, so the kernel of i_* is exactly the subgroup of $\pi_n(W)$ generated by the attaching classes $[g_\beta]$, where $g_\beta\colon S^n \to W$ is the attaching map of the cell e_β (these are the classes that bound a disc in X, namely the cell e_β). Therefore

$$\pi_n(X) = \pi_n(W)/\langle [g_\beta] \rangle = \operatorname{coker}\big(\partial\colon \pi_{n+1}(X, W) \to \pi_n(W)\big)$$

where $\partial[e_\beta] = [g_\beta] \in \pi_n(W)$; the simple connectivity of W (here $n \ge 2$) is what makes $\langle [g_\beta] \rangle$ a subgroup rather than a normal closure with π_1-action.

Step 4: Matching the two cokernels. By Step 1 the Hurewicz map $h_W\colon \pi_n(W) \to H_n(W)$ is an isomorphism carrying $[j_\alpha] \mapsto \iota^\alpha_n$. Under this isomorphism the attaching class $[g_\beta] \in \pi_n(W) = \bigoplus_\alpha \mathbb{Z}[j_\alpha]$ has coordinates $h_W([g_\beta]) = (g_\beta)_*(\iota_n) = \sum_\alpha \deg_{\beta\alpha}\,\iota^\alpha_n = d_{n+1}(e_\beta)$, because the α-coordinate of $(g_\beta)_*(\iota_n)$ is exactly the degree $\deg_{\beta\alpha}$ of the collapsed composite. Thus h_W identifies the relation subgroup $\langle [g_\beta] \rangle \subseteq \pi_n(W)$ with the image of d_{n+1} in $H_n(W)$. By naturality of h (proposition 5.4.3) applied to the inclusion $i\colon W \hookrightarrow X$, the square

$$\begin{tikzcd}
\pi_n(W) \arrow[r, "i_*"] \arrow[d, "h_W", "\cong"'] & \pi_n(X) \arrow[d, "h"] \\
H_n(W) \arrow[r, "i_*"] & H_n(X)
\end{tikzcd}$$

commutes, and passing to the cokernels of the two vertical-then-horizontal relation subgroups, h is the induced map

$$\pi_n(X) = \pi_n(W)/\langle [g_\beta] \rangle \xrightarrow{\bar{h}_W} H_n(W)/\mathrm{im} d_{n+1} = H_n(X)$$

Since h_W is an isomorphism carrying $\langle [g_\beta] \rangle$ onto $\mathrm{im} d_{n+1}$, the induced map $\bar{h}_W$ on quotients is an isomorphism, and it is h. Therefore $h\colon \pi_n(X) \to H_n(X)$ is an isomorphism, completing the proof.

The theorem says homology is the first nonvanishing homotopy group, made abelian and made computable. Above the connectivity threshold the two invariants can diverge: $\pi_4(S^2) \cong \mathbb{Z}/2$ while $H_4(S^2) = 0$. What the theorem secures is a foothold: knowing that X is $(n-1)$-connected, one computes $\pi_n(X)$ by the often-easier homology functor, and this single input feeds the entire machinery of obstruction theory and Postnikov towers developed later in this chapter. The proof's mechanism, comparing the homotopy and homology cokernels of the same attaching maps, is the template for the relative version, which upgrades the statement to a pair:

Theorem 5.4.10: Relative Hurewicz Theorem

Let (X, A) be a pair of path-connected spaces with A nonempty and simply connected, and let $n \geq 2$. Suppose the pair is $(n-1)$-connected, that is $\pi_i(X, A) = 0$ for $i < n$. Then the relative Hurewicz homomorphism

$$h\colon \pi_i(X, A) \longrightarrow H_i(X, A)$$

is an isomorphism for $i = n$; moreover $H_i(X, A) = 0$ for $i < n$. Here h sends the class of a map $(D^n, S^{n-1}) \to (X, A)$ to the image of the relative fundamental class of (D^n, S^{n-1}) under the induced map on relative homology.

***Proof Sketch*:**
The argument runs parallel to theorem 5.4.9 line for line, with the absolute pair (X, x_0) replaced by (X, A) and the relative homotopy long exact sequence proposition 5.1.11 in place of the absolute one; reproducing it in full would repeat the cokernel comparison already carried out above, so the complete details are deferred to [14, Theorem 4.37]. The proof there uses exactly the tools assembled in this chapter: CW approximation (theorem 5.3.9) reduces to a CW pair with no relative cells below dimension n; the compression lemma (lemma 5.3.3) gives $\pi_i(X, A) = 0$ and $H_i(X, A) = 0$ for $i < n$; and the identification of $\pi_n(X, A)$ and $H_n(X, A)$ as the same cokernel of attaching maps proceeds as in Steps 3 and 4 above. The hypothesis that A be simply connected is what guarantees the action of $\pi_1(A)$ on the relative homotopy groups is trivial, so that the cokernel computation is unobstructed; without it the conclusion must be stated in terms of the quotient by this action, as we sought to show.

The relative theorem is the form most often invoked in practice, because pairs come directly from inclusions of subcomplexes and from the mapping cylinder of an arbitrary map. It is the relative theorem that powers the Whitehead theorem (theorem 5.3.5): a weak equivalence between simply connected CW complexes induces isomorphisms on all relative homology, which by relative Hurewicz forces all relative homotopy to vanish, hence the map is a homotopy equivalence. The most immediate consequences, however, are for spheres, where the absolute theorem alone settles the homotopy in the bottom range.

Corollary 5.4.11: Connectivity and Bottom Homotopy of Spheres

For $n \geq 1$ the sphere S^n is $(n-1)$-connected. Consequently

$$\pi_k(S^n) = 0 \quad \text{for } k < n, \qquad \text{and} \qquad \pi_n(S^n) \cong \mathbb{Z}$$

where for $n \geq 1$ the isomorphism $\pi_n(S^n) \cong \mathbb{Z}$ is realized by the Hurewicz map, sending the homotopy class of a map $S^n \to S^n$ to its degree.

***Proof*:**
Step 1: S^n is $(n-1)$-connected. Equip S^n with the CW structure consisting of a single 0-cell and a single n-cell, and give S^k its standard CW structure with k-skeleton equal to all of S^k. For $k < n$, any map $f\colon S^k \to S^n$ is homotopic to a cellular map by the cellular approximation theorem ([14, Theorem 4.8]), and a cellular map sends the k-skeleton S^k of its domain into the k-skeleton of S^n. But the k-skeleton of S^n for $k < n$ is the single 0-cell, so the cellular representative is constant. Hence every map $S^k \to S^n$ with $k < n$ is nullhomotopic, giving $\pi_k(S^n) = 0$ for $k < n$, which is precisely $(n-1)$-connectivity.

Step 2: $\pi_n(S^n) \cong \mathbb{Z}$. For $n \geq 2$, Step 1 shows S^n is $(n-1)$-connected, so the Hurewicz theorem (theorem 5.4.9) applies: the Hurewicz map

$$h\colon \pi_n(S^n) \xrightarrow{\cong} H_n(S^n) \cong \mathbb{Z}$$

is an isomorphism. By definition $h([f]) = f_*(\iota_n) = (\deg f)\,\iota_n$, so under the identification $H_n(S^n) = \mathbb{Z}\iota_n$ the Hurewicz isomorphism sends a class $[f]$ to its degree $\deg f$. For $n = 1$ the same conclusion is the computation $\pi_1(S^1) \cong \mathbb{Z}$ of chapter 2, which agrees with theorem 5.4.5 since $\pi_1(S^1)$ is abelian and $H_1(S^1) \cong \mathbb{Z}$. In all cases $\pi_n(S^n) \cong \mathbb{Z}$ generated by the identity map, of degree 1, completing the proof.

The corollary is the first computation of an infinite family of homotopy groups, and it is the payoff that justifies the entire apparatus. That $\pi_n(S^n) \cong \mathbb{Z}$ with the isomorphism given by degree is the higher-dimensional analogue of the winding-number description of $\pi_1(S^1)$: a self-map of S^n is classified up to homotopy by how many times it wraps the sphere around itself, exactly as a self-map of S^1 is classified by its winding number. The vanishing $\pi_k(S^n) = 0$ for $k < n$ records that a low-dimensional sphere cannot detect a higher-dimensional hole, which is why the interesting homotopy of spheres begins on the diagonal $k = n$ and continues, far less predictably, above it. The contrast with homology, where $\tilde{H}_k(S^n)$ is $\mathbb{Z}$ for $k = n$ and 0 otherwise in *all* degrees, is the sharpest illustration of the slogan that homology is the stabilized, abelianized "shadow" of homotopy (this will be made precise using stable homotopy theory in chapter 8).

Forgetting the basepoint extends the classification from based self-maps of a sphere to all of them.

Corollary 5.4.12: Free Homotopy Classes of Self-Maps of Spheres

For $n \geq 1$ the degree is a bijection

$$\deg\colon [S^n, S^n] \xrightarrow{\cong} \mathbb{Z}$$

from the free homotopy classes of maps $S^n \to S^n$ onto the integers. In particular $[S^1, S^1] \cong \mathbb{Z}$ and $[S^n, S^n] \cong \mathbb{Z}$, and two self-maps of S^n are freely homotopic if and only if they have the same degree.

***Proof*:**
Freely homotopic maps induce the same homomorphism on $H_n(S^n) \cong \mathbb{Z}$ (homology is homotopy-invariant), so they have equal degree and deg is a well-defined function on $[S^n, S^n]$. To see it is a bijection, reduce free classes to based ones by proposition 5.1.15, which gives $[S^n, S^n] \cong \pi_n(S^n)/\pi_1(S^n)$. For $n \geq 2$ the sphere S^n is simply connected by the $(n-1)$-connectivity of corollary 5.4.11, so the quotient is trivial and $[S^n, S^n] = \pi_n(S^n)$. For $n = 1$ the group $\pi_1(S^1) \cong \mathbb{Z}$ is abelian, so by corollary 5.1.16 its conjugacy classes are its elements and again $[S^1, S^1] = \pi_1(S^1)$. In both cases the forgetful map is a bijection onto $\pi_n(S^n)$, and by corollary 5.4.11 the group $\pi_n(S^n) \cong \mathbb{Z}$ with the isomorphism given by degree. Composing, $\deg\colon [S^n, S^n] \to \mathbb{Z}$ is a bijection, completing the proof.

This gives the converse direction of proposition 3.5.2(3) for a self-map of a sphere, deferred there to Hopf's degree theorem: degree is a complete invariant of the free homotopy class, so it determines a self-map of S^n up to homotopy. The route here, through the Hurewicz isomorphism of corollary 5.4.11, uses none of the general-position arguments of the classical proof. Maps between spheres of *different* dimensions are a separate and far harder matter, recorded by the homotopy groups $\pi_m(S^n)$ with $m > n$, which degree cannot see.

Exercise 5.4.1

1. Let X be the wedge $\bigvee_{i=1}^{k} S^1$ of k circles. Using theorem 5.4.5, compute $H_1(X)$, and explain why the answer is independent of the (highly nonabelian) structure of $\pi_1(X)$.
2. Let X be the real projective plane $\mathbb{R}\mathrm{P}^2$. Its fundamental group is $\mathbb{Z}/2$. Compute $H_1(\mathbb{R}\mathrm{P}^2)$ using the dimension-one Hurewicz theorem, and compare with the cellular computation of section 3.5.
3. Show that the Hurewicz map $h\colon \pi_n(X) \to H_n(X)$ is the zero map whenever X is contractible, directly from definition 5.4.2, and reconcile this with theorem 5.4.9.
4. Let $X = S^2 \times S^2$. Using theorem 5.4.9 together with the homotopy groups of a product, identify $\pi_2(X)$ and verify the Hurewicz isomorphism against the known second homology $H_2(S^2 \times S^2) \cong \mathbb{Z}^2$.
5. Suppose X is simply connected and $\tilde{H}_i(X) = 0$ for all i. Use the Hurewicz theorem inductively to prove that $\pi_i(X) = 0$ for all i, that is, X is weakly contractible.
6. Give an example of a path-connected space X for which the Hurewicz map $h\colon \pi_2(X) \to H_2(X)$ is *not* an isomorphism, and explain which hypothesis of theorem 5.4.9 fails.

5.5 Eilenberg-MacLane Spaces and the Representation of Cohomology

A sphere S^n is the cleanest geometric object whose homotopy is concentrated in one place only after the bottom dimension, since $\pi_n(S^n) \cong \mathbb{Z}$ but the higher groups $\pi_i(S^n)$ for $i > n$ are very complicated. This section constructs spaces whose homotopy is concentrated in exactly one dimension and is trivial everywhere else. These are the *Eilenberg-MacLane spaces* $K(G, n)$, and their construction proceeds by the familiar method of attaching cells, first realizing the desired group in degree n via the Moore space of section 3.5, then sterilizing every higher homotopy group by gluing on cells that kill its generators.

The payoff is a structural statement about [ordinary] cohomology. The homotopy-concentration of $K(G, n)$ forces, via the Hurewicz theorem and the universal coefficient theorem, the existence of a distinguished cohomology class ι_n, and pulling ι_n back along a map $f : X \to K(G, n)$ produces a cohomology class $f^*(\iota_n) \in H^n(X; G)$. The representation theorem asserts that this assignment is a bijection between homotopy classes of maps $[X, K(G, n)]$ and the cohomology group $H^n(X; G)$. Cohomology, defined in chapter 4 through cochains and coboundaries, is thereby revealed to be a homotopy functor in the strongest possible sense: it is *represented* by a single space in each degree. This is the seed from which the entire representable viewpoint on cohomology theories is developed in chapter 8.

Definition 5.5.1: Eilenberg-MacLane Space

Let G be a group (abelian if $n \geq 2$) and let $n \geq 1$. An *Eilenberg-MacLane space of type* (G, n) is a path-connected CW complex X with

$$\pi_n(X) \cong G \qquad \text{and} \qquad \pi_i(X) = 0 \quad \text{for all } i \neq n$$

Any such space is denoted $K(G, n)$.

The defining condition isolates all of the homotopy of $K(G, n)$ in the single dimension n. For $n \geq 2$ the group π_n is abelian by theorem 5.1.4, so the restriction to abelian G in that range is forced rather than chosen; for $n = 1$ the group π_1 may be nonabelian, and the construction below produces a $K(G, 1)$ for any group G, abelian or not. The notation $K(G, n)$ is mild abuse: nothing yet guarantees that two spaces satisfying the definition are the same, and indeed they need not be homeomorphic. What theorem 5.5.4 establishes is that they are homotopy equivalent, which is the only sense in which uniqueness can be expected for an object defined purely by its homotopy groups. The reader should think of $K(G, n)$ not as a particular space but as a homotopy type, the way one thinks of "a point" or "a circle" once the topology has been forgotten modulo homotopy.

Example 5.5.2: The Circle as $K(\mathbb{Z}, 1)$

The circle S^1 is a $K(\mathbb{Z}, 1)$. Its fundamental group is $\pi_1(S^1) \cong \mathbb{Z}$, as computed in chapter 2. For the higher groups, the exponential map $p : \mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$, is a covering map with contractible total space $\mathbb{R}$. A covering map induces isomorphisms $\pi_i(\mathbb{R}) \xrightarrow{\cong} \pi_i(S^1)$ for all $i \geq 2$, because the homotopy lifting property lets one lift any map $S^i \to S^1$ together with a null-homotopy through $\mathbb{R}$ (this is the long exact sequence of the fibration $\mathbb{Z} \to \mathbb{R} \to S^1$ of theorem 5.2.9, whose fiber $\mathbb{Z}$ is discrete, hence $\pi_i(\mathbb{Z}) = 0$ for $i \geq 1$). Since $\mathbb{R}$ is contractible, $\pi_i(\mathbb{R}) = 0$ for all $i \geq 1$, so

$$\pi_i(S^1) = 0 \qquad \text{for all } i \geq 2$$

Thus S^1 has $\pi_1 \cong \mathbb{Z}$ and all higher homotopy groups trivial, which is the definition of a $K(\mathbb{Z}, 1)$.

The construction of a general $K(G, n)$ begins where the homology computation of section 3.5 left off. The Moore space $M(G, n)$ built there is a CW complex with the correct homology in degree n; what it lacks is control over its homotopy. The strategy is to start from the Moore space, whose π_n is already G in the simplest cases, and then to attach cells of dimension $n+2$ and higher so as to annihilate every homotopy group above degree n, all without disturbing what has already been achieved in degrees $\leq n+1$. The mechanism for killing a homotopy class is the most basic move in cell attachment: a map $\varphi\colon S^m \to X$ representing a class in $\pi_m(X)$ becomes null-homotopic in $X \cup_\varphi e^{m+1}$, because the attached cell provides exactly the disk needed to contract φ.

Lemma 5.5.3: Killing Homotopy Classes by Cell Attachment

Let X be a path-connected CW complex and let $m \geq 1$. There is a CW complex $Y \supseteq X$, obtained from X by attaching cells of dimension $m+1$, such that the inclusion $X \hookrightarrow Y$ induces an isomorphism $\pi_i(X) \xrightarrow{\cong} \pi_i(Y)$ for $i < m$ and the surjection $\pi_m(X) \twoheadrightarrow \pi_m(Y)$ has $\pi_m(Y) = 0$.

***Proof*:**
Step 1: Attaching the cells. Choose a set of generators $\{[\varphi_\alpha]\}_{\alpha \in A}$ of $\pi_m(X, x_0)$, with representatives $\varphi_\alpha\colon S^m \to X$ that may be taken to be based maps sending the basepoint of S^m to x_0. Form

$$Y \;=\; X \cup_{\{\varphi_\alpha\}} \Big(\coprod_{\alpha \in A} e_\alpha^{m+1} \Big)$$

by attaching an $(m+1)$-cell e_α^{m+1} along φ_α for each α. Since the attaching maps land in X, the space Y is a CW complex containing X as a subcomplex, and X is the m-skeleton-up portion in the sense that Y differs from X only by cells of dimension $m+1$.

Step 2: Lower homotopy is unchanged. For a CW pair (Y, X) in which Y is obtained from X by attaching cells of dimension $\geq m+1$, the inclusion induces an isomorphism $\pi_i(X) \to \pi_i(Y)$ for $i < m$ and a surjection for $i = m$; this is the compression lemma (lemma 5.3.3) applied to the m-connected pair (Y, X). Concretely, any map $S^i \to Y$ with $i \leq m$ is homotopic, rel basepoint, to one whose image avoids the interiors of the attached cells of dimension $\geq m+1$, hence lies in X, because a map of a complex of dimension $i \leq m$ can be pushed off cells of strictly larger dimension; and any homotopy $S^i \times I \to Y$ with $i < m$ (so that $\dim(S^i \times I) = i+1 \leq m$) may likewise be pushed off those cells into X. Thus $\pi_i(X) \to \pi_i(Y)$ is bijective for $i < m$ and surjective for $i = m$.

Step 3: The chosen generators die. For each α, the attached cell e_α^{m+1} is a map $D^{m+1} \to Y$ restricting on $\partial D^{m+1} = S^m$ to φ_α. This map is precisely a null-homotopy of φ_α inside Y: contracting D^{m+1} to its center drags φ_α to a constant. Hence each generator $[\varphi_\alpha]$ maps to 0 in $\pi_m(Y)$. Since the $[\varphi_\alpha]$ generate $\pi_m(X)$ and $\pi_m(X) \to \pi_m(Y)$ is surjective by Step 2, the whole of $\pi_m(Y)$ is generated by the images of the $[\varphi_\alpha]$, all of which vanish. Therefore $\pi_m(Y) = 0$, completing the proof.

The lemma performs surgery on a single homotopy group: it removes π_m while leaving everything below it intact. The price is that attaching $(m+1)$-cells may *create* new elements in π_{m+1}, the very next group up, because each new cell is a potential generator of a higher class. This is why the construction of $K(G,n)$ is an infinite process rather than a single step. One kills π_{n+1}, which may disturb π_{n+2}; one then kills π_{n+2}, which may disturb π_{n+3}; and so on upward forever. Because each stage only adds cells of strictly increasing dimension, the homotopy groups in any fixed range below the current stage stabilize, and the infinite union has the desired groups in every degree. This is the content of the existence half of the next theorem.

Theorem 5.5.4: Existence and Uniqueness of $K(G,n)$

Let $n \geq 1$ and let G be a group, abelian if $n \geq 2$.

1. *(Existence)* There exists a CW complex of type $K(G,n)$.
2. *(Uniqueness)* Any two CW complexes of type $K(G,n)$ are homotopy equivalent. More precisely, if X and X' are both of type $K(G,n)$, then any isomorphism $\pi_n(X) \xrightarrow{\cong} \pi_n(X')$ (of groups, respecting the π_1-action when $n=1$) is induced by a homotopy equivalence $X \to X'$, unique up to homotopy.

***Proof*:**
Step 1: Existence, base case $n \geq 2$. Let G be abelian. The Moore space construction of example 3.5.15 produces a CW complex $M = M(G,n)$ with $\tilde{H}_i(M) = 0$ for $i \neq n$ and $H_n(M) \cong G$. Building M with cells in dimensions n and $n+1$ only, one obtains a space whose $(n-1)$-skeleton is a point (a single 0-cell, no cells in dimensions $1, \ldots, n-1$), so M is $(n-1)$-connected: $\pi_i(M) = 0$ for $i < n$. By the Hurewicz theorem (theorem 5.4.9) the Hurewicz map $\pi_n(M) \xrightarrow{\cong} H_n(M) \cong G$ is an isomorphism, so M has the correct homotopy in degree n. This M is the starting point.

Step 1′: Existence, base case $n=1$. For a possibly nonabelian group G, choose a presentation $G = \langle s_\alpha \mid r_\beta \rangle$. Form the wedge $\bigvee_\alpha S^1_\alpha$ of circles, one for each generator s_α; by Van Kampen's theorem (section 2.2) its fundamental group is the free group on the s_α. For each relator r_β, regarded as a loop, attach a 2-cell e^2_β along it. The resulting 2-complex K^2 has $\pi_1(K^2) \cong G$, again by Van Kampen, since attaching a 2-cell along a loop w imposes the relation $w = 1$ on π_1. Set $M = K^2$; it is 0-connected with $\pi_1(M) \cong G$.

Step 2: Killing the higher groups. Starting from M (in either base case), build an increasing sequence of CW complexes

$$M = X_{n+1} \subseteq X_{n+2} \subseteq X_{n+3} \subseteq \cdots$$

as follows. Given X_m with $\pi_i(X_m) = 0$ for $n < i < m$ and $\pi_n(X_m) \cong G$ and $\pi_i(X_m) = 0$ for $i < n$, apply lemma 5.5.3 with the homotopy group $\pi_m(X_m)$ to obtain $X_{m+1} \supseteq X_m$ by attaching $(m+1)$-cells. By that lemma $\pi_m(X_{m+1}) = 0$, and the inclusion induces isomorphisms $\pi_i(X_m) \xrightarrow{\cong} \pi_i(X_{m+1})$ for all $i < m$. Since $m \geq n+1 > n$, the groups π_n and all groups below it are preserved, so $\pi_n(X_{m+1}) \cong G$ and $\pi_i(X_{m+1}) = 0$ for $i < n$ and for $n < i \leq m$. The induction starts at $m = n+1$ with $X_{n+1} = M$, whose only possibly nonzero homotopy group above n is unconstrained; the inductive hypothesis is met vacuously since there are no integers strictly between n and $n+1$.

Step 3: The union. Set

$$X = \bigcup_{m \geq n+1} X_m$$

with the weak topology, a CW complex. Fix i. Any map $S^i \to X$ has compact image, hence lands in some finite subcomplex, hence in some X_m with $m > i$; likewise any homotopy $S^i \times I \to X$ lands in some X_m with $m > i+1$. Therefore $\pi_i(X) = \varinjlim_m \pi_i(X_m)$, and for $m > i$ the inclusions $X_m \hookrightarrow X_{m+1} \hookrightarrow \cdots$ induce isomorphisms on π_i by Step 2. Consequently $\pi_i(X) \cong \pi_i(X_{i+1})$, which equals G when $i = n$ and equals 0 when $i \neq n$. Thus X is of type $K(G, n)$, proving existence.

Step 4: Uniqueness, reduction and constructing the map. Let X, X' be of type $K(G, n)$ and let $\theta\colon \pi_n(X) \to \pi_n(X')$ be an isomorphism. First reduce to a convenient cell structure on the source. Since X is $(n-1)$-connected, CW approximation (theorem 5.3.9) provides a CW complex Z homotopy equivalent to X with a single 0-cell and no cells in dimensions $1, \ldots, n-1$; constructing the map out of Z and composing with the equivalence $Z \simeq X$ suffices, since homotopy equivalence is what is to be proved. Rename Z as X, so X now has trivial $(n-1)$-skeleton.

Choose basepoints $x_0 \in X$, $x_0' \in X'$ and send the $(n-1)$-skeleton $X^{n-1} = \{x_0\}$ to x_0'. For each n-cell e^n_α of X, its attaching map is null-homotopic (it lands in the point X^{n-1}), so the characteristic map gives an element $[e^n_\alpha] \in \pi_n(X)$; define f on e^n_α by a based map $S^n \to X'$ representing $\theta([e^n_\alpha])$. This defines f on X^n inducing θ on π_n, because $\pi_n(X)$ is generated by the images of the n-cells under the Hurewicz-cellular identification.

Step 5: Uniqueness, extending over higher cells. Extend f over the cells of dimension $> n$ inductively. Suppose f is defined on X^m for some $m \geq n$. An $(m+1)$-cell e^{m+1} of X is attached along $\varphi\colon S^m \to X^m$. To extend f over e^{m+1} amounts to null-homotoping $f \circ \varphi\colon S^m \to X'$, that is, to showing $[f \circ \varphi] = 0$ in $\pi_m(X')$. Now φ is null-homotopic in X (it bounds the cell e^{m+1}), so $[\varphi] = 0$ in $\pi_m(X)$; but for $m > n$ we have $\pi_m(X') = 0$ outright, so $[f \circ \varphi] = 0$ automatically and the extension exists. For $m = n$ the class $[f \circ \varphi] = f_*[\varphi] = \theta([\varphi]) = \theta(0) = 0$ vanishes because $[\varphi] = 0$ in $\pi_n(X)$ and θ is a homomorphism. In every case $f \circ \varphi$ is null-homotopic, and choosing a null-homotopy extends f over e^{m+1}. Inducting over all cells (and taking the weak-topology union) yields $f\colon X \to X'$ with $f_* = \theta$ on π_n, while on π_i for $i \neq n$ the map f_* is the zero map between zero groups.

Step 6: Uniqueness, f is an equivalence. The map f induces an isomorphism on every homotopy group: on π_n it is θ, an isomorphism by hypothesis, and on π_i for $i \neq n$ both groups vanish so f_* is trivially an isomorphism. Thus f is a weak homotopy equivalence (definition 5.3.1). Since X and X' are CW complexes, Whitehead's theorem (theorem 5.3.5) upgrades f to a genuine homotopy equivalence.

Step 7: Uniqueness of the map up to homotopy. If $f, g\colon X \to X'$ both induce θ on π_n, the same skeletal argument shows they are homotopic. They agree up to homotopy on X^n because two based maps $S^n \to X'$ inducing the same element of $\pi_n(X')$ are based-homotopic; and the obstruction to homotoping them over an $(m+1)$-cell with $m > n$ lives in $\pi_m(X') = 0$. Inducting up the skeleta produces a homotopy $f \simeq g$. Hence the homotopy equivalence realizing θ is unique up to homotopy, as we sought to show.

The two halves of the theorem are complementary uses of the same skeletal philosophy. Existence builds a space by attaching cells to enforce a homotopy condition; uniqueness builds a map by extending over cells, the obstruction at each stage living in a homotopy group of the target that the $K(G, n)$ condition has set to zero. The vanishing of the higher homotopy of $K(G, n)$ is what makes the extension and the homotopy unobstructed: there is simply nowhere for an obstruction to live. This is the first appearance of a pattern that the section on obstruction theory will systematize, where cohomology groups with coefficients in homotopy groups measure exactly the failure of such extensions. Here those coefficient groups all vanish, so the extension always succeeds.

Example 5.5.5: $\mathbb{C}\mathrm{P}^\infty$ as $K(\mathbb{Z}, 2)$

The infinite complex projective space $\mathbb{C}\mathrm{P}^\infty = \bigcup_n \mathbb{C}\mathrm{P}^n$ is a $K(\mathbb{Z}, 2)$. Recall from chapter 1 that $\mathbb{C}\mathrm{P}^\infty$ is a CW complex with one cell in each even dimension $0, 2, 4, \ldots$ and none in odd dimensions. Its homotopy groups are computed from the fibration

$$S^1 \longrightarrow S^\infty \longrightarrow \mathbb{C}\mathrm{P}^\infty$$

where $S^\infty = \bigcup_n S^{2n-1}$ is the unit sphere in $\mathbb{C}^\infty$ and the map is the quotient $v \mapsto [v]$ identifying v with λv for $|\lambda| = 1$. This is the infinite analogue of the Hopf fibration (example 5.2.7), and it is a fiber bundle, hence a fibration by theorem 5.2.4. The total space S^∞ is contractible: any map from a compact space into S^∞ lands in some finite S^{2n-1}, and S^{2n-1} is contractible inside $S^{2n+1} \subseteq S^\infty$ by sliding to a missing point, so $\pi_i(S^\infty) = \varinjlim_n \pi_i(S^{2n-1}) = 0$ for every i. The long exact sequence of the fibration (theorem 5.2.9) reads

$$\cdots \to \pi_i(S^1) \to \pi_i(S^\infty) \to \pi_i(\mathbb{C}\mathrm{P}^\infty) \to \pi_{i-1}(S^1) \to \cdots$$

Since $\pi_i(S^\infty) = 0$ for all i, the boundary map $\pi_i(\mathbb{C}\mathrm{P}^\infty) \to \pi_{i-1}(S^1)$ is an isomorphism for every i. The fiber S^1 is a $K(\mathbb{Z}, 1)$ by example 5.5.2, so $\pi_{i-1}(S^1) = \mathbb{Z}$ when $i - 1 = 1$ and 0 otherwise. Therefore

$$\pi_i(\mathbb{C}\mathrm{P}^\infty) \cong \pi_{i-1}(S^1) = \begin{cases} \mathbb{Z} & i = 2 \\ 0 & i \neq 2 \end{cases}$$

which exhibits $\mathbb{C}\mathrm{P}^\infty$ as a $K(\mathbb{Z}, 2)$.

Example 5.5.6: $\mathbb{R}\mathrm{P}^\infty$ as $K(\mathbb{Z}/2, 1)$ and the Classifying Space BG

The infinite real projective space $\mathbb{R}\mathrm{P}^\infty = \bigcup_n \mathbb{R}\mathrm{P}^n$ is a $K(\mathbb{Z}/2, 1)$. The double cover $S^\infty \to \mathbb{R}\mathrm{P}^\infty$ identifies antipodal points and has total space S^∞, which is contractible by the same argument as in example 5.5.5. This is the universal cover of $\mathbb{R}\mathrm{P}^\infty$: the deck group is $\mathbb{Z}/2$ acting by the antipodal map, so $\pi_1(\mathbb{R}\mathrm{P}^\infty) \cong \mathbb{Z}/2$ by the classification of covering spaces of chapter 2. The higher groups vanish because a covering map induces isomorphisms $\pi_i(S^\infty) \xrightarrow{\cong} \pi_i(\mathbb{R}\mathrm{P}^\infty)$ for $i \geq 2$, and $\pi_i(S^\infty) = 0$. Hence

$$\pi_i(\mathbb{R}\mathrm{P}^\infty) = \begin{cases} \mathbb{Z}/2 & i = 1 \\ 0 & i \neq 1 \end{cases}$$

so $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$. The same mechanism produces $K(G, 1)$ for any group G acting freely on a contractible CW complex EG: the quotient $BG = EG/G$ has $\pi_1(BG) \cong G$ and trivial higher homotopy, so $BG = K(G, 1)$. For $G = \mathbb{Z}/2$ one may take $EG = S^\infty$, recovering $BG = \mathbb{R}\mathrm{P}^\infty$; for $G = \mathbb{Z}$ one may take $EG = \mathbb{R}$, recovering $B\mathbb{Z} = S^1$ as in example 5.5.2.

The last example identifies $K(G, 1)$ with the classifying space BG of section 2.4. A $K(G, 1)$ is exactly a space whose universal cover is contractible and whose deck group is G; such spaces are called *aspherical*, and the classifying-space construction $BG = EG/G$ is the canonical functorial recipe for building one. The two descriptions agree on homotopy type by the uniqueness in theorem 5.5.4, since both have $\pi_1 \cong G$ and trivial higher homotopy. This is why the cohomology of the space $K(G,1)$ coincides with the group cohomology of G, the bridge that lets topology compute an algebraic invariant and that chapter 2 and the companion volume on group cohomology both exploit.

With the spaces in hand, attention turns to the cohomology class that makes them useful. The vanishing of homotopy below degree n forces, through Hurewicz and the universal coefficient theorem, a canonical identification of $H^n(K(G,n);G)$ with $\mathrm{Hom}(G,G)$, and the identity homomorphism singles out a distinguished class. Everything in the representation theorem flows from this class, so it deserves a formal name.

Definition 5.5.7: [Eilenberg-MacLane] Fundamental Class

Let G be abelian and $n \geq 1$, and let $K = K(G,n)$. The Hurewicz map gives an isomorphism $h\colon \pi_n(K) \xrightarrow{\cong} H_n(K)$, whence $H_n(K) \cong G$. For $n \geq 2$ the space K is $(n-1)$-connected, so this is the Hurewicz theorem (theorem 5.4.9), which moreover gives $H_i(K) = 0$ for $0 < i < n$. For $n = 1$ the group $\pi_1(K) = G$ is abelian, so by the dimension-one Hurewicz theorem (theorem 5.4.5) the Hurewicz map induces $H_1(K) \cong \pi_1(K)^{\mathrm{ab}} = G$. The universal coefficient theorem (theorem 4.1.8) then gives

$$H^n(K;G) \xrightarrow{\cong} \mathrm{Hom}(H_n(K),G) \cong \mathrm{Hom}(G,G)$$

the Ext term $\mathrm{Ext}(H_{n-1}(K),G)$ vanishing because $H_{n-1}(K) = 0$ when $n \geq 2$, while for $n = 1$ it is $\mathrm{Ext}(H_0(K),G) = \mathrm{Ext}(\mathbb{Z},G) = 0$ since $H_0(K) \cong \mathbb{Z}$ is free. The *fundamental class* $\iota_n \in H^n(K(G,n);G)$ is the class corresponding under this isomorphism to the identity homomorphism $\mathrm{id}_G \in \mathrm{Hom}(G,G)$.

The fundamental class is the universal cohomology class of degree n with coefficients in G: it is the cohomology incarnation of the abstract isomorphism $\pi_n(K) \cong G$. Tracing the identifications, ι_n is the unique class in $H^n(K;G)$ whose evaluation on the Hurewicz image of any $\alpha \in \pi_n(K)$ returns $\alpha \in G$ itself. The choice of id_G is not arbitrary decoration; it is the only homomorphism that makes the class *tautological*, so that pulling it back records a map exactly rather than after composing with some endomorphism of G. The next theorem shows this tautological character propagates: a class on any space X is the pullback of ι_n along the map that the class itself determines.

Example 5.5.8: The Generator of $H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$

For $K(\mathbb{Z},2) = \mathbb{C}\mathrm{P}^\infty$ the fundamental class $\iota_2 \in H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$ is the standard generator of the cohomology ring. The cellular chain complex of $\mathbb{C}\mathrm{P}^\infty$ has one $\mathbb{Z}$ in each even degree and zero in odd degrees, with all boundary maps zero, so $H_i(\mathbb{C}\mathrm{P}^\infty) \cong \mathbb{Z}$ for i even and 0 for i odd. In particular $H_2(\mathbb{C}\mathrm{P}^\infty) \cong \mathbb{Z}$ and $H_1(\mathbb{C}\mathrm{P}^\infty) = 0$. The universal coefficient theorem gives $H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathrm{Hom}(H_2,\mathbb{Z}) \cong \mathbb{Z}$, and ι_2 corresponds to $\mathrm{id}_{\mathbb{Z}}$, hence to the homomorphism $H_2(\mathbb{C}\mathrm{P}^\infty) \to \mathbb{Z}$ sending a chosen generator to 1. Thus ι_2 is a generator of $H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}$. The cohomology ring is the polynomial ring $H^\bullet(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}[\iota_2]$, with ι_2^k generating H^{2k}, matching the cup-product computation of chapter 4: the fundamental class is the polynomial generator.

The representation theorem is the structural endpoint of the section. It says that the assignment $f \mapsto f^*(\iota_n)$, taking a based map into $K(G,n)$ to the pullback of the fundamental class, sets up a bijection between homotopy classes of maps and degree-n cohomology classes. Naturality means the bijection is compatible with maps of the source X, so that $H^n(-;G)$ and $[-,K(G,n)]$ are isomorphic functors. The hypotheses must be stated with care: X is required to have the homotopy type of a CW complex, the homotopy classes are based and the cohomology is reduced, and these based notions agree with the unbased ones because $K(G,n)$ is a simple space (its π_1 acts trivially on all π_n, vacuously so for $n \geq 2$, and abelian π_1 acting on itself by conjugation is trivial for $n = 1$ when G is abelian).

Theorem 5.5.9: Cohomology is Represented by $K(G,n)$

Let G be abelian, $n \geq 1$, and let X be a CW complex (or a space of CW homotopy type) with basepoint. Let $\iota_n \in H^n(K(G,n);G)$ be the fundamental class. The map (where the $*$ represents based homotopies)

$$T\colon [X, K(G,n)]_* \longrightarrow \tilde{H}^n(X;G), \qquad [f] \longmapsto f^*(\iota_n)$$

sending a based homotopy class of maps to the pullback of ι_n is a bijection, natural in X: for any based map $g\colon X' \to X$ the square

$$\begin{array}{ccc} [X, K(G,n)]_* & \xrightarrow{T} & \tilde{H}^n(X;G) \\ {\scriptstyle g^*}\big\downarrow & & \big\downarrow{\scriptstyle g^*} \\ [X', K(G,n)]_* & \xrightarrow{T} & \tilde{H}^n(X';G) \end{array}$$

commutes. Because $K(G,n)$ is a simple space, the based set $[X, K(G,n)]_*$ coincides with the free set $[X, K(G,n)]$, so the bijection holds for free homotopy classes and unreduced cohomology when X is connected as well.

***Proof*:**
Write $K = K(G,n)$ throughout. Naturality is immediate: for $g\colon X' \to X$ and $f\colon X \to K$ one has $(f \circ g)^*(\iota_n) = g^*(f^*(\iota_n))$ by functoriality of $H^n(-;G)$, which is exactly the assertion that the square commutes. The content is that T is a bijection, proved by exhibiting surjectivity and injectivity through cell-by-cell constructions on X.

Step 1: A cohomology class as a cocycle on skeleta. Take cohomology computed cellularly (proposition 4.1.13), legitimate since X is a CW complex. A class $u \in \tilde{H}^n(X;G)$ is represented by a cellular cocycle, a homomorphism $u\colon C_n^{\text{cell}}(X) \to G$ on the free abelian group on the n-cells of X that vanishes on cellular $(n+1)$-boundaries. The strategy is to manufacture a map $f\colon X \to K$ whose pullback $f^*(\iota_n)$ realizes u, by defining f on successive skeleta so that on n-cells it records the value of u.

Step 2: Surjectivity, defining f on X^n. Build a representative K with a single 0-cell and cells only in dimensions $\geq n$ (the cellular model of theorem 5.5.4), with a chosen n-cell e_0^n whose characteristic map represents a generator of $\pi_n(K) \cong G$ corresponding to a chosen generator-system of G. Send the $(n-1)$-skeleton X^{n-1} to the basepoint of K. For each n-cell e_α^n of X, the cocycle assigns a value $u(e_\alpha^n) = g_\alpha \in G$; define f on e_α^n to be a based map $S^n \to K$ representing the element $g_\alpha \in G \cong \pi_n(K)$ (using that the attaching map of e_α^n goes to the basepoint, so the characteristic map factors through $S^n = D^n/\partial D^n$). This defines a map $f\colon X^n \to K$ on the n-skeleton.

Step 3: Surjectivity, f extends over X^{n+1}. An $(n+1)$-cell e_β^{n+1} is attached along $\varphi_\beta\colon S^n \to X^n$.

To extend f over e_β^{n+1} one needs $[f\circ\varphi_\beta]=0$ in $\pi_n(K)\cong G$. Under the identification $\pi_n(K)\cong G$, the class $[f\circ\varphi_\beta]$ equals $\sum_\alpha(\text{degree of } \varphi_\beta \text{ on } e_\alpha^n)\cdot g_\alpha$, which is precisely $u(\partial e_\beta^{n+1})$, the cellular boundary $\partial e_\beta^{n+1}=\sum_\alpha d_{\beta\alpha}e_\alpha^n$ paired with u. Because u is a cocycle, $u(\partial e_\beta^{n+1})=(\delta u)(e_\beta^{n+1})=0$. Hence $[f\circ\varphi_\beta]=0$, the map $f\circ\varphi_\beta$ is null-homotopic, and f extends over each e_β^{n+1}, giving $f\colon X^{n+1}\to K$.

Step 4: Surjectivity, f extends over higher skeleta. For $m\geq n+1$, extending f from X^m over an $(m+1)$-cell attached along $\psi\colon S^m\to X^m$ requires $[f\circ\psi]=0$ in $\pi_m(K)$, and $\pi_m(K)=0$ since $K=K(G,n)$ and $m>n$. So every such extension is unobstructed, and inducting up the skeleta (with the weak-topology union) produces $f\colon X\to K$.

Step 5: Surjectivity, $f^*(\iota_n)=u$. By construction f sends X^{n-1} to the basepoint and sends e_α^n to a representative of $g_\alpha=u(e_\alpha^n)$. The fundamental class ι_n is characterized by evaluating to the identity under $H^n(K;G)\cong\mathrm{Hom}(H_n(K),G)$, that is, ι_n assigns to the generating n-cell e_0^n of K the corresponding element of G. The pullback $f^*(\iota_n)$ is the cellular cocycle whose value on e_α^n is the value of ι_n on $f_*(e_\alpha^n)\in H_n(K^n,K^{n-1})$, which is the element of G represented by $f|_{e_\alpha^n}$, namely $g_\alpha=u(e_\alpha^n)$. Thus $f^*(\iota_n)$ and u agree as cellular cocycles on n-cells, so $f^*(\iota_n)=u$ in $\tilde{H}^n(X;G)$, and T is surjective.

Step 6: Injectivity, the relative construction. Suppose $f_0,f_1\colon X\to K$ satisfy $f_0^*(\iota_n)=f_1^*(\iota_n)$ in $\tilde{H}^n(X;G)$. To show $[f_0]=[f_1]$, apply the surjectivity construction to the CW pair $(X\times I,\ X\times\{0,1\})$. The two maps assemble to a map $F\colon X\times\{0,1\}\to K$ on the subcomplex $X\times\{0,1\}$, and one seeks to extend F to a homotopy $X\times I\to K$, which is exactly a based homotopy $f_0\simeq f_1$. The obstruction to extending F over $X\times I$ is the relative analogue of the cocycle in Step 2-3: extending over the 0- and 1-cells of I is given, and extending over an $(n+1)$-cell $e\times I$ of the product requires a relative cocycle to vanish.

Step 7: Injectivity, the obstruction is the difference of the two classes. The cells of $X\times I$ not in $X\times\{0,1\}$ are of the form $e\times\{(0,1)\}$, the product of a cell e of X with the open 1-cell of I; an n-cell of X contributes an $(n+1)$-cell $e\times I$ to the product. Extending F over $X\times I$ skeleton by skeleton, the first obstruction lives in the cellular cochains $C^{n+1}(X\times I,\ X\times\{0,1\};\pi_nK)$ and equals, under $\pi_nK\cong G$, the difference cocycle $f_1^*(\iota_n)-f_0^*(\iota_n)$ transported to the product. Concretely, on $e\times I$ for an n-cell e of X, the obstruction value is $u_1(e)-u_0(e)$ where $u_j=f_j^*(\iota_n)$ are the cocycles. By hypothesis u_0 and u_1 are cohomologous, $u_1-u_0=\delta v$ for some $v\in C^{n-1}(X;G)$. The freedom in extending F over $X\times I$ at the previous stage is exactly a choice of value in $\pi_nK\cong G$ on each n-cell of $X\times I$ of the form $e\times I$ with e an $(n-1)$-cell of X; these n-cells are indexed by the $(n-1)$-cells of X, so the choices form precisely a cochain in $C^{n-1}(X;G)$. Re-choosing the extension of F over these cells by the cochain v adjusts the obstruction cocycle by δv, replacing u_1-u_0 by $u_1-u_0-\delta v=0$. With the obstruction cocycle now zero, F extends over the $(n+1)$-skeleton of $X\times I$.

Step 8: Injectivity, higher cells. For cells of $X\times I$ of dimension $>n+1$, the obstruction to extending F lies in $\pi_m(K)=0$ for the relevant $m>n$, so all further extensions are unobstructed exactly as in Step 4. Inducting up the skeleta extends F to all of $X\times I$, producing a based homotopy $f_0\simeq f_1$. Hence $[f_0]=[f_1]$ and T is injective.

Step 9: Conclusion and the based-versus-free clause. Steps 2-5 give surjectivity and Steps 6-8 give injectivity, so T is a natural bijection. Finally, $K(G,n)$ is simple: for $n\geq 2$ the action of $\pi_1(K)=0$ on $\pi_n(K)$ is trivial, and for $n=1$ the abelian group $G=\pi_1(K)$ acts on itself by conjugation, which is trivial because G is abelian. For a simple space Y and connected X of CW type, the forgetful map $[X,Y]_*\to[X,Y]$ from based to free homotopy classes is a bijection.

Therefore the bijection T also identifies free homotopy classes with unreduced $H^n(X;G)$ on connected X, completing the proof.

The theorem turns the functor $X \mapsto H^n(X;G)$ into a geometric object. To know all degree-n cohomology of all spaces, with coefficients in G, it suffices to know one space, $K(G,n)$, together with one class ι_n on it; every cohomology class anywhere is a pullback of ι_n, and homotopic maps give the same class while non-homotopic maps give different classes. Cohomology operations, the natural transformations $H^n(-;G) \to H^m(-;G')$, become in this language nothing but homotopy classes of maps $K(G,n) \to K(G',m)$, recovered by the same Yoneda-style argument applied to the universal class (see [6]). The computation of such operations, the Steenrod squares chief among them, rests entirely on this representability.

5.5.10 Foreshadowing: Brown Representability The representation theorem $\tilde{H}^n(X;G) \cong [X, K(G,n)]_*$ is the prototype of a general phenomenon. The proof used only formal features of $\tilde{H}^n(-;G)$: it is a homotopy-invariant functor on CW complexes, it sends wedges to products, and it has a Mayer-Vietoris-type gluing property. Brown's representability theorem, developed in chapter 8, shows that *any* functor with these properties is represented by some space, so every reasonable cohomology theory, not just ordinary cohomology, has its own sequence of representing spaces. The Eilenberg-MacLane spaces are the first instance, and the sequence $\{K(G,n)\}_n$, linked by the loop-space relation $K(G,n) \simeq \Omega K(G,n+1)$, is the first example of a spectrum, the central object of stable homotopy theory.

Exercise 5.5.1

1. Show that for an abelian group G and $n \geq 1$ there is a homotopy equivalence $K(G,n) \simeq \Omega K(G,n+1)$, where ΩY denotes the loop space of Y. (Use the long exact sequence of the path-loop fibration $\Omega Y \to PY \to Y$ from example 5.2.12.)

2. Prove that $K(G \times H, n) \simeq K(G,n) \times K(H,n)$ for abelian G, H and $n \geq 1$. Deduce that $K(\mathbb{Z}^k, n) \simeq (K(\mathbb{Z}, n))^k$ and identify $K(\mathbb{Z}^2, 1)$ explicitly.

3. Using theorem 5.5.9, compute the set of homotopy classes $[S^m, K(G,n)]_*$ for all $m, n \geq 1$, and check the answer against the homotopy groups of $K(G,n)$ directly.

4. Let G be abelian and $n \geq 1$. Show that a cohomology operation, that is, a natural transformation $\theta\colon \tilde{H}^n(-;G) \to \tilde{H}^m(-;G')$ of functors on CW complexes, is the same datum as a single class $\theta(\iota_n) \in \tilde{H}^m(K(G,n);G')$ (this is the Yoneda lemma in a topological setting).

5. The torus $T^2 = S^1 \times S^1$ is a $K(\mathbb{Z}^2, 1)$. Verify directly from example 5.5.6 (the BG description) that T^2 is aspherical, and identify the contractible universal cover and the deck group.

6. Show that $\mathbb{RP}^\infty$ and S^1 are not homotopy equivalent, even though both are Eilenberg-MacLane spaces of degree 1, by exhibiting an algebraic invariant distinguishing them. Why does this not contradict the uniqueness in theorem 5.5.4?

5.6 Postnikov Towers and Obstruction Theory

The homotopy groups $\pi_n(X)$ assembled in section 5.1 record an enormous amount of information about a space, but they do so all at once, with no indication of how the groups in different degrees fit together. A space with $\pi_1 = \mathbb{Z}$ and $\pi_2 = \mathbb{Z}$ could be the torus-like product $S^1 \times \mathbb{CP}^\infty$, or it could be a twisted assembly in which the action of π_1 on π_2 and a cohomology class entangle the two groups:

Example 5.6.1: Two Assemblies of $\pi_1 = \pi_2 = \mathbb{Z}$

The untwisted assembly is the product

$$P \;=\; S^1 \times \mathbb{CP}^\infty.$$

Since $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$ (example 5.5.5) and $S^1 = K(\mathbb{Z}, 1)$, the product formula for homotopy groups (example 5.1.7) gives $\pi_1(P) = \mathbb{Z}$, $\pi_2(P) = \mathbb{Z}$, and $\pi_n(P) = 0$ otherwise.

The twisted assembly is a mapping torus. Let $c\colon \mathbb{CP}^\infty \to \mathbb{CP}^\infty$ be complex conjugation. On each $\mathbb{CP}^1$ it reverses orientation, so on $\pi_2(\mathbb{CP}^\infty) = H_2(\mathbb{CP}^\infty) = \mathbb{Z}$ it acts by $c_* = -1$. Glue the two ends of $\mathbb{CP}^\infty \times [0, 1]$ together along c:

$$M \;=\; \mathbb{CP}^\infty \times [0, 1]/(x, 0) \sim (c(x), 1).$$

This is a fiber bundle $\mathbb{CP}^\infty \hookrightarrow M \to S^1$, hence a Serre fibration (theorem 5.2.4). Its long exact sequence (theorem 5.2.9),

$$\cdots \to \pi_n(\mathbb{CP}^\infty) \to \pi_n(M) \to \pi_n(S^1) \to \pi_{n-1}(\mathbb{CP}^\infty) \to \cdots,$$

collapses using $\pi_n(S^1) = 0$ for $n \geq 2$ and $\pi_n(\mathbb{CP}^\infty) = 0$ for $n \neq 2$: it forces $\pi_1(M) = \pi_1(S^1) = \mathbb{Z}$, $\pi_2(M) = \pi_2(\mathbb{CP}^\infty) = \mathbb{Z}$, and $\pi_n(M) = 0$ for $n \geq 3$. Thus P and M have *identical homotopy groups in every degree*; the list $(\pi_n)_n$ cannot tell them apart.

What separates them is the action of π_1 on π_2 (proposition 5.1.12). For the product this action is trivial. For the mapping torus, transporting the fiber once around the base circle applies the gluing map c, so the generator of $\pi_1(M) = \mathbb{Z}$ acts on $\pi_2(M) = \mathbb{Z}$ by $c_* = -1$. A homotopy equivalence $M \simeq P$ would intertwine the two actions through isomorphisms of π_1 and π_2; but the only automorphisms of $\mathbb{Z}$ are ± 1, and conjugating multiplication-by-(-1) by any of these returns multiplication by -1. The trivial action and the sign action are distinct, hence

$$M \;\not\simeq\; P \qquad \text{even though} \qquad \pi_*(M) \cong \pi_*(P) \text{ degree by degree.}$$

Note this doesn't contradict Whitehead's theorem (theorem 5.3.5) since there is no map $f : M \to P$ where $f_* : \pi_*(M) \to \pi_*(P)$ induces the isomorphism.

The difference is also visible in ordinary homology: a Mayer–Vietoris computation for the mapping torus (theorem 3.4.19) gives $H_2(M) \cong \mathbb{Z}/2$, whereas the Künneth theorem (theorem 4.3.5) gives $H_2(P) \cong \mathbb{Z}$.

The list of homotopy groups cannot distinguish these. A *Postnikov tower* makes that gluing data visible. It resolves a space into a sequence of approximations, each one capturing the homotopy groups up to a fixed degree and nothing above, and it records exactly how each new layer is attached to the ones below by a single cohomology class. The Eilenberg-MacLane spaces and the representability theorem of section 5.5 are the tools that attach layers using a class in a cohomology group.

This same cohomological mechanism governs a second, older problem: given a continuous map defined on part of a space, when does it extend to the whole space? Extending one cell at a time, the obstruction to pushing a map across the cells of a single dimension is a cocycle, and the map extends precisely when its cohomology class vanishes. The two themes are one theme. The k-invariants that build a Postnikov tower and the obstruction classes that extend a map both live in $H^{n+1}(-;\pi_n)$, and both answer a yes/no question about the existence of a map by producing a single algebraic invariant whose vanishing is necessary and sufficient. This section develops both, and shows that cohomology is the natural home of obstructions.

Throughout, X denotes a connected CW complex, and (where homotopy groups of X enter the construction of a tower) a *simple* space in the sense of definition 5.1.13: the action of π_1 on every π_n is trivial. Simplicity is what lets local homotopy-theoretic data be glued into cohomology with untwisted coefficients; without it the same constructions go through with local (twisted) coefficients, not developed here.

The first task is to manufacture, from a space X, an approximation that has the correct homotopy groups in low degrees and is trivial above a chosen degree. The mechanism is to attach cells: to kill an element of $\pi_m(X)$, glue on an $(m+1)$-cell along a representing map, and to kill all of π_m in every degree $m > n$ while leaving π_i for $i \le n$ untouched, attach cells of dimension $m+1 \ge n+2$. Cells of such dimension cannot create or destroy homotopy in degrees $\le n$, so the lower groups survive. First, we give a name to such a construction:

Definition 5.6.2: Postnikov Tower

Let X be a connected CW complex. A *Postnikov tower* for X is a sequence of spaces X_n for $n \ge 1$, together with maps

$$f_n \colon X \to X_n \qquad \text{and} \qquad p_n \colon X_n \to X_{n-1}$$

for $n \ge 2$, such that the following hold.

1. The triangle commutes up to homotopy: $p_n \circ f_n \simeq f_{n-1}$ for all $n \ge 2$.
2. The map f_n induces an isomorphism $\pi_i(X) \xrightarrow{\cong} \pi_i(X_n)$ for all $i \le n$.
3. The space X_n has $\pi_i(X_n) = 0$ for all $i > n$.

The space X_n is the *n-th Postnikov section* of X.

The defining conditions say that X_n is a space with the homotopy groups of X truncated above degree n: it sees $\pi_1(X), \dots, \pi_n(X)$ faithfully and is blind to everything above. The maps p_n organize these truncations into a tower

$$\cdots \to X_n \xrightarrow{p_n} X_{n-1} \xrightarrow{p_{n-1}} \cdots \to X_2 \xrightarrow{p_2} X_1$$

sitting under X by the compatible family f_n. Reading the tower upward, each stage refines its predecessor by installing one more homotopy group; reading the maps f_n, the space X maps compatibly into every stage, and (as the construction will show) the induced map $X \to \varprojlim_n X_n$ into the limit is a weak equivalence. The tower is therefore a method of building X, up to weak equivalence, out of its homotopy groups one degree at a time, in exact analogy with how a CW structure builds X out of cells one dimension at a time.

Example 5.6.3: The Postnikov Tower of S^2 Begins
The sphere S^2 has $\pi_1(S^2) = 0$, $\pi_2(S^2) = \mathbb{Z}$, $\pi_3(S^2) = \mathbb{Z}$ (generated by the Hopf map, example 5.2.7), and nonzero homotopy in infinitely many higher degrees. Its Postnikov tower begins

$$X_1 = *, \qquad X_2 = K(\mathbb{Z}, 2) = \mathbb{C}\mathrm{P}^\infty, \qquad X_3 \to X_2$$

Indeed X_1 must be a space with $\pi_i = 0$ for all i, so $X_1 \simeq *$. The section X_2 must have $\pi_2 = \mathbb{Z}$ and all other homotopy groups zero, so X_2 is an Eilenberg-MacLane space $K(\mathbb{Z}, 2)$, realized concretely as $\mathbb{C}\mathrm{P}^\infty$ (example 5.5.5 identifies $S^\infty \to \mathbb{C}\mathrm{P}^\infty$ as a fibration with contractible total space, forcing $\pi_i(\mathbb{C}\mathrm{P}^\infty) = \pi_{i-1}(S^1)$, which is $\mathbb{Z}$ for $i = 2$ and 0 otherwise). The map $f_2\colon S^2 \to \mathbb{C}\mathrm{P}^\infty$ is the inclusion of the bottom cell, an isomorphism on π_2. The next stage X_3 has $\pi_2 = \pi_3 = \mathbb{Z}$ and nothing else; it is *not* a product $K(\mathbb{Z}, 2) \times K(\mathbb{Z}, 3)$, and the class that measures its failure to be a product is the first nontrivial k-invariant of S^2, computed at the end of this section.

The existence of a Postnikov tower for an arbitrary connected CW complex is not obvious: one must produce, for each n, a space with the prescribed truncated homotopy and fit these into a tower. The construction attaches cells to X itself.

Theorem 5.6.4: Existence of Postnikov Towers

Every connected CW complex X admits a Postnikov tower. Moreover the tower may be built so that each X_n is a CW complex containing X as a subcomplex, with $f_n\colon X \hookrightarrow X_n$ the inclusion, and so that f_n is the inclusion of a subcomplex obtained from X by attaching cells of dimension $\geq n + 2$.

***Proof*:**
Fix $n \geq 1$. The strategy is to enlarge X to a CW complex $X_n \supseteq X$ by attaching cells of dimension at least $n+2$, chosen to kill all homotopy of X in degrees above n while preserving it in degrees $\leq n$.

Step 1: Killing π_{n+1}. Choose a set of generators $\{[\varphi_\alpha]\}_\alpha$ of the group $\pi_{n+1}(X)$, represented by maps $\varphi_\alpha\colon S^{n+1} \to X$. Form

$$Y^{(n+1)} = X \cup \Big(\coprod_\alpha D_\alpha^{n+2} \Big),$$

attaching one $(n+2)$-cell D_α^{n+2} along each φ_α (using $\partial D_\alpha^{n+2} = S^{n+1}$). The inclusion $X \hookrightarrow Y^{(n+1)}$ is built by attaching cells of dimension $n + 2$. Two facts control its effect on homotopy. First, cellular approximation (theorem 5.3.9) shows that attaching cells of dimension $\geq n + 2$ does not change π_i for $i \leq n$: any map $S^i \to Y^{(n+1)}$ with $i \leq n$, and any homotopy $S^i \times I \to Y^{(n+1)}$ with $i \leq n$ (so the homotopy has source dimension $\leq n + 1$), is homotopic rel endpoints into the subcomplex X, since the cells D_α^{n+2} have dimension exceeding both i and $i + 1$. Hence

$$\pi_i(X) \xrightarrow{\cong} \pi_i\big(Y^{(n+1)}\big), \qquad i \leq n$$

is an isomorphism. Second, in degree $n + 1$, each generator φ_α now bounds the attached cell D_α^{n+2}, so $[\varphi_\alpha] = 0$ in $\pi_{n+1}(Y^{(n+1)})$. Since the $[\varphi_\alpha]$ generate $\pi_{n+1}(X)$ and the inclusion is surjective on π_{n+1} (every $(n + 1)$-sphere lands in the $(n + 1)$-skeleton, which lies in X), we conclude $\pi_{n+1}(Y^{(n+1)}) = 0$.

Step 2: Killing all higher homotopy. Repeat the procedure in each higher degree, inductively.

Having built $Y^{(m)}$ with $\pi_i(Y^{(m)}) \cong \pi_i(X)$ for $i \leq n$ and $\pi_i(Y^{(m)}) = 0$ for $n < i \leq m$, choose generators of $\pi_{m+1}(Y^{(m)})$, represent them by maps $S^{m+1} \to Y^{(m)}$, and attach $(m+2)$-cells along them to form $Y^{(m+1)}$. As in Step 1, attaching cells of dimension $m+2 \geq n+2$ leaves π_i unchanged for $i \leq m$ (in particular for $i \leq n$ the groups remain $\pi_i(X)$, and for $n < i \leq m$ they remain 0) and kills π_{m+1}. Set

$$X_n := \bigcup_{m \geq n+1} Y^{(m)} = X \cup \big(\text{cells of dimension} \geq n+2\big),$$

the increasing union, given the weak topology. A map from a sphere S^i into X_n has compact image, hence lands in some finite stage $Y^{(m)}$ with $m \geq i$; the same holds for a nullhomotopy. Therefore

$$\pi_i(X_n) = \varinjlim_m \pi_i\big(Y^{(m)}\big) = \begin{cases} \pi_i(X), & i \leq n, \\ 0, & i > n. \end{cases}$$

The inclusion $f_n \colon X \hookrightarrow X_n$ is the inclusion of a subcomplex and induces the displayed isomorphisms in degrees $\leq n$. This establishes conditions (2) and (3) of definition 5.6.2, and the supplementary claim that X_n is obtained from X by attaching cells of dimension $\geq n+2$.

Step 3: The tower maps. It remains to produce, for each $n \geq 2$, a map $p_n \colon X_n \to X_{n-1}$ with $p_n \circ f_n \simeq f_{n-1}$, establishing condition (1) of definition 5.6.2. The key observation is that X_n is built from X by attaching cells whose attaching maps become nullhomotopic in X_{n-1}, so the map $f_{n-1} \colon X \to X_{n-1}$ extends across those cells.

Make this precise. By Step 2, the section X_n is obtained from X by attaching cells, all of dimension $\geq n+2$. The space X_{n-1} satisfies conditions (2) and (3) (the construction of Steps 1-2 with index $n-1$ in place of n), so $\pi_i(X_{n-1}) = 0$ for all $i \geq n$. Extend $f_{n-1} \colon X \to X_{n-1}$ over the cells of $X_n \setminus X$ one dimension at a time, starting in dimension $n+2$. Suppose a map h extending f_{n-1} has been built over X together with all cells of $X_n \setminus X$ of dimension $< d$ (where $d \geq n+2$), and let e be a d-cell of $X_n \setminus X$, attached by $\varphi \colon S^{d-1} \to X_n^{(d-1)}$. The attaching sphere lands in the domain of h, so the composite $h \circ \varphi \colon S^{d-1} \to X_{n-1}$ represents a class in $\pi_{d-1}(X_{n-1})$, and $d-1 \geq n+1 \geq n$, so $\pi_{d-1}(X_{n-1}) = 0$ and $h \circ \varphi$ is nullhomotopic. A nullhomotopy of an attaching map is exactly the datum needed to extend a map over the cell it attaches, so h extends over e; doing this for every d-cell and letting d increase produces a map

$$p_n \colon X_n \to X_{n-1}, \qquad p_n|_X = f_{n-1},$$

defined on all of X_n. Since p_n restricts to f_{n-1} on X and $f_n \colon X \hookrightarrow X_n$ is the inclusion, $p_n \circ f_n = f_{n-1}$ on the nose, which in particular gives $p_n \circ f_n \simeq f_{n-1}$, condition (1) of definition 5.6.2. Constructing p_n for every n assembles the maps of the tower, completing the proof.

The construction is the homotopy-theoretic analogue of forming the quotient of a group by the subgroup generated by certain elements: to kill a homotopy class one attaches a cell whose boundary is a representative, exactly as one kills a group element by imposing a relation. The price is that X_n is typically infinite-dimensional even when X is a finite complex, since killing homotopy in one degree generically creates new homotopy in the next, which must in turn be killed. The tower trades the finiteness of X for the transparency of having one homotopy group visible per stage. Two further points deserve emphasis. The map $X \to X_n$ is an isomorphism on π_i for $i \leq n$ and the target has no higher homotopy, so X_n is the universal target receiving X among n-truncated spaces; this universality (which one proves by a cell-attaching extension argument: a map from X to any

n-truncated space extends over the killing cells of dimension $\geq n+2$ since their attaching spheres are nullhomotopic in the target) makes the tower functorial up to homotopy. And the limit $\varprojlim_n X_n$ receives a map from X inducing isomorphisms on all π_i, hence a weak equivalence when X is a CW complex, by theorem 5.3.5; the tower genuinely reconstructs X.

With existence settled, attention turns to the structural content of a Postnikov tower, which lies in the nature of the maps $p_n\colon X_n \to X_{n-1}$. The next result identifies them: each is a fibration whose fiber is an Eilenberg-MacLane space $K(\pi_n(X), n)$, and (when X is simple) it is a *principal* fibration, meaning it is pulled back from a path-loop fibration over a $K(\pi_n(X), n+1)$. Principality is what packages the entire map p_n into a single classifying map, the k-invariant.

To set up the statement, recall from section 5.2 that a fibration $F \to E \to B$ has a long exact sequence of homotopy groups (theorem 5.2.9), and from example 5.2.12 that the path space fibration $\Omega K \to PK \to K$ over any space K has contractible total space PK, so $\Omega K \simeq$ the loop space and the connecting map identifies $\pi_i(\Omega K) \cong \pi_{i+1}(K)$. Recall also from definition 5.5.1 that a $K(\pi, n)$ is a space with $\pi_n = \pi$ and all other homotopy groups zero, and from theorem 5.5.4 that it is unique up to homotopy equivalence; combined with the path-loop fibration this gives $\Omega K(\pi, n+1) \simeq K(\pi, n)$, since $\Omega K(\pi, n+1)$ has the homotopy groups of a $K(\pi, n)$ by example 5.2.12 and uniqueness then identifies it.

Proposition 5.6.5: The Tower Maps are Fibrations with Eilenberg-MacLane Fiber

Let X be a connected CW complex with Postnikov tower $\{X_n, p_n\}$ as in theorem 5.6.4. For each $n \geq 2$ the map $p_n\colon X_n \to X_{n-1}$ may be converted (up to homotopy) into a fibration whose fiber F is an Eilenberg-MacLane space $K(\pi_n(X), n)$.

***Proof*:**

Replace p_n by a fibration $\tilde{p}_n\colon \tilde{X}_n \to X_{n-1}$ with $\tilde{X}_n \simeq X_n$, using the standard mapping-path-space construction of definition 5.2.2: set $\tilde{X}_n = \{(x, \gamma) : x \in X_n,\ \gamma\colon I \to X_{n-1},\ \gamma(0) = p_n(x)\}$, with $\tilde{p}_n(x, \gamma) = \gamma(1)$. The inclusion $X_n \hookrightarrow \tilde{X}_n$, $x \mapsto (x, \mathrm{const})$, is a homotopy equivalence, and $\tilde{p}_n$ is a fibration. Let $F = \tilde{p}_n^{-1}(b_0)$ be the fiber over the basepoint.

The long exact sequence of the fibration $F \to \tilde{X}_n \to X_{n-1}$ (theorem 5.2.9) reads

$$\cdots \to \pi_{i+1}(X_{n-1}) \to \pi_i(F) \to \pi_i(X_n) \to \pi_i(X_{n-1}) \to \cdots$$

Insert the known homotopy groups. For $i \leq n-1$, the map $\pi_i(X_n) \to \pi_i(X_{n-1})$ is the isomorphism $\pi_i(X) \to \pi_i(X)$ induced by p_n (both equal $\pi_i(X)$ via f_n, f_{n-1}, and $p_n \circ f_n \simeq f_{n-1}$), so the connecting maps force $\pi_i(F) = 0$ for $i < n$ and the segment around $i = n$ becomes

$$\underbrace{\pi_{n+1}(X_{n-1})}_{=0} \to \pi_n(F) \to \underbrace{\pi_n(X_n)}_{=\pi_n(X)} \to \underbrace{\pi_n(X_{n-1})}_{=0}$$

so $\pi_n(F) \cong \pi_n(X)$. For $i > n$ both $\pi_i(X_n) = 0$ and $\pi_{i+1}(X_{n-1}) = 0$, forcing $\pi_i(F) = 0$. Thus F has $\pi_n(F) = \pi_n(X)$ and all other homotopy groups zero, which is exactly a $K(\pi_n(X), n)$ by definition 5.5.1, as we sought to show.

The proposition says that passing from X_{n-1} to X_n adds precisely the homotopy group $\pi_n(X)$, packaged as the fiber $K(\pi_n(X), n)$ of a fibration. To know X_n given X_{n-1}, it therefore suffices to know how this $K(\pi_n(X), n)$ is fibered over X_{n-1}. The remaining structural input is that this fibration is principal, which converts the fibration data into a single map out of X_{n-1}.

A fibration $F \to E \xrightarrow{p} B$ is *principal* if it is the pullback of a path fibration: there is a space B', a map $k\colon B \to B'$, and a homotopy equivalence over B between E and the pullback $k^*PB' = B \times_{B'} PB'$, where $PB' \to B'$ is the path fibration with fiber $\Omega B'$. When $B' = K(\pi, n{+}1)$, the fiber $\Omega B' \simeq K(\pi, n)$ matches the fiber of the tower map. The classifying map k then carries all the information of the fibration. For Postnikov towers of simple spaces this is always the situation.

Proposition 5.6.6: Principality of the Postnikov Fibrations

Let X be a simple connected CW complex with Postnikov tower $\{X_n, p_n\}$. Then for each $n \geq 2$ the fibration $K(\pi_n(X), n) \to X_n \to X_{n-1}$ of proposition 5.6.5 is principal: there is a map

$$k_n\colon X_{n-1} \to K\big(\pi_n(X),\, n+1\big)$$

and a homotopy equivalence $X_n \simeq k_n^* PK(\pi_n(X), n+1)$ over X_{n-1}, where $PK \to K$ is the path fibration of $K = K(\pi_n(X), n+1)$.

***Proof*:**
Write $\pi = \pi_n(X)$ and $K = K(\pi, n+1)$, so that $\Omega K \simeq K(\pi, n)$ (by example 5.2.12 and the uniqueness of theorem 5.5.4) is the fiber of p_n. Use the based path fibration $PK \to K$, $\gamma \mapsto \gamma(0)$, where $PK = \{\gamma\colon I \to K : \gamma(1) = *\}$ is contractible with fiber ΩK over the basepoint. The simplicity hypothesis guarantees that $\pi_1(X_{n-1})$ acts trivially on $\pi_n(F) = \pi$, so the obstruction classes below live in ordinary (untwisted) cohomology. The proof produces a class k_n, builds the comparison map $X_n \to k_n^* PK$, and shows it is a weak equivalence. It invokes the section form of obstruction theory, developed in theorem 5.6.10 and box 5.6.13 below; there is no circularity, as the proof of theorem 5.6.10 makes no use of the present proposition.

Step 1: The class k_n as the obstruction to a section of p_n. Convert p_n to a fibration $K(\pi, n) \to X_n \to X_{n-1}$ as in proposition 5.6.5, and apply the section form of obstruction theory (theorem 5.6.10, in its lifting/section guise of box 5.6.13) to the problem of constructing a section $s\colon X_{n-1} \to X_n$ of p_n. The obstructions to building s over successive skeleta of X_{n-1} live in $H^m(X_{n-1}; \pi_{m-1}(K(\pi, n)))$. Since the fiber $K(\pi, n)$ is $(n-1)$-connected, $\pi_{m-1}(K(\pi, n)) = 0$ for $m - 1 < n$, so all obstructions in degrees $m \leq n$ vanish and a section exists over the n-skeleton $X_{n-1}^{(n)}$. The primary obstruction to extending it over the $(n+1)$-skeleton is a cocycle $\mathfrak{o} \in C^{n+1}(X_{n-1}; \pi_n(K(\pi, n))) = C^{n+1}(X_{n-1}; \pi)$, and since $\pi_{m-1}(K(\pi, n)) = 0$ for $m - 1 > n$ as well, this is the *only* obstruction to a global section. Define

$$k_n := [\mathfrak{o}] \in H^{n+1}(X_{n-1}; \pi),$$

the complete obstruction to sectioning p_n.

Step 2: k_n as a map and the key vanishing. By representability (theorem 5.5.9), the isomorphism $[X_{n-1}, K] \cong H^{n+1}(X_{n-1}; \pi)$ sends a map $g\colon X_{n-1} \to K$ to $g^*\iota$, where $\iota \in H^{n+1}(K; \pi)$ is the fundamental class (definition 5.5.7). Let g be the map corresponding to k_n, so $g^*\iota = k_n$. The decisive fact is

$$p_n^* k_n = 0 \in H^{n+1}(X_n; \pi) \tag{5.1}$$

which holds because k_n is, by construction, the obstruction to a section of p_n, and the obstruction class is natural: pulling p_n back along itself gives the fibration $q\colon X_n \times_{X_{n-1}} X_n \to X_n$ (projection to the second factor) whose obstruction-to-a-section class is $p_n^* k_n$, by naturality. But q has the diagonal $x \mapsto (x, x)$ as a tautological section, so its obstruction class vanishes, giving $p_n^* k_n = 0$. By representability, (5.1) says exactly that $g \circ p_n \colon X_n \to K$ is nullhomotopic.

Step 3: The comparison map. A nullhomotopy of $g \circ p_n$ is a map $H \colon X_n \times I \to K$ with $H_0 = g \circ p_n$ and $H_1 = *$. For each $x \in X_n$ the path $t \mapsto H(x, 1-t)$ runs from $*$ to $g(p_n(x))$; reversing and recording it as $\gamma_x \in PK$ with $\gamma_x(0) = g(p_n(x))$ and $\gamma_x(1) = *$ defines a map

$$\Psi \colon X_n \to g^* PK = X_{n-1} \times_K PK, \qquad \Psi(x) = (p_n(x), \gamma_x),$$

which is a map over X_{n-1} (it covers p_n). The pullback $g^* PK \to X_{n-1}$ is a fibration whose fiber over b is $\{\gamma : \gamma(0) = g(b),\ \gamma(1) = *\} \cong \Omega K \simeq K(\pi, n)$, the same fiber as p_n.

Step 4: Ψ is a weak equivalence. Compare the long exact homotopy sequences (theorem 5.2.9) of the two fibrations $X_n \to X_{n-1}$ and $g^* PK \to X_{n-1}$ over the common base, with Ψ inducing a map between them that is the identity on X_{n-1} and the canonical fiber identification on fibers:

$$\begin{tikzcd}
\pi_i(\Omega K) \arrow[r] \arrow[d, "\cong"] & \pi_i(X_n) \arrow[r] \arrow[d, "\Psi_*"] & \pi_i(X_{n-1}) \arrow[r] \arrow[d, equal] & \pi_{i-1}(\Omega K) \arrow[d, "\cong"] \\
\pi_i(\Omega K) \arrow[r] & \pi_i(g^* PK) \arrow[r] & \pi_i(X_{n-1}) \arrow[r] & \pi_{i-1}(\Omega K)
\end{tikzcd}$$

The outer vertical maps are isomorphisms on $\pi_i(\Omega K) = \pi_i(K(\pi, n))$ (both fibers are $K(\pi, n)$, and on π_i the restriction of Ψ to fibers is the canonical identification of π_i of the fiber of p_n with $\pi_i(\Omega K)$ furnished by the connecting nullhomotopy), and the base maps are equalities. The five lemma (see [10, Five Lemma]) then forces $\Psi_* \colon \pi_i(X_n) \to \pi_i(g^* PK)$ to be an isomorphism for every i. Hence Ψ is a weak equivalence. This is not yet enough for principality: theorem 5.3.5 would promote Ψ to a *plain* homotopy equivalence $X_n \simeq g^* PK$, whereas principality requires the stronger conclusion that Ψ is a homotopy equivalence *over* X_{n-1}, that is, a fiber homotopy equivalence respecting the projection. The fiberwise upgrade is given by Dold's theorem on fiber homotopy equivalences: a map of fibrations over a common base, covering the identity of that base and inducing a weak equivalence on total spaces, is a fiber homotopy equivalence whenever the fibers have the homotopy type of CW complexes (a full proof is given in [14]). Here Ψ covers $\mathrm{id}_{X_{n-1}}$, both fibers are the Eilenberg-MacLane space $K(\pi, n)$ (which has the homotopy type of a CW complex), and Ψ_* is an isomorphism on all homotopy groups of the total spaces, so the criterion applies and Ψ is a homotopy equivalence over X_{n-1}. Therefore $X_n \simeq g^* PK = k_n^* PK$ over X_{n-1}, establishing principality, completing the proof.

The map $g \colon X_{n-1} \to K(\pi, n+1)$ *is* the fibration: by representability (theorem 5.5.9) it is interchangeable with a cohomology class, and pulling back the contractible path fibration over K along g reconstructs the total space X_n. The whole transition from X_{n-1} to X_n is thus compressed into one cohomology class, the k-invariant.

Definition 5.6.7: k-invariant

Let X be a simple connected CW complex with Postnikov tower $\{X_n, p_n\}$. The *n-th k-invariant* of X is the cohomology class

$$k_n \in H^{n+1}\big(X_{n-1}; \pi_n(X)\big)$$

classifying the principal fibration $K(\pi_n(X), n) \to X_n \to X_{n-1}$, in the sense of proposition 5.6.6: under the representability isomorphism $[X_{n-1}, K(\pi_n(X), n+1)] \cong H^{n+1}(X_{n-1}; \pi_n(X))$ of theorem 5.5.9, k_n corresponds to the classifying map $k_n\colon X_{n-1} \to K(\pi_n(X), n+1)$, and $X_n \simeq k_n^* PK(\pi_n(X), n+1)$ over X_{n-1}.

The k-invariants are the gluing data the homotopy groups alone cannot see. The list $\pi_1(X), \pi_2(X), \ldots$ together with the classes $k_2 \in H^3(X_1; \pi_2)$, $k_3 \in H^4(X_2; \pi_3)$, $\ldots$ determine X up to weak equivalence, because they determine the entire tower $X_2, X_3, \ldots$ inductively and $X \simeq \varprojlim X_n$. A space whose every k-invariant vanishes is, stage by stage, a product of Eilenberg-MacLane spaces: $k_n = 0$ makes the classifying map nullhomotopic, so the pullback $X_n \simeq X_{n-1} \times K(\pi_n, n)$ splits, and by induction $X_n \simeq \prod_{i \le n} K(\pi_i, i)$. Such spaces are called *generalized Eilenberg-MacLane spaces*, and they are exactly the simple spaces with trivial Postnikov k-invariants. A nonzero k-invariant is the precise measure of how far X is from being a product of Eilenberg-MacLane spaces.

Example 5.6.8: The First k-invariant of $\mathbb{R}\mathrm{P}^\infty$

Take $X = \mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}_2, 1)$, the Eilenberg-MacLane space identified in example 5.5.6 via the contractible double cover $S^\infty \to \mathbb{R}\mathrm{P}^\infty$. Its homotopy groups are $\pi_1 = \mathbb{Z}_2$ and $\pi_i = 0$ for $i \ge 2$. The Postnikov tower is therefore

$$X_1 = \mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}_2, 1), \qquad X_n = X_1 \text{ for all } n \ge 1,$$

since there is no higher homotopy to install. Every k-invariant vanishes: $k_n \in H^{n+1}(X_{n-1}; \pi_n) = H^{n+1}(X_{n-1}; 0) = 0$ because $\pi_n(X) = 0$ for $n \ge 2$. The tower is constant from stage 1 onward, reflecting that X is already an Eilenberg-MacLane space. This is the degenerate case: an Eilenberg-MacLane space is its own Postnikov tower, with all k-invariants trivial because there are no nonzero coefficient groups to receive them.

The cohomological machinery just used to build towers solves the extension problem in the same breath. Suppose Y is a simple space (so $\pi_1(Y)$ acts trivially on each $\pi_n(Y)$), X is a CW complex with skeleta $X^{(n)}$, and a map $g\colon X^{(n)} \to Y$ is given on the n-skeleton. The question is whether g extends over the $(n+1)$-skeleton, and then over all of X. The answer is a single cohomology class.

The mechanism is local. Each $(n+1)$-cell e^{n+1}_α of X is attached by a map $\varphi_\alpha\colon S^n \to X^{(n)}$. Composing with g gives a map $g \circ \varphi_\alpha\colon S^n \to Y$, a based map whose homotopy class lies in $\pi_n(Y)$. To extend g over the cell e^{n+1}_α means to fill in a map $D^{n+1}_\alpha \to Y$ agreeing with $g \circ \varphi_\alpha$ on the boundary S^n, which is possible exactly when $g \circ \varphi_\alpha$ is nullhomotopic, that is when $[g \circ \varphi_\alpha] = 0$ in $\pi_n(Y)$. Assembling these obstructions over all cells produces a cellular cochain.

Definition 5.6.9: Obstruction Cochain

Let Y be a simple space, X a CW complex, and $g\colon X^{(n)} \to Y$ a map on the n-skeleton. The *obstruction cochain* of g is the cellular $(n+1)$-cochain

$$\mathfrak{o}(g) \in C^{n+1}\big(X; \pi_n(Y)\big), \qquad \mathfrak{o}(g)(e_\alpha^{n+1}) = [\, g \circ \varphi_\alpha \,] \in \pi_n(Y),$$

assigning to each $(n+1)$-cell e_α^{n+1}, attached by $\varphi_\alpha\colon S^n \to X^{(n)}$, the homotopy class of $g \circ \varphi_\alpha$.

The obstruction cochain records, cell by cell, the local failure of g to extend. Its vanishing on a particular cell means g extends over that cell; its vanishing identically means g extends over the whole $(n+1)$-skeleton. The substance of the theory is that $\mathfrak{o}(g)$ is automatically a cocycle, and that its cohomology class (not the cochain itself) is the honest obstruction: the class vanishes precisely when g can be *modified on the n-skeleton*, without changing it on the $(n-1)$-skeleton, so as to extend over the $(n+1)$-skeleton.

Theorem 5.6.10: The Obstruction to Extension

Let Y be a simple space, X a CW complex, and $g\colon X^{(n)} \to Y$ a map defined on the n-skeleton. Then:

1. The obstruction cochain $\mathfrak{o}(g) \in C^{n+1}(X; \pi_n(Y))$ is a cocycle, so it determines a class $[\mathfrak{o}(g)] \in H^{n+1}(X; \pi_n(Y))$.
2. The restriction $g|_{X^{(n-1)}}$ extends to a map $X^{(n+1)} \to Y$ if and only if $[\mathfrak{o}(g)] = 0$.

Consequently, if $[\mathfrak{o}(g)] = 0$ then g may be altered on the n-skeleton, rel $X^{(n-1)}$, to a map that extends over the $(n+1)$-skeleton.

***Proof*:**
Write $\pi = \pi_n(Y)$. The proof has three movements: $\mathfrak{o}(g)$ is a cocycle; changing g on the n-skeleton changes $\mathfrak{o}(g)$ by a coboundary; and g extends iff the resulting class vanishes.

Step 1: $\mathfrak{o}(g)$ is a cocycle. Recognize $\mathfrak{o}(g)$ as a composite of two homomorphisms, after which the cocycle property reduces to exactness of a pair sequence. Write

$$\partial\colon \pi_{n+1}\big(X^{(n+1)}, X^{(n)}\big) \to \pi_n\big(X^{(n)}\big), \qquad [F] \mapsto [F|_{S^n}],$$

for the connecting boundary of the pair (proposition 5.1.11), and $g_*\colon \pi_n(X^{(n)}) \to \pi_n(Y)$ for the map induced by g. The relative group $\pi_{n+1}(X^{(n+1)}, X^{(n)})$ is free abelian on the classes $[\chi_\alpha]$ of the characteristic maps $\chi_\alpha\colon (D^{n+1}, S^n) \to (X^{(n+1)}, X^{(n)})$ of the $(n+1)$-cells, one generator per cell, with $\partial[\chi_\alpha] = [\varphi_\alpha] \in \pi_n(X^{(n)})$; it is abelian since $n \geq 1$, and identifies with the cellular chain group $C_{n+1}(X)$ in degree $n+1$ (definition 5.1.8, proposition 5.1.11). Simplicity of Y makes $\pi_n(Y)$ abelian with trivial $\pi_1(Y)$-action, so g_* is a well-defined homomorphism independent of basepoint paths. Reading off the cellular basis,

$$(g_* \circ \partial)[\chi_\alpha] = g_*[\varphi_\alpha] = [g \circ \varphi_\alpha] = \mathfrak{o}(g)(e_\alpha^{n+1}),$$

so the obstruction cochain *is* the homomorphism $\mathfrak{o}(g) = g_* \circ \partial\colon C_{n+1}(X) \to \pi_n(Y)$.

Now compute the coboundary on an $(n+2)$-cell e_β^{n+2}, with characteristic map χ_β and attaching map $\Phi_\beta\colon S^{n+1} \to X^{(n+1)}$. The cellular boundary is $\partial_{n+2}[\chi_\beta] = j\,[\Phi_\beta]$, where $[\Phi_\beta] \in$

$\pi_{n+1}(X^{(n+1)})$ is the absolute class of the attaching map and $j\colon \pi_{n+1}(X^{(n+1)}) \to \pi_{n+1}(X^{(n+1)}, X^{(n)})$ is induced by inclusion (the standard identification of the cellular boundary $\partial_{n+2}e^{n+2}_\beta = \sum_\alpha d_{\beta\alpha}\, e^{n+1}_\alpha$ with the degrees $d_{\beta\alpha}$ of Φ_β on the $(n+1)$-cells). Therefore

$$\delta\mathfrak{o}(g)(e^{n+2}_\beta) = \mathfrak{o}(g)\big(\partial_{n+2}[\chi_\beta]\big) = (g_* \circ \partial \circ j)[\Phi_\beta]$$

By exactness of the long exact homotopy sequence of the pair $(X^{(n+1)}, X^{(n)})$ at the term $\pi_{n+1}(X^{(n+1)}, X^{(n)})$, the composite $\partial \circ j = 0$: an absolute class pushed into the relative group has constant boundary restriction $F|_{S^n}$. Hence $\delta\mathfrak{o}(g)(e^{n+2}_\beta) = g_*(0) = 0$ for every β, so $\delta\mathfrak{o}(g) = 0$ and $\mathfrak{o}(g)$ is a cocycle.

Step 2: Recoordinatizing g on n-cells. Fix g on $X^{(n-1)}$ and consider all maps $g'\colon X^{(n)} \to Y$ agreeing with g on $X^{(n-1)}$. Two such, g and g', restrict to maps on each n-cell that agree on its boundary S^{n-1}; gluing the two disks along their common boundary yields a map $S^n \to Y$, whose class is an element $d(g,g')(e^n_\alpha) \in \pi_n(Y)$ for each n-cell e^n_α. This defines the *difference cochain* $d(g,g') \in C^n(X;\pi)$. Every cochain arises this way: given any $c \in C^n(X;\pi)$, one constructs g' agreeing with g on $X^{(n-1)}$ and realizing $d(g,g') = c$, by modifying g on the interior of each n-cell using a representative of $c(e^n_\alpha) \in \pi_n(Y)$ (collapse a small disk in the cell's interior to a point, map the resulting pinch sphere by the representative, and keep the old map on the complement). The relation between difference cochains and obstruction cochains is the coboundary formula

$$\mathfrak{o}(g') - \mathfrak{o}(g) = \delta\, d(g,g') \tag{5.2}$$

which is verified cell by cell: on an $(n+1)$-cell e^{n+1}_α with attaching map φ_α, the classes $[g' \circ \varphi_\alpha]$ and $[g \circ \varphi_\alpha]$ differ by the sum over the boundary n-cells of e^{n+1}_α of the difference classes $d(g,g')$, weighted by the incidence degrees, which is exactly $(\delta d(g,g'))(e^{n+1}_\alpha)$. The identity (5.2) shows that as g' ranges over all modifications of g rel $X^{(n-1)}$, the obstruction cochain $\mathfrak{o}(g')$ ranges over the entire coset $\mathfrak{o}(g) + \delta C^n(X;\pi)$. Therefore the cohomology class $[\mathfrak{o}(g)] \in H^{n+1}(X;\pi)$ depends only on $g|_{X^{(n-1)}}$, and is unchanged by any modification of g on the n-skeleton rel $X^{(n-1)}$.

Step 3: Extension criterion. Suppose first that $g|_{X^{(n-1)}}$ extends to a map $h\colon X^{(n+1)} \to Y$. Then $h|_{X^{(n)}}$ is a map on the n-skeleton agreeing with g on $X^{(n-1)}$, and since h extends over every $(n+1)$-cell, $\mathfrak{o}(h)(e^{n+1}_\alpha) = [h \circ \varphi_\alpha] = 0$ for all α (the map h on the cell is a nullhomotopy of $h \circ \varphi_\alpha$). Thus $\mathfrak{o}(h) = 0$, and by (5.2) with $g' = h|_{X^{(n)}}$,

$$[\mathfrak{o}(g)] = [\mathfrak{o}(h|_{X^{(n)}})] = [0] = 0$$

in $H^{n+1}(X;\pi)$. So extendability forces $[\mathfrak{o}(g)] = 0$.

Conversely, suppose $[\mathfrak{o}(g)] = 0$, so $\mathfrak{o}(g) = \delta c$ for some $c \in C^n(X;\pi)$. By the surjectivity established in Step 2, there is a map $g'\colon X^{(n)} \to Y$ agreeing with g on $X^{(n-1)}$ with $d(g,g') = -c$. Then (5.2) gives

$$\mathfrak{o}(g') = \mathfrak{o}(g) + \delta\, d(g,g') = \delta c - \delta c = 0$$

as cochains. An obstruction cochain that vanishes identically means $[g' \circ \varphi_\alpha] = 0$ for every $(n+1)$-cell, so $g' \circ \varphi_\alpha$ is nullhomotopic and g' extends over each $(n+1)$-cell. These extensions assemble to a map $X^{(n+1)} \to Y$ extending $g' \simeq g$ rel $X^{(n-1)}$. This proves both directions of (2), and the final assertion is the construction of g' just given, completing the proof.

The theorem isolates exactly what cohomology measures in this context. The cochain $\mathfrak{o}(g)$ is a *local* record, one element of $\pi_n(Y)$ per $(n+1)$-cell, of whether the map extends over that cell. The passage to the cohomology class $[\mathfrak{o}(g)]$ quotients out the freedom to modify g on the n-skeleton: a coboundary δc can always be absorbed by recoordinatizing g, so only the class survives as a genuine obstruction. This is why cohomology, and not the cochain group, is the natural home of obstructions: the cochain depends on a choice (the map g on n-cells), while the class depends only on the geometrically meaningful data (g on the $(n-1)$-skeleton). The same coefficient group $\pi_n(Y)$ and the same degree H^{n+1} appear here as in the k-invariant of definition 5.6.7, and for the same reason: building a map across cells of dimension $n+1$ tests π_n of the target, and consistency across cells of dimension $n+2$ produces the cocycle condition, landing the obstruction in H^{n+1}. The k-invariant is the special case where the "map being extended" is the section that would split the Postnikov fibration.

Example 5.6.11: Extending a Map over a Moore Space Cell

Let $Y = K(\pi, n)$ and take the two-cell complex $X = S^n \cup_d e^{n+1}$, the mapping cone of a degree-d self-map of S^n (when $\pi = \mathbb{Z}$ this is the Moore space $M(\mathbb{Z}_d, n)$ of section 3.5). Its n-skeleton is $X^{(n)} = S^n$, with the single $(n+1)$-cell attached by the degree-d map $\varphi\colon S^n \to S^n$. A map $g\colon X^{(n)} = S^n \to K(\pi, n)$ is, by representability (theorem 5.5.9), the same datum as a class in $[S^n, K(\pi, n)] = \tilde{H}^n(S^n; \pi) = \pi$; write $a \in \pi$ for that class, so that $g_*\colon \pi_n(S^n) = \mathbb{Z} \to \pi_n(K(\pi, n)) = \pi$ carries the generator to a.

The obstruction cochain has a single value, on the lone $(n+1)$-cell:

$$\mathfrak{o}(g)(e^{n+1}) = [g \circ \varphi] = g_*([\varphi]) = g_*(d \cdot 1) = d \cdot a \in \pi,$$

using that φ has degree d, so $[\varphi] = d \in \pi_n(S^n) = \mathbb{Z}$. The fixed map g extends over X exactly when $g \circ \varphi$ is nullhomotopic, that is, when the cochain value vanishes:

$$g \text{ extends over } X \iff d \cdot a = 0 \text{ in } \pi$$

the criterion governing this two-cell complex. For $\pi = \mathbb{Z}$ and $d \geq 1$ this forces $a = 0$: only the nullhomotopic g extends. For $\pi = \mathbb{Z}_d$ every a satisfies $d\,a = 0$, so every g extends.

The cohomology class refines this to a statement about the basepoint datum $g|_{X^{(n-1)}}$. The cellular cochain complex of X in the relevant degrees is $C^n(X; \pi) = \pi \xrightarrow{\times d} C^{n+1}(X; \pi) = \pi$, so $H^{n+1}(X; \pi) = \pi / d\pi$ and the coboundaries are exactly $d\pi$. Since $\mathfrak{o}(g) = da \in d\pi$, the class $[\mathfrak{o}(g)] = 0$ for *every* a, reflecting that $g|_{X^{(n-1)}}$ is the basepoint, which always extends (by the constant map) once one is permitted to alter g on the n-cell. The contrast is exactly the content of theorem 5.6.10: the vanishing cochain $\mathfrak{o}(g) = 0$ controls extension of the *fixed* g, while the vanishing class $[\mathfrak{o}(g)] = 0$ controls extension of $g|_{X^{(n-1)}}$ after modification on the n-skeleton. The two coincide precisely when the n-skeleton offers no modification freedom, here visible in $\tilde{H}^n(X; \pi) = \ker(\times d\colon \pi \to \pi)$, which is 0 for $\pi = \mathbb{Z}$ (no freedom, so $a = 0$ is forced) and all of $\mathbb{Z}_d$ for $\pi = \mathbb{Z}_d$ (full freedom).

Returning to the example with which the section began, the obstruction-theoretic computation pins down the first nontrivial k-invariant of S^2, the class that distinguishes its third Postnikov section from a product.

Example 5.6.12: The First k-invariant of S^2

The space S^2 has $\pi_2 = \mathbb{Z}$, $\pi_3 = \mathbb{Z}$ (generated by the Hopf map η, example 5.2.7). Its Postnikov tower has $X_2 = K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$. The first nontrivial k-invariant is

$$k_3 \in H^4\big(X_2; \pi_3(S^2)\big) = H^4(\mathbb{CP}^\infty; \mathbb{Z})$$

using $\pi_3(S^2) = \mathbb{Z}$. The cohomology ring of $\mathbb{C}\mathrm{P}^\infty$ is the polynomial ring $\mathbb{Z}[u]$ with $u \in H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$ a generator (computed in chapter 4 via the cup product on projective space), so $H^4(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = \mathbb{Z}\langle u^2 \rangle$ is infinite cyclic, generated by u^2. The class k_3 must be a multiple $k_3 = m\, u^2$ for some integer m. To identify m, use that X_3 has $\pi_2 = \pi_3 = \mathbb{Z}$ and that the fibration $K(\mathbb{Z}, 3) \to X_3 \to \mathbb{C}\mathrm{P}^\infty$ has its connecting behavior governed by k_3. The transgression in the Serre spectral sequence of this fibration (developed in chapter 6) sends the fundamental class of the fiber $K(\mathbb{Z}, 3)$, namely $\iota_3 \in H^3(K(\mathbb{Z}, 3); \mathbb{Z}) = \mathbb{Z}$, to $k_3 \in H^4(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$, and matching this against the known cohomology of S^2 (which X_3 approximates through degree 3) forces k_3 to generate $H^4(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$ up to sign, that is $m = \pm 1$ and

$$k_3 = \pm\, u^2 \in H^4(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$$

a generator. The nonvanishing of k_3 is the precise statement that $X_3 \not\simeq K(\mathbb{Z}, 2) \times K(\mathbb{Z}, 3)$: the Hopf map $\eta \in \pi_3(S^2)$ is detected by the cup-square $u \mapsto u^2$, and this nontrivial cup product is exactly the obstruction to splitting the third Postnikov section of S^2 as a product of Eilenberg-MacLane spaces.

5.6.13 Remark: Obstructions to lifting and to sections The extension theorem (theorem 5.6.10) also governs the dual problems of *lifting* a map across a fibration and of finding a *section* of a fibration, by the same argument applied fiberwise. Given a fibration $F \to E \xrightarrow{p} B$ with F simple, and a map $g\colon X \to B$, a lift $\tilde{g}\colon X \to E$ with $p\tilde{g} = g$ is built cell by cell: over each cell of X the fibration $g^*E \to X$ is trivial, so extending a partial lift across an $(n+1)$-cell amounts to extending a map of the boundary S^n into the fiber F, exactly the situation of the theorem with $Y = F$. The obstruction cochain assigns to each $(n+1)$-cell the class in $\pi_n(F)$ of the partial lift on its attaching sphere, is a cocycle by the identical pair-sequence argument, and its class in $H^{n+1}(X; \pi_n(F))$ (with local coefficients if F is not simple) is the obstruction to extending the lift over the $(n+1)$-skeleton; a lift over the whole skeleton filtration exists precisely when these classes vanish in turn. A section is the case $X = B$, $g = \mathrm{id}_B$. The Postnikov k-invariant $k_n \in H^{n+1}(X_{n-1}; \pi_n)$ is precisely the obstruction to sectioning the fibration p_n, which is why the same cohomology group hosts both the tower's gluing data and the extension theory's obstructions.

Exercise 5.6.1

1. Let X be a connected CW complex whose only nonzero homotopy group is $\pi_n(X) = \pi$ for a single $n \geq 1$. Show that every Postnikov section X_m for $m \geq n$ is an Eilenberg-MacLane space $K(\pi, n)$, that $X_m \simeq *$ for $m < n$, and that every k-invariant of X vanishes. Conclude that $X \simeq K(\pi, n)$.

2. Let X be a simple connected CW complex. Show that if every k-invariant of the Postnikov tower of X vanishes, then $X_n \simeq \prod_{i=1}^{n} K(\pi_i(X), i)$ for each n, and hence X is weakly equivalent to the product $\prod_{i\geq 1} K(\pi_i(X), i)$. (Such an X is a *generalized Eilenberg-MacLane space.*)

3. Let Y be a simple space and X a CW complex with $H^{n+1}(X; \pi_n(Y)) = 0$ for all $n \geq 1$. Show that every map $g\colon X^{(1)} \to Y$ from the 1-skeleton (sending the basepoint to the basepoint) extends to a map $X \to Y$. Deduce that if X has trivial reduced cohomology in all positive degrees with all relevant coefficients, then $[X, Y] = *$ for every such Y.

4. Using theorem 5.6.10, prove that a fibration $F \to E \xrightarrow{p} B$ over a CW complex B, with

$F = K(\pi, n)$ a single Eilenberg-MacLane space and B simply connected, admits a section if and only if a single class $\theta \in H^{n+1}(B; \pi)$ vanishes. Identify θ as the k-invariant of the fibration.

5. Let $X = S^n$ for $n \geq 2$. Compute the first two stages X_{n-1} and X_n of the Postnikov tower of S^n, and identify the n-th section X_n as an Eilenberg-MacLane space. Determine the lowest degree in which a nonzero k-invariant can occur.

6. Show that the obstruction class $[\mathfrak{o}(g)] \in H^{n+1}(X; \pi_n(Y))$ of theorem 5.6.10 is natural: if $h\colon X' \to X$ is a cellular map and $g\colon X^{(n)} \to Y$, then $[\mathfrak{o}(g \circ h|_{X'^{(n)}})] = h^*[\mathfrak{o}(g)]$ in $H^{n+1}(X'; \pi_n(Y))$. Explain why this naturality is what makes obstruction classes computable.

5.7 The Freudenthal Suspension Theorem

Computing the homotopy groups of spheres has been the hardest problem of this chapter. The Hurewicz theorem (theorem 5.4.9) pins down the first nonzero group, the long exact sequence of the Hopf fibration (example 5.2.13) reaches $\pi_3(S^2) \cong \mathbb{Z}$, and beyond that the groups $\pi_i(S^n)$ resist every direct method. Against this difficulty one regularity stands out. Laid out along diagonals, the groups $\pi_{n+k}(S^n)$ for a fixed k appear to stop depending on n once n is large: the value settles down as one climbs the diagonal. The Freudenthal suspension theorem is the explanation. It studies the suspension homomorphism $\sigma\colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$, which on spheres is exactly the comparison $\pi_{n+k}(S^n) \to \pi_{n+1+k}(S^{n+1})$ moving one step up the diagonal, and identifies the precise range in which σ is an isomorphism. This section defines that homomorphism, proves the theorem, and draws its first consequences. The single hard input is homotopy excision (the Blakers–Massey theorem), whose long technical proof is the one book-length deferral in the argument; everything else is established here. The reward is that the stable value along each diagonal is a genuine, computable invariant, the *stable homotopy group* $\pi_k^s = \varinjlim_n \pi_{n+k}(S^n)$. These groups are the foundation on which the stable homotopy category and the theory of spectra are later built (chapter 8); the present section is what makes them well defined.

The reduced suspension ΣX was introduced in chapter 1 as the quotient of SX collapsing the segment $\{x_0\} \times I$ through the basepoint, homotopy equivalent to the unreduced suspension for a CW complex and equal to the smash product $X \wedge S^1$. The construction needed here presents ΣX as two cones on X glued along their common base, which is the form to which homotopy excision applies.

Definition 5.7.1: Reduced Suspension as a Union of Cones

Let (X, x_0) be a pointed CW complex. The *reduced cone* on X is

$$CX = (X \times I)/\big((X \times \{1\}) \cup (\{x_0\} \times I)\big)$$

the quotient of the cylinder collapsing the top $X \times \{1\}$ to the cone point and the segment $\{x_0\} \times I$ to the basepoint, with X included as the base $X \times \{0\}$. The reduced cone is contractible, the contraction sliding each point up to the cone point. The *reduced suspension* ΣX is the union of two reduced cones glued along their common base copy of X:

$$\Sigma X = C_{-}X \cup_X C_{+}X$$

where $C_{-}X$ and $C_{+}X$ are two copies of CX and the gluing identifies the base $X \times \{0\}$ of one with the base of the other. Equivalently $\Sigma X = (X \times I)/\sim$, collapsing $X \times \{0\}$ to one point, $X \times \{1\}$ to another, and $\{x_0\} \times I$ to the basepoint; the two cones are the images of $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$, meeting in the equatorial copy $X \times \{\frac{1}{2}\}$.

The decomposition $\Sigma X = C_{-}X \cup_X C_{+}X$ presents the suspension as a space covered by two contractible pieces whose intersection is X (more precisely the equatorial copy $X \times \{\frac{1}{2}\}$, which the gluing identifies with the base X). This is the topological analogue of writing a sphere as the union of its two closed hemispheres meeting along the equator, and it is exactly the configuration of an excisive triad to which the homotopy excision theorem will apply. Each cone contributes its base X on the boundary and nothing in its interior up to homotopy, so the relative homotopy of the pair $(\Sigma X, C_{+}X)$ will turn out to record the homotopy of ΣX itself, while the relative homotopy of $(C_{-}X, X)$ records the homotopy of X shifted by one.

Example 5.7.2: Spheres as Iterated Suspensions

Take $X = S^n$ with $n \geq 0$. The two cones $C_{\pm}S^n$ are copies of the disk D^{n+1} (the reduced cone on a sphere is a disk, the cone point being the disk's center and the base S^n its boundary), glued along their common boundary S^n. Two disks glued along their boundary sphere form S^{n+1}, so

$$\Sigma S^n = S^{n+1}$$

recovering the prototypical suspension $S(S^n) = S^{n+1}$ of chapter 1. Iterating, $\Sigma^k S^n = S^{n+k}$. The upper cone $C_{+}S^n = D^{n+1}$ is the closed upper hemisphere of S^{n+1}, the lower cone $C_{-}S^n = D^{n+1}$ the closed lower hemisphere, and their intersection the equator $S^n \subset S^{n+1}$, matching the hemisphere decomposition exactly. This is the case that drives the entire theory: the suspension homomorphism on spheres is the comparison $\pi_i(S^n) \to \pi_{i+1}(S^{n+1})$ whose stabilization is the subject of the theorem.

Suspension is a functor: a based map $f\colon X \to Y$ induces $\Sigma f\colon \Sigma X \to \Sigma Y$ by suspending in the cone coordinate, and a based homotopy of maps suspends to a based homotopy. Applying this to a representative $S^i \to X$ of a homotopy class and using $\Sigma S^i = S^{i+1}$ produces a map $S^{i+1} \to \Sigma X$, hence a homomorphism on homotopy groups. This is the central construction of the section.

Definition 5.7.3: The Suspension Homomorphism

Let (X, x_0) be a pointed space and $i \geq 1$. The *suspension homomorphism*

$$\sigma \colon \pi_i(X, x_0) \longrightarrow \pi_{i+1}(\Sigma X, *)$$

sends the class of a based map $f \colon S^i \to X$ to the class of its suspension $\Sigma f \colon S^{i+1} = \Sigma S^i \to \Sigma X$, where $*$ is the suspension basepoint. The assignment is well defined on homotopy classes because a based homotopy $f \simeq f'$ suspends to a based homotopy $\Sigma f \simeq \Sigma f'$, and it is a homomorphism because Σ carries the pinch map defining the group operation on $\pi_i(X)$ (concatenation of cubes) to the pinch map defining it on $\pi_{i+1}(\Sigma X)$, so $\Sigma(f + g)$ is based homotopic to $\Sigma f + \Sigma g$.

The suspension homomorphism is the algebraic form of the geometric operation of suspending a map of spheres. A class in $\pi_i(X)$ is a way of wrapping an i-sphere through X; suspending wraps an $(i + 1)$-sphere through ΣX by coning off the original wrapping at both poles. Whether this loses information is precisely the question the theorem answers. The composite of suspensions $\sigma \colon \pi_i(X) \to \pi_{i+1}(\Sigma X) \to \pi_{i+2}(\Sigma^2 X) \to \cdots$ is what the stable groups will be built from, and the theorem guarantees that this chain of homomorphisms is eventually constant.

Example 5.7.4: Suspension on the Bottom Class of a Sphere

Take $X = S^n$ and the generator $[\mathrm{id}] \in \pi_n(S^n) \cong \mathbb{Z}$ of corollary 5.4.11, the identity map of degree 1. Its suspension $\Sigma\mathrm{id} = \mathrm{id} \colon S^{n+1} \to S^{n+1}$ is again the identity, of degree 1, so

$$\sigma \colon \pi_n(S^n) \to \pi_{n+1}(S^{n+1}), \qquad \sigma[\mathrm{id}_{S^n}] = [\mathrm{id}_{S^{n+1}}]$$

carries a generator to a generator. More generally a degree-d map $S^n \to S^n$ suspends to a degree-d map $S^{n+1} \to S^{n+1}$, because suspension preserves degree (proposition 3.5.7). Hence $\sigma \colon \pi_n(S^n) \to \pi_{n+1}(S^{n+1})$ is the identity homomorphism $\mathbb{Z} \to \mathbb{Z}$, an isomorphism in every degree $n \geq 1$. This is the first nontrivial instance of the theorem: on the diagonal $i = n$, suspension of $\pi_n(S^n) = \mathbb{Z}$ is always an isomorphism, consistent with the assertion that σ is an isomorphism for $i < 2n - 1$ once $n \geq 2$, and with $\pi_1(S^1) \cong \mathbb{Z} \cong \pi_2(S^2)$ at the boundary $n = 1$.

The whole weight of the theorem rests on a single comparison of relative homotopy groups across the two cones. To set it up cleanly, recall the precise statement of homotopy excision. It is the homotopy-theoretic counterpart of the excision theorem in homology (theorem 3.4.7), but with a sharp limitation: where excision in homology is an exact statement valid for any excisive pair, excision in homotopy holds only in a range of dimensions determined by the connectivities of the pieces. This failure of exactness is the entire reason the homotopy of spheres is hard, and the range it produces is the source of the "stable range" throughout the chapter.

Theorem 5.7.5: Homotopy Excision (Blakers–Massey)

Let X be a CW complex written as a union $X = A \cup B$ of subcomplexes, with $C = A \cap B$ nonempty, so that $(X; A, B)$ is a CW triad. Suppose

1. the pair (A, C) is m-connected, that is $\pi_i(A, C) = 0$ for $i \leq m$, with $m \geq 0$; and
2. the pair (B, C) is n-connected, that is $\pi_i(B, C) = 0$ for $i \leq n$, with $n \geq 0$.

Then the inclusion of pairs $(A, C) \hookrightarrow (X, B)$ induces a homomorphism

$$\pi_i(A, C) \longrightarrow \pi_i(X, B)$$

that is an isomorphism for $i < m + n$ and an epimorphism for $i = m + n$.

***Proof Key ideas*:**
The theorem is the one book-length input of this section, and its proof is deferred. The argument compresses a map of the pair (X, B) off the cells of A not in C by a multiple-general-position argument, controlling the dimensions in which the compression can fail by the connectivities m and n; the combinatorics of the failure dimensions is exactly what produces the bound $i < m+n$. A complete and self-contained account, including the sharper version giving the first obstruction as a Whitehead product, is in [14, §4.2 and §4.J]. The tools it uses (relative homotopy groups, the compression lemma lemma 5.3.3, CW approximation theorem 5.3.9) are all available from this chapter, but the full general-position argument is long and self-contained, so it is cited rather than reproduced, completing the proof.

The theorem says that excision in homotopy holds, but only below the dimension $m + n$ set by the connectivities, and only as a surjection at the borderline degree. The contrast with homology is the heart of the matter: in homology the analogous comparison $H_i(A, C) \to H_i(X, B)$ is an isomorphism in *all* degrees (this is excision, theorem 3.4.7, since $X/B = A/C$ for such a triad), whereas in homotopy the isomorphism degrades to a range. Everything specific to homotopy theory, including the entire difficulty of computing homotopy groups of spheres, flows from this degradation. The two cones on X furnish the triad to which the theorem will be applied, and the connectivity inputs come from the next two identifications.

The first identification reads the relative homotopy of a cone-pair as the homotopy of the base, shifted by one. It is a direct consequence of the long exact sequence of the pair (proposition 5.1.11) together with the contractibility of the cone.

Proposition 5.7.6: Relative Homotopy of the Cone Pair

Let (X, x_0) be a pointed CW complex and let CX be its reduced cone, containing X as its base. Then the boundary homomorphism of the pair (CX, X) is an isomorphism

$$\partial \colon \pi_{i+1}(CX, X, x_0) \xrightarrow{\cong} \pi_i(X, x_0)$$

for every $i \geq 0$. In particular, if X is $(n-1)$-connected then the pair (CX, X) is n-connected.

***Proof*:**
The reduced cone CX is contractible, so $\pi_j(CX, x_0) = 0$ for all $j \geq 0$ (example 5.1.2, applied to any contractible space). The long exact sequence of the pair (CX, X) from proposition 5.1.11 reads

$$\cdots \to \pi_{i+1}(CX) \to \pi_{i+1}(CX, X) \xrightarrow{\partial} \pi_i(X) \to \pi_i(CX) \to \cdots$$

Both flanking absolute groups $\pi_{i+1}(CX)$ and $\pi_i(CX)$ vanish, so exactness forces the boundary map ∂ to be both surjective (its image is the kernel of $\pi_i(X) \to \pi_i(CX) = 0$, hence all of $\pi_i(X)$) and injective (its kernel is the image of $\pi_{i+1}(CX) = 0$, hence trivial). Thus ∂ is an isomorphism for every $i \geq 0$.

For the connectivity statement, suppose X is $(n-1)$-connected, so $\pi_i(X) = 0$ for $i \leq n-1$ (definition 5.4.7). The isomorphism ∂ then gives $\pi_{i+1}(CX, X) \cong \pi_i(X) = 0$ for $i \leq n-1$, that is $\pi_j(CX, X) = 0$ for $j \leq n$. By definition 5.4.7 applied to the pair, the pair (CX, X) is n-connected, as we sought to show.

The proposition is the device that converts a connectivity hypothesis on X into a connectivity hypothesis on a cone pair, exactly the input homotopy excision requires. A cone is contractible, so all of the relative homotopy of (CX, X) is the homotopy of the base X pushed up one degree by the boundary map; an $(n-1)$-connected base therefore yields an n-connected cone pair. The second identification reads the relative homotopy of the other cone pair, $(\Sigma X, C_+X)$, as the homotopy of the suspension.

Proposition 5.7.7: The Suspension Map as a Cone-Pair Inclusion

Let (X, x_0) be a pointed CW complex and write $\Sigma X = C_-X \cup_X C_+X$ as in definition 5.7.1. Then:

1. the map $j_* \colon \pi_{i+1}(\Sigma X, *) \to \pi_{i+1}(\Sigma X, C_+X, *)$ of the long exact sequence of the pair $(\Sigma X, C_+X)$ is an isomorphism for $i \geq 1$; and

2. under the cone isomorphism $\partial \colon \pi_{i+1}(C_-X, X) \xrightarrow{\cong} \pi_i(X)$ of proposition 5.7.6, the composite
$$\pi_i(X, x_0) \xrightarrow{\sigma} \pi_{i+1}(\Sigma X, *) \xrightarrow{j_*} \pi_{i+1}(\Sigma X, C_+X, *)$$
equals the inclusion-of-pairs map $\iota_* \colon \pi_{i+1}(C_-X, X) \to \pi_{i+1}(\Sigma X, C_+X)$ induced by $(C_-X, X) \hookrightarrow (\Sigma X, C_+X)$, precomposed with ∂^{-1}; that is, $j_* \circ \sigma = \iota_* \circ \partial^{-1}$.

***Proof*:**

1. *j_* is an isomorphism.* The upper cone C_+X is contractible (definition 5.7.1), so $\pi_j(C_+X) = 0$ for all j. The long exact sequence of the pair $(\Sigma X, C_+X)$ from proposition 5.1.11 reads
$$\pi_{i+1}(C_+X) \to \pi_{i+1}(\Sigma X) \xrightarrow{j_*} \pi_{i+1}(\Sigma X, C_+X) \xrightarrow{\partial} \pi_i(C_+X)$$
The flanking terms $\pi_{i+1}(C_+X)$ and $\pi_i(C_+X)$ vanish, so j_* has trivial kernel (the image of $\pi_{i+1}(C_+X) = 0$) and is surjective (its cokernel injects into $\pi_i(C_+X) = 0$); hence j_* is an isomorphism for $i \geq 1$.

2. *The composite is the pair inclusion.* A class $[f] \in \pi_i(X)$ has suspension $\sigma[f] = [\Sigma f] \in \pi_{i+1}(\Sigma X)$, represented by the suspended sphere $\Sigma f \colon S^{i+1} = \Sigma S^i \to \Sigma X$. Decompose

$S^{i+1} = C_-S^i \cup_{S^i} C_+S^i$ into its two cones (example 5.7.2); by construction the suspended map carries C_-S^i into C_-X, carries C_+S^i into C_+X, and carries the equator S^i into the equator X by f itself. Now $j_*\sigma[f]$ is $\sigma[f]$ regarded as a class of the pair $(\Sigma X, C_+X)$: it is represented by Σf viewed as a map $(D^{i+1}, S^i) \to (\Sigma X, C_+X)$, where $D^{i+1} = C_-S^i$ is the lower hemisphere (whose boundary sphere Σf sends into C_+X, since the equator and the upper cone both land in C_+X). This is precisely the image, under ι_*, of the relative class represented by $\Sigma f|_{C_-S^i} \colon (C_-S^i, S^i) \to (C_-X, X)$. Under the cone isomorphism ∂ of proposition 5.7.6, that relative class corresponds to $\partial[\Sigma f|_{C_-S^i}] = [f] \in \pi_i(X)$, because ∂ restricts a relative cube to its X-face and the X-face of $\Sigma f|_{C_-S^i}$ is f (definition 5.1.10). Reading the equality backward, $j_*\sigma[f] = \iota_*\big(\partial^{-1}[f]\big)$, which is the claimed identity $j_* \circ \sigma = \iota_* \circ \partial^{-1}$, completing the proof.

The two propositions reduce the suspension homomorphism to the homotopy-excision comparison map. After the isomorphism j_* of part (1) and the cone isomorphism ∂ of proposition 5.7.6, the suspension $\sigma\colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$ becomes the inclusion-induced map $\iota_*\colon \pi_{i+1}(C_-X, X) \to \pi_{i+1}(\Sigma X, C_+X)$, which is exactly the excision comparison of theorem 5.7.5 for the triad $(\Sigma X; C_-X, C_+X)$ in degree $i+1$ (the roles of A, B, C being C_-X, C_+X, X). The theorem then transfers the range in which excision is an isomorphism directly to the range in which suspension is one. Assembling the pieces gives the main result.

Theorem 5.7.8: Freudenthal Suspension Theorem

Let X be an $(n-1)$-connected pointed CW complex with $n \geq 1$. Then the suspension homomorphism

$$\sigma\colon \pi_i(X, x_0) \longrightarrow \pi_{i+1}(\Sigma X, *)$$

is an isomorphism for $i < 2n-1$ and an epimorphism for $i = 2n-1$.

***Proof*:**
Write $\Sigma X = C_-X \cup_X C_+X$ as in definition 5.7.1, with the equatorial intersection $C_-X \cap C_+X = X$. The proof applies homotopy excision to this triad, then translates the conclusion back to σ through the cone identifications.

Step 1: Connectivity of the two cone pairs. The space X is $(n-1)$-connected by hypothesis. By proposition 5.7.6 each cone pair (C_-X, X) and (C_+X, X) is then n-connected: the boundary isomorphism $\pi_j(C_\pm X, X) \cong \pi_{j-1}(X)$ vanishes for $j \leq n$ because $\pi_{j-1}(X) = 0$ for $j-1 \leq n-1$. Thus in the notation of theorem 5.7.5, applied to the triad $(\Sigma X; C_-X, C_+X)$ with $A = C_-X$, $B = C_+X$, $C = X$, both connectivity indices equal n:

$$m = n, \qquad (A, C) = (C_-X, X) \text{ is } n\text{-connected}, \qquad (B, C) = (C_+X, X) \text{ is } n\text{-connected}$$

Step 2: Apply homotopy excision. The triad $(\Sigma X; C_-X, C_+X)$ satisfies the hypotheses of theorem 5.7.5 with both connectivities equal to n. The theorem gives that the inclusion of pairs $(C_-X, X) \hookrightarrow (\Sigma X, C_+X)$ induces

$$\pi_i(C_-X, X) \longrightarrow \pi_i(\Sigma X, C_+X)$$

an isomorphism for $i < n + n = 2n$ and an epimorphism for $i = 2n$.

Step 3: Translate to the suspension homomorphism. By proposition 5.7.7, the suspension homomorphism $\sigma\colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$ is, after the isomorphism $j_*\colon \pi_{i+1}(\Sigma X) \xrightarrow{\cong} \pi_{i+1}(\Sigma X, C_+X)$ of that proposition, identified with the excision comparison map of Step 2 in degree $i+1$. Precisely, the commuting diagram

$$\begin{tikzcd} \pi_i(X) \arrow[r, "\sigma"] \arrow[d, "\partial^{-1}", "\cong"'] & \pi_{i+1}(\Sigma X) \arrow[d, "\cong", "j_*"'] \\ \pi_{i+1}(C_-X, X) \arrow[r] & \pi_{i+1}(\Sigma X, C_+X) \end{tikzcd}$$

has both vertical maps isomorphisms (the left by proposition 5.7.6, the right by proposition 5.7.7 Step 1), and the bottom map is the excision comparison in degree $i+1$. Therefore σ is an isomorphism exactly when the bottom map in degree $i+1$ is, and an epimorphism exactly when the bottom map is.

By Step 2 the bottom map in degree $i+1$ is an isomorphism for $i+1 < 2n$, that is $i < 2n-1$, and an epimorphism for $i+1 = 2n$, that is $i = 2n-1$. Transporting through the vertical isomorphisms, $\sigma\colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$ is an isomorphism for $i < 2n-1$ and an epimorphism for $i = 2n-1$, as we sought to show.

The theorem is the precise measure of how much information the suspension operation preserves. An $(n-1)$-connected space has its first possible homotopy in degree n, and the theorem says that suspension faithfully records all homotopy up to degree just below $2n-1$, weakening to surjectivity at the borderline $i = 2n-1$ and unconstrained above it. The doubling in the bound $i < 2n-1$ traces back to the two cones contributing equal connectivity n to the excision range $m+n = 2n$, with the single unit subtracted by the dimension shift between $\pi_i(X)$ and $\pi_{i+1}(\Sigma X)$. Below the stable range nothing is lost; at and above it, suspension can collapse the group, and the collapse is governed by Whitehead products that the sharper form of theorem 5.7.5 computes.

5.7.9 Remark: The Stable Range For the sphere $X = S^n$, which is $(n-1)$-connected (corollary 5.4.11), the theorem reads: $\sigma\colon \pi_i(S^n) \to \pi_{i+1}(S^{n+1})$ is an isomorphism for $i < 2n-1$ and an epimorphism for $i = 2n-1$. Writing $i = n+k$ for the k-th homotopy group above the diagonal, the isomorphism condition $n+k < 2n-1$ is $k < n-1$, equivalently $n > k+1$. Thus the k-stem $\pi_{n+k}(S^n)$ stabilizes once $n > k+1$, the precise stabilization threshold. The interval $i < 2n-1$ is the *stable range* for X, and a class lying in it suspends isomorphically forever.

The first and most basic consequence is that the colimits defining the stable homotopy groups are well posed and computable: the suspension maps eventually become isomorphisms, so the colimit is attained at a finite stage.

Corollary 5.7.10: Well-Definedness and Stabilization of the Stable Stems

For each $k \geq 0$ the suspension homomorphisms

$$\pi_k(S^0) \xrightarrow{\sigma} \pi_{k+1}(S^1) \xrightarrow{\sigma} \pi_{k+2}(S^2) \xrightarrow{\sigma} \cdots \xrightarrow{\sigma} \pi_{n+k}(S^n) \xrightarrow{\sigma} \pi_{n+1+k}(S^{n+1}) \xrightarrow{\sigma} \cdots$$

are isomorphisms for all $n > k+1$. Consequently the colimit

$$\pi_k^s := \varinjlim_n \pi_{n+k}(S^n)$$

is well defined and equals $\pi_{n+k}(S^n)$ for every $n > k+1$; the colimit is attained at a finite stage, not approached in a limit. In particular $\pi_0^s \cong \mathbb{Z}$, realized by $\pi_n(S^n) \cong \mathbb{Z}$ for all $n \geq 1$.

***Proof*:**
Step 1: The maps stabilize. The sphere S^n is $(n-1)$-connected (corollary 5.4.11), so theorem 5.7.8 applies with the connectivity index n: the map $\sigma\colon \pi_i(S^n) \to \pi_{i+1}(S^{n+1})$ is an isomorphism for $i < 2n-1$. Setting $i = n+k$, the condition $n+k < 2n-1$ rearranges to $n > k+1$. Hence for every $n > k+1$ the suspension $\sigma\colon \pi_{n+k}(S^n) \to \pi_{n+1+k}(S^{n+1})$ is an isomorphism.

Step 2: The colimit is attained. The system $\{\pi_{n+k}(S^n)\}_n$ with suspension transition maps is a sequence of abelian groups in which all maps beyond the stage $n = k+2$ are isomorphisms, by Step 1. A colimit of a sequence of groups whose transition maps are eventually all isomorphisms equals the common value of the stabilized groups: explicitly, the canonical map $\pi_{n+k}(S^n) \to \varinjlim_n \pi_{n+k}(S^n)$ is an isomorphism for each $n > k+1$, since every element of the colimit is represented at some stage and any two representatives become equal at a later stage, both of which are already isomorphisms. Therefore $\pi_k^s = \varinjlim_n \pi_{n+k}(S^n) \cong \pi_{n+k}(S^n)$ for all $n > k+1$.

Step 3: The case $k = 0$. For $k = 0$ the stabilization threshold is $n > 1$, so $\pi_0^s \cong \pi_n(S^n)$ for all $n \geq 2$. By corollary 5.4.11, $\pi_n(S^n) \cong \mathbb{Z}$ generated by the degree-1 identity, and example 5.7.4 showed σ is the identity $\mathbb{Z} \to \mathbb{Z}$ on these groups, so the colimit is $\mathbb{Z}$. The stage $n = 1$ also gives $\pi_1(S^1) \cong \mathbb{Z}$ with σ an isomorphism onto $\pi_2(S^2) \cong \mathbb{Z}$, so in fact the system is constant at $\mathbb{Z}$ from the very first stage. Thus $\pi_0^s \cong \mathbb{Z}$, completing the proof.

The corollary puts the stable stems on a rigorous footing. The stem π_k^s is not a delicate limiting object; it is literally the value $\pi_{n+k}(S^n)$ for any single n past the threshold $n > k+1$, so computing one such group computes the stem, and the colimit notation is bookkeeping for "the eventual value" whose eventuality Freudenthal guarantees. For $k = 0$ it gives $\pi_0^s \cong \mathbb{Z}$, the stable degree of a self-map of a sphere; this is the value the sphere spectrum will carry in degree zero (chapter 8). The next consequence turns to $k = 1$, where the eventual value is more interesting and the borderline behaviour of suspension first appears.

Corollary 5.7.11: The First Stable Stem is $\mathbb{Z}/2$

The first stable stem is $\pi_1^s \cong \mathbb{Z}/2$. It is realized by $\pi_{n+1}(S^n)$ for every $n \geq 3$, and the unstable group $\pi_3(S^2) \cong \mathbb{Z}$ suspends *onto* it: the suspension

$$\sigma\colon \pi_3(S^2) = \mathbb{Z} \longrightarrow \pi_4(S^3) = \mathbb{Z}/2$$

is surjective but not injective, the borderline case $i = 2n-1$ of theorem 5.7.8 with $n = 2$.

***Proof*:**
Step 1: Stabilization threshold for $k = 1$. By corollary 5.7.10 with $k = 1$, the groups $\pi_{n+1}(S^n)$ stabilize once $n > 2$, that is for all $n \geq 3$, and the common value is π_1^s; this much is Freudenthal. The value of that common group is $\mathbb{Z}/2$, generated by the iterated suspensions of the Hopf map $\eta\colon S^3 \to S^2$; this value is the Serre spectral sequence computation carried out later (example 6.3.7), which the stabilization just established licenses us to read off from any single stage $n \geq 3$. Hence $\pi_1^s \cong \mathbb{Z}/2$.

Step 2: The boundary case $n = 2$. For $n = 2$ the sphere S^2 is 1-connected, so $X = S^2$ has connectivity index $n = 2$ in theorem 5.7.8, and the stable range is $i < 2 \cdot 2 - 1 = 3$ with surjectivity at $i = 3$. The group $\pi_3(S^2)$ sits exactly at $i = 3 = 2n-1$, the borderline degree, so the theorem guarantees that

$$\sigma\colon \pi_3(S^2) \to \pi_4(S^3)$$

is an epimorphism, but says nothing about injectivity. By corollary 5.4.11 and the Hopf-fibration computation of example 5.2.13, $\pi_3(S^2) \cong \mathbb{Z}$, generated by the Hopf map η; by Step 1, $\pi_4(S^3) \cong \mathbb{Z}/2$, generated by $\sigma[\eta] = [\Sigma\eta]$. A surjection $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$ from an infinite cyclic group onto a group of order 2 necessarily has nontrivial kernel (the kernel is $2\mathbb{Z}$), so σ is not injective. This is consistent with $\sigma[\eta]$ generating $\mathbb{Z}/2$: the integer Hopf class $[\eta]$ surjects onto its parity, and the multiples $2[\eta], 4[\eta], \ldots$ all suspend to 0.

The borderline behavior is forced: an isomorphism $\mathbb{Z} \to \mathbb{Z}/2$ is impossible, so the boundary degree $i = 2n-1$ is exactly where the theorem must weaken from isomorphism to epimorphism, and the example $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$ shows the weakening is genuine, completing the proof.

What Freudenthal contributes here is the structure, not the value $\mathbb{Z}/2$ itself. The group is stable because $n = 3$ exceeds the threshold $k+1 = 2$, and the unstable $\pi_3(S^2) \cong \mathbb{Z}$ falls below the threshold precisely because $n = 2$ is the borderline degree where suspension is only surjective; the value $\mathbb{Z}/2$ is given separately by the Serre spectral sequence computation of chapter 6, built from the homology of $K(\mathbb{Z}, n)$. The transition $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$ is the sharpest illustration of the theorem's boundary: it is the unique degree in the stem where suspension fails to be an isomorphism, and it fails in exactly the way the epimorphism clause predicts.

Example 5.7.12: The Suspension $\pi_3(S^2) \to \pi_4(S^3)$ in Detail

This is the canonical worked instance of the boundary of the stable range. Take $X = S^2$, so $n = 2$ and X is $(n-1) = 1$-connected. The relevant homotopy groups are

$$\pi_3(S^2) \cong \mathbb{Z} \quad \text{(generated by } \eta, \text{ example 5.2.13)}, \qquad \pi_4(S^3) \cong \mathbb{Z}/2 \quad \text{(example 6.3.7)}$$

The degree in question is $i = 3$. The stable range of theorem 5.7.8 for $n = 2$ is $i < 2n-1 = 3$, so $i = 3$ is exactly the borderline $i = 2n-1$, outside the isomorphism range and inside the epimorphism range. The theorem therefore predicts $\sigma\colon \pi_3(S^2) \to \pi_4(S^3)$ is surjective but not necessarily injective, and indeed

$$\sigma\colon \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2, \qquad [\eta] \mapsto [\Sigma\eta], \qquad \ker\sigma = 2\mathbb{Z}$$

The Hopf map suspends to a generator of $\mathbb{Z}/2$, while twice the Hopf class suspends to zero, so σ realizes the reduction $\mathbb{Z} \to \mathbb{Z}/2$ sending an integer to its parity. Suspending once more lands in the stable range: $\sigma\colon \pi_4(S^3) \to \pi_5(S^4)$ has $n = 3$ and degree $i = 4 < 2 \cdot 3 - 1 = 5$, so it is an isomorphism $\mathbb{Z}/2 \xrightarrow{\cong} \mathbb{Z}/2$, and every further suspension is an isomorphism as well. The integer information of $\pi_3(S^2)$ is destroyed at the first suspension and the stable value $\mathbb{Z}/2$ is reached immediately after, which is why $\pi_1^s = \mathbb{Z}/2$ is computed from any $\pi_{n+1}(S^n)$ with $n \geq 3$ but not from $n = 2$.

5.7.13 Historical Note: Freudenthal's 1937 Theorem The theorem was proved by Hans Freudenthal in 1937, predating both the relative homotopy groups in the form used here and the Blakers–Massey theorem (1949–1953) from which it is derived above. Freudenthal's original argument was a direct general-position analysis of maps of spheres, establishing the stabilization of $\pi_{n+k}(S^n)$ that makes the stable stems π_k^s well defined. Homotopy excision later subsumed the result as a special case, which is the modern route taken here; the derivation from theorem 5.7.5 isolates the geometric content (two cones, equal connectivity) and produces the range $i < 2n - 1$ as the arithmetic $m + n - 1 = 2n - 1$ of the excision bound.

Exercise 5.7.1

1. Let X be a pointed CW complex that is 4-connected. For which i does theorem 5.7.8 guarantee that $\sigma\colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$ is an isomorphism, and for which single i does it guarantee only an epimorphism? Express the stable range in terms of the connectivity.

2. Using corollary 5.7.10, determine the smallest n for which $\pi_{n+2}(S^n)$ is guaranteed to equal the stable stem π_2^s. (You are not asked to compute π_2^s; it is known to be $\mathbb{Z}/2$.) Then state, with reasons, which of $\pi_3(S^1)$, $\pi_4(S^2)$, $\pi_5(S^3)$, $\pi_6(S^4)$ are guaranteed stable.

3. Show that for $n \geq 2$ the suspension $\sigma\colon \pi_n(S^n) \to \pi_{n+1}(S^{n+1})$ is an isomorphism $\mathbb{Z} \to \mathbb{Z}$, directly from theorem 5.7.8 (identify the connectivity and check $i = n$ lies in the stable range), and reconcile this with the degree computation of example 5.7.4.

4. The space $X = S^2 \vee S^2$ is 1-connected. Compute the stable range from theorem 5.7.8, and use $\Sigma(A \vee B) = \Sigma A \vee \Sigma B$ (chapter 1) to describe ΣX. In what degree i is σ only guaranteed to be surjective?

5. Prove that a 1-connected pointed CW complex X has $\sigma\colon \pi_2(X) \to \pi_3(\Sigma X)$ an isomorphism. Deduce that $\pi_3(\Sigma X) \cong \pi_2(X)$, and check this against $X = S^2$, where $\Sigma X = S^3$.

6. Explain why the colimit $\pi_k^s = \varinjlim_n \pi_{n+k}(S^n)$ would not be guaranteed to stabilize at a finite stage if homotopy excision (theorem 5.7.5) gave isomorphisms only in degrees $i < m$ rather than $i < m + n$. Concretely, what would go wrong with the threshold $n > k + 1$ of corollary 5.7.10?

6

Spectral Sequences

Spectral sequences have a reputation for opacity that this chapter is designed to dispel. The reputation is not wholly undeserved: the standard presentations open with a doubly-indexed family of bigraded groups and a forest of differentials, and the reader is asked to take on faith that this apparatus computes anything at all. The development here inverts that order. A spectral sequence is, before it is anything else, a bookkeeping engine for computing the homology or cohomology of an object that has been built up in stages (ex. a chain complex carrying a filtration, a space assembled from a tower of subspaces, a fibration whose total space is glued from fibers over a base). The engine produces a sequence of *pages* $E_1, E_2, E_3, \dots$, each a bigraded group with a differential, and each page is the homology of the page before it. The first page is the crudest approximation to the answer, the associated graded of the filtration; every subsequent page corrects it by detecting differentials of longer and longer range, until the corrections run out and the pages stabilize at a limit page E_∞ that is the associated graded of the (co)homology one set out to compute. Turning the pages is successive approximation, and the limit is the answer assembled one graded piece at a time.

This chapter builds that engine from the ground up and then puts it to work. The present section isolates the purely algebraic mechanism, the *exact couple* of Massey, from which every page and every differential is generated by a single operation (the passage to the derived couple). The two sections that follow construct the Serre spectral sequence, which computes the homology of the total space of a fibration from the homology of base and fiber, and turns the homotopy-theoretic machinery of chapter 5 into a computational tool capable of extracting homotopy groups of spheres. The closing sections develop the multiplicative structure that makes the cohomology spectral sequence a ring, and prepare the spectral-sequence language in which the generalized cohomology theories of the next chapter are organized. The abelian-category generalities (spectral sequences of double complexes, of derived functors, the comparison and convergence theorems in their full generality) are developed in [10]; the narrative here is topological, and overlap with that treatment is deliberate where it keeps the present account self-contained.

6.1 Exact Couples and the Spectral Sequence of a Filtration

The cleanest way to meet a spectral sequence is to watch one compute something by hand, before any of the formal apparatus is in place. The setting is a chain complex equipped with a filtration: a chain complex $C_\bullet$ together with an increasing chain of subcomplexes

$$0 = F_{-1}C_\bullet \subseteq F_0C_\bullet \subseteq F_1C_\bullet \subseteq \cdots \subseteq F_pC_\bullet \subseteq \cdots \subseteq C_\bullet$$

each $F_pC_\bullet$ closed under the boundary map ∂, with $\bigcup_p F_pC_\bullet = C_\bullet$. The homology $H_n(C_\bullet)$ is the object of interest, and the filtration is a "recipe" for building $C_\bullet$ in stages. A spectral sequence is the systematic accounting of how the homology of the stages assembles into the homology of the whole.

To see this in action, take the smallest nontrivial case, a filtration with two steps, and compute the homology directly. Namely, consider a chain complex $C_\bullet$ with a single proper subcomplex $A_\bullet = F_0C_\bullet$, so the filtration is

$$0 \subseteq A_\bullet \subseteq C_\bullet$$

with quotient $B_\bullet = C_\bullet/A_\bullet = F_1C_\bullet/F_0C_\bullet$. The associated graded complex is $\operatorname{gr} C_\bullet = A_\bullet \oplus B_\bullet$, the direct sum of the two filtration layers. A first approximation to $H_n(C_\bullet)$ is the homology of this associated graded,

$$E^1 = H_\bullet(A_\bullet) \oplus H_\bullet(B_\bullet)$$

which is what one would get if the complex genuinely split as a direct sum. It does not, in general, and the discrepancy is governed by the short exact sequence of complexes $0 \to A_\bullet \to C_\bullet \to B_\bullet \to 0$, whose long exact sequence in homology

$$\cdots \xrightarrow{\delta} H_n(A_\bullet) \xrightarrow{\iota_*} H_n(C_\bullet) \xrightarrow{q_*} H_n(B_\bullet) \xrightarrow{\delta} H_{n-1}(A_\bullet) \to \cdots$$

contains the connecting map δ. This δ is the entire correction. Reading the exact sequence, $H_n(C_\bullet)$ is built from the part of $H_n(B_\bullet)$ on which δ vanishes (the image of q_*) and the part of $H_n(A_\bullet)$ not hit by δ (the image of ι_*, isomorphic to $H_n(A_\bullet)/\mathrm{im}\delta$). Concretely, exactness gives a short exact sequence

$$0 \to \frac{H_n(A_\bullet)}{\mathrm{im}\big(\delta\colon H_{n+1}(B_\bullet) \to H_n(A_\bullet)\big)} \to H_n(C_\bullet) \to \ker\big(\delta\colon H_n(B_\bullet) \to H_{n-1}(A_\bullet)\big) \to 0$$

The connecting map δ is playing the role of a differential: forming the kernel-modulo-image of δ on the approximation E^1 produces a second, better approximation, and that second approximation is precisely the associated graded of the true homology $H_n(C_\bullet)$. This two-term computation is a spectral sequence with exactly one nontrivial differential. The page $E^1 = H_\bullet(A) \oplus H_\bullet(B)$ carries the differential δ; its homology is the page E^2; and E^2 is the associated graded of $H_\bullet(C)$. There are no further corrections because there are only two filtration layers, so the pages stabilize at $E^2 = E^\infty$.

The general spectral sequence is this computation iterated. With more filtration layers, the analogue of δ does not capture the whole correction in one step. Instead a differential d_1 on the associated graded E^1 produces E^2; a new differential d_2 on E^2, of longer range, produces E^3; and so on, each d_r detecting the part of the assembly that crosses r filtration layers at once. The pages turn until the differentials become zero in every spot, and the limit page is the associated graded of $H_\bullet(C)$. The machine that generates this entire sequence of pages and differentials from a single piece of data is the exact couple, to which the development now turns.

Example 6.1.1: A Filtered Mapping Cone

Let $f\colon \mathbb{Z} \to \mathbb{Z}$ be multiplication by 2, regarded as a chain map between two copies of $\mathbb{Z}$ concentrated in degree 0. Its mapping cone is the complex $C_\bullet$ with $C_1 = \mathbb{Z}$, $C_0 = \mathbb{Z}$, boundary $\partial = 2$, and $C_n = 0$ otherwise, so $H_0(C_\bullet) = \mathbb{Z}/2$ and $H_1(C_\bullet) = 0$. Filter it by $A_\bullet = (C_0$ in degree 0), the subcomplex $0 \to 0 \to \mathbb{Z}$ with zero boundary, and $C_\bullet$ itself. Then $A_\bullet$ has $H_0(A) = \mathbb{Z}$, and $B_\bullet = C/A$ is $\mathbb{Z}$ in degree 1 with zero boundary, so $H_1(B) = \mathbb{Z}$. The page $E^1 = H_\bullet(A) \oplus H_\bullet(B)$ is $\mathbb{Z}$ in bidegree recording degree 0 (from A) and $\mathbb{Z}$ recording degree 1 (from B).

Placing these in the (p, q)-plane with $E^1_{p,q} = H_{p+q}(\mathrm{gr}_p C)$, the two filtration steps $p = 0$ (from A) and $p = 1$ (from B) contribute the class $H_0(A) = \mathbb{Z}$ at $(0, 0)$ and $H_1(B) = \mathbb{Z}$ at $(1, 0)$. Every other spot is zero, so the entire sequence lives in the row $q = 0$:

$$E^1 : \quad \begin{tikzcd} q{=}1 & 0 & 0 \\ q{=}0 & \mathbb{Z} & \mathbb{Z} \arrow[l, "\times 2"'] \\ & p{=}0 & p{=}1 \end{tikzcd} \qquad d^1 = \delta\colon E^1_{1,0} = \mathbb{Z} \xrightarrow{\times 2} E^1_{0,0} = \mathbb{Z}$$

The connecting map $\delta\colon H_1(B) = \mathbb{Z} \to H_0(A) = \mathbb{Z}$ sends the generator of $H_1(B)$, represented by the cycle $1 \in C_1$, to $\partial(1) = 2 \in C_0$, that is, $\delta = 2$. In the homological convention $d^1\colon E^1_{p,q} \to E^1_{p-1,q}$ decreases the filtration degree by one, so this connecting map is precisely the d^1 differential, the single leftward horizontal arrow drawn above. The next page is its kernel–cokernel, $E^2_{1,0} = \ker(\delta)$ and $E^2_{0,0} = \mathrm{coker}(\delta)$:

$$E^2 = E^\infty : \quad \begin{array}{ccc} q{=}1 & 0 & 0 \\ q{=}0 & \mathbb{Z}/2 & 0 \\ & p{=}0 & p{=}1 \end{array}$$

Since $d^2\colon E^2_{p,q} \to E^2_{p-2,q+1}$ has no room to be nonzero, $E^2 = E^\infty$. Concretely,

$$E^2 = \ker(\delta) \ \oplus \ \mathrm{coker}(\delta) = 0 \ \oplus \ \mathbb{Z}/2.$$

The kernel records $H_1(C) = 0$ and the cokernel records $H_0(C) = \mathbb{Z}/2$, matching the homology computed directly. The single differential $\delta = 2$ is exactly the multiplication that the mapping cone was built to encode, and the spectral sequence has recovered it.

Generalizing this example, Massey observed that the long exact sequences attached to every layer of a filtration can be packaged into a single exact triangle of bigraded groups, and that this triangle regenerates itself under one formal operation, producing every page of the spectral sequence at once. The packaging is the exact couple:

Definition 6.1.2: Exact Couple

An *exact couple* is a pair of (possibly bigraded) abelian groups D and E together with three homomorphisms

$$\begin{tikzcd} D \arrow[rr, "i"] & & D \arrow[dl, "j"] \\ & E \arrow[ul, "k"] & \end{tikzcd}$$

such that the triangle is exact at each of its three corners: $\mathrm{im} i = \ker j$, $\mathrm{im} j = \ker k$, and $\mathrm{im} k = \ker i$. The composite

$$d := j \circ k \colon E \to E$$

is the *differential* of the couple.

The data (D, E, i, j, k) should be read as a compression of a long exact sequence. In the filtered case D collects the homologies of the subcomplexes $F_pC_\bullet$, the group E collects the homologies of the layers $F_pC_\bullet/F_{p-1}C_\bullet$, the map i is induced by the inclusions $F_{p-1} \hookrightarrow F_p$, the map j is the projection to the layer, and k is the connecting homomorphism. Exactness of the triangle is exactly the exactness of all these long exact sequences at once. The group E is the page one can see; the group D is the auxiliary data that remembers how the layers are glued, and it is what lets the page be refined. The first thing to verify is that d deserves the name differential.

Lemma 6.1.3: The Differential Squares to Zero

For any exact couple (D, E, i, j, k), the differential $d = jk$ satisfies $d \circ d = 0$, so (E, d) is a differential group and its homology $H(E, d) = \ker d/\mathrm{im} d$ is defined.

***Proof*:**
Compute the composite directly:

$$d \circ d = (jk)(jk) = j(kj)k$$

The middle composite kj is zero, because $\mathrm{im} j = \ker k$ by exactness of the couple at E, so every element in the image of j lies in the kernel of k. Hence $kj = 0$, and therefore $d^2 = j(kj)k = j \cdot 0 \cdot k = 0$, as we sought to show.

The proof uses only one of the three exactness conditions, the one at the visible corner E. The other two are spent in the construction of the next page, to which the lemma is the gateway: with $d^2 = 0$ established, the homology $H(E, d)$ is a new bigraded group, and the claim is that this new group is again the E-term of an exact couple, built from the same D by a derived process. Producing that new couple is the heart of the mechanism. The new D-term is the image $i(D) \subseteq D$, and the new maps are induced from i, j, k in a way that has to be checked for well-definedness at every step.

Definition 6.1.4: Derived Couple

Let (D, E, i, j, k) be an exact couple with differential $d = jk$. Its *derived couple* (D', E', i', j', k') is defined by

$$D' := i(D) = \mathrm{im} i, \qquad E' := H(E, d) = \frac{\ker d}{\mathrm{im} d}$$

with structure maps

- $i' := i|_{D'} : D' \to D'$, the restriction of i to its image;
- $j' : D' \to E'$ defined by $j'(ia) := [\,ja\,]$ for $a \in D$, the class of ja in $E' = \ker d / \mathrm{im} d$;
- $k' : E' \to D'$ defined by $k'([e]) := ke$ for $e \in \ker d$, the value of k on a cycle.

The two new groups are forced by the structure: the only way to shrink D compatibly with the differential is to pass to $\mathrm{im} i$, and the only sensible refinement of the visible page E is its homology. The subtlety is entirely in the three maps, each of which is defined by a representative and must be shown independent of the choice; j' takes an element $ia \in D'$, lifts it to a, and reads off the class of ja, while k' takes a homology class, lifts it to a d-cycle, and applies k. That these are well-defined and that the resulting triangle is exact is the content of the next result, on which the whole theory rests.

Theorem 6.1.5: The Derived Couple is Exact

Let (D, E, i, j, k) be an exact couple. Then the derived couple (D', E', i', j', k') of definition 6.1.4 is well-defined and is again an exact couple.

***Proof*:**
The argument proceeds in stages: first the maps land where claimed and are independent of choices, then exactness is checked at each of the three corners.

Step 1: j' is well-defined. Let $a \in D$ and consider $j'(ia) = [ja]$. Two checks are required: that ja is a d-cycle, so that its class in $E' = \ker d / \mathrm{im} d$ makes sense, and that the class is independent of the choice of a with the given image ia.

For the first, $d(ja) = jk(ja) = j(kj)(a) = 0$ since $kj = 0$ (exactness at E, used as in lemma 6.1.3). So $ja \in \ker d$.

For the second, suppose $ia = ia'$, so $i(a - a') = 0$ and $a - a' \in \ker i = \mathrm{im} k$ by exactness at D (the corner where k enters i). Write $a - a' = ke$ for some $e \in E$. Then

$$ja - ja' = j(a - a') = jke = de \in \mathrm{im} d$$

so $[ja] = [ja']$ in E'. Thus j' is well-defined on $D' = \mathrm{im} i$.

Step 2: k' is well-defined. Let $e \in \ker d$ and consider $k'([e]) = ke$. Two checks: that $ke \in D' = \mathrm{im} i$, and that the value depends only on the homology class $[e]$.

For the first, $de = 0$ means $jke = 0$, so $ke \in \ker j = \mathrm{im} i$ by exactness at D (the corner where i enters j). Hence $ke \in D'$.

For the second, if $[e] = [e']$ then $e - e' = dc = jkc$ for some $c \in E$, and

$$ke - ke' = k(e - e') = k(jkc) = (kj)(kc) = 0$$

since $kj = 0$. So $ke = ke'$, and k' is well-defined.

Step 3: i' is well-defined. The map $i' = i|_{D'}$ restricts i to $D' = \mathrm{im} i$. Its image is $i(\mathrm{im} i) = \mathrm{im}(i \circ i) \subseteq \mathrm{im} i = D'$, so i' does map D' into D'. No choices are involved.

Step 4: Exactness at D' (the corner $D' \xrightarrow{i'} D' \xrightarrow{j'} E'$). Show $\mathrm{im} i' = \ker j'$.

($\mathrm{im} i' \subseteq \ker j'$): An element of $\mathrm{im} i'$ has the form $i'(ia) = i(ia) = i^2 a$ for some $a \in D$. Then $j'(i^2 a) = j'\big(i(ia)\big) = [j(ia)]$. Now $j(ia) = (ji)(a) = 0$ since $\mathrm{im} i = \ker j$ by exactness at D, so $[j(ia)] = 0$ and $i^2 a \in \ker j'$.

($\ker j' \subseteq \mathrm{im} i'$): Suppose $ia \in D'$ satisfies $j'(ia) = [ja] = 0$, that is, $ja \in \mathrm{im} d = \mathrm{im}(jk)$. Write $ja = jkc$ for some $c \in E$. Then $j(a - kc) = 0$, so $a - kc \in \ker j = \mathrm{im} i$ by exactness at D; write $a - kc = ib$ for some $b \in D$. Applying i,

$$ia - ikc = i^2 b$$

But $ik = 0$ since $\mathrm{im} k = \ker i$ by exactness at D, so $ikc = 0$ and $ia = i^2 b = i'(ib) \in \mathrm{im} i'$. Thus $\ker j' \subseteq \mathrm{im} i'$.

Step 5: Exactness at E' (the corner $D' \xrightarrow{j'} E' \xrightarrow{k'} D'$). Show $\mathrm{im} j' = \ker k'$.

($\mathrm{im} j' \subseteq \ker k'$): An element of $\mathrm{im} j'$ is $j'(ia) = [ja]$ with $ja \in \ker d$ (Step 1). Then $k'([ja]) = k(ja) = (kj)(a) = 0$, so $[ja] \in \ker k'$.

($\ker k' \subseteq \mathrm{im} j'$): Suppose $[e] \in E'$ with $e \in \ker d$ and $k'([e]) = ke = 0$. Then $e \in \ker k = \mathrm{im} j$ by exactness at E; write $e = ja$ for some $a \in D$. Then $[e] = [ja] = j'(ia) \in \mathrm{im} j'$.

Step 6: Exactness at D' for the corner $E' \xrightarrow{k'} D' \xrightarrow{i'} D'$. Show $\mathrm{im} k' = \ker i'$.

($\mathrm{im} k' \subseteq \ker i'$): An element of $\mathrm{im} k'$ is $k'([e]) = ke$ with $e \in \ker d$. Then $i'(ke) = i(ke) = (ik)(e) = 0$ since $\mathrm{im} k = \ker i$. So $ke \in \ker i'$.

($\ker i' \subseteq \mathrm{im} k'$): Suppose $ia \in D'$ with $i'(ia) = i(ia) = i^2 a = 0$. Then $ia \in \ker i = \mathrm{im} k$, so $ia = ke$ for some $e \in E$. To use $[e]$ as an element of E', check $e \in \ker d$: $de = jke = j(ia) = (ji)(a) = 0$ since $ji = 0$ (exactness at D). Thus $e \in \ker d$ and $k'([e]) = ke = ia$, so $ia \in \mathrm{im} k'$.

All three corners are exact and all three maps are well-defined, so (D', E', i', j', k') is an exact couple, completing the proof.

The theorem is the engine that turns a single exact couple into an infinite sequence of pages. Apply it once to (D, E, i, j, k) to obtain (D', E', i', j', k'); apply it again to the derived couple to obtain a second derived couple; and so on indefinitely. Each iteration replaces the visible page E by its homology under the current differential, so the sequence of E-terms is exactly a sequence of pages, each the homology of the previous. The auxiliary group D shrinks at each stage to $\mathrm{im} i$, recording the progressively deeper part of the filtration that survives. Reading off the sequence of E-terms with their differentials gives the formal object the chapter is named for.

Definition 6.1.6: Spectral Sequence

A (homological) *spectral sequence* is a sequence $\{(E_r, d_r)\}_{r \geq r_0}$ of bigraded abelian groups $E_r = \bigoplus_{p,q} E_r^{p,q}$, each equipped with a differential $d_r\colon E_r \to E_r$ satisfying $d_r \circ d_r = 0$, together with isomorphisms

$$E_{r+1} \cong H(E_r, d_r) = \frac{\ker(d_r)}{\operatorname{im}(d_r)}$$

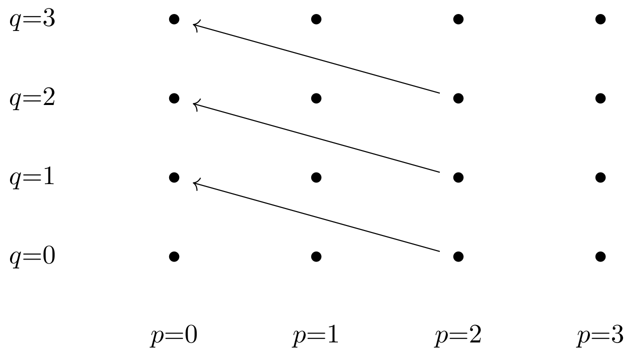
identifying each page as the homology of the page before it. The index r is the *page number*, E_r is the *r-th page*, and d_r is the *r-th differential*. In the bigraded setting the differential has a fixed bidegree: for a homological spectral sequence $d_r\colon E_r^{p,q} \to E_r^{p-r,\,q+r-1}$, lowering the filtration index p by r and preserving the total degree $p+q$.
The spectral sequence *arising from an exact couple* (D, E, i, j, k) is the sequence whose page E_r is the E-term of the $(r-1)$-fold derived couple, with differential d_r the differential of that couple; theorem 6.1.5 guarantees each page is the homology of its predecessor.

The bidegree of d_r encodes the geometry of the construction. Recall the two-step example: the connecting map δ ran from $H_n(B)$, a layer at filtration level 1, to $H_{n-1}(A)$, a layer at level 0, dropping the filtration by 1 and the homological degree by 1. After one derivation the differential reaches one filtration layer further, after two derivations two layers further, and in general d_r on the r-th page connects spots whose filtration indices differ by r. This is why higher differentials detect higher-order gluing: d_1 sees adjacent layers, d_2 sees layers two apart, and a class that survives to the r-th page only to be killed by d_r encodes an extension that spans r filtration steps. Each derivation of the couple advances the reach of the differential by one. The bookkeeping is exact and mechanical, which is the point: once the exact couple is in hand, the entire spectral sequence is determined, page by page, with no further choices.

A picture makes the loop concrete. Strip away the particular groups and follow an abstract page through two turns. Before any differential is drawn, a page is only an array of groups, one at each lattice point (p, q); this is the input the machine receives, drawn here as the second page E_2 with a dot at each spot:

$$E_2\colon \quad \begin{array}{ccccc} q{=}3 & \bullet & \bullet & \bullet & \bullet \\ q{=}2 & \bullet & \bullet & \bullet & \bullet \\ q{=}1 & \bullet & \bullet & \bullet & \bullet \\ q{=}0 & \bullet & \bullet & \bullet & \bullet \\ & p{=}0 & p{=}1 & p{=}2 & p{=}3 \end{array}$$

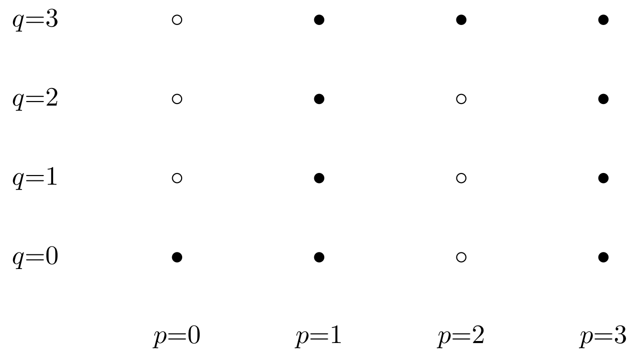
A differential is then drawn on the page. On E_2 it has bidegree $(-2, 1)$, so every arrow points two columns left and one row up, the same slope at each spot:



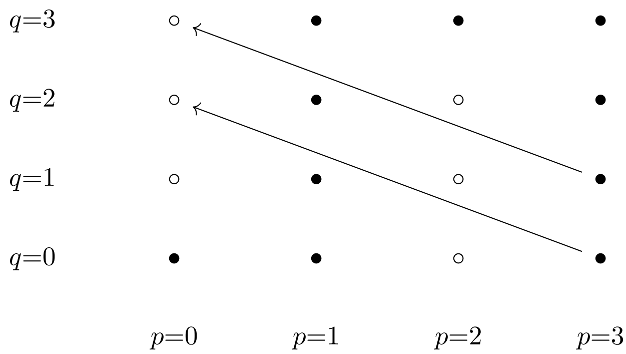
$$d_2\colon E^2_{p,q} \to E^2_{p-2,\,q+1}$$

Turning the page passes to homology: each spot of E_3 is $\ker d_2/\mathrm{im} d_2$ computed there, so a group touched by a d_2 is cut down or killed, and the surviving array is in general smaller. The lattice of spots persists while its entries are refined; the open circles below mark the spots where a d_2 acted, each now a subquotient (possibly zero), and the solid dots are untouched:

E_3:



The new behavior appears when the next differential is drawn. On E_3 it is d_3, of bidegree $(-3, 2)$: three columns left and two rows up. Its arrows are longer than those of d_2 and run at a steeper angle, reaching one filtration layer deeper into the page:



$$d_3\colon E^3_{p,q} \to E^3_{p-3,\,q+2}$$

Each further page tilts the differential once more, d_r running r columns left and $r-1$ rows up, until in any fixed spot the arrows overshoot the quadrant and vanish, leaving the limit page E_∞. The whole computation is this loop: read the page, draw its differential, pass to homology, and watch the next differential swing out to a longer, steeper reach.

Example 6.1.7: The Two-Page Spectral Sequence of a Two-Step Filtration

Return to the two-step filtration $0 = F_{-1} \subseteq F_0 = A_\bullet \subseteq F_1 = C_\bullet$ with quotient layer $B_\bullet = F_1/F_0$, and assemble its exact couple in the indexing of theorem 6.1.10. The auxiliary group collects the homologies of the filtration stages and the page collects the homologies of the layers,

$$D_{p,q} = H_{p+q}(F_p C_\bullet), \qquad E^1_{p,q} = H_{p+q}(F_p C_\bullet / F_{p-1} C_\bullet)$$

so that the only occupied filtration levels are $p = 0$, with layer $F_0/F_{-1} = A_\bullet$, and $p = 1$, with layer $F_1/F_0 = B_\bullet$. The maps are $i = \iota_*$ (induced by $F_{p-1} \hookrightarrow F_p$), $j = \pi_*$ (projection to the layer), and $k = \partial$ (the connecting homomorphism), and exactness of the triangle is the long exact sequence of $0 \to A \to C \to B \to 0$. The first page therefore reads

$$E^1_{0,q} = H_q(A_\bullet), \qquad E^1_{1,q} = H_{1+q}(B_\bullet)$$

with all other spots zero, so the entire first page occupies just two columns, $p = 0$ carrying the A-layer and $p = 1$ carrying the B-layer, with the rows indexed by the complementary degree q. The only possible differential runs from level 1 to level 0, $d_1 \colon E^1_{1,q} \to E^1_{0,q}$, that is, $H_{1+q}(B) \to H_q(A)$; unwinding $d_1 = jk = \pi_* \circ \partial$, the connecting map $\partial = \delta$ carries $H_{1+q}(B)$ into $D_{0,q} = H_q(A)$ and then $\pi_* \colon H_q(A) = H_q(F_0) \to H_q(F_0/F_{-1}) = H_q(A)$ is the identity, so d_1 is the connecting homomorphism itself. Abbreviating it $\delta_q \colon H_{q+1}(B) \to H_q(A)$ at each height q (the map $\delta \colon H_n(B) \to H_{n-1}(A)$ with $n = q + 1$), the first page carries one such arrow per row:

$$E^1 : \quad \begin{tikzcd}
\vdots & \vdots & \vdots \\
q{+}1 & H_{q+1}(A) & H_{q+2}(B) \arrow[l, "\delta_{q+1}"'] \\
q & H_q(A) & H_{q+1}(B) \arrow[l, "\delta_q"'] \\
q{-}1 & H_{q-1}(A) & H_q(B) \arrow[l, "\delta_{q-1}"'] \\
\vdots & \vdots & \vdots \\
 & p = 0 & p = 1
\end{tikzcd} \qquad \begin{array}{c} d_1 = \delta_q : \\ \underset{(1,q)}{H_{q+1}(B)} \longrightarrow \underset{(0,q)}{H_q(A)} \end{array}$$

Passing to homology replaces each row by the kernel and cokernel of its δ_q: the kernel survives at

$(1, q)$ and the cokernel at $(0, q)$, so the second page has no arrows left to draw,

$$E^2 = E^\infty : \quad \begin{array}{cccc}
\vdots & \vdots & \vdots & \\
q{+}1 & \operatorname{coker} \delta_{q+1} & \ker \delta_{q+1} & \\
q & \operatorname{coker} \delta_q & \ker \delta_q & \begin{array}{c} E^2_{1,q} = \ker \delta_q \\ E^2_{0,q} = \operatorname{coker} \delta_q \\ d_2 = 0 \end{array} \\
q{-}1 & \operatorname{coker} \delta_{q-1} & \ker \delta_{q-1} & \\
\vdots & \vdots & \vdots & \\
 & p = 0 & p = 1 &
\end{array}$$

The next differential $d_2 \colon E^2_{p,q} \to E^2_{p-2,q+1}$ would drop the filtration level by 2, from a column $p \in \{0, 1\}$ into the empty columns $p = -2$ or $p = -1$, so its source or target is always zero; hence $d_2 = 0$, every later differential vanishes, and $E^2 = E^\infty$ is already the limit page. Reading up the anti-diagonal $p + q = n$ picks out its two nonzero spots, $E^\infty_{0,n} = \operatorname{coker} \delta_n$ at $(0, n)$ and $E^\infty_{1,n-1} = \ker \delta_{n-1}$ at $(1, n-1)$, explicitly $\operatorname{coker}(\delta \colon H_{n+1}(B) \to H_n(A))$ and $\ker(\delta \colon H_n(B) \to H_{n-1}(A))$, the two successive quotients of the filtration of $H_n(C)$, recovering the short exact sequence

$$0 \to \operatorname{coker}\big(\delta \colon H_{n+1}(B) \to H_n(A)\big) \to H_n(C) \to \ker\big(\delta \colon H_n(B) \to H_{n-1}(A)\big) \to 0$$

found by hand at the start of the section.

The two pages become concrete in the mapping cone of example 6.1.1. There $A_\bullet$ and $B_\bullet$ are copies of $\mathbb{Z}$ concentrated in degrees 0 and 1, so of all the rows above only $q = 0$ survives: the first page has $E^1_{0,0} = H_0(A) = \mathbb{Z}$ at $(0, 0)$ and $E^1_{1,0} = H_1(B) = \mathbb{Z}$ at $(1, 0)$, the differential is $\delta_0 \colon \mathbb{Z} \to \mathbb{Z}$ equal to multiplication by 2, and the second page collapses to $E^\infty_{0,0} = \operatorname{coker} \delta_0 = \mathbb{Z}/2$ and $E^\infty_{1,0} = \ker \delta_0 = 0$, the values $H_0(C) = \mathbb{Z}/2$ and $H_1(C) = 0$. The general grids are that same picture with $H_{p+q}(-)$ left unevaluated.

The example exhibits the general phenomenon that makes spectral sequences finite to compute in practice: in any fixed spot, the differentials eventually run out of room. When only finitely many filtration levels are occupied in each total degree, the differentials d_r for large r must vanish because their sources or targets are zero, and the page stabilizes. The condition that guarantees this uniformly, and that holds for the spectral sequences of the next two sections, is that the spectral sequence be concentrated in a quadrant.

Definition 6.1.8: Convergence and First-Quadrant Spectral Sequences

A spectral sequence $\{(E_r, d_r)\}$ is *first-quadrant* if $E_r^{p,q} = 0$ whenever $p < 0$ or $q < 0$, for every page r (equivalently, for $r = r_0$, since later pages are subquotients). For such a sequence, fix a spot (p, q). The differential d_r into (p, q) comes from $(p + r, q - r + 1)$ and the differential out of (p, q) goes to $(p - r, q + r - 1)$; for $r > \max(p, q + 1)$ both source and target lie outside the first quadrant, so $d_r = 0$ in and out of (p, q). Consequently the groups $E_r^{p,q}$ are equal for all sufficiently large r; this stable value is denoted $E_\infty^{p,q}$.
The spectral sequence *converges* to a graded abelian group $H_\bullet$, written

$$E_{r_0}^{p,q} \Longrightarrow H_{p+q}$$

if $H_\bullet$ carries a filtration $0 = F_{-1}H_n \subseteq F_0H_n \subseteq \cdots \subseteq F_nH_n = H_n$ (in the first-quadrant case, bounded in each degree) and there are isomorphisms

$$E_\infty^{p,q} \cong \frac{F_pH_{p+q}}{F_{p-1}H_{p+q}}$$

identifying the limit page, spot by spot, with the associated graded of $H_\bullet$.

The convergence statement is the payoff: it says the limit page E_∞ is not some abstract terminal object but the associated graded of the very homology one wanted. The spectral sequence does not hand over H_n on a plate - it hands over the successive quotients $F_pH_n/F_{p-1}H_n$ along a filtration, and recovering H_n from these requires solving the extension problems that the differentials leave behind. In favorable cases (a field of coefficients, or a filtration that splits) the extensions are trivial and $H_n \cong \bigoplus_p E_\infty^{p,n-p}$; in general the E_∞-page determines H_n only up to those extensions, which is exactly the residue of information that the factor of 2 in example 6.1.1 encoded. That a first-quadrant spectral sequence stabilizes at every spot, with no convergence hypotheses beyond the quadrant condition, is what makes it a usable computational tool, and the precise statement deserves to be recorded.

Proposition 6.1.9: Stabilization of First-Quadrant Spectral Sequences

Let $\{(E_r, d_r)\}_{r\geq 1}$ be a first-quadrant spectral sequence. Then for each fixed (p, q) there is an integer $R = R(p, q)$ such that the differentials d_r into and out of the spot (p, q) vanish for all $r \geq R$, and consequently the canonical maps

$$E_R^{p,q} = E_{R+1}^{p,q} = \cdots = E_\infty^{p,q}$$

are equalities for all $r \geq R$. Moreover, if the sequence converges to $H_\bullet$ in the sense of definition 6.1.8, then each H_n admits a finite filtration of length $n + 1$ whose successive quotients are the groups $E_\infty^{p,n-p}$ for $0 \leq p \leq n$.

***Proof*:**
Step 1: The differentials vanish for large r. Fix (p, q) with $p, q \geq 0$ (otherwise $E_r^{p,q} = 0$ for all r and there is nothing to prove). The outgoing differential is $d_r \colon E_r^{p,q} \to E_r^{p-r,\,q+r-1}$. Its target lies in the first quadrant only if $p - r \geq 0$, that is, $r \leq p$; for $r > p$ the target group is zero, so the outgoing $d_r = 0$. The incoming differential is $d_r \colon E_r^{p+r,\,q-r+1} \to E_r^{p,q}$; its source lies in

the first quadrant only if $q-r+1 \geq 0$, that is, $r \leq q+1$; for $r > q+1$ the source is zero, so the incoming $d_r = 0$. Set $R = R(p,q) := \max(p, q+1) + 1$. For every $r \geq R$ both differentials touching (p,q) are zero.

Step 2: The page stabilizes. For $r \geq R$, the spot (p,q) has $d_r = 0$ both into and out of it, so at (p,q) one has $\ker(d_r) = E_r^{p,q}$ and $\operatorname{im}(d_r) = 0$. Therefore

$$E_{r+1}^{p,q} = \frac{\ker(d_r\colon E_r^{p,q} \to \cdot)}{\operatorname{im}(d_r\colon \cdot \to E_r^{p,q})} = \frac{E_r^{p,q}}{0} = E_r^{p,q}$$

so the groups are equal for all $r \geq R$. Their common value is by definition $E_\infty^{p,q}$, and the identifications $E_R^{p,q} = E_{R+1}^{p,q} = \cdots = E_\infty^{p,q}$ hold.

Step 3: The filtration of H_n is finite. Suppose the sequence converges to $H_\bullet$ as in definition 6.1.8, so H_n carries a filtration with $E_\infty^{p,n-p} \cong F_pH_n/F_{p-1}H_n$. The spot $(p, n-p)$ lies in the first quadrant only when $0 \leq p$ and $n-p \geq 0$, that is, $0 \leq p \leq n$; outside this range $E_\infty^{p,n-p} = 0$, forcing $F_pH_n = F_{p-1}H_n$ there. Hence the filtration is constant for $p < 0$ (where it is 0) and for $p \geq n$ (where it is all of H_n), so

$$0 = F_{-1}H_n \subseteq F_0H_n \subseteq \cdots \subseteq F_nH_n = H_n$$

is a filtration of length $n+1$ with successive quotients $F_pH_n/F_{p-1}H_n \cong E_\infty^{p,n-p}$ for $0 \leq p \leq n$, as claimed. This is a finite filtration recovering H_n from the limit page up to the extension problems posed by its successive quotients, completing the proof.

The proposition is the structural guarantee that underwrites every computation in the next two sections. It says that for a first-quadrant spectral sequence one never has to worry about convergence as a limiting process: at each spot the value is reached after finitely many pages, and the diagonal $p+q=n$ of the limit page contains, in its $n+1$ nonzero spots, the associated graded pieces of H_n. A computation then reduces to drawing the first page, computing the finitely many differentials, and reading off the stable values along each diagonal. With the abstract machinery in place, the construction promised at the start can be carried out in full: every filtered chain complex produces an exact couple, hence a spectral sequence, converging to its homology.

The setup is a filtered chain complex. Let $C_\bullet$ be a chain complex of abelian groups with an increasing filtration by subcomplexes $F_pC_\bullet$, $p \in \mathbb{Z}$, with $F_pC_\bullet \subseteq F_{p+1}C_\bullet$ and $\partial(F_pC_\bullet) \subseteq F_pC_\bullet$. Assume the filtration is bounded below in each degree, meaning for each n there is an integer $s(n)$ with $F_{s(n)}C_n = 0$, and exhaustive, meaning $\bigcup_p F_pC_\bullet = C_\bullet$. To extract a spectral sequence, set

$$D_{p,q} := H_{p+q}(F_pC_\bullet), \qquad E_{p,q}^1 := H_{p+q}\big(F_pC_\bullet/F_{p-1}C_\bullet\big)$$

where the second group is the homology of the p-th filtration layer, with q recording the complementary degree so that $p+q$ is the total homological degree. The short exact sequence of complexes

$$0 \to F_{p-1}C_\bullet \to F_pC_\bullet \to F_pC_\bullet/F_{p-1}C_\bullet \to 0$$

gives, for each p, a long exact sequence in homology, and these assemble into the maps of an exact couple. The inclusion $F_{p-1} \hookrightarrow F_p$ induces $i\colon D_{p-1,q+1} \to D_{p,q}$; the projection induces $j\colon D_{p,q} \to E_{p,q}^1$; and the connecting map gives $k\colon E_{p,q}^1 \to D_{p-1,q}$. This is the exact couple of the filtration, and its spectral sequence is the object to be constructed.

Theorem 6.1.10: The Spectral Sequence of a Filtered Chain Complex

Let $C_\bullet$ be a chain complex with an increasing filtration $\{F_pC_\bullet\}_{p\in\mathbb{Z}}$ by subcomplexes that is exhaustive and bounded below in each degree. Then there is a spectral sequence $\{(E_r, d_r)\}_{r\geq 1}$, natural in the filtered complex, with first page

$$E^1_{p,q} = H_{p+q}\big(F_pC_\bullet/F_{p-1}C_\bullet\big)$$

the homology of the associated graded complex, and differential d_1 equal to the connecting homomorphism of the layer short exact sequences. The spectral sequence converges to the homology of $C_\bullet$,

$$E^1_{p,q} \Longrightarrow H_{p+q}(C_\bullet)$$

where $H_n(C_\bullet)$ carries the filtration $F_pH_n(C_\bullet) := \operatorname{im}\big(H_n(F_pC_\bullet) \to H_n(C_\bullet)\big)$ induced by the inclusions, and the limit page is identified with the associated graded:

$$E^{p,q}_\infty \cong \frac{F_pH_{p+q}(C_\bullet)}{F_{p-1}H_{p+q}(C_\bullet)}$$

If moreover the filtration is bounded (in each degree $F_pC_n = 0$ for p small and $F_pC_n = C_n$ for p large), the convergence is the stabilization of proposition 6.1.9: each spot reaches E_∞ after finitely many pages.

***Proof*:**
The proof builds the exact couple, reads off the first page, and then identifies the limit page with the associated graded of $H_\bullet(C)$ by tracking the iterated derived couples through the cycle-and-boundary description.

Step 1: The exact couple. For each p the short exact sequence of complexes $0 \to F_{p-1}C_\bullet \to F_pC_\bullet \to F_pC_\bullet/F_{p-1}C_\bullet \to 0$ yields a long exact sequence in homology,

$$\cdots \to H_n(F_{p-1}C) \xrightarrow{\iota_*} H_n(F_pC) \xrightarrow{\pi_*} H_n\big(\tfrac{F_pC}{F_{p-1}C}\big) \xrightarrow{\partial} H_{n-1}(F_{p-1}C) \to \cdots$$

Set $D = \bigoplus_{p,q} D_{p,q}$ with $D_{p,q} = H_{p+q}(F_pC)$ and $E = \bigoplus_{p,q} E^1_{p,q}$ with $E^1_{p,q} = H_{p+q}(F_pC/F_{p-1}C)$, and define $i = \iota_*$, $j = \pi_*$, $k = \partial$ as above. The three exactness conditions of definition 6.1.2 are precisely the exactness of these long exact sequences at the three positions $H_n(F_pC)$, $H_n(F_pC/F_{p-1}C)$, and $H_{n-1}(F_{p-1}C)$. Thus (D, E, i, j, k) is an exact couple, and by definition 6.1.6 it generates a spectral sequence with $E^1 = E$ and $d_1 = jk = \pi_*\partial$, the connecting homomorphism followed by the projection.

Step 2: First-quadrant boundedness. The hypothesis that the filtration is bounded below and exhaustive forces, for each total degree n, only finitely many p to contribute nontrivially in the bounded case, and in the bounded-below case it guarantees that for each (p, q) the incoming and outgoing differentials eventually leave the support of the spectral sequence. Concretely, $E^1_{p,q} = 0$ when $F_pC_{p+q} = F_{p-1}C_{p+q}$, which holds for p below $s(p + q)$ (boundedness below) and, in the bounded case, for p above the top of the filtration in degree $p + q$. The stabilization argument of proposition 6.1.9 applies verbatim once the support lies in a translate of the first quadrant, which the boundedness hypotheses provide; this gives a well-defined $E^{p,q}_\infty$ at each spot.

Step 3: Cycles and boundaries on the r-th page. To identify E_∞, track the iterated derived couples explicitly. Write $Z_n^p = \ker(\partial\colon C_n \to C_{n-1})$ for the cycles and let

$$Z_r^p = \{\, x \in F_pC_n : \partial x \in F_{p-r}C_{n-1} \,\}$$

be the elements of F_pC whose boundary drops r filtration levels (the "approximate cycles" of depth r), and

$$B_r^p = F_pC_n \cap \partial\big(F_{p+r-1}C_{n+1}\big) + F_{p-1}C_n \cap Z_{r-1}^p$$

the boundaries reachable from $r-1$ levels above together with the lower-filtration approximate cycles. A direct induction on r, unwinding the definition of the derived couple, shows that the r-th page of the spectral sequence of the exact couple is

$$E_{p,q}^r \cong \frac{Z_r^p}{B_r^p} \quad \text{in total degree } n = p+q$$

with d_r induced by ∂. The base case $r=1$ is $E_{p,q}^1 = Z_1^p/B_1^p = H_{p+q}(F_pC/F_{p-1}C)$, the homology of the layer, since Z_1^p consists of chains whose boundary already lies in $F_{p-1}C$ (cycles modulo F_{p-1}) and B_1^p of the boundaries from one level up together with F_{p-1}. The inductive step is exactly the passage to the derived couple of theorem 6.1.5: forming $\ker d_r/\mathrm{im} d_r$ replaces Z_r^p/B_r^p by Z_{r+1}^p/B_{r+1}^p, because taking homology with respect to the differential induced by ∂ relaxes the boundary condition by one more filtration level (enlarging Z_r^p to Z_{r+1}^p) and enlarges the boundaries correspondingly. The full verification is the unwinding of the derived-couple definition and is carried out in the abelian-category setting in [10, Explicit Description of the Pages]. The topological content is the identification of E^1 with the layer homology, established here.

Step 4: The limit page is the associated graded. For r large (larger than the filtration length in degree n, available by boundedness, or in the limit using bounded-below exhaustiveness), the condition $\partial x \in F_{p-r}C$ becomes $\partial x = 0$, so Z_r^p stabilizes to $Z_\infty^p = F_pC_n \cap Z_n$, the genuine cycles in filtration $\le p$. Likewise B_r^p stabilizes to $B_\infty^p = F_pC_n \cap \partial C_{n+1} + F_{p-1}C_n \cap Z_n$, the genuine boundaries in filtration $\le p$ together with the lower-filtration cycles. Therefore

$$E_\infty^{p,q} = \frac{F_pC_n \cap Z_n}{F_pC_n \cap \partial C_{n+1} + F_{p-1}C_n \cap Z_n}$$

Now compare with the filtration $F_pH_n(C) = \mathrm{im}(H_n(F_pC) \to H_n(C))$. An element of $F_pH_n(C)$ is the class of a cycle $z \in F_pC_n \cap Z_n$, and two such classes coincide in $F_pH_n/F_{p-1}H_n$ exactly when they differ by a boundary from C_{n+1} or by a cycle of lower filtration. That is precisely the equivalence defining the quotient displayed above, so the natural map

$$E_\infty^{p,q} = \frac{F_pC_n \cap Z_n}{F_pC_n \cap \partial C_{n+1} + F_{p-1}C_n \cap Z_n} \xrightarrow{\cong} \frac{F_pH_n(C)}{F_{p-1}H_n(C)}$$

sending the class of a cycle z to the class of its homology class is a well-defined isomorphism: surjectivity is the definition of $F_pH_n(C)$ as the image of $H_n(F_pC)$, and injectivity is the matching of the two equivalence relations just described. This is the asserted identification $E_\infty^{p,q} \cong F_pH_{p+q}(C)/F_{p-1}H_{p+q}(C)$.

Step 5: Naturality. A filtered chain map $f\colon C_\bullet \to C'_\bullet$ (one with $f(F_pC) \subseteq F_pC'$) induces a map of short exact sequences of layers in every degree, hence a morphism of exact couples $(D, E, \dots) \to (D', E', \dots)$ commuting with i, j, k. By the functoriality of the derived-couple construction (each of i', j', k' is built from i, j, k by operations a couple morphism respects), f induces a map of spectral sequences $E_r(C) \to E_r(C')$ compatible with all differentials and with the filtration on homology, which on E^1 is the map induced by f on layer homology and on E_∞ is the associated graded of $f_*\colon H_\bullet(C) \to H_\bullet(C')$. This is the naturality claim. All assertions of the theorem are established, completing the proof.

The theorem is the prototype every later spectral sequence specializes. The Serre spectral sequence of section 6.2 arises from the filtration of the total space of a fibration by preimages of the skeleta of the base; the cellular filtration of a CW complex by its skeleta recovers, through this same machinery, the cellular chain complex and its homology as the degenerate spectral sequence that collapses at E^2; and the spectral sequence of a double complex, central in homological algebra, is the filtered-complex spectral sequence applied to the total complex filtered by columns. In each case the input is a filtration, the E^1-page is the homology of the layers, the differentials are the higher connecting maps, and the limit is the homology one wanted, filtered by the depth at which each class first appears. The single mechanism of the exact couple and its derived couple generates them all.

Example 6.1.11: The Skeletal Filtration Recovers Cellular Homology

Let X be a CW complex and filter its singular chain complex $C_\bullet(X)$ by the skeleta, $F_pC_\bullet(X) := C_\bullet(X^{(p)})$, the singular chains supported on the p-skeleton. This is increasing, exhaustive ($\bigcup_p X^{(p)} = X$), and bounded below ($X^{(-1)} = \emptyset$). The layer is $F_pC/F_{p-1}C = C_\bullet(X^{(p)})/C_\bullet(X^{(p-1)}) = C_\bullet(X^{(p)}, X^{(p-1)})$, whose homology is the relative homology of the skeletal pair. By the computation of cellular homology (section 3.5), this relative homology is concentrated in a single degree:

$$E^1_{p,q} = H_{p+q}(X^{(p)}, X^{(p-1)}) = \begin{cases} C_p^{\text{cell}}(X), & q = 0, \\ 0, & q \neq 0 \end{cases}$$

the free abelian group on the p-cells, sitting along the row $q = 0$. The differential $d_1 \colon E^1_{p,0} \to E^1_{p-1,0}$ is the connecting map $H_p(X^{(p)}, X^{(p-1)}) \to H_{p-1}(X^{(p-1)}, X^{(p-2)})$, which is exactly the cellular boundary map ∂^{cell} (section 3.5). Since the entire E^1-page lies in the single row $q = 0$, every higher differential d_r for $r \geq 2$ maps into or out of a zero group (the target of d_r on the row $q = 0$ is the row $q = r - 1 \neq 0$), so $d_r = 0$ and the sequence collapses at E^2:

$$E^2_{p,0} = \frac{\ker(\partial_p^{\text{cell}})}{\operatorname{im}(\partial_{p+1}^{\text{cell}})} = H_p^{\text{cell}}(X) = H_p(X)$$

The spectral sequence has degenerated to a single row whose homology is cellular homology, and the convergence statement $E^2_{p,0} \cong F_pH_p(X)/F_{p-1}H_p(X) = H_p(X)$ recovers the agreement of cellular and singular homology established in chapter 3.

The skeletal example is the sanity check that the machine reproduces what is already known: when the filtration is the skeletal one, the spectral sequence collapses immediately and returns cellular homology, with the cellular boundary map appearing as the single differential d_1. The spectral sequence becomes genuinely informative only when the E^1-page spreads over more than one row, so that higher differentials have room to act; that is exactly the situation of a fibration, where the fiber contributes a second grading and the Serre spectral sequence carries differentials on every page. The construction of that spectral sequence, from the filtration of the total space by preimages of base skeleta, is the work of the next section.

Exercise 6.1.1

1. Verify in full the two identities $kj = 0$ and $ik = 0$ used throughout the derived-couple proof, citing the precise exactness condition of definition 6.1.2 each invokes. Then show, conversely, that if (D, E, i, j, k) is a triple of maps forming a triangle with $kj = 0$, $ik = 0$, and $ji = 0$, it need *not* be an exact couple, by exhibiting a triple satisfying all three composite-zero

conditions whose triangle fails exactness at E.

2. Let (D, E, i, j, k) be an exact couple in which i is an isomorphism. Show that $E' = H(E, d) = 0$, so the derived couple has trivial E-term and the spectral sequence collapses to zero from the second page onward. Interpret this for the filtered-complex spectral sequence: which filtrations have i an isomorphism, and why is the homology then concentrated entirely in the limit of D?

3. Consider the chain complex $C_\bullet$ with $C_2 = \mathbb{Z}$, $C_1 = \mathbb{Z} \oplus \mathbb{Z}$, $C_0 = \mathbb{Z}$, boundaries $\partial_2(1) = (3, 0)$ and $\partial_1(a, b) = b$, and $C_n = 0$ otherwise. Filter it by $F_0C_\bullet = (0 \to 0 \to \mathbb{Z}' \to \mathbb{Z})$ where $\mathbb{Z}'$ is the second summand of C_1, and $F_1C_\bullet = C_\bullet$. Compute the E^1-page, the differential d_1, and the $E^2 = E^\infty$-page of the resulting spectral sequence, and check that it converges to $H_\bullet(C_\bullet)$ by computing the latter directly.

4. Let $\{(E_r, d_r)\}$ be a first-quadrant spectral sequence converging to $H_\bullet$, and suppose the E_2-page is concentrated in the two columns $p = 0$ and $p = 1$ (so $E_2^{p,q} = 0$ unless $p \in \{0, 1\}$). Show that all differentials d_r for $r \geq 2$ vanish, that $E_2 = E_\infty$, and write down the resulting filtration of H_n. Conclude that there is a short exact sequence relating H_n to the two columns, and identify the remaining extension problem.

5. Show that for a first-quadrant homological spectral sequence (differentials $d_r \colon E_r^{p,q} \to E_r^{p-r,q+r-1}$ as in definition 6.1.6) the edge spots satisfy: $E_\infty^{n,0}$ is a *subgroup* of $E_2^{n,0}$ (the bottom row emits differentials but receives none), and $E_\infty^{0,n}$ is a *quotient* of $E_2^{0,n}$ (the left column receives differentials but emits none). Identify each as an explicit subquotient in terms of the differentials touching those spots, and write down the two edge homomorphisms relating these groups to H_n. (These edge homomorphisms are developed for the Serre spectral sequence in section 6.2.)

6. A morphism of spectral sequences $f \colon E_\bullet \to E'_\bullet$ is a family of maps $f_r \colon E_r \to E'_r$ commuting with differentials and inducing f_{r+1} on homology. Prove the *mapping lemma*: if f_r is an isomorphism on some page E_r, then f_s is an isomorphism for all $s \geq r$, and (for first-quadrant sequences) f_∞ is an isomorphism on E_∞. Explain why this does *not* by itself imply that the induced map on the abutments $H_\bullet \to H'_\bullet$ is an isomorphism, and state what additional input (convergence plus a filtration-compatibility) closes that gap.

6.2 The Serre Spectral Sequence

The exact couple of section 6.1 is a machine: feed it a filtered chain complex and it returns a spectral sequence converging to the homology of the complex. The Serre spectral sequence is what comes out when the filtered complex is the singular chains of the total space of a fibration $F \to E \to B$, filtered by how far into the base each chain reaches. The base B is taken to be a CW complex, and the filtration is by preimages of its skeleta: $\cdots \subseteq p^{-1}(B^{p-1}) \subseteq p^{-1}(B^p) \subseteq \cdots \subseteq E$. The associated graded pieces of this filtration are controlled by local triviality of the fibration over the cells of B, and that control is exactly what turns the abstract E^1 page into something recognizable: the cellular chains of B with coefficients in the homology of the fiber. The result is a calculational device that, when two of the three homologies $H_*(F)$, $H_*(E)$, $H_*(B)$ are known, constrains the third, often forcing it completely.

The standing hypotheses are that B is a path-connected CW complex and that $\pi_1(B)$ acts trivially on $H_*(F)$. The second condition (automatic when B is simply connected) is what makes the coefficient group $H_q(F)$ a constant coefficient system over B rather than a twisted one; the version without it, recorded in box 6.2.8, replaces ordinary homology of B by homology with local coefficients. Throughout, $p\colon E \to B$ is a Serre fibration (definition 5.2.2) with fiber $F = p^{-1}(b_0)$ over the basepoint, and homology is taken with coefficients in a fixed abelian group, suppressed from the notation when it is $\mathbb{Z}$.

The construction begins with the filtration. Recall that B^p denotes the p-skeleton of the CW complex B, with $B^p = \varnothing$ for $p < 0$.

Definition 6.2.1: Skeletal Filtration of the Total Space

Let $p\colon E \to B$ be a Serre fibration with B a CW complex. The *skeletal filtration* of E is the increasing filtration of singular chain complexes

$$\cdots \subseteq F_{p-1}C_*(E) \subseteq F_pC_*(E) \subseteq F_{p+1}C_*(E) \subseteq \cdots \subseteq C_*(E)$$

where $F_pC_*(E) = C_*\big(p^{-1}(B^p)\big)$ is the subcomplex of singular chains supported in the preimage of the p-skeleton. By convention $F_pC_*(E) = 0$ for $p < 0$, and $\bigcup_p F_pC_*(E) = C_*(E)$ because every singular simplex of E has compact image, hence lands in $p^{-1}(B^p)$ for some finite p.

The filtration is by "base degree": $F_pC_*(E)$ collects the chains of E that project into the p-skeleton of B, regardless of their own dimension. Two gradings are now in play, the total (homological) degree n of a chain in E and the filtration degree p recording how deep into the base it reaches; their difference $q = n - p$ will turn out to be fiber degree. The spectral sequence is the bookkeeping device that disentangles these two gradings, and the whole content of the Serre spectral sequence is the computation of what the associated graded of this particular filtration is.

Example 6.2.2: The Filtration for the Path-Loop Fibration over S^2

Take $B = S^2$ with its standard CW structure: one 0-cell b_0 and one 2-cell, so $B^0 = B^1 = \{b_0\}$ and $B^2 = S^2$. Let $p\colon PS^2 \to S^2$ be the path fibration of example 5.2.12, with fiber the loop space $F = \Omega S^2$. The skeletal filtration of $E = PS^2$ has only two nontrivial steps:

$$F_0C_*(E) = F_1C_*(E) = C_*\big(p^{-1}(b_0)\big) = C_*(\Omega S^2), \qquad F_2C_*(E) = C_*(PS^2)$$

The bottom filtration stage is the chains of the fiber itself, since $p^{-1}(B^0) = p^{-1}(b_0) = \Omega S^2$. The jump from F_1 to F_2 corresponds to the single 2-cell of S^2: a chain in PS^2 lies in F_2 but not F_1 precisely when its projection genuinely uses the 2-cell. With PS^2 contractible, the spectral sequence of this filtration must converge to the homology of a point, and the constraint that everything cancels except in degree 0 is what will compute $H_*(\Omega S^2)$ in section 6.3. Here the point is only structural: the filtration has exactly one step per dimension of B, and the bottom step is the fiber.

Applying the exact couple of a filtered complex (theorem 6.1.10) to the skeletal filtration produces a spectral sequence. The exact couple is assembled from the long exact sequences of the pairs $\big(p^{-1}(B^p), p^{-1}(B^{p-1})\big)$ in homology: setting

$$D^1_{p,q} = H_{p+q}\big(p^{-1}(B^p)\big), \qquad E^1_{p,q} = H_{p+q}\big(p^{-1}(B^p),\, p^{-1}(B^{p-1})\big)$$

the maps $i\colon D^1 \to D^1$ (induced by inclusion $p^{-1}(B^{p-1}) \hookrightarrow p^{-1}(B^p)$), $j\colon D^1 \to E^1$ (the quotient map to relative homology), and $k\colon E^1 \to D^1$ (the connecting homomorphism ∂) form an exact couple in the sense of definition 6.1.2, with bidegrees $|i| = (1, -1)$, $|j| = (0, 0)$, $|k| = (-1, 0)$. The derived couple of definition 6.1.4 then yields the pages E^r, and convergence (definition 6.1.8) holds because the filtration is bounded below and exhausts $C_*(E)$. All of this is the general filtered-complex machinery; the work specific to fibrations is identifying the E^1 and E^2 pages, to which the rest of the section is devoted.

The identification rests on a single geometric input: over each cell of B the fibration is trivial, so the preimage of a cell looks like (cell)$\times F$. This is what converts the relative homology $H_*(p^{-1}(B^p), p^{-1}(B^{p-1}))$ into a sum over the p-cells of B of copies of $H_*(F)$. The precise statement requires the local-triviality lemma below.

Lemma 6.2.3: Homology of a Skeletal Pair

Let $p\colon E \to B$ be a Serre fibration with B a CW complex, F the fiber over the basepoint, and suppose $\pi_1(B)$ acts trivially on $H_*(F)$. Let $\{e^p_\alpha\}_\alpha$ be the p-cells of B, with characteristic maps $\Phi_\alpha\colon (D^p, \partial D^p) \to (B^p, B^{p-1})$. Then there is an isomorphism

$$H_n\big(p^{-1}(B^p),\, p^{-1}(B^{p-1})\big) \;\cong\; \bigoplus_\alpha H_{n-p}(F)$$

natural in the obvious sense, where the summand indexed by e^p_α is the image of $H_n\big(p^{-1}(\Phi_\alpha(D^p)),\, p^{-1}(\Phi_\alpha(\partial D^p))\big)$. Equivalently, $H_n(p^{-1}(B^p), p^{-1}(B^{p-1}))$ is free over $H_{n-p}(F)$ on the p-cells of B, vanishing unless $n \geq p$.

***Proof*:**
Step 1: Reduction to a single cell via the cofiber. The CW pair (B^p, B^{p-1}) has $B^p/B^{p-1} \cong \bigvee_\alpha S^p_\alpha$, a wedge of p-spheres, one per p-cell. The characteristic maps assemble into a map

$$\Phi\colon \Big(\bigsqcup_\alpha D^p_\alpha,\, \bigsqcup_\alpha \partial D^p_\alpha\Big) \longrightarrow (B^p, B^{p-1})$$

which is a relative homeomorphism: it carries $\bigsqcup_\alpha (D^p_\alpha \setminus \partial D^p_\alpha)$ homeomorphically onto $B^p \setminus B^{p-1}$. Pulling the fibration back along Φ and using that excision and the homotopy invariance of fibrations both pass through such a relative homeomorphism (verified in the next steps), the relative homology of the skeletal pair splits as a direct sum over the cells. Concretely, the open cells $e^p_\alpha = \Phi_\alpha(D^p \setminus \partial D^p)$ are disjoint, so a singular chain of $p^{-1}(B^p)$ relative to $p^{-1}(B^{p-1})$ decomposes uniquely according to which open cell its projection meets, giving

$$H_n\big(p^{-1}(B^p), p^{-1}(B^{p-1})\big) \cong \bigoplus_\alpha H_n\big(p^{-1}(\overline{e^p_\alpha}),\, p^{-1}(\partial e^p_\alpha)\big)$$

where the cell-summand is computed by excising the complement of the closed cell. It therefore suffices to compute one summand, that is, to compute $H_n(p^{-1}(D^p), p^{-1}(\partial D^p))$ for the pullback of the fibration along a single characteristic map $\Phi_\alpha\colon D^p \to B$.

Step 2: Triviality of the pulled-back fibration over a disk. Let $q\colon \widetilde{E} \to D^p$ be the pullback of p along Φ_α, a Serre fibration over the disk D^p. The disk is contractible, so the fibration is fiber-homotopy trivial: there is a fiber-homotopy equivalence $\widetilde{E} \simeq D^p \times F$ over D^p, with F the fiber. This is the statement that a fibration over a contractible base is trivial up to fiber homotopy equivalence, which holds for any Serre fibration over a (paracompact, here compact) contractible CW base; the homotopy $\Phi_\alpha \simeq \mathrm{const}_{b_0}$ lifts through the homotopy lifting property of definition 5.2.1 to a deformation of $\widetilde{E}$ onto the fiber over a point, identifying the fiber of q with F up to homotopy. A complete account of fiber-homotopy triviality over a contractible base is given in [14]; in the present setting the conclusion is that the pair $(\widetilde{E}, \widetilde{E}|_{\partial D^p})$ is homotopy equivalent, as a pair, to $(D^p \times F,\, \partial D^p \times F)$.

Step 3: Computing the relative homology of the trivial pair. The quotient of the trivial pair is $(D^p \times F)/(\partial D^p \times F) = (D^p/\partial D^p) \wedge F_+ = S^p \wedge F_+$, the p-fold suspension of F_+ (the fiber with a disjoint basepoint adjoined). Since the pair $(D^p \times F, \partial D^p \times F)$ is a good pair, its relative homology is the reduced homology of the quotient, and the suspension isomorphism gives

$$H_n\big(D^p \times F,\, \partial D^p \times F\big) \cong \widetilde{H}_n\big(S^p \wedge F_+\big) \cong \widetilde{H}_{n-p}\big(F_+\big) = H_{n-p}(F)$$

the shift by p being the p-fold suspension isomorphism $\widetilde{H}_n(\Sigma^p Y) \cong \widetilde{H}_{n-p}(Y)$ applied to $Y = F_+$. Equivalently, by the relative Künneth formula $H_n(D^p \times F, \partial D^p \times F) \cong \bigoplus_{i+j=n} H_i(D^p, \partial D^p) \otimes H_j(F)$, only the $i = p$ term survives because $H_i(D^p, \partial D^p) \cong \widetilde{H}_i(S^p)$ is $\mathbb{Z}$ for $i = p$ and 0 otherwise, yielding $H_{n-p}(F)$; there are no Tor terms since $H_*(D^p, \partial D^p) \cong \mathbb{Z}$ is free. With coefficients in an arbitrary group G the same suspension isomorphism gives $H_{n-p}(F; G)$.

Step 4: Independence of the trivialization and triviality of the π_1-action. The isomorphism of Step 3 depends a priori on the chosen trivialization of $\widetilde{E}$ over D^p, equivalently on a chosen identification of the fiber of q with F. Different trivializations differ by the action of $\pi_1(B, b_0)$ on $H_*(F)$ obtained by transporting the fiber around loops, the monodromy of the fibration. The hypothesis that $\pi_1(B)$ acts trivially on $H_*(F)$ is precisely what makes this identification canonical: any two trivializations induce the same isomorphism $H_{n-p}(\text{fiber of } q) \cong H_{n-p}(F)$ on homology. Summing the single-cell computation of Step 3 over all p-cells via the splitting of Step 1 gives

$$H_n\big(p^{-1}(B^p), p^{-1}(B^{p-1})\big) \cong \bigoplus_\alpha H_{n-p}(F)$$

with the α-summand the image of the closed-cell pair as claimed, and free over $H_{n-p}(F)$ on the p-cells, vanishing for $n < p$ since $H_{n-p}(F) = 0$ there, completing the proof.

The lemma is the geometric heart of the construction. It says that, one base-dimension at a time, the total space looks like the base cells thickened by the fiber, and homology of the thickened pair shifts the fiber homology up by the cell dimension. The role of the trivial π_1-action surfaces exactly where it must: without it the fiber homology groups attached to different cells would be glued by monodromy, and the coefficient system would twist. With it, every cell carries the same untwisted copy of $H_*(F)$, and the E^1 page below becomes ordinary cellular chains of B.

With the pages identified at the level of groups, the boundary d^1 on the E^1 page must be computed. The next theorem assembles the lemma into the full first quadrant statement and identifies d^1 as the cellular boundary of B, which is what produces $E^2_{p,q} = H_p(B; H_q(F))$.

Theorem 6.2.4: Homology Serre Spectral Sequence

Let $p\colon E \to B$ be a Serre fibration with B a path-connected CW complex and fiber F, and suppose $\pi_1(B)$ acts trivially on $H_*(F;G)$ for a fixed coefficient group G. Then there is a first-quadrant homological spectral sequence $\{E^r_{p,q}, d^r\}$, with differentials d^r of bidegree $(-r, r-1)$, whose second page is

$$E^2_{p,q} \;\cong\; H_p\big(B;\, H_q(F;G)\big)$$

the homology of the base with coefficients in the fiber homology, and which converges to the homology of the total space:

$$E^2_{p,q} \;\Longrightarrow\; H_{p+q}(E;G)$$

Convergence means that the filtration of definition 6.2.1 induces on $H_n(E;G)$ a finite filtration $0 = \mathcal{F}_{-1} \subseteq \mathcal{F}_0 \subseteq \cdots \subseteq \mathcal{F}_n = H_n(E;G)$ whose successive quotients are the stable terms: $\mathcal{F}_p/\mathcal{F}_{p-1} \cong E^\infty_{p,\,n-p}$.

***Proof*:**
Step 1: The spectral sequence and its E^1 page. Apply theorem 6.1.10 to the skeletal filtration of $C_*(E;G)$ from definition 6.2.1. This yields a spectral sequence from the exact couple with $D^1_{p,q} = H_{p+q}(p^{-1}(B^p);G)$ and

$$E^1_{p,q} = H_{p+q}\big(p^{-1}(B^p),\, p^{-1}(B^{p-1});G\big)$$

By lemma 6.2.3 (with coefficients in G throughout the fiber homology),

$$E^1_{p,q} \cong \bigoplus_\alpha H_q(F;G) = C^{CW}_p\big(B;\, H_q(F;G)\big)$$

where the last equality comes from the definition of a cellular p-chain, i.e. the cellular p-chains of B with coefficients in $H_q(F;G)$, free on the p-cells. The filtration is bounded below ($F_{-1} = 0$) and exhaustive, so by definition 6.1.8 the spectral sequence converges to $H_*(C_*(E;G)) = H_*(E;G)$ with the stated associated-graded identification. The spectral sequence is first-quadrant because $E^1_{p,q} = 0$ for $p < 0$ (no negative skeleta) and for $q < 0$ ($H_q(F) = 0$).
Step 2: The d^1 differential is the cellular boundary. The differential $d^1\colon E^1_{p,q} \to E^1_{p-1,q}$ is $j \circ k$ in the exact couple, where $k = \partial\colon H_{p+q}(p^{-1}(B^p), p^{-1}(B^{p-1})) \to H_{p+q-1}(p^{-1}(B^{p-1}))$ is the connecting map and j is the quotient to $H_{p+q-1}(p^{-1}(B^{p-1}), p^{-1}(B^{p-2}))$. Under the identification of Step 1, this is exactly the cellular chain differential of B with coefficients twisted by the fiber, by the same diagram that defines the cellular boundary in definition 3.5.10. To see this, fix the fiber degree q and consider the row $E^1_{*,q}$. The triple $(p^{-1}(B^p), p^{-1}(B^{p-1}), p^{-1}(B^{p-2}))$ produces the composite

$$H_{p+q}\big(p^{-1}(B^p), p^{-1}(B^{p-1})\big) \xrightarrow{\partial} H_{p+q-1}\big(p^{-1}(B^{p-1})\big) \xrightarrow{j} H_{p+q-1}\big(p^{-1}(B^{p-1}), p^{-1}(B^{p-2})\big)$$

which is the abstract d^1. For the trivial coefficient case $F = \mathrm{pt}$ this composite is, cell by cell, the degree-counting map of proposition 3.5.14: the attaching map of the cell e^p_α, read through the quotient onto the $(p-1)$-cell e^{p-1}_β, contributes its degree. For general F with trivial π_1-action, the local triviality of lemma 6.2.3 identifies each fiber-summand coherently across the triple, so the composite acts as the cellular boundary $\partial_p^{CW} \otimes \mathrm{id}_{H_q(F;G)}$ on $C_p^{CW}(B; H_q(F;G))$. The naturality clause of the lemma guarantees these identifications commute with ∂ and j.

Step 3: The E^2 page. The differential d^1 on the row $E^1_{*,q} = C_*^{CW}(B; H_q(F;G))$ is the cellular boundary, so its homology is, by the cellular homology theorem (theorem 3.5.11),

$$E^2_{p,q} = H_p\big(C_*^{CW}(B; H_q(F;G)), \partial^{CW}\big) = H_p\big(B;\, H_q(F;G)\big)$$

the singular homology of B with coefficients in the abelian group $H_q(F;G)$. This is the claimed second page.

Step 4: Convergence to $H_*(E;G)$. The general convergence theorem for the spectral sequence of a bounded-below exhaustive filtration (theorem 6.1.10, definition 6.1.8) gives that $E^\infty_{p,q}$ is the associated graded of the filtration $\mathcal{F}_\bullet$ on $H_{p+q}(E;G)$ induced by $F_\bullet C_*(E;G)$, namely $\mathcal{F}_p H_n(E;G) = \mathrm{im}\big(H_n(p^{-1}(B^p);G) \to H_n(E;G)\big)$ and $\mathcal{F}_p/\mathcal{F}_{p-1} \cong E^\infty_{p,n-p}$. The filtration on H_n is finite because $E^1_{p,q} = 0$ for $p > n$ or $q < 0$, so for fixed n only the finitely many spots with $0 \le p \le n$ contribute and the differentials in and out of any fixed spot vanish for r large. This is the convergence asserted, completing the proof.

The theorem packages the whole construction into one statement: a first-quadrant grid whose (p,q)-entry on the second page is the base homology in degree p with fiber-homology coefficients in degree q, and which assembles, after running all the differentials, into the homology of the total space. The two axes have clean meanings: the bottom row $q = 0$ is $H_p(B; H_0(F)) = H_p(B)$ (the fiber being path-connected), the base homology; the left column $p = 0$ is $H_0(B; H_q(F)) = H_q(F)$, the fiber homology. The differentials d^r run leftward and upward, trading base degree for fiber degree, and the surviving E^∞ page records $H_*(E)$ only up to the extension problem of reassembling the filtration quotients. The power of the device is that the E^2 page is computable from $H_*(B)$ and $H_*(F)$ alone, so any partial knowledge of $H_*(E)$ forces relations among the differentials, and conversely.

The same construction carries over verbatim to cohomology, with the filtration directions reversed. Filtering $C^*(E)$ by the subcomplexes of cochains vanishing on chains supported deep in the base produces a cohomological exact couple, and the dual of lemma 6.2.3 identifies its second page as cohomology of the base with fiber-cohomology coefficients. The cohomological version is the one used most in practice, because cohomology carries the cup product, which (as section 6.4 develops) makes the cohomology Serre spectral sequence into a spectral sequence of algebras.

Theorem 6.2.5: Cohomology Serre Spectral Sequence

Let $p\colon E \to B$ be a Serre fibration with B a path-connected CW complex and fiber F, and suppose $\pi_1(B)$ acts trivially on $H^*(F;G)$ for a fixed coefficient group G. Then there is a first-quadrant cohomological spectral sequence $\{E_r^{p,q}, d_r\}$, with differentials d_r of bidegree $(r, 1-r)$, whose second page is

$$E_2^{p,q} \;\cong\; H^p\big(B;\, H^q(F;G)\big)$$

and which converges to the cohomology of the total space:

$$E_2^{p,q} \;\Longrightarrow\; H^{p+q}(E;G)$$

where convergence means $H^n(E;G)$ carries a finite decreasing filtration $H^n = \mathcal{F}^0 \supseteq \mathcal{F}^1 \supseteq \cdots \supseteq \mathcal{F}^{n+1} = 0$ with $\mathcal{F}^p/\mathcal{F}^{p+1} \cong E_\infty^{p,n-p}$.

***Proof*:**
The cochain complex $C^*(E;G)$ carries the decreasing filtration $F^pC^*(E;G) = \ker\big(C^*(E;G) \to C^*(p^{-1}(B^{p-1});G)\big)$, the cochains vanishing on every chain supported in $p^{-1}(B^{p-1})$. This is the filtration dual to the skeletal filtration of definition 6.2.1: it is decreasing and exhaustive, with $F^0 = C^*(E;G)$ (every cochain vanishes on the chains of $p^{-1}(B^{-1}) = \varnothing$) and $\bigcap_p F^pC^*(E;G) = 0$ (a cochain vanishing on the chains of $p^{-1}(B^{p-1})$ for every p vanishes on all of $C_*(E)$, by the exhaustiveness of the skeletal filtration). It is *not* degreewise eventually zero: F^pC^n is in general nonzero for $p > n$, since a singular n-simplex of E projects to a map $\Delta^n \to B$ whose image need not lie in the n-skeleton (a 1-simplex may run through the interior of a 2-cell), so $C_n(p^{-1}(B^{p-1}))$ is a proper subgroup of $C_n(E)$ and cochains vanishing on it persist into high filtration. Convergence will instead come from first-quadrant stabilization on the E_1 page, established below. The associated exact couple (theorem 6.1.10) has

$$E_1^{p,q} = H^{p+q}\big(p^{-1}(B^p),\, p^{-1}(B^{p-1});G\big)$$

the relative cohomology of the same skeletal pairs. Applying $\mathrm{Hom}(-,G)$ to the homology computation of lemma 6.2.3, or equivalently running the cohomological version of its four steps (excision over open cells, fiber-homotopy triviality over each disk, the cohomology Künneth formula for $(D^p \times F, \partial D^p \times F)$, and triviality of the monodromy on $H^*(F;G)$), gives

$$E_1^{p,q} \cong \prod_\alpha H^q(F;G) = C^p_{CW}\big(B;\, H^q(F;G)\big)$$

the cellular p-cochains of B with coefficients in $H^q(F;G)$. In particular the spectral sequence is first-quadrant: $E_1^{p,q} = C^p_{CW}(B; H^q(F;G))$ vanishes for $p < 0$ (there are no negative skeleta) and for $q < 0$ ($H^q(F;G) = 0$). The differential d_1 is the composite $j \circ \delta$ of the connecting map and the restriction, which is the cellular coboundary δ^{CW} on $C^*_{CW}(B; H^q(F;G))$ by the identical triple-diagram argument as in theorem 6.2.4 Step 2, dualized. Its cohomology is

$$E_2^{p,q} = H^p\big(C^*_{CW}(B; H^q(F;G)), \delta^{CW}\big) = H^p\big(B;\, H^q(F;G)\big)$$

by cellular cohomology. Convergence to $H^*(E;G)$ now follows from first-quadrant stabilization (the cohomological form of definition 6.1.8, dual to proposition 6.1.9): being first-quadrant on E_1, the sequence has, for each total degree n, only the finitely many spots $(p, n-p)$ with $0 \leq p \leq n$ nonzero, and the differentials into and out of any fixed spot vanish for r large, so every spot reaches its stable value $E_\infty^{p,q}$ after finitely many pages. The resulting decreasing filtration $H^n = \mathcal{F}^0 \supseteq \cdots \supseteq \mathcal{F}^{n+1} = 0$ on each $H^n(E; G)$, dual to the skeletal filtration, has associated graded $\mathcal{F}^p/\mathcal{F}^{p+1} \cong E_\infty^{p,n-p}$, the stated convergence, completing the proof.

The cohomological version reverses the geometry of the differentials: d_r now raises base degree and lowers fiber degree, of bidegree $(r, 1-r)$, so the arrows on the cohomology page run rightward and downward. The filtration of $H^n(E)$ is decreasing rather than increasing, with the top quotient $\mathcal{F}^0/\mathcal{F}^1 = E_\infty^{0,n}$ a subquotient of the fiber cohomology $H^n(F)$ and the bottom piece $\mathcal{F}^n = E_\infty^{n,0}$ a subquotient of the base cohomology $H^n(B)$. Beyond this bookkeeping difference the two spectral sequences carry the same information, and which one to reach for is dictated by whether the problem at hand gives (co)homology of two of the three spaces and by whether the cup product is needed.

A spectral sequence is only useful when its edges can be read off, because the edges are where the abstract grid touches the actual homology of B, E, and F. The bottom row and left column of the E^2 page survive to E^∞ in part, and the surviving pieces map to or from the homology of the total space through the filtration. These edge maps are the source of every quick application: the Gysin and Wang sequences, the behavior of the spectral sequence for highly connected fiber or base, and the comparison with the maps $F \to E \to B$ on homology.

Theorem 6.2.6: Edge Homomorphisms of the Serre Spectral Sequence

Let $F \xrightarrow{\iota} E \xrightarrow{p} B$ be a Serre fibration as in theorem 6.2.4, with B path-connected and $\pi_1(B)$ acting trivially on $H_*(F)$. In the homology spectral sequence:

1. *(Base edge.)* The bottom-row term $E^\infty_{p,0}$ is a quotient of $E^2_{p,0} = H_p(B; H_0(F)) = H_p(B)$, by the images of the differentials $d^r : E^r_{p,0} \to E^r_{p-r,r-1}$ leaving it, and the composite
$$H_p(E) \twoheadrightarrow \mathcal{F}_p/\mathcal{F}_{p-1} = E^\infty_{p,0} \hookrightarrow E^2_{p,0} = H_p(B)$$
equals the map $p_* : H_p(E) \to H_p(B)$ induced by the projection.

2. *(Fiber edge.)* The left-column term $E^\infty_{0,q}$ is a quotient of $E^2_{0,q} = H_0(B; H_q(F)) = H_q(F)$, by the images of the differentials $d^r : E^r_{r,q-r+1} \to E^r_{0,q}$ entering it, and the composite
$$H_q(F) = E^2_{0,q} \twoheadrightarrow E^\infty_{0,q} = \mathcal{F}_0 H_q(E) \hookrightarrow H_q(E)$$
equals the map $\iota_* : H_q(F) \to H_q(E)$ induced by the fiber inclusion.

Dually, in the cohomology spectral sequence the base edge $H^p(B) = E_2^{p,0} \twoheadrightarrow E_\infty^{p,0} \hookrightarrow H^p(E)$ is p^*, and the fiber edge $H^q(E) \twoheadrightarrow E_\infty^{0,q} \hookrightarrow E_2^{0,q} = H^q(F)$ is the restriction ι^*.

***Proof*:**
Step 1: The bottom row and the base edge. On the bottom row $q = 0$ the entry $E^2_{p,0} = H_p(B; H_0(F)) = H_p(B)$ uses $H_0(F) \cong \mathbb{Z}$ for F path-connected (when F is disconnected the coefficient is $H_0(F)$, the same argument applies). No differential d^r for $r \geq 2$ can leave the bottom row, since d^r has bidegree $(-r, r-1)$ and would land in fiber degree $r - 1 \geq 1$, that is below the bottom row, where there is nothing; but differentials *enter* $E^r_{p,0}$ from $E^r_{p+r,\,1-r}$,

which sits in negative fiber degree and so is zero. Therefore no differential enters or leaves $E^r_{p,0}$ from within the first quadrant after the first page in the relevant direction; precisely, the only differentials hitting $E^r_{p,0}$ come from $E^r_{p+r,1-r} = 0$, so $E^r_{p,0}$ can only shrink by differentials *out of* it, $d^r\colon E^r_{p,0} \to E^r_{p-r,r-1}$. Hence $E^\infty_{p,0}$ is a quotient of $E^2_{p,0} = H_p(B)$. By the convergence of theorem 6.2.4, $E^\infty_{p,0} = \mathcal{F}_p H_p(E)/\mathcal{F}_{p-1}H_p(E)$, and since $\mathcal{F}_p H_p(E) = H_p(E)$ (the top filtration step in total degree p, as $E^\infty_{p',p-p'} = 0$ for $p' > p$), the edge is the composite $H_p(E) \twoheadrightarrow E^\infty_{p,0} \hookrightarrow E^2_{p,0} = H_p(B)$.

To identify this composite with p_*, trace the filtration. The class of a cycle $z \in C_p(E)$ in $E^\infty_{p,0} \subseteq E^2_{p,0} = H_p(B)$ is, unwinding the identification of lemma 6.2.3, represented by the projection $p_* z \in C_p(B)$ read in cellular homology: the bottom-row identification $E^2_{p,0} \cong H_p(B)$ is induced by p itself, because the fiber-degree-zero part of the associated graded is governed by $p^{-1}(B^p) \to B^p$ on H_0 of fibers, which is the augmentation $H_0(F) \to H_0(\mathrm{pt})$ tracking the projection. Thus the composite sends $[z] \in H_p(E)$ to $p_*[z] \in H_p(B)$, that is, the base edge is p_*.

Step 2: The left column and the fiber edge. Since B is path-connected it is homotopy equivalent to a CW complex with a single 0-cell b_0, and the Serre spectral sequence depends only on the fiber-homotopy type of the fibration, so assume $B^0 = \{b_0\}$; then $p^{-1}(B^0) = p^{-1}(b_0) = F$. On the left column $p = 0$ the entry $E^2_{0,q} = H_0(B; H_q(F)) = H_q(F)$ uses $H_0(B) \cong \mathbb{Z}$ for B path-connected. No differential leaves $E^r_{0,q}$, since $d^r\colon E^r_{0,q} \to E^r_{-r,q+r-1} = 0$ lands in negative base degree. Differentials *enter* $E^r_{0,q}$ from $E^r_{r,q-r+1}$ for $r \geq 2$, so $E^\infty_{0,q}$ is a quotient of $E^2_{0,q} = H_q(F)$ by the images of those incoming differentials. By convergence, $E^\infty_{0,q} = \mathcal{F}_0 H_q(E)$, the bottom filtration step $\mathcal{F}_0 H_q(E) = \mathrm{im}(H_q(p^{-1}(B^0)) \to H_q(E)) = \mathrm{im}(H_q(F) \to H_q(E)) = \mathrm{im}(\iota_*)$. The fiber edge is thus the composite $H_q(F) = E^2_{0,q} \twoheadrightarrow E^\infty_{0,q} = \mathrm{im}(\iota_*) \hookrightarrow H_q(E)$.

To see this composite is ι_*, note that the surjection $H_q(F) \twoheadrightarrow E^\infty_{0,q}$ is the map induced by $H_q(F) = H_q(p^{-1}(B^0)) \to \mathcal{F}_0 H_q(E) = \mathrm{im}(\iota_*)$, which by definition of $\mathcal{F}_0$ is the inclusion-induced map ι_* followed by its corestriction onto its image. Composing with $\mathcal{F}_0 H_q(E) \hookrightarrow H_q(E)$ recovers $\iota_*\colon H_q(F) \to H_q(E)$ exactly. Hence the fiber edge is ι_*.

Step 3: The cohomological edges. In cohomology the differentials d_r have bidegree $(r, 1-r)$, so on the bottom row $q = 0$ the differentials leaving $E_r^{p,0} \to E_r^{p+r,1-r} = 0$ vanish (negative fiber degree), while those entering come from $E_r^{p-r,r-1}$ and can be nonzero, making $E_\infty^{p,0}$ a *quotient* of $E_2^{p,0} = H^p(B)$. The decreasing filtration on $H^p(E)$ has bottom step $\mathcal{F}^p H^p(E) = E_\infty^{p,0}$, and the base edge $H^p(B) = E_2^{p,0} \twoheadrightarrow E_\infty^{p,0} = \mathcal{F}^p H^p(E) \hookrightarrow H^p(E)$ is p^* by the dual of Step 1. On the left column $p = 0$, differentials entering $E_r^{0,q}$ come from $E_r^{-r,q+r-1} = 0$ and vanish, while those leaving go to $E_r^{r,q-r+1}$, so $E_\infty^{0,q}$ is a *subgroup* of $E_2^{0,q} = H^q(F)$, cut out as the intersection of the kernels of the outgoing differentials; the fiber edge $H^q(E) \twoheadrightarrow E_\infty^{0,q} \hookrightarrow E_2^{0,q} = H^q(F)$ is the restriction ι^* to the fiber by the dual of Step 2. This establishes all four edge homomorphisms, completing the proof.

The edge homomorphisms are the spectral sequence's interface with the maps in the fibration. The base edge factors p_* through the bottom row, so $p_*\colon H_p(E) \to H_p(B)$ is surjective onto $E^\infty_{p,0}$, the part of $H_p(B)$ that survives all differentials; classes of $H_p(B)$ killed by some d^r are precisely those not hit by p_*. The fiber edge factors ι_* through the left column, so the image of $H_q(F) \to H_q(E)$ is $E^\infty_{0,q}$, the part of $H_q(F)$ that survives, and classes of $H_q(F)$ killed by an incoming differential die in the total space. In cohomology the roles invert: the base edge p^* factors as the surjection $H^p(B) \twoheadrightarrow E_\infty^{p,0}$ followed by the inclusion $E_\infty^{p,0} \hookrightarrow H^p(E)$, so $\mathrm{im}(p^*) = E_\infty^{p,0}$; and the fiber edge ι^* surjects $H^q(E)$ onto $E_\infty^{0,q} \subseteq H^q(F)$, the classes of the fiber that extend over the total space. The simplest consequence, used constantly, is that when the spectral sequence collapses (all differentials zero) the edges become honest injections and surjections relating the three homologies directly.

The product case is the cleanest test of the whole apparatus. When the fibration is trivial, $E = B \times F$, the total space homology is governed by the Künneth theorem, and the Serre spectral sequence must reproduce that answer. It does so by collapsing immediately, which exposes both the meaning of the E^2 page and the role of the differentials as the measure of nontriviality of the fibration.

Example 6.2.7: The Product Fibration and the Künneth Theorem

Let $E = B \times F \xrightarrow{p} B$ be the trivial fibration, with B a path-connected CW complex and F a space with $H_*(F)$ of finite type, and suppose all groups are taken with field coefficients in a field $\Bbbk$ so that Tor terms vanish. The projection p admits the section $s\colon b \mapsto (b, f_0)$ and the fiber inclusion $\iota\colon F = \{b_0\} \times F \hookrightarrow B \times F$ admits the retraction $r\colon (b, f) \mapsto f$. The E^2 page is

$$E^2_{p,q} = H_p\big(B; H_q(F; \Bbbk)\big) \cong H_p(B; \Bbbk) \otimes_{\Bbbk} H_q(F; \Bbbk)$$

the last isomorphism because $H_q(F; \Bbbk)$ is a $\Bbbk$-vector space and homology with vector-space coefficients is $H_p(B; \Bbbk) \otimes_{\Bbbk} H_q(F; \Bbbk)$ by the universal coefficient theorem over a field.

The edges collapse, forcing Künneth in a range. The projection $p\colon E \to B$ admits the section just named, and the fiber inclusion ι admits the retraction r. By theorem 6.2.6 the fiber edge $\iota_*\colon H_q(F) \to H_q(E)$ factors as $H_q(F) = E^2_{0,q} \twoheadrightarrow E^\infty_{0,q} \hookrightarrow H_q(E)$, and the retraction gives $r_*\iota_* = \mathrm{id}$, so ι_* is injective; an injective map whose first factor is the surjection $E^2_{0,q} \twoheadrightarrow E^\infty_{0,q}$ forces that surjection to be an isomorphism, hence $E^\infty_{0,q} = E^2_{0,q}$ and no differential enters the left column. Dually the section makes p_* surjective, so the inclusion $E^\infty_{p,0} \hookrightarrow E^2_{p,0} = H_p(B)$ is onto, hence $E^\infty_{p,0} = E^2_{p,0}$ and no differential leaves the bottom row.

Now restrict to total degree $n \le N$ where N is below the first interior obstruction: concretely, suppose F is $(m-1)$-connected, so the rows $1 \le q \le m-1$ of the E^2 page vanish entirely. Then in total degree $n \le m$ the only nonzero spots are the bottom row $E^2_{n,0} = H_n(B)$ and the left-column spot $E^2_{0,n} = H_n(F)$ (the latter present only for $n = 0$ or $n \ge m$), every other spot $E^2_{p,q}$ with $p + q = n$ and $0 < q < m$ being zero. All differentials in this range have a zero source or target except those on the two edges, which vanish by the previous paragraph. Hence $E^\infty_{p,q} = E^2_{p,q}$ for all (p, q) with $p + q \le m$. Over the field $\Bbbk$ the extension problem is trivial (every short exact sequence of vector spaces splits), so

$$H_n(B \times F; \Bbbk) \cong \bigoplus_{p+q=n} E^2_{p,q} = \bigoplus_{p+q=n} H_p(B; \Bbbk) \otimes_{\Bbbk} H_q(F; \Bbbk) \qquad (n \le m)$$

which is the Künneth theorem over a field in the stated range. The full collapse $E^\infty = E^2$ in all degrees holds as well, but its proof uses the multiplicative structure of the cohomology spectral sequence developed in section 6.4 (the page is generated by the edges as an algebra, and the Leibniz rule propagates the vanishing of edge differentials to the whole page); the present sanity check needs only the edges and the connectivity range. The differentials d^r are thus the corrections that a nontrivial fibration imposes on this product answer.

When $\pi_1(B)$ acts nontrivially on $H_*(F)$, the lemma's identification of the E^2 page breaks: different cells carry copies of $H_*(F)$ glued by the monodromy, and the coefficient group is no longer constant. The repair is to use homology of B with local coefficients in the homology of the fiber, the coefficient system that assigns $H_q(F)$ to the basepoint and the monodromy automorphisms to loops.

6.2.8 Remark: The Local-Coefficients Serre Spectral Sequence Drop the hypothesis that $\pi_1(B)$ acts trivially on $H_*(F)$. The fibration determines an action of $\pi_1(B, b_0)$ on $H_q(F)$ by transporting the fiber around loops (the monodromy), making $H_q(F)$ a $\mathbb{Z}[\pi_1(B)]$-module, equivalently a local coefficient system $\mathcal{H}_q(F)$ on B. The construction of theorem 6.2.4 goes through unchanged except that lemma 6.2.3 Step 4 no longer canonically identifies the cell-summands, so the E^1 row $E^1_{*,q}$ is the cellular chain complex of B with local coefficients in $\mathcal{H}_q(F)$, and its homology is homology with local coefficients:

$$E^2_{p,q} \cong H_p\big(B;\, \mathcal{H}_q(F)\big) \implies H_{p+q}(E)$$

Cohomologically, $E_2^{p,q} \cong H^p(B; \mathcal{H}^q(F)) \Rightarrow H^{p+q}(E)$. When B is simply connected, or more generally when the action is trivial, the local system is constant and this reduces to theorem 6.2.4. The local-coefficient homology $H_*(B; \mathcal{M})$ is the homology of the chain complex $C^{CW}_*(\widetilde{B}) \otimes_{\mathbb{Z}[\pi_1]} \mathcal{M}$ built from the universal cover; a full treatment is in [14].

Exercise 6.2.1

1. Let $F \to E \to B$ be a Serre fibration satisfying the hypotheses of theorem 6.2.4, and suppose B is $(n-1)$-connected, so $\widetilde{H}_p(B) = 0$ for $p < n$. Show that the fiber edge homomorphism $\iota_*\colon H_q(F) \to H_q(E)$ is an isomorphism for $q < n-1$ and surjective for $q = n-1$. (This is the homology analogue of the connectivity statement for fibrations.)

2. Suppose instead the fiber F is $(m-1)$-connected, so $\widetilde{H}_q(F) = 0$ for $q < m$. Show that the base edge $p_*\colon H_p(E) \to H_p(B)$ is an isomorphism for $p < m$ and that there is an exact sequence $H_{m+1}(B) \to H_m(F) \to H_m(E) \to H_m(B) \to 0$ in low degrees. Identify the first map with a single differential of the spectral sequence.

3. Consider the Hopf fibration $S^1 \to S^3 \to S^2$ of example 5.2.7. Write out the E^2 page of the homology Serre spectral sequence, determine the single possibly-nonzero differential, and use convergence to $H_*(S^3)$ to compute that differential. Confirm the answer is consistent with $H_*(S^3)$ being that of a 3-sphere.

4. Let $F \to E \to B$ be a fibration with $B = S^n$ $(n \geq 2)$ simply connected. Using the CW structure on S^n with one 0-cell and one n-cell, show directly from theorem 6.2.4 that the E^2 page has only two nonzero columns, $p = 0$ and $p = n$, each isomorphic to $H_*(F)$. Deduce that the only possibly-nonzero differential is d^n, and write down the resulting long exact sequence relating $H_*(F)$, $H_*(E)$, and the shift $H_{*-n}(F)$.

5. In the cohomology Serre spectral sequence of theorem 6.2.5, suppose E is contractible (for instance the path fibration $\Omega B \to PB \to B$). Explain why $E_\infty^{p,q} = 0$ for $(p, q) \neq (0, 0)$, and describe in words how this forces $H^*(\Omega B)$ to be determined by $H^*(B)$. (You need not compute a specific example; the point is the structural mechanism that section 6.3 exploits.)

6.3 Computations with the Serre Spectral Sequence

The Serre spectral sequence of theorem 6.2.4 and theorem 6.2.5 is an apparatus for transferring homological information across a fibration $F \to E \to B$: knowing the homology of two of the three spaces, one reads off the third from the E_2 page and its differentials. The previous section assembled the machine; this section runs it. Three computations occupy the bulk of what follows, and each is carried out to the last differential. The first computes the homology of the loop space ΩS^n by feeding the path-loop fibration of example 5.2.12 into the cohomology spectral sequence, exploiting the contractibility of the total space PS^n to force every group of $H^*(\Omega S^n)$ into existence. The second extracts two general-purpose exact sequences, the Gysin sequence for a sphere bundle and the Wang sequence for a fibration over a sphere, directly from the shape of the E_2 page when the base or fiber has cohomology concentrated in two degrees. The third returns to the problem that opened the chapter on homotopy theory, the higher homotopy of spheres, and uses the spectral sequence together with the Hurewicz theorem (theorem 5.4.9) to compute $\pi_3(S^2) \cong \mathbb{Z}$ and the first stable stem $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$ for $n \geq 3$.

The governing principle throughout is that a contractible total space, or a base or fiber whose cohomology vanishes outside a narrow band, collapses the spectral sequence so completely that the surviving differentials are forced to be isomorphisms. Every computation below is an instance of this principle: identify the page, locate the one differential that can be nonzero, and use convergence to pin it down.

The first application is the cleanest. The path-loop fibration $\Omega S^n \to PS^n \to S^n$ of example 5.2.12 has contractible total space, so the spectral sequence must converge to the cohomology of a point. The base S^n has cohomology in only two degrees, 0 and n, so the E_2 page is supported on two vertical lines, and a single family of differentials d_n connects them. Forcing the abutment to be trivial in positive degrees determines $H^*(\Omega S^n)$ completely.

Example 6.3.1: The Cohomology of ΩS^n

Fix $n \geq 2$ and consider the path-loop fibration $\Omega S^n \hookrightarrow PS^n \xrightarrow{p} S^n$ of example 5.2.12, with contractible total space PS^n. Because S^n is simply connected for $n \geq 2$, the local coefficient system is trivial, and the cohomology Serre spectral sequence (theorem 6.2.5) has

$$E_2^{p,q} = H^p\big(S^n; H^q(\Omega S^n;\mathbb{Z})\big) = H^p(S^n;\mathbb{Z}) \otimes H^q(\Omega S^n;\mathbb{Z})$$

converging to $H^{p+q}(PS^n;\mathbb{Z})$. Write $F = \Omega S^n$ and abbreviate $H^q = H^q(F;\mathbb{Z})$.

Step 1: The shape of the E_2 page. Since $H^p(S^n;\mathbb{Z})$ is $\mathbb{Z}$ for $p \in \{0,n\}$ and 0 otherwise, the E_2 page is concentrated in the two columns $p = 0$ and $p = n$:

$$E_2^{0,q} = H^q, \qquad E_2^{n,q} = H^q, \qquad E_2^{p,q} = 0 \text{ for } p \notin \{0,n\}$$

A differential $d_r\colon E_r^{p,q} \to E_r^{p+r,\,q-r+1}$ raises the column index by r. With only the columns 0 and n populated, the sole differential that can be nonzero is $d_n\colon E_n^{0,q} \to E_n^{n,\,q-n+1}$, joining the two columns; every other d_r has source or target in an empty column and so vanishes. Hence $E_2 = E_n$ (no earlier differential acts) and $E_{n+1} = E_\infty$ (no later differential acts).

Step 2: Convergence forces d_n to be an isomorphism. The total space PS^n is contractible, so $H^k(PS^n;\mathbb{Z}) = 0$ for $k > 0$ and $\mathbb{Z}$ for $k = 0$. The abutment in total degree $k > 0$ is the associated graded of $H^k(PS^n)$, which is 0; therefore $E_\infty^{p,q} = 0$ for $(p,q) \neq (0,0)$, and $E_\infty^{0,0} = \mathbb{Z}$. The only surviving page-$(n+1)$ groups are the homology of the two-term complexes

$$E_n^{0,q} \xrightarrow{d_n} E_n^{n,\,q-n+1}$$

read across the page: $E_{n+1}^{n,\,q-n+1} = \operatorname{coker} d_n$ and $E_{n+1}^{0,q} = \ker d_n$. For these to vanish for all $(p,q) \neq (0,0)$, the map

$$d_n\colon E_n^{0,q} = H^q \longrightarrow E_n^{n,\,q-n+1} = H^{q-n+1}$$

must be injective (so its kernel, sitting in column 0, dies) and surjective (so its cokernel, sitting in column n, dies) for every $q \geq 1$. Thus $d_n\colon H^q \to H^{q-n+1}$ is an isomorphism for all q with $q \geq 1$, while $E_\infty^{0,0} = \ker(d_n\colon H^0 \to H^{1-n}) = H^0 = \mathbb{Z}$ survives (the target $H^{1-n} = 0$).

Step 3: Solving the recursion. The isomorphism $d_n\colon H^q \xrightarrow{\cong} H^{q-n+1}$ for $q \geq 1$ is a recursion that shifts the degree down by $n-1$. Starting from $H^0 = \mathbb{Z}$ (the loop space is path-connected, since S^n is simply connected, so ΩS^n is connected and $H^0(\Omega S^n;\mathbb{Z}) = \mathbb{Z}$), the isomorphism with $q = n-1$ gives $H^{n-1} \cong H^0 = \mathbb{Z}$; with $q = 2(n-1)$ it gives $H^{2(n-1)} \cong H^{n-1} = \mathbb{Z}$; and inductively

$$H^q(\Omega S^n;\mathbb{Z}) \cong \begin{cases} \mathbb{Z} & q \equiv 0 \pmod{n-1},\ q \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

For any degree q not divisible by $n-1$, the isomorphism chain reaches a degree in the open range $0 < q' < n-1$, where the target $H^{q'-n+1}$ has negative degree and is 0; so $H^{q'} = 0$, and tracing back, $H^q = 0$.

The recursion is a staircase. On the $E_2 = E_n$ page the columns $p = 0$ and $p = n$ each carry $H^\bullet(\Omega S^n)$, a copy of $\mathbb{Z}$ at every height divisible by $n-1$; each transgression d_n is a down-right arrow dropping the degree by $n-1$, forced to be an isomorphism, so the two columns cancel in pairs:

$$\begin{tikzcd}
& \vdots & \vdots \arrow[dr, "\cong"] & \vdots \\
& 2(n-1) & \mathbb{Z} \arrow[dr, "\cong"] & \mathbb{Z} \\
E_2 = E_n\colon & n-1 & \mathbb{Z} \arrow[dr, "\cong"] & \mathbb{Z} \\
& 0 & \mathbb{Z} & \mathbb{Z} \\
& & p = 0 & p = n
\end{tikzcd}
\qquad
\begin{array}{c} E_n^{0,q} = E_n^{n,q} = H^q \\ d_n\colon E_n^{0,q} \xrightarrow{\cong} E_n^{n,\,q-n+1} \end{array}$$

Each arrow $\mathbb{Z} \xrightarrow{\cong} \mathbb{Z}$ pairs a class in column 0 with one in column n; the only class with no partner is the generator of $E_2^{0,0}$, whose d_n lands in negative degree. It alone survives to the limit page,

$$E_\infty: \quad \begin{array}{ccc} \vdots & \vdots & \vdots \\ 2(n-1) & 0 & 0 \\ n-1 & 0 & 0 \\ 0 & \boxed{\mathbb{Z}} & 0 \\ & p=0 & p=n \end{array} \qquad \begin{array}{l} E_\infty^{0,0} = \mathbb{Z} \\ E_\infty^{p,q} = 0 \quad \text{otherwise} \end{array}$$

the associated graded of $H^*(PS^n) = \mathbb{Z}$ concentrated in degree 0, which is the contractibility the argument started from.

Step 4: Reading off homology. The groups just computed are free abelian of rank one in each degree $k(n-1)$, $k \geq 0$, so they are torsion-free in every degree. By the universal coefficient theorem (chapter 4) the integral homology agrees with the cohomology in each degree:

$$H_q(\Omega S^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & q \equiv 0 \pmod{n-1},\ q \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

For $n = 2$ this reads $H_q(\Omega S^2; \mathbb{Z}) = \mathbb{Z}$ in every degree $q \geq 0$, matching the fact that ΩS^2 has the homology of $\mathbb{CP}^\infty$ shifted; for $n = 3$ it gives $\mathbb{Z}$ in every even degree; and in general the homology is concentrated in the arithmetic progression of multiples of $n-1$.

The computation is the spectral-sequence proof of a result first found by Serre, and it is the archetype of every collapse argument: a contractible total space turns the differential into an isomorphism, and an isomorphism that shifts degree by a fixed amount is solved by a geometric recursion. The shape of the answer, one $\mathbb{Z}$ in each multiple of $n-1$, is exactly what one expects from the additive structure of the free loop space, and the multiplicative refinement (that $H^*(\Omega S^n; \mathbb{Z})$ is a divided power algebra on a single generator in degree $n-1$) follows from the multiplicative structure carried by the cohomology Serre spectral sequence, treated in the next section and in full in [14, Section 5.1]. The additive computation above is already enough to drive the homotopy applications later in this section, where ΩS^n appears as the fiber of the path-loop fibration over an Eilenberg-MacLane space.

The next two results are not single computations but reusable exact sequences. Each arises when one of the two spaces in the fibration has cohomology concentrated in exactly two degrees: the Gysin sequence when the *fiber* is a sphere, the Wang sequence when the *base* is a sphere. In both cases the E_2 page occupies two rows or two columns, the only possibly-nonzero differential connects them, and the resulting two-term structure unwinds into a long exact sequence relating the cohomology of base, total space, and the shift induced by that differential.

Begin with the Gysin sequence. An oriented sphere bundle $S^k \to E \to B$ has fiber S^k, whose cohomology is $\mathbb{Z}$ in degrees 0 and k and zero elsewhere. The orientation hypothesis guarantees that the local coefficient system on B is trivial, so the E_2 page is the tensor product $H^*(B) \otimes H^*(S^k)$, supported on the two rows $q = 0$ and $q = k$. The single differential d_{k+1} crossing from the top row to the bottom row is, up to sign, cup product with a distinguished class, the Euler class of the bundle.

Theorem 6.3.2: The Gysin Sequence

Let $S^k \xrightarrow{i} E \xrightarrow{p} B$ be a fiber bundle with fiber the k-sphere ($k \geq 1$), over a path-connected base B, oriented in the sense that the local system $H^k(S^k;\mathbb{Z})$ on B is trivial. Then there is a class $e \in H^{k+1}(B;\mathbb{Z})$, the *Euler class* of the bundle, and a long exact sequence

$$\cdots \to H^m(B) \xrightarrow{\smile e} H^{m+k+1}(B) \xrightarrow{p^*} H^{m+k+1}(E) \xrightarrow{\int_{S^k}} H^{m+1}(B) \xrightarrow{\smile e} H^{m+k+2}(B) \to \cdots$$

with $\mathbb{Z}$ coefficients, where p^* is induced by the projection, $\smile e$ is cup product with the Euler class, and $\int_{S^k}$ is the fiber-integration map lowering degree by k.

***Proof*:**
Step 1: The E_2 page lies in two rows. Since B is path-connected and the system $H^k(S^k;\mathbb{Z})$ is trivial by the orientation hypothesis, the Serre spectral sequence (theorem 6.2.5) has

$$E_2^{p,q} = H^p(B;\mathbb{Z}) \otimes H^q(S^k;\mathbb{Z})$$

The cohomology of the fiber is $H^0(S^k) = \mathbb{Z}$ and $H^k(S^k) = \mathbb{Z}$ with all other groups zero, so only the rows $q = 0$ and $q = k$ are nonzero:

$$E_2^{p,0} = H^p(B), \qquad E_2^{p,k} = H^p(B), \qquad E_2^{p,q} = 0 \text{ for } q \notin \{0, k\}$$

Fix the generator $1 \in H^0(S^k)$ and a generator $\alpha \in H^k(S^k)$ specified by the orientation; then $E_2^{p,0} = H^p(B) \cdot 1$ and $E_2^{p,k} = H^p(B) \cdot \alpha$.

Step 2: Only d_{k+1} is nonzero. A differential d_r shifts (p, q) to $(p + r,\, q - r + 1)$. With nonzero entries only on rows 0 and k, a differential out of row k lands in row $k - r + 1$, which is a nonzero row only when $k - r + 1 = 0$, that is $r = k + 1$. Differentials out of row 0 land in row $1 - r < 0$, always zero. Differentials *into* row 0 come from row $r - 1$, nonzero only for $r - 1 = k$, again $r = k + 1$. Hence the only possibly-nonzero differential is

$$d_{k+1} \colon E_{k+1}^{p,k} \longrightarrow E_{k+1}^{p+k+1,\,0}$$

and $E_2 = E_{k+1}$, $E_{k+2} = E_\infty$, since no earlier or later differential acts. The nonzero part of the page occupies the two rows $q = 0$ and $q = k$, joined by the single slanted differential d_{k+1} running from the top row down into the bottom row, $k + 1$ columns to the right:

$$E_2 = E_{k+1}\colon \quad \begin{tikzcd} q{=}k & \cdots & H^p(B) \arrow[drr, "d_{k+1}"] & \cdots & \\ q{=}0 & \cdots & \cdots & \cdots & H^{p+k+1}(B) \\ & & p & & p{+}k{+}1 \end{tikzcd}$$

Step 3: Naming the Euler class. Define the Euler class by applying d_{k+1} to the fiber generator placed over $1 \in H^0(B)$:

$$e := d_{k+1}(1 \otimes \alpha) \in E_{k+1}^{k+1,0} = H^{k+1}(B)$$

The cohomology Serre spectral sequence carries a multiplicative structure for which each differential d_r is a derivation, $d_r(xy) = d_r(x)y + (-1)^{|x|} x\, d_r(y)$, with the product on E_2 induced by the cup products of base and fiber; the construction of this product and the verification of the Leibniz rule are the content of the next section and are given in full in [14, Section 5.1]. Granting it, d_{k+1} is a derivation, and the classes in row 0 are permanent cycles (they are the image of p^* from the base, so d_{k+1} vanishes on them). For $x \in H^p(B) = E_{k+1}^{p,0}$, the Leibniz rule gives

$$d_{k+1}(x \cdot \alpha) = d_{k+1}\big((x \otimes 1) \smile (1 \otimes \alpha)\big) = (x \otimes 1) \smile d_{k+1}(1 \otimes \alpha) = x \smile e$$

using $d_{k+1}(x \otimes 1) = 0$. Therefore on row k the differential is exactly cup product with e:

$$d_{k+1} \colon H^p(B) \xrightarrow{\smile e} H^{p+k+1}(B)$$

under the identifications $E_{k+1}^{p,k} = H^p(B) \cdot \alpha \cong H^p(B)$ and $E_{k+1}^{p+k+1,0} = H^{p+k+1}(B)$.

Step 4: The E_∞ page and the extension. After d_{k+1} the surviving groups are

$$E_\infty^{p,0} = \operatorname{coker}\big(\smile e \colon H^{p-k-1}(B) \to H^p(B)\big), \qquad E_\infty^{p,k} = \ker\big(\smile e \colon H^p(B) \to H^{p+k+1}(B)\big)$$

In total degree $m + k + 1$ only two entries can be nonzero, namely $E_\infty^{m+k+1,0}$ (on row 0) and $E_\infty^{m+1,k}$ (on row k), so the abutment $H^{m+k+1}(E)$ sits in a short exact sequence

$$0 \to E_\infty^{m+k+1,0} \to H^{m+k+1}(E) \to E_\infty^{m+1,k} \to 0$$

the row-0 piece being the bottom of the filtration (the image of p^*) and the row-k piece the top quotient (the fiber-integration $\int_{S^k}$, evaluation against the fiber generator α). Writing these out,

$$0 \to \operatorname{coker}\big(\smile e \colon H^m(B) \to H^{m+k+1}(B)\big) \xrightarrow{p^*} H^{m+k+1}(E) \xrightarrow{\int_{S^k}} \ker\big(\smile e \colon H^{m+1}(B) \to H^{m+k+2}(B)\big) \to 0$$

Step 5: Splicing into a long exact sequence. The short exact sequence of Step 4, for varying m, splices into the long exact sequence claimed. Concretely, the map $\int_{S^k} \colon H^{m+k+1}(E) \to H^{m+1}(B)$ followed by $\smile e \colon H^{m+1}(B) \to H^{m+2}(B)$ is zero precisely because $\int_{S^k}$ lands in $\ker(\smile e)$; exactness there records that $\operatorname{im}(\int_{S^k}) = \ker(\smile e)$. The map $\smile e \colon H^m(B) \to H^{m+k+1}(B)$ followed by p^* is zero because p^* kills the image of $\smile e$ (that image is exactly the kernel of the projection onto the cokernel), and exactness there is $\ker(p^*) = \operatorname{im}(\smile e)$. Finally p^* followed by $\int_{S^k}$ is zero because the image of p^* is the bottom filtration piece $E_\infty^{*,0}$, which the fiber-integration annihilates, and exactness is $\ker(\int_{S^k}) = \operatorname{im}(p^*)$. The three exactness statements are the three nodes of the asserted sequence, completing the proof.

The Gysin sequence is the spectral-sequence avatar of the Thom isomorphism and the long exact sequence of the pair $(D(E), S(E))$ for the disk bundle of an oriented vector bundle; the Euler class e here is the same Euler class that appears in the theory of characteristic classes in [21]. Its computational force is that it relates the cohomology of the total space E to that of the base B through one algebraic operation, cup product with e. When e is a non-zero-divisor the sequence breaks into short exact pieces and $H^*(E)$ is determined; when $e = 0$ the bundle is cohomologically a product. The most familiar instance is the Hopf fibration, where the base is S^2 and the Euler class is a generator of $H^2(S^2)$, giving the cohomology of S^3 in a single stroke.

Example 6.3.3: The Hopf Fibration via Gysin

Apply theorem 6.3.2 to the Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$ of example 5.2.7, an oriented S^1-bundle, so $k = 1$. The base is $B = S^2$ with $H^0(S^2) = H^2(S^2) = \mathbb{Z}$ and $H^1(S^2) = 0$. The Euler class $e \in H^2(S^2) \cong \mathbb{Z}$ is a generator (the bundle is nontrivial, as $S^3 \neq S^1 \times S^2$). With $k = 1$ the E_2 page occupies the two fiber rows $q = 0$ and $q = 1$, and the single differential $d_2 = \smile e$ crosses from the top row down into the bottom row, two columns to the right; since e generates $H^2(S^2)$ it is an isomorphism:

$$\begin{tikzcd}
 & q{=}1 & \mathbb{Z} \arrow[drr, "\smile e"] & 0 & \mathbb{Z} & \\
E_2\colon & q{=}0 & \mathbb{Z} & 0 & \mathbb{Z} & d_2 = \smile e\colon E_2^{0,1} \xrightarrow{\cong} E_2^{2,0} \\
 & & p{=}0 & p{=}1 & p{=}2 &
\end{tikzcd}$$

The Gysin sequence around the relevant degrees, with $\int = \int_{S^1}$ lowering degree by 1, reads in degree $m = 0$,

$$H^0(S^2) \xrightarrow{\smile e} H^2(S^2) \xrightarrow{p^*} H^2(S^3) \xrightarrow{\int} H^1(S^2) = 0$$

The map $\smile e\colon H^0(S^2) = \mathbb{Z} \to H^2(S^2) = \mathbb{Z}$ sends $1 \mapsto e$, a generator, hence is an isomorphism. Exactness then forces $p^*\colon H^2(S^2) \to H^2(S^3)$ to be the zero map on a surjection, so $H^2(S^3) = 0$. Likewise the segment

$$H^2(S^2) \xrightarrow{\smile e} H^4(S^2) = 0, \qquad 0 = H^3(S^2) \to H^3(S^3) \xrightarrow{\int} H^2(S^2) \xrightarrow{\smile e} H^4(S^2) = 0$$

shows $H^3(S^3) \cong H^2(S^2) = \mathbb{Z}$. Together with $H^0(S^3) = \mathbb{Z}$ from path-connectedness, this recovers $H^*(S^3; \mathbb{Z}) = \mathbb{Z}, 0, 0, \mathbb{Z}$ in degrees $0, 1, 2, 3$, exactly the cohomology of S^3.

The dual situation, where the base rather than the fiber is a sphere, produces the Wang sequence. Here the E_2 page occupies two *columns*, $p = 0$ and $p = n$, because $H^*(S^n)$ is concentrated in those two degrees. The single differential d_n now runs between columns, and the resulting long exact sequence relates the cohomology of the total space to that of the fiber, with the connecting map encoding how the fiber is twisted as one goes around the base sphere.

Theorem 6.3.4: The Wang Sequence

Let $F \xrightarrow{i} E \xrightarrow{p} S^n$ be a fibration over the n-sphere ($n \geq 2$), with path-connected fiber F, so that the simply connected base S^n gives a trivial local system. Then there is a long exact sequence with $\mathbb{Z}$ coefficients

$$\cdots \to H^m(E) \xrightarrow{i^*} H^m(F) \xrightarrow{\theta} H^{m-n+1}(F) \to H^{m+1}(E) \xrightarrow{i^*} H^{m+1}(F) \to \cdots$$

where i^* is restriction to the fiber and $\theta\colon H^m(F) \to H^{m-n+1}(F)$ is a derivation of degree $-(n-1)$ (it satisfies the Leibniz rule $\theta(xy) = \theta(x)\, y + (-1)^{(n-1)|x|} x\, \theta(y)$).

***Proof*:**
Step 1: The E_2 page lies in two columns. As S^n is simply connected for $n \geq 2$, the local system is trivial and the Serre spectral sequence (theorem 6.2.5) has

$$E_2^{p,q} = H^p(S^n; \mathbb{Z}) \otimes H^q(F; \mathbb{Z})$$

Since $H^p(S^n)$ is $\mathbb{Z}$ for $p \in \{0, n\}$ and zero otherwise, only the columns $p = 0$ and $p = n$ are nonzero:

$$E_2^{0,q} = H^q(F), \qquad E_2^{n,q} = H^q(F), \qquad E_2^{p,q} = 0 \text{ for } p \notin \{0, n\}$$

Let $s \in H^n(S^n)$ be the generator, so $E_2^{n,q} = H^q(F) \cdot s$.

Step 2: Only d_n is nonzero. A differential d_r raises the column index by r. From column 0 it reaches column r, which is nonzero only for $r = n$; from column n it reaches column $n + r > n$, always zero. Differentials *into* column n come from column $n - r$, nonzero only for $r = n$. So the only possibly-nonzero differential is

$$d_n \colon E_n^{0,q} = H^q(F) \longrightarrow E_n^{n,\, q-n+1} = H^{q-n+1}(F)$$

and $E_2 = E_n$, $E_{n+1} = E_\infty$. The nonzero part of the page occupies the two columns $p = 0$ and $p = n$, joined by the single differential d_n running from the left column down into the right column, dropping the fiber degree by $n - 1$:

$$\begin{tikzcd}
 & \vdots & \vdots & \vdots \\
 & q & H^q(F) \arrow[dr, "d_n"] & H^q(F) \cdot s \\
E_2 = E_n \colon & q{-}n{+}1 & H^{q-n+1}(F) & H^{q-n+1}(F) \cdot s \\
 & \vdots & \vdots & \vdots \\
 & & p{=}0 & p{=}n
\end{tikzcd}$$

Step 3: The differential is the Wang derivation. Define

$$\theta := d_n \colon H^q(F) = E_n^{0,q} \longrightarrow E_n^{n,\, q-n+1} = H^{q-n+1}(F)$$

where the target identification drops the generator s. By the derivation property of the differentials in the multiplicative cohomology Serre spectral sequence (recorded in the proof of theorem 6.3.2 and proved in full in [14, Section 5.1]), d_n is a derivation: for $x \in H^{|x|}(F) = E_n^{0,|x|}$ and $y \in E_n^{0,|y|}$,

$$\theta(xy) = d_n(x \otimes y) = d_n(x)\, y + (-1)^{|x|}\, x\, d_n(y)$$

Tracking the sign through the identification of $E_n^{n,q-n+1}$ with $H^{q-n+1}(F)$, which moves the class $d_n(x)$ of column n and bidegree shift $-(n-1)$ past y, the Koszul sign becomes $(-1)^{(n-1)|x|}$, giving the stated Leibniz rule $\theta(xy) = \theta(x)y + (-1)^{(n-1)|x|} x\,\theta(y)$.

Step 4: The E_∞ page and the abutment. After d_n,

$$E_\infty^{0,q} = \ker\big(\theta\colon H^q(F) \to H^{q-n+1}(F)\big), \qquad E_\infty^{n,q} = \operatorname{coker}\big(\theta\colon H^{q+n-1}(F) \to H^q(F)\big)$$

In total degree m exactly two entries can survive, $E_\infty^{0,m}$ (column 0) and $E_\infty^{n,m-n}$ (column n), so the abutment $H^m(E)$ fits in

$$0 \to E_\infty^{n,m-n} \to H^m(E) \to E_\infty^{0,m} \to 0$$

The bottom of the filtration is the column-n piece and the top quotient is the column-0 piece $E_\infty^{0,m} = \ker(\theta) \subseteq H^m(F)$; the quotient map $H^m(E) \to E_\infty^{0,m} \hookrightarrow H^m(F)$ is the restriction i^* to the fiber (edge homomorphism along the fiber, theorem 6.2.6). Writing the short exact sequence as

$$0 \to \operatorname{coker}\big(\theta\colon H^{m-1}(F) \to H^{m-n}(F)\big) \to H^m(E) \xrightarrow{i^*} \ker\big(\theta\colon H^m(F) \to H^{m-n+1}(F)\big) \to 0$$

Step 5: Splicing. For varying m these short exact sequences splice into the long exact sequence asserted. The connecting homomorphism $H^m(F) \to H^{m-n+1}(F)$ is θ itself: an element of $H^m(F)$ lifts to $H^m(E)$ exactly when it lies in $\ker\theta$ (by the right-hand surjection above), and its obstruction to lifting is its image $\theta(x) \in H^{m-n+1}(F)$, landing in the cokernel piece $\operatorname{coker}\theta$ that forms the column-n part of $H^{m+1}(E)$. The three exactness conditions ($\operatorname{im} i^* = \ker\theta$, $\operatorname{im}\theta = \ker(\text{into } H^{m+1}(E))$, and $\ker i^* = \operatorname{im}(\text{from } H^m(E))$) are read off the short exact sequences exactly as in the Gysin case, yielding

$$\cdots \to H^m(E) \xrightarrow{i^*} H^m(F) \xrightarrow{\theta} H^{m-n+1}(F) \to H^{m+1}(E) \xrightarrow{i^*} \cdots$$

as we sought to show.

The Wang sequence is the tool of choice whenever a space fibers over a sphere, and it appears throughout the homotopy theory of Lie groups and loop spaces, where total spaces are frequently bundles over spheres. The derivation θ measures the monodromy of the fibration: it records how the cohomology of the fiber is sheared as the base sphere is traversed. When $\theta = 0$ the cohomology of E is additively the tensor product $H^*(F) \otimes H^*(S^n)$, the cohomological signature of a trivial bundle. As a consistency check, the same computation of $H^*(\Omega S^n)$ carried out in example 6.3.1 can be recovered from the Wang sequence applied to the path-loop fibration, reading the contractibility of PS^n as the statement that i^* vanishes and θ is an isomorphism in the relevant degrees.

Example 6.3.5: Cohomology of ΩS^n via Wang

Take the path-loop fibration $\Omega S^n \to PS^n \to S^n$ as a fibration over the base S^n, and apply the Wang sequence theorem 6.3.4 with $F = \Omega S^n$, $E = PS^n$. Since PS^n is contractible, $H^m(PS^n) = 0$ for $m > 0$, and the sequence

$$H^m(PS^n) \xrightarrow{i^*} H^m(\Omega S^n) \xrightarrow{\theta} H^{m-n+1}(\Omega S^n) \to H^{m+1}(PS^n)$$

has its two outer terms vanishing for $m \geq 1$ (the right term $H^{m+1}(PS^n) = 0$ for $m + 1 \geq 1$, the left $H^m(PS^n) = 0$ for $m \geq 1$). Exactness forces $\theta\colon H^m(\Omega S^n) \xrightarrow{\cong} H^{m-n+1}(\Omega S^n)$ to be an isomorphism for every $m \geq 1$. This is the identical recursion of Step 2-3 of example 6.3.1, with θ playing the role of d_n, and it reproduces

$$H^q(\Omega S^n;\mathbb{Z}) = \begin{cases} \mathbb{Z} & (n-1)|q,\ q \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

confirming the earlier computation by an independent route.

The final and most substantial application returns to the homotopy groups of spheres. The fibration long exact sequence of theorem 5.2.9 computed $\pi_n(S^n) \cong \mathbb{Z}$ and, via the Hopf fibration, gave $\pi_3(S^2) \cong \mathbb{Z}$; but it stalls immediately above the diagonal because the relevant fibrations run out. The spectral sequence breaks the impasse through a different mechanism: it computes the *homology* of an Eilenberg-MacLane space $K(\mathbb{Z},n)$, and the Hurewicz theorem (theorem 5.4.9) converts homology back into homotopy. The strategy, due to Serre, is to study the map $S^n \to K(\mathbb{Z},n)$ inducing an isomorphism on π_n, replace it by a fibration, and analyze the fiber.

The central computation is the integral homology of $K(\mathbb{Z},n)$ in the first range above degree n. It is built from the path-loop fibration over $K(\mathbb{Z},n)$, whose fiber is $K(\mathbb{Z},n-1)$, descending an induction on n that starts from the known $K(\mathbb{Z},1) = S^1$ and $K(\mathbb{Z},2) = \mathbb{C}\mathrm{P}^\infty$.

Proposition 6.3.6: Low-Degree Homology of $K(\mathbb{Z},n)$

For $n \geq 3$ the integral homology of $K(\mathbb{Z},n)$ in degrees $\leq n+2$ is

$$H_i(K(\mathbb{Z},n);\mathbb{Z}) \cong \begin{cases} \mathbb{Z} & i = 0 \\ 0 & 0 < i < n \\ \mathbb{Z} & i = n \\ 0 & i = n+1 \\ \mathbb{Z}/2 & i = n+2 \end{cases}$$

For $n = 2$ one has instead $H_i(K(\mathbb{Z},2);\mathbb{Z}) = H_i(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$, which is $\mathbb{Z}$ in every even degree and 0 in every odd degree; in particular $H_3(K(\mathbb{Z},2)) = 0$ and $H_4(K(\mathbb{Z},2)) = \mathbb{Z}$. Thus $H_{n+1}(K(\mathbb{Z},n);\mathbb{Z}) = 0$ for all $n \geq 2$, while $H_{n+2}(K(\mathbb{Z},n);\mathbb{Z}) \cong \mathbb{Z}/2$ holds for $n \geq 3$ and fails at $n = 2$.

***Proof*:**
Step 1: The base case $K(\mathbb{Z},2) = \mathbb{C}\mathrm{P}^\infty$. By example 5.5.5 the homology of $K(\mathbb{Z},2) = \mathbb{C}\mathrm{P}^\infty$ is $\mathbb{Z}$ in every even degree and 0 in every odd degree, read off the cellular chain complex of example 5.5.8. This gives $H_3(K(\mathbb{Z},2)) = 0$ and $H_4(K(\mathbb{Z},2)) = \mathbb{Z}$, as stated, and the cohomology ring is the polynomial ring $H^*(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) = \mathbb{Z}[a]$ with $|a| = 2$ (example 5.5.8). The case $n \geq 3$ is the substance of the proposition.

Step 2: The path-loop fibration over $K(\mathbb{Z},n)$. For $n \geq 3$ use the path-loop fibration of example 5.2.12 applied to $X = K(\mathbb{Z},n)$:

$$\Omega K(\mathbb{Z},n) \hookrightarrow PK(\mathbb{Z},n) \xrightarrow{p} K(\mathbb{Z},n)$$

The total space $PK(\mathbb{Z},n)$ is contractible. The fiber is $\Omega K(\mathbb{Z},n) \simeq K(\mathbb{Z},n-1)$, because $\pi_i(\Omega K(\mathbb{Z},n)) \cong \pi_{i+1}(K(\mathbb{Z},n))$ is $\mathbb{Z}$ for $i+1 = n$, that is $i = n-1$, and 0 otherwise (example 5.2.12

and the loop-space equivalence $K(G,m) \simeq \Omega K(G,m+1)$ of section 5.5). The cohomology Serre spectral sequence (theorem 6.2.5) of this fibration has

$$E_2^{p,q} = H^p\big(K(\mathbb{Z},n);\ H^q(K(\mathbb{Z},n-1);\mathbb{Z})\big)$$

converging to $H^{p+q}(PK(\mathbb{Z},n);\mathbb{Z})$, which is $\mathbb{Z}$ in total degree 0 and 0 in every positive total degree. Both base and fiber are simply connected (each is $(n-1)$- or $(n-2)$-connected with $n \geq 3$), so the local system is trivial and $E_2^{p,q} = H^p(B) \otimes H^q(F)$ with $B = K(\mathbb{Z},n)$, $F = K(\mathbb{Z},n-1)$.

Step 3: The transgression kills the fundamental classes. Both base and fiber have known cohomology below the band of interest: by the connectivity of $K(\mathbb{Z},m)$ and Hurewicz (theorem 5.4.9), $H^q(F) = 0$ for $0 < q < n-1$ and $H^{n-1}(F) = \mathbb{Z}$, while $H^q(B) = 0$ for $0 < q < n$ and $H^n(B) = \mathbb{Z}$. Let $a \in H^{n-1}(F) = \mathbb{Z}$ and $b \in H^n(B) = \mathbb{Z}$ be the generators. The only off-axis E_2 entry in total degree $n-1$ is $E_2^{0,n-1} = \mathbb{Z}\langle a\rangle$, and the only entry in total degree n on the base axis is $E_2^{n,0} = \mathbb{Z}\langle b\rangle$. Convergence to a point forces both to vanish on E_∞. The sole differential out of $E^{0,n-1}$ is the transgression

$$d_n\colon E_n^{0,n-1} \longrightarrow E_n^{n,0}, \qquad d_n(a) = \pm b$$

and since $E_\infty^{0,n-1} = 0$ this d_n must be injective, while $E_\infty^{n,0} = 0$ forces it onto $\mathbb{Z}\langle b\rangle$; hence d_n is an isomorphism. Orient b so that $d_n(a) = b$.

Step 4: Degree $n+1$ gives $H^{n+1}(K(\mathbb{Z},n)) = 0$. The entries of total degree $n+1$ in the populated region are $E_2^{n+1,0} = H^{n+1}(B)$, $E_2^{0,n+1} = H^{n+1}(F)$, and $E_2^{2,n-1} = H^2(B) \otimes H^{n-1}(F) = 0$ (since $H^2(B) = 0$ as $2 < n$). All must die. Examine $E^{n+1,0}$ on the base axis: a differential into it is $d_r\colon E_r^{n+1-r,\,r-1} \to E_r^{n+1,0}$; for the source to be nonzero one needs a nonzero $H^{n+1-r}(B)$ with $0 < n+1-r$, but $H^p(B) = 0$ for $0 < p < n$ and $n+1-r < n$ for $r \geq 2$, while the fiber factor $H^{r-1}(F)$ is nonzero only at $r-1 = 0$ (giving the axis itself) or $r-1 = n-1$, i.e. $r = n$, where the source is $E_n^{1,n-1} = H^1(B) \otimes H^{n-1}(F) = 0$ because $H^1(B) = 0$. So no differential hits $E^{n+1,0}$, and none leaves it into a populated group. Therefore $E_\infty^{n+1,0} = H^{n+1}(B)$, which convergence forces to be 0. By the universal coefficient theorem (the homology of B being torsion-free in degrees $< n+1$), $H^{n+1}(B) = 0$ gives $H_{n+1}(K(\mathbb{Z},n);\mathbb{Z}) = 0$, valid for every $n \geq 2$.

Step 5: The case $n = 3$, where the $\mathbb{Z}/2$ is produced explicitly. Take $n = 3$, fiber $F = K(\mathbb{Z},2) = \mathbb{CP}^\infty$ with $H^*(F) = \mathbb{Z}[a]$, $|a| = 2$, and base $B = K(\mathbb{Z},3)$ with $H^3(B) = \mathbb{Z}\langle b\rangle$, $b = d_3(a)$. The relevant E_2 entries are $E_2^{0,2} = \mathbb{Z}\langle a\rangle$, $E_2^{0,4} = \mathbb{Z}\langle a^2\rangle$, $E_2^{3,2} = \mathbb{Z}\langle ba\rangle$, $E_2^{3,0} = \mathbb{Z}\langle b\rangle$, $E_2^{6,0} = H^6(B)$. The differential d_3 is a derivation (the multiplicative structure cited in theorem 6.3.2, full proof in [14, Section 5.1]), so

$$d_3(a^2) = 2a\, d_3(a) = 2ab = 2\,(ba) \in E_3^{3,2}, \qquad d_3(ba) = d_3(b)\,a \pm b\, d_3(a) = \pm b^2 \in E_3^{6,0} = H^6(B)$$

using $d_3(b) = 0$ (b lies on the base axis, a permanent cycle). Now $b \in H^3(B)$ has odd degree, so graded-commutativity gives $b^2 = -b^2$, whence $2b^2 = 0$: the class $b^2 \in H^6(B)$ is annihilated by 2.

Track the two-term complex on the diagonal of total degree 5, namely $E_3^{0,4} \xrightarrow{d_3} E_3^{3,2} \xrightarrow{d_3} E_3^{6,0}$, which reads

$$\mathbb{Z}\langle a^2\rangle \xrightarrow{\times 2} \mathbb{Z}\langle ba\rangle \xrightarrow{b^2} H^6(B)$$

Convergence to a point forces $E_\infty^{3,2} = 0$ (total degree $5 > 0$). The homology at the middle spot is $\ker(ba \mapsto b^2)/\operatorname{im}(\times 2)$. Since b^2 is 2-torsion, $k\,(ba) \mapsto k\,b^2 = 0$ precisely when k is even, so $\ker(ba \mapsto b^2) = 2\mathbb{Z}\langle ba\rangle$; and $\operatorname{im}(\times 2) = 2\mathbb{Z}\langle ba\rangle$. Hence $E_4^{3,2} = 2\mathbb{Z}/2\mathbb{Z} = 0$, and no later differential is needed: the entry dies exactly because $d_3(a^2) = 2\,ba$ and $d_3(ba) = b^2$ are compatible. Consequently the class $b^2 = d_3(ba)$ is a boundary, so it becomes 0 on $E_4^{6,0}$; but $b^2 \neq 0$ on E_2, and it is exactly the 2-torsion subgroup $\mathbb{Z}/2\langle b^2\rangle \subseteq H^6(B)$ that is hit. This 2-torsion is the Ext-image of $H_5(K(\mathbb{Z},3))$ under the universal coefficient theorem: the torsion subgroup of $H^6(B)$ is $\mathrm{Ext}(H_5(B),\mathbb{Z}) \cong (H_5(B))_{\mathrm{tors}}$, and the $\mathbb{Z}/2$ produced here gives $(H_5(B))_{\mathrm{tors}} = \mathbb{Z}/2$. The free part of $H_5(B)$ vanishes because $H^5(B) = 0$: the base-axis entry $E^{5,0} = H^5(B)$ in total degree 5 receives no differential (its only candidate source $E^{0,4} = \langle a^2\rangle$ already died on E_4, as $\times 2$ is injective on $\mathbb{Z}$, so $E_5^{0,4} = 0$) and supports none, so $E_\infty^{5,0} = H^5(B) = 0$ by convergence, whence $\mathrm{Hom}(H_5(B),\mathbb{Z}) = 0$. Therefore $H_5(K(\mathbb{Z},3);\mathbb{Z}) \cong \mathbb{Z}/2$ is pure torsion, and $H_{n+2}(K(\mathbb{Z},3);\mathbb{Z}) = H_5(K(\mathbb{Z},3);\mathbb{Z}) \cong \mathbb{Z}/2$.

Step 6: Propagation to $n > 3$. The same fibration $K(\mathbb{Z},n-1) \to PK(\mathbb{Z},n) \to K(\mathbb{Z},n)$ propagates the $\mathbb{Z}/2$ upward: granting the inductive hypothesis $H_{(n-1)+2}(K(\mathbb{Z},n-1);\mathbb{Z}) = \mathbb{Z}/2$, the fiber $F = K(\mathbb{Z},n-1)$ carries a $\mathbb{Z}/2$ in degree $n+1$ homology, and the transgression of the fiber's torsion class lands the $\mathbb{Z}/2$ in $H_{n+2}(K(\mathbb{Z},n))$ by the identical two-term-complex mechanism of Step 5, now driven by the transgression theorem for the fundamental class and its square. The integral torsion bookkeeping that carries this through for all n, identifying the surviving differential as multiplication by 2 and ruling out cancellation by other classes in the band, is the content of Serre's computation of $H_*(K(\mathbb{Z},n);\mathbb{Z})$; it uses the transgression theorem developed in the next section in full generality, and the complete argument is the computation of the cohomology of Eilenberg-MacLane spaces in the spectral-sequences chapter of [14]. The conclusion is $H_{n+2}(K(\mathbb{Z},n);\mathbb{Z}) \cong \mathbb{Z}\otimes\mathbb{Z}/2 \cong \mathbb{Z}/2$ for every $n \geq 3$, as we sought to show.

The proposition is the homological engine of the stable computation. Its content is the appearance of 2-torsion in $H_{n+2}(K(\mathbb{Z},n))$ for $n \geq 3$, traced to the derivation rule $d_n(a^2) = 2a\,d_n(a)$ together with the graded-commutative identity $b^2 = -b^2$ for the odd-degree transgressed class. The factor of 2 is the source of the $\mathbb{Z}/2$ in the stable stem, and it is the first place in the book where torsion in a homotopy group is produced by an explicit algebraic mechanism rather than asserted. With the homology of $K(\mathbb{Z},n)$ in hand, the homotopy of spheres in the first degree above the diagonal falls out.

Example 6.3.7: $\pi_3(S^2)$ and the First Stable Stem $\pi_{n+1}(S^n)$

Two computations: first $\pi_3(S^2) \cong \mathbb{Z}$, then $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$ for $n \geq 3$.

Part 1: $\pi_3(S^2) \cong \mathbb{Z}$. The Hopf fibration $S^1 \to S^3 \to S^2$ of example 5.2.7 has long exact homotopy sequence (theorem 5.2.9)

$$\pi_3(S^1) \to \pi_3(S^3) \xrightarrow{p_*} \pi_3(S^2) \xrightarrow{\partial} \pi_2(S^1)$$

Now $\pi_3(S^1) = 0$ and $\pi_2(S^1) = 0$ because $S^1 = K(\mathbb{Z},1)$ has all higher homotopy trivial (example 5.2.10), so $p_*\colon \pi_3(S^3) \to \pi_3(S^2)$ is an isomorphism. By corollary 5.4.11 $\pi_3(S^3) \cong \mathbb{Z}$, generated by the identity, hence

$$\pi_3(S^2) \cong \mathbb{Z}$$

generated by the Hopf map, recovering the result of example 5.2.13.

Part 2: The stable stem $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$ for $n \geq 3$. Fix $n \geq 3$. Choose a map $f\colon S^n \to K(\mathbb{Z},n)$ representing a generator of $\pi_n(K(\mathbb{Z},n)) = \mathbb{Z}$; since $f_*\colon \pi_n(S^n) \to \pi_n(K(\mathbb{Z},n))$ is an isomorphism (both are $\mathbb{Z}$ and f is a generator), and both spaces are $(n-1)$-connected, the map f is an isomorphism on π_i for $i \leq n$. Convert f into a fibration with fiber F; by the long exact homotopy sequence the fiber F is n-connected ($\pi_i(F) = 0$ for $i \leq n$) and

$$\pi_{n+1}(F) \cong \pi_{n+1}(S^n)$$

where the isomorphism is read from the long exact homotopy sequence of the fibration $F \to S^n \to K(\mathbb{Z}, n)$ in two adjacent segments. The segment $\pi_{n+1}(K(\mathbb{Z}, n)) \to \pi_n(F) \to \pi_n(S^n) \to \pi_n(K(\mathbb{Z}, n))$ reads $0 \to \pi_n(F) \to \mathbb{Z} \xrightarrow{\cong} \mathbb{Z}$, forcing $\pi_n(F) = 0$; and the segment $\pi_{n+2}(K(\mathbb{Z}, n)) \to \pi_{n+1}(F) \to \pi_{n+1}(S^n) \to \pi_{n+1}(K(\mathbb{Z}, n))$ reads $0 \to \pi_{n+1}(F) \to \pi_{n+1}(S^n) \to 0$, an isomorphism, using $\pi_{n+1}(K(\mathbb{Z}, n)) = \pi_{n+2}(K(\mathbb{Z}, n)) = 0$ since $K(\mathbb{Z}, n)$ has homotopy concentrated in degree n.

Since F is n-connected, the Hurewicz theorem (theorem 5.4.9) gives

$$\pi_{n+1}(F) \cong H_{n+1}(F; \mathbb{Z})$$

so it remains to compute $H_{n+1}(F)$. Run the *homology* Serre spectral sequence (theorem 6.2.4) of the fibration $F \to S^n \xrightarrow{f} K(\mathbb{Z}, n)$, with base $B = K(\mathbb{Z}, n)$ and total space S^n, to keep all torsion in its native degree. The E^2 page is $E^2_{p,q} = H_p(K(\mathbb{Z}, n); H_q(F))$, converging to $H_{p+q}(S^n)$, which is $\mathbb{Z}$ in degrees 0 and n and 0 otherwise. The base homology, by proposition 6.3.6, is $H_0(B) = \mathbb{Z}$, $H_p(B) = 0$ for $0 < p < n$, $H_n(B) = \mathbb{Z}$, $H_{n+1}(B) = 0$, and $H_{n+2}(B) = \mathbb{Z}/2$; the fiber, being n-connected, has $H_0(F) = \mathbb{Z}$, $H_q(F) = 0$ for $0 < q \le n$, and $H_{n+1}(F)$ the first unknown group.

The cancellation. The class in $H_n(B) = \mathbb{Z}$ survives to E^∞ and maps isomorphically to $H_n(S^n) = \mathbb{Z}$ under the edge homomorphism (theorem 6.2.6), since f induces an isomorphism on H_n in this range; so $E^\infty_{n,0} = \mathbb{Z}$ accounts for $H_n(S^n)$. Now read off total degree $n+1$. The only populated E^2 entries in total degree $n+1$ are $E^2_{n+1,0} = H_{n+1}(B) = 0$ and $E^2_{0,n+1} = H_{n+1}(F)$; convergence to $H_{n+1}(S^n) = 0$ forces $E^\infty_{0,n+1} = 0$. The fiber class in $E^2_{0,n+1} = H_{n+1}(F)$ sits in the leftmost (fiber) column and receives the transgression differential

$$d^{n+2} \colon E^{n+2}_{n+2,0} = H_{n+2}(B) = \mathbb{Z}/2 \longrightarrow E^{n+2}_{0,n+1} = H_{n+1}(F)$$

the only differential touching either entry (all d^r for $r \le n+1$ vanish on these because the intervening rows and columns are empty). On the page, the two live spots sit at opposite corners of an otherwise empty region, joined by the lone transgression running up-left from the base row to the fiber column:

$$E^2 : \quad \begin{tikzcd}
q{=}n{+}1 & H_{n+1}(F) & & & & \\
\vdots & 0 & & & & \\
q{=}0 & \mathbb{Z} & \cdots & \mathbb{Z} & 0 & \mathbb{Z}/2 \arrow[uullll, "d^{n+2}"'] \\
& p{=}0 & \cdots & p{=}n & p{=}n{+}1 & p{=}n{+}2
\end{tikzcd}$$

Convergence forces both ends to die: $E^\infty_{0,n+1} = 0$ requires d^{n+2} surjective, and $E^\infty_{n+2,0} = 0$ (the abutment $H_{n+2}(S^n) = 0$) requires d^{n+2} injective. Hence

$$d^{n+2} \colon H_{n+2}(K(\mathbb{Z}, n); \mathbb{Z}) = \mathbb{Z}/2 \xrightarrow{\cong} H_{n+1}(F; \mathbb{Z})$$

is an isomorphism, giving $H_{n+1}(F; \mathbb{Z}) \cong \mathbb{Z}/2$ directly, with no universal-coefficient shift to track. Therefore

$$\pi_{n+1}(S^n) \cong \pi_{n+1}(F) \cong H_{n+1}(F; \mathbb{Z}) \cong \mathbb{Z}/2 \qquad (n \ge 3)$$

The group $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$ is independent of n for $n \geq 3$: this is the first *stable* homotopy group of spheres, the first stable stem $\pi_1^s \cong \mathbb{Z}/2$, generated by the suspensions of the Hopf map $\eta\colon S^3 \to S^2$. The transition from the unstable $\pi_3(S^2) \cong \mathbb{Z}$ to the stable $\pi_4(S^3) \cong \pi_5(S^4) \cong \cdots \cong \mathbb{Z}/2$ is exactly the passage through one suspension, under which the integer Hopf invariant collapses to its parity.

The two parts of the example exhibit the two regimes of the homotopy of spheres. Below and on the diagonal the fibration sequence alone suffices, as in $\pi_3(S^2) \cong \mathbb{Z}$, where the contractible-fiber mechanism of the Hopf fibration reads the answer off directly. One step above the diagonal the fibration sequence is exhausted and the spectral sequence takes over: the homology of $K(\mathbb{Z}, n)$, computed by collapsing the path-loop fibration, carries a $\mathbb{Z}/2$ that the Hurewicz theorem transports into $\pi_{n+1}(S^n)$. The stabilization, that the answer ceases to depend on n once $n \geq 3$, is the first visible consequence of the Freudenthal suspension theorem and the entire edifice of stable homotopy theory built on it, the subject taken up with spectra in chapter 8.

Exercise 6.3.1

1. Carry out the spectral-sequence computation of example 6.3.1 for $n = 2$ explicitly: identify every nonzero $E_2^{p,q}$ on the page for the fibration $\Omega S^2 \to PS^2 \to S^2$, draw the single family of differentials d_2, and conclude that $H_q(\Omega S^2; \mathbb{Z}) = \mathbb{Z}$ for every $q \geq 0$. Compare with the homology of $\mathbb{C}\mathrm{P}^\infty$ and explain the relationship.

2. Use the Gysin sequence (theorem 6.3.2) for the sphere bundle $S^k \to S^{2k+1} \to \mathbb{C}\mathrm{P}^k$ when $k = 1$ (the Hopf bundle $S^1 \to S^3 \to \mathbb{C}\mathrm{P}^1 = S^2$) to recompute $H^*(S^3; \mathbb{Z})$, and then state what the Gysin sequence gives for the bundle $S^1 \to S^5 \to \mathbb{C}\mathrm{P}^2$, computing $H^*(S^5; \mathbb{Z})$ from $H^*(\mathbb{C}\mathrm{P}^2; \mathbb{Z}) = \mathbb{Z}[\alpha]/(\alpha^3)$.

3. Let $F \to E \to S^n$ be a fibration over a sphere with $n \geq 2$ and fiber F having $H^*(F; \mathbb{Z})$ free abelian. Show using the Wang sequence (theorem 6.3.4) that if the derivation θ is identically zero then $H^*(E; \mathbb{Z}) \cong H^*(F; \mathbb{Z}) \otimes H^*(S^n; \mathbb{Z})$ as graded groups. Give the example $E = F \times S^n$ to confirm $\theta = 0$ for a product.

4. In the computation of $H_{n+2}(K(\mathbb{Z}, n))$ in proposition 6.3.6, the factor of 2 arose from the derivation rule applied to a^2. Verify the rule $d_n(a^2) = 2a\, d_n(a)$ directly from the Leibniz formula $d_n(xy) = d_n(x)y + (-1)^{|x|} x\, d_n(y)$ for $x = y = a$ with $|a| = n - 1$, and explain why the sign works out to give $+2$ rather than 0 precisely when $n - 1$ is even (equivalently n odd) and what happens for n even.

5. Using example 6.3.7, identify $\pi_4(S^3)$, $\pi_5(S^4)$, and $\pi_6(S^5)$, and state which of these are stable (independent of further suspension) and which is the first stable one. Explain in one or two sentences why $\pi_3(S^2) \cong \mathbb{Z}$ does not contradict the stable value $\mathbb{Z}/2$.

6. Combine the homology computation $H_{n+1}(\Omega S^n; \mathbb{Z})$ from example 6.3.1 with the Hurewicz theorem to compute $\pi_{n-1}(\Omega S^n)$ for $n \geq 2$, and reconcile the answer with the dimension-shift isomorphism $\pi_{n-1}(\Omega S^n) \cong \pi_n(S^n)$ from example 5.2.12.

6.4 Multiplicative Structure and Transgression

The Serre spectral sequence of section 6.2 computes the additive structure of $H^*(E)$ from $H^*(B)$ and $H^*(F)$, but a cohomology group is only half of what cohomology offers. The cup product makes

$H^*(E;R)$ into a graded-commutative ring (definition 4.2.5), and the spectral sequence is blind to that ring structure unless the cup product is woven into the machinery from the start. This section installs the multiplicative refinement: every page $E_r^{\bullet,\bullet}$ carries a bigraded-algebra structure, each differential d_r is a derivation with respect to it, and the product on the second page is the cup product of base and fiber transported through the identification $E_2^{p,q} \cong H^p(B; H^q(F))$. The product on the abutment $H^*(E)$ is recovered from the product on E_∞ up to the ambiguity inherent in any filtration, so that the spectral sequence becomes a tool for computing cohomology *rings* and not just groups.

The second theme is a single distinguished differential. A class in the fiber that survives to a torsion-free situation cannot bound on any earlier page, so the first differential that can act on it is forced; the resulting map, the *transgression*, sends a fiber class of degree n to a base class of degree $n+1$, and the transgression theorem identifies it with the connecting behavior already present in the long exact sequences of low-dimensional fibrations. Together, multiplicativity and transgression turn the spectral sequence into a calculating engine: knowing the ring structure of two of $H^*(F)$, $H^*(B)$, $H^*(E)$ frequently *forces* all the differentials, because a derivation is determined by its values on algebra generators and graded-commutativity constrains those values severely. The worked computations at the end re-derive $H^*(\mathbb{CP}^\infty;\mathbb{Z}) \cong \mathbb{Z}[x]$ and compute the ring $H^*(\Omega S^{n+1};\mathbb{Z})$ entirely from this principle.

Throughout, R is a commutative coefficient ring (taken to be $\mathbb{Z}$ or a field unless stated otherwise), and $p\colon E \to B$ is a Serre fibration whose base B is a path-connected CW complex, with fiber F and $\pi_1(B)$ acting trivially on $H^*(F;R)$, so that the cohomology Serre spectral sequence of theorem 6.2.5 has second page

$$E_2^{p,q} \;\cong\; H^p\big(B;\, H^q(F;R)\big)$$

converging to $H^{p+q}(E;R)$, with the abutment carrying the decreasing filtration

$$H^n(E;R) = F^0H^n \supseteq F^1H^n \supseteq \cdots \supseteq F^nH^n \supseteq F^{n+1}H^n = 0$$

whose successive quotients are $F^pH^n/F^{p+1}H^n \cong E_\infty^{p,n-p}$. Recall from section 6.1 that this filtration is the one induced on the cohomology of the total space by the skeletal filtration of the base.

Before stating multiplicativity, the precise sense in which the abutment filtration is multiplicative must be isolated, since it is the compatibility that ties the spectral-sequence product to the cup product on $H^*(E)$.

Definition 6.4.1: Multiplicative Filtration

Let $A = \bigoplus_n A^n$ be a graded R-algebra equipped with a decreasing filtration $A^n = F^0A^n \supseteq F^1A^n \supseteq \cdots$ by R-submodules. The filtration is *multiplicative* if the cup product respects filtration degree:

$$F^pA^m \smile F^{p'}A^{m'} \;\subseteq\; F^{p+p'}A^{m+m'}$$

for all p, p', m, m'. The associated graded object $\operatorname{gr} A = \bigoplus_{p,q} F^pA^{p+q}/F^{p+1}A^{p+q}$ is then a bigraded R-algebra, with product induced by the cup product: the class of $a \in F^pA^{p+q}$ times the class of $b \in F^{p'}A^{p'+q'}$ is the class of $a \smile b \in F^{p+p'}A^{p+q+p'+q'}$ in the quotient $F^{p+p'}/F^{p+p'+1}$.

A multiplicative filtration is the algebraic form of a geometric stratification that is compatible with products. The condition $F^p \smile F^{p'} \subseteq F^{p+p'}$ says that “deeper” classes multiply to “even deeper” ones, so the leading-order behavior of a product depends only on the leading-order behavior of the factors. This is exactly what makes the associated graded a ring: in $\operatorname{gr} A$ one multiplies leading terms and discards everything lower, and the inclusion guarantees the discarded part really is lower. The spectral-sequence product will live on $\operatorname{gr} H^*(E)$, and the gap between $\operatorname{gr} H^*(E)$ and $H^*(E)$ itself is the extension problem, both additive and multiplicative, that the spectral sequence cannot resolve on its own.

Example 6.4.2: The Two-Step Filtration on $H^*(\mathbb{C}\mathrm{P}^1)$

Take $A = H^*(\mathbb{C}\mathrm{P}^1; \mathbb{Z}) = H^*(S^2; \mathbb{Z})$, which is $\mathbb{Z}$ in degrees 0 and 2 and zero elsewhere, with $H^0 = \mathbb{Z}\langle 1 \rangle$ and $H^2 = \mathbb{Z}\langle x \rangle$ where $x^2 = 0$ for dimension reasons. View $\mathbb{C}\mathrm{P}^1 = S^2$ as the base of the trivial fibration $\{*\} \to S^2 \to S^2$, whose abutment filtration on $H^2(S^2)$ is $F^0H^2 = F^1H^2 = F^2H^2 = \mathbb{Z}\langle x \rangle$ and $F^3H^2 = 0$, while on H^0 it is $F^0H^0 = \mathbb{Z}\langle 1 \rangle \supseteq F^1H^0 = 0$. The product $1 \smile 1 = 1$ lands in F^0, and $1 \smile x = x$ lands in $F^2H^2 = F^{0+2}H^2$, consistent with $F^0 \smile F^2 \subseteq F^2$. There is no pair of positive-filtration classes to multiply, so the inclusion holds vacuously in the remaining cases, and the filtration is multiplicative. The associated graded recovers A itself, since each filtration jump is by a single $\mathbb{Z}$, exhibiting the degenerate case where $\operatorname{gr} A \cong A$ as rings.

The product on the spectral sequence is built so that, page by page, it is compatible with the differential and descends to the next page, terminating in a product on E_∞ that agrees with the product on $\operatorname{gr} H^*(E)$. The construction of the product at the cochain level (the cup product on the filtered cochain complex of E pulled back through the skeletal filtration of B) is carried out in the homological-algebra companion; the present section takes the existence of the cochain-level multiplicative structure as input and develops its consequences in full.

Theorem 6.4.3: Multiplicativity of the Cohomology Serre Spectral Sequence

Let $p\colon E \to B$ be a Serre fibration whose base is a path-connected CW complex, with fiber F and trivial action of $\pi_1(B)$ on $H^*(F;R)$, with R a commutative ring. The cohomology Serre spectral sequence $\{E_r^{\bullet,\bullet}, d_r\}$ of theorem 6.2.5 admits products

$$E_r^{p,q} \otimes E_r^{p',q'} \longrightarrow E_r^{p+p',\,q+q'}, \qquad a \otimes b \longmapsto a \cdot b$$

one on each page, with the following properties.

1. *(Bigraded algebra)* On each page the product is R-bilinear, associative, and graded-commutative for the total degree: if $a \in E_r^{p,q}$ and $b \in E_r^{p',q'}$ then

$$a \cdot b = (-1)^{(p+q)(p'+q')}\, b \cdot a$$

and the unit is $1 \in E_r^{0,0}$.

2. *(Leibniz rule)* Each differential d_r is a derivation:

$$d_r(a \cdot b) = d_r(a) \cdot b + (-1)^{p+q}\, a \cdot d_r(b) \qquad \text{for } a \in E_r^{p,q}$$

Consequently the product on $E_{r+1} = H(E_r, d_r)$ induced on cohomology classes is well defined, and the isomorphism $E_{r+1} \cong H(E_r, d_r)$ is one of bigraded algebras.

3. *(Product on E_2)* Under the identification $E_2^{p,q} \cong H^p(B; H^q(F;R))$, the page-two product is the composite

$$H^p(B;H^q(F)) \otimes H^{p'}(B;H^{q'}(F)) \xrightarrow{\smile} H^{p+p'}\big(B;\, H^q(F) \otimes H^{q'}(F)\big) \xrightarrow{\smile_F} H^{p+p'}\big(B;\, H^{q+q'}(F)\big)$$

of the cup product on B with coefficients paired by the cup product $\smile_F$ of the fiber.

4. *(Convergence to the ring)* The abutment filtration on $H^*(E;R)$ is multiplicative in the sense of definition 6.4.1, and the induced product on $\operatorname{gr} H^*(E;R)$ agrees with the product on E_∞ under the isomorphism $E_\infty^{p,q} \cong F^p H^{p+q}/F^{p+1}H^{p+q}$.

The four clauses split the labor between formal algebra and the input from the cochain construction. Clauses (1) and (2) are purely algebraic consequences of having a product on E_1 that is a derivation for d_1: graded-commutativity and the Leibniz rule, once true on one page, propagate to every later page by passing to homology, because the homology of a differential graded algebra is again a graded algebra. Clause (3) identifies the input product concretely, and clause (4) records that the whole apparatus converges to the genuine cup product on $H^*(E)$, up to the filtration. The proof below establishes the propagation (1)-(2) in full from the page-one product, then identifies (3) and (4) by tracing the cochain-level cup product through the constructions of sections 6.1 and 6.2.

***Proof*:**
The construction of section 6.2 realizes the Serre spectral sequence as the spectral sequence of the filtered cochain complex $C^*(E;R)$ filtered by $F^pC^*(E;R) = \ker\big(C^*(E;R) \to C^*(p^{-1}(B^{p-1});R)\big)$, the cochains vanishing on the preimage of the $(p-1)$-skeleton of B (theorem 6.1.10). The cup product on $C^*(E;R)$ is the singular cup product of definition 4.2.1. The proof has two parts: first, that this cup product is compatible with the filtration and with δ, which seeds a

multiplicative structure on E_1 that is a derivation for d_1; second, that derivation-compatibility and graded-commutativity propagate through every page and survive to the abutment.

Step 1: The cup product respects the filtration. The cochain filtration is multiplicative on $C^*(E;R)$:

$$F^pC^m(E) \smile F^{p'}C^{m'}(E) \subseteq F^{p+p'}C^{m+m'}(E)$$

The verification compares the singular cup product on $C^*(E;R)$ with the skeletal filtration pulled back from B, and is the one technical input of the construction that this section does not reprove: it is carried out in detail for the spectral sequence of a filtered differential graded algebra in [10], and specialized to the Serre filtration $F^pC^*(E) = \ker\big(C^*(E) \to C^*(p^{-1}B^{p-1})\big)$ in section 6.2. The conceptual reason is that the cup product on the relative cochain complex of the skeletal filtration is induced by the diagonal $E \to E \times E$, which carries $p^{-1}(B^{p+p'-1})$ into $\big(p^{-1}B^{p-1} \times E\big) \cup \big(E \times p^{-1}B^{p'-1}\big)$ by the cellular approximation of the diagonal of B; a product of cochains vanishing on the two factors therefore vanishes on $p^{-1}(B^{p+p'-1})$, placing $u \smile v$ in $F^{p+p'}$. This inclusion is the cochain-level form of definition 6.4.1, and it is all that the rest of the proof uses.

Step 2: The induced product on E_0 and E_1. A multiplicative filtration on a differential graded algebra (C^*, δ) induces a product on the associated graded $E_0^{p,q} = F^pC^{p+q}/F^{p+1}C^{p+q}$, since by Step 1 the cup product carries $F^p \otimes F^{p'}$ into $F^{p+p'}$ and the lower terms $F^{p+1}, F^{p'+1}$ into $F^{p+p'+1}$, so the product descends to the quotients. The differential d_0 on E_0 is induced by δ, and the Leibniz rule $\delta(u \smile v) = \delta u \smile v + (-1)^{|u|}u \smile \delta v$ of lemma 4.2.2 descends, making d_0 a derivation. Passing to homology, $E_1 = H(E_0, d_0)$ inherits a product (the homology of a DGA is a graded algebra), and d_1 is the derivation induced on this homology. Identifying $E_1^{p,q} \cong C^p_{\mathrm{cell}}(B; H^q(F))$ as in section 6.2, this product is the cellular cup product of B with fiberwise-cup-product coefficients.

Step 3: Propagation to all pages. The key algebraic fact is: if (E_r, d_r) is a bigraded algebra whose differential d_r is a derivation, then $E_{r+1} = H(E_r, d_r)$ is again a bigraded algebra and the induced d_{r+1} is a derivation. The product on E_{r+1} is defined on homology classes by $[a] \cdot [b] = [a \cdot b]$; this is well defined because d_r is a derivation: if a is a cycle ($d_r a = 0$) then $d_r(a \cdot b) = (-1)^{|a|}a \cdot d_r(b)$, so multiplying a cycle by a cycle gives a cycle, and multiplying a cycle by a boundary $b = d_r c$ gives $a \cdot d_r c = \pm d_r(a \cdot c)$, a boundary. Hence the product descends to homology. Associativity, bilinearity, the unit, and graded-commutativity all pass to homology classes verbatim. That d_{r+1} is a derivation is the statement that the differential induced on the homology of a DGA by the next page of a spectral sequence again satisfies Leibniz; this is the standard algebraic lemma on multiplicative spectral sequences, proved in full in [10]. By induction from Step 2, every page E_r ($r \geq 1$) is a bigraded algebra with d_r a derivation, establishing (1) and (2). Graded-commutativity holds on E_1 because it holds at the cochain level by theorem 4.2.4 and descends to homology, hence holds on every page.

Step 4: The page-two product. By Step 2 the product on $E_1 \cong C^*_{\mathrm{cell}}(B; H^*(F))$ is the cellular cup product with coefficient pairing the fiber cup product. Passing to $E_2 = H(E_1, d_1)$ identifies d_1 with the cellular coboundary of B (with the local coefficient system $H^*(F)$, untwisted here by the triviality hypothesis), so $E_2^{p,q} = H^p(B; H^q(F))$ and the induced product is the cup product on $H^*(B)$ paired with the fiber cup product on coefficients. This is exactly the composite displayed in clause (3): cup on the base, then cup on the fiber coefficients. Thus (3) holds.

Step 5: Convergence to the ring structure. By Step 1 the abutment filtration $F^pH^n(E;R) = \mathrm{im}\big(H^n(E, p^{-1}B^{p-1}) \to H^n(E)\big)$ inherits multiplicativity: a class in F^p is represented by a cocycle in $F^pC^*(E)$, a class in $F^{p'}$ by one in $F^{p'}C^*(E)$, and their cup product lies in $F^{p+p'}C^*(E)$ by Step 1, hence represents a class in $F^{p+p'}H^*(E)$. So the abutment filtration is multiplicative in the sense of definition 6.4.1. The product on $\operatorname{gr} H^*(E)$ is induced by the cup product on representatives modulo higher filtration, which is precisely the product on $E_\infty = \operatorname{gr} H^*(E)$ obtained as the limit of the page products (the page products on E_r stabilize to the E_∞ product because for fixed (p,q) the entries $E_r^{p,q}$ stabilize for large r). Therefore the E_∞ product agrees with the $\operatorname{gr} H^*(E)$ product under $E_\infty^{p,q} \cong F^pH^{p+q}/F^{p+1}$, giving (4), and completing the proof.

The theorem is the one used throughout the section. Its practical force is concentrated in clause (2): because d_r is a derivation, it is determined by its values on a set of algebra generators of E_r. If E_2 is generated as an algebra by a few classes, then knowing d_r on those classes determines d_r everywhere, and conversely a single nonzero differential propagates multiplicatively through all products. Clause (3) is what makes E_2 computable: it is the cohomology of the base with coefficients in the cohomology ring of the fiber, with the obvious product. Clause (4) is the bridge back to topology, and it carries the standard caveat of all spectral sequences: E_∞ computes only the associated graded ring, so reconstructing $H^*(E)$ requires solving extension problems, which in the multiplicative setting means determining products of classes from different filtration levels that the spectral sequence sees only separately.

6.4.4 Warning: E_∞ Gives Only the Associated Graded Ring The isomorphism $E_\infty^{p,q} \cong F^pH^{p+q}/F^{p+1}$ is one of *bigraded* algebras, not of the (singly) graded ring $H^*(E)$. Two phenomena can obstruct reading off $H^*(E)$ from E_∞. First, additive extensions: a column of E_∞ with entries $\mathbb{Z}$ and $\mathbb{Z}$ leaves open whether $H^*(E)$ has a $\mathbb{Z} \oplus \mathbb{Z}$ or a $\mathbb{Z}$ (a nontrivial extension cannot occur for free abelian entries, but can for finite ones). Second, multiplicative extensions: even when the additive structure is settled, a product $a \smile b$ may equal a class of strictly higher filtration than its leading term in E_∞, so E_∞ records $a \cdot b$ only modulo higher filtration. Both ambiguities vanish when the spectral sequence collapses with free entries concentrated so that no two basis classes share a total degree, which is what makes the computations below clean.

The most consequential single differential is the one acting on a fiber class for the first time. A class $x \in H^n(F)$ lives, via the edge identification $E_2^{0,n} \cong H^0(B; H^n(F)) \cong H^n(F)$, in the leftmost column of the spectral sequence. The differentials $d_2, \ldots, d_n$ landing on it come from columns to its left, all of which are zero (there is no $E_r^{-r,\,n+r-1}$). The first differential that can carry x out of the fiber column is $d_{n+1} \colon E_{n+1}^{0,n} \to E_{n+1}^{n+1,0}$, landing in the bottom row, which is $H^{n+1}(B)$ via the edge identification. When x survives to the page E_{n+1} (equivalently, when no earlier differential hits or kills it), this $d_{n+1}(x)$ is defined and is the transgression of x.

Definition 6.4.5: Transgression

Let $p\colon E \to B$ be a fibration as above with fiber F. A class $x \in H^n(F;R) \cong E_2^{0,n}$ is called *transgressive* if it survives to the page E_{n+1}, that is, if the differentials $d_2(x), \dots, d_n(x)$ all vanish (equivalently x persists to $E_{n+1}^{0,n}$). For a transgressive class the *transgression* is the differential

$$\tau = d_{n+1}\colon E_{n+1}^{0,n} \longrightarrow E_{n+1}^{n+1,0}$$

evaluated on x, giving $\tau(x) = d_{n+1}(x) \in E_{n+1}^{n+1,0}$. Since $E_{n+1}^{n+1,0}$ is a subquotient of $E_2^{n+1,0} \cong H^{n+1}(B;R)$, the transgression assigns to x a coset $\tau(x)$ in a quotient of $H^{n+1}(B;R)$; one says $\tau(x) \in H^{n+1}(B;R)$ when this coset has a canonical representative, abusing notation by the standard convention.

The transgression is the spectral-sequence avatar of the connecting map. A class on the fiber that has not yet been forced to bound is examined by the longest differential that can reach the base row, and that differential measures the obstruction to extending the fiber class to the total space. In the simplest fibrations, where E is highly connected, every fiber class is transgressive and the transgression is an isomorphism onto the relevant part of $H^{*+1}(B)$, recovering base cohomology from fiber cohomology shifted by one. The bottom row $E_*^{*,0} \cong H^*(B)$ and the left column $E_*^{0,*} \cong H^*(F)$ are called the *edge* of the spectral sequence, and the transgression is the differential connecting the two edges. The qualification “transgressive” is essential: only classes that survive long enough to reach $E_{n+1}^{0,n}$ have a transgression, and the differential is well defined only up to the indeterminacy created by earlier differentials, which is why $\tau(x)$ naturally lives in a quotient.

Example 6.4.6: Transgression in the Path-Loop Fibration over S^{n+1}

Consider the path-loop fibration $\Omega S^{n+1} \to PS^{n+1} \to S^{n+1}$ of example 5.2.12, with $n \geq 1$. The base S^{n+1} has $H^*(S^{n+1};\mathbb{Z}) = \mathbb{Z}\langle 1, s\rangle$ with $|s| = n+1$, so $E_2^{p,q} = H^p(S^{n+1}; H^q(\Omega S^{n+1}))$ is concentrated in the two columns $p = 0$ and $p = n+1$. The total space PS^{n+1} is contractible, so $H^*(PS^{n+1}) = \mathbb{Z}$ in degree 0 and zero otherwise, forcing $E_\infty^{p,q} = 0$ except at $(0,0)$. Let $a \in H^n(\Omega S^{n+1};\mathbb{Z})$ be a generator of the bottom nonzero fiber cohomology (which is $\mathbb{Z}$, since ΩS^{n+1} is $(n-1)$-connected with $\pi_n(\Omega S^{n+1}) \cong \pi_{n+1}(S^{n+1}) \cong \mathbb{Z}$ by example 5.2.12). The class $a \in E_2^{0,n}$ is transgressive: the only nonzero columns are $p = 0$ and $p = n+1$, so all of $d_2, \dots, d_n$ vanish on a for column reasons, and a survives to $E_{n+1}^{0,n}$. Because $E_\infty = 0$ off the origin, a cannot survive to E_∞, so $d_{n+1}(a)$ must be nonzero and must be an isomorphism $E_{n+1}^{0,n} \xrightarrow{\cong} E_{n+1}^{n+1,0} = H^{n+1}(S^{n+1}) = \mathbb{Z}\langle s\rangle$. Thus

$$\tau(a) = d_{n+1}(a) = s$$

up to sign: the bottom fiber generator transgresses to the base generator. This single differential, propagated multiplicatively, will compute the entire ring $H^*(\Omega S^{n+1})$ in example 6.4.9.

The example exhibits the transgression as the engine that kills the fiber cohomology against the base cohomology when the total space is contractible. The transgression theorem makes the link to the connecting maps precise and explains why τ deserves to be called a transgression. The statement requires the relative cohomology of the pair (E, F) and the connecting homomorphism $\delta\colon H^n(F) \to H^{n+1}(E,F)$ of its long exact sequence, together with the projection $p^*\colon H^{n+1}(B, b_0) \to H^{n+1}(E,F)$; the transgression is the relation $\tau = (p^*)^{-1} \circ \delta$ on the classes for which this composite is defined.

Theorem 6.4.7: The Transgression Theorem

Let $p\colon E \to B$ be a Serre fibration as above, with $b_0 \in B$ a basepoint, $F = p^{-1}(b_0)$, and $i\colon F \hookrightarrow E$ the inclusion. Write $\delta\colon H^n(F;R) \to H^{n+1}(E,F;R)$ for the connecting homomorphism of the long exact cohomology sequence of the pair (E,F), and $p^*\colon H^{n+1}(B,b_0;R) \to H^{n+1}(E,F;R)$ for the map induced by p. A class $x \in H^n(F;R)$ is transgressive (in the sense of definition 6.4.5) if and only if $\delta(x)$ lies in the image of p^*, and in that case the transgression $\tau(x) \in E_{n+1}^{n+1,0}$ is represented by the class $\bar{x} \in H^{n+1}(B;R)$ with

$$p^*(\bar{x}) = \delta(x)$$

modulo the indeterminacy of the transgression. Equivalently, transgression is the "zig-zag"

$$\tau\colon H^n(F;R) \xrightarrow{\delta} H^{n+1}(E,F;R) \xleftarrow{p^*} H^{n+1}(B,b_0;R)$$

defined on the classes x with $\delta(x) \in \operatorname{im} p^*$ and valued in the quotient $E_{n+1}^{n+1,0}$ of $H^{n+1}(B;R)$ by the indeterminacy $\ker p^*$ together with the images of the earlier differentials $d_2, \ldots, d_n$ landing in the bottom row.

***Proof*:**

The argument identifies the spectral-sequence differential $d_{n+1}\colon E_{n+1}^{0,n} \to E_{n+1}^{n+1,0}$ with the zig-zag $(p^*)^{-1}\delta$ by tracing representatives through the filtered cochain complex. Recall (section 6.1) that the differential d_{n+1} on a spectral sequence of a filtered complex is computed by the following recipe: a class in $E_{n+1}^{0,n}$ is represented by a cochain $u \in F^0C^n(E)$ with $\delta u \in F^{n+1}C^{n+1}(E)$ (the cochain is a cocycle modulo filtration $n+1$), and $d_{n+1}[u]$ is the class of δu in $E_{n+1}^{n+1,0} = Z_{n+1}^{n+1,0}/(\cdots)$.

Step 1: Represent a transgressive fiber class. Let $x \in H^n(F;R)$ be transgressive. The edge identification $E_2^{0,n} \cong H^n(F)$ realizes x by a cochain $u \in C^n(E;R)$ whose restriction $i^\# u \in C^n(F;R)$ is a cocycle representing x, with $u \in F^0C^n(E)$. That x survives to $E_{n+1}^{0,n}$ means u can be chosen so that $\delta u \in F^{n+1}C^{n+1}(E)$, that is, δu vanishes on simplices of E lying over the n-skeleton B^n; this is exactly the condition that $d_2 x = \cdots = d_n x = 0$, since each earlier differential is the obstruction to pushing δu into the next deeper filtration. Survival to E_{n+1} is therefore equivalent to the solvability of these successive corrections, which is the spectral-sequence translation of $\delta(x) \in \operatorname{im} p^*$, as Step 3 makes precise.

Step 2: The coboundary lands in the base row. Since $\delta u \in F^{n+1}C^{n+1}(E)$, the cochain δu vanishes on all simplices over B^n, so it is determined by its values on simplices over the $(n+1)$-cells of B, that is, it defines a cellular cochain on B with constant ($H^0(F) = R$) coefficients. Concretely, δu represents a class in $E_{n+1}^{n+1,0}$, a subquotient of $H^{n+1}(B;R)$ via the bottom-edge identification $E_2^{n+1,0} \cong H^{n+1}(B;H^0(F)) = H^{n+1}(B;R)$. Call the resulting base class $\bar{x}$; by definition of d_{n+1}, $\tau(x) = d_{n+1}[u] = [\delta u] = \bar{x}$.

Step 3: Compare with the connecting map. The connecting homomorphism $\delta\colon H^n(F) \to H^{n+1}(E,F)$ of the pair (E,F) is computed exactly as here: a class $x \in H^n(F)$ is represented by a cocycle $w \in C^n(F)$; extend w to a cochain $u \in C^n(E)$ with $i^\# u = w$ (possible because the restriction $i^\#\colon C^n(E) \to C^n(F)$ is surjective, the singular chains of F forming a direct summand of those of E); then $\delta u \in C^{n+1}(E)$ vanishes on F (because $i^\#(\delta u) = \delta(i^\# u) = 0$ as $i^\# u$ is a cocycle on F), hence defines a relative cocycle representing $\delta(x) \in H^{n+1}(E, F)$. This is the same cochain δu as in Step 2. On the other hand, $p^*\colon H^{n+1}(B, b_0) \to H^{n+1}(E, F)$ sends a relative cocycle β on (B, b_0) to $p^\# \beta$ on (E, F). The base class $\bar{x}$ of Step 2, pulled back by $p^\#$, gives precisely the relative cocycle δu (since δu is constant on fibers over each cell of B, it is the pullback of its cellular reduction $\bar{x}$). Therefore

$$p^*(\bar{x}) = [\,p^\# \bar{x}\,] = [\delta u] = \delta(x)$$

in $H^{n+1}(E, F; R)$. This is the asserted relation $p^*(\tau(x)) = \delta(x)$.

Step 4: The equivalence. If x is transgressive, Steps 1-3 produce $\bar{x} = \tau(x)$ with $p^*(\bar{x}) = \delta(x)$, so $\delta(x) \in \operatorname{im} p^*$. Conversely, if $\delta(x) \in \operatorname{im} p^*$, choose $\bar{x} \in H^{n+1}(B, b_0)$ with $p^*(\bar{x}) = \delta(x)$; running the cochain computation backward shows that the cochain u representing x can be corrected within its filtration class so that δu represents $\bar{x}$ in the base row, hence the earlier differentials $d_2 x, \dots, d_n x$ vanish and x is transgressive with $\tau(x) = \bar{x}$ modulo the indeterminacy $\ker p^*$ together with the images of earlier differentials into $E_{n+1}^{n+1,0}$. The two descriptions of τ thus coincide, completing the proof.

The transgression theorem says that the longest differential out of the fiber column is the connecting map of the pair (E, F), read in base coordinates. In the path-loop fibration of example 6.4.6 the pair $(PX, \Omega X)$ has PX contractible, so the long exact sequence of the pair gives $\delta\colon H^n(\Omega X) \xrightarrow{\cong} H^{n+1}(PX, \Omega X)$, and p^* identifies $H^{n+1}(X, x_0)$ with $H^{n+1}(PX, \Omega X)$; the transgression is then the suspension isomorphism in disguise, matching the homotopy-theoretic dimension shift $\pi_n(\Omega X) \cong \pi_{n+1}(X)$. More generally the theorem licenses the computation of base cohomology from fiber cohomology whenever the total space is understood, and it is the precise statement underlying the assertion in section 5.6 that the k-invariant of a Postnikov tower is the transgression of the fundamental class of the fiber.

6.4.8 Note: Transgression and the k-Invariant In section 5.6 the first k-invariant of S^2 was identified as $\pm u^2 \in H^4(\mathbb{CP}^\infty; \mathbb{Z})$, with the assertion that the transgression in the Serre spectral sequence of the fibration $K(\mathbb{Z}, 3) \to X_3 \to \mathbb{CP}^\infty$ sends the fundamental class $\iota_3 \in H^3(K(\mathbb{Z}, 3))$ to $k_3 \in H^4(\mathbb{CP}^\infty)$. Theorem 6.4.7 is the theorem behind that assertion: the fiber $K(\mathbb{Z}, 3)$ has $H^3 = \mathbb{Z}\langle \iota_3 \rangle$ as its bottom class, ι_3 is transgressive because the lower columns vanish, and $\tau(\iota_3) = k_3$ is the connecting map of the pair, which is exactly the obstruction class defining the Postnikov fibration. The transgression is thus the spectral-sequence computation of the k-invariant.

The remaining work is to demonstrate the combined force of multiplicativity and transgression on concrete rings. The strategy in each case is identical: identify a transgressive generator, compute its transgression from a degeneration constraint (typically a contractible or known total space), and then let the Leibniz rule propagate the single differential across all products, with graded-commutativity giving the divided-power or polynomial relations among the propagated classes.

Example 6.4.9: Computing Cohomology Rings by Multiplicativity
Two rings are computed, the second illustrating how graded-commutativity forces divided powers.

1. *The ring $H^*(\mathbb{CP}^\infty; \mathbb{Z}) \cong \mathbb{Z}[x]$, $|x| = 2$, from $S^1 \to S^\infty \to \mathbb{CP}^\infty$.* Use the fibration $S^1 \hookrightarrow S^\infty \to \mathbb{CP}^\infty$ of example 5.5.5, with contractible total space S^∞ (so $H^*(S^\infty) = \mathbb{Z}$ in degree 0, zero above). The fiber is S^1 with $H^*(S^1; \mathbb{Z}) = \mathbb{Z}\langle 1, t \rangle$, $|t| = 1$, $t^2 = 0$. The base $\mathbb{CP}^\infty$ is

simply connected, so π_1 acts trivially and theorem 6.2.5 gives

$$E_2^{p,q} = H^p(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \otimes H^q(S^1;\mathbb{Z})$$

concentrated in rows $q = 0, 1$, since $H^q(S^1) = 0$ for $q \geq 2$. Write $H^*(\mathbb{C}\mathrm{P}^\infty)$ provisionally as an unknown graded ring A^* to be determined; the spectral sequence will compute it. The bottom row is $E_2^{*,0} = A^*$ and the top row is $E_2^{*,1} = A^* \cdot t$ (the bottom row tensored with t). The only possibly nonzero differential is $d_2 \colon E_2^{p,1} \to E_2^{p+2,0}$, since d_2 drops the fiber degree by 1 and raises base degree by 2, and all higher d_r $(r \geq 3)$ vanish for bidegree reasons (only two rows). Convergence to $H^*(S^\infty) = \mathbb{Z}$ in degree 0 forces $E_3 = E_\infty = 0$ except at $(0, 0)$, so d_2 must be an isomorphism $E_2^{p,1} \xrightarrow{\cong} E_2^{p+2,0}$ for every $p \geq 0$ where the target is in positive total degree.

Now apply multiplicativity. The class $t \in E_2^{0,1}$ is transgressive with $d_2(t) = x$ for some $x \in E_2^{2,0} = A^2$; since d_2 is an isomorphism $E_2^{0,1} \to E_2^{2,0}$ and $E_2^{0,1} = \mathbb{Z}\langle t\rangle$, the image x generates $A^2 \cong \mathbb{Z}$. The Leibniz rule (theorem 6.4.3(2)) computes d_2 on the whole top row by induction on k. Suppose $A^{2k} = \mathbb{Z}\langle x^k\rangle$ is infinite cyclic on the power x^k and $A^{2k-1} = 0$ (the base case $k = 1$ being $A^2 = \mathbb{Z}\langle x\rangle$, $A^1 = H^1(\mathbb{C}\mathrm{P}^\infty) = 0$ since $\mathbb{C}\mathrm{P}^\infty$ is simply connected). Then $E_2^{2k,1} = A^{2k} \cdot t = \mathbb{Z}\langle x^k t\rangle$, and graded-commutativity gives

$$d_2(x^k\, t) = d_2(x^k) \cdot t + (-1)^{2k} x^k \cdot d_2(t) = 0 + x^k \cdot x = x^{k+1}$$

because $d_2(x^k) = 0$ (the bottom row $q = 0$ consists of permanent cycles, no differential leaving it landing in a nonzero entry). Since $d_2 \colon E_2^{2k,1} \to E_2^{2k+2,0} = A^{2k+2}$ must be an isomorphism (convergence to a point in this total degree), the target A^{2k+2} is infinite cyclic generated by the image $x^{k+1} = d_2(x^k t)$, advancing the first part of the induction hypothesis. For the odd group, A^{2k+1} sits in the bottom row and can only be killed by d_2 from $E_2^{2k-1,1} = A^{2k-1} \cdot t = 0$; receiving no differential, it must vanish for E_∞ to be zero off the origin, so $A^{2k+1} = 0$, advancing the second part. The induction gives $A^{2k} \cong \mathbb{Z}$ generated by x^k and $A^{2k+1} = 0$ for all k. Hence $A^* = \mathbb{Z}[x]$ with $|x| = 2$:

$$H^*(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}[x], \qquad |x| = 2$$

recovering the polynomial ring computed by cup products in chapter 4, now from the fibration alone. The ring structure is forced: multiplicativity requires $x^k = d_2(x^{k-1}t)$ generate H^{2k}, so the powers of x are exactly the generators, with no room for any relation.

2. *The ring $H^*(\Omega S^{n+1};\mathbb{Z})$ as a divided-power algebra.* Use the path-loop fibration $\Omega S^{n+1} \to PS^{n+1} \to S^{n+1}$ of example 6.4.6, $n \geq 1$, with contractible total space. The base has $H^*(S^{n+1}) = \mathbb{Z}\langle 1, s\rangle$, $|s| = n + 1$, so $E_2^{p,q} = H^p(S^{n+1}) \otimes H^q(\Omega S^{n+1})$ is concentrated in columns $p = 0$ and $p = n + 1$. Let $a_1 \in H^n(\Omega S^{n+1})$ be the transgressive generator with $\tau(a_1) = d_{n+1}(a_1) = s$ from example 6.4.6. The fiber ring $H^*(\Omega S^{n+1})$ is unknown; the spectral sequence computes it from the constraint $E_\infty = 0$ off the origin.

   First determine the additive structure. The two-column spectral sequence ($p \in \{0, n+1\}$) has differentials only $d_{n+1} \colon E_{n+1}^{0,q} \to E_{n+1}^{n+1,q-n}$, and contractibility forces $E_\infty = 0$ off the origin. A class in $E_2^{0,q}$ with $0 < q < n$ has target $E^{n+1,q-n} = 0$ (negative fiber degree), so it survives to E_∞ and must therefore vanish: $H^q(\Omega S^{n+1}) = 0$ for $0 < q < n$. For $q \geq n$, the class in $E_2^{0,q}$ must die, which (the only available differential being d_{n+1}) forces $d_{n+1} \colon E^{0,q} \xrightarrow{\cong} E^{n+1,q-n}$ to be an isomorphism; since $E^{n+1,q-n} \cong H^{n+1}(S^{n+1}) \otimes H^{q-n}(\Omega S^{n+1}) \cong H^{q-n}(\Omega S^{n+1})$, this gives $H^q(\Omega S^{n+1}) \cong H^{q-n}(\Omega S^{n+1})$. Starting from $H^0 = \mathbb{Z}$ and iterating the shift by n together with the vanishing in the gaps yields infinite cyclic groups exactly in degrees $0, n, 2n, 3n, \dots$ and zero elsewhere. Write a_k for a generator of $H^{kn}(\Omega S^{n+1}) \cong \mathbb{Z}$ (so $a_0 = 1$, $|a_k| = kn$, and $a_k \in E_2^{0,kn}$). The entries are $E_2^{0,kn} = \mathbb{Z}\langle a_k \rangle$ and $E_2^{n+1,kn} = \mathbb{Z}\langle s\, a_k \rangle$ (column $n+1$ is column 0 shifted by the base class s, by multiplicativity). For E_∞ to vanish off the origin, d_{n+1} must pair off $E_{n+1}^{0,(k+1)n}$ with $E_{n+1}^{n+1,kn}$ isomorphically for each $k \geq 0$, since these are the only nonzero entries and they sit in the right bidegrees: d_{n+1} sends bidegree $(0, (k+1)n)$ to $\big(n+1,\ (k+1)n - n\big) = (n+1, kn)$, which is exactly the bidegree of $s\, a_k$, confirming the pairing $a_{k+1} \leftrightarrow s\, a_k$.

The Leibniz rule now determines the integer. Compute, using $d_{n+1}(s) = 0$ (the base class s is a permanent cycle, lying in the bottom row $q = 0$) and graded-commutativity,

$$d_{n+1}(s\, a_k) = d_{n+1}(s)\, a_k \pm s\, d_{n+1}(a_k) = \pm s \cdot s\, a_{k-1} = 0$$

since $s^2 = 0$ in $H^*(S^{n+1})$ (the base is a single sphere). So $s\, a_k$ is a d_{n+1}-cycle, consistent with its being hit from the left column. To find $d_{n+1}(a_{k+1})$, use the product structure of the fiber: write a_{k+1} in terms of lower classes. Graded-commutativity of the fiber ring constrains a_1^2: if n is odd, $a_1^2 = -a_1^2$ forces $2a_1^2 = 0$, but $H^{2n}(\Omega S^{n+1}) \cong \mathbb{Z}$ is torsion-free, so $a_1^2 = 0$ and a_1^2 *cannot* generate H^{2n}. Hence a_2 is a new generator, not a_1^2. The relation between a_1^2 and a_2 is the divided-power relation: applying d_{n+1} as a derivation to the product $a_1 \cdot a_1$,

$$d_{n+1}(a_1^2) = d_{n+1}(a_1)\, a_1 + (-1)^n a_1\, d_{n+1}(a_1) = s\, a_1 + (-1)^n a_1\, s = (1 + (-1)^n)\, s\, a_1$$

where $a_1\, s = s\, a_1$ because the total degrees satisfy $n(n+1) \equiv 0 \pmod 2$, so graded-commutativity introduces no sign. For n odd this is 0, consistent with $a_1^2 = 0$; for n even it is $2\, s\, a_1$. Meanwhile $d_{n+1}(a_2) = s\, a_1$ (the pairing isomorphism, normalized so $a_2 \mapsto s\, a_1$). Comparing, for n even $d_{n+1}(a_1^2) = 2\, s\, a_1 = 2\, d_{n+1}(a_2)$, so $d_{n+1}(a_1^2 - 2a_2) = 0$; since the only d_{n+1}-cycle in $E_{n+1}^{0,2n}$ that survives is zero (the column-0 entry must die), $a_1^2 - 2a_2$ must itself be a boundary, but $E_{n+1}^{0,2n}$ receives no differential, so $a_1^2 - 2a_2 = 0$, giving

$$a_1^2 = 2\, a_2 \qquad (n \text{ even})$$

Iterate the computation, taking n even first so that $(-1)^{kn} = 1$ and all signs are positive. The Leibniz rule applied to $a_k \cdot a_1$ gives, using $d_{n+1}(a_k) = s\, a_{k-1}$, $d_{n+1}(a_1) = s$, and $a_k\, s = s\, a_k$,

$$d_{n+1}(a_k\, a_1) = s\, a_{k-1}\, a_1 + s\, a_k = s\,(a_{k-1} a_1) + s\, a_k$$

By the inductive recursion $a_{k-1} a_1 = k\, a_k$ this is $k\, s\, a_k + s\, a_k = (k+1)\, s\, a_k = (k+1)\, d_{n+1}(a_{k+1})$. Hence $d_{n+1}\big(a_k a_1 - (k+1) a_{k+1}\big) = 0$, and since the column-0 entry $E_{n+1}^{0,(k+1)n}$ must die (it receives no differential and cannot survive to E_∞), the cycle $a_k a_1 - (k+1) a_{k+1}$ vanishes. Therefore

$$a_1 \cdot a_k = (k+1)\, a_{k+1}, \qquad \text{equivalently} \qquad a_1^k = k!\, a_k \qquad (n \text{ even})$$

so $a_k = a_1^k / k!$ is a genuine class in integral cohomology even though a_1^k is divisible by $k!$. The ring is the *divided-power algebra* $\Gamma_{\mathbb{Z}}[a]$ on one generator $a = a_1$ of degree n, with additive basis $\{a_k = a^{[k]}\}_{k \geq 0}$ and multiplication $a^{[j]} a^{[k]} = \binom{j+k}{j} a^{[j+k]}$:

$$H^*(\Omega S^{n+1}; \mathbb{Z}) \cong \Gamma_{\mathbb{Z}}[a], \qquad |a| = n$$

For n odd the additive computation is unchanged (one $\mathbb{Z}$ in each degree kn), but graded-commutativity now forces $a_1^2 = 0$, so the odd generator a_1 is exterior and the genuine divided powers begin one degree up: the same Leibniz analysis applied to a_2 (of even degree $2n$) shows $a_2^k = k!\, a_{2k}$, so $H^*(\Omega S^{n+1}; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[a_1] \otimes \Gamma_{\mathbb{Z}}[a_2]$ with $|a_1| = n$, $|a_2| = 2n$. In both parities this is the divided-power algebra $\Gamma_{\mathbb{Z}}[a]$ on a degree-n generator, since $\Gamma_{\mathbb{Z}}[a]$ for an odd-degree generator is by definition this exterior-times-divided-power tensor product. Over a field of characteristic zero, $\Gamma_{\mathbb{Q}}[a] \cong \mathbb{Q}[a]$ is a polynomial algebra (on a_1 for n even, on a_2 for n odd); the divided-power structure is visible only integrally, where the divisibility $a^k = k!\, a^{[k]}$ records that ΩS^{n+1} has more integral cohomology than a polynomial algebra would predict.

The two computations exhibit the two outcomes multiplicativity can force. In the first, graded-commutativity placed no constraint (the generator x has even degree, so x^k are all nonzero and independent) and the answer is a free polynomial algebra; the spectral sequence confirms that the powers of x generate, with the differentials computing each power as the transgression of the previous one times the fiber class. In the second, graded-commutativity of an odd-degree class would force $a_1^2 = 0$, which collides with the torsion-free integral cohomology the convergence requires, and the resolution is the divided-power structure: the class a_1^k exists but is $k!$ times a primitive generator. The divided-power phenomenon is the integral fingerprint of the loop-space multiplication, invisible rationally and central to the structure of the homology of iterated loop spaces.

6.4.10 Historical Note: Serre's Thesis and the Method of Killing Homotopy Groups The multiplicative spectral sequence and the transgression were the tools with which Jean-Pierre Serre, in his 1951 thesis, broke open the computation of homotopy groups of spheres. The path-loop fibration over S^{n+1}, computed in example 6.4.9, was Serre's lever: combined with the method of "killing homotopy groups" (the dual of the Postnikov construction of section 5.6), it gave the first proof that $\pi_i(S^n)$ is finite for $i > n$ except for $\pi_{4k-1}(S^{2k})$, and computed many stable stems. The divided-power structure of $H^*(\Omega S^{n+1})$ and its mod-p refinements, where the divided powers interact with the Steenrod operations, remain the computational heart of the subject.

Exercise 6.4.1

1. Let $E_r^{\bullet,\bullet}$ be a multiplicative spectral sequence with d_r a derivation. Suppose $a \in E_r^{p,q}$ satisfies $d_r(a) = 0$ and $b \in E_r^{p',q'}$ satisfies $b = d_r(c)$ for some c. Show directly that $a \cdot b$ is a boundary, and conclude that the product $[a] \cdot [b]$ on $E_{r+1} = H(E_r, d_r)$ is well defined.

2. Using the multiplicative Serre spectral sequence of the trivial fibration $F \to B \times F \to B$ with B a path-connected CW complex and B, F of finite type over a field k, prove the Künneth formula $H^*(B \times F; k) \cong H^*(B; k) \otimes_k H^*(F; k)$ as graded k-algebras. (Show that $E_2 = E_\infty$ and that there are no extension problems.)

3. Compute the cohomology ring $H^*(\mathbb{CP}^n; \mathbb{Z})$ for finite n from the fibration $S^1 \to S^{2n+1} \to \mathbb{CP}^n$, and identify it as $\mathbb{Z}[x]/(x^{n+1})$ with $|x| = 2$. Explain which differentials change relative to the $\mathbb{CP}^\infty$ computation of example 6.4.9 and why the relation $x^{n+1} = 0$ appears.

4. Let n be even. Show that $H^*(\Omega S^{n+1}; \mathbb{Q}) \cong \mathbb{Q}[a]$ is a polynomial algebra on a single generator of degree n, while $H^*(\Omega S^{n+1}; \mathbb{Z})$ is the divided-power algebra $\Gamma_{\mathbb{Z}}[a]$. Exhibit explicitly the

class $a^{[p]} = a^p/p!$ and verify that it is not in the image of $H^*(\Omega S^{n+1};\mathbb{Z}) \to H^*(\Omega S^{n+1};\mathbb{Q})$ from any integral polynomial in a.

5. In the path-loop fibration $\Omega X \to PX \to X$, show that the transgression $\tau\colon H^n(\Omega X;R) \to H^{n+1}(X;R)$ defined by definition 6.4.5 coincides with the cohomology suspension isomorphism on the classes for which it is defined, using theorem 6.4.7 and the contractibility of PX.

6. Let $p\colon E \to B$ be a fibration with B simply connected, $H^*(B;\mathbb{Q})$ and $H^*(F;\mathbb{Q})$ both finite-dimensional, and suppose the rational Serre spectral sequence collapses at E_2 (all differentials vanish). Show that the Poincaré series satisfy $P_E(t) = P_B(t)\cdot P_F(t)$, where $P_X(t) = \sum_i \dim_{\mathbb{Q}} H^i(X;\mathbb{Q})\, t^i$. Give an example where collapse fails and the equality is a strict inequality $P_E(t) \le P_B(t)P_F(t)$ coefficientwise.

6.5 The Leray-Hirsch Theorem

The Serre spectral sequence computes $H^*(E)$ from $H^*(B)$ and $H^*(F)$ only once its differentials are known, and pinning those down can be the hard part of a calculation. One recurring situation settles the outcome in advance, with no differential computed at all: when the cohomology of the fiber is free and every fiber class extends to a global class on the total space. The spectral sequence then collapses for a structural reason, and $H^*(E)$ is a free module over $H^*(B)$ on the extending classes. This is the Leray-Hirsch theorem. It is the last tool this chapter assembles and the one that opens the next: the cohomology of a projective bundle is computed by it directly, and on that computation the characteristic classes of chapter 7 are built.

The reason the hypothesis forces collapse can be given before any proof. A class defined on all of E is a permanent cycle of the spectral sequence, since it already lives in $H^*(E)$ and can be neither the source nor the target of a differential; if such global classes restrict to a basis of the fiber cohomology, multiplicativity spreads their survival across the whole spectral sequence and every differential is forced to vanish. Recall that $p^*\colon H^*(B;R) \to H^*(E;R)$ is a ring homomorphism, so $H^*(E;R)$ is a module over $H^*(B;R)$ through $\alpha\cdot\beta = p^*(\alpha) \smile \beta$ (definition 4.2.5); the theorem identifies this module completely.

Theorem 6.5.1: The Leray-Hirsch Theorem

Let $p\colon E \to B$ be a Serre fibration over a path-connected base with fiber F, and let R be a commutative ring. Suppose that

1. $H^n(F;R)$ is a free R-module of finite rank for every n, and
2. there exist classes $c_j \in H^{q_j}(E;R)$, $j = 1,\dots,k$, whose restrictions $i^*c_1,\dots,i^*c_k$ along the fiber inclusion $i\colon F \hookrightarrow E$ form an R-basis of $H^*(F;R)$.

Then $H^*(E;R)$ is a free $H^*(B;R)$-module with basis $c_1,\dots,c_k$; equivalently, the $H^*(B;R)$-linear map

$$\Phi\colon H^*(B;R)\otimes_R H^*(F;R) \longrightarrow H^*(E;R), \qquad \alpha\otimes i^*c_j \longmapsto p^*(\alpha)\smile c_j$$

is an isomorphism.

***Proof*:**
The spectral sequence that drives the proof, theorem 6.2.5, asks for the base to be a path-connected *CW complex*, while the statement allows an arbitrary path-connected base. The first move closes that gap by CW approximation, following the final paragraph of [14, Theorem 4D.1]; the spectral-sequence argument then runs over the CW model.

Reduction to a CW base. Fix $b_0 \in B$ with $F = p^{-1}(b_0)$ and let $f\colon Z \to B$ be a CW approximation (theorem 5.3.9). Since B is path-connected and f induces a bijection on path components, Z is a path-connected CW complex, and theorem 5.3.9 gives the approximation with $f(z_0) = b_0$ for a 0-cell z_0. Form the pullback $f^*E = \{(z,e) \in Z \times E : f(z) = p(e)\}$ with projection $q\colon f^*E \to Z$ and comparison map $\tilde{f}\colon f^*E \to E$, $(z,e) \mapsto e$; then q is a Serre fibration and $\tilde{f}$ restricts over each $z \in Z$ to a homeomorphism $q^{-1}(z) \to p^{-1}(f(z))$ (proposition 5.2.6), so in particular the fiber of q over z_0 is a homeomorphic copy of F.

The comparison map $\tilde{f}$ is a weak homotopy equivalence. At any point (z,e) of f^*E, the triple $(f, \tilde{f}, \tilde{f}|_{q^{-1}(z)})$ maps the long exact sequence of q (theorem 5.2.9) to that of p; the ladder commutes, the squares involving i_* and the projections because $f \circ q = p \circ \tilde{f}$, and the squares involving ∂ because both connecting maps are the boundary of the relative sequence transported through the lifting isomorphism (proposition 5.2.8), which is natural in the map of pairs $(f^*E, q^{-1}(z)) \to (E, p^{-1}(f(z)))$. The vertical maps on base and fiber terms are isomorphisms, f being a weak homotopy equivalence and the fiber comparison a homeomorphism, so the five lemma (lemma 3.4.22) gives that $\tilde{f}_*\colon \pi_n(f^*E,(z,e)) \to \pi_n(E,e)$ is an isomorphism for $n \geq 2$. At the bottom the chase runs in groups and pointed sets. For injectivity on π_1: a loop γ at (z,e) with $\tilde{f} \circ \gamma$ null-homotopic has $f_*[q \circ \gamma] = p_*[\tilde{f} \circ \gamma] = 0$, so $[q \circ \gamma] = 0$ and exactness puts $[\gamma] = i_*[\delta]$ for a loop δ in the fiber; then $[\delta] \in \ker\big(\pi_1(p^{-1}(f(z))) \to \pi_1(E)\big) = \mathrm{im}\partial_p$, and since f_* is onto $\pi_2(B, f(z))$ and the connecting maps correspond, $[\delta] \in \mathrm{im}\partial_q$, whence $[\gamma] = i_*[\delta] = 0$. For surjectivity on π_1: given a loop ε at $e = \tilde{f}(z,e)$, its projection satisfies $p_*[\varepsilon] = f_*[\zeta]$ for some $[\zeta] \in \pi_1(Z,z)$, and $\partial_q[\zeta] = \partial_p(f_*[\zeta]) = \partial_p(p_*[\varepsilon])$ is the basepoint component (the loop ε is itself a lift of $p \circ \varepsilon$ closing up), so $[\zeta] = q_*[d]$ for some $[d] \in \pi_1(f^*E)$; then $[\varepsilon] \cdot (\tilde{f}_*[d])^{-1} \in \ker p_* = \mathrm{im} i_*$, and adjusting $[d]$ by the corresponding fiber class exhibits $[\varepsilon]$ in the image of $\tilde{f}_*$. On π_0: every point $e' \in E$ is joined, by lifting a path in the path-connected B from $p(e')$ to b_0 through the Serre fibration p, to a point of $F = \tilde{f}\big(q^{-1}(z_0)\big)$, so $\pi_0(\tilde{f})$ is onto; and if a path σ in E joins $\tilde{f}(z,e)$ to $\tilde{f}(z',e')$, choose a path τ in Z from z to z', correct it by a loop at z (using that f_* is onto $\pi_1(B, f(z))$) so that $f \circ \tau \simeq p \circ \sigma$ rel endpoints, and lift the homotopy through p starting from σ: the lift deforms σ, through paths moving its endpoints only inside the fibers over $f(z)$ and $f(z')$, to a path lying over $f \circ \tau$, which together with τ assembles to a path in f^*E from (z,e) to (z',e'), so $\pi_0(\tilde{f})$ is injective. Hence $\tilde{f}$ is a weak homotopy equivalence; it induces isomorphisms on singular homology with any coefficients ([14, Proposition 4.21], used the same way in lemma 5.4.8), hence on cohomology $H^*(-;R)$ by corollary 4.1.10 applied to the induced map of singular chain complexes, and the same holds for f.

The hypotheses transfer to q: hypothesis (1) concerns only F, and the classes $\tilde{f}^*c_j \in H^{q_j}(f^*E; R)$ restrict on the fiber over z_0 to the images of the i^*c_j under the fiberwise homeomorphism, hence to an R-basis. The map Φ is natural for the pair $(f, \tilde{f})$: writing Φ_Z for the map built from q and the $\tilde{f}^*c_j$,

$$\tilde{f}^*\,\Phi\big(\alpha \otimes i^*c_j\big) = \tilde{f}^*\big(p^*(\alpha) \smile c_j\big) = q^*(f^*\alpha) \smile \tilde{f}^*c_j = \Phi_Z\big(f^*\alpha \otimes \tilde{f}^*c_j|_{q^{-1}(z_0)}\big)$$

so Φ and Φ_Z are intertwined by the isomorphisms $f^* \otimes \mathrm{id}$ and $\tilde{f}^*$, and Φ is an isomorphism if and only if Φ_Z is. It therefore suffices to prove the theorem when the base is a path-connected

CW complex, which is assumed for the rest of the proof.

The existence of the global classes forces the fundamental group of the base to act trivially on the fiber cohomology: each restriction i^*c_j is the image of a class defined on all of E, hence is invariant under the monodromy action of $\pi_1(B)$ on $H^*(F;R)$, and by hypothesis (2) the i^*c_j span $H^*(F;R)$, so the action is trivial. Theorem 6.2.5 therefore gives the cohomology Serre spectral sequence, with second page

$$E_2^{p,q} \cong H^p\big(B; H^q(F;R)\big) \cong H^p(B;R) \otimes_R H^q(F;R)$$

converging to $H^{p+q}(E;R)$; the second isomorphism holds because $H^q(F;R)$ is free of finite rank by hypothesis (1), so cohomology with these coefficients is the tensor product with $H^*(B;R)$.

Step 1: The entire fiber column survives. By theorem 6.2.6 the edge homomorphism in the fiber direction is the restriction $i^*\colon H^q(E;R) \to H^q(F;R)$, and it factors as the projection $H^q(E) \twoheadrightarrow E_\infty^{0,q}$ onto the top filtration quotient followed by the inclusion $E_\infty^{0,q} \hookrightarrow E_2^{0,q} \cong H^q(F;R)$; in particular the image of i^* is exactly the surviving subgroup $E_\infty^{0,q} \subseteq E_2^{0,q}$. Each i^*c_j lies in this image, and by hypothesis (2) the classes i^*c_j exhaust a basis of $H^q(F;R)$ as q ranges over all degrees, so i^* is surjective and $E_\infty^{0,q} = E_2^{0,q}$ for every q. Thus no element of the fiber column $E_2^{0,\bullet}$ is killed by a differential. Since the fiber column is the leftmost, no differential maps into it either: a differential d_r into $E_r^{0,q}$ would originate in $E_r^{-r,\,q+r-1} = 0$. Hence every class in the fiber column is a permanent cycle, and d_r vanishes on $E_r^{0,\bullet}$ for all $r \geq 2$.

Step 2: All differentials vanish. The base row $E_2^{\bullet,0} \cong H^\bullet(B;R)$ also consists of permanent cycles: a differential out of $E_r^{p,0}$ lands in $E_r^{p+r,\,1-r} = 0$, so d_r vanishes on the base row. By the multiplicative structure (theorem 6.4.3(3)) the product

$$E_2^{p,0} \otimes_R E_2^{0,q} \longrightarrow E_2^{p,q}$$

is the isomorphism $H^p(B;R) \otimes_R H^q(F;R) \xrightarrow{\sim} E_2^{p,q}$, so E_2 is generated as a bigraded algebra by the base row together with the fiber column. Each differential d_r is a derivation (theorem 6.4.3(2)) that vanishes on both sets of generators by Steps 1 and 2; a derivation vanishing on algebra generators vanishes identically, so $d_2 = 0$ on all of E_2. Then $E_3 = E_2$ with the same generators, and the argument repeats verbatim: inductively $d_r = 0$ and $E_{r+1} = E_r$ for every $r \geq 2$. The spectral sequence collapses, $E_2 = E_\infty$.

Step 3: From collapse to the module isomorphism. Collapse identifies the associated graded of the abutment filtration,

$$\operatorname{gr} H^*(E;R) \cong E_\infty = E_2 = H^*(B;R) \otimes_R H^*(F;R)$$

and it remains to see that Φ realizes this identification. The map Φ is $H^*(B;R)$-linear because p^* is a ring homomorphism and the cup product is associative:

$$\Phi\big((\alpha'\alpha) \otimes i^*c_j\big) = p^*(\alpha'\alpha) \smile c_j = p^*(\alpha') \smile \big(p^*(\alpha) \smile c_j\big) = \alpha' \cdot \Phi(\alpha \otimes i^*c_j)$$

It preserves the filtration: $p^*(\alpha) \in F^{|\alpha|}H^*(E)$ and $c_j \in F^0H^*(E)$, so by multiplicativity of the abutment filtration (theorem 6.4.3(4)) the product $p^*(\alpha) \smile c_j$ lies in $F^{|\alpha|}$. On the associated graded, Φ induces the product map $E_2^{|\alpha|,0} \otimes E_2^{0,q_j} \to E_2^{|\alpha|,q_j}$ of Step 2, which is exactly the isomorphism $E_2 \cong H^*(B) \otimes H^*(F)$; hence $\operatorname{gr}\Phi$ is an isomorphism. In each total degree the abutment filtration is finite, since the spectral sequence is first-quadrant (proposition 6.1.9), so it is bounded and exhaustive; a filtration-preserving map inducing an isomorphism on the associated graded of a finite filtration is itself an isomorphism, by induction up the filtration steps and the five lemma. Therefore Φ is an isomorphism of $H^*(B;R)$-modules, and $c_1, \ldots, c_k$ are a free basis of $H^*(E;R)$ over $H^*(B;R)$, as we sought to show.

The theorem is a collapse criterion wearing the clothes of a module statement: hypothesis (2) is precisely what makes the fiber column survive, multiplicativity does the rest, and no differential is ever computed. The freeness of hypothesis (1) cannot be dropped, since torsion in the fiber cohomology introduces coefficient extensions the collapse argument does not control (exercise 6.5.1 (3)). One hypothesis deserves removal before the theorem is put to work: the base is assumed path-connected, while the characteristic-class constructions of chapter 7 run over bases with no connectivity guarantee. The assumption can be dropped wholesale, because singular cohomology sees the path components of a space separately and a fibration decomposes over the path components of its base. Both hypotheses below are imposed at *every* fiber; over a disconnected base no single fiber can speak for the others. The corollary is stated for a Serre fibration; every fiber bundle, over any base whatsoever, qualifies by theorem 5.2.4.

Corollary 6.5.2: Leray-Hirsch over a Disconnected Base

Let $p\colon E \to B$ be a Serre fibration whose fibers are all nonempty, and let R be a commutative ring. Suppose that

1. for every $b \in B$ the fiber $F_b = p^{-1}(b)$ has $H^m(F_b; R)$ a free R-module of finite rank for every m, and
2. there exist classes $c_j \in H^{q_j}(E; R)$, $j = 1, \ldots, k$, whose restrictions along the inclusion $F_b \hookrightarrow E$ form an R-basis of $H^*(F_b; R)$ for *every* $b \in B$.

Then $H^*(E; R)$ is a free $H^*(B; R)$-module with basis $c_1, \ldots, c_k$: the $H^*(B; R)$-linear map

$$\Phi\colon \bigoplus_{j=1}^{k} H^*(B; R) \longrightarrow H^*(E; R), \qquad (\alpha_j)_j \longmapsto \sum_{j=1}^{k} p^*(\alpha_j) \smile c_j$$

is an isomorphism.

***Proof*:**
Step 1: Cohomology is the product over a saturated partition. Let Y be any space and let $\{Y_c\}_{c \in C}$ be a partition of Y into subsets each of which is a union of path components of Y. Every singular simplex $\sigma\colon \Delta^m \to Y$ has path-connected image, so it lands in a single path component, hence in a single Y_c, and the boundary of a simplex lying in Y_c again lies in Y_c; hence the singular chain complex splits as a direct sum of subcomplexes, $C_*(Y) = \bigoplus_c C_*(Y_c)$. Applying $\operatorname{Hom}_{\mathbb{Z}}(-, R)$ converts the direct sum into a direct product of cochain complexes, $C^*(Y; R) = \prod_c C^*(Y_c; R)$, and cohomology commutes with direct products of cochain complexes, kernels and images being computed coordinatewise. The resulting isomorphism

$$H^*(Y; R) \xrightarrow{\cong} \prod_c H^*(Y_c; R)$$

is induced by the restrictions to the members of the partition. It is natural for partition-preserving maps: if $\varphi\colon Y \to Y'$ carries each member Y_c of such a partition into the member Y'_c of a like partition of Y', then restriction commutes with φ^* coordinate by coordinate. And it is a ring isomorphism, cup products being computed by restriction member by member.

Step 2: The total space partitions over the path components of the base. Let $\{B_c\}$ be the path components of B, a partition of B of the kind Step 1 asks for, and put $E_c = p^{-1}(B_c)$, a nonempty set because the fiber over any point of B_c is nonempty. A path in E projects to a path in B and so cannot leave E_c; hence each E_c is a union of path components of E, and $\{E_c\}$ is a partition of E of the same kind. No connectivity of E_c itself is claimed, and none is needed. The restriction $p_c\colon E_c \to B_c$ is again a Serre fibration: a lifting problem for p_c is one for p whose homotopy lands in B_c, and any solution for p then lands in $p^{-1}(B_c) = E_c$.

Step 3: The connected case on each component, and reassembly. Fix a component B_c and a point $b \in B_c$. The Serre fibration $p_c\colon E_c \to B_c$ has path-connected base and fiber F_b; hypothesis (1) makes $H^m(F_b; R)$ free of finite rank for every m, and hypothesis (2), read at the fiber over b, makes the restrictions $c_j|_{E_c}$ a family of homogeneous classes whose further restriction to F_b is an R-basis of $H^*(F_b; R)$. Theorem 6.5.1 therefore applies to p_c and makes $H^*(E_c; R)$ a free $H^*(B_c; R)$-module with basis the $c_j|_{E_c}$; equivalently, the map Φ_c defined like Φ with B_c, E_c, and $c_j|_{E_c}$ in place of B, E, and c_j is an isomorphism. Now assemble. The maps p and the inclusions $E_c \hookrightarrow E$, $B_c \hookrightarrow B$ all preserve the partitions of Step 2, so the naturality clause of Step 1 applies to them; and restriction commutes with the module structure, $p^*(\alpha)|_{E_c} = p_c^*(\alpha|_{B_c})$, both being induced by the square formed by p and the two inclusions, restriction being a ring map. Under the isomorphisms of Step 1 for B and for E, the map Φ is therefore identified with the composite

$$\bigoplus_{j=1}^{k} \prod_{c} H^*(B_c; R) \;\cong\; \prod_{c} \bigoplus_{j=1}^{k} H^*(B_c; R) \xrightarrow{\prod_c \Phi_c} \prod_{c} H^*(E_c; R)$$

where the first exchange of sum and product is an isomorphism because the sum is finite. A product of isomorphisms is an isomorphism, so Φ is one, and $c_1, \dots, c_k$ are a free basis of $H^*(E; R)$ over $H^*(B; R)$, completing the proof.

The quantifier in clause (2) is what makes the corollary true over a disconnected base, and it cannot be weakened to a single fiber. Two examples show why, and they fail in different ways.

First, take $B = \mathrm{pt} \sqcup \mathrm{pt}$, $E = S^1 \sqcup S^1$ the product bundle, $R = \mathbb{Z}$, and the classes $c_0 = (1, 1)$, $c_1 = (u, 0)$ where u generates $H^1(S^1; \mathbb{Z})$. Clause (1) holds, the fibers being circles. At the first point the restrictions are 1 and u, a basis of $H^*(S^1; \mathbb{Z})$; at the second they are 1 and 0, which are not. So clause (2) holds at one fiber and fails at the other, and the conclusion fails too: $H^*(B; \mathbb{Z})$ is concentrated in degree 0, so the $H^*(B; \mathbb{Z})$-span of c_0 and c_1 misses $(0, u)$ and they are no basis of $H^*(E; \mathbb{Z})$. Checking clause (2) at a single fiber would have admitted this.

The second example shows that clause (1) holding everywhere does not rescue matters. Take $R = \mathbb{Z}$, fiber $F = S^1$, base $B = \mathrm{pt} \sqcup S^1$, and $E = S^1 \sqcup K$ with K the Klein bottle as the circle bundle over S^1 given by the mapping torus of a reflection. Every fiber is a circle, so clause (1) holds. But $H^2(K; \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$, while a free graded module over $H^*(B; \mathbb{Z})$ is torsion-free as an abelian group, so $H^*(E; \mathbb{Z})$ is not free and the conclusion fails. What fails in the hypotheses is clause (2) over the S^1 component: $H^1(K; \mathbb{Z}) = \mathbb{Z}$ restricts to zero on a fiber circle, that circle being 2-torsion in $H_1(K) = \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, so no choice of classes restricts to a basis there. Requiring clause (2) at every b excludes exactly this.

The corollary carries no connectivity hypothesis on base or fibers, which is the strength of the statement recorded in [15, §3.1]; the proof is not given there but in [14, Theorem 4D.1], by an induction over a CW base followed by the same CW-approximation reduction that opens the proof of theorem 6.5.1, and the route through path components taken here replaces that induction. The paradigm application, and the one the next chapter turns into a definition, is the cohomology of a projective bundle.

Example 6.5.3: The Cohomology of a Projective Bundle
Let $E \to B$ be a complex vector bundle of rank n over a paracompact Hausdorff base, and let $\mathbb{P}(E) \to B$ be its projectivization, the fiber bundle whose fiber over b is the space $\mathbb{P}(E_b) \cong \mathbb{C}\mathrm{P}^{n-1}$ of complex lines in E_b. The fiber cohomology $H^*(\mathbb{C}\mathrm{P}^{n-1}; \mathbb{Z}) = \mathbb{Z}[u]/(u^n)$ is free of rank n over $\mathbb{Z}$ with basis $1, u, \ldots, u^{n-1}$, so hypothesis (1) of corollary 6.5.2 holds at every fiber. For hypothesis (2), the projectivization carries a tautological line bundle $\gamma \to \mathbb{P}(E)$, the subbundle of p^*E whose fiber over a line $\ell \in \mathbb{P}(E_b)$ is ℓ itself; set $t = -c_1(\gamma) \in H^2(\mathbb{P}(E); \mathbb{Z})$, the negative first Chern class of γ (example 7.1.3, developed in the next chapter). On each fiber $\mathbb{P}(E_b) = \mathbb{C}\mathrm{P}^{n-1}$ the bundle γ restricts to the tautological line bundle of $\mathbb{C}\mathrm{P}^{n-1}$, whose class t restricts to the generator u of $H^2(\mathbb{C}\mathrm{P}^{n-1}; \mathbb{Z})$, so $1, t, \ldots, t^{n-1}$ restrict to the fiber basis $1, u, \ldots, u^{n-1}$ in every fiber. Corollary 6.5.2, applied to the projective bundle, a Serre fibration with nonempty fibers by theorem 5.2.4, then gives that $H^*(\mathbb{P}(E); \mathbb{Z})$ is a free module over $H^*(B; \mathbb{Z})$ with basis $1, t, \ldots, t^{n-1}$. In particular t^n is an $H^*(B)$-linear combination of the lower powers: there are unique classes $a_i \in H^{2i}(B; \mathbb{Z})$ with

$$t^n + a_1 t^{n-1} + \cdots + a_{n-1} t + a_n = 0$$

and this single relation presents the ring,

$$H^*(\mathbb{P}(E); \mathbb{Z}) \cong H^*(B; \mathbb{Z})[t] \big/ \big(t^n + a_1 t^{n-1} + \cdots + a_n\big)$$

The coefficients a_i are invariants of the bundle E; naming them $a_i = c_i(E)$ is the projective-bundle route to the Chern classes recorded in proposition 7.1.15.

Exercise 6.5.1

1. Let B and F be spaces with $H^*(F; R)$ free of finite rank over a commutative ring R, and consider the trivial fibration $F \to B \times F \xrightarrow{p} B$, which is a Serre fibration with fiber F over every point, so that corollary 6.5.2 applies to any B. Using the projection $\mathrm{pr}_F \colon B \times F \to F$, exhibit classes on $B \times F$ satisfying the hypotheses of corollary 6.5.2, and conclude that $H^*(B \times F; R) \cong H^*(B; R) \otimes_R H^*(F; R)$ as $H^*(B; R)$-modules. Explain how this recovers the Künneth theorem in this case.

2. Show that hypothesis (2) of theorem 6.5.1 holds if and only if $H^*(F; R)$ is free of finite rank and the restriction $i^* \colon H^*(E; R) \to H^*(F; R)$ is surjective. Conclude that Leray-Hirsch applies exactly when the fiber cohomology is free and entirely restricted from the total space.

3. Locate the two places in the proof of theorem 6.5.1 where freeness of $H^*(F; R)$ is used: the identification $E_2^{p,q} \cong H^p(B; R) \otimes_R H^q(F; R)$ and the notion of an R-basis in hypothesis (2). Give a fiber F and ring R for which $H^*(F; R)$ is not free (for instance $F = \mathbb{R}\mathrm{P}^2$, $R = \mathbb{Z}$), and identify which step of the argument breaks.

7

Characteristic Classes

A characteristic class is a rule that attaches to every vector bundle a cohomology class of its base, in a way that respects pullback. The decisive observation of this chapter is that such a rule is the same thing as a single cohomology class on one universal space, the classifying space BG of chapter 2. Every principal G-bundle over a space X is the pullback of one universal bundle along a classifying map $X \to BG$, unique up to homotopy by theorem 2.4.7. A characteristic class is therefore nothing more than an element of $H^*(BG)$, pulled back to X along that map. The whole theory of characteristic classes is, in this reading, the computation of the cohomology rings $H^*(BG)$ for the structure groups that arise in geometry, together with the dictionary that turns the resulting universal classes into invariants of bundles. This is the cohomological route to a subject that also admits a bundle-theoretic route (the Grassmannian and obstruction-theoretic constructions, developed in section 7.1.1 and section 7.4) and an analytic route (the Chern-Weil construction from curvature, developed in the Riemannian-geometry volume [11]). The three agree, and the present chapter develops the first.

The computation draws on the whole apparatus the book has assembled. The classifying spaces themselves come from chapter 2. Their cohomology is a ring, with the cup product of chapter 4 furnishing the multiplicative structure that makes the answer a polynomial algebra rather than a bare sequence of groups. The engine that computes that ring is the Serre spectral sequence of chapter 6, run on the universal sphere bundles relating $BU(n-1)$ to $BU(n)$ and $BO(n-1)$ to $BO(n)$, with the Gysin sequence of theorem 6.3.2 doing the bookkeeping one dimension at a time. The meaning of the resulting classes, in particular the Euler class as the primary obstruction to a nowhere-zero section, is read through the obstruction theory of chapter 5. The chapter thus gathers the threads of everything before it into one computation and one definition, and it prepares the generalized cohomology of chapter 8, where these same classes orient the Thom spectra and drive the cobordism calculations.

7.1 Characteristic Classes and the Cohomology of Classifying Spaces

The classification theorem theorem 2.4.7 states that for a topological group G the isomorphism classes of principal G-bundles over a CW complex X are in natural bijection with the homotopy classes of maps $X \to BG$, the bijection sending a classifying map $f\colon X \to BG$ to the pullback f^*EG of the universal bundle. Naturality means that for a map $g\colon X' \to X$ the classifying map of $g^*(f^*EG)$ is $f \circ g$, so the assignment of bundles to maps converts pullback of bundles into composition of maps. A characteristic class is exactly an invariant that is compatible with this structure: a rule assigning to each bundle a cohomology class of the base, natural under pullback. The point of the section is that naturality forces such a rule to be the pullback of one universal class living on BG, so the classification theorem converts the search for invariants into a cohomology computation on a single space.

Definition 7.1.1: Characteristic Class

Fix a topological group G, an abelian coefficient group A, and an integer $k \geq 0$. A *characteristic class* of degree k for principal G-bundles, with values in A, is a rule c that assigns to every principal G-bundle $P \to X$ over a CW complex X a cohomology class

$$c(P) \in H^k(X; A)$$

subject to *naturality under pullback*: for every map $g\colon X' \to X$ of CW complexes,

$$c(g^*P) = g^*\big(c(P)\big) \in H^k(X'; A)$$

where $g^*P \to X'$ is the pulled-back bundle and $g^*\colon H^k(X; A) \to H^k(X'; A)$ is the induced map on cohomology. A characteristic class for rank-n complex (respectively real) vector bundles is a characteristic class for principal $U(n)$-bundles (respectively $O(n)$-bundles), under the equivalence sending a vector bundle to its principal bundle of unitary (respectively orthonormal) frames and a principal bundle to its associated vector bundle; this correspondence, an instance of the associated-bundle construction, is developed in section 2.4.

The naturality condition is the whole content of the definition, and it is what makes the notion computable. It says that the value of c on a bundle is determined by the value of c on bundles into which the given one maps, and ultimately by the value of c on the one bundle through which all others factor. The vector-bundle version is the form in which characteristic classes are usually met: the structure group of a rank-n complex vector bundle is $U(n)$ once a Hermitian metric is chosen (any two metrics give homotopic, hence isomorphic, structure-group reductions, so the choice does not affect the bundle's classification), and similarly $O(n)$ for a rank-n real bundle with a Euclidean metric. The frame bundle of orthonormal frames is then a principal bundle, and characteristic classes of the vector bundle are by definition those of its frame bundle.

The classification theorem turns definition 7.1.1 into the statement that a characteristic class is a single element of the cohomology of BG.

Proposition 7.1.2: Characteristic Classes are the Cohomology of BG

Let G be a topological group whose classifying space BG has the homotopy type of a CW complex, let A be an abelian group, and fix $k \geq 0$. Evaluation on the universal bundle gives a bijection

$$\{\text{ degree-}k \text{ characteristic classes for } G\text{-bundles with values in } A\,\} \xrightarrow{\cong} H^k(BG; A), \qquad c \longmapsto c(EG)$$

Its inverse sends a class $u \in H^k(BG; A)$ to the characteristic class c_u defined by $c_u(P) = f_P^* u$, where $f_P \colon X \to BG$ is the classifying map of $P \to X$.

***Proof*:**
The two assignments $c \mapsto c(EG)$ and $u \mapsto c_u$ are shown to be mutually inverse.

Step 1: c_u is a characteristic class. Fix $u \in H^k(BG; A)$ and define $c_u(P) = f_P^* u$ for $P \to X$ with classifying map $f_P \colon X \to BG$. The class f_P is well defined up to homotopy by theorem 2.4.7, and homotopic maps induce equal maps on cohomology, so $c_u(P)$ depends only on the isomorphism class of P. For naturality, let $g \colon X' \to X$ and consider $g^*P \to X'$. By the naturality in theorem 2.4.7, a classifying map for g^*P is $f_P \circ g$, so

$$c_u(g^*P) = (f_P \circ g)^* u = g^*\big(f_P^* u\big) = g^*\big(c_u(P)\big)$$

using functoriality of cohomology. Thus c_u satisfies definition 7.1.1.

Step 2: The two maps are inverse. Apply $c \mapsto c(EG)$ to c_u. The universal bundle $EG \to BG$ has classifying map the identity id_{BG} (the identity classifies the bundle one started with), so

$$c_u(EG) = \mathrm{id}_{BG}^* u = u$$

Hence $u \mapsto c_u \mapsto c_u(EG) = u$ is the identity on $H^k(BG; A)$.

Conversely, start with a characteristic class c, set $u = c(EG)$, and form c_u. For any bundle $P \to X$ with classifying map f_P, the bundle P is isomorphic to $f_P^* EG$ by theorem 2.4.7. Naturality of c applied to the map $f_P \colon X \to BG$ and the bundle $EG \to BG$ gives

$$c(P) = c(f_P^* EG) = f_P^*\big(c(EG)\big) = f_P^* u = c_u(P)$$

so $c_u = c$ as functions on bundles. Thus $c \mapsto c(EG) \mapsto c_{c(EG)} = c$ is the identity on characteristic classes. The two assignments are mutually inverse, completing the proof.

The proposition is the organizing principle of the entire chapter. It collapses the problem of finding all natural invariants of G-bundles, which on its face ranges over every base space at once, into the problem of computing the single graded ring $H^*(BG; A)$. The cup product on BG even gives the characteristic classes a ring structure: the product of two characteristic classes, formed by pulling back the product $u \smile u'$, is again natural, and the bijection of proposition 7.1.2 is a ring isomorphism between the characteristic classes (with the obvious product) and $H^*(BG; A)$. The remaining work is computational: identify the rings $H^*(BU(n); \mathbb{Z})$ and $H^*(BO(n); \mathbb{Z}/2)$, name their generators, and record what those generators measure. The hypothesis that BG has CW homotopy type holds for every group considered here: $BU(n)$ is the infinite complex Grassmannian $\mathrm{Gr}_n(\mathbb{C}^\infty)$, hence a CW complex, by example 2.4.9, and $BO(n)$ is the infinite real Grassmannian $\mathrm{Gr}_n(\mathbb{R}^\infty) = \varinjlim_N \mathrm{Gr}_n(\mathbb{R}^N)$

by the same model with $\mathbb{R}$ in place of $\mathbb{C}$, a CW complex with one cell for each Schubert symbol.

A first instance is already in hand and worth stating in the new language, since it fixes the normalization for everything that follows.

Example 7.1.3: The First Chern Class of a Line Bundle
Take $G = U(1)$, the circle group. Its classifying space is $BU(1) = \mathbb{C}\mathrm{P}^\infty$ (example 2.4.5), and the cohomology ring was computed in example 6.4.9 by the Serre spectral sequence of the universal circle bundle $S^1 \to S^\infty \to \mathbb{C}\mathrm{P}^\infty$:

$$H^*(BU(1);\mathbb{Z}) = H^*(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}[c_1], \qquad |c_1| = 2$$

a polynomial ring on one degree-two generator c_1. By proposition 7.1.2 the characteristic classes of complex line bundles are exactly the elements of this ring: the additive ones in degree 2 are the integer multiples $m\,c_1$, and the full ring of characteristic classes is $\mathbb{Z}[c_1]$. The generator c_1 is the *first Chern class*. For a line bundle $L \to X$ with classifying map $f_L\colon X \to \mathbb{C}\mathrm{P}^\infty$, its value is $c_1(L) = f_L^* c_1 \in H^2(X;\mathbb{Z})$. Since $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z},2)$ represents $H^2(-;\mathbb{Z})$ (example 5.5.5), this recovers the statement that complex line bundles over X are classified by $H^2(X;\mathbb{Z})$, with c_1 the complete invariant: two line bundles are isomorphic if and only if their first Chern classes agree. Concretely, over $X = \mathbb{C}\mathrm{P}^1 = S^2$ the tautological line bundle γ^1 (whose fiber over a line $\ell \subseteq \mathbb{C}^2$ is ℓ itself) has $c_1(\gamma^1) = -u$, where $u \in H^2(\mathbb{C}\mathrm{P}^1;\mathbb{Z})$ is the generator dual to the fundamental class, so its Chern number $\int_{\mathbb{C}\mathrm{P}^1} c_1(\gamma^1) = -1$ detects γ^1 as a nontrivial generator of $\mathrm{Prin}_{U(1)}(S^2) \cong \mathbb{Z}$ (its dual, the hyperplane bundle, is the Hopf bundle H of example 5.2.7 with $c_1(H) = +u$).

7.1.1 Vector Bundles over a Paracompact Base

Definition 7.1.1 and proposition 7.1.2 are stated for principal G-bundles over a CW complex, and that hypothesis is not decorative: the classification theorem 2.4.7 on which they rest is proved by extending a map over the skeleta one at a time, and for a general topological group there is no substitute. For *vector* bundles the classification admits an entirely different proof, one that never mentions a cell. A vector bundle over a paracompact Hausdorff base carries a map to $\mathbb{R}^\infty$ that is linear and injective on every fiber, built from a partition of unity, and such a map is the same data as a map to a Grassmannian. This subsection carries out that route, because the multiplicative theory of section 7.4 (naturality, the splitting principle, the Whitney sum formula) is applied to flag bundles, and a flag bundle carries no CW structure that this book constructs. Without the paracompact classification the theory would be defined only on spaces it is not applied to.

Two elementary facts about vector bundles come first. Both serve to split a subbundle off, which is how the flag bundle is built one stage at a time.

Lemma 7.1.4: Inner Products on a Vector Bundle

Let $E \to X$ be a real vector bundle over a paracompact Hausdorff base. Then E admits a *Euclidean inner product*: a continuous map $E \oplus E \to \mathbb{R}$ restricting on every fiber to a positive-definite symmetric bilinear form. A complex vector bundle over such a base admits a *Hermitian inner product*, restricting on every fiber to a positive-definite conjugate-symmetric sesquilinear form.

***Proof*:**
Let $\{U_\alpha\}$ be an open cover of X by sets over which E is trivial, and for each α let $h_\alpha \colon p^{-1}(U_\alpha) \to \mathbb{R}^n$ be the fiber coordinate of a trivialization. Pulling back the standard inner product of $\mathbb{R}^n$ gives a continuous form $\langle v, w\rangle_\alpha = h_\alpha(v) \cdot h_\alpha(w)$ over U_α, symmetric and positive-definite on each fiber there. Let $\{\varphi_\alpha\}$ be a partition of unity subordinate to the cover, which exists by [12, Existence of Subordinate Partitions of Unity], and set

$$\langle v, w\rangle = \sum_\alpha \varphi_\alpha\big(p(v)\big)\, \langle v, w\rangle_\alpha$$

reading a term as 0 off $p^{-1}(\mathrm{supp}\varphi_\alpha)$. The supports are locally finite, so near any point of X the sum has finitely many nonzero terms and is continuous there; continuity is local, so the form is continuous. On the fiber over x it is a combination of symmetric bilinear forms with nonnegative coefficients, hence symmetric and bilinear. For positive-definiteness let $v \neq 0$ lie in that fiber: each α with $\varphi_\alpha(x) > 0$ has $x \in U_\alpha$ and contributes $\varphi_\alpha(x)\, |h_\alpha(v)|^2 > 0$, no term is negative, and at least one such α exists because the φ_α sum to 1. Hence $\langle v, v\rangle > 0$. The complex case is the same construction with $h_\alpha \colon p^{-1}(U_\alpha) \to \mathbb{C}^n$ and the standard Hermitian form: the multipliers φ_α are real, so a nonnegative combination of conjugate-symmetric sesquilinear forms is again one, and the positivity argument is unchanged, as we sought to show.

Proposition 7.1.5: Subbundles Have Complements

Let $E \to X$ be a vector bundle over a paracompact Hausdorff base, real or complex, and let $E_0 \subseteq E$ be a subbundle. Then there is a subbundle $E_0^\perp \subseteq E$ with $E \cong E_0 \oplus E_0^\perp$.

***Proof*:**
Fix an inner product on E by lemma 7.1.4, Euclidean in the real case and Hermitian in the complex case, and let $E_0^\perp \subseteq E$ be the set of vectors orthogonal to E_0 within their own fiber. Each fiber $(E_0^\perp)_x$ is the orthogonal complement of $(E_0)_x$ in E_x, of dimension $n - m$ for n and m the ranks of E and E_0, so the fiberwise sum $E_0 \oplus E_0^\perp \to E$, $(v, w) \mapsto v + w$, is a linear isomorphism on every fiber. It is continuous, being a restriction of the addition of E, so once $E_0^\perp$ is known to be a subbundle this map is an isomorphism of vector bundles.

Local triviality of $E_0^\perp$ is a local question on X, so fix x_0 and work over a neighbourhood on which both E and E_0 are trivial. A trivialization of E_0 supplies sections $s_1, \ldots, s_m$ that form a basis of $(E_0)_x$ for every nearby x. Gram-Schmidt applied to $s_1, \ldots, s_m$ against the inner product yields sections $e_1, \ldots, e_m$ orthonormal in every nearby fiber; the process is a composite of inner products, subtractions and square roots of positive quantities, hence continuous in x, and it is defined because the s_i are independent in each fiber. The fiberwise orthogonal projection onto the complement,

$$\pi(v) = v - \sum_{i=1}^{m} \langle v, e_i(x)\rangle\, e_i(x), \qquad v \in E_x$$

is then continuous over that neighbourhood. Extend $e_1, \ldots, e_m$ to sections $e_1, \ldots, e_n$ spanning each nearby fiber of E, by completing a basis at x_0 and taking the same vectors in nearby fibers under the trivialization of E, which stay independent near x_0 since independence is an open condition. Then $\pi(e_{m+1}), \ldots, \pi(e_n)$ are sections of $E_0^\perp$ spanning each nearby fiber of it: π carries the spanning set $e_1, \ldots, e_n$ of E_x onto a spanning set of $(E_0^\perp)_x$ while killing $e_1, \ldots, e_m$.

As $\dim(E_0^\perp)_x = n - m$, these $n - m$ spanning sections are a basis in every nearby fiber, and $(x, a_{m+1}, \dots, a_n) \mapsto \sum_j a_j \, \pi(e_j)(x)$ is a fiberwise-linear homeomorphism onto the part of $E_0^\perp$ over that neighbourhood. Hence $E_0^\perp$ is locally trivial, completing the proof.

The same inner product names a construction used throughout the Thom and Gysin arguments.

Definition 7.1.6: Sphere Bundle

Let $p\colon E \to B$ be a real vector bundle of rank n over a paracompact Hausdorff base, equipped with an inner product (which exists by lemma 7.1.4). The *sphere bundle* $S(E) \to B$ is the subspace of E consisting of unit vectors,

$$S(E) = \{v \in E \mid \langle v, v\rangle = 1\}$$

with the restricted projection. It is a fiber bundle with fiber the unit sphere $S^{n-1} \subset \mathbb{R}^n$.

Vector bundles and principal bundles are two views of the same object, and the passage between them is what lets the classification of theorem 2.4.7 and the homotopy invariance of lemma 2.4.6, both stated for principal bundles, be applied to vector bundles. The correspondence is recorded here rather than cited, because everything it needs is now available.

Lemma 7.1.7: Vector Bundles and Their Frame Bundles

Let X be paracompact Hausdorff. Sending a rank-n real vector bundle $E \to X$, equipped with a Euclidean inner product, to its bundle $\mathrm{Fr}(E) \to X$ of orthonormal frames, and sending a principal $O(n)$-bundle $P \to X$ to the associated bundle $P \times_{O(n)} \mathbb{R}^n$, are mutually inverse up to isomorphism. Both constructions commute with pullback: $\mathrm{Fr}(f^*E) \cong f^*\mathrm{Fr}(E)$ and $(f^*P) \times_{O(n)} \mathbb{R}^n \cong f^*(P \times_{O(n)} \mathbb{R}^n)$. The same holds for complex bundles with $U(n)$ and a Hermitian inner product.

***Proof*:**
An inner product exists by lemma 7.1.4. The fiber of $\mathrm{Fr}(E)$ over x is the set of orthonormal bases of E_x, on which $O(n)$ acts simply transitively by change of basis, so $\mathrm{Fr}(E)$ is a principal $O(n)$-bundle; a local trivialization of E over U, made orthonormal fiberwise by Gram-Schmidt as in proposition 7.1.5, trivializes $\mathrm{Fr}(E)$ over U. Conversely $P \times_{O(n)} \mathbb{R}^n$, the quotient of $P \times \mathbb{R}^n$ by $(p \cdot g, v) \sim (p, gv)$, is locally $U \times \mathbb{R}^n$ and fiberwise linear, hence a rank-n vector bundle, and it carries the inner product descending from the standard one on $\mathbb{R}^n$, which is $O(n)$-invariant.

The two composites are the identity up to isomorphism. Starting from E, the map $\mathrm{Fr}(E) \times_{O(n)} \mathbb{R}^n \to E$ sending $[(e_1, \dots, e_n), v]$ to $\sum_i v_i e_i$ is well defined, since replacing the frame by g-times-itself replaces v by $g^{-1}v$, and it is continuous, fiberwise linear and bijective, hence an isomorphism. Starting from P, a point of $\mathrm{Fr}(P \times_{O(n)} \mathbb{R}^n)$ is an orthonormal frame of a fiber, and $p \mapsto [(p, e_1)], \dots, [(p, e_n)]$ for the standard basis identifies P with it equivariantly. Both constructions are defined fiberwise from local trivializations, and pullback replaces the fiber over x by the fiber over $f(x)$, which gives the two compatibilities. The complex case replaces orthonormal by unitary throughout, as we sought to show.

The classification itself rests on one construction, a fiberwise linear injection into $\mathbb{R}^\infty$. It is what replaces, for vector bundles over a paracompact base, the skeletal induction that proves theorem 2.4.7.

Lemma 7.1.8: Fiberwise Linear Injections into $\mathbb{R}^\infty$

Let $p\colon E \to X$ be a rank-n real vector bundle over a paracompact Hausdorff base, and let $\mathbb{R}^\infty = \bigcup_N \mathbb{R}^N$ carry the weak topology. Then there is a map $g\colon E \to \mathbb{R}^\infty$ whose restriction to every fiber is linear and injective, and any two such maps are homotopic through maps whose restriction to every fiber is linear and injective. The same statements hold for a rank-n complex vector bundle with $\mathbb{C}^\infty$ in place of $\mathbb{R}^\infty$ and complex-linear restrictions in place of linear ones.

***Proof*:**
Step 1: A countable trivializing cover with a subordinate partition of unity. Let $\{U_\alpha\}$ be an open cover of X by sets over which E is trivial, and let $\{\varphi_\alpha\}$ be a partition of unity subordinate to it, given by [12, Existence of Subordinate Partitions of Unity]: continuous functions with locally finite supports, $\mathrm{supp}\varphi_\alpha \subseteq U_\alpha$, summing to 1. For a finite set S of indices let

$$V_S = \{x \in X : \varphi_\alpha(x) > \varphi_\beta(x) \text{ for all } \alpha \in S \text{ and all } \beta \notin S\}$$

Near any point only finitely many φ_α are nonzero, so near that point the set V_S is cut out by finitely many strict inequalities between continuous functions, and V_S is open. For $\alpha \in S$ the function φ_α is positive on V_S, so $V_S \subseteq \mathrm{supp}\varphi_\alpha \subseteq U_\alpha$, and E is trivial over V_S. If $S \neq S'$ have the same number of elements, choose $\alpha \in S \setminus S'$ and $\beta \in S' \setminus S$; on $V_S \cap V_{S'}$ both $\varphi_\alpha > \varphi_\beta$ and $\varphi_\beta > \varphi_\alpha$ would hold, so $V_S \cap V_{S'} = \varnothing$. Let W_k be the union of the sets V_S over the S with exactly k elements, $k \geq 1$. Each x lies in $V_{S(x)}$ for $S(x) = \{\alpha : \varphi_\alpha(x) > 0\}$, a finite set by local finiteness, so $\{W_k\}_{k\geq 1}$ is a countable open cover of X; each W_k is a disjoint union of open sets over each of which the rank-n bundle E is trivial, and trivializations over disjoint open sets amalgamate, so E is trivial over each W_k. This is the countable-cover argument of [15, Lemma 1.21]. Applying [12, Existence of Subordinate Partitions of Unity] once more, now to the countable cover $\{W_k\}$, gives a partition of unity $\{\varphi_k\}_{k\geq 1}$ indexed by the same countable set with $\mathrm{supp}\varphi_k \subseteq W_k$.

Step 2: The map. For each k fix a trivialization of E over W_k and let $h_k\colon p^{-1}(W_k) \to \mathbb{R}^n$ be its fiber coordinate, a linear isomorphism on every fiber over W_k. Define $g_k\colon E \to \mathbb{R}^n$ by $g_k(v) = \varphi_k\big(p(v)\big)\, h_k(v)$ on $p^{-1}(W_k)$ and $g_k = 0$ off $p^{-1}\big(\mathrm{supp}\varphi_k\big)$; the two prescriptions agree on the overlap, their open domains cover E because $\mathrm{supp}\varphi_k \subseteq W_k$, and so g_k is continuous. Set

$$g = (g_1, g_2, \dots)\colon E \longrightarrow \mathbb{R}^n \oplus \mathbb{R}^n \oplus \cdots = \mathbb{R}^\infty$$

Near any point of X only finitely many φ_k are nonzero, so over a suitable neighbourhood g takes values in a finite partial sum $\mathbb{R}^{nm} \subseteq \mathbb{R}^\infty$ and is continuous into it; continuity is a local property, so g is continuous into the weak topology. On the fiber over x each coordinate g_k is the scalar $\varphi_k(x)$ times a map linear on that fiber, so g is linear there, and choosing k with $\varphi_k(x) > 0$ shows g is injective there, the block g_k already being injective on that fiber.

Step 3: Any two are homotopic through fiberwise linear injections. Let g, g' be two such maps and let $J_{\mathrm{o}}, J_{\mathrm{e}}\colon \mathbb{R}^\infty \to \mathbb{R}^\infty$ be the linear injections into the odd and the even coordinates, $J_{\mathrm{o}}(z_1, z_2, \dots) = (z_1, 0, z_2, 0, \dots)$ and $J_{\mathrm{e}}(z_1, z_2, \dots) = (0, z_1, 0, z_2, \dots)$. The straight line $L_t = (1-t)\,\mathrm{id} + t\,J_{\mathrm{o}}$ consists of injective linear maps: for $0 < t < 1$ and a nonzero z with highest nonzero coordinate z_m, if $m \geq 2$ then coordinate $2m - 1 > m$ of $L_t z$ equals $t\,z_m \neq 0$, while if $m = 1$ the first coordinate of $L_t z$ equals $(1-t)z_1 + tz_1 = z_1 \neq 0$; for $t = 0, 1$ injectivity is clear. Hence $g \simeq J_{\mathrm{o}} \circ g$ through the fiberwise linear injections $L_t \circ g$, and likewise $g' \simeq J_{\mathrm{e}} \circ g'$ through

$L'_t \circ g'$ for $L'_t = (1-t)\,\mathrm{id} + t\,J_\mathrm{e}$. The images of J_o and J_e meet only in 0, so the straight line $(1-t)\,J_\mathrm{o} \circ g + t\,J_\mathrm{e} \circ g'$ is, for each t, linear on every fiber and injective there: a nonzero fiber vector v has $g(v) \neq 0$ and $g'(v) \neq 0$, the odd coordinates of the combination are $(1-t)$ times those of $J_\mathrm{o} g(v)$ and the even ones are t times those of $J_\mathrm{e} g'(v)$, and for every $t \in [0,1]$ at least one of the factors $1-t$, t is nonzero. Continuity of the three homotopies is not the local finiteness of Step 2, which was a property of the *constructed* map g and is unavailable for arbitrary g, g'; it comes from the topology of $\mathbb{R}^\infty$ itself. The product topology on $\mathbb{R}^\infty \times \mathbb{R}^\infty$ agrees with the colimit topology of the subspaces $\mathbb{R}^N \times \mathbb{R}^N$: one inclusion is immediate, each $\mathbb{R}^N \times \mathbb{R}^N$ carrying its product topology as a subspace, and for the other let U meet every $\mathbb{R}^N \times \mathbb{R}^N$ in an open set and let $(a,b) \in U$ with $a, b \in \mathbb{R}^{N_0}$. Choose opens $A_{N_0} \ni a$, $B_{N_0} \ni b$ in $\mathbb{R}^{N_0}$ with compact closures and $\overline{A_{N_0}} \times \overline{B_{N_0}} \subseteq U$, and inductively, the compact product $\overline{A_N} \times \overline{B_N}$ lying in the open set $U \cap (\mathbb{R}^{N+1} \times \mathbb{R}^{N+1})$, the tube lemma [18, §26] and local compactness of $\mathbb{R}^{N+1}$ give opens $A_{N+1} \supseteq \overline{A_N}$, $B_{N+1} \supseteq \overline{B_N}$ in $\mathbb{R}^{N+1}$ with compact closures and $\overline{A_{N+1}} \times \overline{B_{N+1}} \subseteq U$; the unions $A = \bigcup_N A_N$ and $B = \bigcup_N B_N$ are open in the weak topology, since $A \cap \mathbb{R}^M = \bigcup_{N \geq M} (A_N \cap \mathbb{R}^M)$, and satisfy $(a,b) \in A \times B \subseteq U$ by the nesting. Multiplying by the locally compact Hausdorff interval, $\mathbb{R}^\infty \times \mathbb{R}^\infty \times I$ then carries the colimit topology of the subspaces $\mathbb{R}^N \times \mathbb{R}^N \times I$ as well [12, Products of Quotient Maps]. The combination map $(y,z,t) \mapsto (1-t)\,y + t\,z$ is continuous on each $\mathbb{R}^N \times \mathbb{R}^N \times I$ into $\mathbb{R}^N$, hence continuous on the colimit; in particular addition and scalar multiplication are continuous and $\mathbb{R}^\infty$ is a topological vector space in the weak topology. Each of the three homotopies is the composite of this combination map with a map to $\mathbb{R}^\infty \times \mathbb{R}^\infty \times I$ having continuous coordinates, such as $(v,t) \mapsto \big(J_\mathrm{o} g(v),\, J_\mathrm{e} g'(v),\, t\big)$ for the third, hence is continuous, and their concatenation proves the claim.

Step 4: The complex case. Replace $\mathbb{R}^n$ and $\mathbb{R}^\infty$ by $\mathbb{C}^n$ and $\mathbb{C}^\infty$ and linear by complex-linear throughout: the partitions of unity are real-valued scalars, hence complex-linear multipliers, and the interleaving maps $J_\mathrm{o}, J_\mathrm{e}$ are complex-linear. The three steps repeat verbatim, as we sought to show.

A fiberwise linear injection into $\mathbb{R}^\infty$ is the same data as a map to the infinite Grassmannian: send x to the n-plane $g(E_x)$. Since $\mathrm{Gr}_n(\mathbb{R}^\infty)$ is a model for $BO(n)$, by the real analogue of example 2.4.9, the lemma classifies vector bundles over a paracompact base, and this is the form in which the rest of the chapter uses it.

Theorem 7.1.9: Classification of Vector Bundles over a Paracompact Base

Let X be a paracompact Hausdorff space and $n \geq 0$. Pulling back the universal bundle $\gamma^n \to \mathrm{Gr}_n(\mathbb{R}^\infty) = BO(n)$ gives a bijection

$$[X, BO(n)] \xrightarrow{\cong} \mathrm{Vect}^n_{\mathbb{R}}(X), \qquad [f] \longmapsto f^*\gamma^n$$

from free homotopy classes of maps to isomorphism classes of rank-n real vector bundles over X, natural in X: for a map $u \colon X' \to X$ the bundle $u^*(f^*\gamma^n)$ is classified by $f \circ u$. The same holds for rank-n complex bundles with $\mathrm{Gr}_n(\mathbb{C}^\infty) = BU(n)$ and the universal complex bundle.

***Proof*:**
Well-definedness. The assignment must be shown to depend only on the homotopy class of f, that is, that $f_0 \simeq f_1$ implies $f_0^*\gamma^n \cong f_1^*\gamma^n$. Pass to principal bundles: fix a Euclidean inner product on γ^n by lemma 7.1.4 and let $P \to BO(n)$ be its bundle of orthonormal frames, a principal

$O(n)$-bundle whose associated rank-n vector bundle is γ^n again by lemma 7.1.7. Pullback commutes with both constructions by the same lemma, so $f_i^*\gamma^n$ is the vector bundle associated to f_i^*P. By lemma 2.4.6, whose hypothesis is exactly that the base be paracompact, $f_0 \simeq f_1$ gives $f_0^*P \cong f_1^*P$ as principal $O(n)$-bundles, and applying the associated-bundle construction to that isomorphism gives $f_0^*\gamma^n \cong f_1^*\gamma^n$. Hence the map on homotopy classes is defined. (In the complex case replace $O(n)$ by $U(n)$ and the Euclidean inner product by a Hermitian one.)

Surjectivity. Let $E \to X$ be a rank-n real bundle and let $g\colon E \to \mathbb{R}^\infty$ be fiberwise linear and injective, by lemma 7.1.8. Define $f\colon X \to \mathrm{Gr}_n(\mathbb{R}^\infty)$ by $f(x) = g(E_x)$. Over a neighbourhood on which E is trivial, a frame $e_1, \dots, e_n$ gives $f(x) = \operatorname{span}\big(g(e_1(x)), \dots, g(e_n(x))\big)$ with the $g(e_i(x))$ continuous in x and independent in every fiber, so f is continuous into the Grassmannian, whose topology is that of an n-frame up to change of basis. The map $E \to f^*\gamma^n$ sending $v \in E_x$ to $\big(x, g(v)\big)$, legitimate because $g(v)$ lies in the plane $f(x)$, which is the fiber of γ^n over $f(x)$, is continuous and fiberwise linear and injective, hence an isomorphism of vector bundles.

Injectivity. Suppose $f_0^*\gamma^n \cong f_1^*\gamma^n$ for maps $f_0, f_1\colon X \to \mathrm{Gr}_n(\mathbb{R}^\infty)$, and call the common bundle E. Composing each isomorphism $E \cong f_i^*\gamma^n$ with the tautological inclusion of the fiber plane into $\mathbb{R}^\infty$ gives fiberwise linear injections $g_0, g_1\colon E \to \mathbb{R}^\infty$ inducing f_0 and f_1. By the homotopy clause of lemma 7.1.8 there is a homotopy g_t from g_0 to g_1 through fiberwise linear injections; taking images fiberwise, $(x,t) \mapsto g_t(E_x)$ is a homotopy $X \times I \to \mathrm{Gr}_n(\mathbb{R}^\infty)$ from f_0 to f_1, continuous by the frame description above applied to the bundle $E \times I \to X \times I$. Hence $[f_0] = [f_1]$.

Naturality. Pullback is functorial, $u^*(f^*\gamma^n) \cong (f \circ u)^*\gamma^n$, which is the stated compatibility. The complex case replaces $\mathbb{R}$ by $\mathbb{C}$ and linear by complex-linear throughout, on the complex clause of lemma 7.1.8, completing the proof.

Over a base that is both a CW complex and paracompact Hausdorff, theorem 7.1.9 and theorem 2.4.7 classify the same bundles and they agree, both sending a map to the pullback of the universal bundle along it. Nothing below depends on which of the two supplies a classifying map.

The line-bundle case is the seed of the general theory, and the strategy for higher rank is to reduce to it. There are two routes to $H^*(BU(n))$, and both are developed below because each is illuminating. The first inducts on n through the universal sphere bundle, feeding the Gysin sequence of theorem 6.3.2. The second, the projective-bundle route, expresses $H^*(BU(n))$ by the Leray-Hirsch theorem (theorem 6.5.1) and produces the splitting principle directly. Both routes are carried out in full here; the bundle-theoretic development through the Grassmannian model is section 7.1.1 above, which is what makes the classes available over the paracompact bases [21, Convention: Paracompact Includes Hausdorff] works with.

The inductive route rests on one geometric fact about classifying spaces, the universal sphere bundle relating consecutive unitary groups. The quotient $U(n)/U(n-1)$ is the unit sphere S^{2n-1} in $\mathbb{C}^n$, because $U(n)$ acts transitively on that sphere with stabilizer $U(n-1)$ (the subgroup fixing a unit vector). Applying the classifying-space functor to the inclusion $U(n-1) \hookrightarrow U(n)$ produces a fibration of classifying spaces whose fiber is exactly this quotient sphere.

Lemma 7.1.10: The Universal Sphere Bundle for $U(n)$

For $n \geq 1$ the inclusion $U(n-1) \hookrightarrow U(n)$ induces a fibration

$$S^{2n-1} \longrightarrow BU(n-1) \xrightarrow{\rho} BU(n)$$

with fiber the sphere S^{2n-1}, in which ρ is the map of classifying spaces induced by the inclusion. The fibration is the sphere bundle $S(\gamma_n)$ of the universal rank-n complex bundle $\gamma_n \to BU(n)$: a point of $BU(n-1)$ over $b \in BU(n)$ is a unit vector in the fiber $(\gamma_n)_b$, and ρ forgets the vector.

***Proof*:**
Step 1: The homogeneous space is a sphere. The group $U(n)$ acts on the unit sphere $S^{2n-1} \subseteq \mathbb{C}^n$ by its standard action on $\mathbb{C}^n$. This action is transitive, since any unit vector can be carried to the first standard basis vector e_1 by a unitary transformation. The stabilizer of e_1 is the subgroup of unitary matrices fixing e_1, which acts unitarily on the orthogonal complement $e_1^\perp \cong \mathbb{C}^{n-1}$ and so is $U(n-1)$. Hence $U(n)/U(n-1) \cong S^{2n-1}$ as $U(n)$-spaces.

Step 2: Classifying spaces of a subgroup form a sphere bundle. For a closed subgroup $H \subseteq G$ acting on a contractible free G-space EG, the space EG/H is a model for BH, since H acts freely on the contractible EG (definition 2.4.3). The projection $EG/H \to EG/G = BG$ induced by the quotient is then a fiber bundle with fiber G/H: locally over a trivializing open set $V \subseteq BG$ where $EG|_V \cong V \times G$, the preimage is $V \times (G/H)$, and the transition functions act through the G-action on G/H. Applying this with $G = U(n)$, $H = U(n-1)$, and $EG = EU(n)$ gives a fiber bundle

$$EU(n)/U(n-1) \longrightarrow EU(n)/U(n) = BU(n)$$

with fiber $U(n)/U(n-1) = S^{2n-1}$ by Step 1. The total space $EU(n)/U(n-1)$ is a model for $BU(n-1)$, because $EU(n)$ is a contractible free $U(n-1)$-space (it is free even for the larger group $U(n)$, hence free for the subgroup, and it is contractible), so it is a model for $EU(n-1)$ and its quotient by $U(n-1)$ is $BU(n-1)$. This identifies the bundle as $S^{2n-1} \to BU(n-1) \to BU(n)$.

Step 3: Identification with the sphere bundle of γ_n. The universal bundle $\gamma_n \to BU(n)$ is the rank-n complex vector bundle $EU(n) \times_{U(n)} \mathbb{C}^n \to BU(n)$ associated to the standard representation. Its unit sphere bundle $S(\gamma_n)$ has fiber the unit sphere of $\mathbb{C}^n$, namely S^{2n-1}, with $U(n)$ acting as in Step 1, so $S(\gamma_n) = EU(n) \times_{U(n)} S^{2n-1} = EU(n)/U(n-1)$, the last equality because $U(n)$ acts transitively on S^{2n-1} with stabilizer $U(n-1)$. This is the total space $BU(n-1)$ of Step 2, and the bundle projection is ρ. A point of $BU(n-1)$ over $b \in BU(n)$ is thus a unit vector in $(\gamma_n)_b$, and ρ forgets it, completing the proof.

The lemma packages the geometry that drives the induction: passing from $U(n)$ to $U(n-1)$ is the same as passing from the universal rank-n bundle to its unit sphere bundle, and the cohomological cost of that passage is governed by one Euler class through the Gysin sequence. Because the fiber is the odd sphere S^{2n-1}, the Gysin sequence has its differential in degree $2n$, and the Euler class of the universal bundle lives in $H^{2n}(BU(n))$. This single class is the top Chern class c_n, and the induction shows that adjoining it to the lower Chern classes is exactly what builds $H^*(BU(n))$ from $H^*(BU(n-1))$. The base case $n = 1$ is example 7.1.3.

Theorem 7.1.11: The Cohomology of $BU(n)$

For every $n \geq 0$ there is an isomorphism of graded rings

$$H^*(BU(n);\mathbb{Z}) \cong \mathbb{Z}[c_1, c_2, \ldots, c_n], \qquad |c_i| = 2i$$

a polynomial ring on n generators c_i in degree $2i$. The generators may be chosen so that $\rho^*\colon H^*(BU(n)) \to H^*(BU(n-1))$ sends $c_i \mapsto c_i$ for $i \leq n-1$ and $c_n \mapsto 0$, where $\rho\colon BU(n-1) \to BU(n)$ is the map of lemma 7.1.10. In particular the cohomology is concentrated in even degrees and is torsion-free.

***Proof*:**
The argument is induction on n, the inductive step run through the Gysin sequence of the fibration of lemma 7.1.10.

Step 1: Base cases. For $n = 0$ the group $U(0)$ is trivial, $BU(0)$ is a point, and $H^*(\mathrm{pt};\mathbb{Z}) = \mathbb{Z} = \mathbb{Z}[\,]$, the empty polynomial ring. For $n = 1$, example 7.1.3 gives $H^*(BU(1);\mathbb{Z}) = \mathbb{Z}[c_1]$ with $|c_1| = 2$, the case $n = 1$ of the claim.

Step 2: The Gysin sequence of the universal sphere bundle. Assume $n \geq 2$ and that $H^*(BU(n-1);\mathbb{Z}) \cong \mathbb{Z}[c_1, \ldots, c_{n-1}]$ with $|c_i| = 2i$, concentrated in even degrees and torsion-free. The fibration $S^{2n-1} \to BU(n-1) \xrightarrow{\rho} BU(n)$ of lemma 7.1.10 has fiber the odd sphere S^{2n-1}, with $k = 2n-1$. The base $BU(n)$ is simply connected, since $BU(n)$ has $\pi_1 \cong \pi_0(U(n)) = 0$ by the delooping relation of corollary 2.4.8 and the path-connectedness of $U(n)$; hence the orientation hypothesis of theorem 6.3.2 holds and the Gysin sequence applies. With $e \in H^{2n}(BU(n);\mathbb{Z})$ the Euler class, it reads

$$\cdots \to H^m(BU(n)) \xrightarrow{\smile e} H^{m+2n}(BU(n)) \xrightarrow{\rho^*} H^{m+2n}(BU(n-1)) \xrightarrow{\int} H^{m+1}(BU(n)) \xrightarrow{\smile e} H^{m+2n+1}(BU(n)) \to \cdots$$

where $\int = \int_{S^{2n-1}}$ lowers degree by $2n-1$.

Step 3: ρ^* is surjective and $\int$ vanishes. The inductive hypothesis gives $H^*(BU(n-1)) = \mathbb{Z}[c_1, \ldots, c_{n-1}]$ concentrated in even degrees. The fiber-integration $\int$ lowers degree by the odd number $2n-1$, so it carries the even-degree group $H^{m+2n}(BU(n-1))$ into the odd-degree group $H^{m+1}(BU(n))$ whenever m is even, and the even-degree group into the even-degree group only when m is odd. The strategy is to prove by induction on total degree that $H^{\mathrm{odd}}(BU(n)) = 0$ and that ρ^* is surjective; the two statements feed each other through the sequence. Suppose, as an inner induction on d, that $H^j(BU(n)) = 0$ for all odd $j < d$. Consider the segment of the Gysin sequence ending in $H^d(BU(n))$ with d odd:

$$H^{d-2n}(BU(n)) \xrightarrow{\smile e} H^d(BU(n)) \xrightarrow{\rho^*} H^d(BU(n-1))$$

Here $H^d(BU(n-1)) = 0$ because d is odd and the fiber cohomology is even, so ρ^* from $H^d(BU(n))$ is the zero map and its kernel is all of $H^d(BU(n))$. Exactness gives $H^d(BU(n)) = \mathrm{im}(\smile e)$, the image of $\smile e\colon H^{d-2n}(BU(n)) \to H^d(BU(n))$. Since e has even degree $2n$, $\smile e$ sends the odd-degree group $H^{d-2n}(BU(n))$ (note $d-2n$ is odd) into H^d, and $H^{d-2n}(BU(n)) = 0$ by the inner inductive hypothesis ($d-2n < d$ odd). Hence $H^d(BU(n)) = 0$. By induction on d, the cohomology of $BU(n)$ vanishes in all odd degrees.

With $H^{\mathrm{odd}}(BU(n)) = 0$ established, return to the general segment

$$H^{m+2n}(BU(n)) \xrightarrow{\rho^*} H^{m+2n}(BU(n-1)) \xrightarrow{\int} H^{m+1}(BU(n))$$

When $m+2n$ is even (that is m even), the term $H^{m+1}(BU(n))$ is odd-degree, hence zero, so $\int = 0$ on $H^{m+2n}(BU(n-1))$ and exactness forces ρ^* to be surjective onto the even-degree groups. When $m+2n$ is odd, both $H^{m+2n}(BU(n-1))$ (odd, fiber even) and the source $H^{m+2n}(BU(n))$ (odd, just shown zero) vanish, so there is nothing to check. Therefore $\rho^*\colon H^*(BU(n)) \to H^*(BU(n-1))$ is surjective in every degree, and $\int = 0$ throughout.

Step 4: $\smile e$ is injective, and the structure of the ring. With $\int = 0$, the Gysin sequence breaks into short exact sequences

$$0 \to H^m(BU(n)) \xrightarrow{\smile e} H^{m+2n}(BU(n)) \xrightarrow{\rho^*} H^{m+2n}(BU(n-1)) \to 0$$

for every m: exactness at $H^m(BU(n))$ says $\ker(\smile e) = \mathrm{im}(\int) = 0$ (the incoming $\int$ is zero), so $\smile e$ is injective; exactness at $H^{m+2n}(BU(n))$ says $\mathrm{im}(\smile e) = \ker(\rho^*)$; and surjectivity of ρ^* was Step 3. Thus for each m,

$$H^{m+2n}(BU(n)) \cong H^m(BU(n)) \cdot e \;\oplus\; \big(\text{a lift of } H^{m+2n}(BU(n-1))\big)$$

as groups, the first summand the image of $\smile e$ and the second a section of ρ^* (which exists because the groups are free abelian, being torsion-free of finite rank in each degree). Set $c_n := e \in H^{2n}(BU(n))$, and for $i < n$ let $c_i \in H^{2i}(BU(n))$ be any class with $\rho^* c_i = c_i$ (the lower Chern class of $BU(n-1)$); such a lift exists by surjectivity of ρ^*. These choices give a ring homomorphism

$$\Phi\colon \mathbb{Z}[c_1, \ldots, c_n] \longrightarrow H^*(BU(n); \mathbb{Z})$$

sending each polynomial generator to the chosen class. To see Φ is an isomorphism, filter $\mathbb{Z}[c_1, \ldots, c_n]$ by powers of c_n: every element is uniquely $\sum_{j \geq 0} p_j\, c_n^j$ with $p_j \in \mathbb{Z}[c_1, \ldots, c_{n-1}]$. The short exact sequence above says precisely that $H^*(BU(n))$ is a free module over the subring generated by $c_1, \ldots, c_{n-1}$ with basis the powers $\{c_n^j\}_{j \geq 0}$: applying the short exact sequence iteratively, multiplication by $e = c_n$ is injective with cokernel ρ^*-isomorphic to $H^*(BU(n-1)) = \mathbb{Z}[c_1, \ldots, c_{n-1}]$, so

$$H^*(BU(n)) \cong H^*(BU(n-1))[c_n] = \mathbb{Z}[c_1, \ldots, c_{n-1}][c_n] = \mathbb{Z}[c_1, \ldots, c_n]$$

by induction on the power of c_n and dimension count in each degree. This is exactly the statement that Φ is an isomorphism. The relation $\rho^* c_i = c_i$ for $i \leq n-1$ holds by construction, and $\rho^* c_n = \rho^* e = 0$ because the Euler class restricts to zero on the total space of its own sphere bundle (the class $e = d_{2n}(1 \otimes \alpha)$ lies in the kernel of ρ^* as $\mathrm{im}(\smile e) = \ker \rho^*$ contains $e = 1 \smile e$). The induction closes, and the cohomology is concentrated in even degrees and torsion-free, completing the proof.

The theorem is the central computation of the chapter, and its shape repays attention. The cohomology of $BU(n)$ is as simple as a graded ring can be that is not a single copy of $\mathbb{Z}$: a polynomial ring, free of relations, on one generator in each even degree up to $2n$. The generator c_n is the Euler class of the universal bundle, born as the transgression of the fiber generator in the Gysin sequence; the lower generators c_i are inherited from $BU(n-1)$ and ultimately from the line-bundle class of example 7.1.3. The absence of relations is what makes Chern classes independent invariants: no polynomial identity holds among the c_i universally, so a characteristic class of complex rank-n bundles is precisely a polynomial in the c_i with integer coefficients, and two such polynomials that differ give distinct invariants. The even-degree, torsion-free conclusion also means there is no information lost in passing to rational or complex coefficients, which is why the Chern-Weil construction of the Riemannian-geometry volume [11] over $\mathbb{R}$ recovers the same classes.

Taking the colimit over n assembles the stable theory, where the rank is allowed to grow without bound and the polynomial ring acquires infinitely many generators.

Corollary 7.1.12: The Cohomology of BU

Let $BU = \varinjlim_n BU(n)$ be the colimit along the maps ρ of lemma 7.1.10. Then

$$H^*(BU;\mathbb{Z}) \cong \mathbb{Z}[c_1, c_2, c_3, \dots], \qquad |c_i| = 2i$$

the polynomial ring on countably many generators, one in each positive even degree.

***Proof*:**

The space BU is the colimit of the tower $\cdots \to BU(n-1) \xrightarrow{\rho} BU(n) \to \cdots$, and cohomology of a colimit of CW inclusions is the limit of the cohomologies (the cohomology of a colimit over an increasing sequence of CW subcomplexes is computed by the Milnor exact sequence, whose $\varprojlim^1$ term vanishes here because the maps $\rho^*\colon H^k(BU(n)) \to H^k(BU(n-1))$ are surjective in each fixed degree k, by theorem 7.1.11, so the tower satisfies the Mittag-Leffler condition and its $\varprojlim^1$ vanishes by [10, Mittag-Leffler vanishing]; for the Milnor sequence itself see [14, §3.F]). In each fixed degree k, the restriction $H^k(BU(n)) \to H^k(BU(n-1))$ is an isomorphism once $2n > k$, since the only generators of degree $\leq k$ are $c_1, \dots, c_{\lfloor k/2 \rfloor}$, all present and matched by $\rho^* c_i = c_i$ for $i \leq n-1$. The limit of the system $\{\mathbb{Z}[c_1, \dots, c_n]\}_n$, taken degreewise, is therefore $\mathbb{Z}[c_1, c_2, \dots]$ with $|c_i| = 2i$, as the generators stabilize and no new relations appear, completing the proof.

The stable ring $H^*(BU;\mathbb{Z}) = \mathbb{Z}[c_1, c_2, \dots]$ is the universal home of Chern classes: every complex vector bundle of every rank has a classifying map into BU (after stabilizing by adding trivial summands), and pulling back the universal c_i gives the Chern classes of the bundle, independent of the rank used to define them. This stability is what makes the total Chern class, introduced next, multiplicative under Whitney sum, and it is the cohomological counterpart of the fact that BU is the classifying space for the reduced K-theory $\widetilde{K}^0$ of [21, Representability of K^0].

With the ring computed, the named classes can be defined. The Chern classes are the polynomial generators, pulled back to a bundle; the total Chern class packages them into a single inhomogeneous class whose virtue is multiplicativity.

Definition 7.1.13: Chern Classes and the Total Chern Class

Let $E \to X$ be a complex vector bundle of rank n over a paracompact Hausdorff base X, with classifying map $f_E\colon X \to BU(n)$, unique up to homotopy by theorem 7.1.9. For $1 \leq i \leq n$ the i*-th Chern class* of E is
$$c_i(E) := f_E^* c_i \in H^{2i}(X;\mathbb{Z})$$
the pullback of the polynomial generator $c_i \in H^{2i}(BU(n);\mathbb{Z})$ of theorem 7.1.11, and $c_i(E) = 0$ for $i > n$ by convention. The *total Chern class* is the inhomogeneous sum
$$c(E) := 1 + c_1(E) + c_2(E) + \cdots + c_n(E) \in H^*(X;\mathbb{Z})$$
a unit in the cohomology ring (its degree-zero part is 1).

The definition is forced by proposition 7.1.2: the Chern classes are the images of the universal generators, so they are the characteristic classes from which all others (the polynomials in them) are built. The total class is a bookkeeping device whose payoff is the Whitney sum formula $c(E \oplus E') = c(E) \smile c(E')$, proved in theorem 7.4.5 by the splitting principle; the formula says that the total Chern class converts direct sum of bundles into cup product of cohomology classes, which is exactly the multiplicativity one needs to compute Chern classes of bundles built from line bundles. The convention $c_i(E) = 0$ for $i > n$ records that a rank-n bundle has no Chern classes above degree $2n$, since $H^{>2n}(BU(n))$ has no generators not already products of lower ones.

Example 7.1.14: Chern Classes of a Sum of Line Bundles

Let $L_1, \ldots, L_n$ be complex line bundles over a paracompact Hausdorff base X with first Chern classes $x_j = c_1(L_j) \in H^2(X;\mathbb{Z})$, and let $E = L_1 \oplus \cdots \oplus L_n$. The Whitney sum formula (theorem 7.4.5) gives
$$c(E) = \prod_{j=1}^{n} c(L_j) = \prod_{j=1}^{n} (1 + x_j)$$
Expanding the product, the degree-$2i$ part is the i-th elementary symmetric polynomial in the x_j:
$$c_i(E) = \sigma_i(x_1, \ldots, x_n) = \sum_{j_1 < \cdots < j_i} x_{j_1} \cdots x_{j_i}$$
so $c_1(E) = x_1 + \cdots + x_n$, $c_2(E) = \sum_{j<k} x_j x_k$, and the top class $c_n(E) = x_1 \cdots x_n$. This is the computational meaning of the splitting principle: the universal calculation takes place in $H^*((\mathbb{C}\mathrm{P}^\infty)^n;\mathbb{Z}) = \mathbb{Z}[x_1, \ldots, x_n]$, the cohomology of $B(U(1)^n) = (\mathbb{C}\mathrm{P}^\infty)^n$ from example 2.4.5, and the Chern classes of a sum of lines are the elementary symmetric functions of the line classes. Since the diagonal $U(1)^n \hookrightarrow U(n)$ induces a map $(\mathbb{C}\mathrm{P}^\infty)^n \to BU(n)$ under which $c_i \mapsto \sigma_i(x_1, \ldots, x_n)$, the universal Chern class $c_i \in H^*(BU(n))$ is detected by the symmetric polynomial σ_i, recovering the polynomial-generator description of theorem 7.1.11 as the ring of symmetric functions $\mathbb{Z}[x_1, \ldots, x_n]^{S_n} = \mathbb{Z}[\sigma_1, \ldots, \sigma_n]$.

The projective-bundle route to theorem 7.1.11 deserves a statement, both because it produces the splitting principle directly and because it is the route the applications in [21] draw on. The idea is to replace the sphere bundle by the projectivization. The projectivized universal bundle $\mathbb{P}(\gamma_n) \to BU(n)$ has fiber $\mathbb{C}\mathrm{P}^{n-1}$, whose cohomology is $\mathbb{Z}[t]/(t^n)$, and the Leray-Hirsch theorem applies because the fiber classes extend over the total space.

Proposition 7.1.15: The Projective Bundle Formula

Let $E \to X$ be a complex vector bundle of rank n over a CW complex X, and let $\mathbb{P}(E) \xrightarrow{\pi} X$ be its projectivization, the bundle whose fiber over x is the projective space $\mathbb{P}(E_x) \cong \mathbb{C}\mathrm{P}^{n-1}$ of lines in E_x. Let $\gamma_E \to \mathbb{P}(E)$ be the tautological line bundle (whose fiber over a line $\ell \subseteq E_x$ is ℓ itself), and set $t = -c_1(\gamma_E) = c_1(\gamma_E^*) \in H^2(\mathbb{P}(E);\mathbb{Z})$, the first Chern class of its dual. Then $H^*(\mathbb{P}(E);\mathbb{Z})$ is a free module over $H^*(X;\mathbb{Z})$ with basis $1, t, t^2, \ldots, t^{n-1}$, and as a ring

$$H^*(\mathbb{P}(E);\mathbb{Z}) \cong H^*(X;\mathbb{Z})[t]\big/\big(t^n + c_1(E)\,t^{n-1} + \cdots + c_{n-1}(E)\,t + c_n(E)\big)$$

where the $c_i(E)$ are the Chern classes of definition 7.1.13.

***Proof*:**
The class $t = c_1(\gamma_E^*)$ restricts on a fiber $\mathbb{P}(E_x) = \mathbb{C}\mathrm{P}^{n-1}$ to the generator of $H^2(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$, since γ_E restricts to the tautological line bundle γ^{n-1} on each fiber, whose negative first Chern class is the standard generator (example 7.1.3). Hence the powers $1, t, \ldots, t^{n-1}$ restrict to a basis of $H^*(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z}) = \mathbb{Z}[t]/(t^n)$ in each fiber. The projectivization $\mathbb{P}(E) \to X$ is a fiber bundle, hence a Serre fibration with nonempty fibers by theorem 5.2.4, so corollary 6.5.2 applies with no connectivity hypothesis on X and asserts that $H^*(\mathbb{P}(E))$ is the free $H^*(X)$-module on $1, t, \ldots, t^{n-1}$, its hypotheses met because the fiber cohomology is free with the globally-defined basis given by the powers of t. Freeness on $1, t, \ldots, t^{n-1}$ means t^n is an $H^*(X)$-linear combination of the lower powers, say $t^n = -\sum_{i=1}^n a_i\, t^{n-i}$ with $a_i \in H^{2i}(X;\mathbb{Z})$; the relation defining the ring is $t^n + \sum_{i=1}^n a_i t^{n-i} = 0$. That the coefficients a_i are the Chern classes $c_i(E)$ is the projective-bundle characterization of the Chern classes, and it agrees with definition 7.1.13 because both are natural and agree on sums of line bundles. On $E = \bigoplus_j L_j$ with $x_j = c_1(L_j)$, the tautological inclusion $\gamma_E \hookrightarrow \pi^*E$ is a nowhere-zero section of the bundle $\pi^*E \otimes \gamma_E^* = \bigoplus_j \pi^*L_j \otimes \gamma_E^*$, so its top Chern class vanishes:

$$0 = c_n\big(\pi^*E \otimes \gamma_E^*\big) = \prod_{j=1}^{n} c_1\big(\pi^*L_j \otimes \gamma_E^*\big) = \prod_{j=1}^{n}\big(x_j + t\big) = t^n + \sigma_1(x)\,t^{n-1} + \cdots + \sigma_n(x)$$

using $c_1(\pi^*L_j \otimes \gamma_E^*) = x_j + t$ and the Whitney formula; comparing with $t^n + \sum_i a_i t^{n-i} = 0$ identifies $a_i = \sigma_i(x) = c_i(E)$ by example 7.1.14. Applying this with $E = \gamma_n$ over $X = BU(n)$ and reading off the module structure recovers $H^*(BU(n)) = \mathbb{Z}[c_1, \ldots, c_n]$, completing the proof.

The projective-bundle formula is what proves the splitting principle: pulling a bundle back to its full flag bundle splits it as a sum of line bundles after a cohomology-injective base change, reducing every identity among Chern classes to the symmetric-function identities of example 7.1.14. The same formula, specialized to the universal bundle, gives a second proof of theorem 7.1.11; the two proofs compute the same ring by complementary means, the Gysin route building it one Euler class at a time, the projective route building it one tautological class at a time.

The real case runs in parallel, with two changes. Coefficients must be taken in $\mathbb{Z}/2$, because the relevant sphere bundles need not be orientable and torsion appears integrally; and the generators land in every degree, not only the even ones, because the fiber sphere S^{n-1} has its top class in degree $n-1$. The structural mechanism is identical: a universal sphere bundle relates $BO(n-1)$ to $BO(n)$, and the Gysin sequence with $\mathbb{Z}/2$ coefficients adjoins one Stiefel-Whitney class at each stage.

Lemma 7.1.16: The Universal Sphere Bundle for $O(n)$

For $n \geq 1$ the inclusion $O(n-1) \hookrightarrow O(n)$ induces a fibration

$$S^{n-1} \longrightarrow BO(n-1) \xrightarrow{\rho} BO(n)$$

with fiber the sphere S^{n-1}, equal to the sphere bundle $S(\gamma_n^{\mathbb{R}})$ of the universal real rank-n bundle $\gamma_n^{\mathbb{R}} \to BO(n)$.

***Proof*:**
The proof is identical in structure to lemma 7.1.10, replacing $\mathbb{C}$ by $\mathbb{R}$ and U by O. The group $O(n)$ acts transitively on the unit sphere $S^{n-1} \subseteq \mathbb{R}^n$, since any unit vector is carried to e_1 by an orthogonal transformation; the stabilizer of e_1 is the orthogonal group $O(n-1)$ of the complement $e_1^{\perp} \cong \mathbb{R}^{n-1}$, so $O(n)/O(n-1) \cong S^{n-1}$. By the general fact that for a closed subgroup $H \subseteq G$ the projection $EG/H \to EG/G = BG$ is a fiber bundle with fiber G/H (Step 2 of lemma 7.1.10), applied to $H = O(n-1) \subseteq G = O(n)$ with $EG = EO(n)$, the map $BO(n-1) = EO(n)/O(n-1) \to EO(n)/O(n) = BO(n)$ is a sphere bundle with fiber S^{n-1}. The identification with $S(\gamma_n^{\mathbb{R}})$ is Step 3 of that proof verbatim, with the standard representation of $O(n)$ on $\mathbb{R}^n$ in place of the unitary one, completing the proof.

The real sphere bundle differs from the complex one in that its fiber S^{n-1} may be even-dimensional, and over $\mathbb{Z}$ the local coefficient system $H^{n-1}(S^{n-1};\mathbb{Z})$ need not be trivial: monodromy around a loop in $BO(n)$ can reverse the orientation of the fiber sphere, exactly when the bundle is non-orientable. This is why integral coefficients fail to give a clean polynomial answer. Over $\mathbb{Z}/2$ the obstruction evaporates, because $H^{n-1}(S^{n-1};\mathbb{Z}/2) = \mathbb{Z}/2$ has a unique nonzero element and no orientation to reverse, so the local system is automatically trivial and the Gysin sequence of theorem 6.3.2 applies with $\mathbb{Z}/2$ coefficients.

Theorem 7.1.17: The Cohomology of $BO(n)$

For every $n \geq 0$ there is an isomorphism of graded rings

$$H^*(BO(n);\mathbb{Z}/2) \cong \mathbb{Z}/2[w_1, w_2, \ldots, w_n], \qquad |w_i| = i$$

a polynomial ring over $\mathbb{Z}/2$ on n generators w_i in degree i. The generators may be chosen so that $\rho^*\colon H^*(BO(n);\mathbb{Z}/2) \to H^*(BO(n-1);\mathbb{Z}/2)$ sends $w_i \mapsto w_i$ for $i \leq n-1$ and $w_n \mapsto 0$, where ρ is the map of lemma 7.1.16.

***Proof*:**
The argument is induction on n via the Gysin sequence of lemma 7.1.16 with $\mathbb{Z}/2$ coefficients, parallel to theorem 7.1.11 but without the even-degree restriction. All cohomology in this proof is with $\mathbb{Z}/2$ coefficients.

Step 1: Base cases. For $n = 0$, $BO(0)$ is a point and $H^*(\mathrm{pt};\mathbb{Z}/2) = \mathbb{Z}/2$, the empty polynomial ring. For $n = 1$, the group $O(1) = \{\pm 1\} \cong \mathbb{Z}/2$ is discrete, so $BO(1) = B\mathbb{Z}/2 = \mathbb{R}\mathrm{P}^\infty$ (example 5.5.6), and its mod-2 cohomology is

$$H^*(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}/2) \cong \mathbb{Z}/2[w_1], \qquad |w_1| = 1$$

the polynomial ring on a single degree-one class, as computed in chapter 4 (the cup-product structure of $\mathbb{RP}^\infty$). This is the case $n = 1$.

Step 2: The Gysin sequence. Assume $n \geq 2$ and $H^*(BO(n-1); \mathbb{Z}/2) \cong \mathbb{Z}/2[w_1, \dots, w_{n-1}]$ with $|w_i| = i$. The fibration $S^{n-1} \to BO(n-1) \xrightarrow{\rho} BO(n)$ of lemma 7.1.16 has fiber S^{n-1}, so $k = n-1$ in theorem 6.3.2. With $\mathbb{Z}/2$ coefficients the orientation hypothesis holds automatically, as noted above. The Euler class $e \in H^n(BO(n); \mathbb{Z}/2)$ and the Gysin sequence read

$$\cdots \to H^m(BO(n)) \xrightarrow{\smile e} H^{m+n}(BO(n)) \xrightarrow{\rho^*} H^{m+n}(BO(n-1)) \xrightarrow{\int} H^{m+1}(BO(n)) \xrightarrow{\smile e} H^{m+n+1}(BO(n)) \to \cdots$$

with $\int$ lowering degree by $n-1$.

Step 3: $\int = 0$ and ρ^* is surjective in every degree. Unlike the complex case, there is no parity to exploit: the Euler class e has degree n and $\int$ lowers degree by the possibly-even number $n-1$, so the vanishing of H^{odd} is unavailable and a different argument is required. The two statements "$\int = 0$ out of degree d" and "ρ^* is surjective in degree d" are equivalent at each d: the exact segment

$$H^d(BO(n)) \xrightarrow{\rho^*} H^d(BO(n-1)) \xrightarrow{\int} H^{d-n+1}(BO(n))$$

has $\operatorname{im}(\rho^*) = \ker(\int)$ by exactness, so ρ^* is onto $H^d(BO(n-1))$ if and only if $\int = 0$ there. The proof of both, by induction on d, runs through a degree count in the Gysin sequence rather than through ring generation by lower classes.

Low degrees $d < n-1$. The flanking terms of the segment

$$H^{d-n}(BO(n)) \xrightarrow{\smile e} H^d(BO(n)) \xrightarrow{\rho^*} H^d(BO(n-1)) \xrightarrow{\int} H^{d-n+1}(BO(n))$$

vanish for $d < n-1$: the source $H^{d-n}(BO(n)) = 0$ since $d - n < 0$, and the target $H^{d-n+1}(BO(n)) = 0$ since $d - n + 1 < 0$. The segment collapses to $0 \to H^d(BO(n)) \xrightarrow{\rho^*} H^d(BO(n-1)) \to 0$, so ρ^* is an isomorphism, hence surjective, in every degree $d < n-1$, and $\int = 0$ there.

The degree $d = n-1$, where the content sits. Here the source $H^{d-n}(BO(n)) = H^{-1}(BO(n)) = 0$ still, but the target $H^{d-n+1}(BO(n)) = H^0(BO(n)) = \mathbb{Z}/2$ does not vanish, so flank vanishing alone gives only injectivity of ρ^*, not surjectivity. Surjectivity of ρ^* in degree $n-1$ is equivalent, by the exactness $\ker(\smile e) = \operatorname{im}(\int)$ at the node $H^0(BO(n))$, to the injectivity of $\smile e\colon H^0(BO(n)) \to H^n(BO(n))$, that is to $e \neq 0$ in $H^n(BO(n); \mathbb{Z}/2)$. The Euler class is nonzero by a direct restriction. Let $f\colon BO(1)^n \to BO(n)$ classify the external Whitney sum $\gamma_1^{(1)} \oplus \cdots \oplus \gamma_1^{(n)}$ of the tautological line bundles over the n factors $\mathbb{RP}^\infty = BO(1)$, and write $t_i \in H^1(\mathbb{RP}^\infty; \mathbb{Z}/2)$ for the generator of the i-th factor, where $H^*(\mathbb{RP}^\infty; \mathbb{Z}/2) = \mathbb{Z}/2[t]$ is the base case of Step 1 (example 5.5.6). The mod-2 Euler class is natural, and it is multiplicative on a Whitney sum because the sphere bundle of a direct sum is the fiberwise join of the summand sphere bundles, so the mod-2 Thom class of the sum is the smash product of the summands' Thom classes and the Euler class, its restriction to the zero section, is their cup product; for a real line bundle L the mod-2 Euler class equals $w_1(L)$, so

$$f^*e = e\big(\textstyle\bigoplus_{i=1}^n \gamma_1^{(i)}\big) = \prod_{i=1}^n e\big(\gamma_1^{(i)}\big) = \prod_{i=1}^n t_i$$

in $H^*(BO(1)^n; \mathbb{Z}/2) = \mathbb{Z}/2[t_1, \dots, t_n]$, where the ring identification is the Künneth theorem over the field $\mathbb{Z}/2$ applied to $H^*(\mathbb{RP}^\infty; \mathbb{Z}/2) = \mathbb{Z}/2[t]$. The product $t_1 \cdots t_n$ is nonzero, so $f^*e \neq 0$, hence $e \neq 0$. Therefore $\smile e$ is injective on H^0, $\int = 0$ out of degree $n-1$, and ρ^* is surjective in degree $n-1$; choosing lifts $w_i \in H^i(BO(n))$ with $\rho^* w_i = w_i$ for $i \leq n-1$ (possible by surjectivity in degrees $\leq n-1$) realizes ρ^* as a ring map hitting every generator $w_1, \ldots, w_{n-1}$ of $H^*(BO(n-1))$.

High degrees $d \geq n$. Assume inductively that $\int = 0$ and ρ^* is surjective in all degrees $< d$, so $\smile e$ is injective in all degrees $< d - n + 1$ (the equivalence above, read one node lower, shows $\smile e\colon H^j(BO(n)) \to H^{j+n}(BO(n))$ is injective whenever $\int = 0$ out of degree $j + n - 1 < d$). The generators $w_1, \ldots, w_{n-1}$ of $H^*(BO(n-1))$ have degrees $1, \ldots, n-1$, all strictly less than d, and each lies in $\mathrm{im}(\rho^*)$ by the inductive surjectivity (their lifts w_i were fixed at the degree $n-1$ stage). Since ρ^* is a ring homomorphism and $H^d(BO(n-1))$ is spanned over $\mathbb{Z}/2$ by the degree-d monomials in $w_1, \ldots, w_{n-1}$, each a product of generators of degree $< d$ already in $\mathrm{im}(\rho^*)$, the whole group $H^d(BO(n-1))$ lies in $\mathrm{im}(\rho^*)$. Thus ρ^* is surjective in degree d, and $\int = 0$ there. The induction closes: ρ^* is surjective and $\int = 0$ in every degree, and consequently, by exactness $\ker(\smile e) = \mathrm{im}(\int) = 0$ at each node, $\smile e$ is injective in every degree.

Step 4: The structure of the ring. With $\int = 0$ the Gysin sequence breaks into short exact sequences

$$0 \to H^m(BO(n)) \xrightarrow{\smile e} H^{m+n}(BO(n)) \xrightarrow{\rho^*} H^{m+n}(BO(n-1)) \to 0$$

Set $w_n := e \in H^n(BO(n))$ and keep the lifts w_i $(i < n)$ of Step 3. The short exact sequence says $H^*(BO(n))$ is a free module over the subring generated by $w_1, \ldots, w_{n-1}$ on the powers $\{w_n^j\}_{j \geq 0}$, by the same module argument as Step 4 of theorem 7.1.11 ($\smile w_n$ injective with cokernel ρ^*-isomorphic to $H^*(BO(n-1))$). Therefore

$$H^*(BO(n); \mathbb{Z}/2) \cong H^*(BO(n-1); \mathbb{Z}/2)[w_n] = \mathbb{Z}/2[w_1, \ldots, w_{n-1}][w_n] = \mathbb{Z}/2[w_1, \ldots, w_n]$$

The relations $\rho^* w_i = w_i$ for $i \leq n-1$ hold by construction and $\rho^* w_n = \rho^* e = 0$ because $e \in \ker \rho^* = \mathrm{im}(\smile e)$. The induction closes, establishing the polynomial ring on generators w_i of degree i, completing the proof.

The mod-2 cohomology of $BO(n)$ has the same polynomial shape as the integral cohomology of $BU(n)$, with the single change that the generators are spread across all degrees 1 through n rather than confined to even degrees. The reason for the degree shift is the real sphere fiber S^{n-1}, whose top class sits in degree $n-1$ and produces an Euler class in degree n, against the complex fiber S^{2n-1} producing a class in degree $2n$. The reason for the mod-2 coefficients is orientation: integrally the universal real bundle is non-orientable, its sphere bundle carries a nontrivial $\mathbb{Z}$-local system, and the Gysin machinery stalls; over $\mathbb{Z}/2$ orientation is automatic and the computation proceeds. The integral cohomology of $BO(n)$ is correspondingly more intricate, carrying 2-torsion together with the Pontryagin and Euler classes taken up in section 7.3.

Corollary 7.1.18: The Cohomology of BO

Let $BO = \varinjlim_n BO(n)$. Then

$$H^*(BO; \mathbb{Z}/2) \cong \mathbb{Z}/2[w_1, w_2, w_3, \ldots], \qquad |w_i| = i$$

the polynomial ring over $\mathbb{Z}/2$ on countably many generators, one in each positive degree.

***Proof*:**
As in corollary 7.1.12, $H^*(BO;\mathbb{Z}/2)$ is the limit of the system $\{H^*(BO(n);\mathbb{Z}/2)\}_n$ along the surjections ρ^* of theorem 7.1.17, with vanishing $\varprojlim^1$ by the Mittag-Leffler condition (surjectivity of each ρ^* in a fixed degree), for which see [10, Mittag-Leffler vanishing]. In degree k the restriction $H^k(BO(n);\mathbb{Z}/2) \to H^k(BO(n-1);\mathbb{Z}/2)$ is an isomorphism once $n > k$, since the generators of degree $\leq k$ are $w_1, \ldots, w_k$, all present and matched by $\rho^* w_i = w_i$. The degreewise limit is $\mathbb{Z}/2[w_1, w_2, \ldots]$, completing the proof.

The named real classes are now defined exactly as in the complex case, as the polynomial generators pulled back to a bundle.

Definition 7.1.19: Stiefel-Whitney Classes and the Total Class

Let $E \to X$ be a real vector bundle of rank n over a paracompact Hausdorff base X, with classifying map $f_E \colon X \to BO(n)$, unique up to homotopy by theorem 7.1.9. For $1 \leq i \leq n$ the *i-th Stiefel-Whitney class* of E is

$$w_i(E) := f_E^* \, w_i \in H^i(X;\mathbb{Z}/2)$$

the pullback of the generator $w_i \in H^i(BO(n);\mathbb{Z}/2)$ of theorem 7.1.17, with $w_i(E) = 0$ for $i > n$. The *total Stiefel-Whitney class* is

$$w(E) := 1 + w_1(E) + w_2(E) + \cdots + w_n(E) \in H^*(X;\mathbb{Z}/2)$$

a unit in the mod-2 cohomology ring.

The Stiefel-Whitney classes are the mod-2 characteristic classes of real bundles, and by proposition 7.1.2 they generate all of them: every mod-2 characteristic class of real rank-n bundles is a $\mathbb{Z}/2$-polynomial in $w_1, \ldots, w_n$. The total class again satisfies a Whitney sum formula $w(E \oplus E') = w(E) \smile w(E')$ (theorem 7.4.5), which is the source of the name. The lowest class $w_1(E) \in H^1(X;\mathbb{Z}/2)$ is the obstruction to orientability: it is the image of the bundle under $BO(n) \to BO(1) = \mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$ classifying the determinant line bundle, so $w_1(E) = 0$ if and only if the structure group reduces to $SO(n)$, that is, if and only if E is orientable. This is the obstruction-theoretic reading of chapter 5: w_1 is the primary obstruction to a reduction of structure group, living in the cohomology group $H^1(X;\pi_0(O(n)/SO(n))) = H^1(X;\mathbb{Z}/2)$ predicted by obstruction theory.

Example 7.1.20: Stiefel-Whitney Class of the Tautological Bundle over $\mathbb{R}\mathrm{P}^\infty$
Take $X = \mathbb{R}\mathrm{P}^\infty = BO(1)$ and $E = \gamma_1^{\mathbb{R}}$ the universal (tautological) real line bundle, with classifying map the identity. By theorem 7.1.17 with $n = 1$,

$$H^*(BO(1);\mathbb{Z}/2) = H^*(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}/2) = \mathbb{Z}/2[w_1], \qquad |w_1| = 1$$

and $w_1(\gamma_1^{\mathbb{R}}) = \mathrm{id}^* w_1 = w_1$ is the generator of $H^1(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}/2) \cong \mathbb{Z}/2$. The total class is $w(\gamma_1^{\mathbb{R}}) = 1 + w_1$. Restricting to a finite skeleton $\mathbb{R}\mathrm{P}^k \subseteq \mathbb{R}\mathrm{P}^\infty$, the tautological line bundle $\gamma_1^{\mathbb{R}}|_{\mathbb{R}\mathrm{P}^k}$ has $w_1 =$ the generator of $H^1(\mathbb{R}\mathrm{P}^k;\mathbb{Z}/2)$, and this class is nonzero, which is the cohomological proof that the tautological bundle over $\mathbb{R}\mathrm{P}^k$ is non-orientable: $w_1 \neq 0$ obstructs orientability. The same class detects, for instance, that the Möbius band (the total space of $\gamma_1^{\mathbb{R}}|_{\mathbb{R}\mathrm{P}^1}$ over the circle $\mathbb{R}\mathrm{P}^1 = S^1$) is non-orientable, w_1 being the generator of $H^1(S^1;\mathbb{Z}/2) = \mathbb{Z}/2$.

The Euler class, which appeared in both computations as the transgressive Gysin generator, deserves a definition in its own right, in the integral oriented setting where it is most informative. For an oriented bundle the structure group is the special orthogonal group $SO(n)$, and the orientation trivializes the local coefficient system, so the integral Gysin sequence runs and produces an integral Euler class. The home of the universal Euler class is the cohomology of $BSO(n)$.

Definition 7.1.21: Euler Class of an Oriented Bundle

Let $E \to X$ be an oriented real vector bundle of rank n, with classifying map $f_E \colon X \to BSO(n)$ into the classifying space of the special orthogonal group. The oriented universal sphere bundle $S^{n-1} \to BSO(n-1) \to BSO(n)$ has trivial integral local coefficient system (the orientation of E trivializes $H^{n-1}(S^{n-1};\mathbb{Z})$), so the integral Gysin sequence of theorem 6.3.2 produces a universal class

$$e \in H^n(BSO(n);\mathbb{Z}), \qquad e := d_n(1 \otimes \alpha)$$

the transgression of the oriented fiber generator $\alpha \in H^{n-1}(S^{n-1};\mathbb{Z})$. The *Euler class* of E is its pullback

$$e(E) := f_E^* e \in H^n(X;\mathbb{Z})$$

Equivalently, $e(E)$ is the image of the Thom class $u_E \in H^n(\mathrm{Th}(E);\mathbb{Z})$ under the composite $H^n(\mathrm{Th}(E)) \to H^n(E) \xrightarrow{s_0^*} H^n(X)$, where $s_0 \colon X \to E$ is the zero section, the restriction of the Thom class to the zero section.

The Euler class is the primary obstruction to a nowhere-zero section, which is the geometric content that chapter 5 makes precise. A section of E avoiding the zero section is a lift of the classifying map through the sphere bundle $S(E) \to X$, and the primary obstruction to such a lift lives in $H^n(X;\pi_{n-1}(S^{n-1})) = H^n(X;\mathbb{Z})$, the same group as $e(E)$; the obstruction class is exactly the Euler class. Thus $e(E) = 0$ is necessary for a nowhere-zero section to exist, and for the tangent bundle of a closed oriented n-manifold M the evaluation $\int_M e(TM)$ is the Euler characteristic $\chi(M)$, by the Poincaré-Hopf theorem proved in [20, Poincaré-Hopf]. The Thom-class description connects the cohomological definition to the Thom isomorphism that organizes the cobordism theory of chapter 8: the Euler class is the self-intersection of the zero section, and its appearance as the Gysin differential is the spectral-sequence form of that intersection. The reduction mod 2 of the integral Euler class $e(E)$ is the top Stiefel-Whitney class $w_n(E)$, since both are the obstruction to a nowhere-zero section read with the respective coefficients, which links definition 7.1.21 to the $w_n = e$ of theorem 7.1.17.

Example 7.1.22: The Euler Class of a Complex Line Bundle

A complex line bundle $L \to X$ has an underlying oriented real rank-2 bundle $L_{\mathbb{R}}$, the complex structure giving a canonical orientation (the ordered real basis v, iv of each fiber). Its Euler class and first Chern class coincide:

$$e(L_{\mathbb{R}}) = c_1(L) \in H^2(X;\mathbb{Z})$$

Both are pulled back from the universal case: the classifying map factors through $BU(1) = \mathbb{C}\mathrm{P}^\infty = BSO(2)$ (the inclusion $U(1) \hookrightarrow SO(2)$ is an isomorphism, so $BU(1) \simeq BSO(2)$), and under this identification the universal Euler class $e \in H^2(BSO(2);\mathbb{Z})$ matches the universal first Chern class $c_1 \in H^2(BU(1);\mathbb{Z})$, since both are the transgression of the fiber generator of S^1 in the Gysin sequence of the universal circle bundle $S^1 \to S^\infty \to \mathbb{C}\mathrm{P}^\infty$ (example 7.1.3). Pulling back along f_L gives the equality. For $L = \gamma^1$ the tautological line bundle over $\mathbb{C}\mathrm{P}^1 = S^2$, the common value is $-u$

with u the positive generator of $H^2(S^2;\mathbb{Z})$, and $\int_{S^2} e(L_{\mathbb{R}}) = -1$, consistent with the tautological bundle having no nowhere-zero section (a nowhere-zero section would trivialize it, contradicting $c_1 \neq 0$).

Exercise 7.1.1

1. Using theorem 7.1.11, compute the Poincaré series $\sum_k \dim_{\mathbb{Q}} H^k(BU(n);\mathbb{Q})\, t^k$ of $H^*(BU(n);\mathbb{Q})$, and verify that it equals $\prod_{i=1}^n (1-t^{2i})^{-1}$. Then do the same for $H^*(BO(n);\mathbb{Z}/2)$, computing $\sum_k \dim_{\mathbb{Z}/2} H^k(BO(n);\mathbb{Z}/2)\, t^k$ and confirming it equals $\prod_{i=1}^n (1-t^i)^{-1}$.
2. Let $\gamma \to \mathbb{C}\mathrm{P}^\infty$ be the universal line bundle and $E = \gamma \oplus \gamma \to \mathbb{C}\mathrm{P}^\infty$. Using the Whitney sum formula and example 7.1.3, compute the total Chern class $c(E)$ and each $c_i(E) \in H^*(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) = \mathbb{Z}[c_1]$ explicitly. Is E isomorphic to the trivial rank-2 bundle? Justify using a Chern class.
3. Show that the map $\rho^*\colon H^*(BU(n);\mathbb{Z}) \to H^*(BU(n-1);\mathbb{Z})$ of theorem 7.1.11 is surjective but not injective, and identify its kernel as a principal ideal. Interpret the generator of the kernel as a characteristic class and say which bundle invariant it computes.
4. Let $T^n = (S^1)^n$ be the n-torus and consider the real rank-n bundle $E = \bigoplus_{j=1}^n L_j$, where $L_j \to T^n$ is the real line bundle pulled back from the j-th circle factor under the projection $T^n \to S^1 = \mathbb{R}\mathrm{P}^1 \hookrightarrow \mathbb{R}\mathrm{P}^\infty$ classifying a nontrivial line bundle. Compute the total Stiefel-Whitney class $w(E) \in H^*(T^n;\mathbb{Z}/2)$ and determine $w_n(E)$. Is E orientable?
5. Use proposition 7.1.15 to compute $H^*(\mathbb{P}(E);\mathbb{Z})$ as a ring when $E = \underline{\mathbb{C}}^3$ is the trivial rank-3 bundle over a point, and confirm the answer is $H^*(\mathbb{C}\mathrm{P}^2;\mathbb{Z}) = \mathbb{Z}[t]/(t^3)$. Then compute $H^*(\mathbb{P}(E);\mathbb{Z})$ when $E = \gamma \oplus \underline{\mathbb{C}}^2$ over $\mathbb{C}\mathrm{P}^\infty$ with γ the universal line bundle, expressing the relation in terms of $c_1 = c_1(\gamma)$.
6. Prove that $BU(1) \times \cdots \times BU(1) = (\mathbb{C}\mathrm{P}^\infty)^n$ has cohomology $\mathbb{Z}[x_1, \ldots, x_n]$ and that the map $B\iota\colon (\mathbb{C}\mathrm{P}^\infty)^n \to BU(n)$ induced by the diagonal torus $\iota\colon U(1)^n \hookrightarrow U(n)$ sends $c_i \mapsto \sigma_i(x_1, \ldots, x_n)$, the elementary symmetric polynomial. Deduce that $B\iota^*$ is injective and identifies $H^*(BU(n);\mathbb{Z})$ with the ring of symmetric polynomials $\mathbb{Z}[x_1, \ldots, x_n]^{S_n}$.

7.2 The Thom Isomorphism

The Euler class of definition 7.1.21 was produced by the Gysin sequence, as a universal class on $BSO(n)$ pulled back along a classifying map. That construction is efficient but it hides the geometry: it does not say what $e(E)$ measures on the bundle itself. This section gives the missing description. Attached to an oriented bundle is a single cohomology class of its total space rel the complement of the zero section, the Thom class, and cup product with it is an isomorphism raising degree by the rank. The Euler class is what that class becomes when the relative information is discarded, and the two basic properties of e that the Gysin construction does not give, multiplicativity under Whitney sum and 2-torsion in odd rank, both fall out of the Thom class in a line.

The isomorphism is also the computational engine behind the Thom spectra of section 8.5: the identification $H^*(MO;\mathbb{Z}/2) \cong H^*(BO;\mathbb{Z}/2)$ that drives Thom's calculation of $\pi_*(MO)$ is exactly this theorem applied to the universal bundles.

Throughout, $E \to B$ is a real vector bundle of rank n over a paracompact Hausdorff base, and $E_0 \subseteq E$ denotes the complement of the zero section. An orientation of E is a choice, for each b, of a generator of $H^n(E_b, (E_b)_0; \mathbb{Z}) \cong \mathbb{Z}$, varying continuously in the sense that each point of B has a neighbourhood over which the choices come from a single generator under a trivialization.

Definition 7.2.1: Thom Class

Let $E \to B$ be an oriented real vector bundle of rank n. A *Thom class* for E is a class

$$u_E \in H^n(E, E_0; \mathbb{Z})$$

whose restriction to each fiber pair $(E_b, (E_b)_0) \cong (\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ is the generator of $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; \mathbb{Z}) \cong \mathbb{Z}$ determined by the orientation of E_b. For an arbitrary real bundle, a *mod-2 Thom class* $u_E \in H^n(E, E_0; \mathbb{Z}/2)$ is defined by the same condition with $\mathbb{Z}/2$ coefficients, where no orientation is needed because $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; \mathbb{Z}/2) \cong \mathbb{Z}/2$ has a unique nonzero element.

The definition is a normalization condition, one fiber at a time. What is not obvious is that a class satisfying it exists at all, or that it is unique, and both are part of the theorem below. The local model is the following computation, which is the whole content of the trivial case.

Lemma 7.2.2: The Local Model

Let R be a commutative ring, let X be any space, let $n \geq 1$, and let u_n be a generator of the free rank-one R-module $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R)$. Then the cohomology cross product with u_n,

$$H^k(X; R) \longrightarrow H^{k+n}\big(X \times \mathbb{R}^n,\ X \times (\mathbb{R}^n \setminus \{0\}); R\big), \qquad \alpha \longmapsto \alpha \times u_n$$

is an isomorphism for every k.

***Proof*:**
Write $(Y, B_0) = (\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$. The relative Eilenberg-Zilber theorem, the version of theorem 4.3.1 for a product in which one factor carries a subspace, gives a natural chain homotopy equivalence

$$C_*\big(X \times Y,\ X \times B_0; R\big) \ \simeq\ C_*(X; R) \otimes_R C_*(Y, B_0; R)$$

obtained from the absolute equivalence by passing to the quotient complexes, which is legitimate because the Eilenberg-Zilber maps are natural and carry $C_*(X \times B_0)$ into $C_*(X) \otimes C_*(B_0)$.

Take first R a PID; the general case follows at the end. The complex $C_*(Y, B_0; R)$ consists of free R-modules and has homology R concentrated in degree n; a complex of free modules over a PID with free homology splits, so it is *chain homotopy equivalent* to the complex $R[n]$ carrying a single free rank-one module in degree n. This is the point on which the absence of any hypothesis on X turns. Chain homotopy equivalences are preserved both by $\mathrm{Hom}_R(-, R)$ and by $- \otimes_R C_*(X; R)$, so the interchange map

$$C^*(X; R) \otimes_R C^*(Y, B_0; R) \longrightarrow \mathrm{Hom}_R\big(C_*(X; R) \otimes_R C_*(Y, B_0; R),\ R\big)$$

may be checked after replacing $C_*(Y, B_0; R)$ by $R[n]$, where it is an isomorphism outright, since tensoring with a single free rank-one module concentrated in one degree commutes with $\mathrm{Hom}_R(-, R)$. No finite-type condition on X is required, because the finiteness the interchange needs is given entirely by the fiber factor. Invoking the algebraic Künneth theorem theorem 4.3.4 then gives a short exact sequence whose Tor term vanishes, because $H^*(Y, B_0; R)$ is free. The remaining map is the cross product, and it is an isomorphism

$$\bigoplus_{p+q=m} H^p(X; R) \otimes_R H^q(Y, B_0; R) \xrightarrow{\cong} H^m\big(X \times Y, X \times B_0; R\big)$$

Since $H^q(Y, B_0; R)$ vanishes except for $q = n$, where it is free of rank one on u_n, the left side is $H^{m-n}(X; R) \otimes_R R \cong H^{m-n}(X; R)$ and the map is $\alpha \mapsto \alpha \times u_n$. Setting $m = k + n$ gives the statement, completing the proof.

The finiteness that makes the Künneth argument work sits entirely on the fiber factor, which is a single free module in one degree, so no hypothesis whatever is needed on X. That is what lets the induction below be run over the cells of the base without any further condition at each stage. For a general commutative R, apply the case just proved over $\mathbb{Z}$ and tensor: $C_*(Y, B_0; R) \cong C_*(Y, B_0; \mathbb{Z}) \otimes_{\mathbb{Z}} R \simeq R[n]$, and the same replacement argument goes through.

Theorem 7.2.3: The Thom Isomorphism

Let $E \to B$ be an oriented real vector bundle of rank n over a paracompact Hausdorff base, and suppose either that B admits a finite trivializing open cover (for instance B compact, or a finite CW complex) or that B is a CW complex of finite type, meaning one with finitely many cells in each dimension. The latter covers the classifying spaces $BO(n)$, $BSO(n)$ and $BU(n)$, whose Schubert cells are finite in each dimension. Then:

1. a Thom class $u_E \in H^n(E, E_0; \mathbb{Z})$ exists and is unique;
2. the map
$$\Phi\colon H^k(B; \mathbb{Z}) \longrightarrow H^{k+n}(E, E_0; \mathbb{Z}), \qquad \Phi(\alpha) = \pi^*(\alpha) \smile u_E$$
is an isomorphism for every k, where $\pi\colon E \to B$ is the projection.

The same statements hold with $\mathbb{Z}/2$ coefficients for an arbitrary real vector bundle, with no orientation hypothesis.

***Proof*:**
Write $h^k(-)$ for the assignment $U \mapsto H^{k+n}(E|_U, (E|_U)_0; \mathbb{Z})$ on open subsets of B. The argument proves the statement for successively larger classes of base, at each stage producing the Thom class and the isomorphism together.

Step 1: E trivial. If $E|_U \cong U \times \mathbb{R}^n$, then $(E|_U, (E|_U)_0) = (U \times \mathbb{R}^n, U \times (\mathbb{R}^n \setminus \{0\}))$ and lemma 7.2.2 applies with $X = U$ and $R = \mathbb{Z}$. The class $1 \times u_n$ restricts on each fiber to u_n, so it is a Thom class; the isomorphism Φ is the cross product of that lemma, since $\pi^*(\alpha) \smile (1 \times u_n) = \alpha \times u_n$. Uniqueness holds because $H^n(E|_U, (E|_U)_0; \mathbb{Z}) \cong H^0(U; \mathbb{Z})$ by the same isomorphism, and a Thom class corresponds to the element of $H^0(U; \mathbb{Z})$ taking the value 1 on every component, which is a single class.

Step 2: Attaching one cell. The induction is cellular rather than over an open cover, because the Mayer-Vietoris the book gives for a union of subcomplexes is the one for a reduced cohomology theory (proposition 8.2.8), and ordinary reduced cohomology is such a theory by example 8.2.7. Choose a fibre metric on E and pass to the disk-sphere pair, so that $H^{k+n}(E|_W, (E|_W)_0; \mathbb{Z}) \cong$

$\widetilde{H}^{k+n}(\mathrm{Th}(E|_W);\mathbb{Z})$ for every subcomplex $W \subseteq B$, where Th is the Thom space of definition 8.5.4. For subcomplexes $W = W_1 \cup W_2$ with $W_1 \cap W_2 = W_0$, the Thom spaces satisfy $\mathrm{Th}(E|_W) = \mathrm{Th}(E|_{W_1}) \cup \mathrm{Th}(E|_{W_2})$ with intersection $\mathrm{Th}(E|_{W_0})$, all three being based subcomplexes sharing the basepoint at which the sphere bundle is collapsed.

Suppose the conclusion holds over W_1, W_2 and W_0. Then Φ is a map from the Mayer-Vietoris sequence of the triad $(B; W_1, W_2)$ in $H^*(-;\mathbb{Z})$ to that of $(\mathrm{Th}(E|_W); \mathrm{Th}(E|_{W_1}), \mathrm{Th}(E|_{W_2}))$ in $\widetilde{H}^{*+n}(-;\mathbb{Z})$. It commutes with the restriction maps by naturality of the cup product (proposition 4.2.3) together with the fact that the Thom classes over W_1 and W_2 restrict to the one over W_0, by uniqueness there; and it commutes with the connecting map because the relative cup product is compatible with δ up to the sign $(-1)^n$, which does not affect whether a map is an isomorphism. Existence of a Thom class over W follows from exactness: the classes u_{W_1} and u_{W_2} agree on W_0, so the pair lies in the kernel of $j^*_{W_1} - j^*_{W_2}$ and therefore comes from a class over W, which restricts on each fibre to a generator because that may be checked in W_1 or W_2. Uniqueness over W follows because the indeterminacy is the image of $\widetilde{H}^{n-1}(\mathrm{Th}(E|_{W_0})) \cong H^{-1}(W_0;\mathbb{Z}) = 0$. Then lemma 3.4.22 gives that Φ is an isomorphism over W.

Step 3: Finite CW bases. Induct on the number of cells. The empty complex is trivial. If $W = W' \cup e^m$ is obtained by attaching a single m-cell, take $W_1 = W'$ and W_2 a subcomplex carrying the cell; E is trivial over the cell because a cell is contractible, so Step 1 applies to W_2, the inductive hypothesis to W_1, and again to $W_0 = W_1 \cap W_2$, which has fewer cells. Step 2 then gives the conclusion over W. Hence the theorem holds over every finite CW complex, and in particular over every compact base admitting a finite CW structure.

Step 4: A CW base of finite type. Let B be a CW complex with finitely many cells in each dimension, with skeleta $B^{(m)}$. Each $B^{(m)}$ is then a *finite* complex, hence compact, hence covered by Step 3. (Finite type is what this step needs: a skeleton of an arbitrary CW complex may have infinitely many cells and need not admit a finite trivializing cover, so the tower below would not be built from cases already settled.) Choose a fibre metric on E, so that the pair (E, E_0) may be replaced by the disk-sphere pair $(D(E), S(E))$ up to homotopy of pairs, and $H^*(E, E_0;\mathbb{Z}) \cong \widetilde{H}^*(\mathrm{Th}(E);\mathbb{Z})$ for the Thom space of definition 8.5.4. The Thom space is the increasing union of the subcomplexes $\mathrm{Th}(E|_{B^{(m)}})$. Both $\widetilde{H}^*(\mathrm{Th}(E))$ and $H^*(B)$ therefore sit in Milnor exact sequences over these filtrations (theorem 8.2.11, applied to ordinary reduced cohomology, which is a reduced cohomology theory by example 8.2.7), and Φ is a map between the two short exact sequences: it is compatible with the restriction maps by naturality of cup product, and the Thom classes over the skeleta are compatible by uniqueness. By Step 3 it is an isomorphism at every finite stage, hence an isomorphism on both the $\varprojlim$ and the $\varprojlim^1$ terms, and lemma 3.4.22 gives that it is an isomorphism for B. Existence and uniqueness of u_E pass to the limit the same way, the $\varprojlim^1$ obstruction in degree $n-1$ vanishing because the tower of groups $H^{n-1}(B^{(m)};\mathbb{Z})$ has surjective restriction maps once $m \geq n$, so corollary 8.2.12 applies.

Step 5: The mod-2 statement. Every step above used the orientation only to name a preferred generator of H^n of the fiber pair. With $\mathbb{Z}/2$ coefficients that group is $\mathbb{Z}/2$, whose nonzero element is unique and therefore canonical, so no orientation is required and the identical argument runs with $R = \mathbb{Z}/2$ throughout, as we sought to show.

Proposition 7.2.4: The Gysin Sequence from the Thom Isomorphism

Let $E \to B$ be an oriented real vector bundle of rank n over a CW complex of finite type, with sphere bundle $S(E) \to B$ and Thom class u_E. Transporting the long exact sequence of the pair $(D(E), S(E))$ through the Thom isomorphism Φ of theorem 7.2.3 and the homotopy equivalence $z\colon B \to D(E)$ produces a long exact sequence in which the map $H^{m-n}(B;\mathbb{Z}) \to H^m(B;\mathbb{Z})$ is cup product with $z^*j^*(u_E)$. This is the Gysin sequence of theorem 6.3.2 for $S(E) \to B$, and consequently $z^*j^*(u_E) = e(E)$.

***Proof*:**
The zero section z is a homotopy equivalence with inverse π, so $H^*(D(E);\mathbb{Z}) \cong H^*(B;\mathbb{Z})$, and Φ gives $H^m(D(E), S(E);\mathbb{Z}) \cong H^{m-n}(B;\mathbb{Z})$ by theorem 7.2.3. Substituting both into the long exact sequence of the pair yields a long exact sequence relating $H^*(B)$ and $H^*(S(E))$ in which the connecting map lowers degree by $n-1$: this is the shape of theorem 6.3.2.

Unwinding the identifications, the map $H^{m-n}(B) \to H^m(B)$ sends α to $z^*j^*\Phi(\alpha) = z^*j^*(\pi^*\alpha \smile u_E)$. Both z^* and j^* are ring homomorphisms and $z^*\pi^* = (\pi z)^* = \mathrm{id}$, so this equals $\alpha \smile z^*j^*(u_E)$.

Both sequences are long exact sequences relating the same two graded groups $H^*(B)$ and $H^*(S(E))$ through the same maps π^* and restriction to $S(E)$, so they agree term by term, and the remaining arrow is determined: cup product with $z^*j^*(u_E)$ on the one hand and, by theorem 6.3.2, cup product with $e(E)$ on the other. Evaluating both at $\alpha = 1$ gives $z^*j^*(u_E) = e(E)$, as we sought to show.

The theorem says that the cohomology of the pair (E, E_0) is a free module of rank one over the cohomology of the base, generated by u_E. The Thom class is thus the exact analogue for a bundle of a fundamental class for a manifold, and the isomorphism is the analogue of Poincaré duality: it converts a degree shift by the rank into multiplication by a single class. The Euler class is the shadow this leaves on the base.

Corollary 7.2.5: The Euler Class is the Restriction of the Thom Class

Let $E \to B$ be an oriented real vector bundle of rank n over a CW complex of finite type, with Thom class u_E given by theorem 7.2.3. Let $j^*\colon H^n(E, E_0;\mathbb{Z}) \to H^n(E;\mathbb{Z})$ be induced by the inclusion of pairs $(E, \emptyset) \hookrightarrow (E, E_0)$ and let $z\colon B \to E$ be the zero section. Then the Euler class of definition 7.1.21 satisfies

$$e(E) = z^*\big(j^*(u_E)\big)$$

Equivalently, under the isomorphism $\pi^*\colon H^n(B;\mathbb{Z}) \xrightarrow{\cong} H^n(E;\mathbb{Z})$, the class $j^*(u_E)$ corresponds to $e(E)$.

***Proof*:**
This is the last clause of proposition 7.2.4, which identifies $z^*j^*(u_E)$ with the Euler class by matching the long exact sequence of the pair, transported through Φ, against the Gysin sequence that definition 7.1.21 uses to define $e(E)$.

The equivalent formulation follows because z and π are mutually inverse homotopy equivalences, so $z^* = (\pi^*)^{-1}$ on cohomology. The orientation enters both sides through the same choice: definition 7.2.1 uses it to name the generator of H^n of the fibre pair and definition 7.1.21 to trivialize the local system in the Gysin sequence, and proposition 7.2.4 matches the two sequences under that one choice, so no sign discrepancy arises, completing the proof.

This discharges the description asserted in definition 7.1.21, where the Euler class was said to be "equivalently" the image of the Thom class; the equivalence is now a theorem rather than a remark. Two properties of the Euler class follow at once, and neither is visible from the Gysin construction.

Proposition 7.2.6: Naturality of the Euler Class

Let $E \to X$ be an oriented real vector bundle of rank n over a CW complex of finite type, and let $f\colon X' \to X$ be a map with X' paracompact Hausdorff and of the same type, so that both Euler classes are defined. Then, giving f^*E the pulled-back orientation,

$$e(f^*E) = f^*\big(e(E)\big) \in H^n(X';\mathbb{Z})$$

***Proof*:**
The Thom class is natural: the map of pairs $(f^*E, f^*E \setminus 0) \to (E, E \setminus 0)$ covering f carries the pulled-back orientation to the given one fiberwise, so it pulls the Thom class u_E back to the Thom class u_{f^*E}, these being characterized fiberwise by definition 7.2.1. The Euler class is the restriction of the Thom class along the zero section (corollary 7.2.5), and the zero sections commute with the map of pairs, so restricting the identity $\tilde{f}^*u_E = u_{f^*E}$ along them gives $f^*e(E) = e(f^*E)$, completing the proof.

Proposition 7.2.7: Whitney Sum Formula for the Euler Class

Let $E_1, E_2 \to B$ be oriented real vector bundles over a CW complex of finite type, of ranks n_1 and n_2, and give $E_1 \oplus E_2$ the direct-sum orientation. Then

$$e(E_1 \oplus E_2) = e(E_1) \smile e(E_2)$$

***Proof*:**
Write $E = E_1 \oplus E_2$ and let $q_i\colon E \to E_i$ be the two projections. The pair (E, E_0) receives a map from the "external" pair: a point of E lies outside the zero section exactly when at least one of its two components does, so

$$E_0 = q_1^{-1}\big((E_1)_0\big) \cup q_2^{-1}\big((E_2)_0\big)$$

and the relative cup product therefore lands in the right place, giving a class $q_1^*(u_{E_1}) \smile q_2^*(u_{E_2}) \in H^{n_1+n_2}(E, E_0;\mathbb{Z})$. Restricted to a fiber this is the product of the two fiber generators, which is the generator of $H^{n_1+n_2}(\mathbb{R}^{n_1+n_2}, \mathbb{R}^{n_1+n_2} \setminus \{0\};\mathbb{Z})$ determined by the direct-sum orientation, by the definition of that orientation. Hence $u_E = q_1^*(u_{E_1}) \smile q_2^*(u_{E_2})$, by the uniqueness clause of theorem 7.2.3.

Now apply z^*j^*, using corollary 7.2.5. Both z^* and j^* are ring maps on the relevant cohomology, and the zero section of E composed with q_i is the zero section of E_i followed by no further map,

so
$$e(E) = z^*j^*\left(q_1^*u_{E_1} \smile q_2^*u_{E_2}\right) = z^*j^*q_1^*(u_{E_1}) \smile z^*j^*q_2^*(u_{E_2}) = e(E_1) \smile e(E_2)$$
as we sought to show.

Proposition 7.2.8: The Euler Class of an Odd-Rank Bundle is 2-Torsion

Let $E \to B$ be an oriented real vector bundle of odd rank n over a CW complex of finite type. Then $2e(E) = 0$.

***Proof*:**
Let $\sigma \colon E \to E$ be fiberwise multiplication by -1. It is an isomorphism of vector bundles over id_B, so it carries E to E and preserves the zero section, and the induced map on $H^*(B)$ is the identity because σ covers id_B. On a fiber, σ acts on $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; \mathbb{Z})$ by the degree of the antipodal map of S^{n-1}, namely $(-1)^n$, which is -1 because n is odd. So σ reverses the orientation of every fiber, and therefore $\sigma^*(u_E) = -u_E$ by the uniqueness clause of theorem 7.2.3 applied to the orientation-reversed bundle.

Applying z^*j^* and using corollary 7.2.5 together with $\sigma \circ z = z$ gives $e(E) = z^*j^*\sigma^*(u_E) = -e(E)$, hence $2e(E) = 0$, completing the proof.

Example 7.2.9: The Thom Class of the Möbius Bundle

Let $L \to S^1$ be the Möbius line bundle of example 8.5.6, which is non-orientable, so it has no integral Thom class. It does have a mod-2 one, and theorem 7.2.3 predicts $H^{k+1}(L, L_0; \mathbb{Z}/2) \cong H^k(S^1; \mathbb{Z}/2)$, that is $\mathbb{Z}/2$ for $k = 0$ and $k = 1$ and zero otherwise. This is confirmed by the identification $\mathrm{Th}(L) \cong \mathbb{R}\mathrm{P}^2$ of example 8.5.6: the reduced mod-2 cohomology of $\mathbb{R}\mathrm{P}^2$ is $\mathbb{Z}/2$ in degrees 1 and 2 and zero elsewhere, matching the prediction after the shift by the rank $n = 1$. The Thom class is the generator $\alpha \in H^1(\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2)$, and the Thom isomorphism is multiplication by α, which carries the degree-0 generator to α and the degree-1 generator to $\alpha^2 \neq 0$. The nonvanishing of α^2, which is what distinguished $\mathbb{R}\mathrm{P}^2$ from a suspension in example 8.5.6, is exactly the statement that the Thom isomorphism is multiplication by a class whose square is nonzero, that is, that L is nontrivial.

Exercise 7.2.1

1. Let $E \to B$ be an oriented rank-n bundle over a CW complex and suppose E admits a nowhere-zero section s. Show directly from corollary 7.2.5 that $e(E) = 0$. (The section gives a factorization of z through E_0 up to homotopy.)

2. Verify proposition 7.2.8 is sharp: exhibit an oriented bundle of odd rank whose Euler class is nonzero. (Consider TS^2 and explain why the rank being even is what lets e be non-torsion, then find an odd-rank example over a base with 2-torsion in H^n.)

3. Let $E \to B$ be an oriented rank-n bundle over a finite CW complex with $\dim B < n$. Show that $e(E) = 0$, and identify the step at which theorem 7.2.3 is used.

4. Show that the trivial oriented bundle $\underline{\mathbb{R}}^n$ over a finite CW complex has $e(\underline{\mathbb{R}}^n) = 0$ for $n \geq 1$, and deduce from proposition 7.2.7 that $e(E \oplus \underline{\mathbb{R}}) = 0$ for every oriented bundle E. Conclude that the Euler class, unlike the Stiefel-Whitney and Pontryagin classes, is not a stable invariant.

5. Let $E_1 \to B$ and $E_2 \to B$ be oriented bundles with E_2 of odd rank. Deduce from proposition 7.2.7 and proposition 7.2.8 that $2e(E_1 \oplus E_2) = 0$, and give an example where $e(E_1)$ and $e(E_2)$ are both nonzero while their product vanishes.

7.3 Pontryagin Classes

The Stiefel-Whitney classes are mod-2 invariants, and over $\mathbb{Z}$ or $\mathbb{Q}$ they carry no information: reduced from $H^*(BO(n);\mathbb{Z}/2)$, they cannot distinguish real bundles rationally, and in particular cannot give the characteristic numbers that separate smooth manifolds up to cobordism. Real bundles do carry integral characteristic classes, but not through the integral cohomology of $BO(n)$, which is laden with 2-torsion. The classes are obtained instead by *complexifying*: a real bundle E has a complexification $E \otimes_{\mathbb{R}} \mathbb{C}$, a complex bundle whose Chern classes are honest integral classes of E. The odd Chern classes of the complexification are 2-torsion, but the even ones survive, and they are the Pontryagin classes, the fundamental integral characteristic classes of real bundles and, as this section ends by showing, rationally the only ones. They are the invariants behind the Pontryagin numbers, the signature theorem, and the index theorem.

The construction begins with the observation that the complexification of a real bundle is isomorphic to its own conjugate, which is what forces the odd Chern classes to be torsion.

Lemma 7.3.1: Complexification is Self-Conjugate

Let $E \to X$ be a real vector bundle over a paracompact Hausdorff base, with complexification $E_{\mathbb{C}} := E \otimes_{\mathbb{R}} \mathbb{C}$, a complex vector bundle of the same rank. Then $E_{\mathbb{C}}$ is isomorphic to its complex conjugate, $\overline{E_{\mathbb{C}}} \cong E_{\mathbb{C}}$, and consequently its odd Chern classes are 2-torsion:

$$2\, c_{2i-1}(E_{\mathbb{C}}) = 0 \qquad \text{for all } i \geq 1$$

***Proof*:**
The conjugate $\overline{F}$ of a complex bundle F is the same underlying real bundle with the complex structure J replaced by $-J$, so that multiplication by z becomes multiplication by $\overline{z}$. For $E_{\mathbb{C}} = E \otimes_{\mathbb{R}} \mathbb{C}$ the map

$$\varphi\colon E_{\mathbb{C}} \to E_{\mathbb{C}}, \qquad v \otimes z \mapsto v \otimes \overline{z}$$

is a real bundle isomorphism, and it is complex-antilinear: $\varphi\big(z \cdot (v \otimes w)\big) = \varphi(v \otimes zw) = v \otimes \overline{zw} = \overline{z}\,(v \otimes \overline{w}) = \overline{z}\,\varphi(v \otimes w)$. A complex-antilinear self-isomorphism of $E_{\mathbb{C}}$ is the same datum as a complex-linear isomorphism $E_{\mathbb{C}} \xrightarrow{\sim} \overline{E_{\mathbb{C}}}$, so $E_{\mathbb{C}} \cong \overline{E_{\mathbb{C}}}$.

For the Chern classes, first take a complex line bundle L: its conjugate is the dual, $\overline{L} \cong L^*$ (a Hermitian metric gives a conjugate-linear isomorphism $L \to L^*$), so $c_1(\overline{L}) = c_1(L^*) = -c_1(L)$. For a bundle F of any rank the splitting principle (theorem 7.4.3) pulls F back to a sum $L_1 \oplus \cdots \oplus L_r$ of line bundles along a map injective on cohomology; then $\overline{F}$ pulls back to $\overline{L_1} \oplus \cdots \oplus \overline{L_r}$, and with $x_j = c_1(L_j)$ the Whitney formula gives

$$c(\overline{F}) = \prod_{j=1}^{r} \big(1 + c_1(\overline{L_j})\big) = \prod_{j=1}^{r} (1 - x_j)$$

whose degree-i part is $(-1)^i \sigma_i(x_1, \ldots, x_r) = (-1)^i c_i(F)$. Hence $c_i(\overline{F}) = (-1)^i c_i(F)$ for every complex bundle F. Applying this to $F = E_{\mathbb{C}}$ together with $\overline{E_{\mathbb{C}}} \cong E_{\mathbb{C}}$ gives $c_i(E_{\mathbb{C}}) = c_i(\overline{E_{\mathbb{C}}}) = (-1)^i c_i(E_{\mathbb{C}})$. For i odd this reads $c_i(E_{\mathbb{C}}) = -c_i(E_{\mathbb{C}})$, that is $2\,c_i(E_{\mathbb{C}}) = 0$, as we sought to show.

The lemma says the odd-degree Chern data of $E_{\mathbb{C}}$ is invisible rationally: over $\mathbb{Q}$, or after inverting 2, the odd Chern classes vanish and only the even ones $c_2(E_{\mathbb{C}}), c_4(E_{\mathbb{C}}), \ldots$ remain. These are the classes worth naming.

Definition 7.3.2: Pontryagin Classes

Let $E \to X$ be a real vector bundle of rank n. For $1 \le i \le \lfloor n/2 \rfloor$ the *i-th Pontryagin class* of E is

$$p_i(E) := (-1)^i\, c_{2i}(E \otimes_{\mathbb{R}} \mathbb{C}) \in H^{4i}(X; \mathbb{Z})$$

the sign-corrected $2i$-th Chern class of the complexification. The *total Pontryagin class* is

$$p(E) := 1 + p_1(E) + p_2(E) + \cdots + p_{\lfloor n/2 \rfloor}(E) \in H^*(X; \mathbb{Z})$$

with the convention $p_i(E) = 0$ for $2i > n$.

The sign $(-1)^i$ is a normalization: it makes the Pontryagin classes of the tangent bundles of the standard examples come out positive, and it turns the self-intersection sign in the splitting computation below into a plus. The degree jump to $4i$ reflects that a Pontryagin class packages two Chern degrees at once, and only $\lfloor n/2 \rfloor$ classes appear because $c_{2i}(E_{\mathbb{C}}) = 0$ once $2i$ exceeds the complex rank n of $E_{\mathbb{C}}$. The Whitney sum formula holds only up to 2-torsion, which is the price of the definition through the complexification.

Proposition 7.3.3: Naturality and the Whitney Sum Formula for Pontryagin Classes

Let $E, F \to X$ be real vector bundles over a paracompact Hausdorff base.

1. *(Naturality)* For every map $g\colon Y \to X$ of paracompact Hausdorff spaces, one has $p_i(g^*E) = g^*p_i(E)$.
2. *(Whitney sum, modulo 2-torsion)* The total Pontryagin class is multiplicative up to 2-torsion:
$$2\big(p(E \oplus F) - p(E) \smile p(F)\big) = 0$$
In particular $p(E \oplus F) = p(E) \smile p(F)$ in $H^*(X; \mathbb{Q})$ and in $H^*(X; \mathbb{Z}[\frac{1}{2}])$.

***Proof*:**
(1) Complexification commutes with pullback, $(g^*E) \otimes_{\mathbb{R}} \mathbb{C} = g^*(E \otimes_{\mathbb{R}} \mathbb{C})$, and Chern classes are natural (definition 7.1.13), so $p_i(g^*E) = (-1)^i c_{2i}\big(g^*(E_{\mathbb{C}})\big) = (-1)^i g^* c_{2i}(E_{\mathbb{C}}) = g^*p_i(E)$.

(2) Complexification is additive, $(E \oplus F)_{\mathbb{C}} = E_{\mathbb{C}} \oplus F_{\mathbb{C}}$, so the complex Whitney sum formula (theorem 7.4.5) gives $c(E_{\mathbb{C}} \oplus F_{\mathbb{C}}) = c(E_{\mathbb{C}}) \smile c(F_{\mathbb{C}})$. Work modulo 2-torsion, writing $\equiv$ for congruence there; by lemma 7.3.1 the odd Chern classes of every complexification vanish modulo 2-torsion, so

$$c(E_{\mathbb{C}}) \equiv \sum_{i \ge 0} c_{2i}(E_{\mathbb{C}}) = \sum_{i \ge 0} (-1)^i p_i(E)$$

and likewise for F and for $E \oplus F$ (a product of a class with a 2-torsion class is again 2-torsion, so the congruence survives multiplication). Introduce the graded ring automorphism ψ of $H^{4\bullet}(X;\mathbb{Z})$ that multiplies H^{4i} by $(-1)^i$; it is multiplicative because $\psi(a \smile b) = (-1)^{i+j} a \smile b = \psi(a) \smile \psi(b)$ for $a \in H^{4i}$ and $b \in H^{4j}$. By construction $\psi\big(\sum_i (-1)^i p_i(E)\big) = \sum_i p_i(E) = p(E)$. Applying ψ to the congruence $c\big((E \oplus F)_{\mathbb{C}}\big) \equiv c(E_{\mathbb{C}}) \smile c(F_{\mathbb{C}})$ and using that ψ is a ring homomorphism yields

$$p(E \oplus F) \equiv p(E) \smile p(F) \pmod{2\text{-torsion}}$$

which is the first assertion. Over $\mathbb{Q}$ or $\mathbb{Z}[\frac{1}{2}]$ the 2-torsion vanishes and the equality is exact, completing the proof.

A concrete computation shows the classes are nonzero and ties them to the geometry of manifolds.

Definition 7.3.4: Stiefel-Whitney Number

Let M be a closed smooth m-manifold with tangent bundle TM and mod-2 fundamental class $[M] \in H_m(M;\mathbb{F}_2)$ (definition 4.4.8, summed over components when M is disconnected). For a partition $I = (i_1, \ldots, i_r)$ of m, the associated *Stiefel-Whitney number* is

$$w_I[M] = \big\langle w_{i_1}(TM) \smile \cdots \smile w_{i_r}(TM),\ [M]\big\rangle \in \mathbb{F}_2$$

formed from the Stiefel-Whitney classes of definition 7.1.19. Only partitions of m are considered, since the cup product must land in $H^m(M;\mathbb{F}_2)$ to be paired with $[M]$.

Definition 7.3.5: Pontryagin Number

Let M be a closed oriented smooth manifold of dimension $m = 4k$ with tangent bundle TM and integral fundamental class $[M] \in H_m(M;\mathbb{Z})$. For a partition $I = (i_1, \ldots, i_r)$ of k, the associated *Pontryagin number* is

$$p_I[M] = \big\langle p_{i_1}(TM) \smile \cdots \smile p_{i_r}(TM),\ [M]\big\rangle \in \mathbb{Z}$$

formed from the Pontryagin classes of definition 7.3.2. Since $p_{i_j}(TM) \in H^{4i_j}(M;\mathbb{Z})$, the product lands in $H^{4k}(M;\mathbb{Z}) = H^m(M;\mathbb{Z})$ exactly when I partitions k, so Pontryagin numbers exist only when $\dim M \equiv 0 \pmod 4$.

Example 7.3.6: Pontryagin Classes of Spheres and of $\mathbb{C}\mathrm{P}^2$

1. *The tangent bundle of a sphere.* The tangent bundle TS^n is stably trivial: $TS^n \oplus \nu \cong \underline{\mathbb{R}}^{n+1}$, where ν is the trivial normal line bundle of the standard embedding $S^n \subseteq \mathbb{R}^{n+1}$. By proposition 7.3.3, modulo 2-torsion $p(TS^n) \smile p(\nu) = p(\underline{\mathbb{R}}^{n+1}) = 1$, and $p(\nu) = 1$, so $p(TS^n) \equiv 1$. Since $H^*(S^n;\mathbb{Z})$ is torsion-free the congruence is an equality: $p(TS^n) = 1$, all Pontryagin classes of a sphere vanish.

2. *The complex projective plane.* Regard $T\mathbb{C}\mathrm{P}^2$ as a real rank-4 bundle. Its complexification is $(T\mathbb{C}\mathrm{P}^2)_{\mathbb{C}} \cong T' \oplus \overline{T'}$, where $T' = T^{1,0}\mathbb{C}\mathrm{P}^2$ is the holomorphic tangent bundle, a complex rank-2 bundle with total Chern class $c(T') = (1+u)^3 = 1 + 3u + 3u^2$ (from the Euler

sequence, with $u \in H^2(\mathbb{C}\mathrm{P}^2;\mathbb{Z})$ the hyperplane class and $u^3 = 0$). Then

$$c\big((T\mathbb{C}\mathrm{P}^2)_{\mathbb{C}}\big) = c(T') \smile c(\overline{T'}) = (1+3u+3u^2)(1-3u+3u^2) = 1-3u^2$$

using $c(\overline{T'}) = 1-3u+3u^2$ and $u^3 = 0$. Hence $c_2\big((T\mathbb{C}\mathrm{P}^2)_{\mathbb{C}}\big) = -3u^2$ and

$$p_1(T\mathbb{C}\mathrm{P}^2) = (-1)^1 c_2\big((T\mathbb{C}\mathrm{P}^2)_{\mathbb{C}}\big) = 3u^2$$

a generator of $H^4(\mathbb{C}\mathrm{P}^2;\mathbb{Z}) \cong \mathbb{Z}$ tripled. The Pontryagin number is $\langle p_1(T\mathbb{C}\mathrm{P}^2), [\mathbb{C}\mathrm{P}^2]\rangle = 3$, and the relation $\frac{1}{3}p_1[\mathbb{C}\mathrm{P}^2] = 1 = \sigma(\mathbb{C}\mathrm{P}^2)$ to the signature is the first case of the Hirzebruch signature theorem.

The Pontryagin classes exhaust the rational characteristic classes of real bundles, exactly as the Chern classes exhaust the complex ones. The precise statement is the rational cohomology of the classifying spaces, and it is proved by the same splitting philosophy used for $BU(n)$, now applied to the complexification and refined by the involution that conjugation induces on the Chern roots.

Theorem 7.3.7: Rational Cohomology of $BO(n)$ and $BSO(n)$

As graded rings,

$$H^*(BO(n);\mathbb{Q}) \cong \mathbb{Q}[p_1,\dots,p_{\lfloor n/2\rfloor}], \qquad |p_i| = 4i$$

a polynomial ring on the universal Pontryagin classes. For the oriented classifying spaces,

$$H^*(BSO(2m+1);\mathbb{Q}) \cong \mathbb{Q}[p_1,\dots,p_m], \qquad H^*(BSO(2m);\mathbb{Q}) \cong \mathbb{Q}[p_1,\dots,p_{m-1},e]$$

where $e \in H^{2m}(BSO(2m);\mathbb{Q})$ is the Euler class of definition 7.1.21 and $e^2 = p_m$. In every case the rational characteristic classes of real bundles are the polynomials in the Pontryagin classes, together with the Euler class in the oriented even-rank case.

***Proof*:**
All cohomology in this proof is rational. The computation is an induction on n through the oriented universal sphere bundle

$$S^{n-1} \longrightarrow BSO(n-1) \xrightarrow{\rho} BSO(n)$$

of definition 7.1.21, with Euler class $e_n \in H^n(BSO(n);\mathbb{Q})$; the fiber is oriented, so the rational Gysin sequence of theorem 6.3.2 applies, reading

$$\cdots \to H^{k-n}(BSO(n)) \xrightarrow{\smile e_n} H^k(BSO(n)) \xrightarrow{\rho^*} H^k(BSO(n-1)) \xrightarrow{\int} H^{k-n+1}(BSO(n)) \to \cdots$$

with $\int$ lowering degree by $n-1$. Two pullback facts drive the induction. First, ρ classifies $\gamma_{n-1} \oplus \underline{\mathbb{R}}$ (the universal bundle of $BSO(n-1)$ with a trivial line, since $SO(n-1)$ fixes a vector), so $\rho^* p_i = p_i$ by stability of the Pontryagin classes (proposition 7.3.3) and $\rho^* e_n = e(\gamma_{n-1} \oplus \underline{\mathbb{R}}) = 0$, the Euler class of a bundle with a nowhere-zero section. Second, for odd rank the Euler class vanishes rationally: the fiberwise negation $-1\colon \gamma_{2m+1} \to \gamma_{2m+1}$ has fiberwise determinant $(-1)^{2m+1} = -1$, so it is an orientation-reversing automorphism carrying the universal bundle to its orientation-reverse and giving $e_{2m+1} = -e_{2m+1}$, whence $e_{2m+1} = 0$ in $H^*(BSO(2m+1);\mathbb{Q})$.

Base cases. $BSO(1)$ is a point with $H^* = \mathbb{Q}$, and $BSO(2) = BU(1) = \mathbb{C}\mathrm{P}^\infty$ (as $SO(2) = U(1)$)

with $H^*(BSO(2);\mathbb{Q}) = \mathbb{Q}[e_2]$, $|e_2| = 2$, the Euler class equal to c_1 (example 7.1.22). These are the cases $n = 1, 2$.

Even rank $n = 2m$. By induction $H^*(BSO(2m-1);\mathbb{Q}) = \mathbb{Q}[p_1, \ldots, p_{m-1}]$, and the fiber S^{2m-1} is odd. The target ring is generated by the classes $p_i = \rho^* p_i$, so ρ^* is surjective; by the exactness $\mathrm{im}\rho^* = \ker \int$ this forces $\int = 0$, and the Gysin sequence breaks into short exact sequences

$$0 \to H^{k-2m}(BSO(2m)) \xrightarrow{\smile e_{2m}} H^k(BSO(2m)) \xrightarrow{\rho^*} H^k(BSO(2m-1)) \to 0$$

with $\smile e_{2m}$ injective (its kernel is the image of the incoming $\int = 0$). By the module argument of theorem 7.1.11, $H^*(BSO(2m);\mathbb{Q})$ is free over $H^*(BSO(2m-1);\mathbb{Q})$ on the powers of e_{2m}, so

$$H^*(BSO(2m);\mathbb{Q}) \cong \mathbb{Q}[p_1, \ldots, p_{m-1}][e] = \mathbb{Q}[p_1, \ldots, p_{m-1}, e]$$

with $e = e_{2m}$ of degree $2m$. Finally $p_m = e^2$: as real bundles $\gamma_{2m} \otimes_{\mathbb{R}} \mathbb{C} \cong \gamma_{2m} \oplus \gamma_{2m}$, so identifying the top Chern class with the Euler class of the underlying oriented real bundle (theorem 7.4.11) and using multiplicativity of the Euler class gives $c_{2m}(\gamma_{2m} \otimes_{\mathbb{R}} \mathbb{C}) = \pm e^2$, with the sign $(-1)^m$ of definition 7.3.2 fixed precisely so that $p_m = (-1)^m c_{2m}(\gamma_{\mathbb{C}}) = e^2$.

Odd rank $n = 2m+1$. By induction $H^*(BSO(2m);\mathbb{Q}) = \mathbb{Q}[p_1, \ldots, p_{m-1}, e]$ with $|e| = 2m$ and $e^2 = p_m$, and the fiber S^{2m} is even. Here $e_{2m+1} = 0$ by the odd-rank vanishing, so $\smile e_{2m+1} = 0$ throughout, and the Gysin sequence breaks into short exact sequences

$$0 \to H^k(BSO(2m+1)) \xrightarrow{\rho^*} H^k(BSO(2m)) \xrightarrow{\int} H^{k-2m}(BSO(2m+1)) \to 0$$

In particular ρ^* is injective, realizing $H^*(BSO(2m+1);\mathbb{Q})$ as the subring $\mathrm{im}\rho^*$ of $\mathbb{Q}[p_1, \ldots, p_{m-1}, e]$, which contains $\rho^* p_i = p_i$ for $i < m$ and $\rho^* p_m = e^2$, hence contains the polynomial ring $\mathbb{Q}[p_1, \ldots, p_{m-1}, e^2]$ on m algebraically independent classes. To see there is nothing more, compare Poincaré series: the short exact sequence gives $\dim H^k(BSO(2m+1)) = \dim H^k(BSO(2m)) - \dim H^{k-2m}(BSO(2m+1))$, so $P_{2m+1} = P_{2m} - t^{2m} P_{2m+1}$, that is $P_{2m+1} = P_{2m}/(1+t^{2m})$. With $P_{2m} = (1-t^{2m})^{-1} \prod_{i=1}^{m-1} (1-t^{4i})^{-1}$ and the identity $(1-t^{2m})^{-1}(1+t^{2m})^{-1} = (1-t^{4m})^{-1}$,

$$P_{2m+1}(t) = \prod_{i=1}^{m} \frac{1}{1-t^{4i}}$$

the Poincaré series of $\mathbb{Q}[p_1, \ldots, p_m]$ with $|p_i| = 4i$. Since that polynomial ring already injects as $\mathbb{Q}[p_1, \ldots, p_{m-1}, e^2] \subseteq \mathrm{im}\rho^*$ and matches the dimension in every degree, the inclusion is an equality, $H^*(BSO(2m+1);\mathbb{Q}) = \mathbb{Q}[p_1, \ldots, p_m]$.

The unoriented case. The double cover $BSO(n) \to BO(n)$ has deck group $O(n)/SO(n) = \mathbb{Z}/2$, and since 2 is invertible in $\mathbb{Q}$ the transfer gives $H^*(BO(n);\mathbb{Q}) = H^*(BSO(n);\mathbb{Q})^{\mathbb{Z}/2}$, the invariants of an orientation-reversing element. That element fixes each p_i (pulled back from $BO(n)$) and negates the Euler class, $e \mapsto -e$. For odd n there is no Euler class and the invariant ring is all of $\mathbb{Q}[p_1, \ldots, p_m]$; for even $n = 2m$ the $e \mapsto -e$ invariants of $\mathbb{Q}[p_1, \ldots, p_{m-1}, e]$ are $\mathbb{Q}[p_1, \ldots, p_{m-1}, e^2] = \mathbb{Q}[p_1, \ldots, p_m]$. In both cases $H^*(BO(n);\mathbb{Q}) = \mathbb{Q}[p_1, \ldots, p_{\lfloor n/2 \rfloor}]$, completing the proof.

The theorem is the sense in which Pontryagin classes are indispensable: rationally they, with the Euler class, generate every characteristic class of a real bundle, just as the Chern classes generate the complex ones and the Stiefel-Whitney classes generate the mod-2 ones. The three computations

$H^*(BU(n))$, $H^*(BO(n);\mathbb{Z}/2)$, and $H^*(BO(n);\mathbb{Q})$ are the three faces of the same classifying-space philosophy, one for each coefficient regime, and together they account for all characteristic classes of vector bundles. The integral cohomology of $BO(n)$, which interpolates between the mod-2 and rational answers and carries the subtle 2-torsion, is the one piece left uncomputed, here and in [21] alike.

Exercise 7.3.1

1. For bundles over a paracompact Hausdorff base, show that Pontryagin classes are stable: $p(E \oplus \underline{\mathbb{R}}^k) = p(E)$ for a trivial bundle $\underline{\mathbb{R}}^k$. Deduce that Pontryagin classes are invariants of the stable isomorphism class of E, and re-derive $p(TS^n) = 1$ from stable triviality.
2. Let $V \to X$ be a complex vector bundle of rank r over a paracompact Hausdorff base with Chern classes $c_j = c_j(V)$, and let $V_{\mathbb{R}}$ be its underlying real rank-$2r$ bundle. Using $(V_{\mathbb{R}})_{\mathbb{C}} \cong V \oplus \overline{V}$, prove the identity
$$1 - p_1(V_{\mathbb{R}}) + p_2(V_{\mathbb{R}}) - \cdots = (1 - c_1 + c_2 - \cdots)(1 + c_1 + c_2 + \cdots)$$
and deduce $p_k(V_{\mathbb{R}}) = c_k^2 - 2c_{k-1}c_{k+1} + \cdots + (-1)^k 2c_{2k}$ modulo 2-torsion. Specialize to a line bundle to recover $p_1(L_{\mathbb{R}}) = c_1(L)^2$.
3. Compute the total Pontryagin class $p(T\mathbb{C}\mathrm{P}^n)$ for general n from $(T\mathbb{C}\mathrm{P}^n)_{\mathbb{C}} \cong T' \oplus \overline{T'}$ and $c(T'\mathbb{C}\mathrm{P}^n) = (1+u)^{n+1}$, and verify $p(T\mathbb{C}\mathrm{P}^n) = (1+u^2)^{n+1}$ in $H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = \mathbb{Z}[u]/(u^{n+1})$. Read off $p_1(T\mathbb{C}\mathrm{P}^2) = 3u^2$ as a special case and compute $p_1(T\mathbb{C}\mathrm{P}^3)$.

Pairing a product of characteristic classes against the fundamental class of a closed manifold turns the classes into numbers, and those numbers are the form in which the theory reaches cobordism in chapter 8. Both kinds are recorded here, where the classes they are built from have just been constructed.

Both are diffeomorphism invariants, being built functorially from the tangent bundle and the fundamental class. What makes them worth naming is that they are invariants of the far coarser cobordism relation, and in the unoriented case a complete one: that is theorem 8.5.7 and corollary 8.5.9.

A closed formula for the Pontryagin classes of a complex bundle follows from lemma 7.3.1, and it is what makes the Pontryagin numbers of a complex manifold computable from Chern classes.

Corollary 7.3.8: Pontryagin Classes of a Complex Bundle

Let $F \to X$ be a complex vector bundle of complex rank n over a paracompact Hausdorff base, regarded as an oriented real bundle $F_{\mathbb{R}}$. Then $F_{\mathbb{R}} \otimes \mathbb{C} \cong F \oplus \overline{F}$, and for each i,
$$p_i(F_{\mathbb{R}}) = (-1)^i \sum_{r+s=2i} (-1)^s\, c_r(F) \smile c_s(F) \in H^{4i}(X;\mathbb{Z})$$
Equivalently, the total classes satisfy
$$1 - p_1(F_{\mathbb{R}}) + p_2(F_{\mathbb{R}}) - \cdots = \big(1 + c_1(F) + \cdots + c_n(F)\big) \smile \big(1 - c_1(F) + c_2(F) - \cdots\big)$$
in the degrees divisible by 4; the degree-$(4i+2)$ components of the right-hand side are 2-torsion and are discarded.

***Proof*:**
Step 1: The splitting. Let $J\colon F_{\mathbb{R}} \to F_{\mathbb{R}}$ be multiplication by i in the complex structure of F, a real bundle endomorphism with $J^2 = -\mathrm{id}$. Extend J complex-linearly to $F_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$, where it still satisfies $J^2 = -\mathrm{id}$, so its minimal polynomial divides $t^2 + 1 = (t-i)(t+i)$, which has distinct roots in $\mathbb{C}$. Hence $F_{\mathbb{R}} \otimes \mathbb{C}$ splits as the direct sum of the $(+i)$- and $(-i)$-eigenbundles of J, and these are F and $\overline{F}$ respectively: on the $(+i)$-eigenbundle the extended J acts as multiplication by i, which is the original complex structure, while the $(-i)$-eigenbundle carries the conjugate one. Eigenbundle decompositions of a bundle endomorphism with locally constant eigenvalue multiplicities are subbundles, so $F_{\mathbb{R}} \otimes \mathbb{C} \cong F \oplus \overline{F}$.

Step 2: Reading off the classes. By definition 7.3.2, $p_i(F_{\mathbb{R}}) = (-1)^i c_{2i}(F_{\mathbb{R}} \otimes \mathbb{C})$. Applying the Whitney sum formula (theorem 7.4.5) to the splitting of Step 1 gives $c(F_{\mathbb{R}} \otimes \mathbb{C}) = c(F) \smile c(\overline{F})$, and $c_s(\overline{F}) = (-1)^s c_s(F)$ because conjugation reverses the sign of the first Chern class of a line bundle and the splitting principle (theorem 7.4.3) propagates this to the elementary symmetric functions. Extracting the degree-$4i$ component,

$$c_{2i}(F_{\mathbb{R}} \otimes \mathbb{C}) = \sum_{r+s=2i} c_r(F) \smile c_s(\overline{F}) = \sum_{r+s=2i} (-1)^s\, c_r(F) \smile c_s(F)$$

and multiplying by $(-1)^i$ gives the stated formula. The total-class identity is the same statement summed over i, the odd-degree-in-c components being exactly the degree-$(4i+2)$ terms discarded by definition 7.3.2, which are 2-torsion by lemma 7.3.1, completing the proof.

7.4 Characteristic Classes as Obstructions

The previous section built the characteristic classes by hand out of the cohomology of the classifying spaces: the Stiefel-Whitney classes $w_i(E) \in H^i(X;\mathbb{F}_2)$ as the pullbacks of the polynomial generators of $H^*(BO(n);\mathbb{F}_2) = \mathbb{F}_2[w_1,\ldots,w_n]$ along a classifying map $X \to BO(n)$, and the Chern classes $c_i(E) \in H^{2i}(X;\mathbb{Z})$ as the pullbacks of the generators of $H^*(BU(n);\mathbb{Z}) = \mathbb{Z}[c_1,\ldots,c_n]$, all of this developed in section 7.1. That construction is correct but opaque: it produces invariants without saying what they measure. This section gives the meaning along two complementary axes. The first axis is *formal*: a short list of axioms (naturality, the Whitney sum formula, normalization, and the identification of the top class with the Euler class) characterizes the classes uniquely and turns them into a computational calculus. The second axis is *geometric*: each characteristic class is an obstruction, in the precise sense of theorem 5.6.10, to a structure one would like the bundle to carry. The class $w_1(E)$ is the obstruction to orienting E; $w_2(E)$, once E is oriented, is the obstruction to a spin structure; and the top class, the Euler class or the top Stiefel-Whitney class, is the obstruction to a single nowhere-zero section. For the tangent bundle of a closed manifold this last obstruction is the Euler characteristic, closing the loop with Poincaré duality (section 4.5). The vector-bundle framework in which these classes arise is developed in [21], and their analytic incarnation through curvature is the Chern-Weil theory of the Riemannian-geometry volume [11]; the present chapter develops the same theory algebraic-topologically, constructing the classes from the cohomology of classifying spaces and reading each as a cohomological obstruction.

The starting point is the list of properties that pin the classes down. The first is naturality, which is immediate from the classifying-map definition, since a characteristic class is a pullback and pullback is functorial.

Proposition 7.4.1: Naturality of Characteristic Classes

Let $f\colon X \to Y$ be a map of paracompact Hausdorff spaces and $E \to Y$ a real (respectively complex) vector bundle. Then for the pulled-back bundle $f^*E \to X$,

$$w_i(f^*E) = f^*\big(w_i(E)\big) \in H^i(X;\mathbb{F}_2), \qquad c_i(f^*E) = f^*\big(c_i(E)\big) \in H^{2i}(X;\mathbb{Z})$$

for every i. Equivalently, the total classes satisfy $w(f^*E) = f^*w(E)$ and $c(f^*E) = f^*c(E)$.

***Proof*:**
A rank-n real bundle $E \to Y$ is classified by a map $g\colon Y \to BO(n)$ with $E \cong g^*\gamma^n$ for the universal bundle $\gamma^n \to BO(n)$, by theorem 7.1.9; the class $w_i(E)$ is by definition $g^*(w_i)$, where $w_i \in H^i(BO(n);\mathbb{F}_2)$ is the universal class of section 7.1. The pullback f^*E is classified by the composite $g \circ f\colon X \to BO(n)$, since $(g \circ f)^*\gamma^n \cong f^*(g^*\gamma^n) \cong f^*E$ by functoriality of pullback. Therefore

$$w_i(f^*E) = (g \circ f)^*(w_i) = f^*\big(g^*(w_i)\big) = f^*\big(w_i(E)\big)$$

using that pullback on cohomology is contravariantly functorial, $(g \circ f)^* = f^* \circ g^*$. The complex case is identical with $BO(n)$, γ^n, $\mathbb{F}_2$ replaced by $BU(n)$, the universal complex bundle, and $\mathbb{Z}$. Summing over i gives the total-class statement, completing the proof.

Naturality is the reason characteristic classes are computable at all: it reduces every computation to the universal case and lets one transport a known answer from one space to another along a map. The deeper axiom is multiplicativity under direct sum, the Whitney sum formula. It says that the total class converts the Whitney sum of bundles into a cup product of total classes, so that the characteristic class of a sum is determined by the classes of the summands. The proof rests on two facts about the classifying spaces: the cohomology of $BO(n)$ injects into the cohomology of $BO(1)^n$ as the symmetric polynomials in one-dimensional generators, and on line bundles the formula is a direct cohomological computation. This is the *splitting principle*, and it is the central technical device of the whole theory.

The principle is stated over an arbitrary paracompact Hausdorff base, which is exactly the generality section 7.1.1 was built to supply: the flag construction below projectivizes bundles over bases that carry no CW structure, so theorem 2.4.7 is unavailable and theorem 7.1.9 stands in its place. Every statement in this chapter whose base is paracompact requires it to be Hausdorff as well, and the hypothesis is written into each statement rather than left to a convention. This changes nothing against the chapter's sources, both of which build the Hausdorff property into the word paracompact ([15, §1.1], where paracompact means Hausdorff with subordinate partitions of unity for every open cover; [17, §5.8], where it means Hausdorff with locally finite refinements); Hausdorffness is what makes the subordinate partitions of unity of [12, Existence of Subordinate Partitions of Unity] available, and [12, Locally Trivial Maps with Compact Fiber Preserve Paracompactness] propagates both properties up the towers built here.

The projective-bundle input the splitting principle uses comes next: it is the mod-2, real sibling of proposition 7.1.15, stated over a paracompact Hausdorff base with no connectivity hypothesis, and its complex clause extends the integral statement to the same generality. Only the module structure is recorded, because only the module structure is used; the ring presentation of proposition 7.1.15 is not needed here.

Proposition 7.4.2: Projective Bundles over a Paracompact Base

Let $E \to X$ be a rank-n real vector bundle, $n \geq 1$, over a paracompact Hausdorff base. Its projectivization $\mathbb{P}(E) \xrightarrow{\pi} X$, the quotient of the complement of the zero section by fiberwise scalar multiplication, is a fiber bundle with fiber $\mathbb{R}\mathrm{P}^{n-1}$, and there is a class $x_E \in H^1(\mathbb{P}(E); \mathbb{F}_2)$ whose restriction to every fiber $\mathbb{P}(E_x) \cong \mathbb{R}\mathrm{P}^{n-1}$ is the generator of $H^1(\mathbb{R}\mathrm{P}^{n-1}; \mathbb{F}_2)$ (for $n = 1$ the condition is empty). For any such class, $H^*(\mathbb{P}(E); \mathbb{F}_2)$ is a free $H^*(X; \mathbb{F}_2)$-module with basis $1, x_E, \ldots, x_E^{n-1}$; in particular $\pi^*\colon H^*(X; \mathbb{F}_2) \to H^*(\mathbb{P}(E); \mathbb{F}_2)$ is a split injection. For a rank-n complex bundle the same holds with fibers $\mathbb{C}\mathrm{P}^{n-1}$, a class $t_E \in H^2(\mathbb{P}(E); \mathbb{Z})$ restricting on every fiber to a generator of $H^2(\mathbb{C}\mathrm{P}^{n-1}; \mathbb{Z})$ (for $n = 1$ the condition is empty), the module $H^*(\mathbb{P}(E); \mathbb{Z})$ free over $H^*(X; \mathbb{Z})$ with basis $1, t_E, \ldots, t_E^{n-1}$, and π^* a split injection on $H^*(-; \mathbb{Z})$.

***Proof*:**
Step 1: $\mathbb{P}(E)$ is a fiber bundle. A trivialization $p^{-1}(U) \cong U \times \mathbb{R}^n$ of E is fiberwise linear, so it carries the complement of the zero section to $U \times (\mathbb{R}^n \setminus 0)$ compatibly with scalar multiplication and descends to a homeomorphism $\pi^{-1}(U) \cong U \times \mathbb{R}\mathrm{P}^{n-1}$ over U. Hence $\mathbb{P}(E) \to X$ is locally trivial with fiber $\mathbb{R}\mathrm{P}^{n-1}$.

Step 2: The class. Lemma 7.1.8 gives $g\colon E \to \mathbb{R}^\infty$ linear and injective on every fiber. It carries nonzero vectors to nonzero vectors and commutes with scalar multiplication, so it descends to a map of projectivizations

$$\mathbb{P}(g)\colon \mathbb{P}(E) \longrightarrow \mathbb{P}(\mathbb{R}^\infty) = \mathbb{R}\mathrm{P}^\infty$$

Set $x_E = \mathbb{P}(g)^*(a)$, where a is the degree-one generator of $H^*(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2) = \mathbb{F}_2[a]$, as computed in chapter 4. Over a point $x \in X$ the restriction of $\mathbb{P}(g)$ to the fiber $\mathbb{P}(E_x) \cong \mathbb{R}\mathrm{P}^{n-1}$ is the projectivization $\mathbb{P}(u)$ of a linear injection $u\colon \mathbb{R}^n \to \mathbb{R}^\infty$. Applying the homotopy clause of lemma 7.1.8 to the trivial bundle $\mathbb{R}^n$ over a one-point base, u is homotopic to the standard inclusion $\mathbb{R}^n \subseteq \mathbb{R}^\infty$ through linear injections; the homotopy never vanishes on $\mathbb{R}^n \setminus 0$, and $(\mathbb{R}^n \setminus 0) \times I \to \mathbb{R}\mathrm{P}^{n-1} \times I$ is a quotient map since I is locally compact Hausdorff [12, Products of Quotient Maps], so the homotopy projectivizes, and $\mathbb{P}(u)$ is homotopic to the standard inclusion $\mathbb{R}\mathrm{P}^{n-1} \hookrightarrow \mathbb{R}\mathrm{P}^\infty$, under which a restricts to the generator of $H^1(\mathbb{R}\mathrm{P}^{n-1}; \mathbb{F}_2)$ (chapter 4). Hence x_E restricts on every fiber to the generator. Since restriction to a fiber is a ring homomorphism and $H^*(\mathbb{R}\mathrm{P}^{n-1}; \mathbb{F}_2) = \mathbb{F}_2[a]/(a^n)$ (chapter 4), the powers $1, x_E, \ldots, x_E^{n-1}$ restrict on every fiber to the basis $1, a, \ldots, a^{n-1}$; this last conclusion uses only the stated restriction property of the class, not its construction.

Step 3: Leray-Hirsch. By Step 1 the projectivization is a fiber bundle, hence a Serre fibration by theorem 5.2.4, with nonempty fibers $\mathbb{R}\mathrm{P}^{n-1}$; $H^m(\mathbb{R}\mathrm{P}^{n-1}; \mathbb{F}_2)$ is a free $\mathbb{F}_2$-module of finite rank in every degree ($\mathbb{F}_2$ for $0 \leq m \leq n-1$ and 0 above), and the n homogeneous classes $1, x_E, \ldots, x_E^{n-1}$ restrict to a basis in every fiber. Corollary 6.5.2 applies with $R = \mathbb{F}_2$ and makes $H^*(\mathbb{P}(E); \mathbb{F}_2)$ a free $H^*(X; \mathbb{F}_2)$-module with basis $1, x_E, \ldots, x_E^{n-1}$. Freeness on a basis containing 1 splits π^*: the $H^*(X; \mathbb{F}_2)$-linear projection onto the coefficient of the basis element 1 is a left inverse, so π^* is a split injection.

Step 4: The complex case. The complex clause of lemma 7.1.8 gives $g\colon E \to \mathbb{C}^\infty$ complex-linear and injective on every fiber, descending as in Step 2 to $\mathbb{P}(g)\colon \mathbb{P}(E) \to \mathbb{C}\mathrm{P}^\infty$ on the complex projectivizations; set $t_E = \mathbb{P}(g)^*(u)$ for a generator u of $H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) \cong \mathbb{Z}$ (chapter 4). The fibers are $\mathbb{C}\mathrm{P}^{n-1}$, the restriction of $\mathbb{P}(g)$ to a fiber is homotopic to the standard inclusion

$\mathbb{C}\mathrm{P}^{n-1} \hookrightarrow \mathbb{C}\mathrm{P}^\infty$ by the same interleaving homotopies, a generator of $H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$ restricts to a generator of $H^2(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$, and $H^*(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z}) = \mathbb{Z}[u]/(u^n)$ (chapter 4), so the powers $1, t_E, \ldots, t_E^{n-1}$ restrict to a basis of the fiber cohomology in every fiber, free of finite rank over $\mathbb{Z}$ in each degree. The complex projectivization is a fiber bundle by the argument of Step 1, hence a Serre fibration with nonempty fibers, so corollary 6.5.2 with $R = \mathbb{Z}$ gives the freeness, and the same projection argument splits π^*, completing the proof.

Theorem 7.4.3: Splitting Principle

Let $E \to X$ be a rank-n real vector bundle over a paracompact Hausdorff base. There is a space $F(E)$, the flag bundle of E, and a map $\pi\colon F(E) \to X$ such that

1. the induced map $\pi^*\colon H^*(X;\mathbb{F}_2) \to H^*(F(E);\mathbb{F}_2)$ is injective, and
2. the pulled-back bundle π^*E splits as a Whitney sum of real line bundles, $\pi^*E \cong L_1 \oplus \cdots \oplus L_n$.

The same statement holds for a rank-n complex bundle E with $\mathbb{F}_2$ coefficients replaced by $\mathbb{Z}$ coefficients and real line bundles replaced by complex line bundles.

***Proof*:**
The construction is a tower of projectivizations, and the cohomological input is proposition 7.4.2, whose base is an arbitrary paracompact Hausdorff space with no connectivity hypothesis. Form the projective bundle $\mathbb{P}(E) \xrightarrow{\pi_1} X$ whose fiber over x is the projective space of lines in E_x. Over $\mathbb{P}(E)$ the bundle π_1^*E contains a tautological line subbundle L_1 (the line each point of $\mathbb{P}(E)$ names). By proposition 7.1.5, applied over the paracompact Hausdorff base $\mathbb{P}(E)$ and resting on the inner product of lemma 7.1.4 in whichever of the two categories is in play, the subbundle L_1 has a complement: $\pi_1^*E \cong L_1 \oplus E_1$ with E_1 of rank $n-1$. By proposition 7.4.2, $H^*(\mathbb{P}(E);\mathbb{F}_2)$ is a free $H^*(X;\mathbb{F}_2)$-module on the powers $1, x_E, \ldots, x_E^{n-1}$, a basis containing 1, so π_1^* is a split injection. Each stage is a fiber bundle whose fiber is a projective space, hence compact Hausdorff, so [12, Locally Trivial Maps with Compact Fiber Preserve Paracompactness] carries paracompactness and the Hausdorff property from the base of that stage to its total space; inducting up the tower, every intermediate projectivization is again paracompact Hausdorff, so proposition 7.4.2 applies at every stage and the inner product used to split off each complement is available there. Iterating the construction on E_1, then on its rank-$(n-2)$ complement, and so on, produces after $n-1$ steps the flag bundle $F(E)$, over which π^*E has split completely into line bundles $L_1 \oplus \cdots \oplus L_n$, and the composite of split injections is a split injection, which proves clauses (1) and (2) in the real case. The complex case runs identically on the complex clause of proposition 7.4.2: complex projectivizations replace real ones, the tower splits off complex line bundles, and the split injectivity is on $H^*(-;\mathbb{Z})$, completing the proof.

The proof establishes one thing the statement does not record, and downstream arguments need it: every stage of the tower, the flag bundle included, is again paracompact Hausdorff. Without that, the characteristic classes of definition 7.1.13 are not defined on $F(E)$ and the splitting principle could not be used to prove anything about them.

Corollary 7.4.4: The Flag Bundle is Paracompact

Let $E \to X$ be a vector bundle over a paracompact Hausdorff base. Then the flag bundle $F(E)$ of theorem 7.4.3, and every intermediate projectivization in the tower that builds it, is again paracompact Hausdorff. In particular the Stiefel-Whitney and Chern classes of definition 7.1.19 and definition 7.1.13 are defined on $F(E)$.

***Proof*:**
Each stage of the tower is a fiber bundle whose fiber is a real or complex projective space, hence compact Hausdorff. By [12, Locally Trivial Maps with Compact Fiber Preserve Paracompactness], the total space of such a bundle over a paracompact Hausdorff base is again paracompact Hausdorff. Starting from X and inducting up the $n-1$ stages, every intermediate projectivization and $F(E)$ itself are paracompact Hausdorff. The last sentence is then theorem 7.1.9 applied over that base, which is what makes the definitions available, as we sought to show.

The force of the splitting principle is that to prove an identity among characteristic classes, one may pull back to $F(E)$, where the bundle is a sum of line bundles; the identity becomes a statement about line bundles, where it can be checked directly, and the injectivity of π^* carries the verified identity back down to X. The multiplicativity of the total class is the first and most important application.

Theorem 7.4.5: Whitney Sum Formula

Let $E, F \to X$ be vector bundles over a paracompact Hausdorff base, both real or both complex. Then the total Stiefel-Whitney class (in the real case) and the total Chern class (in the complex case) are multiplicative:

$$w(E \oplus F) = w(E)\, w(F) \in H^*(X; \mathbb{F}_2), \qquad c(E \oplus F) = c(E)\, c(F) \in H^*(X; \mathbb{Z})$$

where the products are cup products in the cohomology ring. Degree by degree, $w_k(E \oplus F) = \sum_{i+j=k} w_i(E) \smile w_j(F)$ and likewise for c.

***Proof*:**
The proof is in two movements: the formula for a sum of line bundles, established by the cohomology of $BO(1)^n$ and its image in the symmetric polynomials, and the reduction of the general case to that one by the splitting principle. The real case is written out; the complex case is identical with $\mathbb{Z}$ in place of $\mathbb{F}_2$ and degrees doubled.

Step 1: The total class of a sum of line bundles. For a single real line bundle $L \to X$, classified by a map $X \to BO(1) = \mathbb{R}\mathrm{P}^\infty$, the total Stiefel-Whitney class is $w(L) = 1 + w_1(L)$, since $H^*(BO(1); \mathbb{F}_2) = H^*(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2) = \mathbb{F}_2[w_1]$ has its generator in degree one and a rank-one bundle has no higher classes (section 7.1). Consider the external sum of n line bundles, modeled universally over $BO(1)^n = (\mathbb{R}\mathrm{P}^\infty)^n$: let $a_i \in H^1(BO(1)^n; \mathbb{F}_2)$ be the first Stiefel-Whitney class of the i-th tautological line bundle ℓ_i (the pullback of the tautological line bundle along the i-th projection). The Künneth theorem (theorem 4.3.9) gives

$$H^*\big((\mathbb{R}\mathrm{P}^\infty)^n; \mathbb{F}_2\big) = \mathbb{F}_2[a_1, \ldots, a_n], \qquad |a_i| = 1$$

a polynomial algebra on the n degree-one classes. The map $BO(1)^n \to BO(n)$ classifying the

Whitney sum $\ell_1 \oplus \cdots \oplus \ell_n$ induces, on cohomology, the inclusion

$$H^*(BO(n); \mathbb{F}_2) = \mathbb{F}_2[w_1, \ldots, w_n] \hookrightarrow \mathbb{F}_2[a_1, \ldots, a_n]^{S_n} = \big(\text{symmetric polynomials}\big) \hookrightarrow \mathbb{F}_2[a_1, \ldots, a_n]$$

sending the universal class w_k to the k-th elementary symmetric polynomial $\sigma_k(a_1, \ldots, a_n)$, and this map is injective onto the symmetric polynomials; the identification of $H^*(BO(n); \mathbb{F}_2)$ with $\mathbb{F}_2[a_1, \ldots, a_n]^{S_n}$ via $w_k \mapsto \sigma_k$ is the structural computation of section 7.1 (proved there, and in [17, §7]). Now read off the total class of $\ell_1 \oplus \cdots \oplus \ell_n$. Its k-th Stiefel-Whitney class is the pullback of $w_k = \sigma_k(a_1, \ldots, a_n)$, namely $\sigma_k(a_1, \ldots, a_n)$ itself with $a_i = w_1(\ell_i)$, so the total class is

$$w(\ell_1 \oplus \cdots \oplus \ell_n) = \sum_{k=0}^{n} \sigma_k(a_1, \ldots, a_n) = \prod_{i=1}^{n}(1 + a_i) = \prod_{i=1}^{n} w(\ell_i)$$

the second equality being the generating identity $\sum_k \sigma_k = \prod_i (1 + a_i)$ for elementary symmetric polynomials. Thus for sums of line bundles the total class is the product of the total classes, which is the Whitney formula in this special case.

Step 2: Reduction by the splitting principle. Let E, F be arbitrary, of ranks m, n. Apply the splitting principle (theorem 7.4.3) to $E \oplus F$: there is $\pi\colon Y \to X$ with π^* injective on $\mathbb{F}_2$-cohomology and $\pi^*(E \oplus F) = \pi^*E \oplus \pi^*F$ splitting as a sum of line bundles, say $\pi^*E \cong L_1 \oplus \cdots \oplus L_m$ and $\pi^*F \cong L_{m+1} \oplus \cdots \oplus L_{m+n}$ (one may split E first, then F, building Y as a tower over X). By Step 1 applied to the full collection of line bundles,

$$w(\pi^*E \oplus \pi^*F) = \prod_{i=1}^{m+n} w(L_i) = \Big(\prod_{i=1}^{m} w(L_i)\Big)\Big(\prod_{j=m+1}^{m+n} w(L_j)\Big) = w(\pi^*E)\, w(\pi^*F)$$

where the middle factorization regroups the product and the outer equalities are Step 1 applied to π^*E, π^*F, and their sum separately. By naturality (proposition 7.4.1), $w(\pi^*E) = \pi^* w(E)$ and similarly for F and for $E \oplus F$, so

$$\pi^*\big(w(E \oplus F)\big) = \pi^*\big(w(E)\, w(F)\big)$$

using that π^* is a ring homomorphism. Since π^* is injective by theorem 7.4.3(1), the equality of the pullbacks forces $w(E \oplus F) = w(E)\, w(F)$ in $H^*(X; \mathbb{F}_2)$. The complex case runs identically over $\mathbb{Z}$, giving $c(E \oplus F) = c(E)\, c(F)$, completing the proof.

The Whitney sum formula is the multiplication table of the theory. Combined with naturality and the normalization recorded next, it determines every characteristic class of every bundle assembled from line bundles by sums and pullbacks, which by the splitting principle is effectively every bundle. The normalization axiom fixes the one free constant: it says what the classes do on the universal line bundle.

Proposition 7.4.6: Normalization on the Universal Line Bundle

Let $\gamma^1_{\mathbb{R}} \to \mathbb{R}\mathrm{P}^\infty$ be the tautological real line bundle, and let $\gamma^1_{\mathbb{C}} \to \mathbb{C}\mathrm{P}^\infty$ be the tautological complex line bundle, with dual (hyperplane) bundle $\overline{\gamma^1_{\mathbb{C}}} = (\gamma^1_{\mathbb{C}})^* \to \mathbb{C}\mathrm{P}^\infty$. Then

$$w_1(\gamma^1_{\mathbb{R}}) = a \in H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2), \qquad c_1(\overline{\gamma^1_{\mathbb{C}}}) = u \in H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$$

where a and u are the canonical generators of $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2) = \mathbb{F}_2[a]$ and $H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = \mathbb{Z}[u]$ respectively, with u the positive generator dual to the fundamental class (the convention fixed in section 7.1). Equivalently $c_1(\gamma^1_{\mathbb{C}}) = -u$. In particular $w_1(\gamma^1_{\mathbb{R}}) \neq 0$ and $c_1(\gamma^1_{\mathbb{C}}) \neq 0$.

***Proof*:**
The classifying space of $O(1) = \{\pm 1\}$ is $BO(1) = \mathbb{R}\mathrm{P}^\infty$, with universal bundle $\gamma^1_{\mathbb{R}}$, and $w_1 \in H^1(BO(1); \mathbb{F}_2)$ is by definition the generator of $H^*(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2) = \mathbb{F}_2[a]$ pulled back along the identity classifying map of $\gamma^1_{\mathbb{R}}$, which is a itself. For the complex case, the positive generator $u \in H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = \mathbb{Z}[u]$ is normalized in section 7.1 as the first Chern class of the hyperplane bundle, $c_1(\overline{\gamma^1_{\mathbb{C}}}) = u$, the bundle whose restriction to $\mathbb{C}\mathrm{P}^1 = S^2$ is the Hopf bundle with positive Euler number; this is the same normalization recorded for line bundles in example 7.1.3, where the tautological bundle over $\mathbb{C}\mathrm{P}^1$ has $c_1(\gamma^1_{\mathbb{C}}) = -u$ and its dual has $c_1 = +u$. Since the dual of a line bundle has the opposite first Chern class, $c_1(\gamma^1_{\mathbb{C}}) = -c_1(\overline{\gamma^1_{\mathbb{C}}}) = -u$. Both classes are nonzero generators by the cohomology computations of chapter 4 (the cellular chain complex of $\mathbb{R}\mathrm{P}^\infty$ reduces mod 2 to zero differentials, giving $\mathbb{F}_2$ in every degree; the chain complex of $\mathbb{C}\mathrm{P}^\infty$ has cells only in even degrees, giving $\mathbb{Z}$ in every even degree), as we sought to show.

Two consequences of the multiplicative theory are recorded here, both used in [21] and both statable as soon as the Whitney sum formula is available. The first is the tensor-product rule for line bundles, which is what makes c_1 a homomorphism; the second compares the real and complex theories on one bundle.

Proposition 7.4.7: The First Chern Class is Additive on Tensor Products

Let L, L' be complex line bundles over X.

1. If X is paracompact Hausdorff, then

$$c_1(L \otimes L') = c_1(L) + c_1(L')$$

   and in particular $c_1(L^\vee) = -c_1(L)$.

2. If moreover X is a CW complex, c_1 is an isomorphism of groups from $(\mathrm{Vect}^1_{\mathbb{C}}(X), \otimes)$ to $(H^2(X; \mathbb{Z}), +)$.

***Proof*:**
Part (1). Complex line bundles over a paracompact X are classified by homotopy classes of maps into $\mathrm{Gr}_1(\mathbb{C}^\infty) = BU(1) = \mathbb{C}\mathrm{P}^\infty$ by theorem 7.1.9, and $c_1(L) = f_L^* x$ for a classifying map f_L, which exists by theorem 7.1.9 and defines c_1 by definition 7.1.13, where x generates $H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) \cong \mathbb{Z}$ (example 7.1.3).

The identity is proved once on $\mathbb{C}\mathrm{P}^\infty \times \mathbb{C}\mathrm{P}^\infty$ and transported. Write γ_1, γ_2 for the pullbacks of the tautological line bundle along the two projections, and let $\mu\colon \mathbb{C}\mathrm{P}^\infty \times \mathbb{C}\mathrm{P}^\infty \to \mathbb{C}\mathrm{P}^\infty$ be a classifying map for $\gamma_1 \otimes \gamma_2$; such a map exists because $\mathbb{C}\mathrm{P}^\infty \times \mathbb{C}\mathrm{P}^\infty$ is paracompact, and it is the map Bm induced by the multiplication $m\colon U(1) \times U(1) \to U(1)$. Then $L \otimes L' \cong (f_L, f_{L'})^*(\gamma_1 \otimes \gamma_2)$, since pulling γ_1, γ_2 back along $(f_L, f_{L'})$ gives L, L' and tensor product commutes with pullback, so by theorem 7.1.9 it suffices to compute $\mu^* x$. Nothing beyond theorem 7.1.9 is used of X.

The tensor product of line bundles is the bundle associated to the multiplication homomorphism $m\colon U(1) \times U(1) \to U(1)$, so if L, L' are classified by $f_L, f_{L'}$ then $L \otimes L'$ is classified by the composite

$$X \xrightarrow{(f_L, f_{L'})} \mathbb{C}\mathrm{P}^\infty \times \mathbb{C}\mathrm{P}^\infty \xrightarrow{\mu} \mathbb{C}\mathrm{P}^\infty$$

where $\mu = Bm$. It therefore suffices to compute $\mu^* x$.

Since $H^*(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = \mathbb{Z}[x]$ is concentrated in even degrees and is free, the Künneth theorem gives $H^2(\mathbb{C}\mathrm{P}^\infty \times \mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = \mathbb{Z}x_1 \oplus \mathbb{Z}x_2$, where x_i is the pullback of x along the i-th projection, with no cross term because $H^1(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = 0$. Write $\mu^* x = a\,x_1 + b\,x_2$ with $a, b \in \mathbb{Z}$. Restricting along the inclusion $\iota_1\colon \mathbb{C}\mathrm{P}^\infty \to \mathbb{C}\mathrm{P}^\infty \times \mathbb{C}\mathrm{P}^\infty$, $z \mapsto (z, *)$, the composite $\mu\iota_1$ classifies the tensor product of the tautological bundle with the trivial bundle, which is the tautological bundle itself, so $\mu\iota_1$ is homotopic to the identity. Hence $x = \iota_1^* \mu^* x = a\,x$, giving $a = 1$, and symmetrically $b = 1$. Therefore $\mu^* x = x_1 + x_2$ and

$$c_1(L \otimes L') = (f_L, f_{L'})^* \mu^* x = (f_L, f_{L'})^*(x_1 + x_2) = f_L^* x + f_{L'}^* x = c_1(L) + c_1(L')$$

Part (2). The map c_1 is a bijection $\mathrm{Vect}^1_{\mathbb{C}}(X) \to H^2(X; \mathbb{Z})$ when X is a CW complex, and this is where that hypothesis is needed and part (1)'s is not enough. It takes two inputs: the classification just quoted gives $\mathrm{Vect}^1_{\mathbb{C}}(X) \cong [X, \mathbb{C}\mathrm{P}^\infty]$, and separately $\mathbb{C}\mathrm{P}^\infty = BU(1)$ is an Eilenberg-MacLane space $K(\mathbb{Z}, 2)$, so that $[X, \mathbb{C}\mathrm{P}^\infty] \cong H^2(X; \mathbb{Z})$ by representability of ordinary cohomology on CW complexes (theorem 5.5.9), the isomorphism being $f \mapsto f^* x$. Composing the two is exactly c_1. The displayed identity then makes this bijection a homomorphism, hence an isomorphism of groups. Applying it to $L \otimes L^\vee \cong \underline{\mathbb{C}}$, whose first Chern class vanishes, gives $c_1(L^\vee) = -c_1(L)$, completing the proof.

Proposition 7.4.8: Relating Chern and Stiefel-Whitney Classes

Let $E \to B$ be a complex n-dimensional vector bundle over a paracompact Hausdorff base, and regard it as a $2n$-dimensional real vector bundle $E_{\mathbb{R}}$. Then

$$w_{2i+1}(E_{\mathbb{R}}) = 0$$

and $w_{2i}(E_{\mathbb{R}})$ is the image of $c_i(E)$ under the coefficient homomorphism $\rho_2\colon H^{2i}(B; \mathbb{Z}) \to H^{2i}(B; \mathbb{Z}/2\mathbb{Z})$. Equivalently, $w(E_{\mathbb{R}}) = \rho_2(c(E))$.

***Proof*:**
Step 1: Reduction to line bundles. Let $p\colon F(E) \to B$ be the complex flag bundle of theorem 7.4.3, so that $p^* E$ splits as a sum of complex line bundles $L_1 \oplus \cdots \oplus L_n$. That theorem gives injectivity of p^* on $H^*(-; \mathbb{Z})$; injectivity on $H^*(-; \mathbb{Z}/2\mathbb{Z})$ holds for the same reason, since $F(E) \to B$ is built as a tower of projectivizations and corollary 6.5.2 applies over any commutative coefficient ring R, in particular $R = \mathbb{Z}/2\mathbb{Z}$, making $H^*(F(E); \mathbb{Z}/2\mathbb{Z})$ a free $H^*(B; \mathbb{Z}/2\mathbb{Z})$-module on a basis containing

1. Both $E \mapsto w(E_{\mathbb{R}})$ and $E \mapsto \rho_2(c(E))$ are natural (proposition 7.4.1, proposition 7.4.1) and multiplicative under direct sums (theorem 7.4.5, together with the fact that ρ_2 is a ring homomorphism and that $(E \oplus E')_{\mathbb{R}} = E_{\mathbb{R}} \oplus E'_{\mathbb{R}}$). So both sides pull back to $F(E)$ as products over the line summands, and it suffices to prove the identity for a complex line bundle.

Step 2: The line bundle case. Let $L \to B$ be a complex line bundle, so $L_{\mathbb{R}}$ is a real bundle of rank 2 carrying the canonical orientation of theorem 7.4.11(2). Its first Stiefel-Whitney class vanishes. To see this, pass to the universal case: by theorem 7.1.9 in its complex form with $n = 1$, there is a map $f : B \to \mathrm{Gr}_1(\mathbb{C}^\infty) = \mathbb{C}\mathrm{P}^\infty$ and an isomorphism $L \cong f^*\gamma$ with $\gamma \to \mathbb{C}\mathrm{P}^\infty$ the universal complex line bundle. Realification commutes with pullback, so $L_{\mathbb{R}} \cong f^*(\gamma_{\mathbb{R}})$, and naturality (proposition 7.4.1) gives $w_1(L_{\mathbb{R}}) = f^*w_1(\gamma_{\mathbb{R}})$. Now $\mathbb{C}\mathrm{P}^\infty$ is a path-connected CW complex, with one cell in each even dimension and none in odd dimensions (example 5.5.5), and $\gamma_{\mathbb{R}}$ is oriented by the canonical orientation of a complex line, hence orientable. So theorem 7.4.12, whose hypothesis of a connected CW base is met by $\mathbb{C}\mathrm{P}^\infty$, gives $w_1(\gamma_{\mathbb{R}}) = 0$, and therefore $w_1(L_{\mathbb{R}}) = f^*(0) = 0$. (Applying that criterion to $L_{\mathbb{R}}$ directly would also work, clause (1) of that theorem having exactly the hypothesis B paracompact Hausdorff; the universal route is kept because it computes the class rather than merely detecting its vanishing. The older reason given here, that B is only assumed paracompact, and in the application of Step 1 the base is the flag bundle $F(E)$, which need not be a connected CW complex. What the argument does ask of that base is paracompactness, so that theorem 7.1.9 gives f and definition 7.1.19 defines w_1 there, and that is corollary 7.4.4.) Its top Stiefel-Whitney class is $w_2(L_{\mathbb{R}}) = \rho_2(e(L_{\mathbb{R}}))$ by theorem 7.4.11(1), and $e(L_{\mathbb{R}}) = c_1(L)$ by theorem 7.4.11(2) applied with $n = 1$. Hence $w_2(L_{\mathbb{R}}) = \rho_2(c_1(L))$, and $w_i(L_{\mathbb{R}}) = 0$ for $i > 2$ by definition 7.1.19 since $\mathrm{rk} L_{\mathbb{R}} = 2$. Therefore

$$w(L_{\mathbb{R}}) = 1 + \rho_2(c_1(L)) = \rho_2\big(1 + c_1(L)\big) = \rho_2(c(L))$$

Step 3: Conclusion. By Step 1 the two total classes agree after pulling back along p, and p^* is injective on $H^*(B; \mathbb{Z}/2\mathbb{Z})$, so $w(E_{\mathbb{R}}) = \rho_2(c(E))$ on B. Comparing degrees, the odd-degree components of $\rho_2(c(E))$ vanish because $c_i(E)$ sits in degree $2i$, giving $w_{2i+1}(E_{\mathbb{R}}) = 0$ and $w_{2i}(E_{\mathbb{R}}) = \rho_2(c_i(E))$, as we sought to show.

This normalization is what makes the first Chern class a complete invariant of complex line bundles: the assignment $L \mapsto c_1(L)$ is the bijection $\mathrm{Vect}^1_{\mathbb{C}}(X) \cong H^2(X; \mathbb{Z})$ already recorded in section 2.4, since $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z}, 2)$ and the classifying map of L is the same datum as the cohomology class $c_1(L)$. The analogous statement for real line bundles, with $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$ in place of $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z}, 2)$, is the following.

Theorem 7.4.9: Real Line Bundles are Classified by the First Stiefel-Whitney Class

Let X be a CW complex. The first Stiefel-Whitney class is a bijection

$$w_1 : \mathrm{Vect}^1_{\mathbb{R}}(X) \xrightarrow{\cong} H^1(X; \mathbb{F}_2)$$

natural in X, from the isomorphism classes of real line bundles over X to the first cohomology with $\mathbb{F}_2$-coefficients. Under tensor product of line bundles on one side and addition on the other it is an isomorphism of abelian groups, so a real line bundle is trivial if and only if its first Stiefel-Whitney class vanishes.

***Proof*:**
A real line bundle over X is a principal $O(1)$-bundle, and $O(1) = \{\pm 1\} = \mathbb{Z}/2$, so the classification theorem (theorem 2.4.7) for the group $\mathbb{Z}/2$ gives a natural bijection $\mathrm{Vect}^1_{\mathbb{R}}(X) \cong [X, B\mathbb{Z}/2] = [X, \mathbb{R}\mathrm{P}^\infty]$ sending a line bundle to the homotopy class of its classifying map. Since $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$ (example 5.5.6), representability of cohomology by Eilenberg-MacLane spaces (theorem 5.5.9) gives a natural bijection $[X, \mathbb{R}\mathrm{P}^\infty] \cong H^1(X; \mathbb{F}_2)$ sending a map to the pullback of the generator $a \in H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2)$. That generator is the universal first Stiefel-Whitney class $w_1(\gamma^1_{\mathbb{R}})$ of section 7.1, so the composite sends a line bundle L with classifying map f_L to $f_L^*(a) = w_1(L)$, the class of definition 7.1.19. Both bijections carry the group structures to one another: the set $\mathrm{Vect}^1_{\mathbb{R}}(X)$ is an abelian group under tensor product because $\mathbb{Z}/2$ is abelian, and this operation is the one classified by the H-space multiplication on $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$, which represents the addition on $H^1(-; \mathbb{F}_2)$. Hence w_1 is an isomorphism of abelian groups, and a line bundle is trivial exactly when its class is the identity 0, completing the proof.

A worked instance grounds the normalization in a finite-dimensional computation.

Example 7.4.10: The Tautological Line Bundle on $\mathbb{R}\mathrm{P}^n$ and $\mathbb{C}\mathrm{P}^n$

Let $\gamma^1 \to \mathbb{R}\mathrm{P}^n$ be the tautological real line bundle, whose fiber over a line $\ell \in \mathbb{R}\mathrm{P}^n$ is ℓ itself, viewed as a one-dimensional subspace of $\mathbb{R}^{n+1}$. It is the restriction of $\gamma^1_{\mathbb{R}}$ along the inclusion $\iota\colon \mathbb{R}\mathrm{P}^n \hookrightarrow \mathbb{R}\mathrm{P}^\infty$. The cohomology ring is $H^*(\mathbb{R}\mathrm{P}^n; \mathbb{F}_2) = \mathbb{F}_2[a]/(a^{n+1})$ with $|a| = 1$ (chapter 4), and ι^* sends the generator of $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2)$ to the generator a of $H^1(\mathbb{R}\mathrm{P}^n; \mathbb{F}_2)$. By naturality (proposition 7.4.1) and normalization (proposition 7.4.6),

$$w_1(\gamma^1) = \iota^*\big(w_1(\gamma^1_{\mathbb{R}})\big) = \iota^*(a) = a \neq 0 \in H^1(\mathbb{R}\mathrm{P}^n; \mathbb{F}_2)$$

so the total class is $w(\gamma^1) = 1 + a$. The nonvanishing of $w_1(\gamma^1)$ is the cohomological certificate that γ^1 is nontrivial, indeed (as theorem 7.4.12 below shows) non-orientable: a trivial bundle would have $w_1 = 0$ by naturality applied to the constant classifying map.

For the complex tautological line bundle $\gamma^1 \to \mathbb{C}\mathrm{P}^n$ (fiber over $\ell \in \mathbb{C}\mathrm{P}^n$ the complex line $\ell \subseteq \mathbb{C}^{n+1}$), the restriction $\jmath\colon \mathbb{C}\mathrm{P}^n \hookrightarrow \mathbb{C}\mathrm{P}^\infty$ sends the generator $u \in H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$ to the generator of $H^2(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) = \mathbb{Z}\langle u\rangle$ in the ring $\mathbb{Z}[u]/(u^{n+1})$. By naturality and normalization (proposition 7.4.6), with $c_1(\gamma^1_{\mathbb{C}}) = -u$ on the universal bundle,

$$c_1(\gamma^1) = \jmath^*\big(c_1(\gamma^1_{\mathbb{C}})\big) = \jmath^*(-u) = -u$$

a generator of $H^2(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) \cong \mathbb{Z}$. The total Chern class is $c(\gamma^1) = 1 - u$. The dual (hyperplane) bundle $\overline{\gamma^1}$ then has $c_1(\overline{\gamma^1}) = +u$, the positive generator dual to the fundamental class: this is the topological avatar of the hyperplane class, and the sign convention is the one fixed in section 7.1 and example 7.1.3, where the hyperplane bundle over $\mathbb{C}\mathrm{P}^1 = S^2$ is the Hopf bundle with Euler number $+1$.

The last axiom identifies the top class. For an oriented real bundle there is an integral Euler class $e(E) \in H^n(X; \mathbb{Z})$, defined via the Thom isomorphism or, equivalently, as the obstruction to a nowhere-zero section; its mod-2 reduction is the top Stiefel-Whitney class, and for a complex bundle the top Chern class *is* the Euler class of the underlying oriented real bundle.

Theorem 7.4.11: Top Class is the Euler Class

Let $E \to X$ be a real vector bundle of rank n over a paracompact Hausdorff base.

1. If E is oriented, with Euler class $e(E) \in H^n(X;\mathbb{Z})$, then the top Stiefel-Whitney class is the mod-2 reduction of the Euler class, $w_n(E) = \rho_2\big(e(E)\big)$, where $\rho_2 \colon H^n(X;\mathbb{Z}) \to H^n(X;\mathbb{F}_2)$ is reduction of coefficients.

2. If E is a complex bundle of complex rank n, regarded as an oriented real bundle of real rank $2n$ via the canonical orientation of a complex vector space, then the top Chern class equals the Euler class, $c_n(E) = e(E) \in H^{2n}(X;\mathbb{Z})$.

***Proof*:**
Both statements are verified over the flag bundle of E and pushed down by the injectivity of π^*. The Euler class is natural under pullback by proposition 7.2.6, which is proved from its construction from the Thom class (theorem 6.3.2; the Euler class is the image of 1 under the composite $H^0(X) \to H^n(X)$ of Thom isomorphism and restriction to the zero section), and the characteristic classes are natural by proposition 7.4.1.

1. *Real, oriented case.* The structure group of an oriented rank-n bundle is $SO(n)$, classified by $BSO(n)$. The argument runs the splitting-principle reduction entirely with $\mathbb{F}_2$ coefficients, which sidesteps the fact that the splitting line bundles need not be orientable. For any real line bundle $L \to X$ (orientable or not), the Thom class with $\mathbb{F}_2$ coefficients always exists, since $\mathbb{F}_2$-orientability is automatic, so the *mod-*2 *Euler class*
$$\bar{e}(L) := \rho_2\big(e(L)\big) \in H^1(X;\mathbb{F}_2)$$
is defined even when the integral Euler class $e(L)$ is not, and it equals $w_1(L)$: the mod-2 Euler class of a line bundle is its first Stiefel-Whitney class, both being the common $\mathbb{F}_2$-obstruction to a nowhere-zero section, computed in section 7.1 and [17, §9]. The mod-2 Euler class is multiplicative under Whitney sum, $\bar{e}(E \oplus F) = \bar{e}(E) \smile \bar{e}(F)$, from the multiplicativity of the $\mathbb{F}_2$-Thom classes. Now apply the splitting principle (theorem 7.4.3): writing $\pi^*E = L_1 \oplus \cdots \oplus L_n$ over the splitting space, with π^* injective on $\mathbb{F}_2$-cohomology,
$$\rho_2\big(e(\pi^*E)\big) = \bar{e}(\pi^*E) = \prod_{i=1}^{n} \bar{e}(L_i) = \prod_{i=1}^{n} w_1(L_i) = w_n(\pi^*E)$$
where the first equality is the definition of $\bar{e}$ applied to the (oriented) bundle π^*E, the second is multiplicativity of $\bar{e}$, the third is the line-bundle identity $\bar{e}(L_i) = w_1(L_i)$, and the last is the top term of the Whitney product $w(\pi^*E) = \prod_i (1 + w_1(L_i))$. Every object in this chain is a well-defined $\mathbb{F}_2$-class; no integral Euler class of a non-orientable L_i appears. Injectivity of π^* pushes the identity $\rho_2(e(E)) = w_n(E)$ down to X.

2. *Complex case.* A complex line bundle L, viewed as an oriented real 2-plane bundle, has Euler class $e(L) = c_1(L)$: both are the obstruction to a nowhere-zero section and both are classified by the same map $X \to \mathbb{C}\mathrm{P}^\infty = BU(1) = BSO(2)$, the last identification being the standard isomorphism $U(1) \cong SO(2)$ that matches the complex orientation with the chosen real orientation, so the universal Euler class of $BSO(2)$ and the universal first Chern class of $BU(1)$ are the same generator of H^2 (verified in section 7.1 and [17, §14]).

For a general complex bundle E of rank n, split $\pi^*E = L_1 \oplus \cdots \oplus L_n$ into complex line bundles over a space with π^* injective on $\mathbb{Z}$-cohomology (theorem 7.4.3). The top Chern class is the product of the first Chern classes,

$$c_n(\pi^*E) = \prod_{i=1}^{n} c_1(L_i)$$

by the top term of the Whitney product $c(\pi^*E) = \prod_i (1 + c_1(L_i))$. The Euler class is multiplicative under Whitney sum and equals c_1 on each complex line bundle by the line-bundle case just settled, so

$$e(\pi^*E) = \prod_{i=1}^{n} e(L_i) = \prod_{i=1}^{n} c_1(L_i) = c_n(\pi^*E)$$

Both sides being natural and equal after applying the injective π^*, the identity $c_n(E) = e(E)$ holds in $H^{2n}(X; \mathbb{Z})$, completing the proof.

These four properties (naturality, Whitney sum, normalization, top class equals Euler class) do more than hold for the classes built in section 7.1; they characterize them. The uniqueness statement is that any assignment of cohomology classes to bundles satisfying naturality, the Whitney sum formula, and the normalization on the universal line bundle must coincide with the Stiefel-Whitney classes (respectively Chern classes), because the splitting principle reduces any such assignment to its values on line bundles, where normalization fixes it. The full uniqueness theorem is proved in [17, §7, §14]; the present section uses the axioms as a calculus, deriving consequences rather than re-establishing existence.

With the axioms in place as a formal calculus, the second face of the theory comes into view: the geometric meaning of these classes is that each is an obstruction. The general principle is the obstruction theory of theorem 5.6.10: a structure on a bundle (an orientation, a spin structure, a nowhere-zero section, a reduction of the structure group) is a section of an associated fibration, and the primary obstruction to building such a section lives in a cohomology group $H^{k+1}(X; \pi_k(F))$ for the relevant fiber F. The characteristic classes are exactly these primary obstructions, identified through the homotopy of the fiber. The cleanest case is orientability.

Theorem 7.4.12: First Stiefel-Whitney Class is the Obstruction to Orientability

Let $E \to X$ be a real vector bundle.

1. If X is paracompact Hausdorff and E is orientable, then $w_1(E) = 0 \in H^1(X; \mathbb{F}_2)$.
2. If X is a connected CW complex and $w_1(E) = 0$, then E is orientable.
3. If E is orientable over a paracompact Hausdorff base, its orientations form a torsor under the group $C(X; \mathbb{F}_2)$ of *locally constant* functions $X \to \mathbb{F}_2$. In particular an orientable bundle over a connected base has exactly two orientations. When X is locally path-connected, for instance a CW complex, $C(X; \mathbb{F}_2) = H^0(X; \mathbb{F}_2)$; over a general paracompact base the two differ, since H^0 sees path-components and C sees components.

In particular, over a connected CW complex E is orientable if and only if $w_1(E) = 0$.

The two clauses carry different hypotheses because they need different theorems, and the asymmetry is the one running through this whole section. Clause (1) reads a consequence off a bundle that has been trivialized, and every step of it is available wherever the classes are defined. Clause (2) must produce a trivialization from the vanishing of a cohomology class, and that is the assertion that a map into $\mathbb{R}\mathrm{P}^\infty$ inducing zero on H^1 is nullhomotopic, which is representability (theorem 5.5.9), holding over a CW complex and not over an arbitrary paracompact space. Stating the theorem under the single hypothesis that makes both halves true would discard clause (1) over exactly the bases, flag bundles among them, where the rest of the chapter applies it.

***Proof*:**
The proof identifies orientability with the triviality of a $\mathbb{Z}/2$-bundle, identifies that $\mathbb{Z}/2$-bundle's class with $w_1(E)$, and then reads the two implications off in the two regimes.

Step 1: Orientability as a section of the orientation cover. An orientation of a rank-n bundle E is a continuous choice of orientation of each fiber E_x, that is, a choice of generator of $H_n(E_x, E_x \setminus 0; \mathbb{Z}) \cong \mathbb{Z}$ varying continuously in x. The two generators per fiber assemble into a double cover $\mathrm{Or}(E) \to X$, the *orientation cover*, whose fiber over x is the two-element set of orientations of E_x; this is the bundle associated to the principal $O(n)$-frame bundle of E along the determinant-sign homomorphism $\det\colon O(n) \to \{\pm 1\} = \mathbb{Z}/2$ (on orthogonal matrices $\det$ takes the values ± 1). An orientation of E is precisely a section of this double cover, and a double cover admits a section if and only if it is trivial, with no connectivity hypothesis on X: a section s has clopen image (it is open because s is a section of a covering and closed because its complement is the image of the other sheet, also a section), so the cover splits as the disjoint union of $s(X)$ and its complement, each mapping homeomorphically to X. Conversely a trivial cover has the obvious section. So E is orientable if and only if its orientation cover is trivial.

Step 2: The double cover as a map to $\mathbb{R}\mathrm{P}^\infty$. A double cover of X is the unit-sphere bundle of a real line bundle, namely of the line bundle it induces along $\mathbb{Z}/2 = O(1)$ acting on $\mathbb{R}$, and this correspondence is a bijection between double covers and rank-one real bundles, and the two constructions are mutually inverse. Given a double cover $P \to X$, the associated bundle $P \times_{\mathbb{Z}/2} \mathbb{R}$ is a real line bundle whose unit-sphere bundle for the inner product induced from the standard one on $\mathbb{R}$ is P again, since $P \times_{\mathbb{Z}/2} \{\pm 1\} = P$. Given a line bundle L, fix an inner product by lemma 7.1.4; then $S(L) \times_{\mathbb{Z}/2} \mathbb{R} \cong L$ by $(v, t) \mapsto tv$, which is well defined on the quotient and fiberwise linear and injective. The choice of inner product does not matter: any two are connected by the convex path $(1-t)\langle -, - \rangle_0 + t\langle -, - \rangle_1$, positive-definite at every t, and the resulting sphere bundles are isomorphic over X by normalizing. For the orientation cover the associated line bundle is $\det E = \Lambda^n E$, since both are induced from the frame bundle along $\det\colon O(n) \to \{\pm 1\}$. Over a paracompact Hausdorff base $\mathrm{Vect}^1_{\mathbb{R}}(X) = [X, \mathbb{R}\mathrm{P}^\infty]$ by theorem 7.1.9 with $n = 1$, so the orientation cover has a classifying map $f\colon X \to \mathbb{R}\mathrm{P}^\infty = BO(1)$, and it is trivial exactly when f is nullhomotopic. Set $\omega(E) = f^*(a) \in H^1(X; \mathbb{F}_2)$ for a the generator of $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2)$; by definition $\omega(E) = w_1(\det E)$.

Step 3: Identifying $\omega(E) = w_1(E)$. It remains to see that the class $\omega(E)$ classifying the orientation cover equals the first Stiefel-Whitney class $w_1(E)$. Both are natural in E under pullback (the orientation cover of f^*E is the pullback of the orientation cover of E; w_1 is natural by proposition 7.4.1), and both vanish on trivial bundles, so by the universal property it suffices to compare them on the universal bundle $\gamma^n \to BO(n)$. There $H^1(BO(n); \mathbb{F}_2) = \mathbb{F}_2\langle w_1 \rangle$ is one-dimensional (the degree-one part of $\mathbb{F}_2[w_1, \dots, w_n]$), so $\omega(\gamma^n)$ is either 0 or w_1. It is not 0. Let $\iota\colon BO(1) = \mathbb{R}\mathrm{P}^\infty \to BO(n)$ be induced by the inclusion $O(1) \hookrightarrow O(n)$ as the upper-left block. Then $\iota^*\gamma^n \cong \gamma^1_{\mathbb{R}} \oplus \underline{\mathbb{R}}^{n-1}$: the block inclusion splits $\mathbb{R}^n$ as the tautological line plus a fixed complement, and the restriction of a rank-n bundle along any map has rank n, so it is not $\gamma^1_{\mathbb{R}}$ itself. Taking determinants, $\det(\gamma^1_{\mathbb{R}} \oplus \underline{\mathbb{R}}^{n-1}) \cong \gamma^1_{\mathbb{R}}$, because the top exterior power of a sum with a trivial bundle is the top exterior power of the nontrivial summand. Hence by Step 2 and naturality of det, $\iota^*\omega(\gamma^n) = \omega\big(\iota^*\gamma^n\big) = w_1\big(\det \iota^*\gamma^n\big) = w_1(\gamma^1_{\mathbb{R}}) = a \neq 0$ in $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2)$, the last step being proposition 7.4.6. A class pulling back to something nonzero is nonzero, so $\omega(\gamma^n) \neq 0$. Thus $\omega(\gamma^n) = w_1(\gamma^n)$, and by naturality $\omega(E) = w_1(E)$ for every E.

Step 4: The two implications. For clause (1), let X be paracompact Hausdorff and E orientable. By Step 1 the orientation cover is trivial, so its classifying map f of Step 2 is nullhomotopic, hence constant up to homotopy and $\omega(E) = f^*(a) = 0$; by Step 3, $w_1(E) = 0$. Nothing here identifies $[X, \mathbb{R}\mathrm{P}^\infty]$ with a cohomology group: the argument runs from a trivialization to the vanishing of a pullback, and theorem 7.1.9 supplies the classifying map over any paracompact Hausdorff base.

For clause (2), let X be a connected CW complex with $w_1(E) = 0$. By Step 3, $\omega(E) = f^*(a) = 0$. Since $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$ (example 5.5.6), representability (theorem 5.5.9) makes $[X, \mathbb{R}\mathrm{P}^\infty] \to H^1(X; \mathbb{F}_2)$, $[f] \mapsto f^*(a)$, a bijection, so $f^*(a) = 0$ forces f nullhomotopic. Hence the orientation cover is trivial and E is orientable by Step 1. This is the step that fails over a general paracompact base, where the same map $[X, \mathbb{R}\mathrm{P}^\infty] \to H^1(X; \mathbb{F}_2)$ need not be injective.

Step 5: The torsor clause. Suppose E is orientable, so the orientation cover $\mathrm{Or}(E) \to X$ has a section, and let $\varepsilon \in C(X; \mathbb{F}_2)$ be a locally constant function. Acting on a section s by flipping it over the locus $\{\varepsilon = 1\}$, which is clopen, gives a map that is again continuous and again a section, so $H^0(X; \mathbb{F}_2)$ acts on the set of orientations. The action is free: if ε fixes s then ε vanishes, since flipping changes the sheet wherever $\varepsilon = 1$. It is transitive: given orientations s, s', the function taking the value 1 exactly where they name different sheets is locally constant, because $\mathrm{Or}(E)$ is a covering and the two sections agree or disagree locally, and it carries s to s'. Hence the orientations form a torsor under $C(X; \mathbb{F}_2)$, and when X is connected the only locally constant functions are the two constants, giving exactly two orientations. The identification $C(X; \mathbb{F}_2) = H^0(X; \mathbb{F}_2)$ needs X locally path-connected: $H^0(X; G)$ is the functions from *path*-components (section 5.3), and on a space like the topologist's sine curve, connected with two path-components, there are two locally constant functions and four elements of H^0. This completes the proof.

The theorem casts orientability as the vanishing of a single cohomology class, and it does so by exhibiting orientability as a sectioning problem. The orientation cover is the fibration; its fiber is the discrete two-point space $\mathbb{Z}/2$; the obstruction to a section is the primary (and here only) obstruction of theorem 5.6.10, living in $H^1(X; \pi_0(\mathbb{Z}/2)) = H^1(X; \mathbb{Z}/2)$, and that obstruction is $w_1(E)$. The two-element fiber has $\pi_0 = \mathbb{Z}/2$ and no higher homotopy, so a single obstruction in H^1 controls the entire problem, which is why orientability is governed by one class rather than a tower of them. The next structure up the ladder, a spin structure, has a fiber with nontrivial π_1, and its obstruction lands one degree higher.

Definition 7.4.13: Spin Structure

Let $E \to X$ be an oriented real vector bundle of rank $n \geq 3$ over a paracompact Hausdorff base, with principal $SO(n)$-bundle of oriented orthonormal frames $P_{SO} \to X$. A *spin structure* on E is a principal $\mathrm{Spin}(n)$-bundle $P_{\mathrm{Spin}} \to X$ together with a map $P_{\mathrm{Spin}} \to P_{SO}$ over X restricting on each fiber to the double cover $\mathrm{Spin}(n) \to SO(n)$.

A spin structure is a globally consistent square root of the rotation group acting on the fibers. The local square roots always exist, since $\mathrm{Spin}(n) \to SO(n)$ is a covering; the obstruction to assembling them into one global choice is a mod-2 class in degree two, and the content of the next theorem is that the class is $w_2(E)$.

Theorem 7.4.14: Second Stiefel-Whitney Class is the Obstruction to a Spin Structure

Let $E \to X$ be an oriented real vector bundle of rank $n \geq 3$ over a connected CW complex (so $w_1(E) = 0$). Then E admits a spin structure if and only if $w_2(E) = 0 \in H^2(X; \mathbb{F}_2)$. When spin structures exist, they form a torsor under $H^1(X; \mathbb{F}_2)$; in particular a bundle over a connected base admits $|H^1(X; \mathbb{F}_2)|$ of them when it admits any.

***Proof*:**
A spin structure on an oriented bundle is a lift of its structure group from $SO(n)$ to the connected double cover $\mathrm{Spin}(n) \to SO(n)$, equivalently a lift of the classifying map $g\colon X \to BSO(n)$ across the fibration $B\mathrm{Spin}(n) \to BSO(n)$. The proof identifies the fiber of this fibration, locates the primary obstruction, and identifies it with w_2.

Step 1: The fibration and its fiber. For $n \geq 3$ the group $SO(n)$ has fundamental group $\pi_1(SO(n)) = \mathbb{Z}/2$, and $\mathrm{Spin}(n)$ is its universal (double) cover, a central extension

$$1 \to \mathbb{Z}/2 \to \mathrm{Spin}(n) \to SO(n) \to 1$$

Applying the classifying-space functor B to this extension yields a fibration sequence

$$B\mathbb{Z}/2 = K(\mathbb{Z}/2, 1) \to B\mathrm{Spin}(n) \to BSO(n) \xrightarrow{\beta} B(B\mathbb{Z}/2) = K(\mathbb{Z}/2, 2)$$

where the fiber of $B\mathrm{Spin}(n) \to BSO(n)$ is $B\mathbb{Z}/2 = K(\mathbb{Z}/2, 1)$ (the delooping of corollary 2.4.8 applied to the kernel $\mathbb{Z}/2$), and the map $\beta\colon BSO(n) \to K(\mathbb{Z}/2, 2)$ is the classifying map of the next stage, representing a class in $H^2(BSO(n); \mathbb{Z}/2)$. A spin structure on E is a lift of $g\colon X \to BSO(n)$ to $B\mathrm{Spin}(n)$.

Step 2: The primary obstruction. By the lifting form of obstruction theory (theorem 5.6.10, in its lifting guise box 5.6.13), the obstructions to lifting g across the fibration with fiber $F = K(\mathbb{Z}/2, 1)$ live in $H^{m+1}(X; \pi_m(F))$. Since $F = K(\mathbb{Z}/2, 1)$ has $\pi_1(F) = \mathbb{Z}/2$ and $\pi_m(F) = 0$ for $m \neq 1$, every obstruction group vanishes except in degree $m = 1$, where the single obstruction is a class

$$\theta(E) \in H^2\big(X; \pi_1(F)\big) = H^2(X; \mathbb{Z}/2)$$

This $\theta(E)$ is the complete obstruction: a lift exists over the 2-skeleton if and only if $\theta(E) = 0$, and since there are no obstructions in higher degrees ($\pi_m(F) = 0$ for $m \geq 2$), the lift then extends over all of X. Concretely, $\theta(E) = g^*\beta$ is the pullback of the universal class $\beta \in H^2(BSO(n); \mathbb{Z}/2)$ along the classifying map.

Step 3: $\theta(E) = w_2(E)$. The universal class $\beta \in H^2(BSO(n); \mathbb{Z}/2)$ is the obstruction to spinning the universal oriented bundle. Now $H^2(BSO(n); \mathbb{F}_2)$ is one-dimensional for $n \geq 3$, spanned by w_2: the full computation $H^*(BSO(n); \mathbb{F}_2) = \mathbb{F}_2[w_2, w_3, \ldots, w_n]$ (the mod-2 cohomology of $BO(n)$ from theorem 7.1.17 with w_1 removed by orientability) is carried out in [17, §12], and gives $H^2(BSO(n); \mathbb{F}_2) = \mathbb{F}_2\langle w_2 \rangle$ in degree two for $n \geq 3$. Hence β is either 0 or w_2. It is not 0, because the universal oriented bundle admits no spin structure: the explicit identification $\beta = w_2$ in

$H^2(BSO(n);\mathbb{F}_2)$ is the computation of [17, §12], where w_2 is shown to be exactly the connecting class of the central extension $1 \to \mathbb{Z}/2 \to \mathrm{Spin}(n) \to SO(n) \to 1$ under the delooped fibration. A concrete witness that $\beta \neq 0$ is the bundle $\gamma^1 \oplus \gamma^1$ over $\mathbb{R}\mathrm{P}^2$ stabilized to rank n by adding a trivial bundle, where γ^1 is the tautological line bundle with $w_1(\gamma^1) = a$ (example 7.4.10): it is oriented, since $w_1(\gamma^1 \oplus \gamma^1) = a + a = 0$ by the Whitney formula, yet $w_2(\gamma^1 \oplus \gamma^1) = w_1(\gamma^1)^2 = a^2 \neq 0$ in $H^2(\mathbb{R}\mathrm{P}^2;\mathbb{F}_2) = \mathbb{F}_2\langle a^2\rangle$; this oriented bundle admits no spin structure and so detects β. Granting $\beta = w_2 \in H^2(BSO(n);\mathbb{F}_2)$, naturality gives $\theta(E) = g^*\beta = g^*w_2 = w_2(E)$, so E admits a spin structure if and only if $w_2(E) = 0$, completing the proof.

Step 4: The torsor clause. Suppose E admits a spin structure $\lambda\colon P_{\mathrm{Spin}} \to P_{SO}$, and let $Q \to X$ be a principal $\mathbb{Z}/2$-bundle. Write ζ for the central element of $\mathrm{Spin}(n)$ generating the kernel of $\mathrm{Spin}(n) \to SO(n)$. Twist by Q:

$$P_{\mathrm{Spin}} \cdot Q := \big(P_{\mathrm{Spin}} \times_X Q\big)/(\mathbb{Z}/2)$$

where the generator of $\mathbb{Z}/2$ acts by ζ on the first factor and by the deck transformation on the second. The residual $\mathrm{Spin}(n)$-action on the first factor descends, since ζ is central, and makes $P_{\mathrm{Spin}} \cdot Q$ a principal $\mathrm{Spin}(n)$-bundle; the map λ descends too, because $\lambda(p\zeta) = \lambda(p)$, so $P_{\mathrm{Spin}} \cdot Q \to P_{SO}$ is again a spin structure. This defines an action of $\mathrm{Prin}_{\mathbb{Z}/2}(X)$ on the set of spin structures, associative because twisting by Q then Q' agrees with twisting by the contracted product $Q \otimes Q'$.

The action is transitive. Given two spin structures $\lambda\colon P \to P_{SO}$ and $\lambda'\colon P' \to P_{SO}$, the fiber product $P \times_{P_{SO}} P'$ carries a free action of $\mathrm{Spin}(n)$ acting diagonally, and the quotient $Q(P') := (P \times_{P_{SO}} P')/\mathrm{Spin}(n)$ is a principal $\mathbb{Z}/2$-bundle over X: over a point of P_{SO} the two fibers are $\mathbb{Z}/2$-torsors, so the fiber product is a $(\mathbb{Z}/2 \times \mathbb{Z}/2)$-torsor and the diagonal quotient is a $\mathbb{Z}/2$-torsor. Unwinding the definitions gives $P \cdot Q(P') \cong P'$ over P_{SO}.

The action is free. Because it is transitive it suffices to check one stabilizer: if $P \cdot Q \cong P$ over P_{SO} then $Q \cong Q(P \cdot Q) \cong Q(P)$, and $Q(P)$ is the trivial double cover because $p \mapsto [(p,p)]$ is a section of it. Hence the spin structures form a torsor under $\mathrm{Prin}_{\mathbb{Z}/2}(X)$, which is $H^1(X;\mathbb{F}_2)$ by theorem 2.4.7 and theorem 5.5.9, using $B\mathbb{Z}/2 = \mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2,1)$ (example 5.5.6). It is here, not in the existence criterion, that the CW hypothesis is used a second time.

The pattern is now visible. Each successive structure on a bundle (orientation, then spin, then string, and onward) is a lift of the structure group along a tower of two-fold or higher covers, and the primary obstruction to each lift is a characteristic class one degree higher than the last. Orientation removes w_1; with $w_1 = 0$, the spin obstruction is w_2; the framework continues, with the next obstruction (to a string structure) a class in H^4 detected by a fractional Pontryagin class. The unifying statement is that a characteristic class is the primary obstruction to a reduction of the structure group, the cohomological measurement of how far the bundle is from carrying the desired geometric structure. The most classical of these obstructions is to a single nowhere-zero section, and there the top class enters.

Theorem 7.4.15: Top Class is the Obstruction to a Nowhere-Zero Section

Let $E \to X$ be a real vector bundle of rank n over a connected CW complex.

1. If E is oriented, the Euler class $e(E) \in H^n(X;\mathbb{Z})$ is the primary obstruction to a nowhere-zero section of E: if E has a nowhere-zero section then $e(E) = 0$, and if in addition $\dim X \leq n$ the converse holds, so that E has a nowhere-zero section if and only if $e(E) = 0$.

2. Without orientation hypotheses, the top Stiefel-Whitney class $w_n(E) \in H^n(X;\mathbb{F}_2)$ is the mod-2 primary obstruction: a nowhere-zero section forces $w_n(E) = 0$.

***Proof*:**
A nowhere-zero section of E is a section of the associated sphere bundle $S(E) \to X$ (unit vectors in each fiber, after choosing a metric), whose fiber is S^{n-1}. The obstruction theory of theorem 5.6.10 in its section form (box 5.6.13) governs the existence of such a section.

Step 1: Locating the obstruction. The fiber $F = S^{n-1}$ is $(n-2)$-connected, so $\pi_m(S^{n-1}) = 0$ for $m < n-1$, and $\pi_{n-1}(S^{n-1}) = \mathbb{Z}$ by corollary 5.4.11 (the bottom homotopy of a sphere is $\mathbb{Z}$, generated by the identity). A section of $S(E) \to X$ can therefore be built freely over the $(n-1)$-skeleton, since all obstruction groups $H^{m+1}(X;\pi_m(S^{n-1}))$ with $m < n-1$ vanish. The primary obstruction to extending the section over the n-skeleton is a class

$$\mathfrak{o}(E) \in H^n\big(X;\, \pi_{n-1}(S^{n-1})\big) = H^n(X;\mathbb{Z})$$

when the fiber bundle is oriented so that the coefficient system $\pi_{n-1}(S^{n-1}) = \mathbb{Z}$ is untwisted; with twisting (the general unoriented case) the coefficients become the local system $\mathbb{Z}$ twisted by $w_1(E)$, whose mod-2 reduction is the constant system $\mathbb{Z}/2$.

Step 2: Oriented case and identification with $e(E)$. When E is oriented, $w_1(E) = 0$ by theorem 7.4.12, the local system is the constant $\mathbb{Z}$, and $\mathfrak{o}(E) \in H^n(X;\mathbb{Z})$ is the primary obstruction. This class is precisely the Euler class: the Euler class is defined (theorem 6.3.2, and [17, §12]) as the image of the fundamental class of the fiber under the transgression of the sphere bundle, which is the same connecting-map construction that produces the primary obstruction cocycle in theorem 5.6.10. Thus $\mathfrak{o}(E) = e(E)$. If E has a nowhere-zero section, the section already extends over the n-skeleton, so $\mathfrak{o}(E) = 0$ and $e(E) = 0$. Conversely, when $\dim X \leq n$, the obstruction $\mathfrak{o}(E) = e(E)$ is the *only* obstruction, since higher obstructions would live in $H^{m+1}(X;\pi_m(S^{n-1}))$ with $m \geq n$, hence in degree $> n \geq \dim X$, where cohomology vanishes; therefore $e(E) = 0$ implies the section extends over all of X.

Step 3: Mod-2 case. Reducing coefficients mod 2 collapses the twisting: the local system $\mathbb{Z}$ twisted by $w_1(E)$ becomes the constant $\mathbb{Z}/2$ after reduction (the sign ambiguity disappears in characteristic two), so the mod-2 primary obstruction $\rho_2(\mathfrak{o}(E)) \in H^n(X;\mathbb{F}_2)$ is defined with no orientation hypothesis. By theorem 7.4.11(1), in the oriented case $\rho_2(e(E)) = w_n(E)$; the general identification of the mod-2 obstruction with $w_n(E)$ holds universally on $BO(n)$ and is transported by naturality ([17, §9]). Hence a nowhere-zero section forces $w_n(E) = 0$, completing the proof.

For the tangent bundle of a closed manifold, this section obstruction is the Euler characteristic, by way of the Poincaré-Hopf theorem and Poincaré duality. A nowhere-zero vector field on M is a nowhere-zero section of TM, and its obstruction $e(TM)$ evaluates against the fundamental class to give $\chi(M)$.

Corollary 7.4.16: Euler Class of the Tangent Bundle Computes the Euler Characteristic

Let M be a closed connected oriented smooth n-manifold with fundamental class $[M] \in H_n(M;\mathbb{Z})$ (definition 4.4.8). Then the Euler class of the tangent bundle evaluates to the Euler characteristic:

$$\big\langle e(TM),\, [M]\big\rangle = \chi(M) \in \mathbb{Z}$$

In particular M admits a nowhere-zero vector field if and only if $\chi(M) = 0$.

***Proof*:**
The evaluation $\langle e(TM), [M]\rangle$ is the Euler number of the tangent bundle. By the Poincaré-Hopf theorem, for a vector field V on M with isolated zeros the sum of the local indices of V equals $\chi(M)$; the proof, via the relation of the index sum to the degree of the Gauss map and Poincaré duality (section 4.5), is carried out in [16, Ch. 6], and via the self-intersection of the diagonal in [20, Poincaré-Hopf]. The obstruction-theoretic content is that the primary obstruction $e(TM) \in H^n(M;\mathbb{Z})$ of theorem 7.4.15, paired against $[M]$, counts the zeros of a generic section of TM with sign, which is the index sum. Since $\dim M = n = \mathrm{rank} TM$, theorem 7.4.15(1) applies: TM has a nowhere-zero section if and only if $e(TM) = 0$, and over the top-dimensional M the class $e(TM)$ is detected entirely by its evaluation $\langle e(TM), [M]\rangle$ against the fundamental class, since $H^n(M;\mathbb{Z}) \cong \mathbb{Z}$ is generated by the dual of $[M]$ (Poincaré duality, corollary 4.5.10). Therefore $e(TM) = 0$ if and only if $\langle e(TM), [M]\rangle = \chi(M) = 0$, giving the nowhere-zero vector field criterion, completing the proof.

The corollary is the bridge promised at the start of the section: the Euler class, defined cohomologically and interpreted as a section obstruction, computes the most classical numerical invariant of a closed manifold. It also explains the hairy-ball theorem as a special case. A worked computation makes the mechanism concrete and ties the strands of this section together.

Example 7.4.17: The Total Stiefel-Whitney Class of TS^2

The tangent bundle TS^2 is a rank-2 real bundle over S^2. Its total Stiefel-Whitney class is computed by a stabilization trick: the Whitney sum $TS^2 \oplus \nu$ of the tangent bundle with the normal bundle ν of the standard embedding $S^2 \subseteq \mathbb{R}^3$ is the restriction $T\mathbb{R}^3|_{S^2}$, which is the trivial rank-3 bundle $\underline{\mathbb{R}^3}$ (the tangent bundle of Euclidean space is trivial). The normal bundle ν is the trivial line bundle $\underline{\mathbb{R}}$ (it is spanned by the outward unit normal, a nowhere-zero global section). Therefore

$$TS^2 \oplus \underline{\mathbb{R}} \cong \underline{\mathbb{R}^3}$$

Apply the Whitney sum formula (theorem 7.4.5) and naturality. A trivial bundle has total Stiefel-Whitney class 1 (it is pulled back from a point, so $w = 1$ by proposition 7.4.1), so $w(\underline{\mathbb{R}^3}) = 1$ and $w(\underline{\mathbb{R}}) = 1$. Hence

$$w(TS^2)\, w(\underline{\mathbb{R}}) = w(TS^2 \oplus \underline{\mathbb{R}}) = w(\underline{\mathbb{R}^3}) = 1$$

and since $w(\underline{\mathbb{R}}) = 1$ this forces $w(TS^2) = 1$. So all Stiefel-Whitney classes of TS^2 vanish: $w_1(TS^2) = 0$ and $w_2(TS^2) = 0$ in $H^*(S^2;\mathbb{F}_2)$.

The vanishing $w_1(TS^2) = 0$ records that S^2 is orientable (theorem 7.4.12), as expected for a sphere.

The vanishing $w_2(TS^2) = 0$ does *not* mean TS^2 is trivial: the integral Euler class is nonzero. By corollary 7.4.16, $\langle e(TS^2), [S^2]\rangle = \chi(S^2) = 2$, so $e(TS^2) = 2u' \neq 0$ where u' generates $H^2(S^2;\mathbb{Z}) \cong \mathbb{Z}$. The top Stiefel-Whitney class is the mod-2 reduction $w_2(TS^2) = \rho_2(e(TS^2)) = \rho_2(2u') = 0$ (theorem 7.4.11(1)), consistent with the Whitney computation, and the reduction loses exactly the factor of 2. The conclusion is sharp: TS^2 has no nowhere-zero section, since $e(TS^2) \neq 0$, which is the hairy-ball theorem, while its mod-2 obstruction w_2 vanishes because $2 \equiv 0$. The integral Euler class is the finer invariant here, and the example shows why the oriented theory carries strictly more information than the mod-2 theory.

The companion computation is the non-orientability of the tautological line bundle, which displays the first Stiefel-Whitney class as a genuine, computable obstruction.

Example 7.4.18: Non-Orientability of the Tautological Bundle on $\mathbb{R}\mathrm{P}^n$

The tautological real line bundle $\gamma^1 \to \mathbb{R}\mathrm{P}^n$ has $w_1(\gamma^1) = a \neq 0$ in $H^1(\mathbb{R}\mathrm{P}^n;\mathbb{F}_2) = \mathbb{F}_2[a]/(a^{n+1})$ by example 7.4.10. By theorem 7.4.12, the nonvanishing of w_1 is exactly the statement that γ^1 is *non-orientable*. Geometrically the obstruction is visible already over $\mathbb{R}\mathrm{P}^1 = S^1$: the restriction $\gamma^1|_{\mathbb{R}\mathrm{P}^1}$ is the Möbius band line bundle, whose total space is a Möbius strip, the standard nonorientable line bundle over the circle (the orientation cover is the connected double cover $S^1 \to S^1$, $z \mapsto z^2$, recovered as in example 4.4.5 for the Möbius band). Pulling back along $\mathbb{R}\mathrm{P}^1 \hookrightarrow \mathbb{R}\mathrm{P}^n$ detects $w_1(\gamma^1) = a$ on the bottom cell, confirming the obstruction is nonzero.

This also computes the Stiefel-Whitney classes of $\mathbb{R}\mathrm{P}^n$ itself, that is of its tangent bundle. The standard bundle isomorphism (proved in [17, §4])

$$T\mathbb{R}\mathrm{P}^n \oplus \underline{\mathbb{R}} \cong (\gamma^1)^{\oplus(n+1)} = \gamma^1 \oplus \cdots \oplus \gamma^1 \ (n+1 \text{ copies})$$

combined with the Whitney sum formula gives

$$w(T\mathbb{R}\mathrm{P}^n) = w(T\mathbb{R}\mathrm{P}^n)\, w(\underline{\mathbb{R}}) = w\big((\gamma^1)^{\oplus(n+1)}\big) = \big(w(\gamma^1)\big)^{n+1} = (1+a)^{n+1}$$

in $\mathbb{F}_2[a]/(a^{n+1})$, since $w(\underline{\mathbb{R}}) = 1$. For $n = 2$ this is $(1+a)^3 = 1 + 3a + 3a^2 + a^3 = 1 + a + a^2$ (mod 2, with $a^3 = 0$), so $w_1(T\mathbb{R}\mathrm{P}^2) = a \neq 0$ and $w_2(T\mathbb{R}\mathrm{P}^2) = a^2 \neq 0$; the nonvanishing of $w_1(T\mathbb{R}\mathrm{P}^2)$ reconfirms that $\mathbb{R}\mathrm{P}^2$ is non-orientable, and the nonvanishing of $w_2(T\mathbb{R}\mathrm{P}^2) = a^2$, paired against the fundamental class $[\mathbb{R}\mathrm{P}^2] \in H_2(\mathbb{R}\mathrm{P}^2;\mathbb{F}_2)$, gives the Stiefel-Whitney number $w_2[\mathbb{R}\mathrm{P}^2] = 1$ that detects $\mathbb{R}\mathrm{P}^2$ as the generator of the unoriented bordism group $\mathfrak{N}_2$ (chapter 8). The single computation $w(\gamma^1) = 1 + a$ thus drives both the non-orientability statement and the bordism invariant.

The thread running through both examples is that the structure-group reductions form a hierarchy, and each rung is governed by one characteristic class. Demanding k linearly independent nowhere-zero sections, rather than one, reduces the structure group from $O(n)$ to $O(n-k)$, and the obstruction is a string of top classes of successive quotient bundles.

Proposition 7.4.19: Vanishing of Top Classes is Necessary for Independent Sections

Let $E \to X$ be a real vector bundle of rank n over a connected CW complex. If E admits k everywhere linearly independent sections (a k-frame field), then

$$w_{n-k+1}(E) = w_{n-k+2}(E) = \cdots = w_n(E) = 0 \in H^*(X; \mathbb{F}_2)$$

the top k Stiefel-Whitney classes all vanish.

***Proof*:**
A k-frame field on E is a trivial rank-k subbundle $\underline{\mathbb{R}^k} \hookrightarrow E$ spanned by the k independent sections. Choosing a metric, split E as a Whitney sum

$$E \cong \underline{\mathbb{R}^k} \oplus Q$$

where $Q = E/\underline{\mathbb{R}^k}$ is the quotient bundle, of rank $n-k$. By the Whitney sum formula (theorem 7.4.5) and $w(\underline{\mathbb{R}^k}) = 1$ (a trivial bundle is classified by a constant map, so $w = 1$ by proposition 7.4.1),

$$w(E) = w(\underline{\mathbb{R}^k})\, w(Q) = w(Q)$$

Now Q has rank $r = n-k$, so $w_i(Q) = 0$ for every $i > n-k$. This is the rank bound on Stiefel-Whitney classes: a rank-r bundle is classified by a map $g\colon X \to BO(r)$, and $w_i(Q) = g^*(w_i)$ is the pullback of the universal class $w_i \in H^i(BO(r); \mathbb{F}_2)$. The point is that $H^*(BO(r); \mathbb{F}_2) = \mathbb{F}_2[w_1, \ldots, w_r]$ has polynomial generators only in degrees $1, \ldots, r$, so for $i > r$ there is no universal class w_i to pull back, and the invariant $w_i(Q)$ is by definition zero. (The cohomology ring itself is nonzero in every degree, for instance $w_r^2 \in H^{2r}(BO(r); \mathbb{F}_2)$ is nonzero; what fails to exist past degree r is the single generator w_i, not the whole group.) Therefore $w_i(E) = w_i(Q) = 0$ for all $i > n-k$, that is for $i = n-k+1, \ldots, n$, completing the proof.

The proposition is the necessary direction of the vector-field problem on a manifold: the maximal number of independent vector fields on a sphere, computed by Adams using K-theory and the Stiefel-Whitney (and higher) obstructions, is bounded below by where these classes first fail to vanish. The obstruction picture is now complete: orientation is the w_1 obstruction, spin is the w_2 obstruction, a single nowhere-zero section is the Euler/w_n obstruction, and k independent sections require the top k classes to vanish. Each characteristic class is a primary obstruction in the sense of theorem 5.6.10, measuring the failure of a reduction of the structure group, and the cohomological degree of the class is exactly the connectivity of the fiber of the associated fibration plus one.

Exercise 7.4.1

1. Using the Whitney sum formula (theorem 7.4.5) and the bundle isomorphism $T\mathbb{R}\mathrm{P}^n \oplus \underline{\mathbb{R}} \cong (\gamma^1)^{\oplus(n+1)}$, compute the total Stiefel-Whitney class $w(T\mathbb{R}\mathrm{P}^3)$ in $\mathbb{F}_2[a]/(a^4)$, and determine whether $\mathbb{R}\mathrm{P}^3$ is orientable and whether it admits a spin structure. (Recall w_1 detects orientability by theorem 7.4.12 and, once $w_1 = 0$, w_2 detects spin by theorem 7.4.14.)

2. Let $E \to X$ be a complex line bundle with $c_1(E) = u \in H^2(X; \mathbb{Z})$. Using the Whitney sum formula, compute the total Chern class $c(E \oplus \bar{E})$ of the sum of E with its conjugate $\bar{E}$ (for which $c_1(\bar{E}) = -c_1(E)$), and show that $c_2(E \oplus \bar{E}) = -u^2$. Interpret the sign in terms of the

underlying real rank-4 bundle and its Euler class.

3. Prove that the tangent bundle TS^n is stably trivial for every n (that is, $TS^n \oplus \underline{\mathbb{R}}$ is trivial), and deduce that $w(TS^n) = 1$ for all n. Then explain why this does *not* contradict the fact that S^n admits no nowhere-zero vector field when n is even. (Use corollary 7.4.16.)

4. Let $E \to X$ be an oriented rank-n bundle over a connected CW complex with $\dim X < n$. Using theorem 7.4.15, prove that E admits a nowhere-zero section, regardless of any characteristic class. Conclude that every rank-n bundle over S^m with $m < n$ has a nowhere-zero section, and relate this to the stable range of the homotopy groups of $BO(n)$.

5. The top class detecting a section. Show directly from the Whitney sum formula and example 7.4.18 that the rank-2 bundle $\gamma^1 \oplus \gamma^1 \to \mathbb{R}\mathrm{P}^2$ has $w_1 = 0$ and $w_2 = a^2 \neq 0$, hence is orientable but admits no nowhere-zero section, and contrast this with the trivial bundle $\underline{\mathbb{R}^2} \to \mathbb{R}\mathrm{P}^2$, which has $w = 1$ and does have one. Explain why w_2 distinguishes these two orientable rank-2 bundles even though w_1 cannot. (Hint: compute w_1 of each first.)

6. The splitting principle in action. Let $E \to X$ be a complex rank-2 bundle that splits as $E = L_1 \oplus L_2$ with $c_1(L_i) = x_i$. Compute $c_1(E)$, $c_2(E)$ in terms of x_1, x_2, and verify that they are the elementary symmetric polynomials σ_1, σ_2. Then express the Chern character $\mathrm{ch}(E) = e^{x_1} + e^{x_2}$ through degree 4 in terms of $c_1(E)$ and $c_2(E)$, illustrating how the splitting principle converts a symmetric-function identity into a universal formula. (You may cite the Newton identity $x_1^2 + x_2^2 = \sigma_1^2 - 2\sigma_2$.)

8

Generalized Cohomology Theories and Spectra

The invariants of the preceding chapters were built by hand. Singular homology is manufactured from a chain complex of formal sums of simplices; cohomology dualizes that complex; homotopy groups count maps out of spheres. Each construction is concrete, and each comes with a proof that homotopic maps act identically, that a pair gives a long exact sequence, that excision holds. This chapter takes the opposite stance. It isolates the formal properties that made those invariants useful (homotopy invariance, the long exact sequence of a pair, excision, behaviour on wedges) and treats them as axioms. A functor on spaces satisfying the axioms is a *cohomology theory*, and the question becomes how much the axioms pin down. The answer comes in three movements. The Eilenberg-Steenrod axioms, supplemented by a single dimension axiom asserting that a point has cohomology concentrated in degree zero, determine ordinary cohomology completely: any two theories agreeing on the point agree everywhere. Drop the dimension axiom and the door opens to genuinely new invariants (the *generalized*, or extraordinary, theories) whose value on a point is spread across many degrees. Brown's representability theorem then shows that every cohomology theory, ordinary or not, is represented by a sequence of spaces, and that these sequences are exactly the objects called *spectra*. K-theory and cobordism, the two prototypes obtained by dropping the dimension axiom, are spectra of independent geometric origin, and they organize the most powerful computations in the subject.

This is the promise made in the introduction, where K-theory and cobordism were named as generalized cohomology theories without explanation. The machinery of the previous chapters (the homotopy groups of chapter 5, the representability of ordinary cohomology by Eilenberg-MacLane spaces from section 3.5 onward, the spectral sequences of chapter 6) is exactly what is needed to make the notion precise and to compute with it. Section 8.1 fixes the category the axioms are conditions on, and establishes the facts about it that every later argument uses: the wedge is the coproduct, homotopy classes into a loop space form a group, suspension and looping are adjoint, and cofiber sequences are the sequences a theory is asked to make exact. Section 8.2 then states the axioms in their proper generality, superseding the abbreviated list recorded earlier in chapter 3, proves the uniqueness theorem that gives ordinary cohomology its claim to canonicity, and identifies the coefficient groups whose non-vanishing in many degrees is the signature of an extraordinary theory.

8.1 The Pointed Homotopy Category

An axiom on a cohomology theory is a condition on a functor, and a functor needs a source. That source is the category whose objects are pointed spaces and whose morphisms are based homotopy classes of based maps, introduced for unpointed spaces in definition 1.1.3 and used without comment ever since. This section makes it explicit and establishes the four facts about it that the rest of the chapter uses: which constructions on spaces are the categorical product and coproduct, when a set of homotopy classes carries a group structure, how suspension and looping are related, and what exactness means for a sequence of spaces. All four are statements about the category, not about any particular invariant.

All four hold. Two properties one might expect alongside them do not. Homotopy classes of maps into an arbitrary space form a pointed set carrying no group structure, and the wedge and the product of two circles are not homotopy equivalent, so CW_* admits no biproduct and is not an additive category. A group structure is restored by taking the target to be a loop space, and the loop space is related to the suspension by an adjunction. Suspension itself has no inverse here, and theorem 5.7.8 measures how far it is from having one.

Definition 8.1.1: Pointed Homotopy Category

The *pointed homotopy category* hTop_* has as objects the pointed spaces (X, x_0) and as morphisms from (X, x_0) to (Y, y_0) the set

$$[X, Y]_* = \{\text{based maps } X \to Y\}/\simeq$$

of based homotopy classes of based maps, with composition induced by composition of representatives. Write CW_* for the full subcategory whose objects are the pointed connected CW complexes with basepoint a 0-cell.
Each morphism set $[X, Y]_*$ is a *pointed* set, with distinguished element the class of the constant map $X \to y_0$, written 0.

Composition is well defined because a based homotopy $f \simeq f'$ composed with a based map g on either side is again a based homotopy, so the class of $g \circ f$ depends only on the classes of g and f. The point of passing to homotopy classes is that a functor defined on hTop_* is by construction homotopy-invariant: the requirement $f \simeq g \implies F(f) = F(g)$, which is an axiom when F is defined on spaces and maps, is discharged by the source category when F is defined on hTop_*. Every invariant in this chapter is a functor on CW_* or on its opposite, and the restriction to CW complexes is what makes theorem 5.3.5 available: it fails for general spaces, the Warsaw circle of example 5.3.8 being the witness.

The one-point space $*$ is both initial and terminal in hTop_*: a based map $* \to Y$ must send the point to y_0 and a based map $X \to *$ is the constant map, so $[*, Y]_*$ and $[X, *]_*$ each have a single element. An object that is both initial and terminal is a zero object, and its presence is what makes the distinguished element of $[X,Y]_*$ canonical rather than a choice: the class 0 is the composite $X \to * \to Y$. With a zero object in hand, the next question is which topological constructions realise the categorical product and coproduct.

Proposition 8.1.2: The Wedge is the Coproduct and the Product is the Product

Let $\{X_\alpha\}_{\alpha\in A}$ be a family of pointed CW complexes. In hTop_*:

1. The wedge sum $\bigvee_\alpha X_\alpha$ of definition 1.2.16, together with the inclusions $\iota_\alpha\colon X_\alpha \hookrightarrow \bigvee_\alpha X_\alpha$, is the coproduct. That is, for every pointed space Y the map

$$\Big[\bigvee_\alpha X_\alpha,\ Y\Big]_* \longrightarrow \prod_\alpha [X_\alpha, Y]_*, \qquad [f] \longmapsto \big([f\circ\iota_\alpha]\big)_\alpha$$

   is a bijection.

2. The product space $\prod_\alpha X_\alpha$, based at the tuple of basepoints and equipped with the projections p_α, is the product. That is, for every pointed space Z the map

$$\Big[Z,\ \prod_\alpha X_\alpha\Big]_* \longrightarrow \prod_\alpha [Z, X_\alpha]_*, \qquad [g] \longmapsto \big([p_\alpha\circ g]\big)_\alpha$$

   is a bijection.

***Proof*:**
Step 1: The coproduct, at the level of maps. Write $W = \bigvee_\alpha X_\alpha$ and let $q\colon \bigsqcup_\alpha X_\alpha \to W$ be the quotient map identifying all the basepoints to a single point, which is the basepoint of W. A based map $f\colon W\to Y$ determines the family $f_\alpha = f\circ\iota_\alpha$, and each f_α is based. Conversely, given based maps $f_\alpha\colon X_\alpha \to Y$, the map $\bigsqcup_\alpha f_\alpha\colon \bigsqcup_\alpha X_\alpha \to Y$ is continuous and constant on the fibre $q^{-1}(*)$, which is the set of basepoints, since every f_α sends its basepoint to y_0. By the universal property of a quotient map [12, Universal Property Of Quotient Maps] it descends to a unique continuous $f\colon W\to Y$ with $f\circ\iota_\alpha = f_\alpha$, and f is based. The two assignments are mutually inverse on maps.

Step 2: The coproduct, at the level of homotopies. It remains to check that the correspondence respects based homotopy in both directions. If $H\colon W\times I\to Y$ is a based homotopy then each $H\circ(\iota_\alpha\times \mathrm{id}_I)$ is a based homotopy, so homotopic f give homotopic families. Conversely, let $H^\alpha\colon X_\alpha\times I \to Y$ be based homotopies. Since I is locally compact Hausdorff, $q\times\mathrm{id}_I\colon (\bigsqcup_\alpha X_\alpha)\times I \to W\times I$ is again a quotient map [12, Products of Quotient Maps]. The map $\bigsqcup_\alpha H^\alpha$ is constant on its fibres, each H^α being based and therefore sending (x^0_α, t) to y_0 for every t, so it descends to a continuous $H\colon W\times I\to Y$, which is the required based homotopy. Hence the correspondence is a bijection on homotopy classes, proving (1).

Step 3: The product. A based map $g\colon Z\to\prod_\alpha X_\alpha$ is by the universal property of the product topology the same datum as a family of maps $g_\alpha\colon Z\to X_\alpha$, and g is based exactly when every g_α is. A based homotopy $Z\times I\to\prod_\alpha X_\alpha$ is likewise the same datum as a family of based homotopies $Z\times I\to X_\alpha$, again by the universal property applied to the space $Z\times I$. No quotient is involved and no local compactness is needed. The correspondence is therefore a bijection on homotopy classes, proving (2) and completing the proof.

The proposition is what gives the additivity axiom its shape. A functor that carries coproducts to products is the categorical condition, and part (1) identifies the coproduct in hTop_* as the wedge; the axiom asking that $\widetilde{h}^n(\bigvee_\alpha X_\alpha) \to \prod_\alpha \widetilde{h}^n(X_\alpha)$ be an isomorphism is that condition and nothing more. Part (2) records that products in hTop_* are computed as products of spaces, so no separate verification is needed on the target side.

Having both a zero object and finite products and coproducts raises the question whether they agree, as they do in an additive category, where the canonical map from a finite coproduct to the finite product is an isomorphism [10, Additive Categories: Products and Coproducts]. For pointed spaces there is such a canonical map, the inclusion of the wedge into the product, and it is not an isomorphism.

Example 8.1.3: The Wedge and the Product of Two Circles

Give the torus $T^2 = S^1 \times S^1$ its standard CW structure with one 0-cell, two 1-cells and one 2-cell. Its 1-skeleton is the wedge $S^1 \vee S^1$, and the canonical map

$$j\colon S^1 \vee S^1 \longrightarrow S^1 \times S^1$$

determined by the identity on each factor and the constant map into the other is exactly this skeletal inclusion. On fundamental groups j induces the homomorphism

$$j_*\colon \pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z} \longrightarrow \pi_1(S^1 \times S^1) \cong \mathbb{Z} \times \mathbb{Z}$$

from the free group on two generators a, b (computed in section 2.2) to the free abelian group of rank two, sending each generator to the corresponding generator. The commutator $aba^{-1}b^{-1}$ is a nontrivial reduced word in $\mathbb{Z} * \mathbb{Z}$ and lies in the kernel of j_*, so j_* is not injective. A based homotopy equivalence induces an isomorphism on π_1, so j is not one, and $S^1 \vee S^1$ and $S^1 \times S^1$ are not isomorphic in CW_*.

Consequently CW_* is not an additive category. It has a zero object and it has both finite products and finite coproducts, but the coproduct and the product of two objects are not identified, which is the failure of the biproduct condition.

The failure recorded in the example is the first of the two structural gaps this section locates. A cohomology theory takes values in abelian groups, so its axioms speak of a category in which morphism sets are abelian groups and the two ways of combining two objects agree; CW_* is not such a category, and the morphism sets $[X, Y]_*$ are in general only pointed sets. This is not a defect in the axioms. It is the reason a theory is a functor *out of* CW_* rather than an operation *inside* it, and the reason the values are required to be groups rather than observed to be. What can be recovered is a group structure on $[X, Y]_*$ for targets Y of a restricted kind, and the restriction turns out to be exactly the condition on the spaces of a spectrum.

Proposition 8.1.4: Homotopy Classes into a Loop Space Form a Group

Let (Y, y_0) be a pointed space and ΩY its loop space (definition 5.2.11). For every pointed space X:

1. The set $[X, \Omega Y]_*$ is a group under the operation $[\alpha] + [\beta] = [\mu \circ (\alpha, \beta)]$, where $\mu\colon \Omega Y \times \Omega Y \to \Omega Y$ is concatenation of loops, with identity the class of the constant map and inverse induced by reversal of loops.
2. The operation is natural in X: for a based map $h\colon X' \to X$ the induced map $h^*\colon [X, \Omega Y]_* \to [X', \Omega Y]_*$ is a group homomorphism.
3. The group $[X, \Omega^2 Y]_*$ is abelian, where $\Omega^2 Y = \Omega(\Omega Y)$.

***Proof*:**
Step 1: Loop concatenation makes ΩY a group up to homotopy. Concatenation

$$\mu(\gamma, \delta)(t) = \begin{cases} \gamma(2t), & 0 \le t \le \frac{1}{2} \\ \delta(2t-1), & \frac{1}{2} \le t \le 1 \end{cases}$$

is continuous as a map $\Omega Y \times \Omega Y \to \Omega Y$ in the compact-open topology, and the reparametrisation homotopies constructed for the fundamental group in chapter 2 show that μ is associative up to based homotopy, that the constant loop e is a two-sided unit up to based homotopy, and that $\gamma \mapsto \bar{\gamma}$, $\bar{\gamma}(t) = \gamma(1-t)$, is a two-sided inverse up to based homotopy. These are the same homotopies as there, applied to loops in Y rather than to their classes, so nothing new is required.

Step 2: The induced operation on homotopy classes. For based maps $\alpha, \beta\colon X \to \Omega Y$ set $\alpha + \beta = \mu \circ (\alpha, \beta)$, where $(\alpha, \beta)\colon X \to \Omega Y \times \Omega Y$ is the map with components α and β. If $\alpha \simeq \alpha'$ and $\beta \simeq \beta'$ through based homotopies A and B, then $\mu \circ (A, B)$ is a based homotopy $\alpha + \beta \simeq \alpha' + \beta'$, so the operation descends to $[X, \Omega Y]_*$. Postcomposing the homotopies of Step 1 with the map (α, β, γ) transports associativity, the unit law and the inverse law from ΩY to $[X, \Omega Y]_*$: for instance the based homotopy $\mu \circ (\mu \times \mathrm{id}) \simeq \mu \circ (\mathrm{id} \times \mu)$ composed with $(\alpha, \beta, \gamma)\colon X \to (\Omega Y)^3$ gives $(\alpha + \beta) + \gamma \simeq \alpha + (\beta + \gamma)$. The identity element is the class of the constant map at the constant loop, which is the distinguished element 0 of definition 8.1.1. This proves (1).

Step 2b: Naturality. For $h\colon X' \to X$ based, $(\alpha + \beta) \circ h = \mu \circ (\alpha, \beta) \circ h = \mu \circ (\alpha \circ h, \beta \circ h) = (\alpha \circ h) + (\beta \circ h)$, an equality of maps, hence of classes. So $h^*([\alpha] + [\beta]) = h^*[\alpha] + h^*[\beta]$, proving (2).

Step 3: Commutativity in the double loop space. A map $X \to \Omega^2 Y$ is a map into the loops on ΩY, so $[X, \Omega^2 Y]_*$ carries two operations of the kind just constructed: the operation $+_1$ obtained by applying Step 2 with target $\Omega(\Omega Y)$, which concatenates in the outer loop coordinate, and the operation $+_2$ obtained by applying the functor Ω to the concatenation μ of ΩY, which concatenates in the inner coordinate. Both have the class of the constant map as identity, since concatenating a constant loop in either coordinate is the constant map. The two coordinates are independent, so concatenating in one coordinate commutes with concatenating in the other, giving the interchange law

$$(\alpha +_1 \beta) +_2 (\gamma +_1 \delta) = (\alpha +_2 \gamma) +_1 (\beta +_2 \delta)$$

for all $\alpha, \beta, \gamma, \delta \in [X, \Omega^2 Y]_*$. Writing e for the common identity and applying interchange twice,

$$\alpha +_1 \beta = (\alpha +_2 e) +_1 (e +_2 \beta) = (\alpha +_1 e) +_2 (e +_1 \beta) = \alpha +_2 \beta$$

so the two operations coincide, and

$$\alpha +_1 \beta = (e +_2 \alpha) +_1 (\beta +_2 e) = (e +_1 \beta) +_2 (\alpha +_1 e) = \beta +_2 \alpha = \beta +_1 \alpha$$

so the common operation is commutative. Hence $[X, \Omega^2 Y]_*$ is abelian, proving (3). This is the Eckmann-Hilton argument, and the case $X = S^n$ is theorem 5.1.4, completing the proof.

The proposition isolates the condition under which the morphism sets of CW_* behave like the morphism sets of an additive category: not for arbitrary targets, but for targets that are loop spaces, and abelian for targets that are double loop spaces. A theory taking values in abelian groups therefore has a chance of being represented by maps into a space only if that space is a double loop space, and the requirement that this hold in every degree at once is what the definition of an Ω-spectrum in section 8.4 imposes. The relation between the loop space of the target and the suspension of the source is the next structural fact, and it is the one on which the suspension axiom rests.

Theorem 8.1.5: The Loop-Suspension Adjunction

Let X and Y be pointed spaces, ΣX the reduced suspension of definition 5.7.1 and ΩY the loop space of definition 5.2.11. Then:

1. There is a bijection
$$\Phi\colon [\Sigma X,\, Y]_* \xrightarrow{\cong} [X,\, \Omega Y]_*$$
sending the class of $g\colon \Sigma X \to Y$ to the class of $\widehat{g}\colon X \to \Omega Y$, $\widehat{g}(x)(t) = g[x,t]$, natural in X and in Y.

2. Φ is an isomorphism of groups, where $[X, \Omega Y]_*$ carries the concatenation structure of proposition 8.1.4 and $[\Sigma X, Y]_*$ the structure induced by the pinch map $\Sigma X \to \Sigma X \vee \Sigma X$ collapsing the equatorial copy of X.

***Proof*:**
Write $\pi\colon X \times I \to \Sigma X$ for the quotient map of definition 5.7.1, which collapses $X \times \{0\}$, $X \times \{1\}$ and $\{x_0\} \times I$. Since $(x_0, 0)$ and $(x_0, 1)$ lie in $\{x_0\} \times I$, all three collapsed sets have the same image, the basepoint of ΣX.

Step 1: The correspondence on maps. Let $g\colon \Sigma X \to Y$ be based and put $G = g \circ \pi\colon X \times I \to Y$, a continuous map with $G(x, 0) = G(x, 1) = y_0$ for every x and $G(x_0, t) = y_0$ for every t. Since I is locally compact Hausdorff, the exponential law [12, The Exponential Law] says that G is continuous if and only if its adjoint

$$\widehat{G}\colon X \longrightarrow C(I, Y), \qquad \widehat{G}(x)(t) = G(x, t)$$

is continuous, where $C(I, Y)$ carries the compact-open topology. The conditions $G(x, 0) = G(x, 1) = y_0$ say that $\widehat{G}(x)$ is a loop at y_0, so $\widehat{G}$ takes values in the subspace $\Omega Y \subseteq C(I, Y)$, and the condition $G(x_0, t) = y_0$ says that $\widehat{G}(x_0)$ is the constant loop, so $\widehat{G}$ is based. Set $\widehat{g} = \widehat{G}$.

Conversely let $h\colon X \to \Omega Y$ be based and let $H\colon X \times I \to Y$, $H(x, t) = h(x)(t)$, be its adjoint, continuous by the same exponential law. Every $h(x)$ is a loop at y_0, so H collapses $X \times \{0\}$ and $X \times \{1\}$ to y_0; and $h(x_0)$ is the constant loop, so H collapses $\{x_0\} \times I$ to y_0. Thus H is constant on the fibres of π and descends, by the universal property of a quotient map [12, Universal Property Of Quotient Maps], to a unique based $\check{h}\colon \Sigma X \to Y$ with $\check{h} \circ \pi = H$. The assignments $g \mapsto \widehat{g}$ and $h \mapsto \check{h}$ are mutually inverse, since both are the exponential correspondence read through π.

Step 2: The correspondence on homotopies. A based homotopy of maps $\Sigma X \to Y$ is a map $\Sigma X \times I \to Y$, and $\pi \times \mathrm{id}_I \colon (X \times I) \times I \to \Sigma X \times I$ is a quotient map because I is locally compact Hausdorff [12, Products of Quotient Maps]. Applying Step 1 with the parameter of the homotopy carried along, a based homotopy $g \simeq g'$ corresponds to a based homotopy $\widehat{g} \simeq \widehat{g'}$, and conversely. Hence the bijection descends to homotopy classes, giving Φ.

Step 3: Naturality. For a based map $u\colon X' \to X$ one has $\widehat{g \circ \Sigma u}(x') = \widehat{g}(u(x'))$, that is $\Phi(\,(\Sigma u)^*[g]\,) = u^*\Phi([g])$; for a based map $v\colon Y \to Y'$ one has $\widehat{v \circ g}(x) = v \circ \widehat{g}(x)$ as loops, that is $\Phi(v_*[g]) = (\Omega v)_*\Phi([g])$. So Φ is natural in both variables, proving (1).

Step 4: Compatibility with the group structures. The pinch map $p\colon \Sigma X \to \Sigma X \vee \Sigma X$ collapses the equatorial copy $X \times \{\frac{1}{2}\}$, and the induced operation on $[\Sigma X, Y]_*$ sends $([g],[g'])$ to the class of the map that runs g on the image of $X \times [0, \frac{1}{2}]$ and g' on the image of $X \times [\frac{1}{2}, 1]$, reparametrised linearly in each. Under the correspondence of Step 1 this map has adjoint

$$x \longmapsto \begin{cases} \widehat{g}(x)(2t), & 0 \le t \le \frac{1}{2} \\ \widehat{g'}(x)(2t-1), & \frac{1}{2} \le t \le 1 \end{cases}$$

which is exactly $\mu \circ (\widehat{g}, \widehat{g'})$, the concatenation of proposition 8.1.4. Hence $\Phi([g] + [g']) = \Phi([g]) + \Phi([g'])$, and Φ carries the identity to the identity because the constant map corresponds to the constant loop. So Φ is a group isomorphism, proving (2) and completing the proof.

The adjunction is the mechanism behind every suspension isomorphism in this chapter. It says that raising the degree of the source by suspending and lowering the degree of the target by looping are the same operation on homotopy classes, so a graded functor built from maps into a fixed sequence of spaces will shift degree correctly precisely when consecutive spaces are related by looping. That is the content of the Ω-spectrum condition, and theorem 8.4.6 uses this theorem at exactly the point where the suspension axiom is verified. Testing the adjunction on spheres recovers a relation already in hand.

Example 8.1.6: The Adjunction on Spheres

Take $X = S^n$, so that $\Sigma S^n = S^{n+1}$ by example 5.7.2. Theorem 8.1.5 gives

$$\pi_{n+1}(Y) = [S^{n+1}, Y]_* = [\Sigma S^n, Y]_* \;\cong\; [S^n, \Omega Y]_* = \pi_n(\Omega Y)$$

an isomorphism of groups for $n \ge 1$. This is the dimension shift $\pi_{n+1}(Y) \cong \pi_n(\Omega Y)$ obtained in example 5.2.12 from the long exact sequence of the path-loop fibration, now derived without any fibration: the two routes agree because both send the class of a map $S^{n+1} \to Y$ to the class of its adjoint. Iterating, $\pi_n(Y) \cong \pi_0(\Omega^n Y)$, and proposition 8.1.4(3) applied with $X = S^n$ and $n \ge 0$ recovers the commutativity of $\pi_{n+2}(Y)$.

Three of the four structural facts are now in place: the coproduct, the group structure on homotopy classes into a loop space, and the adjunction relating suspension to looping. The fourth is exactness.

A cohomology theory converts a sequence of spaces into an exact sequence of groups, and for this to be a condition on the theory rather than on the choice of spaces, there must be a distinguished class of sequences in CW_* that every theory is asked to convert. That class is produced by the mapping cone.

Definition 8.1.7: [Topological] Mapping Cone and Cofiber Sequence

Let $f\colon X \to Y$ be a based map of pointed CW complexes and let CX be the reduced cone of definition 5.7.1, containing X as its base $X \times \{0\}$. The *mapping cone* of f is

$$C_f = Y \cup_f CX = \big(Y \sqcup CX\big)\big/\big(x \sim f(x) \text{ for } x \in X \subseteq CX\big)$$

based at the image of the basepoint, and the *cofiber sequence* of f is the pair of maps

$$X \xrightarrow{f} Y \xrightarrow{i} C_f$$

where i is the inclusion of Y.

The mapping cone attaches to Y a cone on X along f, so that the image of X in C_f becomes contractible inside C_f without being contractible in X. For the inclusion of a subcomplex $A \subseteq X$ this recovers the quotient: the pair (X, A) is a CW pair, so CA is a contractible subcomplex of $C_i = X \cup CA$ satisfying the homotopy extension property by proposition 1.3.4, and collapsing it is a homotopy equivalence by theorem 1.3.5, giving $C_i \simeq X/A$. The construction is the homotopy-theoretic replacement for the quotient, defined for an arbitrary map rather than only for an inclusion. Its use is that maps out of C_f measure exactly the failure of maps out of Y to die on X.

Proposition 8.1.8: The Puppe Sequence and its Coexactness

Let $f\colon X \to Y$ be a based map of pointed CW complexes.

1. For every pointed space K the sequence of pointed sets

$$[C_f, K]_* \xrightarrow{i^*} [Y, K]_* \xrightarrow{f^*} [X, K]_*$$

is exact at $[Y, K]_*$, meaning that $\mathrm{im} i^* = (f^*)^{-1}(0)$.

2. Collapsing Y in the mapping cone of $i\colon Y \to C_f$ gives a based homotopy equivalence $C_i \simeq \Sigma X$, under which the cofiber sequence of i becomes

$$Y \xrightarrow{i} C_f \xrightarrow{\partial} \Sigma X$$

with ∂ the map collapsing $Y \subseteq C_f$.

3. Iterating (2) produces the *Puppe sequence*

$$X \xrightarrow{f} Y \xrightarrow{i} C_f \xrightarrow{\partial} \Sigma X \xrightarrow{\Sigma f} \Sigma Y \xrightarrow{\Sigma i} \Sigma C_f \to \cdots$$

in which every three consecutive terms are, up to based homotopy equivalence, the cofiber sequence of the first map. Consequently, applying $[-, K]_*$ yields a sequence of pointed sets that is exact at every term.

Proof:
Step 1: The image of i^* is contained in the kernel of f^*. Let $[g] \in [C_f, K]_*$. The composite $i \circ f\colon X \to C_f$ sends x to the class of $f(x)$, which by the defining relation of definition 8.1.7 is the class of the base point $(x, 0)$ of the cone. The reduced cone CX is contractible, the contraction sliding each (x, s) up to the cone point, and this contraction is a based homotopy inside C_f. Hence $i \circ f$ is based nullhomotopic and $f^* i^*[g] = [g \circ i \circ f] = 0$.

Step 2: The kernel of f^* is contained in the image of i^*. Let $g\colon Y \to K$ be based with $f^*[g] = 0$, and let $H\colon X \times I \to K$ be a based nullhomotopy with $H(x, 0) = g(f(x))$ and $H(x, 1) = k_0$, the basepoint of K, and $H(x_0, t) = k_0$ for every t. The conditions $H(x, 1) = k_0$ and $H(x_0, t) = k_0$ say that H is constant on the sets collapsed by the quotient map $X \times I \to CX$ of definition 5.7.1, so H descends to a continuous $\widetilde{H}\colon CX \to K$ with $\widetilde{H}(x, 0) = g(f(x))$ on the base. The maps g on Y and $\widetilde{H}$ on CX agree on the identified copy of X, since both send x to $g(f(x))$, so they glue to a continuous based $\bar{g}\colon C_f \to K$ with $\bar{g} \circ i = g$. Therefore $[g] = i^*[\bar{g}] \in \mathrm{im} i^*$, proving (1).

Step 3: The cofiber of the inclusion is the suspension. By definition $C_i = C_f \cup_i CY$, which contains CY as a subcomplex. The pair (C_i, CY) is a CW pair, so it satisfies the homotopy extension property by proposition 1.3.4, and CY is contractible, so the collapse $C_i \to C_i/CY$ is a based homotopy equivalence by theorem 1.3.5. Collapsing CY identifies the copy of Y inside C_f with the basepoint, leaving

$$C_i/CY = C_f/Y = CX/X = \Sigma X$$

the last equality because collapsing the base $X \times \{0\}$ of the reduced cone gives the reduced suspension, by the presentation of ΣX in definition 5.7.1. Under this equivalence the inclusion

$C_f \to C_i$ becomes the collapse $\partial\colon C_f \to C_f/Y = \Sigma X$, proving (2).

Step 4: Iteration and exactness throughout. Applying (2) to i in place of f identifies the cofiber of i with ΣX; applying it again to ∂ identifies the cofiber of ∂ with ΣY, and the resulting map $\Sigma X \to \Sigma Y$ is Σf up to composition with the self-equivalence of ΣX reversing the suspension coordinate. That self-equivalence is an isomorphism in CW_*, so it changes no image and no preimage of a distinguished element, and exactness is unaffected by it; the same remark applies at every later stage.[a] Iterating produces the displayed sequence, in which every three consecutive terms are the cofiber sequence of the first map up to based homotopy equivalence. Applying $[-, K]_*$ and invoking part (1) at each such stretch gives exactness at every term, proving (3) and completing the proof.

[a]The reversal is the reason the Puppe sequence is often written with a sign on the suspended maps. Since only exactness is used here, and exactness is invariant under composing a term with an isomorphism, the sign is suppressed throughout.

Coexactness is the property that makes a single short exactness axiom generate a long exact sequence. A theory is asked to convert one cofiber sequence into a three-term exact sequence; the Puppe sequence then gives infinitely many cofiber sequences built from the same two spaces, and the suspension isomorphism reindexes them into a single doubly infinite sequence. Proposition 8.2.3 carries this out for a reduced cohomology theory. The construction is worth grounding in a case where both the mapping cone and the resulting sequence are computable.

Example 8.1.9: The Mapping Cone of a Degree-d Self-Map

Let $d \geq 2$ and let $f\colon S^1 \to S^1$ be a map of degree d. Its mapping cone

$$C_f = S^1 \cup_f CS^1 = S^1 \cup_f e^2$$

is the space obtained by attaching a 2-cell to the circle along a loop that wraps d times, since the reduced cone on S^1 is a 2-cell attached along its boundary circle. This is the Moore space $M(\mathbb{Z}/d, 1)$: its cellular chain complex is $\mathbb{Z} \xrightarrow{d} \mathbb{Z}$ in degrees 2 and 1, giving $H_1(C_f;\mathbb{Z}) \cong \mathbb{Z}/d$ and $H_n(C_f;\mathbb{Z}) = 0$ for $n \neq 0, 1$. For $d = 2$ the cone is $\mathbb{RP}^2$.

Taking $K = K(\mathbb{Z}, 1) = S^1$ (example 5.5.2) in proposition 8.1.8(1) gives the exact sequence of pointed sets

$$[C_f, S^1]_* \xrightarrow{i^*} [S^1, S^1]_* \xrightarrow{f^*} [S^1, S^1]_*$$

which by $[S^1, S^1]_* = \pi_1(S^1) \cong \mathbb{Z}$ reads $[C_f, S^1]_* \to \mathbb{Z} \xrightarrow{\cdot d} \mathbb{Z}$, the second map being multiplication by d because precomposition with a degree-d map multiplies degree by d. Multiplication by d on $\mathbb{Z}$ is injective for $d \geq 2$, so its kernel is trivial and exactness forces $\operatorname{im} i^* = 0$. Since $[C_f, S^1]_* \cong \widetilde{H}^1(C_f;\mathbb{Z})$ by theorem 5.5.9, and $\widetilde{H}^1(C_f;\mathbb{Z}) \cong \operatorname{Hom}(\mathbb{Z}/d, \mathbb{Z}) = 0$ by the universal coefficient theorem (theorem 4.1.8), the group $[C_f, S^1]_*$ is itself trivial, consistent with the computation.

The last structural fact concerns not the objects of CW_* but its functors. Several arguments in this chapter produce a natural isomorphism between two functors of the form $[-, A]_*$ and conclude that the representing spaces are homotopy equivalent. That inference is the Yoneda lemma, applied in CW_* rather than in a category of algebraic objects, and the only topological content is the identification of what an isomorphism in CW_* is.

Proposition 8.1.10: Representing Objects are Unique up to Homotopy Equivalence

Let A and B be pointed CW complexes and let

$$\Phi\colon [-,A]_* \implies [-,B]_*$$

be a natural isomorphism of contravariant functors on CW_*. Then $\epsilon := \Phi_A([\mathrm{id}_A]) \in [A,B]_*$ is represented by a based homotopy equivalence $A \to B$, and $\Phi_X = \epsilon_*$ is postcomposition with ϵ for every X.

***Proof*:**
The Yoneda lemma [6, the Yoneda lemma] applied to the category CW_*, whose morphism sets are the $[X,Y]_*$, states that natural transformations $[-,A]_* \Rightarrow [-,B]_*$ correspond bijectively to elements of $[A,B]_*$, the correspondence sending Φ to $\Phi_A([\mathrm{id}_A])$ and an element ϵ to postcomposition ϵ_*. The hypotheses are met: CW_* is a locally small category and $[-,A]_*$, $[-,B]_*$ are the contravariant hom-functors it represents. Unwinding the correspondence, naturality of Φ along a based map $u\colon X \to A$ applied to $[\mathrm{id}_A] \in [A,A]_*$ gives

$$\Phi_X\big(u^*[\mathrm{id}_A]\big) = u^*\Phi_A([\mathrm{id}_A]), \qquad \text{that is} \qquad \Phi_X([u]) = [\epsilon \circ u]$$

since $u^*[\mathrm{id}_A] = [u]$. So $\Phi_X = \epsilon_*$ for every X.

Because Φ is a natural isomorphism, the same argument applied to $\Psi = \Phi^{-1}$ produces $\delta := \Psi_B([\mathrm{id}_B]) \in [B,A]_*$ with $\Psi_X = \delta_*$. Evaluating $\Psi_A \circ \Phi_A = \mathrm{id}$ at $[\mathrm{id}_A]$ gives $[\delta \circ \epsilon] = [\mathrm{id}_A]$, and evaluating $\Phi_B \circ \Psi_B = \mathrm{id}$ at $[\mathrm{id}_B]$ gives $[\epsilon \circ \delta] = [\mathrm{id}_B]$. Hence ϵ and δ are mutually inverse in CW_*, which is to say that any representative map $A \to B$ of ϵ is a based homotopy equivalence with homotopy inverse a representative of δ, as we sought to show.

The proposition is the reason a representing space, when it exists, is determined by the functor it represents: the general statement that representing objects are unique up to canonical isomorphism [6, Uniqueness of Representing Objects] reads, in CW_*, as uniqueness up to based homotopy equivalence. On CW_* this is also uniqueness up to weak equivalence, since theorem 5.3.5 identifies the two notions for CW complexes, and it is in that form that theorem 8.3.9 states its uniqueness clause.

8.1.11 Remark: What a Homotopy Functor Can Be Asked to Preserve The four facts of this section fix what a condition on a functor $\mathsf{CW}_*^{\mathrm{op}} \to \mathsf{Ab}$ can say. One of them, proposition 8.1.4, governs the values: it says which targets make homotopy classes into a group at all. The other three identify constructions in CW_*, and each supports one condition on the functor. The wedge is the coproduct (proposition 8.1.2), so a functor may be asked to carry coproducts to products. The mapping cone produces cofiber sequences whose induced sequences of homotopy classes are exact (proposition 8.1.8), so a functor may be asked to carry a cofiber sequence to an exact sequence of groups. The suspension is related to the loop space by an adjunction (theorem 8.1.5), so a functor may be asked to carry Σ to a shift in the grading.
The three are not on the same footing. The first two are conditions on how a functor treats a construction that CW_* performs, and a functor either satisfies them or does not. The third asks the functor to convert an operation into an invertible one, and Σ is not invertible in CW_*: the suspension homomorphism of definition 5.7.3 is an isomorphism only in the range given by theorem 5.7.8, and outside that range suspending loses information, as $\pi_3(S^2) \cong \mathbb{Z}$ collapsing to $\pi_4(S^3) \cong \mathbb{Z}/2$ records. A graded functor can be required to shift degree under Σ, and singular cohomology does, but the requirement is imposed on the functor rather than inherited from an invertible operation on spaces.
This asymmetry is where the objects of section 8.4 come from. A category in which Σ is invertible, morphism sets are abelian groups, and the wedge and the product agree would carry all three conditions structurally rather than axiomatically; CW_* is none of these, by example 8.1.3 and the failure of Freudenthal outside the stable range. The spectra of section 8.4 are the objects of such a category, and box 8.4.12 names it.

Exercise 8.1.1

1. Show that the one-point space is a zero object of hTop_* and deduce that for any pointed spaces X and Y the constant class $0 \in [X, Y]_*$ is the unique morphism factoring through a zero object. Show also that $[X, *]_*$ and $[*, Y]_*$ are singletons, and explain why the analogous statement fails in the unpointed category hTop of definition 1.1.3.

2. Prove that $\Sigma(X \vee Y) \cong \Sigma X \vee \Sigma Y$ for pointed CW complexes X and Y, and deduce from proposition 8.1.2 and theorem 8.1.5 that $\Omega(X \times Y) \cong \Omega X \times \Omega Y$. (For the second part, compare the functors $[-, \Omega(X \times Y)]_*$ and $[-, \Omega X \times \Omega Y]_*$ and apply proposition 8.1.10.)

3. Let Y be a pointed space. Show that a natural group structure on the functor $[-, Y]_*$ determines a map $\mu\colon Y \times Y \to Y$ that is unital and associative up to based homotopy, by evaluating naturality at $X = Y \times Y$. Conclude that proposition 8.1.4 has a converse: the targets Y for which $[-, Y]_*$ is naturally a group are exactly those admitting such a μ.

4. Compute $[S^1 \vee S^1, S^1]_*$ and $[S^1 \times S^1, S^1]_*$ using $S^1 = K(\mathbb{Z}, 1)$ and theorem 5.5.9, and show that the map induced by the inclusion j of example 8.1.3 *is* a bijection between them. Explain why this does not contradict example 8.1.3, and name an invariant that does separate the two spaces.

5. Let $f\colon X \to Y$ be a based map of pointed CW complexes. Show that $[f] = 0$ in $[X, Y]_*$ if and only if the inclusion $i\colon Y \to C_f$ admits a based left homotopy inverse. (Both directions are proposition 8.1.8(1) applied with $K = Y$; ask what exactness at $[Y, Y]_*$ says about the class $[\mathrm{id}_Y]$.)

6. Using proposition 8.1.8 with $K = K(\mathbb{Z}/2, 1)$, compute the image of $i^*\colon [C_f, K]_* \to [S^1, K]_*$ for $f\colon S^1 \to S^1$ of degree 2, and compare with the value of $\widetilde{H}^1(\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2)$ obtained in chapter 4.
7. Show that the bijection of theorem 8.1.5 is *not* in general a bijection $[\Sigma X, Y]_* \cong [X, Y]_*$ by exhibiting X and Y with $[\Sigma X, Y]_*$ and $[X, Y]_*$ of different cardinality. Explain which step of the proof fails if ΩY is replaced by Y.
8. Prove that a based homotopy equivalence $A \to B$ induces a natural isomorphism $[-, A]_* \cong [-, B]_*$, so that the correspondence of proposition 8.1.10 is a bijection between based homotopy classes of equivalences $A \to B$ and natural isomorphisms of the represented functors. Where is connectedness of the objects of CW_* used, if anywhere?

8.2 The Eilenberg-Steenrod Axioms and Generalized Theories

The earliest list of axioms in this book, recorded in chapter 3, was a working summary: enough to recognize a homology theory when one meets it, but stated for homology only, with the category of pairs left vague and the suspension behaviour absent. The axioms deserve a proper treatment, because they carry the entire weight of the chapter. Two formulations run in parallel throughout. The *unreduced* formulation works with pairs (X, A) of CW complexes and a coboundary map raising degree; it is the natural home for the long exact sequence and for excision. The *reduced* formulation works with a single based space and replaces excision and the pair sequence by a suspension isomorphism; it is leaner and is the formulation under which representability is cleanest. The two are equivalent, and the translation between them is recorded once and used freely. The development begins with the reduced version, where the axioms are fewest.

The ambient category is CW_* of definition 8.1.1, and ΣX is the reduced suspension of definition 5.7.1, with $\Sigma S^n = S^{n+1}$ by example 5.7.2. The reduced theory assigns to each based CW complex a graded abelian group and turns the operation Σ on spaces into a degree shift on groups.

Definition 8.2.1: Reduced Cohomology Theory

A *reduced cohomology theory* is a sequence of contravariant functors $\widetilde{h}^n$, one for each integer n, from the homotopy category of based CW complexes to abelian groups, together with for each based CW complex X a natural isomorphism (the *suspension isomorphism*)

$$\sigma_X^n : \widetilde{h}^n(X) \xrightarrow{\cong} \widetilde{h}^{n+1}(\Sigma X)$$

natural in X, subject to the following axioms.

1. (*Homotopy*) If $f \simeq g\colon X \to Y$ are based-homotopic, then $f^* = g^* : \widetilde{h}^n(Y) \to \widetilde{h}^n(X)$ for every n.
2. (*Exactness*) For every based CW pair (X, A) with inclusion $i\colon A \hookrightarrow X$ and quotient map $q\colon X \to X/A$, the sequence

$$\widetilde{h}^n(X/A) \xrightarrow{q^*} \widetilde{h}^n(X) \xrightarrow{i^*} \widetilde{h}^n(A)$$

   is exact at $\widetilde{h}^n(X)$, for every n.
3. (*Additivity*) For any indexed family $\{X_\alpha\}_\alpha$ of based CW complexes with wedge sum $\bigvee_\alpha X_\alpha$, the inclusions $X_\alpha \hookrightarrow \bigvee_\alpha X_\alpha$ induce an isomorphism

$$\widetilde{h}^n\Big(\bigvee_\alpha X_\alpha\Big) \xrightarrow{\cong} \prod_\alpha \widetilde{h}^n(X_\alpha)$$

   for every n.

A *reduced homology theory* is the same data with all arrows reversed: covariant functors $\widetilde{h}_n$, a suspension isomorphism $\widetilde{h}_n(X) \cong \widetilde{h}_{n+1}(\Sigma X)$, exactness of $\widetilde{h}_n(A) \to \widetilde{h}_n(X) \to \widetilde{h}_n(X/A)$, and additivity with the product replaced by a direct sum.

The three axioms are the irreducible content of what it means to be a cohomology theory, stripped of any reference to chains or cocycles. Homotopy invariance says the theory sees only the homotopy type. Exactness, applied to the cofiber sequence $A \to X \to X/A$, is the engine that propagates information along inclusions; iterating it (a point explained below) produces the full long exact sequence. Additivity is the compatibility with infinite wedges that distinguishes a cohomology theory in the sense relevant to representability from the weaker notion governing only finite complexes; without it the theory is determined on finite complexes but may behave erratically in the limit. The suspension isomorphism is the structural heart: it makes the graded group $\widetilde{h}^*(X)$ into a single object spread across degrees, linked rung to rung by suspension, and it is exactly this linkage that a spectrum will encode geometrically.

The axioms already determine the value of any reduced theory on every sphere, with no further data than the cohomology of S^0, which is the first concrete computation worth carrying out.

Example 8.2.2: Reduced Cohomology of Spheres from the Axioms

Let $\widetilde{h}^*$ be any reduced cohomology theory. The sphere S^m is the m-fold reduced suspension $\Sigma^m S^0$, since $\Sigma S^k = S^{k+1}$. The suspension isomorphism, applied m times, gives

$$\widetilde{h}^n(S^m) = \widetilde{h}^n(\Sigma^m S^0) \xrightarrow{(\sigma^{-1})^m,\ \cong} \widetilde{h}^{n-m}(S^0)$$

so the cohomology of every sphere is a degree-shifted copy of the cohomology of S^0. Writing $a_k := \widetilde{h}^k(S^0)$ for the graded group $\widetilde{h}^*(S^0)$, the answer is $\widetilde{h}^n(S^m) \cong a_{n-m}$ for every $m \geq 0$ and every n. The graded group $a_* = \widetilde{h}^*(S^0)$ is the only input; everything else is forced by suspension. For singular cohomology $\widetilde{H}^*(-;G)$, the group $a_k = \widetilde{H}^k(S^0;G)$ is G for $k = 0$ and 0 otherwise, recovering $\widetilde{H}^n(S^m;G) = G$ for $n = m$ and 0 elsewhere, the computation of chapter 4. For complex K-theory, $a_k = \widetilde{KU}^k(S^0)$ is $\mathbb{Z}$ in every even k and 0 in odd k, so $\widetilde{KU}^n(S^m)$ is $\mathbb{Z}$ whenever $n - m$ is even, illustrating how an extraordinary theory sees a sphere in infinitely many degrees.

That the short exactness in axiom (2) already forces a long exact sequence is the next thing to verify, since it is what licenses calling exactness an "axiom" in the singular. The mechanism is the iterated cofiber sequence.

Proposition 8.2.3: The Long Exact Sequence of a Cofiber Sequence

Let $\widetilde{h}^*$ be a reduced cohomology theory and (X, A) a based CW pair with inclusion $i\colon A \hookrightarrow X$ and quotient $q\colon X \to X/A$. Then there is a long exact sequence

$$\cdots \to \widetilde{h}^n(X/A) \xrightarrow{q^*} \widetilde{h}^n(X) \xrightarrow{i^*} \widetilde{h}^n(A) \xrightarrow{\delta} \widetilde{h}^{n+1}(X/A) \xrightarrow{q^*} \widetilde{h}^{n+1}(X) \to \cdots$$

natural in the pair, where the connecting map δ is σ composed with $\widetilde{h}^{n+1}$ of the collapse map $X/A \to \Sigma A$.

***Proof*:**
Step 1: The cofiber sequence of a pair. For a based CW pair (X, A), the inclusion $i\colon A \hookrightarrow X$ has mapping cone $C_i = X \cup_A CA$ (definition 8.1.7), and collapsing the contractible subcomplex CA identifies it with X/A up to based homotopy equivalence, as recorded after that definition. Proposition 8.1.8(3) then gives the *Puppe sequence*

$$A \xrightarrow{i} X \xrightarrow{q} X/A \xrightarrow{\partial} \Sigma A \xrightarrow{\Sigma i} \Sigma X \xrightarrow{\Sigma q} \Sigma(X/A) \to \cdots$$

in which each three-term stretch is, up to based homotopy equivalence, the cofiber sequence of the first map, and $\partial\colon X/A \to \Sigma A$ is the map collapsing X.

Step 2: Applying the theory. Each consecutive pair of maps in the Puppe sequence is a cofiber sequence $U \xrightarrow{f} V \to C_f$ with C_f the mapping cone, and the exactness axiom applied to the CW pair (C_f, V) (whose quotient C_f/V is the suspension ΣU) gives exactness of $\widetilde{h}^n(\Sigma U) \to \widetilde{h}^n(C_f) \to \widetilde{h}^n(V)$. Concretely, applying $\widetilde{h}^n$ to the Puppe sequence and reading it as a doubly infinite sequence yields exactness of

$$\cdots \to \widetilde{h}^n(\Sigma A) \to \widetilde{h}^n(X/A) \xrightarrow{q^*} \widetilde{h}^n(X) \xrightarrow{i^*} \widetilde{h}^n(A) \to \cdots$$

at every term, since each three consecutive terms come from a cofiber sequence to which axiom (2) applies.

Step 3: Reindexing by the suspension isomorphism. The terms $\widetilde{h}^n(\Sigma A)$ appearing in Step 2 are identified by the suspension isomorphism with $\widetilde{h}^{n-1}(A)$, and $\widetilde{h}^n(\Sigma(X/A))$ with $\widetilde{h}^{n-1}(X/A)$, naturally. Substituting these identifications converts the sequence of Step 2 into

$$\cdots \to \widetilde{h}^n(X/A) \xrightarrow{q^*} \widetilde{h}^n(X) \xrightarrow{i^*} \widetilde{h}^n(A) \xrightarrow{\delta} \widetilde{h}^{n+1}(X/A) \to \cdots$$

where the connecting map $\delta\colon \widetilde{h}^n(A) \to \widetilde{h}^{n+1}(X/A)$ is the composite of $(\partial)^*\colon \widetilde{h}^{n+1}(\Sigma A) \to \widetilde{h}^{n+1}(X/A)$ with the suspension isomorphism $\widetilde{h}^n(A) \cong \widetilde{h}^{n+1}(\Sigma A)$. Naturality of σ and of the Puppe construction in the pair makes the entire sequence natural, completing the proof.

The proposition is the reason a single short-exactness axiom suffices: the suspension isomorphism recycles it indefinitely. Each degree's worth of information feeds the next through the connecting map δ, and the resulting long exact sequence is the daily tool of the subject. The cost of the leanness of the reduced axioms is that the geometry of excision is hidden inside the homotopy equivalence "mapping cone of i is X/A," which holds for CW pairs because the cofiber is computed homotopy-theoretically. The unreduced formulation puts excision back in the foreground.

The passage from a reduced theory to an unreduced one is mechanical. Given $\widetilde{h}^*$, define for a CW pair (X, A) the group $h^n(X, A) := \widetilde{h}^n(X/A)$, with the convention $X/\emptyset = X_+ = X \sqcup \{*\}$ adjoining a disjoint basepoint, so that $h^n(X) := h^n(X, \emptyset) = \widetilde{h}^n(X_+)$ is the unreduced cohomology of X. The reduced and unreduced groups of a based space differ by the coefficients: the inclusion of the basepoint $x_0 \hookrightarrow X$ admits the constant retraction $X \to x_0$, which splits the restriction map $h^n(X) \to h^n(\mathrm{pt})$, giving $h^n(X) \cong \widetilde{h}^n(X) \oplus h^n(\mathrm{pt})$ with $\widetilde{h}^n(X) = \ker\big(h^n(X) \to h^n(\mathrm{pt})\big)$. This is the cohomological analogue of the splitting $H_0(X) \cong \widetilde{H}_0(X) \oplus \mathbb{Z}$ from definition 3.2.6. The unreduced axioms are the following.

Definition 8.2.4: Eilenberg-Steenrod Cohomology Theory

An *(unreduced) cohomology theory* on the category of CW pairs is a sequence of contravariant functors $h^n\colon (X,A) \mapsto h^n(X,A)$ to abelian groups, one for each integer n, together with a natural transformation (the *coboundary*)

$$\delta\colon h^n(A) \to h^{n+1}(X,A), \qquad h^n(A) := h^n(A,\emptyset)$$

natural in the pair, subject to the following axioms. (Here a map of pairs $f\colon (X,A) \to (Y,B)$ induces $f^*\colon h^n(Y,B) \to h^n(X,A)$.)

1. (*Homotopy*) Homotopic maps of pairs $f \simeq g\colon (X,A) \to (Y,B)$ induce $f^* = g^*$ on every h^n.
2. (*Exactness*) Each pair (X,A) with inclusions $i\colon A \hookrightarrow X$ and $j\colon (X,\emptyset) \hookrightarrow (X,A)$ induces a long exact sequence
$$\cdots \to h^n(X,A) \xrightarrow{j^*} h^n(X) \xrightarrow{i^*} h^n(A) \xrightarrow{\delta} h^{n+1}(X,A) \to \cdots$$
3. (*Excision*) If (X,A) is a CW pair and $U \subseteq A$ is a subcomplex whose closure lies in the interior of A (for CW pairs, any subcomplex U with $\overline{U} \subseteq \mathrm{int}A$, equivalently any pair admitting an excisive decomposition), then the inclusion $(X \setminus U,\, A \setminus U) \hookrightarrow (X,A)$ induces an isomorphism
$$h^n(X,A) \xrightarrow{\cong} h^n(X \setminus U,\, A \setminus U)$$
for every n.
4. (*Additivity*) For a disjoint union $X = \coprod_\alpha X_\alpha$ of CW complexes, the inclusions induce an isomorphism $h^n(X) \xrightarrow{\cong} \prod_\alpha h^n(X_\alpha)$ for every n.

A *homology theory* is the same data with all functors covariant, the coboundary replaced by a boundary $\partial\colon h_n(X,A) \to h_{n-1}(A)$ lowering degree, exactness running $\cdots \to h_n(A) \to h_n(X) \to h_n(X,A) \xrightarrow{\partial} h_{n-1}(A) \to \cdots$, the excision isomorphism in the same form, and additivity with the product replaced by a direct sum.

These axioms are the ones isolated by Eilenberg and Steenrod and used to organize the whole of ordinary homology and cohomology. Homotopy and exactness are shared with the reduced formulation. Excision is the geometric input that makes the theory *local*: it says that cohomology of a pair does not see the part of A that can be cut away, so that $h^*(X,A)$ depends only on a neighbourhood of the part of X outside A. This is what makes Mayer-Vietoris and the cellular computation possible, and it is the axiom that singular cohomology satisfies by the excision theorem of theorem 3.4.7. Additivity, again, controls the passage to infinite complexes. The coboundary δ is the unreduced incarnation of the connecting map δ from proposition 8.2.3; the equivalence of the two formulations matches them up exactly.

That the reduced and unreduced packages carry the same information is worth stating precisely, since the proof of uniqueness and the discussion of representability both move between them without comment.

Proposition 8.2.5: Equivalence of Reduced and Unreduced Theories

The assignments

$$\widetilde{h}^* \rightsquigarrow \big(h^n(X,A) := \widetilde{h}^n(X/A)\big), \qquad h^* \rightsquigarrow \big(\widetilde{h}^n(X) := \ker\big(h^n(X) \to h^n(\mathrm{pt})\big)\big)$$

are mutually inverse bijections between reduced cohomology theories and unreduced cohomology theories, up to natural isomorphism, where the basepoint inclusion $\mathrm{pt} \hookrightarrow X$ provides the splitting $h^n(X) \cong \widetilde{h}^n(X) \oplus h^n(\mathrm{pt})$. Under this correspondence the suspension isomorphism $\widetilde{h}^n(X) \cong \widetilde{h}^{n+1}(\Sigma X)$ corresponds to the composite of the coboundary of the pair (CX, X) with excision, and the long exact sequence of proposition 8.2.3 corresponds to the exactness axiom of definition 8.2.4.

***Proof*:**
Step 1: From unreduced to reduced. Given h^*, set $\widetilde{h}^n(X) := \ker(h^n(X) \to h^n(\mathrm{pt}))$, the kernel of restriction to the basepoint; equivalently $\widetilde{h}^n(X) = h^n(X, \mathrm{pt})$, since the long exact sequence of the pair (X, pt) together with the retraction $X \to \mathrm{pt}$ splits $h^n(X) \cong h^n(X, \mathrm{pt}) \oplus h^n(\mathrm{pt})$ and identifies $h^n(X, \mathrm{pt})$ with the kernel. Homotopy and additivity for $\widetilde{h}^*$ are inherited termwise. For exactness, the cofiber sequence $A \to X \to X/A$ of a based CW pair has $h^n(X/A, \mathrm{pt}) \cong h^n(X, A)$ by excision (collapsing A is an excision of the interior of a mapping cylinder neighbourhood, valid for CW pairs), so the pair sequence of (X, A) becomes the reduced exact sequence $\widetilde{h}^n(X/A) \to \widetilde{h}^n(X) \to \widetilde{h}^n(A)$.

Step 2: The suspension isomorphism from excision. For a based CW complex X, the reduced cone CX is contractible, so $h^n(CX) \cong h^n(\mathrm{pt})$ and the long exact sequence of the pair (CX, X) gives an isomorphism $\delta\colon \widetilde{h}^n(X) = h^n(X, \mathrm{pt}) \xrightarrow{\cong} h^{n+1}(CX, X)$. Excision applied to collapsing the base of one cone in the union $\Sigma X = CX \cup_X CX$ identifies $h^{n+1}(CX, X) \cong h^{n+1}(\Sigma X, CX) \cong \widetilde{h}^{n+1}(\Sigma X)$, the last step because the second cone CX is contractible. Composing yields the suspension isomorphism $\sigma^n_X\colon \widetilde{h}^n(X) \cong \widetilde{h}^{n+1}(\Sigma X)$, natural in X because every map in the chain is natural.

Step 3: From reduced to unreduced and back. Given $\widetilde{h}^*$, set $h^n(X, A) := \widetilde{h}^n(X/A)$ with $X/\emptyset := X_+$. The reduced long exact sequence of proposition 8.2.3 for (X_+, A_+) is the unreduced pair sequence, the wedge axiom gives unreduced additivity (since $(\coprod_\alpha X_\alpha)_+ = \bigvee_\alpha (X_\alpha)_+$), and excision holds because for a CW triad $X = A \cup B$ the collapse $X/A = (B)/(A \cap B)$ realizes the excision isomorphism as an equality of quotient spaces. Starting from h^*, passing to $\widetilde{h}^*$ by Step 1 and back recovers $h^n(X, A) = \widetilde{h}^n(X/A) = h^n(X/A, \mathrm{pt}) \cong h^n(X, A)$ by the excision identification of Step 1, and the two round trips are the identity up to natural isomorphism, as we sought to show.

With the equivalence in hand the two formulations are used interchangeably. The reduced and unreduced groups of a based space sit in the split exact sequence $0 \to \widetilde{h}^n(X) \to h^n(X) \to h^n(\mathrm{pt}) \to 0$, so the only difference is a copy of the cohomology of a point in each degree. The cohomology of a point is therefore the object that controls the bookkeeping, and it earns a name.

Definition 8.2.6: Coefficient Groups and the Dimension Axiom

Let h^* be a cohomology theory. The graded abelian group

$$h^*(\mathrm{pt}) = \{\, h^n(\mathrm{pt}) \,\}_{n\in\mathbb{Z}}$$

is the group of *coefficients* of the theory; $h^n(\mathrm{pt})$ is the *coefficient group in degree n*. The theory satisfies the *dimension axiom* if

$$h^n(\mathrm{pt}) = 0 \quad \text{for all } n \neq 0$$

so that the coefficients are concentrated in degree 0. A cohomology theory satisfying the dimension axiom is called an *ordinary* cohomology theory; one not required to satisfy it is called a *generalized* (or *extraordinary*) cohomology theory. The same terminology applies to homology theories, with $h_*(\mathrm{pt})$ in place of $h^*(\mathrm{pt})$.

The dimension axiom is the single hypothesis that separates the cohomology theory of the previous two chapters from everything else. For an ordinary theory the coefficients reduce to one abelian group $G = h^0(\mathrm{pt})$, and the reduced cohomology of a sphere is forced: applying the suspension isomorphism n times to $\widetilde{h}^*(S^0)$ and using $\widetilde{h}^k(S^0) = \widetilde{h}^k(\mathrm{pt} \sqcup \mathrm{pt}) = h^k(\mathrm{pt})$ gives $\widetilde{h}^m(S^n) \cong h^{m-n}(\mathrm{pt})$, which under the dimension axiom is G when $m = n$ and 0 otherwise. For a generalized theory the coefficients spread across many degrees, the cohomology of a sphere is correspondingly larger, and a sphere can carry nonzero classes in infinitely many dimensions. The non-vanishing of $h^n(\mathrm{pt})$ for $n \neq 0$ is precisely the slack that K-theory and cobordism exploit, as previewed below.

Example 8.2.7: Singular Cohomology as an Ordinary Theory

Singular cohomology with coefficients in an abelian group G, the functor $(X, A) \mapsto H^n(X, A; G)$ of chapter 4, is an ordinary cohomology theory in the sense of definition 8.2.4, and verifying each axiom is a matter of citing the relevant result from that chapter.

1. *Homotopy.* Homotopic maps of pairs induce the same map on $H^n(-; G)$, the cohomological form of the homotopy invariance of singular homology established in chapter 3 and dualized through $\mathrm{Hom}(-, G)$.

2. *Exactness.* The short exact sequence of cochain complexes $0 \to C^\bullet(X, A; G) \to C^\bullet(X; G) \to C^\bullet(A; G) \to 0$ (the dual of the inclusion of singular chains, which is split since $C_\bullet(A) \hookrightarrow C_\bullet(X)$ is a split injection of free groups) yields the long exact cohomology sequence of the pair, with δ the connecting homomorphism.

3. *Excision.* The excision theorem theorem 3.4.7, dualized, gives $H^n(X, A; G) \cong H^n(X \setminus U, A \setminus U; G)$ under the stated condition $\overline{U} \subseteq \mathrm{int} A$.

4. *Additivity.* A singular simplex $\Delta^n \to \coprod_\alpha X_\alpha$ has connected image, hence lands in a single X_α, so $C_\bullet(\coprod_\alpha X_\alpha) = \bigoplus_\alpha C_\bullet(X_\alpha)$; applying $\mathrm{Hom}(-, G)$ turns the direct sum into the product $\prod_\alpha H^n(X_\alpha; G)$.

The coefficient groups are $H^n(\mathrm{pt}; G) = G$ for $n = 0$ and 0 for $n \neq 0$, since a point has a single 0-simplex and no higher simplices, so the dimension axiom holds and the theory is ordinary with $h^0(\mathrm{pt}) = G$. De Rham cohomology of smooth manifolds and cellular cohomology of CW complexes are two further ordinary models with $G = \mathbb{R}$ and $G = \mathbb{Z}$ respectively; both are naturally isomorphic to singular cohomology on the spaces where all three are defined, by the comparison theorems of chapter 4, and the uniqueness theorem below explains why such isomorphisms must exist.

The example is the calibration point: ordinary cohomology is not one theory among many but, as the next theorem shows, the only ordinary theory with a given coefficient group, up to canonical isomorphism. This is what entitles one to speak of "the" cohomology with coefficients in G regardless of which model (singular, cellular, de Rham, sheaf-theoretic with constant coefficients) is used to compute it. The uniqueness theorem makes the claim precise and proves it.

Its proof climbs the skeleta of a CW complex one at a time, and for a finite-dimensional complex the climb terminates. For an infinite-dimensional one it does not, and the theorem then needs an answer to a question the axioms do not settle by themselves: how is $\widetilde{h}^n(X)$ determined by the groups $\widetilde{h}^n(X^m)$ of the skeleta? The naive answer, that it is their limit, is very nearly right and fails by exactly one group. Settling this is the business of the next three results, which are stated here because the uniqueness theorem uses them, and which are used again for the Atiyah-Hirzebruch spectral sequence in section 8.6.

The first ingredient is a Mayer-Vietoris sequence for a reduced theory. Nothing in definition 8.2.1 mentions unions, so the sequence has to be built from the exactness axiom applied twice.

Proposition 8.2.8: Mayer-Vietoris for a Reduced Cohomology Theory

Let $\widetilde{h}^*$ be a reduced cohomology theory and let $X = A \cup B$ be a based CW complex written as the union of two subcomplexes whose intersection $C = A \cap B$ is a subcomplex containing the basepoint. Write $i_A\colon A \hookrightarrow X$, $i_B\colon B \hookrightarrow X$, $j_A\colon C \hookrightarrow A$ and $j_B\colon C \hookrightarrow B$ for the inclusions. Then there is a long exact sequence

$$\cdots \to \widetilde{h}^n(X) \xrightarrow{(i_A^*,\, i_B^*)} \widetilde{h}^n(A) \oplus \widetilde{h}^n(B) \xrightarrow{j_A^* - j_B^*} \widetilde{h}^n(C) \xrightarrow{\delta} \widetilde{h}^{n+1}(X) \to \cdots$$

natural in the triad $(X; A, B)$.

***Proof*:**
Write $\Phi = (i_A^*, i_B^*)$ and $\Psi = j_A^* - j_B^*$.

Step 1: The excision identification. Since $X = A \cup B$ is a CW triad, every cell of X not in B is a cell of A not in C, so collapsing B in X and collapsing C in A produce the same quotient: the map induced by i_A is a based homeomorphism

$$\kappa\colon A/C \xrightarrow{\cong} X/B$$

and by construction $\kappa \circ q_A = q_X \circ i_A$, where $q_A\colon A \to A/C$ and $q_X\colon X \to X/B$ are the quotient maps. Note also that $q_X \circ i_B$ is the constant map to the basepoint, so $(q_X i_B)^*$ factors through $\widetilde{h}^n(\mathrm{pt})$. That group vanishes for any reduced theory: the additivity axiom of definition 8.2.1 applied to the *empty* family of based complexes has $\bigvee_{\alpha\in\emptyset} X_\alpha = \mathrm{pt}$ on the left and the trivial group $\prod_{\alpha\in\emptyset} \widetilde{h}^n(X_\alpha) = 0$ on the right, so $\widetilde{h}^n(\mathrm{pt}) \cong 0$. Hence $(q_X i_B)^* = 0$, and that $i_A j_A = i_B j_B$, both being the inclusion $C \hookrightarrow X$.

Step 2: Two long exact sequences and a ladder. Proposition 8.2.3 applied to the based CW pairs (A, C) and (X, B) gives

$$\cdots \to \widetilde{h}^n(A/C) \xrightarrow{q_A^*} \widetilde{h}^n(A) \xrightarrow{j_A^*} \widetilde{h}^n(C) \xrightarrow{\delta_A} \widetilde{h}^{n+1}(A/C) \to \cdots$$

$$\cdots \to \widetilde{h}^n(X/B) \xrightarrow{q_X^*} \widetilde{h}^n(X) \xrightarrow{i_B^*} \widetilde{h}^n(B) \xrightarrow{\delta_X} \widetilde{h}^{n+1}(X/B) \to \cdots$$

The pair $(i_A, j_B)\colon (A, C) \to (X, B)$ is a map of based CW pairs inducing κ on quotients, so naturality of proposition 8.2.3 makes the diagram

$$\begin{tikzcd}
\widetilde{h}^n(X/B) \arrow[r, "q_X^*"] \arrow[d, "\kappa^*"'] & \widetilde{h}^n(X) \arrow[r, "i_B^*"] \arrow[d, "i_A^*"'] & \widetilde{h}^n(B) \arrow[r, "\delta_X"] \arrow[d, "j_B^*"'] & \widetilde{h}^{n+1}(X/B) \arrow[d, "\kappa^*"] \\
\widetilde{h}^n(A/C) \arrow[r, "q_A^*"] & \widetilde{h}^n(A) \arrow[r, "j_A^*"] & \widetilde{h}^n(C) \arrow[r, "\delta_A"] & \widetilde{h}^{n+1}(A/C)
\end{tikzcd}$$

commute, with the outer verticals isomorphisms by Step 1. Define

$$\delta := q_X^* \circ (\kappa^*)^{-1} \circ \delta_A \colon \widetilde{h}^n(C) \to \widetilde{h}^{n+1}(X)$$

Step 3: Exactness at $\widetilde{h}^n(A) \oplus \widetilde{h}^n(B)$. That $\Psi\Phi = 0$ is the identity $i_A j_A = i_B j_B$ of Step 1. Conversely let $\Psi(a, b) = 0$, so $j_A^* a = j_B^* b =: c$. The ladder gives $\kappa^* \delta_X(b) = \delta_A(j_B^* b) = \delta_A(j_A^* a) = 0$, the last equality by exactness of the first sequence at $\widetilde{h}^n(C)$; as κ^* is injective, $\delta_X(b) = 0$, so exactness of the second sequence at $\widetilde{h}^n(B)$ yields $x \in \widetilde{h}^n(X)$ with $i_B^* x = b$. Then

$$j_A^*(i_A^* x - a) = (i_A j_A)^* x - c = (i_B j_B)^* x - c = j_B^* b - c = 0$$

so exactness of the first sequence at $\widetilde{h}^n(A)$ gives $w \in \widetilde{h}^n(A/C)$ with $q_A^* w = i_A^* x - a$. Put $x' := x - q_X^*\big((\kappa^*)^{-1} w\big)$. Using $q_X i_A = \kappa q_A$ from Step 1,

$$i_A^* x' = i_A^* x - q_A^* \kappa^* (\kappa^*)^{-1} w = i_A^* x - q_A^* w = a$$

and $i_B^* x' = i_B^* x = b$ because $(q_X i_B)^* = 0$. Hence $\Phi(x') = (a, b)$.

Step 4: Exactness at $\widetilde{h}^n(C)$. From the ladder, $(\kappa^*)^{-1} \delta_A j_B^* = \delta_X$, so

$$\delta\Psi(a, b) = q_X^* (\kappa^*)^{-1} \delta_A j_A^* a - q_X^* \delta_X b = 0 - 0 = 0$$

the first term vanishing by exactness of the first sequence at $\widetilde{h}^n(C)$ and the second by exactness of the second at $\widetilde{h}^{n+1}(X/B)$. Conversely suppose $\delta(c) = 0$ and set $u := (\kappa^*)^{-1} \delta_A(c)$. Then $q_X^* u = 0$, so $u = \delta_X(b)$ for some $b \in \widetilde{h}^n(B)$, whence $\delta_A(c) = \kappa^* \delta_X(b) = \delta_A(j_B^* b)$ and $\delta_A(c - j_B^* b) = 0$. Exactness of the first sequence at $\widetilde{h}^n(C)$ produces $a \in \widetilde{h}^n(A)$ with $j_A^* a = c - j_B^* b$, and then $\Psi(a, -b) = j_A^* a + j_B^* b = c$.

Step 5: Exactness at $\widetilde{h}^{n+1}(X)$. Both components of $\Phi\delta$ vanish: $i_B^* \delta = (q_X i_B)^*(\cdots) = 0$, and $i_A^* \delta(c) = q_A^* \kappa^* (\kappa^*)^{-1} \delta_A c = q_A^* \delta_A c = 0$ by exactness of the first sequence at $\widetilde{h}^{n+1}(A/C)$. Conversely let $\Phi(x) = 0$. From $i_B^* x = 0$ and exactness of the second sequence, $x = q_X^* u$ for some $u \in \widetilde{h}^{n+1}(X/B)$. From $i_A^* x = 0$ and $q_X i_A = \kappa q_A$, $q_A^*(\kappa^* u) = 0$, so exactness of the first sequence at $\widetilde{h}^{n+1}(A/C)$ gives $c \in \widetilde{h}^n(C)$ with $\delta_A(c) = \kappa^* u$. Then $\delta(c) = q_X^* (\kappa^*)^{-1} \kappa^* u = q_X^* u = x$.

Naturality in the triad follows from naturality of proposition 8.2.3 and of κ, completing the proof.

The second ingredient replaces an increasing union by a space built out of the stages one cylinder at a time, so that the union can be cut into two pieces and proposition 8.2.8 applied. The construction is the mapping telescope, used informally in the proof of theorem 8.3.9 below and recorded here.

Definition 8.2.9: Mapping Telescope

Let $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \to \cdots$ be a sequence of based CW complexes and based cellular maps, and let M_m denote the mapping cylinder of f_m, that is

$$M_m := \big(X_m \times I \ \sqcup \ X_{m+1}\big)/\big((x,1) \sim f_m(x)\big)$$

so that M_m contains $X_m = X_m \times \{0\}$ at its near end and X_{m+1} at its far end, and deformation retracts onto the far end by sliding the cylinder coordinate. The *mapping telescope* is

$$\mathrm{Tel}(X_\bullet) := \bigcup_{m\geq 0} M_m$$

the union taken by identifying the copy of X_{m+1} at the far end of M_m with the copy at the near end of M_{m+1}, with the basepoint ray collapsed. It is a based CW complex, and $M_m \cap M_{m+1} = X_{m+1}$.

Proposition 8.2.10: The Telescope of an Increasing Union

Let X be a based CW complex written as an increasing union $X = \bigcup_{m\geq 0} X_m$ of based subcomplexes, and let $\mathrm{Tel}(X_\bullet)$ be the telescope of the inclusions. Then the map $p\colon \mathrm{Tel}(X_\bullet) \to X$ collapsing each cylinder coordinate is a based homotopy equivalence.

***Proof*:**
Both spaces are CW complexes, so by theorem 5.3.5 it suffices to show p is a weak homotopy equivalence. Fix $q \geq 0$. A map $S^q \to X$ has compact image, hence meets only finitely many cells and lands in some X_m; likewise a homotopy $S^q \times I \to X$ lands in some X_m. Therefore $\pi_q(X) = \varinjlim_m \pi_q(X_m)$. The same argument applies to $\mathrm{Tel}(X_\bullet)$, whose finite subtelescope $\bigcup_{m\leq M} M_m$ deformation retracts onto X_{M+1} by sliding along the cylinder coordinates, so $\pi_q(\mathrm{Tel}(X_\bullet)) = \varinjlim_m \pi_q(X_m)$ as well. Under these identifications p_* is the identity of the colimit, since on the copy of X_m sitting in the telescope, p restricts to the inclusion $X_m \hookrightarrow X$. Hence p_* is an isomorphism on π_q for every q and every basepoint, the argument being the same at each since every point lies in a finite subtelescope, so p is a weak homotopy equivalence and theorem 5.3.5 makes it a homotopy equivalence.

One upgrade remains, because theorem 5.3.5 as stated concludes only that p is an *unbased* homotopy equivalence, while $\widetilde{h}^*$ is a functor on based complexes (definition 8.2.1). Its second clause gives the based form. Let M_p be the reduced mapping cylinder of the cellular map p, so that $\mathrm{Tel}(X_\bullet)$ and X both sit in M_p as subcomplexes, the retraction $r\colon M_p \to X$ is a based deformation retraction by construction, and $p = r \circ \iota$ for the inclusion $\iota\colon \mathrm{Tel}(X_\bullet) \hookrightarrow M_p$. Since r is a weak equivalence and p is one by the previous paragraph, so is ι; and ι is the inclusion of a subcomplex, so the second clause of theorem 5.3.5 makes $\mathrm{Tel}(X_\bullet)$ a deformation retract of M_p by a homotopy performed rel $\mathrm{Tel}(X_\bullet)$, which is therefore based. Both factors of $p = r \circ \iota$ are based homotopy equivalences, hence so is p, as we sought to show.

Cutting the telescope into its even and odd cylinders and feeding the pieces to proposition 8.2.8

now produces the sequence the uniqueness theorem needs. The point to watch in the proof is that the Mayer-Vietoris map, once additivity has identified the two middle terms, turns out to be the map whose kernel and cokernel define $\varprojlim$ and $\varprojlim^1$ in [10, The Limit and $\varprojlim^1$ of a Tower], up to a sign on alternate coordinates that changes neither. Nothing is transported by analogy: it is that homomorphism, not an analogue of it.

Theorem 8.2.11: The Milnor Exact Sequence

Let $\widetilde{h}^*$ be a reduced cohomology theory and let X be a based CW complex written as an increasing union $X = \bigcup_{m\geq 0} X_m$ of based subcomplexes. For each n the groups $\widetilde{h}^n(X_m)$ form a tower under the restriction maps, and there is a short exact sequence

$$0 \to \varprojlim{}^1_m \widetilde{h}^{n-1}(X_m) \to \widetilde{h}^n(X) \to \varprojlim_m \widetilde{h}^n(X_m) \to 0$$

natural in the filtered complex, where $\varprojlim$ and $\varprojlim^1$ are as in [10, The Limit and $\varprojlim^1$ of a Tower]. The middle map is restriction.
The hypothesis that the X_m be subcomplexes of a single X is used only to identify $\widetilde{h}^*(X)$ with $\widetilde{h}^*(\mathrm{Tel}(X_\bullet))$. For an arbitrary sequence $X_0 \xrightarrow{f_0} X_1 \to \cdots$ of based CW complexes and based cellular maps, the same conclusion holds verbatim with $\mathrm{Tel}(X_\bullet)$ in place of X and the maps f_m^* as the transition maps of the tower.

***Proof*:**
Step 1: Replace X by the telescope. By proposition 8.2.10 the collapsing map $\mathrm{Tel}(X_\bullet) \to X$ is a based homotopy equivalence, so by the homotopy axiom it induces isomorphisms $\widetilde{h}^n(X) \cong \widetilde{h}^n(\mathrm{Tel}(X_\bullet))$ compatible with restriction to each X_m. Write $T := \mathrm{Tel}(X_\bullet)$ and work with T.

Step 2: The even-odd decomposition. Let A be the union of the cylinders M_m with m even and B the union of those with m odd. Then $A \cup B = T$, and since M_m and $M_{m'}$ meet only in the basepoint whenever $|m - m'| \geq 2$, the even cylinders are pairwise disjoint away from the basepoint, and so are the odd ones. That is what makes A and B wedges rather than disjoint unions, which is the form the additivity axiom of definition 8.2.1 requires. Each M_m deformation retracts onto the copy of X_{m+1} at its far end, so

$$A \simeq \bigvee_{m \text{ even}} X_{m+1}, \qquad B \simeq \bigvee_{m \text{ odd}} X_{m+1}$$

as based spaces, and $C := A \cap B = \bigvee_{m\geq 1} X_m$, since X_m for $m \geq 1$ lies in exactly the two cylinders M_{m-1} and M_m, one of each parity, while X_0 lies only in M_0.

Step 3: Additivity identifies the terms. The additivity axiom of definition 8.2.1 turns each wedge into a product. The odd values $m+1$ arising from A and the even ones arising from B together run over every $m \geq 1$ exactly once, so

$$\widetilde{h}^n(A) \oplus \widetilde{h}^n(B) \cong \prod_{m\geq 1} \widetilde{h}^n(X_m), \qquad \widetilde{h}^n(C) \cong \prod_{m\geq 1} \widetilde{h}^n(X_m)$$

Step 4: The Mayer-Vietoris map is the map defining $\varprojlim$ and $\varprojlim^1$. Fix $m \geq 1$ and compute the two restrictions of proposition 8.2.8 to the summand X_m of C. The subcomplex X_m sits at the far end of M_{m-1}, and the deformation retraction of M_{m-1} onto that far end is the identity there, so restricting from the cylinder M_{m-1} is the identity of $\widetilde{h}^n(X_m)$. The same X_m sits at the near end of M_m, and the deformation retraction of M_m onto its far end X_{m+1} restricts on the near end to $f_m\colon X_m \to X_{m+1}$, so restricting from M_m is f_m^*. Under the identifications of Step 3 the Mayer-Vietoris map $j_A^* - j_B^*$ is therefore

$$\prod_{m\geq 1} \widetilde{h}^n(X_m) \to \prod_{m\geq 1} \widetilde{h}^n(X_m), \qquad (u_m)_m \longmapsto \big(u_m - f_m^*(u_{m+1})\big)_m$$

up to a sign depending on the parity of m. Which of the two cylinders M_{m-1} and M_m lies in A is decided by that parity: for m odd the identity contribution comes from A and the f_m^* contribution from B, giving $u_m - f_m^*(u_{m+1})$, while for m even the two are exchanged and the coordinate reads $f_m^*(u_{m+1}) - u_m$. So $j_A^* - j_B^*$ is $\varepsilon \circ d$ for the automorphism ε of $\prod_{m\geq 1} \widetilde{h}^n(X_m)$ negating the coordinates with m even. Post-composing with an automorphism of the target changes neither kernel nor cokernel, so $j_A^* - j_B^*$ and d have the same kernel and the same cokernel and the sign may be dropped.

The map d is precisely the one of [10, The Limit and $\varprojlim^1$ of a Tower] for the tower $\big(\widetilde{h}^n(X_m)\big)_{m\geq 1}$ with transition maps f_m^*, whose kernel is $\varprojlim_m \widetilde{h}^n(X_m)$ and whose cokernel is $\varprojlim^1_m \widetilde{h}^n(X_m)$ by definition. Discarding the single term $m = 0$ changes neither: an element of the kernel is a coherent family, whose entry $u_0 = f_0^*(u_1)$ is determined by the rest, and the equation $u_0 - f_0^*(u_1) = b_0$ can be solved for u_0 whatever u_1 is, so no cokernel is contributed in that coordinate either.

Step 5: Read off the short exact sequence. Proposition 8.2.8 for the triad $(T; A, B)$ gives the exact sequence

$$\widetilde{h}^{n-1}(A) \oplus \widetilde{h}^{n-1}(B) \xrightarrow{d} \widetilde{h}^{n-1}(C) \xrightarrow{\delta} \widetilde{h}^n(T) \to \widetilde{h}^n(A) \oplus \widetilde{h}^n(B) \xrightarrow{d} \widetilde{h}^n(C)$$

Three exactness statements are used, one at each of the three displayed terms. Exactness at $\widetilde{h}^n(A) \oplus \widetilde{h}^n(B)$ says the image of $\widetilde{h}^n(T)$ there is $\ker d$ in degree n, so the right-hand map of the sequence below is onto its stated target. Exactness at $\widetilde{h}^{n-1}(C)$ says $\ker\delta$ is the image of d in degree $n-1$, so δ factors through an injection of coker$d$ in degree $n-1$ into $\widetilde{h}^n(T)$. Exactness at $\widetilde{h}^n(T)$ itself says $\ker\Phi = \mathrm{im}\delta$, which is exactness in the middle of the spliced sequence. Together they give

$$0 \to \mathrm{coker}\big(d \text{ in degree } n-1\big) \to \widetilde{h}^n(T) \to \ker\big(d \text{ in degree } n\big) \to 0$$

which by Step 4 is the displayed sequence. The middle map is restriction because the identification of Step 3 was by restriction to the summands. Naturality follows from naturality of proposition 8.2.8 together with naturality of $\varprojlim$ and $\varprojlim^1$ in the tower, completing the proof.

Corollary 8.2.12: Mittag-Leffler Collapse of the Milnor Sequence

In the situation of theorem 8.2.11, if the tower $\big(\widetilde{h}^{n-1}(X_m)\big)_m$ satisfies the Mittag-Leffler condition, in particular if the restriction maps $\widetilde{h}^{n-1}(X_{m+1}) \to \widetilde{h}^{n-1}(X_m)$ are surjective, then restriction is an isomorphism

$$\widetilde{h}^n(X) \xrightarrow{\cong} \varprojlim_m \widetilde{h}^n(X_m)$$

***Proof*:**
Mittag-Leffler forces $\varprojlim^1_m \widetilde{h}^{n-1}(X_m) = 0$ by [10, Mittag-Leffler vanishing], and a surjective tower satisfies the condition. The left-hand term of theorem 8.2.11 vanishes, leaving the restriction map an isomorphism, as we sought to show.

The sequence is also natural in the *theory*, which is the form theorem 8.2.14 uses. By a transformation $\tau\colon \widetilde{h}^* \to \widetilde{k}^*$ of reduced theories is meant a family of homomorphisms commuting with all induced maps and with the suspension isomorphisms σ. Such a τ carries the Milnor sequence of $\widetilde{h}^*$ to that of $\widetilde{k}^*$, because every map appearing in the construction is of one of three kinds: induced by a map of spaces, hence commuting with τ by definition; a connecting map of proposition 8.2.3, which is built from σ and the Puppe collapse and so commutes with τ for the same two reasons; or an additivity identification, which is induced by the wedge inclusions. The Mayer-Vietoris connecting map $\delta = q_X^*(\kappa^*)^{-1}\delta_A$ of proposition 8.2.8 is a composite of maps of the first two kinds, so it commutes with τ as well.

The corollary is the form in which the sequence is almost always applied, and it explains why the $\varprojlim^1$ term is easy to forget: for the skeletal filtration of a CW complex and an ordinary theory, the restrictions are usually surjective and the naive answer is correct. It is not always correct, and a single example settles the matter.

Example 8.2.13: A Telescope With a Nonvanishing $\varprojlim^1$
Fix a prime p and let T be the telescope of

$$S^1 \xrightarrow{p} S^1 \xrightarrow{p} S^1 \to \cdots$$

where p denotes a degree-p map. These maps are not inclusions of subcomplexes, so it is the second form of theorem 8.2.11 that applies, computing $\widetilde{h}^*(T)$ directly. Take $\widetilde{h}^* = \widetilde{H}^*(-;\mathbb{Z})$, which is an ordinary theory satisfying additivity. A degree-p map induces multiplication by p on $\widetilde{H}^1(S^1;\mathbb{Z}) = \mathbb{Z}$ and the zero map on $\widetilde{H}^0(S^1;\mathbb{Z}) = 0$, so the two towers are

$$\big(\widetilde{H}^1(S^1)\big)_m = \big(\mathbb{Z} \xleftarrow{p} \mathbb{Z} \xleftarrow{p} \cdots\big), \qquad \big(\widetilde{H}^0(S^1)\big)_m = 0$$

The first is the tower computed in [10, A Tower With Nonvanishing $\varprojlim^1$], where its limit is 0 and its $\varprojlim^1$ is $\mathbb{Z}_p/\mathbb{Z}$. Theorem 8.2.11 in degree $n = 1$ therefore gives $\widetilde{H}^1(T;\mathbb{Z}) \cong \varprojlim_m \mathbb{Z} = 0$, and in degree $n = 2$ it gives

$$0 \to \varprojlim{}^1_m \widetilde{H}^1(S^1;\mathbb{Z}) \to \widetilde{H}^2(T;\mathbb{Z}) \to \varprojlim_m \widetilde{H}^2(S^1;\mathbb{Z}) = 0 \to 0$$

so that

$$\widetilde{H}^2(T;\mathbb{Z}) \cong \mathbb{Z}_p/\mathbb{Z}$$

The telescope is a two-dimensional complex whose second cohomology is an uncountable group, all of it contributed by the $\varprojlim^1$ term. The answer can be checked independently: $H_1(T;\mathbb{Z}) = \varinjlim(\mathbb{Z} \xrightarrow{p} \mathbb{Z} \to \cdots) = \mathbb{Z}[1/p]$, Singular homology commutes with the telescope colimit, by the same compactness argument as in proposition 8.2.10, so also $H_2(T;\mathbb{Z}) = \varinjlim H_2(S^1;\mathbb{Z}) = 0$. The universal coefficient theorem (theorem 4.1.8) gives the short exact sequence $0 \to \mathrm{Ext}^1_{\mathbb{Z}}(H_1(T;\mathbb{Z}),\mathbb{Z}) \to H^2(T;\mathbb{Z}) \to \mathrm{Hom}(H_2(T;\mathbb{Z}),\mathbb{Z}) \to 0$, whose right-hand term therefore vanishes, leaving $H^2(T;\mathbb{Z}) \cong \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}[1/p],\mathbb{Z})$, which is $\mathbb{Z}_p/\mathbb{Z}$. The two computations agree.

A natural transformation of cohomology theories is the structure that the theorem compares. Two cohomology theories h^*, k^* on CW pairs being given, a *natural transformation* $\tau\colon h^* \to k^*$ is a family of homomorphisms $\tau_{(X,A)}\colon h^n(X,A) \to k^n(X,A)$, one in each degree, commuting with all induced maps f^* and with the coboundary δ. The condition that τ commute with δ is what allows the five lemma to be applied to the ladder of long exact sequences, and it is the crux of the argument.

The motivating question is whether the value of a theory on a point determines it everywhere. For ordinary theories the answer is yes, and the proof is an induction up the skeleta of a CW complex, using the dimension axiom to compute the relative groups of each skeletal pair and the five lemma to climb from one skeleton to the next.

Theorem 8.2.14: Uniqueness of Ordinary Cohomology

Let h^* and k^* be ordinary cohomology theories on the category of CW pairs, and let $\tau\colon h^* \to k^*$ be a natural transformation commuting with the coboundary maps. If τ induces an isomorphism on coefficients,

$$\tau\colon h^n(\mathrm{pt}) \xrightarrow{\cong} k^n(\mathrm{pt}) \quad \text{for all } n$$

(equivalently, an isomorphism $h^0(\mathrm{pt}) \cong k^0(\mathrm{pt})$, the only possibly nonzero degree), then $\tau_{(X,A)}\colon h^n(X,A) \to k^n(X,A)$ is an isomorphism for every CW pair (X,A) with X finite-dimensional and every n. If in addition both theories satisfy the additivity axiom, then τ is an isomorphism for every CW pair.

***Proof*:**
Write $G := h^0(\mathrm{pt})$, so τ identifies the coefficients of the two theories. The proof reduces the general pair to a single space, climbs the skeleta of that space, and assembles the limit; the additivity hypothesis enters only at the last stage.

Step 1: Reduction to absolute groups. For a CW pair (X,A) the two long exact sequences of definition 8.2.4, for h^* and for k^*, form a commutative ladder

$$\begin{tikzcd}
h^{n-1}(A) \arrow[r, "\delta"] \arrow[d, "\tau"] & h^n(X,A) \arrow[r] \arrow[d, "\tau"] & h^n(X) \arrow[r] \arrow[d, "\tau"] & h^n(A) \arrow[r, "\delta"] \arrow[d, "\tau"] & h^{n+1}(X,A) \arrow[d, "\tau"] \\
k^{n-1}(A) \arrow[r, "\delta"] & k^n(X,A) \arrow[r] & k^n(X) \arrow[r] & k^n(A) \arrow[r, "\delta"] & k^{n+1}(X,A)
\end{tikzcd}$$

whose rows are exact and whose squares commute (the δ-squares by hypothesis on τ, the others by naturality). If τ is an isomorphism on the absolute groups $h^*(X) \to k^*(X)$ and $h^*(A) \to k^*(A)$ for all spaces, the five lemma forces τ to be an isomorphism on $h^n(X,A) \to k^n(X,A)$. It therefore suffices to prove τ is an isomorphism on absolute groups $h^*(X) \to k^*(X)$ for every (finite-dimensional, then arbitrary) CW complex X.

Step 2: Spheres and wedges of spheres. On a point, $\tau\colon h^n(\mathrm{pt}) \to k^n(\mathrm{pt})$ is an isomorphism by hypothesis. For the sphere S^m, the dimension axiom gives, via the reduced theory of definition 8.2.1 and the iterated suspension isomorphism $\widetilde{h}^n(S^m) \cong \widetilde{h}^{n-m}(S^0) \cong h^{n-m}(\mathrm{pt})$, that $\widetilde{h}^n(S^m) = G$ for $n = m$ and 0 otherwise, and likewise for k. The transformation τ commutes with the suspension isomorphism (which it does because, by proposition 8.2.5, suspension is built from δ and excision, with both of which τ commutes), so on $\widetilde{h}^*(S^m)$ it is the iterate of $\tau\colon h^*(\mathrm{pt}) \to k^*(\mathrm{pt})$, hence an isomorphism. For a wedge $\bigvee_\alpha S^m_\alpha$ of m-spheres, additivity (used here for a possibly infinite wedge; for a finite wedge the product is a finite direct sum and additivity is automatic from exactness) gives $\widetilde{h}^n(\bigvee_\alpha S^m_\alpha) \cong \prod_\alpha \widetilde{h}^n(S^m_\alpha)$ compatibly under τ, so τ is an isomorphism on any wedge of equidimensional spheres.

Step 3: The skeletal induction. Let X be a CW complex and X^m its m-skeleton. The quotient X^m/X^{m-1} is a wedge of m-spheres, one for each m-cell, so by excision and proposition 8.2.5,

$$h^n(X^m, X^{m-1}) \cong \widetilde{h}^n(X^m/X^{m-1}) \cong \widetilde{h}^n\Big(\bigvee_\alpha S^m_\alpha\Big)$$

and τ is an isomorphism on these relative groups by Step 2. The claim is that τ is an isomorphism on $h^*(X^m)$ for every m, proved by induction on m. The base case $m = 0$ is a wedge of 0-spheres (a discrete set with a basepoint), handled by Step 2. For the inductive step, the ladder of Step 1 applied to the pair (X^m, X^{m-1}) reads

$$\begin{tikzcd}
h^n(X^m, X^{m-1}) \arrow[r] \arrow[d, "\cong"] & h^n(X^m) \arrow[r] \arrow[d, "\tau"] & h^n(X^{m-1}) \arrow[r, "\delta"] \arrow[d, "\cong"] & h^{n+1}(X^m, X^{m-1}) \arrow[d, "\cong"] \\
k^n(X^m, X^{m-1}) \arrow[r] & k^n(X^m) \arrow[r] & k^n(X^{m-1}) \arrow[r, "\delta"] & k^{n+1}(X^m, X^{m-1})
\end{tikzcd}$$

To apply the five lemma to the term $h^n(X^m)$ one uses the five-term window

$$h^{n-1}(X^{m-1}) \xrightarrow{\delta} h^n(X^m, X^{m-1}) \to h^n(X^m) \to h^n(X^{m-1}) \xrightarrow{\delta} h^{n+1}(X^m, X^{m-1})$$

of the pair sequence, the right half of which is shown in the diagram. The two relative terms $h^n(X^m, X^{m-1})$ and $h^{n+1}(X^m, X^{m-1})$ carry isomorphisms by Step 2, and the two absolute terms $h^{n-1}(X^{m-1})$ and $h^n(X^{m-1})$ carry isomorphisms by the inductive hypothesis. With those four flanking maps isomorphisms, the five lemma forces the middle map $\tau\colon h^n(X^m) \to k^n(X^m)$ to be an isomorphism, completing the induction on m.

Step 4: Finite-dimensional complexes. If X is finite-dimensional, say $\dim X = N$, then $X = X^N$ and Step 3 already gives that $\tau\colon h^n(X) \to k^n(X)$ is an isomorphism for all n. Feeding this back into Step 1 for an arbitrary pair (X, A) with X (hence A) finite-dimensional shows τ is an isomorphism on $h^n(X, A)$, which is the first assertion of the theorem.

Step 5: Infinite-dimensional complexes via additivity. For general X the skeleta $X^0 \subseteq X^1 \subseteq \cdots$ exhaust X, and $h^n(X)$ is related to the inverse system $\{h^n(X^m)\}_m$ by the Milnor exact sequence

$$0 \to \varprojlim\nolimits^1_m h^{n-1}(X^m) \to h^n(X) \to \varprojlim\nolimits_m h^n(X^m) \to 0$$

which is theorem 8.2.11, proved above from the additivity axiom by way of the mapping telescope of the skeleta. The transformation τ maps the Milnor sequence for h^* to that for k^*, compatibly, by the naturality in the theory established after that theorem (applied to the reduced theories associated with h^* and k^* under proposition 8.2.5). By Step 3, τ is an isomorphism on each $h^n(X^m) \to k^n(X^m)$, so it is an isomorphism of towers and hence induces isomorphisms on both the $\varprojlim$ terms and the $\varprojlim^1$ terms of the two sequences, these being functors of the tower. The five lemma applied to the map of Milnor short exact sequences then gives that $\tau\colon h^n(X) \to k^n(X)$ is an isomorphism. Step 1 upgrades this to arbitrary pairs, completing the proof.

The theorem is the foundation of the entire subject's right to speak of "the" ordinary cohomology with coefficients in an abelian group. Any two constructions producing ordinary cohomology theories with isomorphic coefficient groups are isomorphic as theories, provided a natural transformation between them exists carrying one coefficient group to the other. This is exactly the situation for singular, cellular, simplicial, de Rham, and constant-coefficient sheaf cohomology: each is an ordinary theory, the comparison maps between them are natural transformations commuting with coboundary, and on coefficients they all give the identity of G, so all the comparison maps are isomorphisms. The de Rham theorem (singular cohomology with $\mathbb{R}$ coefficients agrees with de Rham cohomology) and the agreement of simplicial with singular homology, proved by separate arguments in their home contexts, are both instances of one uniqueness statement. The dimension axiom is doing essential work: it is what makes $h^n(\mathrm{pt})$ a single group, which is what makes "the coefficient group" an unambiguous datum.

8.2.15 Warning: Naturality of the comparison map is not automatic The uniqueness theorem compares two theories *through a given natural transformation*; it does not assert that any two ordinary theories with isomorphic coefficients are isomorphic in the absence of a comparison map. The hypothesis that τ exist and commute with δ is genuine content. For singular versus cellular cohomology the comparison map is the one induced by the cellular-to-singular chain inclusion; for singular versus de Rham it is integration of forms over chains. Each must be exhibited and checked to commute with coboundary before uniqueness applies. What the theorem removes is the need to check the comparison is an isomorphism cell by cell: once it is the identity on coefficients, it is an isomorphism everywhere.

Dropping the dimension axiom is what produces the theories the chapter is named for. The coefficient groups $h^n(\mathrm{pt})$ of a generalized theory are no longer concentrated in a single degree, and the uniqueness theorem fails: there can be many non-isomorphic generalized theories, and a single theory carries genuine information already on a point. The two prototypes, developed later in the chapter, illustrate how far the coefficients can spread.

Definition 8.2.16: Generalized Cohomology Theory

A *generalized cohomology theory* is a cohomology theory in the sense of definition 8.2.4 (equivalently a reduced theory in the sense of definition 8.2.1) that is *not* required to satisfy the dimension axiom of definition 8.2.6. Its coefficient groups $h^n(\mathrm{pt})$ may be nonzero for infinitely many n. A generalized homology theory is defined dually. When the coefficients are concentrated in degree 0 the theory is ordinary; otherwise it is genuinely extraordinary, and the graded group $h^*(\mathrm{pt})$ is the first invariant distinguishing it from ordinary cohomology.

The definition is a deliberate weakening: every axiom of an ordinary theory is retained except the dimension axiom, so a generalized theory still has long exact sequences, excision, suspension isomorphisms, and additivity, and the entire formalism of the chapter (Mayer-Vietoris, the spectral sequences of chapter 6 in their generalized form, representability) applies verbatim. What is lost is the rigidity: the coefficients are now a graded group, the theory is no longer pinned down by a single abelian group, and the cohomology of a point already records nontrivial information. The two examples that follow are the reason the definition is made, and they are the subject of section 8.5.

Example 8.2.17: The Coefficients of K-theory and Cobordism

Complex topological K-theory KU^*, the generalized cohomology theory built from vector bundles and developed fully in [21] and in definition 8.5.1, has coefficient groups given by Bott periodicity:

$$KU^n(\mathrm{pt}) = \begin{cases} \mathbb{Z} & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

so $KU^*(\mathrm{pt}) \cong \mathbb{Z}[\beta, \beta^{-1}]$ with β the Bott class in degree -2. This is nonzero in infinitely many degrees, so KU is genuinely extraordinary: the dimension axiom fails in every even nonzero degree. Unoriented cobordism MO^*, the theory built from the Thom spectrum of definition 8.5.5, has coefficients

$$MO^*(\mathrm{pt}) \cong \mathbb{F}_2[x_i : i \geq 2,\ i \neq 2^j - 1]$$

a polynomial algebra over $\mathbb{F}_2$ on generators x_i in each degree $i \geq 2$ not of the form $2^j - 1$ (Thom's computation, carried out in section 8.5), again nonzero in infinitely many degrees. A point, which in ordinary cohomology carries a single copy of G, carries in K-theory a copy of $\mathbb{Z}$ in every even degree and in cobordism an entire polynomial ring. The non-vanishing of these coefficients in many degrees is exactly the slack the dimension axiom forbids, and it is what gives the generalized theories their additional discriminating power: a sphere S^m, which ordinary cohomology sees only in degrees 0 and m, is detected by K-theory in every degree of the same parity.

The example exhibits the dimension axiom as the precise dividing line. With it, the coefficients collapse to one group and the theory is rigid and unique; without it, the coefficients can be an arbitrary graded ring and the resulting theories proliferate, each carrying geometry that ordinary cohomology cannot see (vector bundles for K-theory, manifolds up to cobordism for MO). The remaining sections of the chapter explain how every such theory, ordinary or generalized, is represented by a sequence of spaces and how that sequence is the object called a spectrum; K-theory and cobordism are the two extraordinary spectra whose computations close the chapter.

Exercise 8.2.1

1. Let h^* be a reduced cohomology theory in the sense of definition 8.2.1. Using only the axioms, compute $\widetilde{h}^n(S^m)$ for all $m \geq 0$ and all n in terms of the coefficient groups $h^*(\mathrm{pt}) = \widetilde{h}^*(S^0)$. Then specialize to an ordinary theory with $h^0(\mathrm{pt}) = G$ and recover the familiar answer $\widetilde{h}^n(S^m) = G$ for $n = m$ and 0 otherwise.

2. Proposition 8.2.8 gives a Mayer-Vietoris sequence for a *reduced* theory. Deduce the unreduced version from it: for an unreduced theory h^* and a CW triad $X = A \cup B$ with $C = A \cap B$ nonempty, there is a long exact sequence $\cdots \to h^n(X) \to h^n(A) \oplus h^n(B) \to h^n(C) \to h^{n+1}(X) \to \cdots$. (Use proposition 8.2.5 and note that adjoining a disjoint basepoint takes the triad to a based triad.)

3. Verify that singular *homology* $H_*(-; G)$ satisfies the homology version of the Eilenberg-Steenrod axioms of definition 8.2.4, and identify its coefficient groups. State which axiom would fail if one used compactly supported homology (Borel-Moore homology) on non-compact spaces, and why this prevents Borel-Moore homology from being a homology theory in the unrestricted sense.

4. The reduced cohomology of a point. Show that for *any* reduced cohomology theory $\widetilde{h}^*$ one has $\widetilde{h}^n(\mathrm{pt}) = 0$ for all n (where pt is the one-point space, which as a based space is its own basepoint). Explain why this is consistent with the coefficient groups $h^n(\mathrm{pt})$ of the *unreduced* theory being nonzero, by exhibiting the relationship between the two.

5. Let $\tau\colon h^* \to k^*$ be a natural transformation of ordinary cohomology theories that is an isomorphism on coefficients. The uniqueness theorem theorem 8.2.14 concludes τ is an isomorphism on all finite-dimensional CW complexes. Show by a counterexample that the conclusion can fail for *generalized* theories: exhibit two generalized cohomology theories and a coefficient-preserving natural transformation between them that is not an isomorphism on some space. (Hint: take $h^* =$ ordinary cohomology $H^*(-;\mathbb{Z})$ regarded as a generalized theory, $k^* = KU^*$, and the natural Chern-character-type comparison; identify where the coefficient hypothesis fails.)

6. Prove that the suspension isomorphism of a reduced cohomology theory is compatible with the long exact sequence: for a based CW pair (X, A), the connecting map $\delta\colon \widetilde{h}^n(A) \to \widetilde{h}^{n+1}(X/A)$ of proposition 8.2.3 agrees, under the identification $\widetilde{h}^{n+1}(\Sigma A) \cong \widetilde{h}^n(A)$, with $(\partial)^*$ for the collapse $\partial\colon X/A \to \Sigma A$. Deduce that a natural transformation of theories commuting with δ automatically commutes with the suspension isomorphism.

7. Let X be a CW complex of finite dimension N, filtered by its skeleta. Show that the Milnor sequence of theorem 8.2.11 carries no information in that case: for each fixed level the images of the restriction maps are eventually constant, so $\varprojlim^1$ vanishes and restriction is an isomorphism. Which hypothesis of corollary 8.2.12 is being verified, and why does the weaker one suffice?

8. In example 8.2.13 replace the degree-p maps by degree-0 maps and recompute $\widetilde{H}^1(T;\mathbb{Z})$ and $\widetilde{H}^2(T;\mathbb{Z})$; then replace them by degree-1 maps and recompute again. Exactly one of the three telescopes has a nonvanishing $\varprojlim^1$ term. Identify it, and say what distinguishes its tower from the other two.

9. Proposition 8.2.10 assumes every f_m is the inclusion of a subcomplex. Show the conclusion genuinely needs a hypothesis by exhibiting a sequence whose telescope is homotopy equivalent to no single X_m, and explain why this costs theorem 8.2.11 nothing.

8.3 Representable Cohomology and Brown Representability

The axioms of section 8.2 pin down what a cohomology theory *is*, but they say nothing about where it comes from. The representation theorem for ordinary cohomology (theorem 5.5.9) gave one answer in a single case: the functor $\tilde{H}^n(-;G)$ is the set of homotopy classes of maps into a fixed space $K(G, n)$. The aim of this section is to show that this is a structural feature of every cohomology theory in the sense of section 8.2. The vehicle is Brown's representability theorem, which characterises representable functors $[-, Y]$ by two formal properties, a wedge axiom and a Mayer-Vietoris axiom, that every reduced cohomology theory possesses degree by degree.

The argument is constructive. Given a functor F satisfying the two axioms, the representing space Y is built by attaching cells to realise larger and larger pieces of F, exactly as $K(G, n)$ was built in section 5.5 by attaching cells to kill homotopy. What replaces the homotopy-group bookkeeping there is a single element $u \in F(Y)$, the *universal element*, engineered so that pullback along it, $[X, Y] \to F(X)$, $f \mapsto f^*u$, is a bijection. Whitehead's theorem (theorem 5.3.5) certifies that the natural transformation produced is an isomorphism. The payoff, recorded at the end, is that every reduced cohomology theory is represented in each degree by a space Y_n, and the suspension isomorphism assembles the Y_n into a single homotopy-theoretic object, an Ω-spectrum, the bridge to section 8.4.

Throughout, CW_* is the category of definition 8.1.1 and the bracket $[X, Y]$ abbreviates $[X, Y]_*$. A contravariant functor $F\colon \mathsf{CW}_*^{\mathrm{op}} \to \mathbf{Set}_*$ to pointed sets is the object of study; the requirement that F factor through the homotopy category is precisely homotopy invariance, $f \simeq g \implies F(f) = F(g)$.

Before isolating the abstract axioms, the prototype must be restated in the language of the previous section. Ordinary cohomology in a fixed degree is, as a functor on CW_*, exactly $[-, K(G, n)]$.

Theorem 8.3.1: Ordinary Cohomology as a Represented Theory

Let G be an abelian group and $n \geq 1$. On the category CW_* of pointed connected CW complexes there is a natural isomorphism of functors

$$\tilde{H}^n(-; G) \;\cong\; [-,\, K(G, n)]$$

sending a class $u \in \tilde{H}^n(X; G)$ to the homotopy class of the map $f_u\colon X \to K(G, n)$ with $f_u^*(\iota_n) = u$, where $\iota_n \in H^n(K(G, n); G)$ is the fundamental class of definition 5.5.7. Under this isomorphism the coefficient group is recovered as $\tilde{H}^n(S^n; G) \cong [S^n, K(G, n)] \cong \pi_n(K(G, n)) \cong G$, and the suspension isomorphism $\tilde{H}^n(X; G) \cong \tilde{H}^{n+1}(\Sigma X; G)$ corresponds under representability to the loop-suspension adjunction together with the homotopy equivalence $K(G, n) \simeq \Omega K(G, n + 1)$.

***Proof*:**

The natural bijection $T\colon [X, K(G, n)] \to \tilde{H}^n(X; G)$, $[f] \mapsto f^*(\iota_n)$, is theorem 5.5.9; its inverse is $u \mapsto [f_u]$, which is the displayed correspondence. Naturality in X is the commuting square recorded there. It remains to verify the two identifications.

Coefficient group. Taking $X = S^n$, the bijection T gives $[S^n, K(G, n)] \cong \tilde{H}^n(S^n; G)$. By definition of the higher homotopy groups (definition 5.1.1), $[S^n, K(G, n)] = \pi_n(K(G, n))$, which is G because $K(G, n)$ is an Eilenberg-MacLane space (definition 5.5.1). On the other side, $\tilde{H}^n(S^n; G) \cong G$ by the cellular computation of the sphere. The composite isomorphism is the coefficient identification.

Suspension. The reduced suspension ΣX and the loop space ΩY are adjoint: there is a natural bijection $[\Sigma X, Y] \cong [X, \Omega Y]$ by theorem 8.1.5. Applying this with $Y = K(G, n + 1)$ and using the equivalence $K(G, n) \simeq \Omega K(G, n + 1)$ (established in example 5.2.12 from the path-loop fibration, with uniqueness theorem 5.5.4),

$$[\Sigma X, K(G, n + 1)] \;\cong\; [X, \Omega K(G, n + 1)] \;\cong\; [X, K(G, n)]$$

Transporting both ends through the representability isomorphism turns the left side into $\tilde{H}^{n+1}(\Sigma X; G)$ and the right into $\tilde{H}^n(X; G)$, and the composite is the suspension isomorphism. That it agrees with the suspension isomorphism of ordinary cohomology is the statement that both are induced by the same loop-suspension adjunction, which holds because ι_{n+1} transgresses to ι_n under the path-loop fibration, completing the proof.

The theorem reframes theorem 5.5.9 as the assertion that ordinary cohomology, viewed degree by degree, is a sequence of representable functors whose representing spaces are linked by the loop relation $K(G, n) \simeq \Omega K(G, n+1)$. The coefficient group is read off from the sphere, and the suspension axiom of section 8.2 becomes a formal consequence of the loop-suspension adjunction. This is the template Brown's theorem generalises: any functor behaving formally like $\tilde{H}^n(-; G)$ is $[-, Y]$ for some Y, and any cohomology theory furnishes such a functor in each degree.

Example 8.3.2: Degree-One Cohomology and the Circle

For $G = \mathbb{Z}$ and $n = 1$ the representing space is $K(\mathbb{Z}, 1) = S^1$ (example 5.5.2), so theorem 8.3.1 reads

$$\tilde{H}^1(X;\mathbb{Z}) \;\cong\; [X, S^1]$$

for every pointed connected CW complex X. Both sides can be computed for $X = S^1$: the left is $\tilde{H}^1(S^1;\mathbb{Z}) \cong \mathbb{Z}$, and the right is $[S^1, S^1] = \pi_1(S^1) \cong \mathbb{Z}$, the isomorphism sending a map of degree d to d times the generator. For $X = S^1 \vee S^1$ the wedge of two circles, $\tilde{H}^1(S^1 \vee S^1;\mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}$, while $[S^1 \vee S^1, S^1] \cong [S^1, S^1] \times [S^1, S^1] \cong \mathbb{Z} \times \mathbb{Z}$ because a map out of a wedge is a pair of maps out of the summands agreeing at the basepoint. The two computations match, and the matching is exactly the wedge axiom that Brown's theorem will require in the abstract. The class in $\tilde{H}^1(X;\mathbb{Z})$ corresponding to $f\colon X \to S^1$ is $f^*(\iota_1)$, the pullback of the generator $\iota_1 \in H^1(S^1;\mathbb{Z})$, which records the winding of f around the circle.

The example exhibits the two formal properties on which everything turns. A map out of a wedge is determined by its restrictions to the summands, so F applied to a wedge is the product of the values on the summands; and a class on a space glued from two pieces is determined by compatible classes on the pieces. These are the wedge and Mayer-Vietoris axioms. The strength of Brown's theorem is that they suffice: no further structure on F is needed to produce a representing space.

Definition 8.3.3: Brown Functor

A *Brown functor* is a contravariant functor $F\colon \mathsf{CW}_*^{\mathrm{op}} \to \mathbf{Set}_*$ from the homotopy category of pointed connected CW complexes to pointed sets satisfying the two axioms:

1. *(Wedge axiom)* For any family $\{X_\alpha\}_{\alpha \in A}$ of pointed connected CW complexes, the inclusions $X_\alpha \hookrightarrow \bigvee_\alpha X_\alpha$ induce a bijection
$$F\Big(\bigvee_\alpha X_\alpha\Big) \xrightarrow{\cong} \prod_\alpha F(X_\alpha)$$
2. *(Mayer-Vietoris axiom)* If a CW complex $X = A \cup B$ is the union of two subcomplexes with A, B, $A \cap B$ all containing the basepoint and connected, then for any classes $a \in F(A)$ and $b \in F(B)$ whose restrictions to $A \cap B$ agree, there is a class $x \in F(X)$ restricting to a on A and to b on B.

A Brown functor is a contravariant functor that turns disjoint assembly into product and admits gluing along intersections. The wedge axiom is the infinitary version of the statement that F sends finite wedges to finite products, and it forces F of a point to be a single point (the empty product), so the distinguished basepoint of $F(X)$ is the restriction of the unique class on the basepoint. The Mayer-Vietoris axiom is existence-only: it asserts that compatible classes glue, but says nothing about uniqueness of the glued class. This asymmetry is deliberate. Uniqueness fails for general F (the kernel of the difference of restrictions can be nontrivial), and Brown's construction is arranged so that the lack of uniqueness never obstructs the building of Y. Every reduced cohomology theory provides a Brown functor in each degree, as the verification below records, so the theorem applies to all of them at once.

Example 8.3.4: Reduced Cohomology is a Brown Functor

Fix an abelian group G and $n \geq 1$, and let $F = \tilde{H}^n(-;G)$. Then F is a Brown functor. Homotopy invariance makes F a functor on CW_*, and $F(X)$ is a pointed set (an abelian group, in fact) with basepoint 0.

Wedge axiom. For a wedge $\bigvee_\alpha X_\alpha$, reduced cohomology converts the wedge into a product: $\tilde{H}^n(\bigvee_\alpha X_\alpha; G) \cong \prod_\alpha \tilde{H}^n(X_\alpha; G)$. This is the additivity (wedge) axiom for ordinary cohomology, which holds because $\bigvee_\alpha X_\alpha$ is the union of the X_α along the single basepoint, and reduced cellular cochains of a wedge are the product of the reduced cellular cochains of the summands. The induced map is restriction to the summands, exactly the map in definition 8.3.3(1).

Mayer-Vietoris axiom. For $X = A \cup B$ with A, B subcomplexes, the reduced Mayer-Vietoris sequence reads

$$\tilde{H}^n(X;G) \xrightarrow{(\mathrm{res}_A,\, \mathrm{res}_B)} \tilde{H}^n(A;G) \oplus \tilde{H}^n(B;G) \xrightarrow{(a,b) \mapsto a|_{A\cap B} - b|_{A \cap B}} \tilde{H}^n(A \cap B; G)$$

Exactness at the middle term says precisely that a pair (a, b) with equal restrictions to $A \cap B$ lies in the image of $(\mathrm{res}_A, \mathrm{res}_B)$, that is, there is $x \in \tilde{H}^n(X;G)$ restricting to a and b. This is the gluing property of definition 8.3.3(2). Hence $\tilde{H}^n(-;G)$ satisfies both axioms.

The verification is the formal core of why Brown's theorem subsumes ordinary cohomology: the two axioms are exactly the wedge and Mayer-Vietoris properties, both standard for $\tilde{H}^n$. The same verification applies verbatim to any reduced cohomology theory of section 8.2, since the wedge axiom is its additivity axiom and the Mayer-Vietoris sequence is a formal consequence of the exactness axiom, derived from the cofiber long exact sequence. The theorem now produces, from any Brown functor, a representing space. The construction proceeds in two movements: first build a space and a class on it that is surjective for spheres (the universal element on a candidate Y), then enlarge the space so that the induced transformation $[-, Y] \to F$ becomes a bijection on all complexes.

The construction rests on a single notion. A class $u \in F(Y)$ determines, for every X, a map of pointed sets

$$T_u \colon [X, Y] \longrightarrow F(X), \qquad [f] \longmapsto F(f)(u) = f^*u$$

by pulling u back along f. This T_u is automatically a natural transformation, by functoriality of F. The goal is to choose Y and u so that T_u is an isomorphism. The pair (Y, u) achieving this is governed by a representability criterion phrased in terms of spheres.

Definition 8.3.5: Universal Element

Let F be a Brown functor. A pair (Y, u) with Y a pointed connected CW complex and $u \in F(Y)$ is *n-universal* if the map

$$T_u \colon [S^q, Y] \longrightarrow F(S^q), \qquad [f] \longmapsto f^*u$$

is a bijection for $0 < q < n$ and a surjection for $q = n$. It is *universal* (or ∞-universal) if $T_u \colon [S^q, Y] \to F(S^q)$ is a bijection for all $q \geq 1$. The class u is then a *universal element* for F.

A universal element packages the entire functor F, tested on spheres, into a single class on a single space. The restriction to spheres is the decisive simplification: rather than control T_u on every complex at once, one controls it on the cells S^q out of which complexes are built, and the wedge and Mayer-Vietoris axioms then propagate sphere-level control to all complexes. The condition for spheres is a condition on homotopy groups, $T_u\colon \pi_q(Y) \to F(S^q)$, which the cell-attachment machinery of section 5.5 is built to adjust. The next two lemmas carry out the adjustment: the first attaches cells to enlarge $[S^q, Y]$ until T_u surjects, the second attaches cells to shrink it until T_u injects, both without disturbing the lower spheres.

Example 8.3.6: The Fundamental Class as a Universal Element

For $F = \tilde{H}^n(-;G)$, the pair $(K(G,n),\, \iota_n)$ is a universal element, where $\iota_n \in H^n(K(G,n);G) = F(K(G,n))$ is the fundamental class of definition 5.5.7. The map $T_{\iota_n}\colon [S^q, K(G,n)] \to \tilde{H}^n(S^q;G)$ sends $[f]$ to $f^*(\iota_n)$, which is exactly the representability bijection T of theorem 5.5.9 restricted to spheres, hence a bijection for every $q \geq 1$. Concretely, in degree $q = n$ the source is $[S^n, K(G,n)] = \pi_n(K(G,n)) = G$ and the target is $\tilde{H}^n(S^n;G) = G$, and T_{ι_n} is the identity on G under these identifications: a map $f\colon S^n \to K(G,n)$ representing $a \in G = \pi_n(K(G,n))$ pulls ι_n back to the class $a \in \tilde{H}^n(S^n;G)$, by the tautological characterisation of ι_n. For $q \neq n$ both sides vanish, so T_{ι_n} is the bijection $0 \to 0$. Thus ι_n is the universal element that Brown's construction would produce for ordinary cohomology, recovered here directly from theorem 5.5.9; the abstract construction below reconstructs $K(G,n)$ without knowing it in advance.

Lemma 8.3.7: Construction of a Universal Element

For every Brown functor F there exists a pointed connected CW complex Y and a universal element $u \in F(Y)$.

***Proof*:**
The space is built as an increasing union $Y = \bigcup_n Y_n$, each Y_n carrying a class $u_n \in F(Y_n)$ making (Y_n, u_n) n-universal, with u_{n+1} restricting to u_n. The class u is then assembled from the u_n by the wedge axiom in its mapping-telescope form.

Step 1: Realising every element of F on a wedge of spheres. Begin by arranging surjectivity of T_u in every degree on a single space. For each $q \geq 1$ and each element $a \in F(S^q)$, take a sphere S^q_a (a copy of S^q indexed by a), and form the wedge

$$Y_1 \;=\; \bigvee_{q\geq 1}\ \bigvee_{a\in F(S^q)} S^q_a$$

By the wedge axiom (definition 8.3.3(1)), $F(Y_1) \cong \prod_{q,a} F(S^q_a) = \prod_{q,a} F(S^q)$, so there is a unique class $u_1 \in F(Y_1)$ whose restriction to the summand S^q_a is the element $a \in F(S^q)$. Then $T_{u_1}\colon [S^q, Y_1] \to F(S^q)$ is surjective for every $q \geq 1$: the inclusion $\iota_a\colon S^q_a \hookrightarrow Y_1$ of the summand indexed by a satisfies $T_{u_1}(\iota_a) = \iota_a^* u_1 = a$ by construction. Thus (Y_1, u_1) has T_{u_1} surjective on all spheres; in particular it is 1-universal.

Step 2: From n-universal to $(n+1)$-universal by attaching cells. Suppose (Y_n, u_n) is n-universal: T_{u_n} is bijective on S^q for $0 < q < n$ and surjective for $q = n$. The aim is to enlarge Y_n to Y_{n+1}, extending u_n to u_{n+1}, so that $T_{u_{n+1}}$ becomes injective on S^n (hence bijective there) and remains surjective on S^{n+1}, while the lower spheres are undisturbed. Injectivity on S^n is arranged by gluing in cylinders that identify maps not distinguished by T_{u_n}, and surjectivity on S^{n+1} by wedging on $(n+1)$-spheres. The cylinder device replaces the difference-of-maps argument that would require $F(S^n)$ to be a group, which it need not be for a general Brown functor.

Consider first injectivity. Two maps $f_0, f_1\colon S^n \to Y_n$ with $f_0^* u_n = f_1^* u_n$ must become homotopic after enlargement. For each such pair, glue a cylinder $S^n \times I$ to Y_n along the two ends, attaching

$S^n \times \{0\}$ via f_0 and $S^n \times \{1\}$ via f_1. Performing this for every pair (f_0, f_1) with $f_0^* u_n = f_1^* u_n$ produces

$$Y_n' \;=\; Y_n \cup_{\{(f_0,f_1)\}} \Big(\coprod_{(f_0,f_1)} S^n \times I \Big)$$

Each glued cylinder is built from cells of dimension $n+1$ (the cylinder $S^n \times I$ adds an $(n+1)$-cell to the spheres at its ends, which already map into Y_n), so Y_n' is obtained from Y_n by attaching cells of dimension $n+1$. The class u_n extends over Y_n': apply the Mayer-Vietoris axiom to $Y_n' = Y_n \cup C$, where C is the disjoint union of cylinders, glued along $A \cap B = \coprod S^n \times \{0,1\}$. On a cylinder $S^n \times I$ the projection to S^n followed by f_0 and the constancy of T_{u_n} on the two ends ($f_0^* u_n = f_1^* u_n$) make the restriction of u_n to the two boundary spheres equal; hence the class $\mathrm{pr}^*(f_0^* u_n)$ on the cylinder agrees with u_n on the overlap, and the two glue to a class $u_n' \in F(Y_n')$ restricting to u_n on Y_n. In Y_n' the cylinder is a homotopy from f_0 to f_1, so $f_0 \simeq f_1$ there. Thus every pair undistinguished by T_{u_n} becomes a single homotopy class.

Next surjectivity on S^{n+1}. Wedge on a sphere S_b^{n+1} for each element $b \in F(S^{n+1})$ and use the wedge axiom to extend u_n' to a class u_{n+1} on

$$Y_{n+1} \;=\; Y_n' \vee \bigvee_{b \in F(S^{n+1})} S_b^{n+1}$$

restricting to u_n' on Y_n' and to b on each S_b^{n+1}. By construction $T_{u_{n+1}}$ is surjective on S^{n+1}.

Step 3: Checking $(n+1)$-*universality.* The complex Y_{n+1} is obtained from Y_n by attaching cells of dimension $n+1$ only (the cylinders of the injectivity step and the wedged $(n+1)$-spheres). For $q < n$ the inclusion $Y_n \hookrightarrow Y_{n+1}$ induces an isomorphism on $[S^q, -]$ (cells of dimension $\geq n+1$ do not affect π_q for $q < n$, by the cellular approximation argument of lemma 5.5.3), so $T_{u_{n+1}}$ remains bijective on S^q for $0 < q < n$. On S^n, the surjectivity of T_{u_n} passes to $T_{u_{n+1}}$ because every map $S^n \to Y_n$ still maps into Y_{n+1} and u_{n+1} restricts to u_n; injectivity is the new gain: if $f_0, f_1 \colon S^n \to Y_{n+1}$ have $f_0^* u_{n+1} = f_1^* u_{n+1}$, then by cellular approximation each is homotopic in Y_{n+1} to a map into Y_n (since S^n has no cells above dimension n and the new cells are of dimension $n+1$), and the two resulting maps into Y_n have equal u_n-pullbacks, so a cylinder was glued for exactly that pair, making them homotopic in Y_{n+1}. Hence $T_{u_{n+1}}$ is bijective on S^n. On S^{n+1} surjectivity holds by Step 2. Therefore (Y_{n+1}, u_{n+1}) is $(n+1)$-universal.

Step 4: Passing to the union. Set $Y = \bigcup_n Y_n$ with the weak topology. Any map $S^q \to Y$ has compact image, hence lands in some Y_n with $n > q$, and similarly for homotopies, so $[S^q, Y] = \varinjlim_n [S^q, Y_n]$ and $\pi_q(Y) = \varinjlim_n \pi_q(Y_n)$, which stabilises at Y_{q+1}. The classes u_n are compatible under restriction, and the mapping telescope of the Y_n together with the wedge axiom (applied to the telescope, which is homotopy equivalent to Y) furnishes a class $u \in F(Y)$ restricting to each u_n.[a] For each q, the map $T_u \colon [S^q, Y] \to F(S^q)$ is the colimit of the bijections $T_{u_n} \colon [S^q, Y_n] \to F(S^q)$ for $n > q$ (each a bijection by n-universality), hence itself a bijection. Therefore (Y, u) is universal, completing the proof.

[a] Concretely, Y is homotopy equivalent to the mapping telescope $\mathrm{Tel}(Y_n)$, built from $\bigvee_n (Y_n)_+ \wedge I_+$ by gluing; the wedge axiom and the Mayer-Vietoris axiom together produce a class on the telescope restricting to each u_n, and the telescope's projection to Y transports it to u. The compatibility $u_{n+1}|_{Y_n} = u_n$ is what makes the gluing consistent.

The lemma is the heart of the matter. It produces a space whose homotopy groups, tested through F, exactly reproduce F on spheres. The mechanism is the cell-attachment surgery of section 5.5

run in reverse direction of control: there one killed homotopy to sculpt $K(G,n)$, here one kills and creates homotopy to make T_u a sphere-level isomorphism, with the Mayer-ietoris axiom guaranteeing the universal class survives each attachment. What remains is to promote sphere-level control to control on every CW complex, and this is where a relative refinement of the same lemma, combined with Whitehead's theorem, finishes the proof.

Lemma 8.3.8: Relative Construction over a Subcomplex

Let F be a Brown functor, let (X, A) be a CW pair with X connected, and let $u_A \in F(A)$ be a class such that (A, u_A) is universal (the map $T_{u_A}: [S^q, A] \to F(S^q)$ is bijective for all $q \geq 1$). Then for any class $x \in F(X)$ with $x|_A = u_A$ there is a CW complex $Y \supseteq X$, obtained by attaching cells to X away from A, and a universal element $u \in F(Y)$ with $u|_X = x$; in particular $u|_A = u_A$.

***Proof*:**
Run the construction of lemma 8.3.7 relative to X, starting from the class $x \in F(X)$ rather than from a wedge of spheres, and attaching all cells away from A. Set $Y_{(0)} = X$ with its class x. At each stage, to make T surjective on S^q wedge spheres S^q_b $(b \in F(S^q))$ onto $Y_{(k)} \setminus A$ and extend the class by the wedge axiom; to make T injective on S^q glue cylinders $S^n \times I$ as in Step 2 of lemma 8.3.7 along pairs of maps into $Y_{(k)} \setminus A$, extending the class by the Mayer-Vietoris axiom. No attachment is ever made to A, and the hypothesis that (A, u_A) is already universal means none is needed there: a map $S^q \to A$ is already detected isomorphically by T_{u_A}, and since $x|_A = u_A$ the same holds for the part of T coming from A. The colimit Y contains X (hence A) as a subcomplex, carries a universal element u by the colimit argument of Step 4 of lemma 8.3.7, with $u|_X = x$ because x was the starting class and nothing already in X was altered, and so $u|_A = x|_A = u_A$, completing the proof.

The relative lemma is the engine for both surjectivity and injectivity of the natural transformation T_u on arbitrary complexes. Surjectivity uses it to extend any class $x \in F(X)$ to a universal element on an enlargement, which then receives the universal Y; injectivity uses it on the cylinder $X \times I$ to glue two maps into a homotopy. With these pieces assembled, Brown's theorem follows by comparing the universal Y to an arbitrary X through T_u and invoking Whitehead.

Theorem 8.3.9: Brown Representability

Let $F: \mathsf{CW}_*^{\mathrm{op}} \to \mathbf{Set}_*$ be a Brown functor (a homotopy-invariant contravariant functor on pointed connected CW complexes satisfying the wedge and Mayer-Vietoris axioms of definition 8.3.3). Then F is representable: there exist a pointed connected CW complex Y and a universal element $u \in F(Y)$ such that the natural transformation

$$T_u: [X, Y] \longrightarrow F(X), \qquad [f] \longmapsto f^*u$$

is a bijection for every pointed connected CW complex X. The representing pair (Y, u) is unique up to canonical homotopy equivalence: if (Y', u') also represents F, the class u' corresponds under T_u to a homotopy class $h: Y \to Y'$, and h is a homotopy equivalence.

***Proof*:**
Take (Y, u) universal, as furnished by lemma 8.3.7. The natural transformation T_u is defined for all X and is natural by functoriality of F. The content is that T_u is bijective on every connected pointed CW complex; the proof establishes surjectivity and injectivity separately, each by a gluing argument built on lemma 8.3.8.

Step 1: Surjectivity of T_u on every X. Let $x \in F(X)$ be arbitrary. Form the disjoint-then-wedge space $X \vee Y$ (joined at the common basepoint). By the wedge axiom, $F(X \vee Y) \cong F(X) \times F(Y)$, so the pair (x, u) corresponds to a class $w \in F(X \vee Y)$ restricting to x on X and to u on Y. Apply the relative construction lemma 8.3.8 to the pair $(X \vee Y,\ Y)$ with the universal class u on the subcomplex Y: it produces an enlargement $Y' \supseteq X \vee Y$ and a universal element $u' \in F(Y')$ restricting to w on $X \vee Y$, in particular to u on Y and to x on X. Now both (Y, u) and (Y', u') are universal, so T_u and $T_{u'}$ are bijective on all spheres. The inclusion $j\colon Y \hookrightarrow Y'$ satisfies $j^* u' = u'|_Y = u$, so on each sphere S^q the triangle

$$\begin{tikzcd}
{[S^q, Y]} \arrow[rr, "j_*"] \arrow[dr, "T_u"'] & & {[S^q, Y']} \arrow[dl, "T_{u'}"] \\
& F(S^q) &
\end{tikzcd}$$

commutes, and since T_u and $T_{u'}$ are bijections, $j_*\colon \pi_q(Y) \to \pi_q(Y')$ is a bijection for every $q \geq 1$. As Y and Y' are connected, j is a weak homotopy equivalence (definition 5.3.1), hence a homotopy equivalence by Whitehead's theorem (theorem 5.3.5). Let $r\colon Y' \to Y$ be a homotopy inverse, so $r \circ j \simeq \mathrm{id}_Y$. Consider the composite $f = r \circ \iota_X \colon X \hookrightarrow Y' \xrightarrow{r} Y$, where $\iota_X \colon X \hookrightarrow X \vee Y \hookrightarrow Y'$ is the inclusion. Then

$$f^* u = \iota_X^* r^* u$$

and $r^* u = r^*(j^* u') = (j \circ r)^* u'$; since $j \circ r \simeq \mathrm{id}_{Y'}$, this gives $r^* u = u'$, and hence $f^* u = \iota_X^*(r^* u) = \iota_X^* u' = u'|_X = x$. Therefore $T_u([f]) = x$, and T_u is surjective.

Step 2: Injectivity of T_u on every X. Suppose $f_0, f_1 \colon X \to Y$ satisfy $f_0^* u = f_1^* u = x$ in $F(X)$. Write $C = X \times I$ for the cylinder, a CW complex containing $X \times \{0, 1\}$ as a subcomplex, with projection $\mathrm{pr}_X \colon C \to X$. The class $w := \mathrm{pr}_X^* x \in F(C)$ restricts on the two ends to $\mathrm{pr}_X^* x|_{X \times 0} = x = f_0^* u$ and likewise to $x = f_1^* u$ on $X \times 1$ (using $f_0^* u = f_1^* u = x$). Thus w is a class on the cylinder restricting to $f_0^* u$ at one end and $f_1^* u$ at the other.

Now build the comparison map by gluing Y to the cylinder. The maps f_0, f_1 together define $f_0 \sqcup f_1 \colon X \times \{0, 1\} \to Y$ with $(f_0 \sqcup f_1)^* u = w|_{X \times \{0,1\}}$ (the two ends pull u back to $f_0^* u$ and $f_1^* u$, which are the restrictions of w), so the class w on C and the class u on Y agree, after this map, on the two boundary copies of X. Form $Z := C \cup_{f_0 \sqcup f_1} Y$, gluing Y to C along $f_0 \sqcup f_1$ on the ends. By the Mayer-Vietoris axiom, w and u glue to a class $v \in F(Z)$ restricting to w on C and to u on Y. The subcomplex $Y \subseteq Z$ carries the universal u, so lemma 8.3.8 (with ambient Z, subcomplex Y, and starting class v) enlarges Z to a universal (Z', v') with $v'|_Y = u$, and as in Step 1 the inclusion $Y \hookrightarrow Z'$ is a homotopy equivalence with inverse $\rho\colon Z' \to Y$. The composite

$$H\colon X \times I = C \hookrightarrow Z \hookrightarrow Z' \xrightarrow{\rho} Y$$

is a homotopy. At the ends, $H|_{X \times 0} = \rho \circ (\text{inclusion of } X \times 0)$; the inclusion of $X \times 0$ into Z followed by $Y \hookrightarrow Z'$ agrees, by construction of the gluing, with f_0 followed by $Y \hookrightarrow Z'$, so ρ sends it back to f_0 up to homotopy (because $\rho \circ (Y \hookrightarrow Z') \simeq \mathrm{id}_Y$). Thus $H|_{X \times 0} \simeq f_0$ and likewise $H|_{X \times 1} \simeq f_1$, so H is a homotopy from f_0 to f_1. Hence $[f_0] = [f_1]$ and T_u is injective.

Step 3: Uniqueness of the representing pair. Let (Y', u') be another representing pair, so $T_{u'} : [-, Y'] \to F$ is a natural isomorphism. By surjectivity of T_u on Y' the class $u' \in F(Y')$ equals h^*u for some $h : Y' \to Y$; symmetrically $u = k^*u'$ for some $k : Y \to Y'$. Then $(h \circ k)^*u = k^*h^*u = k^*u' = u = \mathrm{id}_Y^* u$, and injectivity of T_u on Y forces $h \circ k \simeq \mathrm{id}_Y$; symmetrically $k \circ h \simeq \mathrm{id}_{Y'}$. Hence h is a homotopy equivalence, canonical in that it is the unique class with $h^*u = u'$, as we sought to show.

The theorem is the converse to the representability already known for ordinary cohomology: where theorem 8.3.1 produced the functor $\widetilde{H}^n(-; G)$ from the space $K(G, n)$, Brown's theorem produces a space from any functor with the two formal axioms. The proof is purely homotopy-theoretic, using no special feature of cohomology. Three ingredients carry it: the wedge axiom, which lets one realise every element of F on a wedge of spheres; the Mayer-Vietoris axiom, which lets the universal class survive every cell attachment and lets compatible classes glue; and Whitehead's theorem, which upgrades the sphere-level isomorphism T_u to a genuine homotopy equivalence and thereby converts sphere-level control into control on all complexes. The role of connectedness is to make Whitehead's theorem applicable basepoint-free; the role of the CW hypothesis is the same as everywhere in this chapter, that weak equivalences of CW complexes are equivalences.

8.3.10 Remark: What the Mayer-Vietoris Axiom Does Not Give The Mayer-Vietoris axiom of definition 8.3.3(2) asserts only that compatible classes *glue*, never that the glued class is unique. This is essential: for a genuine cohomology theory the difference of two glued classes lives in the image of a connecting map, which is generally nonzero, so uniqueness would be too strong an axiom and would exclude the very examples one wants. Brown's construction is arranged to need only existence of gluings: the universal element u is built up by extending over cells, and at no point is it claimed that the extension is unique. The price is that the representing space is determined only up to homotopy equivalence (the uniqueness clause of theorem 8.3.9), not on the nose, which is the correct level of canonicity for a homotopy functor.

The deduction now writes itself. A reduced cohomology theory $\tilde{h}^*$ in the sense of section 8.2 gives, in each degree n, a functor $\tilde{h}^n(-)$ that is homotopy-invariant and satisfies the wedge and Mayer-Vietoris axioms, hence a Brown functor. Brown's theorem represents each one by a space Y_n, and the suspension isomorphism of the theory forces these spaces to loop into one another, producing the structure that the next section names a spectrum.

Corollary 8.3.11: Every Cohomology Theory is Represented by an Ω-Spectrum

Let $\tilde{h}^* = \{\tilde{h}^n\}_{n \in \mathbb{Z}}$ be a reduced cohomology theory on pointed connected CW complexes (definition 8.2.1), with suspension isomorphisms $\sigma^n : \tilde{h}^n(X) \xrightarrow{\cong} \tilde{h}^{n+1}(\Sigma X)$. Then:

1. For each n there is a pointed connected CW complex Y_n and a natural isomorphism $\tilde{h}^n(X) \cong [X, Y_n]$ for all pointed connected CW complexes X.
2. The spaces $\{Y_n\}$ are linked by weak homotopy equivalences $\epsilon_n : Y_n \xrightarrow{\simeq} \Omega Y_{n+1}$, so the sequence $\{Y_n\}$ is an Ω-spectrum.

***Proof*:**
(1) Representability degree by degree. Fix n. The functor $\tilde{h}^n(-)$ is contravariant and homotopy-invariant by the axioms of a cohomology theory, and takes values in abelian groups, hence in pointed sets. It satisfies the wedge axiom because additivity is one of the axioms of definition 8.2.1, and the Mayer-Vietoris axiom because the exactness axiom yields the reduced Mayer-Vietoris sequence (proposition 8.2.3 spliced over a CW triad) whose middle exactness is the gluing statement, exactly as in example 8.3.4. Thus $\tilde{h}^n(-)$ is a Brown functor, and Brown's theorem (theorem 8.3.9) provides a pointed connected CW complex Y_n and a natural isomorphism $\tilde{h}^n(X) \cong [X, Y_n]$.

(2) The loop relation. The suspension isomorphism and the loop-suspension adjunction combine to produce the equivalences. For every X the adjunction gives a natural bijection $[\Sigma X, Y_{n+1}] \cong [X, \Omega Y_{n+1}]$. Chaining with the representability isomorphisms of part (1) and the suspension isomorphism of the theory,

$$[X, Y_n] \;\cong\; \tilde{h}^n(X) \;\xrightarrow[\cong]{\sigma^n}\; \tilde{h}^{n+1}(\Sigma X) \;\cong\; [\Sigma X, Y_{n+1}] \;\cong\; [X, \Omega Y_{n+1}]$$

is a natural isomorphism of functors $[-, Y_n] \cong [-, \Omega Y_{n+1}]$ on CW_*. By proposition 8.1.10 such an isomorphism is induced by a class $\epsilon_n \in [Y_n, \Omega Y_{n+1}]$ represented by a based homotopy equivalence, so $\epsilon_n\colon Y_n \to \Omega Y_{n+1}$ is the required equivalence. Since a homotopy equivalence is in particular a weak homotopy equivalence, the sequence $\{Y_n\}$ with the structure maps ϵ_n is an Ω-spectrum. This is the structure that section 8.4 takes as the definition, completing the proof.

The corollary closes the loop opened by theorem 8.3.1. There ordinary cohomology was seen to be represented in each degree by $K(G, n)$, with the spaces linked by $K(G, n) \simeq \Omega K(G, n+1)$; here the same conclusion is drawn for an arbitrary cohomology theory, with the $K(G, n)$ replaced by whatever spaces Y_n Brown's construction produces. The sequence $\{Y_n\}$ with its loop maps is the universal home for the theory: every cohomology group $\tilde{h}^n(X)$ is a set of homotopy classes $[X, Y_n]$, and the suspension axiom is encoded geometrically in the loop relation. The Eilenberg-MacLane spectrum $\{K(G, n)\}$ representing ordinary cohomology is the first instance; K-theory and cobordism, treated in section 8.5, are the next two, each arising from its own Ω-spectrum. The abstract object $\{Y_n\}$ with maps $Y_n \simeq \Omega Y_{n+1}$ is what section 8.4 formalises as a spectrum, the central object of stable homotopy theory.

8.3.12 Remark: Brown Representability and Derived Functors Brown representability (theorem 8.3.9) is the homotopy-theoretic instance of a categorical principle that the reader will meet again in homological algebra. The principle is this: a contravariant functor on a triangulated category that is homotopy-invariant, sends exact triangles to exact sequences, and carries coproducts (wedges, in the topological case) to products is representable by a single object, recovered from the functor by a universal element exactly as Y is recovered from F in the proof above. The wedge and Mayer-Vietoris axioms of definition 8.3.3 are the topological form of the coproduct and exact-triangle conditions. The stable homotopy category, glimpsed through spectra in chapter 8, is one triangulated category in which the principle operates: there ordinary cohomology is the functor represented by the Eilenberg-MacLane spectrum (definition 8.4.2), which is what corollary 8.3.11 produces degree by degree.
The derived category $D(R)$ of a ring R, constructed in [10, Existence of the Derived Category], is another triangulated category, and the same representability principle there is the engine behind the adjoint functors of homological algebra. It enters as an adjoint-functor theorem: a triangulated functor out of $D(R)$ that preserves coproducts admits a right adjoint, because for each target object the maps out of its image form a contravariant functor of the kind above, hence one representable by the sought adjoint value. This is what guarantees, for a proper morphism f, the existence of the right adjoint $f^!$ underlying Grothendieck duality (see [10]). Thus the two ways the reader has met cohomology as "something derived" are one phenomenon viewed in two triangulated categories. The Ext term in the universal coefficient theorem (theorem 4.1.8), where $\mathrm{Ext}(H_{n-1}(C), G)$ is the derived functor of $\mathrm{Hom}(-, G)$, lives in $D(\mathbb{Z})$; the representability of a cohomology theory established in this section lives in the stable homotopy category. The shared mechanism is corepresentability in a triangulated category, and Brown's theorem is its first appearance in this book. The triangulated framework that makes the comparison precise is developed in [10]; the point here is only that the representability proved above and the derived functors of homological algebra are two faces of it.

Exercise 8.3.1

1. Verify that the wedge axiom of definition 8.3.3 forces $F(*) = *$ (a one-point set) for a Brown functor F, where $*$ is the one-point space. Deduce that the distinguished basepoint of $F(X)$ is $c^*(*_{F(*)})$ pulled back along the constant map $c\colon X \to *$, and explain why this makes T_u a map of *pointed* sets for any $u \in F(Y)$.

2. Let F be a Brown functor and (Y, u) a universal element. Using only the universality on spheres and the wedge axiom, show that $T_u\colon [W, Y] \to F(W)$ is a bijection for every wedge of spheres $W = \bigvee_\alpha S^{q_\alpha}$, without invoking the full Brown theorem. (This is the first nontrivial case beyond a single sphere.)

3. Show that the functor $X \mapsto \tilde{H}^n(X; \mathbb{Z}) \oplus \tilde{H}^m(X; \mathbb{Z})$ is a Brown functor for any $n, m \geq 1$, and identify its representing space in terms of Eilenberg-MacLane spaces. Generalise to a finite direct sum $\bigoplus_i \tilde{H}^{n_i}(X; G_i)$.

4. Let F be the functor sending a pointed connected CW complex X to the set of isomorphism classes of complex line bundles on X, with the basepoint the trivial bundle. Assuming the classification of line bundles by the first Chern class gives a natural isomorphism $F(X) \cong \tilde{H}^2(X; \mathbb{Z})$, identify the representing space of F and the universal element on it. (You may cite theorem 8.3.1.)

5. Explain precisely where the proof of theorem 8.3.9 uses Whitehead's theorem, and why the conclusion would fail if F were defined on *all* topological spaces rather than on CW complexes. Illustrate with the Warsaw circle (cf. the discussion following theorem 5.3.5).

6. Suppose $\tilde{h}^*$ is a reduced cohomology theory whose coefficient groups $\tilde{h}^n(S^0)$ vanish for $n \neq 0$ and equal an abelian group G for $n = 0$. Using corollary 8.3.11 and the uniqueness clause of theorem 8.3.9, identify the representing spaces Y_n with Eilenberg-MacLane spaces and conclude that $\tilde{h}^*$ is ordinary cohomology with coefficients in G. (This is the representable form of the uniqueness theorem theorem 8.2.14.)

8.4 Spectra and Ω-Spectra

The representation theorem of section 5.5 produced, for each abelian group G and each degree n, a single space $K(G, n)$ that represents the functor $\tilde{H}^n(-; G)$, so that a cohomology class is the same datum as a homotopy class of maps into $K(G, n)$. The degrees are not independent. The loop relation $K(G, n) \simeq \Omega K(G, n+1)$ of example 5.2.12, combined with the uniqueness in theorem 5.5.4, threads the representing spaces into a single sequence in which each space is the loop space of the next. This section abstracts that linkage. A *spectrum* is a sequence of pointed spaces tied together by structure maps relating consecutive degrees, and an Ω-*spectrum* is one in which the tie is as tight as possible: each space is, up to weak equivalence, the loop space of its successor. The payoff is twofold. An Ω-spectrum manufactures a reduced cohomology theory out of homotopy classes of maps, with the suspension isomorphism of the theory coming for free from the structure maps; and Brown representability runs the construction in reverse, producing an Ω-spectrum from any cohomology theory. Ω-spectra and reduced cohomology theories are thereby equivalent data. The Eilenberg-MacLane spectrum encodes ordinary cohomology, the suspension spectrum of a point gives the sphere spectrum whose theory is stable cohomotopy, and the whole framework opens onto the stable homotopy category in which spheres of every dimension become invertible.

A spectrum is built from two pieces of data that say the same thing in adjoint forms. One either suspends E_n and maps the result into E_{n+1}, or one maps E_n into the loop space of E_{n+1}; the loop-suspension adjunction $[\Sigma X, Y]_* \cong [X, \Omega Y]_*$ makes these interchangeable. The suspension $\Sigma X = S^1 \wedge X$ used here is the reduced suspension, the smash of X with a circle, which for a CW complex agrees up to homotopy with the unreduced suspension and carries reduced cohomology up one degree by the suspension isomorphism $\tilde{H}^n(X) \cong \tilde{H}^{n+1}(\Sigma X)$ of the axioms in definition 8.2.1.

Definition 8.4.1: [Stable Homotopy] Spectrum

A *spectrum* E is a sequence $(E_n)_{n\geq 0}$ of pointed spaces together with *structure maps*

$$\sigma_n \colon \Sigma E_n \longrightarrow E_{n+1}$$

one for each $n \geq 0$, where $\Sigma X = S^1 \wedge X$ is the reduced suspension. Equivalently, by the loop-suspension adjunction $[\Sigma E_n, E_{n+1}]_* \cong [E_n, \Omega E_{n+1}]_*$, a spectrum is the same datum as a sequence (E_n) together with *adjoint structure maps*

$$\tilde{\sigma}_n \colon E_n \longrightarrow \Omega E_{n+1}$$

A *map of spectra* $f \colon E \to F$ is a sequence of pointed maps $f_n \colon E_n \to F_n$ commuting with the structure maps, that is, $f_{n+1} \circ \sigma_n^E = \sigma_n^F \circ (\Sigma f_n)$ for every n.

A spectrum packages an entire sequence of spaces into one object, with the structure maps recording how each space sits inside the loop space of the next. The two presentations are genuinely the same: passing from σ_n to $\tilde{\sigma}_n$ is the adjunction isomorphism, and a map of spectra is required to respect either presentation (the two compatibility conditions are adjoint and hence equivalent). The indexing here starts at $n \geq 0$ and the structure maps go up in degree; this is the convention adapted to cohomology, where $E^n(X)$ will be read off from the n-th space. Nothing yet requires the structure maps to be equivalences, so the definition is loose enough to admit suspension spectra, where the maps are far from equivalences, alongside the Ω-spectra introduced next, where they are equivalences. The looseness is deliberate: it is the difference between the two that distinguishes a generic spectrum from one that directly represents a cohomology theory.

The first example of a spectrum is the one already implicit in the representation theorem of section 5.5: the Eilenberg-MacLane spaces, strung together by the loop relation. It deserves a name, because it is the spectrum-level incarnation of ordinary cohomology.

Definition 8.4.2: Eilenberg-MacLane Spectrum

Let G be an abelian group. The *Eilenberg-MacLane spectrum* HG is the spectrum with n-th space

$$(HG)_n = K(G, n)$$

the Eilenberg-MacLane space of definition 5.5.1 (with $K(G, 0) = G$ the discrete group, based at 0), and adjoint structure maps

$$\tilde{\sigma}_n \colon K(G, n) \xrightarrow{\simeq} \Omega K(G, n+1)$$

the homotopy equivalences of example 5.2.12 and theorem 5.5.4. The case $G = \mathbb{Z}$ is written $H\mathbb{Z}$.

The Eilenberg-MacLane spectrum is the prototype of the entire theory, and it is already an Ω-spectrum (the next definition makes this precise): each adjoint structure map is not an arbitrary map but a homotopy equivalence. This is exactly the tightness that the representation theorem of section 5.5 encoded degree by degree, now assembled into a single object. The cohomology theory this spectrum represents is ordinary cohomology with G coefficients, the loop relation that builds it being precisely the geometric source of the suspension isomorphism in that theory. The general definition isolates this tightness as a property.

Example 8.4.3: The Structure Maps of $H\mathbb{Z}$ are Equivalences
For HG with $G = \mathbb{Z}$, the adjoint structure map $\tilde{\sigma}_n \colon K(\mathbb{Z}, n) \to \Omega K(\mathbb{Z}, n+1)$ is a homotopy equivalence for every $n \geq 0$, which is the verification that $H\mathbb{Z}$ is well defined as in definition 8.4.2. For $n \geq 1$ the path-loop fibration $\Omega K(\mathbb{Z}, n+1) \to PK(\mathbb{Z}, n+1) \to K(\mathbb{Z}, n+1)$ of example 5.2.12 has contractible total space, so its long exact sequence gives

$$\pi_i\big(\Omega K(\mathbb{Z}, n+1)\big) \cong \pi_{i+1}\big(K(\mathbb{Z}, n+1)\big) \;=\; \begin{cases} \mathbb{Z} & i = n \\ 0 & i \neq n \end{cases}$$

Thus $\Omega K(\mathbb{Z}, n+1)$ has the homotopy groups of a $K(\mathbb{Z}, n)$, and since the loop space of a CW complex is again of CW type, theorem 5.5.4 gives a homotopy equivalence $K(\mathbb{Z}, n) \simeq \Omega K(\mathbb{Z}, n+1)$. For $n = 0$ one checks directly that $\Omega K(\mathbb{Z}, 1) = \Omega S^1$ has $\pi_0(\Omega S^1) \cong \pi_1(S^1) \cong \mathbb{Z}$ and $\pi_i(\Omega S^1) \cong \pi_{i+1}(S^1) = 0$ for $i \geq 1$, so ΩS^1 is homotopy equivalent to the discrete group $\mathbb{Z} = K(\mathbb{Z}, 0)$. The suspended forms $\sigma_n \colon \Sigma K(\mathbb{Z}, n) \to K(\mathbb{Z}, n+1)$ are the adjoints. Hence $H\mathbb{Z}$ is a spectrum whose adjoint structure maps are all equivalences.

Definition 8.4.4: Ω-Spectrum

An *Ω-spectrum* is a spectrum $E = (E_n, \tilde{\sigma}_n)$ in which every adjoint structure map

$$\tilde{\sigma}_n \colon E_n \longrightarrow \Omega E_{n+1}$$

is a weak homotopy equivalence (definition 5.3.1). When each E_n has the homotopy type of a CW complex, this is equivalent, by Whitehead's theorem (theorem 5.3.5), to each $\tilde{\sigma}_n$ being a homotopy equivalence.

The Ω-spectrum condition says that the spectrum loses no information as it climbs in degree: the space E_n recovers, up to weak equivalence, the loop space ΩE_{n+1}, so E_n is determined by E_{n+1} and the whole spectrum is determined by its high-degree behavior. The contrast with a general spectrum is sharp. In an arbitrary spectrum the adjoint map $\tilde{\sigma}_n \colon E_n \to \Omega E_{n+1}$ can collapse or expand homotopy wildly; in an Ω-spectrum it is an isomorphism on every homotopy group. This is the precise property a spectrum needs in order to represent a cohomology theory through homotopy classes of maps, because the loop relation $E_n \simeq \Omega E_{n+1}$ is what makes the suspension isomorphism of the theory hold on the nose. The next theorem is the central construction of the section: it manufactures a reduced cohomology theory from any Ω-spectrum.

Example 8.4.5: The Sphere Spectrum is Not an Ω-Spectrum
Anticipating the suspension spectrum of definition 8.4.8, consider the sequence $E_n = S^n$ with structure maps $\sigma_n \colon \Sigma S^n = S^{n+1} \to S^{n+1}$ the identity, the homeomorphism $S^1 \wedge S^n \cong S^{n+1}$. This is a spectrum, the sphere spectrum $\mathbb{S}$. Its adjoint structure maps are

$$\tilde{\sigma}_n \colon S^n \longrightarrow \Omega S^{n+1}$$

the canonical inclusion of S^n as the constant-speed great-circle loops, which on homotopy groups is the suspension homomorphism $\pi_i(S^n) \to \pi_{i+1}(S^{n+1})$. By the Freudenthal suspension theorem this map is an isomorphism only in the stable range $i < 2n - 1$, and it fails to be an isomorphism for i large: for instance $\tilde{\sigma}_2 \colon S^2 \to \Omega S^3$ is not a weak equivalence, since $\pi_3(S^2) \cong \mathbb{Z}$ while

$\pi_3(\Omega S^3) \cong \pi_4(S^3) \cong \mathbb{Z}/2$ by example 6.3.7, and $\mathbb{Z} \not\cong \mathbb{Z}/2$. Hence $\mathbb{S}$ is a spectrum but not an Ω-spectrum. The failure is exactly the instability of the homotopy of spheres below the stable range.

The example marks the boundary precisely: the sphere spectrum is the cleanest spectrum imaginable, built from spheres with identity structure maps, yet it is not an Ω-spectrum because the suspension homomorphism on spheres is not an isomorphism outside the stable range. This is not a defect to be repaired but the source of the entire subject. The stable homotopy groups, defined below as colimits over suspension, are exactly what the sphere spectrum measures, and the passage from a general spectrum to the theory it defines requires either the Ω-spectrum condition or a stabilization that achieves the same effect. With the contrast in hand, the construction of cohomology from an Ω-spectrum can be carried out in full.

The reduced cohomology theory attached to an Ω-spectrum is defined by the formula one expects from section 5.5: the degree-n cohomology of a pointed space X is the set of based homotopy classes of maps into E_n. The work is to verify that this assignment satisfies the axioms of a reduced cohomology theory from definition 8.2.1: that it is a homotopy-invariant functor, that a cofibration induces an exact sequence, that the suspension isomorphism holds, and that a wedge goes to a product. The Ω-spectrum condition enters at exactly one point, the suspension isomorphism, and the group structure on the cohomology sets comes from the loop-space structure that the Ω-spectrum condition guarantees.

Theorem 8.4.6: An Ω-Spectrum Defines a Cohomology Theory

Let $E = (E_n, \tilde{\sigma}_n)$ be an Ω-spectrum with each E_n of CW homotopy type. For a pointed CW complex X and $n \geq 0$, set

$$E^n(X) \;=\; [X, E_n]_*$$

the set of based homotopy classes of maps $X \to E_n$, and for a based map $g\colon X' \to X$ let $g^*\colon E^n(X) \to E^n(X')$ be precomposition $[f] \mapsto [f \circ g]$. Then:

1. Each $E^n(X)$ is an abelian group, natural in X, so $E^n(-)$ is a contravariant functor from pointed CW complexes to abelian groups.
2. *(Homotopy)* If $g \simeq g'$ as based maps then $g^* = (g')^*$.
3. *(Exactness)* For a based CW pair (X, A) with inclusion $i\colon A \hookrightarrow X$ and quotient map $q\colon X \to X/A$, the sequence

$$E^n(X/A) \xrightarrow{q^*} E^n(X) \xrightarrow{i^*} E^n(A)$$

is exact at $E^n(X)$.

4. *(Suspension)* There is a natural isomorphism $\Sigma\colon E^n(X) \xrightarrow{\cong} E^{n+1}(\Sigma X)$.
5. *(Wedge)* For any family (X_α) of pointed CW complexes the inclusions induce an isomorphism

$$E^n\Big(\bigvee_\alpha X_\alpha\Big) \xrightarrow{\cong} \prod_\alpha E^n(X_\alpha)$$

Consequently the collection $(E^n)_{n \geq 0}$ with these structure maps is a reduced cohomology theory in the sense of definition 8.2.1.

***Proof*:**
Throughout, $[-,-]_*$ denotes based homotopy classes of based maps, and all spaces are pointed CW complexes (or of CW homotopy type), so that Whitehead's theorem (theorem 5.3.5) converts weak equivalences into homotopy equivalences and the loop-suspension adjunction $[\Sigma X, Y]_* \cong [X, \Omega Y]_*$ is available.

Step 1: The abelian group structure on $E^n(X)$. The Ω-spectrum condition gives a homotopy equivalence $\tilde{\sigma}_n\colon E_n \xrightarrow{\simeq} \Omega E_{n+1}$. Postcomposition with $\tilde{\sigma}_n$ is a bijection

$$(\tilde{\sigma}_n)_*\colon [X, E_n]_* \xrightarrow{\cong} [X, \Omega E_{n+1}]_*$$

and the right-hand side carries a group structure, natural in X, by proposition 8.1.4(1) and (2). Transporting it along the bijection $(\tilde{\sigma}_n)_*$ makes $E^n(X) = [X, E_n]_*$ a group. It is moreover abelian: applying $\tilde{\sigma}_n$ a second time identifies $E^n(X)$ with $[X, \Omega^2 E_{n+2}]_*$, which is abelian by proposition 8.1.4(3). Naturality in X holds because precomposition g^* commutes with postcomposition by $\tilde{\sigma}_n$, so g^* is a homomorphism. This proves (1).

Step 2: Homotopy invariance. If $g \simeq g'$ through based maps then for any $[f] \in [X, E_n]_*$ the composites $f \circ g$ and $f \circ g'$ are based homotopic (compose the homotopy $g \simeq g'$ with f), so $[f \circ g] = [f \circ g']$, that is $g^*[f] = (g')^*[f]$. Hence $g^* = (g')^*$, proving (2). In particular a based homotopy equivalence $X \simeq X'$ induces an isomorphism $E^n(X) \cong E^n(X')$, so E^n is an invariant of based homotopy type.

Step 3: Exactness. Fix n and write $K = E_n$. The pair (X, A) is a based CW pair, so proposition 8.1.8 gives the coexact Puppe sequence

$$A \xrightarrow{i} X \xrightarrow{q} X/A \xrightarrow{\partial} \Sigma A \xrightarrow{\Sigma i} \Sigma X \to \cdots$$

in which $X/A \simeq X \cup_A CA$ is the mapping cone, q the collapse, and consecutive maps are a cofibration followed by the collapse of its domain. The defining property of this sequence is that for *any* pointed space K, applying $[-, K]_*$ yields an exact sequence of pointed sets

$$[\Sigma X, K]_* \to [\Sigma A, K]_* \to [X/A, K]_* \xrightarrow{q^*} [X, K]_* \xrightarrow{i^*} [A, K]_*$$

Exactness at $[X, K]_*$ is precisely the statement to be proved: a class $[f] \in [X, K]_*$ satisfies $i^*[f] = [f \circ i] = 0$, that is $f|_A$ is null-homotopic, if and only if f extends over the cone CA, that is f factors up to homotopy through the quotient $q\colon X \to X/A$, which is the statement $[f] \in \operatorname{im} q^*$. Concretely, a based null-homotopy of $f|_A$ is a map $CA \to K$ restricting to $f|_A$ on A, and gluing it to f along A produces a map on $X \cup_A CA = X/A$ whose restriction to X is f; this is the required factorization $[f] = q^*[\bar{f}]$. Conversely if $[f] = q^*[\bar{f}]$ then $i^*[f] = i^*q^*[\bar{f}] = (q \circ i)^*[\bar{f}] = 0$ because $q \circ i$ is the constant map $A \to X/A$ (the subspace A collapses to the basepoint of X/A). Setting $K = E_n$ gives exactness of $E^n(X/A) \xrightarrow{q^*} E^n(X) \xrightarrow{i^*} E^n(A)$ as groups, the maps being homomorphisms by Step 1, proving (3).

Step 4: The suspension isomorphism. The desired isomorphism $\Sigma\colon E^n(X) \to E^{n+1}(\Sigma X)$ is the composite of three natural bijections. First, the adjoint structure map gives, by Step 1,

$$E^n(X) = [X, E_n]_* \xrightarrow[\cong]{(\tilde{\sigma}_n)_*} [X, \Omega E_{n+1}]_*$$

Second, the loop-suspension adjunction gives a natural bijection

$$[X, \Omega E_{n+1}]_* \xrightarrow{\cong} [\Sigma X, E_{n+1}]_* = E^{n+1}(\Sigma X)$$

sending a map $X \to \Omega E_{n+1}$ to its adjoint $\Sigma X \to E_{n+1}$. Both bijections are isomorphisms of groups: the first because the group structure on $E^n(X)$ was *defined* by transporting along $(\tilde{\sigma}_n)_*$, and the second by theorem 8.1.5(2), which identifies loop concatenation on the source with the pinch-map addition on $[\Sigma X, -]_*$. The composite Σ is therefore a natural isomorphism, and it is here, and only here, that the Ω-spectrum hypothesis is used: without $\tilde{\sigma}_n$ being an equivalence, the first arrow would fail to be a bijection. This proves (4).

Step 5: The wedge axiom. Let $(X_\alpha)_{\alpha \in A}$ be pointed CW complexes with wedge $\bigvee_\alpha X_\alpha$, the quotient of $\coprod_\alpha X_\alpha$ identifying all basepoints. A based map $f\colon \bigvee_\alpha X_\alpha \to E_n$ is, by the universal property of the wedge, exactly a family of based maps $f_\alpha\colon X_\alpha \to E_n$ agreeing at the common basepoint (which they do automatically, all sending it to the basepoint of E_n); that is, the restriction maps assemble into a bijection of sets

$$\Big[\bigvee_\alpha X_\alpha, E_n\Big]_* \xrightarrow{\cong} \prod_\alpha [X_\alpha, E_n]_*$$

The same universal property applied to based homotopies (a based homotopy out of $\bigvee_\alpha X_\alpha$ is a family of based homotopies, one out of each X_α) shows the map is well defined and injective on homotopy classes, and surjective because any family of maps glues to a map on the wedge. The bijection is a group homomorphism because the group operation is computed pointwise through loop concatenation in $E_n \simeq \Omega E_{n+1}$, which commutes with restriction to each summand. Hence

$$E^n\Big(\bigvee_\alpha X_\alpha\Big) \xrightarrow{\cong} \prod_\alpha E^n(X_\alpha)$$

is an isomorphism, proving (5).

Step 6: Assembling the cohomology theory. Parts (1)-(5) are exactly the data and axioms of a reduced cohomology theory in definition 8.2.1: a sequence of homotopy-invariant abelian-group-valued functors E^n (parts 1, 2), satisfying exactness for cofiber sequences (part 3), with natural suspension isomorphisms $E^n(X) \cong E^{n+1}(\Sigma X)$ (part 4) and the wedge-to-product property (part 5). The long exact sequence of a pair, required by that definition, is obtained by iterating the exactness of Step 3 along the Puppe sequence: applying $[-, E_n]_*$ to the full coexact sequence $A \to X \to X/A \to \Sigma A \to \Sigma X \to \cdots$ and using the suspension isomorphism to reindex $[\Sigma^k A, E_n]_* \cong E^{n-k}(A)$ converts the Puppe exactness into the long exact sequence

$$\cdots \to E^n(X/A) \to E^n(X) \to E^n(A) \to E^{n+1}(X/A) \to \cdots$$

Thus $(E^n)_n$ is a reduced cohomology theory, completing the proof.

The theorem is what makes the representable viewpoint work. It says that homotopy classes of maps into the spaces of an Ω-spectrum automatically organize themselves into a cohomology theory, with every axiom traced to a structural feature: the group law to the loop space, exactness to the Puppe sequence of a cofibration, the suspension isomorphism to the adjoint structure map, and the wedge axiom to the universal property of the wedge. The dimension axiom, conspicuously absent from the list, is what separates ordinary from generalized theories: it imposes a condition on the homotopy of the spaces E_n that the Eilenberg-MacLane spectrum satisfies and that the sphere spectrum and K-theory do not. The construction is also exhaustive, by Brown representability, every reasonable cohomology theory arises this way; that converse is recorded after the present examples.

Example 8.4.7: The Cohomology Theory of $H\mathbb{Z}$ is Ordinary Cohomology

Apply theorem 8.4.6 to the Eilenberg-MacLane spectrum $H\mathbb{Z}$ of definition 8.4.2, which is an Ω-spectrum because each $\tilde{\sigma}_n \colon K(\mathbb{Z}, n) \to \Omega K(\mathbb{Z}, n+1)$ is a homotopy equivalence (example 8.4.3). The resulting theory is

$$(H\mathbb{Z})^n(X) = [X, K(\mathbb{Z}, n)]_* \;\cong\; \tilde{H}^n(X;\mathbb{Z})$$

the isomorphism being exactly the representation theorem $T\colon [X, K(\mathbb{Z}, n)]_* \xrightarrow{\cong} \tilde{H}^n(X;\mathbb{Z})$, $[f] \mapsto f^*(\iota_n)$, of theorem 5.5.9. The identification is natural in X (naturality of T) and compatible with the suspension isomorphisms: the suspension isomorphism $(H\mathbb{Z})^n(X) \cong (H\mathbb{Z})^{n+1}(\Sigma X)$ built from the structure maps in Step 4 corresponds under T to the suspension isomorphism $\tilde{H}^n(X;\mathbb{Z}) \cong \tilde{H}^{n+1}(\Sigma X;\mathbb{Z})$ of ordinary cohomology, because the adjoint structure map $\tilde{\sigma}_n$ was constructed from the loop relation $K(\mathbb{Z}, n) \simeq \Omega K(\mathbb{Z}, n+1)$ that the fundamental class ι_{n+1} transgresses to ι_n. Hence the cohomology theory represented by $H\mathbb{Z}$ is ordinary singular cohomology with integer coefficients, and replacing $\mathbb{Z}$ by G gives $\tilde{H}^n(-;G)$ from the spectrum HG. To see the values on a concrete space, take $X = S^k$: then

$$(H\mathbb{Z})^n(S^k) = [S^k, K(\mathbb{Z}, n)]_* = \pi_k(K(\mathbb{Z}, n)) = \begin{cases} \mathbb{Z} & k = n \\ 0 & k \neq n \end{cases}$$

matching $\tilde{H}^n(S^k;\mathbb{Z})$, which is $\mathbb{Z}$ for $k=n$ and 0 otherwise.

The Eilenberg-MacLane spectrum is thus the spectrum incarnation of ordinary cohomology: the entire theory $\tilde{H}^*(-;\mathbb{Z})$, with all its naturality and suspension structure, is compressed into the single object $H\mathbb{Z}$. The dimension axiom holds for it because the spaces $K(\mathbb{Z},n)$ have their homotopy concentrated in degree n, so the value on a sphere is concentrated in a single degree. This is the cohomological reading of the slogan that ordinary cohomology is the theory of the Eilenberg-MacLane spectrum, and it is the link between the abstract axioms of definition 8.2.4 and the concrete representing spaces. The next construction goes the other way, producing a spectrum from a space rather than from a theory.

The suspension spectrum is the most economical way to turn a single pointed space into a spectrum: suspend it repeatedly. The structure maps are then identities, since suspending the n-th space $\Sigma^n X$ produces exactly the $(n+1)$-st space $\Sigma^{n+1}X$. This is almost never an Ω-spectrum, as the sphere case in example 8.4.5 already showed, but it is the canonical bridge from unstable to stable homotopy theory and the source of the sphere spectrum.

Definition 8.4.8: Suspension Spectrum and the Sphere Spectrum

Let X be a pointed space. The *suspension spectrum* $\Sigma^\infty X$ is the spectrum with n-th space

$$(\Sigma^\infty X)_n = \Sigma^n X = \underbrace{S^1 \wedge \cdots \wedge S^1}_{n} \wedge X$$

and structure maps $\sigma_n\colon \Sigma(\Sigma^n X) = \Sigma^{n+1}X \to \Sigma^{n+1}X$ the identity. The *sphere spectrum* is

$$\mathbb{S} = \Sigma^\infty S^0$$

whose n-th space is $\Sigma^n S^0 = S^n$ (using $\Sigma S^k = S^{k+1}$ and $S^0 = \{\pm 1\}$), with identity structure maps $S^{n+1} \to S^{n+1}$.

The suspension spectrum is the left adjoint to the forgetful passage from spectra to spaces: it is the free spectrum on a pointed space, built by doing nothing but suspending. The sphere spectrum $\mathbb{S} = \Sigma^\infty S^0$ is the unit of this construction and the most basic spectrum of all; its spaces are the spheres themselves. As example 8.4.5 recorded, $\mathbb{S}$ is not an Ω-spectrum, so the cohomology theory it defines must be extracted by a stabilization rather than read off directly from a single space, and the homotopy it carries is the stable homotopy of spheres. The cohomology theory of the sphere spectrum is stable cohomotopy, and its homotopy groups are the stable stems.

Example 8.4.9: Stable Cohomotopy and the Stable Stems

The sphere spectrum $\mathbb{S}$ defines, by the stabilized form of theorem 8.4.6, the generalized cohomology theory called *stable cohomotopy*, written $\pi_s^n(X)$ or $\mathbb{S}^n(X)$. On a finite pointed CW complex X its value is the colimit

$$\pi_s^n(X) = \mathbb{S}^n(X) = \varinjlim_k \, [\Sigma^k X, S^{k+n}]_*$$

the structure maps of the colimit being suspension $[\Sigma^k X, S^{k+n}]_* \to [\Sigma^{k+1}X, S^{k+1+n}]_*$. The colimit is needed precisely because $\mathbb{S}$ is not an Ω-spectrum: suspending stabilizes the unstable sets $[\Sigma^k X, S^{k+n}]_*$ into a group once k is large, by the Freudenthal suspension theorem. Dually, the *homotopy groups* of the sphere spectrum are

$$\pi_n(\mathbb{S}) = \varinjlim_k \pi_{n+k}(S^k) = \pi_n^s$$

the stable homotopy groups of spheres, the stable stems. For $n = 0$ the colimit $\varinjlim_k \pi_k(S^k) = \varinjlim_k \mathbb{Z}$ along degree-1 suspensions is $\mathbb{Z}$, so $\pi_0(\mathbb{S}) \cong \mathbb{Z}$. For $n = 1$ the colimit

$$\pi_1(\mathbb{S}) = \varinjlim_k \pi_{k+1}(S^k) = \pi_1^s \cong \mathbb{Z}/2$$

is the first stable stem, computed in example 6.3.7: $\pi_{k+1}(S^k) \cong \mathbb{Z}/2$ for all $k \geq 3$, so the colimit, already constant from $k = 3$ onward, is $\mathbb{Z}/2$, generated by the suspensions of the Hopf map η. The unstable group $\pi_3(S^2) \cong \mathbb{Z}$ at $k = 2$ lies below the stable range and collapses to $\mathbb{Z}/2$ under the first suspension, so it does not affect the colimit.

The sphere spectrum is to stable homotopy theory what the integers are to algebra: the initial object whose endomorphisms control everything else. Its homotopy groups, the stable stems π_n^s, are the central computational problem of the subject, finite for $n \geq 1$ (a fact proved by Serre using the spectral-sequence machinery of chapter 6) and known explicitly only in a range. The contrast between $\pi_0(\mathbb{S}) \cong \mathbb{Z}$ and $\pi_1(\mathbb{S}) \cong \mathbb{Z}/2$ already exhibits the two regimes: degree zero is detected by the integer degree of a self-map of a sphere, while degree one is the 2-torsion produced by the Hopf map. The general definition of the stable homotopy groups of an arbitrary spectrum is the same colimit construction, and it is the right notion of homotopy group in the stable world.

Definition 8.4.10: Stable Homotopy Groups of a Spectrum

For a spectrum $E = (E_n, \sigma_n)$ and $n \in \mathbb{Z}$, the *n-th stable homotopy group* is the colimit

$$\pi_n(E) = \varinjlim_k \pi_{n+k}(E_k)$$

where the maps in the colimit are the composites

$$\pi_{n+k}(E_k) \xrightarrow{\Sigma} \pi_{n+k+1}(\Sigma E_k) \xrightarrow{(\sigma_k)_*} \pi_{n+k+1}(E_{k+1})$$

of the suspension homomorphism with the map induced by the structure map.

The stable homotopy groups $\pi_n(E)$ are indexed by all integers, positive and negative, reflecting that a spectrum has no bottom dimension: suspension is invertible in the stable category, so degrees may go below zero. For an Ω-spectrum the colimit stabilizes immediately, since the adjoint structure maps are equivalences and $\pi_{n+k}(E_k) \cong \pi_{n+k+1}(E_{k+1})$ already at each stage, so $\pi_n(E) \cong \pi_{n+k}(E_k)$ for any single large k; for a suspension spectrum the colimit is a genuine limit of suspensions, as in the sphere case. The two presentations of a spectrum, by structure maps or adjoints, give the same colimit because the suspension homomorphism and the loop-space dimension shift are adjoint. These groups are the homotopy-theoretic invariants of a spectrum, and a map of spectra inducing isomorphisms on all π_n is a stable equivalence, the spectrum-level analogue of a weak homotopy equivalence.

The cohomology theory of an Ω-spectrum, the theory of $H\mathbb{Z}$, and the sphere spectrum's stable cohomotopy are three instances of a single correspondence. Brown representability, established in theorem 8.3.9, asserts that every reduced cohomology theory on CW complexes is representable: in each degree n there is a space E_n with $h^n(X) \cong [X, E_n]_*$ naturally, and the suspension isomorphism of the theory upgrades these representing spaces into an Ω-spectrum. Conversely, theorem 8.4.6 produces a cohomology theory from any Ω-spectrum. The two constructions are mutually inverse up to the natural notions of equivalence, giving the correspondence that organizes the entire subject.

Theorem 8.4.11: Cohomology Theories and Ω-Spectra Correspond

There is a correspondence

$$\left\{ \begin{array}{c} \text{reduced cohomology theories} \\ \text{on pointed CW complexes} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \Omega\text{-spectra} \\ \text{(up to weak equivalence)} \end{array} \right\}$$

given in one direction by sending an Ω-spectrum E to the theory $X \mapsto [X, E_n]_*$ of theorem 8.4.6, and in the other by sending a theory (h^n) to the Ω-spectrum whose n-th space represents h^n, which exists by Brown representability (theorem 8.3.9 and corollary 8.3.11). The two assignments are mutually inverse: a theory (h^n) is naturally isomorphic to the theory of its representing Ω-spectrum, and an Ω-spectrum is weakly equivalent to the Ω-spectrum representing its theory.

***Proof*:**
Step 1: From an Ω-spectrum to a theory and back. Let E be an Ω-spectrum and $h^n(X) = [X, E_n]_*$ the cohomology theory of theorem 8.4.6. By construction h^n is represented by E_n: the identity bijection $h^n(X) = [X, E_n]_*$ is the representing isomorphism, natural in X. Brown representability applied to h^n produces a representing space, and the representing space of a representable functor is unique up to weak homotopy equivalence (two spaces Y, Y' with $[X, Y]_* \cong [X, Y']_*$ naturally in X are weakly equivalent, by the Yoneda argument: the natural isomorphism applied to $X = Y$ and $X = Y'$ produces maps $Y \to Y'$ and $Y' \to Y$ whose composites correspond to the identities, hence are weak equivalences). Therefore the Ω-spectrum produced from h^n by Brown representability has n-th space weakly equivalent to E_n, and the equivalences are compatible with the structure maps because both sides realize the same suspension isomorphism of h^*. Hence E is weakly equivalent to the Ω-spectrum representing its own theory.

Step 2: From a theory to an Ω-spectrum and back. Let (h^n) be a reduced cohomology theory. By Brown representability (theorem 8.3.9) each functor h^n is represented by a pointed CW complex E_n, with a natural isomorphism $h^n(X) \cong [X, E_n]_*$. The suspension isomorphism $h^n(X) \cong h^{n+1}(\Sigma X)$ of the theory, combined with the loop-suspension adjunction, gives natural isomorphisms

$$[X, E_n]_* \cong h^n(X) \cong h^{n+1}(\Sigma X) \cong [\Sigma X, E_{n+1}]_* \cong [X, \Omega E_{n+1}]_*$$

natural in X. By the Yoneda uniqueness of representing objects this natural isomorphism is induced by a weak homotopy equivalence $\tilde{\sigma}_n \colon E_n \xrightarrow{\simeq} \Omega E_{n+1}$ (take $X = E_n$ and chase the identity, or $X = \Omega E_{n+1}$). These $\tilde{\sigma}_n$ make (E_n) an Ω-spectrum. Feeding this Ω-spectrum back into theorem 8.4.6 returns the theory $X \mapsto [X, E_n]_* \cong h^n(X)$, naturally isomorphic to the original (h^n), with the suspension isomorphisms matching by construction. Hence (h^n) is naturally isomorphic to the theory of its representing Ω-spectrum, completing the proof.

The correspondence is the organizing principle of stable homotopy theory: to study cohomology theories is to study Ω-spectra, and conversely. Ordinary cohomology corresponds to $H\mathbb{Z}$, K-theory corresponds to a spectrum built from $\mathbb{Z} \times BU$ by Bott periodicity (chapter 8 and [21]), and cobordism corresponds to a Thom spectrum; each generalized theory is a spectrum, and computations in the theory become computations with the spectrum. The dictionary also explains why the dimension axiom is the dividing line: it is the condition that the representing spaces E_n be Eilenberg-MacLane spaces, with homotopy concentrated in a single degree, and dropping it admits all the extraordinary theories at once.

8.4.12 Foreshadowing: The Stable Homotopy Category The maps of spectra of definition 8.4.1 and the stable homotopy groups of definition 8.4.10 are the first ingredients of the *stable homotopy category*, the natural home for spectra. In it the objects are spectra, the morphisms are stable homotopy classes of maps (formed by inverting the maps that induce isomorphisms on all π_n), and the suspension functor Σ becomes invertible: every spectrum has a desuspension $\Sigma^{-1}E$, so degrees range over all of $\mathbb{Z}$. The sphere spectrum $\mathbb{S}$ is the unit for a symmetric monoidal smash product, making the category the stable analogue of the category of abelian groups, with $[\mathbb{S}, E] = \pi_0(E)$ playing the role of global sections. This is the setting in which the Atiyah-Hirzebruch spectral sequence of theorem 8.6.4 computes a generalized theory from ordinary cohomology, and in which the entire apparatus of generalized cohomology becomes a form of stable algebra.

Exercise 8.4.1

1. Verify that the Eilenberg-MacLane spectrum $H\mathbb{Z}$ of definition 8.4.2 is an Ω-spectrum, and compute its stable homotopy groups $\pi_n(H\mathbb{Z})$ for all $n \in \mathbb{Z}$. (Use that each $\tilde{\sigma}_n$ is an equivalence so the colimit stabilizes at any single stage.)

2. Let E be an Ω-spectrum and $E^n(X) = [X, E_n]_*$ its cohomology theory. Compute $E^n(S^k)$ in terms of the stable homotopy groups $\pi_*(E)$ of definition 8.4.10, and check your formula against $E = H\mathbb{Z}$ using example 8.4.7.

3. Show that for any pointed space X the suspension spectrum $\Sigma^\infty X$ has stable homotopy groups

$$\pi_n(\Sigma^\infty X) = \varinjlim_k \pi_{n+k}(\Sigma^k X)$$

   the *stable homotopy groups of* X, written $\pi_n^s(X)$. Deduce $\pi_n(\mathbb{S}) = \pi_n^s(S^0) = \pi_n^s$, the stable stems, and compute $\pi_0(\mathbb{S})$.

4. Prove that the structure-map presentation and the adjoint presentation of a spectrum (definition 8.4.1) carry the same information, by writing out the bijection $[\Sigma E_n, E_{n+1}]_* \cong [E_n, \Omega E_{n+1}]_*$ explicitly and checking that a map of spectra in one presentation is a map of spectra in the other.

5. Explain why the dimension axiom holds for the theory represented by $H\mathbb{Z}$ but fails for the theory represented by the sphere spectrum $\mathbb{S}$. (Compare $H^n(\mathrm{pt})$ with the stable stems; recall the dimension axiom concerns the value of the theory on a point, equivalently on S^0.)

6. Let E be an Ω-spectrum and let $f \colon X \to Y$ be a based map of pointed CW complexes whose cofiber is $C_f = Y \cup_f CX$. Using only the exactness axiom of theorem 8.4.6 and the suspension

isomorphism, derive the long exact sequence

$$\cdots \to E^n(C_f) \to E^n(Y) \xrightarrow{f^*} E^n(X) \to E^{n+1}(C_f) \to \cdots$$

8.5 K-theory and Cobordism as Generalized Theories

The Eilenberg-Steenrod axioms of section 8.2 single out ordinary cohomology by one extra requirement, the dimension axiom: the cohomology of a point is concentrated in degree zero. Drop that axiom and a wider world of cohomology theories opens up, each represented by an Ω-spectrum through the machinery of section 8.3 and section 8.4. This section exhibits the two theories that historically forced the generalization, and that remain its archetypes. The first is topological K-theory, built from vector bundles: $K^0(X)$ measures how far the bundles on X are from being trivial, and Bott periodicity assembles the groups $K^n(X)$ into a 2-periodic cohomology theory whose value on a point is the polynomial ring $\mathbb{Z}[\beta, \beta^{-1}]$, the loudest possible violation of the dimension axiom. The second is cobordism, built from manifolds: two closed n-manifolds are cobordant when together they bound a compact $(n+1)$-manifold, and the Pontryagin-Thom construction identifies the resulting bordism groups with the stable homotopy groups of a Thom spectrum. Both theories trade the rigidity of the dimension axiom for geometric content invisible to ordinary cohomology, and both are represented by spectra: KU for complex K-theory, MO and MU for cobordism. The vector-bundle foundations and Bott periodicity are developed in full in the companion volume [21]; here they are imported and put to work building the spectrum.

The starting point is the set of isomorphism classes of complex vector bundles over a space X. This set is a commutative monoid under direct sum (Whitney sum), but not a group: a nontrivial bundle has no additive inverse among bundles, since $E \oplus E' \cong \underline{0}$ forces both to be zero. The Grothendieck group construction repairs this, formally adjoining inverses exactly as the integers are built from the natural numbers.

Definition 8.5.1: Topological K-theory

Let X be a compact Hausdorff space. Write $\mathrm{Vect}_{\mathbb{C}}(X)$ for the commutative monoid of isomorphism classes of complex vector bundles over X under Whitney sum $\oplus$. The *(complex) K-theory group* $K^0(X)$ is the Grothendieck group of this monoid: the free abelian group on isomorphism classes $[E]$ modulo the relations $[E \oplus F] = [E] + [F]$, so that every element has the form $[E] - [F]$ for bundles E, F, with $[E] - [F] = [E'] - [F']$ if and only if $E \oplus F' \oplus G \cong E' \oplus F \oplus G$ for some bundle G. Tensor product of bundles makes $K^0(X)$ a commutative ring, with unit the class of the trivial line bundle $\underline{\mathbb{C}}$. For a basepoint $x_0 \in X$ the restriction $K^0(X) \to K^0(x_0) = \mathbb{Z}$ (taking the rank of a bundle) is a ring homomorphism; its kernel is the *reduced K-theory* $\tilde{K}^0(X)$, the subgroup of virtual bundles of rank zero.
The *representing space* for K-theory is $\mathbb{Z} \times BU$, where $BU = \varinjlim_n BU(n)$ is the classifying space of the infinite unitary group and the factor $\mathbb{Z}$ records the rank. For every compact Hausdorff (or, more generally, paracompact) space X there is a natural bijection

$$K^0(X) \;\cong\; [X,\, \mathbb{Z} \times BU]$$

where the right side is the set of unbased homotopy classes of maps, and the real analogue replaces BU by $BO = \varinjlim_n BO(n)$, giving real K-theory $KO^0(X) \cong [X,\, \mathbb{Z} \times BO]$.

The two halves of the definition are the algebraic and the homotopical face of the same object. Algebraically, $K^0(X)$ is the universal target for additive invariants of bundles: any function on bundles sending $\oplus$ to $+$ factors through it. Homotopically, $K^0(X) = [X, \mathbb{Z} \times BU]$ records that a rank-n bundle is the pullback of the universal bundle along a classifying map $X \to BU(n)$, and that stabilizing in n (adding trivial summands) corresponds to landing in the colimit BU. The identification of these two descriptions is the representability of vector bundles: a complex vector bundle of rank n over a paracompact base is classified up to isomorphism by a homotopy class of maps into the Grassmannian $BU(n) = \varinjlim_N \mathrm{Gr}_n(\mathbb{C}^N)$ (theorem 7.1.9), and over a connected compact Hausdorff base two bundles are stably equivalent (isomorphic after adding trivial summands, possibly of different ranks) precisely when their classifying maps agree in $[X, BU]$ ([21, Reduced K-Theory Classifies Stable Equivalence, Representability of K^0]). The $\mathbb{Z}$ factor carries the rank, which can be any integer once virtual bundles are allowed. The reduced group $\tilde{K}^0(X) = [X, BU]$ (based classes into the connected space BU) is the part that genuinely measures nontriviality.

Example 8.5.2: K^0 of the Spheres S^2 and S^4

Take $X = S^2$. A complex vector bundle over S^2 is built by clutching: cover S^2 by two disks D_+, D_- meeting in an equatorial S^1, trivialize over each disk, and glue by a clutching function $S^1 \to U(n)$. Isomorphism classes of rank-n bundles correspond to π_0 of the space of such functions up to homotopy, that is to $\pi_1(U(n)) = \mathbb{Z}$ for $n \geq 1$ (and to $\pi_1(GL_n(\mathbb{C})) = \mathbb{Z}$ in the topological category). Thus rank-n bundles over S^2 are classified by an integer, their first Chern number. Passing to the stable range, the reduced group is

$$\tilde{K}^0(S^2) \cong \pi_1(U) \cong \mathbb{Z}$$

where $U = \varinjlim_n U(n)$. A generator is $[H] - [\underline{\mathbb{C}}]$, the class of the tautological (Hopf) line bundle H over $S^2 = \mathbb{C}\mathrm{P}^1$ minus a trivial line; the full ring is $K^0(S^2) = \mathbb{Z} \oplus \tilde{K}^0(S^2) = \mathbb{Z}[t]/(t^2)$ with $t = [H] - 1$ satisfying $t^2 = 0$. The relation $t^2 = 0$ holds in $K^0(\mathbb{C}\mathrm{P}^1)$ because the line bundle H over $\mathbb{C}\mathrm{P}^1$ satisfies $(H-1)^2 = 0$ there: in $K^0(\mathbb{C}\mathrm{P}^n)$ the tautological line bundle obeys $(H-1)^{n+1} = 0$, and for $n = 1$ this reads $H^2 - 2H + 1 = 0$, that is $t^2 = (H-1)^2 = 0$, computed in [21, K-Theory of Complex Projective Space].

For $X = S^4$ the same clutching analysis gives bundles classified by $\pi_3(U(n))$. By Bott periodicity (theorem 8.5.3 below) $\pi_3(U) \cong \pi_1(U) \cong \mathbb{Z}$, so

$$\tilde{K}^0(S^4) \cong \pi_3(U) \cong \mathbb{Z}$$

generated by the class of a rank-2 bundle with second Chern number 1. In both cases the answer is $\mathbb{Z}$, reflecting the fact that $\tilde{K}^0(S^{2k}) \cong \mathbb{Z}$ for every $k \geq 0$ and $\tilde{K}^0(S^{2k+1}) = 0$, the 2-periodicity of K-theory seen on spheres.

The reduced groups $\tilde{K}^0(S^{2k}) \cong \mathbb{Z}$ and $\tilde{K}^0(S^{2k+1}) = 0$ are the homotopy groups $\pi_{2k}(\mathbb{Z} \times BU)$ and $\pi_{2k+1}(\mathbb{Z} \times BU)$ read through the identification $\tilde{K}^0(S^m) = [S^m, \mathbb{Z} \times BU]_* = \pi_m(\mathbb{Z} \times BU)$. Since $\Omega(\mathbb{Z} \times BU) \simeq U$ up to homotopy, these are $\pi_m(\mathbb{Z} \times BU) \cong \pi_{m-1}(U)$, a shift by one of the homotopy of U, whose pattern $\pi_0(U), \pi_1(U), \pi_2(U), \cdots = 0, \mathbb{Z}, 0, \mathbb{Z}, \ldots$ is $\mathbb{Z}$ in odd degrees and 0 in even. Equivalently the homotopy of $\mathbb{Z} \times BU$ itself runs $\mathbb{Z}, 0, \mathbb{Z}, 0, \ldots$ ($\mathbb{Z}$ in even degrees, 0 in odd), and that this pattern repeats with period 2 is the content of Bott's theorem, which is what turns the single group K^0 into a periodic cohomology theory.

The decisive input is Bott periodicity, proved in [21, Bott Periodicity]. The form used here is the space-level statement that the representing space is its own double loop space, which that volume records with references rather than derives ([21, Note: The Loop-Space Form]), and it is exactly the structure needed to extend K^0 to negative degrees and then, by periodicity, to all integers.

Theorem 8.5.3: The K-theory Spectrum KU

Let $Z_0 = \mathbb{Z} \times BU$ be the K-theory representing space. Bott periodicity provides a homotopy equivalence

$$\beta\colon \mathbb{Z} \times BU \xrightarrow{\simeq} \Omega^2(\mathbb{Z} \times BU)$$

Define a sequence of spaces KU_n by $KU_n = \mathbb{Z} \times BU$ for n even and $KU_n = U$ for n odd, with structure maps $KU_n \xrightarrow{\simeq} \Omega KU_{n+1}$ given by Bott periodicity for the even-to-odd step ($\mathbb{Z} \times BU \simeq \Omega U$ from $\Omega^2(\mathbb{Z} \times BU) \simeq \mathbb{Z} \times BU$) and by the standard equivalence $U \simeq \Omega(\mathbb{Z} \times BU)$ for the odd-to-even step. Then:

1. The collection $KU = \{KU_n\}_{n \geq 0}$ is an Ω-spectrum in the sense of definition 8.4.4.
2. The associated generalized cohomology theory $K^n(X) := [X, KU_n]_*$ (reduced) is 2-periodic: there is a natural isomorphism $K^n(X) \cong K^{n+2}(X)$ for every $n \in \mathbb{Z}$ and every based CW complex X, induced by β.
3. The coefficient groups are

$$K^n(\mathrm{pt}) = \tilde{K}^n(S^0) \cong \begin{cases} \mathbb{Z} & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

   and the coefficient ring $K^*(\mathrm{pt}) = \bigoplus_n K^n(\mathrm{pt})$ is the ring of Laurent polynomials $\mathbb{Z}[\beta, \beta^{-1}]$ on the Bott class $\beta \in K^{-2}(\mathrm{pt})$, with $|\beta| = -2$. In particular the dimension axiom of definition 8.2.6 fails: $K^n(\mathrm{pt}) \neq 0$ for infinitely many $n \neq 0$.

***Proof*:**
Step 1: The spaces and structure maps form an Ω-spectrum. An Ω-spectrum (definition 8.4.4) is a sequence of based spaces $\{E_n\}$ together with weak homotopy equivalences $E_n \xrightarrow{\simeq} \Omega E_{n+1}$. Two equivalences are needed, one for each parity step, and both come from Bott's theorem.

First, the odd-to-even structure map. The infinite unitary group $U = \varinjlim_n U(n)$ is the loop space of $\mathbb{Z} \times BU$ up to homotopy: the path-loop fibration $U(n) \to EU(n) \to BU(n)$ has contractible total space (the standard model $EU(n)$ is the infinite Stiefel manifold, contractible by the sliding argument of example 5.2.12), so its long exact sequence (theorem 5.2.9) gives $\pi_i(U(n)) \cong \pi_{i+1}(BU(n))$ for every $i \geq 0$, hence in the colimit $\pi_i(U) \cong \pi_{i+1}(BU) = \pi_{i+1}(\mathbb{Z} \times BU)$ for all $i \geq 0$ (the $\mathbb{Z}$ factor affects only π_0, and $\pi_0(\mathbb{Z} \times BU) = \mathbb{Z}$ plays no role here since the shift lands it at $i = -1$). The right side is $\pi_i(\Omega(\mathbb{Z} \times BU))$ by the loop-space degree shift, so $\pi_i(U) \cong \pi_i(\Omega(\mathbb{Z} \times BU))$ for every i, and these groups realize the equivalence $U \simeq \Omega(\mathbb{Z} \times BU)$, the structure map $KU_n \to \Omega KU_{n+1}$ for n odd.

Second, the even-to-odd structure map. Bott periodicity furnishes $\beta\colon \mathbb{Z} \times BU \xrightarrow{\simeq} \Omega^2(\mathbb{Z} \times BU) = \Omega(\Omega(\mathbb{Z} \times BU)) \simeq \Omega U$ using the equivalence $\Omega(\mathbb{Z} \times BU) \simeq U$ from the previous paragraph. Reading β as a map $\mathbb{Z} \times BU \to \Omega U = \Omega KU_{n+1}$ for n even gives the remaining structure map. Both maps are homotopy equivalences, so $KU = \{KU_n\}$ is an Ω-spectrum, establishing (1).

Step 2: The cohomology theory and its periodicity. By theorem 8.4.6 every Ω-spectrum E defines a reduced generalized cohomology theory $X \mapsto \tilde{E}^n(X) = [X, E_n]_*$, with the suspension isomorphism given by the structure maps:

$$\tilde{E}^n(X) = [X, E_n]_* \cong [X, \Omega E_{n+1}]_* \cong [\Sigma X, E_{n+1}]_* = \tilde{E}^{n+1}(\Sigma X)$$

the middle isomorphism being the loop-suspension adjunction $[X, \Omega Y]_* \cong [\Sigma X, Y]_*$. To extend to negative n and to all integers, set $KU_n := KU_{n+2}$ for $n < 0$ using the equivalence $KU_n \simeq \Omega^2 KU_{n+2}$, which is legitimate because the structure maps are equivalences in both directions; the spectrum is thus a $\mathbb{Z}$-graded family with $KU_n \simeq \Omega^2 KU_{n+2}$ for all n.

Periodicity is now immediate from the Bott equivalence. For any based CW complex X,

$$K^n(X) = [X, KU_n]_* \xrightarrow[\cong]{\beta_*} [X, \Omega^2 KU_{n+2}]_* \cong [X, KU_{n+2}]_* = K^{n+2}(X)$$

where the first isomorphism is post-composition with $\beta\colon KU_n \xrightarrow{\simeq} \Omega^2 KU_{n+2}$ and the second is the identity of spaces $KU_n = KU_{n+2}$ in even degrees (resp. $U = U$ in odd). This is the natural 2-periodicity isomorphism, proving (2).

Step 3: The coefficient groups. The coefficients of a reduced theory are $\tilde{E}^n(S^0) = [S^0, E_n]_* = \pi_0(E_n)$. For the spectrum KU,

$$K^n(\mathrm{pt}) = \pi_0(KU_n) = \begin{cases} \pi_0(\mathbb{Z} \times BU) = \mathbb{Z} & n \text{ even} \\ \pi_0(U) = 0 & n \text{ odd} \end{cases}$$

since BU and U are connected and the $\mathbb{Z}$ factor of KU_n (n even) contributes $\pi_0 = \mathbb{Z}$. The coefficient group $K^n(\mathrm{pt}) = \tilde{K}^n(S^0)$ satisfies $\tilde{K}^{-2k}(S^0) \cong \tilde{K}^0(S^0) \cong \mathbb{Z}$ for every k by periodicity, with the odd groups zero. The Bott class $\beta \in K^{-2}(\mathrm{pt}) \cong \mathbb{Z}$ is the generator corresponding under the suspension isomorphism $\tilde{K}^0(S^2) \cong \tilde{K}^{-2}(S^0) = K^{-2}(\mathrm{pt})$ to the generator $[H] - 1$ of $\tilde{K}^0(S^2)$ computed in example 8.5.2.

Step 4: The coefficient ring is $\mathbb{Z}[\beta, \beta^{-1}]$. The ring structure on $K^*(\mathrm{pt}) = \bigoplus_n K^n(\mathrm{pt})$ comes from the tensor product of bundles, which makes KU a ring spectrum (the multiplication $BU \times BU \to BU$ classifying tensor product, with the unit the trivial bundle). Multiplication by β is the periodicity isomorphism $K^n(\mathrm{pt}) \xrightarrow{\cong} K^{n-2}(\mathrm{pt})$, so β is an invertible element of degree -2, and its inverse $\beta^{-1} \in K^2(\mathrm{pt})$ realizes the reverse isomorphism. The graded ring with $K^0 = \mathbb{Z}$, $K^{-2} = \mathbb{Z}\beta$, $K^2 = \mathbb{Z}\beta^{-1}$, and $\beta\beta^{-1} = 1$, and zero in odd degrees, is precisely the Laurent polynomial ring $\mathbb{Z}[\beta, \beta^{-1}]$ with $|\beta| = -2$. Because $K^n(\mathrm{pt}) = \mathbb{Z} \neq 0$ for every even n, the dimension axiom of definition 8.2.6 (which requires $\tilde{E}^n(S^0) = 0$ for $n \neq 0$) fails maximally, completing the proof.

The theorem is the precise sense in which Bott periodicity converts the bundle-theoretic invariant K^0 into a cohomology theory. The equivalence $\Omega^2(\mathbb{Z} \times BU) \simeq \mathbb{Z} \times BU$ is what lets the single representing space generate an entire Ω-spectrum, and it forces the coefficient ring to be periodic. The contrast with ordinary cohomology is sharp: where $H^*(\mathrm{pt}; \mathbb{Z}) = \mathbb{Z}$ sits in degree zero and nothing else, $K^*(\mathrm{pt}) = \mathbb{Z}[\beta, \beta^{-1}]$ spreads a copy of $\mathbb{Z}$ across every even degree. That spreading is not a defect but the source of K-theory's computational power: classes can be moved up or down by two degrees at will, and invariants like the Chern character $K^0(X) \otimes \mathbb{Q} \cong \bigoplus_k H^{2k}(X; \mathbb{Q})$ become available, relating K-theory to ordinary cohomology through a ring isomorphism after tensoring with $\mathbb{Q}$. Real K-theory KO is built the same way from $\mathbb{Z} \times BO$, but its Bott periodicity has period 8 rather than 2, and its coefficient ring $KO^*(\mathrm{pt})$ contains 2-torsion in degrees 1 and 2; the eightfold pattern $\mathbb{Z}, \mathbb{Z}/2, \mathbb{Z}/2, 0, \mathbb{Z}, 0, 0, 0$ is the real analogue of the twofold $\mathbb{Z}, 0$, recorded (with the real case referred to Atiyah rather than proved) in [21, Bott Periodicity].

The transition to cobordism keeps the same structural goal, a generalized cohomology theory represented by a spectrum, but changes the geometry entirely. Where K-theory measures bundles, cobordism measures manifolds: the relation that two closed n-manifolds are equivalent when they jointly bound a compact $(n+1)$-manifold. The representing object is no longer a Grassmannian but a Thom space, built from the universal bundle over $BO(n)$ by collapsing its boundary at infinity. The first task is to define the Thom space of a vector bundle and to organize the Thom spaces of the universal bundles into a spectrum.

Definition 8.5.4: Thom Space

Let $\pi\colon E \to B$ be a real vector bundle of rank k over a paracompact base, equipped with a fiber metric. The *disk bundle* $D(E)$ is the subspace of vectors of length ≤ 1, and $S(E)$ is the sphere bundle of definition 7.1.6, the vectors of length $= 1$; both are subspaces of vectors of the indicated length respectively. The *Thom space* of E is the quotient

$$\mathrm{Th}(E) \;=\; D(E)/S(E)$$

the disk bundle with its boundary sphere bundle collapsed to a single point, the basepoint ∞. Equivalently, $\mathrm{Th}(E)$ is the one-point compactification of the total space E when B is compact: each fiber $\mathbb{R}^k$ is compactified to S^k and all the points at infinity are identified. The Thom space of the trivial rank-k bundle over B is the k-fold reduced suspension $\mathrm{Th}(\underline{\mathbb{R}}^k) = \Sigma^k(B_+)$, where B_+ is B with a disjoint basepoint adjoined.

In particular, letting $\gamma^k \to BO(k)$ denote the universal rank-k real vector bundle over the classifying space $BO(k)$, the associated universal Thom spaces are denoted

$$MO(k) := \mathrm{Th}(\gamma^k).$$

Let $\gamma^k \to BO(k)$ be the universal rank-k real vector bundle over the infinite Grassmannian $BO(k) = \varinjlim_N \mathrm{Gr}_k(\mathbb{R}^N)$. The Thom spaces $MO(k)$ are linked by structure maps: the inclusion $BO(k) \hookrightarrow BO(k+1)$ classifying $\gamma^k \oplus \underline{\mathbb{R}}$ induces, on Thom spaces, a map $\Sigma\, MO(k) \to MO(k+1)$ (suspension corresponds to adding a trivial line summand, and $\mathrm{Th}(\gamma^k \oplus \underline{\mathbb{R}}) = \Sigma\,\mathrm{Th}(\gamma^k)$).

Definition 8.5.5: Thom Spectrum

Given the above suspension, the resulting sequence

$$MO \;=\; \{MO(k),\ \Sigma\, MO(k) \to MO(k+1)\}_{k\geq 0}$$

is the *Thom spectrum* MO (the unoriented cobordism spectrum). Replacing $BO(k)$ by the complex Grassmannian $BU(k)$ and the universal complex bundle $\gamma^k_{\mathbb{C}}$ (of real rank $2k$) gives the *complex cobordism spectrum* MU, with $MU(k) = \mathrm{Th}(\gamma^k_{\mathbb{C}})$ and structure maps $\Sigma^2 MU(k) \to MU(k+1)$.

The Thom space is the device that turns a bundle into a single pointed space carrying the bundle's twisting in its homotopy type. Collapsing the sphere bundle to a point records exactly the data of how the fibers are glued: a trivial bundle gives a suspension (the simplest possible Thom space), and the deviation of $\mathrm{Th}(E)$ from a suspension measures the nontriviality of E. The structure maps $\Sigma\, MO(k) \to MO(k+1)$ encode that adding a trivial direction to a bundle suspends its Thom space, which is precisely the compatibility a spectrum requires. The reason the universal bundles are used is that every rank-k bundle is pulled back from γ^k, so $MO(k)$ is the universal Thom space, and its homotopy is where all cobordism information collects. Unlike KU, the spectrum MO is not an Ω-spectrum on the nose (the structure maps are not equivalences); it is a suspension-type spectrum whose homotopy groups define the theory, and one passes to an associated Ω-spectrum by the spectrification of section 8.4 when a representing space is wanted.

Example 8.5.6: The Thom Space of the Möbius Bundle

Let $L \to S^1$ be the Möbius line bundle, the nontrivial real line bundle over the circle, pulled back from the universal line bundle $\gamma^1 \to BO(1) = \mathbb{RP}^\infty$ along the inclusion $S^1 = \mathbb{RP}^1 \hookrightarrow \mathbb{RP}^\infty$. Its disk bundle $D(L)$ is the Möbius band, a compact surface with boundary, and its sphere bundle $S(L)$ is the boundary circle (a single circle, double-covering the base, not two circles as for the trivial bundle). The Thom space is

$$\mathrm{Th}(L) = D(L)/S(L) = (\text{Möbius band})/(\text{its boundary})$$

Collapsing the boundary circle of the Möbius band to a point yields the real projective plane:

$$\mathrm{Th}(L) \cong \mathbb{RP}^2$$

Indeed $\mathbb{RP}^2$ is exactly a Möbius band with a disk glued along its boundary, and collapsing the boundary to a point is the same as coning it off, which fills it with a disk that is then crushed; the cell structure $\mathbb{RP}^2 = e^0 \cup e^1 \cup e^2$ matches $\mathrm{Th}(L)$, whose basepoint is e^0, the collapsed boundary contributes e^1, and the open Möbius band contributes e^2.

Contrast the trivial line bundle $\underline{\mathbb{R}} \to S^1$: its disk bundle is the cylinder $S^1 \times [-1, 1]$, its sphere bundle is two circles, and its Thom space is $\mathrm{Th}(\underline{\mathbb{R}}) = \Sigma(S^1_+) = \Sigma(S^1 \sqcup \mathrm{pt}) = S^2 \vee S^1$. The Thom space of the trivial bundle is a suspension, as the definition predicts, while the Thom space $\mathbb{RP}^2$ of the Möbius bundle is not a suspension: its mod-2 cohomology ring $H^*(\mathbb{RP}^2; \mathbb{Z}/2) = \mathbb{Z}/2[\alpha]/(\alpha^3)$ has a nontrivial cup square $\alpha^2 \neq 0$ with $|\alpha| = 1$, whereas the cup product on the reduced cohomology of any suspension ΣY vanishes identically (every reduced class on ΣY is a suspension class, and suspension classes square to zero). The nontriviality of L is visible in the Thom space as exactly this failure to be a suspension.

The example makes the Thom-space construction concrete: a nontrivial bundle produces a nontrivial space ($\mathbb{RP}^2$), and the nontriviality is detected by the cup product structure that distinguishes $\mathbb{RP}^2$ from a suspension. The same phenomenon at the level of the universal bundles is what makes MO a nontrivial spectrum. With the spectrum in hand, the next result states what its homotopy computes: Thom's theorem, which identifies the homotopy groups of MO with the unoriented bordism groups of manifolds. First the bordism relation must be defined.

Two closed (compact, boundaryless) smooth n-manifolds M_0 and M_1 are *cobordant* (or *bordant*) if there is a compact smooth $(n+1)$-manifold W whose boundary is their disjoint union, $\partial W = M_0 \sqcup M_1$. The manifold W is a *cobordism* between them. Cobordism is an equivalence relation: reflexivity uses the cylinder $W = M \times [0, 1]$; symmetry is the symmetry of disjoint union; transitivity glues two cobordisms along their common boundary component (the result is smooth after smoothing the corner). The set of cobordism classes of closed n-manifolds, denoted $\mathfrak{N}_n$, is an abelian group under disjoint union, with identity the class of the empty manifold (equivalently, of any manifold that bounds) and with every element its own inverse, since $M \sqcup M = \partial(M \times [0,1])$ bounds. Thus $\mathfrak{N}_n$ is a vector space over $\mathbb{F}_2$. The graded group $\mathfrak{N}_* = \bigoplus_n \mathfrak{N}_n$ is a ring under Cartesian product of manifolds.

Theorem 8.5.7: Thom's Theorem

For every $n \geq 0$ there is an isomorphism of abelian groups

$$\mathfrak{N}_n \;\cong\; \pi_n(MO) \;=\; \varinjlim_k \pi_{n+k}\big(MO(k)\big)$$

between the unoriented bordism group of closed n-manifolds and the n-th stable homotopy group of the Thom spectrum MO. The isomorphism is multiplicative, identifying the bordism ring $\mathfrak{N}_*$ with the homotopy ring $\pi_*(MO)$. Moreover $\pi_*(MO)$ is a polynomial algebra over $\mathbb{F}_2$,

$$\pi_*(MO) \;\cong\; \mathfrak{N}_* \;\cong\; \mathbb{F}_2[x_i : i \geq 2,\; i \neq 2^j - 1]$$

with one polynomial generator x_i in each degree i that is not of the form $2^j - 1$; the generators in even degrees $i = 2m$ may be taken to be the classes of the real projective spaces $\mathbb{R}\mathrm{P}^{2m}$.

***Proof*:**
The isomorphism $\mathfrak{N}_n \cong \pi_n(MO)$ is the Pontryagin-Thom construction, proved in full as theorem 8.5.8 below; this proof records the two remaining assertions and defers the homotopy-theoretic computation of $\pi_*(MO)$ to a cited source.

Step 1: Multiplicativity. The product on $\mathfrak{N}_*$ is Cartesian product of manifolds, and the product on $\pi_*(MO)$ is induced by the ring-spectrum multiplication $MO \wedge MO \to MO$ coming from the bundle maps $\gamma^j \times \gamma^k \to \gamma^{j+k}$ (Whitney sum of universal bundles, classified by $BO(j) \times BO(k) \to BO(j+k)$). Under the Pontryagin-Thom correspondence of theorem 8.5.8, a closed j-manifold and a closed k-manifold are sent to classes whose smash product is the class of their Cartesian product, because the normal bundle of $M \times N$ in $\mathbb{R}^{j+a} \times \mathbb{R}^{k+b}$ is the external Whitney sum of the normal bundles of M and N, and the collapse map of a product is the smash of the collapse maps. Hence the bijection is a ring isomorphism.

Step 2: The structure of $\pi_*(MO)$ as a polynomial algebra. This is Thom's computation, and it is the one place in the argument where the homotopy theory genuinely outruns the geometry. The method is to compute the mod-2 cohomology $H^*(MO;\mathbb{F}_2)$ and identify it as a module over the Steenrod algebra $\mathcal{A}_2$. The Thom isomorphism $H^*(MO(k);\mathbb{F}_2) \cong H^{*-k}(BO(k);\mathbb{F}_2)$ (cup product with the Thom class $u_k \in H^k(MO(k);\mathbb{F}_2)$, the cohomological avatar of the fiber-integration in theorem 6.3.2) identifies $H^*(MO;\mathbb{F}_2)$ with $H^*(BO;\mathbb{F}_2) = \mathbb{F}_2[w_1, w_2, \ldots]$, a polynomial algebra on the Stiefel-Whitney classes. The action of the Steenrod squares on the Thom class, computed by the Wu formula $\mathrm{Sq}^i(u_k) = w_i \smile u_k$, shows that $H^*(MO;\mathbb{F}_2)$ is a free module over $\mathcal{A}_2$. A free $\mathcal{A}_2$-module structure forces MO to split as a wedge of suspensions of mod-2 Eilenberg-MacLane spectra $H\mathbb{F}_2$, by a theorem of Thom and Milnor; consequently $\pi_*(MO)$ is the polynomial algebra on generators dual to the $\mathcal{A}_2$-module generators, one in each degree i not of the form $2^j - 1$ (the degrees $2^j - 1$ are excluded because the indecomposables w_{2^j-1} are hit by Steenrod operations). The complete computation, including the Wu formula calculation and the ThomMilnor splitting theorem, is carried out in [17, §4 and §§16-17]; reproducing the Steenrod-algebra bookkeeping here would add several pages of mod-2 characteristic-class computation without conceptual gain.

Step 3: The even generators are projective spaces. The class $[\mathbb{R}\mathrm{P}^{2m}] \in \mathfrak{N}_{2m}$ is nonzero and indecomposable for each $m \geq 1$. It is nonzero because the Stiefel-Whitney number $w_{2m}[\mathbb{R}\mathrm{P}^{2m}]$ is odd: the total Stiefel-Whitney class of $\mathbb{R}\mathrm{P}^N$ is $(1+\alpha)^{N+1}$ in $H^*(\mathbb{R}\mathrm{P}^N; \mathbb{F}_2) = \mathbb{F}_2[\alpha]/(\alpha^{N+1})$, so for $N = 2m$ the top class is $w_{2m} = \binom{2m+1}{2m}\alpha^{2m} = (2m+1)\alpha^{2m}$, whose coefficient $2m+1$ is odd, giving $w_{2m}[\mathbb{R}\mathrm{P}^{2m}] = 1 \in \mathbb{F}_2$; and Stiefel-Whitney numbers are cobordism invariants by theorem 8.5.8 (a bounding manifold has all of them zero). It is indecomposable because the characteristic number $s_{2m}[\mathbb{R}\mathrm{P}^{2m}]$, the Newton-polynomial number that vanishes on all products $M^a \times N^b$ with $a, b > 0$, equals $2m+1 \equiv 1 \pmod 2$ for the projective space and so is nonzero; a class with a nonzero s-number cannot be a polynomial in lower generators, hence is indecomposable. The odd projective spaces $\mathbb{R}\mathrm{P}^{2m+1}$ are null-cobordant: $\mathbb{R}\mathrm{P}^{2m+1} = S^{2m+1}/\{\pm 1\}$ is the total space of an S^1-bundle over $\mathbb{C}\mathrm{P}^m$ (the circle action $\lambda \mapsto e^{i\theta}\lambda$ on S^{2m+1} descends to the antipodal-quotient), and this circle bundle is the boundary of its associated disk bundle over $\mathbb{C}\mathrm{P}^m$, so $[\mathbb{R}\mathrm{P}^{2m+1}] = 0$. The polynomial generators $x_{2m} = [\mathbb{R}\mathrm{P}^{2m}]$ therefore fill the even non-(2^j-1) degrees (every even degree avoids the odd numbers $2^j - 1$), as we sought to show.

Thom's theorem is the bridge from differential topology to stable homotopy. The left side, $\mathfrak{N}_n$, is a geometric object: equivalence classes of manifolds under the bounding relation, computable in low dimensions by hand. The right side, $\pi_n(MO)$, is a homotopy-theoretic object: a stable homotopy group of a specific spectrum, computable by the algebraic machine of Steenrod operations. The isomorphism makes each side accessible to the other's tools, and the explicit answer $\mathbb{F}_2[x_2, x_4, x_5, x_6, x_8, \dots]$ means the bordism ring is completely known: $\mathfrak{N}_0 = \mathbb{F}_2$ (a point), $\mathfrak{N}_1 = 0$ (every closed 1-manifold is a union of circles, each bounding a disk), $\mathfrak{N}_2 = \mathbb{F}_2$ generated by $\mathbb{R}\mathrm{P}^2$, $\mathfrak{N}_3 = 0$, $\mathfrak{N}_4 = (\mathbb{F}_2)^2$ generated by $\mathbb{R}\mathrm{P}^4$ and $\mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^2$, and so on. That a manifold's entire unoriented cobordism class is detected by its Stiefel-Whitney numbers, finitely many $\mathbb{F}_2$-valued invariants, is the geometric payoff of the computation.

The heart of the matter, and the result that justifies Thom's theorem, is the Pontryagin-Thom construction. It produces the isomorphism $\mathfrak{N}_n \cong \pi_n(MO)$ directly and geometrically, by turning a manifold into a map and a map into a manifold. The construction rests on two facts: transversality, which guarantees that a generic map can be perturbed to meet a submanifold cleanly, and the collapse map, which compresses a tubular neighborhood into a Thom space.

Theorem 8.5.8: The Pontryagin-Thom Isomorphism

For each $n \geq 0$ the Pontryagin-Thom construction gives mutually inverse bijections

$$\mathfrak{N}_n \underset{\Theta}{\overset{\mathrm{PT}}{\rightleftarrows}} \varinjlim_k \pi_{n+k}(MO(k)) = \pi_n(MO)$$

defined as follows. Given a closed n-manifold M, embed it (by Whitney embedding) in $\mathbb{R}^{n+k}$ for k large, with normal bundle ν of rank k. Let $V \cong$ total space of ν be a tubular neighborhood. The *collapse map*

$$c\colon S^{n+k} = \mathbb{R}^{n+k} \cup \{\infty\} \longrightarrow V/\partial V = \mathrm{Th}(\nu)$$

sends V to $\mathrm{Th}(\nu)$ by the tubular-neighborhood identification and everything outside V to the basepoint ∞. Composing with the classifying map $\mathrm{Th}(\nu) \to MO(k)$ of the normal bundle (which classifies ν as pulled back from γ^k) gives an element $\mathrm{PT}(M) \in \pi_{n+k}(MO(k))$, stable in k. Conversely, given $f\colon S^{n+k} \to MO(k)$, make f transverse to the zero section $BO(k) \subseteq MO(k)$; the preimage $\Theta(f) = f^{-1}(BO(k))$ is a closed n-manifold whose normal bundle is $f^*\gamma^k$. The two constructions are inverse on cobordism and homotopy classes, and the bijection is the isomorphism of theorem 8.5.7.

***Proof*:**
The proof has four stages: the forward map is well defined on cobordism classes; the backward map is well defined on homotopy classes; the two are mutually inverse; and the bijection is additive. The Thom transversality theorem (every smooth map is homotopic, by an arbitrarily small perturbation, to one transverse to a given submanifold, with the homotopy itself takeable transverse) is the technical engine and is used in stages 2 and 3; the full statement and proof, resting on Sard's theorem, are standard differential topology and are given in [16, §§1-7].

Step 1: The forward map PT *is well defined.* Let $M \subseteq \mathbb{R}^{n+k}$ be a closed embedded n-manifold with tubular neighborhood V, identified with the total space of the normal bundle ν (the tubular neighborhood theorem). The one-point compactification $\mathbb{R}^{n+k} \cup \infty = S^{n+k}$ admits the collapse map c that is the identity on V (sending it to $\mathrm{Th}(\nu)$ via $V \cong D(\nu)$ and $\partial V \mapsto \infty$) and sends the complement of V to ∞; this is continuous because the complement of V is closed and maps to the single point ∞, and on $\overline{V}$ the map is the quotient $D(\nu) \to D(\nu)/S(\nu)$. The normal bundle ν of rank k is classified by a map $g\colon M \to BO(k)$ with $g^*\gamma^k \cong \nu$, inducing on Thom spaces $\mathrm{Th}(g)\colon \mathrm{Th}(\nu) \to \mathrm{Th}(\gamma^k) = MO(k)$. The composite

$$S^{n+k} \xrightarrow{c} \mathrm{Th}(\nu) \xrightarrow{\mathrm{Th}(g)} MO(k)$$

is a based map $S^{n+k} \to MO(k)$, hence an element of $\pi_{n+k}(MO(k))$. Increasing the embedding dimension from $\mathbb{R}^{n+k}$ to $\mathbb{R}^{n+k+1}$ replaces ν by $\nu \oplus \underline{\mathbb{R}}$ and the map by its suspension, matching the spectrum structure map $\Sigma MO(k) \to MO(k+1)$; so the class is well defined in the colimit $\pi_n(MO)$.

Independence of the embedding: any two embeddings of M in $\mathbb{R}^{n+k}$ are isotopic for k large (Whitney), and an isotopy carries the tubular neighborhood and collapse map along, giving a homotopy of the resulting maps. Independence of the choice of tubular neighborhood and classifying map is the uniqueness up to homotopy of these choices. Hence $\mathrm{PT}(M) \in \pi_n(MO)$ depends only on M.

Cobordism invariance: suppose M_0 and M_1 are cobordant via $W \subseteq \mathbb{R}^{n+k} \times [0,1]$, a compact $(n+1)$-manifold with $\partial W = M_0 \times \{0\} \sqcup M_1 \times \{1\}$, embedded so that it meets the boundary slabs orthogonally. The normal bundle of W has rank k, and the collapse-map construction applied to W gives a map

$$S^{n+k} \times [0,1] \longrightarrow MO(k)$$

restricting on the two ends to $\mathrm{PT}(M_0)$ and $\mathrm{PT}(M_1)$. This is a homotopy, so $\mathrm{PT}(M_0) = \mathrm{PT}(M_1)$ in $\pi_n(MO)$. Therefore PT descends to a map $\mathfrak{N}_n \to \pi_n(MO)$.

Step 2: The backward map Θ is well defined. Let $f\colon S^{n+k} \to MO(k)$ be a based map. The Grassmannian $BO(k)$ sits inside $MO(k) = \mathrm{Th}(\gamma^k)$ as the zero section (the image of B under $B \to D(E) \to \mathrm{Th}(E)$), a closed submanifold of the manifold part of $MO(k)$ away from the basepoint. By the Thom transversality theorem, f is homotopic to a smooth map f' transverse to $BO(k)$; near the part of S^{n+k} mapping into a neighborhood of $BO(k)$ the map f' is smooth, and transversality makes

$$\Theta(f) := (f')^{-1}(BO(k)) \subseteq S^{n+k}$$

a smooth closed submanifold of dimension $(n+k) - k = n$ (transversality: the preimage of a codimension-k submanifold under a transverse map is codimension k). Its normal bundle in S^{n+k} is $(f')^*(\text{normal bundle of } BO(k) \text{ in } MO(k)) = (f')^*\gamma^k$, since the normal bundle of the zero section B in $\mathrm{Th}(E)$ is E itself. The manifold $\Theta(f)$ lies in $\mathbb{R}^{n+k} = S^{n+k} \setminus \infty$ (the basepoint ∞ maps to the basepoint of $MO(k)$, not into $BO(k)$, so $\infty \notin \Theta(f)$), giving a closed n-manifold embedded in Euclidean space.

Independence of the choice of transverse representative: two maps f', f'' transverse to $BO(k)$ and homotopic to f are homotopic to each other, and the homotopy $H\colon S^{n+k} \times [0,1] \to MO(k)$ can itself be taken transverse to $BO(k)$ (relative transversality), so $H^{-1}(BO(k))$ is a cobordism between $\Theta(f')$ and $\Theta(f'')$. Thus $\Theta(f) \in \mathfrak{N}_n$ depends only on the homotopy class of f, giving a map $\Theta\colon \pi_n(MO) \to \mathfrak{N}_n$.

Step 3: The maps are mutually inverse. First, $\Theta \circ \mathrm{PT} = \mathrm{id}$. Start with a closed n-manifold $M \subseteq \mathbb{R}^{n+k}$, form $\mathrm{PT}(M) = \mathrm{Th}(g) \circ c$. The zero section $BO(k) \subseteq MO(k)$ pulls back under $\mathrm{Th}(g) \circ c$ to exactly M: the collapse map c sends M (the zero section of ν inside V) to the zero section of $\mathrm{Th}(\nu)$, and $\mathrm{Th}(g)$ sends the zero section of $\mathrm{Th}(\nu)$ to the zero section $BO(k)$, while everything off M either lies off the zero section or is sent to the basepoint. The composite is already transverse to $BO(k)$ along M (the collapse map is a diffeomorphism on a neighborhood of M, and g is the classifying map, so the differential is surjective onto the normal directions), so $\Theta(\mathrm{PT}(M)) = M$ as a manifold, hence as a cobordism class.

Second, $\mathrm{PT} \circ \Theta = \mathrm{id}$. Start with $f\colon S^{n+k} \to MO(k)$, taken transverse to $BO(k)$, and set $N = \Theta(f) = f^{-1}(BO(k))$, a closed n-manifold with normal bundle $\nu_N = f^*\gamma^k$. Near N the map f looks like the composite of the projection of a tubular neighborhood of N onto N followed by the bundle map $\nu_N \to \gamma^k$ classifying ν_N (transversality gives a tubular-neighborhood model in which f is fiberwise linear and isomorphic to the normal-bundle data). Away from a tubular neighborhood of N, the map f misses a neighborhood of the zero section $BO(k)$, so it can be homotoped (pushing along the radial direction of the Thom space toward the basepoint) to send the complement of the tubular neighborhood to the basepoint ∞. After this homotopy f is exactly the collapse map of the tubular neighborhood of N composed with the classifying map of ν_N, which is $\mathrm{PT}(N)$. Hence $\mathrm{PT}(\Theta(f)) = [f]$ in $\pi_n(MO)$.

Step 4: Additivity. The group operation on $\mathfrak{N}_n$ is disjoint union, and on $\pi_n(MO)$ it is the sum of homotopy classes (concatenation in the sphere). If $M = M_0 \sqcup M_1$ is a disjoint union, embed

M_0 and M_1 in disjoint half-spaces of $\mathbb{R}^{n+k}$; the collapse map of the union is the based sum of the collapse maps (the two Thom-space images sit in disjoint hemispheres of S^{n+k} and the map factors through the pinch $S^{n+k} \to S^{n+k} \vee S^{n+k}$), so $\mathrm{PT}(M_0 \sqcup M_1) = \mathrm{PT}(M_0) + \mathrm{PT}(M_1)$. Thus PT and Θ are mutually inverse group isomorphisms, and the bijection $\mathfrak{N}_n \cong \pi_n(MO)$ is the isomorphism asserted in theorem 8.5.7, completing the proof.

The Pontryagin-Thom construction is the geometric content of all of cobordism theory in a single picture: a manifold M is encoded by the way a sphere wraps around its normal-bundle Thom space, and conversely a map of spheres into a Thom space cuts out a manifold as the transverse preimage of the zero section. Transversality is what makes the dictionary two-sided. It guarantees that the cutting-out operation $f \mapsto f^{-1}(BO(k))$ produces a manifold and that homotopic maps cut out cobordant manifolds, so that homotopy classes and cobordism classes correspond exactly. The collapse map is the reverse direction, packaging the local geometry of a tubular neighborhood into a single based map. The construction adapts to other tangential structures: replacing the universal bundle γ^k over $BO(k)$ by other classifying spaces yields a Thom spectrum for each, and in each case bordism of so-structured manifolds is the homotopy of that spectrum. The oriented case is carried out in theorem 8.5.11 below; complex structures give MU and complex cobordism, and stable framings give the sphere spectrum, recovering framed bordism. The case of framed cobordism, where the Thom spectrum is the sphere spectrum $\mathbb{S}$, recovers the original Pontryagin theorem $\Omega_n^{\mathrm{fr}} \cong \pi_n^s$ identifying framed bordism with the stable homotopy groups of spheres.

Stiefel-Whitney numbers were used twice above, in Step 3 of the proof of theorem 8.5.7 to detect $[\mathbb{R}\mathrm{P}^{2m}]$ and in the discussion following it, on the strength of their invariance under cobordism. That invariance, together with its converse, is a corollary of Thom's theorem, and it is the form in which the theorem applies to a manifold presented concretely: the entire cobordism class is computed by finitely many mod-2 evaluations of characteristic classes of the tangent bundle.

Corollary 8.5.9: Stiefel-Whitney Numbers Are a Complete Unoriented Cobordism Invariant

Let M be a closed smooth n-manifold and let $w_I[M] \in \mathbb{F}_2$ be its Stiefel-Whitney numbers, indexed by the partitions I of n (definition 7.3.4). Then M bounds a compact smooth $(n+1)$-manifold if and only if $w_I[M] = 0$ for every such I. Equivalently, two closed smooth n-manifolds M and N are cobordant if and only if $w_I[M] = w_I[N]$ for every such monomial.

***Proof*:**
Step 1: A manifold that bounds has all Stiefel-Whitney numbers zero. Let W be a compact smooth $(n+1)$-manifold with $\partial W = M$, and let $j\colon M \hookrightarrow W$ be the inclusion. A collar of the boundary gives a nowhere-zero outward normal field along M, splitting off a trivial line summand:

$$j^*TW \;\cong\; TM \oplus \underline{\mathbb{R}}$$

A trivial bundle has total Stiefel-Whitney class 1 (it is pulled back from a point, so $w = 1$ by proposition 7.4.1), so the Whitney sum formula (theorem 7.4.5) and naturality give $j^*w(TW) = w(TM \oplus \underline{\mathbb{R}}) = w(TM)$. Since j^* is a ring homomorphism, every monomial satisfies $w_I(TM) = j^*\big(w_I(TW)\big)$: the classes being evaluated all restrict from W.

The fundamental class restricts too, in the dual sense. Every manifold is $\mathbb{F}_2$-orientable (definition 4.4.4 with $R = \mathbb{F}_2$, where 1 is the only generator, so the local consistency condition holds automatically), and a compact smooth manifold has a collared boundary, so W carries a relative fundamental class $[W,M] \in H_{n+1}(W,M;\mathbb{F}_2)$ by proposition 4.5.15. The connecting homomorphism $\partial\colon H_{n+1}(W,M;\mathbb{F}_2) \to H_n(M;\mathbb{F}_2)$ of the pair sends $[W,M]$ to $[M]$, and this is checked in the collar. Over $M \times [0,1)$ the pair (W,M) is identified with $(M \times [0,1), M \times \{0\})$, and under that identification $[W,M]$ corresponds to the cross product of $[M]$ with the generator of $H_1([0,1),\{0\};\mathbb{F}_2)$, whose image under ∂ is $[M]$. So $\partial[W,M]$ restricts to the nonzero element of $H_n(M, M \setminus \{x\};\mathbb{F}_2) \cong \mathbb{F}_2$ at every $x \in M$, and the uniqueness clause of definition 4.4.8 identifies it with $[M]$.

Now fix a monomial w_I of degree n and set $x = w_I(TW) \in H^n(W;\mathbb{F}_2)$. With field coefficients the cohomology sequence of the pair is the vector-space dual of the homology sequence, so the two connecting homomorphisms are adjoint under the Kronecker pairing, $\langle y, \partial c\rangle = \langle \delta y, c\rangle$. Applying this to $y = j^*x$ and $c = [W,M]$,

$$w_I[M] \;=\; \big\langle j^*x,\ \partial[W,M]\big\rangle \;=\; \big\langle \delta\big(j^*x\big),\ [W,M]\big\rangle$$

and the composite $H^n(W;\mathbb{F}_2) \xrightarrow{j^*} H^n(M;\mathbb{F}_2) \xrightarrow{\delta} H^{n+1}(W,M;\mathbb{F}_2)$ is zero by exactness of the cohomology sequence of (W,M). Hence $\delta(j^*x) = 0$ and $w_I[M] = 0$.

Step 2: Vanishing Stiefel-Whitney numbers force the class to vanish. This direction is the one that uses Thom's computation, and it uses exactly the splitting recorded in Step 2 of the proof of theorem 8.5.7: the mod-2 cohomology $H^*(MO;\mathbb{F}_2)$ is a free module over the Steenrod algebra $\mathcal{A}_2$, and a Thom spectrum with that property splits as a wedge

$$MO \;\simeq\; \bigvee_j \Sigma^j H\mathbb{F}_2$$

of suspensions of the mod-2 Eilenberg-MacLane spectrum, by the theorem of Thom and Milnor, for which the source is [23]. For such a wedge the mod-2 Hurewicz map is injective: on the summand $\Sigma^j H\mathbb{F}_2$ it is the map $\pi_n(\Sigma^j H\mathbb{F}_2) \to H_n(\Sigma^j H\mathbb{F}_2;\mathbb{F}_2)$, which in the bottom degree $n = j$ is the identity $\mathbb{F}_2 \to \mathbb{F}_2$ and in every other degree has trivial source. So

$$h\colon \pi_n(MO) \longrightarrow H_n(MO;\mathbb{F}_2)$$

is injective, and a class in $\pi_n(MO)$ is determined by its image in mod-2 homology.

It remains to identify that image for $\mathrm{PT}(M)$. Represent $\mathrm{PT}(M)$ by the composite $S^{n+k} \xrightarrow{c} \mathrm{Th}(\nu) \xrightarrow{\mathrm{Th}(g)} MO(k)$ of theorem 8.5.8, with ν the normal bundle of an embedding $M \subseteq \mathbb{R}^{n+k}$ and g its classifying map. The collapse map carries the mod-2 fundamental class of S^{n+k} to the Thom-isomorphism image of $[M]$, because c is a relative homeomorphism onto $\mathrm{Th}(\nu)$ over the tubular neighbourhood. Applying $\mathrm{Th}(g)_*$ and the Thom isomorphism $H_{n+k}(MO(k);\mathbb{F}_2) \cong H_n(BO(k);\mathbb{F}_2)$ then identifies $h(\mathrm{PT}(M))$ with $g_*[M]$, an element of $H_n(BO;\mathbb{F}_2)$ once the colimit in k is taken. Pairing $g_*[M]$ against a monomial in the universal Stiefel-Whitney classes gives $\langle w_I(\nu), [M]\rangle$, a Stiefel-Whitney number of the normal bundle. Since $H^n(BO;\mathbb{F}_2)$ is the degree-n part of the polynomial algebra $\mathbb{F}_2[w_1, w_2, \ldots]$ and $H_n(BO;\mathbb{F}_2)$ is its finite-dimensional dual, these pairings determine $g_*[M]$ completely. Finally $TM \oplus \nu$ is trivial, so $w(\nu) = w(TM)^{-1}$ by theorem 7.4.5, and each normal number is a universal polynomial in the tangential ones and conversely. The tangential Stiefel-Whitney numbers therefore determine $g_*[M]$, hence $h(\mathrm{PT}(M))$, hence $\mathrm{PT}(M)$, hence the cobordism class of M by theorem 8.5.8. In particular all numbers zero forces $\mathrm{PT}(M) = 0$, which says that M bounds.

Step 3: The two formulations are equivalent. A disjoint union has $T(M \sqcup N)$ restricting to TM and to TN and fundamental class $[M] + [N]$, so $w_I[M \sqcup N] = w_I[M] + w_I[N]$. Every element of $\mathfrak{N}_n$ is its own inverse, so M and N are cobordant precisely when $M \sqcup N$ bounds, which by the first formulation happens precisely when $w_I[M] + w_I[N] = 0$ for every I, that is, when $w_I[M] = w_I[N]$ in $\mathbb{F}_2$, as we sought to show.

The corollary is what makes Thom's theorem usable. Deciding whether a given closed n-manifold bounds is a question about all compact $(n+1)$-manifolds, of which there is no list. The corollary replaces that question by the evaluation of finitely many cup products of Stiefel-Whitney classes against the mod-2 fundamental class, each of which is a computation inside $H^*(M;\mathbb{F}_2)$. The two halves of the proof are not of comparable depth. The vanishing direction uses only the collar splitting $TW|_M \cong TM \oplus \underline{\mathbb{R}}$ and exactness of the sequence of the pair, while the converse is a restatement of the wedge decomposition of MO, and no geometric argument for it is known that avoids the homotopy theory.

The construction just carried out ignores orientations, and the group it computes is entirely 2-torsion for a single geometric reason: the cylinder $M \times [0,1]$ has boundary $M \sqcup M$. Requiring the manifolds and the cobordisms between them to be oriented removes that reason, since the oriented cylinder has boundary $M \sqcup (-M)$ instead. The Thom spectrum built from the oriented Grassmannians computes the group that results.

Definition 8.5.10: The Oriented Thom Spectrum and Oriented Bordism

Let $BSO(k)$ be the classifying space of the special orthogonal group, and let $\widetilde{\gamma}^k \to BSO(k)$ be the universal oriented rank-k real vector bundle, the pullback of γ^k along the double cover $BSO(k) \to BO(k)$ carrying its tautological orientation. The associated Thom spaces (definition 8.5.4) are

$$MSO(k) := \mathrm{Th}\big(\widetilde{\gamma}^k\big)$$

The inclusion $BSO(k) \hookrightarrow BSO(k+1)$ classifies $\widetilde{\gamma}^k \oplus \underline{\mathbb{R}}$, oriented by an oriented frame of $\widetilde{\gamma}^k$ followed by the positive generator of the trivial line, and induces on Thom spaces a map $\Sigma\, MSO(k) \to MSO(k+1)$. The resulting sequence

$$MSO \;=\; \{MSO(k),\ \Sigma\, MSO(k) \to MSO(k+1)\}_{k \geq 0}$$

is the *oriented Thom spectrum*, formed exactly as MO was in definition 8.5.5.

For a closed oriented smooth n-manifold N (definition 4.4.4 with $R = \mathbb{Z}$), write $-N$ for the same smooth manifold with the reversed orientation, whose fundamental class is $-[N]$ by theorem 4.4.9. Two closed oriented smooth n-manifolds M and N are *oriented-cobordant* if $M \sqcup (-N)$ is the boundary of a compact oriented smooth $(n+1)$-manifold W, the orientation induced on ∂W agreeing with the given ones. This is an equivalence relation, and the set of equivalence classes is the *oriented bordism group*

$$\Omega_n^{SO}$$

an abelian group under disjoint union whose identity is the class of the oriented manifolds that bound and in which the inverse of the class of M is the class of $-M$, since $M \sqcup (-M) = \partial(M \times [0,1])$ with the boundary orientation. The graded group $\Omega_*^{SO} = \bigoplus_{n \geq 0} \Omega_n^{SO}$ is a ring under Cartesian product with the product orientation.

The structural difference between Ω_*^{SO} and $\mathfrak{N}_*$ is already in the definition. Doubling gives only

$[M]+[-M]=0$, not $2[M]=0$, so there is no reason for Ω_*^{SO} to be an $\mathbb{F}_2$-algebra, and it is not one: the class of $\mathbb{C}\mathrm{P}^2$ has infinite order in Ω_4^{SO}. The invariant that shows this is a Pontryagin number, and its cobordism invariance is Step 1 of corollary 8.5.9 run with integral coefficients. Along the boundary of a compact oriented W one has $j^*TW \cong TM \oplus \underline{\mathbb{R}}$, whose complexification is $(TM)_{\mathbb{C}} \oplus \underline{\mathbb{C}}$, so the complex Whitney sum formula (theorem 7.4.5) with $c(\underline{\mathbb{C}})=1$ gives $j^*p(TW)=p(TM)$ exactly, with no 2-torsion ambiguity. When $M=\partial W$ carries the induced orientation one also has $\partial[W,M]=[M]$ integrally, so the same exactness argument makes every Pontryagin number of M vanish. Since $\langle p_1(T\mathbb{C}\mathrm{P}^2),[\mathbb{C}\mathrm{P}^2]\rangle = 3$ by example 7.3.6, and this pairing is additive under disjoint union and changes sign under orientation reversal, no nonzero multiple of $[\mathbb{C}\mathrm{P}^2]$ bounds.

The Pontryagin-Thom construction adapts to oriented manifolds, and the places where the adaptation changes the argument are the places where Ω_*^{SO} differs from $\mathfrak{N}_*$. Two of the four steps of the proof of theorem 8.5.8 are reused verbatim. The new content sits in Step 1, where the normal bundle inherits an orientation and the classifying map lifts to $BSO(k)$, in Step 2, where the transverse preimage comes out oriented, and in Step 4, where the inverse of a class is no longer the class itself.

Theorem 8.5.11: Oriented Bordism and the Spectrum MSO

Let $n \geq 0$.

1. The Pontryagin-Thom construction gives an isomorphism of abelian groups
$$\Omega_n^{SO} \;\cong\; \pi_n(MSO) \;=\; \varinjlim_k \pi_{n+k}\big(MSO(k)\big)$$
between the oriented bordism group of definition 8.5.10 and the n-th stable homotopy group of the oriented Thom spectrum. The isomorphism is multiplicative, identifying the ring Ω_*^{SO} with $\pi_*(MSO)$.

2. Rationally, the oriented bordism ring is a polynomial algebra on the even complex projective spaces,
$$\Omega_*^{SO} \otimes \mathbb{Q} \;\cong\; \mathbb{Q}\big[[\mathbb{C}\mathrm{P}^2],[\mathbb{C}\mathrm{P}^4],[\mathbb{C}\mathrm{P}^6],\dots\big]$$
with $[\mathbb{C}\mathrm{P}^{2m}]$ in degree $4m$, so that $\Omega_n^{SO}\otimes\mathbb{Q}=0$ unless 4 divides n. Two closed oriented n-manifolds have the same class in $\Omega_n^{SO}\otimes\mathbb{Q}$ if and only if all their Pontryagin numbers agree, so the Pontryagin numbers are a complete rational invariant of oriented cobordism.

***Proof*:**
This establishes clause (1). Clause (2) is treated separately below. The four steps that follow are the four steps of the proof of theorem 8.5.8, each recording what carries over unchanged and what the orientation adds.

Step 1: The normal bundle inherits an orientation, and the forward map lands in $MSO(k)$. Let M be a closed oriented n-manifold, embedded in $\mathbb{R}^{n+k}$ for k large with normal bundle ν of rank k, so that $TM \oplus \nu \cong \underline{\mathbb{R}}^{n+k}$ over M. Here the orientation enters. The orientation of M orients each tangent space T_xM, and the standard orientation of $\mathbb{R}^{n+k}$ orients each fibre of the trivial bundle, so each fibre ν_x receives the unique orientation for which an oriented basis of T_xM followed by an oriented basis of ν_x is an oriented basis of $\mathbb{R}^{n+k}$. Both input orientations are locally constant in x, so the resulting choice is locally constant as well, and ν is an oriented rank-k bundle. Consequently its classifying map lifts across the double cover $BSO(k) \to BO(k)$

to a map $g\colon M \to BSO(k)$ with $g^*\widetilde{\gamma}^k \cong \nu$ as oriented bundles, uniquely up to homotopy once the orientation of ν is fixed, since lifting across a covering is unique given the lift of one point in each component. The collapse map is untouched: it is built from the tubular neighbourhood alone and refers to no orientation. So the composite

$$S^{n+k} \xrightarrow{c} \mathrm{Th}(\nu) \xrightarrow{\mathrm{Th}(g)} MSO(k)$$

is a class $\mathrm{PT}(M) \in \pi_{n+k}(MSO(k))$. Raising the embedding dimension by one replaces ν by $\nu \oplus \underline{\mathbb{R}}$ with precisely the orientation used to define the structure map of MSO in definition 8.5.10, so the class is stable and defines $\mathrm{PT}(M) \in \pi_n(MSO)$. Independence of the embedding is the isotopy argument of theorem 8.5.8, unchanged, since an isotopy carries the orientation of the normal bundle along with the bundle.

Cobordism invariance needs the sign bookkeeping made explicit. Suppose M_0 and M_1 are oriented-cobordant, and realize the cobordism as a compact oriented $W \subseteq \mathbb{R}^{n+k} \times [0,1]$ meeting the two boundary slabs orthogonally, with M_0 at $t = 0$ and M_1 at $t = 1$. With the outward-normal-first convention, the orientation induced by W on its boundary agrees with the given orientation of M_1 and is opposite to that of M_0, because the outward normal points in the $-t$ direction at one end and the $+t$ direction at the other. That is exactly the statement $\partial W = (-M_0) \sqcup M_1$, which is the relation defining oriented cobordism. Orient the normal bundle of W in $\mathbb{R}^{n+k+1}$ by the rule above. At each end the two reversals compensate, and the induced orientation of ν_W restricts to the orientation of ν_{M_i} determined by the given orientation of M_i. The collapse construction applied to W therefore gives a homotopy $S^{n+k} \times [0,1] \to MSO(k)$ from $\mathrm{PT}(M_0)$ to $\mathrm{PT}(M_1)$, and PT descends to $\Omega_n^{SO} \to \pi_n(MSO)$.

Step 2: Transversality is unchanged, and the preimage comes out oriented. The Thom transversality theorem is a statement about smooth maps and smooth submanifolds and takes no account of orientations, so nothing in it needs adjusting: a based map $f\colon S^{n+k} \to MSO(k)$ is homotoped to a map f' transverse to the zero section $BSO(k) \subseteq MSO(k)$, and $\Theta(f) = (f')^{-1}(BSO(k)) \subseteq \mathbb{R}^{n+k}$ is a closed n-manifold with normal bundle $(f')^*\widetilde{\gamma}^k$, by the argument of Step 2 of theorem 8.5.8 with $BO(k)$ replaced by $BSO(k)$ throughout. What is new is the output. The bundle $\widetilde{\gamma}^k$ is oriented, so its pullback is, and reading the rule of Step 1 backwards, the orientation of the normal bundle together with the standard orientation of $\mathbb{R}^{n+k}$ orients $T\Theta(f)$. The transverse preimage is thus a closed *oriented* n-manifold, where in the unoriented setting it was only a manifold, and the orientation is given by the universal bundle rather than chosen. Independence of the transverse representative is the relative-transversality argument as before, with $H^{-1}(BSO(k))$ oriented by the same rule. The ends of that cobordism carry the orientations of $\Theta(f')$ and $\Theta(f'')$ with the boundary signs of the previous paragraph, so it is an oriented cobordism.

Step 3: The two maps are still mutually inverse. Both halves of Step 3 of the proof of theorem 8.5.8 are arguments about manifolds, tubular neighbourhoods, and homotopies, and no orientation appears anywhere in them, so they are reused word for word. What must be added is that the orientations correspond. For $\Theta \circ \mathrm{PT}$, the manifold recovered from $\mathrm{PT}(M)$ is M, and the orientation it acquires in Step 2 is the one determined by the orientation of ν together with that of $\mathbb{R}^{n+k}$. But the orientation of ν was itself defined from the orientation of M by the same rule in Step 1, and the rule is an involution, so the recovered orientation is the original one. For $\mathrm{PT} \circ \Theta$, the homotopy pushing the complement of a tubular neighbourhood to the basepoint moves no orientation data, and the classifying map it exhibits is the oriented classifying map of ν_N by the construction of Step 2.

Step 4: Additivity, and where the inverses now come from. Embedding M_0 and M_1 in disjoint half-spaces of $\mathbb{R}^{n+k}$ makes the collapse map of the disjoint union the based sum of the two collapse maps, exactly as in Step 4 of the proof of theorem 8.5.8, and the orientations of the two pieces are independent of one another. Hence $\mathrm{PT}(M_0 \sqcup M_1) = \mathrm{PT}(M_0) + \mathrm{PT}(M_1)$ and PT is a homomorphism, and by Steps 2 and 3 it is a bijection, so it is an isomorphism $\Omega_n^{SO} \cong \pi_n(MSO)$.

The structural difference between the two theories is located here. In $\mathfrak{N}_n$ the identity $M \sqcup M = \partial(M \times [0,1])$ makes every element its own inverse, which is what makes $\mathfrak{N}_*$ an $\mathbb{F}_2$-vector space. Orienting the cylinder gives $\partial(M \times [0,1]) = M \sqcup (-M)$ instead, so the same manifold now yields only that the inverse of the class of M is the class of $-M$. On the homotopy side no separate argument is needed: PT is a homomorphism and vanishes on oriented boundaries, so

$$\mathrm{PT}(M) + \mathrm{PT}(-M) \;=\; \mathrm{PT}\big(M \sqcup (-M)\big) \;=\; 0$$

which says $\mathrm{PT}(-M) = -\mathrm{PT}(M)$: orientation reversal on the geometric side is negation in $\pi_n(MSO)$.

Multiplicativity is Step 1 of the proof of theorem 8.5.7 with BSO in place of BO. The ring-spectrum multiplication $MSO \wedge MSO \to MSO$ comes from the maps $BSO(j) \times BSO(k) \to BSO(j+k)$ classifying the Whitney sum of the two universal oriented bundles with its product orientation, the normal bundle of $M \times N$ is the external Whitney sum of the two normal bundles with the product orientation, and the collapse map of a product is the smash of the collapse maps. So the bijection is a ring isomorphism, completing the proof of clause (1).

***Proof Sketch*:**
Clause (2) is stated without proof, and the reason is the one that put the computation of $\pi_*(MO)$ outside the argument in Step 2 of the proof of theorem 8.5.7. The route to $\Omega_*^{SO} \otimes \mathbb{Q}$ runs through the rational Hurewicz isomorphism

$$\pi_*(MSO) \otimes \mathbb{Q} \;\cong\; H_*(MSO; \mathbb{Q})$$

which belongs to Serre's theory of classes of abelian groups and to rational homotopy theory, machinery this book does not develop.

Granting that identification, the remainder is characteristic-class bookkeeping already available here. The Thom isomorphism identifies $H_*(MSO;\mathbb{Q})$ with $H_*(BSO;\mathbb{Q})$, and theorem 7.3.7 in its stable limit gives $H^*(BSO;\mathbb{Q}) = \mathbb{Q}[p_1, p_2, \ldots]$ with $|p_i| = 4i$: the Euler class present in $H^*(BSO(2m);\mathbb{Q})$ does not persist in the limit, since the proof of that theorem records that it restricts to 0 in odd rank. So $\Omega_*^{SO} \otimes \mathbb{Q}$ is a polynomial algebra on one generator in each degree $4m$, and in particular vanishes in degrees not divisible by 4. The even projective spaces give the generators: the total Pontryagin class of $T\mathbb{C}\mathrm{P}^N$ is $(1+u^2)^{N+1}$ in $H^*(\mathbb{C}\mathrm{P}^N;\mathbb{Z}) = \mathbb{Z}[u]/(u^{N+1})$, whose Pontryagin roots are $N+1$ copies of u^2, so the Newton-polynomial number in the Pontryagin classes of $\mathbb{C}\mathrm{P}^{2m}$ is $s_m[\mathbb{C}\mathrm{P}^{2m}] = 2m+1 \neq 0$ in $\mathbb{Q}$. Since s_m vanishes on decomposables, a class with nonzero s-number is indecomposable, exactly as for $\mathbb{R}\mathrm{P}^{2m}$ in Step 3 of the proof of theorem 8.5.7, and the classes $[\mathbb{C}\mathrm{P}^{2m}]$ are polynomial generators. The statement about Pontryagin numbers follows from the same identification: under the Thom isomorphism the pairings of $H^*(BSO;\mathbb{Q}) = \mathbb{Q}[p_1, p_2, \ldots]$ against the Hurewicz image of a class are its Pontryagin numbers, and these pairings span the space of $\mathbb{Q}$-linear functionals on $\Omega_n^{SO} \otimes \mathbb{Q}$, so two classes agree rationally exactly when all their Pontryagin numbers agree. The full computation is Thom's [23], and [17] presents it in the characteristic-class language used here, completing the sketch.

The two clauses split the work the same way the unoriented case did. Clause (1) is geometry: an isomorphism between a set of manifolds modulo a relation and a stable homotopy group, proved by an explicit construction in both directions. Clause (2) is homotopy theory, and it is where the answer is computed. The two coefficient rings then separate the theories. The ring $\mathfrak{N}_*$ is a polynomial algebra over $\mathbb{F}_2$ with generators in all degrees not of the form $2^j - 1$, while $\Omega_*^{SO} \otimes \mathbb{Q}$ is a polynomial algebra over $\mathbb{Q}$ with generators only in degrees divisible by 4. Clause (2) describes the rationalization alone, and whatever torsion Ω_*^{SO} carries is invisible to it, since a Pontryagin number is a homomorphism $\Omega_n^{SO} \to \mathbb{Z}$ and therefore kills every torsion class. Low degrees show the difference against $\mathfrak{N}_*$ immediately.

Example 8.5.12: Oriented Bordism in Degrees at Most Four

In degree 0 a closed oriented manifold is a finite set of points each carrying a sign, and the signed count is an isomorphism

$$\Omega_0^{SO} \;\cong\; \mathbb{Z}$$

It is additive under disjoint union, and a signed point set bounds a compact oriented 1-manifold exactly when the count is zero, since such a 1-manifold is a union of circles and arcs and each arc contributes one point of each sign.

In degrees 1, 2 and 3 the groups vanish. Every closed oriented 1-manifold is a union of circles, each bounding a disk; every closed oriented surface bounds a handlebody; and every closed oriented 3-manifold bounds a compact oriented 4-manifold, which is not elementary and is part of Thom's computation [23]. So

$$\Omega_1^{SO} = \Omega_2^{SO} = \Omega_3^{SO} = 0$$

matching clause (2) of theorem 8.5.11, which forces these groups to be torsion, and Thom's integral computation, which shows the torsion is trivial here.

In degree 4, $\Omega_4^{SO} \cong \mathbb{Z}$ generated by $[\mathbb{C}\mathrm{P}^2]$, again by [23]. Clause (2) gives only the rationalization $\Omega_4^{SO} \otimes \mathbb{Q} \cong \mathbb{Q}$, and the absence of torsion is part of Thom's integral computation. The generator is pinned down by the signature. For a closed oriented $4k$-manifold M the cup-product pairing $(x, y) \mapsto \langle x \smile y, [M]\rangle$ on $H^{2k}(M; \mathbb{R})$ is symmetric, and it is nondegenerate by Poincaré duality (theorem 4.5.7) in the form corollary 4.5.6, which identifies the pairing with the Kronecker pairing against the duality isomorphism. Its *signature* $\sigma(M)$ is the number of positive eigenvalues minus the number of negative ones. It is additive under disjoint union, changes sign under orientation reversal, and vanishes on oriented boundaries [23], so it descends to a homomorphism $\sigma\colon \Omega_{4k}^{SO} \to \mathbb{Z}$. On $\mathbb{C}\mathrm{P}^2$ the group $H^2(\mathbb{C}\mathrm{P}^2; \mathbb{R})$ is spanned by the hyperplane class u with $\langle u^2, [\mathbb{C}\mathrm{P}^2]\rangle = 1$, giving a 1×1 form with entry $+1$, so $\sigma(\mathbb{C}\mathrm{P}^2) = 1$. Hence $\sigma\colon \Omega_4^{SO} \to \mathbb{Z}$ is surjective, therefore an isomorphism, and $[\mathbb{C}\mathrm{P}^2]$ generates. The relation $\sigma(\mathbb{C}\mathrm{P}^2) = \frac{1}{3}\langle p_1(T\mathbb{C}\mathrm{P}^2), [\mathbb{C}\mathrm{P}^2]\rangle$ noted in example 7.3.6 is the first case of the Hirzebruch signature theorem, which expresses σ on Ω_{4k}^{SO} as a universal rational polynomial in the Pontryagin numbers.

Against the unoriented groups $\mathfrak{N}_0 = \mathbb{F}_2$, $\mathfrak{N}_1 = 0$, $\mathfrak{N}_2 = \mathbb{F}_2$, $\mathfrak{N}_3 = 0$, $\mathfrak{N}_4 = (\mathbb{F}_2)^2$ the pattern is different in both directions: $\Omega_2^{SO} = 0$ while $\mathfrak{N}_2 = \mathbb{F}_2$, since the generator $\mathbb{R}\mathrm{P}^2$ of the latter is not orientable and so has no class at all in oriented bordism, and $\Omega_4^{SO} = \mathbb{Z}$ is infinite while $\mathfrak{N}_4$ is finite.

The example records the two effects of adding an orientation to the data. It removes manifolds from the theory: the generator $\mathbb{R}\mathrm{P}^2$ of $\mathfrak{N}_2$ is non-orientable, so it represents nothing at all in Ω_2^{SO}, which is zero. It also refines the relation on those that remain, so classes of infinite order appear where before every element had order two. The invariants that come with the refinement are the Pontryagin numbers and, in dimensions divisible by four, the signature that the Hirzebruch theorem expresses in terms of them. Neither is available in $\mathfrak{N}_*$, where the coefficients are $\mathbb{F}_2$ and there is no orientation to pair against. Complex cobordism MU, defined alongside MO in definition 8.5.5, is the next step in the same direction, replacing the orientation of the stable normal bundle by a complex structure on it.

Both theories now stand assembled as generalized cohomology theories obtained by dropping the dimension axiom. K-theory K^* is the cohomology theory represented by the Ω-spectrum KU, with coefficient ring $\mathbb{Z}[\beta, \beta^{-1}]$ periodic of period 2; its violation of the dimension axiom is the periodicity, a copy of $\mathbb{Z}$ in every even degree. Cobordism MO^* (and MU^*) is the cohomology theory represented by the Thom spectrum MO (resp. MU), with coefficient ring $\pi_*(MO) = \mathbb{F}_2[x_i]$ the polynomial bordism ring; its violation of the dimension axiom is the nonvanishing of $\pi_n(MO) = \mathfrak{N}_n$ in infinitely many positive degrees. These are the two prototypes: K-theory the periodic, bundle-theoretic theory whose coefficients are simplest, and cobordism the geometric, manifold-theoretic theory whose coefficients carry the entire classification of manifolds up to bounding. Every generalized cohomology theory is, by Brown representability (theorem 8.3.9), represented by a spectrum, and these two spectra are the templates against which the general theory of section 8.6 measures all others.

Exercise 8.5.1

1. Using the clutching description of example 8.5.2, compute $\tilde{K}^0(S^1)$ and $\tilde{K}^0(S^3)$, and confirm they vanish. Explain how this is consistent with the coefficient pattern $K^n(\mathrm{pt}) = \mathbb{Z}$ for n even and 0 for n odd from theorem 8.5.3, via the identification $\tilde{K}^0(S^m) \cong K^{-m}(\mathrm{pt})$.
2. Verify directly that the Thom space of the trivial rank-k bundle $\underline{\mathbb{R}}^k \to B$ over a compact base is the k-fold reduced suspension $\Sigma^k(B_+)$, where $B_+ = B \sqcup \{\mathrm{pt}\}$. Use this to recompute $\mathrm{Th}(\underline{\mathbb{R}} \to S^1)$ and confirm it agrees with the value $S^2 \vee S^1$ stated in example 8.5.6.
3. Show that every closed 1-manifold is null-cobordant, hence $\mathfrak{N}_1 = 0$, and confirm this against Thom's theorem (theorem 8.5.7): check that the polynomial algebra $\mathbb{F}_2[x_i : i \geq 2,\ i \neq 2^j - 1]$ has no generator in degree 1 and that there is no product of generators landing in degree 1.
4. In the Pontryagin-Thom construction (theorem 8.5.8), explain why transversality of $f\colon S^{n+k} \to MO(k)$ to the zero section $BO(k)$ forces the preimage $f^{-1}(BO(k))$ to have dimension exactly n, and identify its normal bundle. Then explain in one or two sentences why two homotopic transverse maps yield cobordant preimages.
5. Compute the Stiefel-Whitney number $w_2[\mathbb{R}\mathrm{P}^2]$ and use it to show $[\mathbb{R}\mathrm{P}^2] \neq 0$ in $\mathfrak{N}_2$, so that $\mathbb{R}\mathrm{P}^2$ generates $\mathfrak{N}_2 \cong \mathbb{F}_2$. (You may use that the total Stiefel-Whitney class of $\mathbb{R}\mathrm{P}^2$ is $w(\mathbb{R}\mathrm{P}^2) = (1+\alpha)^3 = 1 + \alpha + \alpha^2$ in $H^*(\mathbb{R}\mathrm{P}^2; \mathbb{F}_2) = \mathbb{F}_2[\alpha]/(\alpha^3)$, and that a Stiefel-Whitney number is the evaluation of a top-degree monomial on the fundamental class.)
6. Contrast the failure of the dimension axiom in K-theory and in cobordism. State the coefficient ring of each theory ($K^*(\mathrm{pt})$ and $MO^*(\mathrm{pt}) = \pi_*(MO)$), identify in which degrees each is nonzero, and explain why neither equals the ordinary-cohomology coefficient $H^*(\mathrm{pt}; \mathbb{Z}) = \mathbb{Z}$ concentrated in degree 0.

8.6 The Atiyah-Hirzebruch Spectral Sequence

The Serre spectral sequence of section 6.2 computes ordinary homology of a total space from ordinary homology of base and fiber. The chapter just completed has shown that ordinary cohomology is one member of a larger family: the generalized cohomology theories of definition 8.2.16, each represented by a spectrum (definition 8.4.1, theorem 8.4.6), each satisfying every Eilenberg-Steenrod axiom (definition 8.2.4) but the dimension axiom. K-theory and cobordism, the two theories that opened this book's list of "ways of assigning algebraic information", are the prototypes. The obvious question is whether the computational engine that makes ordinary cohomology tractable extends to these theories. It does, and the extension is almost mechanical: the skeletal filtration of a CW complex, which fed the cellular chain complex into ordinary cohomology, feeds any generalized theory E^* into the exact-couple machine of section 6.1 and produces a spectral sequence converging to $E^*(X)$. Its second page is ordinary cohomology of X with coefficients in the coefficient ring $E^*(\mathrm{pt})$ of the theory. This is the Atiyah-Hirzebruch spectral sequence, and it is the bridge that lets every computation of ordinary cohomology in the preceding chapters be leveraged into a computation in K-theory or cobordism.

The construction parallels the Serre spectral sequence so closely that most of the work has already been done. Where the Serre construction filtered the total space of a fibration by preimages of base skeleta, the present construction filters the space X by its own skeleta, and where the fiber homology $H_*(F)$ gave the coefficient grading, the coefficient groups $E^q(\mathrm{pt})$ of the generalized theory take its place. The single new ingredient is that E^* is not the ordinary cohomology that the exact-couple theorem of theorem 6.1.10 was stated for; the theorem must be re-run with E^* in place of H^*, which is legitimate because the only properties of H^* used in the construction are the Eilenberg-Steenrod axioms, and a generalized theory satisfies all of them save dimension.

Throughout, E^* is a generalized (reduced or unreduced) cohomology theory in the sense of definition 8.2.16, with coefficient groups

$$E^q(\mathrm{pt}), \qquad q \in \mathbb{Z}$$

the values of the unreduced theory on a point. These groups, taken together, form the graded *coefficient ring* $E^*(\mathrm{pt}) = \bigoplus_q E^q(\mathrm{pt})$ when E^* is multiplicative; for the construction of the spectral sequence only the additive structure is needed. Unlike ordinary cohomology, where $H^q(\mathrm{pt}) = 0$ for $q \neq 0$ by the dimension axiom, the coefficient groups of a generalized theory may be nonzero in many degrees, including negative ones, and it is exactly this spread that makes the spectral sequence carry information.

The construction begins with the filtration of the cochains of X by skeleta, exactly as in the Serre case but now applied to the generalized theory. Recall that $X^{(p)}$ denotes the p-skeleton of the CW complex X, with $X^{(p)} = \varnothing$ for $p < 0$.

Definition 8.6.1: The Skeletal Exact Couple of a Generalized Theory

Let E^* be a generalized cohomology theory and X a CW complex with skeleta $X^{(p)}$. The *skeletal exact couple* of E^* on X is the exact couple (definition 6.1.2) assembled from the long exact sequences of the pairs $(X^{(p)}, X^{(p-1)})$ in the theory E^*. Its groups are

$$D_1^{p,q} := E^{p+q}\big(X^{(p)}\big), \qquad E_1^{p,q} := E^{p+q}\big(X^{(p)},\, X^{(p-1)}\big)$$

with structure maps $i\colon D_1^{p,q} \to D_1^{p-1,q+1}$ induced by the inclusion $X^{(p-1)} \hookrightarrow X^{(p)}$, $j\colon D_1^{p,q} \to E_1^{p,q}$ the restriction to the relative group, and $k\colon E_1^{p,q} \to D_1^{p+1,q}$ the connecting homomorphism δ of the pair. The associated spectral sequence is the one generated by this exact couple in the sense of definition 6.1.6.

The exact couple records, in a single bigraded triangle, the long exact sequence of every skeletal pair at once. The auxiliary group $D_1^{p,q}$ collects the value of the theory on the p-skeleton, in total degree $p+q$; the visible group $E_1^{p,q}$ collects the value on the skeletal layer $(X^{(p)}, X^{(p-1)})$; the maps i, j, k are the inclusion, restriction, and connecting maps of the axioms. That this is genuinely an exact couple is immediate, since exactness of the triangle at its three corners is precisely the exactness of the long exact sequence of the pair $(X^{(p)}, X^{(p-1)})$ in E^*, which holds by the exactness axiom for a generalized theory. The work, exactly as for the Serre spectral sequence, is identifying the first two pages.

Example 8.6.2: The Skeletal Exact Couple for K^* on S^2

Take $X = S^2$ with its minimal CW structure: one 0-cell and one 2-cell, so $X^{(0)} = X^{(1)} = \mathrm{pt}$ and $X^{(2)} = S^2$. Let $E^* = K^*$ be complex topological K-theory (definition 8.5.1), whose coefficient groups are $K^q(\mathrm{pt}) = \mathbb{Z}$ for q even and 0 for q odd, by Bott periodicity (proved in [21, Bott Periodicity]). The exact couple of definition 8.6.1 has only two occupied filtration levels, $p = 0$ and $p = 2$. At level 0 the layer is $(X^{(0)}, X^{(-1)}) = (\mathrm{pt}, \varnothing)$, so

$$E_1^{0,q} = K^q(\mathrm{pt}) = \begin{cases} \mathbb{Z} & q \text{ even} \\ 0 & q \text{ odd} \end{cases}$$

The level $p = 1$ is empty, $E_1^{1,q} = K^{1+q}(X^{(1)}, X^{(0)}) = K^{1+q}(\mathrm{pt}, \mathrm{pt}) = 0$, there being no 1-cells. At level 2 the layer is (S^2, pt), whose reduced K-theory is $\widetilde{K}^{2+q}(S^2) = \widetilde{K}^{2+q}(\Sigma^2 S^0) \cong \widetilde{K}^q(S^0) = K^q(\mathrm{pt})$ by the double suspension isomorphism, so

$$E_1^{2,q} = \widetilde{K}^{2+q}(S^2) = \begin{cases} \mathbb{Z} & q \text{ even} \\ 0 & q \text{ odd} \end{cases}$$

The auxiliary group at level 2 is $D_1^{2,q} = K^{2+q}(S^2)$, the full K-theory of the sphere, which the spectral sequence is on its way to computing. The single first differential $d_1\colon E_1^{1,q} \to E_1^{2,q}$ has source 0 and the differential $E_1^{0,q} \to E_1^{1,q}$ has target 0, so $d_1 = 0$ identically: the cellular cochain complex of S^2 with coefficients in $K^*(\mathrm{pt})$ has no 1-cells to carry a coboundary, and the second page already equals the first.

The example is the skeletal exact couple stripped to two layers, where the absence of 1-cells forces $d_1 = 0$ and the page can be read off the two layers at once. In general the layers are not so sparse, and the first differential is the cellular coboundary of X. The next two results compute the E_1 and E_2 pages in general, and they reduce to a single homotopy-theoretic input already used for the Serre spectral sequence: the cofiber of a skeletal pair is a wedge of spheres, on which a generalized theory is computed by the suspension axiom.

Lemma 8.6.3: The First Page is Cellular Cochains with Coefficients in $E^*(\mathrm{pt})$

Let E^* be a generalized cohomology theory and X a CW complex with p-cells indexed by a set J_p. Then for the skeletal exact couple of definition 8.6.1 there is a natural isomorphism

$$E_1^{p,q} \cong \prod_{\alpha \in J_p} E^q(\mathrm{pt}) = C^p_{\mathrm{CW}}\big(X;\, E^q(\mathrm{pt})\big)$$

the cellular p-cochains of X with coefficients in the abelian group $E^q(\mathrm{pt})$, and under this identification the differential $d_1\colon E_1^{p,q} \to E_1^{p+1,q}$ is the cellular coboundary map δ^{CW} of X with coefficients in $E^q(\mathrm{pt})$.

***Proof*:**
Step 1: The relative group of a skeletal pair. The pair $(X^{(p)}, X^{(p-1)})$ is a good CW pair, and the quotient is a wedge of p-spheres, one for each p-cell:

$$X^{(p)}/X^{(p-1)} \cong \bigvee_{\alpha \in J_p} S^p_\alpha$$

one sphere S^p_α per characteristic map $\Phi_\alpha\colon (D^p, \partial D^p) \to (X^{(p)}, X^{(p-1)})$. By the relative-to-reduced identification for a good pair (an Eilenberg-Steenrod axiom for any generalized theory, definition 8.2.4), the relative group is the reduced group of the quotient:

$$E^{p+q}\big(X^{(p)}, X^{(p-1)}\big) \cong \widetilde{E}^{p+q}\Big(\bigvee_\alpha S^p_\alpha\Big)$$

Step 2: Reduced theory of a wedge of spheres. The additivity (wedge) axiom for a generalized theory turns the reduced group of a wedge into a product over the wedge summands,

$$\widetilde{E}^{p+q}\Big(\bigvee_\alpha S^p_\alpha\Big) \cong \prod_{\alpha \in J_p} \widetilde{E}^{p+q}(S^p)$$

and the suspension axiom, applied p times to $S^p = \Sigma^p S^0$, gives the suspension isomorphism

$$\widetilde{E}^{p+q}\big(S^p\big) = \widetilde{E}^{p+q}\big(\Sigma^p S^0\big) \cong \widetilde{E}^q\big(S^0\big) = E^q(\mathrm{pt})$$

the last equality because $\widetilde{E}^q(S^0) = E^q(\mathrm{pt})$ for the reduced theory on the two-point space S^0. Combining,

$$E_1^{p,q} = E^{p+q}\big(X^{(p)}, X^{(p-1)}\big) \cong \prod_{\alpha \in J_p} E^q(\mathrm{pt}) = C^p_{\mathrm{CW}}\big(X; E^q(\mathrm{pt})\big)$$

the cellular p-cochains with coefficients in $E^q(\mathrm{pt})$, free (as a product) on the p-cells. The identification is natural in X because each step (the quotient, the additivity isomorphism, the suspension isomorphism) is natural for cellular maps.

Step 3: The differential d_1 is the cellular coboundary. The differential $d_1 = j \circ k$ is the composite

$$E^{p+q}\big(X^{(p)}, X^{(p-1)}\big) \xrightarrow{\delta} E^{p+q+1}\big(X^{(p+1)}, X^{(p)}\big)$$

formed from the connecting map $k = \delta$ of the pair $(X^{(p+1)}, X^{(p)})$ followed by the restriction j through $X^{(p)}$, that is, the composite associated to the triple $(X^{(p+1)}, X^{(p)}, X^{(p-1)})$. Under the identification of Step 2 this is the connecting homomorphism of the cofiber sequence

$$\bigvee_{\beta \in J_{p+1}} S^p_\beta \longrightarrow X^{(p)}/X^{(p-1)} = \bigvee_{\alpha \in J_p} S^p_\alpha$$

induced by the attaching maps of the $(p+1)$-cells, read on $\widetilde{E}^*$. For ordinary cohomology this connecting map is, cell by cell, the degree of the composite $S^p_\beta \to X^{(p)}/X^{(p-1)} \to S^p_\alpha$ (the attaching map of e^{p+1}_β projected onto the cell e^p_α), which is the integer entry of the cellular coboundary matrix; this is the content of section 3.5 dualized to cochains. For a general theory E^*, the suspension isomorphism is natural and additive, so the connecting map acts on each coefficient copy of $E^q(\mathrm{pt})$ by multiplication by that same degree integer. Thus d_1 is the cellular coboundary δ^{CW} with the integer matrix acting on $E^q(\mathrm{pt})$-coefficients, completing the proof.

The lemma says the first page of the skeletal spectral sequence of any generalized theory is, in each row q, the cellular cochain complex of X with coefficients in the single abelian group $E^q(\mathrm{pt})$, and its differential is the ordinary cellular coboundary. The generalized theory contributes nothing new at the level of the first differential; it enters only through the coefficient groups $E^q(\mathrm{pt})$, which decorate the rows. The whole departure from ordinary cohomology is thus deferred to the higher differentials $d_2, d_3, \ldots$, which are where a generalized theory can differ from ordinary cohomology. Taking homology of each row computes the second page, and the second page is the named object of the theory.

Theorem 8.6.4: The Atiyah-Hirzebruch Spectral Sequence

Let E^* be a generalized cohomology theory and X a CW complex. Then there is a cohomological spectral sequence $\{E_r^{p,q}, d_r\}$, with differentials d_r of bidegree $(r, 1-r)$, natural in X, whose second page is ordinary cohomology of X with coefficients in the coefficient groups of E^*:

$$E_2^{p,q} \cong H^p\big(X;\, E^q(\mathrm{pt})\big)$$

If X is finite-dimensional (or, more generally, if for each n only finitely many of the groups $E_2^{p,n-p}$ are nonzero and the theory satisfies the limit axiom on X), the spectral sequence converges to the generalized cohomology of X:

$$E_2^{p,q} \implies E^{p+q}(X)$$

Convergence means that $E^n(X)$ carries a finite decreasing filtration $E^n(X) = F^0 \supseteq F^1 \supseteq \cdots \supseteq F^{N+1} = 0$, where $F^p = \ker\big(E^n(X) \to E^n(X^{(p-1)})\big)$ and $N = \dim X$, whose successive quotients are the stable terms $F^p/F^{p+1} \cong E_\infty^{p,n-p}$.

***Proof*:**
Step 1: The spectral sequence and its first page. The skeletal exact couple of definition 8.6.1 is an exact couple, since exactness of its triangle at the three corners is the exactness of the long exact sequence of each pair $(X^{(p)}, X^{(p-1)})$ in the theory E^*, which holds by the exactness axiom of definition 8.2.4. By definition 6.1.6 it generates a spectral sequence $\{E_r^{p,q}, d_r\}$, whose first page and first differential are computed by lemma 8.6.3:

$$E_1^{p,q} \cong C^p_{\mathrm{CW}}\big(X; E^q(\mathrm{pt})\big), \qquad d_1 = \delta^{\mathrm{CW}}$$

The bidegree of d_r is $(r, 1-r)$, the cohomological convention of definition 6.1.6: a connecting map of the skeletal pairs raises the filtration degree p by 1 at the first stage and by r after $r-1$ derivations, while preserving the total degree $p+q$, so the second coordinate drops by $r-1$.

Step 2: The second page. The first differential is the cellular coboundary on each row q, by lemma 8.6.3. The row $E_1^{\bullet,q} = C^\bullet_{\mathrm{CW}}(X; E^q(\mathrm{pt}))$ is therefore the cellular cochain complex of X with coefficients in the abelian group $E^q(\mathrm{pt})$, and its cohomology is, by the agreement of cellular and singular cohomology (section 3.5, dualized),

$$E_2^{p,q} = H^p\big(C^\bullet_{\mathrm{CW}}(X; E^q(\mathrm{pt})), \delta^{\mathrm{CW}}\big) = H^p\big(X;\, E^q(\mathrm{pt})\big)$$

the ordinary cohomology of X with coefficients in $E^q(\mathrm{pt})$. This is the claimed second page.

Step 3: Concentration in columns $0 \le p \le \dim X$. For each p, the coefficient group $E_2^{p,q} = H^p(X; E^q(\mathrm{pt}))$ vanishes when $p < 0$ (no negative skeleta, so no negative cellular cochains) and, if X is finite-dimensional of dimension N, when $p > N$ (no cells above dimension N). The rows q, by contrast, are not bounded: $E^q(\mathrm{pt})$ may be nonzero for arbitrarily large positive and negative q. The spectral sequence is therefore not first-quadrant but occupies the vertical strip $0 \le p \le N$, a finite band of columns. This is the structural difference from the Serre spectral sequence and is the one place the dimension axiom would have mattered: dropping it lets the rows extend into negative degrees.

Step 4: Convergence for finite-dimensional X. Fix a total degree n and a spot (p,q) with $p+q=n$. A differential out of (p,q) is $d_r\colon E_r^{p,q} \to E_r^{p+r,q-r+1}$; its target lies in the strip only if $p+r \le N$, that is $r \le N-p$, so for $r > N-p$ the outgoing differential is zero. A differential into (p,q) is $d_r\colon E_r^{p-r,q+r-1} \to E_r^{p,q}$; its source lies in the strip only if $p-r \ge 0$, that is $r \le p$, so for $r > p$ the incoming differential is zero. Setting $R(p,q) := \max(p, N-p)+1$, both differentials touching (p,q) vanish for all $r \ge R(p,q)$, and as in proposition 6.1.9 the groups $E_r^{p,q}$ stabilize to a value $E_\infty^{p,q}$ after finitely many pages. The exact couple is that of a filtration of $E^*(X)$: the auxiliary groups $D_1^{p,q} = E^{p+q}(X^{(p)})$ together with the inverse system of restrictions $E^n(X) \to E^n(X^{(p)})$ define the decreasing filtration

$$F^pE^n(X) := \ker\big(E^n(X) \to E^n(X^{(p-1)})\big)$$

the classes restricting to zero on the $(p-1)$-skeleton. For finite-dimensional X the restriction $E^n(X) \to E^n(X^{(N)}) = E^n(X)$ is the identity (as $X^{(N)} = X$), so $F^{N+1}E^n(X) = \ker(\mathrm{id}) = 0$ and $F^0E^n(X) = \ker(E^n(X) \to E^n(\varnothing)) = E^n(X)$, giving a finite filtration of length $N+1$. The general theory of the spectral sequence of an exact couple (theorem 6.1.10, transcribed from chains to the theory E^*, the transcription being valid because only the Eilenberg-Steenrod axioms were used) identifies the limit page with the associated graded of this filtration:

$$E_\infty^{p,q} \cong \frac{F^pE^{p+q}(X)}{F^{p+1}E^{p+q}(X)}$$

For infinite-dimensional X the filtration is no longer finite, and convergence becomes the statement that the restriction $E^n(X) \to \varprojlim_p E^n(X^{(p)})$ is an isomorphism. That is exactly what theorem 8.2.11 governs: it identifies the kernel and cokernel of that restriction as $\varprojlim_p^1 E^{n-1}(X^{(p)})$ and 0, so the restriction is an isomorphism precisely when the $\varprojlim^1$ term vanishes, which by corollary 8.2.12 holds whenever the tower $\big(E^{n-1}(X^{(p)})\big)_p$ satisfies the Mittag-Leffler condition, in particular whenever the restrictions between skeleta are surjective. Under the finite-dimensional hypothesis or this limit hypothesis, the convergence $E_2^{p,q} \Rightarrow E^{p+q}(X)$ asserted holds, completing the proof.

The theorem is the promised bridge. Its second page is built entirely from ordinary cohomology of X, every group of which the preceding chapters can compute, and its abutment is the generalized cohomology $E^*(X)$, the object of interest. A generalized theory is thereby computed in two stages: first assemble the ordinary cohomology with the coefficient groups $E^q(\mathrm{pt})$ of the theory laid out in rows, then run the differentials $d_2, d_3, \ldots$ and solve the resulting extension problems. When all differentials vanish, the theory $E^*(X)$ is the associated graded of $H^*(X; E^*(\mathrm{pt}))$, so it differs from a direct sum of ordinary cohomology groups only by extensions; the differentials measure the genuine departure of E^* from ordinary cohomology, and they are concentrated in the higher pages because the first differential is always the ordinary cellular coboundary. The edge of the spectral sequence records the relationship between the generalized theory and ordinary cohomology precisely.

Proposition 8.6.5: The Edge Homomorphisms of the Atiyah-Hirzebruch Spectral Sequence

Let E^* and X be as in theorem 8.6.4, with X finite-dimensional and path-connected.

1. *(Coefficient edge.)* On the left column $p = 0$, no differential enters $E_r^{0,q}$, so the surviving term $E_\infty^{0,q}$ is a *subgroup* of $E_2^{0,q} = H^0(X; E^q(\mathrm{pt})) = E^q(\mathrm{pt})$, cut out as the intersection of the kernels of the outgoing differentials, and the composite
$$E^q(X) \twoheadrightarrow F^0E^q(X)/F^1 = E_\infty^{0,q} \hookrightarrow E_2^{0,q} = E^q(\mathrm{pt})$$
is the restriction $E^q(X) \to E^q(\mathrm{pt})$ to a 0-cell.

2. *(Leading-term edge.)* On the bottom row $q = 0$, the surviving term $E_\infty^{n,0}$ is the subquotient $F^nE^n(X)/F^{n+1}E^n(X)$ of $E_2^{n,0} = H^n(X; E^0(\mathrm{pt}))$. If E^* is connective ($E^q(\mathrm{pt}) = 0$ for $q < 0$), then no differential leaves the bottom row, so $E_\infty^{n,0}$ is a *quotient* of $H^n(X; E^0(\mathrm{pt}))$, and there is a natural edge surjection
$$H^n\big(X; E^0(\mathrm{pt})\big) = E_2^{n,0} \twoheadrightarrow E_\infty^{n,0} = F^nE^n(X)/F^{n+1}E^n(X)$$

***Proof*:**
Step 1: The left column receives nothing. On the left column $p = 0$, an incoming differential is $d_r\colon E_r^{-r,q+r-1} \to E_r^{0,q}$, whose source sits in column $-r < 0$; there are no negative skeleta, so $E_2^{-r,\cdot} = 0$ and no differential enters $(0, q)$. The outgoing differential $d_r\colon E_r^{0,q} \to E_r^{r,q-r+1}$ has target in column $r \geq 2 > 0$, possibly nonzero. Hence each page satisfies $E_{r+1}^{0,q} = \ker(d_r|_{E_r^{0,q}})$, a descending chain of subgroups, and $E_\infty^{0,q} = \bigcap_{r\geq 2} \ker(d_r)$ is a subgroup of $E_2^{0,q}$. For path-connected X, $H^0(X; E^q(\mathrm{pt})) = E^q(\mathrm{pt})$. By the convergence of theorem 8.6.4 the bottom

filtration step is $F^0E^q(X) = E^q(X)$ and $F^0/F^1 = E_\infty^{0,q}$; since $F^1E^q(X) = \ker(E^q(X) \to E^q(X^{(0)})) = \ker(E^q(X) \to E^q(\mathrm{pt}))$, the quotient $E^q(X)/F^1$ is exactly the image of the restriction $E^q(X) \to E^q(\mathrm{pt})$. Thus the composite $E^q(X) \twoheadrightarrow E_\infty^{0,q} \hookrightarrow E^q(\mathrm{pt})$ is the restriction to a 0-cell, establishing (1).

Step 2: The bottom row as a subquotient. On the bottom row $q = 0$, an outgoing differential $d_r \colon E_r^{n,0} \to E_r^{n+r,1-r}$ lands in row $1-r \leq -1 < 0$, and an incoming differential $d_r \colon E_r^{n-r,r-1} \to E_r^{n,0}$ comes from row $r-1 \geq 1 > 0$. For a general theory both the negative rows (carrying $E^{1-r}(\mathrm{pt})$) and the positive rows are nonzero, so the bottom row both emits and receives, and $E_\infty^{n,0}$ is a genuine subquotient of $E_2^{n,0} = H^n(X; E^0(\mathrm{pt}))$, identified by convergence with $F^nE^n(X)/F^{n+1}E^n(X)$. This is the contrast with the Serre spectral sequence, whose bottom row carried permanent cycles: there the dimension axiom forced the rows below $q = 0$ to vanish, and here those rows are exactly the negative coefficient groups that the dimension axiom would have killed.

Step 3: The connective case. If $E^q(\mathrm{pt}) = 0$ for $q < 0$, then the target row $1 - r < 0$ of every outgoing differential is zero, so the bottom row emits nothing and only receives. Then $E_{r+1}^{n,0} = E_r^{n,0}/\mathrm{im}(d_r)$ is a quotient at every stage, and $E_\infty^{n,0} = E_2^{n,0}/\sum_{r\geq 2}\mathrm{im}(d_r)$ is a quotient of $H^n(X; E^0(\mathrm{pt}))$. By convergence $E_\infty^{n,0} = F^nE^n(X)/F^{n+1}E^n(X)$, giving the natural edge surjection $H^n(X; E^0(\mathrm{pt})) \twoheadrightarrow F^nE^n(X)/F^{n+1}E^n(X)$, establishing (2) and completing the proof.

The coefficient edge is the device that connects the generalized theory back to its coefficient ring: every class of $E^q(X)$ restricts to a coefficient in $E^q(\mathrm{pt})$, and the spectral sequence records which coefficients are hit, namely the subgroup $E_\infty^{0,q}$ surviving all differentials out of the left column. The leading-term edge, available for connective theories, relates ordinary cohomology to the generalized theory in the bottom degree: a class in $H^n(X; E^0(\mathrm{pt}))$ either survives to contribute the top filtration quotient of $E^n(X)$ or is killed by a differential. The contrast with the Serre spectral sequence, whose bottom row consisted of permanent cycles, is the structural fingerprint of dropping the dimension axiom: the coefficient groups now populate negative rows, and those negative rows can absorb differentials out of the bottom row, so for a non-connective theory the bottom row is a subquotient rather than a subgroup of ordinary cohomology. With the machine assembled, the prototypical computation can be carried out in full.

The cleanest test is complex K-theory on complex projective space, where the answer is known independently from Bott periodicity (computed in [21, K-Theory of Complex Projective Space]) and the spectral sequence must reproduce it. The computation exhibits the central phenomenon of the Atiyah-Hirzebruch spectral sequence: a parity obstruction forces every differential to vanish, so the spectral sequence collapses, and the generalized theory is read off as the associated graded of ordinary cohomology with the Bott-periodic coefficients.

Example 8.6.6: $K^*(\mathbb{C}\mathrm{P}^n)$ via the Atiyah-Hirzebruch Spectral Sequence

Let $E^* = K^*$ be complex topological K-theory, with coefficient groups

$$K^q(\mathrm{pt}) = \begin{cases} \mathbb{Z} & q \text{ even} \\ 0 & q \text{ odd} \end{cases}$$

by Bott periodicity (the coefficient ring is $\mathbb{Z}[\beta, \beta^{-1}]$ with the Bott class $\beta \in K^{-2}(\mathrm{pt})$ a unit; see theorem 8.5.3). Take $X = \mathbb{C}\mathrm{P}^n$, whose ordinary cohomology ring is $H^*(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) = \mathbb{Z}[x]/(x^{n+1})$ with $|x| = 2$, the truncation of the polynomial ring $H^*(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = \mathbb{Z}[x]$ of example 6.4.9; additively $H^p(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) = \mathbb{Z}$ for p even with $0 \leq p \leq 2n$ (generated by $x^{p/2}$) and 0 otherwise.

Step 1: The second page. By theorem 8.6.4,

$$E_2^{p,q} = H^p\big(\mathbb{C}\mathrm{P}^n;\, K^q(\mathrm{pt})\big)$$

This is $\mathbb{Z}$ exactly when both p is even with $0 \le p \le 2n$ (so that $H^p(\mathbb{C}\mathrm{P}^n) = \mathbb{Z}$) and q is even (so that $K^q(\mathrm{pt}) = \mathbb{Z}$), and 0 otherwise:

$$E_2^{p,q} = \begin{cases} \mathbb{Z} & p \text{ even},\ 0 \le p \le 2n,\ q \text{ even} \\ 0 & \text{otherwise} \end{cases}$$

Every nonzero entry of E_2 sits at a spot (p, q) with both coordinates even.

Step 2: The first differential d_3 and why it vanishes. The lowest differential that could be nonzero on the K-theory page is d_3, of bidegree $(3, -2)$ (the differential d_2 has bidegree $(2, -1)$, and any source (p, q) with p, q both even maps under d_2 to $(p+2, q-1)$, where $q-1$ is odd, hence into a zero group; so $d_2 = 0$ for parity reasons). Consider $d_3\colon E_3^{p,q} \to E_3^{p+3,q-2}$. A nonzero source requires p even, but then the target column $p + 3$ is odd, so $E_2^{p+3,q-2} = H^{p+3}(\mathbb{C}\mathrm{P}^n; \cdot) = 0$ (odd-degree cohomology of $\mathbb{C}\mathrm{P}^n$ vanishes). Hence $d_3 = 0$. The same parity count rules out every higher differential: d_r has bidegree $(r, 1 - r)$, sending an even-even source (p, q) to $(p + r, q + 1 - r)$; for the target to be nonzero one needs $p + r$ even (forcing r even, since p is even) *and* $q + 1 - r$ even (forcing $1 - r$ even, that is r odd). No integer r is both even and odd, so for every $r \ge 2$ either the target column is odd or the target row is odd, and $E_2^{p+r,q+1-r} = 0$. Therefore

$$d_r = 0 \quad \text{for all } r \ge 2, \qquad E_2 = E_\infty$$

the spectral sequence collapses at the second page.

Step 3: Reading off $K^*(\mathbb{C}\mathrm{P}^n)$. With $E_\infty = E_2$, the group $K^m(\mathbb{C}\mathrm{P}^n)$ is, up to the extension problem, the direct sum along the diagonal $p + q = m$:

$$K^m(\mathbb{C}\mathrm{P}^n) \cong \bigoplus_{p+q=m} E_\infty^{p,q} = \bigoplus_{\substack{p+q=m \\ p,q \text{ even},\ 0\le p\le 2n}} \mathbb{Z}$$

For $m = 0$ even: the spots are $(p, -p)$ with p even and $0 \le p \le 2n$, namely $p \in \{0, 2, 4, \dots, 2n\}$, a total of $n + 1$ spots, each contributing $\mathbb{Z}$. (Here the row index $q = -p$ runs through the negative even integers, populated precisely because Bott periodicity makes $K^{-p}(\mathrm{pt}) = \mathbb{Z}$; this is where dropping the dimension axiom is felt.) Every group in the filtration is free abelian, so each successive extension $0 \to F^{p+1} \to F^p \to \mathbb{Z} \to 0$ splits, and

$$K^0(\mathbb{C}\mathrm{P}^n) \cong \mathbb{Z}^{n+1}$$

For $m = 1$ odd: the spots $(p, 1 - p)$ on the diagonal need p even and $1 - p$ even, impossible, so every term vanishes and

$$K^1(\mathbb{C}\mathrm{P}^n) = 0$$

These match the values computed in [21, K-Theory of Complex Projective Space]: $K^0(\mathbb{C}\mathrm{P}^n) \cong \mathbb{Z}[t]/(t^{n+1})$, free of rank $n+1$ on $1, t, t^2, \dots, t^n$ for $t = [H] - 1$ the reduced class of the *tautological* line bundle, and $K^1(\mathbb{C}\mathrm{P}^n) = 0$. The spectral sequence recovers the additive structure, and only that. The ring is not read off from it here, and the omission is not an economy: reading products off the abutment would need the filtration to be multiplicative in the sense of definition 6.4.1, the induced product on E_∞ to be identified with the cup product on E_2, and an edge identification carrying $[L]-1$ to $c_1(L)$. Theorem 8.6.4 supplies none of the three, and the leading-term edge of proposition 8.6.5 is gated on connectivity, which K^* fails. Both multiplicative facts are instead proved directly in [21, The Tautological Relation on $\mathbb{C}\mathrm{P}^n$, The Reduced Diagonal on the Top Cell of $\mathbb{C}\mathrm{P}^n$], the relation $t^{n+1}=0$ from an exterior-power identity among bundles over $\mathbb{C}\mathrm{P}^n$ and the generation of the top filtration step from a comparison of the reduced diagonal with the cup product. Once that ring structure is known, the $n+1$ summands above correspond to $1, t, t^2, \ldots, t^n$ in the associated graded.

Consistency check on $S^2 = \mathbb{C}\mathrm{P}^1$. For $n=1$ the formula gives $K^0(\mathbb{C}\mathrm{P}^1) = \mathbb{Z}^2$ and $K^1(\mathbb{C}\mathrm{P}^1) = 0$, that is $K^0(S^2) = \mathbb{Z}\oplus\mathbb{Z}$ and $K^1(S^2)=0$, agreeing with the reduced computation $\widetilde{K}^0(S^2)=\mathbb{Z}$ of example 8.6.2 (the extra $\mathbb{Z}$ being $K^0(\mathrm{pt})$ from the unreduced theory). The collapse here is the same $d_3=0$ parity vanishing in its smallest instance.

The computation is the archetype of an Atiyah-Hirzebruch collapse. The coefficient ring of K-theory is concentrated in even degrees, and the ordinary cohomology of $\mathbb{C}\mathrm{P}^n$ is concentrated in even degrees, so every nonzero entry of the second page sits at an even-even spot; the differentials, which always shift the bidegree by $(r, 1-r)$ with r and $1-r$ of opposite parity, cannot connect two even-even spots, and the page collapses for parity alone. The same parity argument computes K^* of any space whose ordinary cohomology is concentrated in even degrees, which is why K-theory of flag manifolds, Grassmannians, and projective spaces is so readily accessible: the spectral sequence does no work beyond ordinary cohomology and Bott periodicity. The first nonzero differential of the K-theory spectral sequence appears only when the even-degree concentration fails, and the lowest one is d_3, which for general spaces equals a cohomology operation built from the Steenrod squares; that this d_3 is nonzero on suitable spaces is what makes K-theory a genuinely finer invariant than ordinary cohomology. That operation is not computed anywhere in this series, here or in [21]; the exercises below take its form as given.

The Atiyah-Hirzebruch spectral sequence returns to where this book began. The introduction listed homotopy, homology and cohomology, K-theory, and cobordism as four ways of assigning algebraic information to a space, each "creating its own theory". The closing chapter has shown that the last three of these are instances of one notion: a generalized cohomology theory (definition 8.2.16) is a functor satisfying the Eilenberg-Steenrod axioms (definition 8.2.4) except dimension, every such theory is representable by a spectrum (theorem 8.3.9, theorem 8.4.6), ordinary cohomology is the theory whose spectrum is the Eilenberg-MacLane spectrum (definition 8.4.2), and K-theory and cobordism are the theories whose spectra are the K-theory spectrum (theorem 8.5.3) and the Thom spectra (definition 8.5.5, theorem 8.5.7). The Atiyah-Hirzebruch spectral sequence is the computational expression of this unification: it computes any generalized theory E^* from ordinary cohomology with coefficients in $E^*(\mathrm{pt})$, so the ordinary cohomology developed in chapter 4 is not one theory among several but the universal first approximation to all of them. The four ways of assigning algebraic information have become one representable framework, with ordinary cohomology its E_2 page and the spectra of K-theory and cobordism its further reaches.

Exercise 8.6.1

1. Compute $K^*(S^n)$ for every $n \geq 1$ using the Atiyah-Hirzebruch spectral sequence. Write out the E_2 page (using $H^*(S^n;\mathbb{Z})$ and the Bott-periodic coefficients $K^q(\mathrm{pt})$), identify the only potentially nonzero differential, show it vanishes for parity reasons, and conclude $K^0(S^n)$ and $K^1(S^n)$. Distinguish the cases n even and n odd, and confirm the reduced values $\widetilde{K}^0(S^n)=\mathbb{Z}$ for n even and 0 for n odd.

2. Let $E^* = H^*(-;\mathbb{Z})$ be ordinary integral cohomology, regarded as a generalized theory. Show that its Atiyah-Hirzebruch spectral sequence collapses at E_2 with $E_2^{p,q} = 0$ for $q \neq 0$, and that the convergence statement of theorem 8.6.4 reduces to the identity $H^n(X;\mathbb{Z}) = H^n(X;\mathbb{Z})$. Which Eilenberg-Steenrod axiom is responsible for the concentration in the single row $q = 0$, and how does the present collapse compare with the skeletal-filtration collapse of example 6.1.11?

3. Consider the unreduced K-theory K^* of the torus $T^2 = S^1 \times S^1$. Using $H^*(T^2;\mathbb{Z}) = \mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}$ in degrees $0, 1, 2$ and the even coefficient ring $K^*(\mathrm{pt})$, write the E_2 page and determine all differentials. Compute $K^0(T^2)$ and $K^1(T^2)$ as abelian groups, and note that here the odd cohomology $H^1(T^2) = \mathbb{Z}^2$ does contribute, unlike the $\mathbb{CP}^n$ case.

4. The mod-2 cohomology of X supports the Atiyah-Hirzebruch spectral sequence for K-theory with $\mathbb{Z}/2$ coefficients only after reduction, but the integral d_3 of the K-theory Atiyah-Hirzebruch spectral sequence is known to equal the integral Bockstein composed with Sq^2 (an operation $\beta\,\mathrm{Sq}^2\colon H^p(X;\mathbb{Z}) \to H^{p+3}(X;\mathbb{Z})$). Without computing the operation, explain why on any space X with $H^*(X;\mathbb{Z})$ concentrated in even degrees this d_3 must vanish, and give two distinct such spaces (besides $\mathbb{CP}^n$) where the spectral sequence consequently collapses.

5. Let E^* be a generalized theory with $E^q(\mathrm{pt}) = 0$ for all $q < 0$ and $E^0(\mathrm{pt}) = \mathbb{Z}$ (a connective theory). Show that for any finite-dimensional CW complex X the bottom row $q = 0$ of the Atiyah-Hirzebruch E_2 page is $H^p(X;\mathbb{Z})$, that no differential *leaves* the bottom row, and hence that $E_\infty^{n,0}$ is a quotient of $H^n(X;\mathbb{Z})$. Conclude using proposition 8.6.5 that there is a natural edge surjection $H^n(X;\mathbb{Z}) \twoheadrightarrow F^nE^n(X)/F^{n+1}E^n(X)$, the top filtration quotient of $E^n(X)$. Interpret this as the statement that ordinary cohomology is the "bottom" of every connective generalized theory.

6. Explain precisely where the construction of the Atiyah-Hirzebruch spectral sequence in theorem 8.6.4 would break if the dimension axiom were imposed, and conversely identify every step that used a generalized (non-ordinary) feature of E^*. Use this to argue that the Atiyah-Hirzebruch spectral sequence is, structurally, the Serre spectral sequence's analogue with the fiber replaced by a point and ordinary cohomology replaced by E^*.

8.7 *Beyond the Axioms: Modern Directions

The chapter closed on a single identification: a cohomology theory is a spectrum (theorem 8.3.9, theorem 8.4.6), and the Atiyah-Hirzebruch spectral sequence (theorem 8.6.4) computes any theory from ordinary cohomology, which is thereby the universal first approximation to all of them. This endpoint is also a beginning. The identification of a theory with its representing spectrum remembers the homotopy type of that spectrum and little else; the structure that organizes modern stable homotopy theory (a multiplication, a hierarchy among theories, a base to work over, a universal origin, a full calculus of operations) is invisible to the bare axioms. Each direction below restores one of those layers, and in each the objects of this chapter reappear as a special case of something larger. The lesson the chapter has repeated, that ordinary cohomology is a special case of the generalized theories, applies once more to the generalized theories themselves.

The first missing layer is multiplication. The cup product made $H^\bullet(X;R)$ a graded ring (definition 4.2.5), and box 8.4.12 recorded that the stable homotopy category carries a symmetric monoidal smash product with the sphere spectrum $\mathbb{S}$ as its unit. Combining the two, the Eilenberg-

MacLane spectrum $H\mathbb{Z}$ (definition 8.4.2) is not only a spectrum but a *ring spectrum*: a multiplication $H\mathbb{Z} \wedge H\mathbb{Z} \to H\mathbb{Z}$ that induces the cup product, associative and commutative up to coherent homotopy. K-theory and the cobordism spectra MU, MO of section 8.5 carry the same structure, their products coming from tensor product of vector bundles and Cartesian product of manifolds. A spectrum whose multiplication is commutative up to coherent higher homotopy is an E_∞-ring spectrum, and the sphere $\mathbb{S}$ is the initial one, so every ring spectrum is an algebra over $\mathbb{S}$.

This is the entrance to "brave new algebra": commutative algebra carried out over the sphere spectrum in place of a commutative ring, with modules, tensor products, base change, and even Galois theory lifting from ordinary algebra to the stable category. Making the smash product strictly symmetric monoidal at the point-set level, rather than only up to homotopy, was the technical obstacle that held the theory back for decades; it was resolved in the 1990s by the model categories of Elmendorf, Kriz, Mandell, and May (*Rings, Modules, and Algebras in Stable Homotopy Theory*) and by the symmetric spectra of Hovey, Shipley, and Smith. The modern account recasts the whole framework in the language of ∞-categories, where the category of spectra is the stabilization of the category of spaces and a ring spectrum is an algebra object inside it; this is the subject of Lurie's *Higher Algebra*, and the setting for derived and spectral algebraic geometry, in which a ring spectrum plays the role a commutative ring plays in ordinary geometry.

The second layer is a hierarchy among theories. The complex cobordism spectrum MU of section 8.5 carries a universal property discovered by Quillen: its ring of coefficients $MU^\bullet(\mathrm{pt})$ is the universal ring bearing a *formal group law*, the one-variable power series $F(x, y)$ recording how the first Chern class of a tensor product of line bundles is expressed through the Chern classes of the factors. Every complex-orientable theory inherits such a law, and the two theories the reader knows sit at its simplest values: ordinary cohomology carries the additive law $F(x, y) = x + y$, and K-theory the multiplicative law $F(x, y) = x + y + \beta xy$ with β the Bott class. The geometry of formal groups, through the invariant called the *height*, then stratifies the stable homotopy category into layers (the chromatic filtration), each governed by a periodic theory, the Morava K-theories, interpolating between rational information at one extreme and mod-p ordinary cohomology at the other.

This chromatic picture is the organizing principle behind the stable homotopy groups of spheres, whose first nonzero value $\pi_1^s \cong \mathbb{Z}/2$ was computed in section 5.7: the irregular patterns in those groups resolve, layer by layer, into periodic families indexed by the heights. The foundational results are the nilpotence and periodicity theorems of Devinatz, Hopkins, and Smith, and the elliptic refinement has produced the spectrum of topological modular forms tmf, a theory as central to the subject now as K-theory was a generation ago. The standard references are Ravenel's two books, *Complex Cobordism and Stable Homotopy Groups of Spheres* and *Nilpotence and Periodicity in Stable Homotopy Theory*, named already in the closing chapter.

The third layer is a base to work over. Everything in this book takes place over topological spaces, but the axioms of definition 8.2.4 and the representability of theorem 8.3.9 make sense in any setting equipped with a workable notion of homotopy. Morel and Voevodsky gave one for algebraic geometry, replacing the unit interval by the affine line $\mathbb{A}^1$ and constructing, over any base scheme S, a motivic stable homotopy category $SH(S)$ of motivic spectra. In it algebraic K-theory and algebraic cobordism are representable exactly as their topological counterparts are here, and a distinguished ordinary theory, motivic cohomology, plays the role of $H\mathbb{Z}$ while computing the Chow groups of algebraic cycles.

The payoff was arithmetic. Voevodsky's construction of motivic cohomology and the operations acting on it resolved the Milnor conjecture relating Galois cohomology, quadratic forms, and mod-2 cohomology, work recognized with a Fields Medal. For the topologist the point is that the entire apparatus of this chapter (the axioms, the representing spectra, the Atiyah-Hirzebruch spectral sequence) has an algebro-geometric twin, and the two lie close enough that computations pass between them. The foundational reference is Morel and Voevodsky's $\mathbb{A}^1$-*homotopy theory of schemes*; the companion volume [3] develops the underlying scheme theory.

The fourth layer is a universal origin. A single algebraic variety carries several cohomology theories at once (Betti cohomology of its complex points, algebraic de Rham cohomology, ℓ-adic étale cohomology, crystalline cohomology in characteristic p), and they share so much behaviour that Grothendieck conjectured a single universal theory, the theory of *motives*, through which all of them factor. A motive is the cohomological essence of a variety stripped of the choice of which cohomology to compute; each such Weil cohomology is then a realization functor out of the category of motives, and the agreement among the individual theories is the trace of the one motivic object standing above them. This is the exact analogue, one level up, of the chapter's own lesson that ordinary cohomology is the universal first approximation recorded on the second page of the Atiyah-Hirzebruch spectral sequence: motives are the universal cohomology of a variety, and the Weil theories are its realizations.

The theory of pure motives, for smooth projective varieties, rests on still-open conjectures (Grothendieck's standard conjectures), but Voevodsky's triangulated category of mixed motives DM gives an unconditional derived-categorical home for the idea, and it is where the motivic cohomology of the previous paragraph actually lives. A readable entry point is André's *Une introduction aux motifs*.

The fifth and most encompassing layer is a full calculus of operations. The axioms of definition 8.2.4 equip a cohomology theory with pullback f^* along every map, but a manifold or a variety supports far more: an ordinary pushforward f_*, a proper (or compactly supported) pushforward $f_!$, an exceptional pullback $f^!$, the tensor product $\otimes$, and the internal hom $\underline{\mathrm{Hom}}$. These six operations are bound together by base-change and projection formulas and by a duality exchanging $f_!$ with f_* and $f^!$ with f^*. The right adjoint $f^!$ that surfaced in box 8.3.12 as the basis of Grothendieck duality is one of the six. In this language Poincaré duality (section 4.5) becomes the statement that, for a closed oriented n-manifold mapping to a point, $f^!$ applied to the coefficients is a degree shift by n, and the Atiyah and Spanier-Whitehead dualities of stable homotopy theory are the same duality read on the sphere spectrum. The whole formalism, a cohomology theory presented not as a graded functor but as a package of six operations, originates in Grothendieck's étale cohomology (SGA 4) and its Poincaré-Verdier duality, the arithmetic counterpart taken up in [3] and [8].

Making the six operations precise and functorial across all the relevant geometry at once, with the coherence that the ∞-categorical language requires, is an active research program: the formalization through categories of correspondences in Gaitsgory and Rozenblyum's *A Study in Derived Algebraic Geometry*, and the condensed and analytic mathematics of Clausen and Scholze, in which a six-functor formalism becomes the working definition of a cohomology theory and is extended to p-adic and rigid-analytic geometry (Scholze's lecture notes on six-functor formalisms and Mann's p-adic thesis are the current expositions). This is the direction in which the very notion the chapter set out to axiomatize, "a cohomology theory", has most fully outgrown the Eilenberg-Steenrod axioms: what began as four axioms on a graded functor is now a symmetric monoidal functor equipped with six operations and a duality, and the generalized cohomology theories of definition 8.2.16 are what it induces on topological spaces.

Read together, the five directions trace a single arc. The axioms of this chapter pin a cohomology theory to the homotopy type of a spectrum; multiplication makes that spectrum a ring, the height of a formal group sorts the rings into a tower, a base scheme carries the whole picture into algebraic geometry, motives give the universal cohomology that the individual theories realize, and the six operations restore the calculus of pushforward and duality that a graded functor cannot see. At every step the invariants built here by hand, singular cohomology, K-theory, cobordism, reappear as the projection of a structure that remembers more. That the plainest invariant of a space, its ordinary cohomology, should sit at the foot of a hierarchy this tall is the last and largest instance of the pattern the book has followed throughout: an invariant is understood not in isolation but through the structure that produces it.

A

Common Structures and Related Algebraic Structure

This appendix collects, for quick reference, common spaces together with their fundamental groups and (co)homology groups. Every entry can be derived by the methods of the preceding chapters; the tables are meant as a check on those computations and as a map of the small zoo of spaces that recur throughout the subject.

A.1 Δ-Complex and Cell Complexes

The torus T^2, the Klein bottle K, and the projective plane $\mathbb{R}\mathrm{P}^2$ all arise as quotients of a single square with its four edges identified in pairs; the three identification patterns, and the Δ-complex structures they induce, are shown below. Reading the boundary word of each square off its identifications gives the standard presentations $\langle a, b | aba^{-1}b^{-1} \rangle$, $\langle a, b | abab^{-1} \rangle$, and $\langle a | a^2 \rangle$ of chapter 2, and the induced cellular chain complexes compute the homology recorded in appendix A.3.

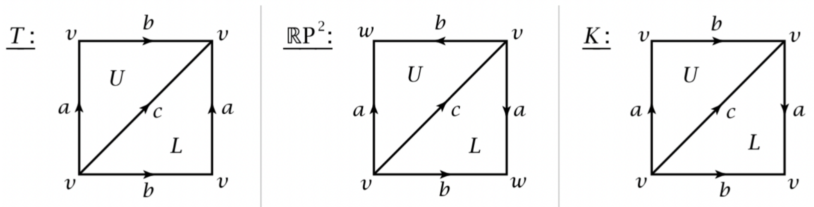


Figure A.1: Edge identifications and Δ-complex structures on the square presenting T^2, K, and $\mathbb{R}\mathrm{P}^2$.

A.2 Fundamental Groups and Covering Spaces

The fundamental groups of the spaces met most often are collected below; each may be computed by the van Kampen and covering-space methods of chapter 2. The abelianisation of each group recovers the first homology H_1 of the corresponding row in the table of appendix A.3.

Space	Fundamental group π_1
S^1	$\mathbb{Z}$
S^n, $n \geq 2$	0
$T^2 = S^1 \times S^1$	$\mathbb{Z}^2$
$T^n = (S^1)^n$	$\mathbb{Z}^n$
$\bigvee_{i=1}^k S^1$	free group F_k of rank k
$\bigvee_{i=1}^k S^n$, $n \geq 2$	0
M_g (orientable surface, genus g)	$\left\langle a_1, b_1, \ldots, a_g, b_g \;\middle\vert\; \prod_{i=1}^g [a_i, b_i] \right\rangle$
N^g (non-orientable, g cross-caps)	$\left\langle a_1, \ldots, a_g \;\middle\vert\; a_1^2 a_2^2 \cdots a_g^2 \right\rangle$
K (Klein bottle $= N_2$)	$\langle a, b \vert abab^{-1} \rangle$
$\mathbb{R}\mathrm{P}^1 = S^1$	$\mathbb{Z}$
$\mathbb{R}\mathrm{P}^n$, $n \geq 2$	$\mathbb{Z}/2\mathbb{Z}$
$\mathbb{C}\mathrm{P}^n$	0
$L(p;q)$ (lens space)	$\mathbb{Z}/p\mathbb{Z}$

A.3 Homologies and Cohomologies

Note that $H_0(X) \cong \tilde{H}_0(X) \oplus \mathbb{Z}$. Hence, $\tilde{H}_0(X)$ shall always be omitted, since we are only working with path-connected spaces. To calculate the cohomology groups, if $T_n \subseteq H_n$ is the torsion part:

$$H^n(C;\mathbb{Z}) \cong (H_n/T_n) \oplus T_{n-1}$$

i.e., the torsion part "moves up" while the torsion-free part stays put.

Shape	Homology	Cohomology with $\mathbb{Z}$
$S^1 \simeq M^1$[1]	$\tilde{H}_n(S^1) = \begin{cases} \mathbb{Z} & n = 1 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(S^1; G) = \begin{cases} G & n = 1 \\ 0 & \text{otherwise} \end{cases}$
S^k	$\tilde{H}_n(S^k) = \begin{cases} \mathbb{Z} & n = k \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(S^k; G) = \begin{cases} G & n = k \\ 0 & \text{otherwise} \end{cases}$
$S^1 \times S^1 = T^2$	$\tilde{H}_n(T^2) = \begin{cases} \mathbb{Z}^2 & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(T^2; G) = \begin{cases} \mathbb{Z}^2 & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & \text{otherwise} \end{cases}$
M_g	$\tilde{H}_n(M_g) = \begin{cases} \mathbb{Z}^{2g} & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(M_g; \mathbb{Z}) = \begin{cases} \mathbb{Z}^{2g} & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & \text{otherwise} \end{cases}$

[1] Möbius Band

K	$\tilde{H}_n(K) = \begin{cases} \mathbb{Z}\oplus\mathbb{Z}_2 & n=1 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(K;\mathbb{Z}) = \begin{cases} \mathbb{Z} & n=1 \\ \mathbb{Z}_2 & n=2 \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{R}\mathrm{P}^2$	$\tilde{H}_n(\mathbb{R}\mathrm{P}^2) = \begin{cases} \mathbb{Z}_2 & n=1 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) = \begin{cases} 0 & n=1 \\ \mathbb{Z}/2\mathbb{Z} & n=2 \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{R}\mathrm{P}^k$, k odd	$\tilde{H}_n(\mathbb{R}\mathrm{P}^k) = \begin{cases} \mathbb{Z}_2 & n \text{ odd},\ 1\le n\le k-2 \\ \mathbb{Z} & n=k \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(\mathbb{R}\mathrm{P}^k;\mathbb{Z}) = \begin{cases} \mathbb{Z}_2 & n \text{ even},\ 2\le n\le k-1 \\ \mathbb{Z} & n=k \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{R}\mathrm{P}^k$, k even	$\tilde{H}_n(\mathbb{R}\mathrm{P}^k) = \begin{cases} \mathbb{Z}_2 & n \text{ odd},\ 1\le n\le k-1 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(\mathbb{R}\mathrm{P}^k;\mathbb{Z}) = \begin{cases} \mathbb{Z}_2 & n \text{ even},\ 2\le n\le k \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{C}\mathrm{P}^2$	$\tilde{H}_n(\mathbb{C}\mathrm{P}^2) = \begin{cases} \mathbb{Z} & n=2 \\ \mathbb{Z} & n=4 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(\mathbb{C}\mathrm{P}^2;\mathbb{Z}) = \begin{cases} \mathbb{Z} & n=2 \\ \mathbb{Z} & n=4 \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{C}\mathrm{P}^k$	$\tilde{H}_n(\mathbb{C}\mathrm{P}^k) = \begin{cases} \mathbb{Z} & n \text{ even},\ 2\le n\le 2k \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(\mathbb{C}\mathrm{P}^k;\mathbb{Z}) = \begin{cases} \mathbb{Z} & n \text{ even},\ 2\le n\le 2k \\ 0 & \text{otherwise} \end{cases}$
N^g	$\tilde{H}_n(N^g) = \begin{cases} \mathbb{Z}^{g-1}\oplus\mathbb{Z}_2 & n=1 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(N^g;\mathbb{Z}) = \begin{cases} \mathbb{Z}^{g-1} & n=1 \\ \mathbb{Z}_2 & n=2 \\ 0 & \text{otherwise} \end{cases}$

A.3.1 Homology and Cohomology with $\mathbb{Z}/2$ Coefficients

Over the field $\mathbb{Z}/2$ the universal coefficient theorem identifies homology and cohomology degree by degree, $\tilde{H}_n(X;\mathbb{Z}/2)\cong\tilde{H}^n(X;\mathbb{Z}/2)$, so a single column records both; each entry is the dimension of that group as a $\mathbb{Z}/2$-vector space. Against the integral table, the $\mathbb{Z}/2$ coefficients see every cell and split all 2-torsion off into extra dimensions, which is why $\mathbb{R}\mathrm{P}^k$ acquires a generator in every degree up to k and the Betti numbers of a non-orientable surface grow.

Shape	$\tilde{H}_n(\,\cdot\,;\mathbb{Z}/2)\cong\tilde{H}^n(\,\cdot\,;\mathbb{Z}/2)$
S^k	$\mathbb{Z}/2$ for $n=k$
T^2	$(\mathbb{Z}/2)^2$ for $n=1$; $\mathbb{Z}/2$ for $n=2$
T^k	$(\mathbb{Z}/2)^{\binom{k}{n}}$ for $1\le n\le k$
K (Klein bottle)	$(\mathbb{Z}/2)^2$ for $n=1$; $\mathbb{Z}/2$ for $n=2$
$\mathbb{R}\mathrm{P}^k$	$\mathbb{Z}/2$ for $1\le n\le k$
$\mathbb{C}\mathrm{P}^k$	$\mathbb{Z}/2$ for n even, $2\le n\le 2k$
N^g (g cross-caps)	$(\mathbb{Z}/2)^g$ for $n=1$; $\mathbb{Z}/2$ for $n=2$

A.3.2 Cohomology Rings

Note that

$$\tilde{H}^\bullet\left(\bigvee_\alpha X_\alpha; R\right) \cong \prod_\alpha \tilde{H}^\bullet\left(X_\alpha; R\right)$$

Giving us some flexibility for finding cohomology rings:

Shape	Ring
S^k	$H^\bullet(S^k;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^2),\ \lvert\alpha\rvert = k$ $H^\bullet(S^k;\mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha]/(\alpha^2),\ \lvert\alpha\rvert = k$
T^2	$H^\bullet(T^2;R) \cong \Lambda_R[\alpha,\beta],\ \lvert\alpha\rvert = \lvert\beta\rvert = 1$
T^k	$H^\bullet(T^k;R) \cong \Lambda_R[\alpha_1,\alpha_2,\ldots,\alpha_k],\ \lvert\alpha_i\rvert = 1$
$\mathbb{R}\mathrm{P}^2$	$H^\bullet(\mathbb{R}\mathrm{P}^2;\mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha]/(\alpha^3)$ $H^\bullet(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(2\alpha,\alpha^2)$
$\mathbb{R}\mathrm{P}^k$	$H^\bullet(\mathbb{R}\mathrm{P}^k;\mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha]/(\alpha^{k+1})$
$\mathbb{R}\mathrm{P}^\infty$	$H^\bullet(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha],\ \lvert\alpha\rvert = 1$ $H^\bullet(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(2\alpha),\ \lvert\alpha\rvert = 2$
$\mathbb{R}\mathrm{P}^{2k}$	$H^\bullet(\mathbb{R}\mathrm{P}^{2k};\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(2\alpha,\alpha^{k+1}),\ \lvert\alpha\rvert = 2$
$\mathbb{R}\mathrm{P}^{2k+1}$	$H^\bullet(\mathbb{R}\mathrm{P}^{2k+1};\mathbb{Z}) \cong \mathbb{Z}[\alpha,\beta]/(2\alpha,\alpha^{k+1},\beta^2,\alpha\beta),$ $\lvert\alpha\rvert = 2,\ \lvert\beta\rvert = 2k+1$
$\mathbb{C}\mathrm{P}^2$	$H^\bullet(\mathbb{C}\mathrm{P}^2;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^3),\ \lvert\alpha\rvert = 2$
$\mathbb{C}\mathrm{P}^k$	$H^\bullet(\mathbb{C}\mathrm{P}^k;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^{k+1}),\ \lvert\alpha\rvert = 2$
$\mathbb{C}\mathrm{P}^\infty$	$H^\bullet(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}[\alpha],\ \lvert\alpha\rvert = 2$

B

What Next?

This book has carried the reader from the fundamental group and covering spaces through singular and cellular homology, cohomology rings and Poincaré duality, the higher homotopy groups, spectral sequences, characteristic classes, and a first encounter with stable homotopy theory through Brown representability and spectra. Each of these is the entrance to a subject far larger than any one chapter can hold. This closing chapter is a map of where to go next, pairing every direction with the companion EYNTKA volume where one exists and with the standard texts a working topologist keeps within reach.

Consolidating the foundations. The closest companions for a second pass over this material are Hatcher's *Algebraic Topology*, the source whose organisation this book most often follows; May's *A Concise Course in Algebraic Topology*, terser and more categorical; tom Dieck's *Algebraic Topology*; and, as a classical reference, Spanier's *Algebraic Topology.* Bredon's *Topology and Geometry* threads the subject together with manifold theory from the outset.

Homotopy theory and the stable category. chapter 5 and chapter 8 only opened the door to homotopy theory and spectra. A systematic treatment of the stable category is the natural sequel: Adams's *Stable Homotopy and Generalised Homology*, Switzer's *Algebraic Topology: Homotopy and Homology*, and the modern *Foundations of Stable Homotopy Theory* of Barnes and Roitzheim. The abstract homotopy theory beneath these (model categories, and past them ∞-categories) is the companion volume *Higher Category Theory* [9], and is built in Hovey's *Model Categories*, Riehl's *Categorical Homotopy Theory*, Goerss and Jardine's *Simplicial Homotopy Theory*, and Lurie's *Higher Topos Theory.* The computational frontier, the stable homotopy groups of spheres whose first nonzero value $\pi_1^s \cong \mathbb{Z}/2$ was identified in section 5.7, is the subject of Ravenel's *Complex Cobordism and Stable Homotopy Groups of Spheres* and *Nilpotence and Periodicity in Stable Homotopy Theory.*

Vector bundles, K-theory, and characteristic classes. chapter 7 computed the cohomology of BO and BU and built the Stiefel–Whitney, Chern, and Euler classes. The reference that develops them in full, together with the Pontryagin classes and the Thom isomorphism, is Milnor and Stasheff's *Characteristic Classes.* The K-theory those invariants feed is taken up in the companion volume

Vector Bundles and K-Theory ([21]), covering topological K^0 and K^1, Bott periodicity, and the splitting principle; its algebraic and operator-theoretic siblings are *Algebraic K-Theory* ([4]: K_0, K_1, K_2, and Quillen's higher K-theory) and *C*-Algebras and K-Theory* ([5]). Atiyah's *K-theory* remains the classic short account.

The differentiable and geometric side. Much of the homotopy-invariant machinery of this book has a smooth-manifold counterpart that is frequently more computable. de Rham cohomology, Morse theory, and the geometry of bundles appear in Bott and Tu's *Differential Forms in Algebraic Topology*, in Milnor's *Morse Theory* and *Topology from the Differentiable Viewpoint* ([16]), and in the companion *Differential Topology* ([20]).

Homological algebra and spectral sequences. The algebra behind chapter 3, chapter 4, and chapter 6, the derived functors of the universal coefficient and Künneth theorems and the machinery of the Serre spectral sequence, is developed for its own sake in the companion *Homological Algebra* ([10]) and in Weibel's *An Introduction to Homological Algebra*; spectral sequences as a practical craft are the subject of McCleary's *A User's Guide to Spectral Sequences.* The categorical language used throughout is laid out in *Category Theory* ([6]).

Cohomology in algebra and arithmetic. The cohomological methods of this book reappear across the series in new settings: sheaf cohomology and the cohomology of schemes in *Algebraic Geometry* ([3]), group and Galois cohomology with its arithmetic duality theorems in *Group and Galois Cohomology* ([8]), and Hodge theory on Kähler manifolds in the Hodge-theory volumes. Milne's *Étale Cohomology* is the standard bridge from the topological theory to its arithmetic counterpart.

A closing suggestion. The surest way to consolidate this material is to compute with it. Re-deriving, without looking, the groups and rings tabulated in appendix A, and pushing one genuinely nontrivial example through each major tool (a long exact sequence, a spectral sequence, a Gysin sequence, an obstruction class), fixes the machinery in a way no second reading can. From there, every direction above is open.

Index

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